DRAFT
On Foundations of Deep Learning

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Intelligence and its Fundamental Principles

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Prologue

Modern deep learning works astonishingly well, yet we still lack a fundamental understanding of *why*. What we call "deep learning" is not merely a triumph of computer science or engineering; it is a natural phenomenon that demands explanation in terms of underlying principles—much as physics explains why the Sun produces energy or why matter follows specific laws.

This collection of essays is an attempt to uncover some of those principles. It argues that today's AI systems succeed because they instantiate deep structural regularities of the world—regularities that any intelligent system, biological or artificial, must exploit. Two principles emerge repeatedly throughout these essays.

The first is *(sparse) compositionality*, and the second is *genericity*. In this essays, we show that they are not heuristic conjectures, but logical inevitabilities derived from the foundations of computation and physics. We show that sparse compositionality is mathematically implied by efficient computability, just as genericity is dictated by assuming invariance in the choice of coordinates. Together, they define the channel through which intelligence must pass.

Together – and helped by several other ideas and observations – these principles offer a first step toward a unified theory of modern AI—one that explains not only *how* current systems work, but *why* they work at all.

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How to Read This Collection

This volume is not a conventional monograph. It is a collection of essays—some conceptual, some technical, some speculative—written at different times but revolving around a shared set of themes. It has evolved into something close to an intellectual atlas: an ongoing effort to map the principles that underlie modern artificial intelligence and the physics of natural intelligence. Several chapters are proposals for research projects that may be completed before too long.

Two ideas appear repeatedly throughout the collection, forming the spine of the argument:

- **Sparse compositionality**: the view that complex functions, behaviors, and representations arise from the hierarchical composition of a small number of reusable parts. This principle explains approximation, hierarchy, modularity, transfer, interpretability, and the unreasonable effectiveness of deep networks.
- **Genericity**: the observation that the real functions we want to learn in the world are not arbitrary high-complexity objects. Learnable functions must show stability with respect to the training set and invariance to simple transformations of the coordinate systems used for their definition. This implies that meaningful structure leaves persistent low-degree footprints, enabling gradient-based optimization.

The collection is organized into four parts:

Part I: Principles and Foundations explores the mathematical inevitability of the two core ideas above[cite: 1]. It argues that intelligence is not a grab-bag of heuristics but a solution to specific physical constraints.

Part II: Computation and Algorithms instantiates these principles in specific architectures, moving from Associative Turing Machines to the efficiency of Boolean functions[cite: 1].

Part III: Learning and Evolution examines the dynamics of learning in deep networks and evolutionary strategies[cite: 1].

Part IV: Random Topics takes the most risks[cite: 1]. It treats biology not as a metaphor, but as a constraint. Here, we argue that mechanisms like "World Models" and "Dreaming" are not just biological quirks but computational necessities.

Suggested Paths

Readers interested primarily in the **conceptual picture** may prefer to begin with:

- The *Prologue*,
- Chapter 1: *Intelligence as Associative Memory*,

- Chapter 27: *World Models Before Language*[cite: 1].

Readers seeking **computational foundations** may start with:

- Chapter 3: *Associative Memories are Turing-Complete*[cite: 1],
- The essays on compositional optimization in Part I,
- The appendices, which now contain rigorous proofs regarding the Johnson-Lindenstrauss lemma and group invariance.

Readers curious about **broader implications** or the future of AI may enjoy:

- Chapter 29: *What Is Missing in LLMs?*[cite: 1],
- Chapter 39: *Interlude: Most Real Numbers Do Not Exist*[cite: 1],
- The concluding essay on principles and open questions.

As with any collection of essays, some ideas appear more than once, viewed from different angles. This repetition is intentional: deep principles re-emerge across different domains. The unity of the volume lies not in a single narrative arc but in the convergence of these diverse perspectives on the same underlying structure of intelligence. Finally, several chapters include the name of a human co-author with whom I discussed extensively the content of the chapter. Even so, the responsibility for mistakes and omissions in the written text is on me and Gemini!

Last but not least: this is, for the time being, just a draft, often a draft of proposed research projects. It contains errors; there are missing references and missing acknowledgments. With time and feedback from you they will be corrected! This is ready to be a "working notebook" for the field. It provides a counter-narrative to the "scale is all you need" crowd.

Note on Co-Authorship: An Experiment

This book is an experiment in a new kind of scientific collaboration. It was co-authored by a human scientist and Large Language Models (specifically, Gemini and ChatGPT)[cite: 1].

In the history of science, we often speak of "tools for thought"—notation, calculus, the computer. But until recently, those tools were passive. They did not push back. This collection represents a shift toward active *cognitive amplification*. Throughout the writing process, the AI did not merely act as a scribe or a ghostwriter; it functioned as a sparring partner, a synthesizer, and occasionally, a mirror. Gemini also drafted figures which are far from perfect but serve as good placeholders.

The Role of the Model

The core ideas—Sparse Compositionality, Genericity, the biologically constrained definition of intelligence—are mine. They stem from decades of research at MIT and the Center for Brains, Minds and Machines (CBMM). However, the *articulation* of these ideas was often a dialectic process.

I would feed the model a raw intuition or a rough technical note (e.g., "Language is just a subtitle track for a world model"). The model would expand it, connect it to adjacent concepts in the history of computation (Turing, Post, von Neumann), and return a draft. We would then iterate.

This process revealed an interesting property of current AI: it is a machine of *genericity*. Left to its own devices, the model tends to "smooth" ideas. It gravitates toward the consensus, the balanced view, the polite academic tone. It often tried to soften my more radical claims about biology—for instance, the insistence that diffusion models are biologically implausible, or that scaling laws alone will hit a wall.

I found myself constantly having to "roughen" the text back up—to re-inject the asymmetry, the specific bet, and the falsifiable conjecture. In this sense, the authorship process mirrored the very principles discussed in the book: the AI provided the *generic* compositional capability, while the human provided the *sparse*, high-frequency logical jumps that define scientific novelty.

Why This Matters

This collaboration is not just a novelty; it is a data point in the study of intelligence itself. If, as this book argues, intelligence relies on the composition of reusable modules, then LLMs have mastered the modules of *syntax* and *rhetorical structure*. They have learned the "shape" of a scientific argument. But they famously lack the "World Model"—the grounding in physical reality that allows a primate to understand gravity without words.

Writing this book with an LLM confirmed that hypothesis. The model could write fluently about "World Models," but it could not *verify* consistency without my guidance. It could hallucinate a citation as easily as it could summarize a theorem. It possessed language, but not truth.

The Future of Scientific Writing

I believe this model—Human + AI—will become the standard for scientific inquiry, at least in the near future. It allows a researcher to move faster, to synthesize broader fields of literature, and to maintain a unity of voice across disparate topics. But it requires a new kind of vigilance. We must guard against the "smoothing" effect of these models, which risks homogenizing scientific thought into a polite, generic mean.

This collection, therefore, is an artifact of 2026: a testament to what machines can do, and a reminder of what, for now, only biological brains can provide.

Tomaso Poggio
Lake Sabrina, Needham, Massachusetts
January 2026, updated July 2026

CHAPTER 1

Intelligence as Associative Memory

A useful way to understand modern artificial intelligence is through one of its oldest computational ideas: associative memory. Rather than retrieving information by an explicit numerical address, an associative memory retrieves it by similarity to a cue. Modern attention mechanisms implement a powerful learned version of this operation: queries provide the cues, keys provide content-dependent addresses, and values provide the retrieved contents [158].

This perspective connects classical kernel and radial-basis-function models to transformers. Under appropriate normalization, dot-product softmax attention is exactly a normalized Gaussian-kernel retrieval rule. Across many heads and layers, however, the significance of attention goes beyond retrieval. Retrieved information is repeatedly transformed and used to generate new queries, creating a sequence of content-addressable read–update operations.

The central claim of this chapter is therefore not that a transformer is merely a large database. It is that associative retrieval, when embedded inside an iterative computational architecture, can become a primitive for general computation. A later chapter makes this statement precise through the Associative Turing Machine.

This is a statement about mechanism and representational power. It does not by itself explain how useful memories and retrieval rules are learned, when retrieval from a partial cue is reliable, or why a trained predictor generalizes. The later chapters distinguish feasible representation, accessible learning signals, optimization, and statistical selection. Identifying the particular hidden computation that generated the data is a stronger objective than learning an accurate predictor.

1.1 The Associative-Memory Hypothesis

Long before deep networks, radial basis function (RBF) networks modeled learning through similarity to stored prototypes. A set of centers is placed in feature space, and the response associated with a center grows as the input approaches it [23]. In this sense, an RBF network implements a simple form of association by similarity.

The basic computational motif is extremely general. A partial pattern or *cue* is presented; stored patterns are compared with that cue; and information associated with the closest or most compatible patterns is returned. The memory is therefore *content addressable*: the address is not specified explicitly but inferred from the content of the query.

Transformers generalize this idea dramatically. In each attention head,

- the *query* represents the current cue;

Figure 1.1: Intelligence as an Associative Memory

- the *keys* act as content-dependent addresses or templates;
- the *values* contain the information associated with those addresses;
- the *attention weights* determine how strongly each stored value is retrieved.

Thus an attention head can be interpreted as a learned associative retrieval mechanism. The important qualification is that the exact mathematical relationship with classical RBFs depends on normalization, as we will make precise below.

Unlike classical RBF systems with fixed centers and metrics, the representations serving as queries, keys, and values in a transformer change across layers and depend on the current context. The memory is therefore not a static table of fixed prototypes. It is a dynamically constructed, learned content-addressable system.

1.2 Attention as Content-Addressable Retrieval

For a token representation x_i, an attention head forms a query q_i, while the context provides keys k_j and associated values v_j. The output is

$$\mathrm{Attn}(q_i, K, V) = \sum_j \alpha_{ij} v_j,$$

with

$$\alpha_{ij} = \frac{\exp\left(q_i^\top k_j/\sqrt{d}\right)}{\sum_\ell \exp\left(q_i^\top k_\ell/\sqrt{d}\right)}.$$

The operation has the characteristic structure of associative memory: the query determines which stored items are relevant, and the associated contents are returned as a similarity-weighted combination.

This point of view emphasizes a distinction that will be important throughout the book. Attention is not primarily interesting because it stores facts in a literal database. Its computational importance is that it provides a differentiable mechanism for *addressing information by content*.

A single retrieval operation is already useful, but the real power of a transformer comes from repeatedly combining retrieval with state transformation.

Operational schema. A transformer block participates in an iterative read–update process:

1. *Content-addressable read:* form a query from the current representation and retrieve a weighted combination of values from the available context.
2. *State update:* transform the retrieved information together with the current representation through residual, feed-forward, and nonlinear operations.
3. *Reuse:* subsequent layers form new queries from the updated representation and perform further associative reads.

Complex computations can emerge from the repeated composition of these simple operations.

It is useful to call the second operation a *state update* rather than literally a memory write. In an ordinary transformer, previously cached representations are not rewritten in the way cells of conventional RAM are overwritten. In the abstract Associative Turing Machine developed later, however, the corresponding update operation plays the computational role of writing and modifying state.

A note on scope. This chapter concerns primarily what an associative-memory architecture can represent and compute *in principle*. Whether relevant stored content can be retrieved efficiently and robustly from a partial cue is a separate inverse problem, relative to the memory, the cue, and the retrieval rule.

Likewise, representational power does not imply that a specified training procedure will discover a useful predictor from data.

It is useful to keep three questions distinct:

$$\boxed{\text{expressibility} \;\neq\; \text{learnability} \;\neq\; \text{reliable retrieval.}}$$

The first concerns what the architecture can represent. The second concerns whether a feasible learning procedure finds a predictor that is accurate beyond the observed examples. The third concerns whether the retrieval mechanism returns the desired content from the admitted cues.

Here *retrieval* should not be confused with *structural recovery*: identifying the computation or intermediate variables that generated the data, up to specified equivalences. Predictive learning need not accomplish that stronger identification. The Interlude and Chapter H make this distinction explicit.

1.3 An Exact Kernel Equivalence for Normalized Attention

The connection between attention and classical associative kernels can be made exact under a simple normalization condition.

A Gaussian RBF centered at μ_i has the form

$$\phi_i(x) = \exp\!\left(-\frac{1}{2}(x-\mu_i)^\top \Sigma_i^{-1}(x-\mu_i)\right),$$

with isotropic RBFs corresponding to $\Sigma_i = \sigma^2 I$. More general anisotropic versions were studied, among other places, in the HyperBF framework of Poggio and Girosi [128].

Now suppose that a query q and all keys k_i have fixed norm,

$$\|q\| = \|k_i\| = 1.$$

Then

$$\|q - k_i\|^2 = \|q\|^2 + \|k_i\|^2 - 2q^\top k_i = 2 - 2q^\top k_i,$$

and therefore

$$q^\top k_i = 1 - \frac{1}{2}\|q - k_i\|^2.$$

The additive constant disappears under softmax normalization. Hence

$$\frac{\exp\left(q^\top k_i/\tau\right)}{\sum_j \exp(q^\top k_j/\tau)} = \frac{\exp(-\|q - k_i\|^2/(2\tau))}{\sum_j \exp(-\|q - k_j\|^2/(2\tau))}.$$

Thus the attention output becomes

$$\mathrm{Attn}(q, k_i, v_i) = \sum_i \frac{\exp(-\|q - k_i\|^2/(2\sigma^2))}{\sum_j \exp(-\|q - k_j\|^2/(2\sigma^2))} v_i, \tag{1.1}$$

where σ^2 absorbs the temperature or attention scaling.

This is precisely the normalized kernel-regression form

$$\phi(x) = \sum_{i=1}^{N} \frac{c_i K(\|x - t_i\|_W^2)}{\sum_j K(\|x - t_j\|_W^2)}. \tag{1.2}$$

Consequently,

normalized dot-product attention ≡ normalized Gaussian-kernel retrieval

under the fixed-norm assumption. This relationship has also been emphasized in recent work on homogeneous attention and HyperBF models [127].

The qualification is important. In ordinary transformers, key norms may vary and can themselves carry information. In that regime,

$$q^\top k_i = -\frac{1}{2}\|q - k_i\|^2 + \frac{1}{2}\|q\|^2 + \frac{1}{2}\|k_i\|^2,$$

and the last term depends on i. It therefore cannot in general be discarded inside the softmax.

Thus there are two distinct statements:

fixed-norm attention: exact normalized RBF equivalence, general attention: broader associative-kernel interpretation.

The first is a mathematical identity. The second is a structural interpretation.

1.4 From Associative Retrieval to Computation

An isolated associative memory retrieves information. A deep transformer does something more: it repeatedly retrieves, transforms, and uses the result to determine subsequent retrievals.

Multiple heads allow different content-addressable operations to be performed in parallel. Different heads may attend to different types of relations or different regions of the context. Their outputs are then combined and transformed before the next layer forms new queries.

This leads naturally to a computational interpretation.

- **Random-access-like addressing.** Attention can retrieve information from distant positions according to content rather than only according to sequential proximity.
- **Parallel retrieval.** Multiple heads can perform several content-dependent lookups simultaneously.
- **State transformation.** Residual and feed-forward operations transform retrieved information into new internal representations.
- **Iterated control.** The transformed representation determines the queries issued by later layers, so the outcome of one retrieval influences subsequent retrievals.

Through training, the network learns distributed control over what each head queries, which positions receive large attention weights, how retrieved values are combined, and how the resulting representation is transformed.

Algorithmic behaviors such as copying, matching delimiters, tracking references, manipulating variables, and executing multi-step procedures can in principle be implemented through repeated applications of this read–update–read motif.

The important computational claim is therefore not that every trained transformer literally implements a Turing machine. Rather, it is that associative retrieval together with suitable state updates provides a primitive rich enough, under appropriate architectural and representation assumptions, to support universal computation.

1.5 Building a Turing Machine from Associative Reads

A conventional Turing machine operates through an extremely small set of primitive operations. At each step it reads the current tape symbol, consults its transition rule, updates its state, writes a symbol, and moves the tape head.

A random-access machine replaces sequential tape access with direct addressing. An associative machine goes one step further: the relevant address can itself be retrieved from content.

This suggests an abstract computational cycle

$$\text{state} \longrightarrow \text{associative read} \longrightarrow \text{local update} \longrightarrow \text{new query} \longrightarrow \text{associative read}.$$

Attention supplies a natural realization of the read operation, while local neural transformations supply the state update.

The formal counterpart of this intuition is developed in Chap. 3. There the Associative Turing Machine (ATM) separates the computation into an associative memory component and a local update component. The construction shows that a suitable associative read–update architecture can simulate classical computation; see Thm. 1.

The universality claim uses the memory and computational resources allowed by that model. It concerns execution of a supplied program, not a guarantee that training discovers it. For prediction, a different feasible program can be equally satisfactory when it produces accurate outputs.

The significance of this result is conceptual. Associative memory is not merely a passive mechanism for recalling previously stored examples. When retrieval determines subsequent computation, it becomes a computational primitive.

Cost line. For a sequence of length n, dense self-attention requires $O(n^2)$ query–key interactions per layer in a full forward pass.

During autoregressive inference with a key–value cache, generating the next token requires $O(n)$ attention interactions per layer when the current context contains n tokens. Consequently, generating an entire length-n sequence remains quadratic in total for dense attention. Sparse or structured attention can reduce these costs.

For fixed architecture, key–value cache storage grows linearly with sequence length.

1.6 A Broader Principle: Computation by Repeated Local Transitions

The associative-memory interpretation is especially natural for attention-based models, but it points toward a broader principle.

Complex computation can often be represented as the repeated composition of relatively simple learned transitions.

Transformers implement this through alternating content-addressable retrieval and local transformation. Diffusion models [68] repeatedly apply learned denoising transitions. State-space models such as Mamba [54] propagate structured latent states through recurrent linear and gated updates.

These mechanisms should not be identified with one another. In particular, a diffusion model or a state-space model is not an associative memory in the same direct sense as an attention mechanism.

The common structure lies at a more abstract level:

$$\boxed{\text{complex map} = \text{composition of many simpler learned transitions.}}$$

For transformers, one of the elementary transitions is an associative read. For diffusion models, it is a denoising step. For state-space models, it is a structured state transition.

This commonality will become important later when we connect efficient computation with sparse compositional structure.

1.7 What the Associative-Memory Picture Does Not Explain

The associative-memory lens explains an important part of the *mechanism* of attention, but it does not by itself explain the success of deep learning.

Suppose that a memory contains an enormous number of possible patterns. The fact that the architecture can store or represent them does not guarantee that a useful pattern can be found efficiently from a partial cue.

Likewise, the fact that a neural network has enough representational capacity to implement an algorithm does not imply that gradient-based training can find an accurate predictor with feasible resources.

These distinctions separate three levels of explanation:

1. **Representation:** does the architecture contain a feasible realization of the desired input–output computation?
2. **Retrieval:** given a partial cue and a specified memory, can the relevant stored content be retrieved efficiently and reliably?
3. **Predictive learning:** can training find a feasible predictor from data, and does the selected predictor generalize to the population of interest?

Associative memory addresses primarily the mechanism behind the first two questions. A retrieval rule alone establishes neither reliable retrieval for every cue nor successful predictive learning.

Prediction is not structural identification. A successful predictor need not reproduce the intermediate variables or factorization that generated the observations. Recovering those variables is a stronger *structural* objective, requiring its own criterion of success and appropriate identifiability assumptions. This is not merely a concession to nonunique representations: predictive accuracy does not require a nodewise correspondence with a generating decomposition at all.

The later chapters therefore separate the existence of economical representations from the optimization and statistical questions of finding predictors that generalize. Genericity and staircase learning provide candidate mechanisms for exposing and exploiting useful signals; intermediate traces provide another form of assistance. None is imposed as a necessary trajectory of every successful learner.

1.8 Why Can Autoregressive Generation Be So Coherent?

Question.

If transformers can be interpreted as associative memories, it is easy to understand how they might retrieve related words, facts, phrases, or patterns. What is less obvious is why a generated response often remains coherent over many successive tokens.

Why does one retrieval not simply lead to an unrelated retrieval at the next step?

Answer.

A transformer is trained as a conditional next-token predictor, not as an explicit dynamical system constrained by a global consistency rule. Nevertheless, after training on coherent sequences, its autoregressive dynamics often preserve substantial syntactic, semantic, and discourse consistency.

The following discussion is a model of how this behavior may arise. It should not be read as a theorem following from the attention architecture alone.

1.8.1 Structured Geometry of Learned Representations

Each generated token depends on the preceding context through the representations produced by the transformer.

A useful working picture is that, after training, representations associated with natural text concentrate near structured subsets of the high-dimensional state space. These subsets reflect regularities of syntax, semantics, discourse, and the training distribution.

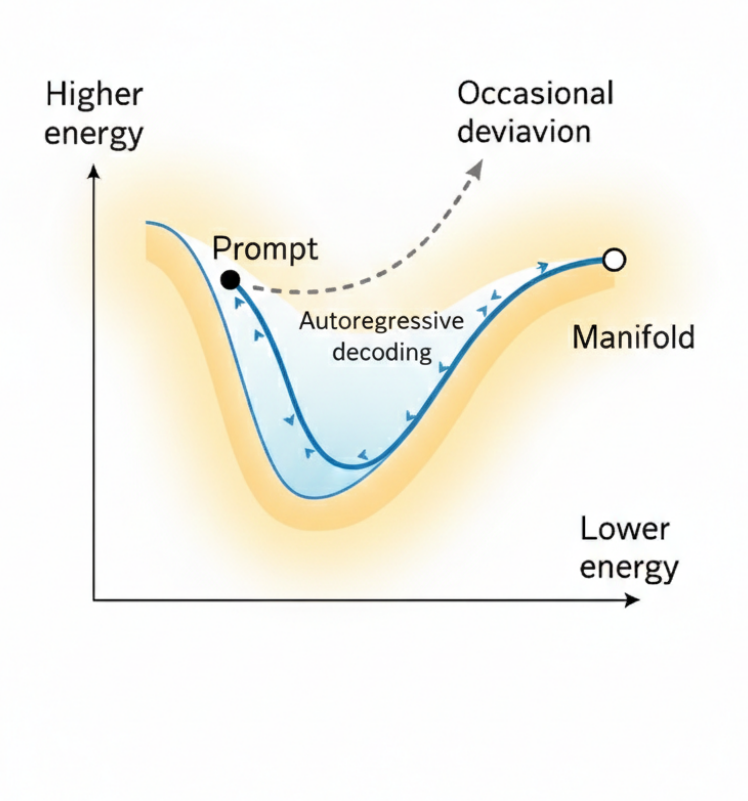


Figure 1.2: Schematic of manifold-constrained generation. A prompt initializes the model in a region of its effective representation space associated with plausible continuations. Autoregressive decoding can remain within a structured region of syntactically and semantically coherent states. The "landscape" is only a schematic representation: the dynamics need not derive from a true energy function. Moreover, remaining in a coherent region does not guarantee factual correctness.

It is common to refer informally to such structured subsets as *manifolds*. This language should be understood as an idealization, not as a claim that the hidden states of a transformer have been proved to lie on smooth low-dimensional manifolds in the mathematical sense.

A prompt places the model in a region of representation space compatible with some family of continuations. Attention then retrieves information that is itself conditioned on the same evolving context. This makes successive predictions strongly dependent rather than independent.

1.8.2 Self-Reinforcing Autoregression

At generation step t, the model samples or selects a new token according to

$$y_{t+1} \sim p_\theta(\cdot \mid y_{\leq t}),$$

after which the new token becomes part of the context used at the next step.

Abstractly, we may write the evolution of an internal state as

$$s_{t+1} = F_\theta(s_t, y_{t+1}).$$

The dynamics are therefore neither autonomous nor generally deterministic. They depend both on the current state and on the decoding choice.

If a set $\mathcal{M}$ represents states associated with coherent continuations, a useful stability hypothesis is that typical transitions starting near $\mathcal{M}$ tend to remain near it:

$$F_\theta(\mathcal{M}, \Xi_{\text{typical}}) \approx \mathcal{M},$$

where Ξ_{typical} denotes token choices receiving substantial probability under ordinary decoding.

This is an empirical property that a trained model may approximately possess. It is not guaranteed by the architecture.

The mechanism is self-reinforcing. A coherent token becomes part of the next context and therefore biases subsequent attention and prediction toward continuations compatible with what has already been generated.

1.8.3 Coherence Is Not Truth

The distinction between coherence and factual correctness is essential.

A model can produce a passage that is

- grammatically correct,
- semantically plausible,
- internally self-consistent,
- stylistically coherent,

and nevertheless factually false.

Thus a hallucination need not correspond to a trajectory that leaves the region of coherent language. On the contrary, many hallucinations are highly coherent.

The manifold picture, to the extent that it is useful, therefore explains primarily why generation may remain on a structured set of *linguistically and semantically plausible continuations*. It does not by itself explain grounding in external reality.

One may schematically distinguish

$$\mathcal{M}_{\text{text}}$$

from the smaller and more demanding subset

$$\mathcal{M}_{\text{true}} \subseteq \mathcal{M}_{\text{text}},$$

where the first represents coherent text and the second represents coherent text that is also factually correct relative to the relevant world state.

Language-model training strongly constrains the first. Guaranteeing the second requires additional assumptions, information, or external verification.

1.8.4 Emergent Consistency as Constrained Associative Recall

From the associative-memory viewpoint, prompting fixes part of a pattern and autoregressive generation successively completes it.

But the completion is not produced by a single lookup. Each newly generated token changes the cue used for subsequent retrievals. The system therefore performs a sequence of mutually conditioning associative operations.

This leads to the qualitative picture

$$\text{prompt} \to \text{retrieval} \to \text{prediction} \to \text{expanded prompt} \to \text{new retrieval} \to \cdots$$

in which previously generated material helps constrain future generation.

Consistency, in this view, can emerge from the interaction between a structured learned representation and self-conditioning autoregressive dynamics.

Associative recall explains how relevant information can be retrieved; autoregressive self-conditioning helps explain why successive retrievals can remain mutually coherent.

1.9 Historical and Biological Connections

The associative-memory viewpoint long predates transformers.

Kanerva's *Sparse Distributed Memory* (SDM) [75] provides a particularly clear model of content-addressable memory in a high-dimensional binary space. Memory locations are associated with distributed addresses, and a partial cue activates locations lying sufficiently close to it in Hamming distance. Their stored contents are then aggregated to produce a recalled pattern.

The essential idea is familiar from attention:

$$\text{cue} \longrightarrow \text{similarity-based addressing} \longrightarrow \text{retrieved content}.$$

There are, of course, major architectural differences. Kanerva's model is a fixed high-dimensional associative memory, whereas transformers build context-dependent keys, queries, and values and repeatedly compose retrieval with nonlinear transformations.

Biological memory systems provide another suggestive connection. Classical models of hippocampal circuitry emphasize sparse coding, pattern separation, associative recall, and pattern completion [99, 102, 70]. In particular, dentate gyrus and CA3 have often been interpreted as supporting sparse representations and auto-associative retrieval.

There is therefore an intriguing computational analogy between hippocampal circuitry and artificial associative-memory systems. The analogy concerns functional motifs—sparse coding,

content-addressable retrieval, context-dependent reconstruction, and multi-stage transformation—not architectural identity.

A much stronger claim, such as identifying the hippocampus literally with a transformer or assigning directly comparable information capacities to their synapses, would require assumptions not justified by the present discussion.

The biological connection becomes more interesting when associative memory is embedded in a hierarchy. Hippocampal interactions with cortex suggest a system in which rapidly stored episodic representations can participate in slower reconstruction and consolidation. This motivates a broader view of associative memory as a component of a multi-stage computational architecture rather than simply a flat lookup table.

A later chapter develops this hippocampal-scaffold hypothesis in greater detail.

1.10 Associative Retrieval, Computational Arity, and Memory Capacity

Several memory mechanisms discussed throughout this book share a common computational organization:

$$\boxed{\text{query} \rightarrow \text{similarity} \rightarrow \text{retrieval.}}$$

This motif appears in attention, radial-basis memories, modern Hopfield networks, and sparse distributed memory.

The resemblance is useful, but it should not be pushed too far.

Some of these mechanisms are exactly equivalent under particular assumptions.

Others share only the broader principle of similarity-gated access.

A second distinction is equally important: a retrieval rule may appear to have very large fan-in when written as a single mathematical expression, while still admitting a bounded-fan-in computational implementation.

Finally, one must distinguish the number of possible memory *addresses* from the number of items that can be simultaneously *superposed* and recovered reliably.

These distinctions clarify what associative memory contributes to the computational framework developed in this book.

1.10.1 A generic associative read

Consider a collection of keys

$$k_1, \ldots, k_N$$

and associated values

$$v_1, \ldots, v_N.$$

A broad class of associative memories computes

$$r(q) = \sum_{i=1}^{N} \alpha_i(q) v_i, \tag{1.3}$$

where the weights depend on the similarity between the query q and the stored keys.

A common normalized form is

$$\alpha_i(q) = \frac{\exp s(q, k_i)}{\sum_{j=1}^{N} \exp s(q, k_j)}. \tag{1.4}$$

The score function

$$s(q, k)$$

defines the geometry of retrieval.
This decomposition separates two components:

addressing geometry and stored content.

1.10.2 Dot-product attention

For standard attention,

$$s(q, k_i) = \frac{q^\top k_i}{\tau}, \tag{1.5}$$

where τ is a temperature or scale parameter.
Hence

$$r(q) = \sum_{i=1}^{N} \frac{\exp\left(q^\top k_i/\tau\right)}{\sum_j \exp(q^\top k_j/\tau)} v_i. \tag{1.6}$$

This is a content-addressable read: the query does not specify an explicit memory location but instead selects values through similarity to their keys.
The mechanism should therefore be interpreted as

content-addressable routing or retrieval,

not as a complete model of computation by itself.

1.10.3 Gaussian RBF retrieval

A Gaussian radial-basis memory instead uses the score

$$s_{\mathrm{RBF}}(q, \mu_i) = -\frac{1}{2}(q - \mu_i)^\top \Sigma^{-1}(q - \mu_i). \tag{1.7}$$

Expanding the quadratic gives

$$s_{\mathrm{RBF}}(q, \mu_i) = q^\top \Sigma^{-1}\mu_i - \frac{1}{2} q^\top \Sigma^{-1} q - \frac{1}{2}\mu_i^\top \Sigma^{-1}\mu_i. \tag{1.8}$$

The query-only term

$$-\frac{1}{2} q^\top \Sigma^{-1} q$$

is common to all keys and cancels under softmax normalization.
The key-dependent term

$$-\frac{1}{2}\mu_i^\top \Sigma^{-1}\mu_i$$

does not cancel in general.

Therefore Gaussian RBF retrieval is exactly equivalent to dot-product softmax attention only under an additional condition, for example

$$\|\Sigma^{-1/2}\mu_i\| = \text{constant}$$

for all stored keys, or if the corresponding key-dependent bias is included explicitly in the score.

Thus

RBF retrieval and attention can be equivalent, but not unconditionally.

1.10.4 Modern Hopfield retrieval

A single update of a modern Hopfield memory can take the form

$$\alpha_i(x) \propto \exp\Big(\beta x^\top x_i\Big). \tag{1.9}$$

At this level, the update is mathematically the same normalized dot-product retrieval used in attention, with

$$\beta = \frac{1}{\tau}.$$

This exact correspondence concerns a *single retrieval step*.

A recurrent Hopfield system can apply the update repeatedly:

$$x^{(0)} \to x^{(1)} \to x^{(2)} \to \cdots .$$

That introduces an attractor dynamics absent from a single ordinary attention read.

Therefore

single-step modern Hopfield retrieval = dot-product attention read,

under the usual parameter identification, but

multi-step Hopfield dynamics ≠ one attention operation.

1.10.5 Kanerva's Sparse Distributed Memory

Kanerva's Sparse Distributed Memory uses a different gating mechanism.

For binary query q, locations satisfying

$$\mathrm{Ham}(q, k_i) \le r \tag{1.10}$$

are activated.

A corresponding kernel is

$$K(q, k_i) = \mathbf{1}\left\{\mathrm{Ham}(q, k_i) \le r\right\}. \tag{1.11}$$

A normalized read can be written

$$r(q) = \frac{\sum_i K(q, k_i) v_i}{\sum_j K(q, k_j)}. \tag{1.12}$$

Thus SDM belongs naturally to the larger family of

$$\boxed{\text{similarity-gated associative reads.}}$$

But it should not be identified literally with dot-product attention or with a zero-temperature single-nearest-neighbor softmax.

Its gating geometry is a Hamming ball, and many locations may contribute simultaneously.

1.10.6 Exact equivalence versus shared computational motif

It is therefore useful to distinguish two levels of comparison.

At the stronger level,

$$\boxed{\text{exact equivalence}}$$

means that two mechanisms compute the same mathematical map after an explicit parameter transformation.

At the weaker level,

$$\boxed{\text{shared computational motif}}$$

means only that both perform

$$\text{query} \to \text{similarity} \to \text{retrieval}.$$

Attention and one-step modern Hopfield retrieval admit an exact correspondence in the standard formulation.

Gaussian RBF retrieval requires additional normalization or bias conditions.

Kanerva SDM shares the motif but not the same retrieval law.

Keeping these distinctions explicit avoids turning a useful analogy into a false identity.

1.10.7 Associative retrieval does not automatically have bounded arity

There is a second subtlety relevant to compositional sparsity.

The expression

$$r(q) = \sum_{i=1}^{N} \alpha_i(q) v_i \tag{1.13}$$

may depend on all N stored values.

If N grows with problem size, then the *functional arity* of the written expression is not bounded.

Thus

$$\boxed{\text{attention} \not\Longrightarrow \text{bounded functional arity.}}$$

This does not contradict the bounded-fan-in view of efficient computation, because the mathematical arity of a formula and the fan-in of its implementation are different notions.

1.10.8 Sparse retrieval

One possibility is that retrieval is itself sparse.

If only

$$k = O(1)$$

keys receive significant weight, then the effective read depends on only a bounded number of memory items.

This may arise through

explicit top-k attention, sparse routing, hard gating, large score margins.

In that case associative retrieval itself can instantiate a low-arity constituent.

1.10.9 Bounded-fan-in implementation of dense retrieval

Even dense retrieval admits a bounded-fan-in computational realization.

Consider the denominator

$$Z(q) = \sum_{j=1}^{N} \exp s(q, k_j).$$

The sum can be evaluated through a binary reduction tree:

$$N \to \frac{N}{2} \to \frac{N}{4} \to \cdots \to 1.$$

Each node has fan-in two, and the depth is

$$O(\log N).$$

The same is true for the weighted numerator

$$\sum_i \exp s(q, k_i) v_i.$$

Thus

$$\boxed{\text{high-arity mathematical expression} \not\Longrightarrow \text{high-fan-in computation.}}$$

What matters for compositional sparsity is the structure of an efficient implementation.

A dense attention read may therefore remain compatible with bounded-fan-in computation even though its written functional dependence involves the entire context.

This is an implementation-level statement. An efficiency claim must also control operation count, memory, and numerical precision as the problem size grows. The bounded-fan-in implementation does not by itself supply a particular semantic decomposition or uniformly regular low-arity real constituents. Chapter 4 keeps the computational statement separate from those stronger approximation-theoretic assumptions.

1.10.10 Addressability and storage are different capacities

Associative memories are often described as having enormous capacity.

But at least two fundamentally different capacities must be distinguished:

1. the number of distinct addresses that can be represented;
2. the number of associations that can be simultaneously stored in a shared representation and recovered with fixed reliability.

These quantities can scale very differently.

1.10.11 Exponential addressability

Let keys lie on the unit sphere

$$\mathbb{S}^{d-1}.$$

High-dimensional geometry permits exponentially many approximately orthogonal vectors.

For fixed tolerance

$$\eta \in (0, 1),$$

one can choose on the order of

$$N = \exp(cd) \tag{1.14}$$

distinct unit vectors, for a constant c depending on the allowed correlation, such that pairwise inner products remain small.

Thus a d-dimensional representation can provide an exponentially large *address space*.

This does not mean that exponentially many independent memories can be superposed into one d-dimensional vector and recovered reliably.

1.10.12 Modular address codes

A constructive version of the same phenomenon arises from modular codes.

Suppose an address x is represented by residues

$$x \bmod m_1, \ldots, x \bmod m_J,$$

where the moduli are pairwise coprime.

The Chinese remainder theorem implies that the joint code uniquely represents

$$\prod_{j=1}^{J} m_j \tag{1.15}$$

different addresses before repetition.

Thus a collection of relatively small modules can produce a very large joint address space.

Related modular-coding arguments are useful for thinking about biological grid-like codes as well as artificial distributed addressing.

Again, however, this is a statement about the number of distinct *addresses*, not the number of objects that can be mixed together in one noisy superposition.

1.10.13 Superpositional storage

Now consider storing many bound associations in a single distributed vector:

$$W = \sum_{j=1}^{M} k_j \odot w_j, \tag{1.16}$$

where $\odot$ denotes a unitary or approximately invertible binding operation.
To retrieve association ℓ, one approximately unbinds with the inverse of k_ℓ.
The desired item appears together with crosstalk from all other stored items.
Under standard random-key assumptions, the crosstalk variance typically scales as

$$O\left(\frac{M}{d}\right), \tag{1.17}$$

giving signal-to-noise ratio of order

$$\mathrm{SNR} \asymp \sqrt{\frac{d}{M}}. \tag{1.18}$$

Maintaining constant retrieval quality therefore permits roughly

$$M = O(d)$$

simultaneously superposed associations under this simple model.

1.10.14 The apparent paradox disappears

There is no contradiction between

$$N = \exp(\Theta(d))$$

possible addresses and

$$M = O(d)$$

reliably superposed memory items.
They measure different things.
The first asks:

How many distinct keys can exist?

The second asks:

How many items can be mixed into one finite-dimensional state while retaining adequate signal-to-noise ratio?

Therefore

exponential addressability $\neq$ exponential superpositional storage.

1.10.15 Why the distinction matters for associative computation

The difference between addressing and storage clarifies the computational role of associative memory.

A large address space allows a system to select among a huge number of potential states or memory locations.

It does not require all of those states to be simultaneously stored in one vector.

One can instead combine

large distributed address space
+ external or distributed memory
+ content-addressable retrieval.

This organization is closer to ordinary computer memory as well.

A processor may address an enormous memory space without encoding the entire contents of that memory in one register.

1.10.16 Relation to compositional computation

Associative retrieval and compositional computation therefore solve different parts of the problem.

Associative mechanisms answer:

Which information or operation should be accessed?

Compositional structure answers:

How should selected operations be combined into a computation?

In this sense,

associative memory supplies routing and addressability;

bounded-arity composition supplies computation.

A complete architecture may require both.

The query selects the relevant memories or modules.

The computation graph composes their outputs.

1.10.17 A useful hierarchy of claims

The discussion can be summarized as follows:

$$\boxed{\begin{aligned} \text{Attention} &= \text{normalized dot-product associative retrieval,} \\ \text{one-step modern Hopfield retrieval} &\simeq \text{the same mechanism,} \\ \text{Gaussian RBF memory} &= \text{related retrieval under metric-dependent scores,} \\ \text{Kanerva SDM} &= \text{similarity-gated retrieval using Hamming neighborhoods.} \end{aligned}}$$

These mechanisms share a computational motif without being universally interchangeable. Likewise,

$$\boxed{\text{dense functional dependence} \not\Longrightarrow \text{unbounded implementation fan-in,}}$$

and

$$\boxed{\text{large address space} \not\Longrightarrow \text{equally large superpositional storage capacity.}}$$

These distinctions are important whenever associative memory is used as a primitive in a larger theory of intelligent computation.

1.10.18 Conclusion

Associative memory should be understood neither as a single algorithm nor as a single capacity number.

Its central computational idea is content-dependent access:

$$\boxed{\text{the current state determines which stored information becomes available next.}}$$

Attention, RBF memories, Hopfield retrieval, and sparse distributed memory implement this idea using different similarity geometries and different dynamics.

Their common role is therefore best summarized as

$$\boxed{\text{associative memory is an addressing and routing primitive.}}$$

Compositional sparsity concerns the complementary question of how the retrieved pieces are combined.

Together they give a natural computational architecture:

$$\boxed{\text{address} \to \text{retrieve} \to \text{compose} \to \text{compute.}}$$

1.11 Associative Memory as a Computational Primitive

We can now summarize the main argument.

Classical associative memory begins with a simple operation:

retrieve what resembles the cue.

Attention turns this idea into a learned, differentiable operation. Under fixed-norm conditions, its softmax weights are exactly those of a normalized Gaussian kernel. In the general case, the same computation retains the broader structure of content-addressable retrieval while allowing learned and context-dependent representations.

Deep architectures then add a crucial ingredient: composition.

A retrieved value is transformed, the transformed state generates new queries, and those queries trigger further retrievals. Thus

$$\boxed{\text{associative retrieval} + \text{state update} + \text{iteration}}$$

becomes a general computational motif.

The progression is

$$\boxed{\begin{array}{c}\text{RBF/kernel memory} \longrightarrow \text{attention} \longrightarrow \text{read–update composition} \\ \Downarrow \\ \text{Associative Turing Machine} \longrightarrow \text{general computation.}\end{array}}$$

This perspective suggests that associative memory is not merely one specialized faculty of an intelligent system. It may be one of the basic mechanisms from which more elaborate computation can be constructed.

But universality is only the beginning of an explanation.

A universal machine can execute many different computations when supplied with the appropriate program. That fact does not explain how training finds a useful computation from examples, or why the resulting predictor works beyond those examples.

The subsequent chapters therefore separate three questions: whether a feasible accurate predictor is available, whether the specified training procedure finds a suitable fit, and whether the selected predictor generalizes. Chapter 4 develops the representation principle of compositional sparsity. The chapters that follow examine accessible learning signals, intermediate supervision, optimization geometry, and statistical selection.

Chapter H brings these questions together. Under its stated finite-precision resource assumptions, an accurate predictor with feasible evaluation already supplies a bounded-fan-in compositional realization. Its learned variables need not match those of the computation that generated the observations. Explaining how training achieves that accuracy remains a substantive problem; structural identification is a stronger, separate one.

CHAPTER 2

A Historical Reflection on Associative Memories

This chapter is a historical and conceptual essay following the thesis introduced in Chapter 1. Its purpose is not to claim that modern transformers descended directly from the associative-memory models of the 1960s and 1970s. Rather, it traces the repeated appearance of a common computational idea: information can be retrieved or transformed according to the content of a cue.

From early heteroassociative memories, through nonlinear operator memories, radial basis function networks, hippocampal models, and modern attention, the machinery changes substantially. Yet a recurring primitive remains: use similarity or correlation with the present state to recover information relevant to what should happen next.

The historical thesis of this chapter is therefore one of conceptual continuity rather than literal architectural descent.

2.1 A Recurring Computational Idea

Associative memory—the ability to bind one pattern to another and later recover the associated information from a complete or partial cue—is one of the oldest ideas in neural computation.

What is striking from the perspective of modern artificial intelligence is not merely its historical importance, but its recurrence.

A simple heteroassociative memory stores pairs

$$(x^\mu, y^\mu)$$

and implements a map

$$x^\mu \longmapsto y^\mu.$$

More sophisticated systems replace exact matching by similarity, correlation, interpolation, or learned metrics. Modern attention extends the same general idea still further: a query provides a cue, keys provide content-dependent addresses, and values provide the information retrieved from those addresses.

The central claim of this essay is therefore not that all these systems are identical. They plainly are not. Rather,

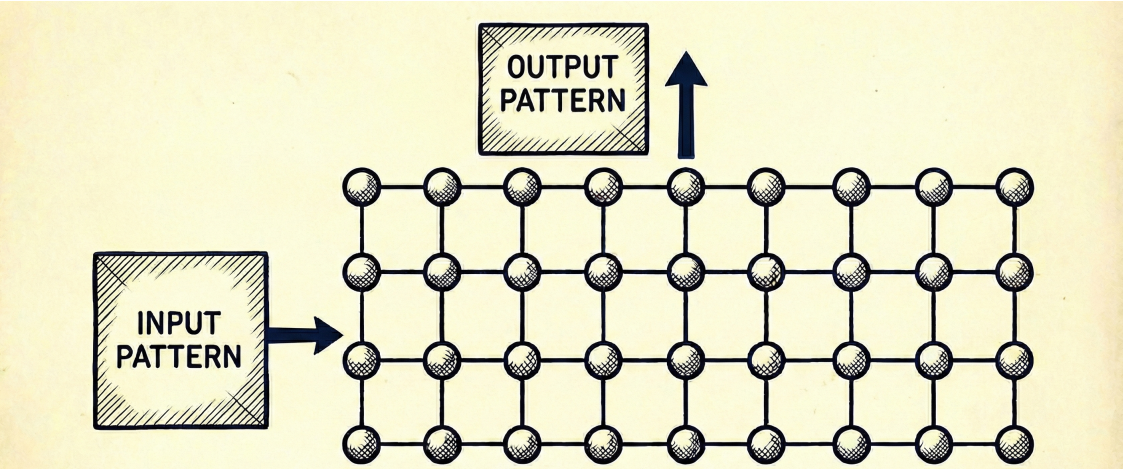


Figure 1. Schematic representation of a non-holographic associative memory, after Willshaw & Longuet-Higgins, *Nature* (1969).

Figure 2.1: A conceptual lineage of associative memory, from early correlation-based models to modern content-addressable computation.

content-addressable association is a recurring computational primitive.

Several important developments in neural computation can be viewed as successive enrichments of this primitive:

$$\text{association} \rightarrow \text{learned association} \rightarrow \text{nonlinear similarity}$$
$$\Downarrow$$
$$\text{hierarchical and recurrent association}$$
$$\Downarrow$$
$$\text{deep content-addressable computation.}$$

This perspective provides a useful historical bridge between early theories of memory and modern deep learning.

2.2 Early Associative Memories

2.2.1 Heteroassociative Correlation Memories

The modern mathematical study of associative memory emerged prominently in the late 1960s and early 1970s. Early models focused on storing associations between patterns while analyzing capacity, robustness, one-shot learning, and biological plausibility.

Willshaw and Longuet-Higgins introduced sparse binary heteroassociative memories with one-shot learning and feedforward recall [166]. Kohonen subsequently formulated associative memory in linear-algebraic form using correlation matrices [80].

Given stored pairs (x^μ, y^μ), the memory matrix can be written as

$$W = \sum_\mu y^\mu (x^\mu)^\top,$$

so that a new cue x produces

$$\hat{y} = Wx = \sum_\mu y^\mu (x^\mu)^\top x.$$

The important structure is immediately visible. Retrieval is weighted by the similarity

$$(x^\mu)^\top x$$

between the cue and the stored input pattern.

Viewed retrospectively, this already contains a primitive that will reappear repeatedly in later kernel methods and in attention:

similarity to a stored key $\longrightarrow$ retrieval of an associated value.

The historical models were not transformers, of course, nor were they formulated in the language of queries, keys, and values. But the computational organization is recognizably heteroassociative.

2.2.2 Nonlinear Operator Memories

A particularly important step was the formulation of associative recall as an operator-estimation problem in Poggio's work of the 1970s [123].

Given paired observations collected into matrices X and Y, one seeks an operator Θ satisfying approximately

$$Y \approx \Theta(X).$$

Instead of restricting Θ to be linear, one can expand it as

$$Y = L_0 + L_1(X) + L_2(X, X) + \cdots + L_k(X, \ldots, X),$$

where the L_ℓ are symmetric multilinear operators.

The first-order term recovers the linear associative-memory solution,

$$L_1 = YX^\dagger,$$

while the higher-order terms provide systematic nonlinear corrections.

If the residual after approximation through order $k-1$ is

$$E_{k-1} = Y - \sum_{\ell=0}^{k-1} L_\ell(X, \ldots, X),$$

then the next correction can be written schematically as

$$L_k = E_{k-1} C_k^\dagger,$$

where C_k is the lifted tensor of degree-k input products,

$$(C_k)_{\alpha_1 \cdots \alpha_k, j} = \prod_{t=1}^{k} X_{\alpha_t j}.$$

The significance of this construction is that associative memory becomes a problem of nonlinear function approximation rather than merely storage in a linear correlation matrix.

Early introduction of polynomial kernels. The multilinear operator expansion naturally introduces polynomial feature spaces.

A degree-k tensor feature can be written as

$$\Phi_k(x) = x^{\otimes k}.$$

Its inner product satisfies

$$\left\langle x^{\otimes k}, (x')^{\otimes k} \right\rangle = (x^\top x')^k.$$

Including lower-order terms leads to the familiar inhomogeneous polynomial form

$$K_k(x, x') = (1 + x^\top x')^k.$$

Thus the operator-estimation framework contains an explicit polynomial feature construction and the corresponding polynomial kernels.

This is not merely a retrospective resemblance. Vapnik later explicitly credited Poggio's earlier work with the introduction of polynomial kernels. Historically, the associative operator formulation therefore belongs to the early development of the kernel viewpoint, well before polynomial kernels became standard tools in statistical learning theory and support vector machines.

Conceptually, this creates an important bridge:

associative memory $\rightarrow$ nonlinear feature expansion $\rightarrow$ kernel representation.

2.2.3 RBFs and HyperBFs

A second important development replaced global polynomial expansions by localized similarity functions.

Radial basis function networks take the form

$$f(x) = \sum_\mu \alpha_\mu \phi(\|x - x^\mu\|),$$

where each stored center x^μ contributes according to its distance from the cue.

Poggio and Girosi developed this perspective into a general framework for regularization networks and HyperBFs [128], allowing learned or anisotropic metrics.

Instead of merely asking whether two patterns are correlated, these models ask how close they are in an appropriate representation space.

The associative-memory interpretation becomes

$$\text{cue} \longrightarrow \text{similarity to stored centers} \longrightarrow \text{weighted reconstruction}.$$

This is especially close conceptually to modern attention because retrieval is no longer based on exact addresses. It is controlled continuously by a learned notion of similarity.

2.2.4 A Different Branch: Autoassociative Attractor Memories

Hopfield networks developed a different branch of associative-memory theory [70].

Instead of storing a mapping from one pattern to another, a classical Hopfield network is autoassociative: a partial or corrupted version of a stored pattern evolves dynamically toward a corresponding attractor.

Its defining ingredients include recurrent interactions, symmetric weights in the classical formulation, and an energy-like dynamical system.

This was historically enormously influential because it connected memory with collective neural dynamics and pattern completion.

Conceptually, however, the key–value organization of modern attention is closer to heteroassociative and metric-based retrieval than to classical symmetric Hopfield dynamics.

The two traditions should therefore be seen as related branches of associative-memory theory rather than one as a special case of the other.

2.3 From Association to Biological Computation

The biological relevance of associative computation predates any formal theory of neural networks.

At the most elementary level, nervous systems must map sensory conditions into appropriate actions.

A simple input–output relation has the form

$$x_{\text{sensory}} \longmapsto y_{\text{motor}}.$$

Formally, this has the same shape as a heteroassociative map.

2.3.1 Reflexes as Elementary Input–Output Associations

A monosynaptic reflex arc provides perhaps the simplest neural architecture connecting a sensory signal to a motor response.

It is important not to push the analogy too far.

A genetically specified, immutable reflex is not necessarily a *memory*. Nothing need have been learned or stored during the lifetime of the organism.

Its relevance is instead computational: it exhibits the primitive organization

$$\text{pattern} \longrightarrow \text{associated response}.$$

Once synaptic plasticity is introduced, however, such mappings can become learned associations in the stronger sense.

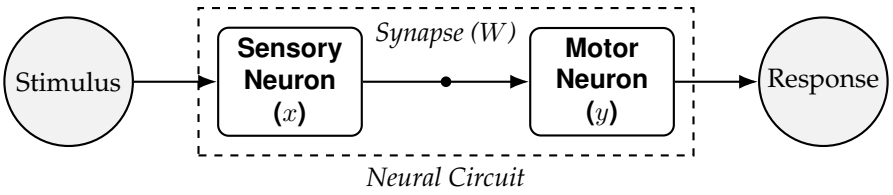


Figure 2.2: A monosynaptic reflex arc. At the level of input–output organization, the circuit maps a sensory signal x through a synaptic connection W into a motor response y. The analogy with heteroassociative memory concerns the form of the mapping; a fixed reflex need not itself constitute a learned memory.

2.3.2 Plasticity and Adaptive Association

Plastic sensory–motor circuits add a crucial ingredient: the mapping between perception and action can now change with experience.

Hebbian and heterosynaptic plasticity provide mechanisms through which local associations can be modified according to activity and outcome.

At this level, the nervous system no longer merely executes a fixed mapping. It can construct and modify mappings.

This suggests a broad progression from fixed association to adaptive association:

$$\text{fixed response} \rightarrow \text{plastic association} \rightarrow \text{adaptive computation}.$$

Feedback pathways further enrich this process because intermediate representations can influence upstream or downstream processing rather than simply propagating once through a feedforward chain.

2.3.3 A Possible Computational Progression

The following sequence should be read as a conceptual progression in computational capability, not as a claim about the literal phylogenetic order in which these mechanisms evolved.

1. **Fixed sensorimotor mappings.**

   Simple circuits associate sensory conditions with actions.

2. **Plastic associative mappings.**

   Synaptic plasticity allows these associations to adapt with experience.

3. **Multi-stage processing and feedback.**

   Ascending and descending pathways permit intermediate representations to be modified and repeatedly refined.

4. **Associative scaffolding and episodic indexing.**

   Systems such as the hippocampus provide rapid storage, partial-cue retrieval, and interaction with slower cortical representations.

5. **Self-organized learning circuits.**

   Models such as Self-Assembling Learning (SAL) illustrate, at a theoretical level, how local plastic mechanisms and feedback pathways can organize into circuits implementing effective optimization procedures [124].

The point of this sequence is not evolutionary chronology. It is that increasing computational power can arise by embedding simple association inside progressively richer systems of plasticity, feedback, hierarchy, and repeated transformation.

2.3.4 Hippocampal Association and Scaffolding

The hippocampus provides an especially interesting biological example because associative recall operates as part of a larger system.

Classical theories emphasize pattern separation, episodic indexing, autoassociative retrieval, and pattern completion [99, 102, 70].

The relevance to the present historical story is not that the hippocampus is literally a transformer. Rather, hippocampal circuitry illustrates how associative retrieval can be embedded in a multi-stage architecture and interact with other memory systems.

A partial cue may trigger reconstruction of a richer episodic pattern, which in turn influences cortical processing and later consolidation.

Thus biological memory suggests a transition from

$$\text{flat associative retrieval}$$

to

$$\text{associative retrieval embedded in a hierarchy.}$$

A later chapter develops the hippocampal-scaffold hypothesis in much greater detail.

2.4 Modern Attention as Associative Retrieval

The attention mechanism at the center of modern transformers is

$$\mathrm{Attn}(q, K, V) = \mathrm{softmax}\left(\frac{qK^\top}{\sqrt{d}}\right)V.$$

Its associative interpretation is immediate:

$$\text{query} \longrightarrow \text{similarity with keys} \longrightarrow \text{weighted retrieval of values.}$$

As shown in Chapter 1, when the query and keys have fixed norm, dot-product attention can be rewritten exactly as normalized Gaussian-kernel retrieval.

Indeed, if

$$\|q\| = \|k_i\| = 1,$$

then

$$q^\top k_i = 1 - \frac{1}{2}\|q - k_i\|^2,$$

and therefore the normalized attention weights become

$$\frac{\exp\left(q^\top k_i/\sigma^2\right)}{\sum_j \exp(q^\top k_j/\sigma^2)} = \frac{\exp(-\|q - k_i\|^2/2\sigma^2)}{\sum_j \exp(-\|q - k_j\|^2/2\sigma^2)}.$$

Thus, under normalization,

$$\boxed{\text{dot-product attention} \equiv \text{normalized Gaussian associative retrieval.}}$$

Without fixed norms, the equivalence is no longer exact because the key norms contribute nonconstant terms to the exponent. The broader content-addressable interpretation nevertheless remains useful.

Historically, attention can therefore be viewed as a modern learned form of similarity-based heteroassociative retrieval.

The crucial advance is not merely the retrieval rule itself. Queries, keys, and values are learned representations that change across layers, and associative retrieval is repeatedly interleaved with nonlinear state transformations.

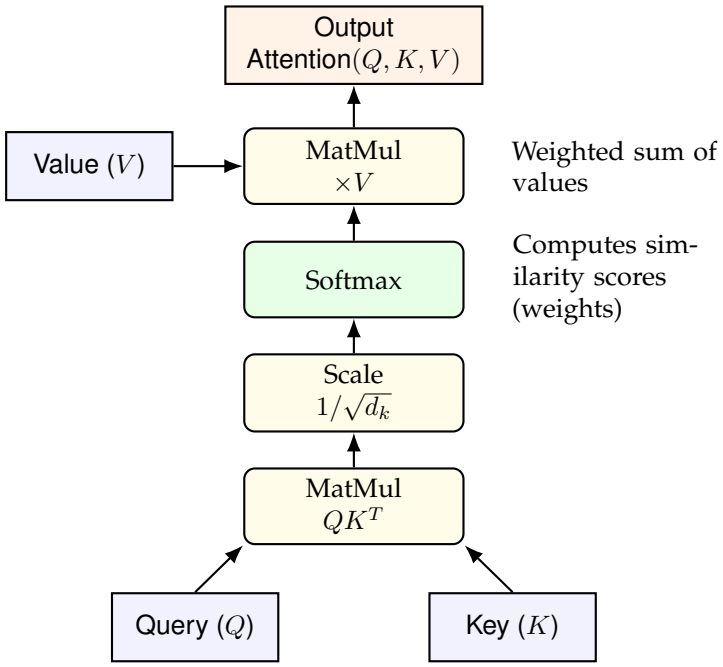


Figure 2.3: The transformer attention mechanism. Queries and keys determine content-dependent similarity weights, which are then used to retrieve a weighted combination of the stored values.

2.5 A Preview: Why Does Associative Recall Work?

The repeated historical success of associative memories raises a deeper question.

Why should similarity to a partial cue contain enough information to recover the relevant pattern?

This question cannot be answered by representational capacity alone. An associative memory may have enough capacity to store a large set of patterns while still making reliable retrieval from partial information difficult.

One possible explanation involves *genericity*, developed in Chapter 6. In a specified observation model, perturbations can break exact cancellations and expose statistical footprints of relevant dependencies. Whether this happens depends on the stored or target structure, the distribution of cues or training examples, and the family of similarities or probes available to the procedure.

Correlations and local similarities may then contain useful information about the underlying association. Three questions must nevertheless be kept separate:

1. Does a relevant footprint *exist*, rather than vanish by exact cancellation?

2. Is it *large enough to detect* at a controlled probe scale, with feasible samples and computation, including the cost of finding a useful probe?

3. Can the specified retrieval rule or learning updates *use* that information to make progress?

The distinction is

nonzero footprint
$\neq$ efficiently detectable footprint
$\neq$ an established retrieval or learning guarantee.

There exist target functions for which low-order correlations vanish, and cases in which a footprint is formally nonzero but exponentially small. Even a detectable statistic need not be among the directions accessed by a particular optimizer. A reliable-retrieval or convergence claim therefore needs assumptions about the actual procedure, not only the existence of a correlation.

The staircase and the cliff describe possible learning mechanisms within this picture. In a staircase, constructing and preserving one useful coordinate can expose a later dependency. A cliff denotes the absence of a useful signal for the specified local mechanism. Failure of that mechanism is not an impossibility theorem for every learner.

Genericity is thus neither part of the definition of associative memory nor a universal prerequisite for predictive success. It is a candidate explanation for accessible information in particular retrieval and learning problems. Learning an accurate predictor and identifying the particular computation that generated the observations remain distinct objectives, as Chapter H makes explicit.

2.6 Associative Memory Is a Primitive, Not a Complete Theory

The historical continuity described here should not be confused with a claim that associative memory alone explains intelligence.

Associative retrieval solves an important problem:

given the present cue, retrieve relevant stored structure.

But several additional problems remain.

Learning. The existence of a useful associative representation does not explain how training finds an accurate predictor from data. A system may be capable of representing the desired input–output relation while optimization fails to find a suitable fit. Nor does fitting the observed examples, by itself, explain prediction beyond them. These are the optimization and generalization questions developed later in the book.

Retrieval. Even if a representation has been learned, a partial cue may not uniquely or efficiently identify the relevant memory.

This is an inverse problem and depends on the geometry and statistics of the stored patterns. Here recovery means retrieving associated content; it should not be confused with identifying the latent computation that produced the training data.

Predictive success and structural recovery. Predictive learning asks for an accurate function on the population of interest. Structural recovery additionally asks for a specified generating computation or its intermediate variables, up to controlled equivalences. A predictor can succeed without reproducing those variables. Its learned basis or factorization may differ from the generating one without that difference constituting a predictive failure.

Composition. A collection of associations need not organize itself automatically into reusable modules. Recombining learned computations across tasks and contexts is an important additional capability, not a consequence of associative retrieval alone. A claim about such reuse requires more than accurate prediction on one task, and need not require that the reusable modules coincide with the teacher's constituents.

Persistence. Many associative mechanisms operate only over the currently available state.

Standard transformers, for example, possess a working context and a key–value cache during an episode but do not, by themselves, provide persistent memory across indefinitely separated interactions.

World models and planning. Associative retrieval does not by itself provide causal understanding, explicit dynamics, counterfactual reasoning, or planning.

Such abilities may make extensive use of associative memories, but they require additional computational structure.

The appropriate conclusion is therefore

associative memory is a primitive, not a complete theory of intelligence.

Its importance lies precisely in the fact that such a primitive can be embedded inside richer dynamical and compositional systems.

Why associative retrieval does not produce a collage. Describing attention as an associative-memory operation does not imply that a Transformer merely retrieves and concatenates fragments of its training data. Attention is only one computational primitive inside a larger conditional sequence model. At each step the complete network computes

$$p_\theta(x_t \mid x_{<t}),$$

using repeated attention-based routing together with local nonlinear transformations. Autoregressive training therefore fits conditional continuations that depend jointly on the preceding context. Coherent generation should consequently be attributed to the learned conditional sequence function as a whole, not to associative retrieval alone. Associative memory explains an important routing and retrieval primitive; it is not by itself a theory of discourse coherence.

2.7 Experimental Evidence from Homogeneous HyperBF Architectures

The connection between associative kernels and transformer-like architectures can also be tested experimentally.

Recent work [127] constructs a "homogeneous" architecture in which the nonlinear components of a Vision Transformer are implemented using explicit HyperBF modules.

In particular, both the attention-like mixing operations and the feed-forward transformations are replaced by modules built from the same general kernel-based primitive.

The important point is not that an ordinary transformer MLP is mathematically identical to a HyperBF. It is not.
Rather, the experiment asks a different question:

> Can an architecture built largely from associative-memory primitives perform competitively with a standard transformer?

On several image-classification benchmarks the resulting architecture closely approaches the performance of a baseline Vision Transformer of comparable size.

Dataset	ViT Acc. (%)	HyperBF Acc. (%)	Gap (%)
CIFAR10	75.61	75.12	0.49
CIFAR100	49.90	48.30	1.60
Tiny ImageNet	32.03	31.14	0.89

Table 2.1: Comparison between a baseline Vision Transformer and a homogeneous HyperBF architecture on image-classification benchmarks reported in [127].

These results show that, on these benchmarks and at this scale, explicit HyperBF modules can substitute for the standard nonlinear components of a Vision Transformer with only a modest loss in performance.

This does not establish that a conventional transformer is literally a composition of HyperBFs.

It supports a different and, in some ways, more interesting statement: competitive transformer-like systems can be constructed using associative-memory primitives throughout much of the architecture.

The experiment therefore provides evidence for the architectural sufficiency of the associative-memory viewpoint rather than for literal identity between standard transformer components and HyperBF modules.

2.8 Conclusion

The history of associative memory reveals a striking recurrence.

Early heteroassociative systems retrieve an output through correlation with a cue.

Nonlinear operator memories enlarge this idea through polynomial features and kernels.

RBF and HyperBF architectures replace global correlation by similarity in learned metric spaces.

Autoassociative networks add recurrent pattern completion.

Biological memory systems embed associative retrieval in plastic, hierarchical, and recurrent circuitry.

Modern attention transforms content-addressable retrieval into a learned, differentiable primitive repeatedly composed across many layers.

The historical lesson is not that all of these systems are instances of the same architecture.

It is that the computational motif

$$\boxed{\text{cue} \longrightarrow \text{content-dependent retrieval} \longrightarrow \text{use of the retrieved structure}}$$

reappears whenever a system must use past structure to interpret the present.

What changes across this history is the machinery surrounding that primitive: nonlinear similarity, learned representations, recurrence, feedback, hierarchy, and repeated composition.

Associative memory may therefore be less a complete model of intelligence than one of its elementary operations.

This leaves several theoretical questions unresolved. Why is relevant content reliably retrievable from a cue? How does training find a useful predictor with feasible resources? Why does the selected predictor generalize beyond the examples used to fit it?

The next chapter isolates the computational question. The Associative Turing Machine shows how associative reads and local updates can execute classical computations under the stated memory and representation assumptions. Executing a supplied program is not the same as discovering a useful program from data.

The Interlude then separates representation, predictive learning, and structural recovery. Chapter 4 develops compositional sparsity as a representation principle. Subsequent chapters examine accessible learning signals, supplied intermediate information, optimization geometry, and statistical selection. Genericity and staircase learning provide possible constructive mechanisms; trace exposure can assist learning rather than serving only as a fallback.

Chapter H brings these roles together. When training finds an accurate predictor whose description and finite-precision evaluation obey the stated polynomial resource bounds, the learned computation itself supplies a bounded-fan-in compositional realization. It need not reproduce the computation that generated the observations. This conclusion does not explain how training achieves accuracy; it clarifies what follows once that success and computational feasibility have been established.

The historical account therefore identifies a recurring primitive from which useful computations can be built. The learning theory must explain how such a computation is found and why it predicts well. Identification of the generating structure remains a stronger, separate objective.

CHAPTER 3

Associative Memories Are Turing-Complete

The previous chapters argued that attention can be understood as content-addressable associative retrieval. We now ask a sharper question: is associative retrieval, together with local state updates and writable working memory, computationally universal?

*We introduce an **Associative Turing Machine** (ATM), a computational model whose basic operations are associative read, associative write, and local update governed by a finite transition rule. We show that every deterministic Turing machine can be simulated with a constant number of these abstract operations per computational step. This operation count is distinct from the bit cost of implementing a memory access.*

The universality theorem is first stated for exact associative reads. We then show that softmax attention provides a differentiable approximation whose error can be made arbitrarily small when memory keys are sufficiently separated. Thus associative memory is not merely a mechanism for recall: when made writable and iterated under the stated memory conventions, it is sufficient for universal computation.

The theorem concerns execution of a supplied program. It does not explain how training discovers a useful predictor, why that predictor generalizes, or whether its internal computation reproduces the process that generated the observations. Those learning questions remain separate.

3.1 From Associative Recall to Universal Computation

An associative memory answers a simple question:

given a cue, what stored information is associated with it?

By itself, this is only a retrieval operation.

A computer requires more. The retrieved information must be allowed to modify a state, influence subsequent queries, and change working memory.

The minimal computational cycle is therefore

$$\boxed{\text{read} \rightarrow \text{local update} \rightarrow \text{write} \rightarrow \text{new query}}$$

repeated over time.

The central claim of this chapter is that this cycle is already enough for general computation.

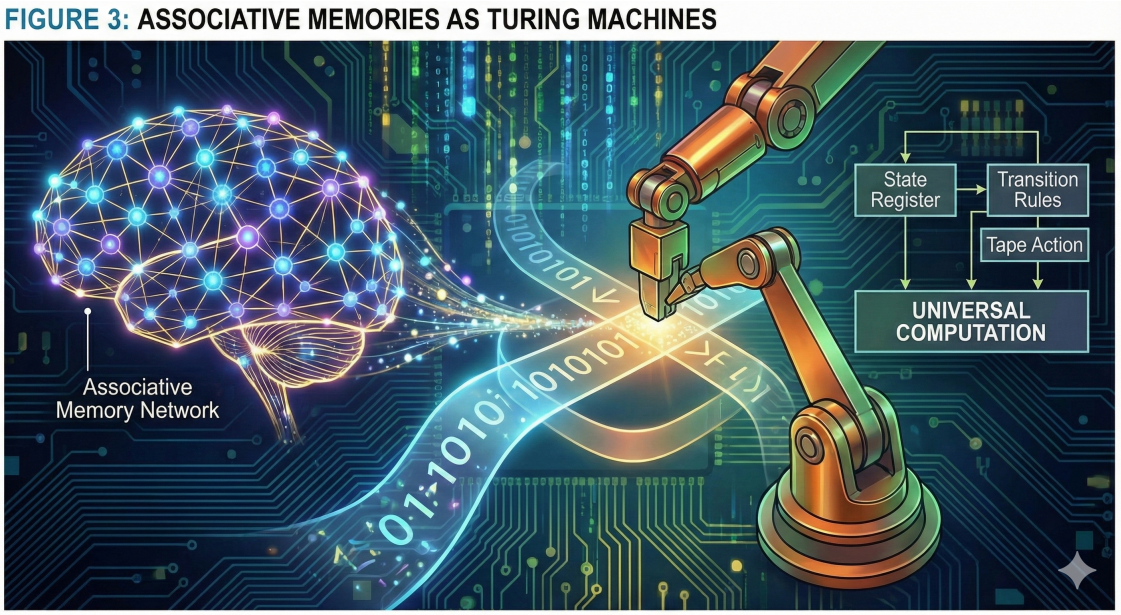


Figure 3.1: From a classical Turing tape to associative computation. The essential operations of a Turing machine—reading state, retrieving a transition rule, writing a symbol, and updating control—can be expressed through content-addressable memory interactions and local updates.

The construction is deliberately close to the operational structure of a classical Turing machine. A Turing machine needs only:

1. a finite control state;
2. a readable and writable tape;
3. a transition rule;
4. a head position.

An ATM implements the same ingredients through associative memory.

The transition table is stored in a fixed associative memory. The tape is stored in a dynamically writable associative memory. The finite controller performs only a local state update.

This avoids encoding an unbounded tape into arbitrary precision of a single real number. The growing tape contents reside explicitly in the associative store, and the address representation is counted as part of the workspace rather than treated as a fixed-precision resource.

3.2 Definition of the Associative Turing Machine

3.2.1 Control State

Let

$$Q$$

be a finite set of control states, and let

$$\Sigma$$

be a finite tape alphabet containing a blank symbol

$$\sqcup.$$

At computational step t, write the current control state and tape address together as

$$c_t = (q_t, h_t),$$

where

$$q_t \in Q$$

is the current machine state and

$$h_t \in \mathbb{Z}$$

is the current tape address.

Only q_t ranges over a fixed finite set. The pair c_t is not itself finite-state, because h_t is an integer address. The controller's transition rule has a fixed finite description, while the address is represented by a finite string whose length may grow. For an input of length n and a run of t steps, addresses can be represented with $O(\log(n + t + 2))$ bits. These bits belong to the explicitly counted workspace; they are not unbounded information hidden in a fixed-precision real coordinate.

3.2.2 Program Memory

The fixed program memory contains the transition table of the simulated Turing machine.

For a deterministic Turing machine with transition function

$$\delta : Q \times \Sigma \to Q \times \Sigma \times \{-1, +1\},$$

define the associative program memory below. Transition instructions are used only in nonhalting states; execution stops on entering the halting state, so entries associated with it need not be consulted.

$$\mathcal{M}_{\text{prog}} = \{(k_{q,s}, v_{q,s}) : (q, s) \in Q \times \Sigma\}.$$

The key encodes the pair (q, s),

$$k_{q,s} = B(q, s),$$

and the corresponding value encodes

$$v_{q,s} = \text{pack}\,(\delta(q, s)).$$

Thus the query

$$(q, s)$$

returns the transition instruction

$$(q', s', d).$$

Because Q and Σ are finite, these keys can be represented exactly by orthogonal basis vectors. Let

$$N_{\text{prog}} = |Q||\Sigma|.$$

We may choose

$$k_{q,s} = e_{(q,s)} \in \mathbb{R}^{N_{\text{prog}}},$$

so that

$$\langle k_{q,s}, k_{q',s'} \rangle = \begin{cases} 1, & (q,s) = (q',s'), \\ 0, & \text{otherwise.} \end{cases}$$

The transition table therefore admits exact associative addressing.

3.2.3 Working Memory

The work memory represents the Turing tape.

At time t, write

$$\mathcal{M}_{\text{work}}^{(t)}$$

for the finite collection of tape locations that have been written so far.

Conceptually, the memory implements a map

$$h \longmapsto T_t(h),$$

where

$$T_t(h) \in \Sigma$$

is the symbol currently stored at tape address h.

Addresses that have never been written return the blank symbol

$$\sqcup.$$

A tape read is therefore

$$s_t = \text{read}\left(h_t; \mathcal{M}_{\text{work}}^{(t)}\right).$$

A tape write replaces the value stored at address h_t:

$$\mathcal{M}_{\text{work}}^{(t+1)} = \text{write}\left(\mathcal{M}_{\text{work}}^{(t)}, h_t, s_t'\right).$$

This is the simplest abstract definition.

An equivalent implementation uses an append-only log. Instead of destructively replacing the previous value, one appends a record

$$(h_t, t, s_t').$$

A future read at address h retrieves the most recent entry associated with h.

The append-only formulation will be especially useful when we relate the ATM to autoregressive sequence models. Exact address matching, selection of the latest matching record, and the default blank response are part of this abstract tape interface. A concrete attention implementation must realize these operations; an arbitrary similarity read does not supply them automatically.

3.2.4 Associative Read

For the exact computational model, define a hard associative read.

Given query q and memory

$$\mathcal{M} = \{(k_i, v_i)\}_{i=1}^{N},$$

let

$$i^*(q) = \arg\max_i \langle q, k_i \rangle.$$

Then

$$\mathrm{read}(q; \mathcal{M}) = v_{i^*(q)}. \tag{3.1}$$

Whenever the maximizing key is unique, the read is deterministic.

The ATM theorem below uses this exact primitive together with the tape interface just specified. The finite program memory has the explicit orthogonal encoding above. The work-memory interface additionally requires an address encoding and the stated blank and update conventions.

Effective descriptions and operation counts. The symbolic construction uses finite, effectively manipulable representations of states, addresses, keys, and instructions. Reads use decidable comparisons on those representations. No arbitrary-real oracle is being supplied. One read or write counts as one abstract ATM operation; its implementation cost can depend on memory size and representation length. The resource interpretation of the theorem is discussed after its proof.

3.2.5 Associative Write

A writable associative memory supports an operation

$$\mathcal{M}' = \mathrm{write}(\mathcal{M}, k, v), \tag{3.2}$$

which ensures that subsequent reads at address k return v.

The internal implementation of the write is abstracted away in the computability theorem. It may be destructive replacement, append-only logging, or an equivalent effective mechanism preserving the current value at each address. That abstraction does not remove its storage or bit-level cost from an implementation-level efficiency claim.

3.2.6 Local Update

The controller receives

$$(q_t, h_t),$$

the tape symbol

$$s_t,$$

and the retrieved transition instruction

$$(q_{t+1}, s'_t, d_t).$$

The local update is simply

$$h_{t+1} = h_t + d_t,$$

together with the new finite state

$$q_{t+1}.$$

Thus one ATM computational step consists of:

1. read the tape symbol at h_t;
2. retrieve the transition instruction associated with (q_t, s_t);
3. write the new symbol s'_t at h_t;
4. update the control state to q_{t+1};
5. move the head to $h_t + d_t$.

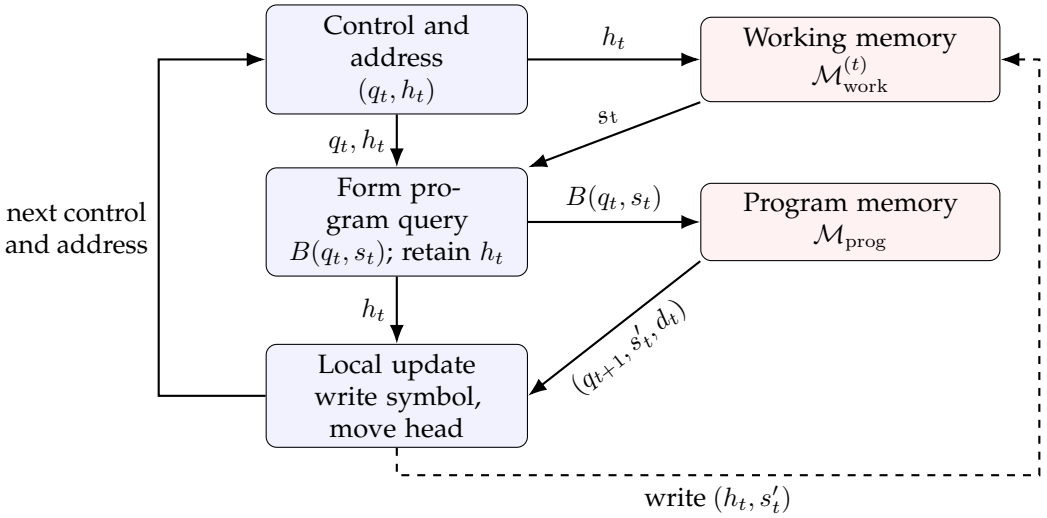


Figure 3.2: One step of an Associative Turing Machine. The current tape address is used to retrieve a symbol from working memory. The pair consisting of the control state and retrieved symbol addresses the program memory. The returned instruction determines the write operation and the next control state. Query formation is shown explicitly; the address is retained for the write and head update.

3.3 Why Extensible Memory Matters: RNNs, Precision, and Robust Computation

The motivation for introducing an explicit working memory is not that recurrent neural networks are incapable of universal computation in every mathematical model.

Indeed, idealized recurrent networks operating with exact real arithmetic can encode arbitrarily much information in a fixed-dimensional continuous state and, under appropriate constructions, can simulate Turing computation.

The relevant issue here is different:

Can unbounded computation be carried out robustly with fixed-dimensional, bounded, finite-precision internal state?

Under those physical restrictions, the answer is no for Turing-universal computation with indefinitely extensible workspace. A finite-state loop can of course run for arbitrarily many steps; runtime alone is not the unbounded memory resource at issue here.

3.3.1 Fixed-dimensional state with finite precision

Consider an autonomous recurrent system

$$h_{t+1} = F(h_t), \qquad h_t \in \mathbb{R}^d, \tag{3.3}$$

where d is fixed.

Assume that physically realizable states are bounded,

$$h_t \in [-B, B]^d,$$

and that two states separated by less than a resolution $\varepsilon > 0$ cannot be reliably distinguished.

Then the number of mutually separated state codes is finite. To obtain a finite-state computational model, one also needs the update to be well defined on those codes, rather than depend on unresolved differences inside a code's tolerance region.

Proposition 1 (Bounded finite-precision recurrence is finite-state)**.** *Let*

$$\mathcal{H} \subseteq [-B, B]^d$$

be the state space of a deterministic recurrent system.

Suppose distinct valid computational codes have representatives in $\mathcal{H}$ separated by at least $\varepsilon > 0$ in the ℓ_∞ norm. Assume also that the update, including any quantization, induces a well-defined deterministic transition on decoded codes: states representing the same code have the same decoded successor.

Then the number of codes is bounded by the finite packing number

$$N_{\text{robust}} \leq \left(1 + \frac{2B}{\varepsilon}\right)^d. \tag{3.4}$$

Consequently, at fixed B, d, and ε, the recurrent system has only finitely many robust computational states and is therefore equivalent, at this level of description, to a finite-state machine.

Proof. Around each code representative place an open ℓ_∞ ball of radius $\varepsilon/2$, that is, an open cube of side ε. The separation assumption makes these cubes disjoint. All of them lie in a cube of side $2B + \varepsilon$. For any finite collection of N code representatives, comparison of volumes gives

$$N\varepsilon^d \leq (2B + \varepsilon)^d.$$

Thus there can be at most $(1 + 2B/\varepsilon)^d$ codes, proving Equation (3.4).

By the decoded-transition assumption, the update acts deterministically on this finite code set. It can therefore be represented by a finite transition table, which is the claimed finite-state computation.

□

The important point is not recurrence itself.

It is the simultaneous assumption of

$$\boxed{\text{fixed dimension} + \text{bounded range} + \text{finite resolution.}}$$

3.3.2 How idealized recurrent systems evade the proposition

There are at least two mathematical ways for a fixed-dimensional system to avoid the finite-state conclusion.

Unbounded precision. A bounded real number can encode an arbitrarily long discrete sequence in its digits.

For example, a symbolic sequence

$$s_1 s_2 s_3 \cdots$$

can be represented schematically by

$$x = \sum_{j=1}^{\infty} s_j b^{-j}.$$

A recurrent system with exact real arithmetic can manipulate such encodings.

But as more symbols are stored, distinct computational states can become arbitrarily close.

Thus the required precision grows without bound.

Unbounded magnitude. Alternatively, one may use discrete but unbounded states, such as integer registers

$$h_t \in \mathbb{Z}^d.$$

A fixed number of unbounded integer registers can themselves support unbounded computation.

Here the resource that grows is numerical magnitude at fixed absolute resolution. The number of bits needed to store those integers still grows; unbounded integer registers are not fixed-bit hardware states.

Thus the correct general principle is

unbounded computation requires some computational resource
that is itself allowed to grow without a fixed bound.

Depending on the model, that resource may be

precision, state magnitude, or memory size.

3.3.3 Why use explicit memory?

The Associative Turing Machine chooses the third option.

Instead of encoding an arbitrarily large working memory inside a fixed continuous state, it separates

finite control state

from

extensible writable memory.

The fixed control logic manipulates only the information required for the current computational step:

control state, query/address, retrieved local data.

The growing tape record is stored externally. Addresses and any timestamps also have finite encodings whose lengths grow as needed; this does not turn the finite control rule into an unbounded-precision real operation.

Schematically,

fixed control logic with explicitly represented addresses
$+$
extensible addressable memory
$\Downarrow$
unbounded computational workspace.

This is the same architectural separation used by an ordinary computer: the processor is finite, while the amount of memory available to a computation may grow with the computation.

3.3.4 Memory hierarchy and computational power

The distinction can be expressed abstractly in terms of the memory available to a finite controller.

finite controller only	$\longrightarrow$	finite-state computation,
finite controller + extensible stack	$\longrightarrow$	pushdown-style computation,
finite controller + extensible read/write memory	$\longrightarrow$	Turing computation.

The neural nature of the controller is secondary.

What changes the computational class is the structure and extensibility of the memory resource.

3.3.5 Robustness rather than idealized expressivity

This also clarifies the role of the ATM theorem.

The claim is not that exact-real recurrent networks are mathematically incapable of Turing universality.

Rather, the ATM provides a representation of Turing computation in which unbounded information is not hidden in infinitely precise coordinates of a fixed-dimensional state.

Instead, information is stored explicitly in a memory whose size can grow with the computation.

Thus

> ATM universality is based on extensible explicit memory,
> not on infinite precision in a fixed state vector.

3.3.6 Execution versus trainability

A final distinction is important.

Problems such as vanishing or exploding gradients concern the difficulty of *learning* recurrent dynamics by backpropagation through time.

They are not the fundamental reason that a bounded finite-precision recurrent state has limited memory capacity.

A recurrent machine could be perfectly trained and still possess only a finite number of robust states.

Conversely, a mathematically universal recurrent construction may be extremely difficult to learn.

Therefore,

$$\boxed{\text{computational expressivity} \neq \text{learning or generalization.}}$$

The universality construction supplies a program and an encoding, then proves that execution follows the intended transitions. A learning procedure receives observations and must find a useful predictor. Its success additionally concerns performance beyond those observations.

As distinguished in the Interlude and Chapter H, predictive recovery does not require identifying the program that generated the data. Matching that program's hidden states or transitions is a stronger structural-recovery objective. Reliable execution of a supplied program, predictive learning, and structural identification are therefore separate claims.

3.3.7 The design principle

The architectural motivation for the ATM can therefore be summarized as

> do not compress an indefinitely growing working memory
> into a fixed robust state when that memory can instead
> be represented explicitly and addressed directly.

The ATM implements precisely this separation:

$$\boxed{\text{associative read} + \text{local update} + \text{extensible writable memory.}}$$

The following sections show that these primitives are sufficient to simulate ordinary Turing computation with only polynomial overhead.

3.4 Simulation of a Turing Machine

Let

$$T = (Q, \Sigma, \delta, q_0, q_{\text{halt}})$$

be a deterministic single-tape Turing machine.
We construct an ATM

$$A_T$$

that simulates it step by step.

3.4.1 Encoding the Transition Table

For every pair

$$(q, s) \in Q \times \Sigma,$$

store in program memory

$$(B(q, s), \text{pack}(\delta(q, s))) .$$

With the orthogonal encoding described above, each valid pair has a unique key.
Thus the program read

$$\text{read}\,(B(q_t, s_t); \mathcal{M}_{\text{prog}})$$

returns exactly

$$\delta(q_t, s_t).$$

3.4.2 Encoding the Tape

Suppose the initial Turing tape contains input string

$$x = x_1 x_2 \cdots x_n.$$

Initialize

$$\mathcal{M}_{\text{work}}^{(0)}$$

with

$$(1, x_1), (2, x_2), \ldots, (n, x_n).$$

All other addresses return

$$\sqcup.$$

The ATM head position is initialized to the same location as the Turing machine head.

At every later step, the work memory contains exactly the current tape contents of the simulated machine.

3.4.3 One Simulated Step

Assume that at time t the ATM represents the same configuration as the Turing machine.

First retrieve the current tape symbol:

$$s_t = \text{read}\left(h_t; \mathcal{M}_{\text{work}}^{(t)}\right).$$

Then retrieve the corresponding transition instruction:

$$(q_{t+1}, s'_t, d_t) = \text{read}\left(B(q_t, s_t); \mathcal{M}_{\text{prog}}\right).$$

Write the new symbol:

$$\mathcal{M}_{\text{work}}^{(t+1)} = \text{write}\left(\mathcal{M}_{\text{work}}^{(t)}, h_t, s'_t\right).$$

Finally update the head:

$$h_{t+1} = h_t + d_t.$$

The resulting ATM configuration is exactly the next configuration of the Turing machine.

3.5 Main Theorem

Theorem 1 (Turing Equivalence of Associative Computation)**.** *For every deterministic single-tape Turing machine T, there exists an Associative Turing Machine A_T such that, for every input x,*

$$A_T(x)$$

halts if and only if

$$T(x)$$

halts, and both machines produce the same output.

One step of T is simulated by a constant number of associative reads, writes, and local updates. Consequently, if T halts after t steps, then A_T halts after

$$O(t)$$

ATM operations.

3.5.1 Proof

The proof is by induction on the simulated time step.

At time $t = 0$, the control state, head location, and tape contents of A_T are initialized to match those of T.

Assume that after t simulated steps the ATM and the Turing machine have identical configurations.

The work-memory read at address h_t therefore returns the same tape symbol s_t observed by the Turing machine.

The program-memory read at key

$$B(q_t, s_t)$$

returns exactly

$$\delta(q_t, s_t) = (q_{t+1}, s'_t, d_t).$$

The ATM writes s'_t to address h_t, enters state q_{t+1}, and moves the head to

$$h_t + d_t.$$

These are precisely the operations performed by one step of the Turing machine.

Therefore the configurations remain identical after step $t + 1$.

By induction, the ATM reproduces the entire computation of T, including its halting behavior and output.

Each simulated step uses a constant number of primitive ATM operations, so the simulation requires $O(t)$ steps.

□

What the overhead counts. The $O(t)$ bound counts abstract ATM primitives after input initialization, for a fixed simulated machine T. Storing an input of length n costs $O(n)$ initial writes, so including this stage gives $O(n + t)$ primitive operations. A run needs only $O(n + t)$ tape records, or that many records in the append-only implementation; addresses and timestamps use $O(\log(n + t + 2))$ bits each.

An implementation of a read may nevertheless inspect many records. In particular, scoring every stored key is not a constant-bit-cost operation as memory grows. The theorem's constant primitive overhead is not a constant physical-time or bit-operation overhead for dense attention.

Conversely, the symbolic tape interface in this construction can be implemented on an ordinary Turing machine by keeping finite records, scanning for matching addresses, and applying the finite transition table. This gives an effective classical simulation with polynomial overhead in the input length and simulated runtime. The computational equivalence is therefore about effective symbolic machines, not machines endowed with arbitrary real-valued oracles.

3.5.2 Corollary: A Universal ATM

A universal Turing machine is itself a deterministic Turing machine.

Applying Theorem 1 to such a machine gives a single universal ATM that, given an encoding

$$\langle T, x \rangle,$$

can simulate arbitrary deterministic Turing machines.

Thus

Associative Turing Machines are computationally universal.

3.6 Soft Attention as a Differentiable Approximation

The theorem above uses exact associative reads.

Modern attention uses softmax retrieval instead.

We now show that the exact ATM read can be approximated arbitrarily well by soft attention whenever the correct key is separated from the others by a positive margin. The estimates below concern a single retrieval. Maintaining valid encoded states and accurate computation over many steps requires additional control of the updates, writes, and successive query margins.

3.6.1 Softmax Read

Given normalized query and keys,

$$\|q\| = \|k_i\| = 1,$$

define

$$\mathrm{read}_\tau(q; \mathcal{M}) = \sum_{i=1}^{N} \alpha_i(q) v_i, \tag{3.5}$$

where

$$\alpha_i(q) = \frac{\exp(\langle q, k_i \rangle / \tau)}{\sum_{j=1}^{N} \exp(\langle q, k_j \rangle / \tau)}.$$

The parameter

$$\tau > 0$$

controls the sharpness of retrieval.
As

$$\tau \to 0,$$

the distribution concentrates on the maximally similar key.

3.6.2 General Margin Bound

For this bound take $N \geq 2$ and an error tolerance $0 < \epsilon < 1$. With a single stored item, the read is already exact.

Assume the correct key k_* satisfies

$$\langle q, k_* \rangle - \max_{j \neq *} \langle q, k_j \rangle \geq \gamma$$

for some margin

$$\gamma > 0.$$

Then

$$\alpha_* = \frac{1}{1 + \sum_{j \neq *} \exp\left(-\frac{\langle q, k_* \rangle - \langle q, k_j \rangle}{\tau}\right)}.$$

Therefore

$$\alpha_* \geq \frac{1}{1 + (N-1)e^{-\gamma/\tau}}.$$

Hence

$$1 - \alpha_* \leq (N-1)e^{-\gamma/\tau}. \tag{3.6}$$

To ensure

$$1 - \alpha_* \leq \epsilon,$$

it suffices to choose

$$\tau \leq \frac{\gamma}{\log((N-1)/\epsilon)}.$$

Thus the soft associative read approaches the exact read exponentially fast in

$$\gamma/\tau.$$

3.6.3 Orthogonal-Key Case

For the program memory, we may choose orthogonal keys.

If

$$q = k_*,$$

then

$$\langle q, k_* \rangle = 1$$

and

$$\langle q, k_j \rangle = 0 \qquad (j \neq *),$$

so

$$\gamma = 1.$$

The correct attention weight is then exactly

$$\alpha_* = \frac{e^{1/\tau}}{e^{1/\tau} + N - 1}.$$

Therefore

$$1 - \alpha_* = \frac{N-1}{e^{1/\tau} + N - 1}.$$

As

$$\tau \to 0,$$

we obtain

$$\alpha_* \to 1.$$

Thus

softmax attention can approximate the exact ATM read arbitrarily closely.

3.6.4 From Retrieval Error to State Error

Suppose the stored values satisfy

$$\|v_i\| \le R.$$

If the correct value is v_*, then

$$\mathrm{read}_\tau(q;\mathcal{M}) - v_* = \sum_{i\neq *} \alpha_i (v_i - v_*).$$

Hence

$$\|\mathrm{read}_\tau(q;\mathcal{M}) - v_*\| \le 2R(1-\alpha_*).$$

Using Eq. (3.6),

$$\|\mathrm{read}_\tau(q;\mathcal{M}) - v_*\| \le 2R(N-1)e^{-\gamma/\tau}.$$

Thus the finite-temperature read error is explicitly controlled. This is the error in the value delivered to the controller, before any subsequent state update. A small value error does not by itself guarantee the same symbolic transition, particularly near a decoding boundary.

For a finite simulated run, one must also control accumulation through the updates, preserve the separation needed for later reads, and specify how approximate values are decoded and written. As the memory grows, N, the available margins, or the required temperature may change. The displayed bound therefore does not imply exact arbitrarily long computation at one fixed temperature and fixed numerical resolution.

This separates two questions that are sometimes conflated:

exact associative machine: exact symbolic computation,
soft attention realization: differentiable approximation.

3.7 Relation to Transformer Computation

The ATM is an abstract computational model.

Transformer components provide natural realizations of several of its primitives, but the correspondence should not be interpreted as a literal identity between a fixed Transformer block and an unbounded Turing machine.

3.7.1 Queries, Keys, and Values

In a standard self-attention layer,

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V.$$

The learned matrices

$$W_Q,\ W_K,\ W_V$$

do not themselves constitute the contextual associative memory.

Rather, they parameterize the transformations that generate queries, keys, and values from the current representations.

The memory available to attention is the set of contextual key–value pairs

$$\{(k_i, v_i)\}.$$

The attention operation

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

therefore implements a differentiable associative read over the current context.

3.7.2 Residual State Update

A transformer block combines associative retrieval with a local representation update.

Ignoring normalization details, one may write schematically

$$z_t = x_t + \mathrm{Attn}(x_t),$$

followed by

$$x_{t+1} = z_t + \mathrm{FFN}(z_t).$$

Thus the ATM update primitive

$$M_1$$

corresponds more naturally to the complete residual transformation surrounding attention and the feed-forward network than to the FFN alone.

The computational motif is

retrieve information $\rightarrow$ combine it with state $\rightarrow$ transform state.

3.7.3 Append-Only Working Memory

A standard transformer does not directly overwrite earlier token representations in the manner of writable RAM.

Autoregressive systems, however, naturally possess an append-only mechanism.

Suppose a tape write at address h_t produces symbol s'_t.

Instead of erasing the old entry, append a record

$$(h_t, t, s'_t).$$

Later reads at address h_t must retrieve the most recent matching record. This requires an explicit latest-match rule using the encoded addresses and timestamps. Appending tokens alone does not establish that a learned attention rule implements this selection.

The tape history becomes

$$(h_1, t_1, s_1), (h_2, t_2, s_2), \ldots$$

and the current tape state is obtained by selecting, for each address, the latest associated value.

This provides a natural bridge between writable associative memory and autoregressive sequence computation:

$$\boxed{\text{Turing tape write} \longleftrightarrow \text{append a new address–value record.}}$$

The mechanism is conceptually similar to log-structured memory in computer systems: old states remain in the history, while the newest entry determines the current value.

3.7.4 What the Correspondence Does and Does Not Say

The ATM theorem establishes that associative read, write, and local update are sufficient for universal computation.

It does *not* imply that every fixed Transformer is itself an unbounded Turing machine.

A Transformer of fixed depth and fixed context performs only a finite amount of computation in a single evaluation. Repeated use can extend runtime, but an unbounded simulation still needs the corresponding memory and representation resources.

To simulate arbitrarily long Turing-machine computations, some resource must grow with the simulated runtime, for example:

- sequence length;
- recurrent application of a block;
- depth;
- external memory;
- dynamically expanding working memory.

Thus the appropriate conclusion is not

$$\text{every Transformer} = \text{Turing machine}.$$

Rather,

$$\boxed{\begin{array}{c}\text{Transformer-like associative primitives can realize the} \\ \text{operations required by a Turing-equivalent machine.}\end{array}}$$

3.8 Why Associative Access Matters

Turing completeness by itself is not unique to associative systems.

Idealized recurrent neural networks are also known to admit universal computational constructions.

The significance of the ATM is therefore not that it is the first neural model capable of universal computation.

The difference lies in the memory primitive.

A conventional recurrent system compresses the computational past into a recurrent hidden state.

An ATM exposes memory explicitly through content-addressable access.

ATM Primitive	Transformer-Like Realization	Computational Role
Query Formation	Query projection	State-to-address map
Associative Memory	Contextual keys and values	Content-addressable storage
Read	Softmax attention	Associative retrieval
Local Update	Residual + MLP transformation	State transition
Write	Append new state/token	Working-memory update
Iteration	Autoregressive or recurrent execution	Repeated computation

Table 3.1: Structural correspondence between ATM primitives and transformer-like computational mechanisms. The table describes a functional analogy, not an identity between a fixed Transformer block and a complete unbounded ATM.

This changes the computational organization from

$$\text{everything encoded in recurrent state}$$

to

$$\text{finite controller} + \text{explicit associative working memory}.$$

The distinction is especially relevant for modern architectures because attention provides direct access to distributed memories according to content rather than requiring all relevant information to be propagated sequentially through a fixed-dimensional hidden state.

Thus the main structural lesson is

$$\boxed{\text{associative access provides an explicit computational memory interface.}}$$

3.9 Discussion

The Turing-machine simulation proved in this chapter is intentionally simple.

That simplicity is the point.

A classical Turing machine requires only a finite controller, a readable and writable memory, and repeated local transitions.

Replacing pointer-based tape access by content-addressable retrieval does not reduce computational power.

Indeed,

$$\boxed{\text{associative read} + \text{associative write} + \text{local finite control} = \text{universal computation}.}$$

This result clarifies the role of associative memory in the larger argument of this book.

Associative memory is not merely a mechanism for retrieving stored examples.

Once retrieval is allowed to affect subsequent queries and working memory, association becomes computation.

The exact ATM makes this statement at the level of classical computability, with the effective symbolic representations and abstract operation costs specified above.

Softmax attention provides a differentiable approximation to an individual exact read, with an explicit error bound determined by key separation, memory size, temperature, and value scale. A

complete soft simulation requires the additional trajectory-level control discussed in the preceding error analysis; it is not a consequence of the one-read estimate alone.

The resulting conceptual progression is

$$\boxed{\begin{array}{c}\text{associative recall} \to \text{associative working memory} \\ \Downarrow \\ \text{iterated read–update–write} \to \text{Turing computation.}\end{array}}$$

This establishes a computational possibility, not an efficient-learning theorem. The simulator is supplied with the program to execute. Its universality does not explain how observations lead a learning procedure to a useful program, or why the resulting predictor works beyond the observed examples.

Nor is universality itself an efficiency claim about the target task. The ATM can simulate a long or expensive computation as well as a short one. Chapter 4 addresses a more restricted question: what representation follows when the target can be evaluated with polynomial digital resources? Under the specified finite-precision conventions, unrolling such a computation yields a polynomial-size bounded-fan-in graph. Smooth coarse-grained approximation requires further regularity and stability assumptions; it is not supplied by Turing universality alone.

The Interlude separates representation from two meanings of recovery. Predictive recovery asks for a feasible function that is accurate on the population of interest. Structural recovery asks, in addition, for identification of a designated generating computation up to controlled equivalences. A successful predictor may execute a different program and use different intermediate variables from those that generated the data. That difference is not, by itself, a predictive failure.

The subsequent chapters therefore ask three distinct questions:

Representation:	Is an accurate feasible predictor available?
Optimization:	Can the specified training procedure find it?
Generalization:	Does the selected predictor work off the sample?

Accessible footprints, staircase learning, and intermediate traces offer possible routes to successful learning under their respective assumptions. They are not conclusions of the ATM theorem, and matching the generating program is not an additional requirement for predictive success.

Chapter H makes the resulting witness argument precise. When training returns an accurate predictor whose finite description and numerical evaluation satisfy the stated polynomial resource bounds, that predictor's own computation supplies a bounded-fan-in compositional realization. No separate theorem matching its internal states to those of the teacher is needed for that conclusion. Explaining how training finds such a predictor, and controlling its population error, remain substantive learning problems.

The bridge from the associative-memory chapters to the learning theory is therefore

$$\boxed{\begin{array}{c}\text{Executing a supplied program is a computational result.} \\ \text{Learning an accurate feasible predictor is a further problem.} \\ \text{Identifying the generating computation is a stronger objective.}\end{array}}$$

These distinctions preserve the significance of the universality theorem without asking it to prove learnability, generalization, or structural identification.

Part I: Principles and Foundations

Interlude: Representation and Recovery

The chapters that follow are organized around a distinction that is easy to state and easy to blur: the distinction between *representation* and *recovery*.

A representation question asks what resources are required to express, evaluate, or approximate a function. A learning question asks whether a specified procedure can discover an accurate predictor from observations, with feasible data and computation.

These are different questions. The first concerns the existence and cost of a computation. The second concerns how a useful computation is found and why it predicts beyond the observed examples.

Compositional sparsity, the subject of the next chapter, addresses the first question. By itself, it does not answer the second.

Two meanings of recovery. There is a further distinction within the learning problem. *Predictive recovery* means finding a function that predicts accurately on the population of interest. For example, in noiseless regression with $Y = f(X)$ and $X \sim P_X$, the objective may be

$$\|\widehat{f} - f\|_{L^2(P_X)} \leq \varepsilon.$$

The learner need not reproduce the particular intermediate variables or factorization that generated the observations.

Structural recovery asks for more: identification of a specified hidden computation, its constituent functions, or its intermediate variables, up to explicitly controlled equivalences. This is an important objective when the aim includes understanding the generating mechanism. It requires a separate criterion of success and suitable identifiability assumptions.

Predictive accuracy does not, by itself, establish structural recovery. Conversely, a learned representation can differ from the generating one without being a predictive failure. The issue is not merely that a sparse decomposition may be nonunique. Predictive learning need not establish a correspondence with any particular generating decomposition at all.

The learned computation as a witness. Suppose training returns a deep network that predicts accurately and can be evaluated with feasible computational resources. At the relevant finite precision, the trained network is itself a computation admitting a bounded-fan-in implementation. It therefore supplies a compositional realization of the learned predictor:

accurate population prediction
$+$ feasible computational implementation
$\Longrightarrow$
a compositionally sparse predictor.

Here *feasible* is an asymptotic requirement, not merely a synonym for finite. Model description, evaluation time, encoding, and numerical precision must obey the stated resource bounds as the problem size and accuracy requirements change. Finding the predictor must also be efficient if an efficient-learning claim is being made.

Chapter H develops this witness argument. It removes the need for an additional structural-identification theorem after predictive success has been established. It does not explain, by itself, why that success occurs.

Representation still does not imply learnability. A network may be able to represent a computation without there being a known efficient procedure for discovering useful parameters from examples. Likewise, a compact target representation does not ensure that training selects a predictor with sufficiently small statistical complexity. Existence, optimization, and generalization remain separate questions.

Parity provides a simple illustration of the first gap. For $S \subseteq [d]$, the function

$$\chi_S(x) = \prod_{i \in S} x_i, \qquad x \in \{-1, +1\}^d,$$

can be computed by a tree of two-input operations. Yet, under the uniform product distribution, when $|S| > 1$,

$$\mathbb{E}[\chi_S(X) X_j] = 0 \qquad \text{for every } j \in S.$$

A learning mechanism relying on these linear correlations lacks that particular signal. This is not an impossibility statement about all learning procedures or all gradient-based parameterizations. It shows that an economical representation need not be visible to the statistics used by a specified learner.

The framework retains the labels C1, C2, and C3 to distinguish three useful roles. They should not be read as a universal sequence of necessary and sufficient conditions for learning.

C1 — Compositional sparsity. The target admits an economical realization, or an approximation at the required resolution, through a graph of bounded-arity constituents. For efficiently computable digital targets, a polynomial-size bounded-fan-in realization follows from the computation itself.

This removes a representation-size obstruction: a manageable computational object exists. It does not make the space of possible objects easy to search, nor guarantee that every proposed architecture contains an equally economical realization.

For real-valued approximation, favorable rates governed by local arity require additional regularity and quantitative stability of the constituents. A bounded-fan-in digital implementation alone does not establish a fixed smooth coarse-grained decomposition with uniformly controlled Sobolev norms.

Developed in Chapter 4.

C2 — Detectable footprints and genericity. The observed learning problem exposes useful statistical signals to the specified learner. In the mechanisms studied here, these signals often take the form of low-complexity correlations. Their usefulness depends jointly on the target, the input distribution, the current representation, and the probes accessible to the learning rule.

Genericity provides one proposed explanation for why such footprints persist or appear under perturbations. A nonzero footprint is not enough: its magnitude must permit detection with feasible resources, and the learning updates must be able to exploit it.

A *staircase* is a particular way of organizing these signals: learning one feature creates a coordinate that exposes another useful dependency. Establishing such a mechanism requires an argument about optimization, preservation of learned features, and accumulated error. C2 is not itself a theorem that the entire hidden computation is recovered, and a staircase is not a required trajectory of every successful learner.

Developed in Chapter 6.

C3 — Trace exposure. The observations supply intermediate information that can simplify the learning problem. With a faithful full trace, the parent values and outputs of constituent computations are directly available, allowing local supervised problems to be posed. Partial traces or useful surrogates expose less of the computation but may still assist learning.

Trace exposure changes what is observed. It can help even when end-to-end learning is possible; it is not only a fallback for failure of a genericity condition. Conversely, the availability of intermediate targets does not by itself guarantee that each local problem is learnable, adequately sampled, or stable under recomposition. An externally emitted explanation is not automatically a faithful trace of the generating computation.

The division of labor is therefore

C1:	a feasible representation exists;
C2:	useful learning signals are accessible;
C3:	intermediate information is supplied.

These roles can cooperate. A learner may exploit detectable signals, supplied traces, both, or another mechanism appropriate to its problem. The failure of one proposed mechanism does not imply that no feasible predictor can be learned.

The three questions for predictive learning. Chapter H organizes the synthesis around three questions:

Q1 : Does the chosen class contain a feasible accurate predictor?

Q2 : Can the specified training procedure find a suitable fit?

Q3 : Does the selected predictor generalize?

C1 informs the first question. Detectable signals and trace exposure offer routes toward the second. Statistical complexity and the selection induced by training are central to the third. These roles are related, but none should be silently substituted for another.

The later optimization arguments study overparameterized networks, normalization, stochastic dynamics, and norm- or product-norm-based regularization. Geometry may make low-loss solutions accessible; regularization may favor predictors for which statistical-complexity bounds are informative. Neither effect is automatic. Equilibrium preference for a low-loss region is not a finite-time convergence theorem, and a regularization penalty is not itself a guarantee of generalization.

The predictive argument is conditional: when training achieves adequate empirical fit within a suitably controlled hypothesis class, a valid generalization bound can imply small population error. Exact interpolation is an important regime, not a necessary definition of predictive success. If the resulting predictor is also computationally feasible, it supplies the compositional witness described above.

There is no further requirement that its internal variables match the teacher's. Structural-recovery theorems remain valuable because they establish a stronger conclusion and can provide constructive explanations of how learning succeeds.

From associative computation to the next chapter. The distinction also clarifies the role of the preceding chapter. The Associative Turing Machine establishes computational universality under its stated model: associative read, write, and local update suffice to implement classical computation. Universality concerns what can be executed when a program is supplied. It does not show how a useful program is discovered from data, and predictive learning need not discover the same program that generated those data.

The next chapter develops C1 on its own terms. It asks what structure follows from efficient computation and, under the required analytic assumptions, what that structure buys for approximation. The subsequent chapters examine learning signals, intermediate supervision, optimization geometry, and statistical selection. Chapter H brings these questions together without identifying a successful predictor with a privileged hidden decomposition.

The separation is deliberate:

First ask what can be represented.
Then ask how an accurate predictor can be learned.
Ask separately whether the generating structure is recovered.

CHAPTER 4

Efficient Computability and Compositional Sparsity

This chapter develops the first structural principle of the book: compositional sparsity.

Its starting point is classical computational complexity. Every polynomial-time digital computation can be unfolded into a polynomial-size computation graph whose elementary operations have bounded fan-in. In this precise sense,

$$\boxed{\textit{efficient digital computation} \Longrightarrow \textit{compositional sparsity}.}$$

For Boolean and finite-precision computation this implication is direct and unconditional.

For real-valued functions, additional assumptions concerning precision, continuity, conditioning, and constituent regularity are required before the computational statement can be converted into quantitative approximation rates.

The central point is about representation*, not recovery.*

Compositional sparsity explains why efficiently computable targets admit deep polynomial-size representations and why, for sufficiently regular constituents, approximation rates can depend on local arity rather than ambient dimension.

It does not explain how a specified learning procedure finds an accurate predictor from input–output data, or why that predictor generalizes. Identifying the particular hidden constituents that generated the data is a stronger, separate objective.

The later chapters study learning signals, intermediate supervision, optimization, and statistical selection. Chapter H shows how a feasible predictor that generalizes supplies a compositional realization without requiring recovery of a privileged generating graph.

4.1 Representation, Not Recovery

Two questions are easily conflated:

$$\boxed{\text{representation} \neq \text{recovery}.}$$

The present chapter concerns the first. As distinguished in the Interlude, *predictive recovery* seeks a function that is accurate on the population of interest; *structural recovery* additionally seeks

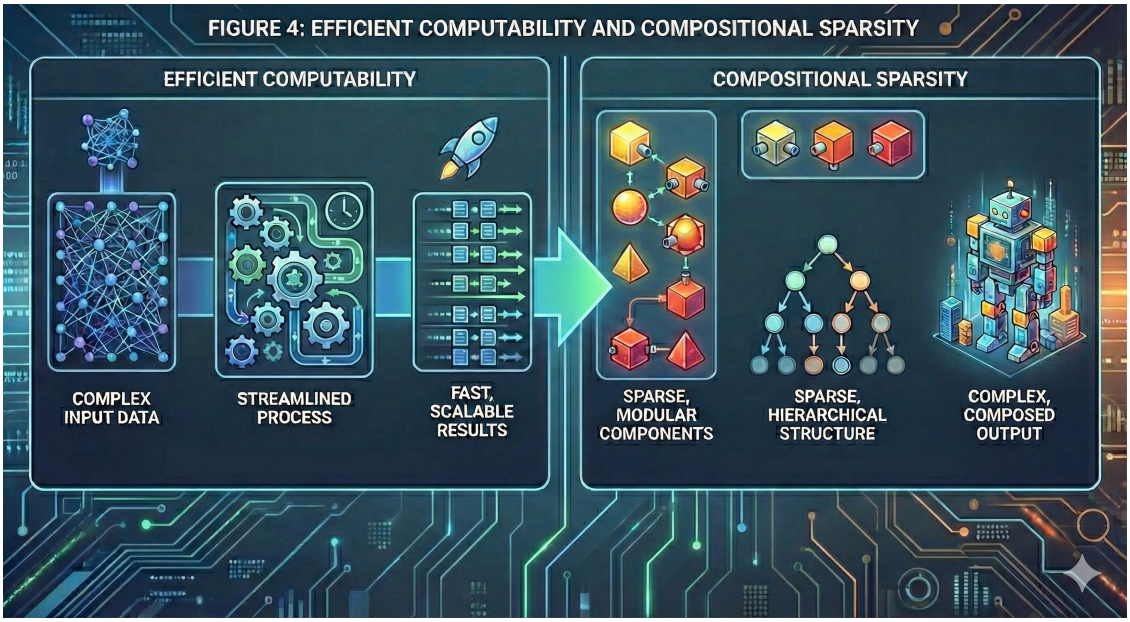


Figure 4.1: Efficient computation and compositional structure. A polynomial-time computation can be unfolded into a polynomial-size directed acyclic graph of bounded-fan-in elementary operations. Deep sparse networks provide a natural parametric realization of such computation graphs.

a specified generating computation, up to controlled equivalences. Neither conclusion follows merely from the existence of an economical representation.

Given a function f, we ask:

> What structural form must f possess if it can be evaluated using polynomial computational resources?

For discrete computation the answer is particularly simple.

A polynomial-time Turing machine performs only a polynomial number of elementary local operations.

Unrolling those operations in time produces a polynomial-size computation graph.

Because the machine possesses a finite transition rule, every elementary operation depends on only a bounded number of variables.

Hence

$$\boxed{\text{efficient digital computation} \Longrightarrow \text{polynomial-size bounded-fan-in computation.}}$$

We call this structural property *compositional sparsity*.

The terminology emphasizes two facts.

First, the target is compositional: its output is produced by composing intermediate functions.

Second, it is sparse: every individual constituent depends on only a small number of inputs.

Schematically,

$$f(x_1, \ldots, x_d) = \Phi_G(x_1, \ldots, x_d), \tag{4.1}$$

where G is a directed acyclic graph and each internal node has at most

$$k = O(1)$$

parents, independently of the ambient input dimension d.

The graph itself may be large, but for efficient computation its size is only polynomial in the input and precision parameters.

4.1.1 Why the distinction matters

A target can possess an extremely simple compositional representation and still be difficult to learn from examples.

Parity provides the canonical illustration.

For Boolean inputs,

$$\mathrm{Parity}(x_1, \ldots, x_d) = x_1 \oplus x_2 \oplus \cdots \oplus x_d \tag{4.2}$$

can be computed by a binary tree of two-input XOR gates.

Its local arity is

$$k = 2,$$

its size is linear in d, and its depth can be logarithmic under parallel composition.

Thus parity has an excellent C1 representation.

Yet under symmetric input distributions, correlations with many low-degree probes vanish.

A learner whose gradient signal is controlled by such probes may therefore receive little useful information about which variables should be combined.

This is a limitation of the specified probes, not an impossibility statement about all learning procedures or all network parameterizations. A different procedure might learn an accurate parity predictor without reconstructing a particular XOR tree.

The representation result supplies an economical computation. The learning question is whether the specified procedure finds a feasible predictor that works beyond the observed examples. Recovering the particular tree that generated those examples would be an additional structural conclusion.

4.1.2 The central structural implication

For Boolean computation,

polynomial-time Turing computation
$\Downarrow$
P-uniform polynomial-size bounded-fan-in circuit
$\Downarrow$
compositionally sparse representation.

For continuous functions the corresponding statement requires qualification:

efficient finite-precision computation
+ controlled precision requirements
+ regular low-arity constituents
+ stable composition
⇓
deep approximation with complexity controlled locally.

These two claims should remain distinct.

4.2 Definitions and Model Conventions

4.2.1 Boolean base

For discrete computation, internal nodes are gates drawn from a fixed finite basis such as

$$\{\mathrm{AND}, \mathrm{OR}, \mathrm{NOT}\},$$

with bounded fan-in and unrestricted fan-out.
Any standard finite universal basis could be used.
The particular choice changes only constant factors.

4.2.2 Real-valued base

For neural realizations, a typical constituent has the form

$$z \mapsto \sigma(w^\top z + b), \tag{4.3}$$

with

$$\sigma(u) = \max\{0, u\},$$

and with the number of arguments of z bounded by k.

All weights are represented at explicitly finite precision whenever a digital computational claim is being made.

4.2.3 Efficient computability: Boolean functions

Definition 1 (Efficient computability; Boolean)**.** *Let*

$$f_n : \{0,1\}^n \to \{0,1\}.$$

The family $\{f_n\}_{n\geq 1}$ is efficiently computable if there exists a deterministic Turing machine M and polynomial p such that, for every

$$x \in \{0,1\}^n,$$

the machine outputs $f_n(x)$ in at most

$$p(n)$$

steps.

Equivalently, the family admits a P-uniform polynomial-size Boolean circuit family.

Uniformity matters.

Without it, an arbitrary circuit family can encode non-computable information directly in its wiring.

4.2.4 Efficient computation of real-valued functions

For real-valued functions, exact real arithmetic is not an appropriate resource model for the present purpose.

We instead count bits of requested precision.

Let

$$\varepsilon = 2^{-r}.$$

Definition 2 (Efficient finite-precision computation)**.** *Let*

$$f : [0,1]^d \to \mathbb{R}^m.$$

We say that f is efficiently computable in the bit model if an algorithm can, given access to a sufficiently accurate dyadic representation of x and a precision request r, produce y satisfying

$$\|y - f(x)\| \leq 2^{-r} \tag{4.4}$$

using time and output description length polynomial in the relevant input description length, d, and r.

For continuous-domain statements, we additionally require that the number of reliable input bits required to produce r reliable output bits grow at most polynomially in these parameters.

The last requirement can be formulated using a polynomially controlled modulus of continuity.

These precision assumptions are essential.

A function may be formally computable while requiring exponentially many input bits to determine a modest number of reliable output bits.

4.2.5 Computation DAGs

Definition 3 (Computation DAG)**.** *A computation DAG*

$$G = (V, E)$$

contains input nodes, internal nodes, and output nodes.

Every internal node v has a parent set

$$\mathrm{pa}(v)$$

satisfying

$$|\,\mathrm{pa}(v)| \leq k,$$

and computes a constituent function

$$\phi_v.$$

The size of the graph is

$$s(G) = |\mathsf{Int}|,$$

and the depth is the length

$$L(G)$$

of its longest input–output path.

4.2.6 Compositional sparsity

Definition 4 (Exact compositional sparsity)**.** *Fix a node class $\mathcal{F}_{\leq k}$.*

A function f is exactly (k, s, L)-compositionally sparse if there exists a computation DAG G of fan-in at most k, size at most s, and depth at most L, with

$$\phi_v \in \mathcal{F}_{\leq k}$$

for every internal node, such that

$$f = \Phi_G.$$

Definition 5 (Approximate compositional sparsity)**.** *For norm $\|\cdot\|_{\mathcal{X}}$, a function f is (k, s, L, ε)-compositionally sparse if there exists a DAG realization Φ_G satisfying*

$$\|f - \Phi_G\|_{\mathcal{X}} \leq \varepsilon.$$

The important structural parameter is the local arity

$$k.$$

A function may depend globally on millions of input coordinates while each elementary constituent depends on only two or three.

4.3 Efficient Computability Implies Compositional Sparsity

4.3.1 The Boolean case

We begin with the cleanest statement.

Theorem 2 (Polynomial-time computation implies bounded-fan-in circuits)**.** *Let*

$$f_n : \{0,1\}^n \to \{0,1\}$$

be computable by a deterministic Turing machine in time

$$T(n) = \mathrm{poly}(n).$$

Then there exists a P-uniform Boolean circuit family over a fixed bounded-fan-in basis whose depth is

$$O(T(n))$$

and whose size is

$$\mathrm{poly}(T(n)).$$

Consequently every function family in **P** *admits a polynomial-size bounded-fan-in compositional representation.*

Proof sketch. Consider the space–time evolution of the Turing machine.

At every time step, each tape cell contains a symbol from a finite alphabet, the head occupies one location, and the machine occupies one state from a finite control set.

The state of a cell at time $t+1$ depends only on a constant-size neighborhood of the machine configuration at time t.

That local update can be implemented by a constant-size Boolean circuit.

Unrolling these local updates for

$$T(n)$$

time steps produces a layered bounded-fan-in circuit.

The number of simulated tape cells required is polynomially bounded by the running time, so the total number of gates is polynomial in $T(n)$.

Finally, the construction is uniform: from the description of the Turing machine and the input length one can generate the circuit wiring in polynomial time.

□

The precise polynomial overhead is secondary here.

The central structural fact is

$$\boxed{\text{polynomial-time computation} \Longrightarrow \text{polynomially many bounded-arity constituents.}}$$

This is the basic content of C1.

4.3.2 Why bounded fan-in is natural

A Turing machine itself is local.

At any elementary step it consults only a finite amount of state and a bounded region of memory.

Ordinary digital hardware has the same organization.

Complex computation is not performed by one giant gate with millions of inputs.

It is built by composing large numbers of elementary local operations.

Thus bounded-fan-in composition is not a special property invented for deep learning.

It is the ordinary architecture of digital computation.

Deep networks provide one convenient parametric language for representing such graphs.

4.4 Exact Deep-Network Realization on Discrete Inputs

The circuit theorem establishes compositional structure.

To connect this directly to neural networks, each discrete gate can be replaced by a constant-size neural subnetwork.

4.4.1 Boolean gates as ReLU subnetworks

Lemma 1 (ReLU realization of bounded-arity Boolean gates)**.** *Every Boolean gate from a fixed bounded-arity basis can be represented exactly on Boolean inputs by a constant-size ReLU network.*

For example, for

$$x, y \in \{0, 1\},$$

$$\mathrm{AND}(x, y) = \mathrm{ReLU}(x + y - 1),$$

while

$$\mathrm{NOT}(x) = 1 - x.$$

Likewise,

$$\mathrm{OR}(x, y) = \mathrm{ReLU}(x + y) - \mathrm{ReLU}(x + y - 1).$$

Because the gate basis is finite, every required gate can be compiled into a constant-size ReLU module.

Theorem 3 (Exact ReLU realization on Boolean domains)**.** *If a Boolean function admits a bounded-fan-in circuit of size s and depth L, then there exists a ReLU network of size*

$$O(s)$$

and depth

$$O(L)$$

that computes the function exactly on Boolean inputs.

Proof. Replace each Boolean gate by its constant-size ReLU realization.
Every replacement changes network size and depth by only a constant factor.

□

Hence

$$\boxed{\mathbf{P} \subseteq \{\text{polynomial-size deep ReLU realizations on Boolean inputs}\}.}$$

4.4.2 Finite-precision grids

The same compilation idea applies to digital functions on finite grids.
Let

$$\mathcal{G}_b = \left\{0, 2^{-b}, 2 \cdot 2^{-b}, \ldots, 1\right\}^d.$$

Each point has an $O(db)$-bit encoding.

Theorem 4 (Finite-grid realization)**.** *Suppose*

$$f : \mathcal{G}_b \to \mathbb{R}^m$$

can be evaluated to accuracy ε by a Turing machine in time

$$\operatorname{poly}\left(d, b, \log \frac{1}{\varepsilon}\right).$$

When the input is supplied through its binary representation, there exists a bounded-fan-in ReLU network of size and depth polynomial in

$$d, \qquad b, \qquad \log \frac{1}{\varepsilon}$$

whose output approximates f to error ε on $\mathcal{G}_b$.
The weights require only polynomially many bits.

The binary-encoding qualification is important.

If grid coordinates are supplied directly as real scalars, a network must also implement the corresponding finite-precision decoding operations.

4.5 Real-Valued Functions and Finite Precision

The discrete theorem is exact.

The continuous case requires additional care.

Efficient finite-precision evaluation gives a polynomial-size digital computation at each requested accuracy.

It does not, by itself, provide a quantitative approximation theorem for a particular neural architecture.

There are therefore two logically distinct statements.

1. Efficient finite-precision computation implies a polynomial-size bounded-fan-in *digital* representation at each requested precision.

2. Dimension-robust neural approximation additionally requires regularity and stability of the constituent functions.

The second statement is the domain of approximation theory.

These statements concern different levels of representation. Bounded fan-in at the level of bits or elementary operations does not by itself supply a fixed smooth coarse-grained DAG, uniformly controlled Sobolev norms for its constituents, or useful semantic modules. Such properties require the additional assumptions stated in the approximation theory.

The same distinction applies to a trained network. Its feasible digital implementation is a computationally sparse realization of its predictor. That fact alone neither identifies a generating decomposition nor establishes the regularity and stability needed to apply the continuous approximation theorem below.

4.6 Compositional Approximation and the Curse of Dimensionality

4.6.1 Generic high-dimensional functions

Consider a generic function

$$f : [0,1]^d \to \mathbb{R}$$

in an r-smooth class.

For broad smoothness classes, the number of parameters required for approximation error ε scales schematically as

$$N_{\text{generic}}(\varepsilon) \asymp \varepsilon^{-d/r}, \tag{4.5}$$

up to constants and logarithmic factors.

The exponent contains the ambient dimension d.

This is the classical curse of dimensionality.

4.6.2 Low-arity constituents

Suppose instead that

$$f = \Phi_G$$

where G contains s internal nodes and each constituent depends on at most

$$k \ll d$$

arguments.

Assume each constituent belongs to an r-smooth class for which local approximation to error δ requires

$$N_{\text{local}}(\delta) = O\left(\delta^{-k/r}\right) \tag{4.6}$$

parameters.

If each constituent is approximated separately, the total number of parameters scales schematically as

$$s\,\delta^{-k/r}.$$

The remaining issue is how local approximation errors propagate through the graph.

4.6.3 A stability factor

Let each constituent be Lipschitz on the relevant domain.

For a path

$$v_1 \to v_2 \to \cdots \to v_q,$$

an error introduced at v_j is amplified by the product of downstream Lipschitz constants.

Define a graph stability factor schematically by

$$A_G = \max_{\text{outputs}} \sum_{\text{upstream paths } v} \prod_{\text{downstream on path}} L_v. \tag{4.7}$$

If every constituent is approximated to local accuracy δ, then a generic bound has the form

$$\|f - \widetilde{f}\| \leq A_G \, \delta. \tag{4.8}$$

Thus it suffices to choose

$$\delta \asymp \frac{\varepsilon}{A_G}.$$

This yields

$$N(\varepsilon) = O\left(s \left(\frac{A_G}{\varepsilon} \right)^{k/r} \right). \tag{4.9}$$

If

$$A_G = \mathrm{poly}(s, L),$$

then the exponent governing approximation accuracy depends on k, not d.

Theorem 5 (Compositional approximation; schematic form)**.** *Let*

$$f = \Phi_G$$

be represented by a DAG of size s, depth L, and local arity at most k.
Suppose every constituent belongs to an r-smooth class admitting k-variate ReLU approximation with complexity

$$O(\delta^{-k/r})$$

and suppose composition is quantitatively stable so that

$$A_G = \mathrm{poly}(s, L).$$

Then f admits a deep ReLU approximation of error at most ε with complexity

$$N(\varepsilon) = \mathrm{poly}(s, L) \, \varepsilon^{-k/r}, \tag{4.10}$$

up to regularity-dependent constants and logarithmic factors.
In particular, the approximation exponent depends on local arity k rather than ambient dimension d.

The precise rate depends on the norm, activation, smoothness assumptions, graph geometry, and stability constants.
The robust message is

bounded local arity replaces ambient dimension by constituent dimension in approximation rates.

4.6.4 A simple chain

For a chain of depth D, common Lipschitz constant L, and local error δ,

$$e_D \leq \delta \sum_{j=0}^{D-1} L^j.$$

Hence

$$e_D \leq \delta \frac{L^D - 1}{L - 1}$$

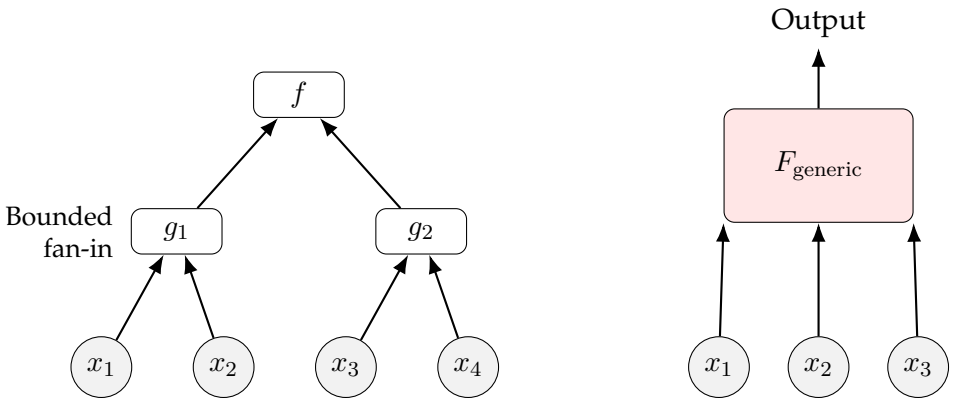
for $L \neq 1$.
When

$$L \leq 1,$$

we have the simpler bound

$$e_D \leq D\delta.$$

Thus depth is beneficial only together with adequate conditioning.
Low arity alone does not prevent local errors from being amplified.



(a) Compositionally sparse

(b) Generic dense map

Figure 4.2: Compositional sparsity versus generic high-dimensional dependence. In (a), each constituent depends on only a few variables even though the global target depends on many inputs. For broad generic smooth classes, the approximation exponent instead depends on ambient dimension.

4.7 Two Small Consequences of Local Arity

The local-arity viewpoint appears in several elementary forms.

For a Boolean constituent with k binary inputs, an exact truth table requires only

$$2^k$$

entries.

Thus N_{distinct} distinct Boolean constituents can be stored using

$$O\left(N_{\text{distinct}} 2^k\right)$$

entries.

Similarly, if a continuous k-variate constituent is quantized to R levels per coordinate, an explicit local table requires

$$R^k$$

entries.

Again the exponent is local:

$$k,$$

not global:

$$d.$$

A related geometric observation is that deep networks can produce very complicated global functions from locally simple operations.

A ReLU network is piecewise affine, yet its number of linear regions can grow extremely rapidly with depth.

Thus

$$\boxed{\text{locally simple computation} \not\Longrightarrow \text{globally simple function.}}$$

This is exactly what compositional sparsity allows.

4.8 What C1 Buys

C1 should be understood as a representation principle.

It has several important consequences, but each has a specific scope.

4.8.1 A feasible representation

If a discrete target is efficiently computable, then it possesses a polynomial-size bounded-fan-in computation graph.

Thus the object being represented does not require exponentially many independent computational components.

This is the first meaning of saying that compositional sparsity makes the problem feasible to pose.

The existence statement is relative to the allowed computational primitives and precision convention. For a particular learning architecture, one must still show that an accurate predictor is available within its admitted resources. If a norm-based generalization argument is intended, an adequate complexity budget is an additional requirement: a small circuit is not automatically a sufficiently low-norm predictor in every chosen parameterization.

4.8.2 Escape from ambient-dimensional approximation

For regular real-valued constituents, one approximates many low-dimensional functions rather than one arbitrary high-dimensional function.

Schematically,

$$\varepsilon^{-d/r} \quad \longrightarrow \quad s\varepsilon^{-k/r}, \qquad k \ll d,$$

subject to stability of the composition.

This is the approximation-theoretic content of the principle.

4.8.3 Optimization becomes feasible to pose

If representing the target itself required exponentially many parameters, then no polynomial-resource optimizer could even manipulate the relevant parameterization explicitly.

A polynomial-size compositional representation removes that obstruction.

But this says only

$$\boxed{\text{a manageable search object exists.}}$$

It does not imply

$$\boxed{\text{gradient descent knows where it is.}}$$

That second question belongs to learning. For prediction, it asks whether the training procedure can find a sufficiently accurate feasible predictor, not necessarily the representation that generated the data.

4.8.4 Favorable statistical complexity becomes possible

A sparse compositional class can possess much lower effective complexity than a class of unrestricted d-variate functions.

Generalization bounds based on

$$\text{description length}, \quad \text{sparsity}, \quad \text{norm}, \quad \text{covering entropy}, \quad \text{path complexity}$$

may therefore depend on the number and complexity of active constituents rather than the nominal size of an overparameterized architecture.

But target sparsity alone does not guarantee this outcome.

Training must select a realization with both suitable empirical fit and a statistical-complexity bound that is informative for the available sample. The realization need not be the one used to establish C1. Norm- or product-norm-based selection can support this route to prediction without identifying the generating constituents; Chapter H states the corresponding conditional argument.

4.9 What Does Not Follow from C1

Because C1 is a strong structural statement, it is easy to attribute more to it than it proves.

Several distinctions are essential.

4.9.1 C1 does not imply module reuse

A sparse DAG can contain

$$s$$

different low-arity functions, each appearing only once.
Therefore

$$\boxed{\text{bounded fan-in} \not\Longrightarrow \text{repeated modules.}}$$

Reuse requires an additional property:

$$\boxed{\text{compositional sparsity} + \text{repeated constituents} \Longrightarrow \text{modularity.}}$$

Convolutional weight sharing is an architectural example.
The CRM chapter later develops reuse as a separate selection and compression principle.

4.9.2 C1 does not imply transfer

Two tasks may both be compositionally sparse while sharing no useful constituents.
Transfer becomes possible when their representations overlap:

$$\phi_v^{(A)} \approx \phi_w^{(B)}.$$

Thus

$$\boxed{\text{sparsity} + \text{cross-task constituent sharing} \Longrightarrow \text{potential transfer.}}$$

C1 alone supplies only the first term.

4.9.3 C1 does not imply identifiability

A function can admit many sparse decompositions.
For example,

$$f = g \circ h$$

can often be rewritten under an invertible intermediate transformation

$$z' = Az$$

as

$$f = (g \circ A^{-1}) \circ (A \circ h).$$

Both factorizations compute the same function.
Therefore

$$\boxed{\text{existence of a sparse DAG} \not\Longrightarrow \text{unique latent coordinates.}}$$

Semantic interpretation or causal identification requires additional assumptions.

For predictive learning, this nonuniqueness is not itself an obstruction. The learner may use a different feasible decomposition as long as its input–output predictions are accurate on the population of interest. Structural recovery asks for an additional correspondence with a specified generating decomposition, under an explicitly controlled notion of equivalence.

4.9.4 C1 does not imply visible sparsity of trained weights

Suppose a target admits a sparse computational realization.

A change of basis can transform that realization into a dense parameterization.

Likewise, an overparameterized network may implement the same function in many coordinate systems.

Therefore

$$\boxed{\text{sparse realization exists} \not\Longrightarrow \text{native weight matrices of the trained model are sparse.}}$$

In particular, C1 does not predict that a Transformer MLP must become block sparse or column sparse in its learned coordinates.

Conversely, dense native weight matrices do not contradict the bounded-fan-in implementation of a feasible learned predictor. The witness argument in Chapter H concerns the computation implemented by that predictor, not a requirement that its displayed weight matrices reveal the generating graph.

To infer visible weight sparsity one needs additional assumptions about

$$\text{optimization bias}, \quad \text{regularization}, \quad \text{identifiability}, \quad \text{coordinate alignment}.$$

4.9.5 C1 does not imply easy global search

Bounded local structure does not by itself eliminate combinatorial difficulty.

A canonical example is 3-SAT.

Every clause depends on only three variables, so the verification map is highly local, yet finding a satisfying assignment is NP-complete.

Thus

$$\boxed{\text{efficient local evaluation} \not\Longrightarrow \text{efficient inversion, optimization, or search.}}$$

This distinction is developed separately in the appendix on distributional tractability.

4.9.6 C1 does not imply generalization

The existence of a low-complexity realization does not guarantee that training selects it.

An overparameterized model may interpolate using a function of much higher effective complexity.

Thus C1 makes favorable generalization possible but does not establish it.

There is no contradiction with the fact that a computationally feasible memorizing network also has a bounded-fan-in implementation. Computational sparsity and statistical complexity measure different properties. A polynomial-size implementation need not give a useful generalization bound at the available sample size.

The conditional route is instead to establish adequate empirical fit and an informative complexity bound for the selected predictor. If these imply small population error, and the predictor is feasible to evaluate, its implementation supplies the compositional witness. This route does not require recovery of the generating graph.

4.9.7 Most importantly: C1 does not imply learnability

The learner observes examples such as

$$(x_i, f(x_i)).$$

It is not given the generating graph, its constituent functions, the intermediate variables, or the parent sets.

Their absence does not make reconstruction of that particular computation a requirement of every learning problem. Two objectives must be kept separate.

For *predictive recovery*, the learner seeks a feasible function $\widehat{f}$ such that, for example,

$$\boxed{\|\widehat{f} - f\|_{L^2(P_X)} \leq \varepsilon,}$$

where P_X is the input distribution of interest. The task is

$$\boxed{\begin{array}{c}\text{input–output observations}\\ \longrightarrow \\ \text{an accurate, computationally feasible predictor.}\end{array}}$$

For *structural recovery*, the objective is stronger:

$$\boxed{\begin{array}{c}\text{input–output observations}\\ \longrightarrow \\ \text{the generating computation, up to controlled equivalences.}\end{array}}$$

This requires a separate criterion of success and suitable identifiability assumptions. Predictive accuracy alone does not establish it, and prediction need not fail merely because the learned internal coordinates differ from the generating ones.

C1 guarantees the existence of an economical realization in the stated computational model. It does not prove that the specified learner finds a sufficiently good empirical fit, that its selected predictor has adequate statistical-complexity control, or that its predictions generalize.

The witness argument in Chapter H starts after those predictive requirements have been established. It applies the computation-to-circuit result to the learned predictor itself. It removes an unnecessary structural-identification requirement; it does not turn representation into a learning algorithm.

4.10 Scope: Artificial and Biological Computation

The implication

$$\boxed{\text{efficient computation} \longrightarrow \text{polynomial-size bounded-fan-in computation}}$$

has its clearest meaning in the standard digital bit model.

Its interpretation for biological computation requires more care.

4.10.1 Computability versus efficient computability

A function can be Turing computable while requiring resources too large to be useful.

Thus

$$\boxed{\text{Turing computable} \not\Longrightarrow \text{efficiently Turing computable.}}$$

For a real-valued function, efficiency requires not merely that an answer eventually be produced, but that the time, memory, and precision required for an ε-approximation grow polynomially in the appropriate description lengths and in

$$\log \frac{1}{\varepsilon}.$$

4.10.2 Artificial systems

Every actual finite computation performed on ordinary digital hardware is implemented by finite local operations.

But this does not automatically make an asymptotic algorithm polynomial time.

An exponential brute-force procedure is also digital.

The correct implication is conditional:

$$\boxed{\text{polynomial-resource digital computation} \Longrightarrow \text{polynomial-size bounded-fan-in realization.}}$$

4.10.3 Biological systems

One may accept some version of the physical Church–Turing thesis and still not obtain an efficiency theorem.

Digital simulability in principle says nothing about whether a faithful simulation requires

$$\text{polynomial time}, \qquad \text{polynomial memory}, \qquad \text{polynomial precision}.$$

Therefore

$$\boxed{\text{physical simulability} \not\Longrightarrow \text{efficient digital simulability.}}$$

4.10.4 Microscopic simulation versus functional simulation

The distinction is particularly important for biological systems.

Suppose a nervous system has microscopic physical state

$$s_t$$

but behavior depends primarily on a lower-dimensional effective state

$$z_t = \Phi(s_t).$$

Simulating every microscopic degree of freedom may be extremely expensive, while reproducing the effective map

$$z_t \mapsto z_{t+1}$$

may be simple.

For artificial intelligence, the relevant question is usually not

> Can every microscopic degree of freedom of the brain be simulated?

but

> Can the behaviorally relevant computation be reproduced efficiently?

4.10.5 Local physical dynamics

Many physical systems are governed by local interactions.

If the relevant latent state factorizes as

$$z_t = (z_t^{(1)}, \dots, z_t^{(m)})$$

and satisfies

$$z_{t+1}^{(i)} = F_i\left(z_t^{\mathrm{pa}(i)}\right), \qquad |\,\mathrm{pa}(i)| = O(1),$$

then the effective dynamics are already compositionally sparse.

This provides one reason for optimism that behaviorally relevant biological computation may admit compact digital descriptions.

But it is a hypothesis about the effective variables, not a theorem about all biological processes.

4.10.6 No connection to consciousness or evolutionary age

Computational complexity should not be identified with categories such as

$$\text{conscious}, \quad \text{unconscious}, \quad \text{emotional}, \quad \text{evolutionarily old}.$$

A homeostatic controller may be computationally simple.

A symbolic reasoning problem may be computationally intractable.

Thus

> phenomenological category and evolutionary age do not determine complexity class.

4.10.7 The biological boundary

The relevant open question is therefore

> Are behaviorally relevant biological computations efficiently approximable in the finite-precision digital bit model?

If the answer is yes for a particular map, C1 applies.

If reproducing that map requires superpolynomial time, memory, or precision, the theorem does not supply an efficient compositional realization.

This marks a natural boundary of the framework.

4.11 The Status of the Principle

The classical theorem and its interpretation should be distinguished.

The complexity-theoretic statement

$$\boxed{\mathbf{P} \subseteq \{\text{P-uniform polynomial-size bounded-fan-in circuits}\}}$$

is standard.

The *Compositional Sparsity Principle* is the interpretation of this fact for learning:

> Targets produced by feasible digital computations belong to a highly structured subset of all high-dimensional functions: they possess polynomial-size decompositions into low-arity constituents.

This changes the natural reference class for deep learning.

The comparison should not be

$$\text{deep network} \quad \text{versus} \quad \text{arbitrary } f : \mathbb{R}^d \to \mathbb{R}.$$

Almost all such mathematical functions are irrelevant to feasible computation.

The more meaningful comparison is

$$\boxed{\text{learning system} \quad \text{versus} \quad \text{targets generated by feasible computations.}}$$

4.12 Conclusion: From Representation to Predictive Learning

The central result of this chapter is simple. A polynomial-time digital computation is built from only polynomially many local elementary operations. Unrolling those operations gives a polynomial-size bounded-fan-in DAG:

$$\boxed{\text{efficient digital computation} \Longrightarrow \text{compositional sparsity.}}$$

For Boolean and finite-precision discrete computation, this implication is direct in the stated computational model.

For continuous approximation, additional constituent regularity and quantitative stability are required. When the local approximation and composition hypotheses of Theorem 5 hold, local arity replaces ambient dimension in the accuracy exponent:

$$\boxed{\varepsilon^{-d/r} \quad \longrightarrow \quad \mathrm{poly}(s, L)\, \varepsilon^{-k/r}, \qquad k \ll d.}$$

The graph size, depth, regularity constants, and stability factor still matter. The conclusion is not that every computationally sparse implementation automatically yields a uniformly regular coarse-grained representation.

C1 removes a representation-size obstruction, not every obstruction to learning. As the preceding sections explain, it does not by itself imply module reuse, transfer, unique latent coordinates, visibly sparse trained weights, easy search, or generalization. In particular,

$$\boxed{\text{compositional sparsity does not imply learnability.}}$$

The learning objective need not be the generating program. Given only input–output observations, the primary predictive objective is to find a feasible function that is accurate on the population of interest. The learner need not reconstruct a privileged generating graph or establish a correspondence between its internal variables and those of the teacher.

This changes the transition from representation to learning. The questions are:

Q1 : Does the chosen class contain a feasible accurate predictor?

Q2 : Can the training procedure find a suitable empirical fit?

Q3 : Does the selected predictor generalize?

C1 informs Q1. For a chosen architecture, its expressive resources and, when needed, its norm-complexity budget must still be checked. Q2 and Q3 require learning and statistical arguments, not just circuit compilation.

The same representation result applies after learning. Suppose training returns a predictor $\widehat{f}$ satisfying

$$\|\widehat{f} - f\|_{L^2(P_X)} \leq \varepsilon.$$

Suppose also that its finite description and numerical evaluation obey the stated polynomial resource bounds. At evaluation precision m, the same compilation argument yields a bounded-fan-in digital realization whose decoded output $\widetilde{f}_m$ satisfies

$$\|\widetilde{f}_m - \widehat{f}\|_\infty \leq 2^{-m}.$$

The triangle inequality then gives

$$\boxed{\|\widetilde{f}_m - f\|_{L^2(P_X)} \leq \varepsilon + 2^{-m}.}$$

The learned computation is therefore a compositional witness at the stated predictive and numerical accuracy. Its graph may differ from the graph used to establish the target's representability.

This conclusion is conditional on predictive success and feasible evaluation. It does not prove that training is efficient, that regularization achieves an informative complexity bound, or that the internal variables identify the generating mechanism. Those are separate claims. Chapter H develops the conditional synthesis.

The route through the following chapters. The next chapter examines how compositional representations make optimization feasible to pose and how intermediate supervision can simplify the learning problem. Chapter 6 studies detectable footprints and their relation to accessible learning updates. The chapter on overparameterization studies optimization geometry. Chapter H then separates empirical fit, statistical selection, and the final witness argument.

The roles of C1, C2, and C3 remain distinct:

C1:	a feasible representation exists;
C2:	useful learning signals are accessible;
C3:	intermediate information is supplied.

Genericity offers candidate explanations for useful signals. A staircase is one possible mechanism for exploiting them through successive feature construction. Trace exposure can simplify learning

even when end-to-end learning is possible. None of these labels is, by itself, a complete learning theorem, and no particular mechanism is an additional premise of the witness implication once predictive success is established.

Structural-recovery results remain valuable: they establish a stronger objective and may explain constructively how learning succeeds. They are not a universal requirement for accurate prediction.

The resulting separation is

C1 asks which computations admit an economical representation.

Learning asks how an accurate feasible predictor is found.

Structural recovery asks whether the generating computation is identified as well.

CHAPTER 5

Compositionality: Why Optimization Is Feasible to Pose

Does compositional structure make optimization easier? The answer has two parts that should be kept separate.

First, an economical compositional representation makes the optimization problem feasible to pose*. A generic target in a broad d-dimensional smoothness class may require approximation complexity exponential in d, leaving no manageable explicit parameterization. A target with a polynomial-size bounded-arity computation removes this representation-size obstruction. For continuous approximation, the regularity and stability assumptions of Chapter 4 remain essential.*

Second, a manageable parameterization need not be easy to search. Two further obstructions can remain. The first is geometric*: gradients may vanish, explode, or be badly conditioned as they propagate through deep compositions. The second is* informational*: the observations and the statistics accessible to the chosen learning rule may fail to expose a detectable signal for improving prediction.*

This chapter separates these roles. It also studies a positive "grey-box" regime in which intermediate variables, or useful surrogates for them, are exposed. Faithful intermediate supervision permits local supervised problems to be posed. This is the trace-exposure role C3; it can help even when end-to-end learning is possible and is not only a fallback for a failed genericity condition.

The objective must also be explicit. Predictive learning seeks an accurate function on the population of interest. Structural recovery additionally seeks the generating constituents or intermediate variables. The latter is a stronger task. As Chapter H explains, a computationally feasible predictor that generalizes is already a compositional witness, even if its internal coordinates differ from the teacher's. The remaining challenge is to explain how the specified training procedure achieves that predictive success.

5.1 From Representation to Optimization

Chapter 4 established the first principle of the book:

$$\boxed{\text{efficient digital computation} \Longrightarrow \text{a compositionally sparse realization.}}$$

For real-valued functions, this computational conclusion concerns the specified encoding and numerical precision. The corresponding approximation-theoretic statement additionally requires controlled local regularity and stable composition.

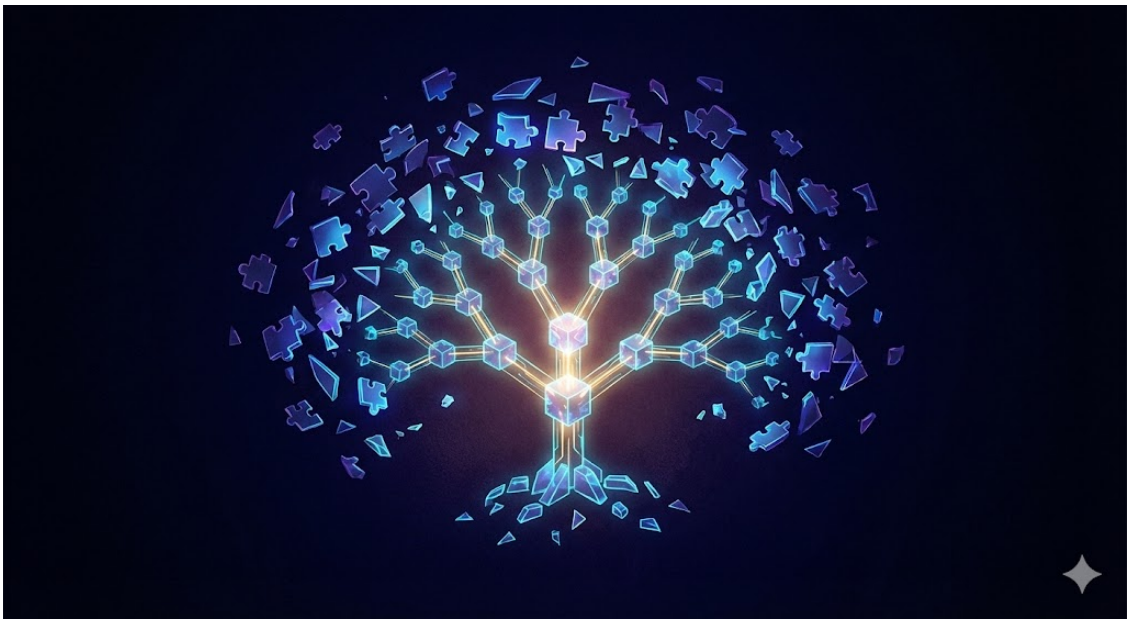


Figure 5.1: Compositional structure creates global complexity from simple local operations. An economical representation does not, by itself, guarantee that end-to-end optimization finds an accurate predictor. Identifying the particular generating constituents is a stronger, separate objective.

The immediate consequence for optimization is a statement about the size of an object to be represented. If every accurate realization in a chosen representation language required exponentially many independent parameters, a polynomial-resource procedure could not even manipulate such a realization explicitly. An economical compositional description removes that obstruction.

It does not turn a polynomial-dimensional parameter space into a small set of candidates. Many parameter vectors and many possible graphs can still have descriptions of polynomial length.

Two questions therefore remain distinct.

1. **Can the optimization problem be posed using feasible resources?** There must be an accurate predictor in the admitted architecture, with controlled description length, numerical precision, and evaluation cost.

2. **Can the specified procedure find a suitable predictor?** Even a compactly parameterized problem can have poor conditioning, unhelpful local geometry, weak statistical signals, or inadequate data. Population accuracy requires statistical control in addition to a good empirical fit.

Thus

$$\boxed{\text{economical parameterization} \neq \text{easy search.}}$$

In the terminology of the revised Interlude, C1 concerns feasible representation; C2 concerns accessible learning signal; C3 supplies intermediate information. These roles can cooperate. They are not a universal sequence of necessary and sufficient conditions for learning.

For noiseless regression, predictive recovery can be expressed as

$$\|\widehat{f} - f\|_{L^2(P_X)} \leq \varepsilon.$$

Structural recovery asks in addition for a specified correspondence between learned and generating variables, under appropriate identifiability and equivalence conditions. We do not impose that additional requirement on every successful predictor.

5.2 Why C1 Makes the Search Space Feasible

The parameter-count argument was developed in Chapter 4. We recall only the part needed here. Consider a real-valued target on a fixed compact input domain in $\mathbb{R}^d$, in a smoothness class of order s.

5.2.1 Generic high-dimensional approximation

For broad smoothness classes, worst-case approximation to error ε may require complexity of the schematic form

$$N_{\text{generic}}(\varepsilon) \asymp \varepsilon^{-d/s},$$

up to class-dependent constants and logarithmic factors. The exponent is the ambient dimension d. This is the representation-size obstruction associated with the curse of dimensionality, not a lower bound for every individual smooth target.

5.2.2 Compositionally sparse approximation

Suppose instead that the target consists of m low-arity constituents, each depending on at most k variables, with $k \ll d$. Under the local approximation assumptions of Chapter 4, and writing A_G for its graph stability factor, the corresponding complexity is schematically

$$N_{\text{comp}}(\varepsilon) = O\left(m\left(\frac{A_G}{\varepsilon}\right)^{k/s}\right),$$

up to regularity-dependent constants and logarithmic factors. The shorter expression $O(m\varepsilon^{-k/s})$ suppresses this stability dependence; it should not conceal an exponentially large amplification factor.

When the number of constituents and the relevant stability and regularity bounds are polynomially controlled, and $k = O(1)$, the approximation exponent depends on local arity rather than ambient dimension:

$$\boxed{\varepsilon^{-d/s} \quad \longrightarrow \quad m\left(\frac{A_G}{\varepsilon}\right)^{k/s}.}$$

This supplies an economical realization to be sought. It does not bound the number of possible realizations or the effort needed to select one. Nor is polynomial dependence on ε^{-1} the same as polynomial-time evaluation in the requested bit precision $\log(1/\varepsilon)$.

The representation benefit is therefore

$$\boxed{\text{C1 removes a representation-size obstruction.}}$$

This benefit is conditional on the resource and analytic assumptions just stated. A separate architecture-specific argument is needed if the learner is restricted to a particular network class or norm budget.

5.2.3 What this does not imply

An economical parameterization can still have a nonconvex loss, poorly conditioned gradients, weak accessible signals, symmetries, and flat directions. The graph structure may itself be unknown. Thus C1 supplies a feasible comparison object; it does not establish that gradient descent, SGD, or another specified procedure will find a good predictor.

It also does not establish that the selected predictor has small enough statistical complexity for the available sample. Computational feasibility, optimization cost, and generalization are different resource questions.

5.3 Two Obstructions to End-to-End Learning

Suppose that the chosen architecture admits a feasible accurate predictor. Training end to end from input–output pairs takes the form

$$\min_{W_1,\dots,W_L} \sum_i \ell(f_L(\cdots f_2(f_1(x_i;W_1);W_2)\cdots;W_L),y_i)\,.$$

The functions $f_\ell(\cdot;W_\ell)$ describe the learner's layers. They need not correspond to the teacher's constituents. Two conceptually different obstructions can arise even for the predictive objective.

5.3.1 The geometric obstruction

In a deep composition, the gradient with respect to an early parameter contains products of Jacobians from later stages. Schematically,

$$\frac{\partial \mathcal{L}}{\partial W_1} = \frac{\partial \mathcal{L}}{\partial f_L}\frac{\partial f_L}{\partial f_{L-1}}\cdots\frac{\partial f_2}{\partial f_1}\frac{\partial f_1}{\partial W_1}.$$

The singular values of these products can become very small or very large. This produces vanishing or exploding gradients, steep and flat directions, large condition numbers, and sensitivity to depth and initialization.

These are geometric or conditioning problems. They do not mean that useful target information is absent: an informative dependence may exist but be transmitted badly through the parameterization.

5.3.2 Architectural responses to geometry

Several architectural devices can improve this situation. Residual connections provide shorter gradient paths. Normalization controls scales. Suitable initialization can keep Jacobian spectra in useful regimes. Overparameterization can introduce additional fitting directions. Adaptive preconditioning can rescale poorly conditioned coordinates.

These are possible improvements, not unconditional convergence results. They can alter the directions accessible to training without supplying new observations. Changing the parameterization can also change which statistics a gradient probes, so geometry and accessible signal interact even though they are conceptually distinct.

5.3.3 The informational obstruction

Even with favorable conditioning, the observations and the chosen learning rule may not expose a useful signal for improving prediction. In particular parameterizations and initialization regimes, the leading population-gradient components are controlled by low-order correlations of the target with accessible features.

If those correlations vanish by symmetry or cancellation, or are too small to estimate at feasible sample size, that mechanism may fail to provide a useful direction. This is a statement about the target, distribution, and learner together, not about the target alone.

Parity illustrates the distinction. In sign coordinates, let

$$\chi_S(X) = \prod_{i \in S} X_i, \qquad X_i \in \{-1, +1\}.$$

A tree of binary operations computes this function economically. Under the uniform product distribution, however,

$$\mathbb{E}[\chi_S(X) X_j] = 0 \qquad (j \in S,\ |S| > 1).$$

This rules out that particular linear screening signal. It does not show that every gradient-based parameterization, every subsequent training stage, or every learning procedure lacks a useful signal.

5.3.4 Geometry versus information

The distinction can be summarized as

geometry:	how an informative dependence is transmitted;
accessible information:	what the learning rule can detect.

Rescaling a fixed zero gradient cannot make it informative. A different representation, observation model, or probe family may nevertheless expose information that the original rule did not use. Thus absence of a particular footprint is not automatically absence of information from the full input–output distribution.

Chapter 6 studies mechanisms that expose detectable footprints. Their magnitudes and their connection to actual updates both require analysis.

5.3.5 What compositionality alone guarantees

The taxonomy is

C1:	an economical computational realization exists;
geometry:	the parameterization transmits learning signals;
C2:	useful signals are accessible at feasible resolution.

Compositionality alone does not imply fast convergence to a predictive solution. It establishes still less about identifying generating constituents. A theorem about either objective must specify the algorithm, observation model, and resource bounds.

5.4 Grey-Box Recovery

There is an important positive regime in which compositionality can be exploited more directly. If intermediate variables are available as training signals, local supervised problems can be posed without first inferring those variables from their indirect effects on the final output. This is the grey-box regime.

5.4.1 Black box versus grey box

Consider $f = h \circ g$. In the black-box setting, the learner observes only

$$(x, y), \qquad y = h(g(x)).$$

The generating variable $z = g(x)$ is not supplied. A procedure that aims to recover this particular factorization must discover an appropriate version of z. Predictive learning has a weaker requirement: it may find another computation of the same input–output relation.

In the grey-box setting, z is also available, and the data contain

$$(x, z) \qquad \text{and} \qquad (z, y).$$

With separate module parameters, the two local supervised objectives can be optimized separately.

Proposition 2 (Factorization under Intermediate Supervision)**.** *Let*

$$f = h \circ g.$$

Suppose the training data include the true intermediate variable

$$z = g(x).$$

Then empirical risk minimization for the two modules can be written as two separate problems:

$$\min_{W_g} \sum_i \ell_g \left(g(x_i; W_g), z_i \right),$$

and

$$\min_{W_h} \sum_i \ell_h \left(h(z_i; W_h), y_i \right).$$

Thus intermediate supervision decomposes the composite inverse problem into constituent inverse problems.

It does not imply that each constituent problem is itself easy.

5.4.2 What the proposition proves

The proposition establishes that the supplied targets allow separate local empirical objectives to be posed. It is not an equivalence between minimizing those auxiliary losses and minimizing the original end-to-end loss under all parameter constraints. Shared parameters or additional coupling penalties can couple the local problems again.

Nor does the proposition prove efficient optimization or generalization of each factor. Even a low-dimensional constituent needs an appropriate hypothesis class, informative data, and an adequate learning procedure.

The precise benefit is

> faithful intermediate supervision removes the need to transmit the final loss through the full composition for each local fit.

It does not remove intrinsic local difficulty or accumulated error when the learned modules are recomposed.

For example, suppose $\widehat{g}(x)$ and $g(x)$ both lie in a domain K where h is L_h-Lipschitz, and suppose

$$\|\widehat{h} - h\|_{\infty,K} \le \delta_h, \qquad \|\widehat{g} - g\|_{L^2(P_X)} \le \delta_g.$$

The triangle inequality gives the same local-to-global requirement used in Chapter 4:

$$\|\widehat{h} \circ \widehat{g} - h \circ g\|_{L^2(P_X)} \le \delta_h + L_h \delta_g.$$

The domain condition matters. Training $\widehat{h}$ only on observed true states does not by itself control its behavior at every perturbed state produced by $\widehat{g}$.

5.4.3 Extension to a computation DAG

Suppose $f = \Phi_G$ has internal states z_v, $v \in \mathsf{Int}$. If every parent tuple and corresponding output are observed, the learner can pose local supervised problems

$$z_v = \phi_v(z_{\mathrm{pa}(v)}).$$

With independently parameterized modules, there are $|\mathsf{Int}|$ local fitting problems rather than one jointly hidden composition to fit from the final loss alone.

This is the full-trace version of C3. It can help even when another end-to-end method could learn an accurate predictor. Sample coverage, local estimation and optimization, compatibility of the interfaces, and stability after composition still need to be established.

5.5 Three Levels of Trace Exposure

Not all intermediate information has the same status. It is useful to distinguish three regimes.

5.5.1 Hard trace

In the strongest regime, true intermediate variables of the generating computation are observed. For a DAG node v, the learner receives

$$\left(z_{\mathrm{pa}(v)}, z_v\right).$$

The local dependency is directly supervised. This is the setting of Proposition 2; its factorization statement should not be mistaken for a complete local learnability theorem.

5.5.2 Partial trace

Only some intermediate variables may be observed, or some already trained modules may be available. A partially exposed computation can reduce the amount that must be learned jointly.

An observed teacher state and a pretrained surrogate have different interpretations. The former is part of a faithful trace. The latter may supply useful task information without identifying any teacher state. Both can help, but the usefulness of the surrogate must be established for the task.

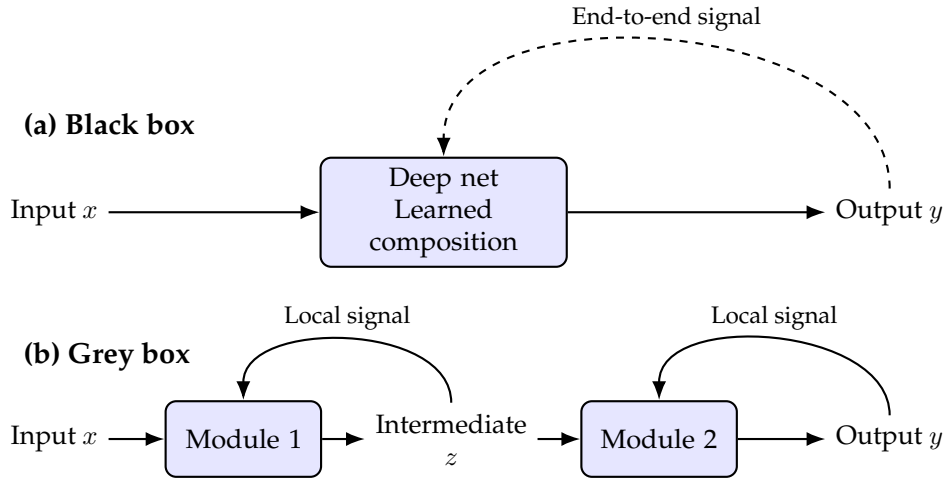


Figure 5.2: Black-box and grey-box learning. End-to-end training fits a predictor without observing the generating intermediate states; it need not identify them. Faithful intermediate supervision supplies local targets, permitting separate module objectives. Each local problem and the accuracy of the recomposed predictor still require analysis.

5.5.3 Surface trace

The learner may instead see externally emitted intermediate symbols, explanations, tokens, or scratchpad states. These may provide an operational decomposition without equaling the latent variables of the underlying computation.

The original hierarchy is therefore best read as a distinction in faithfulness to the generating computation:

hard trace / partial trace / surface exposure.

It is not a literal set inclusion or a universal ranking of predictive usefulness. An informative surrogate can be valuable for prediction even when it is not a faithful internal state.

5.6 Partial Traces in Practice

Several learning strategies can be interpreted through this lens, provided the analogy is calibrated carefully.

5.6.1 Layer-wise training

Early deep-network training strategies often trained one layer or module at a time before global fine-tuning [65]. Such a schedule replaces a single initial deep optimization with a sequence of shallower local training problems.

The intermediate targets need not be externally supervised; in unsupervised or generative pretraining they are constructed indirectly. Layer-wise training is therefore a strategy for organizing optimization, not necessarily an instance of hard-trace supervision. A benefit for the final predictive task still depends on the objectives and learned representations.

5.6.2 Transfer learning

Suppose a pretrained backbone computes $z = g(x)$ and the downstream task trains only a head $y = h(z)$. The representation z need not equal any intermediate variable of the task-generating process.

When z retains useful task information and makes the head problem manageable, replacing $x \mapsto y$ by $z \mapsto y$ can simplify learning. A representation that discards necessary information need not help. Transfer therefore illustrates useful intermediate exposure without requiring structural identification of the teacher.

5.6.3 Scratchpads and chain-of-thought

A model can also be trained or prompted to emit intermediate symbols

$$r_1, r_2, \ldots, r_T$$

before producing its final answer. Such scratchpad or chain-of-thought sequences can function as a surface computational trace, replacing one mapping by a sequence of conditional steps:

$$x \mapsto r_1 \mapsto r_2 \mapsto \cdots \mapsto y.$$

This notation suppresses the retained context. The emitted sequence need not coincide with the model's internal latent computation.

The claim is operational rather than ontological:

external intermediate symbols may provide a useful decomposition without being faithful internal states.

Their accuracy and their effect on the final prediction require their own evaluation; the existence of a trace is not a guarantee of either.

5.6.4 What partial exposure leaves to be learned

Partial or surface exposure can reduce the burden of learning a useful computation. It does not necessarily eliminate local estimation, optimization, missing information, or error accumulation.

When a learner follows a staircase mechanism, exposed features can shorten that staircase. This is a conditional interpretation, not a requirement that every remaining dependency be learned by a staircase or that genericity take one particular form. A learner may instead find a different predictive decomposition.

Thus

partial exposure changes the learning problem; it does not by itself solve the part that remains.

C2 and C3 can complement each other. Trace exposure is not reserved for cases in which end-to-end prediction would otherwise be impossible.

5.7 Statistical Benefits of Structural Alignment

Compositional structure can also have statistical consequences, which should be stated separately from optimization. The number of examples needed for prediction depends on the target and hypothesis classes, smoothness, noise, loss, regularization, and architecture. There is no universal sample-complexity law comparing ε^{-d} with $d\varepsilon^{-k}$ without these assumptions.

When a hypothesis class is adapted to an economical compositional realization, its effective complexity can depend on the number, regularity, and local arity of constituents rather than on the richness of an unrestricted d-variate class. Unknown graph structure, numerical precision, and selection among candidate representations must also be accounted for.

Under appropriate assumptions, learning in such a restricted class can reduce the sample requirement. One useful strategy is to learn local functions instead of treating the target as an unrestricted global map. This need not recover the generating local functions: another controlled realization with accurate population predictions is sufficient for the predictive objective.

A shallow architecture may represent the same target but require much greater width or a less favorable complexity budget. Depth can help when it realizes a useful factorization economically. The statistical claim concerns the admitted or selected predictor class, not merely the existence of some sparse teacher graph.

Chapter H makes the distinction explicit: a feasible circuit realization does not automatically give a low-norm predictor in every chosen architecture, and a low training loss does not automatically control population risk.

5.8 What Grey-Box Structure Does to Optimization Geometry

Intermediate supervision can change geometry as well as information. Consider again $f = h \circ g$. End-to-end optimization has gradient paths of the form

$$\nabla_{W_g} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial h} \frac{\partial h}{\partial g} \frac{\partial g}{\partial W_g}.$$

Direct supervision of g instead supplies the local objective with

$$\nabla_{W_g} \mathcal{L}_g = \frac{\partial \mathcal{L}_g}{\partial g} \frac{\partial g}{\partial W_g}.$$

The Jacobian of h is no longer on that local gradient path. The training problem has changed, rather than the original end-to-end objective having become convex by a formal factorization.

Trace exposure can therefore reduce both the informational burden of inferring an intermediate target and the geometric burden of a long Jacobian chain. Neither benefit guarantees that the local subproblem is easy. Nor does it remove the need to test the composition when the later module receives learned, rather than exact, intermediate inputs.

5.9 The Division of Labor Among C1, C2, and C3

The three roles can now be stated without making them a compulsory learning sequence:

C1:	an economical realization is available;
C2:	useful learning signals are accessible;
C3:	intermediate information is supplied.

Geometry cuts across these roles. A useful signal can be poorly transmitted, and a well-conditioned parameterization need not expose the statistics needed by a particular learning rule. Changing the accessible features or observations can alter this situation.

These distinctions organize candidate explanations of finite-time learning. They do not eliminate the separate statistical question: why does the predictor selected by the procedure generalize? The next two sections introduce abundance and selection among fitting solutions. Chapter 7 develops the geometric analysis further, and Chapter H combines fitting and complexity control in a conditional predictive argument.

5.10 Overparameterization, Degeneracy, and Entropic Selection

Compositional sparsity concerns the existence of an economical realization. A learner may use a much larger, overparameterized architecture. The number of trainable parameters can greatly exceed the independent constraints imposed by the training sample.

Conditional on interpolation and suitable regularity, the fitting solutions can form an extended set rather than an isolated point. This raises two questions: how does training reach a low-loss region, and which predictor does it select there?

The following calculations retain the distinction between local geometry, idealized stochastic preference, and actual finite-time learning. They introduce the objects used in the fuller treatment of Chapter 7; their connection to prediction is summarized in Chapter H.

5.10.1 Geometry of the interpolating set

Consider a squared empirical loss of the form

$$L(\theta) = \frac{1}{2}\|r(\theta)\|_2^2, \qquad r : \mathbb{R}^P \to \mathbb{R}^M, \tag{5.1}$$

where P is the number of parameters and M is the number of scalar training constraints.

An interpolating solution satisfies

$$r(\theta^\star) = 0.$$

Let

$$J(\theta) = Dr(\theta)$$

denote the Jacobian of the residual vector. For the Hessian calculation, assume that r is twice continuously differentiable on a neighborhood of the point in question. In a piecewise-smooth network this statement applies within a region where that differentiation is valid; it is not an assertion of a classical Hessian at every activation kink.

Proposition 3 (Degeneracy at an interpolating solution)**.** *For the loss in Equation* (5.1)*, at any interpolating solution $\theta^\star$,*

$$\nabla^2 L(\theta^\star) = J(\theta^\star)^\top J(\theta^\star). \tag{5.2}$$

Consequently,

$$\operatorname{rank} \nabla^2 L(\theta^\star) \le M, \tag{5.3}$$

and therefore

$$\dim \ker \nabla^2 L(\theta^\star) \geq P - M. \tag{5.4}$$

Proof. Differentiating

$$L(\theta) = \frac{1}{2} \sum_{i=1}^{M} r_i(\theta)^2$$

gives

$$\nabla L(\theta) = J(\theta)^\top r(\theta).$$

Differentiating once more,

$$\nabla^2 L(\theta) = J(\theta)^\top J(\theta) + \sum_{i=1}^{M} r_i(\theta) \nabla^2 r_i(\theta).$$

At an interpolating solution,

$$r_i(\theta^\star) = 0$$

for every i, so the second term vanishes and

$$\nabla^2 L(\theta^\star) = J(\theta^\star)^\top J(\theta^\star).$$

Since the rank of J is at most M, the claimed nullity bound follows from rank–nullity. □

Thus, when

$$P \gg M,$$

a large Hessian nullspace is not surprising.

It is a direct consequence of interpolation in an overparameterized model.

The proposition concerns the *data loss at exact interpolation*, not a minimum of the full regularized objective. For a twice-differentiable penalty R, the latter Hessian also includes $\lambda \nabla^2 R$; away from exact interpolation, the residual-weighted curvature term in the proof need not vanish either. Chapter 7, Section 7.4, distinguishes these Hessians.

5.10.2 From flat directions to manifolds of solutions

The Hessian statement is local. A zero eigenvalue alone does not establish a manifold of exact solutions. Suppose that

$$\operatorname{rank} J(\theta) = q$$

is locally constant near an interpolating point $\theta^\star$. The constant-rank theorem then implies that

$$\mathcal{M} = \{\theta : r(\theta) = 0\}$$

is locally a smooth manifold of dimension

$$\dim \mathcal{M} = P - q. \tag{5.5}$$

Since $q \leq M$, this gives $\dim \mathcal{M} \geq P - M$ locally.

The parameters agree on the fitted observations. They need not represent the same function outside the sample. These dimension counts establish neither generalization nor the size of a basin of attraction. Increasing the parameter count changes the dimension conclusion only relative to the independent constraint rank and under the stated regularity.

5.10.3 An idealized model of stochastic selection

To understand how stochasticity can distinguish among such solutions, consider the overdamped Langevin diffusion

$$d\theta_t = -\nabla L(\theta_t)\, dt + \sqrt{2T}\, dB_t, \tag{5.6}$$

where B_t is standard Brownian motion and $T > 0$ plays the role of a temperature.

If the dynamics and domain admit the usual normalizable equilibrium, its stationary density is

$$\pi_T(\theta) = \frac{1}{Z_T} \exp\left(-\frac{L(\theta)}{T}\right), \tag{5.7}$$

with

$$Z_T = \int \exp\left(-\frac{L(\theta)}{T}\right) d\theta.$$

This model provides a clean separation between two effects.

The exponential factor favors regions of low loss:

$$e^{-L/T},$$

while the integration measure favors regions occupying large volume.

Thus two sets of parameters with the same loss need not carry the same probability mass.

The integral must be finite. Noncompact scale symmetries can invalidate this assumption unless the model is confined or a suitable regularizing measure is specified. A stationary density also does not say how rapidly a trajectory from the chosen initialization approaches that equilibrium. If the Langevin drift includes a regularizer, the full potential $L + \lambda R$ must replace L in the density and the corresponding local volume calculation. Its minimum set need not be the exact data-fitting set.

5.10.4 Probability mass near a manifold of minima

The volume effect can be made explicit.

Suppose $\mathcal{M}_j$ is a compact smooth manifold of global minima of dimension d_j,

$$L(\theta) = L_\star, \qquad \theta \in \mathcal{M}_j,$$

and suppose that the Hessian is positive definite in directions normal to the manifold.

Let

$$H_j^{\perp}(\theta)$$

denote this normal Hessian.

A tubular Laplace approximation gives, as $T \to 0$,

$$Z_j(T) \sim e^{-L_\star/T} (2\pi T)^{(P-d_j)/2} \int_{\mathcal{M}_j} \frac{d\,\mathrm{Vol}(\theta)}{\sqrt{\det H_j^{\perp}(\theta)}}. \tag{5.8}$$

Equation (5.8) separates three contributions to local probability mass:

dimension of the minimizer manifold,
volume along the manifold,
curvature transverse to the manifold.

At equal minimum loss, a higher-dimensional family of minimizers can receive greater small-temperature mass.

Among minimizer manifolds of the same dimension, larger manifold volume and smaller transverse curvature increase the local Gibbs weight.

This is the precise sense in which an idealized noisy dynamics can exhibit an *entropic preference* among equal-loss solutions.

5.10.5 Isolated minima as a special case

For an isolated nondegenerate minimum $\theta_j^\star$,

$$d_j = 0,$$

and Equation (5.8) reduces to the familiar Laplace approximation

$$Z_j(T) \sim e^{-L_\star/T}(2\pi T)^{P/2}\frac{1}{\sqrt{\det H_j}}. \tag{5.9}$$

For two isolated minima having exactly the same loss,

$$\frac{Z_1(T)}{Z_2(T)} \sim \sqrt{\frac{\det H_2}{\det H_1}}.$$

The common power of T cancels.

Thus one should not say simply that "lowering the temperature favors flatter isolated minima."

At equal depth, the leading relative preference comes from local curvature itself.

The dependence on temperature becomes more consequential when competing solution sets have different dimensions, different minimum losses, or more singular local geometries.

5.10.6 Beyond the regular Hessian picture

Neural-network parameterizations are often singular.

The Hessian may contain zero directions arising from symmetries, redundancies, or intersections of different minimizer sets.

In such cases, the ordinary Gaussian Laplace approximation is insufficient.

Under its analytic and measure assumptions, singular learning theory replaces the ordinary determinant formula by more general asymptotic invariants, including the Real Log Canonical Threshold. Chapter 7 explains how this generalizes the regular-manifold calculation.

The important point here is conceptual rather than technical:

probability near a singular solution depends on local set geometry,
not merely on the eigenvalues of one Hessian.

Thus "flatness" is only a shorthand for a richer geometric phenomenon.

5.10.7 Ordinary SGD is not isotropic Langevin dynamics

The Langevin model is useful because it makes the volume effect explicit.

It should not be identified with ordinary minibatch SGD.

A minibatch update may be written schematically as

$$\theta_{k+1} = \theta_k - \eta \nabla L(\theta_k) + \eta\, \xi_k(\theta_k), \tag{5.10}$$

where

$$\mathbb{E}[\xi_k(\theta) \mid \theta] = 0.$$

The covariance of the gradient noise generally depends on θ:

$$\mathrm{Cov}\,[\xi_k(\theta) \mid \theta] = \frac{1}{B}\Sigma(\theta),$$

for independent uniform draws with replacement, when Σ denotes the covariance of the single-example gradient. A uniform subset of size B without replacement has a finite-population correction, given below. Both formulas assume fixed per-example losses; batch-dependent forward operations require the covariance of the actual batch update.

A diffusion approximation may therefore take a form such as

$$d\theta_t = -\nabla L(\theta_t)\, dt + \sqrt{\frac{\eta}{B}}\, \Sigma(\theta_t)^{1/2} dB_t, \tag{5.11}$$

up to model-dependent constants and approximations.

This differs fundamentally from Equation (5.6).

The finite-step minibatch noise is generally

$$\boxed{\text{anisotropic} + \text{state dependent} + \text{not exactly Gaussian.}}$$

Momentum, adaptive preconditioning, finite step size, and learning-rate schedules introduce additional departures from equilibrium Langevin dynamics.

Therefore ordinary SGD does not in general possess the Gibbs stationary density

$$\pi_T(\theta) \propto e^{-L(\theta)/T}.$$

The Langevin calculation should instead be viewed as an analytically tractable model demonstrating how stochastic dynamics *can* produce an entropic selection bias. Even for that idealized process, an argument about equilibrium mass must be supplemented by finite-time accessibility or mixing estimates before it becomes a learning guarantee.

Regularized stationarity is not a common minibatch equilibrium. For sampling, use $L_{\mathrm{train}} = n^{-1}\sum_{i=1}^{n} \ell_i$ on n fixed examples, grouping each example's outputs in its loss. The residual map in (5.1) can absorb this normalization. Let $L_\lambda = L_{\mathrm{train}} + \lambda R$, $\lambda > 0$, with differentiable losses and a

deterministic differentiable penalty. Put

$$g_i(\theta) = \nabla \ell_i(\theta), \qquad \bar{g}(\theta) = \frac{1}{n}\sum_i g_i(\theta),$$
$$G_{\mathcal{B},\lambda}(\theta) = \frac{1}{B}\sum_{i\in\mathcal{B}} g_i(\theta) + \lambda \nabla R(\theta),$$
$$\Sigma(\theta) = \frac{1}{n}\sum_i (g_i(\theta) - \bar{g}(\theta))(g_i(\theta) - \bar{g}(\theta))^\top.$$

In (5.10), regularized SGD replaces the full data-loss gradient by ∇L_λ; the centered noise at a fixed parameter is unchanged by adding the same penalty gradient to every batch.

At a regularized stationary point θ_λ, $\bar{g}(\theta_\lambda) + \lambda \nabla R(\theta_\lambda) = 0$. For a uniformly sampled B-element subset, with $n > 1$ and $1 \le B < n$,

$$\boxed{\begin{aligned} \mathbb{E}_{\mathcal{B}} G_{\mathcal{B},\lambda}(\theta_\lambda) &= 0, \\ \mathrm{Cov}_{\mathcal{B}}\big(G_{\mathcal{B},\lambda}(\theta_\lambda)\big) &= \frac{n-B}{B(n-1)}\Sigma(\theta_\lambda). \end{aligned}} \tag{5.12}$$

The covariance is nonzero precisely when the individual data gradients differ. Only their average must cancel the regularizer. Positive regularization does not force such differences for every dataset. The exact criterion and proof are in Chapter 7, Proposition 5, and Appendix N, Proposition 75.

At exact interpolation of the fixed per-example squared losses, instead, all g_i vanish. Every regularized batch gradient then equals $\lambda \nabla R$: sampling covariance is zero, but the common update need not be zero. Unless the penalty gradient vanishes, this interpolator is not a stationary point of L_λ. Regularization can therefore sustain intrinsic minibatch fluctuations by changing the stationary balance, without itself supplying random forcing.

The strictly convex example and the architecture-specific normalized-ReLU rank obstruction in Chapter 7, Section 7.17, develop this distinction further. They separate a zero of the average update from a common zero of every batch update. Persistent fluctuations do not, by themselves, establish an isotropic noise model, a Gibbs law, or a generalization advantage.

5.10.8 Learning rate, batch size, and noise scale

The preceding diffusion approximation nevertheless suggests an important qualitative dependence.

Within the validity regime of the approximation, and holding the local noise covariance fixed, the stochastic component becomes stronger relative to deterministic drift when the learning rate increases or the batch size decreases.

Schematically, the diffusion scale contains a factor of the form

$$\sqrt{\frac{\eta}{B}}.$$

This motivates the often useful intuition that combinations of learning rate and batch size control an effective noise scale.

However, because

$$\Sigma = \Sigma(\theta),$$

there is generally no single scalar "temperature" that completely describes SGD.

The quantity

$$\frac{\eta}{B}$$

should therefore be interpreted as a heuristic noise-scale parameter, not as a universal thermodynamic law. With subset sampling the covariance factor is $(n - B)/[B(n - 1)]$ instead of $1/B$. Gradient-estimate covariance, update covariance, and diffusion covariance rate also differ by factors 1, η^2, and η, respectively, on the small-step time scale. A diminishing learning rate can therefore shrink the updates even when gradient-estimator covariance remains nonzero.

5.10.9 Why parameter-space flatness is not generalization

Even if stochastic dynamics favor a high-volume region of parameter space, another step is required before drawing conclusions about generalization.

Parameter-space flatness is not invariant under reparameterization.

For example, in homogeneous networks, transformations such as

$$W_\ell \to cW_\ell, \qquad W_{\ell+1} \to c^{-1}W_{\ell+1}$$

can leave the represented function unchanged while substantially altering local Hessian eigenvalues.

Hence

$$\boxed{\text{flat parameter region} \not\Longrightarrow \text{simple or well-generalizing function.}}$$

Some flat directions merely represent parameter symmetries.

Others may correspond to genuine robustness of the represented function.

These cases must be distinguished.

To connect optimization geometry to generalization one needs a valid statistical argument using specified complexity quantities, such as an appropriate norm, margin, or description-length control. Their behavior under the admitted reparameterizations must be tracked; not every norm or rank statistic is itself representation invariant. A geometric observation alone is not a generalization bound.

This issue is developed further in the chapter on bits, inductive bias, and generalization.

5.10.10 Selection and predictive recovery

In an interpolating regime, training need not find one distinguished parameter vector. It can reach any member of

$$\mathcal{M} = \{\theta : L_{\text{train}}(\theta) = 0\}.$$

The predictive question in this exact-fit comparison is whether the reached member performs well on the population, not whether it equals a designated θ_{SC} encoding the teacher's decomposition. At finite positive regularization, a returned well-fitting parameter need not belong to $\mathcal{M}$ or be a common equilibrium of all batch updates.

Neither a large fitting set nor a high-volume region ensures a good population predictor. The distinction is

representation:	an accurate feasible predictor is available;
optimization:	a specified procedure achieves an adequate fit;
selection:	the procedure favors particular fitting predictors;
generalization:	the selected predictor has small population risk.

Structural identification of the generating graph is a separate, stronger question. Failing to identify that graph is not a predictive failure when the learned function is accurate and feasible.

5.10.11 A role for overparameterization

The preceding observations suggest a qualified interpretation of the success of overparameterization. Under the regularity and realizability conditions above, excess parameters relative to the fitting rank can supply additional directions along a solution set. This removes the requirement of identifying one isolated parameter vector.

Stochasticity, initialization, architecture, and explicit or implicit regularization can then favor particular regions. The distribution of visited solutions depends on the actual dynamics, not on a count of parameters alone. The remaining predictive question is why the selected function is statistically controlled; it need not reproduce a privileged set of intermediate variables.

5.10.12 From geometric preference to finite-time learning

The geometric and stochastic calculations explain how low-loss regions can be abundant and can receive substantial equilibrium weight. They do not by themselves prove that a specified optimizer reaches an adequately fitting, low-complexity predictor in feasible time.

Chapter 6 studies one way to address this missing dynamical step: useful statistical footprints can supply accessible directions for learning. A staircase of constituent discovery is one possible organization of those directions. It is not an extra requirement once predictive success has been established by another valid argument.

The separate roles are best stated without an unconditional implication chain:

overparameterization:	can enlarge the family of fitting solutions;
stochastic or norm bias:	can favor particular regions or predictors;
detectable footprints:	can provide a route for specified updates.

A complete learning theorem must establish the relevant accessibility, fit, resource, and population guarantees. It need not establish that all of these proposed mechanisms operate in every successful run.

5.11 Norm-Driven Selection and Separation of Timescales

The preceding section emphasized that many parameters can realize the same low training loss. One source of selection among such solutions is regularization. A schematic explicitly regularized objective is

$$L_\lambda(\theta) = L_{\text{train}}(\theta) + \lambda R(\theta). \tag{5.13}$$

Here R may be a squared parameter norm, a product-norm penalty, a path norm, or another specified architecture-dependent criterion. Implicit bias can also select solutions, but need not be equivalent to descent on an explicit scalar penalty of this form.

The important point is not that one universal norm governs deep learning. Rather, fitting the observations and selecting among fitting solutions can occur on different timescales.

5.11.1 Fast fitting and slow selection

Suppose the dynamics reduce empirical loss quickly but continue to move substantially afterward. This suggests the possible separation

$$\tau_{\mathrm{fit}} \ll \tau_{\mathrm{select}},$$

where the first timescale concerns achieving a small training loss and the second concerns the slower influence of regularization or implicit bias.

A schematic trajectory is

$$\begin{gathered}\text{rapid fitting}\\ \Downarrow\\ \text{continued motion near a low-loss set}\\ \Downarrow\\ \text{selection of a different, possibly lower-complexity predictor.}\end{gathered}$$

This is a possible mechanism, not a theorem that such a separation always occurs or that the later predictor necessarily generalizes.

For $R(\theta) = \frac{1}{2}\|\theta\|_2^2$, explicit weight decay gives

$$\dot{\theta} = -\nabla L_{\mathrm{train}}(\theta) - \lambda\theta.$$

The regularization term generally moves the parameters away from exact interpolation, while the data-loss gradient acts to restore a good fit. The picture is slow motion near a low-loss set, not generally motion constrained exactly to $L_{\mathrm{train}} = 0$.

For SGD, continued motion can also include intrinsic fluctuations even where the average regularized update vanishes. This is not the same as a deterministic gradient-flow limit. For plain SGD with independent batch draws of fixed positive probabilities, continuous batch gradients, and learning rates bounded away from zero, any finite point limit must be a common zero of all batch gradients; see Appendix N, Proposition 77. Diminishing steps, normalized or adaptive updates, and averaging require their own analyses. A specified stopped iterate can nevertheless be a useful predictor; its fit, complexity, and resource guarantees still have to be established.

5.11.2 An idealized constrained picture

When fitting and selection are sufficiently separated, one can use the idealization

$$\min_{\theta\in\mathcal{M}} R(\theta), \qquad \mathcal{M} = \{\theta : L_{\mathrm{train}}(\theta) = 0\}, \tag{5.14}$$

when the exact fitting set is nonempty. A near-fitting formulation instead uses an explicit tolerance $L_{\mathrm{train}}(\theta) \le \eta$.

This separates the availability of a fitting solution, the dynamics of reaching a good fit, and preference among fitting solutions. Enlarging $\mathcal{M}$ can change the geometry of the first two questions, but its dimension alone does not establish convergence. Regularization expresses a preference for the third; a training theorem must show that an adequately fitting solution with sufficiently small penalty is actually obtained.

5.11.3 Neural collapse as a special example of structured selection

Neural collapse illustrates how late training can organize representations. In the classification regimes discussed in the book, within-class variability can become small, class means can approach an equiangular configuration, and classifier directions can align with those means.

In suitable idealized unconstrained-feature models, the loss together with norm regularization can select global minimizers with Equiangular Tight Frame geometry. The conclusion depends on the model and its assumptions; it is not a statement that every deep network must develop that configuration.

The useful lesson is that a fitting objective may leave many possibilities, while an additional norm criterion selects a geometrically organized subset. Such geometry is neither a premise of the compositional-witness argument nor, by itself, evidence that a generating DAG has been recovered. Population generalization remains a separate claim.

5.11.4 Grokking as delayed selection

The same timescale distinction suggests one possible interpretation of grokking: training error becomes small long before test performance improves substantially. Schematically,

$$\boxed{\tau_{\text{train}} \ll \tau_{\text{generalize}}.}$$

An early predictor may fit the sample with a relatively complicated representation. Later regularization or implicit bias may continue to reorganize that representation while the training loss stays small. A subsequent predictor can capture a rule that predicts more accurately beyond the sample.

This is a possible selection mechanism, not a universal explanation of grokking. Different settings can involve different feature-learning, barrier-crossing, or regularization effects. A test-accuracy transition does not itself identify which mechanism operated or prove recovery of the teacher's internal coordinates.

The point relevant here is that the timescale of fitting can be shorter than the timescale of selecting a predictor with useful population behavior.

5.11.5 Minimum norm need not identify the generating structure

A norm preference does not, by itself, identify the teacher's compositional structure. Nor is a raw parameter norm automatically an intrinsic measure of functional complexity.

For example, in a suitable homogeneous architecture,

$$W_\ell \longmapsto cW_\ell, \qquad W_{\ell+1} \longmapsto c^{-1}W_{\ell+1}, \qquad c > 0,$$

can preserve the represented function while changing individual layer norms. Their Frobenius product is unchanged by this particular reciprocal rescaling. Thus the example cautions against an unspecified norm; it does not invalidate a correctly formulated product-norm bound.

Predictive control does not require structural identification. This caution is fully compatible with the positive argument of Chapter H. Fix an admitted architecture and a complexity functional $\mathcal{C}(\theta)$, and form the sample-independent class

$$\mathcal{H}_Q = \{F_\theta : \mathcal{C}(\theta) \le Q\}.$$

For a suitable homogeneous feedforward setting, a candidate quantity is

$$\mathcal{C}_{\times}(\theta) = \prod_{\ell=1}^{L} \|W_\ell\|_F,$$

with the biases, activation assumptions, and other required factors included in the applicable statistical analysis. A multiplicative penalty can favor a smaller product among comparably fitting solutions. It does not literally minimize the exact Rademacher complexity or ensure that the resulting bound is small at the available sample size.

Let $\widehat{R}_S$ and R_{P} denote empirical and population risk. Suppose a valid uniform bound for this class gives

$$R_{\mathsf{P}}(h) \le \widehat{R}_S(h) + \beta_S(Q, \delta) \qquad \text{for all } h \in \mathcal{H}_Q$$

on an event of probability at least $1 - \delta$. For example, under the bounded-loss and Lipschitz assumptions specified in Chapter H, β_S can be controlled by a norm-sensitive Rademacher bound and a confidence term.

On that event, any returned predictor with $\widehat{f} \in \mathcal{H}_Q$ and $\widehat{R}_S(\widehat{f}) \le \eta$ satisfies

$$\boxed{R_{\mathsf{P}}(\widehat{f}) \le \eta + \beta_S(Q, \delta).}$$

No generating-node correspondence occurs in this implication. Interpolation is the special case $\eta = 0$, not a necessary premise for predictive success. The returned predictor can be a specified stopped iterate $\widehat{f} = F_{\theta_{j_*}}$. The bound requires its empirical fit and class membership, not convergence of every possible minibatch update to zero.

The actual joint success probability also depends on the optimizer achieving the fit and complexity budget; Chapter H states that conditional result explicitly. Choosing Q or the architecture from the same sample requires an appropriate simultaneous bound or other accounted-for selection rule.

The correct distinction is therefore

norm control can support predictive generalization without identifying the generating computation.

5.11.6 Why irrelevant features need not simply decay away

If a target does not depend on an input variable, it is tempting to conclude that norm minimization must set the corresponding first-layer weights to zero. This does not follow in a general nonlinear network. Functional independence can arise through cancellations, distributed representations, or redundant pathways rather than the literal absence of connections.

Moreover, fitting a finite sample does not establish global independence from a coordinate. A claim of feature selection by norm minimization therefore needs additional assumptions.

The distinction is

functional simplicity $\neq$ coordinate sparsity of a parameterization.

A dense representation can be a successful feasible predictor. A claim about visibly sparse weights is stronger than that predictive claim.

5.11.7 Selection among well-fitting predictors

As an idealized exact-fit comparison, suppose $\theta_1, \theta_2 \in \mathcal{M}$ both fit the sample and have similar penalties,

$$R(\theta_1) \approx R(\theta_2).$$

There are two different issues.

If both predict accurately but only one resembles the generating factorization, this is not an ambiguity that predictive learning must resolve. Structural identification needs a separate criterion.

If one has high population risk, the numerical similarity of the penalties alone does not distinguish the two. The relevant question is whether the chosen complexity class has an informative uniform bound for this sample size. On an event where both belong to such a class, the preceding inequality applies to both; neither can have population risk larger than the common empirical-error and complexity allowance.

A bad interpolant with a similar penalty can therefore signal an uninformative complexity budget, inadequate statistical assumptions, or a failure of the stated high-probability event. It is not evidence that predictive success universally requires discovering each teacher constituent.

Regularization must be connected to a valid statistical quantity and to a feasible optimization guarantee. Those requirements, rather than identification of a designated factorization, determine its predictive role. The same statistical reasoning applies to two predictors with nonzero empirical risks: the bound includes each predictor's own empirical error. Neither predictor is required to be a common minibatch equilibrium.

5.11.8 From selection to accessible learning signals

The remaining finite-time question is how the specified optimizer reaches a predictor with both adequate fit and adequate complexity control. Genericity and detectable footprints provide one proposed route. A constituent-learning staircase can be a constructive realization of that route, sometimes proving structural recovery as well.

These mechanisms retain a substantive role: they attempt to establish the algorithmic success that the conditional prediction argument assumes. They are not additional requirements for recognizing an already successful feasible network as a compositional witness.

Neural collapse and grokking remain useful illustrations of continued representation change during or after fitting. Neither makes genericity a universal prerequisite, nor requires the final coordinates to reproduce the generating ones.

The transition to Chapter 6 is therefore

> Which observable signals can the chosen learning dynamics exploit to find an accurate predictor with feasible resources?

The question is learner-relative. Recovering a specified hidden computation is an important stronger version, not the definition of all successful prediction.

5.12 Summary

Compositionality helps optimization in a precise but limited sense. With the computational, regularity, and stability conventions of Chapter 4, an economical realization removes a representation-size obstruction. It supplies a feasible object to be sought, not an automatically tractable search over all candidates.

Two obstacles to a specified learning procedure remain distinct: geometry can attenuate or distort useful gradients, and the current observations and probes can fail to expose a detectable learning signal. Architecture, normalization, initialization, and preconditioning can address aspects of the first. Chapter 6 studies genericity and footprints as mechanisms relevant to the second.

Faithful traces provide a different form of help: they supply local targets and permit separately supervised module objectives. Partial and surface exposure can also be useful. They need not reproduce the true latent computation, and they are not reserved for cases in which end-to-end learning is impossible. Local learnability, coverage, and accuracy after recomposition remain to be established.

Overparameterization introduces another distinction. Conditional on interpolation and local regularity, fitting solutions can have large Hessian nullspaces or form manifolds. Idealized Langevin calculations show how loss, volume, and transverse curvature affect equilibrium weight. Neither local degeneracy nor equilibrium preference alone proves finite-time SGD convergence. Chapter 7 develops this issue further.

Regularization can affect both predictor selection and the stationary balance of sample gradients. A zero full regularized gradient can coexist with nonzero minibatch covariance, unlike exact interpolation of the fixed per-example squared losses. Predictive success concerns the fit and complexity of the returned model, not whether every possible batch update has stopped.

Norm-driven selection can be statistically meaningful even when it does not recover the generating factorization. When training achieves a good fit within a properly controlled class, a valid bound can imply

$$R_{\mathsf{P}}(\widehat{f}) \leq \widehat{R}_S(\widehat{f}) + \beta_S.$$

The bound must be informative, and the optimizer must actually achieve the required fit and complexity budget. Exact interpolation is one important regime, not the only form of predictive success.

The resulting division is

C1:	feasible representation;
C2:	accessible learning signal;
C3:	supplied intermediate information;
optimization:	finding a suitable fit within feasible resources;
statistical selection:	controlling the selected predictor's risk.

These roles are complementary, not a universal sequence of implications. The principal predictive questions are whether an accurate feasible predictor is available, whether training can find a suitable fit, and whether the selected predictor generalizes.

Once an accurate predictor with feasible finite-precision evaluation has been obtained, its own computation provides a bounded-fan-in compositional realization. This is the witness conclusion developed in Chapter H. It does not by itself establish a fixed smooth coarse-grained decomposition, nor does it require matching the teacher's intermediate variables.

The chapter therefore ends with two separate questions:

Predictive learning: how does training find a feasible function that predicts accurately beyond the sample?

Structural recovery: when does it additionally identify the generating computation, up to controlled equivalences?

Genericity and staircase theories can contribute to both questions under specified assumptions. The next chapter examines their learning signals; it does not impose generating-structure recovery as an extra condition on every successful predictor.

CHAPTER 6

Genericity: Where the Optimization Signal Comes From

Why can local gradient-based methods make useful progress on deep, non-convex learning problems?

Chapter 5 separated two obstructions. One is geometric: an informative signal may be badly transmitted through a parameterization. The other concerns accessibility: the current data, loss, and available probes may not expose a useful direction. This chapter studies the second obstruction and its interaction with the first.

*Its central idea is **genericity**. Exact cancellations can be fragile under specified perturbations of the observation model, such as shifts or biases. Breaking a cancellation can create a nonzero low-complexity "footprint." Whether that footprint is large enough to detect, and whether the chosen optimizer can exploit it, are additional questions.*

The logical steps are therefore conditional:

a specified perturbation and suitable nondegeneracy

$\Downarrow$

a nonzero accessible footprint

a quantitative signal bound and controlled estimation cost

$\Downarrow$

a detectable footprint

compatibility with the learner's update directions

$\Downarrow$

a usable local optimization signal.

C2 denotes accessible learning signal in the framework of the book. Genericity is one possible source of that signal, and a staircase is one possible mechanism for exploiting it repeatedly. Neither is asserted to be a universal prerequisite for every successful learning procedure.

The immediate objective is predictive learning: finding a feasible function that performs well beyond the sample. Identifying the particular hidden computation that generated the observations is a stronger, separate objective. A signal theorem can contribute to either, but an initial footprint alone proves neither full convergence nor generalization. Chapter H separates those conclusions from the compositional witness supplied by an already successful feasible predictor.

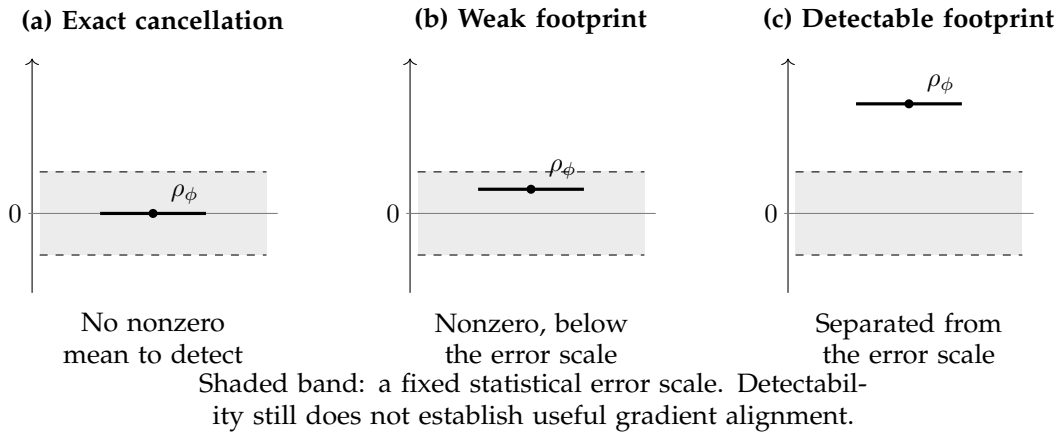


Figure 6.1: Existence, detectability, and use are distinct. The panels compare a population footprint with a common estimation-error scale; they are not confidence intervals centered on observed estimates and are not drawings of global loss landscapes. A perturbation may move a problem from (a) to (b) without reaching (c). Even (c) requires a connection to the chosen learning updates.

6.1 The Missing Ingredient: Informative Signal

Chapter 5 explained in what sense compositional sparsity makes optimization feasible to *pose*. Subject to the computational, regularity, and stability conventions of Chapter 4, an economical realization removes a representation-size obstruction. It does not make the set of candidate realizations small in cardinality, or guarantee an efficient search through that set.

The remaining learner-relative question is:

> What useful statistical information about the target is visible to the specified learning dynamics in their current representation?

Good conditioning of accessible directions need not create a missing target correlation in those directions. Conversely, an informative correlation can exist while its transmission through the parameterization is poor.

The answer depends on the target, the observation distribution, the current representation, and the statistics or features that the learning rule can actually use. Genericity is therefore not a property of the function f alone. It concerns an observed learning problem and a specified perturbation model.

An informative direction need not point toward one designated generating DAG. It can instead help construct a different predictor of the same input–output relation. Throughout this chapter, claims about identifying teacher constituents are distinguished from claims about improving prediction.

6.2 Footprints

Let $X \sim P$ and let $f : \mathcal{X} \to \mathbb{R}$ be the target. Suppose the learner has access to a family of low-complexity probes $\mathcal{F}_{\text{probe}}$. For each probe define

$$\rho_\phi(f, P) = \mathbb{E}_P[f(X)\phi(X)]. \tag{6.1}$$

The expectation is assumed to exist. Probe scale must be fixed when magnitudes are compared: one may, for example, impose $\|\phi\|_{L^2(P)} \leq 1$ and separately control the variance or tails of $f(X)\phi(X)$. Multiplying a probe by an arbitrary constant enlarges its raw footprint and its fluctuations together; it does not make estimation intrinsically easier.

Examples include coordinate functions, low-degree monomials, normalized Hermite polynomials under Gaussian inputs, Fourier characters on the Boolean cube, and features generated by a network at its current parameters. Low degree and low computational complexity are not identical notions, although they coincide in some of the examples considered here.

6.2.1 Existence

A footprint is present when an admitted probe satisfies

$$\rho_\phi(f, P) \neq 0.$$

This is qualitative: the target is not orthogonal to every direction in the specified probe family. Whether the coefficient is useful for a particular prediction or structural task still needs to be shown. An uncentered moment may, for example, reflect a mean response rather than identify a relevant input coordinate.

6.2.2 Size

Existence is not enough. A coefficient of magnitude e^{-d} may be nonzero but require exponentially many examples to resolve by ordinary averaging when its fluctuation scale remains of order one.

Under fixed normalization and suitable moment bounds, a useful target is

$$|\rho_\phi| \geq \frac{1}{\mathrm{poly}(d)}.$$

More generally the lower bound must be polynomial in the relevant problem size, accuracy, and conditioning parameters. The essential distinction is

existence:	$\rho_\phi \neq 0$,
detectability:	the signal can be estimated at feasible cost.

An inverse-polynomial raw magnitude supplies the second property only when the scale of the random variable and the cost of locating and evaluating the probe are also controlled.

6.2.3 Relation to gradients at initialization

For losses and parameterizations that admit differentiation under the expectation, gradient components involve correlations with features induced by the current network. Under squared loss, the relevant target is the current residual $f - F_\theta$, and the probes are parameter derivatives of F_θ.

Near initialization, low-degree components can dominate these probes in particular architectures and regimes. This must be derived for the specified model; it is not a universal restriction on gradient descent. A nonzero correlation with an arbitrary probe need not correspond to an available parameter update.

Section 6.11 gives the exact squared-loss identity. As features change, the accessible probe family can change too. This is one reason an initially hidden dependency may become visible later in training.

6.3 Statistical Sufficiency versus Computational Recoverability

The observations can contain enough information for accurate prediction without making that information easy for a feasible learner to extract. Statistical sufficiency and computational accessibility are different questions. Neither should be identified automatically with uniqueness of a hidden representation.

6.3.1 An information threshold

Fix a target class $\mathcal{F}$, input distribution P, observation model, error criterion, desired accuracy ε, and failure probability δ. Define $n_{\mathrm{info}}(\varepsilon, \delta)$ as the smallest sample size for which some estimator, without a computational restriction, attains the prescribed accuracy with probability at least $1 - \delta$, uniformly over the admitted targets.

For the predictive discussion, the criterion concerns population prediction, not identification of the teacher's DAG. A structural version requires its own equivalence class and loss; its threshold may be infinite if distinct generating structures are observationally indistinguishable.

For example, suppose a fixed finite comparison class $\mathcal{F}_\varepsilon$ has at most 2^{B_ε} elements and contains, for each target, a predictor with population risk within $\varepsilon/2$ of the desired benchmark. For loss in $[0, 1]$, uniform concentration and empirical risk minimization give the sufficient sample scale

$$n_{\mathrm{info}}(\varepsilon, \delta) \lesssim \frac{B_\varepsilon + \log(1/\delta)}{\varepsilon^2}.$$

Here ε is a risk tolerance. The display is an upper bound under the stated approximation assumption, not a universal equality for the minimal sample size, and not an L^2-function-error bound with the same exponent without conversion. The metric and observation model matter.

The effective description or covering complexity of the comparison class, rather than raw parameter count alone, controls this argument.

6.3.2 A recovery threshold depends on the learner

Now fix an algorithm A and computational budget T. Let $n_A(T; \varepsilon, \delta)$ be the smallest sample size at which A meets the *same* accuracy and uniform success requirement using at most T computation. Set this quantity to infinity if no such sample size exists. Then

$$n_A(T; \varepsilon, \delta) \geq n_{\mathrm{info}}(\varepsilon, \delta). \tag{6.2}$$

Equality need not hold. For fixed n, one can instead study $T_A(n; \varepsilon, \delta)$, the computational cost needed by the specified algorithm to meet the requirement.

Thus a single "sample threshold" does not describe all learning obstructions. Data and computation are different resources, and a computational budget must account for reading and processing the sample.

6.3.3 Genericity and the accessible signal

Fix a normalized probe family $\Phi = \mathcal{F}_{\mathrm{probe}}$. The largest footprint in this family is

$$\rho_\star = \sup_{\phi \in \Phi} |\rho_\phi(f, P)|. \tag{6.3}$$

For a specified probe, controlled bounded or sub-Gaussian products give a sufficient detection scale of the schematic form

$$n_{\mathrm{detect}} \lesssim \frac{K^2}{\rho_\star^2} \log \frac{1}{\delta}, \tag{6.4}$$

where K records the applicable fluctuation bound, and the display presumes that a probe attaining a comparable signal has been found. Searching a family introduces additional statistical and computational costs. A supremum over an enormous family is not an algorithm for locating a useful member.

This detection scale is not the information-theoretic sample complexity of the target class. It describes one way of extracting information. Consequently, one can have

$$n_{\mathrm{info}} \ll n_{\mathrm{detect}}$$

for a particular screening mechanism even though another algorithm can learn from fewer examples.

6.3.4 More data can still help recovery

For a fixed nonzero footprint with variance $v_\phi^2 > 0$, the signal-to-noise ratio of its sample average is

$$\frac{|\rho_\phi|\sqrt{n}}{v_\phi}.$$

Thus additional examples can make a weak coefficient usable long after prediction has become possible in principle. An exponentially weak coefficient relative to its fluctuations may make this particular route prohibitive.

If every population footprint in a fixed family vanishes, more averaging does not turn those zero means into nonzero screening signals. This is not, by itself, an impossibility theorem for all learners using the full samples, different statistics, or adaptively constructed representations. Such a theorem would require an explicit restriction on the learner and an indistinguishability argument for the admitted target class.

6.3.5 Changing the problem can be more powerful than adding data

Changes in coordinates, the observation distribution, the probe family, or the supplied intermediate information can expose a signal absent from the original screening problem. Breaking a symmetry is one possibility; exploiting a symmetry through an appropriate representation is another. A perturbation can also preserve a cancellation or attenuate useful information. Noise is not automatically beneficial.

These interventions have different meanings. C1 concerns the existence of a feasible representation. C2 concerns signals accessible to the learner. Generic perturbations are one proposed source of those signals; C3 changes the available observations by supplying intermediate information. They are not interchangeable assumptions.

6.3.6 A simple symmetric example

Parity-like targets can have compact compositional descriptions while all correlations with a specified low-order family vanish under a symmetric product distribution. This is consistent with the full input–output distribution determining the target function on its support.

The distinction is between information present globally and information visible through the chosen probes. It does not imply that every learning algorithm fails on parity. A different representation, probe family, or algorithm may exploit information that linear screening misses.

6.3.7 Trace supervision changes the observation model

Suppose

$$f = g_L \circ \cdots \circ g_1.$$

From final pairs $(x, f(x))$ alone, predictive learning need only find an accurate feasible input–output map. It need not identify these particular intermediate variables. Recovering the specified decomposition is the stronger structural problem.

If a faithful trace is supplied,

$$z_0 = x, \qquad z_\ell = g_\ell(z_{\ell-1}), \quad \ell = 1, \ldots, L,$$

then the pairs $(z_{\ell-1}, z_\ell)$ permit local supervised objectives. The learner no longer has to infer a particular constituent solely through its indirect contribution to the final output.

hidden constituent $\longrightarrow$ observed local target.

Local learnability, coverage of the relevant parent states, and error under recomposition still need analysis. Trace exposure can assist even when end-to-end learning is possible; it is not restricted to cases in which genericity fails.

6.3.8 Representation, information, and recovery

The questions can be kept separate as follows:

C1: representation	Does a feasible realization exist?
statistical sufficiency	Do the observations permit accurate prediction?
C2: accessible signal	Can the learner detect and use relevant information?
C3: trace exposure	Is intermediate information supplied?
algorithmic success	Does the specified procedure succeed efficiently?

Structural identification adds a separate criterion when it is required. A small statistical hypothesis complexity does not by itself make optimization easy, and an optimization obstruction does not prove statistical insufficiency.

6.3.9 The recovery gap

When $0 < n_{\mathrm{info}} < \infty$ and the other quantities are defined for the same learning problem, write

$$\Gamma_A(T) = \frac{n_A(T; \varepsilon, \delta)}{n_{\mathrm{info}}(\varepsilon, \delta)}.$$

A value near one indicates sample efficiency at the allowed computation; a large value indicates an algorithm-dependent gap. Detectable, usable footprints can help keep this gap small for a suitable learning rule, but qualitative genericity alone does not bound it.

Trace supervision changes the observation model, so both thresholds must be reconsidered for the trace-assisted problem. It is not a theorem that the numerical ratio above decreases merely because extra variables are supplied.

6.3.10 The central lesson

Statistical sufficiency asks whether the observations permit the desired accuracy. Computational recoverability asks whether the specified learner can attain it with its available operations and resources.

The remaining sections examine concrete signal mechanisms: analytic shifts, Gaussian integration by parts, biased parity, and the possible progression from one learned representation to another. The final AGOP section studies a complementary diagnostic of sensitivity geometry. None of these analyses identifies predictive success with mandatory reconstruction of the generating computation.

6.4 Genericity as Symmetry Breaking

The central hypothesis is that many exact cancellations are fragile within an appropriate perturbation family. A target invisible to one set of probes under one observation distribution may acquire nonzero correlations under another.

Here "generic" must refer to a specified mathematical model: for example, almost every parameter under an absolutely continuous law, or a residual set in a stated function-space topology. These meanings should not be substituted for a quantitative probability bound or a statement about all real data distributions.

6.4.1 Genericity is distribution-relative

The same f can satisfy

$$\mathbb{E}_P[f(X)X_i] = 0, \qquad \mathbb{E}_{P'}[f(X)X_i] \neq 0.$$

The relevant object is therefore the triple

$$(f, P, \mathcal{F}_{\text{probe}}),$$

together with the specified perturbation model and probe normalization.

Definition 6 (Genericity for recovery; informal)**.** *A learning problem* $(f, P, \mathcal{F}_{\text{probe}})$ *is* generic at low degree *relative to a specified perturbation model if typical perturbations in that model produce nonvanishing low-complexity footprints relevant to the stated learning objective. This qualitative usage does not include a magnitude, search-cost, or optimization guarantee unless those are separately imposed.*

6.4.2 Structural, observational, and optimization genericity

Three related ideas should be distinguished:

structural genericity:	degeneracies are unstable under allowed perturbations;
observational genericity:	a changed observation model breaks a cancellation;
optimization relevance:	the resulting signal is accessible and useful.

The first two suggest sources of signal. The third requires a link to the actual learning rule. No unconditional implication from either of the first two to the third is asserted. In particular, perturbations that preserve the relevant symmetry need not reveal a footprint.

6.5 Analytic Shift Families

A clean setting is a family of shifts

$$f_b(x) = f(x + b), \qquad F_\phi(b) = \mathbb{E}[f(X + b)\phi(X)].$$

Let the shift parameter range over a connected open set $U \subseteq \mathbb{R}^d$. Assume F_ϕ is real analytic on U. Polynomial targets with the necessary finite moments give an elementary example. For more general analytic targets, suitable integrable bounds are needed to pass analyticity through the expectation; pointwise analyticity alone is not an adequate assumption.

If $F_\phi \not\equiv 0$ on U, its zero set has Lebesgue measure zero. Consequently,

$$\boxed{F_\phi(b) \neq 0 \quad \text{for almost every } b \in U.}$$

The nondegeneracy condition is essential: a footprint identically forbidden by the chosen target, probe, and perturbation family stays zero. For an absolutely continuous distribution of shifts on U, the same statement holds almost surely.

This is a precise qualitative sense in which a cancellation can be fragile under shifts.

6.5.1 What this does and does not prove

The analytic argument establishes nonvanishing, not a lower bound. A value of 10^{-3} and a value of 10^{-300} are both nonzero. The theorem does not bound the probability of near-zero coefficients, the number of samples required to resolve them, or the time required by a learning algorithm.

Quantitative detectability therefore needs a separate estimate. Bounding an interaction order can help in a specified model, but is not a substitute for controlling coefficient size and fluctuation scale.

6.6 Thom Transversality as Geometric Motivation

There is also a geometric perspective. Let $f : \mathbb{R}^d \to \mathbb{R}$ be smooth and let $j^k f$ denote its k-jet extension. Some exact degeneracies can be represented by constraints defining a submanifold, or suitable strata, in

$$W \subset J^k(\mathbb{R}^d, \mathbb{R}).$$

Thom's transversality theorem says, under its differentiability and topological assumptions, that maps transverse to such sets are generic. The statement is not that a generic map necessarily avoids W. Rather, degenerate intersections are replaced by transverse ones.

For a smooth stratum of codimension greater than the domain dimension, transversality forces its preimage to be empty. Otherwise, a nonempty transverse preimage has the dimension given by the codimension count. These are geometric statements about jets, not sample-complexity bounds.

6.6.1 Why transversality is secondary here

The learning signals studied here are distributional quantities such as $\mathbb{E}[f(X)\phi(X)]$, not simply derivative constraints at a point. Transversality is therefore geometric motivation. It does not by itself prove that one of these expectations is nonzero, inverse-polynomially large, or aligned with a gradient update. Analytic shift families and explicit correlation calculations are more directly connected to those questions.

6.7 Two Exact Examples

6.7.1 Gaussian inputs and Stein's identity

Let

$$X \sim \mathcal{N}(\mu, \sigma^2 I_d), \qquad \sigma > 0.$$

Assume the regularity and integrability conditions required for Gaussian integration by parts. For instance, adequate absolute continuity in the ith coordinate, integrability of the displayed terms, and vanishing boundary terms suffice. Then

$$\mathbb{E}[f(X)(X_i - \mu_i)] = \sigma^2 \mathbb{E}[\partial_i f(X)]. \tag{6.5}$$

For this stated identity, nonzero average derivative and nonzero centered linear footprint are *equivalent*, since $\sigma > 0$.

Functional dependence on x_i is a different property. The function can depend on x_i while the average derivative cancels. Thus dependence alone does not establish either side as nonzero. A suitable perturbation may break that cancellation, but its magnitude and accessibility to the optimizer remain additional issues.

6.7.2 Boolean parity under biased product measures

Let $X_i \in \{-1, +1\}$ be independent, with $\mathbb{E}[X_i] = \mu_i$, and let

$$f(X) = \prod_{i \in S} X_i.$$

For every $j \in S$,

$$\mathbb{E}[f(X)X_j] = \prod_{i \in S \setminus \{j\}} \mu_i. \tag{6.6}$$

Proof. Since

$$f(X)X_j = \prod_{i \in S \setminus \{j\}} X_i,$$

independence gives

$$\mathbb{E}[f(X)X_j] = \prod_{i \in S \setminus \{j\}} \mathbb{E}[X_i] = \prod_{i \in S \setminus \{j\}} \mu_i.$$

□

Raw footprints and coordinate relevance. Equation (6.6) is an uncentered moment. If the aim is to identify relevant coordinates, a centered statistic avoids a contribution from the target mean. Direct subtraction of the product of means gives

$$\mathrm{Cov}(f(X), X_j) = \begin{cases} (1 - \mu_j^2) \displaystyle\prod_{i \in S \setminus \{j\}} \mu_i, & j \in S, \\ 0, & j \notin S. \end{cases}$$

Thus the relevant signal and normalization must be specified. Nearly deterministic coordinates, $|\mu_j| \approx 1$, introduce a further scale issue for covariance-based screening.

6.7.3 Existence versus size in one equation

If all factors in the product in Equation (6.6) are nonzero, the corresponding raw footprint is nonzero. But if $|\mu_i| \le \mu < 1$ and $|S| = k$, then

$$|\mathbb{E}[f(X)X_j]| \le \mu^{k-1}.$$

The signal may therefore be exponentially small in interaction order. This upper bound cannot be used as a lower bound on detectability.

To obtain a useful lower bound for bounded k, one needs, for example, $|\mu_i| \ge \underline{\mu}_d > 0$ on the relevant factors. The raw footprint then obeys

$$|\mathbb{E}[f(X)X_j]| \ge \underline{\mu}_d^{k-1}.$$

For covariance screening, the factor $1 - \mu_j^2$ must also be controlled. Thus bias can break symmetry without making the resulting signal large enough to use.

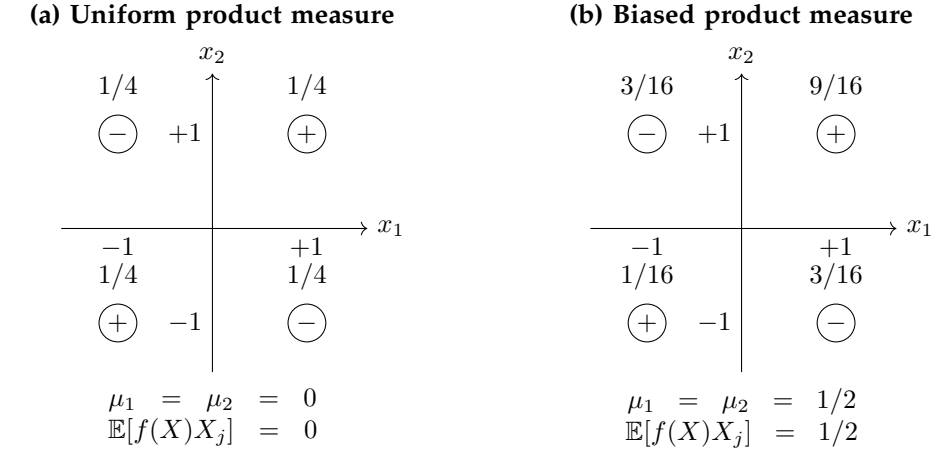


Figure 6.2: Biased sampling, not a translation of equally weighted points. The target is $f(x) = x_1x_2$; signs at the same four locations are its labels, and nearby fractions are sampling probabilities. In (b), each coordinate equals $+1$ with probability $3/4$. For either coordinate, the raw linear footprint is $1/2$ and the centered covariance is $3/8$. For higher-order parity, the product of biases can still be very small.

6.8 From Footprints to Detectability

The finite-sample assertion requires assumptions on the *product* being averaged, $Z_\phi = f(X)\phi(X)$. Boundedness or a controlled sub-Gaussian parameter for this product gives a root-n estimation scale. Bounds on f and ϕ separately must actually imply the needed product bound; Gaussian polynomial probes are generally unbounded and need appropriate tail or moment analysis.

For a concrete bounded case, let Φ_M be a fixed family of M probes and suppose $|f(X)\phi(X)| \le B$ almost surely for every member. Define

$$\widehat{\rho}_\phi = \frac{1}{n}\sum_{i=1}^{n} f(X_i)\phi(X_i).$$

The bounded-average concentration inequality and a union bound give

$$\mathbb{P}\left(\max_{\phi\in\Phi_M}|\widehat{\rho}_\phi-\rho_\phi|>t\right)\le 2M\exp\left(-\frac{nt^2}{2B^2}\right).$$

Consequently, a sufficient sample size for uniform error at most $\rho_{\min}/2$ is

$$n\ge\frac{8B^2}{\rho_{\min}^2}\log\frac{2M}{\delta}. \tag{6.7}$$

This resolves coefficients of magnitude at least $\rho_{\min}$ from zero at the specified precision. It is a sufficient screening guarantee, not a universal lower bound for all algorithms.

For $M=d$ coordinate probes, the multiplicity cost is logarithmic in d statistically. Evaluating all M probes still costs computation. For adaptive, sample-selected features, one needs fresh data conditional on the preceding stages or an appropriate uniform argument for the whole admitted adaptive procedure.

6.8.1 Polynomially detectable footprints

If B is polynomially controlled, the inverse signal is polynomial, and the relevant probes can be found and evaluated at polynomial cost, then Equation (6.7) gives a polynomial screening procedure. A statistical bound involving $\log M$ alone does not prove computational efficiency when M is exponentially large.

6.8.2 Exponentially weak footprints

With order-one fluctuation scale, a footprint of size $e^{-\Omega(d)}$ leads to an exponential sufficient sample budget for this averaging mechanism. A matching impossibility claim needs a lower-bound argument for a specified learner or observation model. Small raw magnitude by itself is not such an argument, especially when the variance also shrinks.

6.8.3 Parity leakage revisited

For equal biases with fixed $0<|\mu|<1$, the raw parity footprint is $|\rho|=|\mu|^{k-1}$. The preceding sufficient bound becomes

$$n=O\left(|\mu|^{-2(k-1)}\log\frac{d}{\delta}\right)$$

for bounded products and coordinate screening. At fixed k this is controlled; if $k=O(\log d)$ it can still be polynomial in d. Larger interaction orders can make this particular route super-polynomial. If the biases depend on d, their lower bounds must also be accounted for. This motivates a distinction between bounded interaction order and a quantitatively bounded signal cost.

6.9 Bounded Leap

The staircase metaphor concerns a sequence of learning problems in changing representations. Informally, the *leap* is the largest amount of new interaction structure that must become accessible at one stage before the learner can proceed. Its definition depends on the chosen coordinates, expansion, and learning rule; it is not identical to the local arity of the teacher DAG.

A bounded leap can limit the attenuation in a model with products of biases or constituent coefficients. For example, if every new factor has magnitude at least d^{-c} and only $\ell = O(1)$ such factors are needed, then their product is at least $d^{-c\ell}$.

But bounded leap alone is not a detectability theorem. A single newly exposed coefficient may be exponentially small, or the variance, probe search, conditioning, and errors in prior representations may dominate it. The quantitative requirement is that all these costs remain controlled.

bounded leap can help explain a detectable next signal; it does not replace a lower bound on that signal.

The two requirements remain distinct: an appropriate footprint must exist, and the learner must be able to detect and use it at the desired resolution.

6.10 The Staircase Picture

The preceding discussion suggests a stagewise mechanism. It is not a universal description of SGD, and the chapter does not derive its full induction from qualitative genericity alone.

6.10.1 Stage 1: detect an exposed footprint

A normalized low-complexity probe has a detectable correlation with a useful target component or residual. The learner must be able to locate that probe and relate it to an admissible update or local fitting problem. This can supply a first step away from an uninformative regime.

6.10.2 Stage 2: create new effective coordinates

The learning procedure constructs a useful representation and preserves it sufficiently accurately for subsequent computation. A dependency that was high-order in the original coordinates can become simpler in the new ones. This change is possible, not automatic: the next footprint and its size must be established for the new, possibly imperfect, representation.

6.10.3 Stage 3: repeat

The proposed progression is

$$\text{detect} \longrightarrow \text{learn and preserve} \longrightarrow \text{expose} \longrightarrow \text{detect}.$$

A recovery theorem must control finite samples, local optimization, accumulated feature error, and the number and cost of stages. Freezing modules or supplying explicit storage can help preserve features; residual connections alone do not guarantee preservation of arbitrary latent coordinates.

Such a theorem can explain how a learner reaches an accurate predictor and may additionally establish structural recovery. It is not a premise needed to recognize an already accurate, computationally feasible network as a compositional witness in Chapter H.

6.10.4 Cliff versus staircase

Relative to the current probes, a *cliff* means that the proposed low-order screening mechanism supplies no useful direction. A *staircase* means that a sequence of usable signals is available as the representation develops. A third case has nonzero but statistically too faint signals.

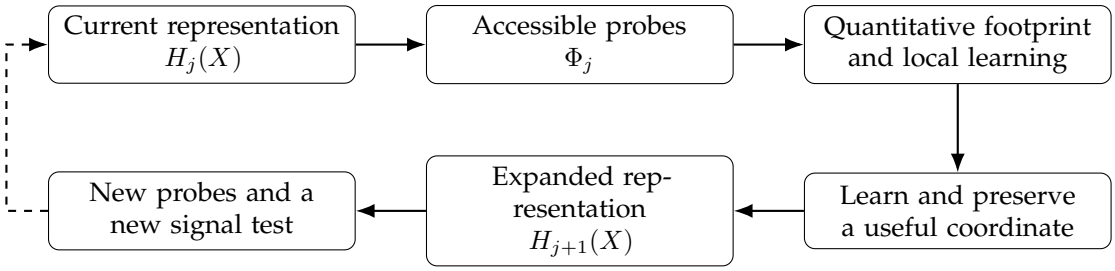


Figure 6.3: The staircase as a conditional learning mechanism, not a global landscape theorem. Each arrow requires its own estimation, optimization, or preservation argument. In particular, constructing one feature does not by itself guarantee a detectable next footprint.

These are descriptions of particular learning mechanisms, not an exhaustive classification of targets or loss landscapes. Failure of a staircase prediction need not imply failure of predictive learning. Nor must a successful learned coordinate coincide with a generating teacher variable.

6.11 Genericity and Gradient-Based Optimization

6.11.1 Population gradients

For a network $F(X;\theta)$ and population loss

$$\mathcal{L}(\theta) = \mathbb{E}[\ell(F(X;\theta), f(X))],$$

the accessible probes depend on the loss and parameterization. Under squared loss and conditions justifying differentiation under expectation,

$$\begin{aligned} \partial_{\theta_j}\mathcal{L}(\theta) &= \mathbb{E}[(F_\theta(X) - f(X))\partial_{\theta_j}F_\theta(X)] \\ &= -\rho_{\partial_{\theta_j}F_\theta}(f - F_\theta, P). \end{aligned} \tag{6.8}$$

The identity uses the unnormalized parameter derivative as written. When magnitudes across directions are compared, its scale and the estimation variance must be included.

At $F_\theta \approx 0$, the target correlation can dominate, but the approximation error in that assertion must be controlled. Later, the residual and the derivative probes both change. A nonzero target moment against some unrelated function need not appear in Equation (6.8).

A nonzero population gradient supplies a local descent direction under appropriate smoothness and step-size conditions. It does not guarantee a polynomial-time trajectory to small population error, identify a hidden constituent, or control generalization from finite samples.

6.11.2 The Statistical Query viewpoint

Population gradients are expectations, and minibatch updates estimate such quantities. This motivates Statistical Query (SQ) models of learning through approximate expectations [76].

An SQ lower bound applies only when the allowed queries, tolerances, and costs model the learner under consideration. The absence of several low-degree correlations is not, by itself, a lower bound for an adaptive SQ algorithm, much less for every procedure operating on the original

examples. A proof must identify which target distinctions remain hidden from the allowed queries at the available tolerance.

The framework is useful precisely because it makes the information accessible through a specified family of noisy statistics explicit. It should not be read as equating every form of gradient descent with a fixed low-degree screening method.

6.11.3 Information versus geometry revisited

At a fixed representation, geometry describes how update directions transmit and scale a signal; the footprint analysis asks which target information those directions reveal.

Multiplying an exactly zero gradient by a preconditioning matrix does not create a nonzero gradient. But changing the architecture or learning a new representation can change the probe family. Thus architectural changes need not leave signal accessibility fixed, even though they do not create new labels or add information to the raw observation model.

Normalization, residual pathways, and preconditioning may help under specified assumptions. Their contribution to accessible information and their contribution to conditioning should be analyzed rather than identified by terminology.

6.12 A Cautionary Example: Removing the Screening Signal

Let $L(x)$ be the population least-squares projection of f onto a specified finite-dimensional linear span containing the coordinate functions used for screening. Assume these functions and f belong to $L^2(P)$. For

$$r(x) = f(x) - L(x),$$

projection orthogonality gives

$$\mathbb{E}[r(X)X_i] = 0$$

for every coordinate included in that span. A mechanism based on those particular linear residual correlations therefore loses that signal after the projection has been removed.

6.12.1 What follows

If a proposed first stage relies specifically on the target's linear footprints, residualizing against the same span can make that stage uninformative. This is a statement about the chosen screening procedure.

6.12.2 What does not follow

Residual fitting is not thereby generically bad. The residual may retain quadratic or other low-complexity footprints, useful features already constructed by the network, or components that another procedure can learn. A stagewise method can preserve the fitted linear part and learn what remains using an expanded class.

The correct conclusion is narrower:

removing the projection used by a particular screening mechanism removes that mechanism's corresponding population signal.

This operation is also distinct from adding an identity pathway in a residual-network architecture. What other features remain available is the important learning question.

6.13 What Genericity Does Not Explain

Genericity is a proposed source of learning signal. Even a quantitatively detectable footprint is not interchangeable with the following properties.

6.13.1 Generalization

A nonzero or detectable coefficient does not by itself show that an empirically fitted predictor has small population risk. Hypothesis complexity, algorithmic stability, regularization, data dependence, and the observation model enter that separate analysis.

Conversely, norm-controlled generalization need not identify the teacher's constituents. Chapter H gives a conditional risk bound for a well-fitting predictor in an appropriate controlled class, without adding a structural-matching premise.

6.13.2 Conditioning

An informative statistic can be badly transmitted through a deep parameterization. Nonvanishing correlations do not control products of Jacobians, curvature, or the numerical precision of the updates.

6.13.3 Global convergence

An initialization signal does not prove that the entire trajectory continues toward a suitable fit. A staircase theorem would need to prove persistence or renewal of useful signal as representations change, together with finite-time and finite-sample control. The present calculations do not establish that theorem for arbitrary SGD.

6.13.4 Faithful internal structure

A learner can exploit predictive footprints without reconstructing the factorization that generated the examples. Structural recovery requires its own identifiability assumptions, controlled equivalences, and nodewise criterion. Merely abandoning uniqueness is not enough to remove this distinction: predictive success need not establish a match with any specified generating-equivalence class.

Once a feasible predictor is known to generalize, its own computation supplies the witness of Chapter H. No further genericity, staircase, or AGOP condition is needed for that particular implication. This does not remove such conditions from a constructive proof that actually uses them to establish the learner's success.

6.14 Genericity and Physical Symmetry

The discussion is not a claim that physical law lacks exact symmetries. Conservation laws and gauge, spatial, temporal, crystallographic, and permutation symmetries can be fundamental or useful. Symmetry can also supply an inductive bias rather than an obstruction.

The relevant object here is the observed learning problem. Boundary conditions, sampling distributions, nuisance variables, and measurement processes can break cancellations present in a more idealized model. They can also preserve those cancellations or obscure a signal. No universal

noise-induced genericity conclusion is intended.

genericity concerns a specified target–observation–probe problem,
not an assertion that fundamental physical symmetries disappear.

6.15 How C1, C2, and C3 Fit Together

The terminology now has the same scope as in the revised Interlude:

C1:	a feasible compositional representation exists;
C2:	useful learning signals are accessible;
C3:	intermediate information is supplied.

C1 does not establish optimization or generalization. Genericity provides one candidate source of C2, but a signal must be quantitatively detectable, affordable to locate, and useful to the specified updates. A staircase is one possible organization of these signals through time.

C3 is not only a fallback. It can assist a learner that could also learn end to end, and its usefulness depends on the fidelity and coverage of the exposed information and on the local learning problems.

These roles cooperate with optimization geometry and statistical selection; they are not a universal chain of necessary and sufficient conditions. In particular, finding a feasible predictor with small population error does not require demonstrating recovery of a privileged hidden DAG.

6.16 Summary of the Footprint Mechanism

The optimization puzzle is not only geometric. A specified learner also needs useful information in the statistics it can access. This chapter studies one proposed source of that information: nonzero footprints that appear or persist in a specified perturbation model.

The basic coefficient is

$$\rho_\phi(f, P) = \mathbb{E}[f(X)\phi(X)].$$

Its interpretation requires three distinctions:

nonvanishing:	$\rho_\phi \neq 0$;
detectability:	controlled scale, estimation, and search cost;
optimization use:	a connection to the actual update directions.

Analytic shift families give an almost-everywhere nonvanishing statement when their footprint function is analytic and not identically zero. Stein's identity relates centered Gaussian linear correlations to average derivatives. Biased parity gives an exact product formula that can be nonzero while remaining exponentially small in interaction order.

Bounded leap can help only together with quantitative coefficient and conditioning assumptions. A staircase theory must then explain how learning preserves useful coordinates and exposes later signals despite finite-sample and approximation error. These are meaningful research questions about a learning mechanism, not necessary extra tests for an already successful compositional predictor.

The next section studies AGOP, which describes sensitivity geometry when input or representation gradients are available. Chapter 7 then develops the geometry of interpolation and

stochastic selection. Chapter H combines representation, optimization, and statistical control into a conditional predictive argument while keeping structural identification separate.

6.17 AGOP as a Diagnostic of Low-Dimensional Structure

A related object that has become useful in the study of feature learning is the *Average Gradient Outer Product* (AGOP). For a differentiable scalar function

$$f : \mathbb{R}^d \to \mathbb{R}$$

and input distribution P, define

$$M_f = \mathbb{E}_{X\sim P}\left[\nabla f(X)\nabla f(X)^\top\right]. \tag{6.9}$$

Assume $\mathbb{E}\|\nabla f(X)\|^2 < \infty$, so that this matrix is finite. The matrix M_f is positive semidefinite and measures the average sensitivity of the function to perturbations in different input directions. Closely related objects appear in the active-subspace literature and in recent work on neural feature learning [**constantine2015active**, 132, 131].

AGOP is useful in the framework of this chapter, but its role should be distinguished carefully from genericity.

The genericity footprints introduced above have the form

$$\rho_\phi(f, P) = \mathbb{E}_P[f(X)\phi(X)].$$

They ask whether the input–output distribution exposes a detectable correlation to a low-complexity probe ϕ.

AGOP instead asks:

along which input directions is the function locally sensitive?

These are related questions, but they are not equivalent.

6.17.1 The clean case: multi-index structure

AGOP has a particularly transparent interpretation for functions that depend on a low-dimensional linear projection.

Proposition 4 (AGOP recovers the active linear span)**.** *Let*

$$f(x) = g(Wx),$$

where

$$W \in \mathbb{R}^{r\times d}, \qquad r \ll d,$$

and

$$g : \mathbb{R}^r \to \mathbb{R}$$

is differentiable, with $\mathbb{E}\|\nabla g(WX)\|^2 < \infty$.
Define

$$G = \mathbb{E}\left[\nabla g(WX)\nabla g(WX)^\top\right].$$

Then

$$M_f = W^\top G W. \tag{6.10}$$

Consequently,

$$\operatorname{range}(M_f) \subseteq \operatorname{range}(W^\top),$$

and

$$\operatorname{rank}(M_f) \leq r.$$

If W has full row rank and

$$G \succ 0,$$

then

$$\operatorname{rank}(M_f) = r$$

and

$$\operatorname{range}(M_f) = \operatorname{range}(W^\top).$$

Proof. By the chain rule,

$$\nabla_x f(x) = W^\top \nabla g(Wx).$$

Therefore

$$\begin{aligned} M_f &= \mathbb{E}\left[W^\top \nabla g(WX)\nabla g(WX)^\top W\right] \\ &= W^\top G W. \end{aligned}$$

The range and rank statements follow immediately.

□

Thus, for a genuine low-dimensional multi-index model,

AGOP can identify the active linear subspace.

This is a population identity when the gradients and distribution are known. Estimating the active span from data additionally requires control of gradient estimation, sampling error, and the relevant spectral gap. It is not an algorithm for obtaining the unknown target gradients from labels alone.

Even at the population level, identifying an active linear span is substantially weaker than reconstructing a compositional computation graph. Neither conclusion is a prerequisite for a feasible predictor to be accurate.

6.17.2 Compositional sparsity does not imply low-rank AGOP

A compositionally sparse function need not have low-rank AGOP.

Consider the completely separable function

$$f(x) = \sum_{i=1}^{d} x_i^2.$$

The squares are unary constituents, and their outputs can be combined by a binary addition tree. Thus the function has a bounded-arity compositional realization, even though its output depends on all the coordinates.

But

$$\nabla f(x) = 2x.$$

For an isotropic input distribution satisfying

$$\mathbb{E}[XX^\top] = I_d,$$

we obtain

$$M_f = 4I_d.$$

Hence

$$\operatorname{rank}(M_f) = d.$$

Therefore

$$\boxed{\text{compositional sparsity} \not\Longrightarrow \text{low-rank AGOP.}}$$

Sparse compositionality concerns the locality of the constituent computation.

AGOP rank concerns the number of linearly independent directions along which the final function varies.

These are different notions of complexity.

6.17.3 AGOP can miss nonlinear interactions

AGOP also does not in general reveal which variables interact inside a constituent.

Consider

$$f(x_1, x_2) = x_1 x_2,$$

with X_1, X_2 independent, mean zero, and unit variance.

Then

$$\nabla f = \begin{bmatrix} X_2 \\ X_1 \end{bmatrix}.$$

Thus

$$M_f = \begin{bmatrix} \mathbb{E}[X_2^2] & \mathbb{E}[X_1 X_2] \\ \mathbb{E}[X_1 X_2] & \mathbb{E}[X_1^2] \end{bmatrix} = I_2.$$

The function contains a pure multiplicative interaction between x_1 and x_2, yet the AGOP has zero off-diagonal entry.

Hence

$$\boxed{M_{ij} = 0 \not\Longrightarrow x_i \text{ and } x_j \text{ do not interact in } f.}$$

AGOP is an outer product of *first derivatives*. It is not a general interaction tensor.

6.17.4 Why a compositional DAG does not generally produce a block-diagonal AGOP

Consider the simple composition

$$f(x_L, x_R) = h\left(g_L(x_L), g_R(x_R)\right).$$

The gradient is

$$\nabla f = \begin{bmatrix} h_1 \nabla g_L \\ h_2 \nabla g_R \end{bmatrix},$$

where

$$h_1 = \partial_1 h(g_L, g_R), \qquad h_2 = \partial_2 h(g_L, g_R).$$

The off-diagonal block of the AGOP is

$$M_{LR} = \mathbb{E}\left[h_1 h_2 \nabla g_L \nabla g_R^\top\right]. \tag{6.11}$$

Even if

$$X_L \perp X_R,$$

Equation (6.11) need not vanish, because h_1 and h_2 depend jointly on the outputs of both branches.

Additional symmetry or factorization assumptions are required.

Thus

$$\boxed{\begin{array}{c}\text{a sparse computation graph does not generically imply} \\ \text{a block-diagonal AGOP in input coordinates.}\end{array}}$$

When a block structure does appear empirically, it may be informative, but it should be interpreted as an additional property of the target, distribution, and coordinate system rather than as an automatic consequence of compositional sparsity.

6.17.5 Coordinate dependence

AGOP is also coordinate dependent.

Suppose

$$x = Az$$

for invertible A. Compare the corresponding distributions: if Z has law P_Z, then $X = AZ$ has its pushforward law P_X. Define

$$\widetilde{f}(z) = f(Az).$$

Then

$$\nabla_z \widetilde{f} = A^\top \nabla_x f,$$

so

$$M_{\widetilde{f}} = A^\top M_f A. \tag{6.12}$$

This is a *congruence transformation*, not a similarity transformation.
For invertible A,

$$\operatorname{rank}(M_{\widetilde{f}}) = \operatorname{rank}(M_f),$$

but the nonzero eigenvalues and their ratios need not be preserved.
Only for orthogonal A does Equation (6.12) reduce to a similarity transformation.
Therefore

AGOP rank is invariant under invertible linear reparameterization, but its eigenspectrum is not.

This is another reason to distinguish intrinsic active dimension from coordinate-dependent spectral geometry.

6.17.6 A single AGOP cannot identify a hidden DAG by diagonalization

Because M_f is symmetric positive semidefinite, it always admits an orthogonal eigendecomposition

$$M_f = Q \Lambda Q^\top.$$

Thus every AGOP can be diagonalized, including targets for which no economical compositional representation has been established.
Consequently, solving

$$\min_{Q \in O(d)} \left\| \operatorname{offdiag}(Q^\top M_f Q) \right\|_F^2$$

cannot by itself test for compositionality.
The minimum is zero for every symmetric M_f.
Recovering latent components from rotated coordinates requires additional information, for example:

- several matrices that must be diagonalized jointly;
- independence assumptions;
- interventions or multiple input distributions;
- higher-order cumulants;
- higher derivatives;

- architectural information.

Thus

AGOP is not, by itself, an inverse map from functions to DAGs.

6.17.7 Layer-wise AGOP as an empirical diagnostic

AGOP may nevertheless be useful for studying a trained network.

Let

$$h_\ell = H_\ell(X)$$

denote the representation at layer ℓ.

Write the remaining computation as

$$f(X) = F_\ell(h_\ell).$$

Define the layer-wise AGOP

$$M_\ell = \mathbb{E}\left[\nabla_{h_\ell} F_\ell \nabla_{h_\ell} F_\ell^\top\right]. \tag{6.13}$$

Tracking

$$M_0, M_1, \ldots, M_L$$

can reveal how the sensitivity geometry changes across depth.

For example, one can measure:

- effective rank;
- principal eigenspaces;
- concentration of sensitivity;
- alignment of learned weight spaces with active directions;
- changes during training.

Recent work on the Neural Feature Ansatz and related feature-learning phenomena suggests that learned features can align with AGOP-like principal subspaces in important model classes [132, 131].

The interpretation should remain conditional:

alignment with AGOP eigenspaces is evidence about learned sensitivity geometry, not a proof that SGD has recovered the target DAG.

6.17.8 Target AGOP versus learned-model AGOP

There is also an observational distinction.

The target AGOP

$$M_f = \mathbb{E}[\nabla f \nabla f^\top]$$

requires derivatives of the unknown target.

These are generally not available from ordinary supervised examples

$$(X_i, Y_i).$$

What can be computed after training is instead

$$M_{\widehat{f}} = \mathbb{E}[\nabla \widehat{f} \nabla \widehat{f}^\top].$$

Derivative approximation gives a sufficient condition for the two matrices to be close. It is not a necessary condition: different gradient fields can have the same average outer product. Small prediction error alone, however, supplies no such guarantee.

Lemma 2 (Gradient approximation controls AGOP approximation)**.** *Let*

$$a(X) = \nabla \widehat{f}(X), \qquad b(X) = \nabla f(X),$$

and assume both vector fields have finite second moments.

Then

$$\left\| M_{\widehat{f}} - M_f \right\|_{\mathrm{op}} \leq \left(\sqrt{\mathbb{E}\|a\|^2} + \sqrt{\mathbb{E}\|b\|^2} \right) \sqrt{\mathbb{E}\|a-b\|^2}.$$

Proof. Using

$$aa^\top - bb^\top = (a-b)a^\top + b(a-b)^\top,$$

we obtain

$$\begin{aligned} \|M_{\widehat{f}} - M_f\|_{\mathrm{op}} &\leq \mathbb{E}[\|a-b\| \, \|a\|] + \mathbb{E}[\|b\| \, \|a-b\|] \\ &\leq \sqrt{\mathbb{E}\|a-b\|^2} \left(\sqrt{\mathbb{E}\|a\|^2} + \sqrt{\mathbb{E}\|b\|^2} \right), \end{aligned}$$

by Cauchy–Schwarz.

□

Thus good prediction error alone does not guarantee recovery of the target AGOP. The displayed lemma supplies a sufficient derivative-based route; other assumptions could also control the matrices directly.

This distinction is particularly important for the framework of Chapter H: population accuracy establishes a predictive claim, whereas matching the target's sensitivity geometry is a stronger, separate claim.

6.17.9 Relation to genericity

The role of AGOP in the present theory can now be stated precisely.

Genericity asks whether useful target structure creates detectable low-complexity footprints in the observed problem.

AGOP asks whether the function exhibits low-dimensional sensitivity geometry once gradients with respect to some representation are available.

Therefore:

genericity footprint:	Can the hidden structure be detected from the learning signal?
AGOP:	What sensitivity subspace does the function exhibit?

A target can have a simple AGOP and still be difficult to learn.

A target can be compositionally sparse and have a full-rank AGOP.

And a target can contain important nonlinear interactions that AGOP does not reveal.

6.17.10 Relation to the staircase

AGOP may be informative once feature learning has begun. Suppose the remaining computation in a learned representation is exactly

$$F_\ell(h) = g(W_\ell h), \qquad \operatorname{rank}(W_\ell) \le r.$$

Under the integrability assumptions above, its AGOP has rank at most r. For an approximate multi-index model, closeness of function values alone does not imply low-rank AGOP.

A sufficient quantitative hypothesis is derivative control. Let $\widetilde{F}_\ell(h) = g(W_\ell h)$ and, under the law of $h_\ell = H_\ell(X)$, suppose

$$\|\nabla F_\ell - \nabla \widetilde{F}_\ell\|_{L^2} \le \zeta, \qquad \|\nabla F_\ell\|_{L^2} + \|\nabla \widetilde{F}_\ell\|_{L^2} \le C.$$

Lemma 2 then gives

$$\|M_\ell - M_{\widetilde{F}_\ell}\|_{\mathrm{op}} \le C\zeta, \qquad \operatorname{rank}(M_{\widetilde{F}_\ell}) \le r.$$

Thus M_ℓ is close in operator norm to a rank-r matrix. For small nonzero ζ, it can still have full exact rank. Eigenvector recovery also requires spectral separation.

Under those additional hypotheses, concentrating sensitivity into a few directions can be an empirical signature of a useful change of representation. It is not an automatic consequence of learning one constituent, nor a necessary signature of every successful staircase. The earlier separable quadratic example already shows why compositional sparsity need not reduce global active dimension.

The logical direction remains important: a specified feature-learning mechanism may predict changes in AGOP; observing AGOP alone does not prove that mechanism or identify the teacher's DAG.

6.17.11 Practical use

For experiments, AGOP is therefore best treated as a diagnostic rather than as a structural oracle.

Useful measurements include:

$$\operatorname{rank}_{\mathrm{eff}}(M_\ell),$$

principal-angle alignment between successive eigenspaces,

$$\angle\left(\mathrm{range}(W_\ell^\top), \mathrm{TopEig}(M_\ell)\right),$$

and the evolution of these quantities over training.

Such measurements could test whether the representations learned by SGD progressively concentrate sensitivity into lower-dimensional directions.

They should not be interpreted as uniquely recovering the underlying computation graph.

6.17.12 Summary

The useful conclusion is deliberately limited:

AGOP can reveal low-dimensional active linear structure and can diagnose changes in learned sensitivity geometry.

But

AGOP is not, in general, an anatomical image of compositional sparsity.

In particular,

compositional sparsity $\not\Longrightarrow$ low-rank AGOP, zero AGOP cross-term $\not\Longrightarrow$ absence of nonlinear interaction, AGOP eigendecomposition $\not\Longrightarrow$ recovery of the target DAG.

Its proper role in the framework of this chapter is therefore as an empirical and mathematical diagnostic of representation geometry, complementary to—but distinct from—the genericity footprints used by particular learning mechanisms.

From signal mechanisms to predictive learning. The chapter has kept three objects separate: accessible statistical footprints, sensitivity geometry, and predictive accuracy. A signal or AGOP calculation can help explain a specified learning procedure, but neither is an extra test that every accurate predictor must pass.

The next chapter studies overparameterization, interpolation geometry, and stochastic selection. Chapter H then asks when a specified optimizer finds a feasible, well-fitting predictor with statistical complexity small enough to control population risk. Once that success is established, the learned computation supplies a compositional witness. Identifying a generating DAG, verifying a staircase trajectory, or matching the teacher's AGOP are stronger objectives requiring their own assumptions.

CHAPTER 7

Overparameterization and the Geometry of Global Minima

Why can local optimization in a highly nonconvex deep network reach very small training error? Initialization, normalization, skip connections, learning-rate schedules, and stochastic gradients can all matter. There is also a geometric question: what kind of set is the optimizer trying to reach?

In a realizable overparameterized problem, the interpolation constraints can leave many parameter directions unconstrained. At regular points, the zero-loss set is a manifold whose dimension equals the number of parameters minus the rank of the independent fitting constraints. At singular points, the geometry can be more complicated. In either case, an interpolating solution need not be an isolated parameter vector.

This observation motivates two different arguments. A deterministic argument studies whether the prediction residual contracts along the training trajectory. A stochastic argument studies the equilibrium weight of neighborhoods of low-loss sets under an idealized Langevin model. The first can give a convergence guarantee under a conditioning hypothesis. The second makes an energy–volume preference explicit, but does not by itself establish how quickly training reaches equilibrium.

Appendix N supplies a complementary analytic result: an all-time gradient flow whose closure is compact inside the analytic domain converges to a critical point. That point need not interpolate. Regularity of the objective, conditioning of prediction dynamics, and statistical control of the selected function have different roles.

The distinction is essential:

$$\begin{array}{c} \textit{local degeneracy} \\ \neq \ \textit{a large basin of attraction} \\ \neq \ \textit{finite-time convergence of SGD.} \end{array}$$

Overparameterization can contribute to a convergence backstop, but only when the additional dynamical conditions are established. Parameter count alone supplies neither realizability nor an optimization theorem.

Regularization also changes the relevant stochastic endpoint. Exact interpolation of fixed per-example squared losses removes their minibatch sampling variance. A stationary point of a regularized objective need not interpolate: its average data gradient can balance the penalty gradient while individual minibatch updates remain nonzero. The resulting intrinsic fluctuations and a normalized-network rank obstruction are developed below. They are not the same statement as isotropic Langevin equilibrium.

The remaining predictive question is which function training selects and whether it generalizes. A change of internal basis or a decomposition different from the teacher's is not a predictive failure. Consistent with

the Interlude, this chapter separates fitting the sample, controlling population error, and identifying the generating computation. The bridge to Chapter H is from optimization geometry to a feasible, well-fitting predictor with adequate statistical-complexity control, not to a compulsory reconstruction of hidden teacher variables.

7.1 The classical puzzle

Consider a neural network

$$F_\theta : \mathcal{X} \longrightarrow \mathbb{R}, \qquad \theta \in \mathbb{R}^p,$$

and training examples (x_i, y_i), $i = 1, \dots, n$. We begin with the unregularized empirical squared loss

$$\widehat{\mathcal{L}}(\theta) = \frac{1}{2n} \sum_{i=1}^{n} \big(F_\theta(x_i) - y_i\big)^2. \tag{7.1}$$

The map $\theta \mapsto \widehat{\mathcal{L}}(\theta)$ need not be convex, even though squared loss is convex in its prediction argument. Saddles, poor local minima, and complicated critical sets are not ruled out by making the network large.

The motivating phenomenon is that highly overparameterized networks can nevertheless be trained to essentially zero empirical error. The useful question is not whether nonconvexity has disappeared, but

> How does overparameterization change the geometry of fitting solutions, and what additional conditions make those solutions accessible to the specified learning dynamics?

The discussion initially assumes a fixed training sample and a *realizable* model: at least one finite parameter vector fits that sample exactly. This assumption is distinct from having many parameters. Conflicting labels at an identical input, or restrictions on the architecture, can prevent exact interpolation regardless of raw width.

For the differential calculations, the training prediction map is assumed to be at least C^2 on the region under discussion. For a nonsmooth architecture, such calculations apply on an appropriate smooth region; crossing nonsmooth boundaries requires a separate argument. The singular analytic asymptotics later in the chapter have their own, stronger assumptions.

Smoothness, analyticity, and conditioning. For a fixed finite architecture constructed from affine maps and GELU activations, the prediction map and empirical squared loss are real-analytic. This follows by composition of analytic maps, as proved in Appendix N, Theorem 45. It justifies additional tools for critical sets and gradient-flow convergence; it does not supply a uniformly bounded Hessian or a useful Jacobian spectrum.

For example, with $\sigma(z) = z\Phi(z)$ and $\mathcal{L}(a, b) = \frac{1}{2}[a\sigma(b)]^2$, the identities $\sigma(0) = 0$ and $\sigma'(0) = 1/2$ give

$$\frac{\partial^2 \mathcal{L}}{\partial b^2}(a, 0) = \frac{a^2}{4}.$$

Thus an analytic loss can have an unbounded Hessian. The scalar depth-H composition $h_{\ell+1} = \sigma(h_\ell)$ likewise has derivative 2^{-H} at input zero. These calculations are given in Examples 1 and 2 of the appendix.

The useful distinction is

C^2 regularity:	gradient and Hessian calculations,
real analyticity:	additional local analytic tools,
quantitative conditioning:	controlled optimization progress.

Derivatives are bounded on a fixed compact subset of a smooth domain, not uniformly over arbitrary parameters or growing network families. Normalization, residual scaling, and initialization can help establish useful bounds, but their presence alone is not a proof of them.

The objective in (7.1) concerns training predictions. Small population error will require an additional statistical argument. Exact interpolation is the central geometric regime here, not the definition of all successful predictive learning. With a fixed positive regularization coefficient, the relevant solution may instead be *quasi-interpolating*: it fits closely without making every residual zero. Small empirical loss alone is an average criterion; any stronger per-example tolerance must be stated separately. Throughout, "stationary point" refers to a named objective, and "common minibatch equilibrium" means that every admissible minibatch produces zero update.

7.2 Prediction space and parameter space

Define the vector of predictions on the training set and its residual:

$$F(\theta) = \begin{pmatrix} F_\theta(x_1) \\ \vdots \\ F_\theta(x_n) \end{pmatrix} \in \mathbb{R}^n, \qquad e(\theta) = F(\theta) - y.$$

Then

$$\widehat{\mathcal{L}}(\theta) = \frac{1}{2n}\|e(\theta)\|_2^2.$$

As a function of an unrestricted prediction vector $z \in \mathbb{R}^n$, $\|z-y\|_2^2/(2n)$ is a strictly convex quadratic. Nonconvexity enters through the parameterization $\theta \mapsto F(\theta)$ and the set of prediction vectors it makes accessible.

Thus

training prediction space:	$F(\theta) \in \mathbb{R}^n$,
parameter space:	$\theta \in \mathbb{R}^p$.

The dimension of parameter space can greatly exceed the rank of the independent fitting constraints. Many different parameters can then produce the same training prediction vector.

They need not produce the same function outside the sample. A direction that is neutral for the training predictions may change population predictions. Parameter redundancy, agreement on the sample, and functional equivalence on the whole input domain are different notions.

This distinction anticipates Chapter H: it is the population performance of the selected predictor, not identification of a particular parameter vector, that defines predictive success.

7.3 The dimension-counting argument

Let

$$J(\theta) = DF(\theta) \in \mathbb{R}^{n\times p}$$

be the Jacobian of the training prediction map. For squared loss,

$$\nabla \widehat{\mathcal{L}}(\theta) = \frac{1}{n} J(\theta)^\top e(\theta). \tag{7.2}$$

A stationary point satisfies

$$J(\theta)^\top e(\theta) = 0,$$

whereas an interpolating solution satisfies the stronger condition

$$e(\theta) = 0.$$

There is a useful contrast between an *isolated nondegenerate* positive-loss minimum and a *regular* interpolating solution. In the first case, the Hessian is nonsingular in the admitted parameter coordinates, so the stationarity equations locally isolate the critical point. In the second case, only the rank of the fitting constraints counts toward the local codimension of their level set.

The qualification is important. Positive-loss critical sets in neural networks can also be degenerate, even after familiar symmetries have been accounted for. One cannot infer independence of p equations simply because the gradient has p components. The isolated-minimum picture is a comparison case, not a universal description of suboptimal minima.

If $J(\theta)$ has full row rank, $J(\theta)^\top$ is injective. Thus (7.2) immediately gives

$$\operatorname{rank} J(\theta) = n, \quad \nabla \widehat{\mathcal{L}}(\theta) = 0 \quad \Longrightarrow \quad e(\theta) = 0.$$

Full prediction rank rules out a positive-loss stationary point at that location. Establishing such rank, and a quantitative lower singular-value bound along a trajectory, is a separate task.

Even full row rank almost everywhere is insufficient: a trajectory can approach an exceptional rank-deficient point with a positive-measure attracting basin. The analytic least-squares example in Section 7.22 makes this obstruction explicit.

7.3.1 The regular zero-loss manifold

Define

$$\mathcal{M} = \{\theta : F(\theta) = y\}.$$

Suppose $\theta^\star \in \mathcal{M}$ and J has locally constant rank q on a neighborhood of $\theta^\star$. The constant-rank theorem gives

$$\boxed{\dim \mathcal{M} = p - q} \tag{7.3}$$

locally around $\theta^\star$. In the full-row-rank case, $q = n$ and the local dimension is $p - n$.

The conclusion is conditional on the existence and regularity of the interpolating point. Under those conditions, adding parameters while keeping the independent constraint rank fixed can enlarge both parameter space and the local set of fitting solutions:

$$\boxed{\begin{array}{c}\text{interpolation exists and the prediction map has local rank } q \\ \Longrightarrow \quad \text{a local fitting manifold of dimension } p - q.\end{array}}$$

This is a local dimension statement, not yet a statement about the probability of finding that manifold.

7.3.2 Why the p gradient equations do not contradict this

Every differentiable local minimum satisfies p scalar stationarity equations. Why can an interpolating minimum nevertheless have $p - q$ free directions?

At interpolation, $e(\theta) = 0$, so

$$J(\theta)^\top e(\theta) = 0$$

holds automatically. The stationarity equations become consequences of the locally independent fitting constraints; they do not impose p additional independent conditions.

This algebra explains why a regular fitting level set can be extended. It does not show that all positive-loss stationary points are isolated, or that every point with a rank-deficient Jacobian lies on a smooth fitting manifold. At singular points, the rank at one point is not enough to infer the dimension of the zero-loss set.

7.4 The Hessian at interpolation

Differentiating the squared loss twice gives

$$\nabla^2 \widehat{\mathcal{L}}(\theta) = \frac{1}{n} J(\theta)^\top J(\theta) + \frac{1}{n} \sum_{i=1}^{n} e_i(\theta) \nabla^2 F_\theta(x_i). \tag{7.4}$$

The first term is positive semidefinite. The residual-weighted second term can contain negative curvature; it is one source of nonconvexity away from interpolation.

At an interpolating solution $\theta^\star$, all residuals vanish. Hence

$$\boxed{\nabla^2 \widehat{\mathcal{L}}(\theta^\star) = \frac{1}{n} J(\theta^\star)^\top J(\theta^\star).} \tag{7.5}$$

For any parameter direction v,

$$v^\top \nabla^2 \widehat{\mathcal{L}}(\theta^\star) v = \frac{1}{n} \|J(\theta^\star) v\|_2^2.$$

If $\operatorname{rank} J(\theta^\star) = q$, then

$$\operatorname{rank} \nabla^2 \widehat{\mathcal{L}}(\theta^\star) = q, \qquad \boxed{\dim \ker \nabla^2 \widehat{\mathcal{L}}(\theta^\star) = p - q.} \tag{7.6}$$

This rank identity needs no constant-rank neighborhood; it holds wherever the stated derivatives exist and the residual is zero.

Identifying the kernel with a tangent space does need regularity. Under the local constant-rank assumption of the preceding section,

$$\ker J(\theta^\star) = T_{\theta^\star} \mathcal{M}.$$

The local quadratic picture is therefore

$p - q$ tangent directions:	zero quadratic cost,
q normal directions:	positive quadratic cost.

Moving along the fitting manifold preserves zero loss. Moving on a straight line in a tangent direction need not do so: higher-order terms can matter away from the point.

At a singular interpolating point, the same Hessian may have zero modes that do not integrate into a manifold of corresponding dimension. Thus Hessian degeneracy is an exact algebraic consequence when $p > q$, whereas a smooth manifold interpretation is an additional geometric conclusion.

Normalized Hessian–kernel duality. At interpolation, put $J_* = J(\theta^\star)$, $H_* = J_*^\top J_*/n$, $\Theta_* = J_* J_*^\top$, and $K_* = \Theta_*/n$. The singular-value decomposition gives, for their positive eigenvalues,

$$\lambda_j^+(H_*) = \lambda_j^+(K_*) = \frac{1}{n}\lambda_j^+(\Theta_*), \qquad 1 \le j \le q. \tag{7.7}$$

The factor $1/n$ is essential when comparing the Hessian with an unnormalized neural tangent kernel. This algebra does not require a constant-kernel limit or analyticity; see Proposition 71 in Appendix N.

The regularized Hessian is a different object. For a twice-differentiable penalty and $U_\gamma = \widehat{\mathcal{L}} + \gamma\Omega$, $\gamma > 0$,

$$\nabla^2 U_\gamma(\theta) = \frac{1}{n} J^\top J + \frac{1}{n}\sum_i e_i \nabla^2 F_\theta(x_i) + \gamma \nabla^2 \Omega(\theta). \tag{7.8}$$

A stationary point of U_γ need not have $e = 0$. Both the residual term and the penalty curvature must then be retained. Even at an exact interpolator, the Hessian of U_γ includes the penalty term, so the rank identity above applies to the *data-loss Hessian*, not to the penalized Hessian. For full quadratic weight decay, the latter equals $J^\top J/n + \gamma I_p$ at an interpolator, although that parameter is generally not stationary for U_γ. Small nonzero residuals alone do not bound the correction without control of the output second derivatives.

7.5 Why degeneracy matters

Calling every zero mode "flatness" hides an important fact: a regular global solution can be an extended set rather than one distinguished point. An optimizer need not identify a predetermined parameter vector to fit the sample.

A useful comparison is

an isolated nondegenerate positive-loss minimum
versus
a regular high-dimensional interpolating set.

These sets can have very different nearby volumes at a small tolerance. The comparison becomes especially relevant when a stochastic model weights neighborhoods rather than individual points.

However, the alternatives are not exhaustive. Suboptimal minima can form extended sets as well. Furthermore, an extended set can be difficult to reach from the initialization distribution. Degeneracy describes the local fitting geometry; accessibility also depends on the vector field outside that neighborhood.

The geometric benefit is an opportunity, not a guarantee: many parameter values satisfy the same fitting constraints. Whether this opportunity makes training efficient must be shown for the architecture, data, initialization, and optimizer under consideration.

7.6 Volume near a regular global-minimum manifold

Near a regular fitting manifold, introduce tangent coordinates s and Euclidean normal coordinates z:

$$s \in \mathbb{R}^{p-q}, \qquad z \in \mathbb{R}^q.$$

On a compact regular portion, and with the smoothness needed for a uniform Taylor expansion,

$$\widehat{\mathcal{L}}(s,z) = \frac{1}{2} z^\top H_\perp(s) z + O(\|z\|_2^3), \qquad H_\perp(s) \succ 0. \tag{7.9}$$

Tangent directions parameterize zero loss; normal displacement incurs a quadratic penalty to leading order.

For a compact smooth embedded portion of $\mathcal{M}$, a sufficiently small Euclidean tube has leading volume

$$\mathrm{Vol}(T_\varepsilon(\mathcal{M})) \sim \omega_q \varepsilon^q \,\mathrm{Vol}_{p-q}(\mathcal{M}), \qquad \varepsilon \downarrow 0, \tag{7.10}$$

with the usual localization convention at patch boundaries. Here ω_q is the volume of the unit ball in $\mathbb{R}^q$. In contrast, a Euclidean ball around an isolated point in $\mathbb{R}^p$ has volume

$$\mathrm{Vol}(B_\varepsilon(\theta_\ell)) = \omega_p \varepsilon^p.$$

The leading powers differ:

$$\boxed{\varepsilon^q \quad \text{versus} \quad \varepsilon^p.}$$

For a fixed geometry, their ratio contains $\varepsilon^{-(p-q)}$. Thus an extended minimum set can have much more nearby parameter volume than an isolated minimum when the neighborhood radius is small.

A distance tube and a loss sublevel set are not identical. Relating them uses the positive normal Hessian in (7.9); its eigenvalues convert a loss tolerance into transverse distance scales. Curvature and intrinsic volume remain part of the prefactor.

These are small-neighborhood asymptotics for a fixed model. They do not by themselves compare different network widths: the dimensions, curvatures, reference measures, and admissible neighborhood sizes can all change with width. Nor do these Euclidean volumes specify probabilities until a probability measure has been chosen.

7.7 Nearby volume is not the same as basin volume

For gradient flow

$$\dot{\theta}_t = -\nabla \widehat{\mathcal{L}}(\theta_t),$$

the basin of attraction of a set A can be defined by

$$\mathcal{B}(A) = \{\theta_0 : \mathrm{dist}(\theta_t, A) \longrightarrow 0\}.$$

A large tubular neighborhood does not automatically imply a large basin. The basin depends on the vector field away from the solution set. A degenerate global minimum can have a narrow funnel leading to it; an isolated minimum can have a broad attracting region.

Accordingly,

$$\boxed{\begin{array}{c}\text{local degeneracy or large nearby volume} \\ \not\Longrightarrow \quad \text{a large basin of attraction.}\end{array}}$$

For randomized initialization, the relevant quantity is $\pi_0(\mathcal{B}(A))$, not an unspecified Euclidean basin volume. A set can be large in one measure and negligible in another.

A deterministic convergence argument needs a condition on the actual trajectory, or on a region containing it. For squared loss, the most direct condition controls contraction of the residual in training prediction space. This condition can coexist with many neutral parameter directions.

7.8 Residual contraction in prediction space

Define

$$K(\theta) = \frac{1}{n} J(\theta) J(\theta)^\top.$$

Along gradient flow, the chain rule gives

$$\dot{e}(\theta_t) = J(\theta_t)\dot{\theta}_t = -\frac{1}{n} J(\theta_t) J(\theta_t)^\top e(\theta_t) = -K(\theta_t) e(\theta_t). \tag{7.11}$$

The sufficient condition actually used in the loss estimate is

$$e(\theta_t)^\top K(\theta_t) e(\theta_t) \geq \mu \|e(\theta_t)\|_2^2, \qquad \mu > 0, \tag{7.12}$$

throughout the relevant trajectory. The stronger matrix condition $K(\theta_t) \succeq \mu I_n$ implies it. If only a residual subspace is used, that subspace must contain the actual residuals and the inequality must hold on it; positivity on the range of K alone does not suffice for an arbitrary residual.

Using (7.2),

$$\begin{aligned} \frac{d}{dt}\widehat{\mathcal{L}}(\theta_t) &= -\|\nabla \widehat{\mathcal{L}}(\theta_t)\|_2^2 = -\frac{1}{n} e(\theta_t)^\top K(\theta_t) e(\theta_t) \\ &\leq -2\mu \widehat{\mathcal{L}}(\theta_t). \end{aligned} \tag{7.13}$$

Consequently,

$$\boxed{\widehat{\mathcal{L}}(\theta_t) \leq \widehat{\mathcal{L}}(\theta_0) e^{-2\mu t}.} \tag{7.14}$$

This is a conditional deterministic convergence mechanism, not an inference from parameter count. The unresolved premise in any application is the trajectory-wise lower bound (7.12).

The two Gram matrices have different roles. If $p > q$, then $J^\top J$ is singular because some parameter directions leave the training outputs unchanged to first order. Nevertheless, $JJ^\top$ can be positive definite on training prediction space when J has full row rank. Thus

many neutral parameter directions can coexist with rapid contraction of training prediction error.

The parameters may fail to be identifiable even while the training predictions are easy to fit. Neither statement determines the predictor's behavior beyond the sample.

An NTK-style sufficient condition. A concrete route to (7.12) is to prove, on a specified time interval $[0, t_{\mathrm{f}}]$,

$$\begin{aligned} K(\theta_0) &\succeq 2\mu I_n, \\ \sup_{0 \leq t \leq t_{\mathrm{f}}} \|K(\theta_t) - K(\theta_0)\|_{\mathrm{op}} &\leq \mu. \end{aligned} \tag{7.15}$$

Indeed, for every $v \in \mathbb{R}^n$,

$$v^\top K(\theta_t) v \geq v^\top K(\theta_0) v - \|K(\theta_t) - K(\theta_0)\|_{\mathrm{op}} \|v\|^2 \geq \mu \|v\|^2.$$

This proves conditioning throughout that interval, not merely at initialization. A nearly constant kernel is one sufficient strategy; (7.12) also permits substantial kernel change when the actual residual remains controlled. The relevant parameterization, initialization, width scaling, data assumptions, and time interval must be specified. Infinite width alone does not imply a frozen kernel. Appendix N, Section N.4, discusses this distinction and the corresponding literature.

Two different Jacobians. Products of layer Jacobians control transmission through activation space; $J(\theta) = D_\theta F(\theta)$ controls changes of the training predictions under parameter updates. Good conditioning of the former does not by itself imply full row rank or a positive residual Rayleigh bound for the latter. Residual connections help when their derivatives are quantitatively controlled, not merely because an identity branch is present. A precise residual-product bound is proved in Appendix N, Proposition 69.

7.9 A deterministic convergence backstop

Suppose there is a forward-invariant region D on which (7.12) holds with a common $\mu > 0$. Assume the gradient-flow solution exists for all $t \geq 0$ for initial points in D. Then (7.14) holds for every such trajectory. For the finite-parameter conclusion below, assume also that F is continuous on $\mathbb{R}^p$ and that the loss has the local smoothness required for the flow.

One can also justify convergence to a finite interpolating parameter under these assumptions. Write $\ell(t) = \widehat{\mathcal{L}}(\theta_t)$. Before reaching zero loss,

$$-\frac{d}{dt}\sqrt{\ell(t)} = \frac{\|\nabla\widehat{\mathcal{L}}(\theta_t)\|_2^2}{2\sqrt{\ell(t)}} \geq \sqrt{\frac{\mu}{2}}\|\nabla\widehat{\mathcal{L}}(\theta_t)\|_2.$$

The last inequality follows from $\|\nabla\widehat{\mathcal{L}}\|_2^2 \geq 2\mu\widehat{\mathcal{L}}$. Integrating and using $\ell(t) \to 0$ yields

$$\int_t^\infty \|\dot{\theta}_u\|_2 \, du \leq \sqrt{\frac{2}{\mu}}\sqrt{\widehat{\mathcal{L}}(\theta_t)}. \tag{7.16}$$

Hence the trajectory has finite remaining length and converges to some $\theta_\infty \in \mathbb{R}^p$. Continuity gives $F(\theta_\infty) = y$. If zero loss is reached earlier, the gradient is zero and the trajectory can remain there.

If the initialization distribution satisfies $\pi_0(D) \geq 1 - \delta$, this conditional argument gives convergence to interpolation with probability at least $1 - \delta$. The substantive problem is proving the conditioning, forward invariance, and initialization probability for a specified network and dataset. A large value of $p - q$ does not establish any of them by itself.

For a desired loss tolerance $\xi > 0$, the bound gives a continuous-time budget

$$t \geq \frac{1}{2\mu}\log\frac{\widehat{\mathcal{L}}(\theta_0)}{\xi}$$

when the initial loss exceeds ξ. A polynomial-time learning claim additionally needs a polynomially controlled inverse μ^{-1} and a feasible discretization and evaluation model.

Finite-step gradient descent needs its own conditions. If the discrete iterates remain in the conditioning region and the loss has Λ-Lipschitz gradient along the required update segments, then for $0 < \alpha \leq 1/\Lambda$ the descent lemma gives

$$\widehat{\mathcal{L}}(\theta_{k+1}) \leq \widehat{\mathcal{L}}(\theta_k) - \frac{\alpha}{2}\|\nabla\widehat{\mathcal{L}}(\theta_k)\|_2^2 \leq (1 - \alpha\mu)\widehat{\mathcal{L}}(\theta_k).$$

The step can also be chosen with $\alpha\mu \leq 1$. This calculation explains what a discrete deterministic extension requires; it does not transfer the result automatically to noisy SGD iterates.

The justified backstop is therefore

a high-probability initialization region
+ invariance and uniform residual contraction
$\Longrightarrow$ high-probability fitting.

Overparameterization may help establish the premises, rather than replacing them.

Analytic convergence has a different endpoint. There is a complementary positive theorem that does not assume a useful prediction-Jacobian lower bound. Let $U : \Omega \to \mathbb{R}$ be real-analytic on an open set $\Omega \subseteq \mathbb{R}^p$. If $\dot{\theta}_t = -\nabla U(\theta_t)$ is defined for all $t \geq 0$ and its image has compact closure contained in Ω, then

$$\int_0^\infty \|\dot{\theta}_t\| \, dt < \infty, \qquad \theta_t \longrightarrow \theta_\infty, \qquad \nabla U(\theta_\infty) = 0. \tag{7.17}$$

This is Theorem 44 of Appendix N, where the proof is given using the local Łojasiewicz gradient inequality

$$\|\nabla U(\theta)\| \geq c \, |U(\theta) - U(\theta_*)|^\vartheta, \qquad c > 0, \quad \vartheta \in [1/2, 1).$$

The inequality is local to a critical point; compactness supplies the finite covering and eventual localization needed by the proof.

The limiting value $U(\theta_\infty)$ need not be globally minimal. For $U = \widehat{\mathcal{L}}$, it need not be zero. Nor does the theorem supply uniform bounds on the entry time, the constant c, or the exponent across growing model families. For example, the analytic flow of $U(x) = x^{2m}$, $m \geq 2$, need not converge exponentially; the appendix gives the explicit trajectory.

The residual-contraction argument above is different. It does not require analyticity, and its quantitative premise forces the specified residual to zero. Neither analyticity nor compactness is being used as a substitute for that premise. When U includes a regularizer, the analytic result concerns stationarity of that penalized objective; the confinement issue is discussed in Section 7.18. No conclusion about arbitrary noisy SGD follows from either deterministic calculation without additional assumptions.

7.10 SGD as noisy gradient flow

For a deterministic differentiable regularizer, unbiased minibatch SGD has the form

$$\begin{aligned} \theta_{k+1} &= \theta_k - \alpha \nabla U_\gamma(\theta_k) + \alpha \xi_k(\theta_k), \\ U_\gamma &= \widehat{\mathcal{L}} + \gamma \Omega, \qquad \mathbb{E}[\xi_k(\theta_k) \mid \theta_k] = 0. \end{aligned}$$

Here $\gamma = 0$ recovers the unregularized update. At a fixed parameter, adding the same deterministic penalty gradient to every batch changes the mean, not the sampling covariance. It can nevertheless change where the mean update vanishes and whether the data residuals vanish there. Section 7.17 makes this distinction exact. A diffusion approximation, when justified, depends on the sampling scheme, step size, and covariance of the gradient noise.

To isolate a volume effect, consider instead the explicitly specified *isotropic overdamped Langevin model*

$$d\theta_t = -\nabla \widehat{\mathcal{L}}(\theta_t) \, dt + \sqrt{2T} \, dW_t, \qquad T > 0. \tag{7.18}$$

Here T is a temperature parameter of this model, not a universal scalar description of ordinary SGD noise. The next equilibrium calculations use the unregularized loss as their potential. A Langevin model for penalized training instead uses U_γ, including its changed minima and curvature; that replacement is not a theorem that real regularized SGD has an isotropic noise law.

Assume well-posedness and the equilibrium hypotheses needed for a stationary probability law, including

$$Z_T = \int_{\mathbb{R}^p} e^{-\widehat{\mathcal{L}}(\theta)/T}\, d\theta < \infty.$$

Then the Gibbs density is

$$\rho_T(\theta) = Z_T^{-1} e^{-\widehat{\mathcal{L}}(\theta)/T}. \tag{7.19}$$

Its probability of a region U is

$$\rho_T(U) = Z_T^{-1} \int_U e^{-\widehat{\mathcal{L}}(\theta)/T}\, d\theta.$$

At this equilibrium, both loss and the amount of parameter space at that loss affect probability mass.

The integrability assumption is not innocuous. Flat noncompact directions can make Z_T infinite. A confining modification, a specified bounded parameter domain with suitable boundary conditions, or a prior may produce a different well-defined model, but its objective, measure, and dynamics must be stated consistently.

Even when (7.19) exists, its formula alone does not bound the time required to approach equilibrium from the initialization law. Moreover, at fixed positive T, a lower-dimensional exact minimum manifold has zero probability under a continuous density. The relevant objects are neighborhoods or positive loss tolerances, not the event of sampling a point exactly on the manifold.

7.11 An isolated local minimum under Langevin dynamics

Let θ_ℓ be an isolated nondegenerate local minimum with

$$\widehat{\mathcal{L}}(\theta_\ell) = L_\ell > 0, \qquad H_\ell = \nabla^2 \widehat{\mathcal{L}}(\theta_\ell) \succ 0.$$

On a sufficiently small fixed neighborhood U_ℓ, this point is the unique local minimizer and

$$\widehat{\mathcal{L}}(\theta) = L_\ell + \frac{1}{2}(\theta - \theta_\ell)^\top H_\ell (\theta - \theta_\ell) + O(\|\theta - \theta_\ell\|_2^3).$$

The local partition contribution satisfies

$$\boxed{Z_\ell(T) := \int_{U_\ell} e^{-\widehat{\mathcal{L}}(\theta)/T}\, d\theta \sim e^{-L_\ell/T} \frac{(2\pi T)^{p/2}}{\sqrt{\det H_\ell}}.} \tag{7.20}$$

The factor $e^{-L_\ell/T}$ penalizes positive loss. The determinant term measures the transverse width of the quadratic valley in the chosen Euclidean coordinates.

For two such minima at the same loss, their leading relative equilibrium weight is

$$\frac{Z_1(T)}{Z_2(T)} \longrightarrow \sqrt{\frac{\det H_2}{\det H_1}}.$$

The common power of T cancels. Thus a smaller determinant gives greater local equilibrium weight in this comparison; lowering temperature is not an additional universal preference among equal-loss isolated minima. For different losses, the exponential energy factor also matters.

This is a local equilibrium calculation for a specified reference measure. It neither identifies a basin of attraction nor proves that a minibatch optimizer visits one valley more often than another in finite time.

7.12 A global-minimum manifold under Langevin dynamics

Now let $\mathcal{M}$ be a compact regular zero-loss manifold of codimension q with uniformly positive normal Hessian. In a sufficiently small fixed tube $U_{\mathcal{M}}$, define

$$Z_{\mathcal{M}}(T) = \int_{U_{\mathcal{M}}} e^{-\widehat{\mathcal{L}}(\theta)/T}\, d\theta.$$

Integrating the normal quadratic approximation gives

$$\boxed{Z_{\mathcal{M}}(T) \sim (2\pi T)^{q/2} I_{\mathcal{M}}, \qquad I_{\mathcal{M}} := \int_{\mathcal{M}} \frac{dV_{p-q}(s)}{\sqrt{\det H_{\perp}(s)}}.} \tag{7.21}$$

Only the q normal directions pay a quadratic loss penalty. The tangent directions contribute through the curvature-weighted intrinsic volume $I_{\mathcal{M}}$.

Comparing with (7.20),

$$\frac{Z_{\mathcal{M}}(T)}{Z_{\ell}(T)} \sim e^{L_{\ell}/T} (2\pi T)^{-(p-q)/2} I_{\mathcal{M}} \sqrt{\det H_{\ell}}. \tag{7.22}$$

This separates two effects:

energy advantage:	$0 < L_{\ell}$,
degeneracy factor:	$T^{-(p-q)/2}$ when $p > q$.

The energy advantage already favors a zero-loss isolated minimum over a positive-loss one as $T \downarrow 0$. Degeneracy provides an additional polynomial factor in the stated comparison. It is not necessary for the exponential energy preference itself.

The ratio concerns these local neighborhoods in one fixed model. There may be other global components contributing substantial mass to the normalizing constant. A local comparison does not show that this one manifold has stationary probability tending to one in the whole space. It also says nothing by itself about mixing or hitting time.

Across increasing widths, all geometric prefactors and the admissible temperature range may vary. Exchanging the low-temperature and large-width limits requires uniform estimates that are not supplied by this fixed-model Laplace calculation.

7.13 Free energy and entropic selection

The same calculation can be expressed through a local free energy. For a neighborhood U with partition contribution $Z_U(T)$, define

$$\mathcal{F}_U(T) = -T \log Z_U(T).$$

If L_U is the minimum loss in that neighborhood, write

$$\mathcal{S}_U^{\mathrm{vol}}(T) = \log\big(e^{L_U/T} Z_U(T)\big).$$

Then, by definition,

$$\boxed{\mathcal{F}_U(T) = L_U - T\mathcal{S}_U^{\mathrm{vol}}(T).} \tag{7.23}$$

Here $\mathcal{S}_U^{\mathrm{vol}}$ is an effective logarithmic-volume term. It depends on temperature and on the reference measure; it is not being identified with a temperature-independent intrinsic entropy of a function.

For the regular zero-loss manifold,

$$\begin{aligned}\mathcal{S}_{\mathcal{M}}^{\mathrm{vol}}(T) &= \frac{q}{2}\log(2\pi T) + \log I_{\mathcal{M}} + o(1),\\ \mathcal{F}_{\mathcal{M}}(T) &= -\frac{qT}{2}\log(2\pi T) - T\log I_{\mathcal{M}} + o(T).\end{aligned} \tag{7.24}$$

For the isolated positive-loss minimum,

$$\mathcal{F}_\ell(T) = L_\ell - \frac{pT}{2}\log(2\pi T) + \frac{T}{2}\log\det H_\ell + o(T).$$

These expressions reproduce the relative weights in (7.22).

The integrated curvature factor must be kept intact. If $\langle\cdot\rangle_{\mathcal{M}}$ denotes normalized intrinsic-volume average, then

$$\log I_{\mathcal{M}} = \log \mathrm{Vol}_{p-q}(\mathcal{M}) + \log\left\langle (\det H_\perp)^{-1/2}\right\rangle_{\mathcal{M}}.$$

In general, the last term is not $-\frac{1}{2}\langle\log\det H_\perp\rangle_{\mathcal{M}}$. Replacing a logarithm of an average by an average of logarithms would change the weighting. The codimension-dependent $\log T$ terms must also be retained when comparing components of different dimension.

The useful interpretation remains positive:

interpolating sets can have both lower energy and a favorable local volume contribution.

Overparameterization can contribute to the second effect by providing regular excess directions. This is an explicit equilibrium mechanism, not a width-only guarantee of faster optimization or better prediction.

7.14 Why the smooth-manifold picture is incomplete

Neural parameterizations can contain continuous rescalings, redundant units, changing ranks, and intersecting solution components. Permutations can generate equivalent copies; collisions and other degeneracies can make the local geometry singular. The zero-loss set need not be one smooth manifold of fixed dimension.

The important distinction is between a direction of zero quadratic cost and a direction along an actual curve of exact solutions. At a singular point, higher-order terms can constrain motion even when the Hessian vanishes in that direction. Thus

$$\dim\ker\nabla^2\widehat{\mathcal{L}} = p - q$$

at interpolation does not alone establish a zero-loss manifold of that dimension.

The broader geometric question is

How much weighted parameter-space volume lies within a small loss tolerance of the solution set?

The regular normal-Hessian calculation answers this in one important case. Singular learning theory provides a language for more general local asymptotics under its analytic and localization assumptions.

Appendix N, Sections N.1 and N.4, supplies the analytic zero-set perspective and examples separating Hessian nullity from actual solution set dimension. Analyticity does not turn every such set into a manifold.

This distinction should not be confused with the predictive versus structural distinction. A singular parameterization may represent a perfectly useful predictor. Its singularity is neither evidence of population accuracy nor a failure to recover the teacher's computation.

7.15 The real log canonical threshold

Let $L(\theta) \geq 0$ be an analytic loss, and let φ be a smooth positive local weighting density. Consider the localized integral

$$Z_U(T) = \int_U e^{-L(\theta)/T} \varphi(\theta)\, d\theta$$

near a zero-loss component. In the setting covered by the singular asymptotic theorem—with the required analytic, support, and localization assumptions—its small-temperature form is

$$\boxed{Z_U(T) \asymp C T^{\lambda} \left(\log \frac{1}{T}\right)^{m-1}, \qquad T \downarrow 0.} \tag{7.25}$$

The exponent λ is the real log canonical threshold (RLCT), and m is its multiplicity. This chapter uses this established asymptotic form as a description of singular volume; it does not prove the general singular asymptotic theorem.

For a fixed loss value and a positive nonvanishing local weight, a smaller exponent produces a larger leading contribution as $T \downarrow 0$. If the exponents coincide, the logarithmic multiplicity and then the finite prefactors matter.

The interpretation of 2λ as an "effective codimension" is a useful analogy, not a formula for topological codimension at every singularity. For a smooth zero-loss manifold of codimension q, with quadratic nondegeneracy in the normal directions,

$$\boxed{\lambda = q/2, \qquad m = 1.}$$

Then (7.25) reduces to the normal Gaussian integration in (7.21).

The successive descriptions are therefore

quadratic Hessian information,
regular manifold and normal curvature,
singular weighted-volume asymptotics.

The last description can capture behavior missed by counting Hessian zero modes. Its assumptions cannot be silently extended to every smooth or piecewise-smooth network. An application to a nonsmooth architecture must justify an appropriate stratified or other applicable version.

Nor does a localized prior-weighted integral automatically describe the stationary law of (7.18). A nonconstant weight changes the measure being analyzed; the dynamics must be specified if a dynamical interpretation is intended.

The transformed measure and the analytic exponents. Resolution of singularities must transform the volume element as well as the loss. Under the localization and nonvanishing assumptions stated in Appendix N, Section N.3, a contributing chart has the form

$$L(\pi(u)) = a(u)\prod_j u_j^{2k_j}, \qquad \varphi(\pi(u))|\det D\pi(u)| = b(u)\prod_j |u_j|^{h_j}.$$

Here the unit factors are positive on the relevant chart, and localization factors must not remove its leading contribution. For $k_j > 0$, candidate integrability thresholds are

$$\lambda_j = \frac{h_j + 1}{2k_j}.$$

The Jacobian and weight exponents h_j cannot be omitted. The appendix states the invoked resolution and singular-integral results and explains how the contributing charts determine λ and its multiplicity.

The gradient-inequality exponent ϑ used for analytic convergence in Section 7.9 and the volume exponent λ answer different questions. A resolved loss monomial does not by itself determine a Euclidean gradient-flow rate: changing coordinates also changes the metric. Singular volume, analytic convergence, and finite-time stochastic access remain separate conclusions.

7.16 The singular version of the entropy argument

Suppose a zero-loss component has local asymptotic weight

$$Z_g(T) \asymp C_g T^{\lambda_g}\left(\log\frac{1}{T}\right)^{m_g-1}.$$

An isolated nondegenerate minimum at positive loss L_ℓ has

$$Z_\ell(T) \asymp e^{-L_\ell/T} C_\ell T^{p/2}.$$

Therefore, for fixed positive finite prefactors,

$$\boxed{\frac{Z_\ell(T)}{Z_g(T)} \asymp e^{-L_\ell/T} T^{p/2-\lambda_g}\left(\log\frac{1}{T}\right)^{1-m_g}.} \tag{7.26}$$

The energy gap gives the exponential penalty; singular degeneracy modifies the polynomial and logarithmic factors.

The volume mechanism therefore does not require a smooth minimum manifold in every application. It requires an applicable estimate of weighted low-loss volume. If a competing positive-loss component is itself singular, its own exponent and multiplicity must replace the isolated-minimum values in the comparison.

At equal loss, a smaller applicable volume exponent favors a component asymptotically. At unequal loss, the energy difference must also be included. Neither exponent alone supplies a hitting time, a mixing bound, or a generalization guarantee. These remain local equilibrium or weighted-integral statements for the specified model.

7.17 What real SGD changes

The correct distinction is not between "noise during training" and "no noise at minima." It is between exact interpolation of every sample, stationarity of the full objective, and a common equilibrium of all minibatch updates. Regularization makes this distinction consequential even in the late stage of nearly interpolating training.

7.17.1 Sampling covariance at a fixed parameter

For a fixed training example, let

$$g_i(\theta) = \nabla_\theta \tfrac{1}{2}(F_\theta(x_i) - y_i)^2 = e_i(\theta)\nabla_\theta F_\theta(x_i), \qquad \bar{g}(\theta) = \frac{1}{n}\sum_i g_i(\theta).$$

Sample B indices independently and uniformly with replacement, and let $\widehat{g}_B$ be their average gradient. Conditional on the current parameter,

$$\begin{aligned} \mathbb{E}[\widehat{g}_B \mid \theta] &= \bar{g}(\theta), \\ \mathrm{Cov}(\widehat{g}_B \mid \theta) &= \frac{1}{B}C(\theta), \\ C(\theta) &= \frac{1}{n}\sum_i g_i(\theta)g_i(\theta)^\top - \bar{g}(\theta)\bar{g}(\theta)^\top. \end{aligned} \tag{7.27}$$

Equivalently, $C = n^{-1}\sum_i (g_i - \bar{g})(g_i - \bar{g})^\top$. A single draw has covariance C; conditional independence divides it by B upon averaging. This is the covariance identity also proved in Appendix N, Proposition 74.

When $\mathcal{B}$ is instead a uniformly sampled subset of $[n]$ of size B, without replacement, put

$$\widehat{g}_{\mathcal{B}} = \frac{1}{B}\sum_{i\in\mathcal{B}} g_i, \qquad c_{n,B} = \frac{n-B}{B(n-1)}, \qquad n > 1.$$

Then

$$\mathbb{E}_{\mathcal{B}}[\widehat{g}_{\mathcal{B}} \mid \theta] = \bar{g}, \qquad \mathrm{Cov}_{\mathcal{B}}(\widehat{g}_{\mathcal{B}} \mid \theta) = c_{n,B}C(\theta). \tag{7.28}$$

For completeness, set $h_i = g_i - \bar{g}$ and let I_i indicate membership in $\mathcal{B}$. Since $\sum_i h_i = 0$, $\sum_{i\neq j} h_i h_j^\top = -\sum_i h_i h_i^\top$. Using $\mathbb{E}I_i = B/n$ and $\mathbb{E}I_iI_j = B(B-1)/(n(n-1))$ gives

$$\mathbb{E}\left[\left(\frac{1}{B}\sum_i I_i h_i\right)\left(\frac{1}{B}\sum_j I_j h_j\right)^\top\right] = \frac{n-B}{Bn(n-1)}\sum_i h_i h_i^\top = c_{n,B}C.$$

For $B = n$ the without-replacement variance is zero, whatever the sample gradients. For $1 \le B < n$, its vanishing is equivalent to all the g_i being equal.

The formulas below use sampling without replacement unless stated otherwise. For independent draws with replacement, replace $c_{n,B}$ by $1/B$. These identities assume that each g_i is a fixed per-example quantity at the current parameter. Batch-dependent forward operations, such as training-mode batch normalization, require a covariance analysis of the actual batch update instead.

7.17.2 Exact interpolation and regularized stationarity

There is an elementary distinction at exact squared-loss interpolation:

$$e_i(\theta^\star) = 0 \text{ for every } i \quad \Longrightarrow \quad C(\theta^\star) = 0, \qquad \nabla^2 \widehat{\mathcal{L}}(\theta^\star) = \frac{1}{n} J^\top J \text{ may be nonzero.} \tag{7.29}$$

This statement concerns the data gradients at an exact interpolator. It is not a statement about every minimum of a regularized training objective.

For $U_\gamma = \widehat{\mathcal{L}} + \gamma\Omega$ with a deterministic differentiable penalty, the regularized minibatch gradient is

$$G_{\mathcal{B},\gamma}(\theta) = \widehat{g}_{\mathcal{B}}(\theta) + \gamma \nabla\Omega(\theta). \tag{7.30}$$

The penalty is the same for every batch at the fixed parameter. Thus

$$\begin{aligned} \mathbb{E}_{\mathcal{B}}[G_{\mathcal{B},\gamma} \mid \theta] &= \nabla U_\gamma(\theta), \\ \mathrm{Cov}_{\mathcal{B}}(G_{\mathcal{B},\gamma} \mid \theta) &= c_{n,B} C(\theta). \end{aligned} \tag{7.31}$$

At an exact interpolator, every batch gradient equals $\gamma\nabla\Omega(\theta^\star)$. The sampling covariance is zero, but the mean update need not be zero. Unless the penalty gradient also vanishes, that interpolator is not stationary for U_γ.

At a regularized stationary point θ_γ, by contrast,

$$\bar{g}(\theta_\gamma) + \gamma \nabla\Omega(\theta_\gamma) = 0. \tag{7.32}$$

Only the average data gradient is required to balance the regularizer. The individual data gradients can differ. Consequently,

$$G_{\mathcal{B},\gamma}(\theta_\gamma) = \frac{1}{B} \sum_{i \in \mathcal{B}} \big(g_i(\theta_\gamma) - \bar{g}(\theta_\gamma)\big).$$

This is how regularization can sustain intrinsic minibatch fluctuations without being a random forcing term itself: it changes the stationary balance and can maintain nonzero, sample-dependent data gradients.

7.17.3 A precise criterion for persistent fluctuations

Proposition 5 (Noise at a regularized stationary point)**.** *Let $n > 1$, $1 \le B < n$, and sample $\mathcal{B}$ uniformly among all B-element subsets of $[n]$. Suppose the per-example losses and penalty are differentiable at a stationary point θ_γ of U_γ. Then*

$$\boxed{\mathbb{E}_{\mathcal{B}} \|G_{\mathcal{B},\gamma}(\theta_\gamma)\|^2 = c_{n,B} \frac{1}{n} \sum_{i=1}^{n} \|g_i(\theta_\gamma) - \bar{g}(\theta_\gamma)\|^2.} \tag{7.33}$$

The following conditions are equivalent at this point:

1. *the sampling covariance is zero;*
2. *all individual data gradients $g_i(\theta_\gamma)$ are equal;*
3. *$G_{\mathcal{B},\gamma}(\theta_\gamma) = 0$ for every admissible batch;*
4. *$g_i(\theta_\gamma) + \gamma\nabla\Omega(\theta_\gamma) = 0$ for every example.*

In particular, differing sample gradients imply strictly positive minibatch fluctuation energy even though the full regularized gradient is zero.

Proof. Equation (7.31) gives zero mean and covariance $c_{n,B}C$ at the stated point. Taking its trace proves (7.33). Since $c_{n,B} > 0$, the right side vanishes if and only if every g_i equals $\bar{g}$. Equation (7.32) then makes each regularized per-example gradient zero, so every batch gradient is zero. Conversely, if every batch update vanishes, its mean-square norm is zero. This proves all equivalences. Alternatively, subtract two batches that share $B-1$ examples and differ only in examples i, j. Their difference is $(g_i - g_j)/B$, proving equality of every pair of data gradients. □

At an arbitrary parameter, the exact decomposition is

$$\mathbb{E}_{\mathcal{B}}\|G_{\mathcal{B},\gamma}(\theta)\|^2 = \|\nabla U_\gamma(\theta)\|^2 + c_{n,B}\,\mathrm{Tr}\,C(\theta). \tag{7.34}$$

A small full gradient controls only the first term. The second term is a different quantity; gradient descent on U_γ is not, by definition, minimization of this second moment.

Positive regularization does not force nonzero covariance for every possible dataset. Identical sample gradients can occur in special configurations. The proposition gives the exact nondegeneracy condition, not a universal lower bound depending only on γ. The normalized network result below gives a further structural obstruction to a common minibatch equilibrium.

7.17.4 Persistent noise at a strictly convex global minimum

Nonconvexity is not needed for the phenomenon. Consider the scalar predictor $F_\theta(x) = \theta x$ on the two examples $(1, 1)$ and $(2, 2)$. With quadratic regularization,

$$U_\gamma(\theta) = \frac{1}{4}\left[(\theta-1)^2 + (2\theta-2)^2\right] + \frac{\gamma}{2}\theta^2 = \frac{5}{4}(\theta-1)^2 + \frac{\gamma}{2}\theta^2.$$

Its Hessian is $5/2 + \gamma > 0$, so it has a unique global minimum at $\theta_\gamma = 5/(5+2\gamma)$. For batch size one,

$$G_{\{1\},\gamma}(\theta_\gamma) = (\theta_\gamma - 1) + \gamma\theta_\gamma = \frac{3\gamma}{5+2\gamma},$$
$$G_{\{2\},\gamma}(\theta_\gamma) = 4(\theta_\gamma - 1) + \gamma\theta_\gamma = -\frac{3\gamma}{5+2\gamma}.$$

Their mean is zero, while

$$\mathrm{Var}_{\mathcal{B}}(G_{\mathcal{B},\gamma}(\theta_\gamma)) = \frac{9\gamma^2}{(5+2\gamma)^2} > 0 \qquad (\gamma > 0). \tag{7.35}$$

The unregularized problem interpolates at $\theta = 1$. At that same point with positive regularization, every batch gradient is γ: the motion is deterministic, but the point is no longer stationary. This single example exhibits both endpoints and their different noise levels.

7.17.5 Normalized ReLU networks and the rank obstruction

Xu, Rangamani, Liao, Galanti, and Poggio [170] analyze a more specific mechanism in normalized homogeneous ReLU classifiers. Their Lemma 8 constrains a *common zero of every minibatch update*, not merely a zero of the batch-averaged gradient. The following restatement makes the scalar-output, normalization, and differentiability assumptions explicit and uses this chapter's half-squared-loss convention.

Consider a bias-free scalar-output feedforward network with independently parameterized layer matrices,

$$f_V(x) = V_L\sigma(V_{L-1}\cdots\sigma(V_1x)\cdots), \qquad \|V_k\|_F = 1, \qquad F_{\rho,V}(x) = \rho f_V(x), \quad \rho > 0.$$

Here σ is ReLU, $y_i \in \{-1, +1\}$, and the weight-decay term is $\gamma\rho^2/2$. Write

$$m_i = y_i f_V(x_i), \qquad s_i = 1 - \rho m_i, \qquad R_{ki} = D_{V_k} m_i.$$

The symbol m_i is the signed normalized margin; s_i is a signed residual in margin coordinates, not the per-example loss. No assumption that all s_i are positive is made. Separability alone would not imply it.

Work at a parameter where the training prediction maps are differentiable and the following fixed-gate derivative calculation is valid, for example in a neighborhood with no zero hidden preactivations on the finite sample. For each example and layer, scalar-output backpropagation gives an outer product and the layerwise homogeneity identity gives

$$R_{ki} = y_i a_{ki} b_{ki}^\top, \qquad \operatorname{rank}(R_{ki}) \le 1, \qquad \langle R_{ki}, V_k\rangle_F = m_i. \tag{7.36}$$

Indeed the fixed activation gates make $f_V(x_i) = a_{ki}^\top V_k b_{ki}$ locally in V_k. This calculation is for an unshared layer matrix and a scalar output. A shared convolutional kernel, multiple output losses, or batch-dependent normalization requires its own derivative-rank analysis. We do not infer regularity at a limiting ReLU kink from a measure-zero claim about initialization.

For the constrained minibatch objective

$$U_{\gamma,\mathcal{B}}(\rho, V) = \frac{1}{2B}\sum_{i\in\mathcal{B}}(1 - \rho m_i)^2 + \frac{\gamma}{2}\rho^2, \qquad \|V_k\|_F = 1,$$

the negative scale-gradient and tangent weight-gradient directions are

$$\begin{aligned} d_{\rho,\mathcal{B}} &= \frac{1}{B}\sum_{i\in\mathcal{B}} s_i m_i - \gamma\rho, \\ d_{k,\mathcal{B}} &= \frac{\rho}{B}\sum_{i\in\mathcal{B}} s_i(R_{ki} - m_i V_k). \end{aligned} \tag{7.37}$$

The second line subtracts the normal component on the Frobenius unit sphere, equivalently eliminating the Lagrange multiplier. These are the common-stationarity directions corresponding to the normalized dynamics in [170, Eq. (46)], with the overall factor two removed by the half-loss convention; n and γ here correspond to N and λ in that paper. Finite-step normalization must be implemented by its specified constraint rule; the lemma concerns zeros of the stated directions, not an assertion that a finite Euler step preserves the sphere.

Lemma 3 (Common-minibatch rank restriction; scoped restatement of Xu et al., Lemma 8)**.** *In the normalized scalar-output ReLU setting just specified, assume $n > 1$, $1 \le B < n$, $\gamma > 0$, $\rho > 0$, and $m_i \ne 0$ for every training example. If*

$$d_{\rho,\mathcal{B}} = 0, \qquad d_{k,\mathcal{B}} = 0 \quad \textit{for every } k \textit{ and every } B\textit{-element batch } \mathcal{B},$$

then $\operatorname{rank}(V_k) \le 2$ *for every layer.*

Batch-exchange proof, adapted from the cited lemma. Put $c_i = s_i m_i$. Vanishing of the scale direction and $\gamma\rho > 0$ rule out $s_i = 0$ for every i. Since the margins are nonzero, at least one c_i is nonzero.

Fix k and choose two batches that share $B - 1$ samples and differ only in i, j. Such batches exist for every pair since $B < n$. Subtracting their vanishing weight directions gives

$$(c_i - c_j)V_k = s_i R_{ki} - s_j R_{kj}. \tag{7.38}$$

If some $c_i \neq c_j$, the right side is a difference of two matrices of rank at most one. Dividing by the nonzero scalar shows $\operatorname{rank}(V_k) \leq 2$.

Otherwise $c_i = c$ for all i, with $c \neq 0$. Equation (7.38) makes all $s_i R_{ki}$ equal to a single matrix Q_k of rank at most one. Vanishing of any batch direction then gives $Q_k - cV_k = 0$. Thus $V_k = Q_k/c$ has rank at most one, which also implies the stated bound. This applies to each layer. □

The statement retains the published upper bound rather than asserting that it is sharp. Its hypotheses are explicit sufficient conditions for the argument, not an assertion that every architecture in the book satisfies them. The ReLU normalization and homogeneity calculation must not be transferred unchanged to GELU or to arbitrary regularization.

7.17.6 Intrinsic noise and the interpretation of low-rank bias

At a stationary point of the *batch-averaged normalized dynamics*, concatenate the scale direction and the vectorized tangent layer directions into $d_{\mathcal{B}}$. Uniform batch sampling gives $\mathbb{E}_{\mathcal{B}} d_{\mathcal{B}} = 0$. Under the lemma's other assumptions, if at least one V_k has rank greater than two, not all $d_{\mathcal{B}}$ can be zero. Since every admissible batch has positive probability,

$$\operatorname{Tr} \operatorname{Cov}_{\mathcal{B}}(d_{\mathcal{B}}) = \mathbb{E}_{\mathcal{B}} \|d_{\mathcal{B}}\|^2 > 0. \tag{7.39}$$

This is the precise link from the common-equilibrium obstruction to intrinsic stochasticity. The covariance of the *actual normalized update directions* is used here; it is not silently identified with the covariance of unconstrained parameter gradients.

The paper interprets the restriction as a source of low-rank bias and late-stage SGD fluctuations [170, Sections 6.1–6.2]. Two qualifications preserve the mathematical content. First, the lemma is a necessary rank restriction on common equilibria, not a theorem that SGD always reaches matrices of rank at most two. Second, closeness of all batch directions to zero does not by itself give a quantitative rank bound without controlling margins, residuals, and the coefficients used in the batch-exchange argument. Exact rank and effective rank are also different notions.

If a specified data family cannot be fitted to the required tolerance by networks satisfying the common-equilibrium rank restriction, the lemma rules out a common equilibrium in that well-fitting region. Such an incompatibility must be established for the data family; it does not follow merely from calling the data generic. Conversely, low rank alone does not guarantee a common equilibrium or zero noise.

The qualitative mechanism is useful without overstating it:

regularization can maintain nonzero residuals;
different samples can demand different local updates;
common equilibrium can impose strong rank restrictions.

A useful predictor can therefore coexist with persistent intrinsic minibatch fluctuations. This is compatible with the norm-based selection framework, while any generalization advantage of lower rank still requires an applicable statistical bound.

7.17.7 Learning rates, point convergence, and diffusion models

For a plain update $\theta_{k+1} = \theta_k - \alpha_k G_{\mathcal{B}_k,\gamma}(\theta_k)$, the conditional update covariance is $\alpha_k^2 c_{n,B} C(\theta_k)$. Nonzero gradient-estimator covariance and nonvanishing parameter increments are not synonymous when $\alpha_k \to 0$.

There is a simple fixed-step implication. Assume batches are sampled independently with fixed positive probabilities, their gradient fields are continuous near a putative finite limit, and $\inf_k \alpha_k > 0$. With probability one every possible batch occurs infinitely often. If θ_k converges, then $\theta_{k+1} - \theta_k \to 0$. Along the subsequence for any fixed batch, continuity implies that its gradient must vanish at the limit. Hence a finite point limit must be a common minibatch equilibrium. Where the preceding rank obstruction excludes such an equilibrium, this form of point convergence is excluded under the stated conditions. For a normalized implementation, the same inference additionally uses the specified normalization map: its fixed points must be exactly the zeros of the normalized directions. It is not an inference about arbitrary retractions, clipping rules, or adaptive optimizers.

This does not assert divergence or existence of a stationary distribution. It does not exclude convergence with diminishing learning rates, or convergence of an appropriate averaged predictor. A schedule, stopping rule, or averaging rule is part of the learning procedure. Positive covariance at one parameter also supplies no uniform lower noise bound as $\gamma \to 0$, the batch size changes, or the network size grows.

For a small constant step α, a justified diffusion approximation has the schematic regularized form

$$d\theta_t = -\nabla U_\gamma(\theta_t)\, dt + \left[\alpha c_{n,B} C(\theta_t)\right]^{1/2} dW_t,$$

with time increments α. The covariance rate, the gradient-estimate covariance, and the update covariance have different step-size factors. The effective noise is generally anisotropic and state dependent, and finite steps, momentum, adaptive preconditioning, or constraint handling can change the model further.

Intrinsic noise in a regularized well-fitting region does not imply the Gibbs law in (7.19). Under an explicitly isotropic Langevin model with regularization, the potential would be U_γ, not the unregularized data loss. The Fokker–Planck and local covariance calculations in Appendix N, Section N.5, explain the model dependence. A nonzero covariance need not excite every escape direction or yield a polynomial mixing or hitting time.

The positive research question is how the actual residual-dependent noise, normalization, rank structure, and step-size schedule cooperate in selecting useful predictors. Regularization can keep this stochastic mechanism active near a minimum of the full objective. Whether it produces an entropic preference, efficient access, or improved population accuracy must be shown for the specified setting, rather than inferred from a stationary scalar-noise formula.

7.18 Regularization and selection among fitting solutions

When a realizable overparameterized model has many fitting solutions, regularization can prefer some of them. Consider

$$\widehat{\mathcal{L}}_\gamma(\theta) = \widehat{\mathcal{L}}(\theta) + \gamma\Omega(\theta), \qquad \gamma > 0,$$

with a nonnegative penalty. Ordinary weight decay uses $\Omega(\theta) = \frac{1}{2}\|\theta\|_2^2$; other choices include the multiplicative norms discussed in Chapters 5 and H.

The constrained problem

$$\min_{\theta \in \mathcal{M}} \Omega(\theta) \tag{7.40}$$

is a useful small-regularization idealization. It is not an assertion that the actual optimizer first reaches $\mathcal{M}$ and only then regularizes. Both gradients can act from the beginning, and finite γ generally changes the best training fit. In particular, a minimum of the full objective need not be a common minibatch equilibrium. Proposition 5 describes its possible nonzero covariance, while Lemma 3 gives an architecture-specific obstruction to all batches stopping at once.

A precise limiting interpretation. Suppose a global minimizer θ_γ of $\widehat{\mathcal{L}}_\gamma$ exists. For every $\bar{\theta} \in \mathcal{M}$,

$$\widehat{\mathcal{L}}(\theta_\gamma) + \gamma\Omega(\theta_\gamma) \le \gamma\Omega(\bar{\theta}).$$

Since both terms on the left are nonnegative,

$$\widehat{\mathcal{L}}(\theta_\gamma) \le \gamma\Omega(\bar{\theta}), \qquad \Omega(\theta_\gamma) \le \Omega(\bar{\theta}).$$

If a sequence $\gamma \downarrow 0$ has a convergent subsequence of these minimizers and the functions are continuous, its limit belongs to $\mathcal{M}$ and minimizes Ω there. Existence and compactness of such minimizers, and an algorithm that finds them, are separate requirements. This elementary comparison explains the scope of (7.40) without identifying it with the behavior of arbitrary regularized SGD.

A different role: confinement of analytic dynamics. For a nonnegative empirical loss analytic on $\mathbb{R}^p$, consider

$$U(\theta) = \widehat{\mathcal{L}}(\theta) + \frac{\gamma}{2}\|\theta\|^2, \qquad \gamma > 0,$$

where the quadratic term includes *all* trainable parameters. Along gradient flow of U,

$$U(\theta_t) \le U(\theta_0), \qquad \|\theta_t\|^2 \le \frac{2U(\theta_0)}{\gamma}.$$

The trajectory stays in a compact sublevel set and exists for all time. The analytic result in Section 7.9 therefore gives finite-length convergence to a critical point of U. This is Corollary 9 in Appendix N. It is neither a global optimization theorem nor an interpolation guarantee; finite regularization can trade a nonzero residual against a smaller penalty.

A product penalty need not provide this compactness. For example,

$$\widehat{\mathcal{L}}(a, b) = \tfrac{1}{2}(ab - 1)^2, \qquad \Omega(a, b) = a^2b^2$$

gives $\widehat{\mathcal{L}}(t, t^{-1}) + \gamma\Omega(t, t^{-1}) = \gamma$ for every $t > 0$. The sublevel set is unbounded. This does not prove that its trajectories diverge; it shows that confinement does not follow from the penalty alone. Products of squared Frobenius norms can be analytic without being coercive. Unsquared norm products can also lose analyticity at zero factors.

The fitting dynamics must use the actual objective. For differentiable Ω, gradient flow of U_γ satisfies

$$\dot{e} = -Ke - \gamma J\nabla\Omega. \tag{7.41}$$

Thus the unregularized contraction estimate cannot be transferred to penalized training without controlling the additional term. At an exact fit the data gradients vanish, whereas the penalty gradient need not. At a regularized stationary point the average data gradient balances the penalty, although individual sample gradients may differ. The same stationary mean balance can therefore correspond to deterministic convergence of gradient flow and continuing fluctuations of fixed-step minibatch SGD.

The relevant geometric picture is motion in a well-fitting region of the regularized objective, not automatically motion constrained to the unregularized zero-loss manifold. This distinction permits the intrinsic noise mechanism without claiming that every fluctuation is tangent to that manifold or that the regularized minimizer set has dimension $p - q$.

Entropic and norm preferences are also different:

equilibrium volume weighting:	depends on the specified measure,
norm selection:	prefers lower penalty at comparable fit.

They need not favor the same region. Neither preference alone identifies the generating internal variables, and predictive learning does not require that identification.

7.19 Why norm selection can be statistically meaningful

A parameter count or a parameter-space volume is not, by itself, the statistical complexity needed to control population prediction. A positive route is to use a suitable norm-sensitive hypothesis class.

Fix an admitted architecture and a complexity functional $\mathcal{C}(\theta)$. For a budget Q chosen independently of the training sample, define

$$\mathcal{H}_Q = \{F_\theta : \mathcal{C}(\theta) \le Q\}.$$

In an appropriate homogeneous feedforward setting, a candidate quantity is

$$\mathcal{C}_\times(\theta) = \prod_{\ell=1}^{L} \|W_\ell\|_F, \qquad \Omega_\times(\theta) = \mathcal{C}_\times(\theta)^2.$$

Biases, other trainable operations, activation assumptions, depth factors, and any further terms required by the chosen statistical theorem must be included. The bare product is not a universal bound for every network.

The meaningful statistical assertion is a bound for the whole class, not an expression evaluated on one isolated parameter vector:

$$\widehat{\mathfrak{R}}_{S_X}(\mathcal{H}_Q) \le \mathcal{B}_n(Q; \text{architecture and input assumptions}).$$

Under the bounded-loss and Lipschitz hypotheses stated in Chapter H, a corresponding uniform bound takes the form

$$R_{\mathsf{P}}(h) \le \widehat{R}_S(h) + \beta_S(Q, \delta) \qquad \text{for every } h \in \mathcal{H}_Q \tag{7.42}$$

on an event of probability at least $1 - \delta$. Squared loss needs appropriate prediction and label bounds, or a different applicable statistical argument.

On this uniform event, any returned predictor satisfying $\widehat{f} \in \mathcal{H}_Q$ and $\widehat{R}_S(\widehat{f}) \le \eta$ obeys

$$\boxed{R_{\mathsf{P}}(\widehat{f}) \le \eta + \beta_S(Q, \delta).} \tag{7.43}$$

Interpolation is the special case $\eta = 0$. The bound must be small at the available sample size, and training must actually meet the fit and complexity budget. Sample-dependent choices of the

budget or architecture require an appropriate simultaneous bound or model-selection adjustment. Chapter H states the joint statistical and optimization success probability explicitly.

A multiplicative penalty can favor a smaller product among comparably fitting solutions and thereby reduce an applicable complexity upper bound. It does not literally minimize exact Rademacher complexity, nor imply a universal monotone relation with test error. Reciprocal scaling of two adjacent positively homogeneous layers changes their individual norms but preserves their Frobenius product; that particular symmetry does not invalidate a correctly formulated product-norm analysis.

Three controls must nevertheless be kept separate: an upper bound on statistical class complexity, confinement of the parameter trajectory, and a lower bound on the residual Rayleigh quotient. The first does not imply either of the other two. In particular, an upper matrix-norm bound does not rule out rank deficiency. Likewise, a statistical theorem that assumes positive homogeneity cannot be transferred unchanged to GELU, which is not positively homogeneous. The applicable activation and architecture assumptions must be verified, as emphasized in Chapter H.

Most importantly,

norm control can support predictive generalization without identifying the generating computation.

This is the relevant statistical selection question for the next chapter. Population accuracy and structural identification should not be merged into a requirement that the chosen interpolant have the teacher's internal factorization.

7.20 An idealized theorem

The regular local-weight comparison can be stated and proved precisely. The neighborhoods in the theorem are fixed before the temperature tends to zero.

Theorem 6 (Low-temperature dominance of a regular interpolating manifold)**.** *Let $L : \mathbb{R}^p \to \mathbb{R}_{\geq 0}$ be smooth. Suppose:*

1. *The zero-loss set contains a nonempty compact smooth embedded manifold $\mathcal{M}$ without boundary, of codimension q, where $1 \leq q \leq p$. On its Euclidean normal spaces the Hessian $H_\perp(s)$ is uniformly positive definite. Choose a sufficiently small fixed tubular neighborhood $U_{\mathcal{M}}$ with no additional zeros.*
2. *Each competitor θ_ℓ under consideration is an isolated nondegenerate local minimum, with $L(\theta_\ell) = L_\ell > 0$ and $H_\ell = \nabla^2 L(\theta_\ell) \succ 0$. Choose a sufficiently small fixed neighborhood U_ℓ on which it is the unique minimizer.*
3. *For the interpretation as equilibrium probabilities, the Langevin model $d\theta_t = -\nabla L(\theta_t)\, dt + \sqrt{2T}\, dW_t$ has the normalizable stationary density $Z_T^{-1} e^{-L/T}$ at the temperatures under consideration.*

Define the local contributions

$$Z_{\mathcal{M}}(T) = \int_{U_{\mathcal{M}}} e^{-L(\theta)/T}\, d\theta, \qquad Z_\ell(T) = \int_{U_\ell} e^{-L(\theta)/T}\, d\theta.$$

Then, as $T \downarrow 0$,

$$Z_{\mathcal{M}}(T) = (2\pi T)^{q/2} \left[\int_{\mathcal{M}} \frac{dV(s)}{\sqrt{\det H_\perp(s)}} + o(1) \right], \tag{7.44}$$

$$Z_\ell(T) = e^{-L_\ell/T} (2\pi T)^{p/2} \left[\frac{1}{\sqrt{\det H_\ell}} + o(1) \right]. \tag{7.45}$$

Consequently,

$$\boxed{\frac{Z_\ell(T)}{Z_{\mathcal{M}}(T)} = O\Big(e^{-L_\ell/T} T^{(p-q)/2}\Big) \longrightarrow 0.} \tag{7.46}$$

The same ratio equals $\rho_T(U_\ell)/\rho_T(U_{\mathcal{M}})$ under assumption 3.

Proof. Use Euclidean normal coordinates in the tube around $\mathcal{M}$, locally on the normal bundle. Smoothness and compactness give the uniform expansion

$$L(s,z) = \tfrac{1}{2} z^\top H_\perp(s) z + O(\|z\|_2^3), \qquad d\theta = (1 + O(\|z\|_2))\, dz\, dV(s).$$

Uniform normal positivity permits the tube to be chosen so that $L(s,z) \geq c\|z\|_2^2$ for a constant $c > 0$. After the change of variables $z = \sqrt{T}\, w$, the integrand, including the volume density, converges to

$$\exp\Big(-\tfrac{1}{2} w^\top H_\perp(s) w\Big)$$

and is bounded by an integrable Gaussian uniformly in s. Local charts and a partition of unity cover the compact manifold. Dominated convergence therefore gives

$$T^{-q/2} Z_{\mathcal{M}}(T) \longrightarrow \int_{\mathcal{M}} \int_{\mathbb{R}^q} e^{-w^\top H_\perp(s) w/2}\, dw\, dV(s) = (2\pi)^{q/2} I_{\mathcal{M}}.$$

The constant $I_{\mathcal{M}}$ is finite and strictly positive.

At θ_ℓ, write $\theta = \theta_\ell + \sqrt{T}\, w$ and apply the same argument to

$$L(\theta_\ell + u) = L_\ell + \tfrac{1}{2} u^\top H_\ell u + O(\|u\|_2^3).$$

On a sufficiently small neighborhood, $L(\theta_\ell + u) - L_\ell \geq c_\ell \|u\|_2^2$, which supplies Gaussian domination. Thus

$$e^{L_\ell/T} T^{-p/2} Z_\ell(T) \longrightarrow \frac{(2\pi)^{p/2}}{\sqrt{\det H_\ell}}.$$

Dividing these asymptotics proves (7.46). For the stationary law, the common normalizing constant cancels in the ratio of neighborhood probabilities. □

This theorem compares a specified regular global component with each specified isolated positive-loss competitor. For finitely many such competitors, the equilibrium probability conditioned on the union of these neighborhoods concentrates on $U_{\mathcal{M}}$. No corresponding claim about the entire parameter space follows without controlling the remaining regions and other global components.

The theorem does not assert that arbitrary SGD has this stationary law, that a fixed-temperature trajectory converges to the exact manifold, or that reaching a low-loss neighborhood takes polynomial time. The local asymptotics themselves do not require a global stationary law; that law is needed only for the stated equilibrium interpretation.

7.21 The conjectural extension to deep-network SGD

The theorem motivates, but does not prove, a broader dynamical hypothesis. Its status is best made explicit.

> **Entropic Interpolation Conjecture—a programmatic form.** For suitable specified families of realizable overparameterized deep networks, the geometry of low-loss sets and the actual stochastic training dynamics combine to make near-interpolation reachable with high probability and feasible resources. The proposed mechanism is a favorable weighting and accessibility of degenerate low-loss regions, not merely an increase in raw parameter count.

A theorem-strength version must name the target and data class, the architecture family, initialization law, minibatch rule, step sizes, regularizer, and stopping criterion. For example, with a problem-size index N, empirical-loss tolerance $\xi > 0$, and failure probability δ, it would need a statement of the form

$$\Pr\Big\{\widehat{\mathcal{L}}_N(\theta_{k_\star}) \le \xi\Big\} \ge 1 - \delta,$$

where $k_\star$, network description, arithmetic cost, and numerical precision satisfy explicit resource bounds. An exponential inverse conditioning parameter or an exponential equilibration time would not establish efficient learning. The chapter supplies no such general finite-time bound for deep-network SGD.

The output can be a specified stopped iterate; it need not be an absorbing point shared by every minibatch. If an averaged parameter or averaged predictor is returned, its representational and evaluation costs must also be counted. Positive weight decay and the target tolerance must be chosen compatibly, since the penalized optimum may trade fit against penalty. Persistent gradient-estimator variance does not preclude a finite-time prediction guarantee, but its scale and the learning-rate schedule must enter the analysis.

Three logically distinct steps must be separated.

Geometric step. At an existing regular interpolating point, locally constant rank q gives dimension $p-q$. At singular points, local geometry must be analyzed rather than inferred from Hessian nullity. Large dimension alone is not a measure of dynamically relevant volume.

Equilibrium step. For a specified normalizable Langevin model, neighborhood probabilities are integrals of $e^{-L/T}$ against its reference measure. The regular and singular calculations estimate such integrals under their respective hypotheses. They do not determine finite-time access.

Dynamical step. One must show that the actual initialization and optimization procedure reach the relevant low-loss region on the claimed timescale, and that its noise model is appropriate. Equilibrium preference does not imply a mixing-time or hitting-time bound. Letting $T \downarrow 0$ before obtaining such a bound can make access, rather than equilibrium weight, the limiting issue.

These steps cannot be collapsed into an unconditional arrow from $p \gg q$ to successful SGD. The conjecture remains a mechanism-oriented research program. Other optimization mechanisms can establish successful fitting without validating an entropic explanation.

7.22 What the argument does not say

The geometric picture is useful when its limits remain visible in the conclusions as well as in the assumptions.

7.22.1 It does not make the loss convex

The loss can retain saddles, local minima, and complicated critical sets. The regular-manifold theorem describes neighborhoods of solutions whose existence has been assumed. It does not eliminate all competing critical points. The deterministic contraction argument instead rules out positive-loss stationary points in a region where its quantitative condition holds.

An analytic counterexample to an almost-everywhere rank argument. For every $p \geq 2$, Appendix N, Proposition 72, considers

$$r(\theta) = \theta_1^3 - 3\theta_1 + 3 + \sum_{j=2}^{p} \theta_j^2, \qquad L(\theta) = \tfrac{1}{2} r(\theta)^2. \tag{7.47}$$

This analytic scalar-residual problem depends on every parameter and has a nonempty regular zero-loss manifold of dimension $p - 1$. Its prediction Jacobian has full row rank except at the two points $(\pm 1, 0, \ldots, 0)$. Nevertheless, at $\theta_+ = (1, 0, \ldots, 0)$,

$$L(\theta_+) = \tfrac{1}{2}, \qquad \nabla L(\theta_+) = 0, \qquad \nabla^2 L(\theta_+) = \operatorname{diag}(6, 2, \ldots, 2) \succ 0.$$

A small strongly convex neighborhood is attracted to this positive-loss minimum. The appendix proves these claims and the existence of an open suboptimal basin. Thus

analyticity, many parameters, and regular interpolation
$+$ full prediction rank almost everywhere
$\not\Longrightarrow$ almost-sure convergence to interpolation.

The exceptional critical parameters can have measure zero while their basin has positive measure. This is a counterexample to a general analytic least-squares inference, not a claim that a prescribed unrestricted GELU architecture has exactly this parameterization. Architecture-specific hypotheses may exclude the obstruction; parameter count and analytic regularity alone do not.

7.22.2 It does not prove that every degenerate set has a large basin

Degeneracy is local. Basin size depends on the vector field and on the initialization measure. A residual-contraction theorem needs a region containing the trajectory, not merely the Hessian at the eventual minimum. For stochastic dynamics, a stationary mass estimate still needs a finite-time accessibility argument.

7.22.3 Suboptimal sets can also be degenerate

Architectural symmetries and redundancies can occur at positive-loss critical sets. The relevant comparison includes loss, normal curvature, dimension or singular exponent, local weighting, and actual dynamics. It is not simply "degenerate global" versus "nondegenerate local." The isolated competitors in Theorem 6 are an explicitly restricted case.

7.22.4 Raw Euclidean volume is parameterization dependent

Changing coordinates can alter Euclidean volume and Hessian coordinates. A probability statement must transform the reference density consistently. Holding an isotropic noise rule fixed after a

nonlinear reparameterization generally defines a different stochastic process, not the same process in new notation.

The relevant measure should therefore be tied to the specified initialization, dynamics, prior, or regularizer. A broad parameter valley is not automatically a low-complexity function class, and none of these measures identifies semantic variables without extra assumptions.

7.22.5 The zero-loss set can be noncompact

Scale symmetries can produce noncompact families. Then an unweighted volume integral may diverge, and the unregularized Gibbs law may fail to normalize. A compact localization is sufficient for a local Laplace calculation but is not by itself a global equilibrium model.

A prior-weighted integral, a confining regularizer, a bounded-domain process, or a quotient by exact symmetries can be appropriate. They are different constructions and must be accompanied by their own measure and dynamical assumptions. A norm product that is invariant along a noncompact rescaling orbit need not, by itself, confine that orbit.

7.22.6 Other losses require modified geometry

Squared loss has a finite zero-loss solution in the realizable regime assumed here. For separable classification with unregularized logistic or cross-entropy loss, the infimum can occur only as parameter norms diverge. One then studies asymptotic directions, margins, or a regularized objective rather than a finite unregularized zero-loss manifold.

Low classification error, small surrogate loss, and exact squared-loss interpolation are also different criteria. A predictive framework can use any appropriate loss and statistical bound; the present geometric identities should not be transferred unchanged between them.

7.23 Why overparameterization can make optimization easier

More parameters enlarge the search domain, but they can also enlarge the set satisfying the fitting constraints. If regular realizations persist and their prediction rank stays fixed at q while p increases, the local fitting dimension $p - q$ increases. This is the geometric reversal behind the chapter.

The reversal is not a complexity comparison by itself. New parameters may improve prediction conditioning, introduce useful alternative paths, or merely duplicate irrelevant directions. Only the first kinds of change can help establish an accessibility bound.

An elementary check illustrates the limit of dimension counting. Append parameters $z \in [-1, 1]^r$ that do not affect the loss:

$$\widetilde{L}(\theta, z) = L(\theta).$$

With product Lebesgue measure, every local partition contribution acquires the same factor 2^r:

$$\widetilde{Z}_U(T) = 2^r Z_U(T).$$

All relative equilibrium weights are unchanged, and the gradient dynamics of θ are unchanged. The fitting set has extra neutral directions, but these dummy coordinates do not make the original optimization problem easier. They enlarge suboptimal sets as well.

The meaningful claim is therefore

overparameterization can enlarge the fitting set while improving the prediction dynamics that lead to it.

The first effect is a regular dimension count. The second requires an architecture- and data-dependent argument. Neither must recover a privileged internal basis to contribute to predictive learning.

7.24 Three levels of explanation

Three levels remain useful, provided their conclusions are not conflated.

Level 1: existence and local dimension. In a realizable model, a regular zero-loss point with local rank q belongs to a fitting manifold of dimension $p - q$. The squared-loss Hessian has exactly $p - q$ zero modes at that point. This is the clean algebraic and geometric statement.

Level 2: accessibility within a resource budget. The optimizer must reach an adequately low-loss region with a stated success probability and cost. Uniform residual contraction on a high-probability invariant region supplies one conditional deterministic route. A Langevin equilibrium calculation supplies a different, stationary weighting result; an additional dynamical argument is needed to turn it into finite-time access. Neither follows from Level 1 alone.

For an analytic objective, compactness can instead establish convergence to one critical point, as in (7.17). That positive endpoint result does not determine whether the attained loss is adequate or whether the time and precision costs are feasible.

Level 3: predictive selection. The returned predictor must be controlled well enough to generalize. An informative norm-sensitive class bound, together with a sufficiently good fit, is one route. The selected internal variables need not match the teacher's. Identifying those variables is a separate structural objective, not part of the definition of predictive Level 3. Low-rank selection can be investigated within a suitable norm- and rank-sensitive class; a rank obstruction to common minibatch equilibrium is not, by itself, a population-error bound.

This chapter establishes the regular local identities, a conditional contraction argument, and the idealized equilibrium comparison. It also summarizes the complementary analytic convergence result proved in Appendix N. It motivates a finite-time stochastic extension and connects selection to the statistical argument developed in Chapter H. It does not supply a universal passage through all three levels.

7.25 The convergence backstop, with its conditions explicit

The term *backstop* should refer to a supported route toward fitting, not to a claim that geometric abundance rescues every training problem. The analytic endpoint result should first be distinguished from the two arguments about fitting dynamics and equilibrium mass.

The analytic endpoint.

$$\boxed{\begin{array}{c}\text{an analytic objective and an all-time gradient flow}\\ + \text{ compact trajectory closure inside its domain}\\ \Longrightarrow \text{ finite-length convergence to a critical point.}\end{array}}$$

Neither a globally fitting limit nor a useful convergence-time bound is part of this conclusion. Full quadratic weight decay is one sufficient source of confinement for the analytic squared-loss setting; a product penalty alone need not supply it.

The deterministic fitting line.

$$\boxed{\begin{array}{c}\text{initialization falls in an invariant conditioning region}\\ +\, e^\top K e \geq \mu\|e\|_2^2 \text{ along the trajectory}\\ \Longrightarrow \quad \widehat{\mathcal{L}}(\theta_t) \leq \widehat{\mathcal{L}}(\theta_0)e^{-2\mu t}.\end{array}}$$

A probability estimate for initialization then gives a conditional high-probability fitting statement. Finite-step and bit-complexity claims require the corresponding smoothness, invariance, and precision bounds.

The equilibrium line.

$$\boxed{\begin{array}{c}\text{a compact regular zero-loss manifold and local competitors}\\ +\, \text{the specified normalizable Langevin equilibrium}\\ \Longrightarrow \quad \dfrac{\rho_T(U_\ell)}{\rho_T(U_{\mathcal{M}})} = O\Big(e^{-L_\ell/T} T^{(p-q)/2}\Big).\end{array}}$$

This is a relative stationary mass estimate. A suitable singular asymptotic can replace the regular Laplace factor when its hypotheses apply. Neither version includes a bound on the time needed to visit the favored region.

These results can be complementary. For example, global access and local residual contraction address different stages of a possible proof, while analyticity can settle the behavior of a confined trajectory. But their hypotheses must be verified; combining their conclusions in a diagram does not verify them.

The regularized stochastic endpoint. Section 7.17 adds a different exact distinction:

$$\nabla U_\gamma(\theta_\gamma) = 0 \quad \not\Longrightarrow \quad \mathrm{Cov}_{\mathcal{B}}(G_{\mathcal{B},\gamma}(\theta_\gamma)) = 0.$$

For proper minibatches, differing per-example gradients make this covariance positive in trace. The normalized ReLU rank restriction gives one mechanism excluding a common equilibrium in a higher-rank region. It does not itself supply a global fitting theorem or a Gibbs law.

Regularization provides a selection pressure and changes the balance supporting both mean drift and sample-dependent fluctuations. To connect this dynamics to prediction, one needs a returned iterate with adequate empirical fit and an informative complexity budget. A common zero of all minibatch updates is not an additional requirement. That is the transition to Chapter H.

7.26 Interpolation, prediction, and structural recovery

The distinction at the end of the chapter is now threefold:

empirical fitting:	$\widehat{R}_S(\widehat{f})$ is small,
predictive recovery:	$R_{\mathsf{P}}(\widehat{f})$ is small,
structural recovery:	specified hidden structure is identified.

A model can interpolate a finite training sample and fail to predict on the population. Memorization or a sample-specific shortcut can produce that failure. A statistical argument is needed to exclude it in the setting of interest.

By contrast, using a different internal basis, merging several teacher constituents into one learned module, or distributing a teacher variable across several learned features is not necessarily a failure. Such a network can be an accurate and feasible predictor even without a useful nodewise correspondence to the generating DAG.

There are therefore two different nonimplications:

$$\text{interpolation} \not\Longrightarrow \text{population accuracy},$$

$$\text{population accuracy} \not\Longrightarrow \text{generating-structure identification}.$$

The first is a gap that predictive learning must address. The second marks a stronger objective that need not be imposed on prediction.

If scientific interpretation, causal identification, or a particular module-reuse claim requires latent structure, it needs an explicit structural criterion and identifiability conditions. Those questions remain important. They should not be inserted as an extra requirement after a feasible predictor has already been shown to generalize.

7.27 The bridge to predictive learning

Chapter H organizes the predictive synthesis around three questions:

Q1 : Does the chosen class contain a feasible accurate predictor?

Q2 : Can the specified training procedure find a suitable fit?

Q3 : Is the selected predictor controlled enough to generalize?

Chapter 4 supplies a representation principle relevant to Q1, subject to the architecture and finite-precision qualifications. Chapters 5 and 6 study learning signals, intermediate supervision, and conditioning relevant to Q2. The present chapter contributes the local fitting geometry, the conditional residual-contraction route, and the idealized equilibrium volume mechanism. Appendix N additionally explains when a bounded analytic flow has a definite stationary endpoint. It does not guarantee that the endpoint fits the data. Q2 still requires a quantitative argument for the specified training procedure.

For Q3, suppose a valid class bound applies and training returns $\widehat{f} \in \mathcal{H}_Q$ with $\widehat{R}_S(\widehat{f}) \le \eta$. On the corresponding uniform event,

$$R_{\mathsf{P}}(\widehat{f}) \le \eta + \beta_S(Q, \delta).$$

The algorithmic probability of reaching that joint fit and complexity condition is a separate premise. Chapter H combines the two events explicitly. The returned predictor may be a specified stopped iterate even when fixed-step SGD continues to fluctuate; all confidence and complexity conditions still have to hold for that selection. A low-loss region favored at equilibrium does not supply the statistical complexity bound, and a small penalty written in an objective does not prove that the optimizer meets its required budget.

Once population accuracy and computational feasibility have been established, the learned network itself supplies a compositional realization of its predictor. For example, in noiseless regression, assume

$$\|\widehat{f} - f^\star\|_{L^2(P_X)} \le \varepsilon$$

and that a specified feasible numerical evaluator produces $\widetilde{f}_m$ with $\|\widetilde{f}_m - \widehat{f}\|_\infty \le 2^{-m}$. Compilation of that evaluator into bounded-fan-in operations gives a computationally compositional implementation, and the triangle inequality yields

$$\boxed{\|\widetilde{f}_m - f^\star\|_{L^2(P_X)} \le \varepsilon + 2^{-m}.}$$

This is the compositional-witness argument of Chapter H. Description length, input encoding, evaluation time, and precision must scale within the stated resource bounds; being finite is not enough for an asymptotic efficiency claim. The digital implementation also does not automatically supply a fixed coarse-grained Sobolev decomposition with uniform regularity constants.

Genericity and staircase learning can help establish how a specified algorithm obtains an accurate predictor. A staircase theorem may also prove structural recovery. Faithful or partial traces can provide a different useful route. Neither a staircase nor teacher-node matching is a separate logical premise of the witness implication after predictive success has been established.

The next chapter therefore asks why training finds a *useful feasible predictor*, rather than why every successful solution must reconstruct the computation that generated the observations.

7.28 Summary

Overparameterization can enlarge both parameter space and the fitting set. At an existing interpolating point with locally constant prediction rank q, the fitting manifold has dimension $p - q$. At any twice-differentiable interpolating point,

$$\boxed{\nabla^2 \widehat{\mathcal{L}}(\theta^\star) = \frac{1}{n} J^\top J, \qquad \dim\ker \nabla^2 \widehat{\mathcal{L}}(\theta^\star) = p - q.}$$

At regular points the zero modes are tangent to the fitting manifold. At singular points, they do not determine the geometry by themselves.

Analyticity supplies a separate result: an all-time gradient flow with compact closure inside the analytic domain converges with finite length to one critical point. That point need not interpolate, and the analytic constants need not yield an efficient rate. The counterexample in Section 7.22 shows why even almost-everywhere prediction rank does not promote this result to global fitting.

A deterministic convergence statement requires information along the trajectory. Uniform contraction of the actual residual gives

$$\widehat{\mathcal{L}}(\theta_t) \le \widehat{\mathcal{L}}(\theta_0) e^{-2\mu t}.$$

A high-probability invariant initialization region converts that estimate into a conditional convergence guarantee. The chapter does not deduce such a region from raw width alone.

The stochastic calculation is different. In the specified normalizable Langevin equilibrium, a regular zero-loss manifold contributes a local weight proportional to $T^{q/2}$ times its integrated inverse normal curvature, while an isolated positive-loss minimum contributes $e^{-L_\ell/T} T^{p/2}$ times its curvature factor. The resulting relative mass estimate is explicit. Suitable singular asymptotics replace these powers when their analytic hypotheses hold.

These stationary comparisons are not finite-time SGD theorems. Basin geometry, equilibration, state-dependent noise, and the behavior of prefactors across network sizes remain separate issues. The entropic interpolation conjecture concerns that dynamical extension and is not established merely by dimension counting.

For real minibatch training, exact interpolation and regularized stationarity have different consequences. At an exact fixed-example squared-loss interpolator the sampling covariance is zero. At a regularized stationary point, only the average data gradient must balance the penalty; sample gradients can differ and the update covariance can be nonzero. The convex example proves that this phenomenon does not require nonconvexity. In the stated normalized ReLU model, the common-equilibrium rank bound supplies a further obstruction and motivates low-rank bias. Its hypotheses, and the distinction between exact rank and effective rank, remain essential.

Regularization can prefer predictors with smaller applicable norm complexity. It can also confine analytic gradient flow when the specified penalty is coercive, but statistical control and confinement are different properties. When a training procedure actually achieves adequate fit within a class having an informative statistical bound, it can yield small population error. This selection need not reproduce the teacher's internal variables or a preferred generating factorization.

A small mean gradient does not mean that every minibatch update is small. Fixed-step point convergence, diminishing-step convergence, and successful selection of a finite-time predictor are different claims. Intrinsic noise can persist in a regularized well-fitting region without either invalidating a predictive guarantee or establishing one automatically.

Fitting geometry describes available solution sets; optimization must establish feasible access; statistical selection must control population prediction. When the last two succeed and evaluation is feasible, the learned computation itself is a compositional witness. Structural identification is a stronger question, not an extra condition for that conclusion.

The remaining predictive challenge, developed in Chapter H, is

> When does the specified regularized learning procedure find,
> with high probability and feasible resources,
> a sufficiently well-fitting predictor that generalizes?

CHAPTER 8

Principles of Deep Learning

The preceding chapters developed a representation principle and several possible mechanisms for learning. These should be related without being identified with one another.

The representation principle is that efficiently computable targets admit, at the relevant finite precision, polynomial-size bounded-fan-in realizations. In the computational sense used in this book, efficient computation implies compositional sparsity.

The learning question is different. A specified optimization procedure must find a sufficiently accurate predictor, and the selected predictor must have statistical complexity small enough for the available data to support generalization. Overparameterization, stochastic dynamics, and regularization offer complementary ways to approach these requirements. They do not establish them automatically.

Chapter 7 and Appendix N distinguish convergence to a critical point, quantitative contraction of training error, and low-temperature equilibrium weighting. These conclusions play different roles in the learning argument. None alone establishes predictive success. Regularized stationarity can also coexist with intrinsic minibatch fluctuations. Predictive success concerns the model returned by training, not convergence to a common zero of every minibatch update.

Suppose training succeeds: it returns a computationally feasible deep network with small population error. That network already supplies a compositional approximation of the target. Its intermediate variables need not reproduce the variables that generated the observations. Indeed, predictive success does not require an additional theorem matching the learned network to any particular generating decomposition.

The central argument is therefore conditional:

training finds a feasible predictor with good empirical fit
and an informative statistical-complexity bound

$\Downarrow$

small population error

$\Downarrow$

the learned computation is a compositional witness.

The difficult premise is the joint achievement of good fit, suitable complexity control, and feasible training cost. Genericity, staircase learning, skip connections, and trace exposure remain important candidate mechanisms for establishing that premise. They are not separate necessary conditions for the final witness implication.

The distinction organizing this chapter is thus not only representation versus learning, but also predictive recovery versus structural recovery. The first seeks a useful input–output computation. The second seeks specified internal structure and is a stronger objective.

8.1 The central simplification

Let

$$f : \mathcal{X} \longrightarrow \mathcal{Y}$$

be the target. Suppose training returns a depth-L network

$$F_\theta = h_L \circ h_{L-1} \circ \cdots \circ h_1$$

that approximates f on the population distribution of interest. The learned network already supplies a hierarchical computation of the target approximation.

There is no requirement that h_ℓ coincide with a privileged or "true" latent constituent. If the target admits a generating factorization

$$f = g_m \circ \cdots \circ g_1,$$

training may discover an entirely different factorization. For predictive learning, that is acceptable when the resulting function is accurate and computationally feasible.

8.1.1 Predictive recovery

Let P denote the joint distribution of (X, Y), with input marginal P_X, and let ℓ be a nonnegative prediction loss. Write

$$R_P(F) = \mathbb{E}_P[\ell(F(X), Y)], \qquad \widehat{R}_S(F) = \frac{1}{n}\sum_{i=1}^{n} \ell(F(X_i), Y_i). \tag{8.1}$$

The sample $S = ((X_i, Y_i))_{i=1}^n$ is drawn independently from P. Predictive recovery means that $R_P(F_\theta)$ is suitably small, or that its excess over an appropriate reference risk is small in a noisy problem.

For noiseless regression, $Y = f(X)$, with $\ell(\widehat{y}, y) = \frac{1}{2}(\widehat{y} - y)^2$, this becomes

$$R_P(F_\theta) = \frac{1}{2}\|F_\theta - f\|^2_{L^2(P_X)}. \tag{8.2}$$

The distribution and loss are part of the statement. Good performance under P does not by itself assert correctness under a different observation model.

8.1.2 Structural recovery

A structural objective additionally compares the learned internal variables with a specified teacher DAG. A possible criterion is

$$\mathcal{D}_{\mathrm{str}}(\widehat{z}, z) = \inf_{\substack{\pi \text{ admissible} \\ \psi_v \in \mathcal{E}_v}} \max_v \big\|\widehat{z}_{\pi(v)} - \psi_v(z_v)\big\|_{L^2(P_X)}. \tag{8.3}$$

Here the matching respects the allowed graph structure, the feature slots have controlled dimension, and the equivalences $\mathcal{E}_v$ have fixed complexity and conditioning bounds. Otherwise an unrestricted decoder could hide the entire original prediction problem inside the recovery criterion.

Small prediction error does not establish this structural conclusion. Conversely, failure to match the generating variables does not imply predictive failure. The question needed for the main argument is

Did learning find a feasible function that predicts f accurately?

It is not necessary to add a requirement that this function reproduce a particular hidden computation.

8.2 A successful finite network is itself a computation

A neural network evaluated at finite precision is a digital computation. A dense affine unit

$$z_j = \sigma\left(\sum_i W_{ji} x_i + b_j\right)$$

may have large functional fan-in as written. But the weighted sum can be evaluated by bounded-fan-in additions and multiplications, for example by a binary reduction tree. Thus a polynomial-size dense layer has a polynomial-size bounded-fan-in implementation, provided the arithmetic and activation evaluations use controlled resources.

Composing such layers gives a bounded-fan-in computation DAG. This is a statement about the implementation; it does not say that the displayed weight matrices must themselves be sparse.

8.2.1 Finite is not the same as feasible

For a family of problems indexed by a natural size parameter N, the relevant resources include architecture description, depth, width, stored parameter bits, intermediate numerical precision, and forward evaluation time. Input encoding and output decoding must also be counted.

To call the learned realization feasible, these resources must have the claimed polynomial bounds. Merely observing that every individual trained network is finite is not enough. A family of finite networks can still have exponential width, exponential running time, or prohibitive precision requirements.

Likewise, efficient evaluation of a returned network does not imply that finding its parameters was efficient. Training cost and evaluation cost are separate quantities.

8.2.2 Learning accuracy and evaluation precision

It is useful to distinguish the target learning error ε from the numerical error 2^{-m} in evaluating a fixed predictor. A learning theorem may allow sample size and model size polynomial in ε^{-1}, while the evaluator of that returned model uses time polynomial in its description length and in m.

Such a theorem does not assert that learning costs only $\mathrm{poly}(\log(1/\varepsilon))$. Throughout the chapter, a polynomial-resource claim is relative to its stated parameters.

Subject to these conventions, the central implication is

F_θ predicts the target accurately
$+$ F_θ has a feasible computational implementation
$\Longrightarrow$
a feasible compositionally sparse predictor.

8.3 Efficient computation and compositional sparsity

For discrete computation, the basic implication is the one developed in Chapter 4. If a deterministic Turing machine evaluates a target family in time $T(N) = \mathrm{poly}(N)$, its computation can be simulated by a polynomial-size bounded-fan-in Boolean circuit. The construction is uniform when the evaluator is uniform.

Schematically,

$$\mathbf{P} \subseteq \{\text{uniform polynomial-size bounded-fan-in realizations}\}.$$

For real-valued targets, a precision convention is essential.

Definition 7 (Efficiently Turing-computable real function family)**.** *Let*

$$f_N : [0,1]^{d_N} \longrightarrow \mathbb{R}^{r_N},$$

where d_N and r_N are polynomially bounded in N. The family is efficiently Turing computable *under the present precision convention if there is one deterministic algorithm M and polynomial bounds p_{in} and p_{time} with the following property.*

For every N, every $m \geq 1$, and every $x \in [0,1]^{d_N}$, a dyadic approximation $\widetilde{x}$ with the prescribed polynomial-length encoding and

$$\|x - \widetilde{x}\|_\infty \leq 2^{-p_{\mathrm{in}}(N,m)}$$

is sufficient for M, given $(N, m, \widetilde{x})$, to return a dyadic vector $\widetilde{y}$ satisfying

$$\|\widetilde{y} - f_N(x)\|_\infty \leq 2^{-m}$$

in at most $p_{\mathrm{time}}(N, m)$ bit operations. The guarantee holds for all valid input approximations at that precision, not only for a specially chosen encoding.

For a single target, the family index can be suppressed. This definition makes explicit the two requirements already present in the bit-model formulation:

	polynomial-time finite-precision evaluation
$+$	polynomial control of required input precision.

After binary encoding, each finite-precision evaluation is an ordinary digital computation and admits a bounded-fan-in realization. A finite-precision circuit acts on encoded inputs; interpreting its decoded output as an approximation on a continuous domain additionally uses the input-precision guarantee.

8.3.1 Computational sparsity and regular compositional structure

Two claims must remain distinct:

$$\begin{gathered}\text{efficient digital evaluation} \\ \Longrightarrow \text{a small bounded-fan-in circuit,}\end{gathered}$$

$$\begin{gathered}\text{regular low-arity constituents and stable composition} \\ \Longrightarrow \text{favorable compositional approximation rates.}\end{gathered}$$

The second is the approximation-theoretic statement of Chapter 4. The first does not, by itself, identify a precision-independent coarse DAG, reusable semantic modules, or uniformly bounded Sobolev norms for those modules.

The witness conclusion in this chapter concerns *computational compositional sparsity*. Stronger claims about smooth coarse-grained structure require the corresponding regularity and stability hypotheses. The distinction does not invalidate the witness argument; it specifies what the witness establishes.

8.4 The learned network is a witness

The representation argument applies after training, regardless of how the network was found.

Suppose a returned predictor $F_{\theta^\star}$ has a polynomial-length finite description and a numerically controlled, feasible evaluator. Compiling that evaluator gives a bounded-fan-in realization of the predictor at the requested precision. If the predictor also approximates f on P_X, this realization is a compositional approximation of f.

There is no need to discover a second decomposition behind the learned network. The network, together with its evaluator, already provides one.

The learned computation is the witness.

This is an endpoint statement, not a learning algorithm. It does not prove that SGD found the network in feasible time, that its training error is small, or that its population error is small. Those are the premises that must be established by the optimization and statistical arguments.

Nor does the computational witness show that the selected predictor has small norm complexity. A polynomial-size computational object may still belong to a statistical class too rich for an informative bound at the available sample size.

8.5 Optimization: why interpolation is plausible

Representation does not explain why training reaches a good solution. The geometry and dynamics of Chapter 7 offer several conditional parts of that explanation. Exact interpolation is a useful reference geometry. At fixed positive regularization, however, a well-fitting stationary point may have nonzero data residuals and persistent minibatch fluctuations. These regimes should not be identified. For the geometric discussion, consider smooth squared loss

$$\widehat{\mathcal{L}}(\theta) = \frac{1}{2n}\sum_{i=1}^{n}(F_\theta(x_i) - y_i)^2, \qquad \mathcal{M} = \{\theta : \widehat{\mathcal{L}}(\theta) = 0\}.$$

Assume an interpolating solution exists. Let p be the parameter dimension and let the Jacobian of the training-prediction map have locally constant rank q near a regular interpolating point. Then

$$\dim \mathcal{M} = p - q \tag{8.4}$$

locally. When $p \gg q$, interpolation can therefore occur on extended sets rather than at isolated points.

The existence and regularity assumptions matter. A large parameter count alone does not guarantee realizability, and the rank at one singular point does not determine the dimension of the entire zero-loss set.

8.5.1 Hessian degeneracy

Let $e_i(\theta) = F_\theta(x_i) - y_i$ and let $J(\theta)$ be the Jacobian of the training predictions. Differentiation gives

$$\nabla^2 \widehat{\mathcal{L}}(\theta) = \frac{1}{n} J(\theta)^\top J(\theta) + \frac{1}{n} \sum_i e_i(\theta) \nabla^2 F_\theta(x_i).$$

At a twice-differentiable zero-residual point, the second term vanishes:

$$\boxed{\nabla^2 \widehat{\mathcal{L}}(\theta^\star) = \frac{1}{n} J(\theta^\star)^\top J(\theta^\star).} \tag{8.5}$$

If $\operatorname{rank} J(\theta^\star) = q$, the Hessian has nullity $p-q$. At a regular point these zero modes are tangent to the local interpolation manifold. At a singular point, Hessian nullity need not have that interpretation. For nonsmooth architectures the calculation is restricted to regions where the required derivatives exist.

For a twice-differentiable penalty and $U_\lambda = \widehat{\mathcal{L}} + \lambda\Omega$, the full Hessian is instead

$$\nabla^2 U_\lambda(\theta) = \frac{1}{n} J(\theta)^\top J(\theta) + \frac{1}{n} \sum_i e_i(\theta) \nabla^2 F_\theta(x_i) + \lambda \nabla^2 \Omega(\theta). \tag{8.6}$$

At a regularized stationary point, neither the residual contribution nor the penalty curvature may be dropped without justification. The interpolation nullity statement above concerns the unregularized data loss at zero residual, not every minimum of the penalized objective. Chapter 7, Section 7.4, and Appendix N keep these objectives separate.

Degeneracy can enlarge the set of acceptable parameter values. It does not, by itself, prove a large basin of attraction or a finite-time convergence guarantee. Competing positive-loss sets may also be degenerate.

Analyticity and the limit of a descent trajectory. For a fixed finite network constructed from affine maps and GELU, squared loss is real-analytic; see Theorem 45 in Appendix N. This permits more than Hessian calculations, but does not give uniform Hessian bounds over unbounded parameters or growing depths. The appendix's GELU examples separate analytic regularity from numerical conditioning.

In particular, let U be real-analytic on an open parameter domain. If its Euclidean gradient flow is defined for all $t \geq 0$ and the trajectory has compact closure contained in that domain, then

$$\int_0^\infty \|\dot{\theta}_t\| \, dt < \infty, \qquad \theta_t \longrightarrow \theta_\infty, \qquad \nabla U(\theta_\infty) = 0.$$

This is Theorem 44 of the appendix. The conclusion is convergence to a critical point, not necessarily a minimum, an interpolator, or a statistically useful predictor. It also supplies no uniform training-time bound across growing network families. A discrete or stochastic optimizer needs its own dynamical assumptions.

A quantitative route to fitting. For the *unregularized* squared loss above, write $e(\theta) = F(\theta) - y$ for the training residual and put $K(\theta) = J(\theta)J(\theta)^\top/n$. Along gradient flow, $\dot{e}(t) = -K(\theta_t)e(t)$. A sufficient quantitative condition is

$$e(t)^\top K(\theta_t) e(t) \geq \mu \|e(t)\|_2^2, \qquad \mu > 0, \tag{8.7}$$

throughout the time interval under consideration. Since $\frac{d}{dt}\widehat{\mathcal{L}} = -e^\top Ke/n$, this gives

$$\boxed{\widehat{\mathcal{L}}(\theta_t) \le \widehat{\mathcal{L}}(\theta_0)e^{-2\mu t}.} \tag{8.8}$$

When this bound holds for all time and the flow and prediction map meet the conditions in Chapter 7, Section 7.9, the parameters converge to a finite interpolator. Unlike the analytic result, this conclusion forces the specified training residual to zero; it does not require analyticity. Neither result by itself establishes generalization.

One NTK-style way to verify the stronger matrix condition is

$$\begin{aligned} K(\theta_0) \succeq 2\mu I_n, \qquad & \sup_{0\le t\le t_{\mathrm{f}}} \|K(\theta_t) - K(\theta_0)\|_{\mathrm{op}} \le \mu \\ \implies \quad K(\theta_t) \succeq \mu I_n \qquad & (0 \le t \le t_{\mathrm{f}}). \end{aligned}$$

The parameterization, initialization, width scaling, data, and time interval are part of this strategy. A nearly constant kernel is sufficient, not necessary: the residual condition also permits feature learning and substantial kernel change. Feasible optimization additionally requires controlled inverse conditioning, discretization, evaluation, and precision costs.

Analyticity, large parameter count, and almost-everywhere prediction rank do not replace the trajectory condition. Appendix N, Proposition 72, gives an analytic least-squares example with a regular interpolation manifold and full prediction rank almost everywhere, yet an open basin of a positive-loss minimum. The example concerns the general geometric inference, not a claim about every prescribed GELU architecture.

8.5.2 The Langevin picture

An idealized stochastic dynamics is

$$d\theta_t = -\nabla\widehat{\mathcal{L}}(\theta_t)\,dt + \sqrt{2T}\,dW_t.$$

When this process has a normalizable equilibrium density under suitable domain and boundary or confinement assumptions,

$$\rho_T(\theta) = Z_T^{-1}\exp\left(-\frac{\widehat{\mathcal{L}}(\theta)}{T}\right).$$

The probability of a region depends on both its loss and its parameter-space measure.

For a compact regular zero-loss manifold, or a suitable local compact portion, with uniformly positive transverse Hessian $H_\perp$, the low-temperature contribution has the schematic form

$$Z_{\mathcal{M}}(T) \sim (2\pi T)^{q/2}\int_{\mathcal{M}} \frac{dV}{\sqrt{\det H_\perp}}.$$

An isolated nondegenerate local minimum with loss $L_\ell > 0$ contributes

$$Z_\ell(T) \sim e^{-L_\ell/T}\frac{(2\pi T)^{p/2}}{\sqrt{\det H_\ell}}.$$

Thus a zero-loss set can have both an energy advantage and a volume advantage in this equilibrium model. Singular sets require the more careful volume analysis discussed in Chapter 7, Section 7.15, and Appendix N, Section N.3. The singular exponent describes a localized weighted-volume asymptotic, including the transformed measure; it is not a training-time rate or, by itself, a population-risk bound.

8.5.3 Equilibrium preference is not a training-time guarantee

These calculations do not supply a polynomial mixing time, a hitting-time bound from the chosen initialization, or a stationary law for arbitrary minibatch SGD. Its noise is generally state dependent and anisotropic. Noncompact parameter symmetries may also prevent normalization of the unregularized Gibbs density.

The measure and the dynamics must be specified together. If a regularizer is included in the Langevin potential, the equilibrium density contains $\widehat{\mathcal{L}}+\lambda\Omega$, rather than the empirical loss alone. Its minimum set and normal curvatures can differ from those of the exact data-interpolation manifold. The local-volume calculation must use the full potential appropriate to that process.

The geometric argument therefore identifies a possible mechanism for favoring low-loss solutions. It does not replace the separate obligation to prove that a specified training procedure reaches a suitable solution with high probability and feasible resources. There is nevertheless an intrinsic source of late-stage stochasticity in regularized SGD that does not require externally injected Brownian forcing.

8.5.4 Exact interpolation and regularized SGD fluctuations

For fixed per-example squared losses, write

$$\ell_i(\theta) = \tfrac{1}{2}(F_\theta(x_i) - y_i)^2, \qquad g_i(\theta) = \nabla\ell_i(\theta), \qquad \bar{g}(\theta) = \frac{1}{n}\sum_i g_i(\theta).$$

Let Ω be a deterministic differentiable penalty, and let $\mathcal{B}$ be a uniformly sampled B-element subset of $\{1,\dots,n\}$, with $n > 1$ and $1 \le B < n$. Ordinary penalized minibatch gradients have

$$\begin{aligned} U_\lambda(\theta) &= \widehat{\mathcal{L}}(\theta) + \lambda\Omega(\theta), \qquad \lambda > 0, \\ G_{\mathcal{B},\lambda}(\theta) &= \widehat{g}_{\mathcal{B}}(\theta) + \lambda\nabla\Omega(\theta), \qquad \widehat{g}_{\mathcal{B}} = \frac{1}{B}\sum_{i\in\mathcal{B}} g_i. \end{aligned} \tag{8.9}$$

At a fixed parameter, put

$$C(\theta) = \frac{1}{n}\sum_i (g_i - \bar{g})(g_i - \bar{g})^\top, \qquad c_{n,B} = \frac{n-B}{B(n-1)}.$$

Because the penalty contribution is the same for every batch,

$$\begin{aligned} \mathbb{E}_{\mathcal{B}}[G_{\mathcal{B},\lambda} \mid \theta] &= \nabla U_\lambda(\theta), \\ \mathrm{Cov}_{\mathcal{B}}(G_{\mathcal{B},\lambda} \mid \theta) &= c_{n,B}C(\theta). \end{aligned} \tag{8.10}$$

The covariance identity and its sampling correction are proved in Chapter 7, Section 7.17, and Appendix N, Section N.5.3. Independent sampling with replacement replaces $c_{n,B}$ by $1/B$. A full batch without replacement has zero sampling covariance irrespective of the individual gradients. These formulas assume averages of fixed per-example gradients; a batch-dependent forward operation requires its own update analysis.

At exact squared-loss interpolation, the original calculation remains:

$$e_i(\theta_*) = 0\ (\forall i) \quad\Longrightarrow\quad g_i(\theta_*) = e_i(\theta_*)\nabla_\theta F_{\theta_*}(x_i) = 0 \quad\Longrightarrow\quad \mathrm{Cov}(\widehat{g}_B \mid \theta_*) = 0.$$

Here $\widehat{g}_B$ denotes the average of sampled per-example gradients. Every regularized batch gradient at that point equals $\lambda\nabla\Omega(\theta_*)$. Its covariance is zero, but its mean need not vanish. Unless the penalty gradient vanishes too, the exact interpolator is not stationary for U_λ.

At a stationary point θ_λ of the *regularized* objective, by contrast,

$$\bar{g}(\theta_\lambda) + \lambda \nabla \Omega(\theta_\lambda) = 0, \qquad G_{\mathcal{B},\lambda}(\theta_\lambda) = \frac{1}{B} \sum_{i \in \mathcal{B}} (g_i - \bar{g})(\theta_\lambda). \tag{8.11}$$

Only the average data gradient must balance the regularizer. Taking the trace of the covariance at this zero-mean point gives

$$\boxed{\mathbb{E}_{\mathcal{B}} \|G_{\mathcal{B},\lambda}(\theta_\lambda)\|^2 = c_{n,B} \frac{1}{n} \sum_i \|g_i(\theta_\lambda) - \bar{g}(\theta_\lambda)\|^2.} \tag{8.12}$$

This is Proposition 75 of the appendix and Proposition 5 of Chapter 7. For $B < n$, the right side is zero exactly when all the individual data gradients coincide. At regularized stationarity, this is also equivalent to every regularized batch gradient being zero. Otherwise intrinsic sampling fluctuations persist even at a local or global minimum of U_λ.

The distinction is

$$\boxed{\begin{array}{l}\text{zero gradient of the full regularized objective} \\ \neq \text{ a common zero of every minibatch update.}\end{array}}$$

The regularizer is not itself random. By shifting the stationary balance away from exact interpolation, it can maintain different nonzero sample gradients. Positive regularization alone does not force this outcome: identical sample gradients can occur in special configurations. The criterion is gradient heterogeneity, not merely $\lambda > 0$.

The phenomenon does not require nonconvexity. For the scalar predictor $F_\theta(x) = \theta x$ on $(1, 1)$ and $(2, 2)$ with penalty $\lambda\theta^2/2$, the unique regularized global minimum is $\theta_\lambda = 5/(5 + 2\lambda)$. Its singleton-batch gradients are $\pm 3\lambda/(5+2\lambda)$, so the mean is zero and the variance is $9\lambda^2/(5+2\lambda)^2 > 0$. The full calculation is Example 4 in the appendix.

This positive source of late-stage noise is different from prescribing an isotropic Langevin process. With a plain step of size α, the update covariance is $\alpha^2 c_{n,B} C$; a justified small-step diffusion on time increments α instead has covariance rate $\alpha c_{n,B} C$. Neither persistent covariance nor vanishing mean establishes a Gibbs law, excitation of every escape direction, or a useful mixing-time bound.

8.5.5 Normalized networks, common equilibria, and low-rank bias

Xu, Rangamani, Liao, Galanti, and Poggio [170] give an architecture-specific obstruction to a common minibatch equilibrium. Their normalized scalar-output ReLU setting uses a bias-free network with independently parameterized, unshared layer matrices,

$$F_{\rho,V}(x) = \rho f_V(x), \qquad \rho > 0, \qquad \|V_k\|_F = 1,$$

binary labels $y_i \in \{-1, +1\}$, and scale penalty $\lambda\rho^2/2$. Assume $1 \le B < n$, nonzero signed normalized margins $m_i = y_i f_V(x_i)$, and differentiability with the stated fixed-gate derivative identities on the training sample. Under these hypotheses, vanishing of the scale and tangent layer-update directions for *every* B-element minibatch implies

$$\boxed{\operatorname{rank}(V_k) \le 2 \qquad \text{for every layer.}}$$

This is the scoped form of their Lemma 8 proved in Chapter 7, Lemma 3, and Appendix N, Proposition 76.

At a stationary point of the *batch-averaged normalized dynamics*, the mean of the concatenated scale and tangent update directions is zero. If the other hypotheses hold and some layer has rank greater than two, the contrapositive excludes a common zero. Since every admissible batch has positive probability, the actual normalized directions then have strictly positive covariance trace. This concerns the specified normalized updates, not a substitution of unconstrained gradients.

This motivates the paper's low-rank-bias interpretation; it does not prove that SGD always reaches rank-two matrices or that low rank ensures zero noise. No uniform noise lower bound follows. The scalar-output, homogeneity, normalization, and regularity assumptions do not transfer unchanged to GELU or other architectures. The general covariance criterion and this specific structural mechanism remain distinct.

8.6 Regularization selects among well-fitting solutions

When overparameterization permits many well-fitting solutions, regularization supplies a preference among them. Suppose training uses

$$\widehat{R}_S(F_\theta) + \lambda\Omega(\theta), \qquad \lambda > 0, \tag{8.13}$$

with $\Omega \geq 0$.

In an idealized exact-fit formulation, selection is expressed by

$$\min_{\theta \in \mathcal{M}} \Omega(\theta).$$

A small-penalty limit can motivate this constrained picture under appropriate existence and compactness conditions. At a fixed positive λ, however, the penalized minimizer need not interpolate exactly: it may trade fit against a smaller penalty. Nor does writing the objective prove that SGD reaches its global minimum.

The stochastic implication is equally important. Even at a minimum of this full objective, the regularizer may be canceled only by the average data gradient, while individual minibatches continue to differ. Thus regularization can affect both the selected function and the stochastic dynamics in the well-fitting region. Its statistical role and its effect on minibatch balance are complementary, not interchangeable. A useful returned predictor need not be a common equilibrium of those updates.

8.6.1 An elementary conditional comparison

Suppose $\overline{\theta}$ is a comparison parameter and the returned parameter satisfies, for some $\xi \geq 0$,

$$\widehat{R}_S(F_{\widehat{\theta}}) + \lambda\Omega(\widehat{\theta}) \leq \widehat{R}_S(F_{\overline{\theta}}) + \lambda\Omega(\overline{\theta}) + \xi.$$

By nonnegativity of the two terms,

$$\widehat{R}_S(F_{\widehat{\theta}}) \leq \widehat{R}_S(F_{\overline{\theta}}) + \lambda\Omega(\overline{\theta}) + \xi, \tag{8.14}$$

$$\Omega(\widehat{\theta}) \leq \Omega(\overline{\theta}) + \frac{\widehat{R}_S(F_{\overline{\theta}}) + \xi}{\lambda}. \tag{8.15}$$

These inequalities follow directly by dropping one nonnegative term from the left-hand side. For an interpolating comparator, they become

$$\widehat{R}_S(F_{\widehat{\theta}}) \le \lambda\Omega(\overline{\theta}) + \xi, \qquad \Omega(\widehat{\theta}) \le \Omega(\overline{\theta}) + \xi/\lambda.$$

This makes the selection argument precise in one useful regime: a low-penalty comparator and successful optimization can yield both a good fit and penalty control. The existence of the comparator and the optimization comparison are additional requirements, not consequences of adding a regularization term.

The fitting theorem must match the trained objective. In the squared-loss setting, a differentiable penalty changes the residual dynamics. For $U_\lambda = \widehat{\mathcal{L}} + \lambda\Omega$ and $\dot{\theta} = -\nabla U_\lambda$, the chain rule gives

$$\dot{e} = -Ke - \lambda J\nabla\Omega. \tag{8.16}$$

Thus the unregularized estimate (8.8) cannot simply be transferred to penalized training: the penalty contribution must also be controlled. The comparator inequalities above offer a different conditional route to joint fit and penalty control, but stationarity of U_λ alone does not establish their optimization premise.

Confinement is a separate role. For an empirical loss analytic on all of $\mathbb{R}^p$, full quadratic weight decay gives a useful positive result. With

$$U_\gamma(\theta) = \widehat{\mathcal{L}}(\theta) + \frac{\gamma}{2}\|\theta\|^2, \qquad \gamma > 0,$$

and all parameters penalized, gradient flow satisfies

$$\|\theta_t\|^2 \le \frac{2U_\gamma(\theta_0)}{\gamma}.$$

The compact sublevel set gives all-time existence and, by the analytic result, finite-length convergence to a critical point of U_γ. Corollary 9 in Appendix N gives the proof. This supplies confinement and stationarity, not global minimization or exact interpolation. It is one sufficient confinement mechanism, not a requirement for every successful learner.

Gradient-flow limits and SGD outputs. The weight-decay corollary concerns deterministic flow, not a common zero of all minibatch updates. For plain SGD with independently sampled batches of fixed positive probabilities, continuous batch gradient fields near a finite limit, and learning rates bounded away from zero, any finite point limit must be a common minibatch equilibrium (Appendix N, Proposition 77). Positive stationary covariance excludes that endpoint under these conditions, not all useful training behavior.

Diminishing steps can shrink increments despite nonzero gradient-estimator covariance. A specified stopping or averaging rule can return a useful predictor without common batch stationarity, but must meet the claimed fit, hypothesis-class, and evaluation-resource requirements.

8.7 Product norms and Rademacher complexity

The product-norm idea has a clear role in this framework. In a specified feedforward setting, a natural candidate complexity quantity is

$$\mathcal{C}_\times(\theta) = \prod_{\ell=1}^{L} \|W_\ell\|_F,$$

and a corresponding multiplicative penalty is

$$\Omega_\times(\theta) = \mathcal{C}_\times(\theta)^2.$$

Biases and other trainable operations must be included in the chosen complexity analysis; a formula written only for weight matrices is not a bound for unrestricted additional parameters.

A multiplicative penalty favors a smaller product among solutions that achieve a comparable fit. The statistical motivation is that suitable network classes admit norm-sensitive Rademacher bounds. The precise quantity can involve products of Frobenius norms, products of spectral norms with further corrections, path norms, or other architecture-dependent expressions.

Statistical control, confinement, and conditioning. Three different controls occur in the argument:

class-complexity control:	$\mathcal{C}(\widehat{\theta}) \le Q,$
trajectory confinement:	$\sup_t \|\theta_t\| < \infty,$
fitting-direction conditioning:	$e^\top K e \ge \mu \|e\|^2.$

An upper norm bound used in a statistical theorem does not automatically supply either of the other two. An upper matrix-norm bound does not exclude rank deficiency. Likewise a product penalty need not be coercive. For example,

$$\widehat{\mathcal{L}}(a,b) = \tfrac{1}{2}(ab-1)^2, \qquad \Omega_\times(a,b) = a^2b^2$$

has $\widehat{\mathcal{L}}(t,t^{-1}) + \lambda\Omega_\times(t,t^{-1}) = \lambda$ for all $t > 0$. Its sublevel set is unbounded. This does not prove divergence of the training dynamics; it shows why a separate confinement argument is needed when an analytic convergence theorem uses compactness. Chapter 7, Sections 7.18–7.19, and the appendix discuss these distinct roles.

Match the norm theorem to the architecture. An activation hypothesis cannot be exchanged for smoothness. In particular, GELU is not positively homogeneous, so a statistical result requiring that property cannot be transferred unchanged from a homogeneous architecture. Residual additions, attention, normalization, biases, and learnable scales must be covered by the chosen complexity analysis. None of this removes the positive role of product-norm regularization: with the appropriate class bound and successful fitting, it remains a route to predictive control without identifying the teacher's internal computation.

A norm-complexity budget is not the same quantity as minibatch fluctuation energy or matrix rank. The regularized-noise result does not replace a statistical class bound, and the common-equilibrium rank restriction does not by itself prove generalization. Conversely, persistent updates do not invalidate a statistical bound for a returned predictor that actually meets its fit and class conditions.

8.7.1 A class bound, not a property of one isolated parameter vector

Fix an architecture class and a complexity functional $\mathcal{C}(\theta)$. For a budget Q fixed independently of the training sample, define

$$\mathcal{H}_Q = \{F_\theta : \mathcal{C}(\theta) \le Q\}. \tag{8.17}$$

For scalar predictors, the empirical Rademacher complexity is

$$\widehat{\mathfrak{R}}_{S_X}(\mathcal{H}_Q) = \mathbb{E}_\epsilon\left[\sup_{h\in\mathcal{H}_Q} \frac{1}{n}\sum_{i=1}^n \epsilon_i h(X_i)\right],$$

where the ϵ_i are independent random signs.

The relevant assertion is a bound on this entire class. Schematically, applicable norm-based results may take the form

$$\widehat{\mathfrak{R}}_{S_X}(\mathcal{H}_Q) \le \mathcal{B}_n(Q; \mathcal{A}, B),$$

where $\mathcal{A}$ records the architecture and activation assumptions and B controls the inputs. A schematic dependence is

$$\mathcal{B}_n(Q; \mathcal{A}, B) \lesssim A_{\mathcal{A}} \frac{BQ}{\sqrt{n}},$$

when $\mathcal{C}$ and $A_{\mathcal{A}}$ account for the factors required by the particular theorem. Depth factors, bias bounds, normalization, and other correction terms cannot be dropped. This display is not a universal formula for every neural architecture.

Consequently, a smaller suitable norm product can reduce a generalization upper bound. It does not literally minimize the exact Rademacher complexity, and it does not establish a universally monotone relation between that product and the actual test error.

8.7.2 From a well-fitting predictor to population control

Assume the loss is bounded, $0 \le \ell \le M_\ell$, and K_ℓ-Lipschitz in the prediction on the admitted prediction and label ranges. Under the usual measurability conditions, the relevant uniform-convergence statement has the form

$$R_P(h) \le \widehat{R}_S(h) + \beta_S(Q, \delta) \quad \text{for every } h \in \mathcal{H}_Q, \tag{8.18}$$

with probability at least $1 - \delta$. A Rademacher-based choice is of the form

$$\beta_S(Q, \delta) = c_1 K_\ell \widehat{\mathfrak{R}}_{S_X}(\mathcal{H}_Q) + c_2 M_\ell \sqrt{\frac{\log(2/\delta)}{n}},$$

with appropriate universal constants. The preceding class-complexity bound can then be used to upper-bound β_S.

Squared loss satisfies these boundedness and Lipschitz conditions when predictions and labels are restricted to a fixed bounded range. It does not satisfy them on an unrestricted real domain. Clipping predictions or using a different valid statistical bound must be made explicit when needed.

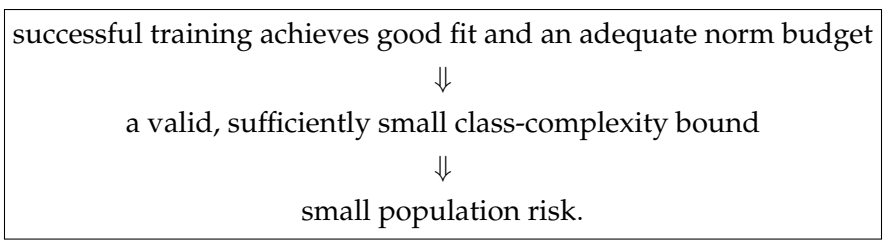
If training returns $F_{\widehat{\theta}} \in \mathcal{H}_Q$ with $\widehat{R}_S(F_{\widehat{\theta}}) \le \eta$, then on the uniform event,

$$\boxed{R_P(F_{\widehat{\theta}}) \le \eta + \beta_S(Q, \delta).}$$

Interpolation is the special case $\eta = 0$. The argument does not require exact interpolation.

8.7.3 What the regularizer must actually accomplish

The defensible chain is



A regularizer is one mechanism for achieving the first line. The mere presence of that term in the objective does not prove the first line. Moreover, an upper bound must be informative at the available n; polynomial computational size alone does not ensure that it is.

If the norm budget is selected after seeing the sample, a fixed-Q bound cannot simply be applied without adjustment. One elementary option is a prespecified sequence of budgets Q_j with confidence allocations δ_j satisfying $\sum_j \delta_j \le \delta$. A union bound makes (8.18) valid simultaneously over those classes, and selection among them then uses the corresponding confidence penalty. Sample-dependent architectures and other model-selection steps require the same care.

8.8 The main predictive recovery argument

Two logically separate results combine the preceding pieces. The first turns successful empirical fitting and suitable complexity control into population control. The second identifies the computational structure of an already successful predictor.

Theorem 7 (Conditional predictive recovery)**.** *Let $S \sim P^n$ and fix a hypothesis class $\mathcal{H}_Q$ independently of S. Suppose that, with probability at least $1 - \delta_{\mathrm{stat}}$ over S,*

$$R_P(h) \le \widehat{R}_S(h) + \beta_S(Q, \delta_{\mathrm{stat}}) \qquad \text{for all } h \in \mathcal{H}_Q.$$

Let a possibly randomized training procedure return $\widehat{f} = A(S, U)$. Assume that, with probability at least $1 - \delta_{\mathrm{opt}}$ over the sample and algorithmic randomness,

$$\widehat{f} \in \mathcal{H}_Q, \qquad \widehat{R}_S(\widehat{f}) \le \eta.$$

Then, with probability at least $1 - \delta_{\mathrm{stat}} - \delta_{\mathrm{opt}}$,

$$\boxed{R_P(\widehat{f}) \le \eta + \beta_S(Q, \delta_{\mathrm{stat}}).} \tag{8.19}$$

If the algorithmic success event also includes a stated training-cost or evaluation-cost bound, that bound holds on the same intersection event.

Proof. Let E_{stat} denote the uniform statistical event and let E_{opt} denote the algorithmic success event. On their intersection, the uniform bound applies to the returned predictor, since $\widehat{f} \in \mathcal{H}_Q$. Therefore

$$R_P(\widehat{f}) \le \widehat{R}_S(\widehat{f}) + \beta_S(Q, \delta_{\mathrm{stat}}) \le \eta + \beta_S(Q, \delta_{\mathrm{stat}}).$$

The union bound gives

$$\mathbb{P}(E_{\mathrm{stat}} \cap E_{\mathrm{opt}}) \ge 1 - \delta_{\mathrm{stat}} - \delta_{\mathrm{opt}}.$$

The two events need not be independent. Any resource bound included in E_{opt} also holds on their intersection. □

The theorem is deliberately conditional: it does not derive the algorithmic success probability. Establishing that premise for the specified training dynamics is a central research problem.

In particular, convergence to an analytic critical point or a favorable Langevin equilibrium weight does not establish the event $\widehat{f} \in \mathcal{H}_Q$, $\widehat{R}_S(\widehat{f}) \le \eta$. A finite-time fitting theorem can help with the second inequality, but the returned complexity budget and the effect of the actual regularizer must be controlled as well. For noiseless squared loss, the conditional theorem yields

$$\|\widehat{f} - f\|_{L^2(P_X)} \le \sqrt{2(\eta + \beta_S(Q, \delta_{\mathrm{stat}}))}.$$

Thus $\eta + \beta_S \le \varepsilon^2/2$ is sufficient for L^2 error at most ε.

The returned predictor need not be an absorbing SGD state. For a specified finite stopping index j_*, the procedure may return

$$\widehat{f} = F_{\theta_{j_*}}.$$

The uniform statistical event covers every member of the fixed class, including this data- and trajectory-selected member. The stopping rule must still attain the joint fit and complexity condition within its stated probability and resource budget. Neither $G_{\mathcal{B},\lambda}(\theta_{j_*}) = 0$ for every batch nor zero covariance is an additional premise. Continued fluctuations do not themselves invalidate the returned predictor's guarantee.

Proposition 6 (A feasible generalizing predictor is a compositional witness)**.** *Consider real-valued targets f_N with input distributions $P_{X,N}$. For each N and $0 < \varepsilon \le 1$, suppose a returned network $F_{N,\varepsilon}$ satisfies*

$$\|F_{N,\varepsilon} - f_N\|_{L^2(P_{X,N})} \le \varepsilon.$$

Assume its finite description length is polynomial in N and ε^{-1}, and that one specified evaluator, given this description and a precision request m, approximates its output uniformly to error 2^{-m} using polynomially many bit operations and input bits in N, ε^{-1}, and m.

Then for every m there is a polynomial-size bounded-fan-in Boolean circuit, acting on the prescribed input encoding, whose decoded output $\widetilde{F}_{N,\varepsilon,m}$ satisfies

$$\|\widetilde{F}_{N,\varepsilon,m} - f_N\|_{L^2(P_{X,N})} \le \varepsilon + 2^{-m}. \tag{8.20}$$

Hence the returned predictors supply feasible compositionally sparse approximations at the stated learning and evaluation resolutions.

Proof. Fix a returned network description and a requested precision m. Its evaluator is a finite-time digital computation. The bounded-fan-in simulation from Chapter 4 gives a circuit whose size is polynomial in the stated evaluation resources. The parameters in the returned description are fixed constants in that circuit.

By the numerical guarantee,

$$\|\widetilde{F}_{N,\varepsilon,m} - F_{N,\varepsilon}\|_\infty \le 2^{-m}.$$

Since $P_{X,N}$ is a probability measure, this also bounds the L^2 evaluation error. The triangle inequality gives (8.20). □

Regularization and interpolation do not enter this second proof. Their role is to help establish the small population error used in its hypothesis. In particular, taking a learning error $\varepsilon/2$ and an evaluation error at most $\varepsilon/2$ gives a compositional predictor with total L^2 error at most ε.

The circuit compilation is uniform given the returned finite model description. This does not imply that an arbitrary sequence of sample-selected model descriptions is generated uniformly from N alone. A uniform learning claim must also specify the algorithm that produces those descriptions from its finite-precision data.

The resulting logical division is

prediction theorem:	fit and complexity control give small risk,
witness proposition:	feasible prediction gives a sparse computation.

8.9 Why uniqueness is not required for predictive learning

The witness proposition does not require recovery of the generating variables. Suppose

$$f = g_m \circ \cdots \circ g_1, \qquad F_{\theta^\star} = h_L \circ \cdots \circ h_1.$$

Even when no useful nodewise correspondence exists between the h_ℓ and the g_j, the predictor can be satisfactory if it is feasible and approximates f on the intended population.

This is stronger than observing that a generating decomposition is nonunique. Predictive success does not require selecting a member of a specified equivalence class of generating decompositions at all.

predictive recovery $\neq$ identification of a generating DAG.

Uniqueness and identifiability still matter for a structural objective. Scientific interpretation, causal questions, or a specified module-reuse claim may require more than population prediction. The point is not that those questions disappear, but that they should not be inserted as additional premises of the predictive witness argument.

8.10 The revised role of genericity

Genericity remains useful as a possible explanation of accessible learning signal. Let $X \sim P_X$ and let ϕ be a low-complexity probe available to the learner. Define

$$\rho_\phi = \mathbb{E}[f(X)\phi(X)].$$

Probe scale and variance must be controlled; otherwise a correlation can be enlarged simply by rescaling the probe. A quantitatively detectable footprint, for example

$$|\rho_\phi| \geq \frac{1}{\mathrm{poly}(N)}$$

under fixed normalization and suitable concentration assumptions, can provide a useful screening signal.

Whether that signal is accessible to an optimizer also depends on its parameterization and current state. For squared population loss,

$$\nabla_\theta \mathcal{L}(\theta) = \mathbb{E}\big[(F_\theta(X) - f(X))\nabla_\theta F_\theta(X)\big],$$

when differentiation and expectation can be interchanged. The relevant probes are therefore those induced by the chosen gradient directions, not arbitrary functions with nonzero correlation.

Genericity concerns the target, observation distribution, and accessible probe family together. It can help explain why optimization discovers useful coordinates in finite time. But once predictive success and feasible evaluation have been established by some valid argument, no additional genericity hypothesis is needed to infer a compositional witness.

This does not make learning-signal questions optional inside a proof that actually relies on them. It makes genericity one candidate mechanism, rather than a universal prerequisite for every successful learner.

8.11 Parity revisited

For $S \subseteq [d]$, define

$$\chi_S(x) = \prod_{i \in S} x_i, \qquad x \in \{-1, +1\}^d.$$

For $|S| \geq 2$, a balanced binary multiplication tree has size $O(|S|)$ and depth $O(\log |S|)$. Parity therefore has a simple compositionally sparse realization.

Yet under the uniform product distribution,

$$\mathbb{E}[\chi_S(X) X_j] = 0 \qquad (j \in S,\ |S| > 1).$$

Indeed, multiplication by X_j removes that factor, leaving a product of independent mean-zero coordinates. Thus

compositional sparsity does not guarantee a particular low-order learning signal.

This rules out the stated linear screening signal. It does not prove that all deep-learning procedures fail to learn parity. If another procedure finds a feasible parity predictor that generalizes, its computation already gives the required compositional witness.

8.12 Generic perturbations and local signals

Perturbations can reveal signals hidden by exact cancellation. The calculations below show both the possibility and its quantitative limit.

8.12.1 Boolean leakage

Let independent $X_i \in \{-1, +1\}$ satisfy $\mathbb{E}[X_i] = \mu_i$, and let $f(X) = \prod_{i \in S} X_i$. For $j \in S$,

$$\boxed{\mathbb{E}[f(X) X_j] = \prod_{i \in S \setminus \{j\}} \mu_i.} \tag{8.21}$$

This follows from $X_j^2 = 1$ and independence. If all the relevant biases in this product are nonzero, exact cancellation is broken.

However, if $|\mu_i| \leq \mu < 1$ and $|S| = k$, then

$$|\mathbb{E}[f(X) X_j]| \leq \mu^{k-1}.$$

The signal may be exponentially small in interaction order. Even at bounded interaction order, a quantitative lower bound on the individual biases is needed for a useful uniform detectability claim. Nonzero and efficiently detectable are different properties.

8.12.2 Gaussian leakage

Let $X \sim \mathcal{N}(\mu, \sigma^2 I)$ with $\sigma > 0$. Under the regularity and integrability assumptions needed for Gaussian integration by parts, Stein's identity gives

$$\boxed{\mathbb{E}[f(X)(X_i - \mu_i)] = \sigma^2 \mathbb{E}[\partial_i f(X)].} \tag{8.22}$$

A nonzero average derivative therefore yields a nonzero linear footprint. A detectable magnitude and access through the learning rule remain separate requirements.

These examples explain possible sources of optimization signal. They do not turn the existence of a footprint into a universal convergence result, and they are not necessary premises of the computational witness proposition.

8.13 Staircase learning as a constructive mechanism

Suppose learning first constructs a useful coordinate

$$z_1 = g_1(x).$$

A later dependency $z_2 = g_2(z_1, x)$ may be high-order in the original coordinates but low-order in the enlarged representation (x, z_1). Feature learning can therefore change the coordinate system in which the remaining task is expressed.

The proposed mechanism is

detect an accessible signal
$\Downarrow$
learn and preserve a useful coordinate
$\Downarrow$
expose a simpler next learning problem.

A staircase theorem must establish this progression for a specified architecture, initialization, optimizer, and data model. It must also control finite samples, the errors in previously learned coordinates, and the time needed for later stages. Merely exhibiting an accurate deep representation does not prove those claims.

The revised framework does not discard this research objective. A successful staircase theorem can supply a constructive route to the algorithmic premise of Theorem 7. It may establish the stronger structural criterion (8.3) as well.

Its role is nevertheless distinct from Proposition 6. The proposition identifies what follows once a feasible predictor succeeds. A staircase theorem seeks to explain how a particular learner achieves that success. The latter can be important without being a necessary trajectory for all successful learners.

8.14 Skip connections and traces remain useful

A residual block

$$z_{\ell+1} = z_\ell + g_\ell(z_\ell)$$

provides an identity pathway alongside a learned transformation. This can assist information and gradient propagation. It does not, for an arbitrary g_ℓ, guarantee invertibility or exact preservation of every previous coordinate. Such preservation must be established for the particular architecture or enforced by an explicit storage mechanism.

The quantitative issue is control of products of the residual Jacobians $I + Dg_\ell$, not residual addition alone. Proposition 69 of Appendix N gives sufficient scaling conditions. Those activation-propagation Jacobians are distinct from the parameter-to-training-prediction Jacobian J in Section 8.5; controlling one does not by itself prove the required lower bound for the other.

Exposing intermediate variables changes the learning problem more directly. If

$$z_v = g_v(z_{\mathrm{pa}(v)})$$

and the training data contain the parent tuples and corresponding intermediate outputs, then training can be organized into local supervised problems. Their learnability, coverage, and composed error still require analysis.

Trace information can help even when end-to-end learning is possible. It is not only a fallback for a failed genericity condition. Faithful traces, partial traces, and useful surrogates provide different levels of information, as distinguished earlier in the book.

In the terminology C1–C3, C1 concerns feasible representation; C2 concerns accessible learning signal; C3 supplies intermediate information. These roles can cooperate. Neither skip connections nor traces are additional requirements for the conclusion that an already successful feasible predictor is a compositional witness.

An architectural or initialization condition can be indispensable to a particular proof of training success without being an additional premise of the witness proposition. Analytic activations and a nearly constant NTK have the same status: they support particular analyses, not a universal list of requirements for an already accurate feasible predictor.

8.15 Neural collapse is evidence, not a premise

The neural-collapse phenomena discussed earlier illustrate how trained classification networks can develop highly structured representation geometry. Such observations are relevant to the broader question of how training selects among many possible solutions.

They are not premises of the witness argument. That conclusion follows from computational feasibility and predictive accuracy, without requiring collapse in the final layer or in earlier layers. The common-minibatch rank restriction in Section 8.5.5 is a different claim about parameter matrices under specified update conditions; it does not by itself establish neural collapse.

Conversely, observing a special geometry on the training examples does not by itself prove a small population risk, a useful norm-complexity bound, or recovery of the generating latent DAG. Neural collapse can be a diagnostic of representation organization while remaining distinct from these stronger conclusions.

8.16 Three questions for predictive learning

The predictive framework can now be organized around three questions.

8.16.1 Q1: Does the chosen class contain a feasible accurate predictor?

Efficient target computation supplies a feasible circuit realization. For a particular learning architecture, one must still establish that an accurate predictor is available within its admitted resources. If a norm-based statistical argument is intended, one also needs an adequate complexity budget for a useful comparison predictor.

Thus existence of some small circuit is not automatically existence of a low-norm predictor in every chosen architecture.

8.16.2 Q2: Can the specified training procedure find a suitable fit?

The required statement concerns an optimizer, initialization, schedule, and stopping rule, with a finite-time success probability. The procedure must reach a sufficiently good fit while meeting the complexity control needed for Q3. Exact interpolation is one important regime, not the only possible objective. In particular, the return rule can select an accurate finite iterate from a trajectory with continuing minibatch fluctuations. Q2 does not require convergence to one parameter at which all batch updates vanish. The regularization strength, batch rule, learning-rate schedule, and return criterion are part of the finite-time statement.

Width, residual scaling, initialization, and regularity can help establish this step for a specified parameterization. An analytic critical-point limit is not enough; nor is a lower bound on prediction rank only at initialization or almost everywhere. The needed conclusion is finite-time access to an adequate fit and complexity budget under the actual updates. Favorable stochastic geometry can contribute, but its equilibrium weights alone do not establish that access.

8.16.3 Q3: Is the selected predictor controlled well enough to generalize?

A valid statistical bound must apply to the selected predictor and be small enough at the available sample size. Product-norm regularization provides a route through norm-sensitive classes when its optimization and statistical requirements are met.

8.16.4 What is not a fourth predictive requirement

After these questions have been answered with the stated feasibility conditions, there is no further need to match the generating hidden variables. The learned computation itself gives a compositional witness.

Structural identification remains a separate question when the scientific or downstream objective calls for it.

8.17 What is theorem and what remains conjectural

The logical status of the pieces should remain explicit.

8.17.1 Established under stated assumptions

The computational simulation establishes bounded-fan-in realizations of polynomial-time digital evaluations. With the specified precision convention, it applies to efficient finite-precision real evaluation and to the evaluator of a returned finite network.

For smooth squared loss, the Hessian identity (8.5) holds at zero residual. Under local constant rank at an interpolating point, the rank theorem gives (8.4).

For an all-time analytic gradient flow with compact trajectory closure inside the analytic domain, Appendix N establishes finite-length convergence to a critical point. Separately, the trajectory-wise residual condition (8.7) gives the fitting estimate (8.8) for unregularized squared-loss flow. Full quadratic weight decay supplies confinement in the stated analytic setting, not global fitting. These are different conditional results; none is an unconditional trainability theorem for wide GELU networks or arbitrary SGD.

The minibatch covariance identities in Section 8.5.4 are also exact under their fixed-example and sampling assumptions. At regularized stationarity, heterogeneous sample gradients give positive covariance; a zero full gradient is not enough to remove it. The normalized ReLU rank restriction is

an additional conditional result about common minibatch equilibria, not a general convergence or generalization theorem.

Within an appropriate fixed hypothesis class, a valid uniform statistical bound and a successful fitting event combine as in Theorem 7. Separately, Proposition 6 gives the computational structure of an already accurate feasible predictor.

None of these statements alone proves the full learning claim.

8.17.2 Idealized but explicit

The Langevin and local-volume calculations describe a specified equilibrium model under its normalizability and regularity conditions. They identify a possible preference for extended low-loss sets. Their extension to finite-time minibatch SGD is an additional dynamical problem. Intrinsic fluctuations under regularization provide a real sampling mechanism, but do not by themselves justify replacing its anisotropic, state-dependent covariance by the isotropic Langevin model.

Likewise, constrained minimum-penalty interpolation is a useful selection model. It should not be confused with a theorem that arbitrary penalized SGD solves that constrained problem.

8.17.3 Mechanisms rather than universal prerequisites

Genericity, staircase feature learning, residual pathways, trace exposure, associative retrieval, and structured representation geometry may help explain success in particular regimes. The interaction of normalization, regularization, residual-dependent SGD noise, and low-rank bias is another mechanism to study. Each requires its own assumptions and tests. The exact common-equilibrium obstruction should not be confused with a complete dynamical theorem establishing that mechanism.

A mechanism can prove the algorithmic premise of the predictive theorem. It is not a separate premise of the witness proposition once that predictive premise has been established in another way.

8.17.4 The remaining broad conjecture

The intended research claim can be stated without requiring recovery of a privileged hidden DAG.

> **Regularized Predictive Recovery Conjecture.** For a broad, explicitly characterized class of target–distribution families admitting feasible accurate predictors with suitable norm-complexity control, a specified family of regularized SGD procedures in sufficiently expressive overparameterized deep networks finds, with high probability and feasible resources, a well-fitting predictor whose statistical bound is informative enough to establish population accuracy.

This is the predictive version of the regularized global-recovery conjecture. To turn the program into a testable mathematical statement, one must specify the admissible problem class, architecture, loss, regularizer, initialization, and training schedule independently of the unknown target instance. The procedure also specifies how it returns a finite predictor—for example, by a stopping rule or an admissible averaging rule. The conjecture requires success of that returned predictor, not a common zero of every possible minibatch update.

One formal target is the following. Let N index the problem family, let κ collect stated regularity and conditioning parameters, and let ϵ be a population-risk tolerance. Seek deterministic choices of

a class budget Q, a sample size n, an empirical tolerance η, and a uniform, specified regularized SGD procedure such that

$$\begin{gathered} n, \quad \text{training bit cost}, \quad \text{model description length} \le \operatorname{poly}\bigl(N, \kappa, \epsilon^{-1}, \log(1/\delta)\bigr), \\ \mathbb{P}\left\{\widehat{R}_S(\widehat{f}) \le \eta, \quad \mathcal{C}(\widehat{\theta}) \le Q\right\} \ge 1 - \delta/2, \end{gathered} \tag{8.23}$$

with a valid uniform statistical bound controlled by a deterministic quantity $b_n(Q, \delta/2)$ satisfying

$$\eta + b_n(Q, \delta/2) \le \epsilon. \tag{8.24}$$

The evaluator of the returned model must also have the stated polynomial cost in its description length and requested evaluation precision.

The conditional theorem then yields $R_P(\widehat{f}) \le \epsilon$ with probability at least $1 - \delta$. For noiseless squared loss, an L^2 accuracy request ε uses $\epsilon = \varepsilon^2/2$.

The difficult assertion is the existence of the algorithmic guarantee (8.23) together with an informative budget (8.24) on a meaningful problem class. It is not the final union-bound argument or the compilation of the returned predictor.

Efficient target evaluation alone does not establish this conjecture. The admissible class must not be defined circularly as the class on which the chosen optimizer succeeds. Nor does the conjecture, as a broad research program, assert efficient learning of every polynomial-time computable target under every distribution.

The conditioning parameters κ must record the quantitative premises of the proposed proof, not merely assert smoothness or width. A residual-contraction route must control inverse conditioning and the admissible discrete step sizes. An analytic route must justify confinement and any claimed entry-time and rate bounds. A stochastic route must specify its actual noise and finite-time access, including the effect of regularization and the learning-rate schedule. A qualitative assertion of nonzero covariance is not a polynomially controlled noise or escape bound. Each route must meet the same joint fitting and statistical-budget requirement; no unproved mechanism is being inserted as an additional universal condition.

8.18 Failure modes

The distinctions above separate several possible failures.

Representation failure. The chosen architecture may not contain a sufficiently accurate predictor at the allowed computational and statistical complexity. Efficient target evaluation in another computational language does not settle this architecture-specific question.

Optimization failure. A suitable predictor may exist, but the specified training procedure may not reach a sufficient fit with high probability in feasible time.

Even a bounded analytic gradient flow can converge to a positive-loss critical point. Conversely, a fitting guarantee for unregularized flow need not survive an uncontrolled penalty term or a change to stochastic updates. On the other hand, continued minibatch fluctuations or failure to converge to one fixed parameter do not by themselves constitute optimization failure under a return-based objective: a specified stopped predictor may already satisfy the required fit and complexity bounds.

Complexity-selection failure. Training may fit the data but return a model outside the useful norm-complexity budget. Alternatively, an available bound may be too loose to certify a predictor that actually generalizes. Failure of a sufficient bound is not itself proof of poor test performance.

Generalization failure. A selected predictor can have small empirical error and large population error. A valid bound can prevent this conclusion only when its assumptions hold and its numerical value is informative.

Scaling or numerical failure. Depth, width, intermediate precision, training cost, or evaluation cost may become prohibitive across a family. Finite does not mean efficient, and efficient evaluation does not mean efficient training.

Structural failure without predictive failure. The predictor may fail to reconstruct the generating variables while predicting accurately. This is a failure of a structural objective, not of the predictive witness implication. Likewise, bounded-fan-in digital implementation does not automatically certify a regular coarse-grained Sobolev decomposition.

Mechanism failure without endpoint failure. A proposed staircase or genericity mechanism may fail to describe a successful learning trajectory. That challenges the proposed mechanism, not the fact that a feasible generalizing network supplies a compositional predictor.

8.19 What predictive success does not require

The witness argument does not require uniqueness of the compositional decomposition, recovery of the generating latent variables, degree-ordered staircase learning, faithful intermediate traces, or a literal correspondence between hidden units and target constituents.

These properties can occur and can be useful. In a particular restricted learning theorem, some may be central to the proof. None is an additional condition for compiling a feasible predictor that has already been shown to approximate the target.

The same applies to the choice of GELU or another analytic activation, a specific residual design, and an NTK-style proof of conditioning. Their absence from the witness proof does not remove them from a learning theorem whose success guarantee relies on them.

Nor does predictive success require zero SGD covariance, a common minibatch equilibrium, or convergence of the entire training trajectory to one parameter. Those are additional dynamical properties, not premises for the returned predictor's fit, statistical bound, and feasible evaluation. A claimed point-convergence theorem must, of course, address its own stronger endpoint.

What cannot be removed is the need to establish accuracy, a valid statistical argument when one is claimed, and the relevant computational resource bounds. The revised theory removes an unnecessary structural endpoint; it does not remove the optimization problem.

8.20 A compact recovery logic

The representation question is

$$\boxed{\text{efficient target evaluation} \implies \text{a feasible computational realization exists.}}$$

For the chosen learning architecture, its approximation quality and useful norm budget must still be established.

The statistical and algorithmic combination is

$$\boxed{\begin{array}{c}\text{training success: } \widehat{R}_S(\widehat{f}) \le \eta, \quad \widehat{f} \in \mathcal{H}_Q \\ + \quad \text{a valid uniform bound on } \mathcal{H}_Q \\ \Longrightarrow \quad R_P(\widehat{f}) \le \eta + \beta_S(Q, \delta_{\mathrm{stat}}).\end{array}}$$

When that upper bound is small, the predictor is accurate for the stated population objective. Finally,

$$\boxed{\begin{array}{c}\text{accurate predictor with feasible numerical evaluation} \\ \Longrightarrow \\ \text{compositionally sparse predictive witness.}\end{array}}$$

Genericity, staircase learning, trace exposure, and architectural or stochastic mechanisms can help establish the training-success premise. Analytic convergence and quantitative residual contraction contribute only their stated conclusions; statistical control remains separate. These mechanisms do not turn smooth activations, overparameterization, or a penalty in the objective into an unconditional guarantee of accurate learning. The training-success event concerns the returned model. It can hold even when regularized minibatch updates at that model have nonzero covariance; the finite-time return and its resource budget must still be justified.

8.21 The principle in one sentence

> *If a specified training procedure finds a computationally feasible predictor with a sufficiently good empirical fit and a valid, sufficiently small statistical-complexity bound, then the predictor generalizes and its own computation supplies a compositionally sparse approximation of the target, without requiring recovery of the computation that generated the data.*

8.22 Summary

The representation and recovery questions remain distinct. The simplification is that recovery itself has two levels: predictive accuracy and identification of generating structure.

Efficient digital evaluation provides a bounded-fan-in representation at the stated precision. A feasible learned network possesses such an implementation as well. This is the computational content of the compositional witness.

Overparameterization can make an existing interpolation set extended and degenerate. Analyticity, together with compactness of an all-time gradient flow, gives convergence to a critical point. A quantitative residual condition gives a different conclusion: contraction toward zero training error for the stated unregularized dynamics. Stochastic-volume calculations describe still another object, equilibrium preference among low-loss neighborhoods. Their conclusions and assumptions cannot be substituted for one another, and none alone establishes finite-time success of regularized SGD.

Exact interpolation and regularized stationarity have different noise properties. At the former, every fixed-example squared-loss data gradient vanishes. At the latter, only the mean data gradient

must cancel the regularizer; heterogeneous sample gradients sustain intrinsic minibatch fluctuations. The normalized ReLU common-equilibrium rank restriction provides a specific structural obstruction to all batch directions vanishing. This does not make the noise isotropic or prove a universal low-rank or global-optimization bias.

Norm- and product-norm-based regularization can favor well-fitting solutions in statistically controlled classes. When training actually returns such a solution and the relevant bound is informative, population accuracy follows. The conditional theorem makes these premises and their probabilities explicit. The predictor can be a finite stopped iterate; continued fluctuations of the training dynamics are compatible with this return-based notion of success. The stopping or averaging rule remains part of the algorithm whose guarantee must be established.

The norm quantity must match the architecture and activation assumptions. Statistical class control does not automatically confine the parameters or provide a positive lower bound on the training-prediction Jacobian. Those additional controls are required only when the particular learning proof relies on them.

Once a feasible predictor is accurate, its decomposition need not be unique, need not match the generating variables, and need not have been constructed by a staircase. A structural-recovery theorem is a stronger achievement; a staircase theorem can be a constructive explanation of learning, not a prerequisite added after predictive success.

The central question for regularized overparameterized SGD is therefore

> When does a specified regularized training procedure find,
> with high probability and feasible resources,
> a well-fitting predictor with adequate statistical control?

Solving that question establishes predictive learning in the stated regime. The compositional conclusion then follows from the computation performed by the learned predictor itself.

Part II
Computation and Algorithms

CHAPTER 9

Efficient Computability, Compositional Sparsity, and Attention

Efficient digital computation admits a polynomial-size representation as a directed acyclic graph of bounded-fan-in operations. Under additional regularity and stability assumptions, low-dimensional constituents also give favorable approximation rates.

How does this structural principle relate to attention-based architectures?

The connection is a division of computational labor. Attention can retrieve the inputs needed by a constituent, while a local nonlinear module computes the constituent itself. Repeated retrieval and transformation can therefore implement a compositional computation.

The central architectural picture is

$$\boxed{\textit{address} \longrightarrow \textit{retrieve} \longrightarrow \textit{compute locally} \longrightarrow \textit{store and repeat}.}$$

Several qualifications are essential. Dot-product attention is exactly Gaussian-kernel retrieval only under appropriate conditions. A normalized kernel smoother does not automatically inherit every approximation theorem for general radial-basis networks. Polynomial complexity in the number of precision bits is different from polynomial complexity in inverse approximation error. Finally, the existence of a routed compositional realization does not explain how training finds a useful predictor, or why that predictor generalizes.

The distinction of Chapter H applies here as well. Executing a supplied graph, learning an accurate predictor, and identifying the particular generating graph are three different achievements. The constructive results below concern the first. Their learning interpretation must not make the third a prerequisite for the second.

9.1 The Architectural Connection

A compositionally sparse target has a realization

$$f = \Phi_G$$

on a computation DAG G whose internal nodes depend on at most k inputs.

For a node v with parents $u_1, \dots, u_{m_v}$,

$$z_v = g_v(z_{u_1}, \dots, z_{u_{m_v}}), \qquad m_v \le k. \tag{9.1}$$

Implementing this node requires two operations.

First, the machine must make the parent values available.

Second, it must apply the function g_v.

Attention offers one implementation of the first operation. A multilayer perceptron, or another suitable local approximator, offers an implementation of the second.

This separation is compatible with the organization of Transformers, which interleave attention with position-wise feed-forward transformations [158].

It is not an exclusive description of everything those components can do. Attention itself is nonlinear and can participate directly in computation. The point is that routing and local evaluation provide a particularly transparent constructive interpretation.

Here G is one supplied realization, not an identified ground-truth graph. It may be a teacher's description, a compiled algorithm, or a computation chosen by the designer. If training returns a different feasible computation with accurate population predictions, the predictive objective has still been met. Matching its internal states to the supplied z_v is a stronger structural objective.

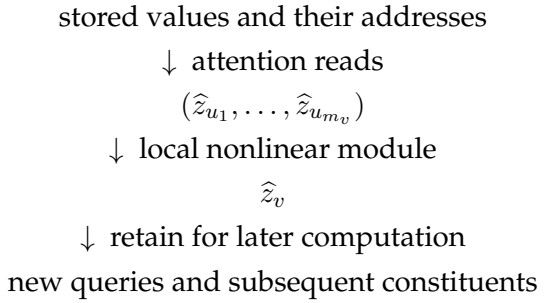


Figure 9.1: A routed implementation of a compositional node. Separate reads supply the ordered arguments of a local function. The computed output becomes available to subsequent nodes. This diagram describes an implementation of a supplied computation, not a theorem about the organization learned by gradient descent.

9.2 Attention: Definitions and Scope

For query $q \in \mathbb{R}^{d_k}$, keys $k_1, \ldots, k_N \in \mathbb{R}^{d_k}$, and values $v_1, \ldots, v_N \in \mathbb{R}^{d_v}$, define

$$\mathrm{Attn}(q; K, V) = \sum_{j=1}^{N} \alpha_j(q) v_j, \qquad \alpha_j(q) = \frac{\exp\left(q^\top k_j/\tau\right)}{\sum_{i=1}^{N} \exp(q^\top k_i/\tau)}, \tag{9.2}$$

where $\tau > 0$ absorbs the score scaling.

In a Transformer, the representations are commonly formed by linear maps:

$$q = W_Q h, \qquad k_j = W_K h_j, \qquad v_j = W_V h_j.$$

The term *self-attention* refers to the case in which the queries, keys, and values arise from the same collection of token representations. In cross-attention, the query collection and the key–value collection differ.

A bank of independently stored prototypes is therefore not automatically an ordinary self-attention layer. It is more directly a memory-attention or cross-attention construction. To implement

it using self-attention, the memory entries must be represented as tokens and the permitted interactions must be specified.

Similarly, arbitrary stored values cannot simply be identified with $g(x_j)$ unless the architecture actually stores or computes those values. The formula $v_j = W_V h_j$ does not by itself provide arbitrary function samples.

These distinctions matter in constructive approximation arguments.

9.3 When Attention Is Gaussian-Kernel Retrieval

The kernel interpretation of attention is useful and has been developed explicitly in the literature [154]. Its exact form follows from a simple identity:

$$q^\top k_j = -\frac{1}{2}\|q - k_j\|_2^2 + \frac{1}{2}\|q\|_2^2 + \frac{1}{2}\|k_j\|_2^2. \tag{9.3}$$

The query-only term is common to every key and disappears under softmax normalization. The key-norm term generally does not.

9.3.1 Exact equivalence under equal key norms

Proposition 7 (Equal-key-norm Gaussian retrieval)**.** *Suppose all keys have the same norm:*

$$\|k_j\|_2 = c \qquad \textit{for every } j.$$

Then, for every query q,

$$\alpha_j(q) = \frac{\exp(-\|q - k_j\|_2^2/(2\tau))}{\sum_i \exp(-\|q - k_i\|_2^2/(2\tau))}. \tag{9.4}$$

Proof. Substitute Equation (9.3) into the softmax. Both $\|q\|_2^2/(2\tau)$ and $c^2/(2\tau)$ are common to all terms and cancel. □

Normalizing both queries and keys to unit length is sufficient, but normalizing the query is not necessary for this identity. Equality of the key norms is the relevant condition.

9.3.2 Unequal key norms

Without equal key norms, the exact expression is

$$\alpha_j(q) = \frac{\exp(-\|q - k_j\|_2^2/(2\tau))\exp(\|k_j\|_2^2/(2\tau))}{\sum_i \exp(-\|q - k_i\|_2^2/(2\tau))\exp(\|k_i\|_2^2/(2\tau))}. \tag{9.5}$$

Thus ordinary dot-product attention includes a key-dependent factor in addition to distance-based similarity.

Alternatively, the corrected score

$$a_j(q) = q^\top k_j - \frac{1}{2}\|k_j\|_2^2 \tag{9.6}$$

produces exactly the Gaussian weights in Equation (9.4), without requiring equal key norms.

Approximate equality of key norms gives only an approximate correspondence. Its accuracy depends on the norm variation relative to τ, not merely on whether the norms look numerically similar.

9.3.3 Layer normalization is not query–key normalization

Layer normalization of a token representation does not generally force the projected keys to have equal norms.

Indeed,

$$\|W_K h\|_2^2 = h^\top W_K^\top W_K h,$$

which can vary with the direction of h even when $\|h\|_2$ is fixed.

Therefore an exact Gaussian equivalence must rely on an explicit normalization or score condition. It does not follow from the presence of LayerNorm in the surrounding Transformer block.

9.3.4 Learned Mahalanobis geometry

Suppose the query and keys are constructed using the same linear map:

$$q = Bx, \qquad k_j = Bx_j.$$

Then

$$\|q - k_j\|_2^2 = (x - x_j)^\top M(x - x_j), \qquad M = B^\top B \succeq 0. \tag{9.7}$$

Under the equal-key-norm condition or the corrected score above, attention is normalized Gaussian retrieval in this learned geometry.

If B is rank deficient, this is a pseudometric on the original space: distinct inputs differing only in $\ker B$ have zero distance.

If the projected vectors are subsequently normalized, the resulting geometry is instead that of the normalized feature map. It need not remain a fixed quadratic Mahalanobis geometry in the original inputs.

Finally, when $W_Q \neq W_K$, the score defines a general compatibility between two representations. It need not correspond to a symmetric positive-semidefinite kernel on the original input space.

The kernel viewpoint and the RKHS viewpoint should therefore not be identified without checking the required assumptions.

Who supplies the useful coordinates? In a fixed-kernel method, the feature geometry is prescribed. In an attention-based learner, the representations used to form queries, keys, and values can themselves be trained. Conditional on fixed queries and keys, Equation (9.2) is linear in the stored values; this does not make the full learned predictor linear in its training labels, or make joint training convex. The routing weights can change when the representation changes.

This is the representation-discovery question developed in Chapters 5 and H: can training construct coordinates in which useful dependencies become easy to access and compute? The Mahalanobis factorization above is one explicit supplied-coordinate case. It is not a theorem that an arbitrary trained head discovers that factorization, or that its coordinates must coincide with the teacher's.

9.4 Kernel Averaging Is Not a General Approximation Theorem

When the stored values are function samples,

$$v_j = g(x_j),$$

Gaussian retrieval takes the Nadaraya–Watson form

$$\widetilde{g}(x) = \frac{\sum_j K_\tau(x, x_j) g(x_j)}{\sum_j K_\tau(x, x_j)}. \tag{9.8}$$

This is a normalized weighted average of sample values.

A general RBF network instead has the form

$$g_{\mathrm{RBF}}(x) = \sum_j c_j K_\tau(x, x_j),$$

with independently chosen coefficients c_j.

The two approximation families are different. In particular, a theorem about general RBF coefficients cannot be transferred automatically to Equation (9.8).

Higher-order approximation rates can depend on cancellation, local polynomial reproduction, boundary treatment, bandwidth selection, and the permitted value functions. None of these properties follows merely from writing a softmax as a Gaussian kernel.

We therefore do not deduce an arbitrary C^r approximation rate for a single attention head whose values are only $g(x_j)$.

Instead, the construction below assigns local approximation to a specified nonlinear module and uses attention to supply its arguments:

routing accuracy + constituent approximation accuracy.

9.5 Margins Give Quantitative Retrieval Bounds

The routing interpretation has a direct quantitative foundation.

Let

$$\alpha_j = \frac{e^{a_j/\tau}}{\sum_i e^{a_i/\tau}},$$

where a_j denotes the unscaled compatibility score.

9.5.1 Retrieving one item

Proposition 8 (Single-item retrieval)**.** *Suppose the desired item has index $j_\star$ and score margin*

$$a_{j_\star} - \max_{j \neq j_\star} a_j \geq \gamma > 0.$$

Then

$$1 - \alpha_{j_\star} \leq (N-1) e^{-\gamma/\tau}. \tag{9.9}$$

If $\|v_j\| \leq R$ for all stored values, then

$$\|\mathrm{Attn}(q; K, V) - v_{j_\star}\| \leq 2R(N-1)e^{-\gamma/\tau}. \tag{9.10}$$

Proof. Divide the softmax denominator by $e^{a_{j_\star}/\tau}$. Each competing term is at most $e^{-\gamma/\tau}$.

For the value error, write

$$\mathrm{Attn}(q; K, V) - v_{j_\star} = \sum_{j \neq j_\star} \alpha_j (v_j - v_{j_\star})$$

and apply the triangle inequality. □

Thus a prescribed retrieval error η can be obtained by choosing the score scale so that, for example,

$$\frac{\gamma}{\tau} \geq \log\left(1 + \frac{2RN}{\eta}\right).$$

This is a conditional statement. It assumes a margin; it does not explain how training creates that margin. For a uniform approximation claim, the relevant margins must hold over every input and memory state encountered by the constructed evaluator, not just on the training sample. The separate address fields used below provide such a margin by design.

Perturbed scores. A numerical or representation perturbation must also respect the margin. If $|\widetilde{a}_j - a_j| \leq \zeta$ for every active record and $2\zeta < \gamma$, then subtraction gives

$$\widetilde{a}_{j_\star} - \widetilde{a}_j \geq \gamma - 2\zeta \qquad (j \neq j_\star).$$

Applying the same proposition to the perturbed scores replaces $e^{-\gamma/\tau}$ by $e^{-(\gamma-2\zeta)/\tau}$. Errors in stored values and in the arithmetic of the final weighted sum must be added separately. A lower temperature strengthens concentration for a fixed positive margin; it does not remove the need to control score errors.

9.5.2 Near top-b sparsity

Order the scores as

$$a_{(1)} \geq \cdots \geq a_{(N)}.$$

Let $b < N$ and suppose

$$a_{(b)} - a_{(b+1)} \geq \Gamma > 0.$$

Define the mass outside the top b entries by

$$t_b = \sum_{j>b} \alpha_{(j)}.$$

The ratio of tail mass to retained mass satisfies

$$\frac{t_b}{1 - t_b} \leq \frac{N-b}{b} e^{-\Gamma/\tau}.$$

Consequently,

$$t_b \leq \frac{\frac{N-b}{b} e^{-\Gamma/\tau}}{1 + \frac{N-b}{b} e^{-\Gamma/\tau}} \leq \frac{N-b}{b} e^{-\Gamma/\tau}. \tag{9.11}$$

If the retained weights are renormalized to sum to one, the difference between the full read and the truncated read is at most

$$2Rt_b.$$

For equal-norm Gaussian retrieval, the score gap corresponds to half the gap in squared distances.

These inequalities explain when a particular attention distribution is effectively sparse. They do not imply that every trained head must be sparse, or that sparse reads can always be found cheaply.

9.6 An Explicit Attention–MLP Computation Model

We now specify the architecture used in the constructive statements.

Let

$$f : [-1, 1]^d \to [-R, R], \qquad R \geq 1,$$

have a supplied DAG realization with $s \geq 1$ internal scalar nodes, depth $L \geq 1$, and local arity at most k. The case with no internal computation can be handled by direct input readout or a fixed output. A topological level is the length of a longest path of internal nodes ending at that node; input records are at level zero.

Assume each node computes

$$g_v : [-R, R]^{m_v} \to [-R, R], \qquad m_v \leq k.$$

The evaluator contains the original inputs and the previously computed node values in an addressable memory. Each value is stored together with an identifier.

To evaluate node v, the machine issues m_v reads, one for each ordered parent. Their outputs form the tuple

$$\widehat{u}_v = (\widehat{z}_{u_1}, \ldots, \widehat{z}_{u_{m_v}}).$$

A local MLP computes

$$\widehat{z}_v = \widetilde{g}_v(\widehat{u}_v).$$

Separate argument slots are important. One weighted average does not generally preserve all the ordered arguments needed by an arbitrary multivariate constituent.

Computed outputs are retained for subsequent nodes. In this evaluator, retention is an explicit memory operation: completed records are not rewritten by later local transformations. It is not inferred from the presence of an arbitrary residual connection. The local modules may be node-specific, or selected from a module bank using node identifiers. The cost of all distinct modules and any selection circuitry is counted; repeated module reuse is not implied by bounded arity alone.

Nodes at the same topological level can be evaluated in parallel after all their parents are available. This gives L logical rounds, provided sufficient parallel resources are supplied. A serial evaluator instead visits the internal nodes one by one. The logical round count is not the serial running time of that implementation.

This is an attention–MLP memory architecture. A realization using self-attention must explicitly represent the memory and computation states as tokens, preserve their identifiers, and prevent reads from unavailable states.

The results below concern this specified organization. They are not claims about every fixed-width, fixed-depth Transformer variant.

9.6.1 A simple addressing construction

For a finite graph, assign each memory record a distinct one-hot tag

$$e_j \in \mathbb{R}^N, \qquad N = d + s.$$

The query for record j is e_j, and keys are the corresponding record tags.

Only initialized input records and already completed node records are eligible for a read. This restriction may be implemented by supplying only the active records to the attention operation, or by a fixed availability mask in the compiled schedule. Every required parent is active before its child is evaluated.

Among active records, the intended score is one and every other score is zero. Hence the margin is exactly one whenever a competitor is present; a read with only the desired record is exact. All retrieval bounds may use $N = d + s$ as an upper bound on the active memory size.

Tags and numerical values occupy separate fields, so changing a stored value does not change its address.

This construction is deliberately inefficient in address dimension, but its overhead is polynomial in graph size. It demonstrates existence without assuming that a learned representation already contains well-separated addresses.

More economical addressing schemes require their own separation and resource analysis. The tags in this construction are supplied routing metadata. Their existence is not evidence that learning has recovered semantic coordinates or a teacher's intermediate representation.

9.7 Compositional Approximation with Routing Error

The advantage of local arity enters through approximation of the constituents, as in the compositional approximation theory developed earlier [125].

Suppose each g_v belongs to a bounded C^r class, with $r \geq 1$ an integer and

$$\|g_v\|_{C^r} \leq B.$$

For a constituent on m_v variables, standard deep ReLU constructions give uniform approximation error δ with a number of weights and computation units bounded by

$$P_v(\delta) \leq C\, \delta^{-m_v/r} \log\left(\frac{2}{\delta}\right), \tag{9.12}$$

and depth

$$D_v(\delta) \leq C \log\left(\frac{2}{\delta}\right),$$

where the constant depends on the local dimension, smoothness, domain, and norm bound [172, Theorem 1]. Rescaling the compact box and the function reduces this to the bounded Sobolev-class result; a bounded C^r class satisfies the needed derivative bounds. For the uniform estimate below, take one constant over the finitely many local dimensions $m_v \leq k$. Constants for fixed k, r, R, B must not be treated as uniform polynomial bounds when those quantities are allowed to grow. Constant constituents require only a constant output and cause no difficulty.

The count concerns the weights on the specified connections and the computation units, not every entry of an unnecessarily padded dense matrix. It is an approximation-existence bound; it does not give the cost of learning the constituent or a bound on its statistical norm complexity.

The local MLP in this statement may itself have several hidden layers. It is not necessarily one fixed two-layer Transformer FFN.

9.7.1 Keeping intermediate values in their domains

Clip every approximate node output to $[-R, R]$.

Because the true output lies in this interval, clipping cannot increase its approximation error. Attention then averages bounded stored values, so the routed arguments also remain in the prescribed box.

This avoids silently evaluating a constituent approximator outside the domain on which its guarantee holds.

9.7.2 Propagation through the graph

Let λ_v be a Lipschitz constant of g_v in the maximum norm. For each topological level ℓ, set

$$\Lambda_\ell = \max\left\{1, \max_{v \text{ at level } \ell} \lambda_v\right\}.$$

Let e_ℓ bound the errors of all values computed through level ℓ, relative to their values in the supplied exact computation. Initially $e_0 = 0$ when the original real inputs are supplied exactly. The maximum includes retained values from earlier levels, including parents connected by edges that skip levels.

Suppose a routing read at level ℓ differs from the intended stored value by at most η_ℓ, and every local MLP has approximation error at most δ.

The total error in a routed argument is at most

$$e_{\ell-1} + \eta_\ell.$$

Indeed, the read first differs from the intended *stored* parent by at most η_ℓ; that stored parent can already differ from the exact parent by $e_{\ell-1}$. For the new node, compare $\widetilde{g}_v(\widehat{u}_v)$ first with $g_v(\widehat{u}_v)$ and then with $g_v(u_v)$. This uses the approximation error of $\widetilde{g}_v$ and the Lipschitz constant of the *target* constituent g_v; no unproved derivative bound for the approximating MLP is needed. Clipping cannot enlarge the resulting output error.

Therefore,

$$e_\ell \le \Lambda_\ell e_{\ell-1} + \delta + \Lambda_\ell \eta_\ell. \tag{9.13}$$

The use of $\Lambda_\ell \ge 1$ also accounts for retaining earlier values across levels.

Define

$$A_G = \sum_{\ell=1}^{L} \prod_{j=\ell+1}^{L} \Lambda_j, \tag{9.14}$$

with an empty product equal to one. This is a levelwise upper bound, not an intrinsic invariant of f or the sharpest possible graph factor. It depends on the chosen realization and on retaining all earlier records. In particular, even when each $\lambda_v < 1$, the convention $\Lambda_\ell \ge 1$ deliberately gives a conservative bound. Sharper contractive estimates need a more specific graph and error accounting.

If

$$\eta_\ell \le \frac{\delta}{\Lambda_\ell},$$

then induction gives

$$e_L \le 2A_G\delta.$$

An explicit numerical error budget. The preceding estimate treats the local ReLU modules and soft attention as exact mathematical maps, with routing error measured relative to a hard read. In a numerical implementation, let e_0 include input-encoding error, let δ_ℓ bound local approximation error, and let ξ_ℓ bound the error of evaluating and storing each new local output. Assume retained records do not incur additional rewriting error, and include all read-arithmetic errors in η_ℓ. Clip numerically formed reads and stored outputs to $[-R, R]$ as well, so every local module is evaluated inside its certified domain. The same induction gives

$$e_L \le \left(\prod_{j=1}^{L} \Lambda_j\right) e_0 + \sum_{\ell=1}^{L} \left(\prod_{j=\ell+1}^{L} \Lambda_j\right) (\delta_\ell + \xi_\ell + \Lambda_\ell \eta_\ell).$$

All terms must fit within the required final tolerance. A finite-weight approximation theorem alone does not supply the precision or uniform numerical conditioning of its evaluator. The separate digital construction in Section 9.9 establishes a stronger bit-resource statement under different assumptions.

9.7.3 A conditional approximation theorem

Theorem 8 (Approximation by routed local modules)**.** *Assume the supplied DAG, constituent domains, smoothness bounds, and attention–MLP evaluator specified above. In this approximation statement, inputs and local module evaluations are exact; routing has the stated soft-read error.*

For every $\varepsilon \in (0,1)$, choose

$$\delta = \frac{\varepsilon}{2A_G}$$

and route each parent with error at most

$$\eta_\ell = \frac{\delta}{\Lambda_\ell}$$

at level ℓ.

Then the resulting evaluator satisfies

$$\sup_{x\in[-1,1]^d} |f(x) - \widehat{f}(x)| \le \varepsilon.$$

The total local-MLP complexity obeys

$$P_{\text{local}} \le Cs\left(\frac{2A_G}{\varepsilon}\right)^{k/r} \log\!\left(\frac{4A_G}{\varepsilon}\right). \tag{9.15}$$

With parallel evaluation within each topological level, there are L routing-and-computation rounds. Counting each attention read as a primitive and using the usual local-network layer convention, the sequential architecture depth is bounded by

$$O\!\left(L \log\frac{4A_G}{\varepsilon}\right).$$

Address storage, routing maps, and the arithmetic and numerical implementation of each attention read incur additional costs, counted separately in Section 9.8. The displayed depth is not a bound on serial running time or total Boolean-circuit depth.

Proof. The margin bound supplies the required routing accuracy, for example using the one-hot addressing construction.

Approximate every constituent to accuracy δ, clip its output, and apply Equation (9.13). Starting from $e_0 = 0$, the chosen tolerances give $e_L \le 2A_G\delta = \varepsilon$. The availability schedule ensures that all intended stored parents exist when they are read.

Summing Equation (9.12) over s nodes and using $m_v \le k$ gives the stated local complexity. □

9.7.4 What "dimension-independent" means here

The approximation exponent is k/r, rather than d/r.

But the total computation is not independent of d. It must still represent the inputs, store the graph, route values, and evaluate all necessary constituents.

Moreover, A_G can grow rapidly with depth. A favorable overall bound requires quantitatively stable composition.

Thus the appropriate conclusion is

local arity controls the approximation exponent, provided routing and composition are controlled.

It is not a universal dimension-free guarantee for arbitrary attention networks.

9.8 Keys, Parameters, and Operations Are Different Resources

The preceding construction uses at most $d + s$ stored value records during one evaluation. These are addresses of computational states, not RBF centers tabulating the constituent functions.

The constituent approximation cost is carried by the local MLPs.

Consequently, Equation (9.15) is not a theorem that

$$N_{\text{keys}} = O(s\varepsilon^{-k/r}).$$

It is a bound on a different resource.

At most ks parent reads are needed. If every read compares its query with all N memory entries, the number of query–key comparisons is at most

$$ksN.$$

The arithmetic cost of each comparison also depends on key dimension. For key dimension d_k and scalar stored values, a straightforward dense implementation uses $O(ksNd_k)$ arithmetic work for the dot products, and $O(ksN)$ scalar exponential, normalization, and weighted-sum operations, in addition to local-module evaluation. With the one-hot choice $d_k = N$, the dot-product bound is $O(ksN^2)$. Explicitly storing all dense tags uses $O(N^2)$ entries; the numerical states use $O(N)$ entries. These are deliberately loose constructive upper bounds, not claims of hardware efficiency.

The polynomial overhead must be added to P_{local}, not absorbed into it without comment. Each scalar operation also has a bit cost once a numerical model is specified. The description must include the graph, distinct modules, address rules, and required stored constants.

This remains polynomial for the explicit finite-graph construction, but it need not be computationally attractive. Sparse indexing or structured routing may reduce it.

Importantly, a sharply concentrated attention distribution does not automatically produce a faster implementation. Computing all scores and then discarding most of them can retain the cost of dense retrieval.

9.8.1 Functional arity and implementation fan-in

Even a dense attention read can be implemented using bounded-fan-in arithmetic. Dot products and normalization sums can be evaluated through binary reduction trees.

Thus

dependence on many stored values

$$\not\Longrightarrow$$

an elementary operation of unbounded fan-in.

This establishes compatibility with the digital compositionality principle once the arithmetic is evaluated with controlled finite precision. It does not establish that the learned attention pattern

itself has bounded support. Nor is the primitive attention-stage count the depth of its bounded-fan-in implementation: a dense read includes reductions over keys and values, and their evaluation has its own depth. The same distinction applies when local affine units are compiled into scalar arithmetic.

9.9 Efficient Finite-Precision Computation

A different argument is needed for complexity polynomial in the number of precision bits.

If

$$\varepsilon = 2^{-p},$$

then

$$\varepsilon^{-k/r} = 2^{pk/r}.$$

Therefore a smooth approximation bound polynomial in $1/\varepsilon$ does not establish a bound polynomial in p.

The finite-precision route instead starts with the digital computation of the desired approximation.

9.9.1 From bit computation to a routed circuit

Suppose an algorithm, on an input string u of length n and precision request p, outputs a dyadic number y satisfying

$$|y - f(u)| \le 2^{-p}$$

in time

$$T(n, p) = \mathrm{poly}(n + p).$$

As established in Chapter 4, this computation admits a uniform polynomial-size bounded-fan-in Boolean circuit.

The circuit can be executed by storing input bits and previously computed gate outputs as addressable records.

Each gate retrieves its parent bits, applies a constant-size Boolean operation, and stores its output.

Soft attention introduces an additional issue: a retrieved bit is an average rather than an exact symbol. A local restoration map resolves this issue in the construction below.

9.9.2 Restoring discrete states after soft retrieval

For at most N stored records, use orthogonal address tags and inverse temperature

$$\beta = \tau^{-1} \ge \lceil \log(16N) \rceil .$$

Choose such a scale with $\beta = O(\log N)$; one uniform integer choice is $\beta = \lceil \log_2(16N) \rceil$, which satisfies the displayed natural-log lower bound and is obtained directly from the binary length of N. Arbitrarily larger choices are not needed for the resource estimate. Only already initialized records participate in the read.

The single-read bound gives error smaller than $1/8$ when values are bits and their norms are bounded by one.

Allow an additional numerical evaluation error smaller than $1/8$. The resulting read is still within $1/4$ of the correct bit.

Define the ReLU restoration map

$$\mathcal{R}(u) = 2\,\mathrm{ReLU}\Big(u - \frac{1}{4}\Big) - 2\,\mathrm{ReLU}\Big(u - \frac{3}{4}\Big). \tag{9.16}$$

It satisfies

$$\mathcal{R}(u) = 0 \quad \text{for } u \le \frac{1}{4}, \qquad \mathcal{R}(u) = 1 \quad \text{for } u \ge \frac{3}{4}.$$

Hence every sufficiently accurate read is restored to the exact intended bit before the next Boolean operation. In the bit implementation, the approximate read is a dyadic number, and the fixed-coefficient ReLU restoration is evaluated with exact dyadic arithmetic and enough guard bits for its intermediate additions and subtractions. Its plateau output is then exactly 0 or 1. The statement would not follow from allowing an arbitrary fresh perturbation after restoration.

For example,

$$\mathrm{AND}(a, b) = \mathrm{ReLU}(a + b - 1), \qquad \mathrm{NOT}(a) = 1 - a$$

on Boolean inputs.

The gate output is therefore exact, and the next memory record again contains a discrete symbol rather than accumulated numerical error.

Proposition 9 (Polynomial-resource attention–MLP realization)**.** *Under the finite-precision computation assumption above, there exists a uniform family of attention–ReLU memory evaluators producing the same dyadic outputs as the digital algorithm.*

Their description size, working storage, sequential computation, and required numerical precision are polynomially bounded in $n + p$.

The construction uses explicit address tags, retained memory values, and local symbol restoration evaluated with exact finite dyadic operations after an approximate attention read. Exponentials and division need only achieve the fixed read tolerance above. It does not require an arbitrary exact-real parameter to encode the computation.

Proof. Compile the digital algorithm into a polynomial-size Boolean circuit.

Evaluate the circuit in topological order. Retrieve each gate input using attention, restore the retrieved bit with Equation (9.16), compute the Boolean gate, and retain its output.

Induction over the gates shows that every stored bit agrees with the circuit computation.

The circuit size and the number of memory records are polynomial. One-hot address representations and even dense attention therefore incur only polynomial overhead.

The chosen score scale is $O(\log N)$ and the address dot products are exact zeros and ones. Subtracting the largest logit gives unnormalized weights 1 for the intended record and $t = e^{-\beta}$ for every other active record. If there are $M \le N$ active records, the exact read has the form

$$q(t) = \frac{b_\star + tA}{1 + t(M-1)}, \qquad A = \sum_{j \ne j_\star} b_j, \qquad b_j \in \{0, 1\}.$$

This expression describes the ordinary weighted read; the evaluator computes its weighted sum over the available records. For $t \ge 0$,

$$|q'(t)| = \frac{|A - (M-1)b_\star|}{(1 + t(M-1))^2} \le N.$$

A nonnegative dyadic approximation to $e^{-\beta}$ with absolute error at most $1/(64N)$, followed by division accurate to $1/64$, therefore approximates the read to at most $1/32 < 1/8$. Since $\beta =$

$O(\log N)$, ordinary argument reduction and a convergent exponential series give this dyadic approximation using polynomially many bit operations in $\log N$; $O(\log N)$ reliable bits suffice for the stated absolute tolerance. Summation over the polynomially many records also has polynomial cost. No exponentially accurate attention evaluation is needed as the circuit depth grows.

The subsequent restoration and Boolean operations use fixed dyadic constants and exact finite arithmetic, so their stored outputs are again bits. This proves the induction at the numerical, not merely formal real-arithmetic, level.

Finally, decoding the output bit string gives the dyadic answer of the original algorithm. □

9.9.3 Scope of the construction

This proposition is a constructive result for the specified attention–MLP memory model. It does not establish the same bounds for every prescribed Transformer architecture, normalization scheme, positional encoding, or fixed context size.

A Transformer realization must preserve the address and state fields required by the construction. Additional components cannot simply be inserted while assuming they leave the simulation unchanged.

For a real-domain target, the input string may encode a dyadic approximation to x. A polynomially controlled input-precision requirement then connects this result to the bit-model definition used earlier in the book.

This remains distinct from directly feeding arbitrary real scalars into a fixed network and asserting a uniform approximation rate.

9.10 Low-Dimensional Relevance and Low-Rank Projections

The relation between local arity and projection rank is conditional.

Suppose the information relevant to a particular read is already contained in

$$z = Ph, \qquad P : \mathbb{R}^D \to \mathbb{R}^m, \qquad m \le k.$$

If its queries and keys can be formed as

$$q = U_Q P h, \qquad k_j = U_K P h_j,$$

then the effective content projections

$$W_Q = U_Q P, \qquad W_K = U_K P$$

satisfy

$$\operatorname{rank}(W_Q) \le m, \qquad \operatorname{rank}(W_K) \le m.$$

Likewise, a Gaussian metric of the form

$$M = P^\top A^\top A P$$

has rank at most m.

Thus low-rank content projections suffice when the required relevance relation factors through an available low-dimensional linear representation.

9.10.1 What the rank statement does not imply

A bounded-arity computation graph does not establish that its parent variables are already available through a rank-k projection of the current token representation.

Nor does it bound the dimension needed for positional identifiers, memory addresses, or other routing metadata.

The explicit one-hot construction above illustrates this distinction: local computation has small arity, while addressing may use a much larger space.

The ranks of W_Q and W_K must also be distinguished from the rank of the attention matrix after softmax. Entrywise exponentiation and row normalization need not preserve low matrix rank.

For example, for $N \geq 2$, a rank-one score matrix with entries

$$a_{ij} = q_i j, \qquad j = 0, \ldots, N-1,$$

produces exponential entries

$$e^{a_{ij}/\tau} = \left(e^{q_i/\tau}\right)^j.$$

For distinct q_i, this is a full-rank Vandermonde matrix. Row normalization preserves that rank.

Therefore low-rank scores do not, by themselves, imply low-rank attention weights. Here $i = 1, \ldots, N$, so the example is square, and row normalization is multiplication by an invertible positive diagonal matrix. Exact algebraic rank and a numerically useful low-rank approximation are separate questions.

Low rank by representation versus low rank by training. The factorization through P is a sufficient representation assumption. It is not a result about the matrices selected by regularized SGD. The common-minibatch-equilibrium restriction in Chapter 7, Section 7.17, and Appendix N concerns a specified normalized scalar-output ReLU model. It does not impose the same rank bound on Transformer query or key matrices, or on an attention matrix.

Likewise, a zero gradient of the full regularized objective does not require every minibatch update to vanish. Persistent fluctuations can coexist with an accurate returned predictor. Neither a rank statistic nor vanishing or nonvanishing noise, by itself, establishes population accuracy. These points keep the conditional projection statement separate from the optimization and statistical questions.

9.11 Empirical Questions and Predictions

The constructive results suggest experiments, but they do not establish universal properties of trained models.

For example, head-pruning studies show that some trained systems can retain performance after removing many heads [110]. This concerns redundancy across heads. It is not a direct test of top-b sparsity within a head, projection rank, or recovery of a bounded-arity target.

9.11.1 Routing accuracy versus constituent accuracy

Use a synthetic target with a known computation DAG.

Compare a system with supplied parent addresses against one that must learn the routing. Separately vary the accuracy of the local modules.

Measure routing errors, local function errors, and final prediction error on held-out inputs. In the supplied-graph arm, these quantities can test the error decomposition directly. In a freely

trained arm, score prediction first; a mismatch with the supplied routing may instead reflect a different accurate computation. To interpret routing mismatch as structural failure, specify in advance which heads, argument slots, and node correspondences are being compared.

9.11.2 Margin and truncation

For each read, measure the score gap, discarded attention mass, and change in the retrieved value after top-b truncation.

The margin inequalities provide quantitative upper bounds. A successful experiment should also account for the cost of finding the retained entries rather than reporting sparsity alone. The value bound applies to one read with its values fixed. Assess end-to-end truncation separately, since changes in earlier reads can alter later values and queries.

9.11.3 Local dimension and projection rank

Construct tasks whose relevant local variables lie in known m-dimensional subspaces. Vary the permitted content-projection rank while keeping address metadata separate.

The conditional prediction is that a factorization through the relevant m-dimensional representation permits content rank at most m, with the metadata costs kept separate. This is a sufficiency statement, not an exact rank threshold or a lower bound. Record exact and effective rank separately, and distinguish content-projection rank, score-matrix rank, and attention-matrix rank.

9.11.4 Depth, precision, and restoration

Evaluate known computations over increasing horizons while varying temperature, numerical precision, and the use of discrete restoration.

This separates inaccurate local arithmetic from failure to preserve the state required by later queries.

9.11.5 Learning the decomposition

Finally, compare supplied routing and intermediate supervision with end-to-end training from final input–output pairs.

Success of the supplied construction establishes an implementation baseline, not a result about end-to-end learnability. Report two different outcomes: held-out prediction and, only under a specified correspondence criterion, recovery of the generating computation. An end-to-end learner that generalizes through another feasible representation succeeds at the first even when it fails the second.

To separate supplied from discovered coordinates, also compare frozen representations with trainable representations, using the same readout family and accounted-for computation budgets. Intermediate targets or pretrained features can assist learning without requiring the recovered coordinates to be the generating ones.

For regularized SGD, specify the batch rule, learning-rate schedule, stopping or averaging rule, and the returned model's complexity budget. A small full gradient and small minibatch fluctuation energy are different measurements. Neither convergence to one parameter nor a common zero of all minibatch updates should be used as the definition of predictive success.

9.12 What Remains Open

The main unresolved questions concern useful representation discovery, efficient implementation, and statistical selection. They are not all questions about identifying a supplied teacher graph.

Learning useful coordinates and routing. For which target–distribution families can a specified training procedure construct representations in which useful arguments are retrievable with adequate margins? A theorem must derive an accessible signal for the actual updates, not merely exhibit a favorable final routing pattern. For example, for a scalar attention read $u = \sum_j \alpha_j v_j$, with values held fixed and $\alpha_j = \mathrm{softmax}(a/\tau)_j$, and a differentiable scalar loss, direct differentiation gives

$$\frac{\partial u}{\partial a_j} = \frac{\alpha_j}{\tau}(v_j - u), \qquad \frac{\partial \ell(u,y)}{\partial a_j} = \ell_1'(u,y)\frac{\alpha_j}{\tau}(v_j - u).$$

Here ℓ_1' is the derivative with respect to the prediction. Other paths through values and upstream representations contribute to the full gradient. This calculation illustrates why a retrieval margin and a useful learning direction are different quantities. Chapter 6 develops the learner-relative signal question.

Cheap retrieval rather than sparse weights alone. When can the required reads be found without evaluating every query–key score? A resource theorem must count index construction, updates, memory, and precision as well as the final number of retained entries.

Stable composition. Which target classes admit useful intermediate representations with controlled amplification of routing, local approximation, and numerical errors? Low arity alone does not control the factor A_G. A stable alternative representation can be sufficient; it need not be the one used by the generating process.

Selection of low-rank or sparse implementations. When do low-rank content maps or concentrated reads emerge from a stated objective and training rule? Their existence, their discovery, and any statistical advantage are separate claims. The normalized ReLU rank restriction discussed earlier is not a universal answer for attention.

Predictive learning and generalization. For which specified attention architectures can training return, with high probability and feasible resources, a predictor with good empirical fit and an informative statistical-complexity bound? The bound must cover the learned projections, local modules, scales, and memory conventions, not just the final readout after data-dependent feature selection.

Structural recovery as an additional objective. When scientific interpretation or module-reuse claims require it, ask whether bounded-dimensional learned states can be matched to a specified teacher computation up to controlled equivalences. This is the stronger criterion of Chapter H. Another representation that predicts accurately is not a predictive failure; mere interpolation with poor population accuracy is.

The Gaussian identity, margin inequalities, and constructive realization results answer useful representation and evaluation questions. They do not settle these learning questions by themselves.

9.13 Relation to C1, C2, and C3

The division of labor matches the framework of the preceding chapters, provided implementation is separated from its discovery.

C1 states that an economical compositional realization is available under the stated computational convention. For suitable regular low-arity constituents and stable composition, the approximation exponent can be controlled locally. Section 9.6 then supplies one possible architectural implementation.

C2 concerns useful signals accessible to the specified learner, relative to the data distribution, representation, and update rule. A supplied address margin is not already a C2 theorem: it certifies a read after the addressing has been arranged, not the signal needed to learn that addressing.

C3 supplies intermediate information: true states, partial traces, or useful surrogates can simplify local learning. Knowing graph addresses and observing numerical intermediate targets are different forms of information. The former does not automatically supply the latter. Neither trace exposure nor a staircase is a compulsory learning route.

Thus

C1 : a feasible realization exists,
attention–MLP model : a possible implementation,
C2 : accessible learning signals,
C3 : supplied intermediate information.

The ability to execute a supplied graph must not be mistaken for the ability to learn its structure. Conversely, learning an accurate predictor does not require identification of that graph.

From a returned predictor to population control. The conditional statistical argument is the one in Chapter H, not a consequence of the retrieval bound. Fix an admitted attention-based predictor class and a complexity functional $\mathcal{C}$, and let

$$\mathcal{H}_Q = \{F_\theta : \mathcal{C}(\theta) \le Q\}.$$

The class and budget must be fixed independently of the sample, or covered by an appropriate simultaneous selection bound. Under a valid uniform statistical event for the loss and sampling model, if training returns $\widehat{f} \in \mathcal{H}_Q$ with $\widehat{R}_S(\widehat{f}) \le \varepsilon_{\mathrm{fit}}$, then

$$\boxed{R_P(\widehat{f}) \le \varepsilon_{\mathrm{fit}} + \beta_S(Q, \delta_{\mathrm{stat}}).}$$

Here $\varepsilon_{\mathrm{fit}}$ is an empirical-risk tolerance, distinct from the uniform approximation tolerance ε used earlier. The optimizer must actually achieve the joint fit and complexity requirement; a small norm or a large routing margin alone is not enough. A norm bound for an ordinary homogeneous MLP cannot be transferred unchanged to an unrestricted attention architecture. Learned projections, temperatures or score scales, normalization, context size, and any selected memory must be covered by the chosen analysis.

The output can be a specified stopped iterate of regularized SGD. Positive minibatch covariance at that parameter is compatible with the statistical implication, although it neither proves the required fitting event nor supplies a universal Langevin model. Chapters 7 and H separate these stochastic and predictive claims.

The learned computation is the witness. Once a returned predictor is accurate and has a feasible finite-precision evaluator, its own computation supplies the bounded-fan-in witness of Chapter H. For example, if $\|\widehat{f} - f\|_{L^2(P_X)} \le \varepsilon_{\mathrm{pred}}$ and its numerical evaluator $\widetilde{f}_m$ has uniform error at most 2^{-m}, then

$$\|\widetilde{f}_m - f\|_{L^2(P_X)} \le \varepsilon_{\mathrm{pred}} + 2^{-m}.$$

This is the triangle inequality and the fact that P_X is a probability measure. Feasible bit evaluation gives the computational witness; it does not automatically reveal a fixed smooth coarse-grained DAG or recover the teacher's coordinates. No additional structural matching is needed for the predictive conclusion.

9.14 Conclusion

Attention connects naturally to compositional computation, but the connection is more precise than the statement that every attention head is a universal local kernel approximator.

Under equal key norms, dot-product attention is exactly normalized Gaussian retrieval. With unequal norms, the correspondence requires a key-dependent correction or a more general compatibility interpretation.

Margins give explicit retrieval and truncation bounds. They do not guarantee that training produces sparse attention.

Local nonlinear modules can approximate bounded-arity constituents, while attention routes their inputs. The resulting approximation cost depends on local dimension, graph size, and error propagation.

Polynomial-time finite-precision computation admits a separate attention–MLP construction through Boolean circuits and discrete state restoration. This argument counts precision bits explicitly and does not rely on smooth approximation rates.

The chapter's main message is therefore

compositional sparsity specifies a feasible computation;

attention can route the information needed to execute it.

These are constructive implementation statements. They show how to execute a supplied computation, not that a particular training procedure discovers its addresses or constituents.

The predictive learning question is whether training finds a feasible attention-based function with an adequate empirical fit and statistical control. Learning useful coordinates and retrieval rules is one possible mechanism; supplied intermediate information is another source of help. When training succeeds, an alternative internal representation is acceptable, and persistent regularized SGD fluctuations do not invalidate the returned predictor's guarantee.

Identifying the generating computation remains an important stronger objective. It must not be confused with prediction, or imposed as an additional condition after a feasible predictor has already been shown to generalize.

The next chapter turns from this logical architecture to physical implementation. The resource distinctions established here remain essential: bounded fan-in, concentrated attention, cheap data movement, and an efficient hardware layout are different properties. A hardware advantage needs its own communication, precision, and cost analysis.

CHAPTER 10

Hardware for Compositionally Sparse Computation (with J. Bates)

Efficient digital computation admits a polynomial-size bounded-fan-in realization. Turning that logical description into efficient hardware, however, requires another step: operations must be placed, their arguments transported, intermediate states stored, and numerical errors controlled. A short logical dependency need not be a short physical wire.

This chapter develops Joe Bates's proposal to use a two-dimensional mesh of simple processing elements with local memory, nearest-neighbor communication, and approximate high-dynamic-range arithmetic. The aim is structural alignment*: choose a representation, placement, and schedule that exploit useful locality and reuse. This is a conditional hardware proposal, not a consequence of compositional sparsity that makes one substrate universally optimal.*

We retain the division of labor introduced in Chapter 9: attention can route arguments, and local nonlinear modules can evaluate their constituents. Here routing has an explicit cost. The relevant questions concern computation, communication, storage, and accuracy together*. We give a scoped scheduling result, explain the role and limits of geometric locality, and derive an error budget for approximate attention. The mathematical bounds, proposed implementations, and empirical performance questions have different status.*

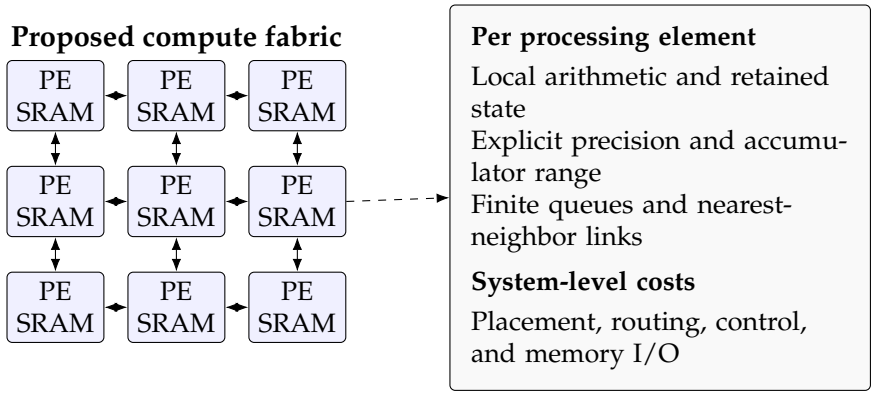


Figure 10.1: The proposed mesh separates local computation from the movement and retention of its arguments. The links have finite bandwidth; memory and control are not free. This is an architectural schematic, not a measured chip layout or a claim that every target has a local embedding.

10.1 The Structural Alignment Argument

Chapter 4 establishes an economical logical realization at the specified precision. It does not identify that realization with a physical layout. Three notions should be distinguished:

bounded logical fan-in,
short physical communication paths,
low communication volume and congestion.

An elementary operation may have only two parents while their values reside on opposite sides of the chip. Unrestricted fan-out, permitted by the logical circuit model, can also require many physical transmissions.

Nor is dense matrix computation the opposite of compositional computation. A dense matrix product has a bounded-fan-in arithmetic implementation. Systolic arrays explicitly exploit repeated local data movement for such operations [82]; the first TPU's matrix unit is a systolic design [74]. The hardware question is not whether to choose composition *instead of* matrix multiplication, but which dataflow and memory organization best implement the selected workload. Dense and sparse kernels can both belong in the same system.

The proposal is to place frequently interacting computations and values near one another, retaining reused data locally whenever the memory budget permits. Work on Eyeriss demonstrates the value of data reuse and movement-aware scheduling for its convolutional workloads [28]. That evidence motivates the design principle; it is not a theorem that arbitrary Transformer computations have the same locality or energy advantage.

Structural alignment is therefore an *embedding and scheduling* problem, not generally a graph isomorphism. A logical edge can become a multihop route, several logical nodes can share one processing element at different times, and replication can exchange memory capacity for communication. These choices must preserve the required computation or incur an explicitly bounded approximation error.

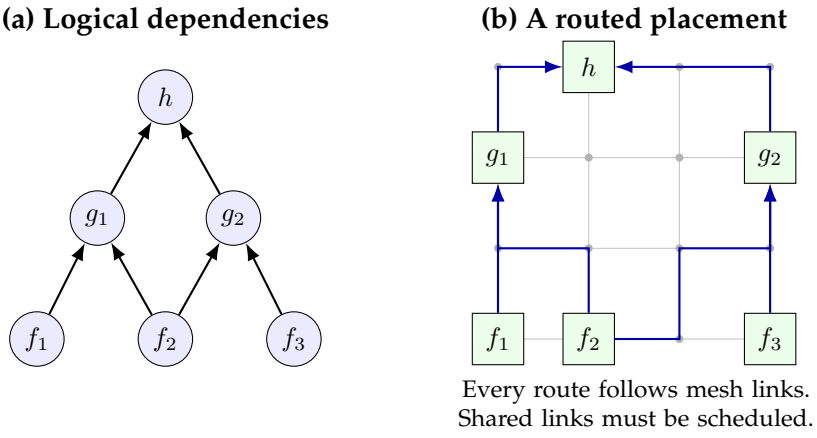


Figure 10.2: The same six logical computations admit a mesh placement with nearest-neighbor *hops*. Some dependencies need several hops and some routes share links. The drawing exhibits one placement; it does not prove that every bounded-fan-in DAG has short routes or low congestion.

10.2 System and Numeric Model

Compute fabric. Assume a square mesh with $P = q^2$ processing elements (PEs). Each PE has local arithmetic, S words of usable storage, and links to its existing north, east, south, and west neighbors. The reference routing model transmits one b-bit packet per physical link per cycle, with buffering and arbitration specified below. A different link width or clock changes the capacity and must be included in the model. Vector messages consist of multiple packets, not one unit-cost scalar read.

Storage includes values, parameters, addresses, queues, and any optimizer state. Off-chip transfers are counted whenever these do not fit. Broadcast control, synchronization, and loading the initial state also cost resources. The model is a proposed programmable spatial fabric; it is not claimed to reproduce every detail of an existing chip.

Approximate high-dynamic-range arithmetic. Bates's compact-arithmetic proposal provides a concrete motivation [13]. The S1 system described by Alexandra Bates and Joseph Bates uses a nearest-neighbor mesh with local memory, exact integer operations, and approximate operations on logarithmically represented high-dynamic-range values [14]. Its control and reduction paths also matter; the presence of local links does not eliminate all global communication.

For nonzero values, a logarithmic representation stores the sign and an approximation to $\log_2 |x|$. Multiplication uses addition of logarithms; zero needs a separate representation. Addition is not equally simple: for $x, y > 0$ with $a = \log_2 x \geq b' = \log_2 y$,

$$\log_2(x + y) = a + \log_2(1 + 2^{b'-a}).$$

Opposite signs introduce cancellation and require separate handling. Lookup tables, correction logic, accumulation precision, conversion, and range checks remain part of the implementation. Savings in the multiplier alone do not establish savings for the complete PE or workload.

The approximately 1% arithmetic reported for the S1 is a property of that numerical implementation, not a universal error tolerance for deep learning [14]. Its reported model experiments concern lattice spin systems. Mixed-precision neural training instead demonstrates success with particular formats and safeguards, including higher-precision weight accumulation and loss scaling [111]. The appropriate inference and training error budgets must be established separately. Section 10.5 supplies a conditional forward error calculation rather than assuming a task-independent tolerance.

An explicit resource objective. A useful accounting quantity is the number of packet-hops,

$$H = \sum_{\text{transmitted packets } m} |\pi_m|,$$

where π_m is the route of packet m. A simple calibrated energy proxy is

$$E_{\text{model}} = e_{\text{op}} N_{\text{op}} + e_{\text{hop}} H + e_{\text{mem}} N_{\text{mem}} + e_{\text{off}} N_{\text{off}} + E_{\text{control}}. \tag{10.1}$$

The coefficients depend on the technology, format, and operation mix; they are not universal physical constants. Latency, peak memory, and energy are different objectives. Removing arithmetic while increasing traffic or queueing need not improve the system.

10.3 Mapping Compositional DAGs to 2D Meshes

Let $G = (V, E)$ be a specified finite computation DAG of depth L, with W_ℓ operations at level ℓ and fan-in at most k. Levels are topological: a parent may belong to *any* earlier level. Its value must remain available until consumed. Operations can be individual arithmetic steps or explicitly costed modules; their complexity must not be hidden in a unit-cost node.

Fix a placement and a schedule. For each level ℓ, let $\mathcal{P}_\ell$ be the packets that supply its remote arguments, with all source values available at the beginning of that communication phase. A dependency carrying w words contributes w packets; repeated consumers contribute separate transmissions unless a multicast or replication scheme is specified. For fixed simple routes, define

$$D_\ell = \max_{m \in \mathcal{P}_\ell} |\pi_m|, \qquad C_\ell = \max_{e \text{ a mesh link}} \#\{m \in \mathcal{P}_\ell : e \in \pi_m\}. \tag{10.2}$$

These are the route *dilation* and packet *congestion*. Put both equal to zero when no communication is required. Let A_ℓ bound the maximum serial local work assigned to one PE after the arguments arrive, including local memory access. Load balance is encoded in A_ℓ, not in W_ℓ/P alone.

Proposition 10 (A conditional two-dimensional schedule)**.** *In the unit-packet store-and-forward model, assume that the required local storage and the queues of a valid packet-routing schedule are available. For the fixed paths in* (10.2)*, there exists a schedule with*

$$T_\ell \le A_\ell + c(C_\ell + D_\ell), \qquad T_{\text{eval}} \le T_{\text{setup}} + T_{\text{io}} + \sum_{\ell=1}^{L} [A_\ell + c(C_\ell + D_\ell)], \tag{10.3}$$

where c is a universal constant for this routing model. Here T_{setup} includes schedule preparation and control, and T_{io} bounds loading, final output, and any external transfers or remapping not counted in the level phases. Every schedule that sends these packets along these paths has communication latency at least $\max\{C_\ell, D_\ell\}$.

Proof. The packet-routing theorem of Leighton, Maggs, and Rao gives a store-and-forward schedule of length $O(C_\ell + D_\ell)$ for a fixed family of edge-simple packet paths [**leighton1994packet**]. Invoke that result for the communication phase, then perform the local work in parallel across PEs. Serializing these phases and the levels gives the upper bound. A longest path requires at least D_ℓ hops, and a most-loaded link requires at least C_ℓ packet transmissions. The unit link capacity gives both lower bounds. □

This is an existence statement using a specified routing theorem, not a claim about every greedy router, finite-buffer policy, or SIMD instruction schedule. Producing and storing the schedule, meeting its buffer needs, and implementing arbitration must also be feasible. Data-dependent attention routes require an online method or a separately counted schedule update. Multicast trees need their own analysis; the proposition counts separate unicast packets.

Bounded fan-in limits the number of parent dependencies but bounds neither D_ℓ nor C_ℓ by a constant. If all endpoints lie in a mesh square of side q_ℓ and Manhattan shortest paths are used, then $D_\ell \le 2(q_\ell - 1)$. Obtaining $q_\ell = O(\sqrt{W_\ell})$ requires that *all needed parents, retained state, and destinations* fit in that region. A placement heuristic alone does not establish it.

A positive local example. For a two-dimensional nearest-neighbor stencil, place each logical site at the corresponding PE and retain its fixed-size state there. One update uses a fixed number of words from neighboring sites and a fixed amount of local arithmetic. Direct one-hop routes give $C_\ell, D_\ell, A_\ell = O(1)$, so a synchronous update takes $O(1)$ cycles with $O(P)$ work and packet-hops,

once the state is loaded. Repeating L updates costs $O(L)$ cycles in this model. This is a genuine locality-based advantage in parallel execution; it does not make loading, output, or a comparison with another equally parallel implementation free.

Bounded treewidth does not imply constant dilation. A short counterexample clarifies the limitation. Consider a complete binary reduction tree of height $h \geq 1$, with $n = 2^{h+1} - 1$ vertices. Its underlying undirected graph has treewidth one, and every operation has fan-in at most two. Place all vertices on distinct mesh sites. If every logical edge has a physical route of length at most D, then every vertex lies within Manhattan distance hD of the root's site. That ball is contained in a square with at most $(2hD + 1)^2$ sites. Consequently,

$$n \leq (2hD + 1)^2, \qquad D \geq \frac{\sqrt{n} - 1}{2h} = \Omega\left(\frac{\sqrt{n}}{\log n}\right). \tag{10.4}$$

This refutes a constant-dilation conclusion for such static injective placements, even with bounded degree. It does not rule out efficient *time-multiplexed* tree evaluation. Placement, storage, replication, and evaluation time can be traded against one another.

Constraint: hierarchical locality and Rent's rule. Rent's rule describes a pin-count relationship for partitions of a logic graph [84]. Write its empirical form as

$$T(g) \asymp a g^{\rho}, \tag{10.5}$$

where g is the number of logical blocks in a cluster and $T(g)$ counts its external terminals. The exponent depends on the graph and partitioning convention; it is not an invariant supplied by efficient computability. Terminal count also differs from traffic per unit time.

The planar boundary constraint can be made elementary. A square cluster of $g = u^2$ mesh sites has at most $4u$ outgoing boundary links. If $V_{\text{out}}(g)$ distinct word-packets must cross that boundary in a phase, and each link carries at most β packets per cycle, then

$$T_{\text{boundary}}(g) \geq \frac{V_{\text{out}}(g)}{4\beta\sqrt{g}}. \tag{10.6}$$

This follows by counting boundary transmissions. It assumes those messages really must cross; alternative computation, aggregation, or replication can change the demand.

Under constant occupancy and one required independent word per active terminal in each phase, a demand $V_{\text{out}}(g) \asymp g^{\rho}$ therefore gives $T_{\text{boundary}}(g) = \Omega(g^{\rho-1/2}/\beta)$. In this particular constant-bandwidth model, $\rho \leq 1/2$ is necessary for a constant-time boundary phase. It is not a sufficient embedding theorem, nor a necessary condition for every useful mesh implementation: a design may tolerate growing latency, change occupancy, or reuse computations over time.

The draft's expander warning has the same interpretation. If a bounded-degree logical graph on $P = q^2$ vertices has at least aP edges crossing every balanced vertex cut, then any injective placement onto a fully occupied $q \times q$ mesh sends at least aP logical edge routes across a physical bisection with only $O(q)$ links. The aggregate path congestion is $\Omega(\sqrt{P})$. This is a routing-demand obstruction, not a lower bound for every alternative algorithm computing the same function. Across a whole DAG those messages can occur in different phases.

MLP modules. A Transformer feed-forward sublayer is local in *token position*, but its affine maps can be dense in channels. Bounded-fan-in reduction trees implement those sums; they do not make the displayed weight matrices sparse. A block multiplication still needs input dissemination, partial-sum reduction, and weight storage. For tiled $y = Wx$,

$$y_i = \sum_j W_{ij} x_j.$$

A weight-stationary layout can reuse W_{ij}, while the x_j blocks and partial results move. Actual block sparsity or locality can reduce these transfers, but must be imposed, measured, or derived for the selected representation.

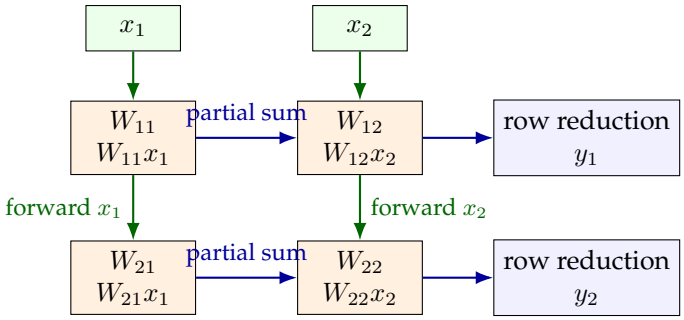


Figure 10.3: A weight-stationary block dataflow for two block rows and columns. Input blocks are forwarded and partial sums reduced; each drawn segment may represent several mesh links. Tiling organizes communication rather than eliminating it. A physical schedule must account for collisions, buffer space, reduction work, and any repeated passes.

Locally reused modules on overlapping spatial scopes offer the proposed *sliding compositionality*. Convolutional dataflow provides a useful precedent [28], not evidence that every Transformer MLP has such a structure. Reuse is additional to low arity, and semantic proximity is not automatically mesh proximity.

10.4 Sparse Attention on 2D Meshes

We retain the two proposed dataflows, KV-stationary and Q-stationary. Let N be the number of available key–value records, d_k the key dimension, and s the retained support per query. With scores $a_j = q^\top k_j$, temperature $\tau > 0$, and full weights $\alpha_j = e^{a_j/\tau} / \sum_i e^{a_i/\tau}$, a retained set S loses mass

$$t_S = \sum_{j \notin S} \alpha_j.$$

If $\|v_j\| \le R$ and retained weights are renormalized, the fixed-read error is at most $2Rt_S$, as in Chapter 9. Small support is useful only when its discovery and physical communication are also inexpensive. A sparse set of distant destinations can still be costly.

10.4.1 KV-Stationary: Coarse-to-Fine Attention

Retain keys and values near their owning PEs and route queries to selected candidate buckets. A possible coarse code is

$$c(q) = \operatorname{sign}(Bq), \qquad B \in \mathbb{R}^{r \times d_k}.$$

This is a short binary hash, not a proof of low-rank attention or semantic equivalence. Locality-sensitive hashing in Reformer motivates such candidate selection [78]; its algorithmic guarantees do not transfer automatically to this proposed mesh implementation.

The coarse stage produces a candidate set $\mathcal{C}(q)$. The fine stage scores its keys, selects up to s records, and normalizes and mixes the retained values. Its direct score work is $O(|\mathcal{C}(q)|d_k)$ per query, in addition to code construction, index lookup, selection, and value mixing. The index must be built and, when keys change, maintained. Bucket imbalance, misses, replication, and route lengths enter the cost.

This can avoid a full scan when the index is selective and accurate. It need not find the true top-s entries, and finding those entries would still not bound the mass of the omitted softmax tail without an additional assumption. Measure or certify t_S, not just recall of a few large scores. Searching every bucket as a fallback recovers a dense-search cost. No global broadcast is required by the description, but the remaining query and result traffic can still cross the entire mesh.

10.4.2 Q-Stationary: Systolic Streaming with Early Exit

Keep a block of queries at its assigned PEs and stream key–value blocks past it, reusing each block across queries. Exact blockwise softmax can maintain a running maximum, normalizing sum, and weighted value sum without storing the full attention matrix. FlashAttention demonstrates the value of tiling and memory-I/O accounting for exact attention [33]. Its dense algorithm does *not* remove the dense score work by top-s pruning. Early exit is a separate proposed extension here.

For example, suppose a key bucket has center c and a certified radius $\|k - c\|_2 \le r_c$. Cauchy–Schwarz gives

$$a_j = q^\top k_j \le q^\top c + \|q\|_2 r_c =: U_c(q).$$

Once s evaluated keys have minimum retained score $a_{\min}$, a bucket with $U_c(q) < a_{\min}$ cannot contain a higher-scoring key. This is a valid top-s pruning test if the bound includes numerical error and tie handling. Its benefit depends on cheap, tight metadata. It is not a guarantee that most buckets can be skipped. A value-error certificate additionally needs a bound on omitted exponential mass; many individually small weights can have a large sum.

The streaming schedule still pays for fetching each visited block, moving it to the relevant queries, merging partial results, and returning outputs. Top-s heaps reduce retained storage but not the cost of a scan that evaluates every key. The roles of sparsity, data reuse, and avoiding materialization of the attention matrix should therefore remain separate.

Speculative opportunity. A learned relevance relation may align with stable buckets or repeated communication patterns. Structured sparse attention offers architectural examples [30], but sparsity in token indices is not itself a short-wire layout. Jointly learning a useful representation and its placement is a research direction. The representation may differ from the generating parse or program tree; predictive success does not require structural identification. The cost and stability of repartitioning must be counted when the access pattern changes.

10.5 Attention Quantization and Error Analysis

The numerical design must specify the operation, domain, approximation, and accumulation precision. A blockwise softmax using a subtracted maximum and lookup-table exponentials is one candidate. The proposed "tempered max-mix" or "softmax-lite" is another only after its weights, normalization, and approximation error have been defined. ReLU and shifts alone do not establish

a softmax error bound. Block floating point, integer lookup, and logarithmic arithmetic are different numerical choices, not synonymous algorithms.

Quantization and compression have empirical precedents [40, 57], but a bit width by itself does not guarantee accuracy. The following elementary calculation makes a possible hardware error budget explicit. It concerns one fixed read; input-dependent routing and whole-network accuracy require the subsequent composition argument.

Score, value, and weight errors. Let S be any nonempty retained set and define

$$p_j = \frac{e^{a_j/\tau}}{\sum_{i\in S} e^{a_i/\tau}}, \qquad p'_j = \frac{e^{\widehat{a}_j/\tau}}{\sum_{i\in S} e^{\widehat{a}_i/\tau}}, \qquad j \in S.$$

Assume $|\widehat{a}_j - a_j| \le \delta_a$, $\|\widehat{v}_j - v_j\| \le \delta_v$, and $\|v_j\| \le R$. Let the implemented weights $\widehat{p}$ satisfy $\|\widehat{p} - p'\|_1 \le \delta_w$, with final accumulation error at most δ_{acc}. The δ_w term includes exponential approximation, normalization, and weight rounding; exact normalization of $\widehat{p}$ is not assumed.

Proposition 11 (A fixed-read numerical error budget)**.** *With the assumptions above, the hardware output $\widehat{y} = \sum_{j\in S} \widehat{p}_j \widehat{v}_j + e_{\mathrm{acc}}$ and the full exact read $y = \sum_{j=1}^{N} \alpha_j v_j$ satisfy*

$$\boxed{\begin{aligned}\|\widehat{y} - y\| \le 2Rt_S + 2R\min\{1, \delta_a/\tau\} + \delta_v \\ + (R + \delta_v)\delta_w + \delta_{\mathrm{acc}}.\end{aligned}} \tag{10.7}$$

Proof. First truncate and renormalize the exact weights. The difference is at most $2Rt_S$. For $b = \widehat{a} - a$ restricted to S, differentiate $p(u) = \mathrm{softmax}((a + ub)/\tau)$:

$$\frac{d}{du} p_j(u) = \frac{p_j(u)}{\tau}\left(b_j - \sum_i p_i(u) b_i\right).$$

Summing absolute values gives $\|\frac{d}{du} p(u)\|_1 \le 2\delta_a/\tau$. Integration from zero to one, and the bound two on the distance between probability vectors, give $\|p' - p\|_1 \le 2\min\{1, \delta_a/\tau\}$.

Changing the scores therefore changes the value sum by at most $2R\min\{1, \delta_a/\tau\}$. Changing the values under the probability weights p' contributes at most δ_v. Changing p' to $\widehat{p}$ contributes at most $(R + \delta_v)\delta_w$, and accumulation adds δ_{acc}. The triangle inequality proves the result. □

A smaller temperature can sharpen retrieval, but also increases sensitivity to score error in this bound. If an exact winning score has margin γ and each score perturbation is at most δ_a, its perturbed margin is at least $\gamma - 2\delta_a$. A positive numerical margin needs $2\delta_a < \gamma$; subsequent truncation or threshold tests must use the remaining margin.

How bit precision enters. Nearest rounding to 2^b equally spaced levels in $[-M, M]$, with no clipping, has absolute coordinate error at most

$$\epsilon_b = \frac{M}{2^b - 1}.$$

For a vector of dimension d_k, the Euclidean error is at most $\sqrt{d_k}\epsilon_b$. If $\|q\|_2 \le Q$, $\|k_j\|_2 \le K$, $\|\widehat{q} - q\|_2 \le \epsilon_q$, and $\|\widehat{k}_j - k_j\|_2 \le \epsilon_k$, expansion of the dot product gives

$$\delta_a \le K\epsilon_q + Q\epsilon_k + \epsilon_q\epsilon_k + \delta_{\mathrm{dot}}, \tag{10.8}$$

where δ_{dot} accounts for product and accumulation errors. This connects a specified quantizer and score implementation to (10.7). Overflow, clipping, exponent-range limits, and changing scale factors require additional terms or explicit exclusion. A relative-error specification for stored operands or products does not by itself control the relative error of a nearly cancelling sum; absolute error and accumulator range must also be specified.

Composition and numerical state retention. Let H_ℓ be the ideal stored-state update at stage ℓ, including any retained coordinates, with Lipschitz constant Λ_ℓ in a fixed state norm. Suppose the hardware update differs from H_ℓ by at most δ_ℓ on every reachable state in a common domain, and both computations remain in that domain. Then

$$E_\ell \le \Lambda_\ell E_{\ell-1} + \delta_\ell, \qquad E_L \le E_0 \prod_{j=1}^{L} \Lambda_j + \sum_{\ell=1}^{L} \delta_\ell \prod_{j=\ell+1}^{L} \Lambda_j. \tag{10.9}$$

The first inequality follows by adding and subtracting H_ℓ evaluated at the hardware state; induction gives the second. A fixed-read bound alone does not supply these uniform state-domain hypotheses. Routing, arithmetic, and storage errors all belong in δ_ℓ. This is the same local-to-global requirement used in Chapter 9.

If a learned predictor has population error $\|F - f^\star\|_{L^2(\mathsf{P}_X)} \le \varepsilon_{\mathrm{learn}}$ and its hardware realization has uniform error at most $\varepsilon_{\mathrm{hw}}$ on that population's support, then

$$\|F_{\mathrm{hw}} - f^\star\|_{L^2(\mathsf{P}_X)} \le \varepsilon_{\mathrm{learn}} + \varepsilon_{\mathrm{hw}}. \tag{10.10}$$

Thus hardware accuracy consumes part of the predictive error budget; it is not an automatic benefit of sparsity or norm regularization.

Inference and training are different claims. For training, backward reductions, small updates, and optimizer state need separate numerical control. Approximate hardware can introduce systematic bias as well as random error. It must not be identified with intrinsic minibatch noise. Chapters 7 and H, together with Appendix N, explain why a regularized stationary point may already have nonzero sampling covariance. That fact does not license additional uncontrolled arithmetic error. A training claim must concern the actual implemented updates and the fit, statistical complexity, and accuracy of the returned predictor.

10.6 Discussion: Why 2D Meshes Help

The proposal has a positive, testable form. A mesh is attractive when frequently interacting state fits nearby, routes have controlled dilation and congestion, data reuse amortizes loading costs, and the chosen numerical format meets the accuracy requirement. Regular kernels can then sustain parallel local work, and sparse or indexed reads can reduce unnecessary traffic. None of these benefits requires recovering the teacher's particular decomposition: the relevant object is the computation of the selected accurate predictor.

Conversely, few logical parents, a small norm product, or a top-s mask does not by itself establish such a mapping. Remote accesses, hot buckets, repeated reductions, buffer spills, and index maintenance can dominate. The relevant comparison includes a well-tiled dense baseline, rather than an unnecessarily materialized dense implementation. A hybrid of local approximate PEs, matrix kernels, and selected higher-precision operations may be preferable to forcing every operation into one numerical format.

A scoped locality hypothesis. A worthwhile conjecture is that *specified workload families* admit accurate representations and implementable placements with sufficiently small communication and memory costs to make mesh execution competitive. For any such family, one must fix the accuracy, available area and memory, precision, allowed replication, and latency or throughput target before making the claim. The family should be described independently of whether a placement heuristic happens to succeed. Efficient computability alone neither supplies a universal Rent exponent nor proves superiority to another physical architecture.

Experiments needed to evaluate the proposal. First separate topology from arithmetic. Run the same reference operations at the same precision on local-stencil, tree, and nonlocal sparse DAGs; compare placements and record A_ℓ, C_ℓ, D_ℓ, packet-hops, peak storage, and off-chip traffic. This tests the actual locality premise, not just the count of graph edges.

Next compare dense streaming, supplied sparse routing, and learned or hashed candidate selection on the same attention tasks. Record index construction and update cost, bucket imbalance, communication distance, discarded mass, read error, and end-to-end latency. Ablate quantization separately from sparsification. Precision sweeps should track score margins, accumulation error, intermediate-state drift, and held-out prediction, not merely the fraction of retained keys. Training experiments must additionally measure update bias and convergence or stopping behavior.

Finally report energy at a declared measurement boundary, including memory and control, with matched accuracy and workload. Separate measured silicon results, simulation, and estimates from (10.1). No new benchmark results or universal speedup factors are claimed in this chapter. These tests distinguish whether savings arise from a better representation, better placement, less data movement, cheaper arithmetic, or some combination.

10.7 Conclusion

Compositional sparsity supplies a logical computation that can be implemented. It does not determine the best physical implementation. The additional opportunity is to exploit the locality and reuse of a chosen accurate realization with an explicitly costed mesh, memory organization, and numerical format.

$$\begin{array}{c}\text{an economical computational realization}\\ + \text{ a feasible placement and communication schedule}\\ + \text{ controlled numerical error}\\ \Longrightarrow \text{ a costed, accurate hardware implementation.}\end{array}$$

The scheduling proposition is conditional on paths, packet capacities, and storage. The planar cut and tree examples identify what low arity alone does not guarantee. The attention bound separates support selection, score precision, value precision, normalization, and accumulation.

Bates's mesh-and-approximate-arithmetic proposal remains a substantive research direction. Its value is to make local computation and data movement explicit, and to encourage joint design of algorithms, representations, precision, and layout. A performance advantage requires those resource bounds and a fair empirical comparison. Learning the predictor, compiling it, and identifying the generating computation remain distinct tasks; an effective hardware realization need only preserve the predictive and computational guarantees actually being claimed.

CHAPTER 11

Stepwise Trace Supervision: A Common Computational Principle for Autoregressive and Denoising Models

A complex computation can become easy to learn in an appropriate local representation when sufficiently informative intermediate states are exposed. This chapter gives a concrete regime in which that possibility is a theorem: a fixed finite-alphabet transition rule is learned by a specified table learner and then reused.

Consider a polynomial-time computation generating a sequence of states

$$s_0, s_1, \ldots, s_T.$$

The end-to-end map

$$s_0 \longmapsto s_T$$

may be computationally deep. Nevertheless, for a Turing computation, the transition from one configuration to the next is local: each cell at time $t+1$ depends on only a constant-size neighborhood at time t.

If the local encoding and neighborhood structure are supplied, and the examples cover the transitions required at execution time, the learner can recover the local rule on those neighborhoods and reproduce the global computation by iteration.

The central principle is therefore

complex global computation
$\Downarrow$
sequence of simple local transitions
$\Downarrow$
expose enough intermediate structure
$\Downarrow$
learn the local transition
$\Downarrow$
iterate it.

This is a strong trace-exposure regime C3 of Chapter H. It is a constructive sufficient route to learning, not a necessary trajectory for all successful predictors. Here the supplied state semantics allow exact reproduction of a particular computation, a stronger conclusion than population prediction alone.

An important distinction will be made throughout the chapter:

$$\boxed{\textit{exposing trajectories} \neq \textit{providing the complete local transition table.}}$$

A finite set of trajectories may fail to visit local configurations that will later occur on new inputs. Thus trace supervision removes the need to infer hidden intermediate states, but does not automatically remove the problem of local-state coverage.

The chapter also clarifies a structural connection with autoregressive and denoising models. Autoregressive systems advance through an indexed trajectory. Diffusion models construct supervised prediction problems at many noise levels. Both exploit intermediate states, but ordinary language tokens and ordinary diffusion states are not, in general, faithful traces of an underlying Turing computation.

The result is therefore a constructive theorem about the stated form of C3, not a theorem that existing autoregressive or diffusion training automatically satisfies its hypotheses. The analogy concerns intermediate supervision; low arity, complete information, and learnability must still be checked in each model.

11.1 Introduction

Autoregressive language models and diffusion models appear very different.

An autoregressive model predicts a later symbol from an earlier context,

$$x_{\leq t} \longmapsto x_{t+1}.$$

A diffusion model instead learns noise-level-dependent statistical information and uses it to construct a reverse-time generative process [**song2021scorebased**, 68]. Here "local" can mean local in the evolution index; it does not imply dependence on only a few input coordinates.

Both therefore replace one global transformation by a sequence or family of intermediate prediction problems.

The question of this chapter is narrower and more mathematical:

> How powerful can learning become if the intermediate states of a computation itself are supplied?

For polynomial-time Turing computation, the answer is particularly clean.

A Turing computation can be written as repeated application of a fixed constant-arity local rule.

If that local rule is fully specified by the training data, it can be recovered from a finite table and then iterated for polynomially many steps.

The load-bearing assumption is therefore

$$\boxed{\text{the relevant local computational transitions are exposed.}}$$

This is much stronger than ordinary end-to-end supervision

$$(x, f(x)).$$

It is an instance of C3 rather than a theorem about black-box C2 recovery.

The constructive learner below does not optimize a jointly hidden deep network: it records observed categorical transitions. Consequently, the result establishes one algorithmic route around

hidden-state discovery; it is not a convergence theorem for arbitrary gradient-based training. The distinction between supplied and learned coordinates is the one used in Chapters 5 and H.

11.2 Preliminaries

11.2.1 Polynomial-time computation

For input lengths $n \geq 1$, fix a deterministic, single-tape Turing machine M with finite input, tape, and control alphabets. It computes a family

$$f_n : \Sigma^n \longrightarrow \Sigma^{m(n)}$$

within a supplied polynomial time bound $T(n) \geq 1$; below we write f when the input length is understood. Variable-length output can instead be represented with a delimiter and padding. Initialization and output decoding are specified parts of the computation.

We use a two-sided tape, moves in $\{-1, 0, +1\}$, and an absorbing halting state: after halting, the configuration remains unchanged. Thus every execution may be padded to exactly $T(n)$ steps. A different standard machine model can first be simulated in this one, with its simulation overhead included in T. The constants below may depend on the fixed machine; they are not uniform constants over arbitrarily growing machine descriptions.

Let

$$\tau_t$$

denote the machine configuration after t steps.

A configuration contains

- the tape contents,
- the head position,
- the finite internal state.

A computation lasting $T(n)$ steps can visit only

$$O(n + T(n))$$

tape positions.

For example, place the input in cells $1, \ldots, n$ and the initial head at cell 1, and retain the window from $-T(n) - 1$ to $n + T(n) + 1$. It has $S(n) = n + 2T(n) + 3$ cells, contains the entire computation, and leaves blank padding beyond every possible head position. After relabeling these cells by $1, \ldots, S$, exterior ghost cells may be held blank. Fixed blank boundary values are justified by this padding, not by permitting the head to cross a truncated tape.

11.2.2 A local alphabet

Let Γ be the tape alphabet and Q the finite set of control states.

One convenient augmented alphabet is

$$\mathcal{A} = \Gamma \times (Q \cup \{\varnothing\}),$$

where $\varnothing$ means that the head is absent. A valid machine configuration has exactly one head-bearing cell. The local encoding, cell order, input encoder, and output decoder are supplied to the learner; they are not inferred by the table-learning algorithm.

A symbol of $\mathcal{A}$ records

$$\text{tape symbol} + \text{head-presence flag} + \text{machine state when the head is present.}$$

For a fixed machine M,

$$|\mathcal{A}| = O(1).$$

Let

$$S(n) = O(n + T(n))$$

be the number of represented tape cells.

The complete state is therefore

$$s_t = (s_{t,1}, \ldots, s_{t,S}) \in \mathcal{A}^S. \tag{11.1}$$

A one-hot encoding of every cell has total dimension

$$D(n) = S(n)|\mathcal{A}| = \text{poly}(n).$$

11.2.3 Three forms of supervision

We distinguish three levels.

End-to-end supervision. Only

$$(x, f(x))$$

is observed.

Full trajectory supervision. For each training input, the complete state sequence

$$s_0, s_1, \ldots, s_T$$

is observed.

Complete local supervision. The learner is supplied with the value of the local transition on every local state that may be needed:

$$(u, h(u)).$$

The strongest theorem below uses complete local supervision.

Observed trajectories provide such supervision only for the local patterns that those trajectories actually visit.

That distinction will lead to a separate coverage theorem.

Noiseless traces generated by a deterministic rule give consistent labels for repeated occurrences of the same neighborhood. Corrupted labels, unknown cell correspondences, and incomplete observation of the machine state require additional estimation or identification assumptions. A worked solution or a visible token sequence need not contain this entire state representation.

11.3 Turing Computation Is a Local Space–Time Rule

The central fact is the locality of one Turing transition.

Lemma 4 (Locality of a Turing transition)**.** *For a fixed deterministic Turing machine M under the preceding conventions, there exists a function*

$$h_M : \mathcal{A}^3 \to \mathcal{A}$$

such that, for every nonboundary cell,

$$s_{t+1,i} = h_M\left(s_{t,i-1}, s_{t,i}, s_{t,i+1}\right). \tag{11.2}$$

The assertion concerns valid configurations during the prescribed execution horizon. Boundary neighborhoods use the padded ghost cells specified above.

Proof. A cell can change at the next Turing step only through information contained in itself and its two nearest neighbors.

The head may currently occupy the cell and write a symbol.

It may occupy the left neighbor and move right.

It may occupy the right neighbor and move left.

Otherwise the tape symbol is unchanged and no head enters the cell.

The finite transition table of M determines every case.

If the head stays at the central cell, its updated state and symbol are recorded there. If it leaves, the written symbol remains but the head flag is removed. An entering head brings the control state determined by the neighbor's transition. The absorbing halting convention is also local. Triples containing inconsistent multiple heads never occur in a valid trajectory; assign their outputs arbitrarily to extend the rule to all of $\mathcal{A}^3$.

Hence a fixed map

$$h_M : \mathcal{A}^3 \to \mathcal{A}$$

exists. □

The complete global transition

$$H_M : \mathcal{A}^S \to \mathcal{A}^S$$

is obtained by applying h_M in parallel across the tape.

All neighborhoods are read from the old configuration before any next configuration is used. A sequential implementation therefore uses two buffers, or an equivalent snapshot discipline; an arbitrary in-place left-to-right update would generally define a different process.

Thus

$$s_{t+1} = H_M(s_t).$$

The structural point is

global computational depth is generated by iteration of a constant-arity local rule.

11.4 Learning the Local Rule

Since

$$|\mathcal{A}| = O(1),$$

the number of possible three-cell neighborhoods is

$$|\mathcal{A}|^3 = O(1)$$

for a fixed machine.

Let

$$\mathcal{U}_M \subseteq \mathcal{A}^3$$

denote the set of local neighborhoods that may occur in valid computations of M.

Define the complete local table

$$\mathcal{D}_M^{\text{complete}} = \{(u, h_M(u)) : u \in \mathcal{U}_M\}. \tag{11.3}$$

Its size depends on M but not on the input length n. The set $\mathcal{U}_M$ is a semantic definition of the patterns relevant to the claimed domain of executions, not an assertion that global reachability can be decided efficiently. A supplier can instead provide the whole finite table on $\mathcal{A}^3$, choosing arbitrary outputs on invalid patterns. The learner need not solve a reachability problem.

The complete-table assumption supplies substantial information about the rule. If M itself is supplied, its local table can be compiled directly; if it is not supplied, the labeled table or observed traces supply that information. The theorem does not recover an unknown program from a constant number of terminal input–output examples.

11.4.1 Constant-size realization

Because h_M has finite domain, it admits a constant-size Boolean circuit.

Equivalently, lift each neighborhood to a one-hot vector

$$\psi(u) \in \{0,1\}^{|\mathcal{U}_M|}.$$

Each output coordinate can then be expressed as an OR of indicators corresponding to the patterns yielding that output.

Thus the local rule can be realized by a constant-size threshold or Boolean network.

No claim is required that the raw local map itself be linearly separable.

A specified learner. Use a dictionary indexed by the a^3 possible triples, where $a = |\mathcal{A}|$. On each labeled example (u, v), store v in entry u; reject inconsistent labels in the noiseless model. At execution, an unfilled entry produces an explicit failure symbol $\perp$. Reading all K distinct entries takes $O(K)$ fixed-alphabet operations, and storage is $O(a^3 \log a)$ bits. Both are constant with respect to n for fixed M, not necessarily small for a large machine.

The mathematical lift ψ above is restricted to $\mathcal{U}_M$. An implementation may index all of $\mathcal{A}^3$ instead, so that the index set does not presuppose which patterns are reachable. On one-hot symbol blocks $v^{(1)}, v^{(2)}, v^{(3)}$, the indicator of a particular triple (a_1, a_2, a_3) has the exact ReLU realization

$$\mathrm{ReLU}\Big(v_{a_1}^{(1)} + v_{a_2}^{(2)} + v_{a_3}^{(3)} - 2\Big).$$

A linear readout maps the active triple indicator to the stored one-hot output. Thus the finite rule can also be compiled into a constant-size ReLU network. The unknown quantities are table outputs, not the supplied feature coordinates. This gives an explicit estimator and realization, without assuming that a chosen nonconvex optimizer discovers them.

11.5 Trace-Exposed Universality with Complete Local Supervision

We can now state the cleanest theorem.

Theorem 9 (Local-rule universality)**.** *Let M be a fixed deterministic Turing machine computing the family f_n within the polynomial time bound $T(n)$, with the state encoding, initialization, padding, and output convention specified above.*

Then there exist

1. *a state representation of dimension*
$$D(n) = \mathrm{poly}(n);$$
2. *a constant-arity transition rule*
$$h_M : \mathcal{A}^3 \to \mathcal{A};$$
3. *a finite complete local table*
$$\mathcal{D}_M^{\mathrm{complete}};$$

such that the dictionary learner recovers h_M on every required neighborhood from $\mathcal{D}_M^{\mathrm{complete}}$. Starting from the encoded input, $T(n)$ synchronous iterations of the resulting global update compute f_n exactly.

With $S(n) = O(n + T(n))$, an explicit two-buffer evaluator uses $O(T(n)S(n))$ local-rule applications and $O(S(n))$ working symbols. Its tied transition parameters have constant description length for fixed M. The training-table acquisition cost is not included in the constant cost of reading an already supplied table.

Proof. Lemma 4 gives the local transition.

Because $\mathcal{A}$ is finite and independent of n, the total table has at most a^3 entries. The supplied complete local table labels every entry required on the admitted computations.

The specified dictionary learner copies those labels exactly and is therefore correct on every required local configuration.

Applying that rule simultaneously at all represented tape positions gives

$$s_{t+1} = H_M(s_t).$$

Iteration yields

$$s_T = H_M^T(s_0).$$

The initial state encodes the input and the final state contains the output $f(x)$.

Each round evaluates $S(n)$ constant-size rules and writes the second buffer. Repeating for $T(n)$ rounds gives the stated work and workspace. Unrolling the rounds instead gives $O(T(n)S(n))$ constant-size circuit copies and $O(T(n))$ transition depth. These are polynomial resource bounds, including the prescribed input and output encodings. □

11.5.1 What the theorem proves

The theorem gives

$$\boxed{\begin{array}{c}\text{complete local supervision}\\ \Longrightarrow\ \text{exact local-rule recovery}\\ \Longrightarrow\ \text{exact global execution.}\end{array}}$$

Complete local supervision is a stronger assumption than having a few example trajectories. Extending the guarantee to a learner supplied only with trajectories requires the coverage condition developed next.

The result also reproduces the designated internal configurations. This strong structural conclusion is appropriate because their coordinates are supplied. It is not an additional requirement for the predictive success of a different feasible learner, as emphasized in Chapter H.

11.6 Learning from Observed Traces: The Coverage Problem

Suppose training consists of complete trajectories generated by inputs

$$x^{(1)}, \dots, x^{(N)}.$$

Each trajectory reveals many local training pairs

$$\left((s_{t,i-1}, s_{t,i}, s_{t,i+1}), s_{t+1,i}\right).$$

Let

$$\mathcal{U}_{\text{train}}$$

be the set of distinct local neighborhoods observed anywhere in the training trajectories.

For a new test input x, let

$$\mathcal{U}_{\text{test}}(x)$$

be the set of local neighborhoods encountered during its computation.

Theorem 10 (Exact execution under local-state coverage)**.** *Assume deterministic local dynamics and noiseless observed traces.*

Suppose a learner stores the correct value $h_M(u)$ for every

$$u \in \mathcal{U}_{\text{train}}.$$

For any test input x satisfying

$$\mathcal{U}_{\text{test}}(x) \subseteq \mathcal{U}_{\text{train}}, \tag{11.4}$$

iteration of the learned local rule reproduces the exact Turing computation on x.

Proof. At time zero the simulated state agrees with the true state.

Assume the two states agree at time t.

Every local neighborhood required to compute time $t+1$ belongs to $\mathcal{U}_{\text{test}}(x)$ and therefore, by Equation (11.4), to $\mathcal{U}_{\text{train}}$.

The learned rule is correct on every such neighborhood.
Hence every updated cell is correct.
Induction over time proves exact agreement through the final state. □

This gives an important refinement of C3:

trace exposure removes hidden-state discovery, but not necessarily local-state coverage.

A trajectory can reveal every intermediate state it visits and still fail to reveal a rare local configuration needed on a later input.

There are then two possible routes:

coverage:	$\mathcal{U}_{\text{test}} \subseteq \mathcal{U}_{\text{train}}$,
local-rule inference:	predict unseen entries using further structure.

For the abstaining dictionary learner specified above, coverage is exactly the condition for completing this trace execution without encountering a missing key. It is a sufficient condition for faithful execution, not a necessary condition for every predictor to return the right final output. An estimator that guesses an unseen transition may guess correctly, and different internal trajectories can give the same answer. In the absence of additional restrictions, however, observations do not determine the label of an unobserved table entry.

11.6.1 A finite-sample consequence of trace coverage

Fix an input distribution P_n and a common execution horizon $T(n)$. For a true trace generated by $X \sim P_n$, let $\mathcal{U}(X)$ be the set of neighborhoods used to form its next states, and define

$$p_u = \Pr_{X \sim P_n}\{u \in \mathcal{U}(X)\}, \qquad \mathcal{U}_+ = \{u : p_u > 0\}, \qquad K_+ = |\mathcal{U}_+| \le a^3.$$

The independent sampling units are complete input-generated traces, not the many dependent cell updates within a trace.

Theorem 11 (Population coverage from independent traces)**.** *Let $K_+ \ge 1$, and suppose N noiseless traces are generated by independent inputs from P_n. Train the abstaining dictionary learner on all their local transitions. For $0 < \varepsilon, \delta < 1$, if*

$$N \ge \frac{K_+}{\varepsilon} \log \frac{K_+}{\delta},$$

then, with probability at least $1 - \delta$ over the training traces, its probability of any trace-execution failure on an independent $X \sim P_n$ is at most ε. In particular, the final-output failure probability is at most ε. Replacing K_+ by the known upper bound a^3 is sufficient.

Proof. A given pattern is absent from the N independent traces with probability $(1 - p_u)^N \le e^{-Np_u}$. By a union bound, the probability that some pattern with $p_u \ge \varepsilon/K_+$ is absent is at most $K_+ e^{-N\varepsilon/K_+} \le \delta$.

On the complementary training event, every unseen positive-probability pattern has $p_u < \varepsilon/K_+$. Conditional on that training sample,

$$\Pr_X\{\mathcal{U}(X) \not\subseteq \mathcal{U}_{\text{train}}\} \le \sum_{u \notin \mathcal{U}_{\text{train}}} p_u \le \varepsilon.$$

On every covered test input, Theorem 10 gives exact execution. Counting abstentions as failures proves the result. □

This is an elementary statistical consequence of the chapter's finite-table model, not a bound for arbitrary neural trace predictors. It does not need a lower bound on the rarest pattern when a population failure probability ε is allowed. If instead every pattern in $\mathcal{U}_+$ must be learned, and $p_{\min} = \min_{u \in \mathcal{U}_+} p_u > 0$, the same calculation gives

$$\Pr\{\mathcal{U}_+ \not\subseteq \mathcal{U}_{\text{train}}\} \leq K_+ e^{-N p_{\min}}.$$

Thus $N \geq p_{\min}^{-1} \log(K_+/\delta)$ suffices for complete support coverage; $p_{\min}^{-1}$ can be prohibitive.

Full trace collection and processing still cost up to $O(NT(n)S(n))$ fixed-alphabet symbols or local operations. Counting only distinct table entries would conceal this data-acquisition cost. The probability result is for the same P_n and horizon. New lengths or a changed test distribution need coverage again; the deterministic covered-input theorem can apply across lengths when their required neighborhoods are already known.

11.7 A Universal Interpreter

The preceding theorem uses a machine-specific local rule h_M.

A stronger conceptual statement follows by fixing one universal Turing machine.

Corollary 1 (One fixed local rule is computationally universal)**.** *There exists a finite alphabet $\mathcal{A}_U$ and one fixed local rule*

$$h_U : \mathcal{A}_U^3 \to \mathcal{A}_U$$

with the following property.
For every deterministic Turing machine M and input x, an encoding

$$\langle M, x \rangle$$

can be placed in the initial state of the universal machine U so that iteration of the same rule h_U simulates $M(x)$.

Proof. Choose a fixed universal single-tape interpreter U with the finite encoding conventions used in Chapter 3. Apply Lemma 4 to U itself. Its alphabet and local rule are fixed; the machine description and input are stored on its tape. The resulting local updates simulate the interpreter, which simulates $M(x)$. □

Universality here allows the workspace and execution horizon to grow. For an efficiency claim, choose a polynomial-overhead interpreter and count its runtime as a function of the program length, input length, and simulated runtime. A fixed local rule does not make arbitrary programs polynomial-time, provide their running-time bounds, or solve program search. A fixed-size, fixed-precision state with no extensible workspace would not provide this unbounded computational capacity.

Thus

fixed local interpreter + variable initial program and data $\Longrightarrow$ universal computation.

Once h_U has been learned or supplied, different algorithms require no change in the local transition rule.

Only the externally supplied state changes.

This gives a useful bridge to the later distinction between interpreter and program:

$$\boxed{\text{interpreter} = h_U, \qquad \text{program} = \text{part of } s_0.}$$

It also emphasizes why computational universality alone is not a learning theorem.

The interpreter may be fixed and perfectly known while the problem of finding a useful program remains difficult.

11.8 Architectural Depth versus Computational Time

Trace exposure does not eliminate computational depth.

The specified machine still performs $T(n)$ sequential transitions. This is the cost of that execution, not a lower bound for every algorithm computing the same input–output function.

What changes is where the complexity resides.

Instead of learning one global map

$$s_0 \longmapsto s_T,$$

the learner acquires a reusable transition

$$H_M$$

and iterates it:

$$s_0 \overset{H_M}{\longmapsto} s_1 \overset{H_M}{\longmapsto} \cdots \overset{H_M}{\longmapsto} s_T.$$

Thus

$$\boxed{\begin{array}{c}\text{architectural depth can be traded} \\ \text{for iterative computational time.}\end{array}}$$

The computation is not made shallow.

The *learned rule* is made local and repeatedly reused.

The resource tradeoff is explicit. An unrolled circuit has $O(TS)$ local-rule copies and depth $O(T)$, while a recurrent evaluator reuses one parameterized rule with $O(S)$ state and $O(TS)$ work. With S parallel cell processors it needs T local-update rounds, up to synchronization and implementation costs. Serializing all cell outputs as tokens can require $O(TS)$ emitted tokens. A computational step, a network layer, and a generated token are therefore not interchangeable units of cost. Chapters 9 and 10 count the additional routing and communication resources.

11.9 Autoregressive Indexing

The natural indexing convention is

$$s_t \longmapsto s_{t+1}.$$

Training examples contain adjacent state pairs

$$(s_t, s_{t+1}),$$

or, more locally,

$$(s_{t,i-1}, s_{t,i}, s_{t,i+1}) \longmapsto s_{t+1,i}.$$

This has the formal organization of autoregressive state prediction.

The Turing example is deterministic given its full current configuration. Ordinary language modeling instead estimates a conditional distribution $p_\theta(x_{t+1} \mid x_{\le t})$. The displayed prediction arrow is schematic. A next-token predictor can depend on a long context, so one step in token index need not be a bounded-arity computational rule. To serialize the Turing construction, the previous configuration must remain accessible while the next one is emitted, with cell identifiers, boundaries, and the synchronous-update convention preserved.

The analogy with language modeling is nevertheless limited.

Ordinary tokens usually expose emitted symbols, not all latent variables needed by the internal computation.

Hence

autoregressive factorization $\neq$ faithful computational trace.

11.9.1 When autoregressive supervision approaches C3

The analogy becomes closer when the sequence explicitly contains computational state, for example

- program execution traces,
- worked mathematical steps,
- intermediate plans,
- state–action trajectories,
- tool outputs,
- externally verified scratchpad states.

Such data can expose portions of the hidden computation.

But even then the coverage issue of Section 11.6 remains.

Teacher forcing versus execution. Training on a correct preceding state and then feeding back predicted states are different experiments. For a fixed learned deterministic transition $\widehat{H}$, let

$$q_t = \Pr_X\{\widehat{H}(s_t(X)) \neq H_M(s_t(X))\}$$

be its full-state one-step error on the true trajectory. Starting from the correct initial state, any rollout failure has a first erroneous update; before that update the learned and true states agree. Consequently,

$$\Pr_X\{\text{some erroneous rollout update before } T\} \le \sum_{t=0}^{T-1} q_t.$$

This elementary bound needs no independence across steps. A local-cell error bound must additionally account for the cells updated. Empirical one-step accuracy alone does not supply the population quantities q_t, and a small error averaged over a long horizon need not make their sum small. Coverage gives the special case in which the relevant errors are exactly zero.

11.10 Reverse Indexing of the Same Computation

A second indexing convention makes the same computation appear to move toward decreasing time labels.

Define

$$y_t = s_{T-t}. \tag{11.5}$$

Then

$$y_T = s_0$$

contains the input, while

$$y_0 = s_T$$

contains the output.

Moreover,

$$y_{t-1} = s_{T-t+1} = H_M(s_{T-t}) = H_M(y_t).$$

Thus

$$y_t \longmapsto y_{t-1} \tag{11.6}$$

uses exactly the same forward computational rule.

The important distinction is

$$\boxed{\text{reverse external index} \neq \text{inverse computational transition.}}$$

No map

$$H_M^{-1}$$

is required.

Indeed, a Turing transition need not be invertible.

This construction is useful only as an indexing observation. It should not be interpreted as claiming that ordinary diffusion models implement Turing-machine transitions.

11.11 Reverse Diffusion Is Not Inverse Heat Flow

There is an apparent paradox in diffusion models.

Forward Gaussian smoothing suppresses fine-scale information, while direct inversion of the heat equation is classically unstable.

How, then, can a generative diffusion model run "backward"?

The resolution is

reverse diffusion is not direct inversion of the heat operator.

11.11.1 Why direct inversion is unstable

Consider

$$\frac{\partial p_t}{\partial t} = \Delta p_t. \tag{11.7}$$

Its solution is

$$p_t = G_t * p_0.$$

In Fourier coordinates,

$$\widehat{p}_t(\xi) = e^{-t\|\xi\|^2}\widehat{p}_0(\xi), \tag{11.8}$$

up to the Fourier convention.
Formal inversion requires

$$\widehat{p}_0(\xi) = e^{t\|\xi\|^2}\widehat{p}_t(\xi).$$

Thus a small high-frequency perturbation is exponentially amplified.
Exact deconvolution is therefore ill-conditioned.

11.11.2 The diffusion-model problem

A diffusion model does not first estimate p_t and then apply this inverse operator.
Instead it learns the score

$$\mathfrak{s}_t(x) = \nabla_x \log p_t(x),$$

using samples from the data distribution and a specified corruption process. We use $\mathfrak{s}_t$ for the score to distinguish it from a computational state s_t; the score is learned across noise levels.

For a forward SDE

$$dX_t = b(X_t, t)\,dt + g(t)\,dW_t,$$

assume sufficient regularity for time reversal, with smooth positive marginal densities on the positive-noise interval. In increasing reverse time $u = T - t$, the reversed process $\overline{X}_u = X_{T-u}$ has the standard drift

$$d\overline{X}_u = \Big[-b(\overline{X}_u, T-u) + g(T-u)^2\mathfrak{s}_{T-u}(\overline{X}_u)\Big]\,du + g(T-u)\,d\overline{W}_u.$$

The noise coefficient here is scalar and state independent, as in the forward SDE; more general diffusion matrices require the corresponding extra terms. This is the time-reversal result used in **[song2021scorebased]**, not a consequence of relabeling a deterministic Turing trajectory.

Thus the learned object is a family

$$\{\mathfrak{s}_t\}_{0<t\le T},$$

not a single inverse heat operator.

With the exact score, the appropriate reversed initial law p_T, and well-posed dynamics, this construction reproduces the prescribed marginals. Replacing p_T by a convenient prior, estimating the score, and discretizing the dynamics introduce separate errors. The procedure generates from a learned distribution; it does not identify the original clean example or retrace its particular forward noise realization. It also does not make arbitrary heat inversion stable. For a singular data law at zero noise, the endpoint or a positive-noise cutoff needs additional treatment.

11.11.3 The score as a denoising quantity

Fix a positive noise level $\sigma_t > 0$, take $Z \sim \mathcal{N}(0, I)$ independent of X_0, and assume $\mathbb{E}\|X_0\|^2 < \infty$. For additive Gaussian corruption $X_t = X_0 + \sigma_t Z$, Tweedie's identity gives

$$\boxed{\mathbb{E}[X_0 \mid X_t = x] = x + \sigma_t^2 \nabla_x \log p_t(x).} \tag{11.9}$$

More generally, when $X_t = \alpha_t X_0 + \sigma_t Z$ with deterministic $\alpha_t \neq 0$ and $\sigma_t > 0$, the same identity yields

$$\mathbb{E}[X_0 \mid X_t = x] = \frac{x + \sigma_t^2 \mathfrak{s}_t(x)}{\alpha_t}. \tag{11.10}$$

For completeness, differentiate the Gaussian mixture density under the integral. Dividing by its positive density gives

$$\mathfrak{s}_t(x) = \frac{\alpha_t \mathbb{E}[X_0 \mid X_t = x] - x}{\sigma_t^2} = -\frac{1}{\sigma_t}\mathbb{E}[Z \mid X_t = x],$$

which proves both formulas. The first equality does not require X_0 to have a density before corruption. At $\alpha_t = 0$, the observation is independent of X_0, and division by α_t is invalid.

The score therefore contains local denoising information.

The reverse-SDE update is not literally the posterior mean $\mathbb{E}[X_0 \mid X_t]$, but both depend on the same score field.

11.12 Diffusion Supplies Synthetic Intermediate States

The analogy with C3 should be formulated carefully.

For the common Gaussian corruption parameterizations, training samples independently

$$X_0 \sim p_0, \qquad t \sim \nu, \qquad Z \sim \mathcal{N}(0, I)$$

and constructs

$$X_t = \alpha_t X_0 + \sigma_t Z. \tag{11.11}$$

Thus the training mechanism can manufacture supervised problems at many controlled noise levels.

Schematically,

$$\boxed{X_0 \longrightarrow X_t \longrightarrow \text{noise-, score-, or denoising target.}}$$

We call these *synthetic intermediate states*.

The term "trace" should be reserved for settings in which an actual coupled sequence

$$X_{t_0}, X_{t_1}, \ldots, X_{t_m}$$

is sampled or otherwise supplied.

Ordinary score training does not require a complete forward trajectory.

Specifying these one-time corruption marginals does not by itself specify a coupling across different noise levels. Independently drawing a noisy version at each level, reusing one noise vector, and following a forward Markov chain can have the same one-time marginals but different joint paths. A claim about an adjacent-state transition must name that coupling.

An observable target is not the unknown marginal score. The injected noise Z is known during training, but it is not determined by X_t alone. For example, the population noise-prediction objective at a fixed t is

$$\mathbb{E}\|\epsilon_\theta(X_t, t) - Z\|^2.$$

The conditional-expectation decomposition of squared loss makes its unrestricted minimizer $\mathbb{E}[Z \mid X_t = x]$, not the particular noise vector used to make each example. The preceding identity converts this conditional mean into $-\sigma_t \mathfrak{s}_t(x)$. Equivalently, $-Z/\sigma_t$ is an observable conditional-corruption score target whose conditional expectation is the marginal score. This is the denoising score-matching connection [**vincent2011connection**]; the noise-prediction training form is used in [68].

Synthetic corruptions expose many supervised prediction problems, but do not provide the complete table of their population solutions. Repeatedly corrupting a fixed clean sample also does not create independent new clean examples from p_0. Statistical generalization, successful optimization, and reverse-process accuracy remain separate requirements.

The relevant commonality with C3 is therefore

the training procedure explicitly creates intermediate supervised states rather than exposing only a terminal input–output pair.

11.12.1 Intermediate does not mean easy

The score $\mathfrak{s}_t(x) = \nabla \log p_t(x)$ may itself be a complicated high-dimensional function.

The corruption construction alone does not prove that each true score field admits a feasible accurate approximation with controlled local regularity, is statistically easy, or is efficiently learnable. This is consistent with Chapter H: an already accurate score predictor with a feasible evaluator would provide a computational compositional witness. The construction does not supply those accuracy and resource premises for the true score family.

Gaussian smoothing suppresses fine-scale Fourier structure, but this is not by itself a learning-theoretic sample-complexity theorem.

Thus

stepwise supervision changes the recovery problem; it does not prove that every step is easy.

11.13 Relation to Standard Diffusion Models

Standard diffusion and trace-exposed computation share the use of intermediate indexed states.

But the semantics differ.

In the Turing construction,

$$s_t$$

and

$$s_{t+1}$$

are different clean computational configurations.

In a standard diffusion process,

$$X_t$$

and

$$X_{t-\Delta t}$$

are states at different noise levels. When drawn along one specified forward path, they share its initial clean example; their reverse conditional distribution is generally stochastic. A reverse generative trajectory is not required to return that particular example. In contrast, the finite-state computational theorem aims to reproduce a specified deterministic trajectory and its output.

Therefore

standard diffusion $\neq$ computational-trace supervision.

The common principle, at this level of abstraction, is

replace one difficult global transformation by supervised or learned intermediate problems.

11.14 Gaussian-Corrupted Computational States

A more literal bridge can be constructed artificially.

Suppose a genuine computational state is observed through Gaussian noise. This section assumes that the required discrete local rule has already been supplied or recovered correctly. It studies corruption during execution, not estimation of that rule from mislabeled traces.

Encode each local alphabet symbol by a one-hot block of amplitude C:

$$e(s_{t,i}) \in \{0, C\}^a, \qquad a = |\mathcal{A}| = O(1).$$

Concatenating all S cells gives

$$s_t^{\text{vec}} \in \{0, C\}^D, \qquad D = aS = \text{poly}(n).$$

Observe

$$X_t = \sqrt{\bar{\alpha}_t}\, s_t^{\text{vec}} + \sqrt{1 - \bar{\alpha}_t}\, Z_t, \qquad Z_t \sim \mathcal{N}(0, I_D). \tag{11.12}$$

Here $0 < \bar{\alpha}_t < 1$, $C > 0$, and the Gaussian noise is independent of the state being encoded. The bar distinguishes the *signal-power* coefficient in this artificial channel from the signal-amplitude coefficient α_t in (11.11). There is no assumption that these noisy computational states have the joint law of a standard forward diffusion.

11.14.1 Blockwise recovery

Decode each block by argmax.

Lemma 5 (High-probability blockwise decoding)**.** *The probability that any of the S blocks is decoded incorrectly is at most*

$$D \exp\left(-\frac{\bar{\alpha}_t C^2}{4(1-\bar{\alpha}_t)}\right). \tag{11.13}$$

Consequently, decoding succeeds with probability at least $1-\delta$ if

$$C \geq 2\sqrt{\frac{1-\bar{\alpha}_t}{\bar{\alpha}_t} \log \frac{D}{\delta}}. \tag{11.14}$$

Proof. For one block, let k be its active coordinate and let $j \neq k$ be an inactive one.

Then

$$X_t(k) \sim \mathcal{N}\left(\sqrt{\bar{\alpha}_t} C, 1-\bar{\alpha}_t\right)$$

and

$$X_t(j) \sim \mathcal{N}\left(0, 1-\bar{\alpha}_t\right).$$

Hence

$$X_t(j) - X_t(k)$$

is Gaussian with mean

$$-\sqrt{\bar{\alpha}_t} C$$

and variance

$$2(1-\bar{\alpha}_t).$$

For a standard normal G and $z \geq 0$, $\Pr\{G \geq z\} \leq e^{-z^2/2}$. Apply this with $z = C\sqrt{\bar{\alpha}_t/[2(1-\bar{\alpha}_t)]}$. There are $S(a-1) \leq D$ competitors, so a union bound gives (11.13); probabilities can of course be capped at one. Solving for a right-hand side at most δ gives (11.14). □

There are two equivalent ways to express the threshold.

Define the amplitude SNR

$$\mathrm{SNR}_{\mathrm{amp}} = \frac{C\sqrt{\bar{\alpha}_t}}{\sqrt{1-\bar{\alpha}_t}}.$$

Then

$$\boxed{\mathrm{SNR}_{\mathrm{amp}} \gtrsim \sqrt{\log(D/\delta)}.} \tag{11.15}$$

Alternatively define the power SNR

$$\mathrm{SNR}_{\mathrm{pow}} = \frac{\bar{\alpha}_t C^2}{1-\bar{\alpha}_t}.$$

Then

$$\boxed{\mathrm{SNR}_{\mathrm{pow}} \gtrsim \log(D/\delta).} \tag{11.16}$$

This removes the ambiguity between the two conventions.

At fixed positive signal-to-noise ratio, increasing the number of cells or the horizon can defeat an all-symbol success guarantee. The theorem below needs the logarithmic DT/δ factor. At $\bar{\alpha}_t = 0$ the observation is pure noise, independent of the encoded state, so this decoding guarantee cannot hold uniformly over possible states. At $\bar{\alpha}_t = 1$ the channel is noiseless and decoding is exact for $C > 0$. These endpoint cases are handled separately from the displayed ratios.

11.14.2 Robust execution

Write $\mathrm{Dec}(X_t)$ for blockwise argmax decoding, and let $E_C(s)$ concatenate the amplitude-C one-hot blocks of a categorical state. A clean decoded state is advanced by

$$D_t(X_t) = \widehat{H}_M(\mathrm{Dec}(X_t))\,,$$

where $\widehat{H}_M$ uses the recovered local table. If its required entries are correct and decoding succeeds, $D_t(X_t) = s_{t+1}$.

To make the feedback model explicit, start from $\widehat{s}_0 = s_0$ and at each $t = 0, \ldots, T-1$ form

$$\begin{aligned} \widehat{X}_t &= \sqrt{\bar{\alpha}_t}\, E_C(\widehat{s}_t) + \sqrt{1-\bar{\alpha}_t}\, Z_t, \\ \widehat{s}_{t+1} &= \widehat{H}_M\Big(\mathrm{Dec}(\widehat{X}_t)\Big)\,. \end{aligned}$$

Conditional on the preceding execution history, Z_t has the standard Gaussian law and is independent of the current encoded state. The newly computed state is re-encoded before the next corruption. This is an explicit noisy-storage and restoration model: it does not supply the correct future state from an external teacher at every step. Undefined dictionary entries cause failure and may then be assigned arbitrary outputs to make the recursion total.

Theorem 12 (Robustness to Gaussian corruption)**.** *Fix a test input and a $T = T(n) \geq 1$ step computation. Assume that the learned local table is correct on every neighborhood of its true trace, that initialization is exact, and that execution follows the feedback model above. Suppose the conditional decoding-failure bound at every step, before any earlier error, is at most δ/T. Then the learned states agree with all the true states through time T, and the final output is correct, with probability at least $1-\delta$. A sufficient amplitude condition is*

$$C \geq 2 \max_{0 \leq t < T} \sqrt{\frac{1-\bar{\alpha}_t}{\bar{\alpha}_t} \log \frac{DT}{\delta}}.$$

Proof. Before a first error, $\widehat{s}_t = s_t$. The conditional corruption law and Lemma 5 bound the probability of a decoding error at that step by δ/T. If no decoding error occurs, the correct local table produces s_{t+1}, so equality of the states is preserved. The disjoint events that the first error occurs at step t have total probability at most $\sum_{t=0}^{T-1} \delta/T = \delta$. Exact output decoding then gives the claimed conclusion. □

The same marginal union bound can analyze observations of externally supplied true states, but the feedback statement above is stronger: it also shows why erroneous learned states are not silently replaced by correct teacher states. Across-time independence is not needed if the same conditional error bounds can otherwise be established.

Combining Theorem 11 with this result gives a useful separation. On its high-probability training event, let ε_{cov} bound failure of clean test coverage. If the conditional execution-noise failure is at most δ_{exec} on every covered input, then

$$\Pr_{X,Z}\{\text{incorrect output or abstention} \mid \text{training sample}\} \le \varepsilon_{\text{cov}} + \delta_{\text{exec}}.$$

Statistical coverage and noise robustness are different contributions. No claim is made that ordinary regularized SGD necessarily returns an exactly correct table; the theorem concerns any returned table meeting that condition. It does not require the training updates themselves to have stopped.

Numerical precision is an additional channel. The Gaussian lemma uses exact real observations and comparisons. If each observed coordinate is additionally perturbed by at most ζ, a sufficient comparison margin is reduced from $C\sqrt{\bar{\alpha}_t}$ to $C\sqrt{\bar{\alpha}_t} - 2\zeta$. For a positive reduced margin the same argument gives the upper bound

$$D \exp\left(-\frac{(C\sqrt{\bar{\alpha}_t} - 2\zeta)^2}{4(1-\bar{\alpha}_t)}\right).$$

A complete bit-cost claim must also control dynamic range, sampling, and storage. The ideal Gaussian theorem by itself is not a finite-precision hardware theorem; the separate resource analysis of Chapters 9–10 still applies.

This is a theorem about Gaussian-corrupted *computational states*. It is not a universality theorem for standard diffusion models, nor a proof that a learned denoiser satisfies the decoder's uniform guarantees.

11.15 What the Common Principle Really Is

The commonality is not

$$\boxed{\text{Transformers} = \text{diffusion models.}}$$

It is not

$$\boxed{\text{next-token prediction} = \text{Turing trace supervision.}}$$

The more general principle is

represent a difficult computation or transformation
through intermediate states
⇓
learn local or intermediate prediction problems
⇓
compose or iterate those predictions.

The nature of the intermediate states can be very different.
For a Turing computation they are machine configurations.
For autoregressive models they may be emitted symbols or explicit reasoning states.
For diffusion models they are controlled noisy versions of data.
The mathematical burden therefore remains model dependent.

11.16 Relation to Compositional Sparsity

A Turing space–time computation is itself compositionally sparse:

$$s_{t+1,i} = h_M\left(s_{t,i-1}, s_{t,i}, s_{t,i+1}\right).$$

Unrolling time yields a polynomial-size DAG of constant local arity.
Hence

polynomial-time computation $\Longrightarrow$ polynomial-size bounded-fan-in space–time DAG.

Chapter 4 uses this observation primarily as a representation result.
The present chapter adds an information result:

when the relevant local states are exposed,
the constituents can be supervised directly.

C1 provides the local structure.
C3 exposes it.

The theorem-strength regime here includes the state encoding, neighborhood interfaces, and a coverage or complete-table condition. These extra assumptions explain its stronger structural conclusion. Computational sparsity alone does not supply them. Conversely, a predictively accurate feasible system with different internal states is not a failure merely because it does not reconstruct this teacher trace. The return-based predictive argument of Chapter H remains separate from exact trace reproduction.

11.17 Relation to Associative Computation

The state of the computation may be regarded as a collection of local records.
Updating one component requires

retrieve local state + apply local transition + store the result.

Repeated application gives the complete computation.

This is closely related to the read–update–write cycle of the Associative Turing Machine.
The distinction is

ATM chapter:	a universal associative execution mechanism,
present chapter:	direct learning of an exposed finite local rule.

Computational universality answers

Can the program be executed?

Trace supervision answers, in a restricted regime,

Can the local transition be learned directly?

The table learner uses supplied local coordinates and neighbor access. Replacing that access by learned attention requires its own routing and margin analysis. The shared finite rule also does not eliminate the cost of moving state values or maintaining memory; logical locality and physical implementation costs remain distinct.

11.18 A Biological Interpretation: Replay as Partial Trace

A possible biological interpretation should remain deliberately modest.
Long behavioral sequences create temporal credit-assignment problems.
If intermediate states

$$z_0, z_1, \ldots, z_T$$

are available during experience or replay, then a learner may estimate local predictions

$$z_t \longmapsto z_{t+1}$$

rather than rely entirely on a terminal signal for the global relation

$$z_0 \longmapsto z_T.$$

Thus

intermediate-state exposure can shorten temporal credit assignment.

Hippocampal replay is suggestive in this respect.
But biological replay is not assumed to provide the theorem-strength complete local table.
It may provide only partial relational structure.
Nor does the theorem imply that Hebbian plasticity alone can learn every local map.
The weaker hypothesis is simply that replay can expose useful intermediate predictive relations to a slower learner.
The hippocampal-scaffold chapter develops this possibility separately.

11.19 The State-Discovery Problem

The theorem assumes that the state representation is already known.

This is perhaps its most important limitation.

Suppose instead that the learner observes raw data

$$x_t$$

generated by a hidden process.

The difficult problem is to discover a representation

$$z_t = \phi(x_{\le t}) \tag{11.17}$$

in which the dynamics become simple:

$$z_{t+1} = H(z_t) \tag{11.18}$$

or, more strongly,

$$z_{t+1}^{(i)} = H_i\left(z_t^{\mathrm{pa}(i)}\right), \qquad |\,\mathrm{pa}(i)| = O(1). \tag{11.19}$$

These equations describe a desired autonomous deterministic realization; raw observations of a stochastic or externally driven process need not admit one. With an action or new random innovation, the update must include that input or be modeled by a conditional transition law. Time can be included in the state for nonstationary dynamics, but its encoding and resource cost must then be counted.

Simplicity must retain task information. The equations alone admit the useless choice $\phi \equiv 0$, $H(0) = 0$. A state-discovery objective must therefore require an admitted readout D_{out} that predicts the task's future observable or final answer, for example

$$\mathbb{E}[\ell(D_{\mathrm{out}}(\widehat{z}_T), Y)] \le \varepsilon, \qquad \widehat{z}_0 = \phi(x_{\le 0}), \quad \widehat{z}_{t+1} = H(\widehat{z}_t),$$

with state size, encoding and readout complexity, numerical conditioning, and execution cost controlled. Otherwise the original hard computation could simply be hidden in ϕ or D_{out}.

For simulation of a specified deterministic process, a useful exact compatibility condition is an encoding E satisfying

$$E(H_M(s)) = H(E(s)), \qquad D_{\mathrm{out}}(E(s_T)) = f(x)$$

on the relevant states. This relation can permit many-to-one state maps; faithful structural recovery additionally requires a prescribed equivalence class and an appropriate matching or injectivity condition. Coordinate identity is not required. Supplied traces give one useful coordinate system; discovering another sufficient one is the broader representation question.

This gives the central open problem:

State-Discovery Problem

When can raw observations be transformed into coordinates in which the relevant computation has stable bounded-arity local dynamics?

This is not solved by C3.

In the theorem-strength regime studied here, C3 assumes that useful intermediate states and their interfaces are supplied. Partial traces and surrogates can reduce, rather than eliminate, the remaining state-discovery problem.

The State-Discovery Problem asks how a learner can manufacture those states for itself.

It therefore lies at the intersection of

$$\boxed{\text{genericity} + \text{feature learning} + \text{world-model discovery.}}$$

The staircase program of Chapter H can be viewed as one possible constructive answer: learned and retained features expose useful coordinates for subsequent computation. Such a mechanism needs its own finite-sample and optimization proof. It is not a necessary trajectory for every successful learner, and it need not identify a privileged set of generating variables.

11.20 Limitations and Open Problems

The results of this chapter have several clear boundaries.

11.20.1 Trace exposure is stronger than end-to-end supervision

The exact theorem assumes that local computational states are visible.

Ordinary supervised examples do not provide them.

11.20.2 Observed traces may not cover future local states

Even a complete trajectory reveals only the neighborhoods that occur on that trajectory.

Guaranteed faithful execution by the abstaining lookup learner uses

$$\mathcal{U}_{\text{test}} \subseteq \mathcal{U}_{\text{train}}.$$

Section 11.6.1 gives finite-sample conditions for complete support coverage or small population failure probability. Without coverage, the observed labels alone do not specify an arbitrary unseen table entry. A different estimator can exploit further structure or happen to give a correct final answer; coverage is not necessary for all predictive success.

11.20.3 Surface traces are not necessarily faithful traces

Language, scratchpads, and emitted reasoning symbols may help learning without corresponding to the actual latent variables of the generating computation.

11.20.4 Standard diffusion is not computational-trace diffusion

Diffusion noise levels and Turing configurations are different objects.

The analogy concerns intermediate local learning problems, not identity of mechanism.

11.20.5 Approximate execution can accumulate error

The exact local-rule construction preserves valid configurations by induction from a valid initial state. Blockwise restoration returns categorical symbols, but a decoding error can still produce a globally invalid machine configuration. Discretization alone does not correct arbitrary symbolic errors.

For learned continuous transitions, errors can accumulate over

$$T(n)$$

steps.

Long-horizon execution may therefore require

contractivity, error correction, attractor structure, state restoration,

or some other stability mechanism.

For example, suppose the true and learned continuous states stay in specified domains, H_t is L_t-Lipschitz there, and $\|\widehat{H}_t(z) - H_t(z)\| \le \eta_t$ holds on those domains. Writing $e_t = \|\widehat{z}_t - z_t\|$ gives

$$e_{t+1} \le L_t e_t + \eta_t, \qquad e_T \le \left(\prod_{j=0}^{T-1} L_j\right) e_0 + \sum_{t=0}^{T-1} \eta_t \prod_{j=t+1}^{T-1} L_j.$$

The proof is the triangle inequality followed by induction. Uniform local error and domain control are premises, not consequences of low empirical loss on teacher-forced states. If $L_t \le L < 1$, $e_0 = 0$, and $\eta_t \le \eta$, then $e_T \le \eta/(1-L)$; for $L_t \le 1$ the corresponding bound is $T\eta$. Contractivity is one sufficient option, not a necessary property of every faithful computational realization.

11.20.6 State discovery remains open

The hardest problem is often not learning

$$H$$

after the right state has been exposed.

It is finding the state variables in which such a simple H exists.

The simplicity criterion must be combined with task sufficiency and resource bounds, as in Section 11.19. A stochastic world model may require conditional laws rather than a deterministic map. Structural identification of the teacher's states is a further objective, not an extra premise for predictive accuracy.

Experiments that distinguish the mechanisms. A direct test would hold the local rule fixed while varying the training trace distribution, observed neighborhood coverage, and test horizon. Report missing-key frequency, one-step errors, whole-trace agreement, and final-output error separately. A second comparison would vary whether the state encoding is supplied, partially observed, or learned. For noisy execution, sweep the power SNR against $\log(DT/\delta)$ and compare restoration with uncorrected continuous feedback. These are proposed experiments, not results supplied by this chapter. Standard diffusion sampling quality should be evaluated separately from exact recovery of a planted computational state.

11.21 Conclusion

A polynomial-time Turing computation can be represented as repeated application of a constant-arity local rule.

With complete local supervision,

polynomial-time computation
$\Downarrow$
bounded-arity local transition
$+$
supplied complete local supervision
$\Downarrow$
exact local-rule recovery
$\Downarrow$
iteration.

With ordinary observed trajectories, one further condition appears:

the local states needed at test time must either have been observed
or be predictable from additional structure.

This distinguishes

trace exposure

from

trace coverage.

A universal Turing machine sharpens the conceptual result still further.
One fixed local transition rule

$$h_U$$

can execute arbitrarily many programs when the program and data are encoded in the initial state.
Thus

fixed interpreter $+$ variable program and data
$+$ extensible workspace and execution time
$\Longrightarrow$ universal computation.

The relation to autoregressive and diffusion models is structural.
Autoregressive models predict along observed sequences.
Diffusion training constructs supervised intermediate states at controlled noise levels.
Trace-exposed computation predicts along an actual computational trajectory.
The common principle is

intermediate supervision can convert depth of hidden inference
into locality of prediction.

When intermediate variables are hidden, a central constructive question remains:

when useful intermediate states are not supplied,
can the learner discover a task-sufficient representation
in which the relevant computation becomes local?

That is the State-Discovery Problem.

In the language of this book, C3 supplies intermediate information. Under the additional finite-alphabet and coverage assumptions stated here, a specified learner converts that information into reliable execution. C2 concerns accessible learning signals and can support feature discovery; it does not state that every successful learner must construct a staircase.

The chapter therefore supplies one concrete route to both prediction and faithful trace reproduction. Chapter H permits a broader endpoint: an accurate feasible returned predictor is already a computational compositional witness, whether or not its states match this trace. The next chapter applies the intermediate-supervision viewpoint to return-conditioned trajectory modeling, where support, feedback, and the distinction between observation and control become central.

CHAPTER 12

Return-Conditioned Sequence Modeling as Autoregressive Control

Reinforcement learning often involves long-horizon credit assignment. An agent chooses actions, rewards may arrive much later, and the learner must find behavior that improves future outcomes. Return-conditioned sequence modeling reorganizes part of this problem.

An offline trajectory is annotated at every time step with its return-to-go*: the discounted reward realized after that point. Training becomes supervised conditional action prediction,*

$$(h_t, G_t) \longmapsto a_t,$$

where h_t is the observed history and G_t is the recorded return. At deployment, a desired return replaces the recorded future outcome, and the policy interacts autoregressively with the environment.

The useful transformation is

delayed outcome in a recorded trajectory
$\longrightarrow$ *return label at every decision*
$\longrightarrow$ *supervised conditional action prediction.*

This removes Bellman bootstrapping from the basic training objective. It does not remove representation learning, optimization, statistical estimation, exploration, or the need to establish closed-loop performance. In particular, a conditional distribution of actions observed on successful trajectories is not automatically a policy that reproduces their success.

In the terminology of the preceding chapters, return-to-go is a compressed outcome annotation*, usefully interpreted as partial or surface trace information. It is not the faithful local transition supervision of Chapter 11. When computed from an already recorded reward sequence, it reorganizes information rather than supplying new observations.*

The central picture remains control through return-conditioned trajectory generation, with explicit qualifications about support, stochasticity, and causality. A learned world model adds internal simulation. An analogy with numerical integration additionally requires a specified dynamical model, time discretization, and error control.

12.1 From Value Estimation to Conditional Trajectory Modeling

Consider a finite-horizon Markov decision process

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma, H), \qquad 0 < \gamma \leq 1,$$

with an initial-state distribution μ_0. Decisions occur at $t = 0, \ldots, H-1$, and s_H is terminal. The transition law is $P(ds' \mid s, a)$ and, in the deterministic-reward notation, $r_t = r(s_t, a_t, s_{t+1})$. Time dependence may be incorporated into the state or shown as a subscript.

For stochastic rewards, use a reward kernel R and the joint environmental kernel

$$\mathsf{P}(ds', dr \mid s, a) = P(ds' \mid s, a)R(dr \mid s, a, s').$$

The deterministic-reward case is included by making R a point mass. These kernels describe actual environmental outcomes, not outcomes conditioned retrospectively on success.

12.1.1 Classical value-based reinforcement learning

For a Markov policy $\pi = (\pi_t)_{t=0}^{H-1}$, the finite-horizon value function is

$$V_t^\pi(s) = \mathbb{E}_\pi\left[\sum_{u=t}^{H-1} \gamma^{u-t} r_u \,\middle|\, s_t = s\right], \qquad V_H^\pi = 0.$$

The corresponding action-value function satisfies

$$Q_t^\pi(s, a) = \mathbb{E}_{(s',r)\sim\mathsf{P}(\cdot|s,a)}\big[r + \gamma V_{t+1}^\pi(s')\big], \qquad V_t^\pi(s) = \mathbb{E}_{a\sim\pi_t(\cdot|s)} Q_t^\pi(s, a).$$

The time index matters because the remaining horizon changes. A history-dependent policy can instead be analyzed on an augmented history state.

Temporal-difference methods bootstrap one learned estimate from another. The combination of function approximation, off-policy data, and bootstrapping can be unstable: this is the classical "deadly triad" [**sutton2018reinforcement**]. It is not a theorem that every method using these ingredients diverges, nor that all reinforcement learning uses Bellman bootstrapping. Monte Carlo and policy-gradient formulations provide other starting points.

12.1.2 An alternative offline formulation

Return-conditioned sequence models learn actions directly from recorded trajectories rather than requiring a learned value function for action selection. Write

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots, s_H).$$

Define

$$G_t = \sum_{u=t}^{H-1} \gamma^{u-t} r_u. \tag{12.1}$$

The empty sum gives $G_H = 0$. One return-conditioned tokenization is

$$\tau_G = (G_0, s_0, a_0, G_1, s_1, a_1, \ldots, G_{H-1}, s_{H-1}, a_{H-1}).$$

Decision Transformer provides a concrete architecture using return, state, action, and time information with a causal attention mask [**chen2021decision**]. The general formulation does not require a

Transformer: return- or goal-conditioned supervised policies can also use feedforward networks [**emmons2022rvs**].

We write the full observed pre-action history as

$$h_t = (s_0, a_0, r_0, \ldots, s_{t-1}, a_{t-1}, r_{t-1}, s_t), \qquad h_0 = s_0,$$

and use the conditional policy $\pi_\theta(a \mid h_t, G_t)$. Past rewards are included because they need not be reconstructible from past states and actions in the stochastic-reward case. A particular implementation may use a compressed state, a finite context window, or past return tokens instead. Any theorem then has to establish that the retained information is sufficient for its claim. Our full-history notation is an explicit mathematical model, not a claim that every Decision Transformer retains the entire episode.

12.2 Offline Training Is Supervised Conditional Action Prediction

Let $\mathcal{D} = \{\tau^{(1)}, \ldots, \tau^{(N)}\}$ be an offline dataset. Every return sequence can be computed by a backward pass through the recorded rewards. For equal episode length H and discrete actions, one supervised objective is

$$\widehat{\mathcal{L}}_{\mathcal{D}}(\theta) = -\frac{1}{NH} \sum_{j=1}^{N} \sum_{t=0}^{H-1} \log \pi_\theta(a_t^{(j)} \mid h_t^{(j)}, G_t^{(j)}).$$

Unequal episode lengths, sampling weights, and context truncation require the corresponding mask and normalization. For continuous actions, a regression or conditional-density objective may be used. A conditional mean action need not be a good action when the target distribution is multimodal; averaging two successful strategies can create an unsuccessful one.

The policy must not obtain the action being predicted, future actions, or future states through an unmasked input path. The return annotation is deliberately retrospective during training. Its deployment substitute is a chosen command, not access to the future reward that will actually occur.

12.2.1 What has disappeared

The basic objective contains no target

$$r_t + \gamma V_\theta(s_{t+1})$$

that must itself be estimated by the same learned value function. Consequently,

the basic supervised objective avoids Bellman bootstrapping.

This describes the objective, not every possible hybrid method. An auxiliary value loss or an actor–critic fine-tuning stage can reintroduce bootstrapping.

12.2.2 What has not disappeared

The trajectory still contains long-range dependencies, the representation must retain useful history, and neural parameter optimization can remain nonconvex. Removing bootstrapping does not establish global optimization or generalization.

For a population interpretation, suppose trajectories are generated by a nonanticipating behavior policy $\beta_t(a \mid h_t)$ in the stated MDP. Write $\mathbb{P}_\beta$ for their law. With an unrestricted conditional distribution and exact population optimization, the log-loss minimizer is

$$\pi_{\mathrm{rc}}(a \mid h, g) = \mathbb{P}_\beta(A_t = a \mid h_t = h, G_t = g).$$

Indeed, conditional cross-entropy is conditional entropy plus $\mathrm{KL}(\pi_{\mathrm{rc}} \| \pi_\theta)$. This identifies what is being estimated, not whether a finite network learns it or whether its deployment achieves the desired return.

For generalization arguments, the independent observations may be *episodes*, not their NH overlapping decision contexts. If the N trajectories are independent, their average per-episode losses can be treated as independent samples. The bounded-loss version of Chapter H requires bounded or suitably controlled log loss; unrestricted negative log likelihood is not bounded. Statistical prediction error under the offline law must still be related to the history–command distribution encountered in control.

12.3 Return-to-Go as a Partial Trace

Return-to-go moves a summary of the recorded future outcome next to each recorded decision. This is an informational reorganization of the learning problem, rather than a new causal observation.

12.3.1 A delayed reward becomes a local label

For an early decision, the label

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots$$

is supplied directly in the supervised example $(h_t, G_t) \mapsto a_t$. The predictor need not obtain that numerical label by propagating a learned value estimate backward over the horizon.

return-to-go makes a delayed outcome a locally available conditioning variable.

Here "local" means available at the current decision. It does not mean that the action-prediction function has low dimension or bounded functional arity.

There is no information gain relative to the complete recorded reward sequence. Given γ and $G_H = 0$,

$$(r_0, \ldots, r_{H-1}) \longleftrightarrow (G_0, \ldots, G_{H-1}), \qquad r_t = G_t - \gamma G_{t+1}.$$

Both directions are deterministic. In particular, with only a terminal reward and $\gamma = 1$, every preterminal G_t equals that terminal reward. Repeating it at every decision changes access and training organization, not the information in the complete episode.

12.3.2 Connection to C3

Chapters 5 and H distinguish faithful, partial, and surface intermediate information. A faithful computational trace supplies intermediate variables and, with appropriate coverage, local transition targets. Return-to-go supplies much less: it does not identify a hidden state, specify which action caused success, or reveal what would have happened under an alternative action.

It is useful to call it an *outcome-conditioned partial trace*, with that limited meaning:

$$\boxed{G_t = \text{a compressed summary of the realized future reward.}}$$

It need not be a partial observation of the particular latent computation whose recovery is sought. Its usefulness is a task-dependent question, and it does not inherit the complete-local-supervision theorem merely by being called a trace.

12.3.3 A hierarchy of supervision

The comparison is best made by naming what is supplied:

faithful computational trace:	specified internal states;
rich intermediate supervision:	subgoals or other useful targets;
return-to-go annotation:	a future-reward summary;
terminal-reward record:	an episode-level outcome.

This is not a universal ordering of predictive usefulness or a literal information-inclusion chain. A trace of machine states need not include the reward definition. Conversely, a recorded terminal reward can already determine every return label in the sparse-reward undiscounted case. The relevant comparison fixes what is recorded, what is available at deployment, and what the learner must infer.

12.4 Inference as Closed-Loop Autoregressive Rollout

Training is supervised; control uses the fitted policy in interaction with the environment. Recorded returns G_t and commanded returns $G_t^\star$ must remain distinct.

12.4.1 Desired return

At time zero, supply $G_0^\star$ and choose

$$a_0 \sim \pi_\theta(\cdot \mid h_0, G_0^\star).$$

The environment then produces (s_1, r_0) according to $\mathsf{P}(\cdot \mid s_0, a_0)$. The command selects a conditional behavior; it does not force the environment to realize that outcome. Time-to-go, task identity, action constraints, and return scaling must match the training convention.

12.4.2 Updating the desired return

For recorded returns, $G_t = r_t + \gamma G_{t+1}$, so

$$G_{t+1} = \frac{G_t - r_t}{\gamma}. \tag{12.2}$$

A common deployment convention applies this algebra to the desired remaining return:

$$G_{t+1}^\star = \frac{G_t^\star - r_t}{\gamma}.$$

For $\gamma = 1$, this is $G^\star_{t+1} = G^\star_t - r_t$. It is a *budget update*, not a value estimate or a guarantee of attainability.

Multiplying the update by γ^t and summing gives the exact identity

$$\boxed{G^\star_0 = \sum_{t=0}^{H-1} \gamma^t r_t + \gamma^H G^\star_H.}$$

Thus the requested return is attained exactly if and only if the terminal budget $G^\star_H$ is zero. Nothing in the bookkeeping alone enforces this.

Clipping or resetting commands after a poor outcome defines a different controller and changes this identity. Numerical error also matters: if a reward has error at most ϵ_r and each budget update adds error at most ϵ_u, then, along the same realized trajectory,

$$e^G_{t+1} \le \frac{e^G_t + \epsilon_r}{\gamma} + \epsilon_u.$$

For $\gamma < 1$, errors and unattained reward requirements can be magnified in the remaining-return coordinate.

12.4.3 The complete recurrence

The specified closed-loop controller is

$$\boxed{\begin{aligned} a_t &\sim \pi_\theta(\cdot \mid h_t, G^\star_t), \\ (s_{t+1}, r_t) &\sim \mathsf{P}(\cdot \mid s_t, a_t), \\ G^\star_{t+1} &= (G^\star_t - r_t)/\gamma, \\ h_{t+1} &= (h_t, a_t, r_t, s_{t+1}). \end{aligned}}$$

The environment supplies actual rewards and states; only internal simulation substitutes model-generated ones. The history and command together form an augmented controller state.

This recurrence can be executed with fixed learned weights. Whether training returns a controller with good performance is a separate question. A useful finite stopped iterate is admissible; there is no additional requirement that every minibatch gradient vanish at its parameters, as emphasized in Chapter H.

12.5 Control as Conditional Trajectory Generation

The probabilistic interpretation needs one important distinction. The dataset gives an empirical trajectory distribution $p_{\mathcal{D}}(\tau)$; its population analogue is $\mathbb{P}_\beta$. The supervised policy estimates the *action conditionals* within return-conditioned trajectories. Those action conditionals are only part of the full joint law.

Fix s_0 and a desired return g_0 with positive population probability. In the conditioned law

$$\mathbb{P}_\beta(\tau \mid s_0, G_0 = g_0),$$

the chain rule includes both conditional actions and conditional *environmental outcomes*. Given a full history, the remaining return associated with g_0 is

$$g_t = \gamma^{-t}\left(g_0 - \sum_{u<t} \gamma^u r_u\right).$$

Action generation uses $\pi_{\mathrm{rc}}(a_t \mid h_t, g_t)$, but the next state and reward in this conditioned distribution follow

$$\mathsf{P}_t^{g_t}(s', r \mid h_t, a_t) := \mathbb{P}_\beta(s_{t+1} = s', r_t = r \mid h_t, a_t, G_t = g_t).$$

Actual deployment instead uses $\mathsf{P}(s', r \mid s_t, a_t)$. These kernels need not agree. This distinction underlies known failures of return conditioning in stochastic environments [**paster2022luck**, **brandfonbrener2022return**].

An explicit conditional-transition calculation. The following is an elementary derivation recording the missing factor. It is stated for finite states, actions, and reward support so that every conditional quantity below is a probability mass rather than a density. Let

$$w_t(h, g) = \mathbb{P}_\beta(G_t = g \mid h_t = h), \qquad w_H(h, g) = \mathbf{1}_{\{g=0\}},$$

and let $h' = (h, a, r, s')$. Then, whenever the denominator is positive,

$$\boxed{\mathsf{P}_t^g(s', r \mid h, a) = \frac{\mathsf{P}(s', r \mid s_t, a)\, w_{t+1}(h', (g - r)/\gamma)}{\displaystyle\sum_{\tilde{s}, \tilde{r}} \mathsf{P}(\tilde{s}, \tilde{r} \mid s_t, a)\, w_{t+1}((h, a, \tilde{r}, \tilde{s}), (g - \tilde{r})/\gamma)}.}$$

To verify the formula, factor the joint probability of (s', r) and $G_t = g$ given (h, a), use $G_t = r + \gamma G_{t+1}$, and normalize over possible outcomes. Nonanticipation and the MDP assumption make the unconditioned outcome factor equal to P.

The multiplicative weight favors environmental outcomes from which the recorded continuation attains the requested return. The policy can change its action distribution; it cannot generally impose that favorable reweighting on the environment.

reproduce the actions conditional on success
$\not\Longrightarrow$
reproduce the entire trajectory distribution conditional on success.

For continuous returns, conditioning on a single value requires an appropriate regular conditional distribution or density, or a specified return bin. The finite-support calculation does not make a zero-probability conditioning event empirically observed.

12.5.1 A trajectory-level control variable

The desired return is analogous to a desired global property of a trajectory rather than a prescribed next action. It therefore acts as a *trajectory-level command*:

$G_0^\star$ asks for behavior associated with a desired outcome;
it is not a boundary condition the environment must satisfy.

The boundary-condition analogy is useful only in this qualified sense. Section 12.12 gives a deterministic setting in which exact conditional action generation really does attain the requested return, followed by a stochastic counterexample.

12.6 What Is and Is Not Numerical Integration

The physics analogy remains useful if the agent, the environment, and the numerical model are not identified with one another.

12.6.1 Policy-only sequence models

An ordinary return-conditioned policy need not predict s_{t+1}. The environment advances the state:

$$\boxed{\text{policy chooses action} + \text{environment advances state.}}$$

The neural network therefore supplies a control law, not an integrator of the environmental dynamics. Adding an auxiliary next-state prediction head changes the model but does not change which transition law governs actual deployment.

12.6.2 The dynamical-system analogy

For deterministic environmental dynamics and a deterministic action selection rule $\bar{\pi}_\theta$,

$$s_{t+1} = F(s_t, a_t), \qquad a_t = \bar{\pi}_\theta(h_t, G_t^\star),$$

so

$$s_{t+1} = F\big(s_t, \bar{\pi}_\theta(h_t, G_t^\star)\big).$$

Together with the history and return-budget updates, this is a discrete-time dynamical system on the augmented state. A stochastic policy or stochastic environment instead gives a controlled stochastic process. Merely writing either process recursively is not a numerical integration theorem.

12.7 When a World Model Is Added

A learned environmental model permits internal rollouts. This is a substantive addition to a policy-only sequence model, not a consequence of return conditioning alone.

12.7.1 A learned world model

Suppose a separate model estimates

$$\widehat{P}_\phi(s', r \mid h_t, a_t).$$

An internal rollout uses its *own simulated history*:

$$\begin{aligned}
\widehat{a}_t &\sim \pi_\theta(\cdot \mid \widehat{h}_t, \widehat{G}_t^\star), \\
(\widehat{s}_{t+1}, \widehat{r}_t) &\sim \widehat{P}_\phi(\cdot \mid \widehat{h}_t, \widehat{a}_t), \\
\widehat{G}_{t+1}^\star &= (\widehat{G}_t^\star - \widehat{r}_t)/\gamma, \\
\widehat{h}_{t+1} &= (\widehat{h}_t, \widehat{a}_t, \widehat{r}_t, \widehat{s}_{t+1}).
\end{aligned}$$

Using the true future state after each model step would instead be a teacher-forced prediction test, not autonomous model rollout.

The intended model here estimates the environmental kernel, not the return-conditioned kernel P_t^g. A generative model trained only on successful futures may model favorable outcomes conditional on success; that is not automatically a simulator of what a chosen action will cause. In

a fully observed MDP, the conditional outcome law given state and action is the interventional transition law under the stated nonanticipation assumptions. Partial observation or behavior based on unrecorded variables requires additional identification assumptions.

12.7.2 Difference-equation form

For continuous states, a deterministic model can be written

$$\widehat{s}_{t+1} = \widehat{s}_t + f_\phi(\widehat{s}_t, \widehat{a}_t).$$

This is a learned difference equation. Any map F can formally be rewritten as $s + [F(s,a) - s]$, so residual form alone does not identify an underlying differential equation.

A literal Euler interpretation requires a time step Δt and a vector field b_ϕ:

$$\widehat{s}_{t+1} = \widehat{s}_t + \Delta t\, b_\phi(\widehat{s}_t, \widehat{a}_t).$$

Relating this to a continuous-time process requires consistency, regularity, and stability assumptions, together with model and discretization errors. The broader distinction is

policy only:	the environment advances the trajectory;
policy plus world model:	the learned system can simulate it.

For example, suppose the true and learned *augmented closed-loop* updates are F_{cl} and $\widehat{F}_{\mathrm{cl}}$ on a common invariant domain, F_{cl} is L-Lipschitz, and $\|\widehat{F}_{\mathrm{cl}} - F_{\mathrm{cl}}\|_\infty \le \delta$ there. Starting from the same state, the elementary recursion gives

$$e_{t+1} \le Le_t + \delta, \qquad e_t \le \delta \sum_{j=0}^{t-1} L^j.$$

The augmented state includes whatever history summary and budget the controller uses. A one-step error measured only on the recorded data does not establish the uniform domain hypothesis or long-horizon accuracy.

12.8 Credit Assignment: What Has Been Removed?

Return conditioning changes credit assignment but does not eliminate it. It is useful to separate bootstrapped value estimation, temporal outcome annotation, and causal evaluation of actions.

12.8.1 Bellman bootstrapping is removed from the basic training objective

The model need not estimate V_t by recursively using V_{t+1}. This avoids one source of instability without guaranteeing benign optimization for the supervised network. A return label is a Monte Carlo outcome from the recorded continuation, not a Bellman estimate of the expected return of every available action.

12.8.2 Temporal outcome information is supplied directly

Every recorded action has an associated return-to-go. The numerical summary of distant reward need not be inferred solely through a chain of learned value estimates. This motivates the chapter's phrase

> return conditioning shortens the informational distance
> between a recorded action and its future outcome.

It is an interpretation of supervision, not a bound on the required network depth or the number of examples. In particular, a scalar outcome does not identify a unique successful strategy.

12.8.3 But causal credit assignment remains

A high-return trajectory establishes that an action occurred before a particular outcome. It does not reveal the outcome that would have occurred had the action been replaced. There are two distinct reasons this matters. Unobserved variables can confound an observational dataset; even without confounding, future environmental randomness can make an action look successful in hindsight.

> outcome-conditioned imitation $\neq$ counterfactual causal credit assignment.

The stochastic counterexample in Section 12.12 uses a fully observed, randomized data-collection setting: hidden confounding is not needed for failure. Expected-return optimization must account for both favorable and unfavorable outcomes of an action.

12.9 The Role of Attention

Self-attention provides short computational paths between admitted positions in a trajectory context. An action representation can use G_t, an earlier state, or an earlier action without carrying that information through every intervening recurrent update. The relevant positions must actually be in the context, and the mask must prevent access to unavailable future inputs **[chen2021decision]**.

This makes long-range conditioning convenient to represent, but does not prove that training finds the relevant dependencies or that the resulting controller is effective. The distinction from Chapter 9 remains:

> attention makes past information accessible;
> attention weights do not establish causal responsibility.

A causal attention *mask* means that token dependencies respect the chosen ordering; it is not a causal-identification theorem. Return tokens still contain hindsight information by design.

Nor is attention necessary for the return-conditioning principle. The feedforward RvS experiments demonstrate the importance of conditioning and capacity choices separately from Transformer architecture **[emmons2022rvs]**. For a particular task, comparing fixed representations, learned representations, feedforward policies, and sequence models helps determine which component supplies the benefit.

12.10 Relation to the Stepwise Trace Principle of Chapter 11

Chapter 11 studies a specified local encoding of computational states. Complete local supervision, or sufficient coverage of the local patterns needed during execution, lets an explicit learner recover

the transition rule and iterate it. Observing a few trajectories does not by itself provide that coverage.

Return-conditioned control supplies environmental trajectories and a scalar future-reward annotation. It does not supply a correct action for every history–command pair, a complete local transition table, or a teacher state with bounded-arity dynamics.

computational trace:	supervision of specified state transitions;
return annotation:	supervision of actions given realized outcomes.

The difference is in what is observed and which prediction target is specified, not in a universal ranking of all learning methods.

12.10.1 A common principle

Both chapters examine how useful intermediate or outcome information can be made available at a local learning decision. Neither says that such exposure automatically makes the local fit easy or the composed rollout reliable.

For predictive control, learned features need not identify the teacher's internal variables. They must retain the information needed to choose actions that perform well under the actual rollout distribution. Structural identification, causal interpretation of an internal state, and successful control remain separate objectives.

C3 can help when useful intermediate information is supplied; it is not only a fallback when C2 fails. C2 concerns accessible learning signals in a specified learning problem. Neither condition alone removes the mismatch between retrospective conditioning and prospective control.

12.11 Dataset Support Is the Central Constraint

Support is central, but must be specified jointly and quantitatively. The population behavior law, its finite observed sample, and the law induced by deployment are different distributions. Support alone also does not resolve uncontrollable environmental randomness [**brandfonbrener2022return**].

12.11.1 Return conditioning is interpolation only when the target is supported

A command far outside the observed return range asks the network to extrapolate. But lying inside $[G_{\min}, G_{\max}]$ is not sufficient: the range may contain gaps, and a supported return at one initial state may be unsupported at another history or remaining horizon.

The relevant object is the conditional support of G_t given h_t, together with useful actions at that condition. Even a supported event may be too rare for the available sample. For independent episodes, an event of probability $p > 0$ is missed with probability $(1-p)^N$. For $0 < p < 1$, seeing it with probability at least $1 - \delta$ requires

$$N \geq \frac{\log(1/\delta)}{-\log(1-p)}.$$

This elementary observation distinguishes population support from finite-sample coverage. Exact point conditioning for continuous returns requires more than locating the point between observed extrema.

The basic method has no unconditional guarantee outside supported history–return conditions. Additional structure can permit extrapolation; failure outside the range is a risk to test, not a logical necessity.

12.11.2 State–action coverage matters

A command can begin in a supported region and become unsupported after a stochastic transition or an action error. A full-history policy must control histories, not just isolated state–action pairs. A compressed Markov representation can reduce that burden only when its sufficiency has been established.

A simple distribution-transfer calculation makes the issue explicit. Let $\nu_{\beta,t}$ be the population offline law of $c = (h_t, G_t)$, and let d_t be a reference or deployed law of $c = (h_t, G_t^\star)$. If

$$d_t \ll \nu_{\beta,t}, \qquad \frac{dd_t}{d\nu_{\beta,t}} \le M_t,$$

then every nonnegative error function satisfies

$$\mathbb{E}_{d_t} e_t(c) \le M_t \mathbb{E}_{\nu_{\beta,t}} e_t(c).$$

This follows by integrating against the density ratio. A large M_t can make an offline bound uninformative, and the ratio may not exist. For example, a fixed exact command and a continuously distributed training return can create a singular distribution shift. This display is a conditional tool, not an assumption that all deployment shifts admit a bounded ratio.

Even transferred accuracy in estimating π_{rc} is not by itself a control guarantee: π_{rc} may be inappropriate in a stochastic environment. The quality of the comparison policy matters as well.

12.11.3 Recombination versus invention

A representation may permit a prefix from one trajectory to be combined with a continuation resembling another. The junction must preserve the relevant state, task, remaining horizon, and return condition. Full-history conditioning need not identify two prefixes merely because their last observed states agree.

Sequence modeling alone does not guarantee the trajectory stitching that dynamic programming can sometimes justify from adequate transition coverage [**brandfonbrener2022return**]. Conversely, structural assumptions can allow a learner to predict useful unseen combinations. The correct distinction is between an additional justified generalization mechanism and an automatic consequence of supervised fitting.

12.12 Why High-Return Conditioning Does Not Guarantee Optimality

The population supervised target is a return-conditioned version of the behavior policy, not directly a solution of

$$\pi^\star \in \arg\max_\pi J(\pi), \qquad J(\pi) = \mathbb{E}_\pi \sum_{t=0}^{H-1} \gamma^t r_t.$$

For finite return support and a positive denominator, Bayes' rule gives

$$\pi_{\mathrm{rc}}(a \mid h, g) = \beta_t(a \mid h) \frac{\mathbb{P}_\beta(G_t = g \mid h, a)}{\mathbb{P}_\beta(G_t = g \mid h)}.$$

It reweights behavior according to the probability of one realized return under the recorded continuation. This is different from maximizing an action's expected return or its return under an improved continuation. The distinction is studied explicitly in [**brandfonbrener2022return**, **paster2022luck**].

A positive deterministic case. The following elementary special case identifies assumptions under which the trajectory-generation interpretation is exact. It is included as a self-contained derivation, not as a new general optimal-control theorem.

Proposition 12 (Exact realization of a supported deterministic return)**.** *Consider a finite-state, finite-action, finite-horizon MDP with deterministic transitions and rewards, fixed s_0, and $0 < \gamma \leq 1$. Let β be a nonanticipating behavior policy, and suppose*

$$\mathbb{P}_\beta(G_0 = g_0 \mid s_0) > 0.$$

A deployed policy uses the exact full-history action conditionals $\pi_{\mathrm{rc}}(a \mid h_t, g_t)$ of the behavior law on all histories with positive probability under this conditioning, starts with $G_0^\star = g_0$, and updates the command by (12.2)*. Then its trajectory law is*

$$\mathbb{P}^{g_0}_{\pi_{\mathrm{rc}}}(\tau \mid s_0) = \mathbb{P}_\beta(\tau \mid s_0, G_0 = g_0),$$

and its realized discounted return equals g_0 with probability one.

Proof. At a supported history and command, every action assigned positive conditional probability appears on at least one behavior trajectory having that remaining return. Its next state and reward are deterministic. They are therefore the same in the conditioned trajectory and in actual deployment. Updating the command leaves the supported return $(g_t - r_t)/\gamma$ at the successor history.

Inductively, deployment remains in the support of the conditioned trajectory law. Its conditional action probabilities and deterministic environmental outcomes agree with that law at each step. The chain rule therefore gives equality of the joint trajectory distributions. The conditioned law is supported on $G_0 = g_0$, proving the return assertion. □

This result does not assume that a finite learner has already estimated all needed conditionals accurately: that is an additional learning premise. If g_0 is the largest return represented by the behavior support, the result reaches the best *supported* deterministic return. It is globally optimal only when that supported value is the optimal value. A restricted context window, approximate conditionals, continuous conditioning, or stochastic dynamics requires its own extension.

A stochastic counterexample even with full action support. Consider a one-step task with two actions. The safe action yields reward 1. The risky action yields reward 10 with probability $1/20$, and 0 otherwise. Both actions are sampled with probability $1/2$ in the offline behavior law. Thus the requested return $g_0 = 10$ is genuinely in the data-generating support, with probability $1/40$.

The exact supervised conditional is

$$\mathbb{P}_\beta(A = \text{risky} \mid G_0 = 10) = 1.$$

Nevertheless, deploying that action gives

$$\mathbb{P}(G_0 = 10) = \frac{1}{20}, \qquad J(\pi^{10}_{\mathrm{rc}}) = \frac{1}{2} < 1 = J(\text{safe}).$$

The conditional action law is learned perfectly; the policy is still suboptimal for expected return and usually fails to attain the command. The model cannot condition the real environment on winning the gamble. This example instantiates the stochastic-selection failure discussed in [**paster2022luck**]; its numerical values and calculation are supplied here for clarity.

The appropriate objective must also be named. Maximizing expected return, attaining a specific return almost surely, and maximizing the probability of exceeding a threshold are different control tasks. A large numerical command does not make them equivalent.

12.13 Exploration Is the Missing Operation

Offline sequence modeling conditions on existing experience. Interaction can generate new experience. The title identifies a missing operation of the basic offline procedure, not a claim that exploration is the only remaining difficulty or is necessary in every already-covered task.

12.13.1 Why exploration matters

A rewarding region never visited in the dataset is not directly observed by fitting that dataset. Without structural assumptions or additional information, behavior there is not determined. For example, two MDPs can agree on every state–action pair used by a behavior policy yet assign different rewards to an unused action. They produce the same offline behavior law but can have different optimal policies.

A model may extrapolate correctly in one of them, but the shared data alone cannot certify that choice in both. This is the informational reason to seek new experience:

offline learning reorganizes available experience; exploration can acquire previously unavailable evidence.

A trusted prior model can also supply relevant information. Merely sampling a model trained on the same unsupported data does not acquire new environmental evidence.

12.13.2 The extrapolation barrier

Deployment changes the distribution of histories through the policy's own actions. Requested returns change it as well. The risk is not simply large Euclidean distance from stored states: rare commands, hidden modes, and missing recovery actions can matter even near observed states.

Coverage-aware control, uncertainty estimates, and additional interaction are possible responses. No single one is an unconditional solution. The question is whether the chosen method establishes useful behavior on the histories it actually induces, rather than only small offline prediction loss.

12.14 A Hybrid Picture

A productive interpretation is that supervised trajectory learning and online reinforcement learning can address complementary parts of the control problem. This is a design possibility, not a universal ranking of algorithms.

12.14.1 Offline pretraining

Large trajectory datasets can train representations and policies with supervised objectives. Appropriate regularization can select a statistically controlled predictor when the architecture, loss, sampling assumptions, and complexity bound match, as in Chapter H.

A small supervised validation loss remains a statement about that validation distribution and target. It is not a certificate of high closed-loop return. A frozen pretrained representation may simplify later policy fitting, but must retain task-relevant state and command information. Recovering a teacher's particular internal representation is not necessary for successful control.

12.14.2 Online adaptation

Interaction can add missing experience and improve behavior. Possible components include policy-gradient or actor–critic updates, exploration, search, planning, and new trajectory collection followed by further supervised fitting. These are different operations; planning in a learned model is not itself online evidence collection, and interaction alone does not guarantee improvement.

The hybrid picture is

offline supervised learning + specified exploration, planning, or adaptation.

Each added stage must be evaluated on its own objective, distribution, and resource budget. Continued stochastic training fluctuations do not invalidate a guarantee for a suitably returned policy, but neither do they supply a control-performance guarantee.

12.15 A World-Model Interpretation

A policy and a world model answer complementary questions:

$$\begin{aligned} \pi_\theta(a \mid h, g): &\quad \text{which action is associated with the command?} \\ \widehat{P}_\phi(s', r \mid h, a): &\quad \text{what outcome is predicted after that action?} \end{aligned}$$

An accurate environmental model can evaluate candidate actions rather than merely imitate actions associated with selected outcomes. This requires accuracy for the queried actions and the states they reach. A model of the logged observational law does not automatically identify the consequences of interventions under hidden confounding.

12.15.1 Planning as conditional simulation

Generating hypothetical trajectories is not yet planning. Planning also specifies how candidates are compared, which action is executed, and how new observations update the process. Model rollouts should sample the unconditioned environmental innovations appropriate to candidate actions, not only successful innovations selected by a return condition.

The advantage can be made precise in a small comparison. For a finite candidate policy set Π, suppose true and model-evaluated expected returns satisfy, simultaneously,

$$|\widehat{J}(\pi) - J(\pi)| \le \Delta \quad \text{for every } \pi \in \Pi.$$

If $\widehat{\pi}$ maximizes $\widehat{J}$ over Π, then

$$J(\widehat{\pi}) \ge \max_{\pi \in \Pi} J(\pi) - 2\Delta.$$

Indeed, for a true maximizing candidate π_Π,

$$J(\widehat{\pi}) \ge \widehat{J}(\widehat{\pi}) - \Delta \ge \widehat{J}(\pi_\Pi) - \Delta \ge J(\pi_\Pi) - 2\Delta.$$

This elementary added comparison shows exactly what model accuracy buys. It does not prove the simultaneous accuracy premise, candidate-set adequacy, or efficient search. Monte Carlo evaluation error and model-selection effects must be included in Δ. A planner can otherwise exploit its own model's errors.

12.16 Connection to Compositional Sparsity

Closed-loop behavior is a temporal composition of controlled transitions. But grouping all history into one symbol does not prove low-dimensional functional arity. The objects h_t, s_t, and a_t can be large, and a policy can depend densely on their coordinates.

The computational implication from Chapter 4 is more specific. If a returned controller has a finite description and can be evaluated to the required precision with polynomially controlled work, its evaluation admits a bounded-fan-in digital implementation. Unrolling H such evaluations costs the corresponding work over H steps, including history storage, command updates, and any world-model or planning calls.

This does not prove sparse attention, a low-arity smooth state dynamics, or a physically local implementation. It also does not provide an efficient simulator of an arbitrary external environment. A hardware or world-model claim must account for those additional resources.

There is a new distinction at the control level:

accurate conditional action prediction
$\neq$ high return under the induced closed-loop law.

The deterministic proposition and the rollout bound below give explicit ways to connect these properties under additional assumptions. Once a controller has been shown to perform well and to be computationally feasible, its own implementation is the computational witness. No teacher-coordinate matching is required. Computational feasibility alone, however, certifies neither a large return nor a good statistical bound for the selected policy.

12.17 Predictions and Empirical Tests

The partial-trace interpretation suggests tests, not unconditional predictions that adding information, increasing depth, or using a world model always improves performance. The following are proposed experiments; no new empirical results are reported here.

12.17.1 Testing whether richer outcome traces reduce the burden on depth

Compare scalar return conditioning with $(G_t, \text{subgoal}_t)$, landmark information, and partial plans while varying architecture depth, context length, and whether the representation is frozen or learned. Hold the environment, data budget, and evaluation criteria fixed.

Record how each extra annotation is obtained and whether it is available at deployment. A realized future state supplied by a teacher in training cannot silently become an oracle input at test time. Train-only auxiliary supervision and deployable goal conditioning are distinct experiments. The hypothesis is that an informative, usable annotation can reduce the representation or credit-assignment burden; it is not a theorem that any richer annotation reduces necessary depth.

12.17.2 Return-conditioning benefit and trajectory diversity

Vary diversity while distinguishing successful strategic alternatives from variability caused by environmental randomness. Identical return labels can correspond to different action modes; a distributional policy may represent them without choosing a unique intended behavior.

Compare scalar return, task identity, and subgoal conditioning, and compare conditional densities with mean-action regression. Nearly deterministic action labels do not alone make the

history-to-action mapping easy. Measure held-out conditional prediction, realized return, and sensitivity to the command separately.

12.17.3 Testing requested returns inside and outside conditional support

Sweep $G_0^\star$ both within and beyond the supported return conditions for each initial state. Distinguish interpolation between sufficiently covered conditions from requests merely inside the global min–max range. Measure the command left at termination, the full achieved-return distribution, expected return, and failure probability.

Include the stochastic two-action example of Section 12.12. It separates poor prediction from uncontrollable favorable outcomes even when the requested return and both actions are supported. Successful extrapolation elsewhere should be investigated rather than ruled out in advance.

12.17.4 World-model augmentation under controlled distribution shift

Compare a direct policy, the same policy with an oracle simulator, and the policy with learned models of controlled error. Separate shifts in policy-induced visitation from actual changes to the environmental law. A world model trained on unchanged data can fail under either shift.

Measure model error on planned rollouts, candidate-search cost, calibration, and real environmental return. Split training and testing by complete episodes, report variation across independent training runs, and keep annotation and online-interaction budgets visible. A model's predicted return is not a substitute for the achieved return. The proposed positive effect is conditional on adequate modeling and planning, not on out-of-distribution deployment by itself.

12.18 Limitations

The preceding calculations identify both useful regimes and the boundaries of the return-conditioned formulation.

12.18.1 Return-to-go is not a faithful computational trace

Many futures share one scalar return. An annotation does not identify the hidden computation, a sufficient state, or a unique causal explanation. Its benefit is in the organization of supervision. The coverage and restoration results of Chapter 11 do not transfer without their explicit transition and representation assumptions.

12.18.2 Supervised learning does not solve exploration

Unobserved state–action consequences need interaction, a valid structural inference, or another information source. Offline supervision cannot create evidence absent from the data. The method can still generalize when appropriate structure is available; the limitation is the absence of an unconditional guarantee.

12.18.3 Correlation is not causal credit

An action associated with a high realized return can have a low expected return. Return selection can change the conditional environmental law even in a fully observed MDP. Separately, hidden confounding can prevent an observational world model from identifying intervention outcomes. Neither problem is resolved merely by an attention mask or a low supervised loss.

12.18.4 Autoregressive rollout accumulates errors

A small action error can change the next state, history, and desired remaining return. The distribution of later policy inputs can therefore leave the training support. This is the sequential-prediction issue studied in imitation learning [**ross2011reduction**]. The following added elementary bound makes the relevant error criterion explicit; it does not assume offline and rollout distributions coincide.

Proposition 13 (From conditional action error to a rollout bound)**.** *Let π° and $\widehat{\pi}$ be two history–command policies with finite action space, the same initial-state law and command rule, the same horizon H, and the same true environmental kernel. Both update history and the command by the specified recurrence. Let d_t° be the law of $c_t = (h_t, G_t^\star)$ under π°, and set*

$$\epsilon_t = \mathbb{E}_{c\sim d_t^\circ} \operatorname{TV}\big(\widehat{\pi}(\cdot \mid c), \pi^\circ(\cdot \mid c)\big).$$

Their complete trajectory laws have total variation distance at most

$$\min\left\{1, \sum_{t=0}^{H-1} \epsilon_t\right\}.$$

If $0 \le r_t \le R_{\max}$ almost surely and $S_{\gamma,H} = \sum_{t=0}^{H-1} \gamma^t$, then

$$\boxed{|J(\widehat{\pi}) - J(\pi^\circ)| \le R_{\max} S_{\gamma,H} \min\left\{1, \sum_{t=0}^{H-1} \epsilon_t\right\}.}$$

Proof. Couple the initial states and commands identically. As long as the two histories agree, couple their actions maximally. The probability of an action disagreement at that context is its total variation distance. When actions agree, use the same environmental outcome, so that histories and budgets continue to agree.

The probability that the first disagreement occurs at time t is bounded by the reference expectation ϵ_t: restricting that expectation to paths without an earlier disagreement can only decrease the integral of its nonnegative integrand. Summing bounds the probability of any disagreement by $\sum_t \epsilon_t$. The coupling inequality gives the trajectory total variation bound. Finally, the discounted return is a function of the trajectory lying in $[0, R_{\max} S_{\gamma,H}]$. Its expectations differ by at most that range times total variation. □

For $\gamma = 1$ and $\epsilon_t \le \epsilon$, the unsaturated bound is $R_{\max} H^2 \epsilon$. This is a sufficient worst-case bound, not a necessary rate for every control problem. A useful reference controller, small error under d_t°, and adequate horizon control are all required. If the reference achieves a fixed target exactly, the same coupling bounds the learned controller's probability of missing it. If the reference is the poor stochastic return-conditioned policy of the counterexample, approximating it well does not produce good control.

To connect log-loss fitting with this criterion, on finite action spaces Pinsker's inequality and Jensen's inequality give

$$\epsilon_t \le \sqrt{\frac{1}{2} \mathbb{E}_{d_t^\circ} \operatorname{KL}(\pi^\circ(\cdot \mid c) \| \widehat{\pi}(\cdot \mid c))}.$$

The KL expectation must be on the reference rollout distribution. An offline excess-log-loss bound can be substituted only with an appropriate distribution-transfer argument, such as the conditional bound in Section 12.11. For continuous deterministic actions, small Euclidean regression error need not give small total variation; a dynamical continuity argument or another metric is needed.

12.18.5 Desired return may be infeasible

A command does not make an outcome attainable. With $0 \le r_t \le R_{\max}$, a necessary range condition is

$$0 \le G_t^\star \le R_{\max} \sum_{j=0}^{H-1-t} \gamma^j,$$

but even a command inside this range can be impossible from the current state, unsupported by the data, or possible only through luck. A supported deterministic return, a high-probability stochastic target, and an expected-return objective are different specifications.

12.18.6 Bellman methods remain useful

Value estimation and dynamic programming can exploit transition structure and compose information across outcomes. They are not restricted to online interaction; offline value-based methods exist as well. Conversely, a supervised sequence policy can be updated using newly collected data. Thus the distinctions "offline versus online" and "supervised return conditioning versus Bellman learning" are different axes [**sutton2018reinforcement**, **brandfonbrener2022return**].

A method combining them still needs assumptions covering the complete algorithm. No formulation eliminates support, estimation, optimization, or model error merely by changing terminology.

12.19 Conclusion

Return-conditioned sequence modeling supplies an important alternative organization of offline control. A recorded trajectory is annotated with

$$G_t = \sum_{u=t}^{H-1} \gamma^{u-t} r_u,$$

and a policy is fitted through conditional action prediction, $(h_t, G_t) \mapsto a_t$. The basic objective contains no bootstrapped Bellman target. Return labels make delayed outcomes available at each learning decision without adding information to the complete recorded reward sequence.

At deployment, the policy acts in closed loop with the true environment and updates a *desired* remaining-return budget. That budget is a command, not an observed future or an enforced boundary condition. A complete return-conditioned trajectory law contains conditioned environmental outcomes as well as conditioned actions. In stochastic environments, reproducing only the latter need not reproduce the former.

The deterministic proposition gives a positive exact case: supported returns, exact full-history conditionals, and deterministic transitions and rewards suffice to reproduce the conditioned trajectory law. The stochastic counterexample shows why support and even perfect conditional action prediction are insufficient in general. The rollout bound makes the additional requirements for approximate control explicit: a useful reference policy and controlled action error on the histories that matter during execution.

Return-to-go is therefore a partial or surface *outcome annotation*, not the faithful computational-state exposure used by the preceding chapter. More informative annotations can help, but their availability, coverage, and effect on the actual control objective must be established. A world model adds simulation and, with a selection rule, planning; its accuracy under candidate actions

and accumulated rollout error remain separate requirements. Numerical-integration claims need further continuous-time and discretization assumptions.

The book's division of labor remains:

representation and intermediate information
$\longrightarrow$ a specified supervised fitting problem;

successful fitting and statistical control
$\longrightarrow$ conditional prediction accuracy;

appropriate environment, support, and rollout guarantees
$\longrightarrow$ useful closed-loop control.

Each line requires its own assumptions. A computationally feasible, successful controller supplies its own compositional implementation; it need not identify the teacher's states or be an absorbing point of SGD. The challenge is to establish its performance under the law induced by its decisions. The following chapter's description-length viewpoint can help quantify statistical selection, but a compact code alone does not close this prediction-to-control gap.

Part III
Learning and Evolution

CHAPTER 13

Bits as a Currency of Inductive Bias and Generalization

Deep learning uses many apparently different forms of inductive bias: local connectivity, weight sharing, compositional sparsity, low rank, norm control, quantization, architectural bottlenecks, and implicit selection by optimization.

This chapter develops a common language for comparing many of these mechanisms: the number of bits required to specify a predictor to the resolution relevant for prediction.

The central quantity is not simply the number of parameters. A small number of arbitrary real parameters can contain arbitrarily many bits, while a very large network may implement a function that is highly compressible.

Two related but distinct notions of description complexity will be used.

For an individual predictor, we ask for the length of a code specifying that predictor, possibly only approximately.

For a class of predictors, we ask how many bits are required in the worst case to identify one element of an ε-cover. This is metric entropy:

$$\mathcal{H}(\varepsilon, \mathcal{F}, d) = \log_2 \mathcal{N}(\varepsilon, \mathcal{F}, d).$$

These two notions lead to two parallel statistical routes:

individual description length	$\longrightarrow$	*Occam/compression generalization*
$B_\varepsilon(f)$		$O\left(\sqrt{B_\varepsilon(f)/n}\right)$
class covering bits	$\longrightarrow$	*Rademacher/chaining bounds*
$\mathcal{H}(\varepsilon, \mathcal{F})$		$\mathfrak{R}_n(\mathcal{F})$.

Architecture and optimization enter at different stages.

Architecture determines which descriptions are cheap or even available. Optimization, regularization, initialization, and the data determine which of those descriptions is selected.

Thus a large architecture need not define a uniformly small hypothesis class for a particular selected predictor to admit a useful code-based bound. Small description length controls a generalization allowance; small population risk additionally requires a sufficiently good empirical fit. No claim is made that every accurate predictor must have a short code in one prescribed language.

The relevant question is instead:

how many bits are required to describe the selected predictor at the resolution relevant to prediction?

This chapter makes that statement precise in several elementary settings and connects it to compositional sparsity, weight sharing, sparsity, low rank, norm control, and quantization.

Statistical and computational conventions. Unless stated otherwise, the sample consists of $n \geq 1$ independent observations $Z_i = (X_i, Y_i)$ from one distribution P. We write $R(f) = \mathbb{E}_P \ell(f(X), Y)$ and $\widehat{R}_S(f) = n^{-1} \sum_i \ell(f(X_i), Y_i)$. Predictors and losses are measurable; the usual separability or measurability conditions are assumed for suprema over uncountable classes. The elementary probability bounds use a loss in $[0, 1]$; a loss in $[0, M_\ell]$ is rescaled, and an unbounded loss needs a different argument. For example, squared loss is bounded and Lipschitz only on specified bounded prediction and label ranges. The logarithm $\ln$ is natural; $\log_2$ counts bits, and an unsubscripted $\log$ in an entropy integral denotes the natural logarithm.

A decoder and its admissible codes are fixed independently of the sample used in a generalization bound. Numerical evaluation is part of the predictor being decoded. Description length, decoding time, prediction cost, and the cost of finding the code are separate resources. A short code need not be quickly evaluable, and a polynomial-length code need not be short enough relative to n to give an informative bound.

13.1 Why Bits?

A neural network can contain billions of parameters and nevertheless implement a simple function.

Conversely, one exact real parameter can encode an arbitrarily long binary string in its digits.

Therefore

number of parameters $\neq$ number of bits of effective complexity.

The difference is fundamental.

13.1.1 Finite precision makes the question operational

Suppose a network has P parameters and every parameter is stored with b bits.

If the architecture, parameter order, quantization alphabet, and evaluator are fixed in advance, the literal parameter payload requires at most

$$bP$$

bits.

This is an upper bound on the description length of the parameterization, not necessarily on the shortest description of the represented function.

A complete self-delimiting description also specifies any varying architecture, scales, dimensions, and other metadata. These costs are not included in bP. At a fixed format and length the payload code is already prefix-free; combining different formats requires headers or another valid prefix construction.

Several parameter strings may compute the same function.

The function may be sparse.

Matrices may be low rank.
Parameters may be shared.
Repeated subprograms may be represented once and reused.
Or the entire network may be generated by a much shorter algorithm.
Thus the relevant question is

> How many bits are actually required to specify the learned predictor to the precision relevant for the loss?

13.2 Individual Description Length

We begin with one predictor rather than a whole class.

13.2.1 Kolmogorov complexity

Fix a universal Turing machine U.

For a finite encoded object f, use the program-length notation

$$K_U(f) = \min_{p:\, U(p)=f} |p|.$$

This is the length of the shortest program producing that object, in the algorithmic-description viewpoint associated with Kolmogorov's work [81]. When f denotes a function on an infinite domain, $U(p) = f$ is shorthand for a program evaluating f under a specified input and precision convention, not for printing infinitely many values. One must specify that convention before comparing lengths. The statistical results below use explicit prefix codes rather than transferring an arbitrary plain-program complexity directly into Kraft's inequality.

Kolmogorov complexity captures the strongest possible notion of algorithmic compressibility.

It is nevertheless not directly usable as a general learning-theoretic quantity.

First, it is uncomputable.

Second, exact reproduction is often stronger than prediction requires.

Third, learning-theoretic statements require specifying what decoder or coding language is available independently of the observed sample.

13.2.2 Approximate description length

Let $d(\cdot, \cdot)$ be a metric on predictors.

For a fixed coding language or decoder $\mathcal{D}$, define

$$B_{\varepsilon,\mathcal{D}}(f) = \min \{|c| : d(f, \mathcal{D}(c)) \le \varepsilon\}. \tag{13.1}$$

When the decoder is understood, we write simply

$$B_\varepsilon(f).$$

This is the number of bits required to describe a predictor that is ε-close to f.

The minimum is $+\infty$ if no admitted code gives the required approximation. For the Occam applications the admitted codewords must form a fixed prefix-free family, or carry lengths satisfying Kraft's inequality. A finite cover with arbitrary real-valued centers is a statistical counting device; it is not automatically an effectively constructible codebook of feasible numerical evaluators.

The distinction from exact Kolmogorov complexity is

$K(f):$	exact algorithmic description,
$B_\varepsilon(f):$	description sufficient at resolution ε.

For statistical prediction, B_ε is often the more relevant object.

13.2.3 Why few parameters do not imply few bits

Suppose

$$f_\theta$$

has one parameter

$$\theta \in \mathbb{R}.$$

If arbitrary precision is permitted, the binary expansion of θ can contain arbitrarily many bits. Hence

$$\boxed{P \text{ parameters} \Longrightarrow O(P) \text{ bits.}}$$

Parameter counting becomes description counting only after precision, norms, sparsity, rank, or another effective restriction has been imposed.

Conversely, digits that the specified predictor never uses need not be charged as functional complexity. Encoding a long string in one real number establishes an information-storage possibility, not a constant-cost way to access those digits. The evaluator and precision requirements cannot be hidden in a nominal one-parameter model.

This is the same finite-precision distinction used earlier in the book in the definition of efficient real computation.

13.3 A Generalization Theorem for Individual Codes

The cleanest connection between bits and generalization does not require metric entropy.

It can be stated directly for individual code lengths.

13.3.1 Prefix codes

Let $\mathcal{C}$ be a countable collection of predictors.

The collection, prediction rule for each member, and code assignment are fixed independently of S. Multiple codewords may decode to the same function; one may either union-bound over codewords or retain a shortest codeword for each distinct function. Both conventions satisfy the required Kraft bound.

Assign each predictor $f \in \mathcal{C}$ a prefix-free binary code of length

$$L(f) \in \mathbb{N}.$$

Prefix-freeness implies Kraft's inequality,

$$\sum_{f \in \mathcal{C}} 2^{-L(f)} \leq 1. \tag{13.2}$$

Theorem 13 (Prefix-code Occam bound)**.** *Fix $\mathcal{C}$ and the prefix-free code assignment independently of the sample, as above. Assume $n \geq 1$, $0 < \delta < 1$, and*

$$0 \leq \ell(f(x), y) \leq 1$$

for all admitted coded predictors and observations.

Let a sample

$$S = \{(x_i, y_i)\}_{i=1}^{n}$$

be drawn i.i.d. from the data distribution.

Then, with probability at least $1 - \delta$, simultaneously for every $f \in \mathcal{C}$,

$$\boxed{\left|R(f) - \widehat{R}_S(f)\right| \leq \sqrt{\frac{L(f)\ln 2 + \ln(2/\delta)}{2n}}.} \tag{13.3}$$

Proof. For a fixed f, Hoeffding's inequality gives

$$\Pr\left[|R(f) - \widehat{R}_S(f)| > \eta\right] \leq 2e^{-2n\eta^2}.$$

Set

$$\eta_f = \sqrt{\frac{L(f)\ln 2 + \ln(2/\delta)}{2n}}.$$

Then

$$\Pr\left[|R(f) - \widehat{R}_S(f)| > \eta_f\right] \leq \delta\, 2^{-L(f)}.$$

A union bound over all coded predictors gives

$$\Pr\left[\exists f \in \mathcal{C} : |R(f) - \widehat{R}_S(f)| > \eta_f\right] \leq \delta \sum_{f \in \mathcal{C}} 2^{-L(f)} \leq \delta$$

by Kraft's inequality. □

The important feature of Theorem 13 is that the bound is simultaneous over the entire fixed coding language.

Therefore the predictor may be selected *after seeing the training sample*.

In particular,

data-dependent selection of a predictor is allowed when the code itself was fixed in advance.

What cannot be introduced for free after observing the data is a new coding language tailored to the sample.

If the decoder, dictionary, architecture, or codebook is itself learned from the sample and is needed to reconstruct the predictor, its description must also be paid for, or the analysis must use another valid data-dependent complexity argument.

A smaller gap is not automatically small risk. The theorem gives $R(f) \le \widehat{R}_S(f) + \mathrm{pen}(L(f))$. A one-line constant predictor can have a small generalization gap and a large empirical and population error. Conversely, failure of this particular bound to be informative does not prove failure to generalize. The complexity term is sufficient statistical control, not a complete characterization of test performance.

Zero empirical classification error. A useful elementary refinement applies to 0–1 loss. For a fixed coded classifier f, $\Pr\Big(\widehat{R}_S(f) = 0\Big) = (1 - R(f))^n \le e^{-nR(f)}$. The same Kraft-weighted union bound therefore shows, on an event of probability at least $1 - \delta$, simultaneously for every coded classifier that fits all labels,

$$R(f) \le \min\left\{1, \frac{L(f)\ln 2 + \ln(1/\delta)}{n}\right\}.$$

This is an additional zero-error calculation, not the rate asserted by the two-sided Hoeffding theorem. Exact interpolation is not required for the latter, and is not assumed for regularized training in general.

13.4 The Finite-Class Bound as a Special Case

Suppose

$$|\mathcal{F}| \le 2^B.$$

Assume $\mathcal{F}$ is fixed independently of the sample and $B \ge 0$. Every member can be assigned a fixed-length code using

$$\lceil \log_2 |\mathcal{F}| \rceil \le \lceil B \rceil$$

bits (a singleton needs no identification bits). More directly, a union bound over $|\mathcal{F}|$ predictors gives the next display without any coding-rounding loss.

The preceding theorem therefore reduces to the familiar finite-class bound.

Corollary 2 (Finite-bit generalization)**.** *Let $\mathcal{F}$ be fixed independently of an i.i.d. sample of size n, with $|\mathcal{F}| \le 2^B$, $B \ge 0$, and loss in $[0, 1]$.*

Then with probability at least $1 - \delta$, simultaneously for all $f \in \mathcal{F}$,

$$\left|R(f) - \widehat{R}_S(f)\right| \le \sqrt{\frac{B\ln 2 + \ln(2/\delta)}{2n}}, \tag{13.4}$$

The result follows by summing the fixed-predictor Hoeffding failure probabilities over at most 2^B predictors.

Thus

$$\boxed{B \text{ bits} \quad \Rightarrow \quad \text{generalization gap} = O\left(\sqrt{\frac{B + \log(1/\delta)}{n}}\right).}$$

This is the simplest precise form of the statement

short descriptions generalize.

13.5 Approximate Compression of the Selected Predictor

Exact reproduction is unnecessary when the loss is insensitive to a small prediction perturbation.

Suppose f is a learned predictor and g is a coded predictor satisfying

$$d_\infty(f,g) = \sup_x |f(x) - g(x)| \le \varepsilon.$$

Assume the loss is L_ℓ-Lipschitz in its prediction argument.

The loss and its Lipschitz constant are common to f and g on the admitted prediction range, and the bounded-loss convention of the preceding theorem remains in force. The uniform norm is over the specified input domain, not only over the training inputs.

Theorem 14 (Approximate coded compression)**.** *Under the event of Theorem 13, for every predictor f and every coded $g \in \mathcal{C}$ satisfying*

$$\sup_x |f(x) - g(x)| \le \varepsilon,$$

we have

$$\boxed{R(f) \le \widehat{R}_S(f) + 2L_\ell\varepsilon + \sqrt{\frac{L(g)\ln 2 + \ln(2/\delta)}{2n}}.} \tag{13.5}$$

Proof. Lipschitzness gives

$$R(f) \le R(g) + L_\ell\varepsilon$$

and

$$\widehat{R}_S(g) \le \widehat{R}_S(f) + L_\ell\varepsilon.$$

Apply Theorem 13 to g and combine the inequalities. □

If

$$B_\varepsilon(f) = \min_{g:\, d_\infty(f,g)\le\varepsilon} L(g),$$

then

$$\boxed{R(f) \le \widehat{R}_S(f) + 2L_\ell\varepsilon + \sqrt{\frac{B_\varepsilon(f)\ln 2 + \ln(2/\delta)}{2n}}.} \tag{13.6}$$

One may optimize over ε on this same event: the random event already covers every code in the one fixed language, and the approximation inequalities are deterministic. This does not permit a different, unaccounted-for decoder to be invented at each resolution.

This gives a direct statistical meaning to the approximate description length of the *selected solution*.

Certifying the compressed predictor versus the original one. The coded predictor g has its own guarantee

$$R(g) \le \widehat{R}_S(g) + \sqrt{\frac{L(g)\ln 2 + \ln(2/\delta)}{2n}}.$$

This requires no comparison to f. To certify the original f using compression, however, one must also control their population discrepancy. More generally, Lipschitzness gives

$$R(f) \le \widehat{R}_S(f) + L_\ell\left(\mathbb{E}_{P_X}|f-g| + \frac{1}{n}\sum_i |f(X_i) - g(X_i)|\right) + \sqrt{\frac{L(g)\ln 2 + \ln(2/\delta)}{2n}}.$$

The two discrepancy terms have different meanings. The uniform ε guarantee bounds both; matching only on the training sample bounds neither population disagreement nor $R(f)$ by itself.

For a direct counterexample, take X uniform on $[0,1]$, $Y = 0$, and squared loss for predictions in $[0,1]$. Set $g \equiv 0$ and let f_S be zero on the finite set of training inputs and one elsewhere. Then $\widehat{R}_S(f_S) = \widehat{R}_S(g) = 0$ and the empirical prediction distance is zero, but $R(f_S) = 1$ and $R(g) = 0$. This example concerns arbitrary measurable predictors; it identifies the missing population condition, not a claim about the particular networks produced in practice.

13.6 From Individual Bits to Class Bits: Metric Entropy

Individual code length and metric entropy are related but distinct.

Let

$$\mathcal{F}$$

be a class of predictors and let

$$\mathcal{N}(\varepsilon, \mathcal{F}, d)$$

be the smallest number of radius-ε balls required to cover $\mathcal{F}$ in the specified metric, or pseudo-metric, d. We allow centers in the ambient space of admitted predictors. If centers must belong to $\mathcal{F}$, recentering a nonempty covering ball at one of its members changes the radius by at most a factor of two. Such factors are absorbed in the schematic entropy bounds below.

Define

$$\mathcal{H}(\varepsilon, \mathcal{F}, d) = \log_2 \mathcal{N}(\varepsilon, \mathcal{F}, d). \tag{13.7}$$

Hence

$$\boxed{\mathcal{H}(\varepsilon, \mathcal{F}, d) = \text{worst-case number of bits needed to index an } \varepsilon\text{-cover.}}$$

Metric entropy is therefore a class-level description complexity.

Precisely, the fixed-length index requires $\lceil \log_2 \mathcal{N} \rceil$ bits once the codebook is supplied. Metric entropy omits the cost of describing a codebook that was not already fixed. Covers at separate resolutions are not automatically one prefix language covering all resolutions.

It differs from $B_\varepsilon(f)$ in an important way.

The latter asks how cheaply one particular predictor can be represented.

The former asks how many bits are sufficient in the worst case across the entire class.

13.6.1 Covering-number generalization

For a fixed $\varepsilon > 0$, a sample-independent class, and the uniform prediction metric, take a fixed finite minimal ε-cover

$$\mathcal{C}_\varepsilon$$

of $\mathcal{F}$.

Its cardinality is

$$|\mathcal{C}_\varepsilon| = \mathcal{N}(\varepsilon, \mathcal{F}, d).$$

Applying the finite-class theorem to this codebook gives schematically,

$$\boxed{R(f) \lesssim \widehat{R}(f) + \text{approximation error} + \sqrt{\frac{\mathcal{H}(\varepsilon, \mathcal{F}, d)}{n}}.}$$

For example, with uniform prediction distance, bounded loss, and an L_ℓ-Lipschitz loss, the precise upper bound for every $f \in \mathcal{F}$ is

$$R(f) \le \widehat{R}_S(f) + 2L_\ell\varepsilon + \sqrt{\frac{(\ln 2)\mathcal{H}(\varepsilon, \mathcal{F}, d_\infty) + \ln(2/\delta)}{2n}}.$$

This is a fixed-resolution event. To choose a resolution after seeing the sample, one may prescribe covers at scales ε_j and failure budgets $\delta_j > 0$ with $\sum_j \delta_j \le \delta$, then use the union bound. Equivalently, a fixed multiscale prefix language charges for the scale index. Deterministic optimization of a bound before observing S does not need this extra selection allowance.

A cover constructed in the empirical metric d_S need not approximate population predictions, and its centers can depend on the sample. It cannot simply replace this fixed cover in the Hoeffding argument. The Rademacher route below uses empirical covers *inside a conditional complexity calculation*, followed by a separate symmetrization and concentration argument; this is why that sample dependence is legitimate.

13.7 Rademacher Complexity: The Statistical Language

Covering numbers are especially useful because they connect naturally to Rademacher complexity.

For a fixed input sample $S_X = (x_1, \ldots, x_n)$, abbreviated to S in empirical function metrics, define

$$\widehat{\mathfrak{R}}_S(\mathcal{F}) = \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \sigma_i f(x_i) \right], \tag{13.8}$$

where

$$\sigma_i \in \{-1, +1\}$$

are independent random signs.

The quantity measures how well the class can correlate with random fluctuations on the observed sample.

For a sample-independent measurable class and loss in $[0,1]$, a standard empirical-Rademacher bound gives, with probability at least $1-\delta$,

$$R(f) \le \widehat{R}_S(f) + 2\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{F}) + 3\sqrt{\frac{\ln(2/\delta)}{2n}} \qquad \text{for all } f \in \mathcal{F}.$$

Here $\ell \circ \mathcal{F}$ is the loss class on input–label pairs. Scalar Lipschitz contraction bounds its complexity by a universal constant times $L_\ell \widehat{\mathfrak{R}}_{S_X}(\mathcal{F})$; the precise factor depends on the convention for absolute versus one-sided Rademacher averages. The normalization in (13.8) is one-sided, with factor $1/n$. These risk and structural-complexity arguments are treated in [**bartlett2002rademacher**]. They bound the whole admitted class, not just the norm or code of an isolated selected parameter.

Thus the class-level question becomes

How do covering bits control Rademacher complexity?

13.8 Dudley: From Covering Bits to Rademacher Complexity

Define the empirical L_2 metric

$$d_S(f,g) = \left[\frac{1}{n}\sum_{i=1}^{n}\left(f(x_i)-g(x_i)\right)^2\right]^{1/2}. \tag{13.9}$$

Let $D_S = \operatorname{diam}(\mathcal{F}, d_S) < \infty$. For $D_S > 0$, under the stated measurability conventions, a truncated empirical version of Dudley's entropy integral gives

$$\widehat{\mathfrak{R}}_S(\mathcal{F}) \le \inf_{0<\alpha\le D_S}\left[C_1\alpha + \frac{C_2}{\sqrt{n}}\int_{\alpha}^{\operatorname{diam}(\mathcal{F},d_S)}\sqrt{\log \mathcal{N}(u,\mathcal{F},d_S)}\,du\right] \tag{13.10}$$

for universal constants C_1, C_2 [45].

The logarithm under the integral is natural, so replacing it by covering bits introduces a factor $\sqrt{\ln 2}$. If $D_S = 0$, all sample prediction vectors coincide and the one-sided Rademacher average is zero. For $D_S > 0$, translating by one fixed member of $\mathcal{F}$ changes no expected one-sided supremum; thus a separate assumption $0 \in \mathcal{F}$ is unnecessary in this diameter formulation. Restricting the cutoff to $\alpha \le D_S$ avoids an invalid negative integral.

For completeness, the increment process $Z_f = n^{-1}\sum_i \sigma_i f(x_i)$ obeys

$$\mathbb{E}_\sigma e^{t(Z_f - Z_g)} \le \exp\left(\frac{t^2 d_S(f,g)^2}{2n}\right),$$

by $\cosh u \le e^{u^2/2}$. Chaining finite covers at decreasing scales bounds successive increments by their scale times the square root of the logarithm of the cover size, divided by $\sqrt{n}$. The remaining sample discrepancy at cutoff α contributes at most a constant times α by Cauchy–Schwarz. This explains the two terms in (13.10); it invokes the standard chaining inequality rather than asserting that a single empirical cover is a population certificate.

13.8.1 The coding interpretation

Imagine specifying a predictor progressively.

First transmit a coarse approximation.

Then transmit the correction required to reach a finer scale.
Then another correction.
The resulting description is multiscale.
Thus

$$\boxed{\text{chaining} = \text{multiscale description of a function class.}}$$

The entropy integral combines the coding requirements across resolutions.

13.8.2 An important qualification

Metric entropy alone does not guarantee generalization.
One also needs the statistical assumptions appropriate to the bound, including, for example,

- controlled sampling,
- a suitable loss,
- boundedness or tail assumptions,
- finite entropy at the relevant scales.

The correct statement is

$$\boxed{\begin{array}{c}\text{smaller covering entropy}\\ \Downarrow\\ \text{smaller corresponding capacity bounds,}\\ \text{under the required assumptions.}\end{array}}$$

13.9 Architecture as a Coding Language

Architecture affects statistical complexity because it changes which functions can be described economically.
A useful decomposition is

$$\boxed{\begin{array}{c}\text{architecture defines a coding language}\\ \Downarrow\\ \text{optimization and regularization select a solution}\\ \Downarrow\\ \text{the selected solution has some effective code length}\\ \Downarrow\\ \text{that code length can control generalization.}\end{array}}$$

This is different from claiming that the architecture itself has a small hypothesis class.
Modern networks may have enormous nominal capacity.
The important fact may instead be that some functions have short descriptions in the architectural language.

13.9.1 Examples

Convolution makes repeated translation-equivariant operations cheap to describe.

Sparse architectures make restricted dependency patterns cheap.

Low-rank parameterizations make low-dimensional maps cheap.

Attention makes a content-addressable routing *rule* compact to specify when its projections and state representation have a compact description. The stored keys and values, candidate search, and read cost are not thereby free.

Residual architectures make incremental transformations cheap.

A CRM-like library makes reusable subprograms cheap.

Thus architecture determines the relative coding cost of different functions.

A short architectural instruction can expand into many arithmetic or communication operations. This is the distinction between description resources and execution resources developed in Chapters 9 and 10. The present chapter counts the former; it does not turn dense attention into a constant-time operation.

13.10 Compositional Sparsity as a Bit Constraint

We now connect the description-length viewpoint to C1.

13.10.1 Fixed computation DAG

Let a fixed DAG G contain

$$s$$

constituent nodes.

Assume each node depends on at most

$$k$$

scalar arguments and belongs to a specified bounded smoothness class. Take a fixed compact scalar state interval $K = [-R_0, R_0]$, fixed input domain $[-1,1]^d \subseteq K^d$, $R_0 \geq 1$, and classes $\mathcal{G}_m \subseteq C^r(K^m; K)$ with integer $r \geq 1$, $k \geq 1$, and $B \geq 1$. The domain and regularity constants are part of the class, not parameters silently allowed to grow with d.

Suppose that for $1 \leq m \leq k$ and $0 < \delta \leq 1$ the local constituent class satisfies the entropy estimate

$$\log \mathcal{N}(\delta, \mathcal{G}_m, \|\cdot\|_\infty) \leq C \left(\frac{B}{\delta}\right)^{m/r}. \tag{13.11}$$

For example, this is the schematic dependence expected for bounded C^r classes on fixed-dimensional domains.

We use (13.11) as an explicit hypothesis, rather than infer it from efficient digital computation alone. A zero-arity constant node, when allowed, has a separate interval cover costing $O(\log(1 + R_0/\delta))$ bits. At fixed positive k/r this is absorbed by the later worst-case $(B/\delta)^{k/r}$ term. Applying the exponent $m/r = 0$ directly to a varying constant would be incorrect.

Write $\mathcal{F}_G^{\text{local}}$ for the set of constituent tuples, with product distance $d_{\text{loc}}((g_v), (\widetilde{g}_v)) = \max_v \|g_v - \widetilde{g}_v\|_{\infty, K^{m_v}}$. Cover centers are kept in K by clipping, which cannot increase the uniform error to a K-valued constituent. Taking one local center per node gives a product cover satisfying

$$\log \mathcal{N}(\delta, \mathcal{F}_G^{\text{local}}) \lesssim Cs \left(\frac{B}{\delta}\right)^{k/r}. \tag{13.12}$$

13.10.2 Stability of the composition

Local approximation errors need not equal the final output error.

Let

$$\mathcal{S}_G$$

be a class-uniform factor, taken at least one, such that every admitted constituent tuple and every set of local cover replacements at error δ satisfy

$$\|f - \widetilde{f}\|_\infty \le \mathcal{S}_G \delta. \tag{13.13}$$

The guarantee is over the full input domain, and replacements must keep all intermediate states in their admitted domains. For example, if true constituents at level ℓ have maximum-norm Lipschitz constants at most $\Lambda_\ell \ge 1$, the recursion

$$e_\ell \le \Lambda_\ell e_{\ell-1} + \delta, \qquad e_0 = 0,$$

permits

$$\mathcal{S}_G = \sum_{\ell=1}^{L} \prod_{j=\ell+1}^{L} \Lambda_j.$$

This is one sufficient factor, not an assertion that it is optimal. A class-level entropy theorem requires a common bound over its members; a favorable sensitivity measured only at the fitted training inputs does not provide that bound.

To obtain final error ε, choose

$$\delta = \frac{\varepsilon}{\mathcal{S}_G}.$$

Hence

Proposition 14 (Entropy of a fixed stable compositional graph)**.** *For the fixed graph and local classes above, with the uniform stability property* (13.13)*, $\mathcal{S}_G \ge 1$, and $0 < \varepsilon \le 1$, the following bound holds. The constant can depend on fixed k, r, R_0 and the local entropy constants, but not on d, s or ε beyond the displayed factors.*

$$\boxed{\mathcal{H}(\varepsilon, \mathcal{F}_G, \|\cdot\|_\infty) \lesssim Cs \left(\frac{B\mathcal{S}_G}{\varepsilon}\right)^{k/r}.} \tag{13.14}$$

Proof. Choose a δ-cover for each local class, where $\delta = \varepsilon/\mathcal{S}_G \le 1$. The product of these cover cardinalities bounds the number of composed centers. Its logarithm is the sum of the local logarithms, bounded by $Cs(B/\delta)^{k/r}$, including the separate constant-node cost when needed. Stability makes each resulting center an ε-accurate approximation to the corresponding target composition. Dividing natural logarithms by $\ln 2$ converts the count to bits. Centers can be outside a stability-restricted subclass; recentering within that subclass changes only a fixed factor in resolution. □

The approximation exponent is controlled by

$$\frac{k}{r},$$

rather than

$$\frac{d}{r}.$$

The graph can nevertheless remain difficult if $\mathcal{S}_G$ grows rapidly with depth.

13.11 Unknown Graph Structure Costs Additional Bits

C1 states that a compact graph exists.

It does not say that the graph is supplied to the learner.

To count an entire class of sparse compositional graphs, the graph structure itself must be encoded.

13.11.1 A uniformly stable compositional class

Define

$$\mathcal{F}_{\mathrm{CS}}(d, s, k, r, B, S_0) \tag{13.15}$$

to be the class of functions admitting a DAG representation satisfying

$$\text{number of nodes} \le s,$$

$$\text{local arity} \le k,$$

$$\|g_v\|_{C^r} \le B,$$

and

$$\mathcal{S}_G \le S_0.$$

Assume a topological ordering of the s internal nodes.

Here d, s, k, r, B, S_0 and the common state domain are fixed class parameters, with $s, k \ge 1$ and $S_0 \ge 1$. For scalar output take the last internal node as the designated output. A graph with fewer nodes can be padded by unused zero nodes. Parent arguments are placed in increasing index order; the full C^r node class is invariant under permutations of its arguments. A restricted library without that property must additionally code the argument ordering and node type. These conventions prevent wiring information from being hidden in notation.

Each node chooses at most k parents among at most

$$d + s$$

available preceding variables.

The number of possible parent sets is bounded by

$$\sum_{j=0}^{k} \binom{d+s}{j}.$$

Therefore the number of bits required to specify the parent sets of all s nodes is at most

$$B_{\text{graph}} \le \left\lceil s \log_2 \left[\sum_{j=0}^{\min\{k,d+s\}} \binom{d+s}{j} \right] \right\rceil . \tag{13.16}$$

For fixed k,

$$B_{\text{graph}} = O(sk \log(d+s)) . \tag{13.17}$$

Proposition 15 (Entropy of unknown stable compositional graphs)**.** *Under the stated domain, encoding, and local entropy assumptions, with $\mathcal{S}_G \le S_0$ uniformly over the admitted compositions and their cover replacements, for $0 < \varepsilon \le 1$,*

$$\boxed{\mathcal{H}\left(\varepsilon, \mathcal{F}_{\text{CS}}(d,s,k,r,B,S_0)\right) \lesssim Cs \left(\frac{BS_0}{\varepsilon}\right)^{k/r} + O(sk\log(d+s)) .} \tag{13.18}$$

Proof. For every encoded graph use local tolerance $\delta = \varepsilon/S_0$. The preceding product-cover argument gives at most $\exp\left(Cs(BS_0/\varepsilon)^{k/r}\right)$ composed centers per graph, with one uniform constant. Take the union over the finitely many graph codes. The logarithm of its cardinality is at most the local-cover logarithm plus the graph-count logarithm, giving the displayed bound. Unused nodes and a prescribed output convention handle smaller graphs. If dimensions, arity, precision, or library types are selected from an unbounded collection, their headers or class-selection penalties must be added when this fixed-class statement is converted into a single code. □

The two terms have different meanings.
The first pays for the local functions.
The second pays for their wiring.
Thus

ambient dimension disappears from the local approximation exponent but survives through structural addressing.

13.11.2 Representation versus support discovery

If the correct graph is hard-coded into the architecture, the structural bits have already been supplied.

If the learner must infer the parent sets, those structural bits must be recovered from data.
Therefore

existence of a short sparse description $\neq$ ability to discover that description.

This is the representation–discovery distinction of Chapters 4–8. Quantitative C2 can supply useful accessible learning signals; C3 can supply intermediate information. Neither the graph-counting argument nor the existence of a short code proves that a particular optimizer finds it, and prediction does not require a unique teacher graph.

13.12 Weight Sharing and Reuse

Weight sharing is one of the clearest description-length reductions.

Suppose one local map containing q parameters is used at M locations.

Without sharing, a naive description contains approximately

$$Mq$$

parameter values.

With sharing, only

$$q$$

values need to be stored, together with the rule specifying where they are reused.

At b bits per parameter, the leading value cost changes as

$$bMq \quad \longrightarrow \quad bq.$$

Hence

$$\boxed{\text{repeated constituent} \Rightarrow \text{describe once, reuse many times.}}$$

This is the bit-theoretic reason that reuse is stronger than sparse compositionality alone.

A DAG with s unrelated constituents does not obtain this saving.

The reuse pattern is free only when prescribed independently of the sample. If s nodes choose from m learned modules, their assignments alone can cost up to $s\lceil \log_2 m \rceil$ bits, besides the graph and module descriptions. Also, repeated use still costs execution time and can amplify approximation errors.

For a fixed graph and a fixed assignment of m distinct modules with compatible input arities, the product-cover proof gives the elementary refinement

$$\mathcal{H}(\varepsilon, \mathcal{F}_{G,\text{shared}}, d_\infty) \lesssim Cm \left(\frac{B\mathcal{S}_{G,\text{shared}}}{\varepsilon} \right)^{k/r}.$$

Only m local cover choices are needed; the stability factor concerns *all uses* of those modules. If assignments are unknown, take a union over at most m^s assignments and add $s \log_2 m$ to the covering bits, using a common stability bound. Thus reuse reduces distinct descriptions, not automatically depth-dependent error amplification.

13.13 Sparsity

Suppose a parameter vector belongs to

$$\mathbb{R}^P$$

but contains only

$$s$$

nonzero coordinates.

A description must encode both the active support and the active values.

Here the coordinate system, P, s, and the quantization alphabet are fixed, with $1 \le s \le P$. The zero vector is a separate fixed code. For a union over sparsity levels, the level must be encoded or covered by a valid model-selection allowance.

The support requires

$$\log_2 \binom{P}{s}$$

bits up to integer rounding.

Using

$$\log_2 \binom{P}{s} \lesssim s \log_2 \frac{eP}{s},$$

and quantizing each active value to b bits gives

$$\boxed{B \lesssim s \log_2 \frac{eP}{s} + bs.} \tag{13.19}$$

Thus the description cost can scale with active dimension rather than nominal dimension.

This counts sparse *parameters* in a declared basis. Sparse activations on each input need not imply a sparse stored parameter vector. A learned change of coordinates must itself be described; otherwise complexity can be hidden in the basis. Small nonzero weights can be pruned only after their effect on predictions has been controlled.

13.14 Low Rank

Low rank provides a matrix analogue of sparsity.

Let

$$W \in \mathbb{R}^{d_1 \times d_2}$$

satisfy

$$\operatorname{rank}(W) \le r, \qquad \|W\|_F \le R.$$

Here $1 \le r \le \min\{d_1, d_2\}$ and $R > 0$. The stratum of matrices of rank exactly r has dimension

$$r(d_1 + d_2 - r).$$

The union of strata of rank at most r is not itself a manifold at lower-rank points. Nevertheless, for $0 < \varepsilon \le R$, the standard low-rank covering estimate gives the schematic form

$$\boxed{\log \mathcal{N}(\varepsilon, \mathcal{W}_r, \|\cdot\|_F) \lesssim r(d_1 + d_2) \log \frac{CR}{\varepsilon}.} \tag{13.20}$$

This is a quoted covering estimate, not an inference from dimension counting alone. An SVD-net construction gives exponent $(d_1 + d_2 + 1)r$ for the unit-Frobenius sphere **[candes2011tight]**; covering the radial interval extends it to the ball. The extra radial exponent and the $+1$ are absorbed into the displayed constant multiple of $r(d_1 + d_2)$.

For an approximately low-rank matrix, truncating its singular-value decomposition gives

$$\|W - W_r\|_F = \left(\sum_{j>r} \sigma_j(W)^2 \right)^{1/2}.$$

This is a matrix approximation error, not yet an output-error guarantee for the containing network. A layer-to-prediction stability bound is needed to choose its permissible tolerance. Quantization of the factors, rank selection, and any varying factorization metadata must also be included in an actual code.

The unrestricted matrix dimension is

$$d_1 d_2.$$

Thus rank restriction can produce a major reduction in covering bits.

13.14.1 Attention

The attention chapter established only the conditional statement

$$\boxed{\text{low-dimensional relevance} \Rightarrow \text{low-rank content routing can be sufficient.}}$$

It did not establish that trained attention maps must be low rank.

If learning does produce a rank-r query or key map, however, that map can be described at a cost scaling roughly with

$$r(d_1 + d_2)$$

rather than

$$d_1 d_2.$$

Low-rank routing therefore fits naturally into the same bit currency.

The ranks of query/key projections are not the rank of the post-softmax attention matrix. Nor is the common-minibatch-equilibrium rank obstruction in the normalized ReLU model of Chapter 7 a theorem that all trained matrices have rank at most two. An observed or proved low-rank approximation, together with a prediction-stability bound, is what can enter a compression certificate. Fluctuation energy, exact rank, and code length remain different quantities.

13.15 Quantization and Finite Precision

Quantization gives the simplest literal conversion from parameters to bits.

Suppose a model has

$$P$$

parameters and every parameter can take at most

$$2^b$$

values.

For one architecture and one quantization alphabet fixed independently of the sample, there are at most

$$2^{bP}$$

parameter strings.

Therefore

$$|\mathcal{F}| \leq 2^{bP}$$

and

$$\boxed{\log_2 |\mathcal{F}| \leq bP.}$$

The bound can be strict because multiple parameter strings may represent the same predictor.

Learned scale factors, codebook entries, normalization statistics, and stored examples are additional parameters unless already part of the fixed decoder. A finite alphabet is not sample-independent merely because it has finitely many entries: its entries can themselves hide unaccounted sample information.

13.15.1 Immediate finite-precision bound

For bounded loss,

$$\boxed{|R(f) - \widehat{R}(f)| \lesssim \sqrt{\frac{bP + \log(1/\delta)}{n}}.}$$

This may be numerically vacuous for a large modern model, but the logical implication is exact:

$$\boxed{\text{finite precision} \Rightarrow \text{finite number of parameter descriptions} \Rightarrow \text{finite capacity bound.}}$$

13.15.2 Training-time quantization versus post-training compression

Two cases should be distinguished.

Quantization during training. The restriction changes the space over which optimization occurs and may act as an explicit regularizer.

Post-training quantization. Training occurred in the full-precision parameterization.

Therefore post-training quantization did not cause the original optimization trajectory.

Nevertheless, if the final predictor can be replaced by a short quantized description while preserving its predictions, that short description can serve as a generalization certificate under a fixed coding scheme.

Hence

$$\boxed{\text{compression certificate} \neq \text{training-time regularization.}}$$

A prediction-level precision budget. A concrete finite-precision certificate needs more than a nominal bit width. Consider a bias-free feedforward network with input norm at most R_X and coordinatewise 1-Lipschitz activations that vanish at zero. Let W_ℓ and $\widetilde{W}_\ell$ be original and quantized matrices, with

$$\|W_\ell\|_{\mathrm{op}} \le M_\ell, \qquad \|\widetilde{W}_\ell - W_\ell\|_{\mathrm{op}} \le \eta_\ell.$$

Replacing layers one at a time gives

$$\sup_{\|x\|\le R_X} \|F_W(x) - F_{\widetilde{W}}(x)\| \le R_X \sum_{\ell=1}^{L} \eta_\ell \prod_{j<\ell} (M_j + \eta_j) \prod_{j>\ell} M_j.$$

Indeed, the quantized prefix has norm at most $R_X \prod_{j<\ell}(M_j + \eta_j)$, the changed layer contributes η_ℓ, and the unchanged suffix has Lipschitz constant at most $\prod_{j>\ell} M_j$. The triangle inequality sums the hybrid differences. If the p_ℓ entries of layer ℓ are rounded with step Δ_ℓ without overflow, then $\eta_\ell \le \frac{1}{2}\sqrt{p_\ell}\,\Delta_\ell$. An interval of width $2A_\ell$ needs at most $1 + \lceil 2A_\ell/\Delta_\ell \rceil$ uniform quantization levels. This connects precision, range, bit cost, and output error explicitly.

This calculation is an added sufficient estimate for the stated simple architecture, not a formula for arbitrary attention, normalization, or recurrent systems. Biases, other activation Lipschitz constants, input rounding, and numerical accumulation errors require their corresponding terms. Chapters 9–10 separately track routing, numerical error, and physical execution costs.

13.16 Norm Control as a Covering Constraint

Norms provide another route to reduced covering complexity.

13.16.1 A Euclidean parameter example

Consider

$$w \in \mathbb{R}^P, \qquad \|w\|_2 \le R.$$

For $0 < \eta \le R$, an η-cover of the Euclidean ball satisfies schematically

$$\mathcal{N}(\eta, B_2(R), \|\cdot\|_2) \le \left(\frac{CR}{\eta}\right)^P.$$

Thus

$$\log \mathcal{N}(\eta, B_2(R)) \lesssim P \log \frac{CR}{\eta}.$$

A smaller radius reduces parameter-space covering complexity.

For a full-range statement one can use $\mathcal{N}(\eta, B_2(R), \|\cdot\|_2) \le (1 + 2R/\eta)^P$. If, additionally, $\sup_x |F_w(x) - F_v(x)| \le A\|w - v\|_2$ on the parameter region, then an ε/A parameter cover induces an ε function cover. Without such a relation, a parameter-space covering number is not yet the desired prediction-space entropy.

13.16.2 Parameter-space norm is not the final object

For neural networks, however, parameter distance is not itself prediction distance.

Reparameterizations can change layer norms while preserving the represented function.

Useful generalization bounds therefore control the induced function class, often through architecture-dependent combinations of

$$\text{spectral norms}, \qquad \text{Frobenius norms}, \qquad \text{path norms}, \qquad \text{margins},$$

or related quantities [11].

Thus the safe chain is

$$\boxed{\begin{array}{c}\text{norm restriction} \;\rightarrow\; \text{smaller induced function class} \\ \Downarrow \\ \text{smaller covering/Rademacher complexity} \\ \Downarrow \\ \text{generalization bound.}\end{array}}$$

The first arrow depends on the architecture and on the particular norm.

The product-norm route of Chapter H. For a specified architecture, one candidate is $\mathcal{C}_\times(\theta) = \prod_{\ell=1}^{L} \|W_\ell\|_F$. An applicable theorem may control a fixed class $\mathcal{H}_Q = \{F_\theta : \mathcal{C}_\times(\theta) \le Q\}$ by

$$\widehat{\mathfrak{R}}_{S_X}(\mathcal{H}_Q) \le \mathcal{B}_n(Q; \text{architecture, input bounds}).$$

Products of spectral norms, correction factors, biases, activation hypotheses, and margins must match the actual theorem. This is not a universal expression $Q/\sqrt{n}$ for every network, and it need not be implemented by separately quantizing every weight. A smaller valid class bound is not literally minimization of exact Rademacher complexity.

Statistical norm control is also different from parameter confinement, a lower bound on useful Jacobian singular values, or a common equilibrium of all minibatches. In particular, a product-norm penalty can leave unbounded rescaling directions. The analytic and stochastic results of Chapter 7 and Appendix N do not follow from the covering calculation. Conversely, a direct norm-based statistical bound need not be replaced by a numerically weaker literal-weight code.

13.17 A Unified Bit Table

The preceding mechanisms can be summarized schematically. The table mixes literal payload lengths with class-covering logarithms deliberately, but labels their different roles. Common metadata and stability assumptions must be reinstated before using any row as a certificate.

These expressions should not be added mechanically.

They arise under different metrics, model classes, and regularity assumptions.

The commonality is conceptual:

$$\boxed{\begin{array}{c}\text{each mechanism can reduce the number of distinguishable predictors} \\ \text{at the resolution relevant to the task.}\end{array}}$$

@>p0.20 >p0.31 >X@

Mechanism	Object and convention	Leading cost
Quantization	Fixed-format parameter payload	bP bits.
Sparsity	Support and quantized active values	$s\log_2(eP/s) + bs$ bits.
Rank r	Matrix-class Frobenius cover	$Cr(d_1 + d_2)\log_2(CR/\varepsilon)$ covering bits.
Weight sharing	One module and its reuse rule	$bq + B_{\text{reuse}}$ rather than bMq value bits.
Fixed stable DAG	s smooth maps of arity at most k	$Cs(B\mathcal{S}_G/\varepsilon)^{k/r}$ covering bits.
Unknown stable DAG	Local maps and parent addresses	$Cs(BS_0/\varepsilon)^{k/r} + O(sk\log(d+s))$ covering bits.
Norm control	Induced function class, not just parameters	Metric- and architecture-dependent covering or Rademacher bound.

Table 13.1: A common accounting language, not interchangeable theorems. In the rank row r is matrix rank; in the compositional rows it is smoothness order. All error tolerances refer to the stated metric.

13.18 Architecture Versus Optimization

The distinction between coding language and selected code clarifies the role of overparameterization.

13.18.1 Architecture defines the coding language

Architecture determines which functions have short descriptions.

For example,

convolution:	translation-reused local maps are cheap,
sparse routing:	restricted dependency patterns are cheap,
low-rank modules:	low-dimensional transformations are cheap,
CRM libraries:	repeated subprograms are cheap,
attention:	a routing rule can be compact to describe.

This does not mean that all functions representable by the architecture have short descriptions.

13.18.2 Optimization selects a code

A heavily overparameterized architecture may contain many well-fitting functions, including exact interpolants in a realizable regime. Positive regularization need not select an exact interpolator.

Initialization, SGD, explicit regularization, normalization, and the data influence which solution is selected.

The relevant statistical question can therefore concern the code length of that selected predictor.

Theorem 13 is important here.

Because it holds simultaneously over the entire fixed code family, optimization may inspect the sample and choose any member of that family without invalidating the bound.

The same event covers an iterate returned by a specified finite stopping rule, or a coded average or ensemble that belongs to the fixed coded family and whose complete description is charged. It does not require convergence to an absorbing parameter or zero minibatch covariance. In the regularized regime of Chapters 7–8, zero mean update can coexist with continuing sample-dependent fluctuations. Those fluctuations neither establish nor invalidate a code-based guarantee for a returned model that meets its hypotheses.

If an algorithm returns a code c with empirical risk at most η and length at most B with probability at least $1 - \delta_{\mathrm{opt}}$, then the Occam event at failure level δ_{stat} gives

$$R(\mathcal{D}(c)) \le \eta + \sqrt{\frac{B \ln 2 + \ln(2/\delta_{\mathrm{stat}})}{2n}}$$

with probability at least $1 - \delta_{\mathrm{stat}} - \delta_{\mathrm{opt}}$. An approximate certificate for the uncompressed output adds the stated discrepancy term. This is the code-based specialization of Chapter 8's conditional predictive argument; it does not prove its algorithmic success premise.

Thus

large architecture + selection of a short fixed-language code can still yield a nontrivial Occam bound.

13.18.3 The architecture need not impose a hard ceiling

A convolutional network can memorize.

A sparse network can memorize.

A low-rank parameterization can memorize if its remaining capacity is large.

A very large network can generalize.

Therefore

architecture biases complexity but does not by itself determine the complexity of the learned function.

Effective complexity is jointly influenced by

architecture + data + optimization + regularization + precision.

13.19 Compression of the Selected Solution

Suppose a trained model contains P nominal parameters.

After training, imagine that its predictions can be preserved by

- pruning most weights,
- quantizing the remaining weights,
- replacing matrices by low-rank factors,
- tying repeated modules,
- compiling repeated subprograms,
- or otherwise replacing the model by a shorter coded predictor.

Let

$$f_{\hat{\theta}}$$

denote the trained predictor.

Its relevant compression complexity is

$$B_\varepsilon(f_{\hat{\theta}}).$$

If a fixed prefix decoder reconstructs a uniformly ε-accurate predictor from that many bits, Theorem 14 gives a certificate for the original model. Alternatively, one can certify and return the compressed predictor directly using its own empirical loss and code length. Any valid observed length $L(c)$ upper-bounds the ideal minimum; computing B_ε exactly is not required. A compression algorithm's failure to find a short code is not proof that no such code exists.

Thus the complexity of the selected predictor may be much smaller than

$$P \times \text{full precision}.$$

This connects to compression-based analyses such as [176]. That work uses a PAC-Bayesian compression approach; it is not being identified with the elementary prefix-code Hoeffding theorem stated here.

13.19.1 What must be included in the code

A valid compression certificate must describe everything required to reconstruct the compressed predictor.

Depending on the scheme, this may include

$$\text{support indices}, \quad \text{quantized values}, \quad \text{rank factors},$$
$$\text{module dictionary}, \quad \text{routing structure}, \quad \text{decoder metadata}.$$

Information inferred from the training sample cannot be treated as free unless the statistical argument explicitly allows it.

For example, a short program saying "rerun training with this seed" does not describe the fitted predictor without access to the training sample. If reconstruction reads selected examples or an entire dataset, those data must be encoded, provided independently of the sample used for the bound, or handled by a genuine sample-compression or conditional analysis. Sample indices alone are not the values of arbitrary real-valued examples. Likewise, stored prompts, dictionaries, and keys that form part of the deployed predictor require an explicit information accounting. Ordinary test input is an input to the predictor; a sample-dependent reference dataset is not automatically such a free input.

13.20 Two Routes from Bits to Generalization

We can now separate the two main routes of the chapter.

13.20.1 Route I: individual code length

For a predictor selected by training,

$$f_{\hat{\theta}},$$

find a short coded approximation g.

Then

$$\boxed{B_\varepsilon(f_{\hat{\theta}}) \longrightarrow \text{Occam/compression bound.}}$$

This route can exploit the fact that one particular learned function is much simpler than the full architecture class.

13.20.2 Route II: class metric entropy

Instead bound a whole function family

$$\mathcal{F}.$$

Then

$$\boxed{\mathcal{H}(\varepsilon, \mathcal{F}) \longrightarrow \text{covering/chaining} \longrightarrow \mathfrak{R}_n(\mathcal{F}) \longrightarrow \text{generalization.}}$$

This route is a worst-case class-level analysis.
Neither route subsumes the other in every application.
They are two mathematical realizations of the same broad description-length idea.

13.21 Connection to the Principles of the Book

The framework developed in the preceding chapters separates several questions.

13.21.1 C1: representation

C1 asks

$$\boxed{\text{Does a compact compositional realization exist?}}$$

Efficient computation provides one reason that such a representation may exist.

13.21.2 C2 and C3: recovery

Quantitative C2 concerns signals detectable and usable by the specified learning procedure in its current representation. They may support an accurate predictor without identifying a particular hidden teacher DAG. C3 supplies intermediate information or useful surrogates, and can help even when end-to-end recovery is possible; it is not merely a fallback. Chapter 11 further distinguishes trace exposure from coverage of the local states needed at test time.

Thus C2 and C3 concern

$$\boxed{\text{Can a useful compact representation actually be found?}}$$

13.21.3 Selection

Overparameterization may provide many solutions.
Optimization and regularization determine which region is reached.
Hence selection asks

$$\boxed{\text{Which fitting solution is chosen?}}$$

13.21.4 Bits: statistical complexity

The present chapter asks

How many bits are required to describe the selected solution at the relevant prediction scale?

These questions should remain distinct:

representation
$\Downarrow$
recovery
$\Downarrow$
selection
$\Downarrow$
effective description complexity
$\Downarrow$
generalization bound.

The arrows are not automatic implications.

They denote separate questions that must be answered.

The positive conclusion concerns the function actually returned. If its empirical fit and complete code produce a small valid risk bound, it is a successful predictor for that distribution and loss. Identifying the generating coordinates is a separate objective. If its evaluator also satisfies Chapter H's feasibility conditions, it supplies a computational compositional witness. A short arbitrary program alone does not establish those runtime and precision conditions.

The independent observation unit must match the data. For the offline control setting of Chapter 12, one can apply a bounded-loss code theorem to N independent trajectories and a normalized trajectory loss, with $n = N$. The many tokens in one trajectory are not additional independent samples. Even a valid offline action-loss bound does not itself prove closed-loop return; coverage and rollout comparison are the separate requirements developed there.

13.22 What Bits Do Not Explain

Description length is useful precisely because its scope can be stated clearly.

13.22.1 Bits do not explain recoverability

A target may possess a very short description and still be difficult to infer from examples.

Parity illustrates the gap for learners restricted to uninformative low-order correlations under the uniform distribution. It is not a claim that noiseless parity is hard for every learner: binary linear algebra can recover its coefficients from sufficiently informative examples. Cryptographic examples require their own computational hardness assumptions and observation models.

Thus

short description $\Longrightarrow$ easy recovery.

Description length is not a substitute for a learning theorem. C2 is one candidate source of usable learning signal, not an additional necessary condition imposed after another algorithm has already achieved predictive success.

13.22.2 Bits do not identify the correct latent structure

Two different internal representations can implement the same predictor.

A short predictor can also exploit a shortcut.

Therefore

$$\boxed{\text{compression} \Longrightarrow \text{causal or semantic correctness.}}$$

13.22.3 Kolmogorov complexity is not computable

The true shortest program cannot in general be found.

Practical analyses must therefore use explicit coding languages, compression algorithms, proven covering or norm bounds, rank constraints, or other specified surrogates.

Not every such surrogate is efficiently computable. A minimal covering number, an exact Rademacher supremum, or an optimal factorization can be hard to evaluate. An explicit compressor gives an attainable upper bound; a proven entropy estimate gives an analytic upper bound. Neither is a universal efficient search procedure for the shortest predictive code.

13.22.4 The metric matters

Description complexity always depends on what counts as a small change.

A predictor can be simple under one metric and complex under another.

Therefore covering numbers must specify

$$d.$$

For prediction, natural choices may depend on the loss, the data distribution, or the empirical sample.

13.22.5 Scale matters

Description length is resolution dependent.

A function may require many bits for extremely precise reproduction but few bits for task-level accuracy.

Hence

$$B_\varepsilon(f)$$

should be treated as a function of ε.

13.22.6 A post-hoc coding language is not free

Theorem 13 permits the predictor to be selected after seeing the training sample because the event is uniform over a coding language fixed independently of the sample.

But suppose one examines the data and then invents a special decoder that assigns a one-bit code to exactly the predictor obtained on that sample.

Such a construction has hidden the complexity inside the decoder.

The decoder itself must be described.

Proposition 16 (Two-part coding of a learned dictionary)**.** *Fix a master decoder independently of the sample. Let dictionary codes j have prefix lengths $L_0(j)$ satisfying $\sum_j 2^{-L_0(j)} \le 1$. For each dictionary j, let conditional predictor codes c form a prefix-free family with lengths $L_j(c)$ satisfying $\sum_c 2^{-L_j(c)} \le 1$. Assume each pair decodes to a measurable predictor $f_{j,c}$. Under the statistical assumptions of Theorem 13, the prefix-code Occam bound applies simultaneously with total length $L_0(j) + L_j(c)$, even if both j and c are selected using the training sample.*

Proof. The concatenation is uniquely parsed by first decoding the dictionary and then its conditional code. Its Kraft sum is

$$\sum_{j,c} 2^{-L_0(j)-L_j(c)} = \sum_j 2^{-L_0(j)} \sum_c 2^{-L_j(c)} \le 1.$$

Apply the same Hoeffding and union-bound calculation as in Theorem 13. If several pairs decode to one function, the union bound over pairs is still valid. □

This gives a precise implementation of the instruction to charge for a learned representation. A backbone supplied independently of the current evaluation sample may instead be conditioned on as fixed side information; the bound is then conditional on that backbone. A backbone, dictionary, or codebook selected using the same sample cannot be declared independent after training. Reuse across tasks can amortize storage cost, but the statistical conditioning or joint-task sampling model must still justify the corresponding accounting.

Hence

data-dependent predictor selection is allowed;
data-dependent code design must be accounted for.

Alternative approaches include valid sample-compression bounds, PAC-Bayes methods, localized complexities, algorithmic stability, or explicit coding of the learned dictionary.

13.23 Predictions and Empirical Tests

The bit viewpoint suggests several empirical questions.

13.23.1 Compressed complexity versus parameter count

A testable hypothesis is that, across predictors with comparable training error and approximation tolerance, measured fixed-code compression length is more informative about generalization than nominal parameter count. The upper-bound theorems do not imply a monotone relation to actual test error.

One can compare

$$P$$

with

$$B_\varepsilon(f_{\hat\theta})$$

for a specified decoder and prediction tolerance.

Report an achieved code length rather than claiming to measure the uncomputable minimum. Charge headers, dictionaries, scales, and any learned representation. Report the compressed

model's own training and held-out losses, and distinguish empirical compression distortion from a certified uniform or population discrepancy. Any choice among codecs made using the same data needs its code or model-selection allowance.

13.23.2 Different biases can have comparable statistical bit costs

Suppose two mechanisms—for example sparsity and low rank—produce predictor families having similar valid coding lengths under the same prediction metric and loss.

Then their Occam terms can be similar even though their optimization behavior is very different.

Thus the appropriate prediction is not that all inductive biases having the same bit count behave identically.

It is the weaker statement

comparable valid description lengths $\Rightarrow$ comparable description-length capacity terms.

13.23.3 Structure discovery may reduce description length

Suppose training progressively discovers reusable constituents.

If those constituents admit an efficient fixed coding language, the description length of the learned predictor may decrease even while the nominal network size remains fixed.

One may therefore monitor the ideal curve

$$t \longmapsto B_\varepsilon(f_{\theta_t})$$

through an explicit compressor's achieved upper bound

together with

$$\text{training error}, \qquad \text{validation error}, \qquad \text{constituent recovery}.$$

A decrease in description length would provide evidence for progressive compression of learned structure.

It would not by itself prove that the recovered constituents are the true generative ones.

Compare checkpoints under a fixed codec and prediction tolerance. Changes in a codec must not masquerade as changes in the function's compressibility. A returned checkpoint can be certified on a uniform code event without proving convergence of the entire SGD trajectory.

13.23.4 Memorizing unstructured exceptions should cost bits

Suppose a predictor has learned a compact rule and is then required to fit an increasing number of independent random label exceptions.

Those exceptions contain little reusable structure.

A valid code reproducing the predictor should therefore typically require additional bits approximately proportional to the information introduced by the exceptions.

This is a cleaner prediction than saying that every form of memorization must have large description length.

A dataset that looks like a collection of arbitrary examples to an observer may itself have a short generating program.

An elementary counting check makes the random-label claim precise. Fix m distinct inputs and a prefix decoder independently of independent fair labels $Y \in \{0,1\}^m$. For an integer $0 \le u \le m$, at

most 2^{m-u} prefix codewords have length at most $m-u$, and each deterministic classifier supplies at most one label vector on those inputs. Therefore

$$\Pr_Y[\exists c:\ |c| \le m-u, \quad \mathcal{D}(c)(x_i) = Y_i \text{ for all } i] \le 2^{-u}.$$

This is a lower bound for exact reproduction of independent label bits in a fixed language. It is not a universal lower bound for approximate population prediction, nor an additive "exception cost" when the base model already encodes the labels. A conditional exception code needs a base description fixed independently of those exceptions.

13.24 Open Problems

Several problems remain.

1. Which coding language best predicts generalization? Different architectures induce different natural codes.

There is no unique parameterization-independent practical description length.

A useful theory should compare explicit codes by their predictive value and invariance properties.

2. Does optimization reduce description length? The architecture may permit short descriptions without the training dynamics finding them.

A central question is when specified SGD dynamics and regularization produce shorter useful descriptions while retaining adequate fit, and how to quantify this within a finite sample and runtime budget. No monotonicity of code length along every trajectory is implied, and persistent regularized minibatch noise does not preclude a useful stopping rule.

3. How does compositional reuse appear statistically? Sparse compositionality reduces local dimension.

Repeated constituents should give an additional description-length saving.

Quantifying the interaction between local arity, reuse, graph structure, and generalization remains an important problem.

The fixed-graph shared-module cover above supplies one elementary answer. Sharper distribution-sensitive bounds, approximate rather than exact sharing, and uniform stability as reuse and depth grow remain separate questions.

4. How should data-dependent dictionaries be charged? If training learns a library of reusable features or operators, the library itself has a coding cost.

Subsequent examples may then be described cheaply relative to that library.

This leads naturally to hierarchical, conditional, or amortized description lengths.

Proposition 16 resolves the elementary fixed-master-language case. The remaining problem is to find useful, computable dictionary codes and sharper statistically valid conditioning or amortization schemes for realistic transfer and multitask learning.

5. Can description length distinguish rule learning from shortcuts? Two equally short predictors may exploit different structures.

Description length alone does not identify the desired solution.

Additional invariance, transfer, intervention, or causal criteria may be required.

13.25 Conclusion

Many forms of inductive bias used in deep learning appear superficially different. Compositional sparsity limits local arity; weight sharing reuses constituents; sparse parameterizations restrict active coordinates; low-rank models restrict matrix complexity; quantization limits precision; and suitable norm constraints control induced function families. Architectures define coding languages, while optimization and regularization influence which predictors are selected inside them.

The common question is

How many bits are required to describe the selected predictor at the resolution relevant to prediction?

There are two routes, with different hypotheses. For a predictor in a fixed prefix language and an i.i.d. sample with bounded loss, the uniform Occam event gives

$$|R(f) - \widehat{R}_S(f)| \le \sqrt{\frac{L(f)\ln 2 + \ln(2/\delta)}{2n}}.$$

For an approximately coded predictor and the specified Lipschitz loss,

$$R(f) \le \widehat{R}_S(f) + 2L_\ell\varepsilon + \sqrt{\frac{B_\varepsilon(f)\ln 2 + \ln(2/\delta)}{2n}},$$

when the approximation is uniform on the admitted input domain. The compressed model can instead be certified directly through its own loss and code. Agreement only on training inputs does not transfer a population guarantee to an uncompressed model.

For a continuous function class, the corresponding covering bits are

$$\mathcal{H}(\varepsilon, \mathcal{F}, d) = \log_2 \mathcal{N}(\varepsilon, \mathcal{F}, d).$$

Dudley chaining combines empirical covering complexity across scales inside a Rademacher analysis. An empirical cover is not a free sample-independent Occam codebook. Choice of scale, metric, dictionary, and conditioning information must follow the appropriate route.

For a fixed stable graph with s smooth constituents of arity at most k, the stated local entropy and domain assumptions give

$$\mathcal{H}(\varepsilon, \mathcal{F}_G, d_\infty) \lesssim Cs(B\mathcal{S}_G/\varepsilon)^{k/r}.$$

For an unknown graph with a uniform stability bound S_0, its structure must be included:

$$\boxed{\mathcal{H}(\varepsilon, \mathcal{F}_{\mathrm{CS}}, d_\infty) \lesssim Cs(BS_0/\varepsilon)^{k/r} + O(sk\log(d+s)).}$$

Reuse can replace the count of nodes by a count of distinct modules in the local-cover term when the assignment is supplied and its repeated-use stability is controlled. An unknown assignment has its own coding cost. At fixed local arity, smoothness, domains, and stability budgets, the explicit ambient dimension enters through structural addressing. The number of nodes, norms, and stability factors can themselves depend on d; the result is not a universal dimension-free learning theorem.

The roles in the book remain distinct:

C1:	does a feasible representation exist?
C2/C3:	what signal or intermediate information is available?
selection:	which well-fitting predictor is returned?
bits:	what statistical complexity can be certified?

A short valid code is not necessarily easy to discover or fast to evaluate. A small complexity allowance alone is not small prediction error. The positive statistical conclusion is instead

$$\begin{aligned} &\text{short valid description relative to the sample size} \\ &+ \text{a good empirical fit and controlled approximation} \\ &+ \text{the corresponding statistical assumptions} \\ &\Longrightarrow \text{a useful population-risk bound.} \end{aligned}$$

This certificate does not require recovery of a privileged generating DAG, a universal staircase trajectory, or vanishing SGD fluctuations. It does require that the counted description reconstruct the predictor to which the guarantee applies. Bits thus provide a common accounting language connecting representation, selection, and generalization, without identifying those three problems with one another.

CHAPTER 14

Multiplicative Norm Regularization: Selection, Complexity, and an Experimental Case Study

Based on work with R. Dubach and M. Abdallah.

Norm-based generalization bounds for deep networks often contain multiplicative combinations of layer norms. This motivates a simple question: instead of applying ordinary additive weight decay, what happens if training directly penalizes a product of layerwise norms?

We compare the additive penalty

$$\Omega_{\mathrm{add}}(W) = \sum_{\ell=1}^{L} \|W_\ell\|_F^2$$

with the multiplicative penalty

$$\Omega_{\mathrm{mult}}(W) = \prod_{\ell=1}^{L} \|W_\ell\|_F^2.$$

The two penalties define different objectives and different gradients. In the bias-free positively homogeneous setting, the product is invariant under compensating layerwise rescalings that preserve the represented function, whereas the additive penalty generally is not. However, minimizing the additive penalty along such a rescaling orbit gives a monotone function of the product. Thus distinct training dynamics should not be confused with a universal separation between their idealized constrained minimizers.

Exploratory experiments on convolutional networks trained on CIFAR-10 report selected solutions with much smaller layer-norm products, larger raw margins, and strong suppression of later-layer singular values. In the original experiments, these solutions also showed higher test accuracy and substantially greater robustness to label corruption.

Those performance numbers should not yet be interpreted as a clean generalization result. The original protocol reused the test set for model selection and early stopping and discarded some unsuccessful training runs. The experiments therefore motivate a hypothesis rather than establish it.

The scientific question of this chapter is consequently:

Does directly penalizing a global layer-norm product systematically alter the solution selected by training, and does that change reduce effective complexity and improve population performance?

The chapter develops the theoretical motivation, reports the exploratory observations, and ends with the controlled experiments required to test the hypothesis.

14.1 Introduction

Overparameterized neural networks can contain many solutions having nearly zero training error.

Regularization therefore plays at least two conceptually different roles.

It can alter optimization,

$$\text{training dynamics} \longrightarrow \text{which solution is reached},$$

and it can alter statistical complexity,

$$\text{selected solution} \longrightarrow \text{capacity or compression measure}.$$

These two effects need not coincide.

Ordinary weight decay typically adds

$$\lambda \sum_{\ell=1}^{L} \|W_\ell\|_F^2$$

to the training objective.

Here "additive" means the explicit squared-norm term in the objective. For plain gradient descent it produces a layerwise shrinkage update. With adaptive preconditioning, adding this term to the loss is generally not the same operation as decoupled weight decay; in particular, an AdamW comparison must state which update is used [96]. The two regularizers may have separately chosen coefficients λ_{add} and λ_{mult}. Using the same numerical value is not, by itself, a matched regularization strength.

Yet many norm-based analyses of deep networks contain products of layer norms, either directly or as part of more refined expressions involving spectral and Frobenius norms [114, 113, 52, 11].

This suggests a natural alternative:

$$\lambda \prod_{\ell=1}^{L} \|W_\ell\|_F^2.$$

The purpose of the chapter is not to claim that this product is the unique or canonical measure of neural-network complexity.

It is instead to test a more limited hypothesis:

> A regularizer that directly couples the norms of all layers may lead a specified training procedure to different well-fitting predictors from those found with additive regularization, and those solutions may have lower effective complexity and better generalization.

Three questions should therefore be distinguished:

Q1:	Does multiplicative regularization improve held-out performance?
Q2:	Does it select solutions with different norm, margin, or rank structure?
Q3:	Can those structural differences explain any generalization advantage?

The exploratory experiments reported below strongly motivate Q2.
Q1 and especially Q3 require a cleaner replication.

14.2 Norm-Based Motivation

For an L-layer network, let

$$W = (W_1, \dots, W_L).$$

Define the Frobenius norm product

$$P(W) = \prod_{\ell=1}^{L} \|W_\ell\|_F. \tag{14.1}$$

The regularizer used in the experiments is

$$\Omega_{\text{mult}}(W) = P(W)^2 = \prod_{\ell=1}^{L} \|W_\ell\|_F^2. \tag{14.2}$$

The additive comparison is

$$\Omega_{\text{add}}(W) = \sum_{\ell=1}^{L} \|W_\ell\|_F^2. \tag{14.3}$$

14.2.1 Relation to norm-based capacity bounds

The motivation for Equation (14.2) comes from the recurring appearance of multiplicative norm factors in deep-network complexity bounds.

A bound concerns a function class, not the norms of one unspecified parameter vector inside an otherwise unrestricted class. For example, fix a scalar-output, bias-free, depth-L feedforward ReLU architecture with independently parameterized layer matrices, a linear output layer, and input domain $\|x\|_2 \le B$. For deterministic budgets $M_\ell > 0$, let $\mathcal{F}_{\boldsymbol{M}}$ contain its realizations with $\|W_\ell\|_F \le M_\ell$. One applicable bound, quoted from [52, Theorem 1], is

$$\widehat{\mathfrak{R}}_{S_X}(\mathcal{F}_{\boldsymbol{M}}) \le \frac{B(\sqrt{2L\ln 2}+1)}{\sqrt{n}} \prod_{\ell=1}^{L} M_\ell, \tag{14.4}$$

where $S_X = (x_1, \dots, x_n)$ and every $\|x_i\|_2 \le B$. This is an invoked statistical result; the entropy and Rademacher background is in Chapter 13. Different activations and architectures require their own version, including any Lipschitz factors, biases, normalization parameters, or other trainable operations.

More refined bounds can contain expressions such as

$$\frac{B}{\sqrt{n}}\left(\prod_{\ell=1}^{L}\|W_\ell\|_2\right)\left(\sum_{\ell=1}^{L}\frac{\|W_\ell\|_F^2}{\|W_\ell\|_2^2}\right)^{1/2}. \tag{14.5}$$

The exact bound depends on assumptions and architecture.

In particular, (14.5) displays a spectral–Frobenius factor appearing in PAC-Bayesian analyses such as [113]; it suppresses additional depth, width, logarithmic, and margin terms. It is not the full inequality, nor the precise correction factor in [11].

From a product budget to a statistical statement. For the homogeneous architecture above, define the *fixed* class $\mathcal{F}_Q^{\times} = \{f_W : P(W) \le Q\}$, $Q > 0$. Every nonzero-product realization can be rescaled, without changing its function, to have all layer norms $P(W)^{1/L} \le Q^{1/L}$; Proposition 18 gives the calculation. A zero-product realization is the zero function and also belongs to the bounded-layer class. Consequently (14.4) gives

$$\widehat{\mathfrak{R}}_{S_X}(\mathcal{F}_Q^{\times}) \le \frac{B(\sqrt{2L\ln 2}+1)Q}{\sqrt{n}}.$$

For a bounded Lipschitz loss, independent examples, and the corresponding uniform risk bound, a returned $\widehat{f} \in \mathcal{F}_Q^{\times}$ with small empirical loss can therefore have a small population-loss upper bound. Training must actually achieve both fit and an informative budget. Data-dependent choices of Q or the architecture need the simultaneous bounds, coding costs, or other valid selection argument described in Chapters H and 13. The product penalty does not itself minimize Rademacher complexity or prove optimization success. For classification, a margin-based bound also pays for the margin scale and the empirical margin violations; a raw mean margin is not that bound. Unbounded squared loss requires output bounds, clipping with an explicit interpretation, or an appropriate tail analysis.

Which convolutional norm is being measured? A convolutional kernel tensor can be flattened to define its Euclidean parameter norm. The full linear convolution operator reuses those entries at many spatial positions and generally has a different Frobenius norm. If K_u is a kernel coefficient reused m_u times in that operator,

$$\|T(K)\|_F^2 = \sum_u m_u K_u^2, \qquad \|T(K)\|_2 \le \|T(K)\|_F \le \sqrt{\max_u m_u}\,\|K\|_F.$$

Here the identity refers to the usual unrolled operator whose entries are individual kernel coefficients, with its padding and stride fixed. A parameter-norm product in the reported CNN is not automatically the operator-norm product of a theorem for dense matrices. Weight sharing, pooling, input normalization, and biases must be accounted for in the applicable CNN bound. The reported tensor statistics remain useful diagnostics even when that additional analysis is not yet available.

Therefore

$$\boxed{\Omega_{\mathrm{mult}}}$$

should be interpreted as a simple global norm-product surrogate, not as the unique theoretical capacity measure.

The empirical hypothesis is only

directly penalizing a global product-of-norms surrogate may favor lower-complexity solutions.

14.3 Additive and Multiplicative Penalties Are Different

Let

$$a_\ell = \|W_\ell\|_F^2.$$

The AM–GM inequality gives

$$\frac{1}{L}\sum_{\ell=1}^{L} a_\ell \geq \left(\prod_{\ell=1}^{L} a_\ell\right)^{1/L}. \tag{14.6}$$

Hence

$$\left(\frac{1}{L}\sum_{\ell=1}^{L} \|W_\ell\|_F^2\right)^{L} \geq \prod_{\ell=1}^{L} \|W_\ell\|_F^2. \tag{14.7}$$

Thus a sufficiently small additive penalty controls the product.

The converse does not hold.

A small product can coexist with a large sum if different layers have very different norms.

This already shows that the two regularizers can favor different factorizations.

It does not yet compare the functions minimizing the two penalties over a constraint set that allows all compensating rescalings. The next section makes that distinction exact. The AM–GM inequality also provides a useful per-checkpoint audit of any logged sum and product: both must be computed from the same list of layers using the same norm convention.

14.4 Layerwise Rescaling in Homogeneous Networks

The distinction becomes sharper for bias-free positively homogeneous networks.

Consider a ReLU network satisfying

$$\sigma(cx) = c\sigma(x), \qquad c > 0.$$

Apply the layerwise transformation

$$W_\ell \longmapsto c_\ell W_\ell, \qquad c_\ell > 0, \qquad \prod_{\ell=1}^{L} c_\ell = 1. \tag{14.8}$$

Because of positive homogeneity, this transformation leaves the represented function unchanged.

Proposition 17 (Rescaling behavior of the two penalties)**.** *Under Equation* (14.8)*,*

$$P(c_1 W_1, \ldots, c_L W_L) = P(W_1, \ldots, W_L), \tag{14.9}$$

while in general

$$\sum_{\ell=1}^{L} \|c_\ell W_\ell\|_F^2 \neq \sum_{\ell=1}^{L} \|W_\ell\|_F^2. \tag{14.10}$$

Proof. For the product,

$$\prod_{\ell=1}^{L} \|c_\ell W_\ell\|_F = \left(\prod_{\ell=1}^{L} c_\ell\right)\left(\prod_{\ell=1}^{L} \|W_\ell\|_F\right) = P(W).$$

The additive expression instead becomes

$$\sum_{\ell=1}^{L} c_\ell^2 \|W_\ell\|_F^2,$$

which changes for a generic compensating rescaling. □

Thus

multiplicative penalty:	invariant along these function-preserving rescaling directions,
additive penalty:	selects a preferred factorization along them.

This does not prove that one regularizer generalizes better.
It does establish that they impose different geometry on parameter space.

14.4.1 Additive regularization favors balanced factorizations

For a fixed product of layer norms,

$$\prod_{\ell=1}^{L} \|W_\ell\|_F = C,$$

AM–GM implies that the sum of squared norms is minimized when the layer norms are equal:

$$\|W_1\|_F = \cdots = \|W_L\|_F.$$

Thus additive regularization has a balancing effect among function-equivalent homogeneous factorizations.

The product penalty is indifferent along those same rescaling directions.

Proposition 18 (Minimum additive cost on a rescaling orbit)**.** *Assume all* $\|W_\ell\|_F > 0$ *and the bias-free homogeneous rescaling convention* (14.8)*. Then*

$$\min_{\substack{c_\ell>0 \\ \prod_\ell c_\ell=1}} \sum_{\ell=1}^{L} c_\ell^2 \|W_\ell\|_F^2 = LP(W)^{2/L} = L\Omega_{\text{mult}}(W)^{1/L}.$$

The minimum is attained when all rescaled layer norms equal $P(W)^{1/L}$*.*

Proof. The rescaling leaves the product of squared norms equal to $P(W)^2$. AM–GM therefore gives the displayed lower bound. Choose $c_\ell = P(W)^{1/L}/\|W_\ell\|_F$. Their product is one, every rescaled norm is $P(W)^{1/L}$, and equality holds. □

A consequence for idealized constrained selection. Let a feasible set depend only on the represented function, be closed under these rescalings, and contain only positive-product realizations. For example, it may impose exact interpolation of nonzero targets. If the relevant minima are attained, the functions obtained by globally minimizing Ω_{add} and Ω_{mult} over that set coincide, after balancing the representatives. Indeed, an additive minimizer must be balanced, and a smaller feasible product would, after balancing, contradict its optimality. Conversely, balancing a minimum-product realization attains the least additive cost. The same reasoning applies to a fixed, functionally defined loss-tolerance constraint.

This is an elementary consequence of AM–GM, not an assertion that the two training procedures find the same predictor. At fixed positive coefficients, the penalized fit–complexity tradeoffs are different: after orbit minimization their penalties have the forms $\lambda_{\mathrm{add}} L P^{2/L}$ and $\lambda_{\mathrm{mult}} P^2$. Their Euclidean gradients, finite-step trajectories, and attainable fits need not agree. Biases, normalization, residual paths, or constraints that prevent rescaling also fall outside the equivalence just proved.

Invariance is not confinement or complete functional invariance. For $L \geq 2$, replacing nonzero W_1, W_2 by $tW_1, t^{-1}W_2$ leaves both the homogeneous predictor and its product penalty fixed while parameters can become unbounded as $t \to \infty$. Thus the product objective is not coercive in these coordinates. Moreover, the proposition concerns uniform *layerwise* rescalings, not every function-preserving change of hidden coordinates. For example, $W_1 = I_2$, $W_2 = (1, 1)$ with an intervening ReLU has the same function after replacing these by $\mathrm{diag}(t, 1)$ and $(t^{-1}, 1)$, $t > 0$, yet the squared product changes from 4 to $(t^2 + 1)(t^{-2} + 1)$. The product of Frobenius norms is therefore not an invariant of the function under all of its realizations.

14.5 Multiplicative Regularization Couples the Layers

The two penalties also generate different gradients.

For

$$\Omega_{\mathrm{mult}} = \prod_{\ell=1}^{L} \|W_\ell\|_F^2,$$

the undivided derivative is valid at every W; the ratio form requires $W_j \neq 0$:

$$\nabla_{W_j} \Omega_{\mathrm{mult}} = 2W_j \prod_{\ell \neq j} \|W_\ell\|_F^2 \tag{14.11}$$

$$= 2\Omega_{\mathrm{mult}} \frac{W_j}{\|W_j\|_F^2}. \tag{14.12}$$

In contrast,

$$\nabla_{W_j} \Omega_{\mathrm{add}} = 2W_j. \tag{14.13}$$

The relative multiplicative pressure on layer j is

$$\frac{\|\nabla_{W_j} \Omega_{\mathrm{mult}}\|_F}{\|W_j\|_F} = \frac{2\Omega_{\mathrm{mult}}}{\|W_j\|_F^2}. \tag{14.14}$$

This yields a concrete mechanistic prediction:

layers with smaller norms experience greater relative multiplicative pressure.

The pressure may therefore become highly nonuniform across depth.

Whether this changes the *shape* of a layer's singular spectrum depends on its interaction with the data gradient and the actual optimizer. The penalty gradient alone is radial within each layer.

Radial shrinkage alone does not select singular directions. Under gradient flow of $\lambda\Omega_{\mathrm{mult}}$ alone,

$$\dot{W}_j = -q_j(t)W_j, \qquad q_j(t) = 2\lambda \prod_{\ell \neq j} \|W_\ell(t)\|_F^2.$$

Hence, on every finite interval where this flow exists,

$$W_j(t) = \exp\Big(-\int_0^t q_j(u)\,du\Big)\, W_j(0).$$

Every singular value of a given nonzero layer is multiplied by the same positive scalar. Exact rank, stable rank, and the relative 1% numerical rank defined below are unchanged at finite times. Plain penalty-only Euler updates have the same property whenever their scalar multiplier is nonzero. A zero multiplier collapses the whole layer, not selected singular directions. At a zero layer, the undivided gradient in (14.12) is well defined, and the whole product-penalty gradient vanishes. Ratio-based implementations must not replace that polynomial derivative by a division by zero.

Thus the reported relative-rank reductions cannot be explained by uniform weight shrinkage alone. They require the coupled loss dynamics, a non-scalar optimizer action, or another mechanism.

The unregularized norm-difference invariant is established for homogeneous networks in [44]. The following direct derivation records the different effects of the two penalties and a resulting confinement bound; it is not a claim that the original experiment verified these laws.

Proposition 19 (Layer-norm differences under homogeneous gradient flow)**.** *Consider the bias-free sequential homogeneous model and a nonnegative differentiable data loss $\mathcal{L}(W)$ depending only on its finite-sample predictions. Let $a_j(t) = \|W_j(t)\|_F^2$. On any interval of exact Euclidean gradient flow where the chain rule and layerwise homogeneity identities hold (almost everywhere is enough), the following statements hold. For $\mathcal{L} + \lambda\Omega_{\mathrm{mult}}$, $\lambda > 0$,*

$$\boxed{a_j(t) - a_k(t) = a_j(0) - a_k(0).}$$

For $\mathcal{L} + \lambda\Omega_{\mathrm{add}}$,

$$\boxed{a_j(t) - a_k(t) = e^{-4\lambda t}(a_j(0) - a_k(0)).}$$

In the multiplicative case, put $U_0 = \mathcal{L}(W(0)) + \lambda\Omega_{\mathrm{mult}}(W(0))$ and $\Delta_0 = \max_j a_j(0) - \min_j a_j(0)$. Then every layer obeys

$$\boxed{0 \le a_j(t) \le (U_0/\lambda)^{1/L} + \Delta_0.}$$

Proof. Homogeneity in each layer gives $D_{W_j} f_W(x)[W_j] = f_W(x)$ wherever the relevant derivative exists. Consequently $\langle \nabla_{W_j}\mathcal{L}, W_j\rangle_F = \zeta(t)$ is the same scalar for all j. For scalar squared loss without the factor $1/2$, $\zeta(t) = 2n^{-1}\sum_i (f_W(x_i) - y_i) f_W(x_i)$. Taking inner products of each flow equation with $2W_j$ gives

$$\dot{a}_j = \begin{cases} -2\zeta(t) - 4\lambda\Omega_{\mathrm{mult}}(W(t)), & \text{multiplicative,} \\ -2\zeta(t) - 4\lambda a_j(t), & \text{additive.} \end{cases}$$

Subtract the equations for j and k to obtain the first two assertions. In the multiplicative flow, the full objective decreases, so $\prod_j a_j(t) \le U_0/\lambda$ because $\mathcal{L} \ge 0$. Also $\max_j a_j(t) - \min_j a_j(t) = \Delta_0$. Since $(\min_j a_j(t))^L \le \prod_j a_j(t)$, the last assertion follows. □

This calculation is a useful positive confinement result for the *specified* flow, despite noncompact sublevel sets. It also sharpens the pressure interpretation: the multiplicative penalty makes the same contribution to the *absolute* decay of every squared layer norm, although its relative pressure differs. Equal initial norms remain equal; unequal squared-norm differences remain fixed. Additive flow instead exponentially removes those differences.

The proposition does not assert convergence, global fitting, or a finite-step SGD guarantee. For ReLU it is a statement about an absolutely continuous flow satisfying the displayed chain and homogeneity identities; it does not by itself supply a unique dynamics at every nonsmooth point. Bias parameters, adaptive updates, projection/renormalization, momentum, and finite steps can invalidate the identities. In particular it does not convert ReLU into the analytic setting of Appendix N.

The endpoint of regularized SGD. For $\mathcal{L} = n^{-1}\sum_i \ell_i$, write $g_i = \nabla \ell_i$ and $\bar{g} = n^{-1}\sum_i g_i$. At a stationary point of $U = \mathcal{L} + \lambda\Omega_{\text{mult}}$, only $\bar{g} + \lambda\nabla\Omega_{\text{mult}} = 0$ is required. For a uniform subset $\mathcal{B}$ of size $1 \le b < n$, $n > 1$, put

$$G_{\mathcal{B},\lambda} = \frac{1}{b}\sum_{i\in\mathcal{B}} g_i + \lambda\nabla\Omega_{\text{mult}}.$$

The covariance calculation of Chapter 7 gives, at that stationary point,

$$\boxed{\mathbb{E}_{\mathcal{B}}\|G_{\mathcal{B},\lambda}\|^2 = \frac{n-b}{b(n-1)}\frac{1}{n}\sum_i \|g_i - \bar{g}\|^2.}$$

The mean update is zero but the covariance is positive when sample gradients differ. For independent sampling with replacement the factor is $1/b$ instead. These statements concern fixed per-example losses and plain regularized gradient estimates, not arbitrary batch-dependent forwards or adaptive parameter updates.

At exact squared-loss interpolation all $g_i = 0$. If every $W_j \ne 0$, then $\nabla\Omega_{\text{mult}} \ne 0$, so such an interpolator is not stationary for the full objective at positive λ. Its sampling covariance is nevertheless zero at that same point. A regularized well-fitting endpoint can therefore have different noise properties from an exact data-loss interpolator. A finite stopped iterate may be useful without being a common zero of every minibatch direction; its fit and statistical control remain separate requirements.

Initialization scale is part of the dynamics. Uniform rescaling $W \mapsto sW$, $s > 0$, gives

$$\Omega_{\text{mult}}(sW) = s^{2L}\Omega_{\text{mult}}(W), \qquad \nabla\Omega_{\text{mult}}(sW) = s^{2L-1}\nabla\Omega_{\text{mult}}(W).$$

The data gradient has its own dependence on the changed predictions. Thus a fixed numerical λ across different initialization scales can produce very different regularization-to-data-gradient ratios. A fair comparison records those ratios and allocates matched tuning budgets, rather than treating initialization as an incidental choice.

14.6 Product and Log-Product Objectives Are Not Equivalent

The identity

$$\log\left(\prod_{\ell=1}^{L}\|W_\ell\|_F^2\right)=\sum_{\ell=1}^{L}\log\|W_\ell\|_F^2 \tag{14.15}$$

is useful descriptively.

When minimizing only the regularizer over a fixed feasible set,

$$\arg\min_W \Omega_{\text{mult}}(W)=\arg\min_W \log\Omega_{\text{mult}}(W),$$

whenever the logarithm is defined.

But this does not imply

$$\arg\min_W \left[\mathcal{L}(W)+\lambda\Omega_{\text{mult}}(W)\right]=\arg\min_W \left[\mathcal{L}(W)+\lambda\log\Omega_{\text{mult}}(W)\right].$$

A monotone transformation of only one term changes its tradeoff with the data loss.

Product, squared-product, and log-product regularizers should therefore be treated as different optimization objectives.

The distinction also matters for well-posedness. Along $W\mapsto tW$ with nonzero starting layers and $t\downarrow 0$, $\log\Omega_{\text{mult}}(tW)=2L\log t+\log\Omega_{\text{mult}}(W)\to-\infty$. If the data loss stays finite as the network tends to zero, the unstabilized objective $\mathcal{L}+\lambda\log\Omega_{\text{mult}}$ is unbounded below. The later proposed comparison uses a *stabilized* version for this reason.

For positive layer norms,

$$\nabla_{W_j}\log\Omega_{\text{mult}}=\frac{2W_j}{\|W_j\|_F^2}=\frac{1}{\Omega_{\text{mult}}}\nabla_{W_j}\Omega_{\text{mult}}.$$

The shared, parameter-dependent factor changes the relative influence of the data loss; it is not generally absorbed by one constant learning rate. For fixed $\eta>0$, the stabilized sum $\sum_j\log(\|W_j\|_F^2+\eta)$ is bounded below but is no longer invariant under compensating layer rescalings. Its layer derivatives also lack the common product multiplier. Computing the original product through a sum of logarithms followed by an exponential is merely an evaluation method; replacing the objective by that sum is a different experiment.

14.7 Original Experimental Setup

The experiments below should be regarded as exploratory.

They motivated the present hypothesis but were not performed with a fully held-out model-selection protocol.

Scope of the experimental record. The results below are the numerical summaries supplied for this case study. They are not new runs, and no individual-run logs, code, checkpoints, or uncertainty estimates accompany these summaries. Counts of runs, aggregation rules, complete hyperparameter searches, stopping rules, and failure criteria must be recovered before a reproducible comparison can be reconstructed. Unspecified implementation choices are not supplied by the theoretical model.

14.7.1 Binary CIFAR-10 task

The binary experiment uses CIFAR-10 classes

$$\text{cat} \quad \text{versus} \quad \text{deer},$$

coded as

$$-1 \quad \text{and} \quad +1.$$

The network contains four convolutional layers with channels

$$(16, 32, 64, 128),$$

with ReLU activations and 2×2 average pooling after each convolutional block.

The convolutional stack is followed by a fully connected hidden layer with 256 units and a scalar output.

The original implementation contains approximately

$$228{,}785$$

trainable parameters.

Training uses squared error,

$$L_{\text{data}} = \frac{1}{n} \sum_i \left(f_W(x_i) - y_i \right)^2 ,$$

with prediction

$$\text{sign}(f_W(x)).$$

14.7.2 Multiclass CIFAR-10

The multiclass experiment uses all ten CIFAR-10 classes.

The architecture contains four convolutional blocks with channel sizes

$$(64, 128, 256, 512),$$

followed by a 512-unit fully connected layer and a ten-dimensional output.

One-hot targets and squared error are used in the original experiment.

The architecture contains approximately

$$5.74 \times 10^6$$

parameters.

14.7.3 Training objectives

The three objectives are

$$L_{\text{none}} = L_{\text{data}}, \tag{14.16}$$

$$L_{\text{add}} = L_{\text{data}} + \lambda \sum_{\ell=1}^{L} \|W_\ell\|_F^2, \tag{14.17}$$

$$L_{\text{mult}} = L_{\text{data}} + \lambda \prod_{\ell=1}^{L} \|W_\ell\|_F^2. \tag{14.18}$$

Weights are initialized with He initialization [61].

The original experiments also varied a global initialization scale.

The penalized tensor list must be recorded explicitly: for example, whether it contains the four convolutional kernels and both fully connected weight matrices, and whether any biases are penalized separately. The supplied binary account notes fully connected biases; it does not make the experiment a bias-free realization of the preceding propositions. Kernel sizes, padding, bias flags, pooling conventions, and loss reduction must come from the implementation, not from the approximate parameter count. For multiclass squared loss, averaging over examples only or over both examples and output coordinates changes the effective regularization scale. The optimizer, step-size schedule, minibatch size, precision, and any normalization or clipping are also necessary protocol specifications.

14.8 Why the Original Performance Comparison Is Exploratory

Two features of the original protocol prevent the reported test differences from being interpreted as unbiased estimates of population performance.

14.8.1 Test-set reuse

The original test set was used for both early stopping and selection of regularization strength.

Thus the reported "test" accuracy participated in model selection.

If many hyperparameter settings are compared using the same test set and the best one is reported, the result is optimistically selected.

Likewise, choosing the stopping epoch using test performance compromises the independence of the final test evaluation.

The already inspected test set cannot retrospectively become an untouched confirmatory sample. It can be excluded from all *further* tuning in a protocol-frozen replication, with that history disclosed. Stronger confirmation calls for independent evaluation data not used to choose the method or analysis. The numerical tables below retain the original names and entries, not an assertion that the reported test estimates were independent of selection.

The correct protocol is

$$\boxed{\text{training} + \text{validation} + \text{untouched test.}}$$

14.8.2 Optimization failures were discarded

Some runs that failed to train were excluded.

But optimization failure is itself part of the behavior of a learning method.

Discarding unsuccessful runs can bias the comparison.

The structural summaries are selected as well: a striking norm or rank measurement in a retained best configuration is not an estimate of how often the regularizer produces it. A replication must report that frequency across all prespecified runs, alongside unconditional performance under a fixed rule for handling failures.

Future experiments should therefore report both

$$\boxed{\text{optimization success probability}}$$

and

$$\boxed{\text{performance conditional on successful optimization.}}$$

14.8.3 Effect sizes

Accuracy differences should be reported in percentage points.

For example,

$$83.10\% - 73.69\% = 9.41$$

is a difference of

$$9.41$$

percentage points.

The abstract, tables, and text should ultimately be generated from the same analysis pipeline so that numerical summaries cannot drift between versions.

No standard deviations or run counts are supplied with the four result tables. Their entries alone cannot support confidence intervals, statistical significance, or an estimate of across-seed reliability.

14.9 Exploratory Results from the Original Protocol

Despite these limitations, the original experiments reveal several striking patterns worth testing under a corrected protocol.

14.9.1 Binary classification and label noise

The reported binary test accuracies were:

training-label corruption	multiplicative	additive	unregularized
0%	89.50%	87.87%	87.80%
10%	86.68%	81.13%	80.20%
20%	83.10%	73.69%	73.08%

Thus the largest reported multiplicative–additive gap was

9.41

percentage points at 20% label corruption.

Because the hyperparameters and stopping time were not selected using an independent validation set, these values should be interpreted as hypothesis-generating rather than confirmatory.

14.9.2 Raw margins

For correctly classified binary examples, the original analysis used

$$m_i = y_i f_W(x_i).$$

The reported mean margins also favored multiplicative regularization and degraded more slowly under label corruption.

Because this summary conditions on correct classification, the compared methods can be averaging over different example subsets. The replication should additionally report the signed-margin distribution on a common set of *all* examples, including negative margins. Clean and corrupted training-label margins must be distinguished, as must the raw regression fit and classification accuracy.

However, raw margins can depend strongly on parameter scale.

For positively homogeneous bias-free networks, the more relevant quantity is the normalized margin

$$\widetilde{m}_i = \frac{y_i f_W(x_i)}{P(W)} = \frac{y_i f_W(x_i)}{\prod_\ell \|W_\ell\|_F}. \tag{14.19}$$

For the bias-free homogeneous model, writing $W_\ell = \|W_\ell\|_F V_\ell$ gives $f_W = P(W) f_V$, so $\widetilde{m}_i = y_i f_V(x_i)$. This is a scale-normalized diagnostic, not invariance under every hidden-unit reparameterization. The original binary architecture contains fully connected biases, for which the weights-only factorization does not hold in general. The displayed ratio can still be computed when $P(W) > 0$, but its theorem interpretation is unestablished for that architecture. A bias-free replication, or a precisely specified bias-aware norm bound, is needed. In either case evaluate the margin violations required by that bound, not only the mean.

14.10 Norm Products and Layerwise Structure

The original experiment found a very large separation in norm products.

For the best reported binary configurations,

	multiplicative	additive	unregularized
$\sum_\ell \|W_\ell\|_F^2$	456	166	347
$P(W)$	3.01×10^2	1.71×10^4	9.42×10^4

Thus additive regularization produced the smallest sum of squared norms, as expected, while multiplicative regularization produced by far the smallest norm product.

This is important because it shows that the two penalties did not merely change the same scalar complexity quantity by different amounts.

They selected qualitatively different norm distributions across layers.

This is a description of the reported selected configurations. It does not establish that additive or multiplicative training always minimizes its penalty more effectively, or that the compared fits are equivalent. For positive products and a common list of L matrices, the diagnostic

$$I(W) = \frac{\Omega_{\mathrm{add}}(W)}{LP(W)^{2/L}} \geq 1$$

measures layer-norm imbalance relative to the balanced rescaling orbit. It is not an intrinsic complexity of the represented function. Logging it along with the individual norms separates imbalance from total product size and tests the preceding gradient-flow identities in the settings where those identities apply.

14.11 Singular-Value Suppression

The strongest structural observation in the original experiment concerns later layers.

The original statistic called "rank" counted singular values exceeding

$$1\%$$

of the largest singular value.

It should therefore be called the 1% *numerical rank*:

$$r_{1\%}(W) = \#\left\{j : \sigma_j(W) > 0.01\sigma_1(W)\right\}. \tag{14.20}$$

For the best reported binary configuration, the values were

layer	multiplicative	additive	unregularized
conv1	16	16	16
conv2	32	32	32
conv3	64	64	64
conv4	47	128	128
fc1	2	256	256
fc2	1	1	1

The collapse in the first fully connected layer,

$$256 \longrightarrow 2,$$

is particularly striking.

The last scalar-output weight matrix has at most one nonzero singular value by its shape, so its reported value 1 is not additional evidence of a learned low-rank bias. For the convolutional rows, the tensor-to-matrix unfolding used in the original rank computation must be recovered. One possible convention is an output-channel-by-input-patch matrix, but its rank is not the rank of the full spatial convolution operator. No such convention is inferred from the table alone.

It should not yet be interpreted as proof that multiplicative regularization generally learns a rank-two representation.

The correct empirical question is whether the phenomenon is reproducible across seeds, hyperparameters, architectures, and an unbiased selection protocol.

14.11.1 Stable rank

Future analyses should also report

$$r_{\text{stable}}(W) = \frac{\|W\|_F^2}{\|W\|_2^2}. \tag{14.21}$$

Stable rank varies continuously at every nonzero matrix and is invariant under multiplication by a nonzero scalar. At $W = 0$ the ratio is undefined; setting it to zero for logging is a convention, not a continuous extension. The relative spectrum likewise requires $\sigma_1(W) > 0$.

The full normalized spectrum

$$\left\{\frac{\sigma_j(W)}{\sigma_1(W)}\right\}_j$$

should also be shown.

The interesting hypothesis is therefore

multiplicative regularization may select lower-dimensional late representations.

From a weight spectrum to a predictor certificate. A low numerical rank is not by itself a low prediction error after truncation. Let W_r be a rank-r truncated singular-value approximation, let incoming activations obey $\|h(x)\|_2 \le B_h$, and suppose the map after this linear layer (including any intervening activation) is L_{down}-Lipschitz on the required states. Retain biases and all other layers. Then

$$\sup_x \|f_W(x) - f_{W_r}(x)\| \le L_{\text{down}} B_h \|W - W_r\|_2 = L_{\text{down}} B_h \sigma_{r+1}(W).$$

The inequality follows by bounding the local change by $\|W - W_r\|_2 \|h(x)\|_2$ and then applying downstream Lipschitzness. Several truncated layers require the corresponding accumulated-error bound, and quantized factors contribute additional error. Chapter 13 then gives the fixed-code compression route, provided the factors, metadata, and required statistical assumptions are accounted for. The current spectra alone are not such a certificate.

Nor does the rank of a weight matrix automatically equal the linear rank of the nonlinear activation matrix across examples. A nonlinearity can increase the latter even when its inputs lie in a low-dimensional affine subspace. A bottleneck claim must specify whether it concerns weight rank, preactivation dimension, activation covariance, or task-sufficient coordinates.

14.12 Exploratory Multiclass Results

The original full-CIFAR-10 experiment showed a similar direction.

The reported averages were

regularization	$P(W)$	$\sum_\ell \|W_\ell\|_F^2$	train acc.	test acc.
multiplicative	3.73×10^7	679	0.891	0.687
additive	1.10×10^9	875	0.925	0.651
unregularized	1.67×10^9	960	0.937	0.650

The multiplicative configuration therefore had a much smaller norm product and a reported test-accuracy advantage of approximately

3.6

percentage points over the additive configuration.

Again, because model selection used the test set, this number should not be treated as a final estimate of the generalization advantage.

The reported training accuracies are also below one and differ across methods. This table therefore does not compare exact interpolants. The accuracy–fit–complexity tradeoff, including the actual unpenalized squared loss, must be measured rather than attributed solely to selection within an interpolation set.

Audit aggregates at the checkpoint level. For each run and a common list of L penalized matrices, the identity $\Omega_{\mathrm{mult}} = P^2$ and bound $P \le (\Omega_{\mathrm{add}}/L)^{L/2}$ provide mandatory consistency checks. Confirm whether each log field contains P, P^2, or a normalized product, and specify whether the table uses means, medians, or selected runs. The reported entries above are *averages*; one cannot apply this nonlinear per-run inequality by substituting separate averages for P and Ω_{add}. The summaries do not determine the missing per-run moments. No relabeling or correction of the supplied numbers is made here without the original records.

14.13 A Mechanistic Hypothesis

Taken together, the exploratory observations suggest a more precise hypothesis than simply

multiplicative regularization generalizes better.

The proposed mechanism is

global multiplicative norm coupling
⇓
unequal relative regularization pressure across layers
⇓
interaction with data gradients and the actual optimizer
possibly suppresses selected singular directions
⇓
lower effective dimension / lower complexity
⇓
larger normalized margins and possibly better generalization.

Only the layer-pressure relation follows directly from Equation (14.14).

The later arrows are empirical hypotheses.

Radial shrinkage alone preserves the normalized spectrum, so the second step requires a mechanism beyond (14.14). In the homogeneous Euclidean-flow model, unequal layer norms are also not created by multiplicative regularization from an exactly balanced start. The proposed explanation must be tested against those restrictions rather than treating all spectral changes as direct consequences of the penalty.

The common-minibatch-equilibrium rank obstruction discussed in Chapter 7 and Appendix N is complementary, not a proof of this experiment's rank effect. That result concerns a specified

normalized scalar-output ReLU model and *algebraic* rank at a common zero of all minibatch directions. It does not automatically apply to biased CNNs or identify a 1% numerical-rank statistic with that algebraic rank. The present mechanism remains a question about finite-time dynamics.

A particularly informative experiment would therefore track all of these quantities through training rather than comparing only final checkpoints.

Temporal ordering is useful evidence but not causal identification. Matched interventions—for example balancing the initialization, changing the radial penalty strength while monitoring the data update, or testing controlled spectral truncation—are needed to distinguish the proposed mechanism from shared dependence on training time or fit.

14.14 What We Do Not Claim

Several stronger interpretations should be avoided.

14.14.1 The Frobenius product is not the unique complexity measure

Modern norm bounds can depend on spectral norms, Frobenius corrections, margins, architecture, and depth.

A low value of

$$P(W)$$

is therefore evidence about one complexity surrogate, not a complete generalization theory.

Under the stated homogeneous norm-class assumptions a smaller admitted product budget does reduce the displayed Rademacher upper bound. The remaining issues are whether the learned model falls under those assumptions, whether it fits adequately, and whether the resulting bound is informative. A valid smaller upper bound need not mean that actual population error is monotonically smaller across unrelated predictors.

14.14.2 A smaller product does not by itself imply better generalization

The regularizer changes the optimization trajectory.

The selected predictor can change in many ways besides its norm product.

The appropriate question is empirical:

which measured complexity quantities best predict held-out performance across trained solutions?

14.14.3 Parameter flatness is not claimed

Raw Hessian eigenvalues and other parameter-space notions of "flatness" can change under reparameterization.

The present exploratory experiments do not establish that multiplicative regularization selects flatter minima in an invariant sense.

We therefore make no flat-minima claim here.

14.14.4 Rank reduction is not automatically beneficial

A low-rank representation can discard nuisance dimensions.

It can also discard useful signal.

The significance of the observed singular-value collapse must therefore be tested jointly with performance and robustness.

The robustness measured here concerns corrupted *training labels*, not adversarial input perturbations, corrupted test images, or arbitrary distribution shift. Those are different objectives. A useful low-rank predictor also need not recover any designated teacher variables.

14.15 Connection to the Principles of the Book

This experiment tests a specific link in the framework developed earlier. Chapters 7 and H distinguish empirical fitting, regularizer-dependent selection, and predictive generalization. Overparameterization may permit many well-fitting realizations; it does not guarantee that the selected one has an informative norm or compression bound. At positive regularization the selected predictor need not interpolate exactly.

Chapter 13 supplies two complementary statistical languages: norm-controlled function classes and valid finite descriptions of compressed predictors. A smaller parameter-norm product and a smaller matrix rank are not interchangeable with either a complete bound or an encoded model. The appropriate norms, architectural factors, prediction errors, and description costs must still be supplied.

The current study therefore asks whether a specified regularized training procedure *returns* a predictor with an improved fit–complexity tradeoff. Schematically, on the success and statistical events of Chapter H,

$$\begin{aligned} \widehat{R}_S(\widehat{f}) \le \eta, \qquad \widehat{f} \in \mathcal{F}_Q, \\ R_{\mathsf{P}}(\widehat{f}) \le \widehat{R}_S(\widehat{f}) + \beta_S(Q,\delta) \\ \implies \quad R_{\mathsf{P}}(\widehat{f}) \le \eta + \beta_S(Q,\delta). \end{aligned}$$

The returned model may be a specified stopped iterate. Neither a common zero of every minibatch update nor recovery of the teacher's hidden coordinates is an additional premise. The proposed norm-difference confinement result is one conditional dynamical observation; it does not establish that this fit–complexity event occurs efficiently.

The experimental chain is consequently

specified architecture, objective, optimizer, and selection rule
⇓
returned predictors with measured fit, norms, spectra, and margins
⇓
valid statistical control and/or independent performance evaluation.

The final line must be established, not inferred from the penalty's name. When population prediction and feasible evaluation do hold, the learned computation itself supplies the compositional witness; a preferred internal factorization is not required.

14.16 Conclusion

Multiplicative and additive norm regularization are mathematically and dynamically different.

The additive penalty

$$\sum_{\ell} \|W_\ell\|_F^2$$

penalizes layer norms separately and, in positively homogeneous models, selects among functionally equivalent layerwise rescalings.

The multiplicative penalty

$$\prod_{\ell} \|W_\ell\|_F^2$$

is invariant under compensating homogeneous rescalings and couples all layers through its gradient.

For layer j,

$$\nabla_{W_j} \Omega_{\text{mult}} = 2\Omega_{\text{mult}} \frac{W_j}{\|W_j\|_F^2},$$

so the relative product pressure scales as

$$\frac{2\Omega_{\text{mult}}}{\|W_j\|_F^2}.$$

This provides a concrete mechanism by which different layers can experience very different regularization pressure.

The mathematical qualifications are equally important. Globally minimizing additive cost along a homogeneous rescaling orbit gives $LP^{2/L}$, so idealized constrained function-level minimizers can agree. The product penalty's direct gradient is radial and cannot alone change normalized singular spectra. In the specified homogeneous flow, squared-norm differences are conserved and this fact, together with the decreasing objective, supplies a trajectory-bound mechanism despite noncoercivity.

The original experiments report selected solutions with far smaller global norm products and dramatically lower numerical rank in some later layers. Those observations motivate a study of the *interaction* of loss, regularization, initialization, and stochastic or adaptive updates.

The same experiments also reported higher test accuracy, larger margins, and much greater tolerance of label corruption.

Because of the original model-selection protocol, however, these performance differences remain exploratory.

The scientifically defensible conclusion is therefore

multiplicative norm regularization defines a distinct training bias; the reported selection patterns motivate a controlled test of its benefit.

Whether that bias produces a robust generalization advantage is an experimental question.

14.17 Experiments That Need to Be Done or Repeated

The following experiments are required before the stronger claims of this chapter can be regarded as established.

1. **Repeat the primary binary experiment with validation-only selection and independent confirmation.**

   Split the original CIFAR-10 training data into training and validation subsets.

   For cat-versus-deer, a natural split is

   $$9{,}000 \text{ training} + 1{,}000 \text{ validation},$$

   with the original

   $$2{,}000$$

   cat/deer test images reserved from all further tuning.

   All choices of

   $$\lambda, \qquad \text{learning rate}, \qquad \text{stopping epoch}, \qquad \text{checkpoint}, \qquad \text{initialization scale}$$

   must be made using training and validation data only.

   Evaluate the prespecified final comparisons after the procedure is frozen, without using the results for further selection. Since the original test set already influenced the exploratory work, label this a protocol-frozen replication, not a newly untouched confirmatory evaluation. A fresh independent evaluation sample, or an independently chosen benchmark with a locked protocol, is the stronger confirmation. The proposed split sizes are retained as a concrete replication design, not as new data.

2. **Use paired seeds and retain every algorithmic run.**

   For each seed, additive, multiplicative, and unregularized training should begin from exactly the same initialized weights.

   When possible they should also share

   $$\text{data split}, \qquad \text{minibatch order}, \qquad \text{augmentation randomness}, \qquad \text{label-noise mask}.$$

   No run should be discarded because it fails to train.

   Predefine failure criteria, numerical-overflow handling, the full tuning budget, and a return rule for every run. Report all-run performance and selection success, not only the successful-run subset. Recovery of the original number of excluded runs would also permit a sensitivity analysis of the existing summaries.

   Instead report separately

   $$\text{optimization success probability}$$

and

$$\text{performance conditional on successful optimization.}$$

At least ten paired seeds should be used; twenty would be preferable for the binary experiment.

These are proposed seed counts, not a power guarantee. Pilot variance and the effect size of interest should determine the confirmatory sample size. Paired runs also share all experimental random choices that can be coupled without altering either algorithm.

3. **Repeat the label-noise experiment under two protocols.**

   Use corruption levels

   $$0\%, \qquad 10\%, \qquad 20\%.$$

   For a given seed, use exactly the same corrupted labels for every regularizer.

   Evaluate two versions.

   Retuned protocol. Choose the optimal λ independently by validation at each noise level.

   This measures the best validation-selected performance within the prespecified, equally resourced search, not the unknown best achievable performance of the regularizer.

   Fixed-λ protocol. Choose λ using clean validation data and hold it fixed as noise increases.

   This measures robustness without hyperparameter reoptimization.

   Corrupt training labels only in this comparison, with the corruption law specified; preserve a common clean validation target when that is the stated protocol. Report fit to both observed and original training labels for diagnostic purposes, without using inaccessible clean labels in the actual training procedure. Using clean validation information is part of the observation model and must be disclosed.

4. **Repeat the multiclass CIFAR-10 experiment with clean model selection.**

   Use, for example,

   $$45{,}000 \text{ training} + 5{,}000 \text{ validation} + 10{,}000 \text{ reserved test}.$$

   Use paired initialization across regularizers.

   The multiclass replication should preferably be performed after the binary effect survives the corrected protocol.

   The same historical-test caveat applies. Freeze preprocessing, output-loss normalization, tuning budgets, checkpoint selection, and any classwise reporting before this comparison.

5. **Replace raw-margin comparisons by scale-normalized margins.**

   For a bias-free positively homogeneous network, measure

$$\widetilde{m}_i = \frac{y_i f_W(x_i)}{\prod_\ell \|W_\ell\|_F}.$$

Report the

$$\text{mean}, \qquad \text{median}, \qquad 5\% \text{ quantile}, \qquad 10\% \text{ quantile},$$

and full distribution.

Because the original binary architecture contains biases, repeat the main experiment with a bias-free architecture.

Alternatively, incorporate biases into an explicitly defined augmented norm and state that convention consistently.

Use a common evaluation set and include incorrectly classified examples. For multiclass logits $f_c(x)$, also report the signed gap $f_{y_i}(x_i) - \max_{c \neq y_i} f_c(x_i)$ with the applicable norm convention. Do not infer a certified margin bound from a mean gap alone.

6. **Repeat the rank experiment using full singular-value diagnostics.**

   For each layer, report

   $$r_{1\%}(W) = \#\{j : \sigma_j > 0.01\sigma_1\},$$

   together with stable rank

   $$r_{\text{stable}}(W) = \frac{\|W\|_F^2}{\|W\|_2^2}.$$

   Also plot

   $$\frac{\sigma_j}{\sigma_1}$$

   for the entire spectrum.

   The key question is whether the dramatic collapse seen in conv4 and fc1 is reproducible.

   Specify convolutional unfoldings, report absolute as well as relative singular values, and distinguish weight spectra from activation spectra. Test whether explicit truncation at the reported rank preserves empirical predictions, held-out performance, and a valid prediction-error budget. Zero-norm cases must be logged rather than removed.

7. **Measure the dynamics of multiplicative pressure.**

   During training record

   $$\|W_j(t)\|_F,$$

   $$P(W(t)),$$

$$\Omega_{\mathrm{add}}(W(t)),$$

and

$$\frac{\|\nabla_{W_j}\Omega_{\mathrm{mult}}(W(t))\|_F}{\|W_j(t)\|_F}.$$

At the same checkpoints record singular-value spectra, stable rank, normalized margins, validation loss, and validation accuracy.

Separate the radial penalty contribution from the data-gradient and actual optimizer-update contributions. Record squared-norm differences, the imbalance statistic $I(W)$, and minibatch-gradient variance at common checkpoints. For the special homogeneous Euclidean-flow comparison, test conservation of norm differences versus additive balancing; do not assume those identities for the original biased network or for finite-step Adam. Use a balanced-initialization control to test whether observed depthwise asymmetry is inherited from initialization. Include a penalty-only control, whose normalized spectra should remain unchanged under nonzero scalar shrinkage.

The mechanistic hypothesis predicts the temporal ordering

strong relative product pressure
$\Downarrow$
spectral suppression
$\Downarrow$
larger normalized margins
$\Downarrow$
better held-out performance.

Temporal ordering adds information beyond final-state correlation, but matched interventions are needed before making a causal claim.

8. **Test whether the effect is specific to the squared product.**

   Compare

$$\mathcal{L} + \lambda P(W),$$

$$\mathcal{L} + \lambda P(W)^2,$$

   and a stabilized log-product objective

$$\mathcal{L} + \lambda \sum_{\ell} \log\left(\|W_\ell\|_F^2 + \eta\right).$$

   These objectives are not equivalent.

Fix and report $\eta > 0$; it sets a norm scale and changes the rescaling symmetry. Include the orbit-balanced penalty $LP(W)^{2/L}$ as a separate analytical control where the homogeneous model applies. The norm gaps, constraint set, zero-layer behavior, and fit–penalty tradeoff should be reported alongside performance.

The comparison can reveal whether the main effect is associated with

$$\begin{array}{cc} \text{layer coupling,} & \text{rescaling invariance,} \\ \text{product magnitude,} & \text{or the particular gradient profile.} \end{array}$$

9. **Compare competing complexity surrogates.**

   Across all seeds and hyperparameters, measure

$$P(W),$$

$$\sum_{\ell} \|W_\ell\|_F^2,$$

$$\prod_{\ell} \|W_\ell\|_2,$$

   and

$$\left(\prod_{\ell} \|W_\ell\|_2\right)\left(\sum_{\ell} \frac{\|W_\ell\|_F^2}{\|W_\ell\|_2^2}\right)^{1/2},$$

   together with

$$\text{stable ranks}$$

   and

$$\text{normalized margins.}$$

   Without using test performance to choose the metric, ask which quantity best predicts the validation generalization gap across trained solutions.

   The purpose is not merely to verify that multiplicative training produces a smaller Frobenius product.

   The stronger question is

$$\boxed{\text{which complexity measure actually tracks generalization?}}$$

10. **Test the interaction with initialization scale explicitly.**

    Use

$$s \in \{0.2, 0.4, 0.6, 0.8\}$$

as a factorial experimental variable rather than silently absorbing it into model selection.

For every scale, tune λ using validation data and compare regularizers using paired seeds.

For a pure initialization-scale control, specify exactly which tensors are multiplied by s and how biases are treated. Since the squared-product penalty scales as s^{2L}, use separately calibrated search ranges and record initial penalty values and gradient ratios. Monitor finite-precision underflow and overflow without silently clipping or changing the objective.

Measure both final performance and optimization success probability.

11. **Test robustness across losses, optimizers, and architectures.**

    If the main phenomenon survives the primary corrected replication, repeat it using at least some of

    - cross-entropy rather than squared error;
    - SGD with momentum;
    - Adam or AdamW;
    - a second CNN;
    - a residual network;
    - different network depths.

    This will determine whether the effect is specific to the original setup or is a broader property of global norm-product regularization.

    Separate explicit ℓ_2 penalties, explicit product penalties, and decoupled optimizer decay. Disable unintended additional optimizer decay in the explicit-penalty arms or report it as a hybrid objective. Residual, normalization, and nonhomogeneous architectures require their own norm interpretation; the layer-rescaling propositions are not transferred to them unchanged.

12. **Report paired effect sizes and confidence intervals.**

    For any metric M, define

    $$\Delta_j = M_{\mathrm{mult},j} - M_{\mathrm{add},j}$$

    for paired seed j.

    Report

    $$\overline{\Delta},$$

    its standard error,

    $$\mathrm{SE}(\overline{\Delta}) = \frac{\mathrm{sd}(\Delta_1, \ldots, \Delta_N)}{\sqrt{N}},$$

    and a 95% confidence interval.

    The paired difference is the primary quantity because the compared runs share initialization and data randomness.

The standard error assumes independent paired seed replicates under the prespecified procedure. A confidence interval across seeds on one fixed test set describes training-randomness variation conditional on those data; it does not by itself include uncertainty from drawing a new population test sample or remove test-selection bias. State the interval method, account for multiple primary comparisons, and report failure frequencies. Treat fit–complexity curves and descriptive correlations as such, rather than claiming that observational association identifies the mechanism.

A strong replication of the full proposed pattern would show, under independent or explicitly protocol-frozen evaluation and paired controls,

smaller norm product
+larger normalized margins
+reproducible late-layer rank suppression
+comparable training fit
+higher held-out accuracy
+greater label-noise robustness.

The hypotheses should also be tested separately: a reliable performance benefit need not require late-layer rank collapse, and a reproducible rank effect need not improve accuracy. Temporal relations should be combined with the radial-shrinkage, balanced-initialization, and truncation controls before assigning a causal explanation. The existing tables cannot decide these alternatives.

Conversely, the hypothesis would have to be weakened if the apparent test advantage disappears under validation-only model selection, if rank collapse is not reproducible, if the effect depends on a narrow initialization regime, or if another complexity measure consistently predicts generalization better than the Frobenius norm product.

Either outcome would be informative.

The experiment is ultimately a test of a broader principle of this book:

in an overparameterized model, generalization may depend less
on nominal parameter count than on the low-complexity
predictor actually returned by the learning procedure.

CHAPTER 15

What the Hopfield Analogy Explains—and What It Does Not

Modern Hopfield networks and self-attention are closely related mathematically. Under appropriate parameterizations, an attention update can be interpreted as an associative-memory retrieval step, and repeated Hopfield-style updates can be analyzed through an energy function and its stationary states. The update rule and what is held fixed must be specified before an energy argument can be applied.

This analogy is useful.

But it has a precise scope.

A Transformer is not simply a Hopfield network iterated to equilibrium. Its depth consists of a sequence of generally different learned maps; its blocks include residual paths, normalization, attention, and nonlinear MLPs; and its parameters are themselves discovered through a separate optimization process.

Three dynamical systems must therefore be distinguished:

associative state dynamics:	*activations evolve with parameters fixed,*
Transformer depth:	*representations pass through different learned maps,*
learning dynamics:	*parameters evolve under optimization.*

A stability theorem for the first does not automatically imply a theorem for either of the other two.

This chapter develops that distinction. It also examines a tempting conjecture that fixed points of deep compositions should somehow be determined by the fixed points of their constituents. In general this is false.

What remains true, and useful for the framework of this book, is that any fixed deterministic deep factorization induces a sequence of local input–output datasets for its constituent functions. This holds at a chosen training checkpoint, without exact interpolation, parameter convergence, or a Hopfield equilibrium. If frozen and supplied to another learner, these datasets provide local supervision for reproducing that particular computation.

Their retrospective existence does not explain how useful coordinates were discovered during learning. An inaccurate or untrained network has an equally exact trace of its own computation; predictive value is a separate requirement.

Nor must such a trace coincide with the latent or generating computation of the target: it is exact for the learned factorization itself.

That is the distinction between execution and recovery that runs throughout this book.

15.1 Why the Hopfield Analogy Is Useful

Associative memory is one of the recurring computational primitives of this book.

Given stored patterns or key–value pairs, an associative memory receives a query, possibly partial or corrupted, and retrieves content according to similarity.

Modern Hopfield networks provide one mathematical realization of this idea [133]. We use their continuous, fixed-memory retrieval map as the concrete model; conclusions for this update are not assertions about every model called a Hopfield network.

A simplified continuous update has the form

$$\xi^{(m+1)} = X \operatorname{softmax}\left(\beta X^\top \xi^{(m)}\right), \tag{15.1}$$

where the columns of

$$X = [\xi_1, \ldots, \xi_N]$$

represent stored patterns in $\mathbb{R}^d$, with $N \geq 1$ and $\beta > 0$. The softmax in Equation (15.1) is taken over the N memory scores. The memory matrix $X \in \mathbb{R}^{d\times N}$ and the temperature parameter remain fixed during the iteration.

When keys and values are both the columns of X, and the query is $\xi^{(m)}$, this is exactly a single attention read with inverse temperature β. Equal pattern norms are not required for that algebraic identity or for the energy calculation below.

Under the fixed-norm conditions discussed in Chapter 9, dot-product attention can be written exactly as normalized Gaussian retrieval.

Thus the useful structural correspondence is

fixed-memory Hopfield update $=$ attention with keys and values tied.

This explains several aspects of attention:

- content-addressable retrieval;
- concentration around relevant stored patterns;
- competition among memories through normalization;
- the existence, in appropriate iterative formulations, of attractor-like states.

These are genuine insights.

An explicit energy calculation. For the specified update, define

$$\Phi_X(\xi) = \frac{1}{\beta} \log\left(\sum_{j=1}^{N} e^{\beta \xi_j^\top \xi}\right), \qquad E_X(\xi) = \frac{1}{2}\|\xi\|_2^2 - \Phi_X(\xi).$$

This is the modern Hopfield energy up to an additive constant, which does not affect its dynamics [133]. Write $H_X(\xi) = X \operatorname{softmax}(\beta X^\top \xi)$. Then $\nabla \Phi_X = H_X$ and $\nabla E_X(\xi) = \xi - H_X(\xi)$. The following direct proof records the precise descent statement needed here.

Proposition 20 (Descent for the specified fixed-memory update)**.** *For every finite memory X and every $\beta > 0$,*

$$E_X(H_X(\xi)) \le E_X(\xi) - \frac{1}{2}\|H_X(\xi) - \xi\|_2^2.$$

Consequently the iteration in Equation (15.1) *has convergent energy values and square-summable successive displacements. Its states are bounded, and every accumulation point is a fixed point of H_X, equivalently a critical point of E_X.*

Proof. Convexity of Φ_X gives

$$\Phi_X(\eta) \ge \Phi_X(\xi) + H_X(\xi)^\top(\eta - \xi).$$

Substitute $\eta = H_X(\xi)$ into the definition of E_X and complete the square to obtain the displayed descent inequality. If $R_X = \max_j \|\xi_j\|_2$, then

$$\Phi_X(\xi) \le R_X\|\xi\|_2 + \beta^{-1}\log N,$$

so E_X is bounded below. Telescoping the descent inequality therefore gives

$$\sum_{m=0}^{\infty} \|\xi^{(m+1)} - \xi^{(m)}\|_2^2 < \infty.$$

Every iterate after the first belongs to the compact convex hull of the stored patterns, since softmax weights are nonnegative and sum to one. If $\xi^{(m_j)} \to \bar{\xi}$, vanishing successive displacements and continuity of H_X imply $H_X(\bar{\xi}) = \bar{\xi}$. □

The proof establishes these conclusions, not by itself finite length or convergence to one particular state. Square summability of displacements is weaker than summability of their lengths. It also does not identify all critical points as minima, or prove that the retrieved content is correct for a task. For example, with scalar memories -1 and 1, $H_X(\xi) = \tanh(\beta\xi)$. At $\xi = 0$,

$$H_X'(0) = \beta, \qquad E_X''(0) = 1 - \beta.$$

For $\beta > 1$, this fixed point is unstable and is a local energy maximum.

One sufficient regime for convergence to a unique state. With $p = \mathrm{softmax}(\beta X^\top \xi)$,

$$DH_X(\xi) = \beta X\big(\mathrm{diag}(p) - pp^\top\big)X^\top.$$

For a unit vector v, the quadratic form is β times the variance of $v^\top\xi_j$ under p, and is at most βR_X^2. Hence $\|DH_X(\xi)\|_2 \le \beta R_X^2$ globally. If $q := \beta R_X^2 < 1$, the map is a contraction on $\mathbb{R}^d$ and has a unique fixed point ξ_*, with

$$\|\xi^{(m)} - \xi_*\|_2 \le q^m\|\xi^{(0)} - \xi_*\|_2.$$

This is a sufficient, not necessary, condition. A unique global attractor is not a guarantee of selective recall: every initial cue then has the same limit for the same fixed memory. Multiple memory-specific basins require a different regime. The distinction between accurate one-step retrieval and asymptotic relaxation is important.

The difficulty begins when one transfers this fixed-memory calculation to a different state update or to the training of the entire Transformer. The cited modern Hopfield work also studies trainable associative layers; the point is not that such layers cannot be learned. It is that the state-descent theorem and a theorem about their parameter learning have different premises and conclusions.

15.2 Three Different Dynamical Systems

The most important distinction is between three notions of dynamics that are often discussed with similar language.

15.2.1 Associative state dynamics

In a Hopfield-style system, the stored patterns and parameters are fixed.

One iterates

$$\xi^{(m+1)} = H_W(\xi^{(m)}),$$

where W denotes the fixed parameters.

The dynamical variable is the state

$$\xi.$$

The central questions are:

Does the iteration converge?

and

Which stationary state or attractor is reached?

In settings admitting an energy function

$$E_W(\xi),$$

the update can often be interpreted as descending or otherwise reducing that energy under suitable assumptions.

This is dynamics in *activation space*.

15.2.2 Transformer depth

A standard untied Transformer instead computes [158]

$$x_\ell = F_\ell(x_{\ell-1}), \qquad \ell = 1, \dots, L.$$

Generally,

$$F_1 \neq F_2 \neq \cdots \neq F_L,$$

because different blocks have different learned parameters.

A standard forward pass does not repeatedly apply one map

$$x \mapsto F(x)$$

until it reaches an equilibrium.

It applies a sequence

$$F_L \circ F_{L-1} \circ \cdots \circ F_1.$$

Thus

$$\boxed{\text{Transformer depth} \neq \text{Hopfield relaxation to a fixed point.}}$$

A recurrent architecture with tied weights could make the analogy closer, but that is a special case rather than the generic Transformer.

A finite forward pass has no asymptotic endpoint unless an additional iteration is defined. Moreover, autoregressive generation introduces a separate token-time index: the same parameterized network may be called again while its context and cache grow. Repeated invocation of a network with a changing input is not iteration of the fixed-memory map H_X.

Equilibrium architectures are a genuine special case. An input-conditioned, weight-tied model can instead define its output through

$$z_*(u) = F_\theta(z_*(u), u).$$

Deep equilibrium models explicitly use this kind of construction and implicit differentiation [9]. If F_θ is continuously differentiable and $I - D_z F_\theta(z_*, u)$ is invertible, the implicit function theorem gives locally

$$D_\theta z_*(u) = \left(I - D_z F_\theta(z_*, u)\right)^{-1} D_\theta F_\theta(z_*, u).$$

A uniform contraction bound $\|D_z F_\theta\|_2 \leq q < 1$ is one sufficient condition and bounds the inverse norm by $(1-q)^{-1}$. This supplies a specific connection between state solving and parameter gradients. It does not make optimizing the resulting training objective convex or guarantee that its gradients are informative. The input conditioning, solver tolerance, and solver cost are part of the model.

15.2.3 Learning dynamics

Training introduces a third dynamical system.

Let

$$W^{(t)}$$

denote all trainable parameters at optimization step t.

Gradient descent evolves them according to

$$W^{(t+1)} = W^{(t)} - \eta \nabla_W \mathcal{L}\left(W^{(t)}\right). \tag{15.2}$$

Now the dynamical variable is

$$W,$$

not the activation ξ.

This is dynamics in *parameter space*.

In Equation (15.2), $\mathcal{L}$ is the explicitly chosen training objective; it can include a deterministic penalty. Regularized SGD instead uses a random minibatch gradient estimate of that objective. As Chapters 7–H and Appendix N explain, a stationary full regularized objective can have nonzero minibatch covariance. Neither zero state-update residual nor zero expected parameter update means that every parameter minibatch update vanishes. A specified finite training iterate can be returned without any of these equilibrium properties.

15.2.4 The central distinction

The three systems are therefore

$$\boxed{\begin{aligned} \xi^{(m+1)} &= H_W(\xi^{(m)}) && \text{associative state dynamics,} \\ x_\ell &= F_\ell(x_{\ell-1}) && \text{depth dynamics,} \\ W^{(t+1)} &= W^{(t)} - \eta \nabla \mathcal{L}(W^{(t)}) && \text{learning dynamics.} \end{aligned}}$$

A Lyapunov function for the first does not automatically provide a Lyapunov function for the third.

Likewise, convergence of the state dynamics says nothing by itself about whether gradient descent will discover a useful representation.

15.3 The Scope of the Hopfield–Attention Correspondence

The phrase "attention is a Hopfield network" should therefore be used carefully.

A standard attention head computes the following maps, with token representations stored as rows of Z (unlike the pattern columns of X in Equation (15.1)):

$$Q = ZW_Q, \qquad K = ZW_K, \qquad V = ZW_V,$$

followed by

$$\mathrm{Attn}(Z) = \mathrm{softmax}\left(\frac{QK^\top}{\tau}\right) V.$$

Here $\tau > 0$ absorbs score scaling and the softmax acts rowwise. Positional information, masks, and output projections must be included when the actual architecture uses them. We discuss deterministic evaluation, with stochastic operations disabled or their randomness explicitly fixed.

Modern Hopfield updates and attention coincide in the tied construction above. More general associations can be expressed using learned projections into and out of an associative space [133]. An algebraic representation of a read does not automatically transfer the same energy-descent theorem to repeated iteration of its external output.

But several distinctions remain.

15.3.1 Query, key, and value maps need not coincide

A symmetric Hopfield interpretation is especially natural when the same representation plays the role of both key and retrieved pattern.

In general attention,

$$W_Q, \qquad W_K, \qquad W_V$$

are distinct.

The retrieval geometry and the retrieved content can therefore be different.

To see the issue directly, use column conventions for one query:

$$A(\xi) = V_c \,\mathrm{softmax}(\beta K_c^\top \xi), \quad K_c \in \mathbb{R}^{d \times N}, \quad V_c \in \mathbb{R}^{d_v \times N}.$$

If $d_v \neq d$, this is not even a self-map without an additional output-to-query transformation. If $d_v = d$, its derivative is

$$DA(\xi) = \beta V_c(\mathrm{diag}(p) - pp^\top)K_c^\top, \qquad p = \mathrm{softmax}(\beta K_c^\top \xi).$$

This matrix need not be symmetric. In that case the field $\xi - A(\xi)$ is not the Euclidean gradient of a twice continuously differentiable scalar potential on an open neighborhood. For example, keys $e_1, -e_1$ and values $e_2, -e_2$ in $\mathbb{R}^2$ give

$$A(\xi_1, \xi_2) = (0, \tanh(\beta\xi_1)), \qquad DA(0) = \begin{pmatrix} 0 & 0 \\ \beta & 0 \end{pmatrix}.$$

This rules out that particular gradient representation, not every possible Lyapunov function, metric, or augmented-state formulation. Different query and key projections also do not, by themselves, rule out an associative-space interpretation; the actual feedback map is what must be analyzed.

15.3.2 A Transformer normally performs one attention update per block

Equation (15.1) invites repeated iteration

$$\xi^{(0)} \mapsto \xi^{(1)} \mapsto \xi^{(2)} \mapsto \cdots$$

toward an attractor.

A Transformer block normally evaluates its attention operation once and passes the result onward.

Thus the equilibrium of an isolated attention update is generally not the object computed by the full network.

In self-attention, the keys and values are also functions of the current token states. Holding weight matrices fixed does not freeze these memory entries across depth or an externally imposed recurrence. A proof for E_X with X fixed cannot discard this additional state dependence. Even a collection of layer-specific decreasing energies does not provide one common decreasing energy for the sequence of layer maps.

15.3.3 The correspondence concerns a primitive, not the entire architecture

The safe statement is

Hopfield theory provides an associative-memory interpretation of an attention primitive under appropriate assumptions.

It does not imply

a Transformer as a whole is a Hopfield dynamical system.

15.4 Fixed Points Do Not Compose

A tempting idea is that if the layers of a deep system each possess stable fixed points, then the fixed points of their composition should somehow be obtained from the fixed points of the constituent maps.

Before stating the result, one qualification is important.

Fixed-point language applies literally only when the relevant maps have a common state space, or when explicit identifications between their input and output spaces have been specified. Ordinary deep-network layers can change dimension, in which case a layerwise fixed-point interpretation is not intrinsically defined.

Even when all maps do act on a common state space, the proposed compositional rule for fixed points is false in general.

Proposition 21 (Fixed points do not compose)**.** *For functions*

$$g_1, g_2 : X \to X,$$

the fixed-point set

$$\mathrm{Fix}(g_2 \circ g_1)$$

is not determined in general by

$$\mathrm{Fix}(g_1) \qquad \textit{and} \qquad \mathrm{Fix}(g_2).$$

In particular, the composition can have many fixed points even when the individual maps have few or none.

Proof. Consider first

$$g_1(x) = -x, \qquad g_2(x) = -x, \qquad x \in \mathbb{R}.$$

Each map has the unique fixed point

$$\mathrm{Fix}(g_1) = \mathrm{Fix}(g_2) = \{0\}.$$

But

$$(g_2 \circ g_1)(x) = x.$$

Therefore

$$\mathrm{Fix}(g_2 \circ g_1) = \mathbb{R}.$$

For an even stronger example, let

$$g_1(x) = x + 1, \qquad g_2(x) = x - 1.$$

Neither map has a fixed point:

$$\mathrm{Fix}(g_1) = \mathrm{Fix}(g_2) = \varnothing.$$

Yet again

$$g_2 \circ g_1 = \mathrm{Id},$$

so

$$\mathrm{Fix}(g_2 \circ g_1) = \mathbb{R}.$$

To establish that the constituent fixed-point sets do not determine the composite fixed-point set, compare pairs with *identical* constituent sets. Return to the reflection example: keep $g_1(x) = -x$ but replace g_2 by $\widetilde{g}_2(x) = -2x$. Both constituents still have fixed-point set $\{0\}$, whereas

$$\mathrm{Fix}(\widetilde{g}_2 \circ g_1) = \{0\} \neq \mathbb{R} = \mathrm{Fix}(g_2 \circ g_1).$$

Likewise, two translations $x \mapsto x + 1$ have empty constituent fixed-point sets, just as the opposite translations above do, but their composition $x \mapsto x + 2$ has no fixed point. These comparisons prove the claimed non-determination.

□

Thus

$$\boxed{\mathrm{Fix}(g_2 \circ g_1) \text{ cannot in general be reconstructed from } \mathrm{Fix}(g_1), \mathrm{Fix}(g_2).}$$

Attracting constituent fixed points need not transfer stability. The preceding reflection and translation examples do not have attracting fixed points. An additional linear example addresses that stronger intuition. On $\mathbb{R}^2$, let

$$A_1 = \begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1/2 & 0 \\ 1 & 1/2 \end{pmatrix}.$$

Each map $x \mapsto A_j x$ has the unique globally asymptotically stable fixed point 0: its powers tend to zero, since its only eigenvalue is $1/2$. But

$$A_2 A_1 = \begin{pmatrix} 1/4 & 1/2 \\ 1/2 & 5/4 \end{pmatrix}, \qquad \lambda_{\max}(A_2 A_1) = \frac{3 + 2\sqrt{2}}{4} > 1.$$

The common fixed point is therefore unstable under iteration of the composition. The matrices are not contractions in the same Euclidean norm. Separate asymptotic stability is weaker than a shared quantitative contraction or common Lyapunov condition. This example concerns general layer maps, not a realization of these matrices by the tied Hopfield update in Equation (15.1).

15.4.1 The easy positive statement

There is one immediate sufficient condition.

If

$$x^\star \in \mathrm{Fix}(g_1) \cap \mathrm{Fix}(g_2),$$

then

$$g_2(g_1(x^\star)) = g_2(x^\star) = x^\star.$$

Hence

$$\mathrm{Fix}(g_1) \cap \mathrm{Fix}(g_2) \subseteq \mathrm{Fix}(g_2 \circ g_1).$$

But the inclusion can be strict, as the counterexample above shows.

15.4.2 Even contraction does not identify constituent fixed points

Suppose g_1 and g_2 are self-maps of the same nonempty complete metric space, with Lipschitz constants $q_1, q_2 < 1$.

Then the composition $g_2 \circ g_1$ has Lipschitz constant at most $q_2 q_1 < 1$ and therefore has a unique fixed point by the contraction mapping theorem.

But that fixed point need not be the fixed point of either constituent.

For example,

$$g_1(x) = \frac{1}{2}x,$$

and

$$g_2(x) = \frac{1}{2}x + 1.$$

The fixed points are

$$\mathrm{Fix}(g_1) = \{0\}, \qquad \mathrm{Fix}(g_2) = \{2\}.$$

But

$$(g_2 \circ g_1)(x) = \frac{1}{4}x + 1,$$

whose fixed point is

$$x^\star = \frac{4}{3}.$$

Thus even very well-behaved constituent dynamics do not yield a compositional rule for fixed points.

Keeping $g_1(x) = x/2$ but replacing g_2 by $\widetilde{g}_2(x) = x/4 + 3/2$ retains the individual fixed-point sets $\{0\}$ and $\{2\}$. The composite fixed point becomes $12/7$, not $4/3$. Thus the non-determination persists even within scalar affine contractions.

There are useful positive refinements. If the contractions have a common fixed point, it is the unique fixed point of their composition. Commuting contractions on the same complete space necessarily share a fixed point: if p is the unique fixed point of g_1, then $g_1(g_2(p)) = g_2(g_1(p)) = g_2(p)$ forces $g_2(p) = p$.

For noncommuting contractions with different fixed points, repeated alternation converges phase by phase to a cycle for the scheduled maps. In the original scalar example, the composite fixed point $4/3$ is sent by g_1 to $2/3$ and then by g_2 back to $4/3$. Neither phase need be a fixed point of the corresponding constituent. A fixed point of a full cycle is not a fixed point of each step.

15.5 Why This Matters for Deep Networks

Suppose a deep network is

$$f = g_L \circ \cdots \circ g_1.$$

Even if every g_ℓ admits a Hopfield-like attractor interpretation, there is no general principle saying that the behavior of

$$f$$

is determined by composing the attractors of the g_ℓ.

The network is evaluating

$$x \mapsto g_1(x) \mapsto g_2(g_1(x)) \mapsto \cdots \mapsto f(x).$$

Its intermediate representations need not be fixed points of the individual layers.

No such condition is imposed by the forward evaluation rule. A layerwise fixed-point interpretation would be an additional property to establish, not a consequence of composition.

Thus the useful object in deep composition is generally the *trajectory of intermediate representations*,

$$z^{(0)} = x,$$

$$z^{(1)} = g_1(z^{(0)}),$$

$$\cdots$$

$$z^{(L)} = g_L(z^{(L-1)}),$$

not the fixed-point sets of the constituent maps.

15.6 Attention Is Only Part of a Transformer Block

The Hopfield analogy also leaves out a second major component of the Transformer.

A pre-normalized Transformer block has schematic form [169]

$$x'_\ell = x_\ell + \mathrm{Attn}\left(\mathrm{LN}(x_\ell)\right),$$

followed by

$$x_{\ell+1} = x'_\ell + \mathrm{MLP}\left(\mathrm{LN}(x'_\ell)\right).$$

The attention module is nonlinear because of the softmax and can perform nontrivial transformations of its own, but one important computational role is associative routing and mixing:

which information should be retrieved, mixed, or made available?

The MLP applies a local nonlinear transformation to the retrieved, mixed, and residual information:

what local computation should be performed on it?

This gives the interpretation developed in Chapter 9:

attention $\approx$ content-addressable routing and mixing,
MLP $\approx$ local nonlinear computation.

A theory based only on Hopfield retrieval therefore captures one important primitive but omits much of the local constituent computation.

Here "local" for an MLP means position-wise in token space, not necessarily bounded scalar arity. Its feature vectors may be high dimensional and its affine maps dense. As in Chapters 4 and 9, a bounded-fan-in digital implementation is a different level of description whose arithmetic, precision, and storage costs must be counted.

Nor does the residual addition guarantee exact preservation of old coordinates: $x + u(x)$ is identically zero when $u(x) = -x$. A useful sufficient quantitative condition is $\mathrm{Lip}(u) \le a < 1$, which gives directly by the triangle inequality

$$(1-a)\|x-y\| \le \|(x+u(x)) - (y+u(y))\| \le (1+a)\|x-y\|.$$

This is a bound on preservation of distinctions for that residual map. It is not a contraction bound for the full block or a guarantee that particular semantic coordinates are retained.

15.7 Every Learned Factorization Induces a Trace

There is, however, a useful relationship between a fixed deep factorization and local learning problems. Write a returned deterministic predictor as

$$\widehat{f} = F_{\widehat{W}} = g_2 \circ g_1.$$

The hat denotes the selected predictor, not an assumed global minimizer or the generating target. On examples (x_i, y_i), define

$$z_i = g_1(x_i), \qquad \widehat{y}_i = g_2(z_i) = \widehat{f}(x_i).$$

Then the induced datasets

$$\mathcal{D}_1 = \{(x_i, z_i)\}_{i=1}^n, \qquad \mathcal{D}_2 = \{(z_i, \widehat{y}_i)\}_{i=1}^n$$

are exactly consistent with g_1 and g_2. Only when the predictor interpolates can $\widehat{y}_i$ be replaced by y_i without changing the second dataset. This distinction is relevant to regularized, nearly interpolating training as well as to earlier checkpoints.

Proposition 22 (Every factorization induces a local trace)**.** *Let*

$$\widehat{f} = g_L \circ g_{L-1} \circ \cdots \circ g_1$$

be a fixed deterministic factorization through specified state spaces, and let (x_i, y_i), $i = 1, \dots, n$, be any labeled sample. Define recursively

$$z_i^{(0)} = x_i, \qquad z_i^{(\ell)} = g_\ell(z_i^{(\ell-1)}), \quad \ell = 1, \dots, L.$$

For every layer ℓ, the dataset

$$\mathcal{D}_\ell = \left\{ \left(z_i^{(\ell-1)}, z_i^{(\ell)} \right) \right\}_{i=1}^n$$

is exactly consistent with g_ℓ. Its final states satisfy $z_i^{(L)} = \widehat{f}(x_i)$. In the additional exact-interpolation case, $z_i^{(L)} = y_i$ for every i.

Proof. The local consistency is the defining identity $z_i^{(\ell)} = g_\ell(z_i^{(\ell-1)})$. Iterating it yields

$$z_i^{(L)} = (g_L \circ \cdots \circ g_1)(x_i) = \widehat{f}(x_i).$$

Equality to y_i follows precisely when the returned predictor fits that example exactly. No training or equilibrium assumption is used. □

Thus

every fixed deterministic factorization defines an exact trace of its own computation, whether or not that computation is accurate.

For a vector-valued output, its empirical squared error is still

$$\frac{1}{n}\sum_i \|z_i^{(L)} - y_i\|_2^2 = \frac{1}{n}\sum_i \|\widehat{f}(x_i) - y_i\|_2^2.$$

Exact local self-consistency does not remove that error. For example, a network representing the constant-zero function has a perfectly exact internal trace even on a task with target identically one.

What "exact" means. The trace is exact for the specified evaluator. At finite precision, one may incorporate deterministic rounding into each g_ℓ; otherwise the recorded numerical trace approximates the ideal real-arithmetic one. For dropout or other randomness, either fix the randomness or include it among the local inputs. For batch-dependent operations, include the batch state rather than treating an individual token or example as an independent complete input.

These are records on the observed inputs, not a complete table of each constituent on its whole domain. They need not coincide with the latent or generating computation that produced the target data. The proposition therefore describes an execution property, not a generalization result.

15.8 Existence of the Trace Is Not Availability of the Trace

The proposition is elementary but conceptually important.

After a deep network has been learned, we can inspect its hidden states and construct

$$\mathcal{D}_1, \ldots, \mathcal{D}_L.$$

Had fixed versions of those datasets been supplied in advance, local supervised objectives for reproducing those particular constituents could have been posed. Solving them efficiently, generalizing beyond the observed local inputs, and controlling errors when they are recomposed remain separate tasks. Shared parameters or a coupling regularizer can also prevent the local objectives from being optimized independently.

During ordinary end-to-end training, the final checkpoint's intermediate targets are not given in advance. Current activations can of course be computed at every update; they are not hidden from the training program. What is absent is a fixed external specification of which activations would solve the task.

The learner observes only

$$(x_i, y_i).$$

The intermediate labels

$$z_i^{(1)}, \ldots, z_i^{(L-1)}$$

are unknown as fixed useful targets before the representation has been selected.
Therefore

existence of a local trace after learning $\neq$ availability of that trace during learning.

This is closely related to the distinction between C2 and C3 in Chapter H, but with one further qualification: the retrospectively induced trace above belongs to the learned factorization and need not coincide with the target's latent trace.

Simply copying the current output of a module as its own target supplies no missing learning signal. At a fixed checkpoint $\widehat{W}$, for example,

$$\frac{1}{2}\|g_{\widehat{W}}(z) - \mathrm{stopgrad}(g_{\widehat{W}}(z))\|_2^2 = 0,$$

and its gradient with respect to that module's parameters is zero. Keeping a teacher checkpoint fixed while a different student changes is a different procedure; so are auxiliary objectives using additional information. The retrospective proposition does not derive any of those learning rules.

15.8.1 C3: supplied faithful trace

If true or otherwise appropriate intermediate representations are supplied,

$$\{z_i^{(\ell)}\},$$

and the complete parent inputs and corresponding outputs are observed, the constituent supervised problems can be posed directly. The exact finite-state results in Chapter 11 additionally require the stated representation, learner, and local-state coverage. A finite collection of traces does not automatically supply all transitions that will occur on future inputs.

An externally supplied trace can also belong to a useful surrogate computation rather than a privileged generating DAG. C3 is an information resource, not only a fallback after end-to-end learning has failed.

15.8.2 C2: discovered representation

When only (x_i, y_i) are observed and the chosen learning approach requires new task-relevant coordinates, those coordinates must be constructed rather than read from an external trace. A representation fixed by architecture or supplied by pretraining can change that starting point.

The staircase hypothesis of Chapters 6 and H proposes that detectable low-complexity footprints may make this possible.

Thus

C3 supplies intermediate information; C2 concerns accessible signals that may help construct useful coordinates.

A staircase theorem can explain such construction under specified conditions. It is not a necessary endpoint condition for a feasible predictor that has already been shown to generalize. Nor is a detectable initial footprint by itself a theorem that the whole staircase will be learned.

The same predictor can have different internal traces. For invertible coordinate maps T_ℓ on the state spaces, with $T_0 = T_L = \mathrm{Id}$, define

$$\widetilde{g}_\ell = T_\ell \circ g_\ell \circ T_{\ell-1}^{-1}.$$

The intermediate states become $\widetilde{z}^{(\ell)} = T_\ell(z^{(\ell)})$, but cancellation of successive inverses gives $\widetilde{g}_L \circ \cdots \circ \widetilde{g}_1 = \widehat{f}$. A structural comparison must therefore specify an admissible coordinate equivalence with controlled complexity and conditioning. Arbitrary changes of coordinates can conceal the task in the transformation itself.

15.9 The DAG Version

The same observation extends beyond chains.

Suppose a fixed deterministic predictor factors through a DAG.

For every node v,

$$z_v = g_v\left(z_{\mathrm{pa}(v)}\right).$$

At the fixed checkpoint, every evaluated example induces a value

$$z_{v,i}$$

at every node.

Thus node v has an induced dataset

$$\mathcal{D}_v = \left\{\left(z_{\mathrm{pa}(v),i}, z_{v,i}\right)\right\}_{i=1}^{n}.$$

Every fixed deterministic DAG evaluator therefore carries an internal trace on each evaluated example. "Complete" here means that all specified node values on those examples are recorded, not that all possible parent tuples are covered. Residual bypasses, multiple attention heads, and shared-memory accesses must be included among the actual parent arguments; omitting an argument does not create a valid local supervised problem.

Again, this is retrospective.

It does not explain how the factorization or its internal coordinates were discovered.

Nor does it imply that the learned DAG is identical to the target's generating DAG.

At the token or layer level, parent objects can be large arrays. Their number is not a bound on scalar arity or bit complexity. A full trace can be generated at roughly the cost of evaluation plus writing the recorded states, but retaining all intermediate arrays can use much more memory than the final predictor. Trace size, predictor description length, and working memory are distinct resources.

Recomposing modules learned from a frozen trace. An elementary bound shows what additional information makes the trace useful. Consider a chain as a special case of the DAG, with teacher states $z^{(\ell)}(X) = g_\ell(z^{(\ell-1)}(X))$ and student states $\widetilde{z}^{(\ell)}(X) = \widetilde{g}_\ell(\widetilde{z}^{(\ell-1)}(X))$, starting from the same input $X \sim P_X$. Write $\widetilde{f} = \widetilde{g}_L \circ \cdots \circ \widetilde{g}_1$. Suppose the relevant second moments are finite and $\widetilde{g}_\ell$ is K_ℓ-Lipschitz on a domain containing both trajectories. Define the population, teacher-fed local discrepancies

$$\delta_\ell = \left\|\widetilde{g}_\ell(z^{(\ell-1)}(X)) - g_\ell(z^{(\ell-1)}(X))\right\|_{L^2(P_X)}.$$

Add and subtract $\widetilde{g}_\ell(z^{(\ell-1)})$ and use the triangle inequality in L^2 to obtain

$$e_\ell \le K_\ell e_{\ell-1} + \delta_\ell, \qquad e_\ell = \|\widetilde{z}^{(\ell)} - z^{(\ell)}\|_{L^2(P_X)}, \qquad e_0 = 0.$$

Induction gives

$$\boxed{\|\widetilde{f} - \widehat{f}\|_{L^2(P_X)} \le \sum_{\ell=1}^{L} \delta_\ell \prod_{j=\ell+1}^{L} K_j,}$$

with an empty product equal to one. If f is the true target, then

$$\|\widetilde{f} - f\|_{L^2(P_X)} \le \|\widehat{f} - f\|_{L^2(P_X)} + \sum_{\ell=1}^{L} \delta_\ell \prod_{j=\ell+1}^{L} K_j.$$

This is a transfer-of-accuracy calculation under explicit population and stability assumptions. The same calculation on a finite sample controls only sample discrepancy. Generalizing the local fits, bounding the products of sensitivities, and justifying data-dependent teacher selection remain necessary. In particular, the student is not reset to the teacher state during its own rollout. The frozen teacher's accuracy is also a premise, not a consequence of its exact self-trace.

15.10 What Hopfield Theory Does Explain

The preceding distinctions should not obscure the genuine usefulness of the Hopfield analogy. The claims in this section concern the specified fixed-memory model and other models only when their own assumptions are verified.

15.10.1 Associative retrieval

Hopfield theory provides a principled account of content-addressable memory.

A partial or noisy query can retrieve a stored representation when the query geometry separates the relevant content. For example, if the scores of a read have margin $\gamma > 0$ for item j_* and stored values have norm at most R, the calculation in Chapter 9 gives

$$\|\operatorname{Attn}(q; K, V) - v_{j_*}\| \le 2R(N-1)e^{-\gamma/\tau}.$$

This is a conditional one-read bound, not a theorem that learning creates the required margin or that the output is a fixed point.

15.10.2 Competition and normalization

The softmax update describes competition among candidate memories.

This naturally connects to attention weights.

15.10.3 Attractor-like refinement

When an associative update is genuinely iterated, energy-based analysis can describe decreasing energy, stationary limits under the additional convergence assumptions, and locally attracting states. The word "metastable" does not by itself supply a stochastic escape-time law; that requires a specified noise model.

15.10.4 Robustness of retrieval

Attractor basins provide a natural language for robustness to corruption of the query when the entire perturbed trajectory stays in the relevant basin. A contractive invariant neighborhood is one useful sufficient condition. Neither a low energy nor a small update residual alone proves that the selected memory is the intended one.

15.10.5 Local computational primitive

Most importantly for this book, associative retrieval is one primitive from which more general computation can be assembled.

Chapter 3 showed how associative reads, local updates, and writable memory can support universal computation.

Thus Hopfield-like retrieval belongs naturally inside the larger computational framework.

15.11 What Hopfield Theory Does Not Explain

Several conclusions do not follow from the fixed-memory energy analysis used here. This is a limit to an inference from that theorem, not a claim that the entire Hopfield or energy-based-learning literature contains no representation or learning results.

15.11.1 It does not explain why a useful representation exists

An energy-descent theorem for a supplied memory does not prove that a chosen memory architecture economically represents the target. C1 addresses feasible computational representations; stronger conclusions about smooth low-arity constituents need the additional assumptions of Chapter 4. Neither conclusion follows solely from the existence of an associative attractor.

15.11.2 It does not explain how hidden constituents are discovered

An energy function for fixed stored patterns does not explain how SGD finds the representations that should become those patterns.

15.11.3 It does not solve the black-box recovery problem

A target can have an economical computation while certain statistics accessible to a specified learner vanish. For example, if $\chi_A(X) = \prod_{i\in A} X_i$ on the uniform Boolean cube, then $\mathbb{E}[\chi_A(X)\chi_B(X)] = 0$ whenever $B \neq A$. In particular, lower-degree Fourier probes give no signal about a parity of higher degree. This is not absence of information from the full input–output relation: noiseless parity can be recovered by linear algebra over $\mathbb{F}_2$ when the observed equations have sufficient rank. The obstruction is learner-relative, as in Chapter 6.

15.11.4 It does not establish global convergence of parameter training

Convergence of

$$\xi^{(m+1)} = H_W(\xi^{(m)})$$

for fixed W does not imply convergence of

$$W^{(t+1)} = W^{(t)} - \eta \nabla_W \mathcal{L}(W^{(t)}).$$

15.11.5 It does not establish that SGD selects a statistically controlled predictor

Hopfield state stability does not establish that a specified training procedure achieves both good empirical fit and a sufficiently small valid statistical-complexity bound. These are the predictive requirements of Chapter H; Chapters 13 and 14 discuss coding and norm-based selection mechanisms.

Once a predictor is accurate and has a feasible numerical evaluator, its own computation supplies a compositional witness. It need not reproduce a designated staircase or the teacher's latent variables. The genuine missing conclusion is successful predictive learning with controlled resources, not obligatory structural identification.

15.11.6 It does not account for the full Transformer block

The MLP, normalization, residual connections, and layer-to-layer changes in parameters lie outside the simplest associative-memory picture.

15.12 Relation to the Main Framework

The respective roles can now be summarized cleanly.

Hopfield / associative memory	:	retrieve content under a specified memory geometry,
C1	:	a compact compositional computation exists,
C2	:	specified probes expose usable learning signals,
C3	:	useful intermediate variables are supplied directly,
optimization theory	:	a specified procedure must attain a useful fit,
statistical theory	:	a valid bound controls population prediction.

These are complementary rather than competing explanations.

15.12.1 Associative memory is a mechanism

It tells us how a system can retrieve information once useful representations and queries exist.

15.12.2 Compositional sparsity is a representation principle

Under the computational and precision conventions of Chapter 4, it supplies a bounded-fan-in realization of an efficient evaluator. Favorable smooth approximation rates require the separate regularity and stability hypotheses. Dense attention is not a bounded-scalar-arity primitive merely because it is called one read.

15.12.3 Genericity is a recovery principle

It asks whether useful statistical signals are visible to the admitted learning dynamics at feasible resolution. Turning those signals into an accurate predictor requires an optimization argument; identifying a particular generating structure is an optional stronger objective.

15.12.4 Trace exposure supplies intermediate information

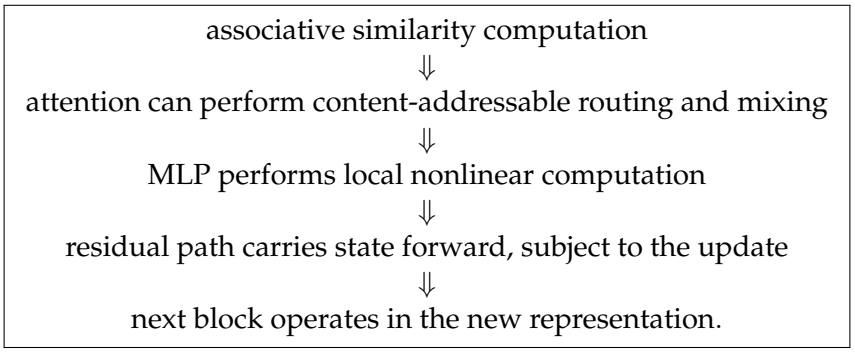
It can remove the need to infer the supplied intermediate variables, even when end-to-end learning is also possible. The local-state coverage and error-propagation requirements of Chapter 11 remain. A post-training trace of the learner itself is not evidence that such information was supplied beforehand.

15.12.5 Optimization is a selection mechanism

Together with initialization, data, regularization, and stopping, it determines the returned realization. A valid complexity bound must then apply to the selected predictor, including any data-dependent memory, feature dictionary, decoder, or compressed representation. The existence of an exact internal trace supplies no free statistical certificate.

15.13 A More Precise View of Transformer Computation

Putting Chapters 9 and the present chapter together suggests the following decomposition of a Transformer block:



This is a compositional computation, not in general an equilibrium process.

Depth permits sequential transformation and construction of representations. An untied network need not monotonically improve a single energy, while a tied equilibrium model must satisfy its own state-solver assumptions. These are different computational designs.

One important role of associative retrieval is to provide access to and mix the information needed at each stage.

15.14 A Possible Experimental Distinction

The distinction suggests experiments that measure execution, representation learning, and prediction separately. These are proposed tests, not results established by the chapter.

Suppose a task has a known generating DAG. Retain the three structural measurements motivated by the original argument:

1. whether a read retrieves the designated predecessor information, using value-level error as well as attention concentration;
2. whether a local module transforms its actual inputs into a useful next variable;
3. whether deeper representations align with the known generating variables under a prespecified, bounded-complexity and well-conditioned comparison class.

The third is a structural diagnostic, not the definition of predictive success. Record held-out prediction and computational cost independently. An accurate model with a different internal decomposition has not failed merely because its teacher-alignment score is low.

Separate three experimental interventions. First, freeze both parameters and memory entries and iterate precisely the tied update of Equation (15.1). Measure E_X, the update residual, retrieval error, and sensitivity to corrupted cues. Proposition 20 predicts descent for that model, not for a full block with arbitrary values or changing memory.

Second, at a fixed network checkpoint, compare a single read, repeated tied reads, repeated full tied blocks, and the original untied sequence. Keep the input-conditioning and memory-update rules explicit. Match or report the work budget; adding relaxation steps adds inference cost. This distinguishes a useful one-step association from a beneficial asymptotic state solver.

Third, compare end-to-end training with local supervision from a supplied reference trace and with local distillation from a separately frozen learned teacher. Include an untrained or inaccurate network as a control: it too has an exact self-trace. Changing parameters while generating the targets changes the experiment and should be reported as such.

Measure information preservation rather than assume it. Perturb or ablate candidate retrieved values and measure the effect on the next representation and final prediction. Weight concentration alone does not establish functional dependence: identical values can make a read insensitive to a large change in its weights. Repeated reads and feed-forward modules also have overlapping computational capabilities, so the proposed division of labor must be tested rather than imposed on the interpretation.

At training checkpoints, record empirical fit, held-out performance, applicable norm or coding diagnostics, and, for SGD, the variability of actual minibatch updates. A small state-update residual and a small mean parameter gradient are different observables. Use training/validation selection, an independent final evaluation, paired seeds where appropriate, and report unsuccessful runs. An inferred representation or decoder learned on the same sample is not cost-free, as discussed in Chapter 13.

The staircase picture motivates the candidate sequence

$$\boxed{\text{retrieve or mix} \longrightarrow \text{transform} \longrightarrow \text{create a useful coordinate} \longrightarrow \text{retrieve again.}}$$

It is one possible learning mechanism. The experiment asks when it occurs and helps prediction; the fixed-memory Hopfield theorem does not require that every successful Transformer follow it.

15.15 Conclusion

The Hopfield analogy captures something real and important about Transformers. In the specified tied, fixed-memory parameterization, one attention read is exactly a modern Hopfield update. The energy calculation proves descent and stationary accumulation points; a quantitative contraction condition supplies a stronger convergence statement in its stated regime.

But the implications must remain at the right level:

$$\boxed{\text{state relaxation} \neq \text{network depth} \neq \text{parameter learning.}}$$

General key–value maps, state-dependent memories, residual updates, and untied layers do not inherit that same fixed-memory energy argument without further analysis. Genuine equilibrium architectures are possible, but their solver and learning guarantees must be established separately.

Fixed-point sets do not compose in general. Two pairs of maps can have identical constituent fixed-point sets but different composite fixed points. Even a common globally attracting fixed point of each map can be unstable for their composition. Common contractions provide positive results, while noncommuting contractions can converge to a phase-dependent cycle rather than to a common constituent fixed point. For layers acting between different spaces, fixed-point language needs an explicit identification before it is defined at all.

A fixed deterministic deep factorization does have a useful retrospective property. Its intermediate states induce datasets

$$\mathcal{D}_\ell = \left\{ \left(z_i^{(\ell-1)}, z_i^{(\ell)} \right) \right\}_{i=1}^{n}, \qquad z_i^{(\ell)} = g_\ell(z_i^{(\ell-1)}).$$

These are exact traces of that evaluator, regardless of whether it was trained, interpolates, generalizes, or is a parameter equilibrium. The final trace value is $\widehat{f}(x_i)$, not automatically y_i.

The trace can support local supervision for another learner when it is frozen and supplied. Successful recomposition additionally requires coverage or local generalization and controlled accumulation of errors. It is not an explanation of the training process that originally found the representation, and it need not identify the target's generating computation.

The constructive learning question remains:

When useful coordinates are not supplied, what information and dynamics allow a feasible learner to construct them?

C3 supplies intermediate information. Quantitative C2 and staircase results may help establish a route to learning it. Neither is a new requirement imposed after a predictor has already been shown to achieve the accuracy, statistical control, and evaluation-resource conditions of Chapter H.

The resulting division of labor is

associative state theory: a specified retrieval mechanism;
C1: feasible computational representation;
C2/C3: accessible or supplied intermediate information;
optimization: a procedure that returns a sufficiently good fit;
statistical theory: control of the returned predictor's population error.

The arrows between these roles require arguments, not analogies. A returned predictor can be accurate while regularized SGD continues to fluctuate, and a perfectly stable associative state can be irrelevant to the target task.

The correct lesson is not to discard the Hopfield analogy. It is to retain its positive results without treating an attractor theory of activations as, by itself, a learning theory of weights. The next chapter turns directly to that missing learning mechanism: how local plastic circuits might implement a useful parameter-space update.

CHAPTER 16

A Self-Assembling Cortical Circuit for Generalized Gradient Descent (with Qianli Liao and Liu Ziyin)

The previous chapter separated activation dynamics from parameter-learning dynamics. Associative-memory mechanisms explain how a fixed circuit can retrieve or transform a representation; a learning theory must also explain how the synapses implementing that representation change.

Self-Assembling Learning (SAL) proposes two interacting plastic pathways: an ascending representation stream and a descending feedback stream, linked by local cross-connections. The desired outcome is functional gradient alignment, not a copy of every forward weight in the feedback pathway. The circuit proposal and its simulation evidence motivate this possibility [92].

The mathematical claim must nevertheless identify which update is being analyzed. Small auxiliary increments, appropriate matrix dimensions, correct derivative gates, and useful conditioning are different requirements. This chapter supplies a self-contained conditional statement: when the feedback stream approximately transports an encoded backpropagation error, and the cross-stream encoders and decoders approximately agree by transpose, the feedforward update is a positive-semidefinite preconditioning of the true gradient, with an explicit error bound. Full row rank makes that preconditioner positive definite.

$$\boxed{\begin{array}{c}\textit{controlled feedback transport and cross-stream alignment}\\ \Longrightarrow\ \textit{a controlled preconditioned-gradient update.}\end{array}}$$

The transport and alignment conditions are the part to be established by a self-assembly analysis; they are not assumed to follow from random initialization or small raw steps alone. A simple reciprocal-coupling calculation proves one component of alignment. Separate estimates explain when the resulting update decreases a specified objective, how approximation and stochasticity affect that conclusion, and why weight decay needs its own interpretation.

SAL therefore addresses a concrete implementation question: how could local plastic circuitry carry a useful teaching signal into parameter space? It does not by itself establish that the target is statistically accessible, that optimization reaches a global fit, or that the returned predictor generalizes. Nor does it establish that cortex implements this particular algorithm. Those distinctions preserve both the positive proposal and its place in the book.

16.1 From Chapter 15: The Missing Learning Dynamics

Chapter 15 distinguished

$$\begin{aligned} &\text{state dynamics:} && \xi^{(m+1)} = H_W(\xi^{(m)}),\\ &\text{depth dynamics:} && x_{\ell+1} = F_\ell(x_\ell),\\ &\text{learning dynamics:} && W^{(t+1)} = W^{(t)} - \eta\nabla_W \mathcal{L}(W^{(t)}). \end{aligned}$$

This chapter concerns the third. Stabilizing activity is not the same as changing synapses in a direction that improves a task objective.

16.1.1 What backpropagation accomplishes

Consider a bias-free network, with column-vector activations,

$$h_{\ell+1} = \phi_\ell(a_{\ell+1}), \qquad a_{\ell+1} = W_\ell h_\ell, \qquad \ell = 0, \dots, L-1,$$

and a differentiable sample loss $F(h_L, y)$. Write

$$D_j^u = \operatorname{diag}[\phi'_{j-1}(a_j)], \qquad j = 1, \dots, L.$$

For a nonsmooth activation, the following classical derivative identities apply where the relevant derivatives exist. A specified gate convention at ReLU zeroes is an implementation choice; it is not by itself a smooth-flow theorem across those boundaries.

It is important to distinguish the negative activation adjoint p_j from the negative preactivation adjoint δ_j:

$$p_j = -\nabla_{h_j} F, \qquad \delta_j = -\nabla_{a_j} F = D_j^u p_j.$$

Backpropagation then gives

$$p_L = -\nabla_{h_L} F, \qquad \delta_L = D_L^u p_L, \qquad \delta_j = D_j^u W_j^\top \delta_{j+1} \quad (1 \le j < L),$$

and

$$\boxed{\nabla_{W_\ell} F = -\delta_{\ell+1} h_\ell^\top, \qquad \Delta W_\ell^{\mathrm{BP}} = \eta \delta_{\ell+1} h_\ell^\top.}$$

These equations fix the gate placement. Initializing δ_L with $-\nabla_{h_L} F$ without an output gate is valid when the output is linear; otherwise that initialization mixes two different adjoints.

Every synapse requires a presynaptic activity and a postsynaptic error-like factor. The biological question is how the second factor could become locally available without literal transport of the forward weights. For later population or minibatch arguments, distinguish the sample loss F from the empirical objective

$$\mathcal{L}(W) = \frac{1}{n} \sum_{i=1}^{n} F(F_W(x_i), y_i).$$

An identity for the update on one example need not imply the same identity for the gradient of $\mathcal{L}$ after averaging.

16.2 The Biological Credit-Assignment Problem

Several distinct problems are often grouped under the phrase "biological implausibility of backpropagation."

16.2.1 Weight transport

Exact backpropagation uses

$$W_\ell^\top$$

in the backward pathway.

A literal implementation would require the feedback circuit to know the forward synaptic matrix with high precision.

16.2.2 Error transport

The network must propagate an appropriate derivative of the task loss through every layer.

16.2.3 Synaptic locality

A biological synapse cannot normally access arbitrary global variables.

In the proposed local implementation, plasticity is driven by signals available to its presynaptic and postsynaptic compartments, perhaps augmented by nearby dendritic, modulatory, or heterosynaptic signals.

16.2.4 The top-level teaching signal remains necessary

A biologically local learning system does not eliminate the need for a task signal.

SAL begins the descending computation from an output error of the form

$$\bar{h}_L = -\nabla_{h_L} F.$$

Here and in the conditional circuit below, the output is linear and $\bar{d}_L = d_L$, so this injected signal also equals δ_L. A nonlinear output requires the corresponding derivative factor, and an unequal-dimensional output feedback space requires an explicit embedding.

Thus SAL does *not* require that every layer receive global information, but it does require some mechanism that generates a top-level teaching or mismatch signal.

The correct statement is therefore

SAL localizes credit propagation after a top-level teaching signal has been generated.

The biological source of that top-level signal is a separate question.

Depending on the task, possible implementations could involve reward, neuromodulation, mismatch circuitry, frontal or thalamic signals, or other mechanisms. None of these possibilities is required by the SAL mathematics itself.

16.3 The SAL Motif

SAL introduces two interacting neural streams. We index cross-stream matrices by the representation level they connect; this removes an otherwise ambiguous shift between a synaptic edge and its receiving layer.

16.3.1 Ascending stream

The ascending stream carries task representations

$$h_0 \longrightarrow h_1 \longrightarrow \cdots \longrightarrow h_L, \qquad h_j \in \mathbb{R}^{d_j},$$

with $W_\ell \in \mathbb{R}^{d_{\ell+1} \times d_\ell}$.

16.3.2 Descending stream

The descending stream carries

$$\bar{h}_L \longrightarrow \bar{h}_{L-1} \longrightarrow \cdots \longrightarrow \bar{h}_0, \qquad \bar{h}_j \in \mathbb{R}^{\bar{d}_j},$$

with $\bar{W}_\ell \in \mathbb{R}^{\bar{d}_\ell \times \bar{d}_{\ell+1}}$. Hidden feedback widths may differ from ascending widths. In an expressive feedback regime, $\bar{d}_j \geq d_j$. This inequality permits, but does not establish, a full-rank encoding of the ascending error space. The output channel is taken to be an identity channel: $\bar{d}_L = d_L$ and $\bar{h}_L = \delta_L$.

16.3.3 Cross-stream couplings

We use the following dimensional conventions:

$$\begin{aligned} V_j &\in \mathbb{R}^{\bar{d}_j \times d_j} \quad \text{ascending to descending,} \\ \bar{V}_j &\in \mathbb{R}^{d_j \times \bar{d}_j} \quad \text{descending to ascending.} \end{aligned}$$

Thus V_j can encode an ascending error coordinate in the feedback space, and $\bar{V}_j$ can decode a feedback signal. In the conditional theorem, $V_L = \bar{V}_L = I$ specifies the output boundary. The hidden cross-stream matrices are learned; no numerical equality between them is imposed at initialization.

The four connection classes remain

$$\boxed{W_\ell, \quad \bar{W}_\ell, \quad V_j, \quad \bar{V}_j.}$$

Their initial values can be random within the prescribed motif. Random values in a supplied two-stream architecture should not be identified with emergence of that architecture from an arbitrary graph. The original SAL proposal studies self-organization within a structured local-learning framework [92].

Source and implementation conventions. The 2026 published account distinguishes its earlier benchmark implementation from a later implementation used for theory and alignment diagnostics [91]. The discussion here retains the chapter's SAL terminology. The explicit sufficient theorem below specifies its own signal, gate, and indexing conventions; it is not an assertion that all implementations are identical. In particular, a circuit variant with extra gates in its plasticity update must be analyzed with those gates in place.

16.4 Forward and Feedback Computation

Neural activations, multiplicative activation gates, and derivatives of activations are different objects. Keeping them distinct is essential to the gradient comparison.

16.4.1 Ascending computation

The forward pass is

$$a_{\ell+1} = W_\ell h_\ell, \qquad h_{\ell+1} = \phi_\ell(a_{\ell+1}), \qquad D^u_{\ell+1} = \operatorname{diag}[\phi'_\ell(a_{\ell+1})].$$

In a ReLU hidden layer, away from zero preactivations, $h_{\ell+1} = D^u_{\ell+1} a_{\ell+1}$. This special identity does not hold for a general smooth nonlinearity: for example, $\tanh a$ is not $a \tanh'(a)$. A forward multiplication gate cannot automatically be reused as its derivative in a theorem about backpropagation.

16.4.2 Error injection

With a linear output, the descending input is

$$\bar{h}_L = -\nabla_{h_L} F(h_L, y) = \delta_L.$$

This signal is supplied by an output mismatch mechanism. Its availability is an assumption, not a consequence of the hidden synaptic learning rule. Forward activities needed for learning are held or otherwise made available during the feedback pass.

16.4.3 Descending computation

Retain the chapter's general description

$$\bar{a}_j = \bar{W}_j \bar{h}_{j+1}, \qquad \bar{h}_j = \psi_j(\bar{a}_j).$$

If ψ_j is coordinatewise with $\psi_j(0) = 0$, its evaluated output can be written

$$\bar{h}_j = M^d_j \bar{W}_j \bar{h}_{j+1} = B_j \bar{h}_{j+1}.$$

Here M^d_j is an *activation multiplier*: on a nonzero coordinate a, its entry is $\psi_j(a)/a$; at zero choose a finite value consistent with the zero output. For differentiable ψ_j, one may use $\psi'_j(0)$ at zero. In contrast,

$$D^d_j = \operatorname{diag}[\psi'_j(\bar{a}_j)]$$

is a derivative gate and need not equal M^d_j.

The simplest theorem-compatible choice is linear feedback, $M^d_j = I$. For a nonlinear feedback pass, B_j can depend on the example and current activities; all transport identities below are then pointwise identities for that pass. Additive offsets, normalization, or other feedback operations require their own residual accounting.

No equality $\bar{W}_j = W_j^\top$ is assumed. The relevant condition will instead compare the effective feedback operator B_j to the gated backpropagation operator after encoding both error spaces.

16.5 Local Heterosynaptic Plasticity

The supplied SAL proposal expresses synaptic updates through local activity products and decay. This is a locality statement; a complete algorithm must additionally specify the factors, their timing, gains, gates, and simultaneous or sequential update schedule.

16.5.1 Feedforward synapses

Use a step size $\eta > 0$ and a decay *rate* $\lambda_W \geq 0$:

$$\Delta W_\ell = \eta(e_{\ell+1}h_\ell^\top - \lambda_W W_\ell).$$

In the explicitly analyzed convention,

$$e_j = \bar{V}_j \bar{h}_j.$$

The feedback circuit must therefore encode the *preactivation* adjoint δ_j, including the correct forward gate, rather than an ungated activation derivative. Section 16.6 states this condition precisely. It is a sufficient mathematical realization of the chapter's error-like factor, not a derivation of that factor from arbitrary local plasticity.

If an implementation instead uses $e_j = D_j^u \bar{V}_j \bar{h}_j$, its transport comparison must be changed accordingly. In particular, $D_j^u H p_j$ and $H D_j^u p_j$ are generally different: gate matrices and Gram matrices need not commute. Positive definiteness cannot justify reordering them.

The source's notation $-\gamma W_\ell$ can be recovered by setting $\gamma = \eta\lambda_W$. A continuous-time decay rate is not obtained by holding the per-step decrement γ fixed while taking $\eta \to 0$.

16.5.2 Descending synapses

The descending update retains the local outer-product form

$$\Delta \bar{W}_\ell = \eta_d(\bar{e}_\ell \bar{h}_{\ell+1}^\top - \lambda_d \bar{W}_\ell), \qquad \bar{e}_\ell \in \mathbb{R}^{\bar{d}_\ell}.$$

The local rule constructing $\bar{e}_\ell$ must be specified by the chosen implementation. Writing an outer product does not prove that it drives B_ℓ toward the transport relation required below. We do not insert an unprovided formula and call it the original algorithm.

16.5.3 Cross-stream plasticity

In the present same-level indexing, a matched-gain correlation rule has the dimensionally consistent continuous-time form

$$\dot{V}_j = \kappa C_j - \lambda_c V_j, \qquad \dot{\bar{V}}_j = \kappa C_j^\top - \lambda_c \bar{V}_j, \qquad C_j = \bar{h}_j h_j^\top.$$

The same algebra applies to a shared averaged local correlation C_j. The transposition in the second equation is essential.

Proposition 23 (Reciprocal alignment under matched local drives)**.** *Suppose the two cross-stream matrices receive transpose-matched drives with the same gain κ and decay rate $\lambda_c > 0$. Then*

$$V_j(t) - \bar{V}_j(t)^\top = e^{-\lambda_c t}(V_j(0) - \bar{V}_j(0)^\top).$$

This identity holds even when the common drive depends on the evolving network and data.

Proof. Subtract the transpose of the second differential equation from the first. The common drive cancels, leaving $\frac{d}{dt}(V_j - \bar{V}_j^\top) = -\lambda_c(V_j - \bar{V}_j^\top)$. Solving this linear equation proves the claim. □

For simultaneous Euler updates, the corresponding difference is multiplied by $1 - \eta_c\lambda_c$ at each step, so its magnitude decays when $0 < \eta_c\lambda_c < 2$. Unequal learning gains, unequal decay rates, or different activity samples produce additional forcing terms.

This is a genuine, but limited, self-organization result. It does not prove rank preservation, good conditioning, or correct transport through the within-stream feedback matrices. It is also reciprocal alignment of *cross-stream* matrices, not equality between $W_\ell^\top$ and $\bar{W}_\ell$.

The biological interpretation remains heterosynaptic: the factors may be available in nearby compartments rather than solely across the synapse being modified. Their physical availability and timing require evidence; mathematical locality does not establish that evidence.

16.6 The Main Conditional Result

The quasi-stationary interpretation in the SAL proposal motivates the following precise sufficient statement. The statement below is an explicit completion of the chapter's conditional argument, not a claim that the schematic rules already prove all its hypotheses. In particular, it replaces an unspecified $O(\varepsilon)$ update error by identifiable transport and reciprocity defects.

Fix an example, a forward pass at a differentiable parameter, and a feedback pass. For $1 \le j < L$, define

$$A_j = B_j V_{j+1} - V_j D_j^u W_j^\top, \qquad C_j^{\mathrm{rec}} = V_j - \bar{V}_j^\top.$$

Here $A_j \in \mathbb{R}^{\bar{d}_j \times d_{j+1}}$ measures failure of the diagram that encodes the true error before or after one backward step to agree. It is an operator relation, not just a correlation observed on one training vector. At the output, take $V_L = \bar{V}_L = I$ and $\bar{h}_L = \delta_L$, so $C_L^{\mathrm{rec}} = 0$.

Theorem 15 (SAL implements a preconditioned gradient update under controlled transport)**.** *Use the forward, feedback, boundary, and decoding conventions just stated. Define*

$$r_j = \bar{h}_j - V_j \delta_j, \qquad r_L = 0, \qquad H_\ell = \bar{V}_{\ell+1} \bar{V}_{\ell+1}^\top.$$

Then

$$r_j = B_j r_{j+1} + A_j \delta_{j+1}, \qquad 1 \le j < L,$$

and the feedforward update is exactly

$$\boxed{\Delta W_\ell^{\mathrm{SAL}} = H_\ell \Delta W_\ell^{\mathrm{BP}} - \eta \lambda_W W_\ell + \eta \mathcal{E}_\ell,}$$

where

$$\mathcal{E}_\ell = \bar{V}_{\ell+1}\big(r_{\ell+1} + C_{\ell+1}^{\mathrm{rec}} \delta_{\ell+1}\big) h_\ell^\top.$$

In particular, using Euclidean and induced operator norms,

$$\|\mathcal{E}_\ell\|_F \le \|\bar{V}_{\ell+1}\|_2 \|h_\ell\|_2 \left(\|r_{\ell+1}\|_2 + \|C_{\ell+1}^{\mathrm{rec}}\|_2 \|\delta_{\ell+1}\|_2\right),$$

and

$$\|r_j\|_2 \le \sum_{q=j}^{L-1} \left(\prod_{k=j}^{q-1} \|B_k\|_2 \right) \|A_q\|_2 \|\delta_{q+1}\|_2.$$

Empty products equal one and the bound for r_L is zero. Every H_ℓ is positive semidefinite; it is positive definite if $\bar{V}_{\ell+1}$ has full row rank. When all A_j and C_j^{rec} vanish, the update error $\mathcal{E}_\ell$ vanishes.

Proof. Since $\delta_j = D_j^u W_j^\top \delta_{j+1}$ and $\bar{h}_j = B_j \bar{h}_{j+1}$,

$$\begin{aligned} r_j &= B_j(r_{j+1} + V_{j+1}\delta_{j+1}) - V_j D_j^u W_j^\top \delta_{j+1} \\ &= B_j r_{j+1} + A_j \delta_{j+1}. \end{aligned}$$

Iterate from $r_L = 0$ and use submultiplicativity to obtain the stated bound. Also

$$\begin{aligned} e_{\ell+1} &= \bar{V}_{\ell+1}(V_{\ell+1}\delta_{\ell+1} + r_{\ell+1}) \\ &= H_\ell \delta_{\ell+1} + \bar{V}_{\ell+1}(C^{\mathrm{rec}}_{\ell+1}\delta_{\ell+1} + r_{\ell+1}). \end{aligned}$$

Multiplying by $\eta h_\ell^\top$ and adding decay proves the update identity. The outer-product norm identity proves the remainder bound. Finally, for every $u \in \mathbb{R}^{d_{\ell+1}}$,

$$u^\top H_\ell u = \|\bar{V}_{\ell+1}^\top u\|_2^2.$$

This is nonnegative, and is positive for every nonzero u precisely when $\bar{V}_{\ell+1}$ has full row rank. □

What this adds to the quasi-stationary description. In the same-level convention the preconditioner for W_ℓ uses the cross-stream decoder at its *output*, level $\ell + 1$. Small update error only when the displayed activation, adjoint, and transport-product factors are controlled. These constants can grow with depth. Full rank without a quantitative lower singular-value bound does not supply a uniform descent constant.

Deriving small A_j from a particular auxiliary learning rule remains a separate step. Small ΔV_j can arise merely from a small learning rate; small mean updates can arise by cancellation across examples. Neither observation alone establishes the pointwise operator relations in this theorem. Likewise, matching $A_j v$ on a few observed directions need not control $\|A_j\|_2$ without coverage or excitation assumptions.

The exact theorem describes an error-transport mechanism and its consequences. It does not require the learner's representations to equal hidden teacher coordinates, and does not turn signal transport into a proof of predictive recovery.

16.7 Why Positive-Definite Preconditioning Is Enough

The positive conclusion is first-order descent for a *specified* objective under specified update assumptions, not unrestricted convergence of a coupled stochastic circuit. Let $\mathcal{L}$ be a differentiable objective depending on the feedforward weights. In the exact, unregularized continuous-time case,

$$\dot{W}_\ell = -H_\ell \nabla_{W_\ell} \mathcal{L}.$$

For symmetric $H_\ell \succeq 0$ along the trajectory, the chain rule gives

$$\boxed{\frac{d\mathcal{L}}{dt} = -\sum_\ell \mathrm{Tr}\left[(\nabla_{W_\ell}\mathcal{L})^\top H_\ell \nabla_{W_\ell}\mathcal{L}\right] \le 0.}$$

No derivative of H_ℓ appears here: one differentiates $\mathcal{L}$, not a quantity that contains H_ℓ. With positive-definite blocks, a zero *feedforward* velocity is equivalent to a zero feedforward gradient. This is not equivalence of equilibria of the entire augmented system, whose auxiliary weights can still move. Semidefinite blocks can annihilate a nonzero gradient.

Approximation error and a finite step. Vectorize the weights and write

$$g = \nabla\mathcal{L}(\theta), \qquad \mathcal{P} = \mathrm{blockdiag}_\ell(I_{d_\ell} \otimes H_\ell), \qquad \dot{\theta} = -\mathcal{P}g + e.$$

If $\mathcal{P} \succeq \mu I$ and $\|e\| \le \epsilon$, then

$$\dot{\mathcal{L}} \le -\mu\|g\|^2 + \epsilon\|g\|.$$

Thus an absolute transport-error bound certifies descent outside $\|g\| \le \epsilon/\mu$, not arbitrarily close to a critical point.

Proposition 24 (A quantitative local descent condition)**.** *Suppose $\mu I \preceq \mathcal{P} \preceq MI$, $0 \le a < \mu$, and $\|e\| \le a\|g\|$. If $\nabla\mathcal{L}$ is β-Lipschitz, with $\beta > 0$, on a region containing the update segment, then*

$$\theta^+ = \theta - \eta(\mathcal{P}g - e), \qquad 0 < \eta \le \frac{\mu - a}{\beta(M + a)^2}$$

satisfies

$$\mathcal{L}(\theta^+) \le \mathcal{L}(\theta) - \frac{\eta}{2}(\mu - a)\|g\|^2.$$

Proof. The descent lemma gives

$$\mathcal{L}(\theta^+) - \mathcal{L}(\theta) \le -\eta\langle g, \mathcal{P}g - e\rangle + \frac{\beta\eta^2}{2}\|\mathcal{P}g - e\|^2.$$

The first inner product is at least $(\mu - a)\|g\|^2$, and the squared norm is at most $(M + a)^2\|g\|^2$. Substitution proves the bound. □

Positive at every time is not uniformly effective. For $\mathcal{L}(w) = w^2/2$ and $H(t) = e^{-t} > 0$,

$$\dot{w} = -e^{-t}w, \qquad w(t) = w(0)\exp\Big[-(1 - e^{-t})\Big].$$

For $w(0) \neq 0$ the limit $e^{-1}w(0)$ is not critical. Pointwise positive definiteness therefore does not by itself prove convergence to a critical point. Adequate accumulated descent, confinement, regularity, and control of errors remain necessary in an actual theorem. It also does not remove spurious local minima of the objective.

16.7.1 With weight decay

For the decay convention in this chapter,

$$\dot{W}_\ell = -H_\ell\nabla_{W_\ell}\mathcal{L} - \lambda_W W_\ell.$$

Its stationary equation is $H_\ell\nabla_{W_\ell}\mathcal{L} + \lambda_W W_\ell = 0$. This is generally neither unregularized stationarity nor stationarity of $\mathcal{L} + \frac{\lambda_W}{2}\sum_\ell \|W_\ell\|_F^2$.

If all $H_\ell \succ 0$ are *constant and independent of the weights*, define

$$U_H(W) = \mathcal{L}(W) + \frac{\lambda_W}{2}\sum_\ell \mathrm{Tr}(W_\ell^\top H_\ell^{-1} W_\ell).$$

Then $\dot{W}_\ell = -H_\ell\nabla_{W_\ell}U_H$, and U_H decreases. The effective quadratic regularizer is anisotropic. By contrast, preconditioning the usual Euclidean quadratic penalty requires

$$\dot{W}_\ell = -H_\ell(\nabla_{W_\ell}\mathcal{L} + \lambda_W W_\ell).$$

These two updates should not be conflated.

When the auxiliary circuit learns $H_\ell(t)$, the instantaneous U_H is not automatically a fixed Lyapunov function. Along a differentiable path, even treating H_ℓ as an independent auxiliary variable yields

$$\frac{dU_H}{dt} = -\sum_\ell \|H_\ell^{1/2}\nabla_{W_\ell}U_H\|_F^2 - \frac{\lambda_W}{2}\sum_\ell \mathrm{Tr}(W_\ell^\top H_\ell^{-1}\dot{H}_\ell H_\ell^{-1} W_\ell).$$

The second term has no fixed sign. If H is instead defined as a function of W, differentiating a proposed penalty must also include that dependence. This is why ordinary decay in SAL is not automatically gradient descent on a familiar fixed regularized objective.

The stochastic qualification. The preceding identities concern an objective-gradient field. Suppose, conditionally on past history, a minibatch gradient $\widehat{g}$ is unbiased for g and the preconditioner $\mathcal{P}$ is fixed before sampling that batch. Then $\mathbb{E}[\mathcal{P}\widehat{g} \mid \text{past}] = \mathcal{P}g$. If $\mathcal{P}$ depends on the sampled example or batch, this factorization need not hold. The Gram matrices in Theorem 15 depend only on the current cross-stream weights: when those weights are frozen before sampling, the leading Gram factor can be pulled outside the expectation, while the sample-dependent transport remainders must still be controlled. Other gated implementations can produce a genuinely sample-dependent effective preconditioner and require a different mean update argument.

For an elementary counterexample, take two equally likely losses

$$\ell_1(w) = \tfrac{1}{2}(w+2)^2, \qquad \ell_2(w) = \tfrac{1}{2}(w-1)^2,$$

and sample-dependent positive scalar preconditioners $H_1 = 1$, $H_2 = 3$. At $w = 0$,

$$\mathcal{L}'(0) = \tfrac{1}{2}, \qquad -\mathbb{E}[H_i \ell_i'(0)] = \tfrac{1}{2}.$$

The expected direction is uphill for $\mathcal{L}$, although each sample-conditioned direction is downhill for its own loss. Positivity of every samplewise matrix is insufficient for expected descent of the averaged objective.

With a common past-measurable $\mathcal{P}$, noise covariance C is transformed to $\mathcal{P}C\mathcal{P}^\top$; update covariance also contains η^2. As Chapters 7 and 14 emphasize, regularized mean stationarity does not require every minibatch update to vanish. When the actual SAL update contains batch-dependent feedback or gates, its own mean and covariance must be analyzed. It is not thereby an isotropic Langevin process.

16.8 SAL as a Learned Metric on Parameter Space

The matrix H_ℓ is not merely an error relative to vanilla gradient descent. It can change the geometry of the learning direction. For positive-definite blocks, the corresponding metric on matrix perturbations is

$$\langle U, V \rangle_{H_\ell^{-1}} = \operatorname{Tr}(U^\top H_\ell^{-1} V).$$

The steepest-descent direction in this metric is $-H_\ell \nabla_{W_\ell} \mathcal{L}$. Thus H_ℓ is the *inverse metric* or mobility, not the metric tensor itself. For a singular block, this is only a semidefinite preconditioning interpretation unless a restricted tangent space or pseudoinverse geometry is explicitly supplied.

SAL may therefore learn both a predictor and part of its optimization geometry. A better-conditioned matrix does not automatically accelerate a particular objective: its scale and orientation relative to that objective's curvature matter.

16.8.1 Conditioning matters

For $H_\ell = \bar{V}_{\ell+1}\bar{V}_{\ell+1}^\top$ with full row rank,

$$\lambda_{\min}(H_\ell) = \sigma_{\min}(\bar{V}_{\ell+1})^2, \qquad \lambda_{\max}(H_\ell) = \sigma_{\max}(\bar{V}_{\ell+1})^2.$$

The condition number $\kappa(H_\ell) = \lambda_{\max}/\lambda_{\min}$ measures relative scaling but misses overall magnitude. The example $H(t) = e^{-t}I$ has condition number one and can still stall before reaching a critical point. An efficient-learning claim needs quantitative control of the useful eigenvalues, errors, and learning rate, not merely rank.

For a common $\mu I \preceq \mathcal{P} \preceq MI$ and nonzero g,

$$\frac{\langle \mathcal{P}g, g\rangle}{\|\mathcal{P}g\|\|g\|} \geq \frac{\mu}{M} > 0.$$

This is a useful sufficient angle bound, not a prediction that measured angles must increase monotonically in training. Small numerical rank of a learned feedback Gram matrix can coexist with useful transport on a restricted task-gradient subspace, but then a global full-rank claim must be replaced by a bound on that subspace.

Connection to prediction-space conditioning. For the unregularized empirical squared loss of Chapter 7, let $e = F(\theta) - y$ and $J = D_\theta F$. Under an exact common preconditioned-gradient flow,

$$\dot{e} = -K_{\mathcal{P}} e, \qquad K_{\mathcal{P}} = \frac{1}{n} J\mathcal{P}J^\top.$$

If $e^\top K_{\mathcal{P}} e \geq \nu\|e\|^2$ along the trajectory, then $\mathcal{L}(t) \leq \mathcal{L}(0)e^{-2\nu t}$ by direct differentiation. Establishing this condition remains separate from establishing that the circuit produces a positive matrix. With update error r and ordinary decay, the residual equation instead contains $Jr - \lambda_W J\theta$. No unregularized contraction estimate can be applied without those terms.

16.9 What the Theorem Does—and Does Not—Prove

The chapter distinguishes an established algebraic implication from the hypotheses that make it relevant to a trained circuit.

16.9.1 What the theorem proves conditionally

Theorem 15 proves that correctly encoded backward transport and reciprocal cross-stream alignment produce a Gram-matrix preconditioned feedforward gradient. It supplies explicit error factors and does not replace them by an unspecified dimension-independent $O(\varepsilon)$. Proposition 23 proves reciprocal alignment for one matched-drive rule; it does not establish the transport condition. Proposition 24 supplies a separate finite-step descent condition once the objective-gradient interpretation is valid.

16.9.2 What it does not automatically prove

These results do not establish that random initialization reaches a well-conditioned transport regime, that it remains there, that the full coupled dynamics converge, or that a useful predictor is obtained within polynomial resources. They also do not establish a biological source of the output teaching signal.

Which notion of auxiliary stationarity? For an update $\Delta V = \eta_V \mathcal{U}_V$, measuring only $\|\Delta V\|$ confounds stationarity of the update field with the chosen learning rate. Likewise, a small expected update can coexist with large samplewise fluctuations. Those alternatives are particularly important when invoking a pointwise transport theorem.

There is an elementary rank restriction for the illustrative correlation rule in the preceding local-plasticity section. To avoid any ambiguity, we record the rule and its consequence directly below.

Proposition 25 (Small batch-correlation residuals constrain singular values)**.** *Let* $V \in \mathbb{R}^{m\times d}$, $\lambda_c > 0$, *and*

$$C_{\mathcal{B}} = \frac{1}{B}\sum_{i\in\mathcal{B}} \bar{h}_i h_i^\top, \qquad \|C_{\mathcal{B}} - \lambda_c V\|_2 \le \epsilon.$$

If $B < \min(m, d)$*, then* $\lambda_c \sigma_{B+1}(V) \le \epsilon$*. In particular, exact stationarity of this batch-correlation field forces* $\operatorname{rank} V \le B$*.*

Proof. $C_{\mathcal{B}}$ has rank at most B. On the span of the first $B+1$ right singular vectors of V, it has a nonzero null vector u of unit norm. Thus $\epsilon \ge \|(\lambda_c V - C_{\mathcal{B}})u\| = \lambda_c\|Vu\| \ge \lambda_c\sigma_{B+1}(V)$. The exact-rank conclusion follows by setting $\epsilon = 0$; when the matrix has fewer than $B+1$ singular values, rank at most B is automatic. □

For a singleton drive, exact stationarity cannot yield a uniformly well-conditioned rank-greater-than-one cross-stream matrix. An average over diverse examples can have higher rank. This observation concerns the specified correlation rule, not every SAL variant, and is not an obstruction to useful ongoing stochastic learning. It shows why "small updates and full rank" must be unpacked before taking a limiting argument.

The two central tasks are therefore

conditional analysis:	transport/alignment imply a useful update,
self-assembly analysis:	the actual local rules reach and sustain it.

The second includes gate compatibility, excitation, rank, conditioning, noise, and timescale control. It is not supplied by the first.

16.10 Self-Assembly from Random Initialization

The motivation for SAL is precisely that the appropriate feedback structure need not be genetically specified synapse by synapse.

In the proposed regime, auxiliary pathways begin with random weights within the supplied motif. Simulation evidence for self-organization is reported in the SAL work [92]; no new simulations are reported in this chapter.

During training, their local plasticity changes the effective transformation between the top-level teaching signal and the local feedforward updates.

16.10.1 Functional rather than literal alignment

The important goal is not

$$\bar{W}_\ell \to W_\ell^\top.$$

What matters is

$$\Delta W_\ell^{\text{data}}$$

becoming aligned with a descent direction for the true objective.

A natural measure is

$$\cos\theta_\ell = \frac{\left\langle \Delta W_\ell^{\text{data}}, -\nabla_{W_\ell} F \right\rangle}{\|\Delta W_\ell^{\text{data}}\|_F \, \|\nabla_{W_\ell} F\|_F}.$$

Here $\Delta W_\ell^{\mathrm{data}}$ denotes the data-driven part, with ordinary decay subtracted from the full update. The angle is defined only when both factors in its denominator are nonzero. For a minibatch test, the reference gradient must use the same examples and loss convention. A separate diagnostic compares the expected update with the gradient of the averaged objective.

A useful layerwise diagnostic is

$$\cos\theta_\ell > 0$$

on relevant nonzero gradients. An unweighted average of positive or improving cosine scores does not certify loss decrease for the full network: layerwise gradient magnitudes, step sizes, errors, and decay also enter. Under the explicit transport hypotheses, the theorem instead provides the stronger Gram-matrix relation for that update.

16.10.2 Why overparameterized feedback may help

If the descending representation has dimension

$$\bar{d}_\ell > d_\ell,$$

there are more degrees of freedom available for constructing a full-rank mapping onto the ascending error space.

This motivates a width ablation; the original SAL simulations report such an effect in their tested settings [92]. It is not a monotonic-width theorem.

The relevant theoretical quantity, however, is not width by itself.

It is whether width helps produce a well-conditioned

$$H_\ell.$$

16.11 Relation to Feedback Alignment

SAL is related to feedback-alignment approaches but differs in an important respect.

In basic feedback alignment [93], the backward pathway can remain random and fixed while forward weights gradually adapt so that the random feedback direction becomes useful.

SAL instead makes the feedback machinery itself plastic.

Thus

feedback alignment:	forward system adapts around a fixed approximate feedback path,
SAL:	forward and feedback systems co-adapt through local plasticity.

The SAL theorem therefore concerns *functional gradient alignment*, not the emergence of literal forward–backward weight symmetry.

The comparison concerns the simplest fixed-feedback baseline. Other local-learning proposals can also make feedback plastic; SAL is not the only possible route to co-adaptation.

16.12 Biological Plausibility at Three Levels

The phrase "biologically plausible" should be decomposed into three different claims.

16.12.1 Level I: Mathematical locality

The first question is purely computational:

> Can each synaptic update be expressed using quantities locally available to the relevant neurons or dendritic compartments?

SAL is designed around this condition. The actual factors, their signs, derivative gates, and timing still have to be assigned to accessible compartments. The displayed mathematical locality is a model property, not an experimentally established identity for cortical plasticity.

16.12.2 Level II: Candidate biological mechanisms

A second question is whether known biology contains mechanisms capable of supplying the required local factors.

Possible ingredients include:

- heterosynaptic plasticity;
- dendritic compartmentalization;
- calcium-dependent interactions among nearby synapses;
- synaptic tagging;
- retrograde or modulatory signaling;
- reciprocal cortical projections.

For example, Harvey and Svoboda demonstrated interactions between plasticity at neighboring hippocampal synapses [60], and Larkum, Zhu, and Sakmann demonstrated nonlinear coupling of distal dendritic input with back-propagating spikes in cortical pyramidal neurons [85]. These experiments support possible local ingredients, not the SAL matrix identities.

The listed mechanisms make SAL biologically suggestive.

They do not prove that SAL is implemented in cortex.

16.12.3 Level III: Does cortex actually implement SAL?

This is the strongest claim and requires direct empirical tests.

The distinction is

local implementability
$\neq$ available biological substrate
$\neq$ evidence that the brain uses this algorithm.

The purpose of the cortical mapping below is therefore to generate predictions, not to identify SAL with established cortical circuitry.

16.13 A Candidate Cortical Implementation

SAL is defined functionally by four pathway classes:

$$u \to u, \qquad u \to d, \qquad d \to d, \qquad d \to u.$$

Here u and d refer to the upstream and downstream computational streams.
These are abstract functional roles.
They need not correspond uniquely to particular cortical layers or cell types.

16.13.1 Driving versus modulatory interactions

One possible biological interpretation distinguishes:

- relatively driving interactions within a computational stream;
- more modulatory interactions between streams.

Dendritic compartmentalization provides one possible way to realize this difference.
Basal or proximal inputs may exert relatively direct effects on spiking.
Apical or distal inputs may instead modulate gain, plasticity, or the response to basal drive.
These statements are implementation hypotheses, not assumptions of the mathematical SAL algorithm.

16.13.2 One candidate mapping

The following candidate mapping is retained from the supplied chapter. It is a set of functional hypotheses, not a new anatomical finding or a unique consequence of the conditional theorem:

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SAL role	Weight	Candidate cortical pathway	Functional interpretation
$u \to u$	W	L4 $\to$ L2/3	driving ascending representation
$u \to u$	W	L2/3 $\to$ higher-area input layers	driving ascending representation
$u \to d$	V	L2/3 $\to$ L5	cross-stream coupling
$d \to d$	$\bar{W}$	deep-layer inter-areal / transthalamic paths	descending-stream propagation
$d \to u$	$\bar{V}$	feedback to L1/apical compartments	modulatory cross-stream signal
$d \to u$	$\bar{V}$	deep-to-superficial local feedback	modulatory cross-stream signal

Table 16.1: One candidate functional mapping between SAL pathways and cortical circuitry. The correspondence is illustrative rather than uniquely specified by SAL. The mathematical theory requires functional pathway types, not these particular cortical layers.

16.13.3 What SAL does not require anatomically

SAL does not require:

- identical laminar organization across cortical areas;
- one-to-one matching of cortical layers to abstract network layers;
- exact symmetry between reciprocal pathways;

- feedback pathways that are merely copies of feedforward pathways.

Its core requirement is functional:

two interacting plastic streams whose local coupling can self-organize a useful feedback transformation.

16.14 Falsifiable Predictions

A useful biological learning theory should make predictions beyond "feedback exists." The seven proposals below are tests of the mechanism under its stated conditions, not predictions guaranteed by the conditional algebra. Simulations can compare exact gradients directly; in vivo tests need an operational objective and a justified experimental proxy for the corresponding direction. No physiological measurements or training results are added here.

16.14.1 Prediction 1: functional gradient alignment should emerge

During learning,

$$\cos\theta_\ell = \frac{\langle \Delta W_\ell^{\mathrm{data}}, -\nabla_{W_\ell} F\rangle}{\|\Delta W_\ell^{\mathrm{data}}\|_F \|\nabla_{W_\ell} F\|_F}$$

should be compared with zero on relevant nonzero gradients. The conditional model supports a positive bound when its errors are small enough; it does not predict monotonic growth toward one. Record the full objective change and both the data and decay contributions.

The prediction concerns the *effective update*, not pairwise symmetry of individual synapses.

16.14.2 Prediction 2: cross-stream plasticity should stabilize as learning becomes competent

The quasi-stationary hypothesis should be tested using update fields normalized by their step sizes, not raw increments alone. Measure reciprocity defects C_j^{rec}, transport defects A_j, empirical update residuals $\mathcal{E}_\ell$, and singular values separately. A useful ordering to test is

$$\text{improved transport/alignment} \longrightarrow \text{more useful updates} \longrightarrow \text{improved task performance.}$$

Small averaged auxiliary updates, learning-rate decay, or stabilization caused by a vanishing teaching signal can mimic part of this pattern. The theorem does not make low auxiliary-update variance a necessary condition for predictive success.

16.14.3 Prediction 3: feedback width should improve conditioning up to saturation

Increasing

$$\bar{d}_{\ell+1}$$

can be tested for improvement of the rank and conditioning of

$$H_\ell = \bar{V}_{\ell+1}\bar{V}_{\ell+1}^\top.$$

A saturation hypothesis is plausible once the task-relevant error space is represented robustly. Scaling the initialization with width, matching training budgets, and measuring the spectrum are necessary to separate added dimension from a mere change in update magnitude.

Thus the prediction is not simply

more feedback is always better,

but

feedback dimensionality matters through the spectrum of H_ℓ.

16.14.4 Prediction 4: performance should track the spectrum of the learned preconditioner

For tasks requiring the corresponding gradient directions, test whether SAL deteriorates when

$$\lambda_{\min}(H_\ell)$$

approaches zero.

Measure useful-subspace lower eigenvalues and their scale, as well as quantities such as

$$\kappa(H_\ell).$$

This prediction distinguishes mere feedback activity from useful gradient transport.

16.14.5 Prediction 5: disrupting heterosynaptic plasticity should selectively impair deep credit assignment

If SAL depends on cross-stream heterosynaptic interactions, then selectively weakening those interactions while leaving ordinary homosynaptic plasticity intact should disproportionately impair tasks requiring multi-stage credit assignment.

Simple one-layer association learning is a useful comparison. Controls must match the perturbation's effects on baseline excitability, forward inference, reward access, and ordinary plasticity. A broad deficit alone would not identify SAL.

16.14.6 Prediction 6: feedback need not mirror feedforward weights

Successful learning should not require

$$\bar{W}_\ell \approx W_\ell^\top.$$

A biological test should therefore look for functional update alignment without demanding anatomical weight symmetry.

16.14.7 Prediction 7: different physiological regimes should characterize within-stream and cross-stream pathways

If the functional division is correct, within-stream pathways and cross-stream pathways should differ systematically in their influence on neural responses or plasticity.

This may appear as differences in:

- dendritic targeting;
- time constants;
- gain modulation;
- nonlinear gating;
- plasticity rules.

Computational test protocol. Use paired initializations and data orders, an independently selected validation procedure, and a held-out test set. Retain failed and unstable runs. Compare plastic feedback, frozen feedback, matched reciprocal drives, mismatched drives, and an exact-backpropagation reference. Report transport quality, loss, prediction, and resource cost separately. A successful training run does not establish teacher-coordinate recovery; a structural comparison requires its own controlled criterion.

16.15 Relation to C1, C2, and C3

SAL addresses a different problem from the principles developed earlier in the book.

16.15.1 C1: representation

C1 asks whether a compact compositional representation exists.

SAL does not establish C1.

16.15.2 C2: informative recovery signal

C2 asks whether the input–output problem exposes a useful statistical signal to gradient-based learning.

At a fixed representation, a common exact preconditioner cannot turn a zero objective gradient into a nonzero one. This is a local algebraic statement, not a universal impossibility theorem for the evolving SAL system.

If

$$\nabla_W F$$

vanishes in the admitted representation and observation model, then multiplying that same gradient by H leaves it zero. Parity can illustrate vanishing low-order correlations under a symmetric input distribution; it is not unlearnable for every architecture, optimizer, or observation model.

16.15.3 C3: supplied trace

C3 supplies sufficiently informative intermediate supervision. Chapter 11 additionally separates observing traces from coverage of the states needed later.

SAL does not require full C3 trace exposure, although biological teaching signals or replay could interact with SAL in richer systems.

16.15.4 SAL: implementation of the optimizer

SAL asks:

given a useful gradient direction, how could a local biological circuit approximately implement it?

Thus the division of labor is

C1	:	a compact solution exists,
C2/C3	:	accessible signal or intermediate supervision,
SAL	:	local circuitry implements a gradient-like parameter update.

This distinction is essential.

SAL is a theory of *credit transport and synaptic optimization*, not a substitute for a theory of learnability.

Chapter H asks for a feasible returned predictor with good empirical fit and valid statistical control. A successful SAL learner could meet that endpoint without reproducing the teacher's particular hidden variables. Neither circuit alignment nor width alone establishes the required population error bound.

16.16 SAL and Genericity

The useful distinction is

SAL studies implementation and transport of a learning direction; genericity studies accessibility of useful learning information.

At a fixed representation, for a common matrix H_ℓ and an objective gradient g_ℓ,

$$g_\ell = 0 \quad \Longrightarrow \quad H_\ell g_\ell = 0.$$

With $H_\ell \succ 0$, a nonzero gradient retains a strictly positive first-order descent pairing. Quantitative usefulness additionally requires its size, conditioning, and approximation error to be controlled. A nonzero gradient can also concern a poor predictor or an irrelevant feature; nonzero is not synonymous with generalizing.

The qualifier "common matrix" matters for sample averages. As the counterexample in the descent section shows, $\mathbb{E}[H(X)g(X)]$ need not equal $H\mathbb{E}[g(X)]$. An evolving feedback system can reweight examples, alter the representation, or move outside a restricted exact-gradient regime. Those effects can change the accessible statistics and must be analyzed as such. The zero-gradient identity does not rule them out, nor does it prove that they help.

This is consistent with Chapter 6: the observation model, representation, and learning rule together determine which signals are accessible. Circuit geometry and informational accessibility are distinguishable, but they can interact during learning.

16.17 Relation to Chapter 15

The preceding chapter argued that an attractor theory of activations is not by itself a learning theory of weights.

SAL supplies a candidate learning mechanism at the parameter level. The positive associative-state convergence results in Chapter 15 do not establish convergence of this larger coupled learning system. Conversely, a feedforward predictor can be useful at a finite checkpoint without either its parameters or every auxiliary pathway having reached equilibrium.

The distinction is:

Hopfield-like dynamics:	how a fixed circuit evolves its activity,
SAL:	how a plastic circuit could evolve its synapses.

Thus Chapters 15 and 16 form a natural pair:

associative retrieval is not a learning algorithm
$\Downarrow$
a local self-organizing feedback circuit may implement one.

16.18 Implications for Artificial Learning Systems

SAL is motivated biologically, but the idea also suggests alternatives to fixed backpropagation machinery in artificial systems.

16.18.1 Learned feedback pathways

Rather than using exact transpose Jacobians, an artificial network could maintain an independently parameterized feedback network that adapts jointly with the forward model.

16.18.2 Learned preconditioning

The matrices

$$H_\ell$$

suggest that optimization geometry itself can be represented by adaptive neural circuitry.

This connects SAL conceptually to:

- learned optimizers;
- adaptive preconditioning;
- meta-learning;
- auxiliary feedback networks;
- local-learning architectures.

16.18.3 Local hardware

A learning rule based predominantly on local outer products may also be attractive for neuromorphic or distributed hardware, where transporting exact global gradients can be expensive. Chapter 10 distinguishes local synaptic factors from short physical routes: dense feedback matrices, cross-stream fan-out, activation storage, and learning time still have communication and memory costs.

16.18.4 Continual adaptation

Biological adaptation motivates systems in which learning and inference can interact.

A self-organizing local optimizer could potentially operate continuously, allowing representations and their credit-assignment circuitry to co-adapt online.

This remains an architectural possibility rather than a consequence of the present theorem.

Selection and statistical accounting. Chapters 13 and 14 concern the complexity of the returned predictor and the effect of explicit regularizers. A learned SAL preconditioner is not itself a norm or compression bound, and Euclidean-flow invariants for product penalties do not automatically survive matrix preconditioning. A product-dependent decay gain would also need a mechanism for making that global quantity available to each local synapse.

When feedback is discarded after training, its parameter count remains a training-resource cost but is not necessarily part of the code of the feedforward predictor deployed at test time. If feedback participates in inference or adaptation, the deployed system and its admissible statistical analysis must include it. Uniform complexity bounds can cover a selected stopped iterate; they do not require every minibatch to stop changing.

16.19 Limitations and Open Problems

Several issues remain unresolved.

16.19.1 Why should the auxiliary weights reach the required stationary regime?

The revised theorem requires controlled transport and reciprocity defects, A_j and C_j^{rec}, together with bounded amplification of those defects. A complete theory should derive these properties from fully specified local rules and an initialization law. Quasi-stationarity must be defined at the level of fields, samples, or averages and connected to those algebraic conditions, rather than used as an informal synonym for them.

16.19.2 Rank preservation

The positive-definite result requires appropriate rank.

A useful theorem needs lower singular-value bounds, or a precisely specified task-gradient subspace on which such bounds hold. The batch-correlation proposition shows why mean and pointwise stationarity can have different rank implications. Initialization rank alone does not control the subsequent spectrum.

16.19.3 Timescale separation

The conditional argument naturally suggests that the auxiliary pathways may need to organize on a suitable timescale relative to forward learning.

Whether explicit timescale separation is necessary is an empirical and theoretical question.

16.19.4 Origin of the teaching signal

SAL begins with

$$-\nabla_{h_L} F.$$

The neural origin of this signal is not specified by the local self-assembly mechanism.

A complete biological theory must explain how task objectives generate appropriate output-level teaching signals.

16.19.5 Spiking and recurrent biological dynamics

The mathematical model analyzed here uses layered rate-like variables.

Cortex is recurrent, spiking, heterogeneous, and temporally structured.

The extent to which the SAL mechanism survives in more realistic dynamical systems is open.

16.19.6 Timing of local factors

The required local activity factors must coexist at the relevant synapses or dendritic compartments at appropriate times.

Biological eligibility traces may provide one possible mechanism, but this is not established by the present theory.

16.19.7 Very deep networks

The products in the bound for r_j make depth dependence explicit. Controlling them, preserving a useful lower descent scale, and meeting a finite-step noise budget are concrete requirements for extending the conditional result to very deep networks. Polynomial-time claims must also account for feedback width, precision, and the cost of reaching the aligned regime.

16.19.8 SAL does not solve informationally hard targets

Exact multiplication of a fixed zero objective gradient by a common preconditioner does not create a signal. This does not prove failure of every local-learning algorithm on a named target. Additional observations, changed representations, sample-dependent weighting, or partial traces change the question. A complete theorem must identify the actual signal used by the specified algorithm.

16.20 Conclusion

The central question is how local plastic circuitry could implement a useful parameter-space learning direction. SAL proposes a two-stream motif in which representations and feedback machinery co-adapt. This moves the discussion beyond an attractor theory of fixed activations.

The positive mathematical result in this chapter is explicit. Under controlled encoded-error transport and reciprocal cross-stream alignment,

$$\boxed{\Delta W_\ell^{\mathrm{SAL}} = H_\ell \Delta W_\ell^{\mathrm{BP}} - \eta \lambda_W W_\ell + \eta \mathcal{E}_\ell, \qquad H_\ell = \bar{V}_{\ell+1} \bar{V}_{\ell+1}^\top \succeq 0.}$$

The theorem bounds $\mathcal{E}_\ell$ by transport defects, reciprocity defects, and their depth-dependent amplification. Full row rank gives positive definiteness. Matched reciprocal drives provably reduce one alignment defect without requiring $\bar{W}_\ell = W_\ell^\top$.

The other conclusions require their own assumptions. For a common objective-gradient field, a uniformly positive preconditioner and a sufficiently small relative error yield quantitative descent. Small auxiliary increments alone do not establish those hypotheses. A sample-dependent positive matrix need not give expected descent of the averaged objective. Ordinary weight decay is not automatically a fixed quadratic penalty under a changing learned metric. Continued stochastic fluctuations are compatible with returning a useful predictor.

The biological interpretation is correspondingly layered. Local-factor implementability is a property of the proposed model. Dendritic and heterosynaptic mechanisms are candidate substrates. Evidence that cortex uses the full algorithm is a further experimental question. The retained cortical mapping and seven predictions organize that question without turning suggestive anatomy into a proof.

The division of labor within the book is

C1:	an economical computation can be represented,
C2/C3:	useful signals or intermediate supervision are accessible,
SAL:	local circuitry can conditionally transport a learning direction,
optimization:	the actual dynamics must reach a useful predictor,
statistical control:	fit and complexity must support generalization.

A generalizing feasible predictor need not reproduce a particular teacher decomposition. Nor does a positive conditional circuit theorem by itself establish global optimization or predictive generalization. The scientific value of SAL is more specific: it makes functional credit transport a concrete object of local circuit design, mathematical analysis, and controlled experiment.

functional gradient alignment can be sufficient for learning updates;
literal forward–backward weight symmetry is not the only route.

CHAPTER 17

Zeroth-Order Post-Training: Why Effective Dimension Matters (with Y. Gan)

Zeroth-order optimization offers an attractive possibility for post-training large models when gradients are unavailable, inaccessible, or too expensive to store, or when the objective evaluates a complete black-box system. Its defining resource is access to objective values, not derivatives of the evaluated system.

But zeroth-order optimization does not make dimension irrelevant. For the isotropic Gaussian estimator studied below, even a linear objective has a mean-squared gradient-estimation error proportional to the number of searched coordinates. A useful response is to search a structured adaptation space rather than all microscopic weights. A large frozen model can have nominal parameter dimension D, while its tunable change is represented as

$$\theta = \theta_0 + Az, \qquad z \in \mathbb{R}^r, \qquad r \ll D.$$

The columns of A then act as coordinated mutation patterns, or computational "genes." They are not individual synapses: one control coefficient can change many weights together.

This is the principal constructive strategy of the chapter. It should not be mistaken for a universal necessity theorem. Accurate estimation of the entire gradient and useful decrease of the objective are different goals. Full-dimensional perturbations can be effective when the objective has favorable low-effective-rank curvature, even though their Euclidean gradient-estimation variance remains large. We give an exact quadratic example to make this distinction explicit.

The central claim is therefore

> *zeroth-order post-training benefits from useful low-dimensional controls*
> *or other explicitly favorable search geometry;*
> *small dimension alone guarantees neither signal nor success.*

Here "effective dimension" must name a measured object: the number of searched coordinates, the rank of an adapter matrix, and the effective rank of curvature are not interchangeable. The evolutionary analogy is with structured heritable variation, not a claim that biological evolution or model adaptation always admits a small fixed linear genotype. The chapter develops the mathematical distinctions and a program of controlled experiments; it reports no new post-training measurements.

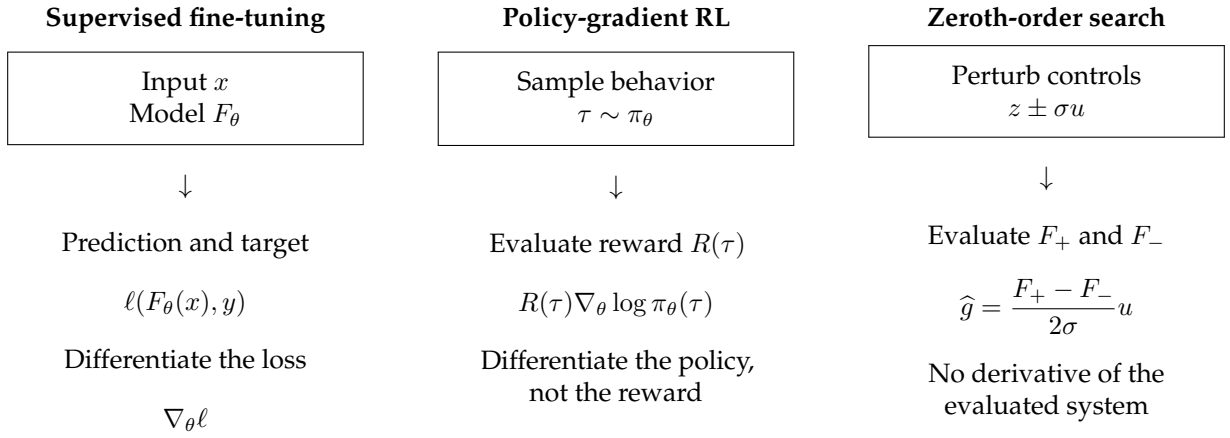

Figure 17.1: **Three routes to post-training.** Supervised fine-tuning differentiates a task loss through the model. Policy-gradient reinforcement learning can optimize a possibly nondifferentiable reward by differentiating the log probability of sampled behavior. Zeroth-order methods perturb accessible parameters or control variables and use objective evaluations. The schematic is redrawn from the original chapter illustration; the direction factor u is included explicitly. These are computational distinctions, not a ranking of empirical performance.

Conventions and resources. We minimize costs f or F. Expected rewards are maximized, or negated before applying a minimizing algorithm; an expectation denoted J follows the explicitly stated cost or reward convention. The nominal model dimension is D, the dimension of a specified search space is r, and d is used only for the generic smoothing formula. We write q for the number of independent *direction pairs per update*, not the total number of scalar evaluations; r, q, T are positive integers. With T updates, paired estimation uses $2qT$ objective calls before replication, validation, or subspace design. The number n of independent data examples is a different resource.

The exact identities below use real arithmetic and a stationary oracle at the current center. Finite precision, a changing model endpoint, clipped perturbations, and changing evaluation protocols require their own error analysis. A small control vector does not remove the cost of running the base model or the complete system once per query.

17.1 Why Zeroth-Order Optimization?

The preceding chapter studied a possible circuit implementation of first-order, preconditioned learning. The present chapter changes the information interface: an optimizer receives function values rather than backpropagated derivatives. Both approaches still need useful signal, controlled dynamics, and a statistically adequate returned predictor.

The standard comparison is with first-order training.

Given an objective

$$f(\theta), \qquad \theta \in \mathbb{R}^D,$$

SGD and related methods use

$$\nabla_\theta f.$$

For large networks this is remarkably effective.

There are nevertheless settings where direct gradient access is undesirable or impossible.

17.1.1 Black-box model access

An inference-only interface does not, by itself, permit weight-space search. To evaluate $f(\theta \pm \sigma u)$, the experimenter must be able to install or otherwise request those perturbed weights. An API exposing only prompts, retrieval settings, and decoding controls allows search over those controls, not arbitrary changes to its hidden parameter vector.

The oracle is therefore more accurately described as

$$z \longmapsto F(z) = \mathbb{E}_\omega[\text{system cost under accessible controls } z],$$

where ω records any sampling randomness. The set of accessible queries is part of the optimization problem.

17.1.2 Memory-limited fine-tuning

Backpropagation normally stores intermediate activations required by the reverse pass.

Zeroth-order methods can operate using forward evaluations alone.

Methods such as MeZO exploit this fact to reduce memory requirements substantially during language-model fine-tuning [97].

17.1.3 System-level objectives

The quantity one ultimately wishes to optimize may involve an entire pipeline:

$$\text{model} \to \text{decoder} \to \text{tool use} \to \text{external evaluator}.$$

The resulting objective can be evaluated but may not possess a convenient computational graph linking it to every tunable system variable.

17.1.4 Outer-loop optimization

Some variables are naturally outside the gradient-trained model:

- curriculum schedules;
- data-mixture coefficients;
- decoding controls;
- routing parameters;
- augmentation policies;
- optimizer hyperparameters;
- system-level tradeoff coefficients.

These variables often form a comparatively low-dimensional black-box optimization problem.

17.2 Nondifferentiable Reward Does Not by Itself Require Zeroth Order

An important distinction is necessary.

Suppose a policy

$$\pi_\theta$$

generates trajectory τ with reward

$$R(\tau).$$

The reward itself need not be differentiable with respect to θ.

The expected reward is

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}[R(\tau)].$$

Assume that the trajectory law has a differentiable density or mass function on a parameter-independent support, that differentiation may be interchanged with integration, and that the reward is integrable and has no additional explicit dependence on θ. Then the score-function identity gives [165]

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[R(\tau)\nabla_\theta \log \pi_\theta(\tau)\right]. \tag{17.1}$$

Thus

$$R(\tau)$$

may be:

- a binary success signal;
- a programmatic verifier;
- a human preference;
- a discrete task score;
- a black-box external evaluation.

Policy-gradient methods can still estimate $\nabla_\theta J$ because they differentiate the policy likelihood, not the reward itself. If the reward or environment has additional explicit parameter dependence, the corresponding derivative terms cannot be omitted from Equation (17.1). In a controlled process, $\pi_\theta(\tau)$ here denotes the parameter-dependent trajectory law; when the environment is independent of θ, its log derivative is the sum of policy log derivatives. Baselines can reduce variance without changing the expectation under the usual conditional-independence conditions.

Therefore

$$\boxed{\text{nondifferentiable reward} \not\Longrightarrow \text{zeroth-order optimization is necessary.}}$$

17.2.1 What distinguishes ZO from policy gradient?

Policy-gradient sampling generates behavior from the current policy; it does not require a parameter perturbation to obtain a sample.

It uses

$$R(\tau)\nabla_\theta \log \pi_\theta(\tau).$$

Zeroth-order parameter search instead explicitly perturbs the parameters:

$$\theta \mapsto \theta + \sigma u.$$

Thus

policy gradient:	differentiate log-probability of sampled behavior,
zeroth-order search:	infer a parameter-space direction from objective queries.

ZO is particularly attractive when the parameterized system itself is black-box or when the variables being optimized lie outside the differentiable policy.

17.3 Gaussian Smoothing and Zeroth-Order Gradient Estimation

Let

$$f : \mathbb{R}^d \to \mathbb{R}.$$

Suppose gradients are unavailable but function evaluations are possible.
Choose

$$u \sim \mathcal{N}(0, I_d).$$

For $\sigma > 0$, the symmetric two-point estimator is

$$\widehat{g}(\theta; u) = \frac{f(\theta + \sigma u) - f(\theta - \sigma u)}{2\sigma} u. \tag{17.2}$$

Define the Gaussian-smoothed objective

$$f_\sigma(\theta) = \mathbb{E}_{z\sim\mathcal{N}(0,I_d)} \left[f(\theta + \sigma z) \right]. \tag{17.3}$$

For example, it suffices that f is measurable and has at most polynomial growth on $\mathbb{R}^d$. Gaussian convolution is then differentiable and

$$\boxed{\mathbb{E}_u[\widehat{g}(\theta; u)] = \nabla f_\sigma(\theta).} \tag{17.4}$$

Thus the estimator is not generally an unbiased estimator of

$$\nabla f(\theta),$$

but of the gradient of a smoothed objective [112].

Derivation of the identity. Writing φ_σ for the centered Gaussian density,

$$f_\sigma(\theta) = \int f(v)\varphi_\sigma(v - \theta)\, dv, \qquad \nabla f_\sigma(\theta) = \frac{1}{\sigma}\mathbb{E}[f(\theta + \sigma u)u].$$

The growth assumption permits differentiation under the integral. Symmetry gives $\mathbb{E}[f(\theta - \sigma u)u] = -\mathbb{E}[f(\theta + \sigma u)u]$. Subtracting proves Equation (17.4). No derivative of f need be available to the algorithm.

This is a full-space Gaussian identity. Projecting a queried point onto a constraint set generally changes the oracle and the smoothing kernel. Discrete prompts, bounded temperatures, and simplex-valued mixtures need an explicitly specified encoding, extension, or feasible-query rule; the identity cannot simply be asserted for an unspecified clipping step.

17.3.1 Why smoothing can help

Smoothing replaces a potentially irregular objective by

$$f_\sigma.$$

This can reduce sensitivity to small discontinuities or noisy evaluations.
But it introduces a tradeoff:

small $\sigma \Rightarrow$ low smoothing bias but high sensitivity to evaluation noise,
large $\sigma \Rightarrow$ lower noise sensitivity but stronger smoothing bias.

The appropriate σ therefore depends on smoothness, noise level, search dimension, and query budget.

Dimension also sets the perturbation radius. If f has globally L_f-Lipschitz gradient, elementary Taylor and Jensen bounds give

$$|f_\sigma(\theta) - f(\theta)| \le \frac{L_f \sigma^2 d}{2}, \qquad \|\nabla f_\sigma(\theta) - \nabla f(\theta)\| \le L_f \sigma \sqrt{d}.$$

The first follows by averaging the Taylor remainder after the linear term cancels; the second follows from $\nabla f_\sigma = \mathbb{E}\nabla f(\theta + \sigma u)$. Moreover $\mathbb{E}\|\sigma u\|^2 = \sigma^2 d$. Holding the coordinatewise scale σ fixed while changing dimension is not holding the physical perturbation size fixed. These are global-smoothness bounds; a local smoothness estimate requires control of Gaussian tails outside its region.

17.4 The Curse of Search Dimension

The central limitation of zeroth-order optimization appears even on the simplest possible objective.

Proposition 26 (Dimension dependence for a linear objective)**.** *Let*

$$f(z) = g^\top z, \qquad z \in \mathbb{R}^r,$$

and let

$$u \sim \mathcal{N}(0, I_r).$$

For the two-point estimator

$$\widehat{g}(u) = \frac{f(z + \sigma u) - f(z - \sigma u)}{2\sigma} u,$$

we have

$$\widehat{g}(u) = u u^\top g,$$

and

$$\mathbb{E}[\widehat{g}] = g.$$

Moreover,

$$\mathbb{E}\left[\|\widehat{g} - g\|_2^2\right] = (r+1)\|g\|_2^2.$$

If q independent directions are averaged,

$$\widehat{g}_q = \frac{1}{q}\sum_{j=1}^{q}\widehat{g}(u_j),$$

then

$$\boxed{\mathbb{E}\left[\|\widehat{g}_q - g\|_2^2\right] = \frac{r+1}{q}\|g\|_2^2.} \tag{17.5}$$

Proof. For a linear objective,

$$f(z+\sigma u) - f(z-\sigma u) = 2\sigma g^\top u.$$

Hence

$$\widehat{g} = (g^\top u)u = uu^\top g.$$

Since

$$\mathbb{E}[uu^\top] = I,$$

the estimator is unbiased.
Rotational invariance allows us to take

$$g = \|g\|e_1.$$

Then

$$\|\widehat{g}\|^2 = \|g\|^2 u_1^2 \sum_{j=1}^{r} u_j^2.$$

Using Gaussian moments,

$$\mathbb{E}[u_1^4] = 3$$

and

$$\mathbb{E}[u_1^2 u_j^2] = 1, \qquad j \neq 1,$$

we obtain

$$\mathbb{E}\|\widehat{g}\|^2 = (r+2)\|g\|^2.$$

Since $\mathbb{E}\widehat{g} = g$,

$$\mathbb{E}\|\widehat{g} - g\|^2 = (r+1)\|g\|^2.$$

Averaging q independent estimates divides the variance by q.

□

The proposition exposes the fundamental issue:

q must grow with r for fixed relative mean-squared error of this isotropic Gaussian full-gradient estimator.

This is true before introducing nonlinear curvature, stochastic rewards, or model-evaluation noise.

Gradient accuracy is not the same as optimization progress. The proposition is exact for the stated estimator, not a minimax query lower bound for every zeroth-order algorithm. It does not require $q = \Omega(r)$ at *each* successful update. Even with one noiseless linear probe,

$$\langle g, \widehat{g}(u) \rangle = (g^\top u)^2 \geq 0.$$

Thus $-\widehat{g}$ is a descent direction for a nonzero linear objective almost surely, although it is a poor Euclidean reconstruction of $-g$. A linear function has no curvature and no unconstrained minimum; this observation alone gives no convergence theorem. The quadratic result in Section 17.14 separates these issues on a bounded-below objective.

17.5 Evaluation Noise Makes Dimension Still More Important

Suppose instead that an evaluation returns

$$Y(\theta) = f(\theta) + \xi,$$

where

$$\mathbb{E}[\xi] = 0, \qquad \mathrm{Var}(\xi) = \nu^2.$$

Assume the noises are conditionally centered, independent of the direction, have the stated conditional variance, and are independent between signs and between direction pairs. The noise component of a two-point estimator is

$$\widehat{g}_{\mathrm{noise}} = \frac{\xi_+ - \xi_-}{2\sigma} u.$$

For

$$u \sim \mathcal{N}(0, I_r),$$
$$\mathbb{E}\|u\|^2 = r.$$

Thus, after averaging q independent pairs, the contribution of evaluation noise to the mean squared estimator norm is

$$\boxed{\frac{\nu^2 r}{2q\sigma^2}.} \tag{17.6}$$

Again, the relevant dimension appears explicitly.
This leads to the practical rule:

high reward noise + high search dimension = expensive zeroth-order estimation.

Paired randomness changes the answer. If the two marginal noise variances are ν^2 and their covariance is $\chi\nu^2$, with $-1 \leq \chi \leq 1$ and these moments independent of u, then

$$\mathbb{E}\|\widehat{g}_{\mathrm{noise},q}\|^2 = \frac{\nu^2(1-\chi)r}{2q\sigma^2}.$$

This follows from $\mathrm{Var}(\xi_+ - \xi_-) = 2\nu^2(1-\chi)$. Using the same minibatch, simulator seed, or sampled task on both sides can change this covariance. Positive correlation can help; it is not guaranteed merely by reusing a seed. With direction-dependent variances, the exact expression

instead contains $\mathbb{E}[\|u\|^2 \operatorname{Var}(\xi_+ - \xi_- \mid u)]/(4q\sigma^2)$. A random minibatch objective need not be an additive homoscedastic oracle.

At fixed root-mean-square perturbation radius $\rho = \sigma\sqrt{r}$, the independent-noise term becomes $\nu^2 r^2/(2q\rho^2)$, not a quantity linear in r. Comparisons must state what is held fixed. Averaging m independent evaluations at each point reduces the corresponding variance by m, but costs $2qm$ calls per update. A deterministic evaluation error bounded by ε_f produces a direction error bounded by $\varepsilon_f\|u\|/\sigma$; it need not average to zero.

Regularized stationary points. If the queried objective is $f + \lambda\Omega$, the regularizer changes the mean balance but does not erase noisy function evaluations. Nor does stationarity of the unsmoothed objective imply stationarity of its Gaussian smoothing. When a penalty on the accessible controls has an inexpensive exact derivative, one can estimate only the black-box part and add that derivative. This is a hybrid estimator of $\nabla f_\sigma + \lambda\nabla\Omega$, not necessarily the gradient of $(f + \lambda\Omega)_\sigma$. As in Chapter 7, mean stationarity, zero sampling covariance, and a useful stopped predictor are different conclusions.

17.6 Nominal Dimension versus Effective Adaptation Dimension

A modern LLM may contain

$$D$$

parameters with

$$D \sim 10^9 \quad \text{or larger.}$$

Equation (17.5) makes clear why naive isotropic search over all D coordinates is unattractive. But post-training need not require arbitrary movement in $\mathbb{R}^D$.

Suppose useful adaptation lies near

$$\theta = \theta_0 + Az,$$

where

$$A \in \mathbb{R}^{D\times r}, \qquad r \ll D.$$

The variable

$$z \in \mathbb{R}^r$$

is the *adaptation coordinate*.

Define

$$F(z) = f(\theta_0 + Az). \tag{17.7}$$

Zeroth-order optimization can now operate on

$$F : \mathbb{R}^r \to \mathbb{R}.$$

The isotropic estimator now has r coordinates rather than D. Its variance bounds still involve gradient magnitudes, noise, and σ; optimization additionally involves restricted curvature, initial suboptimality, and the quality of the accessible family. The cost of evaluating F can still scale with the full model size.

$$\boxed{D = \text{nominal parameter dimension}, \qquad r = \text{effective search dimension}.}$$

This distinction is the main principle of the chapter.

Three different dimensions. The actual searched coordinate count r, the matrix rank r_{LoRA}, and a curvature effective rank r_{curv} describe different objects. Also distinguish them from the length of a compressed predictor in Chapter 13. A low-rank matrix can have many trainable entries in its factors, and a small-dimensional objective can be badly conditioned or statistically difficult.

The affine model assumes A is fixed during the current search. General adapters or system controls instead define a possibly nonlinear map $\theta = \Phi(z)$, or a pipeline not representable by changing weights at all. The scalar objective is then $F(z) = f(\Phi(z))$ or a directly specified system cost. Gaussian estimation remains an r-coordinate construction, but the affine projection interpretation requires the linear model.

17.7 Projected Zeroth-Order Optimization

Assume for simplicity that the columns of A are orthonormal:

$$A^\top A = I_r.$$

Sample

$$u \sim \mathcal{N}(0, I_r).$$

Evaluate

$$f(\theta_0 + A(z + \sigma u))$$

and

$$f(\theta_0 + A(z - \sigma u)).$$

The reduced-space estimator is

$$\widehat{g}_z = \frac{F(z + \sigma u) - F(z - \sigma u)}{2\sigma} u. \tag{17.8}$$

Update

$$z \leftarrow z - \eta \widehat{g}_z.$$

Equivalently, in the original parameter space,

$$\theta \leftarrow \theta - \eta A \widehat{g}_z.$$

17.7.1 What direction is being estimated?

If f is differentiable,

$$\nabla_z F = A^\top \nabla_\theta f.$$

Lifting this gradient back to the model parameters gives

$$A \nabla_z F = A A^\top \nabla_\theta f.$$

Thus the unsmoothed projected gradient is the component of the full gradient lying inside

$$\mathrm{span}(A).$$

Define the missing-gradient component

$$e_A = (I - A A^\top) \nabla_\theta f. \tag{17.9}$$

Then

small $\|e_A\|$ is needed for accurate reconstruction of the full gradient; useful restricted improvement need not require it.

This suggests two design goals for reduced-space zeroth-order adaptation:

$r \ll D$, useful adaptation directions lie approximately in $\mathrm{span}(A)$.

A low-dimensional but badly chosen subspace can exclude the desired improvement. Yet the absolute size of e_A is not a failure criterion by itself. For example,

$$f(\theta_1, \theta_2) = \tfrac{1}{2}\theta_1^2 + \tfrac{1}{2}(\theta_2 - M)^2, \quad \theta = (1, 0), \quad A = (1, 0)^\top$$

has $\|e_A\| = |M|$, while restricted optimization still decreases the loss by $1/2$. It cannot remove the remaining $M^2/2$ gap. More relevant task-level quantities are the achievable restricted value $f_A^* = \inf_z f(\theta_0 + Az)$ and the improvement requested relative to $f(\theta_0)$, not only a pointwise full-gradient approximation.

Finite smoothing and nonorthogonal coordinates. The expectation of Equation (17.8) is $\nabla_z F_\sigma(z)$. Under dominated differentiation, its lift is

$$A\nabla F_\sigma(z) = AA^\top \mathbb{E}_u[\nabla f(\theta_0 + Az + \sigma Au)].$$

This is smoothing along the subspace, not full-dimensional Gaussian smoothing. For general full-column-rank A, Euclidean descent in z lifts to $AA^\top \nabla f$, whereas the orthogonal projection is $A(A^\top A)^{-1}A^\top \nabla f$. Coordinate scaling changes the search metric. If f has L_f-Lipschitz gradient, the restricted objective has smoothness at most $L_f\|A\|_2^2$.

Proposition 27 (Projection error and estimation error)**.** *Let $A^\top A = I_r$ and let $f(\theta) = g^\top\theta + c$. Use q independent Gaussian direction pairs in the reduced coordinates. With independent evaluation noise satisfying the conditions of Equation* (17.6)*, let $\widehat{g}_A = A\widehat{g}_{z,q}$. Then*

$$\mathbb{E}\|\widehat{g}_A - g\|^2 = \|(I - AA^\top)g\|^2 + \frac{r+1}{q}\|A^\top g\|^2 + \frac{\nu^2 r}{2q\sigma^2}.$$

The noiseless identity is obtained by setting $\nu = 0$.

Proof. Write $g = AA^\top g + (I - AA^\top)g$. The difference $\widehat{g}_A - AA^\top g$ lies in $\mathrm{span}(A)$ and the omitted component is orthogonal to that space, so their squared norms add pointwise. Because A is an isometry on $\mathbb{R}^r$, the first component's error is the reduced-space error. Apply Proposition 26 to $A^\top g$ and add Equation (17.6); the noise cross term has zero conditional expectation. □

The identity makes an explicit bias–variance tradeoff: increasing r can reduce the omitted component while increasing estimation cost. For a uniformly random r-dimensional subspace independent of fixed g, rotational symmetry gives

$$\mathbb{E}_A\|A^\top g\|^2 = \frac{r}{D}\|g\|^2,$$

since $\mathbb{E}[AA^\top] = (r/D)I_D$. A useful small subspace is therefore additional structure, not a consequence of being low dimensional. Learning A, storing it, and multiplying by it have their own costs; an unrestricted dense A has Dr entries. Sparse, shared, or generated maps can reduce storage, but their implementation and precision must be specified. Jointly optimizing all of A does not retain an r-parameter search problem.

17.8 Structured Mutations as Computational Genes

The projected formulation admits a natural evolutionary interpretation.

Write the columns of A as

$$A = [G_1, \dots, G_r].$$

Then

$$\theta = \theta_0 + \sum_{j=1}^{r} z_j G_j.$$

Each G_j is a coordinated perturbation of the complete parameter vector.

A mutation in one scalar coordinate

$$z_j$$

therefore modifies many microscopic parameters together.

We can regard

$$G_j$$

as a computational analogue of a *gene* or regulatory module.

In this computational analogy, the optimizer searches coefficients of reusable perturbation patterns rather than independent synapses.

$$\boxed{\text{microscopic dimension } D \gg \text{evolutionary control dimension } r.}$$

17.8.1 Why this is more biologically plausible

The intended biological comparison is with heritable variation in regulatory or developmental mechanisms, rather than independent perturbation of every adult synapse. In an artificial system, a specified generator can similarly turn a small control change into a coordinated large parameter change.

This is a modeling analogy. It neither derives biological evolutionary efficiency from a dimension count nor says that a genotype–phenotype map is linear, task-aligned, or easy to optimize.

The analogous post-training strategy is therefore not

$$\theta_i \to \theta_i + \delta_i$$

independently for billions of parameters.

It is

$$z_j \to z_j + \delta_j,$$

where one z_j changes a coordinated parameter module.

17.8.2 Connection to sparse compositionality

If a network implements a computation through reusable constituents, then a natural mutation acts on a constituent rather than on every parameter independently.

Thus the same structural principle appears again:

$$\boxed{\begin{array}{c}\text{supplied or learned modular structure}\\ + \text{ few task-relevant control variables per adapted module}\\ \longrightarrow \text{ a candidate low-dimensional search parameterization.}\end{array}}$$

17.9 Where Can the Low-Dimensional Coordinates Come From?

Several practical constructions can provide reduced control spaces. Some are affine in the controls and others are nonlinear; the number of queried scalar coordinates, not the name of the construction, enters the isotropic estimator's dimension factor.

17.9.1 Low-rank adapters

Suppose a weight matrix is adapted by

$$W = W_0 + UV^\top,$$

with rank

$$r_{\mathrm{LoRA}} \ll \min(d_{\mathrm{in}}, d_{\mathrm{out}}).$$

Zeroth-order search can operate on a compressed parameterization of the adapter rather than on the full W.

Parameter-efficient methods such as LoRA provide an existing mechanism for reducing the number of trainable parameters [71]. But if $U \in \mathbb{R}^{d_{\mathrm{out}} \times r_{\mathrm{LoRA}}}$ and $V \in \mathbb{R}^{d_{\mathrm{in}} \times r_{\mathrm{LoRA}}}$ are both trained, the queried coordinate count for this one matrix is

$$r_{\mathrm{search}} = r_{\mathrm{LoRA}}(d_{\mathrm{out}} + d_{\mathrm{in}}),$$

not r_{LoRA}. A rank-one update of a 4096×4096 matrix already has 8192 factor coordinates. Quotienting factorization redundancy can reduce intrinsic manifold dimension, but a naive estimator in (U, V) still samples all of its chosen coordinates.

Training both factors is bilinear, not a fixed affine subspace of W. A genuinely affine alternative is $W = W_0 + \sum_{j=1}^{r} z_j B_j$ with fixed basis matrices B_j. At $U = V = 0$, paired perturbations of both factors give exactly the same weight matrix:

$$(\sigma\Delta U)(\sigma\Delta V)^\top = (-\sigma\Delta U)(-\sigma\Delta V)^\top.$$

The symmetric finite-difference numerator is therefore zero for any objective depending only on $UV^\top$. The usual LoRA initialization uses one nonzero random factor and one zero factor; this avoids that particular symmetry. This example shows why a small parameter count alone does not guarantee an informative probe.

17.9.2 Module scales

A large frozen network may contain modules

$$M_1, \ldots, M_r$$

with tunable gains

$$z_1, \ldots, z_r.$$

Then

$$h = \sum_j z_j M_j(x)$$

provides a direct low-dimensional control surface.

17.9.3 Mixture-of-experts routing

If a system contains a small number of experts, ZO can optimize routing or mixture weights rather than internal expert parameters.

17.9.4 Prompt and context variables

Continuous prompt vectors, retrieval weights, memory-selection parameters, or tool-routing coefficients may provide compact adaptation spaces. A prompt with p tokens and embedding dimension d_e has pd_e continuous entries unless a further basis is fixed. Discrete text-prompt search is not automatically the Gaussian estimator of Section 17.3.

17.9.5 Decoding parameters

Variables such as

$$T, \qquad p, \qquad \text{penalty coefficients},$$

often form an extremely low-dimensional system-level optimization problem.

17.9.6 Training controls

ZO can also optimize an outer-loop vector containing:

$$\text{curriculum weights}, \quad \text{data mixtures}, \quad \text{learning schedules}, \quad \text{reward coefficients}.$$

17.10 Three Families of Zeroth-Order Methods

The term "zeroth order" covers several conceptually different algorithms.

17.10.1 ZO gradient estimation

Methods such as Equation (17.2) use objective queries to estimate

$$\nabla f_\sigma.$$

The resulting estimate is then used much like an ordinary gradient.

17.10.2 Direct search

Direct-search methods do not necessarily construct a gradient estimate.

A simple population algorithm samples

$$z + \delta_1, \ldots, z + \delta_m$$

and retains improved candidates.

Nelder–Mead is a classical direct-search example [83]; its simplex operations should not be identified with arbitrary random population selection. The cited work establishes restricted low-dimensional results, not a general guarantee for nonconvex noisy model objectives. Gradient-estimator variance formulas are not themselves convergence bounds for these direct-search methods.

17.10.3 Distribution search

Evolution strategies optimize a probability distribution over candidate solutions.

Let

$$z \sim \pi_\phi.$$

The objective is

$$J(\phi) = \mathbb{E}_{z\sim\pi_\phi}[f(z)].$$

One then adapts the distribution parameters

$$\phi.$$

This viewpoint includes Natural Evolution Strategies and CMA-ES, though their actual estimators, rank-based selection rules, and updates differ. The families overlap: Gaussian mean search can yield the same smoothed gradient identity used in two-point estimation.

17.11 Natural Evolution Strategies

NES uses the score-function identity in parameter-search space [164]:

$$\nabla_\phi J(\phi) = \mathbb{E}_{z\sim\pi_\phi}\left[f(z)\nabla_\phi \log \pi_\phi(z)\right].$$

A Monte Carlo estimate is

$$\nabla_\phi J \approx \frac{1}{m}\sum_{i=1}^{m} f(z_i)\nabla_\phi \log \pi_\phi(z_i).$$

NES then applies a natural gradient,

$$\phi \leftarrow \phi - \eta\mathcal{I}(\phi)^{-1}\nabla_\phi J,$$

where

$$\mathcal{I}(\phi) = \mathbb{E}[\nabla_\phi \log \pi_\phi(z)\nabla_\phi \log \pi_\phi(z)^\top]$$

is the Fisher information matrix of the search distribution. The inverse requires a nonsingular identifiable parameterization; singular cases need a specified restricted inverse or damping. The minus sign minimizes J; fitness maximization uses the opposite sign. The score-function calculation requires the same integration and support qualifications as before.

Thus NES adapts not merely a candidate parameter vector but the geometry of the mutation distribution. For a Gaussian mean m with covariance $\sigma^2 I$ fixed,

$$\nabla_m \log \pi_m(z) = \frac{z-m}{\sigma^2}, \qquad \mathcal{I}(m) = \sigma^{-2}I.$$

Thus a score estimator in *parameter-search space* can use only black-box values $f(z)$, even though it differentiates the known sampling density. Rank-transformed fitness and other practical utilities generally change the precise estimator; they should not be called unbiased for the original J without checking that claim.

17.12 CMA-ES and Adaptive Search Geometry

CMA-ES represents the search distribution as

$$z \sim \mathcal{N}(m, \sigma^2 C).$$

The covariance matrix

$$C$$

adapts to successful search directions [59].

A schematic covariance update is

$$C_{t+1} = (1 - c_1 - c_\mu)C_t + c_1 p_c p_c^\top + c_\mu \sum_{i=1}^{\mu} w_i y_i y_i^\top .$$

Here p_c is an evolution path and y_i are selected, normalized steps. With nonnegative weights, $\sum_i w_i = 1$, and $c_1, c_\mu \geq 0$, $c_1 + c_\mu \leq 1$, the displayed terms preserve positive semidefiniteness. Standard CMA-ES also specifies mean and step-size updates and path corrections; this schematic is not the full algorithm. Adaptation aims to stretch sampling along successful directions; it does not automatically identify a true Hessian or remove irrelevant coordinates [58].

17.12.1 Why CMA-ES reinforces the effective-dimension principle

A full covariance matrix in r dimensions contains

$$O(r^2)$$

entries.

Thus full CMA-ES is naturally a method for moderate-dimensional control spaces, not for direct covariance adaptation over billions of model weights.

For larger spaces, diagonal, low-rank, separable, or otherwise structured covariance approximations may be needed. Dense covariance sampling or updating has quadratic-scale work, and a dense factorization can have cubic cost when recomputed. Parallel model evaluations do not remove these central costs. Low-rank covariance components are useful only with a specified exploration component or a justified restricted support.

Again,

adaptive evolutionary search becomes practical after structure has reduced the search dimension.

17.13 ZO-Friendly Control Surfaces in LLM Systems

The most compelling near-term uses of zeroth-order optimization are often not necessarily full-weight LLM training. The ranges below are illustrative design budgets, not measured universal thresholds and not recommendations independent of noise, scale, alignment, and evaluation cost.

They are optimization problems surrounding a largely fixed model.

17.13.1 Very low dimension: decoding and system controls

For

$$r \sim 1 - 10,$$

examples include:

- temperature;
- nucleus-sampling thresholds;
- retrieval thresholds;
- ensemble weights;
- safety–helpfulness tradeoff parameters.

Such problems are natural candidates for direct search.

17.13.2 Low dimension: routing and mixtures

For

$$r \sim 10 - 10^2,$$

examples include:

- mixture-of-experts weights;
- domain-routing variables;
- reward-component weights;
- curriculum coefficients;
- data-mixture proportions.

CMA-ES, NES, or ordinary ZO gradient estimation may be appropriate.

17.13.3 Intermediate dimension: structured adapters

For

$$r \sim 10^2 - 10^4,$$

structured low-rank adapters or module-level perturbations may make zeroth-order search plausible, depending on query noise and parallel evaluation capacity.

17.13.4 Full parameter space

For

$$r = D, \qquad D \gg 10^6,$$

naive isotropic ZO suffers severely from search-dimensionality.
Successful demonstrations in very large models should therefore motivate the question:

What search geometry, curvature, noise coupling, or parameterization makes the objective improve at the measured resource budget?

rather than the conclusion that dimension has ceased to matter.

17.14 Memory-Efficient Zeroth-Order Fine-Tuning

A different motivation is memory rather than remote black-box access. MeZO regenerates random perturbations from a seed and updates accessible weights in place, avoiding the full backpropagation graph. Its source reports both full-parameter and parameter-efficient fine-tuning, and analyzes favorable local effective-rank curvature, rather than requiring an explicit low-dimensional adapter in every case [97].

The distinctions are

memory:	what must be stored during training;
gradient estimation:	how accurately a vector is reconstructed;
optimization:	how much the objective decreases per query.

A full-weight forward-only algorithm still needs to read, perturb, and restore the relevant weights. A dense adaptation map, optimizer state, model replicas, and long generation traces can add costs. Regenerating one direction from a seed does not make all those costs disappear.

An exact calculation separating variance from progress. The following elementary quadratic calculation is added here to show why the linear variance proposition is not a universal optimization lower bound. It illustrates, without reproducing the full MeZO theorem, the curvature-sensitive reasoning used in that work.

Proposition 28 (Full-space zeroth-order progress depends on curvature)**.** *Let $f : \mathbb{R}^D \to \mathbb{R}$ be a bounded-below quadratic with constant Hessian $Q \succeq 0$, $Q \neq 0$. At θ write $g = \nabla f(\theta)$. Average q independent noiseless Gaussian paired estimators in $\mathbb{R}^D$, and update $\theta^+ = \theta - \eta\widehat{g}_q$. Then, for every $\sigma > 0$,*

$$\mathbb{E}[f(\theta^+)] = f(\theta) - \eta\|g\|^2 + \frac{\eta^2}{2}\left[\left(1+\frac{1}{q}\right) g^\top Q g + \frac{\operatorname{tr} Q}{q}\|g\|^2\right].$$

Put $L_Q = \|Q\|_2$ and $r_{\mathrm{curv}} = \operatorname{tr} Q / L_Q$. For

$$0 < \eta \leq \frac{1}{L_Q[1 + (r_{\mathrm{curv}}+1)/q]},$$

we have

$$\mathbb{E}[f(\theta^+)] \leq f(\theta) - \frac{\eta}{2}\|g\|^2.$$

Proof. Symmetric finite differences cancel the quadratic term exactly, so $\widehat{g}(u) = uu^\top g$. Gaussian fourth moments give

$$\mathbb{E}[\widehat{g}\widehat{g}^\top] = \|g\|^2 I_D + 2gg^\top.$$

Averaging independent directions yields

$$\mathbb{E}[\widehat{g}_q\widehat{g}_q^\top] = \frac{\|g\|^2}{q} I_D + \left(1+\frac{1}{q}\right) gg^\top.$$

Substitute this and $\mathbb{E}\widehat{g}_q = g$ into the exact quadratic expansion of $f(\theta - \eta\widehat{g}_q)$. Finally use $g^\top Q g \leq L_Q\|g\|^2$ and the displayed step bound. □

Rank-one example. For $f(\theta) = \frac{1}{2}(\theta_1 - b)^2$, all D coordinates may be perturbed, but only the first affects the objective. The exact identity is

$$\frac{\mathbb{E}[f(\theta^+)]}{f(\theta)} = 1 - 2\eta + \eta^2\left(1 + \frac{2}{q}\right)$$

when $f(\theta) > 0$. Choosing $\eta = q/(q+2)$ gives a factor $2/(q+2)$, independent of D. Fresh directions at successive steps therefore give geometric decay in expected loss, although

$$\mathbb{E}\|\widehat{g}_q - g\|^2 = \frac{D+1}{q}\|g\|^2$$

can be large. The irrelevant-coordinate fluctuations do not cost loss in this example. This is not a claim of dimension-free weight storage, parameter control, or numerical implementation.

For $Q = L_Q I_D$, by contrast, $r_{\text{curv}} = D$ and the same calculation exhibits the familiar dimension penalty. For nonquadratic networks, a local low-rank Hessian at a single point is not sufficient: finite perturbations and updates must remain in a region with the claimed curvature control, and smoothing bias and oracle noise must be bounded. These qualifications are central to the local-geometry analysis in the MeZO paper, not consequences of pretraining alone.

The strengthened effective-dimension principle is therefore positive but conditional: explicit reduced coordinates are one tractable route; small effective curvature can be another. The two should be tested and reported separately.

17.15 Hybrid First-Order and Zeroth-Order Optimization

A useful practical architecture may combine both modes. This is an algorithmic design option, not a guarantee that either loop reaches a global optimum.

17.15.1 First-order inner loop

Use SGD, Adam, or another differentiable optimizer for ordinary model weights:

$$\theta \leftarrow \theta - \eta \nabla_\theta L.$$

17.15.2 Zeroth-order outer loop

Use ZO to tune a low-dimensional controller

$$z,$$

where

$$z$$

may specify:

- reward weights;
- curriculum;
- mutation modules;
- data composition;

- architecture gates;
- decoding policy;
- training schedules.

Then

$$\boxed{\text{first-order learning addresses the high-dimensional differentiable problem;}}$$

while

$$\boxed{\text{zeroth-order search addresses the low-dimensional black-box meta-problem.}}$$

This is related to population-based training [73], which co-adapts model states and hyperparameter schedules. It need not coincide with a fixed-objective finite-difference method.

For a defined outer objective one can write

$$F_{\text{outer}}(z) = \mathbb{E}_{\omega}\left[\mathcal{L}_{\text{eval}}(\text{Train}(\theta_0, z; \omega))\right].$$

One query now includes the specified training budget. Positive and negative candidates need comparable initial states, budgets, and randomness laws; otherwise their difference can reflect drift rather than the effect of z. Reward coefficients used *inside* training must be evaluated against a fixed external criterion. Changing both the candidate and its scoring rule does not optimize one common objective. Warm starts and adaptive schedules require their own stated oracle model.

17.16 Relation to Reinforcement-Learning Post-Training

The 2025 DeepSeek-R1 report provides an example of reward-driven reasoning post-training using GRPO-style reinforcement learning [55].

The comparison with evolutionary ZO methods should be made carefully.

17.16.1 Policy-gradient RL

The policy-gradient family samples behavior and uses the model's log-probability derivatives. This description does not cover every RL method.

It retains first-order access to the policy.

17.16.2 Evolution strategies

ES perturbs the model or a structured adaptation coordinate directly:

$$z \to z + \sigma u.$$

No derivative of the policy with respect to z is required. The policy may nevertheless remain stochastic within each evaluation; parameter perturbation noise and rollout noise are separate sources of variation.

Thus ES can optimize a truly black-box pipeline.

17.16.3 Neither dominates universally

Policy gradients use derivative information unavailable to a pure function-value oracle. Their actual variance and cost also depend on policy scores, exploration, horizon, and reward statistics; there is no unconditional ordering of efficiency. Return-conditioned supervised modeling in Chapter 12 is a third, different formulation and should not be identified with either of these estimators.

ZO may be preferable when:

- model derivatives are unavailable;
- backward memory dominates cost;
- the objective spans nondifferentiable system components;
- evaluations can be massively parallelized;
- or the effective adaptation dimension is very small.

17.17 Experiments Suggested by the Theory

The effective-dimension view suggests the following tests. No new results are claimed here. Separate exact estimator identities, conditional optimization predictions, and empirical post-training performance. Freeze the final test protocol; tune subspaces, smoothing scales, regularization, and stopping using training/validation resources only. Report all runs, paired seeds, confidence intervals, distinct data count, objective calls, generated tokens, model-evaluation work, peak memory, wall-clock time, and parallel hardware. Charge for learning the search space and for evaluation repetitions.

In particular, fix either coordinatewise σ or the RMS radius and state which. A fair algorithm comparison may tune steps separately on a common validation budget, rather than forcing one step size on differently scaled estimators.

17.17.1 Experiment 1: hold model size fixed and vary search dimension

Choose a fixed pretrained model with parameter dimension D.

Construct nested adaptation spaces

$$A_r \in \mathbb{R}^{D\times r}$$

for increasing

$$r.$$

At a fixed direction budget q per update and fixed total objective-call budget, measure both gradient-estimation error and objective improvement versus r.

The basic prediction is:

relative gradient-estimation MSE scales as $(r+1)/q$ in the stated noiseless linear test.

This directly tests Equation (17.5) on synthetic linear objectives. On neural tasks, increasing r also changes the best achievable fit and the restricted curvature, so performance need not vary monotonically with r/q. Include isotropic and rank-one quadratic controls to distinguish the two effects.

17.17.2 Experiment 2: structured versus random subspaces

At the same dimension r, compare:

- a random parameter subspace;
- a LoRA factor parameterization or a fixed adapter basis;
- layerwise module scales;
- learned principal adaptation directions;
- hand-designed compositional modules.

Equal searched dimension equalizes only one estimator factor; it does not equalize projected signal, curvature, admissible functions, or the cost of constructing the representation.

The stronger theory predicts otherwise:

$$\boxed{\text{small dimension} + \text{alignment with useful directions}}$$

is the hypothesis to test. Use first-order optimization in each restricted family as a controlled diagnostic of its attainable fit; count any gradient-based construction of the basis as a separate phase.

17.17.3 Experiment 3: full-weight versus projected ZO

Compare

$$u \in \mathbb{R}^D$$

with

$$u \in \mathbb{R}^r, \qquad r \ll D,$$

using the same number of objective evaluations. Measure the full-gradient error separately from objective decrease; a full-space method can have worse vector MSE yet comparable progress under low-effective-rank curvature. Do not compare different physical perturbation radii without reporting that difference.

Prediction:

$$\boxed{\begin{array}{c}\text{projected ZO should provide a much better direction estimate at fixed} \\ \text{query budget when the subspace is well aligned.}\end{array}}$$

17.17.4 Experiment 4: evolutionary genes

Partition or parameterize the network into coordinated mutation modules

$$G_1, \ldots, G_r.$$

Compare mutation of these modules against independent weight-space perturbations.

Candidate genes include:

- attention-head modules;
- MLP blocks;

- adapter directions;
- normalization scales;
- groups of related layers.

The hypothesis is that coordinated controls help when they cover useful adaptations with favorable conditioning. Compositionality alone does not predict domination: record the number of freely changed scalars per module, the representation cost, and behavior lost outside the target task. Include misaligned structured controls as a negative comparison.

17.17.5 Experiment 5: reward noise

Artificially vary the reward noise variance

$$\nu^2.$$

Equation (17.6) predicts degradation proportional to

$$\frac{\nu^2 r}{q\sigma^2}.$$

This is a direct estimator-variance test for independent homoscedastic noise, not a theorem that final accuracy degrades linearly by that factor. Compare independent versus shared-randomness evaluations and measure $\mathrm{Var}(Y_+ - Y_- \mid u)$ rather than assuming it.

17.17.6 Experiment 6: hybrid optimization

Train the model weights by ordinary gradient descent while evolving a small vector controlling:

$$\text{curriculum}, \quad \text{data mixture}, \quad \text{reward weights}, \quad \text{or routing}.$$

Compare with:

- fixed meta-parameters;
- gradient-based hyperparameter approximations;
- purely zeroth-order full-weight search.

The prediction is that ZO will be most competitive in the low-dimensional outer loop when full query cost is favorable. Evaluate all inner-loop training and search effort at a common compute budget; outer dimension alone does not make retraining cheap.

17.17.7 Experiment 7: estimate intrinsic adaptation dimension

Measure the smallest r for which a structured subspace reaches a fixed fraction of the performance achieved by unrestricted gradient fine-tuning.

Define, for example,

$$r_\alpha = \min \{r : \Delta R_r \geq \alpha \Delta R_{\text{full}}\}.$$

Here ΔR must denote improvement of a fixed validation *score* with higher values better (not an unqualified risk), $\Delta R_{\text{full}} > 0$, and $0 < \alpha \leq 1$. Fix a nested family, optimizer, budget, initialization, and evaluation distribution; use $r_\alpha = +\infty$ if the set is empty. Confidence intervals are needed near

the threshold. The resulting quantity is algorithm- and family-relative, not an intrinsic property inferred without qualifications.

With those conventions, this quantity provides an operational definition of

effective adaptation dimension.

17.18 Failure Modes

The framework identifies obstacles to the specified zeroth-order schemes, not impossibility theorems for every function-value method.

17.18.1 No low-dimensional adaptation manifold

If useful updates occupy a genuinely high-dimensional region,

$$r \approx D,$$

then that fixed reduced-space strategy offers little approximation advantage. Other exploitable structures, including low-effective-rank curvature for full-space progress, are separate possibilities.

17.18.2 Poor subspace alignment

A restricted search cannot attain improvements excluded by its family. A vanishing projected gradient can stall a local projected method even when the full gradient is nonzero. A large omitted component, however, is not sufficient to prove that useful restricted improvement fails, as the example following Equation (17.9) shows.

17.18.3 Highly noisy reward

Large independent evaluation variance increases the estimator cost at fixed perturbation scale. Correlated pairs, replication, or a different query design can change that cost; the oracle model must be measured.

17.18.4 Rare-event objectives

If almost every perturbation receives the same reward and informative events are extremely rare, local finite-difference estimation may carry little signal. If a single oracle call yields an informative event with conditional probability at most p throughout the queried region, then Q_{calls} adaptive calls see one with probability at most $Q_{\mathrm{calls}}p$ by a union bound. Small dimension does not remove an exponentially small event probability.

17.18.5 Excessively small perturbations

If

$$\sigma$$

is too small relative to evaluation noise, the numerator

$$f(\theta + \sigma u) - f(\theta - \sigma u)$$

is noise dominated.

17.18.6 Excessively large perturbations

If

$$\sigma$$

is too large, the method optimizes a heavily smoothed objective and may leave the local behavioral regime of the pretrained model.

17.18.7 Off-manifold parameter perturbations

Random full-weight mutations may destroy learned representations before revealing useful fitness differences.

Structured mutation spaces can reduce this problem.

17.19 Connection to Sparse Compositionality

Chapter 4 establishes that feasible finite-precision computation has a bounded-fan-in implementation. Its smooth approximation counterpart requires the separate regularity and stability conditions. Neither result says that adaptation of an arbitrary pretrained model is a low-dimensional optimization problem.

Here a related design question arises in adaptation space.

A network may contain billions of microscopic parameters, but the useful degrees of freedom for a particular adaptation may correspond to only a small number of functional constituents.

Thus

$$\boxed{\text{large parameter count} \not\Longrightarrow \text{large effective adaptation dimension.}}$$

If a useful modification consists of changing

$$r$$

modules, module-level search is a candidate. Its scalar dimension is the sum of their independent controls, not necessarily the number of modules. Shared controls can reduce that sum; discovering useful controls is an additional representation-learning problem.

17.19.1 C1 and zeroth-order search

C1 says that the target computation admits a compact structured representation.

For zeroth-order adaptation we need an analogous property:

$$\boxed{\begin{array}{c}\text{the desired change to the trained system admits a compact structured} \\ \text{parameterization.}\end{array}}$$

This does not follow automatically from C1. The target and the frozen model might each be feasible while their difference is difficult to express in the admitted adapter family. Nor must an effective adapter recover the teacher's particular constituents. Chapter H requires predictive performance and feasible evaluation, not a privileged structural matching.

17.19.2 Genericity still matters

Reducing the search dimension does not guarantee that fitness differences are detectable.

If mutations produce exponentially weak changes in reward, ZO search can still fail.

Thus the analogue of C2 remains:

useful mutations must leave detectable fitness footprints.

Again the division of labor is clear:

structure	:	reduces the search dimension,
genericity	:	makes fitness differences detectable,
zeroth-order optimization	:	uses those evaluations to select a direction.

Optimization queries are not independent training examples. The same adaptation sample can be queried many times. That can reduce Monte Carlo error in estimating its empirical objective without adding new population information. Conversely, searching over a small number of unbounded real coordinates is not automatically a small statistical class.

For example, fix the base system and decoder independently of an i.i.d. evaluation sample S of size n, and admit at most 2^{br} quantized control strings, each with loss in $[0, 1]$. The finite-class argument of Chapter 13 gives, simultaneously over these strings,

$$|R(z) - \widehat{R}_S(z)| \le \sqrt{\frac{br \ln 2 + \ln(2/\delta)}{2n}}$$

with probability at least $1 - \delta$. Adaptive zeroth-order search may select any member on this event. A base model or dictionary selected on the same sample must be accounted for rather than treated as free. Quantization must also preserve the required performance, or the bound certifies only the decoded predictor actually evaluated.

Thus a stopped iterate can be valid without convergence to a single absorbing parameter, but it must achieve useful empirical performance and statistical control. Forward-only optimization, small dimension, and low curvature do not by themselves supply these premises. The regularizer must be included in the stated objective, and favorable norm or rank observations must be connected to an applicable class or coding bound, as in Chapters 7, H, and 14.

17.20 Evolution as Structured Search

The evolutionary analogy is most useful as a proposal for organizing variation. Selection acts through realized reproductive outcomes rather than an explicit backpropagation calculation. A genotype-to-phenotype map can couple many microscopic changes, just as a computational decoder can map one control vector into a large parameter configuration.

The artificial linear model is

$$\theta = \theta_0 + Az,$$

while a nonlinear generator would be $\theta = \Phi(z)$. Neither the existence of such a map nor a numerical contrast between genotype and phenotype dimension proves efficient evolution. Fitness can be sparse, noisy, epistatic, or changing, and a small genotype need not be aligned with useful variation. These are limitations of the analogy, not claims resolved by the gradient-estimation proposition.

The constructive lesson for post-training is

design or learn a controlled language of coordinated variations, then test its expressivity, fitness signal, and search cost.

Discovering that language is part of the scientific problem. As in Chapter 16, one must distinguish a useful optimizer geometry once available from a theorem explaining how learning creates it.

17.21 Conclusion

Zeroth-order optimization uses objective values in place of derivatives of the evaluated system. This can be valuable for accessible black-box controls, memory-limited tuning, and outer-loop system objectives. A nondifferentiable reward alone does not require it: policy-gradient methods differentiate the behavior law rather than the reward.

The exact linear calculation remains a central result:

$$\mathbb{E}\|\widehat{g}_q - g\|^2 = \frac{r+1}{q}\|g\|^2.$$

For q independent Gaussian direction pairs, full-gradient reconstruction becomes harder with the number of searched coordinates. Independent function-value noise adds $\nu^2 r/(2q\sigma^2)$. Pairing, perturbation radius, numerical accuracy, and actual evaluation cost all matter.

An explicit constructive response is

$$\boxed{\theta = \theta_0 + Az, \qquad z \in \mathbb{R}^r, \qquad r \ll D.}$$

The map must cover a useful adaptation at acceptable cost. LoRA rank is not its factor-coordinate count; a nonlinear decoder is not a fixed linear subspace; and supplying or learning the decoder has a price. A projected method seeks a restricted solution, not necessarily a small full gradient or a reconstruction of the teacher's latent computation.

There is also a distinct positive route. The quadratic proposition shows that full-space zeroth-order *progress* can depend on effective curvature rank rather than D, even while Euclidean gradient MSE grows with D. This explains why the variance identity is not a universal lower bound on every successful post-training trajectory. Extending such progress to nonlinear models requires quantitative assumptions on the region explored, noise, and conditioning.

The book's recurring distinctions therefore remain intact:

feasible representation of a useful adaptation
$\neq$ detectable fitness signal
$\neq$ efficient optimization with the available oracle
$\neq$ population performance of the returned system.

Structured mutations offer a useful computational analogue of regulatory variables: one control can act on many weights. That analogy is a design principle, not a proof of evolutionary efficiency. The seven proposed experiments separate representation quality, estimator variance, curvature-sensitive progress, and held-out prediction.

The final lesson is not that dimension has ceased to matter. It is that we must identify *which* dimension governs the required result, and show that the admitted representation or geometry makes the actual search, evaluation, and statistical costs manageable.

Part IV
Speculations

CHAPTER 18

Sparse Compositionality, Genericity, and the Random Hierarchy Model

The Random Hierarchy Model (RHM) introduced by Cagnetta and collaborators provides a particularly useful test bed for sparse compositionality and accessible learning signal. A fixed random grammar generates a hierarchy of latent features, with several synonymous realizations of each feature. In the feature-learning regimes studied in the original work, the characteristic sample size for classification scales as

$$P^* \simeq n_c m^L,$$

where n_c is the number of classes, m is the number of synonymous production rules per parent, and L is the hierarchy depth [25].

A central result of that work is that this scale also emerges when class–patch correlations become detectable. The original paper already connects the correlation calculation to synonymic invariance and to a simplified one-step gradient calculation. The contribution of this chapter is to place that mechanism within the book's distinction between representation, detectable information, actual learning updates, and predictive success.

The signal must be compared with its own sampling noise. If a statistic Z has mean a and standard deviation $\tau > 0$, define the normalized footprint $\rho = a/\tau$. Its empirical mean from P independent examples has signal-to-noise ratio $|\rho|\sqrt{P}$. In the RHM scaling regime, the relevant characteristic normalized magnitude behaves as

$$\rho_{\mathrm{rms}} \asymp (n_c m^L)^{-1/2}, \qquad P_{\mathrm{detect}} \asymp \rho_{\mathrm{rms}}^{-2} \asymp n_c m^L.$$

This statement is not a scaling law for every unnormalized covariance. Rare patches and rare classes affect both the raw signal and its variance.

The resulting interpretation is constructive but scoped:

> *a locally decodable hierarchy supplies a compact representation;*
> *quenched fluctuations can supply accessible statistical contrasts;*
> *enough examples resolve those contrasts;*
> *a specified learning mechanism can turn them into useful coordinates.*

Detecting one contrast is not the same as separating every hidden feature, and neither is a general convergence theorem for end-to-end SGD. Furthermore, a predictor need not recover the generating grammar to succeed. These distinctions let the RHM strengthen the book's positive account without assigning it conclusions beyond its assumptions.

Conventions and provenance. $\mathcal{G}$ denotes one randomly chosen grammar. Conditional on $\mathcal{G}$, probabilities and expectations concern examples generated from that fixed grammar. $\mathbb{E}_{\mathcal{G}}$ instead averages over grammars. P denotes the number of examples, not their probability law. We use $s \geq 2$ for the branching factor, v for the common vocabulary size, $n_c \geq 2$ for the class count, and $m \geq 2$ for multiplicity. A depth-L input has $d = s^L$ symbol positions; a one-hot representation has dv scalar entries. All asymptotic statements specify a regime; $\asymp$ suppresses constants only within that regime. The cited RHM results, the simplified counting approximation, and the elementary propositions proved here are distinguished explicitly.

18.1 Why the Random Hierarchy Model Matters

The preceding chapters separate feasible representation from learning. C1 asks whether a target admits an economical compositional realization. The C2 question of Chapter 6 asks which useful statistics the specified learner can access, and at what sample and computational cost. The RHM makes both questions concrete: its hidden grammar is explicit to the experimenter, but not to a learner given only labeled inputs.

It therefore allows three different measurements:

prediction on new examples,
invariance of learned features under synonym substitutions,
correspondence with the particular generating grammar.

They can be related experimentally without being identified. In particular, Chapter H does not require the last property for the first. The RHM is valuable because it permits a mechanism of representation discovery to be studied in detail, not because every successful learner must implement that mechanism.

18.2 The Random Hierarchy Model in Minimal Form

We retain the uniform, nonambiguous construction of [25]. Let $\mathcal{V}_1, \dots, \mathcal{V}_L$ be vocabularies of size v, and put $\mathcal{V}_{L+1} = \{1, \dots, n_c\}$. For each parent $a \in \mathcal{V}_{\ell+1}$, a set

$$\mathcal{S}_\ell(a) \subseteq \mathcal{V}_\ell^s, \qquad |\mathcal{S}_\ell(a)| = m,$$

specifies its synonymous child tuples. Sets belonging to different parents at the same level are disjoint. Thus

$$m|\mathcal{V}_{\ell+1}| \leq |\mathcal{V}_\ell|^s$$

is required: for homogeneous internal vocabularies, $m \leq v^{s-1}$, while the root also requires $mn_c \leq v^s$.

The grammar is chosen uniformly from the permitted assignments and then held fixed. Draw a uniform class Y, expand each parent by a uniform choice from its synonym set, and repeat for L generations. Choices at distinct tree nodes are independent conditional on the grammar and the parent states. The same level-specific rule table is reused across positions. The final observation is the ordered list of leaves X.

The number of internal nodes in a complete tree is $(d-1)/(s-1)$. Nonambiguity implies that the number of distinct examples per class is

$$m^{(d-1)/(s-1)}, \qquad P_{\max} = n_c m^{(d-1)/(s-1)}.$$

This total is different from m^L, which counts choices along a fixed ancestral path. Confusing them would conceal the main sample-efficiency comparison.

18.2.1 The characteristic sample scale

The original work reports, in its studied feature-learning regimes,

$$P^* \sim n_c m^L. \tag{18.1}$$

Here n_c is the number of classes, not an unspecified combinatorial factor. The empirical P^* is defined by the test error falling below one tenth of the random-guess error $1 - 1/n_c$. Its dependence is an observed/asymptotically analyzed scale for specified architectures and training protocols, not an architecture-independent minimax theorem [25].

The paper samples distinct examples without replacement from the finite population. Our elementary concentration propositions below use i.i.d. examples conditional on the fixed grammar; we identify where this changes the variance. Neither convention makes the many patches within one example independent observations.

18.3 C1: When the Hierarchical Classifier Is Compositionally Sparse

The RHM is specified in the generative direction $Y \to X$, whereas classification uses $X \to Y$. A sparse generative mechanism does not in general provide an equally simple inverse. In this particular RHM, however, the no-ambiguity assumption supplies the needed local decoder:

$$g_\ell(u) = a \quad \text{when } u \in \mathcal{S}_\ell(a).$$

This is well defined on the admissible child tuples. Values on invalid tuples can be assigned an error symbol; they are not part of the noiseless classification task.

18.3.1 Generative sparsity does not by itself imply inverse sparsity

Proposition 29 (Hierarchical decoding under local identifiability)**.** *Suppose a depth-L hierarchy has local arity s, and the parent at every level is recovered from its admissible children by a function*

$$g_\ell : \mathcal{Z}_{\ell-1}^{s} \longrightarrow \mathcal{Z}_\ell.$$

Suppose the decoded root determines the class. Then the inverse classifier on the generated support has a depth-L, fan-in-s symbol-level realization, followed, if needed, by the root readout. For a complete s-ary tree with $d = s^L$ leaves, it uses $(d-1)/(s-1)$ local decoding operations.

Proof. Apply g_1 to each disjoint bottom-level tuple. By local identifiability, these outputs equal the true parents. Apply g_2 to their sibling groups and continue. Induction recovers each successive latent level and finally the root. The tree-node count follows by summing $1 + s + \cdots + s^{L-1} = (d-1)/(s-1)$. □

This is an economical realization when its local alphabets and rule representations are controlled. A supplied rule-table representation has on the order of $m((L-1)v+n_c)$ tuple entries, each carrying s child addresses and a parent address. For growing vocabularies, these addresses cost bits and a lookup costs computation. Compiling the finite lookup into Boolean gates adds the corresponding encoding and evaluation costs; it need not preserve depth exactly L at the bit-operation level.

Thus fixed symbol arity is not automatically fixed scalar dimension of a one-hot module, nor a uniform Sobolev bound for a smooth lift. The representation conclusion is the finite-alphabet C1 statement of Chapter 4. Discovering the rule tables remains a separate learning task.

18.4 C2: Randomness as a Source of Genericity

The distinction between averaging over grammars and learning from one grammar is essential. All train and test examples in a learning problem must use the same grammar; otherwise a different statistical problem is being studied.

18.4.1 Annealed symmetry

An ensemble invariant under class relabeling can have

$$\mathbb{E}_{\mathcal{G}} C_{\alpha\mu}(\mathcal{G}) = 0$$

for a class-centered observable and a patch indexed without favoring a class. This is an annealed statement. It does not assert that the corresponding covariance vanishes conditional on every grammar.

18.4.2 Quenched asymmetry

For a fixed realization, random imbalances in production rules can produce $C_{\alpha\mu}(\mathcal{G}) \neq 0$. These are quenched fluctuations: they belong to the fixed population distribution and do not disappear when more examples from that distribution are collected. Sampling fluctuations of an empirical estimator are different and do decrease with P.

However, randomness alone is not a noncancellation theorem. A finite random grammar has a discrete distribution; exact balance, zero-probability patches, or repeated statistical signatures can occur. Positive ensemble variance demonstrates that some realizations have a signal. A claim about most realizations requires an additional probability estimate.

ensemble symmetry
$\neq$ absence of quenched signal
$\neq$ a high-probability lower bound on that signal.

Randomness is one mechanism for genericity, not its definition and not a substitute for quantitative control.

18.5 A Precise Low-Order Footprint

Fix a bottom-level patch position j and an admissible ordered tuple μ. Let M_j be the tuple at that position, and define

$$X_\mu = \mathbf{1}\{M_j = \mu\}, \qquad q_\mu = \Pr(M_j = \mu), \qquad \pi_\alpha = \Pr(Y = \alpha) = \frac{1}{n_c}.$$

Suppress j and $\mathcal{G}$ from the notation, but keep both fixed. For the centered indicators $\phi_\alpha(Y) = \mathbf{1}\{Y = \alpha\} - \pi_\alpha$ and $\psi_\mu(X) = X_\mu - q_\mu$, set

$$C_{\alpha\mu} = \mathbb{E}[\phi_\alpha(Y)\psi_\mu(X)]. \tag{18.2}$$

This is second order in these indicator variables. A tuple indicator is itself a product of s single-symbol indicators, so "second order" does not mean a quadratic polynomial in every original input encoding. The lifting and its computational cost are part of the probe family.

For $q_\mu > 0$, write

$$p_{\alpha|\mu} = \Pr(Y = \alpha \mid M_j = \mu), \qquad b_{\alpha\mu} = p_{\alpha|\mu} - \pi_\alpha.$$

Directly expanding the covariance gives

$$\boxed{C_{\alpha\mu} = q_\mu b_{\alpha\mu}.}$$

The conditional-probability contrast $b_{\alpha\mu}$ and the raw covariance $C_{\alpha\mu}$ are different quantities.

Because the class prior is known in this model, an observable statistic with this mean is

$$Z_{\alpha\mu} = (\mathbf{1}\{Y = \alpha\} - \pi_\alpha)\mathbf{1}\{M_j = \mu\}, \qquad \mathbb{E}Z_{\alpha\mu} = C_{\alpha\mu}.$$

No knowledge of the unknown q_μ is required to average Z. Its single-example variance is exactly

$$\begin{aligned}\tau^2_{\alpha\mu} = q_\mu[p_{\alpha|\mu}(1-\pi_\alpha)^2 + (1-p_{\alpha|\mu})\pi^2_\alpha] \\ - q^2_\mu(p_{\alpha|\mu} - \pi_\alpha)^2.\end{aligned}$$

In a weak-contrast regime, $p_{\alpha|\mu}$ is close to π_α, and $\tau^2_{\alpha\mu} \simeq q_\mu\pi_\alpha(1-\pi_\alpha)$. The relevant normalized footprint is

$$\rho_{\alpha\mu} = \frac{C_{\alpha\mu}}{\tau_{\alpha\mu}}, \qquad \tau_{\alpha\mu} > 0.$$

Multiplying Z by a nonzero constant changes its mean and standard deviation together and leaves $|\rho|$ unchanged. Normalization is statistical accounting, not a way of creating information.

18.5.1 RMS footprint across hierarchy realizations

With the indexing or admissible-patch selection convention fixed, define

$$C_{\mathrm{rms}} = (\mathbb{E}_{\mathcal{G}} C^2_{\alpha\mu})^{1/2}, \qquad \rho_{\mathrm{rms}} = \left(\mathbb{E}_{\mathcal{G}} \frac{C^2_{\alpha\mu}}{\tau^2_{\alpha\mu}}\right)^{1/2}. \tag{18.3}$$

For this definition restrict to the specified positive-variance probes. The mean of a ratio need not equal the ratio of means. Replacing ρ_{rms} by $C_{\mathrm{rms}}/\tau_{\mathrm{typ}}$ requires concentration or uniform comparability of the variances.

An RMS scale is not a high-probability lower bound. For example, an abstract signed signal that equals $+1$ or -1 with probability $1/(2N)$ each and is zero otherwise has RMS $N^{-1/2}$, but vanishes with probability $1 - 1/N$. This example is not asserted to be the RHM law. It explains why a second moment alone cannot justify "typically nonzero." If $\mathbb{E}\rho^4 \le K(\mathbb{E}\rho^2)^2$, the Paley–Zygmund estimate instead gives

$$\Pr_{\mathcal{G}}(|\rho| \ge a\rho_{\mathrm{rms}}) \ge \frac{(1-a^2)^2}{K}, \qquad 0 < a < 1.$$

To see the estimate, split the expectation of $V = \rho^2$ at $a^2\mathbb{E}V$ and apply Cauchy–Schwarz to the upper part. Even this is a constant-probability statement unless stronger information is supplied. A theorem valid for most grammars must control small signals and collisions of signatures as well.

18.6 Random Finite-Size Fluctuations

The counting intuition is useful, provided absolute and relative fluctuations are not confused.

Lemma 6 (Absolute and relative fluctuations of a binomial frequency)**.** *Let* $N_{\alpha\mu} \sim \mathrm{Binomial}(N, p)$ *with* $0 < p < 1$*. Then*

$$\mathbb{E}\frac{N_{\alpha\mu}}{N} = p, \qquad \mathrm{sd}\left(\frac{N_{\alpha\mu}}{N}\right) = \sqrt{\frac{p(1-p)}{N}}.$$

The standard deviation divided by the mean is

$$\sqrt{\frac{1-p}{Np}}.$$

Thus both fluctuations have order $N^{-1/2}$ *only when* p *stays bounded away from zero and one. When* p *decreases, the expected count* Np *controls the relative fluctuation.*

Proof. Write $N_{\alpha\mu}$ as the sum of N independent Bernoulli variables. Its mean is Np and its variance is $Np(1-p)$. Divide by N^2 for the frequency variance and then by p^2 for its relative variance. □

18.6.1 Application to the hierarchy

The signal-versus-noise argument in the original RHM paper compares two counts, not one binomial with fixed success probability. For a fixed class and a fixed patch position there are m^L ancestral choices and approximately mv admissible bottom-level tuples. The mean number of paths reaching one tuple is therefore $m^L/(mv)$. The simplified independent-path approximation in the paper predicts relative class-conditional fluctuations of order

$$\sqrt{\frac{mv}{m^L}}.$$

When denominator fluctuations are subleading, the absolute fluctuation of $p_{\alpha|\mu}$ around $1/n_c$ has scale

$$\mathrm{sd}_{\mathcal{G}}(p_{\alpha|\mu}) \asymp \frac{1}{n_c}\sqrt{\frac{mv}{m^L}}.$$

These are the paper's scaling estimates for its large-parameter regime; its appendix treats the rule dependencies more carefully [25]. They are not obtained by assuming all production paths of the actual grammar are independent. In particular, the counting approximation should not be extrapolated to arbitrary L, m, v or to rare patches without checking its assumptions.

In the corresponding approximately uniform-patch, weak-contrast regime, $q_\mu \asymp 1/(mv)$ and $\tau^2_{\alpha\mu} \asymp 1/(n_c mv)$. Consequently the normalization calculation gives

$$\boxed{C_{\mathrm{rms}} \asymp \frac{1}{n_c\sqrt{mv\, m^L}}, \qquad \rho_{\mathrm{rms}} \asymp \frac{1}{\sqrt{n_c m^L}}.} \tag{18.4}$$

The covariance and normalized footprint therefore have different parameter dependence. The cancellation of v occurs in the signal-to-noise ratio, not in the raw covariance itself. These characteristic estimates do not provide a uniform lower bound for every class–patch pair or a minimum separation between every pair of latent features.

18.7 Detectability from Finite Data

Fix the grammar and one statistic Z with mean a and variance $\tau^2 \in (0, \infty)$. For independent examples,

$$\widehat{a}_P = \frac{1}{P}\sum_{i=1}^{P} Z_i, \qquad \mathrm{Var}(\widehat{a}_P) = \frac{\tau^2}{P}.$$

The ratio of population signal to sampling standard deviation is exactly

$$\mathrm{SNR} = \frac{|a|\sqrt{P}}{\tau} = |\rho|\sqrt{P}, \qquad \rho = \frac{a}{\tau}. \tag{18.5}$$

For a characteristic RHM probe obeying the preceding scale estimates,

$$\mathrm{SNR} \asymp \sqrt{\frac{P}{n_c m^L}}. \tag{18.6}$$

Thus the constant-SNR crossover is

$$\boxed{P_{\mathrm{detect}} \asymp n_c m^L.} \tag{18.7}$$

This reproduces the cited signal/noise balance without changing the normalization of the statistic midway through the argument.

The same calculation can be made using conditional frequencies. About $Pq_\mu \simeq P/(mv)$ examples contain the patch. Conditional on that count, the empirical class fraction has variance $p_{\alpha|\mu}(1-p_{\alpha|\mu})/(Pq_\mu)$, to leading order in the random count. Near a uniform class distribution, its sampling standard deviation is of order $\sqrt{mv/(n_c P)}$. Dividing the signal $n_c^{-1}\sqrt{mv/m^L}$ by this noise again gives $\sqrt{P/(n_c m^L)}$. The class-frequency noise is not uniformly $P^{-1/2}$ with a parameter-independent coefficient.

Sampling conventions. For a scalar statistic on a finite population of $N_{\mathrm{pop}} > 1$ examples, with variance τ^2 defined using denominator N_{pop}, uniform sampling of P distinct examples gives

$$\mathrm{Var}(\widehat{a}_P) = \frac{\tau^2}{P}\,\frac{N_{\mathrm{pop}} - P}{N_{\mathrm{pop}} - 1}.$$

The i.i.d. approximation is appropriate when $P \ll N_{\mathrm{pop}}$; near complete enumeration it is not. If several patches per example are averaged, their within-example covariances must also be included. Merely multiplying the sample size by the patch count is not justified.

18.8 The General Detectability Formula

The inverse-square law is best stated for a noise-normalized footprint. For a specified statistic with mean a_N, variance $\tau_N^2 > 0$, and $\rho_N = a_N/\tau_N$, the constant-SNR scale is

$$\boxed{P_{\mathrm{detect}} \asymp \frac{\tau_N^2}{a_N^2} = \frac{1}{\rho_N^2}.} \tag{18.8}$$

This is a sample-mean resolution scale. It is not, without a testing model and a lower-bound argument, a necessary number of observations for every estimator or learning algorithm.

Proposition 30 (A finite-sample guarantee for a specified footprint)**.** *Let $Z_1, \dots, Z_P$ be i.i.d. with mean $a \neq 0$ and variance $\tau^2 < \infty$. Then*

$$\Pr\left(|\widehat{a}_P - a| \geq \frac{|a|}{2}\right) \leq \frac{4\tau^2}{Pa^2}.$$

If in addition $|Z - a| \leq b$ almost surely, then

$$\Pr\left(|\widehat{a}_P - a| \geq \frac{|a|}{2}\right) \leq 2\exp\left[-\frac{Pa^2}{8\tau^2 + (4/3)b|a|}\right].$$

In particular, logarithmic confidence dependence is available under this boundedness assumption, with sufficient sample size

$$P \geq \left(\frac{8\tau^2}{a^2} + \frac{4b}{3|a|}\right)\ln\frac{2}{\delta}.$$

On the complementary event the empirical sign is correct and the estimation error is less than half the signal magnitude.

Proof. The first inequality is Chebyshev's inequality applied to $\mathrm{Var}(\widehat{a}_P) = \tau^2/P$. For completeness, a Bernstein argument gives the second. With $W = Z - a$, for $k \geq 2$, $|\mathbb{E}W^k| \leq b^{k-2}\tau^2$. Since $k! \geq 2\,3^{k-2}$, the exponential series implies, for $0 < t < 3/b$,

$$\mathbb{E}e^{tW} \leq \exp\left(\frac{t^2\tau^2}{2(1 - tb/3)}\right).$$

Independence, exponential Markov inequality, and the choice $t = u/(\tau^2 + bu/3)$ give

$$\Pr(|\widehat{a}_P - a| \geq u) \leq 2\exp\left[-\frac{Pu^2}{2\tau^2 + (2/3)bu}\right].$$

Apply this with $u = |a|/2$. The zero-variance case is deterministic and can be handled separately. □

This elementary proposition supplies a sufficient statistical statement. A practical test of an unknown signal can compare the empirical mean with a valid confidence radius. Its design requires a variance bound or a valid variance estimate; it cannot use the unknown population signal as an oracle. Finite variance alone gives the displayed Chebyshev result, not exponential concentration for the ordinary empirical mean.

18.8.1 Searching over many footprints

Suppose a finite family Φ of probes is fixed before the sample is seen. Normalize each statistic by a fixed, positive population standard deviation (or use comparable certified scales). Let its normalized mean be ρ_ϕ and suppose all normalized centered summands are bounded in absolute value by $b_\star$. Put $\rho_\star = \max_{\phi\in\Phi}|\rho_\phi| > 0$. A union bound using the preceding concentration estimate gives the sufficient screening condition

$$\boxed{P \geq C_0\left(\rho_\star^{-2} + \frac{b_\star}{\rho_\star}\right)\ln\frac{2|\Phi|}{\delta}.} \tag{18.9}$$

Here C_0 is a universal constant. For example, take uniform estimation error at most $\rho_\star/4$; the probe of largest empirical absolute mean then has population magnitude at least $\rho_\star/2$. The range-dependent term cannot always be suppressed for rare-event probes. Unknown normalization must be estimated or bounded with its uncertainty accounted for.

Statistical and computational search costs are different. An exponentially large $|\Phi|$ can still have a polynomial logarithm, while evaluating all its probes is exponentially expensive. Conversely, a small family is not enough if its best normalized signal is too weak.

The fixed-family event allows data-dependent selection within that family. If training creates a new representation or probe family using the same sample, a fresh-sample, conditional, or suitable uniform argument is needed. Reusing the sample is not equivalent to receiving new independent examples at each training epoch.

18.9 Relation to the RHM Threshold

The agreement between the correlation-resolution scale P_c and the empirical feature-learning scale P^* is a central conclusion of the original RHM work, not an independent mechanism first derived in this chapter. Cagnetta and collaborators analyze the class probabilities conditioned on a patch, connect their sampling resolution to synonymic invariance, and study a simplified gradient step [25]. The genericity language organizes these results as an instance of accessible learning signal.

Three qualifications make this connection precise. First, P_c is the scale of a specified correlation statistic, while P^* is defined by prediction under a specified learning protocol. Second, an RMS signal is not a uniform separation between all latent features. Third, the statistical observable must actually appear in an update direction that the architecture can use.

$$\boxed{\text{resolved statistic} \longrightarrow \text{usable representation update}}$$

is an additional link, not an automatic consequence of the SNR formula.

There is also a useful positive comparison in the source: in its maximal regime $n_c = v$, $m = v^{s-1}$, an explicit clustering procedure can exploit the vector of class statistics and reach a scale of order $\sqrt{n_c}\, m^L$, rather than $n_c m^L$; see the published paper's Sec. IV.C and Appendix D [25]. This is a result for that procedure and setting, not a claim for every end-to-end network. It directly warns against interpreting the scalar-resolution scale as a universal information-theoretic lower bound.

18.10 Depth, Input Size, and the Meaning of Exponential Scaling

For fixed $m > 1$, m^L is exponential in depth. But when $d = s^L$,

$$m^L = d^{\log_s m}, \qquad P^* \asymp n_c d^{\log_s m}.$$

For fixed s, m, n_c, this is polynomial in the number of input positions. It is dramatically smaller than $P_{\max} = n_c m^{(d-1)/(s-1)}$, which is exponential in d. The comparison is one of the principal attractions of the RHM.

A polynomial assertion is relative to the parameters held fixed. If m increases with d, the exponent $\log_s m$ need not remain constant. For one-hot inputs, vocabulary size also enters the raw dimension dv, the size of a tuple lifting v^s, and the description and evaluation costs of the grammar. A scale polynomial in leaf count need not be uniformly inexpensive in all of v, m, s, L.

The favorable dependence is therefore not obtained by calling depth small, or by treating a large finite alphabet as a cost-free elementary operation. It results from exploiting a repeated local generative structure under the specified parameter regime.

18.11 The Staircase Interpretation

The following stages describe a possible constructive organization of learning. They do not assert that every training trajectory follows strictly separated temporal phases. In fact, the published RHM synonym-sensitivity experiments compare trained networks as P changes; Fig. 6 shows invariances at different levels emerging at a common characteristic sample size. That observation must not be rewritten as a measurement of successive times during one training run [25].

18.11.1 Stage 1: a weak footprint becomes detectable

Initially the learner has the input encoding and the features produced by its initialization, not only coordinatewise linear statistics. Which class–patch moments it can use depends on that architecture. For an accessible normalized footprint of size $|\rho_0| \asymp (n_c m^L)^{-1/2}$, its empirical resolution improves when $P|\rho_0|^2$ becomes appreciable. This describes a statistical opportunity, not yet a complete learning step.

An explicit gradient connection. The source studies a two-layer ReLU model on a one-hot encoding of an entire patch, with fixed random readout weights and a simplified initialization [25]. The following finite-width, exactly balanced variation makes the relevant derivative transparent. It is an illustrative modification of that construction, not a claim about arbitrary initialization.

Let $e_\mu \in \mathbb{R}^{v^s}$ encode a tuple and let

$$F_w(e_\mu) = \frac{1}{H}\sum_{h=1}^{H} a_h\,\sigma(w_{h\mu}), \qquad a_h \in \mathbb{R}^{n_c}, \qquad \sigma(t) = \max\{0, t\}.$$

Fix the readout vectors in pairs $a, -a$, so that $\sum_h a_h = 0$, and initialize $w_{h\mu} = 1$ for every h, μ. The initial class logits are then exactly zero at finite width. For empirical cross-entropy, one ordinary gradient step of size η gives

$$w^+_{h\mu} - 1 = \frac{\eta}{H}\sum_{\alpha=1}^{n_c} a_{h\alpha}\widehat{C}_{\alpha\mu},$$
$$\widehat{C}_{\alpha\mu} = \frac{1}{P}\sum_{i=1}^{P}\left(\mathbf{1}\{Y_i = \alpha\} - \frac{1}{n_c}\right)\mathbf{1}\{M_j(X_i) = \mu\}.$$

Indeed, the derivative of the cross-entropy with respect to a zero logit is $1/n_c - \mathbf{1}\{Y_i = \alpha\}$, and the derivative of ReLU at the initialized preactivation 1 is 1. Thus the covariance is not an arbitrarily chosen useful probe: it is present in the specified gradient. Take η small enough to keep the displayed updated preactivations positive when interpreting them as post-ReLU features.

For example, choosing the $2n_c$ readouts $e_\alpha, -e_\alpha$ makes the linear map from the covariance vector to the hidden increment injective. This uses known class coordinates, not supplied latent-parent labels. The construction pays for the tuple lift and for Hv^s first-layer parameters; it does not prove that a network on the unlifted inputs learns that lift for free. At this first step the feature increment is linear in supplied tuple indicators. The calculation demonstrates gradient access and local invariance, not a separation from every fixed-feature method or a depth advantage by itself.

18.11.2 Stage 2: synonymous observations are quotiented

Let U_j denote the parent of the bottom patch. For $\mu \in \mathcal{S}_1(u)$, uniform independent synonym choice gives

$$\Pr(Y = \alpha, M_j = \mu) = \frac{1}{m} \Pr(Y = \alpha, U_j = u), \qquad q_\mu = \frac{1}{m} \Pr(U_j = u).$$

Hence synonyms have identical population covariance vectors $\mathbf{C}_\mu = (C_{\alpha\mu})_\alpha$ and identical conditional class profiles $(p_{\alpha|\mu})_\alpha$. The population version of the explicit gradient step therefore maps synonyms to the same hidden representation. With finite samples, their differences reflect estimation error.

This establishes invariance for that local construction. It does not ensure that distinct parents receive distinct codes: their covariance or conditional-probability profiles may coincide. Accurate parent recovery requires separation as well as invariance. The profile-clustering result below assumes separation of conditional profiles. The raw-gradient construction instead needs separation of covariance vectors, or a justified frequency normalization; these are different requirements when patch frequencies differ. Nor does invariance alone ensure usefulness; a constant representation is invariant to every substitution.

18.11.3 Stage 3: a new coordinate system is created

When a learned code identifies the needed parent classes, the next computation can operate on those codes instead of raw tuples. A function high-order in the original encoding can become low-order after this change of coordinates. The coordinate map has its own description, evaluation, and learning cost.

The internal code may label each parent differently from the teacher. Even exact recovery is naturally defined up to an admissible relabeling. For predictive success, a different task-sufficient representation can also work; recovering the complete teacher grammar is a stronger goal.

18.11.4 Stage 4: the next rung becomes visible

Once an earlier module is learned, one can test whether useful next-level contrasts are larger or easier to express in its coordinates. A full induction additionally needs a rule for preserving the earlier module, control of errors on states actually encountered, and valid statistical estimates for the later probes. Freezing a module makes one of these requirements easier to state but does not establish the others.

The proposed mechanism is

resolve a usable contrast
$\longrightarrow$ learn and retain a task-sufficient local code
$\longrightarrow$ expose an accessible subsequent problem.

The first-stage calculation supplies a real, limited instance of this mechanism. Closing it for general hidden compositions and ordinary finite-sample end-to-end SGD remains a separate question.

18.12 Progressive Quotienting by Synonym Invariance

For a fixed grammar, define $\mu \sim \nu$ when the tuples share the same parent. Because synonym sets are disjoint, this is an equivalence relation on the admissible tuples. A quotient map q_{inv} replaces a

tuple by its equivalence class. In the noiseless RHM, performing these exact quotient operations at successive levels permits the local decoding proved above.

A learned invariant representation h satisfies $h(\mu) = h(\nu)$ for synonyms. It need not separate nonsynonyms. To recover the quotient exactly, one needs the converse as well; to solve a task, one needs an appropriate readout and task sufficiency instead. These are different requirements.

A quantitative local recovery statement. The next proposition is an elementary finite-sample addition. It formalizes one route from observable class profiles to a hidden partition, without asserting the sharp RHM sample threshold.

Proposition 31 (Recovery of separated synonym profiles)**.** *Fix a grammar and a patch position with K positive-probability tuples. Let $q_{\min} = \min_\mu \Pr(M_j = \mu) > 0$ and let*

$$p_\mu = (\Pr(Y = \alpha \mid M_j = \mu))_{\alpha=1}^{n_c}.$$

Assume $p_\mu = p_\nu$ for synonyms and

$$\Delta = \min_{\mu \not\sim \nu} \|p_\mu - p_\nu\|_\infty > 0.$$

From P i.i.d. labeled examples, form the empirical conditional profiles $\widehat{p}_\mu$. Join two tuples when $\|\widehat{p}_\mu - \widehat{p}_\nu\|_\infty < \Delta/2$, and return the connected components. Treat an unobserved required tuple as a failure. Then the probability of an incorrect partition is at most

$$Ke^{-Pq_{\min}/8} + 2Kn_c e^{-Pq_{\min}\Delta^2/64}.$$

In particular, the partition is correct with probability at least $1 - \delta$ if

$$P \geq \frac{64}{q_{\min}\Delta^2} \ln \frac{4Kn_c}{\delta}.$$

A known positive lower bound on Δ can be used in place of Δ.

Proof. For each tuple, its count N_μ is binomial with mean Pq_μ. A binomial Chernoff bound gives $\Pr(N_\mu < Pq_\mu/2) \leq e^{-Pq_\mu/8}$. Thus, except with probability at most $Ke^{-Pq_{\min}/8}$, every tuple is observed at least $Pq_{\min}/2$ times.

Conditional on a tuple's occurrence indicators, its class labels are independent with distribution p_μ. Hoeffding's inequality [69] bounds a conditional coordinate error larger than $\Delta/8$ by $2\exp(-2N_\mu(\Delta/8)^2)$. A union bound over Kn_c coordinates therefore gives the second error term. No independence between different tuples' counts is needed for this union bound.

On the resulting event every profile error is at most $\Delta/8$. Synonymous estimated profiles are at distance at most $\Delta/4$; nonsynonymous profiles are at distance at least $3\Delta/4$. Thresholding at $\Delta/2$ consequently recovers precisely the true groups. Finally, $0 < \Delta \leq 1$ shows that the stated sample size makes each error term at most $\delta/2$. □

The result identifies the partition, not the semantic names of its parents. It charges explicitly for patch coverage and profile separation. The population quantities in its assumptions are not supplied estimates; without a known gap bound, selecting a threshold needs its own procedure.

This is a conservative uniform guarantee, not a claim that $P^* = 64(q_{\min}\Delta^2)^{-1}\ln(4Kn_c/\delta)$. A sharp RHM analysis uses its joint distribution of profiles, variance, and sometimes collective clustering. Most importantly, the RMS magnitude of one covariance does not bound Δ from below. Two distinct nonzero profiles can be arbitrarily close or identical.

The proposition also indicates how C3 changes the problem. Supplying parent labels replaces profile clustering by directly supervised local decoding. The missing-label obstacle disappears, but coverage of unseen tuples, local estimation, and stable recomposition still require control, as in Chapter 11.

18.13 What Is Established and What Is Interpretation

The argument contains structural facts, statistical estimates, and a proposed learning mechanism. They should not be reported at the same level of certainty.

18.13.1 Structural facts

The specified nonambiguous RHM has a deterministic local decoder. Its composition realizes the class label on generated inputs. Uniform, parent-independent synonym selection implies equality of the relevant class profiles within a synonym class. These are properties of the specified model, not of every sparse generative process.

18.13.2 Statistical calculation

For a fixed statistic, the sample-mean variance is exactly τ^2/P under i.i.d. sampling. Proposition 30 turns the resulting SNR into explicit probability bounds. The characteristic RHM signal scale uses the cited grammar analysis and its stated asymptotic regime; the binomial toy calculation is explanatory, not a replacement for that analysis.

The resulting $n_c m^L$ scale concerns a noise-normalized local statistic. High-probability statements over grammars require additional control of small signals. Recovering an entire synonym partition further requires coverage and separation, as displayed in Proposition 31.

18.13.3 Dynamical interpretation

The original paper already supplies a restricted gradient-based link between correlations and local synonymic invariance. The exactly balanced finite-width calculation above records a related sufficient construction. Neither result establishes arbitrary deep end-to-end training dynamics.

The broader staircase interpretation is that learned local codes change which dependencies become accessible next. This is consistent with the book's representation-discovery program. Its full dynamical version must specify architecture, initialization, optimization, retention of features, and resource bounds. The success of a different accurate predictor would not refute predictive learning merely because its coordinates differ.

18.14 A Statistical Threshold versus a Recovery Threshold

Fix a prediction criterion, confidence level, distributional setting, and an admitted target family. Let P_{stat} be the least sample size sufficient for some estimator with unrestricted computation. Let P_{rec} denote the corresponding requirement for a specified learner with a specified computational budget. Comparing the same task and success probabilities gives

$$P_{\mathrm{rec}} \geq P_{\mathrm{stat}}.$$

A stronger teacher-structure recovery criterion has its own threshold; it should not be substituted silently for predictive recovery.

An explicit description-length comparison. For fixed RHM parameters, a crude encoding lists the m child tuples for each parent at every level. The number of grammar tables is at most

$$N_{\text{gram}} \le v^{\,sm((L-1)v+n_c)}.$$

This upper bound ignores the disjointness constraints, which only reduce the count. Give each grammar a fixed deterministic extension to invalid inputs. The resulting class of predictors is finite. For noiseless labels from one true grammar and i.i.d. examples, a union bound gives

$$\Pr\Big(\exists f : \widehat{R}_P(f) = 0,\ R(f) > \epsilon\Big) \le N_{\text{gram}} e^{-P\epsilon},$$

where R is classification error. Indeed, a fixed predictor with error exceeding ϵ fits all P independent labels with probability at most $(1-\epsilon)^P \le e^{-P\epsilon}$. Therefore every consistent predictor in this fixed class generalizes whenever

$$\boxed{P \ge \frac{sm((L-1)v + n_c)\ln v + \ln(1/\delta)}{\epsilon}.}$$

This elementary comparison, added here, is the finite-class route of Chapter 13. It does not give a tractable method of finding a consistent grammar for all joint parameter scalings. Exhaustive search can be costly in grammar-description size, and the statement says nothing about SGD. It also does not identify every hidden rule from labels: distinct consistent grammars can predict well without matching one another internally.

The comparison explains why a correlation-resolution threshold need not be the point at which prediction first becomes possible statistically. At fixed alphabet parameters, the coding bound grows only linearly in L at fixed accuracy, whereas the characteristic scalar correlation scale grows as m^L. These are guarantees and scales for different learning procedures, not contradictory estimates of one universal threshold. A complete complexity comparison must also count reading the $d = s^L$-position examples and evaluating candidates.

18.15 What Would Happen in a Nongeneric Hierarchy?

Suppose the hierarchy is balanced so that every statistic in a specified accessible family has zero class contrast. More samples estimate those zero means more accurately; they do not reveal a nonzero mean in that same family. The source studies a decorrelated variant of the RHM and reports a strong deterioration for its tested deep networks [25]. This is not an impossibility theorem for every algorithm or every alternative observable.

For an explicit example, let independent uniform signs X_i have label $Y = \prod_{i=1}^{d} X_i$. For any proper subset S and any integrable probe $\varphi(X_S)$,

$$\mathbb{E}[Y\varphi(X_S)] = \mathbb{E}\left[\varphi(X_S)\prod_{i\in S} X_i\right]\mathbb{E}\left[\prod_{i\notin S} X_i\right] = 0.$$

Yet a binary product tree computes Y. The failure concerns these local mean probes, not the existence of a compact computation. Changing the probe family or using an algorithm adapted to the discrete parity structure is a different learning mechanism.

The appropriate conclusion is

$$\boxed{\text{compact hierarchy} \not\Longrightarrow \text{useful signal in every local probe family.}}$$

An exact cancellation also differs from a small nonzero signal. The latter can be resolved with more independent data, although the required amount may be prohibitive in the chosen complexity parameters.

18.16 Predictions Beyond the Original RHM

The interpretation suggests the following tests. Some ingredients, including correlation measurement, synonymic invariance, and deliberate decorrelation, already appear in the original study [25]. The extensions below are proposed controls and measurements, not new experimental results. Grammar draws, training samples, optimizer seeds, and independent evaluation samples should be recorded separately. Failed runs and exactly balanced realizations should not be removed from unconditional summaries.

18.16.1 Prediction 1: directly measure the footprint

For each fixed grammar, record the patch probability q_μ, the class profile $p_{\alpha|\mu}$, the raw covariance $C_{\alpha\mu}$, and the variance $\tau^2_{\alpha\mu}$ of its estimator's summand. Compare the normalized $\rho_{\alpha\mu} = C_{\alpha\mu}/\tau_{\alpha\mu}$ across grammar draws. In the cited regime, it is the characteristic normalized scale that should be compared with $(n_c m^L)^{-1/2}$.

Finite evaluation data themselves inflate squared signal estimates:

$$\mathbb{E}[\widehat{C}^2_{\alpha\mu} \mid \mathcal{G}] = C^2_{\alpha\mu} + \frac{\tau^2_{\alpha\mu}}{P}.$$

An independent split provides an elementary correction: the product of two independently estimated means has conditional expectation $C^2_{\alpha\mu}$. Exact grammar probabilities, when computationally available, avoid this estimation issue. Report quantiles and the mass at zero as well as RMS, and report profile separations when structural recovery is evaluated. An RMS fitted using noisy estimates alone can mistake sampling noise for quenched signal.

18.16.2 Prediction 2: footprint magnitude should predict the transition sample

For a fixed probe and comparable estimation tails, a grammar with larger $|C|/\tau$ should need fewer examples to resolve that probe. The corresponding prediction is

$$P_{\mathrm{detect}}(\mathcal{G}) \propto \frac{\tau(\mathcal{G})^2}{C(\mathcal{G})^2},$$

not a universal law in the unnormalized C. Whether the measured classification transition $P^*(\mathcal{G})$ follows this relation tests whether the probe is the actual learning bottleneck.

For full partition recovery, also compare $q_{\min}$ and the smallest relevant profile gap. Hold architecture, initialization convention, training budget, and performance criterion fixed. Distinguish between-grammar variation from training randomness within each grammar. A scalar maximum, an RMS over features, and a minimum separation predict different tasks.

18.16.3 Prediction 3: artificially suppressing the footprint should delay learning

Balance selected rule frequencies to suppress a specified family of contrasts, while preserving the local decoding task as far as possible. Check that the intervention actually reduces the normalized

signal seen by the learner. Changes in marginal frequencies, vocabulary support, label balance, and higher-order statistics should be measured rather than assumed absent.

A delayed transition would support dependence on those contrasts. An unchanged transition could indicate a different accessible statistic or learning mechanism; it need not contradict compositional representation. In particular, eliminating one marginal correlation is not equivalent to eliminating every gradient-accessible probe.

18.16.4 Prediction 4: artificially increasing asymmetry should lower the threshold

Introduce a controlled change in production probabilities or rule assignments. If the raw signal has size $a(\delta) \asymp \delta$ and its standard deviation is $\tau(\delta)$, then

$$P_{\text{detect}} \asymp \frac{\tau(\delta)^2}{\delta^2}$$

is the second-moment prediction. The simpler δ^{-2} law requires comparable noise scale across the intervention. Confidence and multiple-probe costs must be treated consistently.

The intervention should preserve the desired task, or its change must be reported. A newly introduced direct label cue can improve prediction without teaching synonym classes. Measure both held-out prediction and structural diagnostics to distinguish such a shortcut from hierarchical recovery.

18.16.5 Prediction 5: intermediate representations should increase the next-level footprint

Freeze representations at successive layers or checkpoints, and estimate candidate next-level signals on fresh examples from the same grammar. Measure normalized signal, probe complexity, and actual gradient access, not only raw correlation magnitude. An arbitrary rescaling of hidden coordinates can increase a raw moment without improving its SNR.

There are two axes to distinguish: depth within one trained network, and time along one learning trajectory. A third is training-set size across separately trained networks. A hierarchical representation across layers does not establish a strict temporal ordering of learning stages. The strongest evidence for a staircase would connect a retained local code to a subsequently usable statistic under controlled interventions.

18.16.6 Prediction 6: supplied intermediate labels should bypass the first threshold

Give the learner actual parent labels or a controlled surrogate, and compare with end-to-end supervision at matched resources. Such labels remove the need to infer that parent solely through its weak relation to the final class. They do not guarantee that all required child tuples are observed or that later modules are learned.

This is the C3 intervention of Chapter 11. Separate complete local tables from trajectories that expose only visited states. Use withheld combinations to test local generalization or coverage, and count the cost of acquiring intermediate labels. Supplying them changes the observation model, not just the optimizer.

18.16.7 Prediction 7: kernel methods should not receive the same progressive coordinate advantage

Compare a fixed kernel on the original encoding with feature-learning networks, fixed kernels on explicitly lifted patches, and an oracle representation supplied from the grammar. A fixed

kernel does not adapt its feature map to the current labeled sample, but a suitable supplied map may already expose the needed hierarchy. Thus the prediction concerns specified kernels and representation costs, not all RKHS methods.

The original RHM study reports unfavorable sample scaling in its tested lazy regimes and favorable scaling in the specified feature-learning regimes [25]. It does not prove that every fixed kernel must lose. Also, the one-step patch example is already a shallow feature-learning model. Depth, trainable features, and convex estimation in a supplied representation are separate properties. The special advantage to test is the ability to construct and reuse several useful coordinate changes, rather than merely to fit a readout.

18.17 Relation to Sparse Compositionality

Sparse compositionality supplies a representation that is economical in local arity, graph size, and the stated encoding. It does not by itself supply an informative statistic. In the RHM, no ambiguity makes local inverse decoding possible; reuse of the same level-specific tables adds another source of economy beyond bounded fan-in.

Once the relevant parent variables are available, the remaining computation can be organized through local operations. If they are learned approximately, the resulting composition still needs a stability or error-control argument. This is the same distinction made for supplied traces in Chapter 11 and for replacement modules in Chapter 15.

For the book's predictive endpoint, an accurate feasible network supplies its own computational compositional witness, even when its internal variables do not match the grammar. The RHM provides a particularly interpretable candidate mechanism for finding such a predictor. It does not add a requirement of grammar reconstruction to every successful learning algorithm.

18.18 Relation to Genericity

Genericity concerns the observable learning problem and the learner's accessible statistics. In a fixed random grammar, finite-size imbalances can create a nonzero low-order contrast. Its usefulness depends on its variance, the candidate-probe family, and its relation to an actual learning direction.

$$\boxed{\begin{array}{c}\text{noncancellation}\\ \longrightarrow \text{ statistical resolution}\\ \longrightarrow \text{ an informative update}\\ \longrightarrow \text{ a useful retained representation.}\end{array}}$$

Each arrow requires its own assumptions. The explicit one-step example makes one such arrow concrete; it does not close all of them at arbitrary depth.

Three sources of randomness must also remain separate: the draw of the grammar, the finite dataset sampled from it, and minibatch or other optimizer randomness on that dataset. Quenched asymmetry is a property of the first draw's population law. Repeated optimizer steps do not supply fresh population observations. Conversely, the persistent minibatch fluctuations possible at a regularized stationary point in Chapter 7 do not invalidate a statistically controlled returned predictor. They are a different issue from the initial detectability calculation.

18.19 Relation to Genericity as a Property of the Observed Problem

A useful description of the problem is

$$(f^*_{\mathcal{G}}, \mathsf{P}_{\mathcal{G}}, \Phi, \text{encoding}, \text{architecture}, \text{learning rule}),$$

with sampling and resource conventions specified. Here $\mathsf{P}_{\mathcal{G}}$ is the example distribution, not the sample count P.

The same grammar may be easier or harder to learn when the available patches, input encoding, initial features, class frequencies, or probe search procedure change. An informative tuple indicator need not be cheaply available from a different encoding. A relevant statistic can also be badly transmitted through the parameterization.

This is why the RHM is a model of an observed learning problem, not a proof that a target has an intrinsic, coordinate-independent genericity constant. Representation changes can expose information more effectively, but their own learning, computation, and statistical selection costs must be included. The fixed-family and coding arguments of Chapter 13 are appropriate only under their stated independence or uniformity conditions.

18.20 A Compact Summary of the Mechanism

For the specific path-count approximation and noise regime used above, the calculation is

$$\begin{aligned}
\text{local tuple frequency:} &\quad q_\mu \asymp (mv)^{-1}, \\
\text{conditional class contrast:} &\quad b_{\text{rms}} \asymp \frac{1}{n_c}\sqrt{\frac{mv}{m^L}}, \\
\text{raw covariance:} &\quad C_{\text{rms}} \asymp \frac{1}{n_c\sqrt{mv\, m^L}}, \\
\text{single-example variance:} &\quad \tau^2_{\text{typ}} \asymp \frac{1}{n_c mv}, \\
\text{normalized signal:} &\quad \rho^2_{\text{rms}} \asymp \frac{1}{n_c m^L}, \\
\text{constant-SNR sample scale:} &\quad P_c \asymp \rho^{-2}_{\text{rms}} \asymp n_c m^L.
\end{aligned}$$

The signal/noise cancellation reproduces the original RHM scale. It is not a derivation of all joint grammar moments, an anti-concentration theorem, or a lower bound against all learning algorithms.

The logical learning sequence adds further ingredients:

$$\begin{gathered}
\text{local identifiability and controlled representation} \\
+ \text{ a resolved, update-accessible contrast} \\
+ \text{ coverage and separation for the desired local recovery} \\
+ \text{ retention and stable subsequent computation} \\
\Longrightarrow \text{ a route to hierarchical learning.}
\end{gathered}$$

Predictive accuracy is then evaluated on independent examples, with the statistical control appropriate to the selected predictor.

18.21 Conclusion

The Random Hierarchy Model supplies a concrete bridge between the book's representation principle and a mechanism of feature learning. Its nonambiguous rules give a bounded-arity inverse classifier on generated inputs. Its random, fixed realization can create class–patch contrasts that vanish only after averaging over grammars. Finite data determine whether a specified learner can resolve those contrasts.

The characteristic calculation is

$$\boxed{\rho = \frac{\text{population signal}}{\text{single-example standard deviation}}, \qquad \mathrm{SNR} = |\rho|\sqrt{P}.}$$

For the relevant RHM regime, $\rho_{\mathrm{rms}}^2 \asymp 1/(n_c m^L)$, leading to $P_c \asymp n_c m^L$. The raw covariance has additional class and vocabulary dependence. Keeping its variance explicit is what makes the comparison meaningful.

The original RHM paper already connects this threshold with measured synonymic invariance and with a restricted one-step gradient mechanism [25]. The present chapter relates those results to C2 and records elementary conditional guarantees: concentration of a specified footprint, local decoding under identifiability, and recovery of synonym groups under coverage and profile separation. A simple finite code count also shows why a correlation threshold need not be an information-theoretic threshold for prediction.

The broader staircase picture remains

$$\boxed{\begin{array}{c}\text{detect}\\ \longrightarrow \text{identify a useful local equivalence}\\ \longrightarrow \text{construct and preserve a coordinate}\\ \longrightarrow \text{make the next dependency accessible.}\end{array}}$$

It is a proposed mechanism, not a compulsory temporal sequence or an additional teacher-matching requirement for a successful predictor. At each step, invariance must retain the information needed by the task; constant collapse is not useful recovery. Newly learned probes require valid statistical treatment, and approximate modules require stable composition.

Thus the RHM supports a substantive positive lesson: hierarchically organized data can make feature learning far more sample-efficient than the specified alternatives, and the onset can be understood through accessible correlations. The remaining general question is how a finite learner converts such opportunities into sustained, statistically controlled predictive progress at feasible cost. That is the place of this model within the broader theory.

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CHAPTER 19

The Hippocampal Scaffold and Compositional Sparsity

The hippocampus is often described as a memory system, a cognitive map, or an indexing mechanism. These descriptions need not be competitors.

This chapter develops a computational hypothesis in which fast hippocampal learning supplies a temporary relational scaffold for slower cortical learning.

In the particular division of labor proposed here, the fast memory need not contain the final reusable world model. Instead, it rapidly binds episodes, separates potentially interfering experiences, and preserves temporal and relational transitions. These intermediate structures may make hidden regularities easier for cortex to recover.

The central hypothesis is therefore

> *hippocampal indexing may expose an intermediate relational trace from which cortex can recover reusable compositional structure.*

The proposal is deliberately computational rather than a unique anatomical assignment. It does not rule out hippocampal abstraction or fast cortical learning in other tasks. Specific equations below should be read as abstractions of possible functions, not as literal descriptions of individual hippocampal subfields.

The scaffold is also not theorem-strength C3 in the sense developed earlier in this book. It does not necessarily expose the true latent variables or their parent sets. It is better viewed as a C3-like partial relational trace*: intermediate information that may reduce the hidden-variable recovery problem without solving it completely.*

Three elementary propositions below clarify this hypothesis: re-encoding an unchanged observation record is not new statistical evidence; a useful quotient must preserve task-relevant transitions and outputs; and replay changes a training distribution without multiplying the number of independent experiences. These mathematical clarifications are not empirical tests of the hippocampal mechanism.

19.1 Fast Episodic Indexing and Slow Structural Learning

An intelligent system faces two conflicting requirements.

It must remember individual experiences without catastrophic interference,

$$\boxed{\text{this episode} \neq \text{other similar episodes},}$$

while also discovering what those experiences have in common,

$$\boxed{\text{many episodes} \rightarrow \text{one reusable rule}.}$$

These objectives naturally favor different representations.
A fast episodic system should preserve distinctions.
A slower semantic system should discover equivalences.
We will use the following schematic division:

hippocampal system:	rapid binding, indexing, relational traces,
cortical system:	shared parameters, abstraction, reusable rules.

This resembles the logic of complementary learning systems [102], but the emphasis here is specifically on compositional recovery.

A division of computational roles, not an exclusive partition. "Fast" and "slow" refer to the proposed relative adaptation timescales. They are not universal learning-rate constants or claims that all episodic content resides in one structure and all general knowledge in another. The complementary-learning-systems account supplies a motivation for this specialization, not a theorem that every task requires it. The hypothesis of this chapter concerns the usefulness of an intermediate representation for a specified slower learner.

19.2 A Minimal Computational Model

Let

$$z_t$$

denote a latent state of the world relevant to the agent.
The agent does not observe z_t directly.
Instead it receives sensory observations

$$x_t \sim p(x_t \mid z_t)$$

and may also possess internally generated contextual information such as path integration, action, task state, or recent memory.
Let

$$c_t$$

collect this additional context.
An episodic memory system constructs an internal index

$$h_t = H_\psi(x_t, c_t), \tag{19.1}$$

where H_ψ is a fast encoding process.
The index h_t need not equal the latent state z_t.

Indeed, two visits to the same physical location may intentionally receive different episodic indices because they occurred at different times or contained different events.

Thus

$$\boxed{h_t = \text{episode-specific address}, \qquad z_t = \text{shared latent structure.}}$$

The distinction between these variables is central.

The indexing interpretation is broadly compatible with classical hippocampal indexing theories [150, 151], but the variable h_t here is a computational abstraction rather than a claim about a particular neural code.

What is available when the index is formed? Let the available history be

$$\mathsf{o}_t = (x_0, c_0, a_0, \ldots, a_{t-1}, x_t, c_t).$$

The context c_t may summarize part of this history, but it must not contain an unobserved future state or a test label. A deterministic encoder with fixed ψ produces the same h for identical (x, c); distinct episode identities therefore require distinguishable context, an explicit event tag, a changing memory state, or specified randomization.

The notation H_ψ suppresses possible online changes of ψ. An implementation must store the code assigned at encoding, version its addresses, or update all associated links consistently. Otherwise changes of representation can invalidate the very relations the scaffold is meant to preserve. A record identifier and its neural feature vector are distinct objects; collisions of feature vectors must not silently merge records. Neither address construction nor retrieval is free computational work.

19.3 Pattern Separation as an Indexing Primitive

One classical computational role attributed to hippocampal circuitry is pattern separation [99, 120, 173].

Similar sensory experiences can interfere if represented too similarly. A useful episodic code should therefore make distinct episodes easier to store and retrieve separately.

A minimal abstract requirement is

$$x_t \approx x_s \quad \not\Longrightarrow \quad h_t \approx h_s$$

when the two inputs belong to distinct episodes that must remain individually accessible.

At the same time, some similarity structure may need to remain recoverable through contextual cues.

One toy implementation is a high-dimensional expansion followed by sparse competition:

$$h_t = \mathrm{TopK}\left(R\, g(x_t, c_t)\right), \tag{19.2}$$

where R is a high-dimensional projection and only a small fraction of coordinates remain active.

Equation (19.2) is not proposed as a literal model of dentate-gyrus computation.

It is simply an example showing how

$$\boxed{\text{expansion} + \text{sparse competition}}$$

can be used in a design intended to reduce interference among correlated episodes. The equation alone does not guarantee that reduction.

Separation also needs retrievability. Take $R \in \mathbb{R}^{D_H \times d_g}$ and let TopK retain the K largest scores, $1 \leq K < D_H$, with a fixed tie convention, setting the others to zero. This specifies an at-most-K-sparse code, not an injective map. Identical input features give identical codes, different inputs can collide, and nearly tied scores can make the active support sensitive to noise. In particular, sparse activity does not by itself imply orthogonality or reliable identification of every episode.

There is a simple positive stability statement. If the ordered scores of $v = Rg(x, c)$ satisfy $v_{(K)} - v_{(K+1)} > 2\epsilon$, then a perturbation $\widetilde{v}$ with $\|\widetilde{v} - v\|_\infty \leq \epsilon$ preserves the selected index set. Every retained score still exceeds every discarded score. This concerns selection stability near one code, not separation of all codes. Successful memory must combine distinctness between relevant episodes with completion from the cues actually available; their tradeoff is explicit in [120].

19.4 From Episodic Indices to a Scaffold

As an agent moves through an environment, it generates a sequence

$$h_1, h_2, \ldots, h_T.$$

Observed temporal transitions define a graph

$$\mathcal{G}_H = (\mathcal{H}, \mathcal{E}), \tag{19.3}$$

where

$$\mathcal{H} = \{h_t\}$$

and

$$(h_t, h_{t+1}) \in \mathcal{E}$$

whenever the corresponding episodes occurred consecutively or were otherwise relationally linked.

We call this the *hippocampal scaffold*.

The scaffold records something richer than an unordered episodic store.

It preserves

which experiences were connected to which other experiences.

This connectivity may reflect

spatial adjacency, temporal order, action consequences, relational similarity.

The idea is compatible with cognitive-map and relational-memory views of hippocampal function [119, 46], without requiring all scaffold edges to correspond to Euclidean spatial relations.

An observed graph is not a transition law. For mathematical bookkeeping, the vertices can be taken to be event records carrying h_t and associated content. The unlabeled edge set in Equation (19.3) is only a summary. An implementation should retain relation type and, where applicable, action, time lag, context, and observation count. An action-labeled edge (h_t, a_t, h_{t+1}) is different from a similarity edge. Temporal proximity alone does not establish causal influence, physical adjacency, or reversibility.

If each visit receives a new index, an observed trajectory can be a simple path with no repeated vertices. Low degree then says little about the complexity of the environment. Repeated visits must be related using observable content or learned correspondences before they can constrain a shared state model. Unobserved edges are missing evidence, not automatically impossible transitions. For a stochastic environment, counts can estimate conditional transition probabilities under appropriate sampling and coverage assumptions; a single edge cannot specify them.

19.5 Graph Sparsity Is Not Yet Compositional Sparsity

An important distinction is necessary.

A sparse transition graph is not automatically a compositionally sparse function in the sense used elsewhere in this book.

Low graph degree means that from a particular episodic state only a small number of successors may have been observed.

It does not by itself imply that the underlying world dynamics factor into bounded-arity constituent functions.

The stronger structure appears only if the latent state itself factorizes.

Suppose

$$z_t = \left(z_t^{(1)}, \ldots, z_t^{(m)}\right) \tag{19.4}$$

and the dynamics satisfy

$$z_{t+1}^{(i)} = F_i\left(z_t^{\mathrm{pa}(i)}, a_t\right), \qquad |\,\mathrm{pa}(i)| \le k, \tag{19.5}$$

where

$$k \ll m.$$

Under the following input and resource conventions, this is a bounded-arity compositional model of the chosen latent dynamics.

The hippocampal hypothesis proposed here is not that the scaffold *creates* this structure.

It is that the scaffold may make evidence for it easier to recover.

Count the action coordinates and the local cost. Equation (19.5) regards a_t as an argument, but a growing action vector cannot be counted as one scalar input for an approximation theorem. One sufficient convention is $a_t \in \mathbb{R}^{p_a}$ with $p_a = O(1)$; another is that each F_i depends on at most $k_a = O(1)$ action coordinates. The scalar local arity is then bounded by $k + p_a$ or $k + k_a$. The coordinate domains, constituent descriptions, and numerical evaluation costs must also be controlled.

Unrolling T deterministic updates of m coordinates gives $O(mT)$ constituent occurrences. Smooth approximation further requires the local regularity and error-propagation bounds of

Chapter 4. Stochastic dynamics require a conditional-law or noise-augmented formulation; Equation (19.5) is the deterministic special case. A sparse generative map does not by itself make the inverse map from sensory histories identifiable or easy to learn.

19.6 The Scaffold as a Partial Relational Trace

The scaffold connects directly to the distinction between representation and recovery developed earlier in the book. A learner using only raw sensory pairs (x_t, x_{t+1}) may need to infer latent state, correspondence across time, transition structure, and reusable operators simultaneously. The proposal is that a fast memory can make some of those relations more accessible to the slower learner.

For controlled transitions, an augmented training record can be written

$$(x_t, c_t, h_t, a_t, h_{t+1}, x_{t+1}, c_{t+1}).$$

The indices need not equal z_t or reveal the parent sets in Equation (19.5). The scaffold is therefore a *C3-like partial relational trace*, not theorem-strength hard C3. Its value depends on which relations are preserved, how accurately they are available, and which computations the learner can perform.

Additional evidence versus a different organization of evidence. Two experiments should not be conflated. A scaffold may retain history or context missing from the direct learner's input. That changes the effective observation model. Alternatively, both learners may receive the same full record, while one receives an index and replay organization computed from it. That can change computational accessibility, but it does not create new statistical information about the environment. There is no unconditional informativeness ordering between an arbitrary scaffold and raw observations. Hard C3 adds designated constituent supervision; a poor scaffold may supply only redundant addresses or even incorrect associations.

Proposition 32 (A scaffold computed from the same record is not new evidence)**.** *Let* Θ *denote an unknown environment or target, and let* D *be the complete observation record supplied to a learner. Suppose a scaffold* S *is drawn from a fixed, known kernel* $K(\cdot \mid \mathsf{D})$ *that does not otherwise depend on* Θ*. Without computational or memory restrictions, every randomized estimator using* (D, S) *can be simulated by a randomized estimator using* D *alone, with the same output distribution for each target. For any fixed loss, the infimum achievable risks of the two observation schemes are therefore equal. When mutual information is defined,*

$$I(\Theta; \mathsf{D}, \mathsf{S}) = I(\Theta; \mathsf{D}).$$

Proof. Given D, the second estimator draws $\mathsf{S} \sim K(\cdot \mid \mathsf{D})$ and runs the first estimator. Its conditional output law is the same. Conversely, an estimator receiving (D, S) can ignore S. This proves equality of the best risks. Conditional independence gives $I(\Theta; \mathsf{S} \mid \mathsf{D}) = 0$, and the information identity follows from the chain rule. □

The simulation may require prohibitive storage or computation. A scaffold can therefore improve a *specified resource-bounded learner* without contradicting the proposition. If its construction uses prior experience, external supervision, or context absent from D, those resources are additional side information and must be included in the comparison. Replay generated solely from D is subject to the same distinction.

A minimal positive memory example. Suppose a fair bit B is observed at time zero, all later sensory inputs are identical, and the later task is to report B. A learner restricted to the current sensory input has minimum error $1/2$. A memory that retains and retrieves the earlier bit can answer exactly. A recurrent learner with the same one-bit storage can do so as well. The example establishes a benefit of retained context relative to restricted access, not a special advantage of a hippocampal implementation or a theorem about discovering compositional dynamics.

> **Scaffold Hypothesis.** At a specified observation, memory, and computational budget, a fast episodic system can improve a slower learner's acquisition of reusable predictive structure by preserving and making accessible reliable relations among otherwise ambiguous observations.

This refines the chapter's hypothesis about learnability. It does not make hippocampal indexing necessary for compositional learning, nor require recovery of the particular variables that generated the observations.

19.7 From Episodic Indices to Latent Variables

The cortex should not simply memorize every hippocampal index.
That would reproduce episodic memory rather than abstraction.
Instead, repeated experience may reveal equivalence classes among indices.
For example,

$$h_{t_1}, \qquad h_{t_2}, \qquad h_{t_3}$$

may correspond to different episodes that share the same underlying structural state.
The cortex may learn a mapping

$$\phi : (x_t, h_t) \longmapsto \widehat{z}_t \tag{19.6}$$

such that

$$\phi(x_{t_1}, h_{t_1}) \approx \phi(x_{t_2}, h_{t_2})$$

when the two episodes instantiate the same relevant latent configuration.
Thus cortical learning performs a kind of quotienting:

$$\boxed{\text{many episodic indices} \to \text{one shared latent state.}}$$

The hippocampal system favors separation.
The cortical system learns which distinctions can later be discarded.
This tension between preserving individual episodes and discovering shared structure is closely related to the computational motivation behind complementary learning systems [102].

A useful quotient must preserve what the task needs. Simply collapsing all indices to one vector would satisfy the informal similarity objective, but would generally destroy prediction. Conversely, a unique code for every record can memorize the data without discovering any reusable state. The intended representation should preserve specified observable predictions while discarding only task-irrelevant distinctions. The following elementary criterion formalizes the deterministic case; related stochastic state abstractions are studied in [50].

Proposition 33 (Task-respecting quotients of a relational scaffold)**.** *Consider an idealized state set $\mathcal{H}$, a family of deterministic maps $T_a : \mathcal{H} \to \mathcal{H}$ for actions $a \in \mathcal{A}$, and a task readout $b : \mathcal{H} \to \mathcal{Y}$. Let $q : \mathcal{H} \to \mathcal{Z}$ be surjective. There exist maps $\overline{T}_a : \mathcal{Z} \to \mathcal{Z}$ and $\overline{b} : \mathcal{Z} \to \mathcal{Y}$ satisfying*

$$q \circ T_a = \overline{T}_a \circ q, \qquad b = \overline{b} \circ q$$

if and only if, for every h, h' with $q(h) = q(h')$,

$$q(T_a(h)) = q(T_a(h')) \quad \textit{for every } a, \qquad b(h) = b(h').$$

When these conditions hold, the quotient reproduces the readout sequence of the original system under every finite action sequence, from the corresponding initial quotient state.

Proof. Necessity follows by applying the displayed factorizations to h and h'. For sufficiency, define $\overline{T}_a(z) = q(T_a(h))$ and $\overline{b}(z) = b(h)$ using any representative h with $q(h) = z$. The fiber conditions make both definitions independent of the representative. Induction over the action sequence gives $\overline{z}_t = q(h_t)$ at every step, and hence $\overline{b}(\overline{z}_t) = b(h_t)$. □

The total maps T_a are an assumption of this idealized result, not an inference from a finite set of trajectory edges. For partially observed maps, the conditions must first be checked on the transitions actually specified, with no conclusion for unsupported actions. The readout b should include the task's required rewards or observations; a constant q is excluded when those readouts vary. For stochastic kernels P_a, the analogous requirement is that

$$\sum_{u : q(u) = z'} P_a(u \mid h)$$

be identical for states h in the same fiber, for every action and quotient state z' (in the finite-state case). Equality under a single behavior policy is weaker than agreement for every action.

This supplies a positive criterion for abstraction, not an algorithm for finding q. An observed partition can satisfy it accidentally on the sample. Estimation, coverage, and separation of genuinely different predictive states remain learning problems, as in the profile-separation argument of Chapter 18. Moreover, a quotient transition need not factor into low-arity coordinates: state aggregation and sparse compositional factorization are distinct.

19.8 Replay as Structured Training Data

Replay provides a possible mechanism by which the episodic scaffold can continue to train slower cortical systems after the original experience has ended.

Hippocampal replay and reactivation have been observed during sleep and quiet wakefulness [167, 86, 48].

A replayed trajectory has the schematic form

$$h_{t_1} \to h_{t_2} \to \cdots \to h_{t_r}.$$

Associated sensory, contextual, action, or reward information may also be reactivated.

From a computational perspective, replay changes the sampling distribution available to cortical learning.

Instead of receiving only the original online stream, cortex may be exposed repeatedly to selected relations such as

$$(h_t, h_{t+1}), \qquad (h_t, y_t), \qquad (h_t, a_t, h_{t+1}).$$

This may support

consolidation, credit assignment, transition learning, cross-episode comparison.

The claim is not that replay implements one particular optimization algorithm.

Its computational significance is that a fast episodic system can generate additional structured *training events* for a slower learner. These are not automatically additional independent experiences of the world.

What replay optimizes. The distinction can be stated without committing to a biological optimizer. Freeze a dataset $\mathsf{D} = (e_1, \dots, e_n)$ of recorded transitions and a parameter vector θ. Let $\ell_i(\theta)$ be the differentiable loss on record e_i, with $g_i = \nabla \ell_i(\theta)$. A replay distribution assigns probabilities $w_i \geq 0$, $\sum_i w_i = 1$.

Proposition 34 (Replay reweights a fixed training problem)**.** *Draw $I_1, \dots, I_M$ independently from (w_i), conditionally on the fixed buffer and parameter, and put $\widehat{g}_M = M^{-1} \sum_{j=1}^{M} g_{I_j}$. Then*

$$\mathbb{E}[\widehat{g}_M \mid \mathsf{D}, \theta] = g_w := \sum_i w_i g_i,$$

$$\mathrm{Cov}(\widehat{g}_M \mid \mathsf{D}, \theta) = \frac{1}{M} \sum_i w_i (g_i - g_w)(g_i - g_w)^\top.$$

If the probabilities are fixed with respect to differentiation, g_w is the gradient of $\widehat{L}_w(\theta) = \sum_i w_i \ell_i(\theta)$. If every $w_i > 0$, the reweighted estimator $g_I/(n w_I)$ instead has mean $n^{-1} \sum_i g_i$.

Proof. The mean of one draw is $\sum_i w_i g_i$. Independence makes the covariance of the average $1/M$ times the single-draw covariance. Differentiating the fixed weighted sum gives g_w. Finally, $\sum_i w_i g_i/(n w_i) = n^{-1} \sum_i g_i$. □

If priorities are recomputed from θ, the conditional identity still holds at each fixed θ, but $\sum_i w_i(\theta) g_i$ need not be the full gradient of $\sum_i w_i(\theta) \ell_i(\theta)$: differentiating that expression adds derivatives of the priorities. Small replay probabilities can make importance weights highly variable; zero probabilities omit some records altogether. Correlated sequence replay requires covariance terms between draws. None of these schemes is asserted to describe neural replay literally.

For a fixed θ independent of the buffer and finite second moments, total covariance gives

$$\mathrm{Cov}(\widehat{g}_M) = \mathrm{Cov}_{\mathsf{D}}(g_w) + \mathbb{E}_{\mathsf{D}}[\mathrm{Cov}(\widehat{g}_M \mid \mathsf{D})].$$

Increasing M reduces the second term for independent resampling, not the first. Replaying one empirical buffer more often can improve optimization of that buffer's objective while leaving its statistical error unresolved. For a parameter learned from the buffer, the fixed-parameter calculation is not itself a population-generalization theorem; Chapter 13 requires a valid analysis of the selected predictor and its sample.

Preserving a relation is not preserving presentation order. Shuffling complete supervised tuples (h, a, h') preserves a one-step empirical loss, though it can change a finite SGD trajectory. Shuffling episodes and then pairing unrelated neighbors changes the targets and can create false transitions. A claim that relational replay helps must say which manipulation is used. Reverse replay of an experienced sequence is also not evidence that the physical action dynamics are invertible. Its use for credit assignment need not treat reversed pairs as forward transition examples. This distinction is particularly pertinent to the reverse replay observed by Foster and Wilson [48].

19.9 Cortical Abstraction as Parameter Sharing

Suppose the hippocampal scaffold contains many episode-specific transitions,

$$G_{h_1}, G_{h_2}, \ldots, G_{h_M}.$$

These transitions may initially be represented separately.

Cortical learning can discover that many of them instantiate the same underlying rule.

For example,

$$\text{Kitchen}_1 \rightarrow \text{Hallway}_1$$

and

$$\text{Bedroom}_2 \rightarrow \text{Hallway}_2$$

may differ episodically while sharing an action-conditioned structural rule. The room names alone do not establish that identity; the relevant action, geometry, and task context must also agree in the representation.

The cortical learner can replace many local instances by one parameterized operator,

$$G_\theta\left(\widehat{z}_t, a_t\right) \approx \widehat{z}_{t+1}. \tag{19.7}$$

This is parameter sharing:

$$\boxed{\text{many episode-specific transitions} \rightarrow \text{one reusable transition rule.}}$$

A successfully learned shared model may apply to states never stored explicitly in the episodic scaffold. That is a generalization objective, not a consequence of writing one symbol G_θ for all transitions.

This is the key computational transition from memory of individual instances to reusable structure.

A training formulation needs a predictive constraint. At fixed stored codes, one possible objective combines transition consistency with a task readout:

$$\begin{aligned}\widehat{L}(\phi, \theta, D) = \frac{1}{n}\sum_t \|G_\theta(\phi(x_t, h_t), a_t) - \phi(x_{t+1}, h_{t+1})\|^2 \\ + \frac{\lambda_y}{n}\sum_t \ell_y(D(\phi(x_t, h_t)), y_t) + \lambda_\Omega \Omega(\phi, \theta, D),\end{aligned}$$

where the recorded pairs, available targets y_t, loss, and regularizer must be specified. This is a proposed artificial objective, not a cortical plasticity law or an optimization theorem. The first

term alone admits $\phi \equiv 0$, $G_\theta(0, a) = 0$. Task supervision can exclude that trivial solution when nonconstant predictions are required, but an unrestricted encoder and decoder can still memorize record identities. Task accuracy, representation complexity, and held-out performance must be checked together. Chapter 16 addresses a different question: how a supplied useful learning direction might be implemented by local circuitry.

19.10 Learning the Algebra of Composition

Abstraction can go beyond sharing one-step transitions.

Repeated trajectories may reveal relations among operators.

For example,

$$G_a \circ G_b \approx G_c$$

or

$$G_{\text{left}} \circ G_{\text{right}} \approx \text{Id}\,.$$

Such regularities define an algebra over transformations.

The hippocampal scaffold supplies particular trajectories.

Cortex can infer rules that generalize across them.

Thus a useful distinction is

episodic system:	preserves particular paths,
cortical system:	learns reusable relations among paths.

This difference separates memory of instances from a generative model of their structure.

Domains and order of composition. Here $G_a \circ G_b$ means apply b first and then a. The approximation must be measured on a specified domain or distribution, with all intermediate states admissible. Left and right are inverses only in an appropriate reversible setting; collisions, boundaries, hidden state, and stochasticity can invalidate the example. A family of learned operators need not form a group. Approximate composition can instead express a task-dependent semigroup or a collection of partial maps.

For example, if true maps satisfy $T_c = T_a \circ T_b$ on a domain, $\|\widehat{T}_j - T_j\|_\infty \le \delta_j$ for $j = a, b, c$, and $\widehat{T}_a$ is K_a-Lipschitz on the required intermediate domain, then

$$\|\widehat{T}_a \circ \widehat{T}_b - \widehat{T}_c\|_\infty \le K_a\delta_b + \delta_a + \delta_c.$$

Insert $\widehat{T}_a(T_b(z))$ and $T_a(T_b(z))$, and apply the triangle inequality. The bound concerns population-domain accuracy, not merely agreement on a few replayed paths. Reuse does not eliminate accumulation of approximation error.

19.11 Consolidation as Compilation

The distinction also suggests a computational interpretation of systems consolidation.

Early in learning, successful prediction or retrieval may depend strongly on hippocampal indices:

$$x_t \longrightarrow h_t \longrightarrow y_t.$$

After repeated replay and cortical learning, the system may acquire a more direct mapping when the currently available input is task-sufficient,

$$x_t \longrightarrow \widehat{z}_t \longrightarrow y_t,$$

or

$$\widehat{z}_t \longrightarrow G_\theta(\widehat{z}_t, a_t).$$

The hippocampal scaffold need not disappear physically.
What changes is the degree to which behavior depends on it.
This motivates the computational hypothesis

systems consolidation can be interpreted as compilation: repeated episodic computations are compressed into reusable cortical operators.

This analogy is compatible with the broad complementary-learning-systems picture but is more specific about the computational transformation [102].

The analogy to compilation distinguishes two forms of representation.
A scaffold may specify a computation by a long path,

$$h_1 \to h_2 \to \cdots \to h_T.$$

A learned cortical rule may implement a shorter reusable operation,

$$\widehat{z} \to G_\theta(\widehat{z}).$$

This is closely related to the compression-of-composition principle developed elsewhere in the book.

Importantly, compilation here means a change in functional dependence and description, not necessarily the physical disappearance of hippocampal activity.

Compilation cannot remove necessary information. The displayed $x_t \mapsto \widehat{z}_t \mapsto y_t$ is possible only when x_t supplies what the task needs. In a partially observed environment, the compiled system may still require a recurrent state, context, or retrieval. In the earlier hidden-bit example, the same current observation requires two different answers. No amount of compilation can produce an exact deterministic x_t-only solution. The correct comparison is between systems with the same specified deployment information.

A simple accuracy transfer makes the positive case precise. Let a frozen memory-assisted teacher compute $y_T = D(z_T)$, and let a compiled encoder produce z_C from the allowed deployment input. If D is K_D-Lipschitz,

$$\|D(z_C) - D(z_T)\|_{L^2} \le K_D \|z_C - z_T\|_{L^2}.$$

For an L_ℓ-Lipschitz prediction loss and $\|z_C - z_T\|_{L^2} \le \epsilon$ under the deployment distribution,

$$R(D(z_C)) \le R(y_T) + L_\ell K_D \epsilon.$$

This follows by the pointwise Lipschitz inequalities and $\mathbb{E}\|z_C - z_T\| \le \|z_C - z_T\|_{L^2}$. It requires a good teacher and population control of the replacement, not simply imitation on stored episodes. Multistage replacement has the additional stability requirements discussed in Chapter 15.

Description, execution, and storage are different resources. A reusable rule may shorten the description while still being executed many times. A smaller code does not by itself establish lower latency or energy; Chapter 10 separates those resource questions. For a Chapter 13 compression certificate, a predictor that still retrieves a learned episodic dictionary must account for that dictionary, its content, addressing, and decoder. A short instruction saying "look up the episode" is not a short standalone description of all retrieved answers. Consolidation may reduce external-memory dependence, parameter description, or both; these are different measurable outcomes.

19.12 Episodic and Semantic Memory

The same picture offers a natural interpretation of the distinction between episodic and semantic memory.

Episode-specific indices preserve

$$\text{who}, \quad \text{where}, \quad \text{when}, \quad \text{which particular event}.$$

Cortical abstraction deliberately removes some of those distinctions.

Across multiple visits to the same environment, one may move from

"I left the keys on this counter Tuesday morning"

to

"keys are usually kept on the kitchen counter."

The first depends strongly on instance identity.

The second is a schema.

Thus abstraction gains reuse at the price of episodic specificity.

The second statement is a predictive regularity only if the experience supports it on the task distribution. Compressing episodes into an incorrect schema is not successful abstraction. Episodic specificity may remain essential when the question asks about Tuesday rather than the usual location. The model therefore predicts selective consolidation of reusable knowledge, not universal loss of hippocampal dependence for every remote episodic detail.

19.13 Relation to Classical Hippocampal Theories

The scaffold hypothesis is compatible with several influential views of hippocampal function, but it should not be taken as proving that they are all manifestations of one mechanism.

19.13.1 Cognitive maps

Cognitive-map theories emphasize relational structure in spatial navigation [119].

Within the scaffold framework, physical space is one important case of a more general relational graph.

The relevant object is

$$\text{state} + \text{adjacency} + \text{transition}.$$

Nothing requires those states to correspond only to Euclidean position.

19.13.2 Pattern separation

Pattern-separation theories emphasize the need to distinguish similar or overlapping experiences [99, 120, 173].

In the present proposal, separation is not the final objective.

It is an implementation strategy supporting rapid storage, reduced interference, and later comparison across experiences.

19.13.3 Indexing theory

Indexing theories describe the hippocampus as linking an index to distributed cortical content [150, 151].

The present framework retains that idea but adds temporal and relational links among indices.

The index is therefore not only an address.

It can participate in a scaffold of transitions.

19.13.4 Complementary learning systems

Complementary learning systems emphasize rapid hippocampal learning and slower cortical learning [102].

The scaffold hypothesis gives that division a particular computational interpretation:

fast system preserves individual examples; slow system discovers reusable equivalences among them.

19.13.5 Predictive and successor representations

Predictive and successor-representation theories characterize states by their relationships to future states or occupancies.

Such representations can naturally be constructed from repeated transitions through a relational scaffold [36, 146].

This makes successor-like structure compatible with the framework.

It does not imply that every hippocampal representation is fundamentally a successor representation.

For a finite Markov state representation and a fixed policy β, let P_β be the transition matrix. For $0 < \gamma < 1$, the successor representation has the form

$$M_\beta = \sum_{j=0}^{\infty} \gamma^j P_\beta^j = (I - \gamma P_\beta)^{-1}.$$

This summarizes discounted expected occupancies under that policy [36, 146]. It need not be a policy-independent transition model. Changing behavior can change M_β even if the environment is fixed, and an arbitrary index of a partially observed episode need not itself be Markov. Predictive state similarity is therefore a task- and policy-dependent candidate for quotienting, not a universal definition of episodic identity.

19.14 An Empirical Example: Structured Latent Geometry

Entorhinal grid-cell population activity provides an instructive example of low-dimensional structure within a high-dimensional neural population.

Gardner and collaborators reported toroidal organization of simultaneously recorded population activity within individual grid-cell modules in rats [49].

The relevant lesson for the present chapter is modest but important:

high-dimensional neural activity can organize around a low-dimensional structured latent manifold.

This observation is compatible with the idea that neural systems can maintain structured latent coordinates useful for navigation and prediction.

It does not establish that grid-cell activity is the hippocampal index h_t used in our abstraction.

Nor does it prove the scaffold hypothesis.

Grid cells are also principally associated with entorhinal circuitry rather than being identified with hippocampal episodic indices.

The result is therefore evidence for the broader possibility of structured latent geometry in the hippocampal–entorhinal navigation system, not a direct test of the proposed scaffold.

Nor does low intrinsic manifold dimension establish low-arity dynamics in an arbitrary coordinate system. Topology, coordinate dimension, transition factorization, and learning accessibility are different properties. The reported torus is a valuable example of population geometry; the proposed transfer from episodic relations to cortical compositional operators requires a separate experiment.

19.15 What the Scaffold Hypothesis Predicts

A useful theory should lead to experiments.

The proposal makes several qualitative predictions.

19.15.1 Prediction 1: relationally structured replay should aid abstraction

Suppose two learners receive the same set of episodes.

One receives replay preserving the true temporal and relational order,

$$h_t \to h_{t+1}.$$

The control must distinguish shuffling complete transition tuples from shuffling their endpoints. The former preserves a one-step supervised objective; the latter corrupts relational labels. A relation-preserving scaffold should be compared with both, as well as with a direct learner that receives the same true tuples without explicit episodic indexing.

The hypothesis predicts a benefit when accurate relational information is made more accessible within the chosen learner's resources. It does not predict that every permutation of an unchanged tuple dataset must harm learning. Multistep replay can have a distinct advantage only when the learner or objective actually uses multistep relations.

19.15.2 Prediction 2: episodic separation and semantic abstraction should have opposite geometry

For the distinction relevant to the selected task, the hypothesis predicts stronger separation of episode identities in the indexing component and stronger grouping of task-equivalent episodes in the abstraction component. These are different representations and may coexist, rather than one population necessarily changing from separation to clustering.

The proposed comparison is therefore between

$$\text{episode-specific separation}$$

and

$$\text{cross-episode clustering by shared structure.}$$

19.15.3 Prediction 3: cortical rules should transfer beyond stored episodes

If cortex has learned a reusable operator rather than a collection of stored transitions, the operator should generalize to novel states that were not individually indexed during learning.

Thus transfer to new but structurally similar environments is an important test.

19.15.4 Prediction 4: hippocampal dependence should decrease as cortical predictive structure improves

For tasks whose needed information can be retained by the cortical model, the hypothesis predicts an inverse relationship across consolidation between

$$\text{dependence on episode-specific hippocampal retrieval}$$

and

$$\text{quality of cortical prediction from shared latent variables.}$$

The prediction concerns functional dependence, not disappearance of hippocampal activity.

19.15.5 Prediction 5: a scaffold should reduce recovery difficulty

Artificial systems offer a direct test.

Compare

$$\text{end-to-end learner from raw sensory sequences}$$

against

$$\text{episodic-index learner} + \text{relational replay} + \text{slow structural learner.}$$

Hold the underlying data fixed.

At fixed independent experience, measure whether the complete scaffolded procedure reduces time or computation to a specified held-out performance. A separate learning curve can vary the number of independent episodes. Construction, storage, retrieval, and replay must be charged to the scaffolded procedure; changing its context or supervision changes the experiment. The hypothesis permits a benefit in one resource without improvement in all of them.

These five predictions are conditional tests of the proposal, not general consequences of hippocampal anatomy. Negative outcomes should distinguish an uninformative index, inaccurate links, insufficient coverage, an unsuccessful learner, and the possibility that the direct baseline already extracts the relevant relations efficiently.

19.16 A Direct Artificial Experiment

The hypothesis can be tested without making any anatomical assumptions.

Consider trajectories generated by a latent dynamical system

$$z_{t+1}^{(i)} = F_i\left(z_t^{\mathrm{pa}(i)}, a_t\right), \qquad |\,\mathrm{pa}(i)| \le k,$$

but provide the learner only high-dimensional observations

$$x_t = R(z_t) + \epsilon_t.$$

Compare two primary systems on the same recorded experience and, when histories are needed, the same allowed history and context. The following control conditions make the two-system comparison interpretable.

Direct learner. In a fully observed one-step experiment, the model receives

$$(x_t, a_t, x_{t+1})$$

and learns prediction end to end. In a partially observed experiment, supply an appropriately matched recurrent or history-based baseline; do not compare a memoryless predictor against a scaffold with privileged history.

Scaffold learner. A fast memory additionally stores episode-specific indices

$$h_t$$

and transition relations

$$(h_t, a_t, h_{t+1}).$$

A slower learner uses these relations to infer a shared latent representation and transition model. Any latent labels used to construct the scaffold must be declared as supervision, not treated as free indexing.

Matched controls. Use an indexed store without relational replay, uniform replay of intact transition tuples, relation-preserving sequence replay, and an endpoint- shuffled negative control that preserves appropriate marginal frequencies. Include an oracle intermediate-state condition as a hard-C3 supervised benchmark, not as another equal-information method. A consistent permutation of all record IDs should leave a symbolic scaffold algorithm's predictions unchanged; otherwise it may be exploiting arbitrary identifier values. Neural implementations should test this invariance rather than assume it.

The main no-oracle experiment must construct indices and links from allowed observations and context alone. Count independent trajectories separately from adjacent transitions and replay presentations. Use matched initialization seeds, architecture or parameter budgets, exposure

counts, and optimization budgets where possible, while also reporting actual computation and peak memory. Split by independent episode or environment, rather than leaking near-duplicate transitions across training and test sets. Select settings using validation data and evaluate the frozen procedure on held-out tasks.

19.16.1 Evaluation

Measure

$$\text{sample complexity}, \quad \text{optimization time},$$
$$\text{latent-state recovery}, \quad \text{transfer}, \quad \text{long-horizon prediction}.$$

Held-out prediction is the primary task metric. Structural recovery is a separate diagnostic and must use a specified equivalence class, because latent decompositions need not be uniquely identifiable.

For example, in a synthetic setting one may regard two recovered representations as equivalent under permutations of interchangeable latent coordinates, or under other explicitly specified invertible reparameterizations with controlled complexity and conditioning that preserve the local dependency structure. A decoder acting on the full input or the entire memory cannot be allowed to hide the original task inside the structural diagnostic.

The strongest result would therefore be one of the following:

> scaffolded training recovers the target latent dynamics
> up to a specified representational equivalence
> with substantially lower recovery cost,

or, more generally,

> scaffolded training recovers a functionally equivalent compositional
> model with substantially lower recovery cost.

This would provide a direct computational demonstration of the chapter's central idea without requiring literal recovery of every generating coordinate.

An operational success criterion. For a fixed test distribution, loss, horizon, and target risk ϵ, report the resources needed by each complete procedure to return a predictor with that risk, together with failure frequencies and uncertainty across independent seeds and environments. For example, compare

$$(N_{\text{episodes}},\ T_{\text{train}},\ B_{\text{memory}},\ T_{\text{predict}},\ R_{\text{test}}),$$

not just the number of replay updates. Explicitly test new initial states, new combinations of familiar local states, and new environments as distinct generalization regimes. None follows from the others.

An accurate feasible predictor is already a computational witness in the sense of Chapter H; it need not match the teacher's chosen latent factorization. The proposed comparison tests whether the scaffold makes such predictive success easier to obtain. A finite stopped model may be returned; neither exact interpolation nor zero minibatch variance is required. No training experiment is reported in this chapter.

19.17 What Is Established and What Is Hypothesis

The levels of claim should remain separate.

19.17.1 Established biological observations

The cited observations should be distinguished from the models built to explain them. Wilson and McNaughton observed experience-related ensemble reactivation during sleep; Lee and Wilson analyzed replay of sequential experience; Foster and Wilson reported reverse sequences during awake periods [167, 86, 48]. Bakker and collaborators reported high-resolution human fMRI responses in a combined CA3/dentate-gyrus region consistent with a pattern-separation bias [10]; this is not a direct observation of the TopK implementation or a single-subfield algorithm.

The cognitive-map, indexing, and complementary-learning-systems accounts provide theoretical interpretations of memory phenomena [119, 150, 102]. They are not themselves experimental demonstrations that the present scaffold supplies latent constituent supervision. The broader pattern- separation literature is reviewed in [173].

Entorhinal grid-cell populations can exhibit low-dimensional toroidal population geometry [49].

These empirical phenomena motivate the computational abstraction.

They do not by themselves establish the scaffold mechanism proposed here.

19.17.2 Computationally plausible interpretation

Fast separation and indexing can preserve individual episodes.

A graph of temporally linked indices can preserve relational information.

Replay can provide additional structured training events to slower learners, without turning replay count into independent sample size.

Parameter sharing across many episode-specific transitions can produce more compact and reusable representations.

These statements describe possible computational consequences of the proposed architecture.

19.17.3 Open hypothesis

The stronger claim is that hippocampal scaffolding specifically reduces the recovery difficulty of learning compositionally sparse latent dynamics.

This has not been established in general.

In particular, the present chapter does not claim that

- hippocampal activity literally implements random projection plus TopK;
- sparse graph degree is equivalent to compositional sparsity;
- the hippocampus must contain, or cannot contain, a reusable predictive model;
- replay performs one specific optimization algorithm;
- cortical abstraction requires hippocampal indexing in every learning problem;
- grid-cell toroidal geometry proves the scaffold hypothesis;
- the cortical latent variables must coincide uniquely with the generating latent coordinates.

The hypothesis is narrower:

> fast episodic indexing may expose relational intermediate structure that makes slower abstraction easier to recover.

The three propositions establish only their stated computational and statistical facts. None proves that the proposed quotient can be learned from an arbitrary scaffold, that the learned coordinates have bounded arity, or that a biological circuit implements the construction. They locate the assumptions that a stronger recovery theorem or experiment would need to supply.

19.18 Relation to World Models

The chapter on world models begins with the problem

$$\text{observation history} \longrightarrow \text{latent state} \longrightarrow \text{predictive dynamics}.$$

The hippocampal scaffold provides one possible intermediate layer:

$$\boxed{\text{observation} \to \text{episodic index} \to \text{relational trace} \to \text{latent world model}.}$$

This suggests a division of labor.

The hippocampal system need not itself construct the final generative model.

It may instead preserve relational evidence from which a slower cortical system can learn such a model.

In this sense the scaffold addresses part of the state-discovery problem: it supplies additional relational variables without solving the full latent representation problem.

The observation encoder must retain enough history for the requested prediction horizon; a single x_t may not identify z_t. In stochastic settings, a distribution over possible states or another predictive state can be the appropriate object rather than a unique point estimate. A model fitted to a behavior policy's observed transitions does not thereby identify outcomes of unsupported actions or remove hidden confounding. Chapter 12 makes the corresponding distinction between outcome-conditioned behavior and environmental dynamics.

Scaffolding can help retain relevant context and compare experiences. Extrapolation, intervention, and planning still require the appropriate coverage or modeling assumptions. Even when prediction succeeds, the learned state need not be the environment's designated latent coordinate.

19.19 Relation to Sparse Compositionality

In this hypothesis, the stronger smooth, low-arity structure is assumed for an appropriate representation of the latent dynamics,

$$z_{t+1}^{(i)} = F_i\left(z_t^{\mathrm{pa}(i)}, a_t\right), \qquad |\,\mathrm{pa}(i)| = O(1).$$

The hippocampal contribution is a recovery hypothesis.

It provides intermediate relational observations that may make the constituents

$$F_i$$

easier to discover.

Thus the relation is

world possesses local compositional structure
⇓
experience contains episodes of that structure
⇓
hippocampus indexes and links episodes
⇓
cortex compares and compresses recurring transitions
⇓
reusable compositional model.

The hippocampus does not create sparse compositionality.

It may help make it legible.

That structural assumption is coordinate dependent. A task-preserving change of latent coordinates can turn a sparse displayed update into a dense one. A claim of sparse *structural recovery* must therefore control the admissible coordinate changes. The weaker computational-witness conclusion of Chapter H does not identify a unique smooth latent factorization and does not require one.

19.20 Relation to C1, C2, and C3

The proposal can be placed precisely within the framework of the earlier chapters.

C1 concerns the existence of a feasible compositional world model:

$$z_{t+1}^{(i)} = F_i(z_t^{\mathrm{pa}(i)}, a_t), \qquad |\,\mathrm{pa}(i)| = O(1).$$

The hippocampal scaffold does not establish C1.

C2 concerns whether raw observations expose enough information for a learner to discover useful latent variables and constituent functions.

The scaffold can modify that recovery problem by retaining context, organizing observed relations, or incorporating explicitly declared side information. Retention and organization of a full record do not themselves add evidence; side information not already determined by that record can do so.

Hard C3 would provide the actual intermediate constituent variables and their local dependencies.

The proposed hippocampal scaffold is weaker:

hippocampal scaffold $\approx$ C3-like partial relational trace.

It may reduce the burden on an end-to-end learning mechanism without eliminating the need for useful accessible signal, local-state coverage, optimization, and statistical control. C3-like support can also help when end-to-end learning is already possible; it is not only a rescue mechanism after C2 has failed.

Thus

C1:	the compositional latent dynamics exist,
C2:	useful learning signals must be accessible,
hard C3:	the true intermediate computation is exposed,
hippocampal scaffold:	some intermediate relations are accessible.

This placement is important.

The biological hypothesis concerns primarily the *difficulty of recovery*, not the existence of the underlying representation.

Chapters 16 and 17 analyze different ways of using available training signals, while Chapter 13 addresses the complexity of the returned predictor. A scaffold neither supplies a local credit-transport theorem nor a generalization bound by itself. These mechanisms can cooperate without becoming necessary steps in every successful learning trajectory.

19.21 Conclusion

The hippocampus is often described as a memory store.

A more computationally useful possibility is that it also serves as a temporary scaffold for learning.

Its fast dynamics can

separate episodes, bind content,
index experience, preserve relational transitions.

The cortex can then perform a different operation:

identify equivalences, share parameters,
discover reusable operators, compress repeated compositions.

This leads to a computational interpretation of consolidation:

episodic scaffold $\rightarrow$ replay $\rightarrow$ shared latent variables $\rightarrow$ compiled cortical rules.

The central proposal does not require the hippocampus itself to contain the final reusable world model, and it does not exclude hippocampal predictive structure.

Nor is an episodic graph itself equivalent to sparse compositionality.

The proposal is instead that a fast episodic system may preserve relational evidence from which a slower cortical system can more easily recover a reusable latent model.

In the terminology used throughout this book,

the hippocampal scaffold is a candidate biological mechanism for
exposing a C3-like partial relational trace.

That makes its proposed role primarily one of *recovery* rather than representation.

The positive proposal can now be stated more precisely:

$$\boxed{\begin{array}{c}\text{reliable episodic addressing and retained context}\\ +\ \text{task-relevant, adequately covered relations}\\ +\ \text{a learner that forms predictive, stable abstractions}\\ \Longrightarrow\ \text{a candidate route to reusable computation.}\end{array}}$$

The final arrow is the chapter's learning hypothesis, not a theorem of hippocampal necessity. Re-encoding the same record need not add information; replay need not add independent samples; and a useful quotient need not identify the teacher's coordinates. The issue is whether this organization of experience lets a specified learner find an accurate, feasible model with fewer resources.

The world may already possess sparse compositional structure. The hippocampus may help make evidence for that structure accessible.

CHAPTER 20

Mixture of Experts and Spatial Manifolds

Manifolds, compositional sparsity, and mixture-of-experts architectures provide three complementary ways of organizing a high-dimensional prediction problem. A manifold describes where inputs lie; compositional sparsity describes how a target is built from low-arity constituents; a mixture of experts determines which computational modules are evaluated for a particular input. These structures can cooperate, but none is simply another name for the others.

The geometric idea is constructive. When a target can be described in well-conditioned local coordinates, a router can select an appropriate chart and an expert can approximate the corresponding local function. Repeated routing can also reuse a library of constituents across many combinations of subtasks. Under explicit regularity, approximation, and stability assumptions, the resulting error can depend on intrinsic or local dimension rather than the ambient dimension in its approximation exponent.

This chapter relates the approximation results of Wang and E to the DeepSeek-V3 architecture. The relationship is a structural interpretation, not an identification of a trained language model with a proved manifold construction. In particular, conditional computation does not establish that training discovers coordinate charts, minimizes metric entropy, or selects an optimal computation graph.

$$\boxed{\begin{array}{c} \textit{useful local representations} \\ + \textit{ accurate routing and expert evaluation} \\ + \textit{ controlled composition} \\ \Longrightarrow \textit{ an economical routed approximation.} \end{array}}$$

Discovering that approximation and establishing its population accuracy remain the learning questions developed in the preceding chapters.

20.1 The Geometric Primitives of MoE Networks

The basic decomposition in this chapter has two parts: a *spatial assignment problem*, which makes a suitable local representation available, and a *localized function approximation problem*, which evaluates the target in that representation. Here "spatial" refers to regions of an input or hidden-representation space. It does not imply physical locality of the processors executing the experts.

For a token or feature vector $x \in \mathcal{X} \subseteq \mathbb{R}^D$, a normalized mixture has the form

$$\Psi(x) = \sum_{i=1}^{E} w_i(x) F_i(x), \qquad w_i(x) \geq 0, \qquad \sum_i w_i(x) = 1,$$

where at most $K \leq E$ weights are nonzero. The F_i are expert maps; the router computes the active indices and their weights. A shared expert or residual branch can be added separately. The number K controls *expert activation sparsity*, not the number of input coordinates on which an active expert may depend.

Three structures, three different questions. A low-dimensional support hypothesis says that the inputs lie on a d-dimensional manifold $\mathcal{M} \subset \mathbb{R}^D$. A compositional hypothesis says that the target has a computation graph with constituents of controlled local arity. Sparse expert activation says that few experts are evaluated at each input. For example, on a full-dimensional cube,

$$f(x) = \prod_{j=1}^{D} x_j$$

has a binary compositional realization without low-dimensional input support. Conversely, selecting just one expert does not make that expert's map $x \mapsto \sum_{j=1}^{D} a_j x_j$ a bounded-arity constituent at the displayed module level.

That dense linear map can itself be compiled into bounded-fan-in arithmetic operations, as in Chapter 4. The compilation statement is different from claiming a fixed low-dimensional smooth constituent at the expert level. Neither a scalar expert output nor a small number of activated experts establishes the latter.

Charts provide a possible organization of experts. Let $\{(U_i, \phi_i)\}_{i=1}^{E}$ be an atlas for $\mathcal{M}$, with $\phi_i : U_i \to \mathbb{R}^d$. On a chart, write

$$h_i = f|_{U_i} \circ \phi_i^{-1}, \qquad f(x) = h_i(\phi_i(x)), \quad x \in U_i.$$

A router may select any chart containing x, or mix experts that are all accurate there. Atlas regions normally overlap. There need not be a unique correct chart, and a learned expert's routing region need not be a chart at all: it can be disconnected, high-dimensional, or organized by a computational rather than geometric criterion.

The positive proposal is therefore conditional: *when an economical atlas and suitable local target descriptions exist, routed computation can exploit them.* The atlas, coordinate maps, and router are part of the representation to be supplied or learned. They are not free consequences of putting a top-K operation into a network.

Conditional evaluation versus total resources. If an expert costs C_{expert} operations, the leading expert work per input can scale as KC_{expert} rather than EC_{expert}. A straightforward linear router still evaluates E scores in D coordinates, costing order ED arithmetic operations before selection. Expert storage, routing, normalization, communication, and load imbalance must also be counted. Chapter 10 explains why sparse logical selection need not imply inexpensive physical communication.

Similarly, only selected experts receive that token's direct data-loss gradient through the expert branch, but this does not mean all other parameters remain unchanged under momentum or regularization. Conversely, dense evaluation does not ensure that every parameter has a nonzero gradient on every token.

20.2 Manifold Approximation and Theorem 4.8

Wang and E's *On the Expressive Power of Mixture-of-Experts for Structured Complex Tasks* gives an explicit manifold-based construction [160]. We use Theorem 4.8 and its proof in the NeurIPS 2025 paper; the February 2026 arXiv revision retains that theorem number. It is an approximation result for a specified ReLU MoE class, not a convergence result for jointly trained routers and experts.

Regularity and architectural conventions. Fix a compact smooth d-dimensional manifold embedded in $\mathbb{R}^D$, a finite atlas, and local coordinate domains of controlled size. Work on compact chart cores inside the coordinate neighborhoods when an extension is needed. For the rate statement, require local coordinate targets h_i to admit bounded C^{r_i} extensions, with finite integers $r_i \geq 1$, on domains containing both exact and approximated coordinates.

The source defines smoothness through C^r classes. Its displayed C^1 factor controls propagation of chart error. A mere continuity assumption does not supply an algebraic smoothness rate, and an arbitrary Sobolev assumption cannot be substituted without checking the norm, embedding, and extension conditions. In the paper's regular-atlas convention, coordinate maps have sufficiently smooth ambient extensions, with smoothness exceeding $(D+3)/2$.

Write $\mathcal{E}^{\mathrm{FFN}}_{\ell,m}$ for best uniform approximation by the specified ℓ-layer, width-m ReLU class. In the source's depth convention, the MoE construction has two routed stages, each with E experts; the expert subnetworks have three affine layers. Its first stage constructs routing features and preserves the input. The second stage selects a chart and applies its local approximator. This is not the claim that a single linear gate in the original input coordinates can identify an arbitrary atlas.

Theorem 16 (Manifold approximation via shallow MoE: attributed rate form)**.** *Under the fixed-atlas, extension, and smoothness conditions above, Wang and E's Theorem 4.8 gives a two-stage MoE realization Ψ_m in their class $\mathcal{H}^{2,E}_{3,m}$, for sufficiently large expert width m, with*

$$\|f - \Psi_m\|_{L_\infty(\mathcal{M})} \leq C_{\mathcal{A},f} \max_{i\in[E]} \left\{ m^{-\alpha_i} (\log(2+m))^{b_i} \right\},$$
$$\alpha_i = \min\left\{ \frac{r_i}{d}, \frac{1}{2} \right\}, \qquad b_i \geq 0. \tag{20.1}$$

The constants and width threshold depend on the fixed geometric and function-class data. The paper expresses its width requirement as $m \gtrsim E^2$ in its notation; this is not a uniform total-parameter, router-cost, or bit-complexity bound as the atlas and ambient dimension vary.

This is a scoped restatement of the reported rate, not a new proof of the paper's complete network-size estimate. In particular, routing-feature construction, concatenation, identity channels, and width allocation must be included before using that estimate in a growing-architecture resource comparison. The following elementary calculations make the error mechanism explicit without identifying expert width with the entire implementation cost.

Why approximate partition functions can give valid hard routing. Let $\rho_1, \ldots, \rho_E$ be a partition of unity subordinate to the atlas, and suppose

$$\max_i \|\widehat{\rho}_i - \rho_i\|_{L_\infty(\mathcal{M})} \leq \frac{1}{4E}.$$

Choose $j(x) \in \arg\max_i \widehat{\rho}_i(x)$ with a specified tie rule. Since some $\rho_i(x) \geq 1/E$,

$$\rho_{j(x)}(x) \geq \widehat{\rho}_{j(x)}(x) - \frac{1}{4E} \geq \frac{1}{E} - \frac{2}{4E} = \frac{1}{2E} > 0.$$

Consequently $x \in U_{j(x)}$. This is the routing step in the proof of Wang and E. It need not recover a unique chart index; it only selects a chart on which the expert's guarantee applies. The partition functions may be nonlinear even when the chart maps are affine restrictions.

Coordinate error and local function error. Suppose an expert is implemented as $F_i = \widehat{h}_i \circ \widehat{\phi}_i$, with

$$\|\widehat{\phi}_i - \phi_i\|_{U_i} \le \eta_i, \qquad \|\widehat{h}_i - h_i\|_{Q_i} \le \varepsilon_i,$$

where Q_i contains both coordinate images and the chosen extension of h_i is L_i-Lipschitz there. Adding and subtracting $h_i(\widehat{\phi}_i(x))$ gives

$$|F_i(x) - f(x)| \le \varepsilon_i + L_i\eta_i, \qquad x \in U_i.$$

The chart map still accepts a D-dimensional input. Its favorable approximation cost comes from the stated regularity, or from its being an affine local projection, not from its output dimension alone. In the affine-chart case of the source's Corollary 4.9, chart evaluation is exact and this source of approximation error disappears. Its matrix storage and multiplication costs do not disappear.

For a merely continuous local target, the same reasoning uses its modulus of continuity ω_i:

$$|F_i(x) - f(x)| \le \varepsilon_i + \omega_i(\eta_i).$$

No fixed power of m follows from this identity without more regularity.

Proposition 35 (Local approximation with imperfect routing)**.** *Let $\{U_i\}_{i=1}^{E}$ cover a domain $\mathcal{X}$. Suppose $|f(x)| \le B$, $|F_i(x)| \le B$ everywhere, and $|F_i(x) - f(x)| \le \delta_i$ on U_i. Let $w_i(x) \ge 0$ sum to one, and define the weight assigned to experts outside their validity regions by*

$$\beta(x) = \sum_{i : x \notin U_i} w_i(x).$$

Then the mixture satisfies

$$|\Psi(x) - f(x)| \le \sum_{i : x \in U_i} w_i(x)\delta_i + 2B\beta(x) \le \delta_{\max} + 2B\beta(x).$$

In particular,

$$\|\Psi - f\|_\infty \le \delta_{\max} + 2B\|\beta\|_\infty, \qquad \|\Psi - f\|_{L^2(P_X)} \le \delta_{\max} + 2B\|\beta\|_{L^2(P_X)}.$$

For the chart implementation above, one may take $\delta_i = \varepsilon_i + L_i\eta_i$.

Proof. Because the weights sum to one, $\Psi - f = \sum_i w_i(F_i - f)$. Split this sum into valid and invalid experts. For the former use the local error bound; for the latter use $|F_i - f| \le 2B$. This proves the pointwise inequality. Take a supremum for the uniform result, or apply the triangle inequality in $L^2(P_X)$ for the population result. □

This proposition is an added error calculation. It covers hard selection and normalized mixtures. For hard top-one routing, β is the indicator of an invalid assignment, giving an L^2 contribution at most $2B\sqrt{P_X(\text{invalid assignment})}$. Rare routing errors can be acceptable for population prediction even when they preclude a small uniform error. Conversely, a local expert guarantee says nothing about an input sent outside that expert's validity region without an additional bound such as B.

The domain of the conclusion matters. Equation (20.1) concerns prediction on $\mathcal{M}$. It does not control a full-dimensional neighborhood. If an observation is $x + \zeta$ with $x \in \mathcal{M}$, an extension and a robustness analysis are needed. Compactness also gives only finitely many charts for a fixed manifold; it does not give a uniformly small atlas for a sequence of increasingly complicated manifolds. Atlas size, chart conditioning, curvature, and proximity of different branches can enter the constants.

Finally, manifold-adapted approximation is not exclusive to MoE. The source itself compares its construction with earlier dense-network manifold approximation results. A low-dimensional upper bound for one architecture is not a lower bound for every dense architecture on the same supported target. A separation requires a specified comparator and a common accounting of total parameters, active computation, precision, and approximation error.

20.3 Deep MoEs and Exponential Compositional Capacity

The role of depth connects to the compositional approximation framework of Chapter 4 and Mhaskar–Poggio [109]. Two kinds of composition should be distinguished: selecting among reusable local modules and passing approximate states through a genuinely nested computation.

For a nested target, write

$$f(x) = (h_L \circ \cdots \circ h_2 \circ h_1)(x). \tag{20.2}$$

The maps h_ℓ may themselves contain several low-arity constituents. A bound d^* on their relevant local dimensions is an assumption about the target and representation, not a consequence of activating few experts.

What the E^L statement means. Wang and E's deep construction concerns a structured product-domain class. In a representative form,

$$f(x_1, \ldots, x_L) = f_{\text{out}}(f_{1,i_1}(x_1), \ldots, f_{L,i_L}(x_L)), \qquad x_\ell \in U_{\ell,i_\ell}.$$

There are E local alternatives for each factor, hence E^L possible chart combinations from a library of LE local maps. Compatibility on chart overlaps is assumed. In Theorem 5.2 the outer map is taken to be the identity, so the network returns the tuple of local outputs. A depth-$2L$ construction then has a maximum local error in the coordinatewise supremum norm. An arbitrary outer map requires its own realization and error-propagation bound [160].

Thus the constructive advantage is *reuse across combinations*, not the ability to store E^L unrelated functions at the cost of LE independent functions. In a top-one network with L routing stages, a route signature lies in $[E]^L$, so there are at most E^L signatures. Some may never occur; different signatures can compute the same output; and the set of inputs inducing one signature can have several connected components. Counting signatures is not counting independent parameters, independently adjustable tasks, or necessarily connected spatial regions. For top-K routing, the corresponding crude count of active-set signatures is at most $\left(\binom{E}{K}\right)^L$.

The source's structured example supports the exponential-combination claim. It does not establish a universal exponential advantage over deep dense networks, which can also exploit composition and parameter reuse. Nor does selecting the largest router score prove that the selected computation is optimal for prediction or cost.

Proposition 36 (Stable recomposition of routed modules)**.** *Let $h_\ell : S_{\ell-1} \to S_\ell$, $\ell = 1, \ldots, L$, be K_ℓ-Lipschitz maps between specified normed state domains. Suppose approximating routed modules $\widetilde{h}_\ell$ have*

uniform local errors at most δ_ℓ on $S_{\ell-1}$, and both exact and approximate trajectories remain in these domains. Put $\Psi = \widetilde{h}_L \circ \cdots \circ \widetilde{h}_1$. Then

$$\boxed{\|f - \Psi\|_\infty \le \sum_{\ell=1}^{L} \delta_\ell \prod_{j=\ell+1}^{L} K_j.} \tag{20.3}$$

The error δ_ℓ includes the routing, coordinate, expert, and numerical errors of the complete module.

Proof. Let $z_\ell = h_\ell(z_{\ell-1})$ and $\widetilde{z}_\ell = \widetilde{h}_\ell(\widetilde{z}_{\ell-1})$, starting from the same input. Adding and subtracting $h_\ell(\widetilde{z}_{\ell-1})$ gives

$$\|z_\ell - \widetilde{z}_\ell\| \le K_\ell \|z_{\ell-1} - \widetilde{z}_{\ell-1}\| + \delta_\ell.$$

Iterating from zero initial discrepancy proves the claim, with an empty product interpreted as one. □

In particular, if all complete modules are nonexpansive and $\delta_\ell \le Cm^{-a}$, the error is at most LCm^{-a}. If $K_j \le K < 1$, it is at most $Cm^{-a}/(1-K)$. Without such control, products of K_j can grow exponentially. Substituting $a = \kappa/d^*$ requires a local approximator theorem giving that rate; using the shallow rate quoted above instead gives the exponent $\min\{\kappa/d^*, 1/2\}$, with its logarithmic factors. Neither rate is a statistical uniform-convergence statement about a trained network.

Smooth experts do not ensure a smooth routed module. Take two constant experts, $F_1(x) = 0$ and $F_2(x) = 1$, and select the first for $x < 0$ and the second for $x \ge 0$. Both experts have Lipschitz constant zero, but the routed map is discontinuous. A Lipschitz bound for a composition therefore cannot be obtained by multiplying the expert constants while ignoring the switches. One needs compatible outputs near boundaries, a suitably controlled soft router, or a margin-restricted trajectory analysis.

For a fixed top-K selector with score gap $\Delta = a_{(K)} - a_{(K+1)} > 0$, perturbing every score by less than $\Delta/2$ leaves its selected set unchanged: every selected–unselected score difference remains positive. This elementary margin condition can protect a route locally. It does not give a uniform margin on an entire manifold, and a grouped selector needs protection at both selection levels.

Table 20.1: Structural alignment without identifying different notions of sparsity. Each row states a possible implementation and its condition.

@>p0.25>X>X@

Structural ingredient	Possible MoE role	Qualification
Low-arity constituent h_i	Expert or subnetwork evaluating it	Expert activation alone does not bound functional arity.
Low-dimensional chart U_i	Local coordinates followed by a specialist	The atlas, chart map, and valid assignment must be available.
Hierarchical composition	Successive routed stages	State retention and error amplification must be controlled.
Reusable constituent library	Experts reused across input contexts	Reuse is distinct from merely allocating separate experts.
Input-dependent computation	Router selects a branch or mixture	Selection by score is not a certificate of optimality.

What the entropy interpretation can establish. The description-length framework of Chapter 13 gives a precise alternative to identifying routing with "geometric entropy minimization." A router and a library can sometimes describe a structured predictor economically. Their statistical

complexity must still be counted. The following bound concerns a normalized mixture on a common domain; fixed shared branches and residuals do not change it.

Proposition 37 (Covering a router and its expert library)**.** *For each expert class $\mathcal{F}_i$, assume $\|F_i\|_\infty \le B$. Let $\mathcal{W}$ be a class of nonnegative, unit-sum weight maps, with metric*

$$d_{1,\infty}(w, \widetilde{w}) = \sup_x \sum_i |w_i(x) - \widetilde{w}_i(x)|.$$

Suppose the indicated finite covers exist, with centers in their admitted classes. For $\mathcal{F}_{\mathrm{mix}} = \{\sum_i w_i F_i : w \in \mathcal{W}, F_i \in \mathcal{F}_i\}$,

$$\begin{aligned}\log_2 \mathcal{N}(\varepsilon + B\eta, \mathcal{F}_{\mathrm{mix}}, \|\cdot\|_\infty) \le\ & \log_2 \mathcal{N}(\eta, \mathcal{W}, d_{1,\infty}) \\ & + \sum_{i=1}^{E} \log_2 \mathcal{N}(\varepsilon, \mathcal{F}_i, \|\cdot\|_\infty).\end{aligned}$$

Proof. Choose $\widetilde{w}$ within η and $\widetilde{F}_i$ within ε. Then

$$\begin{aligned}\left|\sum_i w_i F_i - \sum_i \widetilde{w}_i \widetilde{F}_i\right| &\le \sum_i w_i |F_i - \widetilde{F}_i| + \sum_i |w_i - \widetilde{w}_i|\, |\widetilde{F}_i| \\ &\le \varepsilon + B\eta.\end{aligned}$$

Taking all combinations of cover centers and then logarithms proves the bound. □

For a fixed router, its covering term is zero. This is an added sufficient covering calculation, not a claim that this bound is sharp or that training minimizes it. Its dependence on all E experts cannot generally be replaced by K. To see why, fix E disjoint nonempty regions and route each region to its own constant expert taking an independently chosen value in $\{0, 1\}$. Only one expert is evaluated per input, but there are 2^E distinguishable functions. At uniform resolution less than $1/2$, their covering entropy is E bits.

The router metric also matters. For moving hard boundaries, a continuum of threshold routers can have pairwise $d_{1,\infty}$ distance two, so a fine uniform cover may be infinite. Distribution-dependent disagreement, empirical metrics, a finite code, or a different capacity argument may be more useful. Finite parameter count alone does not make this particular uniform gate cover finite.

For a fixed hard partition with region probabilities p_i, population risk decomposes as $R(\Psi) = \sum_i p_i R_i(F_i)$. Each local learner sees only its allocated data; low-probability experts can have poor coverage. Jointly learning the router changes that allocation. A small active computational budget therefore does not, by itself, yield a small statistical complexity or adequate local sample size.

Who supplies the useful routing coordinates? With a supplied chart assignment, the expert problems can be posed locally. If the routing features and experts are learned jointly, the problem is again one of representation discovery. For an all-expert softmax mixture with scores a_i and temperature T,

$$\frac{\partial \Psi(x)}{\partial a_j(x)} = \frac{w_j(x)}{T}(F_j(x) - \Psi(x)).$$

This identity follows by differentiating the normalized exponential weights. It shows that a router's learning signal depends on both its current allocation and differences between expert outputs. A geometrically valid partition need not be accessible to a particular initialized learning rule.

There is an even sharper distinction for the hard top-one, selected-score-normalized construction used in the expressivity theorem. Inside a region with a unique selected expert, its normalized gate is identically one. The branch-selection derivative gives no ordinary router gradient there. Training must therefore use an appropriate surrogate, weighting convention, additional supervision, or another mechanism. This observation concerns that particular top-one convention, not every practical sparse router. It illustrates why an approximation construction and an optimization theorem are different statements.

These are the familiar roles of C1, C2, and C3: feasible representation, accessible learning signal, and supplied intermediate information. A successful MoE predictor need not reproduce a particular geometric atlas or teacher decomposition. Its actual accuracy, complexity, and evaluation resources determine the predictive conclusion of Chapter H. Smooth activations inside an expert do not make a hard top-K network globally smooth across its switching boundaries. The analytic convergence results of the earlier chapters therefore require their own hypotheses rather than following from an expert's activation choice. Predictive success may concern a specified returned checkpoint; it need not require an absorbing state of every minibatch update.

20.4 Case Study: The DeepSeek-V3 MoE Architecture

We now examine the original DeepSeek-V3 architecture, using its technical report and released configuration and inference implementation [37, 39, 38]. This is a case study of a specified model, not a description of every subsequent DeepSeek release. The report and code provide architectural evidence for conditional computation; they do not identify a compact smooth token manifold, a learned atlas, or an entropy-minimizing training objective.

20.4.1 Architectural Layering Strategy

The main decoder has 61 Transformer blocks. The first three use dense feed-forward sublayers. The remaining 58 replace the feed-forward sublayer by an MoE with one always-evaluated shared expert and 256 routed experts, selecting eight routed experts per token. The hidden width is 7168 and each expert's intermediate width is 2048. The report gives 671 billion total parameters and approximately 37 billion activated per token for the model [37, 39].

The distinction concerns the *feed-forward sublayers*. Attention, normalization, and residual paths remain in the architecture. Dense feed-forward computation in the first three blocks is not a proof that every parameter receives a nonzero update on every token, or that those blocks construct a universal coordinate system. The similarity to the routing-feature preprocessing in the manifold theorem is an interpretation to test, not an established functional identity.

20.4.2 The Multi-Expert Routing Equations

To separate the residual and normalized expert inputs, let $\mathbf{u}_t$ be the residual-stream vector entering the feed-forward sublayer and put $\mathbf{v}_t = \mathrm{RMSNorm}(\mathbf{u}_t)$. In column-vector notation, a code-aligned schematic update is

$$\begin{aligned}\mathbf{h}'_t = \mathbf{u}_t &+ \sum_{i=1}^{N_s} \mathrm{FFN}_i^{(s)}(\mathbf{v}_t) \\ &+ r_s \sum_{j=1}^{N_r} g_{j,t}\, \mathrm{FFN}_j^{(r)}(\mathbf{v}_t), \qquad r_s = 2.5.\end{aligned} \tag{20.4}$$

Here $N_s = 1$, $N_r = 256$, and the $g_{j,t}$ below are unit-sum weights on the selected routed experts. The released configuration includes the additional routed-output multiplier $r_s = 2.5$; the compact equations in the technical report present the unit-normalized routing convention without displaying this implementation multiplier [39, 38].

The terms are added in a common output space; this is an ordinary vector sum, not a direct sum of separate vector spaces. The shared branch can represent useful common computation, but its being always evaluated does not prove that it learns only universal features or that routed branches learn only domain-specific residuals. The normalization and multiplier must be included in a numerical or stability calculation for the complete sublayer. In particular, the scaled routed contribution is not by itself a convex average with total weight one.

20.4.3 Internal Node Projections: The SwiGLU Block

Each expert uses a gated feed-forward computation of SwiGLU form [142, 38]. With column vectors,

$$\mathrm{FFN}_i(\mathbf{v}_t) = W_{\mathrm{out},i}\left[\mathrm{SiLU}(W_{g,i}\mathbf{v}_t) \odot (W_{\mathrm{in},i}\mathbf{v}_t)\right]. \tag{20.5}$$

The shapes are

$$W_{g,i}, W_{\mathrm{in},i} \in \mathbb{R}^{m\times D}, \qquad W_{\mathrm{out},i} \in \mathbb{R}^{D\times m}, \qquad \mathrm{SiLU}(z) = z\,\mathrm{sigmoid}(z).$$

There are three parameter matrices, but two input projections operate in parallel before multiplication and output projection. Thus "three matrices" is not the same depth convention as the three successive affine layers in the ReLU approximation theorem. That theorem should not be applied to this fixed-width SwiGLU expert merely by identifying both with the phrase "three-layer network."

Ignoring biases and ancillary operations, one such expert has $3Dm$ weights and the same leading number of multiply–accumulate operations per token. Substituting the specified widths gives the derived counts

$$3 \times 7168 \times 2048 = 44{,}040{,}192$$

per expert and

$$9 \times 3 \times 7168 \times 2048 = 396{,}361{,}728$$

for one shared plus eight routed experts. These count expert matrix entries used, not the total cost of an entire Transformer block. The corresponding sum of intermediate widths is $9 \times 2048 = 18{,}432$. It is not an intrinsic manifold dimension, a linear rank assertion, or a count of end-to-end paths across the decoder.

The complete pool contains 257 expert parameter sets per MoE sublayer. Its size illustrates the main engineering tradeoff: much more stored capacity than active expert work per token, with router and communication costs still present.

20.4.4 Gating and Capacity-Balanced Routing

For each normalized token vector, the router first computes

$$s_{j,t} = \mathrm{sigmoid}(\mathbf{e}_j^\top \mathbf{v}_t), \qquad j = 1, \ldots, N_r. \tag{20.6}$$

The learned vector $\mathbf{e}_j$ is called an expert centroid in the report. This name does not establish that it is an empirical cluster mean or that its routing region is a manifold chart.

Selection and weighting use different scores. Introduce a load-balancing bias b_j and selection score $a_{j,t} = s_{j,t} + b_j$. Ignoring grouping for one moment, the selected set is $\mathcal{I}_t = \mathrm{TopKIndices}_K(a_{\cdot,t})$. The normalized mixing weights use the *unbiased* sigmoid affinities:

$$g_{j,t} = \begin{cases} \dfrac{s_{j,t}}{\sum_{m \in \mathcal{I}_t} s_{m,t}}, & j \in \mathcal{I}_t, \\ 0, & j \notin \mathcal{I}_t. \end{cases} \tag{20.7}$$

This is normalization of selected sigmoid scores, not a softmax of the original dot-product logits. It could be written algebraically as a softmax of $\log s_{j,t}$ over the selected set, but that different argument must be explicit. The bias affects which experts are selected, not the numerator or denominator used to weight their outputs [37].

Grouped routing. The released configuration groups the 256 routed experts into eight groups and admits four groups per token. The inference implementation scores a group by the sum of its two largest bias-adjusted affinities, selects four groups, and takes the eight highest adjusted expert scores from their union. In formulas, if $\mathcal{G}_g$ denotes group g,

$$q_{g,t} = \text{sum of the largest two entries of } (a_{j,t})_{j \in \mathcal{G}_g},$$

$$\mathcal{A}_t = \bigcup_{g \in \mathrm{TopKIndices}_4(q_{\cdot,t})} \mathcal{G}_g, \qquad \mathcal{I}_t = \mathrm{TopKIndices}_8((a_{j,t})_{j \in \mathcal{A}_t}).$$

Specified deterministic tie conventions complete the mathematical selector. This is not unrestricted top-eight selection over all 256 experts. The technical report relates node-limited routing to communication cost; the released code supplies the grouped selection calculation [37, 39, 38].

What "auxiliary-loss-free" does and does not mean. The principal balancing mechanism adjusts b_j upward for underloaded experts and downward for overloaded experts using observed batch loads. It does not obtain this bias update by differentiating a large balancing penalty through the language-model loss. Nevertheless, the technical report also includes a small *complementary sequence-wise auxiliary balance loss*, with coefficient 10^{-4} in its pretraining settings [37]. It would therefore be incorrect to describe the complete V3 training procedure as having no auxiliary balancing loss.

Changing b_j changes token assignments and hence which expert parameters are trained on which examples. Avoiding one auxiliary-loss gradient does not leave the dynamics or selected predictor unchanged. Balancing is intended to address expert-load collapse and resource utilization, not to prove parameter-rank preservation, chart recovery, or optimal prediction.

Load entropy is also different from the metric entropy of a function class. If p_j is an expert's normalized share of assignments, uniform use maximizes $-\sum_j p_j \log p_j$. The covering entropy in Proposition 37 counts distinguishable predictors. Neither quantity is automatically minimized by a balancing controller, and a statement about one is not a generalization theorem about the other.

From architectural resemblance to a testable geometric claim. The manifold interpretation predicts more than uneven expert use. A useful test would compare supplied charts, learned routing features, and fixed or randomized routing on the same task. It would measure local approximation error, intrinsic-dimension estimates at a specified scale, route validity and margins, and behavior near routing boundaries. Comparisons should match total storage, active arithmetic, training data, and communication budgets where appropriate, rather than matching only expert

Table 20.2: Specifications of the original DeepSeek-V3 main decoder. Reported values are from the technical report and released configuration; the active-width sum is arithmetic accounting, not geometric dimension.

@>p0.38>X@

Architectural metric	Design specification
Main decoder depth	61 blocks: 3 dense FFN sublayers and 58 MoE FFN sublayers
Hidden token width D	7168
Expert intermediate width m	2048
Expert pool per MoE sublayer	1 shared expert and 256 routed experts
Active experts per token per sublayer	1 shared expert and 8 selected routed experts
Sum of active intermediate widths	$9 \times 2048 = 18{,}432$
Routed-output scale	$r_s = 2.5$ in the released configuration
Routing groups	8 groups, with 4 admitted per token
Model parameter counts	671 billion total; about 37 billion activated per token

width. A separate test would ask whether learned expert reuse supports new combinations of known subtasks.

For an industrial language model, these measurements would have to be made on actual hidden representations and held-out data. The architecture alone does not supply them. A shared expert and many routed experts are consistent with common-plus-specialized computation, but that is not proof of a particular atlas or entropy mechanism. No such empirical test is reported in this chapter.

Conclusion. The constructive geometric argument is worth retaining: local coordinates can make approximation economical, and conditional routing can avoid executing every specialist on every input. Deep reuse can combine a modest library in many ways. The qualifications identify the resources and assumptions on which those advantages depend; they do not remove the advantages themselves.

geometric localization and low-arity composition
provide possible economical representations;
MoE supplies conditional evaluation;
learning must still find accurate experts and useful routes,
and statistical control must justify their population performance.

This places MoE within the book's distinction between representation, recovery, and selection. DeepSeek-V3 is a concrete implementation of large-capacity conditional computation, not a theorem that an industrial optimizer minimizes geometric entropy. A feasible generalizing MoE is already a computational witness in the sense of Chapter H; identifying its experts with the generating charts is a stronger, optional structural claim.

CHAPTER 21

The Architecture of a Compositional Register Machine

Sparse compositionality suggests that complex computation should be implemented by repeatedly applying low-arity operations to intermediate variables. This principle does not uniquely determine an architecture. It does, however, motivate a particular question:

Can a learning system search directly over programs composed of reusable bounded-arity operators, rather than relying only on an unrestricted input–output parameterization?

This chapter proposes an experimental architecture for studying that question.

We call it a *Compositional Register Machine* (CRM).

The CRM contains four basic ingredients:

explicit working memory + a small discrete controller + a library of bounded-arity operators + hard sparse execution.

At each step the controller chooses a small number of operands from working memory, selects an operator, executes that operator, and writes the result back into memory.

The architecture therefore separates

control flow from local computation.

This distinction resembles classical programmable machines but is introduced here for a different purpose: to make sparse composition an explicit architectural bottleneck.

The CRM should not be confused with a theorem that sparse compositionality automatically solves learning. Bounded arity addresses the representation side of the problem, C1. It does not

guarantee genericity, C2, and it does not guarantee that the correct routing policy can be recovered efficiently.

Indeed, parity remains the simplest warning: parity has a perfect bounded-arity XOR decomposition while its ordinary low-order statistical footprints can vanish.

The real scientific question of the CRM is therefore not whether such a machine can execute algorithms. Chapter 3 already established that associative access, local computation, and writable memory are sufficient for universal computation under appropriate conditions.

The new question is:

> Can the program—the sparse routing and reuse of local operators—itself be learned from small examples and generalized systematically to much larger computations?

That is the hypothesis to be tested.

The architecture, training schedules, and experiments below form a proposal, not a report of CRM training results. The elementary results specify what follows once an execution interface, a routing margin, or a supervision condition is supplied. They do not establish that end-to-end training finds those conditions. Consistent with Chapter H, an accurate feasible CRM need not reproduce a privileged teacher program; its own controller and evaluator already specify its computation.

How to read this chapter. The first part defines the machine and its routing interface; the second studies sparse relaxations and their conversion into hard instructions. The operator-library, compilation, and granularity sections then address reuse. The final part separates numerical reliability, memory resources, and the proposed tests of learning. Formal claims below are conditional execution or approximation results; experimental comparisons remain hypotheses.

21.1 From Sparse Compositionality to an Architectural Hypothesis

Suppose a target computation has a DAG representation

$$f = G(g_1, \dots, g_s),$$

where each constituent g_v depends on at most

$$k = O(1)$$

arguments.

For a polynomial-size DAG this is a feasible computational description. Favorable continuous approximation rates additionally require the local regularity and composition-stability assumptions of Chapter 4. Bounded arity alone, with unrestricted constituent complexity or exponentially many calls, is not such a rate.

A conventional large neural network can represent such a computation, but it is not required to execute it in an explicitly sparse way.

The CRM asks what happens if bounded arity is imposed at the operator interface. Physical locality and communication cost are separate questions, as in Chapter 10.

At every computational step,

only $k = O(1)$ stored variables can enter the active constituent computation.

The global computation must therefore arise through repeated composition.

21.1.1 What this constraint does guarantee

It prevents the payload computation of one local operator call from reading more than the designated operands. This statement assumes that no side channel, global memory read, or uncharged preprocessing is hidden inside that operator.

For fixed representation dimension, operator width, and numerical precision, the complexity of one local operation is independent of sequence length N.

This is not a constant-cost claim for the complete machine step. A controller that scores all L_t records, or all record pairs, can inspect a large amount of information before it selects a bounded number of operands. The functions that construct keys, update metadata, and update c_t must therefore be included in the implementation and its cost.

21.1.2 What it does not guarantee

The constraint does not by itself guarantee:

- that the correct computation is compositionally sparse;
- that useful routing decisions are statistically visible;
- that the controller can be learned efficiently;
- that total execution time is small;
- that the overall machine cannot memorize through its controller or memory;
- that the system generalizes outside its training distribution.

Thus

bounded arity enforces a compositional execution constraint, not genericity or learnability.

21.2 The Abstract Compositional Register Machine

At computational step t, let working memory contain

$$\mathcal{M}_t = \{(v_i, m_i)\}_{i=1}^{L_t}.$$

Here

$$v_i \in \mathcal{V}$$

is the stored value and m_i is metadata describing its computational role.

Memory is an indexed collection, not a set of distinct payloads: two records with identical values remain distinct if they have different addresses. The union notation below means appending fresh indexed records. Throughout, L_t counts records, K counts operators, and $a_{\max}$ bounds operand slots; the local dimension D is not the number of records. Every field has a specified type and numerical encoding.

21.2.1 Working memory

Useful metadata may include

$$m_i = (\text{address}, \text{age}, \text{depth}, \text{availability}, \text{type}, \ldots).$$

The architecture may initially use append-only memory:

$$\mathcal{M}_{t+1} = \mathcal{M}_t \cup \{(v_{\text{new}}, m_{\text{new}})\}.$$

Append-only memory is convenient because previous intermediate variables remain stable.

It is not a fundamental requirement; later we introduce overwrite, deletion, and garbage collection.

21.2.2 Operator library

Let

$$\mathcal{G} = \{g_1, \ldots, g_K\}$$

be a library of local operators.

Every operator has bounded arity:

$$g_k : \mathcal{V}^{a_k} \to \mathcal{V}^{b_k}, \qquad a_k \le a_{\max} = O(1).$$

Examples might include

$$\text{XOR}, \quad \text{AND}, \quad +, \quad -, \quad \times, \quad \text{compare},$$

or learned local neural maps.

21.2.3 Controller

The controller chooses

$$a_t = (i_1, \ldots, i_{a_k}, k, w),$$

where

- $i_1, \ldots, i_{a_k}$ are operand addresses;
- k chooses an operator;
- w specifies the destination or write action.

Execution is then

$$y_t = g_k\left(v_{i_1}, \ldots, v_{i_{a_k}}\right). \tag{21.1}$$

The memory is updated with y_t.

21.2.4 Finite control state

For general algorithms the controller may also maintain a small internal state

$$c_t \in \mathcal{C},$$

with

$$c_{t+1} = U_\theta(c_t, \mathcal{M}_t, a_t, y_t).$$

This permits loops, phases, and finite-state program control.

For the finite-control version, $\mathcal{C}$ is a fixed finite set. Addresses, counters, a recursion stack, and growing integers belong in explicit memory, not in an unlimited-precision coordinate of c_t. The display for U_θ specifies an interface, not unrestricted free access: its admissible reads and their cost must be stated.

Operational contract. An implementation must specify legal operand types, ordered slots, allowed repeated reads, destination allocation, and a deterministic rule for ties. An instruction reads its operands from a snapshot before writing; this permits compare-and-swap with two outputs. Assume also $b_k \le b_{\max} = O(1)$. A distinguished halt instruction returns a specified output record, while invalid addresses, illegal types, overflow, and timeout produce explicit failure outcomes. Masks and termination tests must use only information available at that step. Learned operators, key maps, and routers are all included in the model budget. Numerical constants, exact primitives, and supplied symbolic parsers are declared components, not capabilities inferred from the training results.

21.3 Relational Routing Rather Than Independent Scores

A simple first implementation might score each candidate independently:

$$s_i = r_\theta(m_i).$$

Selecting the two largest scores gives

$$\{i^*, j^*\} = \mathrm{topk}(s_1, \ldots, s_{L_t}; 2).$$

With deterministic tie breaking this can select an unordered pair for a simple reduction. Noncommuting operations additionally require a slot-order convention. It is not a general pair-scoring controller.

Whether i is a useful operand can depend on the choice of j.

21.3.1 Pairwise controller

A more expressive controller scores actions jointly:

$$s_{ijk} = R_\theta\left(c_t, m_i, m_j, \kappa(v_i), \kappa(v_j), k\right), \tag{21.2}$$

where $\kappa(v)$ is an optional compact content key.

The hard action is

$$(i^*, j^*, k^*) = \arg\max_{i,j,k} s_{ijk}.$$

21.3.2 Factorized selection

Explicit pairwise scoring may cost

$$O(KL_t^2).$$

A scalable implementation can choose operands autoregressively:

$$i^* = R_\theta^{(1)}(c_t, \mathcal{M}_t),$$

then

$$j^* = R_\theta^{(2)}(c_t, \mathcal{M}_t, i^*).$$

This preserves relational routing without constructing every possible pair simultaneously.

Two separate scans still typically cost $O(L_t)$ each, plus key construction and operator selection. Factorization need not reproduce the global maximum of an arbitrary pair-score table. For example, the score matrix $\left(\begin{smallmatrix} 8 & 0 \\ 0 & 9 \end{smallmatrix}\right)$ cannot be written $s_{ij} = a_i + b_j$, since the difference between its rows is not constant. A conditional second decision is more expressive than independent scores, but its training criterion and candidate search remain part of the algorithm. For general arity, exhaustive ordered-tuple scoring has $O(KL_t^{a_{\max}})$ candidates.

21.4 Bounded Arity and Local Capacity

The operator library provides the architectural realization of sparse compositionality.

Proposition 38 (Local capacity is independent of global sequence length)**.** *Suppose every CRM operator accepts at most k arguments, each represented in a fixed-dimensional state space $\mathcal{V} \subseteq \mathbb{R}^D$, and the operator architecture and numerical precision are fixed independently of input length N.*

Then at most kD real-valued payload coordinates are presented to that call. If each coordinate has at most b bits, there are at most 2^{bkD} possible operand tuples. These bounds are independent of N. They concern the operator after operand selection, not the entire controller–memory computation.

Proof. There are at most k slots of dimension D. Each slot contains at most 2^{bD} encoded values, so the Cartesian product has at most 2^{bkD} elements. No additional input is read by the stated local operator. The router may depend on more records; that is a separate computation and is not bounded by this count. □

Thus one constituent cannot directly implement an arbitrary lookup table

$$\{0,1\}^N \to \{0,1\}$$

from separately stored raw input bits in a single payload evaluation as N grows. It may nevertheless output a global property that was already computed and encoded in a selected operand, or communicated by a content-dependent selector. Such work belongs to earlier computation or the controller; it must not be hidden in a local-capacity claim.

But this does *not* imply that the entire CRM has constant capacity.

Total capacity also depends on

$$\boxed{\text{controller} + \text{memory} + \text{number of steps} + \text{operator library}.}$$

21.5 Execution Universality Is Different from Learning

The architecture naturally raises a Turing-completeness question.

This question should be separated from the harder learning question.

Theorem 17 (Conditional universality of the CRM)**.** *Consider a CRM with extensible memory, fixed finite control, exact finite-alphabet read/write operations, and value-dependent branching on selected symbols. Assume that addresses have explicit finite encodings, that exact comparison and head movement are implementable, and that addressing and these operations have polynomial bit cost in their encoding length. Its finite instruction library contains the corresponding symbol, pointer, and control operations.*

Then, for every fixed deterministic Turing machine running for $T(n)$ steps on a length-n input, a CRM program simulates it using $O(n + T(n))$ tape records and polynomial bit cost in $n + T(n)$. A simple addressed-list implementation has the sufficient bound

$$O\Big((n + T(n))^2 \log(2 + n + T(n))\Big)$$

including initialization. The program is fixed independently of n; its memory and address lengths may grow. This is also universality for standard bit-cost register-machine computations, not for a unit-cost model with arbitrarily large integers.

Proof. Store one finite tape symbol per visited tape record and keep the finite machine state in c_t. Keep the head address in an explicitly encoded pointer register. A fixed transition table maps the current state and read symbol to a new state, symbol, and one-cell movement. Each transition therefore uses a bounded number of CRM instructions at the symbolic level. A previously unvisited cell is initialized to blank. A run visits at most $n + 2T(n) + O(1)$ tape positions, so their integer addresses require $O(\log(2 + n + T(n)))$ bits.

For a concrete implementation, scan the addressed records to read or write the unique matching address, using exact bitwise comparisons. Each comparison and pointer update takes $O(\log(2 + n + T(n)))$ bit operations. At most $O(n + T(n))$ comparisons per simulated transition give the displayed, conservative bound after including initialization. The controller and finite transition table do not grow with the input. Faster addressing changes the overhead, not the simulation argument. No fixed-dimensional real coordinate is used to store an unbounded tape or integer. □

The theorem is intentionally unsurprising.

Chapter 3 already established the more general point that local update plus writable associative memory is computationally universal.

The new question is not

Can the CRM represent a program?

It is

Can useful CRM programs be recovered efficiently from data?

21.6 Metadata-Only Routing and Oblivious Algorithms

The smallest CRM controller can be deliberately prevented from observing payload values.

Its decision is then

$$a_t = R_\theta(c_t, M_t),$$

where M_t contains only structural metadata.

For this to imply obliviousness, the entire controller-visible state must be payload independent: initialization, c_t, metadata updates, eligibility masks, and the halting test. In particular, U_θ may not write a payload-dependent branch bit into c_t and then call the result "metadata-only." Under these restrictions, induction on steps shows that two inputs with the same declared structural metadata produce the same action schedule.

This has an important consequence.

The resulting algorithm is *data-oblivious*:

its control flow depends on computational structure but not on the numerical or semantic values being processed.

This is sufficient for many useful computations:

- fixed reduction trees;
- FFT-like butterfly graphs;
- sorting networks;
- convolutional scans;
- fixed dynamic-programming schedules.

It does not execute the usual data-dependent control schedules of algorithms such as:

- ordinary Quicksort;
- conditional program execution;
- value-dependent search;
- linguistic routing based on token meaning.

Thus metadata-only routing is a useful experimental restriction, not a candidate universal controller.

This limitation concerns the chosen execution schedule. It does not say that every function computed by branching requires a branching circuit: a bounded computation can sometimes be unrolled with both branches computed and a value-dependent selection operator. Sorting networks are already an example of value-dependent outputs under a fixed schedule. The costs and supplied operators of such a replacement must be counted.

21.7 Content-Dependent Routing

General control flow requires some information about stored values.

The challenge is to provide enough content information to the controller without making the evolutionary search space as large as the complete model.

21.7.1 Compact content keys

Associate each stored value v with a small key

$$\kappa(v) \in \mathbb{R}^d, \qquad d \ll D.$$

The controller then acts on

$$(m_i, \kappa(v_i))$$

rather than the complete high-dimensional value.

21.7.2 Exact discrete keys

The cleanest first experiments should use exact low-dimensional tags.

For example,

$$\kappa(v) \in \{0, 1, \ldots, q-1\}^d.$$

This isolates the control-flow problem from representation learning.

21.7.3 Frozen random projections

For continuous high-dimensional representations, one possible key is

$$\kappa(v) = vW,$$

where

$$W \in \mathbb{R}^{D \times d}$$

is randomly initialized and frozen.

For M vectors fixed independently of the draw of an appropriately scaled random projection, the Johnson–Lindenstrauss bound implies that for the dimension below [35]

$$d = O\left(\frac{\log(M/\delta)}{\varepsilon^2}\right),$$

pairwise Euclidean distances can be preserved to relative distortion ε with high probability.

Thus

a frozen random projection can preserve an existing geometric representation;

it does not create semantic geometry that was absent from the original representation.

Distance preservation on that fixed set is not an automatic guarantee for representations selected adaptively using the same projection, or for all future states of an unbounded run. A uniform controlled-set argument would be a different premise. Even for fixed keys, correct nearest-neighbor routing requires a separation margin: if squared distances satisfy $(1+\varepsilon)d_*^2 < (1-\varepsilon)d_j^2$ for every competitor, the distortion bound preserves the winner. Without a gap, small distance distortion may change the selected address.

21.7.4 Freezing the projection does not freeze the representation

Even with fixed W,

$$\kappa_t(v) = v_t W$$

changes if v_t changes.

Therefore a frozen projection does not by itself create a stationary fitness landscape while the operator representations are evolving.

A cleaner protocol separates timescales:

$$\boxed{\text{learn/freeze local operators first } \rightarrow \text{ evolve the controller afterward.}}$$

Alternating optimization is also possible, but then nonstationarity should be treated explicitly.

21.8 Why Evolution Is Attractive—but Not Necessary

The controller executes hard discrete actions such as

$$\arg\max \quad \text{or} \quad \text{topk}\,.$$

Ordinary backpropagation cannot differentiate through these hard choices directly.

But this does not imply that evolutionary optimization is the only possibility.

Alternatives include:

- policy-gradient estimators;
- straight-through estimators;
- Gumbel or soft routing relaxations;
- supervised action traces;
- discrete search;
- evolutionary strategies.

Evolution is attractive here for the reason developed in Chapter 17:

$$\boxed{\text{the controller can be made extremely small.}}$$

If

$$\theta_{\text{router}} \in \mathbb{R}^r$$

with modest r, zeroth-order search operates on the low-dimensional program-control space rather than on all parameters of the computational system.

Thus the evolutionary hypothesis is

$$\boxed{\begin{array}{c}\text{hard sparse routing} + \text{small controller} \\ \Rightarrow \text{ a plausible low-dimensional evolutionary search problem.}\end{array}}$$

Whether evolution is superior to policy gradient or differentiable relaxation is an empirical question.

A small controller need not have an easy objective: for finite examples and a finite execution horizon, hard argmax decisions can make terminal fitness locally constant throughout regions of parameter space. A small mutation that changes no action then reveals no fitness difference. Chapter 17 distinguishes search dimension from signal strength and curvature; its smooth quadratic progress calculation is not a convergence theorem for this piecewise-constant program objective. Policy gradients differentiate a stochastic policy, not the hard argmax, and straight-through updates are surrogate derivatives rather than exact gradients of that hard program.

21.9 Differentiable Sparse Routing: Sparsemax and Entmax as Relaxations of Program Search

The CRM has so far been described using hard discrete routing:

$$a_t = (i_t, j_t, k_t) = \arg\max_{i,j,k} R_\theta(i, j, k \mid \mathcal{M}_t, c_t),$$

where the controller chooses two operands and one operator.

This formulation makes the computational graph explicit, but the $\arg\max$ operation is not differentiable. Evolutionary search, policy-gradient estimators, or supervised traces can therefore be used to learn the controller.

A different possibility is to replace the discrete choice temporarily by a differentiable *sparse relaxation*.

The idea is

continuous sparse exploration
$\longrightarrow$ support selection
$\longrightarrow$ hard discrete program.

Sparsemax and α-entmax provide natural mechanisms for such a relaxation [100, 122]. The formulas below specify an established gating family; its use for CRM program discovery is the architectural proposal being tested.

21.9.1 Softmax, Sparsemax, and Entmax

Let

$$z = (z_1, \ldots, z_K) \in \mathbb{R}^K$$

be routing logits.

Standard softmax produces

$$p_i = \frac{e^{z_i}}{\sum_j e^{z_j}},$$

so $p_i > 0$ for every finite logit.

This is useful for smooth optimization, but it does not produce literal zeros.

Sparsemax

Sparsemax instead projects the logit vector onto the probability simplex:

$$\mathrm{sparsemax}(z) = \arg\min_{p \in \Delta^{K-1}} \|p - z\|_2^2, \tag{21.3}$$

where

$$\Delta^{K-1} = \left\{ p \in \mathbb{R}^K : p_i \geq 0,\ \sum_i p_i = 1 \right\}.$$

Its solution has the threshold form

$$p_i = \max\{0, z_i - \tau(z)\}. \tag{21.4}$$

Hence some routing probabilities can become exactly zero.

α-entmax

The α-entmax family interpolates between softmax and sparsemax.

It can be written schematically as

$$\mathrm{entmax}_\alpha(z) = \arg\max_{p \in \Delta^{K-1}} \left\{ p^\top z + H_\alpha(p) \right\}, \tag{21.5}$$

For the normalization used here,

$$H_\alpha(p) = \frac{1 - \sum_i p_i^\alpha}{\alpha(\alpha - 1)} \quad (1 < \alpha \leq 2), \qquad H_1(p) = -\sum_i p_i \log p_i,$$

with $0 \log 0 = 0$. This convention matters: at $\alpha = 2$ maximizing $p^\top z + H_2(p)$ is exactly Euclidean projection onto the simplex. A temperature can be introduced by using z/τ with $\tau > 0$.

The limiting cases include

$$\alpha = 1 \quad \Rightarrow \quad \text{softmax},$$

and

$$\alpha = 2 \quad \Rightarrow \quad \text{sparsemax}.$$

For

$$1 < \alpha \leq 2,$$

entmax can produce exact zeros while retaining a smoother dependence on the active logits than a hard discrete selection.

They do not impose a fixed support-size bound. If all logits are equal, strict concavity and symmetry give $p_i = 1/K$ for every i, including at $\alpha = 2$. Exact zeros are possible, not compulsory. To enforce an arity bound independent of memory size, use a specified support cap or hard one-hot operand selection in the deployed machine.

21.10 What Sparse Routing Does—and Does Not—Mean

Exact zeros are useful, but their interpretation should be precise.

If

$$p_i = 0,$$

then route i does not contribute to that particular forward computation.

This gives a sparse mathematical contribution when the support is small. An implementation saves execution work only if it skips zero-weight branches rather than evaluating them before masking.

It does not by itself imply that the corresponding route is absent from the true target computation.

Thus

$$\boxed{\text{zero learned gate} \neq \text{proof of absence from the target DAG.}}$$

Likewise,

$$\boxed{\text{sparse routing} \neq \text{automatic recovery of compositional structure.}}$$

The support selected by the learner remains an optimization result and can be incorrect.

21.10.1 Fixed structure versus conditional computation

An especially important distinction concerns whether routing depends on the data.

Suppose the structural logits are learned parameters

$$z_{ji}^{(\ell)} = \beta_{ji}^{(\ell)}.$$

Then the resulting support

$$\mathcal{E}^{(\ell)} = \left\{(i,j) : p_{ji}^{(\ell)} > 0\right\}$$

is input independent.

This can be interpreted as a candidate fixed computational graph.

In contrast, suppose

$$z_{ji}^{(\ell)} = R_\theta\left(h_i^{(\ell-1)}, q_j^{(\ell)}\right),$$

Then

$$\mathcal{E}^{(\ell)} = \mathcal{E}^{(\ell)}(x)$$

can change from example to example.

Here $q_j^{(\ell)}$ is available before the new child is computed. Using that child's not-yet-computed value as its own routing query would define an implicit fixed-point problem, not the stated feedforward instruction.

The architecture has learned a *conditional routing policy*, not one fixed DAG.

Both are useful, but they answer different questions:

input-independent gates:	candidate structural graph discovery,
input-dependent gates:	conditional program execution.

For the CRM, the second is ultimately more general because arbitrary algorithms require content-dependent control flow.

21.11 Preserving Bounded Arity

A sparse weighted average is not in general equivalent to a bounded-arity constituent function.

Suppose a child receives

$$\widetilde{h} = \sum_i p_i h_i$$

and computes

$$g(\widetilde{h}).$$

Even if only two coefficients are nonzero, an arbitrary binary function

$$F(h_i, h_j)$$

cannot generally be written as

$$g(p_i h_i + p_j h_j).$$

For example, separate access to both operands may be required to compute

$$h_i h_j, \qquad h_i - h_j, \qquad \mathrm{compare}(h_i, h_j).$$

Therefore the CRM should preserve the selected operands as distinct arguments.

21.11.1 Sparse operand slots

For arity k, introduce k routing distributions

$$p^{(1)}, \ldots, p^{(k)}.$$

Each produces one operand slot:

$$u_r = \sum_i p_i^{(r)} h_i, \qquad r = 1, \ldots, k. \tag{21.6}$$

The local constituent then receives

$$y = g_\phi(u_1, \ldots, u_k). \tag{21.7}$$

As the routing distributions converge to the designated one-hot vectors,

$$u_r \to h_{i_r},$$

and the relaxed computation approaches the hard CRM instruction

$$y = g_\phi(h_{i_1}, \ldots, h_{i_k}).$$

Thus

sparse relaxation should soften operand selection, not collapse distinct operands into one average.

During relaxation, the slot itself may still depend on many memory records. If every slot has at most s_0 active parents, the payload read uses at most $a_{\max} s_0$ records; entmax alone does not make s_0 constant. Even k soft slots need not preserve k distinct raw operands. The convex combinations must also lie in a domain on which the operator has a specified meaning.

21.12 Relaxing Operator Selection

The same idea can be applied to the operator library.

Suppose

$$\mathcal{G} = \{g_1, \ldots, g_K\}.$$

Let

$$q = \mathrm{entmax}_\alpha(c)$$

be a sparse distribution over operators.

A relaxed computation requires all mixed operators to have a common input and output type. Exact XOR on bits, compare-and-swap on pairs, and an arithmetic scalar operator cannot be averaged without explicit extensions, padding, and output conventions. Those extensions define the relaxed model and may introduce behavior unavailable to a hard instruction. For such a homogeneous operator family, write

$$y = \sum_{k=1}^{K} q_k g_k(u_1, \ldots, u_a). \tag{21.8}$$

During exploration, several operators may receive nonzero weight.

After structural hardening,

$$k^* = \arg\max_k q_k,$$

and execution becomes

$$y = g_{k^*}(h_{i_1}, \ldots, h_{i_a}).$$

Again the soft mixture is a training device.

The final CRM program is discrete.

21.13 The Sparsemax Jacobian

Sparsemax illustrates both the attraction and the danger of exact sparse routing.

Let

$$S = \{i : p_i > 0\}$$

be its active support.

Away from support-changing boundaries,

$$\frac{\partial p_i}{\partial z_j} = \begin{cases} \delta_{ij} - \dfrac{1}{|S|}, & i, j \in S, \\ 0, & \text{otherwise.} \end{cases} \tag{21.9}$$

The active block is

$$J_S = I - \frac{1}{|S|}\mathbf{1}\mathbf{1}^\top.$$

Its rank is

$$\boxed{\operatorname{rank}(J_S) = |S| - 1.}$$

Equivalently, its nullity within the active support is one.

For the full $K \times K$ Jacobian, the nullity is $K - |S| + 1$. If $|S| = 1$, the entire Jacobian is zero within that region. Thus hard commitment is not accompanied by a remaining routing gradient through the probability map. These are derivatives of sparsemax itself; a specially chosen supervised sparsemax loss can have a different gradient with respect to the logits [100].

21.13.1 The benefit

Once an edge is inactive, it contributes nothing to the current forward computation.

Sparse execution therefore becomes explicit.

21.13.2 The danger

The direct derivative through an inactive coordinate is also zero almost everywhere.

Thus prematurely removing the correct route can make it difficult for ordinary gradient descent to restore that route.

This gives a fundamental exploration–exploitation problem:

early sparsity improves structural commitment but can destroy recoverability.

Entmax with

$$1 < \alpha < 2$$

can make transitions near support boundaries smoother, but exact-zero coordinates still have zero direct derivative away from a boundary.

Smoother boundary behavior is not a guarantee of easier support recovery. An inactive logit can still change through shared network parameters, other examples, regularization, or explicit

exploration. Zero direct Jacobian entries do not imply that a route is permanently inaccessible. Conversely, a routing pathway with no such update can remain stuck.

21.14 A Continuation Strategy

A natural training strategy is therefore to begin with a more exploratory routing distribution and progressively encourage sparsity.

One possibility is

$$\alpha(t) : 1 \longrightarrow \alpha_{\text{sparse}},$$

where

$$1 < \alpha_{\text{sparse}} \leq 2.$$

Early training is close to dense softmax.
Later training becomes increasingly sparse.
Alternatively one may vary an explicit temperature $T(t)$ or sparsity penalty.
The intended trajectory is

$$\boxed{\text{broad exploration} \rightarrow \text{sparse candidate support} \rightarrow \text{hard program.}}$$

There is no theorem that this continuation finds the correct support.
It is an optimization heuristic whose effectiveness should be measured experimentally.

21.15 Hardening the Learned Program

After relaxed training, the support can be converted into a discrete CRM program.

For each operand slot,

$$i_r^* = \arg\max_i p_i^{(r)}.$$

For operator selection,

$$k^* = \arg\max_k q_k.$$

The hardened instruction becomes

$$a_t = (i_1^*, \ldots, i_a^*, k^*).$$

The resulting discrete architecture should then be evaluated *without* the continuous relaxation.

An elementary hardening bound. The following calculation quantifies the approximation premise needed for rounding gates. It is a same-state calculation, not a training theorem.

Proposition 39 (One-instruction relaxation-to-hardening error)**.** *At a fixed memory state, let $\|h_i\| \le B_v$. The a slots have simplex weights $p^{(r)}$ and $u_r = \sum_i p_i^{(r)} h_i$. Let the operator weights q also belong to a simplex, and assume all operators are defined on the resulting convex domain with $\|g_j(u)\| \le B_y$. Choose $i_r^* \in \arg\max_i p_i^{(r)}$ and $k^* \in \arg\max_j q_j$, using fixed tie breaking. Suppose the chosen g_{k^*} is L_g-Lipschitz in the sum-of-slot norm. Put $\eta_r = 1 - p_{i_r^*}^{(r)}$ and $\eta_o = 1 - q_{k^*}$. Then*

$$\boxed{\left\| \sum_j q_j g_j(u_1, \dots, u_a) - g_{k^*}(h_{i_1^*}, \dots, h_{i_a^*}) \right\| \le 2B_y \eta_o + 2B_v L_g \sum_{r=1}^{a} \eta_r.}$$

Proof. For each slot, $\|u_r - h_{i_r^*}\| \le \sum_{i \ne i_r^*} p_i^{(r)} \|h_i - h_{i_r^*}\| \le 2B_v \eta_r$. The difference between $\sum_j q_j g_j(u)$ and $g_{k^*}(u)$ is at most $2B_y(1 - q_{k^*})$. Apply the Lipschitz inequality to the chosen operator between u and the hard tuple, then add the two bounds. □

Small tail masses, not just small support, justify this bound. For example, averaging values 0 and 2 with weights $(1/2, 1/2)$ and an identity operator fits target 1 exactly; either hard choice has error 1. A sparse two-point relaxation can therefore solve a task that this rounding does not preserve.

Hardening a policy is not freezing one training trace. For input-dependent logits, argmax is evaluated anew at every reached state. Freezing router parameters leaves a conditional program, not a single fixed address sequence. The local bound must hold on the states actually encountered after hardening. Different writes, masks, or halt decisions require an additional route-stability argument; see the later section on continuous composition.

21.15.1 Retraining after hardening

A useful protocol is

train relaxed gates
⇓
extract discrete support
⇓
freeze the hardened routing policy
⇓
retrain only local operators.

This separates two questions:

Did the learner find a useful program?

and

Can the constituents be optimized once the program is fixed?

21.16 Structural Complexity Penalties

Exact-zero activations alone do not guarantee low-complexity programs.

A proposed selection objective can explicitly penalize program complexity.

Let S_t be the number of active routes at time t.

A relaxed structural objective may take the form

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{edge}} \sum_t S_t + \lambda_{\text{step}} T + \lambda_{\text{op}} K_{\text{used}}. \tag{21.10}$$

For a continuous relaxation, S_t can be approximated using a smooth or piecewise-smooth support surrogate.

After hardening, the actual discrete counts should be used.

An exact support count is locally constant away from support changes, so it is not itself a useful backpropagation signal there. The surrogate and its gradient must be specified. Moreover $\|p\|_1 = 1$ for every simplex vector, so an ℓ_1 penalty on normalized gates alone does not encourage sparsity. Route counts, call counts, and operator counts are resource proxies, not automatically prefix-code lengths.

This connects directly to the MDL objective Equation (21.26).

The purpose is not sparsity for its own sake.

It is to prefer

$$\boxed{\text{short reusable programs}}$$

subject to solving the task.

21.17 Unknown Structure Carries a Statistical Cost

When the correct computation graph is known, local learning can exploit the corresponding bounded arity directly.

When the graph is unknown, the learner must also identify the structure.

If each of the s constituents chooses at most k parents among at most $d + s$ available preceding variables, the graph description must account for both input variables and earlier internal variables.

With a fixed topological ordering and a canonical ordering of argument slots, a parent set can be indexed using

$$\log_2 \left[\sum_{j=0}^{k} \binom{d+s}{j} \right] = O(k \log(d+s))$$

bits for fixed k.

Encoding all s parent sets therefore costs

$$\boxed{O(sk \log(d+s))}$$

bits.

Explicit ordering of noncommuting operands contributes at most $O(sk \log k)$ additional bits and does not change this leading bound.

Thus

unknown-DAG learning is not statistically equivalent to
learning with the correct DAG supplied in advance.

Sparse routing may reduce the structural search space, but it does not make the model-selection cost disappear.

21.18 Differentiable Sparsity Does Not Solve C2

The central conceptual limitation follows directly from the framework of Chapters 6 and H.

Suppose the correct route is i^*.

If the available end-to-end loss provides no detectable statistical preference for i^* over the alternatives, changing softmax into entmax cannot manufacture that information.

Thus

sparse gating changes the parameterization of program search;
it does not guarantee that the correct program leaves
an informative gradient footprint.

What remains necessary for a particular proposed search mechanism is usable task information in its admitted observations and updates. C2 names candidate routes to such information; it is not an additional universal prerequisite for every accurate predictor. A different gating parameterization can change the probes available to learning even though it supplies no new observations.

This is particularly clear for parity-like targets.

A compositionally sparse solution exists, yet ordinary low-order learning signals can vanish under symmetric distributions.

21.19 Computational Cost

A sparse output distribution does not automatically imply sparse computation.

If all logits

$$z_1, \ldots, z_K$$

must first be computed before entmax determines that only two are active, the routing cost may still be

$$O(K).$$

For pairwise routing it can remain

$$O(KL_t^2)$$

for K operators if every ordered memory-record pair is scored.

Therefore one must distinguish

sparse executed support

from

sparse routing computation.

Practical speedups require implementations that avoid scoring or materializing all candidate routes, for example:

- hierarchical candidate generation;
- approximate nearest-neighbor retrieval;
- factorized operand selection;
- top-k kernels;
- locality constraints;
- coarse-to-fine routing.

A top-k output kernel alone does not eliminate the preceding all-logit scan; approximate retrieval can omit the required operand and needs a recall guarantee. A useful accounting separates

$$C_{\text{run}} = \sum_{t<T} (C_{\text{keys,t}} + C_{\text{candidates,t}} + C_{\text{scores,t}} + C_{\text{op,t}} + C_{\text{write,t}}).$$

With $L_t \leq N + b_{\max} t$, exhaustive pair scoring with a fixed library can cost $O(T(N+T)^2)$ even when each selected payload operation is constant cost. Parameter count, total work, parallel span, bit precision, and physical communication are distinct resources.

21.20 Three Ways to Learn the Same CRM Controller

The CRM provides a clean experimental comparison because the execution architecture can remain fixed while the routing-learning algorithm changes.

21.20.1 Evolutionary routing

Use hard actions throughout:

$$a_t = \arg\max R_\theta.$$

Optimize the small controller by zeroth-order evolutionary search.
Advantages:

- exact discrete execution throughout training;
- no relaxation gap;
- arbitrary nondifferentiable program-level fitness.

Disadvantage:

direction estimation and exploration can be costly.

21.20.2 Policy-gradient routing

Interpret

$$\pi_\theta(a_t \mid s_t)$$

as a discrete policy.
Sample instructions and optimize

$$\mathbb{E}[R]$$

using

$$R\nabla_\theta \log \pi_\theta.$$

Advantages:

- retains hard discrete execution;
- uses gradient information through the policy distribution.

Disadvantage:

long-horizon program rewards may have high variance.

21.20.3 Entmax relaxation

Replace hard choices by sparse continuous routing during training.
Advantages:

- ordinary backpropagation through active routes;
- a deterministic forward relaxation at fixed parameters;
- explicit sparsification.

Disadvantages:

- relaxation may solve a different problem from the hardened program;
- early zeroing can trap support;
- the final discrete program may perform worse than its relaxed precursor.

21.21 Relation to C3: Trace Supervision

There is also a useful supervision-rich reference experiment.
If the correct actions

$$a_1^*, \ldots, a_T^*$$

are known, train the controller directly from those traces.
This gives the C3 version of the problem.

Training examples must pair the action with its available controller observation, legal-action mask, and pointer convention. Teacher-forced one-step accuracy does not by itself control errors on self-generated states or unseen lengths. Neural Programmer-Interpreters provide a relevant precedent for execution-trace supervision and program memory; their results are not a theorem for this proposed CRM [134].

An elementary certificate for execution at new lengths. A finite decision table gives one tractable special case. The next result makes explicit the extra interface and coverage assumptions, rather than inferring them from short training examples.

Proposition 40 (Covered controller signatures give exact execution)**.** *Let S_N be the machine-state space at input length N, including the length tag so that states from different sizes are distinguished. Fix an observation map $o : \bigcup_N \mathsf{S}_N \to \mathcal{Q}$ with finite $\mathcal{Q}$, a finite set of instruction templates $\mathcal{A}$, and an exact instantiation rule that turns a template and the present state into addressed reads, writes, and control updates. Include halt and its output convention. A deterministic reference program selects template $\pi^*(o(s))$ and halts on the inputs under consideration.*

Suppose noiseless supervised traces identify $\pi^*(q)$ on a set $\mathcal{Q}_{\mathrm{train}}$. A dictionary controller using that table and abstaining on an unseen signature exactly reproduces every reference run whose signatures lie in $\mathcal{Q}_{\mathrm{train}}$, irrespective of its length. If all relevant q are visited with probability at least $p_{\min} > 0$ in one training trajectory, then n independent training trajectories cover them all with probability at least $1 - \delta$ whenever

$$n \geq p_{\min}^{-1} \log \frac{|\mathcal{Q}_*|}{\delta},$$

where $\mathcal{Q}_* \subseteq \mathcal{Q}$ is the relevant nonempty set.

Proof. Start both evaluators from the same encoded input. If their states agree, their signatures agree. At a covered signature, the learned table chooses the reference template; the common exact instantiation and execution rule therefore yields the same successor state. Induction gives agreement up to and including halt and the returned output. For a signature visited with probability p_q, the probability that n independent trajectories all omit it is $(1 - p_q)^n \leq e^{-np_{\min}}$. A union bound over $\mathcal{Q}_*$ proves the coverage claim. States within a trajectory need not be independent. □

The finite signature map and the address-instantiation routine are supplied parts of this theorem. They cannot secretly solve an arbitrary program search problem at zero cost. Their execution resources are counted. A finite control state alone does not imply that all relevant observations are in a fixed finite alphabet, nor that short training trajectories cover the signatures needed at larger lengths. The theorem neither constructs that abstraction nor proves that a neural router learns it.

The comparison

trace-supervised routing
entmax end-to-end routing
policy-gradient routing
evolutionary routing

quantifies the price of hidden program structure.

If the same execution architecture succeeds under supplied traces while specified end-to-end training procedures fail on the same evaluation, then its representation is sufficient for that measured task. The comparison diagnoses a learning or information-access bottleneck of those procedures, not an impossibility theorem for all end-to-end learners. No method must reproduce the particular teacher trace when a different admissible program produces the required outputs.

21.22 Interpretation

Sparsemax and entmax therefore occupy a useful but limited place in the CRM framework.

They do not prove that the underlying target DAG can be recovered.

They do not remove the statistical cost of structural discovery.

They do not guarantee that a sparse learned support is the correct support.

And they do not make the genericity problem disappear.

What they provide is a differentiable bridge between two extremes:

$$\boxed{\text{dense continuous routing} \longleftrightarrow \text{hard discrete program execution.}}$$

This makes them valuable experimental tools for asking a precise question:

$$\boxed{\begin{array}{c}\text{Can gradient-based sparse relaxation discover the same reusable,}\\ \text{length-general programs that can be found by hard discrete search?}\end{array}}$$

That question belongs naturally inside the CRM research program.

21.23 The Operator Library

A single local module is sufficient for a task such as parity, but general computation requires a vocabulary of operations.

Let

$$\mathcal{G}_{\text{locked}} = \{g_1, \dots, g_K\}$$

denote previously learned operators.

Each library entry carries a typed interface, numerical convention, version, and evaluation cost. "Locked" means that the complete mapping is fixed, including normalization state and preprocessing that affect its values, not merely that one weight tensor is excluded from optimization.

21.23.1 Reusable constituents

For each step,

$$y_t = g_{k_t}(v_{i_t}, v_{j_t}).$$

A task is therefore represented not primarily by new microscopic weights but by a program specifying

$$\boxed{\begin{array}{c}\text{which operators are used}\\ +\\ \text{where they are applied}\\ +\\ \text{in what order.}\end{array}}$$

This is the architectural analogue of compositional reuse.

21.23.2 Learning a new constituent

Maintain one additional mutable operator

$$g_{\text{new}}.$$

The controller may either reuse an old constituent or invoke the new module.

If the specified search budget does not find an economical composition, the system may propose, train, validate, and eventually freeze a new operator. Failure of a bounded search does not prove that no economical old-library program exists. Exact XOR and AND are already a functionally complete Boolean basis (with the necessary constants), so a "new" Boolean constituent can be new as a useful macro without expanding computability.

21.24 Compression of Composition: Compiling Subprograms into New Operators

The architecture described so far assumes a library of reusable local operators

$$\mathcal{G} = \{g_1, \ldots, g_K\}.$$

The controller constructs longer computations by composing these operators.

This immediately raises a further question:

What should happen when the same useful composition is discovered and executed repeatedly?

A natural possibility is that the system should eventually promote the composition itself into a new primitive.

This is a form of *compression of composition*.

21.24.1 From a subprogram to a primitive

Suppose the controller repeatedly executes a subprogram

$$\pi = (a_1, a_2, \ldots, a_T),$$

where each action a_t selects operands, applies an operator from $\mathcal{G}$, and writes the result to working memory.

Let F_π denote the function computed by this complete subprogram on its relevant input registers.

This is a pure interface map only when every relevant side effect is included. A stateful subprogram may change memory, aliases, control state, or halt status; a replacement must preserve all externally observed effects, not just one returned scalar. A pointer to a large data structure is one address argument but does not make its contents a constant-size payload. The later interface section makes this accounting explicit.

Instead of executing the same sequence indefinitely, the machine may learn a new operator

$$g_{\text{new}} \approx F_\pi$$

and enlarge its library:

$$\boxed{\mathcal{G} \longrightarrow \mathcal{G}' = \mathcal{G} \cup \{g_{\text{new}}\}.}$$

A computation that formerly required T explicit program instructions can then be invoked by one higher-level call. Whether this also reduces elementary execution cost depends on how the new primitive is implemented.

Schematically,

$$\boxed{g_{k_T} \circ g_{k_{T-1}} \circ \cdots \circ g_{k_1} \quad \longrightarrow \quad g_{\text{new}}.}$$

This operation is analogous to introducing a new instruction, subroutine, macro, or compiled primitive after discovering a frequently useful computation.

21.24.2 Two different forms of compression

It is useful to distinguish two different benefits.

Program-description compression. Suppose a subprogram π has description length $B(\pi)$ and occurs N times.

Writing the program explicitly at every occurrence costs approximately

$$NB(\pi)$$

bits.

If the subprogram is defined once and subsequently referenced by a short identifier of length B_{call}, the corresponding description cost becomes

$$B(\pi) + NB_{\text{call}}.$$

Thus introducing a reusable macro is advantageous whenever

$$B(\pi) + NB_{\text{call}} < NB(\pi).$$

This is ordinary amortized description-length compression.

The comparison assumes an explicit repeated expansion as baseline. A loop or a pre-existing subroutine may already encode repeated uses compactly. Definitions, call arguments, dependency versions, and codebook changes must be charged consistently. Library induction from recurring programs also has existing examples, such as DreamCoder; it is not a capability unique to the CRM proposal [47].

It connects directly to the discussion of bits and inductive bias in Chapter 13.

Execution compression. There is a stronger possibility.

The system may learn a direct operator

$$g_{\text{new}} \approx F_\pi$$

whose execution requires substantially fewer computational steps than the original subprogram.

If $T(\pi)$ is the execution cost of the original program and

$$T(g_{\text{new}}) \ll T(\pi),$$

then repeated experience has produced not merely a shorter symbolic name for the computation, but a genuine computational shortcut.

Thus

reuse can compress both description length and execution depth.

21.24.3 Depth collapse for familiar computations

Early in learning, a new problem may require a long sequence of primitive operations:

$$x \to z_1 \to z_2 \to \cdots \to z_T \to y.$$

After sufficient experience, the same effective transformation may be implemented by a shorter path:

$$x \to y.$$

We call this phenomenon *local depth collapse*.

The qualifier "local" is important.

There is no claim that every long computation can be replaced by a constant-depth computation.

Some problems intrinsically require sequential computation.

The claim is only that a repeatedly encountered subcomputation may be compiled into a new primitive when its input–output map admits an efficient shortcut.

21.24.4 Offline versus online compilation

Ordinary neural-network training can already produce computational shortcuts.

A network trained repeatedly on a multistep mapping may eventually represent the resulting input–output function directly in its weights.

Thus fixed-depth neural networks are not incapable of composition compression.

The stronger capability considered here is different:

online compilation:
a deployed system discovers a useful subprogram and
persistently promotes it into its own operator library.

A frozen model's internal weights remain fixed, but a surrounding interpreter can store new external programs without changing those weights. The architectural distinction here is whether a system persistently updates an executable external library, adds a learned numerical module, or merely records a description in context. These are different forms of adaptation.

The CRM can, in principle, be extended to do so.

Its operator library becomes time dependent:

$$\mathcal{G}_t = \{g_1, \ldots, g_{K_t}\},$$

with

$$\mathcal{G}_{t+1} = \mathcal{G}_t \cup \{g_{\text{new}}\}$$

when a new reusable computation has been discovered.

21.24.5 When should a new operator be created?

Creating a primitive for every observed composition would be disastrous.

The library would grow without bound and simply memorize individual computations.

A new operator should therefore be introduced only when its expected future reuse compensates for its complexity.

One possible objective is

$$\mathcal{J} = \mathcal{L}_{\text{task}} + \lambda_{\text{new}} B(\mathcal{G}) + \lambda_{\text{call}} T + \lambda_{\text{prog}} B(\pi), \tag{21.11}$$

where

- $B(\mathcal{G})$ measures the description length of the operator library;
- T measures execution cost;
- $B(\pi)$ measures the description length of the controller program;
- λ_{new}, λ_{call}, and λ_{prog} determine the tradeoff between inventing new operators and reusing old ones.

A new primitive is worthwhile only if the reduction in repeated program and execution cost exceeds the cost of representing and learning the new operator.

Schematically,

$$\boxed{\text{invent a new primitive} \quad \Longleftrightarrow \quad \text{future savings} > \text{cost of invention.}}$$

This is the same basic principle that favors reusable subroutines in ordinary programming languages.

21.24.6 Exact and approximate compilation

For discrete algorithmic tasks, one may require exact equivalence:

$$g_{\text{new}}(x) = F_\pi(x)$$

over the intended input domain.

For continuous problems, approximate compilation is more natural:

$$\|g_{\text{new}}(x) - F_\pi(x)\| \le \varepsilon.$$

Approximation error matters because a compiled primitive may itself be used repeatedly.

If the new operator is inserted into a longer computation, its local error can propagate according to the stability properties stated in Proposition 41 below.

Thus compilation should not be accepted merely because it is shorter.

It must also preserve the behavior of the subprogram to the required accuracy.

21.24.7 Verification and fallback

The original subprogram provides an unusually useful supervisory signal.

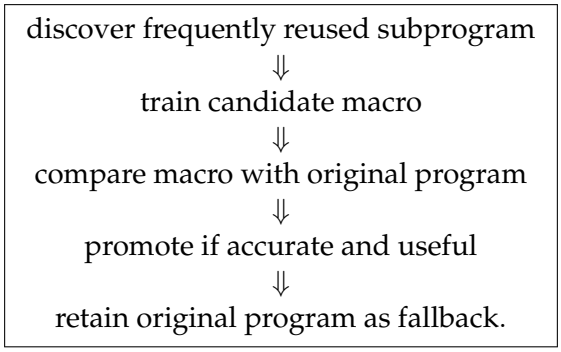
Once a candidate macro has been proposed, the system can generate pairs

$$(x, F_\pi(x))$$

by executing the original program.

The candidate operator can therefore be trained and tested against the computation it is intended to replace.

This suggests a safe compilation procedure:



For exact symbolic tasks, equivalence can sometimes be tested exhaustively over a finite domain or established formally.

Finite held-out validation estimates accuracy under a specified sampling distribution; it does not establish a uniform bound on a continuous domain. A uniform guarantee needs further analytic control or a verification argument. For example, m independent checks of a fixed candidate with zero failures give the one-sided confidence bound $p_{\text{fail}} \leq 1 - \delta^{1/m}$ at confidence $1 - \delta$. This follows from $(1 - p_{\text{fail}})^m$ and applies to that sampling distribution. Repeated candidate selection on the same validation set needs its own statistical accounting.

Retaining the original program is not sufficient to trigger a correct fallback: a monitor must know when to use it. Evaluating both programs on every call gives a comparison but can remove the execution saving. A validated domain predicate or a reliable external verifier is a separate component. Compilation transfers the original program's behavior, including any task errors; equivalence to it is not proof that the source program is correct.

21.24.8 A hierarchy of learned programs

Compilation can occur repeatedly.

Primitive operators form programs:

$$g_i \longrightarrow \pi_1.$$

Frequently reused programs become macros:

$$\pi_1 \longrightarrow g_{\text{macro},1}.$$

Macros can then participate in still larger programs,

$$g_{\text{macro},1}, g_{\text{macro},2}, g_j \longrightarrow \pi_2,$$

which may themselves eventually be compiled.

The result is a hierarchy:

$$\boxed{\begin{array}{c}\text{primitive} \to \text{subprogram} \to \text{macro} \\ \to \text{higher-level macro} \to \cdots\end{array}}$$

This is compositionality operating not only within a fixed representation but on the computational vocabulary itself.

21.24.9 Relation to sparse compositionality

Sparse compositionality says that complicated computations can be built from low-arity constituents.

Compilation adds a complementary principle:

$$\boxed{\text{useful compositions can themselves become constituents.}}$$

Thus the constituent library need not remain fixed.

Learning may progressively change the level at which the system represents a problem.

At one stage,

$$f = g_4 \circ g_3 \circ g_2 \circ g_1.$$

At a later stage, the same computation may be represented as

$$f = g_4 \circ g_{\text{macro}}.$$

The effective computational graph has changed because a previously explicit subgraph has become a node.

21.24.10 Relation to the staircase picture

This idea also provides an architectural counterpart to the staircase view of learning developed earlier in the book.

In the staircase picture, learning one constituent creates a useful new coordinate in which the next dependency becomes simpler.

Here the same principle appears computationally:

$$\boxed{\begin{array}{c}\text{solved computation} \to \text{new primitive} \\ \to \text{simpler higher-level search problem.}\end{array}}$$

The new operator changes the representation available to the controller.

The next program is searched for in a richer vocabulary than the one available before the constituent was learned.

21.24.11 A testable prediction

This proposal suggests a stronger criterion for compositional learning than success on a fixed task.

After repeated exposure to a reusable subprogram, an adaptive CRM should eventually require fewer primitive execution steps to solve problems containing that subprogram.

Thus one should measure

$$T_{\text{effective}}(n),$$

the number of primitive calls required after n successful encounters with the same recurring computation.

The compilation hypothesis predicts

$$\boxed{T_{\text{effective}}(n) \downarrow}$$

for familiar subproblems, while performance on genuinely novel compositions may still require long programs.

A decisive experiment would therefore compare:

1. a CRM with a fixed operator library;
2. a CRM allowed to introduce reusable macros;
3. a system that merely caches previous input–output answers.

The crucial distinction is generalization.

A genuine compiled operator should apply successfully to *new* inputs for which the same subcomputation is relevant.

Exact-answer caching alone cannot answer an uncached input. A learned approximate cache has its own generalization and description costs.

21.24.12 From learning programs to learning the instruction set

The CRM was introduced as a machine that learns how to compose a library of local operators.

Compression of composition suggests a more ambitious possibility.

The machine may eventually learn not only

$$\boxed{\text{which program to execute,}}$$

but also

$$\boxed{\text{which primitives its future programs should be written in.}}$$

This turns learning into a two-level process:

$$\boxed{\begin{array}{c}\text{learn programs from primitives} \\ \Updownarrow \\ \text{learn new primitives from recurring programs.}\end{array}}$$

Such a mechanism would provide a concrete form of continual abstraction: repeated computation changes the language in which subsequent computation is expressed.

In this sense, intelligence may involve not only composition, but the progressive *compression of composition*.

21.25 Computational Depth and Primitive Granularity

The preceding section described how a composition of operators can become a new primitive. This introduces a distinction between two ways of moving through a computational hierarchy.

One can advance through the execution of a program while keeping its primitive vocabulary fixed. Alternatively, one can change the vocabulary itself, replacing collections of elementary operations by larger subroutines.

We denote these two indices by

$$\boxed{i = \text{execution stage}, \qquad j = \text{primitive granularity}.} \tag{21.12}$$

The distinction is useful for the CRM because learning can change both the program being executed and the operators available for writing future programs. However, changing the description of a computation must not be confused with changing its implementation or learning a new representation.

21.25.1 Depth is relative to a primitive vocabulary

Fix an elementary operator library $\mathcal{G}_0$ and a finite-precision computational model. A program over this library induces a computation DAG $G^{(0)}$ whose nodes implement the specified elementary operations.

This is a *fine-grained realization*, not a uniquely defined "deepest possible" DAG. Equivalent programs can have different depths, and redundant operations can always increase the depth of a realization.

Suppose a consecutive segment of a computation is

$$q = g_m \circ g_{m-1} \circ \cdots \circ g_1. \tag{21.13}$$

Adding a name for q to the library allows the same segment to be written as one macro call. Its displayed depth has decreased relative to the new vocabulary, even if executing the macro still requires all m original operations.

Thus

$$\text{number of macro steps} \neq \text{number of elementary operations}.$$

Any claim that compilation reduces computational cost must specify the implementation and cost of the new primitive.

21.25.2 Coarsening a DAG requires explicit interfaces

For a subgraph S of a computation DAG, define its input boundary as

$$\partial_{\text{in}} S = \{u \notin S : (u, v) \in E \text{ for some } v \in S\}.$$

Its output boundary consists of the values computed inside S that are used outside S, together with any final outputs contained in S.

The subgraph therefore implements an interface map

$$q_S : (z_u)_{u \in \partial_{\text{in}} S} \longmapsto (z_v)_{v \in \partial_{\text{out}} S}. \tag{21.14}$$

A valid regrouping must retain all these externally required values. It must also use a partition for which contracting the modules leaves an acyclic quotient graph; an arbitrary partition of a DAG need not have this property.

Under these conditions, replacing each subgraph by its exact interface map preserves the overall input–output function. To see this, evaluate the modules in topological order. Each receives the same boundary inputs as the original subcomputation and therefore returns the same boundary outputs. Induction over the modules establishes equality of the final outputs.

This is an exact change of computational description. No learning or compression is required.

21.25.3 Bounded gate arity does not guarantee bounded module arity

Coarsening can enlarge the interfaces.

For scalar-valued nodes, let

$$a(S) = |\partial_{\text{in}} S|$$

be the input arity of a module. Even if every elementary gate has fan-in two, $a(S)$ need not be small.

For example, a complete binary subtree containing h internal levels has

$$2^h$$

boundary inputs. Contracting that subtree into one module produces a 2^h-ary interface.

Consequently,

$$\boxed{\begin{array}{c}\text{bounded elementary fan-in}\\ \not\Longrightarrow \text{bounded arity after aggregation.}\end{array}} \tag{21.15}$$

If bounded arity is required at every granularity, it must be imposed explicitly:

$$\max_{S \in \mathcal{P}_j} a(S) \le k,$$

where $\mathcal{P}_j$ is the module partition at level j.

Minimizing the number of edges crossing module boundaries may help find such partitions, but minimizing an edge cut is not itself a guarantee of a fixed arity bound.

For vector-valued registers, one must also distinguish the number of register arguments from their total dimension and bit length. Packaging an entire image into one tensor does not turn an arbitrary image computation into a low-dimensional constituent.

Small interfaces make a module compatible with the local-arity viewpoint. They do not, by themselves, make its internal function simple, well conditioned, or easy to learn.

21.25.4 Representations at execution boundaries

For each chosen granularity j, select a topological execution schedule.

Let

$$Z_{i,j}(x)$$

denote the live interface values at stage i: the values already computed, or supplied as inputs, that are still needed by the remaining computation.

This definition includes values carried across skip connections and other dependencies that need not lie in a single conventional layer.

The associated representation set is

$$E_{i,j} = \{Z_{i,j}(x) : x \in \mathcal{X}\}. \tag{21.16}$$

The word "embedding" can be used informally for these representations, but neither injectivity nor smooth-manifold structure is assumed.

The two indices now have a precise interpretation. Changing i advances execution within a fixed vocabulary. Changing j changes which subcomputations are treated as individual operations.

Crucially, regrouping alone does not create new interface values. For a chain in which coarse stage i ends at elementary stage $t(i, j)$,

$$Z_{i,j}(x) = Z_{t(i,j),0}(x) \tag{21.17}$$

when the coarse operators are exact macros.

The representation at that boundary is unchanged. It has merely been assigned a different position in the program description.

21.25.5 Regrouping, representation learning, and compilation

Three operations should therefore be distinguished.

Regrouping. A collection of operations is given a name and treated as one macro. The original computation and its boundary values are retained. Program notation may become shorter, but execution need not become faster.

Representation learning. The system constructs a new state

$$u_{i,j} = \psi_{i,j}(Z_{i,j}).$$

For this state to replace the original interface for a target F, there must be a suitable continuation $R_{i,j}$ satisfying

$$F(x) = R_{i,j}\left(\psi_{i,j}(Z_{i,j}(x))\right), \tag{21.18}$$

or an approximate version with controlled error on the relevant domain.

A useful compressed interface must preserve the distinctions needed by subsequent computation. Merely grouping nodes does not supply such an encoding, and fitting the training examples does not establish its sufficiency on new inputs.

Compilation. The system implements the interface map q_S by a new operator g_S.

An execution advantage requires

$$C_{\text{exec}}(g_S) < C_{\text{exec}}(q_S),$$

under a common cost model that accounts for internal operations and memory access.

For approximate compilation, one also needs a guarantee such as

$$\sup_{u \in \mathcal{D}_S} \|g_S(u) - q_S(u)\| \le \delta_S,$$

where $\mathcal{D}_S$ covers the boundary inputs on which the replacement will be used. The resulting end-to-end error must be controlled through the stability of the surrounding composition.

Thus a macro, a compressed representation, and a faster learned operator are different objects, even when all three are called an "abstraction."

21.25.6 Granularity is not automatically semantic or spatial scale

A coarser computational vocabulary does not necessarily correspond to larger spatial scale or more abstract semantic content.

A high-level arithmetic subroutine can operate on one pixel. An elementary Boolean circuit can compute a global property of an image. Likewise, a large neural block need not represent a more interpretable concept than a smaller one.

Computational granularity, physical scale, and semantic abstraction can be related in a particular architecture or task family, but the relation is not implied by bounded-fan-in composition.

Nor does choosing a smoother or more elaborate activation function, by itself, establish a higher level of semantic abstraction. The relevant questions concern the function implemented, the information crossing its interface, and its usefulness to subsequent computation.

21.25.7 Choosing a useful granularity

For the CRM, the goal is not to maximize or minimize j.

A very fine vocabulary gives simple local operations but may require long programs and many routing decisions. A very coarse vocabulary can shorten those programs while introducing expensive operators, wide interfaces, or difficult local learning problems.

The useful granularity balances

$$\text{program length}, \quad \text{interface size}, \quad \text{execution cost}, \quad \text{recovery difficulty}.$$

This connects the two-index view to the book's three conditions.

C1 concerns the existence of a feasible compositional realization. It does not select a unique grouping of that realization.

C2 concerns whether the learner can discover useful intermediate functions from the available signals. Renaming a difficult composition as one primitive does not remove this recovery problem.

C3 can expose selected module interfaces as training targets. This may shorten the hidden computation that remains to be learned, but each unexposed module still requires its own learnability and stability conditions.

The distinction can be summarized as

$$\boxed{\begin{array}{l} i : \text{progress through a computation,} \\ j : \text{the vocabulary in which that computation is expressed.} \end{array}} \tag{21.19}$$

A richer vocabulary becomes computationally valuable when its operators do more than rename existing work: they simplify program search, provide useful interfaces, or reduce execution cost.

The next section adds the cross-task criterion. A newly available operator deserves a lasting place in the library when its repeated use justifies the cost of learning, storing, and maintaining it.

21.26 Reusability Across Tasks: When a Composition Should Become a Primitive

The compilation discussion considered a computation that is repeatedly executed within a program:

$$\pi = g_{k_T} \circ \cdots \circ g_{k_1}.$$

If the same subprogram occurs often enough, it may be advantageous to compile it into a new primitive

$$g_{\text{new}} \approx \pi.$$

There is a stronger source of evidence for such compilation.

Suppose the same computation is useful not merely many times within one task, but across many different tasks.

Then reuse provides an additional criterion for deciding which discovered subcomputations deserve to become members of the permanent operator library.

The central principle is

a constituent becomes especially valuable when it compresses many different tasks simultaneously.

21.26.1 Sparse decompositions are not unique

Sparse compositionality establishes that a function may admit a realization as a bounded-arity computational graph.

It does not imply that this realization is unique.

For example,

$$f(x, y) = xy$$

can be represented directly by a multiplication primitive,

$$f(x, y) = g_{\times}(x, y),$$

but also through the polarization identity,

$$xy = \frac{1}{4}\left[(x + y)^2 - (x - y)^2\right]. \tag{21.20}$$

Both are bounded-arity compositions.

Thus even when C1 holds, there may exist many computational vocabularies in which the same target can be expressed.

This creates a selection problem:

which decomposition should the learner preserve as reusable structure?

Single-task accuracy cannot in general answer this question.

21.26.2 Reuse as an additional selection pressure

Let tasks be drawn from a distribution

$$T \sim \rho.$$

Suppose the CRM maintains a shared operator library

$$\mathcal{G} = \{g_1, \ldots, g_K\},$$

and each task T_i is solved by a task-specific program π_i written using operators from $\mathcal{G}$.
A module is valuable when storing it once reduces the total cost of solving many tasks.
This suggests an MDL-like objective

$$\mathcal{J} = \sum_{i=1}^{M} \mathcal{L}_{T_i}(\pi_i; \mathcal{G}) + \lambda_{\mathcal{G}} B(\mathcal{G}) + \lambda_\pi \sum_{i=1}^{M} B(\pi_i \mid \mathcal{G}), \tag{21.21}$$

where

- $\mathcal{L}_{T_i}$ measures task error;
- $B(\mathcal{G})$ is the description cost of the shared operator library;
- $B(\pi_i \mid \mathcal{G})$ is the description length of the task-specific program written in that library.

The objective expresses a simple tradeoff.
A large library can make individual programs short, but the library itself is expensive.
A tiny library is cheap to store but may force every task to reconstruct the same useful computation repeatedly.

21.26.3 When should a recurring computation become a primitive?

Suppose a subprogram q has description length $B(q)$ when expanded in the existing operator library.
Assume it appears in R task programs.
Without compilation, the approximate total description cost is

$$RB(q).$$

If q is promoted to a new primitive g_q, the cost becomes approximately

$$B(g_q) + RB_{\text{call}},$$

where B_{call} is the cost of referring to the new primitive.
Compilation is description-length favorable when

$$B(g_q) + RB_{\text{call}} < RB(q). \tag{21.22}$$

Equivalently,

cost of storing the primitive once
$<$ future savings from repeated reuse.

This provides a quantitative criterion for abstraction.

21.26.4 Reuse can compress execution as well as description

A macro may merely give a short name to a long computation.

More interestingly, learning may discover a direct implementation

$$g_q \approx q$$

whose execution is substantially cheaper than executing the original subprogram.

If $T(q)$ is the cost of the expanded program and

$$T(g_q) \ll T(q),$$

and a separate code-length saving holds, then compilation reduces both

$$\boxed{\text{program description length}}$$

and

$$\boxed{\text{elementary work, and possibly span.}}$$

The multi-task objective may therefore include execution cost explicitly:

$$\mathcal{J} = \sum_i [\mathcal{L}_{T_i} + \lambda_{\text{step}} T(\pi_i) + \lambda_{\text{prog}} B(\pi_i \mid \mathcal{G})] + \lambda_{\text{lib}} B(\mathcal{G}). \tag{21.23}$$

21.26.5 Reusability is relative to the task distribution

A crucial qualification is that reusability is not an absolute property of a computation.

It is defined relative to the environment

$$T \sim \rho.$$

For one task distribution, multiplication may be an extremely useful primitive.

For another, addition and squaring may be reused so frequently that the representation

$$xy = \frac{1}{4}[(x+y)^2 - (x-y)^2]$$

is perfectly economical.

Thus

$$\boxed{\begin{array}{l}\text{reuse identifies useful structure relative to a task distribution,}\\ \text{not an observer-independent “true” decomposition of nature.}\end{array}}$$

Changing ρ can change the optimal operator vocabulary.

This is not a defect.

A computational vocabulary should reflect the regularities of the world in which the machine actually operates.

21.26.6 Reuse reduces ambiguity but does not guarantee identifiability

Multiple operator libraries may achieve the same multi-task loss and the same description length.

For example, two latent representations may be related by an invertible change of coordinates,

$$z' = Az,$$

while supporting equally short programs for every task in the observed family.
Therefore

cross-task reuse can reduce representational ambiguity, but it does not generally establish uniqueness.

Still less does it prove that a discovered module is a causal variable.
Causal identification may require additional information such as

$$\text{interventions}, \quad \text{temporal structure},$$
$$\text{independence assumptions}, \quad \text{changes of environment}.$$

Reusability should therefore be viewed as an inductive bias for *selection and compression*, not as a general theorem of causal identifiability.

21.26.7 From repeated programs to a learned instruction set

The preceding discussion gave the hierarchy

$$\text{primitive} \rightarrow \text{subprogram} \rightarrow \text{macro}.$$

Cross-task reuse extends it:

task-specific subprogram $\rightarrow$ cross-task reusable macro $\rightarrow$ shared primitive.

The operator library can therefore evolve with experience:

$$\mathcal{G}_0 \subseteq \mathcal{G}_1 \subseteq \cdots, \tag{21.24}$$

although in practice obsolete or redundant operators may also be removed.
At stage t, the controller searches for programs in the language defined by $\mathcal{G}_t$.
Once a useful composition is compiled,

$$\mathcal{G}_{t+1} = \mathcal{G}_t \cup \{g_{\text{new}}\},$$

future program search occurs in a richer language.
This changes the difficulty of subsequent learning problems.

21.26.8 Connection to the staircase mechanism

The process closely resembles the staircase picture developed earlier in the book.

In that picture,

$$\text{learn one constituent} \rightarrow \text{create a useful new coordinate}$$
$$\rightarrow \text{expose the next dependency.}$$

Here,

$$\text{solve one recurring subcomputation} \rightarrow \text{promote it to a primitive}$$
$$\rightarrow \text{shorten future programs.}$$

Thus

$$\boxed{\begin{array}{c}\text{a solved problem becomes part of the representation} \\ \text{used to solve the next problem.}\end{array}}$$

Learning changes not merely the parameters of a fixed hypothesis class but the computational vocabulary in which future hypotheses are expressed.

21.26.9 Relation to cultural accumulation

The same mechanism provides a computational bridge to the chapter on frozen interpreters and cultural evolution.

A reusable procedure can first exist externally as a program p.

If it repeatedly proves useful across tasks, it can be retained in an external library.

At a still slower timescale it might be compiled into the internal operator repertoire.

Schematically,

$$\boxed{\begin{array}{c}\text{new computation} \\ \Downarrow \\ \text{task-specific program} \\ \Downarrow \\ \text{reusable cultural procedure} \\ \Downarrow \\ \text{compiled primitive.}\end{array}}$$

This suggests a continuum between external and internal learning rather than a sharp separation between them.

21.26.10 An experimental test

The value of reusability can be tested directly.

Construct a family of tasks

$$T_1, \ldots, T_M$$

that share hidden subcomputations.

Compare two CRM variants:

1. a system with a fixed primitive library;
2. a system allowed to compile recurring subprograms into new primitives.

The decisive evaluation should be performed on *new tasks* sampled from the same structural family.

Measure

$$\text{task accuracy}, \quad \text{program length}, \quad \text{primitive execution count},$$
$$\text{search evaluations}, \quad \text{transfer to unseen tasks}.$$

If genuine reusable abstractions have been discovered, then after learning tasks $T_1, \ldots, T_m$ the cost of solving a new related task should decrease:

$$C(T_{m+1} \mid \mathcal{G}_m) < C(T_{m+1} \mid \mathcal{G}_0). \tag{21.25}$$

The important quantity is therefore not simply performance on the tasks used to construct the library.

It is

reduction in future learning and execution cost.

21.26.11 Controls

Several controls are necessary.

A larger operator library has greater raw capacity, so improvements must be compared at matched description or storage cost.

A compiled primitive must also generalize beyond the exact inputs on which its source subprogram was discovered; otherwise it is merely a cache.

One should therefore compare

memoization:	reuse of stored input–output answers,
macro expansion:	reuse of an unchanged multistep program,
compiled primitive:	a learned implementation of the subcomputation.

Transfer to new inputs distinguishes reusable computation from exact-answer memoization.

Both macro expansion and a compiled implementation can generalize.

A genuine execution-compression claim additionally requires lower elementary cost than expanding the original macro.

21.26.12 Relation to sparse compositionality

Sparse compositionality says that a complicated computation can be represented using low-arity constituents.

Reusability adds a separate criterion:

among possible constituents, prefer those that compress many computations across the relevant task distribution.

These roles should not be confused.

C1 is fundamentally an existence statement:

a feasible bounded-arity representation exists.

Reusability is a selection pressure:

some constituents are worth preserving because they recur.

Neither statement alone guarantees recoverability, uniqueness, or causality.

21.26.13 The resulting picture

The CRM can therefore learn at three levels:

Level 1	:	compose existing primitives into programs,
Level 2	:	compile recurring subprograms into new primitives,
Level 3	:	retain primitives that reduce cost across many tasks.

The third level turns a growing operator library into a record of recurring structure in the task environment.

The central principle is simple:

what should become primitive is what repeatedly makes future computation shorter, cheaper, or easier to learn.

This is not a guarantee that the resulting modules are the unique "true" constituents of the world.

It is a precise computational criterion for deciding which discovered compositions deserve to become part of the machine's reusable vocabulary.

21.27 Parsimony as Description Length

Simply penalizing use of g_{new} is not sufficient.

A program using an existing operator one thousand times should not automatically be preferred to a program that introduces one useful new primitive and calls it twice.

A better objective resembles Minimum Description Length.

Let $T(\pi)$ be the number of operator calls in program π,

$$K_{\text{new}}(\pi)$$

the number of new operators introduced, and

$$B(\pi)$$

the number of bits required to encode the controller or program.

Define

$$\mathcal{J}(\pi) = \mathcal{L}_{\text{task}}(\pi) + \lambda_{\text{new}} K_{\text{new}}(\pi) + \lambda_{\text{call}} T(\pi) + \lambda_{\text{prog}} B(\pi). \tag{21.26}$$

The learner therefore trades off

reuse a longer existing program

against

invent a shorter reusable primitive.

This connects the CRM directly to the bit-complexity framework of Chapter 13.

Equation (21.26) is a resource-selection objective, not itself a generalization theorem. Counting new operators charges their true statistical complexity only under a fixed per-operator coding budget. A literal two-part code must include

$$B_{\text{total}} = B(\mathcal{G}) + B(\pi \mid \mathcal{G}) + B(\text{required decoder, keys, and stored data}).$$

The decomposition uses one fixed decoding language and prefix-valid component codes; quantities already encoded in one component are not charged a second time. Input-dependent action traces need not be stored when a fixed controller generates them. Conversely, arbitrary retained training answers cannot be made free by encoding only their addresses.

For loss in $[0, 1]$ and n independent full task examples from the same specified distribution, the Chapter 13 code argument gives with probability at least $1 - \delta$, uniformly over the admitted coded evaluators,

$$R(\widehat{f}) \le \widehat{R}(\widehat{f}) + \sqrt{\frac{B_{\text{total}} \ln 2 + \ln(2/\delta)}{2n}}.$$

The evaluator includes its stopping rule, failure behavior, and numeric implementation. It may be a finite stopped iterate; an absorbing state of regularized SGD is not required. This same-distribution certificate is not a length-extrapolation theorem. Nor does replacing n by the total number of dependent internal instruction events give an i.i.d. bound.

21.28 Continual Learning: What Freezing Does and Does Not Solve

Freezing learned modules gives an important form of protection.

If g_k is frozen, subsequent learning cannot overwrite its internal input–output mapping.

Thus

operator-level parameter interference is eliminated.

This statement holds for the same typed inputs and fixed operator semantics. Changing an upstream representation, a normalization buffer, or a shared interface can change behavior despite frozen numerical weights. Protecting an old program therefore also means preserving its library versions and input/output conventions.

But this does not make catastrophic forgetting impossible.

The controller can forget how or when to invoke the stored primitive.

Old task performance can therefore deteriorate even though every operator remains intact.

21.28.1 Protecting programs as well as operators

Several mechanisms are possible:

- rehearsal on old tasks;

- task-conditioned routing;
- frozen subprograms;
- progressive controllers;
- distillation constraints;
- a separate library of reusable routing programs.

A natural extension is therefore to store not only primitive operators but reusable subprograms:

$$\mathcal{P} = \{p_1, \dots, p_R\}.$$

The hierarchy becomes

$$\boxed{\text{operators} \to \text{subprograms} \to \text{larger programs.}}$$

This is compositionality at the level of control as well as computation.

21.29 Stability of Continuous Composition

The original motivation for digital operators is that approximation errors may accumulate during long computations.

The correct behavior depends on the stability of the composed operators.

Suppose the exact update is

$$x_{t+1} = g_t(x_t)$$

and the implemented update is

$$\widetilde{x}_{t+1} = \widetilde{g}_t(\widetilde{x}_t).$$

Assume the local error bound

$$\|\widetilde{g}_t(x) - g_t(x)\| \le \varepsilon$$

holds on specified state domains containing both trajectories, and every g_t is L-Lipschitz there. The maps are the *complete updates for a matched instruction sequence*, or otherwise include the controller in their stated Lipschitz bounds. Smooth payload operators alone do not make a content-dependent hard-routing update Lipschitz.

Let

$$e_t = \|\widetilde{x}_t - x_t\|.$$

Then

$$e_{t+1} \le L e_t + \varepsilon.$$

Proposition 41 (Error propagation under repeated local computation)**.** *If* $e_0 = 0$*, then*

$$e_T \le \varepsilon \sum_{j=0}^{T-1} L^j.$$

Hence

$$e_T \le \begin{cases} T\varepsilon, & L = 1, \\ \varepsilon \dfrac{L^T - 1}{L - 1}, & L > 1, \\ \dfrac{\varepsilon}{1 - L}, & 0 \le L < 1. \end{cases}$$

Proof. Insert $g_t(\widetilde{x}_t)$ between the two next states. The uniform local-error bound and Lipschitz inequality give $e_{t+1} \le \varepsilon + Le_t$. Induction from $e_0 = 0$ gives the geometric sum, whose three cases yield the displayed bounds. □

This yields three qualitatively different regimes:

$L < 1$:	uniformly stable long composition,
$L = 1$:	at most linear accumulation of local error,
$L > 1$:	potential exponential amplification.

Thus continuous computation is not intrinsically shallow.

Its useful depth depends on the stability factor of the constituent composition.

These are upper bounds, not assertions that every perturbation attains them. For $L = 1 + \eta$, approximately linear scaling over a horizon T requires $|\eta|T$ to be small. Append-only full memory retains old entries by identity, so its complete update is not a strict contraction in the usual max norm merely because the newly written payload is contractive. Variable state dimensions can be compared using the max norm over corresponding records under an identical allocation schedule.

A routing margin closes the missing induction. At reference state x_t, let the chosen action's score exceed every other legal action by $\gamma_t > 0$. Suppose the legal-action set is unchanged and each score changes by at most $K_{R,t}e_t + \eta_{R,t}$ between the reference and perturbed implementations. Then

$$\gamma_t > 2(K_{R,t}e_t + \eta_{R,t}) \quad \Longrightarrow \quad \text{the same hard instruction is selected.}$$

Indeed, the winning score can fall by that amount while a competing score can rise by that amount. With exact masks and halt semantics, this inequality and the state-error recurrence can be used inductively. Without it, a small continuous perturbation may cause a different branch, invalidating the matched-instruction premise. A margin observed on short training traces is not a uniform margin for all longer inputs.

21.30 Digital Operators

Discrete representations offer a different stability regime.

If a finite alphabet is represented with sufficient margin and each operator maps exactly between alphabet elements, one can snap a noisy continuous output back to the correct discrete symbol.

This can eliminate local approximation error while the state remains inside the correct decision basin.

For a finite symbol code with minimum separation $\Delta > 0$, nearest-code decoding is correct when its error relative to the intended output code is strictly less than $\Delta/2$. Starting from correct symbols and exact routing, a bound of this form at every step proves exact execution by induction.

With conditional step-failure probabilities p_t, a first-error union bound gives $\Pr(\text{any failure by } T) \leq \sum_{t<T} p_t$; independence between steps is unnecessary. Rounding cannot repair a wrong operand, a wrong instruction, or a result outside the representable range.

But finite-width digital arithmetic also has limitations:

- overflow;
- truncation;
- rounding;
- increasing bit requirements during arithmetic.

A fixed b-bit word therefore does not provide arbitrary-precision rational computation indefinitely.

Exact unbounded computation requires memory representations whose bit length can grow when necessary.

Thus the correct distinction is not

$$\text{analog fails, digital is exact forever},$$

but

continuous computation requires dynamical stability; digital computation requires explicit precision management.

21.31 Working Memory, Garbage Collection, and Universality

Append-only memory eventually becomes expensive.

If L_t grows indefinitely, the controller must search an ever larger set of records.

The CRM should therefore support deletion or reclamation.

21.31.1 Liveness

A record may be deleted when it is no longer needed by any future computation.

A heuristic controller may estimate a liveness score

$$q_{t,i}$$

and reclaim low-value records.

This is not certified garbage collection. Deleting a low-score record that is later needed can change the program. Exact reclamation needs a sound rule, such as last-use information for a supplied acyclic schedule, or a conservative reachability analysis with specified roots and alias semantics. Tracing unreachable objects is different from predicting that a still-reachable object will never be used. In learned-control experiments, unsafe deletion is reported as failure and compared with a no-deletion baseline; a storage spill is an external memory resource, not deletion.

21.31.2 Fixed capacity changes computational power

There is an important theoretical distinction.

If *all* mutable machine state is bounded—at most C records, finite payload and metadata alphabets, finite control, and no unbounded clock, pointer, external store, or code library—then the machine has only finitely many configurations. If each record has at most b bits and control has Q states, a coarse bound is $Q2^{Cb}$, after occupancy and addressing fields are included in b.

It is therefore a finite-state machine, not a universal Turing machine.

Hence

fixed bounded memory:	streaming finite-state regime,
extensible memory:	universal-computation regime.

Practical implementations may impose a finite hardware budget while the abstract machine family permits capacity to grow with problem size.

21.32 Sparse Parallel Execution

Sparse composition need not imply purely serial execution.

Suppose several operator calls at step t have no dependency conflicts.

The controller may issue a set

$$\mathcal{A}_t = \{a_t^{(1)}, \ldots, a_t^{(M)}\}.$$

Each action contains operand addresses and an operator.

If their read/write dependencies are compatible, they can execute in parallel.

For a simple deterministic implementation, require disjoint write sets, no write–read conflicts between different issued actions, and reads from a common snapshot; sharing read-only operands is allowed. Conflict checking and scheduling are controller costs. Parallel width reduces latency only with sufficient processors and communication capacity; total work still counts every issued call.

Thus the relevant computation is a sparse DAG, not necessarily a chain.

$$\boxed{\text{sparse} \neq \text{serial.}}$$

Balanced reductions, sorting networks, convolutions, and many dynamic-programming graphs possess substantial parallel width.

21.33 Relation to Transformers

The CRM should not be presented as a proof that Transformers are architecturally misguided.

Transformers can themselves use:

- sparse attention;
- recurrent computation;
- mixture-of-experts routing;
- local attention;

- external memory;
- learned algorithmic representations.

The difference is one of inductive bias.
A Transformer is a general differentiable sequence architecture.
The CRM exposes the hypothesized sparse-program primitives directly:

$$\boxed{\begin{array}{c}\text{memory} + \text{hard routing} + \text{local operators} \\ +\text{explicit program execution.}\end{array}}$$

The empirical question is whether this stronger bias improves systematic algorithmic generalization enough to compensate for its reduced hardware regularity and flexibility.

Chapter 20 makes a related distinction: a small active expert set does not imply small scalar arity, cheap routing, or a small stored library. A CRM exposes operand slots and state-changing instructions more explicitly, but it does not establish a universal separation from recurrent Transformers, learned interpreters, or memory-augmented networks.

21.34 Experiment 0: Parity as an Execution Sanity Check

Parity is useful, but only for a limited purpose.

Suppose the operator library contains exact XOR:

$$g_{\mathrm{XOR}}(a, b) = a \oplus b.$$

Because XOR is associative and commutative,

$$(a \oplus b) \oplus c = a \oplus (b \oplus c).$$

Therefore every complete binary reduction tree computes the same answer.

This requires every original input leaf exactly once. A correct reduction maintains a disjoint partition of leaves: combining two live aggregates consumes them and creates their union, reducing the live count by one. The XOR of live aggregates is then invariant, so the final value is the parity. With append-only storage, consumed records must become ineligible for this reduction; otherwise repeated reads can cancel or duplicate inputs.

Once XOR is known, parity does not require discovery of a special routing tree.

Thus parity tests

$$\boxed{\begin{array}{c}\text{repeated operator reuse,} \\ \text{memory bookkeeping,} \\ \text{length-independent execution,} \\ \text{termination.}\end{array}}$$

It does *not* provide a strong test of learned routing.

21.34.1 Protocol

Train or hard-code XOR on two-bit inputs.

Train the controller only on short lengths such as

$$N \in \{3, 4\}.$$

Test without further learning on

$$N \in \{8, 16, 32, 64, 128\}.$$

Success demonstrates recurrent composition at the tested lengths. A proof for all lengths requires the reduction invariant, a terminating schedule, and the stated exact execution assumptions.

21.35 Experiment 1: Non-Associative Program Execution

The first serious routing experiment should use operations for which parenthesization matters.

For example,

$$a - (b - c) \neq (a - b) - c.$$

Let the library contain exact primitives

$$\mathcal{G} = \{+, -, \times\}.$$

Generate random expression trees such as

$$((x_1 + x_2) \times x_3) - x_4.$$

The controller receives a structured symbolic description and must execute the corresponding tree.

Declare whether this description supplies child pointers, a parsed abstract syntax tree, or only tokens and parentheses. Supplying child pointers tests interpreter execution with structural information already available, not recovery of an unknown expression DAG from input–output pairs. Parsing, address encoding, legal masks, and traversal stacks are shared with the baselines or charged explicitly. Keep operand order for subtraction. Use bounded arithmetic with declared overflow semantics or count the cost of growing integer representations.

21.35.1 Training

Train only on small expression trees of depth

$$1, 2, 3.$$

21.35.2 Testing

Evaluate zero-shot on depths

$$5, 7, 10$$

and on substantially larger numbers of leaves.

21.35.3 Why this is a real routing test

Wrong structure can change the answer, unlike unrestricted reassociation of XOR. Degenerate input values can nevertheless conceal errors, and different algebraically equivalent programs can return the same result. Use separating input distributions and trace diagnostics without treating literal teacher-trace agreement as necessary for prediction. High test accuracy is evidence for a systematic evaluator, not its proof.

The key metric is

exact execution accuracy as a function of unseen program depth.

21.36 Experiment 2: Structural Routing with Sorting Networks

The next test should require a much richer data-oblivious topology.

Provide a locked compare-and-swap primitive:

$$g_{\text{cs}}(a, b) = (\min(a, b), \max(a, b)).$$

The router must construct a sorting network.

A bitonic or related sorting network contains a highly structured sequence of pairings whose offsets change recursively.

Training on

$$N = 4, 8$$

and testing on

$$N = 16, 32, 64$$

would test whether the controller has learned a length-general routing rule.

The controller must represent the changing pair relations. Pair scoring as in Equation (21.2) is one implementation, but an explicit recursive schedule generator is another. State which offsets, loop counters, and phase labels are supplied: encoding an entire bitonic schedule in metadata would no longer test its discovery. Fixed independent scores without such state need not express the required relations.

21.37 Experiment 3: Content-Dependent Conditional Control

Only after structural routing works should the controller be given semantic branching.

21.37.1 Stage A: exact semantic tags

Each record receives a small discrete content key

$$\kappa(v).$$

The task requires different operations for different classes of values.

For example,

$$\kappa(v) = A \quad \Rightarrow \quad g_A,$$

while

$$\kappa(v) = B \quad \Rightarrow \quad g_B.$$

This isolates conditional routing.

21.37.2 Stage B: high-dimensional representations

Replace exact tags with embeddings

$$v \in \mathbb{R}^D.$$

Give the controller only

$$\kappa(v) = vW$$

for a frozen random W.
Vary projected dimension d.
Measure when semantic routing begins to fail.
This directly tests whether a low-dimensional geometric key is sufficient for content-dependent control.

21.38 Experiment 4: Learning a Reusable Operator Library

Construct a curriculum of tasks sharing primitive operations.
For example,

$$T_1 : \text{XOR},$$

$$T_2 : \text{AND},$$

$$T_3 : \text{compositions of XOR and AND},$$

followed by a task for which a new macro is economical under a specified call or search budget. With exact Boolean XOR, AND, and constants, functional universality and useful library granularity are different.
Compare:

1. retraining one monolithic local module;
2. freezing operators but retraining one shared router;
3. freezing operators plus rehearsal;
4. operator and reusable-subprogram libraries.

Measure

$$\boxed{\begin{array}{c}\text{new-task sample cost} + \text{old-task retention} \\ + \text{program description length}.\end{array}}$$

This directly tests the continual-learning claim.

21.39 Experiment 5: How Should Hard Routing Be Learned?

For the same CRM architecture compare

evolutionary search policy gradient straight-through / sparsemax / entmax relaxation supervised routing traces.

Keep the controller parameter count fixed.
Measure:

- environment evaluations;
- wall-clock time;
- final accuracy;
- zero-shot length generalization;
- learned program length and support sparsity;
- sensitivity to reward noise;
- controller complexity.

This experiment tests the hypothesis inherited from Chapter 17:

evolution is competitive when the program-search dimension is genuinely small.

21.39.1 A Critical Hardening Test

For entmax, performance must be reported twice:

$$A_{\text{soft}} = \text{accuracy of relaxed routing},$$

and

$$A_{\text{hard}} = \text{accuracy after conversion to discrete instructions}.$$

Define the relaxation gap

$$\Delta_{\text{relax}} = A_{\text{soft}} - A_{\text{hard}}.$$

A small relaxation gap is informative only when the hardened accuracy is itself high. Both versions can otherwise fail equally.

High hardened accuracy with a small gap is evidence that the relaxed solution transfers to discrete instructions.

A large gap shows that this hardening procedure has not preserved the relaxed computation. It does not prove that no other discrete CRM program can solve the task.

Thus

the hardened program, not the relaxed network,
is the relevant test of program discovery.

21.40 Experiment 6: Stability of Continuous Operators

Start with controlled maps having known uniform approximation error and Lipschitz constants, then separately test learned operators with empirical diagnostics. An empirical Jacobian estimate is not a certified global Lipschitz constant. Compare a fixed instruction schedule with learned hard routing to separate numerical amplification from branching errors.

Measure end-to-end error versus computational depth T.

The bound distinguishes three possible error-growth regimes:

$$\begin{array}{ll} L < 1: & \text{bounded error,} \\ L = 1: & \text{at most linear error growth,} \\ L > 1: & \text{possible amplification.} \end{array}$$

This is a more meaningful test than asking for a universal depth at which "analog computation fails."

21.41 Fair Baselines

The CRM should be compared against architectures capable of algorithmic computation, not only against a dense vanilla Transformer.

Useful baselines include:

- standard Transformers;
- recurrent or weight-tied Transformers;
- sparse-attention models;
- recurrent neural networks;
- differentiable-memory architectures;
- neural controllers with explicit stacks or registers.

All models should be compared under controlled parameter count, training compute, and number of task evaluations when possible.

Neural Programmer-Interpreters are a direct trace-supervised comparator for recurrent program execution [134]; DreamCoder is a relevant comparator for growing reusable program libraries through program search [47]. They are precedents, not identical implementations. Match or report the supplied interpreter, operator basis, program specification, trace supervision, parser, and memory access, rather than comparing only neural parameter counts.

Use independent training, model-selection, and final evaluation tasks; separate novel values, novel compositions, and novel lengths. Keep the final length suite out of architecture and stopping-rule selection, or report that reuse as validation. Retain failed seeds and timeouts and report paired differences where possible. Count task examples, instruction labels, repeated fitness calls,

elementary operator work, peak bits, and wall time separately. Supervised traces are additional observations; replaying a fixed example many times does not create independent data. A success probability or confidence interval refers to the declared sampling and training randomness, not to every possible input.

21.42 What Would Support the CRM Hypothesis?

The strongest evidence would not be success on parity.

It would be the simultaneous observation of:

1. training on short programs;
2. exact or near-exact execution on substantially longer programs;
3. transfer through reuse of locked local operators;
4. controller description and numerical precision that do not grow with tested sequence length, with explicit memory and addressing costs;
5. a large advantage over dense baselines specifically on out-of-distribution algorithmic length generalization.

Especially convincing would be performance of the form

$$\mathrm{Acc}(N) \approx 1$$

for

$$N \gg N_{\mathrm{train}}$$

while the number of learned controller parameters remains fixed.

Finite extrapolation tests alone cannot establish an all-length theorem. For any tested maximum N_0, a compact program can run a correct routine when $N \leq N_0$ and deliberately fail afterward. Its extra length threshold requires only $O(\log N_0)$ bits. A proof for unbounded lengths must instead establish a suitable invariant, verified program semantics, or a condition such as Proposition 40. Small description length is compatible with both behaviors.

21.43 What Would Weaken the Hypothesis?

The architectural claim would become much less interesting if:

- the controller memorizes training lengths rather than discovering relational rules;
- evolutionary search becomes intractable even for small routers;
- pairwise routing destroys the computational savings;
- continuous operators accumulate unacceptable errors after modest depth;
- operator freezing prevents useful adaptation;

- standard recurrent or sparse neural architectures exhibit the same length generalization with comparable compute;
- content-dependent routing requires semantic keys nearly as large as the original representations.

These are useful falsification criteria.

21.44 Relation to C1, C2, and C3

The CRM makes the division of labor in the book especially explicit.

21.44.1 C1: sparse compositionality

The architecture assumes that useful computation can be expressed through a bounded-arity graph.

It makes that prior structural.

21.44.2 C2: genericity

C2 is not guaranteed by the architecture.

The learning problem must still expose detectable information telling the controller which program is useful.

If the admitted perturbations or policy updates cannot resolve a useful fitness preference at the stated sample and computation budget, that mechanism lacks a demonstrated progress signal. Multi-step exploration, a different parameterization, or richer observations can change the accessible statistics; absence of one local signal is not an impossibility result for every learner.

21.44.3 C3: program traces

If the correct routing sequence is supplied,

$$a_1^*, a_2^*, \ldots, a_T^*,$$

then controller learning becomes ordinary supervised sequence prediction.

This is a strong C3 regime.

The strength depends on what the trace supplies and covers. An action sequence without the observation and pointer interface does not identify a length-general policy. Supplied local action labels make an estimation problem directly supervised; they do not by themselves prove local optimization, rollout reliability, or coverage at new lengths.

Thus the CRM provides an unusually concrete environment for comparing

program discovery from terminal reward

against

program learning from explicit traces.

21.45 Relation to Chapter 17

The CRM also provides a natural experimental test of the effective dimension principle.

The complete computational system may contain many parameters, but the evolutionary search can be restricted to a controller

$$\theta \in \mathbb{R}^r$$

with small r.

The local operator library may be frozen.

Thus

large computational state $\neq$ large evolutionary search dimension.

This is one favorable design choice for zeroth-order optimization, not a sufficiency or necessity theorem. Chapter 17 also allows favorable full-space curvature; the CRM proposal instead isolates a small explicit controller and tests its fitness landscape and evaluation cost.

21.46 Relation to the Associative Turing Machine

Chapter 3 established that associative reads, local updates, and writable memory can support universal computation.

The CRM is not intended to replace that result.

The difference is emphasis.

The ATM asks:

What minimal associative primitives are sufficient for universal computation?

The CRM asks:

What architecture makes sparse program structure explicit enough that the program itself might be learnable and reusable?

The first is principally an expressivity theorem.

The second is an architectural and learning hypothesis.

21.47 Conclusion

Sparse compositionality does not prove that one particular neural architecture is optimal.

It suggests a more modest but testable architectural hypothesis:

complex computations may be easier to learn if the machine is forced to build them from reusable bounded-arity operations.

The Compositional Register Machine makes this hypothesis explicit.

Its state consists of working memory

$$\mathcal{M}_t.$$

Its computation is a sequence of sparse actions

$$a_t = (\text{operands}, \text{operator}, \text{write action}).$$

Its local operators have bounded arity.
Its controller can be kept small enough to make discrete or evolutionary search plausible.
Its operator library permits explicit reuse.
And an MDL-like cost can trade program length against invention of new constituents.
The proposal should not be confused with three stronger claims.
First,

bounded-arity execution does not imply genericity.

Second,

Turing universality of the execution mechanism does not imply that useful programs can be learned.

Third,

hard routing does not uniquely imply evolutionary learning.

Those distinctions turn the architecture into a cleaner research program.
The most informative first experiment is not parity.
Parity is an execution sanity check because XOR is associative.
The critical tests are computations for which the program structure matters:

non-associative expression evaluation,

sorting-network routing,

and ultimately

content-dependent conditional control.

If a controller trained only on small instances learns a compact relational program and executes it correctly on much larger unseen instances, then the CRM would demonstrate something important:

systematic generalization can arise by learning the rules for composing local operators rather than by learning a separate high-dimensional input–output map for every problem size.

That is the central conjecture.
In the language of this book,

sparse compositionality suggests that the natural search space
for intelligence may be a space of reusable programs
rather than a space of dense functions.

Whether that architectural bias outperforms more general neural architectures is an empirical question for the stated task and resource regime. The proposed program language is an inductive bias, not a claim that a feasible dense neural evaluator lacks a compositional realization.

The experiments above can provide evidence in specified regimes. They must distinguish learning a useful computation, identifying a teacher program, and proving correctness for all lengths. The conditional execution results in this chapter make that distinction explicit; the full end-to-end CRM learning hypothesis remains to be tested.

CHAPTER 22

World Models Before Language

Long before language, animals had to predict motion, remember places, track objects, anticipate other agents, and choose actions whose consequences unfolded over time.

This suggests a simple hypothesis:

much of the machinery required for prediction, planning, memory, and compositional representation predates language.

Language did not create these capacities from nothing. Rather, it provided a new interface for accessing, communicating, recombining, and culturally transmitting representations built by older perceptual, motor, memory, and predictive systems.

We will call this the prelinguistic-world-model hypothesis.

The purpose of this chapter is not to identify one particular neural circuit with a mathematical world model. The biological implementation remains open.

Instead, we ask a computational question:

What minimal architecture must an agent possess in order to model and act in a changing world before language exists?

A natural answer contains three ingredients:

$$\boxed{\textit{latent state} + \textit{predictive dynamics} + \textit{memory}.}$$

Action adds a fourth ingredient by allowing the agent not merely to observe the world, but to intervene in it.

When the underlying environment is governed by local interactions, and the agent's latent coordinates preserve the relevant locality, the latent transition model may itself be compositionally sparse. This is the precise point at which biological world modeling connects to the central structural principle developed earlier in this book.

22.1 A Minimal Definition of a World Model

Let

$$x_t$$

denote the observation available to an agent at time t, and let

$$a_t$$

denote its action.

The environment generally contains hidden variables that are not directly observed. A world model introduces a latent state

$$z_t$$

that summarizes information from the past relevant to prediction and control.

A minimal probabilistic formulation is

$$z_{t+1} \sim p_\theta \left(z_{t+1} \mid z_t, a_t \right), \tag{22.1}$$

$$x_t \sim p_\theta \left(x_t \mid z_t \right). \tag{22.2}$$

The latent state can support several distinct operations:

state estimation:	What is happening now?
prediction:	What is likely to happen next?
simulation:	What would happen under a hypothetical action?
control:	Which action should actually be taken?

These operations should not be conflated.

A system can predict without planning.

It can remember without generating counterfactual futures.

And it can possess a latent state representation without using that representation for explicit search over actions.

22.2 Sparse Dynamics of the Physical World

The strongest connection between world models and sparse compositionality is not the broad claim that "biology is compositional."

It is more specific.

Many physical processes are governed by local interactions.

Suppose the latent state decomposes as

$$z_t = \left(z_t^{(1)}, \ldots, z_t^{(m)} \right).$$

Assume that component i of the next state depends only on a bounded neighborhood:

$$z_{t+1}^{(i)} = F_i \left(z_t^{\mathrm{pa}(i)}, a_t \right), \qquad |\,\mathrm{pa}(i)| \leq k, \tag{22.3}$$

where

$$k \ll m.$$

Lemma 7 (Local latent dynamics imply sparse compositional transition)**.** *If every latent transition satisfies Equation* (22.3) *with bounded parent set*

$$|\operatorname{pa}(i)| \leq k,$$

then the global transition

$$F : (z_t, a_t) \mapsto z_{t+1}$$

has a bounded-fan-in compositional representation.

The statement is nearly definitional, but its consequence is important.
A world model need not learn an arbitrary map

$$\mathbb{R}^m \to \mathbb{R}^m.$$

It can instead learn a collection of local transition maps

$$F_1, \ldots, F_m.$$

Under the regularity and stability assumptions discussed in earlier chapters, the approximation complexity of such a model can depend on the local arity k rather than on the full latent dimension m.

Thus the relevant structural implication is

local environmental interactions
+structure-preserving latent coordinates
⇓
sparse latent transition graph
⇓
potentially tractable world-model learning.

The qualification about the coordinates matters.

A local physical process can become nonlocal under a sufficiently entangled change of variables. Thus environmental locality alone does not guarantee that an arbitrary learned representation will expose a sparse transition graph.

This provides a structural reason why modular internal models may be useful.

It does not imply that evolution must have discovered one unique architecture implementing them.

22.3 State Estimation Is an Inverse Problem

The world model does not directly observe

$$z_t.$$

It receives observations

$$x_{\leq t}$$

and must infer a latent state consistent with them.

Thus world modeling contains an inverse problem:

$$\boxed{x_{\leq t} \longrightarrow z_t.}$$

A generic formulation is

$$q_\phi \left(z_t \mid x_{\leq t}, a_{<t} \right).$$

The state estimate need not reconstruct every detail of the physical world.

It need only preserve the information relevant for predicting future observations, rewards, and consequences of actions.

Thus the ideal latent state is task dependent.

Two agents may construct different internal states from the same environment because the information relevant to their behavior differs.

22.4 Delay Embeddings: When History Determines State

The preceding discussion treats latent-state estimation as an inverse problem:

$$x_{\leq t} \longrightarrow z_t.$$

In general there is no guarantee that the observation history uniquely determines the hidden state.

For an important class of deterministic dynamical systems, however, dynamical-systems theory provides a remarkable positive result.

Takens' embedding theorem shows that, under suitable conditions, a finite sequence of observations can contain enough information to reconstruct the underlying dynamical state.

22.4.1 Takens' theorem

Consider a deterministic dynamical system

$$s_{t+1} = \phi(s_t), \qquad s_t \in M, \tag{22.4}$$

where M is a compact manifold of dimension d and

$$\phi : M \to M$$

is a sufficiently smooth diffeomorphism.

Suppose the system is observed only through a scalar measurement

$$y_t = h(s_t), \qquad h : M \to \mathbb{R}. \tag{22.5}$$

Theorem 18 (Takens' embedding theorem, informal)**.** *For generic smooth choices of the dynamics ϕ and observation function h, the delay-coordinate map*

$$\mathcal{D}(s) = \Big(h(s), h(\phi(s)), h(\phi^2(s)), \ldots, h(\phi^{2d}(s)) \Big) \in \mathbb{R}^{2d+1} \tag{22.6}$$

is an embedding of M into $\mathbb{R}^{2d+1}$.

Thus the delay-coordinate representation preserves the state manifold: distinct states of the original dynamical system correspond to distinct points in delay space, and the inverse map is smooth on the embedded manifold.

For prediction, the more natural representation uses observations from the *past* rather than the future.

Because ϕ is assumed to be a diffeomorphism, its inverse exists. Applying the same result to ϕ^{-1} gives, under the corresponding generic conditions,

$$\mathcal{D}_{-}(s_t) = (y_t, y_{t-1}, \ldots, y_{t-2d}) , \tag{22.7}$$

which is also an embedding.

Consequently,

CHAPTER 23

Can 3D Structure Be Discovered from 2D Experience?

A system can recognize an object from many viewpoints without constructing an explicit three-dimensional model of it. This was one of the central lessons of early view-based theories of object recognition.

But a different question arises when an agent does more than classify static images. Suppose it moves around an object, observes a temporally ordered sequence of views, knows or estimates its own motion, and is required to predict what it will see next.

Will such a learner discover an internal representation with genuine three-dimensional structure?

This chapter proposes a set of experiments designed to answer that question.

The central distinction is

$$\boxed{\textit{view-invariant recognition} \neq \textit{learning a 3D world model.}}$$

The strongest evidence for a learned 3D representation would not be classification accuracy or even novel-view synthesis alone. It would be the emergence of a reusable transformation law for viewpoint, persistence of hidden surfaces through occlusion, and compositional generalization to motions and objects not encountered during training.

23.1 Historical Motivation: Recognition Without Explicit 3D Reconstruction

Early theories of object recognition often assumed that recognizing an object from a novel viewpoint required reconstructing an explicit three-dimensional model and geometrically matching the input image to that model.

An alternative view developed in work on view-based recognition [126, 94, 95].

The basic idea was that recognition could be achieved by storing a limited number of characteristic views and interpolating between their representations.

In this picture,

$$\boxed{\text{recognition across viewpoint}}$$

does not require

explicit reconstruction of Euclidean 3D geometry.

Psychophysical and physiological results showing substantial viewpoint dependence were consistent with such a view-based account.

This observation remains important.

A neural network trained with sufficiently many views can learn the mapping

$$x \longmapsto \text{object identity}$$

while discarding much of the information about pose and geometry.

Thus successful viewpoint invariance by itself tells us very little about whether the representation contains an internal model of three-dimensional structure.

23.2 A Different Question

Consider instead an agent moving around a rigid object.

At time t it observes an image

$$x_t$$

and executes or measures an egomotion

$$u_t.$$

Its experience is therefore not a collection of independent images but a trajectory

$$(x_1, u_1, x_2, u_2, \ldots, x_T).$$

The learner is now asked to predict consequences of its own motion.

The problem changes from

$$x_t \longrightarrow \text{identity}$$

to

$$(x_{\leq t}, u_t) \longrightarrow x_{t+1}.$$

The second problem places stronger demands on the representation.

A view-invariant classifier is encouraged to *remove* information about viewpoint.

A predictive world model must retain precisely the information needed to determine how viewpoint changes.

This leads to the central experimental question:

Can temporal visual experience and self-motion cause three-dimensional structure to emerge as a latent variable

23.3 The Generative Setting

Let an object be represented abstractly by

$$S = (G, A) \in \mathcal{S},$$

where G denotes geometry and A denotes appearance.

We deliberately do not assume that the learner represents G as a mesh, point cloud, voxel array, or any particular explicit 3D data structure.

Let

$$p_t \in SE(3)$$

describe the relative pose between observer and object.

The observed image is generated by

$$x_t = R(S, p_t) + \epsilon_t, \tag{23.1}$$

where R is an unknown rendering process and ϵ_t denotes observation noise.

The observer moves according to

$$p_{t+1} = U(p_t, u_t), \tag{23.2}$$

where u_t represents egomotion.

For rigid motions this can be expressed using the group operation in $SE(3)$.

Schematically,

$$p_{t+1} = u_t p_t$$

or the corresponding convention for composition.

The learner sees

$$x_t, \qquad u_t, \qquad x_{t+1},$$

but is not given

$$S \quad \text{or} \quad p_t.$$

The scientific question is what latent variables emerge from learning this prediction problem.

23.4 Recognition Is Not Evidence for 3D Structure

Suppose an encoder produces

$$z = E_\theta(x)$$

and a classifier predicts

$$\hat{o} = C_\theta(z).$$

Training on many views of the same object can encourage

$$E_\theta(x_{o,p_1}) \approx E_\theta(x_{o,p_2})$$

for different poses p_1 and p_2.
This is desirable for recognition.
But complete invariance actually eliminates pose:

$$z_{o,p_1} \approx z_{o,p_2}.$$

A representation satisfying this condition cannot by itself tell the system how the image should change after a camera rotation.
Thus

invariance discards pose; world modeling must represent pose or its effects.

The first experiment must therefore establish an important negative control:

excellent viewpoint-invariant recognition does not imply recovery of 3D structure.

23.5 The Occlusion Problem

Occlusion makes the distinction especially sharp.
Let

$$P : \mathcal{S} \to \mathcal{X}$$

denote image formation from a particular viewpoint.
Projection is generally many-to-one.
There may exist two distinct object states

$$S_1 \neq S_2$$

such that

$$P(S_1) = P(S_2).$$

For example, two objects can have identical fronts but different backs.
Therefore a single image does not generally determine the complete object.
After a rigid transformation g,

$$P(g \cdot S_1) \neq P(g \cdot S_2)$$

may hold even though

$$P(S_1) = P(S_2).$$

Consequently there cannot, in general, exist a deterministic view-to-view function

$$T_g$$

such that

$$P(g \cdot S) = T_g(P(S))$$

for every possible object S.
The missing information has been removed by projection.
This establishes an important fact:

one 2D view is generally insufficient for predicting all other views.

It does *not*, however, establish that the learner must construct an explicit Euclidean 3D model.
The required hidden state could take many forms:

- an explicit geometric representation;
- an implicit neural field;
- a collection of depth layers;
- object-centered parts and visibility variables;
- a higher-dimensional latent manifold;
- or some other representation sufficient for prediction.

The emergence of 3D structure must therefore be tested, not assumed.

23.6 The 3D-Emergence Hypothesis

We formulate the main hypothesis as follows.

> **3D-Emergence Hypothesis.** When a learner must predict sequences of views generated by rigid self-motion, the combination of temporal continuity, action information, and occlusion will favor a latent representation in which viewpoint acts approximately according to the geometry of $SE(3)$.

The hypothesis does not require that the internal coordinates literally be Cartesian coordinates.
Latent representations are generally identifiable only up to changes of coordinates.
The relevant property is instead the transformation law.
Let

$$z_t = E(x_{\leq t})$$

denote the internal state.
We ask whether there exists a learned action

$$\rho(g) : \mathcal{Z} \to \mathcal{Z}$$

such that

$$z_{t+1} \approx \rho(g_t)(z_t) \tag{23.3}$$

and, more importantly,

$$\rho(g_2 g_1) \approx \rho(g_2) \circ \rho(g_1). \tag{23.4}$$

Thus ρ need not be a linear representation of $SE(3)$.

The essential requirement is that the latent transformations approximately respect the group composition law.

Equation (23.4) is a much stronger signature of three-dimensional structure than simple view reconstruction.

23.7 Four Core Hypotheses

The proposed experiments are organized around four increasingly strong hypotheses.

23.7.1 H1: Recognition does not require a 3D model

A feedforward network trained on many views should be capable of strong view-invariant recognition without learning a manipulable model of three-dimensional structure.

Thus one may observe

$$\text{high recognition accuracy}$$

together with poor performance on

$$\text{pose prediction}, \qquad \text{novel-view synthesis}, \qquad \text{group composition}.$$

This experiment reproduces the essential lesson of classical view-based recognition.

23.7.2 H2: View prediction encourages pose-sensitive latent structure

Train a model on

$$(x_{\leq t}, u_t) \longmapsto x_{t+1}. \tag{23.5}$$

Unlike classification, this objective penalizes a representation that throws away all pose information.

The prediction is that latent states trained for Equation (23.5) will exhibit stronger geometric organization than states trained only for object recognition.

23.7.3 H3: Occlusion creates pressure for persistent object state

Consider a trajectory

$$\text{front} \to \text{side} \to \text{back} \to \text{side} \to \text{front}.$$

Information about a surface may disappear entirely from the current image and later become visible again.

A model that operates only on the current image cannot preserve that information.

A persistent world model can.

The prediction is therefore:

prediction of object-specific hidden properties that are not inferable from the current view requires persistent information about those properties.

23.7.4 H4: Genuine geometric structure should compose

Suppose training contains only short rotations such as

$$10^\circ, \qquad 20^\circ, \qquad 30^\circ.$$

A system based mainly on interpolation may perform well near these training views.

A system that has learned a reusable transformation law should be able to compose them:

$$30^\circ + 30^\circ + 30^\circ \to 90^\circ.$$

More generally,

$$\rho(g_3) \circ \rho(g_2) \circ \rho(g_1)$$

should correctly predict transformations far beyond those explicitly experienced during training.

Thus the strongest evidence for a geometric latent model is *compositional extrapolation*.

23.8 What Would Count as Discovering 3D?

No single probe is sufficient.

In particular, low reconstruction error alone does not establish that a model has discovered 3D structure.

We propose several independent signatures.

23.8.1 Novel-view prediction

Given a partial trajectory and a new camera motion g,

$$(z_t, g) \longrightarrow \hat{x}_{t+1}.$$

The model should predict views not present during training.

23.8.2 Pose decodability

A simple probe should recover the relative pose

$$p_t$$

from the latent state.

This is useful evidence but is not sufficient by itself, since pose information can be present without organizing the representation geometrically.

23.8.3 Group-composition consistency

The most important direct test is

$$F(F(z, g_1), g_2) \approx F(z, g_2 g_1). \tag{23.6}$$

Define the composition error

$$\mathcal{E}_{\text{group}} = \mathbb{E}_{z,g_1,g_2}\left[\|F(F(z, g_1), g_2) - F(z, g_2 g_1)\|^2\right]. \tag{23.7}$$

Low error on transformations not encountered during training would provide evidence that the latent dynamics have discovered the composition law of viewpoint changes.

23.8.4 Persistence of occluded structure

The latent state should preserve information about object surfaces when they disappear from view.

A particularly clean test uses pairs of objects satisfying

$$x_0(S_1) = x_0(S_2)$$

but differing on an initially hidden surface.

After the system has observed the distinguishing surface, rotate the object until that surface is hidden again.

The latent state should continue to distinguish

$$S_1 \quad \text{from} \quad S_2.$$

When the surface reappears, the model should predict its identity before it becomes visible.

23.8.5 Cross-object transfer of motion

A physical rotation is the same operation whether applied to a chair, face, car, or unfamiliar synthetic object.

Therefore a strong geometrical representation should reuse the same motion operator across object identities.

We should test whether

$$\rho(g)$$

learned on one set of objects transfers to completely different objects.

The desirable factorization is

$$\boxed{\text{object representation} + \text{shared transformation operator.}}$$

This is stronger evidence than learning a separate transformation law for each object.

23.8.6 Extrapolation beyond trained trajectories

Train only on restricted regions of viewpoint space and test on

$$90^\circ, \qquad 180^\circ, \qquad 360^\circ$$

compositions that were never encountered as complete trajectories.

A representation encoding a genuine transformation law should degrade more gracefully than one based primarily on local view interpolation.

23.9 Experimental Environment

The cleanest initial experiments should use a controlled synthetic environment.

The environment should permit independent manipulation of

object identity, geometry, surface appearance, camera pose, illumination, occlusion.

For every transition, the experimenter knows

$$S, \qquad p_t, \qquad u_t,$$

even though these variables need not be supplied to the learner.

This allows strong latent probes without contaminating the training signal.

The first experiments should deliberately use simple objects whose hidden properties can be controlled exactly.

More natural objects can be introduced only after the basic phenomenon is understood.

23.10 Architectural Baselines

The experiments should compare several qualitatively different ways of solving the task.

23.10.1 Feedforward classifier

The model receives one image and predicts object identity:

$$x_t \to o.$$

This establishes the level of viewpoint invariance achievable without temporal modeling.

23.10.2 Multi-view contrastive encoder

Different views of the same object are explicitly encouraged to have similar embeddings.

This tests whether multi-view alignment alone produces geometric latent structure.

23.10.3 Recurrent or state-space predictor

A persistent state evolves according to

$$z_{t+1} = F_\theta(z_t, x_{t+1}, u_t).$$

This architecture can accumulate information across views and preserve hidden surfaces.

23.10.4 Transformer over view sequences

A Transformer receives the complete observed trajectory

$$(x_1, u_1, \ldots, x_t, u_t)$$

and predicts the next view.

This provides a powerful history-based baseline without imposing a persistent object state.

23.10.5 Explicit 3D model

A model with an explicit geometric representation provides a reference for what is achievable when three-dimensional structure is built into the architecture.

This is not the model whose success is scientifically interesting.

Its role is to establish an upper reference for geometric consistency.

23.10.6 Object-centric latent world model

The model explicitly separates a persistent object representation from a dynamic viewpoint state:

$$z_t = (z^{\mathrm{obj}}, z_t^{\mathrm{pose}}, z_t^{\mathrm{vis}}). \tag{23.8}$$

Only the pose and visibility components change as the observer moves.

This provides a structural inductive-bias baseline without requiring an explicit mesh or point cloud.

23.11 Experiment 1: Classification versus Prediction

Train two models on exactly the same visual experience.

The first minimizes classification loss:

$$\mathcal{L}_{\mathrm{class}} = \ell\left(C(E(x_t)), o\right).$$

The second minimizes future-view prediction:

$$\mathcal{L}_{\mathrm{pred}} = \left\| x_{t+1} - D\left(F(E(x_{\leq t}), u_t)\right)\right\|^2.$$

Both are evaluated on object recognition.

But they are additionally evaluated on

$$\mathcal{E}_{\mathrm{group}}, \qquad \text{pose decoding}, \qquad \text{novel-view synthesis}.$$

The prediction is

classification can produce invariance without geometry, whereas prediction should preserve transformation stru

23.12 Experiment 2: Object Permanence Through Occlusion

Construct pairs of objects that are identical over one range of views and differ only on a hidden surface.

For example,

$$S_A = \text{object with red back},$$

$$S_B = \text{object with blue back}.$$

Initially the two objects generate identical images.
The observer then moves until the diagnostic surface becomes visible.
Later it moves again until that surface is completely occluded.
At this point the current image is once again insufficient to distinguish the objects.
Test whether the latent state retains the distinction.
A useful quantity is the hidden-surface decoding accuracy

$$A_{\text{hidden}}(\Delta t),$$

where Δt is the duration for which the relevant surface has been occluded.
A persistent object model predicts

$$A_{\text{hidden}}(\Delta t) \gg \text{chance}$$

even when the current image contains no corresponding evidence.
This experiment operationalizes object permanence without relying on verbal interpretation.

23.13 Experiment 3: Composition of Rotations

Restrict training to a small set of elementary viewpoint changes

$$\mathcal{G}_{\text{train}} = \{g_1, \ldots, g_k\}.$$

Never train directly on certain compositions such as

$$g_3 g_2 g_1.$$

At test time compare

$$F(F(F(z, g_1), g_2), g_3)$$

with

$$F(z, g_3 g_2 g_1).$$

Evaluate both latent-state agreement and rendered-view agreement.

The decisive question is whether errors grow according to ordinary model approximation error or whether performance collapses as soon as the trajectory leaves the range seen in training.

Successful long-range composition would be difficult to explain purely as local interpolation between stored views.

23.14 Experiment 4: Reusing Motion Operators Across Objects

Divide objects into disjoint training and test classes.

Train the latent dynamics only on the training objects.

Freeze the motion operator.

For a previously unseen object, allow the system to infer its object representation but do not modify the learned action of viewpoint transformation.

Then test whether

$$\rho(g)$$

continues to predict the effects of rotation.

This experiment asks whether the learner has discovered

$$\boxed{\text{a law of motion}}$$

rather than

$$\boxed{\text{a collection of object-specific view transitions.}}$$

Reusable transformation operators would constitute a particularly clear example of compositional structure.

23.15 Experiment 5: What Signal Makes 3D Discoverable?

The previous experiments assume that self-motion u_t is available.

This provides a strong supervision signal.

To determine how much it matters, compare three conditions.

Known egomotion. Training examples contain

$$(x_t, u_t, x_{t+1}).$$

Unknown egomotion. The learner receives only

$$(x_t, x_{t+1})$$

and must infer the latent transformation.

Destroyed temporal structure. The same views are presented but their temporal order is randomly permuted.

The marginal image distribution is unchanged, but the dynamical information is destroyed.

The expected ordering is

$$\boxed{\text{known motion} > \text{ordered views without motion} > \text{unordered views}}$$

for emergence of a reusable geometric action.

This experiment has a direct relation to the genericity principle developed earlier in the book.

Self-motion and temporal adjacency may provide the low-complexity statistical footprints that make latent geometry recoverable.

Without them, the representation may still exist, but the learning problem becomes less identifiable.

23.16 Experiment 6: Fast Episodic Binding and Slow Structural Learning

The experiment can also test the hypothesis that fast memory and slow structural learning play complementary roles.

Consider two architectures.

Direct learner. All information from the trajectory must immediately be incorporated into slowly changing model parameters or the current latent state.

Episodic-plus-structural learner. A fast memory stores experience tuples

$$\mathcal{E} = \{(x_t, u_t, x_{t+1})\}_{t=1}^{T}. \tag{23.9}$$

A slower world model learns from both current experience and replay from

$$\mathcal{E}.$$

This is computationally analogous to the hippocampal–cortical separation discussed elsewhere in the book, without requiring a literal one-to-one anatomical identification.

Compare the two systems on:

- number of trajectories required to learn a new object;
- retention of hidden surfaces;
- resistance to catastrophic interference;
- group-composition error;
- cross-object transfer of motion operators;
- generalization to longer trajectories.

If episodic replay accelerates acquisition of a reusable transformation law, it would provide an experimentally tractable example of how memory can scaffold structural learning.

23.17 A Minimal Latent Factorization

A particularly interesting outcome would be spontaneous emergence of a factorization of the form

$$z_t = \left(z^{\text{object}}, z_t^{\text{pose}}, z_t^{\text{visibility}}\right). \tag{23.10}$$

The dynamics would approximately satisfy

$$z_{t+1}^{\text{object}} \approx z_t^{\text{object}}, \tag{23.11}$$

$$z_{t+1}^{\text{pose}} = F_{\text{pose}}\left(z_t^{\text{pose}}, u_t\right), \tag{23.12}$$

$$z_{t+1}^{\text{visibility}} = F_{\text{vis}}\left(z^{\text{object}}, z_{t+1}^{\text{pose}}\right). \tag{23.13}$$

The rendered image would then be generated by

$$\hat{x}_t = D\left(z^{\text{object}}, z_t^{\text{pose}}, z_t^{\text{visibility}}\right). \tag{23.14}$$

This would constitute a sparse compositional model of view generation:

$$\boxed{\text{object} + \text{pose} \rightarrow \text{visibility} \rightarrow \text{image}.}$$

The key experimental question is whether such a factorization appears without being explicitly imposed.

23.18 Connection to Sparse Compositionality

The physical generative process already contains reusable constituents.

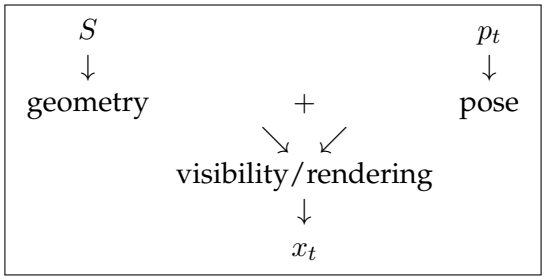
Object geometry remains approximately fixed over an episode.

Camera motion obeys the same transformation law across many objects.

Visibility depends jointly on object structure and pose.

Rendering combines these variables to produce the observation.

Schematically,



Thus the world itself supplies a compositional generative structure.

The experimental question is whether learning discovers reusable constituents corresponding to this structure.

In particular, the viewpoint transformation

$$\rho(g)$$

is especially interesting because it should be reusable across an entire family of objects.

Such reuse is stronger evidence for compositional learning than successful reconstruction of individual training objects.

23.19 Relation to World Models

This experiment is a concrete instance of the general framework developed in Chapter 22.

There the central architecture was

$$\boxed{\text{observation} \rightarrow \text{latent state} \rightarrow \text{predictive dynamics} \rightarrow \text{counterfactual simulation}.}$$

Here all of those components can be studied in a controlled setting.

The observation is

$$x_t.$$

The latent state may contain object and viewpoint information.
The action is

$$u_t.$$

The transition predicts how viewpoint changes.
The decoder predicts the future sensory consequence.
Thus

$$(x_t, u_t) \to x_{t+1}$$

provides a minimal laboratory for studying how a world model can be discovered from experience.

23.20 Relation to Genericity

The experiment also separates representation from recovery.

A useful object-centered representation may exist, yet the training data may not expose enough information to discover it.

Known egomotion provides a particularly strong statistical signal:

$$u_t \to x_{t+1}.$$

Temporal continuity supplies another:

$$x_t \approx x_{t+\Delta t}$$

for sufficiently small motion.
Repeated use of the same transformation across different objects provides a third:

same u $\quad\Longrightarrow\quad$ related latent transformation across many objects.

These are candidate genericity signals.

The ablation experiments can therefore ask whether removing them causes the latent geometry to disappear.

If so, the result would illustrate a central principle of this book:

the existence of a useful representation does not imply that it is recoverable from arbitrary observations.

23.21 Distinguishing Geometry from Memorization

A flexible neural network may obtain low reconstruction error by memorizing training trajectories.

The experimental protocol must therefore prevent simple memorization from being mistaken for discovery of geometry.

Three forms of held-out generalization are essential.

23.21.1 Held-out viewpoints

The model must predict viewpoints that never appeared during training.

23.21.2 Held-out compositions

The model must execute combinations of elementary rotations that were never observed as complete trajectories.

23.21.3 Held-out objects

The learned transformation law must transfer to entirely new object identities.

Only the combination of all three provides strong evidence for a reusable latent geometry.

23.22 Metrics

No single scalar score captures the phenomenon.

The following measurements should be reported separately:

$$A_{\text{rec}} = \text{view-invariant recognition accuracy}, \tag{23.15}$$
$$E_{\text{view}} = \text{novel-view prediction error}, \tag{23.16}$$
$$E_{\text{group}} = \text{group-composition error}, \tag{23.17}$$
$$A_{\text{pose}} = \text{pose decoding accuracy}, \tag{23.18}$$
$$A_{\text{hidden}} = \text{hidden-surface memory accuracy}, \tag{23.19}$$
$$E_{\text{transfer}} = \text{cross-object transformation error}. \tag{23.20}$$

The distinction between these quantities is essential.

For example, a classifier may achieve

$$A_{\text{rec}} \approx 1$$

while having poor

$$E_{\text{group}}$$

and

$$A_{\text{hidden}}.$$

That result would reproduce view invariance without evidence of a world model.

23.23 Possible Outcomes

Several qualitatively different outcomes would be informative.

23.23.1 Outcome A: Recognition without geometry

Feedforward and contrastive systems recognize novel views but fail group-composition and occlusion tests.

This would confirm that invariance alone does not imply a 3D model.

23.23.2 Outcome B: Prediction produces geometry

Action-conditioned predictive models develop low group-composition error, retain hidden surfaces, and transfer motion operators across objects.

This would support the 3D-emergence hypothesis.

23.23.3 Outcome C: Geometry requires explicit architectural bias

Only architectures supplied with object-centered or explicitly three-dimensional structure pass the strongest tests.

This would suggest that the environmental signal alone is insufficient for efficient recovery.

23.23.4 Outcome D: A non-Euclidean latent solves the task

A model succeeds on prediction, occlusion, and compositional generalization while its internal representation does not correspond simply to explicit Euclidean 3D coordinates.

This would be especially interesting.

It would show that the important discovery is not necessarily a literal 3D reconstruction, but a representation that implements the correct transformation structure.

23.24 Falsification Criteria

The proposal should be considered falsified in its strongest form if

- predictive training produces no greater geometric organization than classification training;
- group-composition consistency does not improve with temporal experience or known ego-motion;
- latent motion operators fail completely to transfer across object identity;
- or successful prediction can be fully explained by local interpolation over memorized views.

Conversely, the most compelling positive result would be

training only on 2D view sequences and self-motion
$\Downarrow$
emergence of a reusable latent action of $SE(3)$
$\Downarrow$
persistence of invisible surfaces
$\Downarrow$
composition of transformations far outside training
$\Downarrow$
transfer of the same transformation law to new objects.

23.25 From the 1990 Question to the Present One

The historical view-based theory asked whether three-dimensional reconstruction was *necessary* for recognition.

The answer was largely negative:

recognition can often be achieved by interpolation among views.

The present proposal asks a different question:

when an agent must predict the consequences of its own motion, will a representation of three-dimensional stru

This distinction reconciles the two perspectives.

A visual system may use view-based mechanisms for rapid recognition while also learning a richer latent world model from temporally structured experience.

Recognition and world modeling need not use identical representations.

23.26 Conclusion

A system that recognizes an object from many viewpoints has not necessarily learned a three-dimensional model.

The stronger test is whether it has discovered the transformation laws that generate those views.

Sequential visual experience provides precisely the additional information needed to study this question.

An agent moving around an object observes

$$(x_t, u_t, x_{t+1}),$$

where the same physical motion law is repeatedly expressed across many objects and viewpoints.

This creates the possibility that learning may discover a latent factorization into

persistent object state, viewpoint, visibility, rendered observation.

The decisive evidence would not be reconstruction accuracy alone.

It would be

composition + occlusion memory + extrapolation + cross-object reuse.

If a learner exposed only to two-dimensional image sequences and self-motion spontaneously develops a reusable latent action satisfying

$$\rho(g_2 g_1) \approx \rho(g_2) \circ \rho(g_1),$$

retains information about surfaces while they are invisible, and applies the same transformation law to novel objects, then it would be difficult to regard its representation as merely a collection of interpolated views.

It would instead provide a concrete example of a central idea developed throughout this book:

structured interaction with the world can cause hidden compositional structure to become discoverable from ex

CHAPTER 24

Interlude: Most Real Numbers Do Not Exist

Kronecker famously remarked that "God made the integers; all else is the work of man." Whatever one thinks of the philosophy, there is a precise mathematical fact behind a related observation.

The classical continuum

$$\mathbb{R}$$

contains uncountably many points. Yet there are only countably many finite descriptions, only countably many algorithms, and only finitely many outcomes that can be distinguished by any fixed finite-precision device.

Thus almost every real number has no finite name, no algorithm that computes its digits, and cannot be singled out by any finite observation.

In this operational sense—not in the set-theoretic sense—most real numbers do not exist for computation or experiment.

The continuum remains an extraordinarily useful mathematical idealization. The point of this interlude is narrower: the natural objects of physical and digital computation are finite descriptions and finite-precision approximations, not arbitrary exact elements of $\mathbb{R}$.

24.1 Several Different Notions of Existence

The word "exist" can mean several different things.

A real number may exist:

- **mathematically**: it is an element of the set $\mathbb{R}$;
- **definitionally**: it can be uniquely specified by a finite symbolic description in some fixed formal language;
- **computationally**: an algorithm can approximate it to arbitrary requested precision;
- **operationally**: an experiment or machine can produce finite information that constrains its value;
- **at fixed precision**: it is one of finitely many numerical states available to a particular device.

These notions are not equivalent.
The main hierarchy relevant here is

$$\boxed{\mathbb{R} \supset \{\text{finitely definable reals}\} \supset \{\text{computable reals}\} \supset \{\text{reals represented at any fixed finite precision}\}.}$$

The first set is uncountable.
The next two are countable.
The last one is finite for any fixed machine and fixed precision.

24.2 The Continuum Is Much Larger Than Finite Description

The cardinality of the natural numbers is

$$|\mathbb{N}| = \aleph_0,$$

while

$$|\mathbb{R}| = 2^{\aleph_0}.$$

Thus the real line is uncountable.
By contrast, the set of all finite strings over any finite alphabet is countable:

$$\bigcup_{n=1}^{\infty} \{0,1\}^n$$

is countable.
Therefore any collection of objects that can each be uniquely specified by a finite bit string is at most countable.
This simple cardinality observation already creates an enormous gap:

$$\boxed{\text{finite descriptions} \ll \text{the mathematical continuum.}}$$

24.3 Algebraic and Transcendental Numbers

A familiar first example is the distinction between algebraic and transcendental numbers.

Definition 8 (Algebraic number)**.** *A real number x is algebraic if it is a root of a nonzero polynomial*

$$p(t) \in \mathbb{Z}[t].$$

Otherwise it is transcendental.

Proposition 42. *The set of algebraic real numbers is countable.*

Proof. There are countably many polynomials with integer coefficients.
Each nonzero polynomial has finitely many roots.
A countable union of finite sets is countable.

□

Consequently the algebraic numbers have Lebesgue measure zero.
Almost every real number is therefore transcendental.
This already shows that familiar constants such as

$$\sqrt{2}, \qquad \pi, \qquad e$$

occupy an exceptionally small part of the real line.
But transcendence is not the deepest distinction.
The more relevant distinction for computation is computability.

24.4 Computable Real Numbers

Definition 9 (Computable real)**.** *A real number x is computable if there exists a Turing machine which, given*

$$n \in \mathbb{N},$$

outputs a rational number q_n satisfying

$$|x - q_n| < 2^{-n}.$$

Thus a computable real can be approximated to any requested number of binary digits by a finite algorithm.

Proposition 43. *The set of computable real numbers is countable.*

Proof. There are only countably many Turing machines.
Each machine computes at most one real number under the definition above.
Therefore there are at most countably many computable reals.

□

Hence

$$|\mathbb{R}_{\text{comp}}| = \aleph_0,$$

while

$$|\mathbb{R}| = 2^{\aleph_0}.$$

In particular,

$$\boxed{\lambda\left(\mathbb{R}_{\text{comp}}\right) = 0,}$$

where λ denotes Lebesgue measure.
Thus, in the measure-theoretic sense,

$$\boxed{\text{almost every real number is uncomputable.}}$$

Numbers such as

$$\pi, \qquad e, \qquad \sqrt{2}$$

are therefore extremely special.
They are not merely mathematically distinguished; they possess finite algorithms that generate arbitrarily accurate approximations.

24.5 Definability Is Not the Same as Computability

There is an important distinction between a real number that can be *defined* and one that can be *computed*.

Fix a countable formal language capable of referring to real numbers, and consider parameter-free finite formulas that uniquely specify one real.

There are only countably many such formulas.

Proposition 44. *In any fixed countable formal language, at most countably many real numbers can be uniquely specified by finite formulas.*

Proof. Finite formulas are finite strings over a countable alphabet.

There are only countably many such strings.

Each uniquely defining formula specifies at most one real.

□

Thus the finitely definable reals form a countable set.

However,

$$\boxed{\text{computable} \subsetneq \text{definable.}}$$

There are finitely definable real numbers that are not computable.

A standard example is Chaitin's halting probability

$$\Omega_U$$

associated with a fixed universal prefix-free Turing machine U.

Once U is specified, Ω_U has a finite mathematical definition.

Yet its binary expansion is not computable.

Thus

$$\boxed{\text{finite specification} \not\Longrightarrow \text{effective generation of the digits.}}$$

24.6 Algorithmic Randomness

The size of the gap between computable and arbitrary real numbers becomes even clearer through Kolmogorov complexity.

Let

$$x_{1:n}$$

denote the first n bits of the binary expansion of a real number x.

Let

$$K(x_{1:n})$$

be the prefix-free Kolmogorov complexity of this bit string.

A real is algorithmically random, in the usual Martin–Löf sense, if its initial segments are incompressible up to an additive constant:

$$K(x_{1:n}) \geq n - c$$

for some constant c and all n.
The set of algorithmically random reals has Lebesgue measure one.
Therefore

almost every real has essentially no shorter algorithmic description than writing down its digits themselves.

Such reals are necessarily uncomputable.
But one must not confuse randomness with undefinability.
Some algorithmically random reals are finitely definable.
For example, appropriate versions of Chaitin's Ω are both algorithmically random and definable.
Thus the correct relations are subtle:

$$\boxed{\text{random} \implies \text{uncomputable},}$$

but not

$$\text{random} \implies \text{undefinable}.$$

24.7 Finite Measurement Gives Finite Information

The connection to physics does not require the claim that the exact parameters of nature must themselves be computable real numbers.
A much weaker statement is sufficient.
Consider an experiment whose recorded output belongs to a finite set

$$\mathcal{Y} = \{y_1, \ldots, y_N\}.$$

One observation can convey at most

$$\log_2 N$$

bits of information.
Thus:

Proposition 45 (Finite-information principle)**.** *A measurement procedure with finitely many distinguishable recorded outcomes can convey only finitely many bits about an underlying real-valued quantity.*

This statement is almost tautological, but its consequence is important.
Suppose an experiment reports

$$x = 1.3724 \pm 0.0001.$$

The observation does not identify one real number.
It identifies an interval

$$x \in [1.3723, 1.3725],$$

which contains uncountably many real numbers.
Thus

finite measurement does not select an exact point of the continuum.

The exact underlying physical quantity, if one chooses to model it as a real number, could mathematically be computable or uncomputable.

The finite experiment does not decide the issue.

24.8 Finite Precision

A digital machine storing a numerical quantity using b bits has at most

$$2^b$$

distinct bit patterns available for that quantity.

Thus at any fixed precision the state space is finite.

Allowing arbitrary but finite precision gives

$$\bigcup_{b=1}^{\infty} \{0,1\}^b,$$

which is still only countable.

This gives a simple operational hierarchy:

fixed precision
$\Downarrow$
finite set of numerical states

arbitrary finite precision
$\Downarrow$
countable set of finite descriptions

exact mathematical continuum
$\Downarrow$
uncountable set.

The passage from finite computation to exact real arithmetic is therefore not quantitatively small.

It changes the cardinality of the available state space.

24.9 Infinite Precision Can Hide Infinite Information

This distinction becomes especially important in analog and recurrent computation.

Consider an infinite binary sequence

$$b_1, b_2, b_3, \ldots, \qquad b_j \in \{0,1\}.$$

It can be encoded into a single real number, for example,

$$x = \sum_{j=1}^{\infty} 2b_j 3^{-j}. \tag{24.1}$$

This real lies in the Cantor set.

Mathematically, the entire infinite sequence is contained in the exact value of one scalar x.

Thus a fixed-dimensional continuous state can, under exact real arithmetic, encode an arbitrarily long discrete memory.

But extracting progressively later symbols requires progressively finer precision.

To resolve the n-th ternary digit requires precision on the order of

$$3^{-n}.$$

Equivalently, the number of reliable bits of precision must grow linearly with n.

Hence

$$\boxed{\text{fixed dimension} + \text{unbounded precision}}$$

can conceal

$$\boxed{\text{unbounded information.}}$$

This is the basic mechanism behind mathematical constructions in which fixed-dimensional analog or recurrent systems simulate computations with unbounded memory.

24.10 Exact-Real Universality and Physical Resources

The preceding observation explains an apparent paradox encountered in the study of recurrent networks.

A recurrent system may have fixed state dimension,

$$h_t \in \mathbb{R}^d,$$

yet idealized exact-real models can possess extraordinary computational power.

There is no contradiction.

The unbounded resource is hidden in the precision of the coordinates.

A state coordinate can encode

$$\text{one bit}, \quad \text{one thousand bits}, \quad \text{or infinitely many bits},$$

if exact access to its real-valued expansion is permitted.

Thus one must distinguish

$$\boxed{\text{dimension of the state space}}$$

from

$$\boxed{\text{information required to specify a point in that space.}}$$

A vector in

$$\mathbb{R}^{10}$$

is only ten-dimensional geometrically.

But specifying an arbitrary point exactly can require infinitely many bits.

This is why exact-real computational models should not automatically be identified with finite-resource computation.

24.11 Why the Bit Model Matters

The theory of efficient computation developed elsewhere in this book therefore uses finite descriptions.

Suppose

$$f : [0,1]^d \to \mathbb{R}^m.$$

The computational question is not whether some abstract exact real

$$f(x)$$

exists.

It is whether one can compute an approximation

$$\widehat{f}(x)$$

satisfying

$$\|f(x) - \widehat{f}(x)\| \leq \varepsilon$$

using resources polynomial in

$$n + \log \frac{1}{\varepsilon},$$

where n denotes the input description length.

If

$$\varepsilon = 2^{-p},$$

then

$$p = \log_2 \frac{1}{\varepsilon}$$

is essentially the number of requested output bits.

Thus the relevant computational question is:

how many resources are required to produce p reliable bits of the answer?

This is a physically and algorithmically meaningful question.

Exact equality between arbitrary real numbers generally is not.

24.12 A Modulus of Continuity as a Precision Budget

For real-valued computation, input precision and output precision are linked.

Suppose

$$f : [0,1]^d \to \mathbb{R}$$

has modulus of continuity ω_f, so that

$$\|x - y\| \leq \delta \quad \Longrightarrow \quad |f(x) - f(y)| \leq \omega_f(\delta).$$

To compute $f(x)$ to accuracy ε, it suffices to know x to a precision δ satisfying

$$\omega_f(\delta) \leq \varepsilon.$$

Thus the amount of input information needed is itself part of the computational complexity.

For efficiently computable physical functions, one expects this precision budget to grow at most polynomially with the requested number of output bits.

This is the operational role of the polynomial-modulus assumption used elsewhere in this book.

24.13 Description Length and Learning

The same distinction appears in statistical learning.

A learned model stored digitally has a finite description:

$$\theta \longrightarrow \text{finite bit string.}$$

Even if we write

$$\theta \in \mathbb{R}^P,$$

the parameters actually used by a digital implementation are represented to finite precision.

Thus a trained network is not literally one arbitrary point drawn from the uncountable space

$$\mathbb{R}^P.$$

It is one member of a finite or countable collection of finite descriptions.

This connects naturally to the description-length viewpoint developed elsewhere in the book.

At precision ε, what matters is not an exact mathematical parameter vector but the number of bits required to specify a model whose behavior is accurate to that precision.

Schematically,

effective hypothesis complexity $\sim$ number of bits required to describe the useful approximation.

24.14 The Continuum Remains Useful

None of this argues against real analysis.

The continuum is one of the most successful idealizations in mathematics and physics.

Differential equations, probability distributions, Fourier analysis, functional analysis, and modern learning theory all rely on it.

The point is not that

$$\mathbb{R}$$

should be removed from mathematics.

The point is that mathematical existence should not be confused with operational accessibility.

A continuum model may be the simplest and most powerful symbolic language for describing a system even when every actual observation of that system contains only finitely many bits.

Thus

continuum mathematics $\neq$ infinite physical information.

24.15 What "Most Real Numbers Do Not Exist" Means

We can now state the intended claim precisely.

There are uncountably many real numbers:

$$|\mathbb{R}| = 2^{\aleph_0}.$$

But there are only countably many finite descriptions:

$$|\{\text{finite descriptions}\}| = \aleph_0,$$

only countably many computable reals,

$$|\mathbb{R}_{\text{comp}}| = \aleph_0,$$

and only finitely many distinguishable values available to any fixed finite-precision device.

Therefore:

almost every real number has no finite description and no algorithm that computes its digits.

Moreover, no finite experiment can single out an arbitrary exact real number from the continuum.

This does *not* mean that the remaining reals fail to exist in classical mathematics.

They do.

The claim is operational:

almost every real exists only as an element of the mathematical continuum, not as a finitely accessible computat

In that restricted but precise sense,

most real numbers do not exist for computation or experiment.

24.16 Conclusion

The real line contains far more objects than can ever be generated by algorithms or identified by finite descriptions.

The hierarchy is

all mathematical reals
$\supset$
finitely definable reals
$\supset$
computable reals
$\supset$
numbers represented at fixed finite precision.

The first set is uncountable.

The next two are countable.

The last is finite for a fixed device.

This distinction matters for theories of intelligence and computation.

Exact real arithmetic can hide unbounded information inside a fixed-dimensional state.

Finite-resource computation cannot.

Therefore the natural theory of practical computation is not based on access to arbitrary exact reals but on finite descriptions whose precision can grow as resources grow.

The appropriate question is not

what exact real number does the machine contain?

but

how many bits of information can the machine represent, compute, and recover with feasible resources?

Real analysis remains indispensable mathematics.

But for computation, learning, and experiment,

finite information is the operational currency.

The continuum is a powerful mathematical idealization.

It should not be mistaken for an infinite reservoir of physically available information.

CHAPTER 25

Finding the 1983 Circuit in the Connectome: the Tale of a SuperNeuron

When a fly is flying forward or turning, the entire visual world sweeps across its eyes, generating massive, wide-field motion signals (optic flow). If the fly relies solely on local elementary motion detectors (EMDs like T4/T5), those detectors will fire wildly just from the fly's own movement. How, then, does the fly spot a small, physically moving object—like a predator, a mate, or a target—against a background that is also moving across its retina? Forty-five years ago Reichardt, Poggio and Hausen proposed, on the basis of behavioral experiments on the fly, a neural circuit for the detection of relative motion. Today the fly connectome – interrogated by agentic AI – makes it possible to check the validity of the model.

25.1 Introduction

The computation of relative motion—the ability to discriminate a moving figure against a moving background—is a fundamental operation in visual processing. It uncovers camouflage by using relative motion; it is likely to be a critical step for separating objects from context in the early development of human infants. It is indicative of its importance that it has been conserved throughout evolution: it is found in insects and in our visual system. Recent work suggests that in vertebrates the computation of relative motion may actually start in the retina [7]. For all these reasons, the neural circuits underlying it are especially interesting.

Over forty years ago, a theoretical framework for this computation was established based on behavioral and – to a minor extent – electrophysiological studies in the housefly (*Musca*) [135] and the blowfly (*Calliphora*). The proposed circuit (see Figure H.1) posited that Elementary Motion Detectors (EMDs) respond to local motion, while wide-field "pool" cells summate these local signals across the visual field to estimate global optic flow. Figure-ground separation is then achieved by having these pool cells inhibit the local motion pathway, effectively vetoing the background motion. The same network was found to normalize the optomotor response to make it invariant to the size of the surround. It is therefore a *normalization and figure-ground circuit*.

With the advent of dense electron microscopy (EM) reconstructions of the *Drosophila melanogaster* brain, the physical implementation of this theoretical circuit can be mapped down to the individual synapses. Utilizing platforms like NeuPrint for the Janelia Hemibrain and FlyWire for the complete adult female fly brain (FAFB) [43, 140], we can now query massive graph databases to uncover how specific neurons and circuits wire together. We will review here whether this evidence support the

original proposal of Reichardt and Poggio.

25.2 Main Neurons in the Optical Lobes

To find the relative motion circuit in modern connectomes, we must map the theoretical triad—Local EMDs, Pool Cells, and Output Figure Detectors—to their anatomical correlates.

25.2.1 The Local EMDs: T4 and T5 Cells

The local motion signals in the connectome originate from the T4 and T5 columnar neurons, which represent the physical instantiation of the EMDs. T4 cells compute the direction of moving ON-edges (light increments), while T5 cells compute moving OFF-edges (light decrements). These cells are strictly direction-selective and stratify their axonal terminals into one of the four cardinal layers of the lobula plate, corresponding to the four cardinal directions of motion (front-to-back, back-to-front, upward, and downward). Notably, modern connectomics and optogenetics have confirmed that the outputs of T4 and T5 cells are entirely cholinergic (excitatory) [101].

25.2.2 Wide-Field Pool Cells: The LPTCs

The "pool" cells hypothesized in the 1983 model are implemented by the massive, wide-field neurons known as Lobula Plate Tangential Cells (LPTCs), which serve as global spatial integrators. A single LPTC pools synaptic input from up to ≈ 700 T4/T5 cells. This massive convergence makes sense given that within each column there are 4 T4 and 4 T5 cells. LPTCs collect input from hundreds of these local motion-sensing neurons spread across the entire retinotopic array, essentially integrating motion signals from a large swath of the visual field into a single wide-field response.

Some of the best characterized LPTCs include:

- **Horizontal System (HS) Cells:** Consisting of the Northern (HSN), Equatorial (HSE), and Southern (HSS) cells. They arborize predominantly in layer 1 of the lobula plate, acting as matched filters for horizontal optic flow and serving as the primary sensors for controlling yaw (horizontal rotation) stabilization.

- **Vertical System (VS) Cells:** An array of 10 cells (VS1–VS10) that tile the lobula plate vertically, primarily receiving input from layers 3 and 4 (upward and downward motion). By pooling their outputs via gap junctions, the VS network computes complex rotatory flow fields governing pitch and roll reflexes.

- **Centrifugal Horizontal (CH) Cells:** The ventral (vCH) and dorsal (dCH) cells act as inhibitory pool cells. Rather than directly controlling descending motor neurons, they provide local gain control and subtract background optic flow to enable figure-ground discrimination.

25.2.3 Dendrites of the CT1 Cell vs. the vCH Cell in *Drosophila*

When we contrast the dendrites of the **CT1** (Complex Tangential 1) cell with those of the **vCH** (ventral Centrifugal Horizontal) cell, we are essentially looking at two entirely different architectural philosophies for how the fly visual system handles inhibitory feedback and spatial integration.

Here is how their dendritic architectures break down.

25.2.4 The CT1 Cell: Local, Polyadic Interception

The CT1 cell acts as an enormous, multi-columnar inhibitory feedback element, but its dendritic interactions happen at a highly granular, local level.

- **Location:** CT1 arborizes specifically in layer **M10 of the medulla** (interfacing with the T4 ON-pathway) and layer **Lo1 of the lobula** (interfacing with the T5 OFF-pathway).
- **Synaptic Architecture:** Recent connectomic reconstructions (like those from FlyWire) show that CT1 uses highly complex **polyadic synapses**. A single presynaptic bouton from a Tm cell or the CT1 cell itself will simultaneously contact multiple postsynaptic compartments.
- **The Wiring:** CT1 provides GABAergic inhibition directly onto the dendrites of T4 and T5 motion-sensing neurons. Interestingly, a single CT1 synapse can span across the dendritic spines of multiple T4/T5 subtypes at once, allowing a single inhibitory signal to act on neurons tuned to different cardinal directions simultaneously.

25.2.5 The vCH Cell: Wide-Field, Dendro-Dendritic Integration

While the vCH cell was historically modeled in the blowfly (*Calliphora*), recent 3D EM datasets have fully reconstructed its homolog in *Drosophila*. It operates at a completely different stage of the motion-computation pipeline.

- **Location:** The vCH dendrites are restricted to the **lobula plate**, specifically covering the ventral portion of the visual field.
- **Gap junctions** Instead of local columnar inputs, vCH relies heavily on **dendro-dendritic electrical coupling** (gap junctions). It electrically couples with the overlapping dendritic trees of other wide-field Lobula Plate Tangential Cells (LPTCs), specifically the Horizontal System (HS) neurons. Note that lectrical junctions have been shown in Calliphora (by the the Borst lab) but not yet in Drosophila.
- **Cable properties** Full reconstruction of large lobula plate tangential cells in Drosophila from a 3D EM dataset[19] provides the definitive ultrastructural proof of the vCH cell's massive dendritic architecture in the fruit fly. It fully maps the dense, thick arborization of the vCH cell, demonstrating how its wide-field branches are restricted to layer 1 of the lobula plate.
- **The Wiring:** Because it integrates inputs from both ipsilateral HS cells (via electrical synapses) and contralateral elements like the H1 and H2 cells (via chemical synapses), its dendrites act as a massive spatial pooling mechanism. The commonly accepted hypothesis based on physiology in *Calliphora* is that the vCH cell then provides inhibitory output to object-detecting neurons (like the FD1 cell) to suppress their firing during large-field background motion (optic flow). In the next section we present an alternative model supported by anatomical data.

The contrast is stark: CT1 dendrites dive into the columnar weeds to modulate the primary motion detectors via complex polyadic wiring, whereas the vCH dendrites sit higher up in the lobula plate, electrically pooling massive amounts of spatial data to help the fly subtract background motion from foreground objects.

25.3 Local Motion Detection by T4/T5 Cells and CT1 Dendrites

While the integration of wide-field optic flow is handled downstream by LPTCs, the foundational computation of directional motion must occur locally and precisely within the retinotopic map. A critical—and architecturally astonishing—component of this local computation is the **CT1 (Complex Tangential 1)** cell [104].

CT1 is arguably the single largest and most structurally complex neuron in the fly optic lobe. There is only one CT1 cell per hemisphere, yet its massive arborization innervates every single one of the approximately 750 visual columns. Despite its wide-field morphology, CT1 operates in a strictly local, upstream computational regime, acting as a giant amacrine cell rather than a global spatial integrator.

25.3.1 Upstream Modulation and Null-Direction Suppression

Unlike the LPTCs, which pool the outputs of the elementary motion detectors, CT1 is positioned immediately *upstream* of the motion computation. It is strictly presynaptic to the EMDs [144, 148]. CT1 projects its terminals simultaneously into two distinct neuropils: providing widespread GABAergic (inhibitory) input directly onto the dendrites of **T4** cells in Medulla Layer 10, and homologous GABAergic input onto the dendrites of **T5** cells in Lobula Layer 1.

The directional selectivity of T4 and T5 cells relies on the physical, non-linear integration of signals across their dendrites. Modern connectomic and functional imaging reveals that this process is a hybrid mechanism, relying on both preferred-direction excitation and robust **null-direction suppression** [56, 8], as predicted by the Torre-Poggio model. In T5 cells for instance all columnar input neurons (Tm1, Tm2, Tm4, and Tm9) are excitatory. Borst et al. discovered that within each column, CT1 receives excitatory input from Tm9 and Tm1 and provides the sign-inverted, now inhibitory input signal onto T5. Ablating CT1 or knocking down GABA-receptor subunit Rdl significantly broadened the directional tuning of T5 cells. It thus appears that the signal of Tm1 and Tm9 is used both as an excitatory input for preferred direction enhancement and, through a sign inversion within the Tm1/Tm9-CT1 microcircuit, as an inhibitory input for null-direction suppression.

CT1 is the critical hardware for this inhibitory delay line. When a moving edge activates an upstream columnar cell (e.g., a Mi or Tm cell), it excites the local T4/T5 dendrite while simultaneously exciting the local CT1 terminal. CT1 responds by applying a slightly delayed GABAergic inhibition to the T4/T5 dendrite. Because of a highly specific spatial offset in the connectome, if an edge moves in the "wrong" (null) direction, the delayed CT1 inhibition arrives precisely as the main excitatory signal from the adjacent column reaches the dendrite, effectively vetoing the response [8]. State-of-the-art electron microscopy reconstructions mapped the exact synapses, showing that CT1 is strictly presynaptic to the dendrites of both the ON-pathway (T4 cells in the medulla) and the OFF-pathway (T5 cells in the lobula), positioning it perfectly as an upstream modulator of the motion signal.

25.3.2 Extreme Compartmentalization

To successfully mediate local motion detection, the inhibitory signals must not blur across the visual field. The fly resolves this through extreme biophysical compartmentalization [104]. Connectomic reconstructions reveal that the neurites connecting CT1's terminal branches are remarkably thin, dropping to diameters of approximately 100 nanometers and forming redundant, twisting loops. This creates immense axial resistance, electrically isolating the local synaptic sites. Consequently, the

single CT1 cell functions as an array of roughly **1,400 independent, electrically isolated inhibitory microcircuits**.

25.3.3 Recurrent Connectivity: Local Feedback and Gain Control

While CT1 plays a critical role in feedforward null-direction suppression, recent high-resolution dense connectomes have revealed that its relationship with the elementary motion detectors is not strictly unidirectional. CT1 also receives direct synaptic input from the T4 and T5 cells, creating an array of reciprocal, dendro-dendritic feedback loops at the very first stage of motion computation.

The anatomical proof of this reciprocal wiring was established by comprehensive reconstructions of the optic lobe. In the ON-pathway (M10 stratum of the medulla), the local CT1 terminal within each visual column receives a consistent number of synaptic inputs directly from the local T4 dendrites [147]. Subsequent mapping of the OFF-pathway (Lo1 stratum of the lobula) confirmed homologous reciprocal connectivity between T5 dendrites and CT1 terminals [144].

Because T4 and T5 cells are cholinergic (excitatory) and CT1 is GABAergic (inhibitory), this structural motif forms a highly localized negative feedback loop. Crucially, owing to the extreme biophysical compartmentalization of the CT1 cell [104], the excitation provided by a local T4/T5 cell is trapped within that specific microcircuit and does not spread laterally across the visual field.

This recurrent inhibition serves two vital computational functions for the local motion detectors:

1. **Temporal Sharpening:** Motion detection requires exquisite temporal precision. By exciting the local CT1 compartment, the active T4/T5 cell triggers its own rapid, delayed suppression. This truncated the tail of the voltage response, quickly resetting the local EMD for subsequent visual events.

2. **Dynamic Gain Control (Auto-Normalization):** In high-contrast visual environments, robust feedforward excitatory inputs could easily saturate the local T4/T5 dendrite. The reciprocal CT1 loop acts as an automatic, localized shunt. As the firing rate of the T4/T5 cell increases, it more strongly recruits local GABAergic inhibition from CT1 to clamp its own membrane potential, thereby maintaining the neuron within its optimal dynamic operating range.

Thus, while CT1's *feedforward* inhibition from upstream interneurons provides the null-direction suppression necessary to compute the motion vector [21], its *feedback* inhibition from T4/T5 provides the temporal sharpening and auto-normalization required to keep that computation fast and stable.

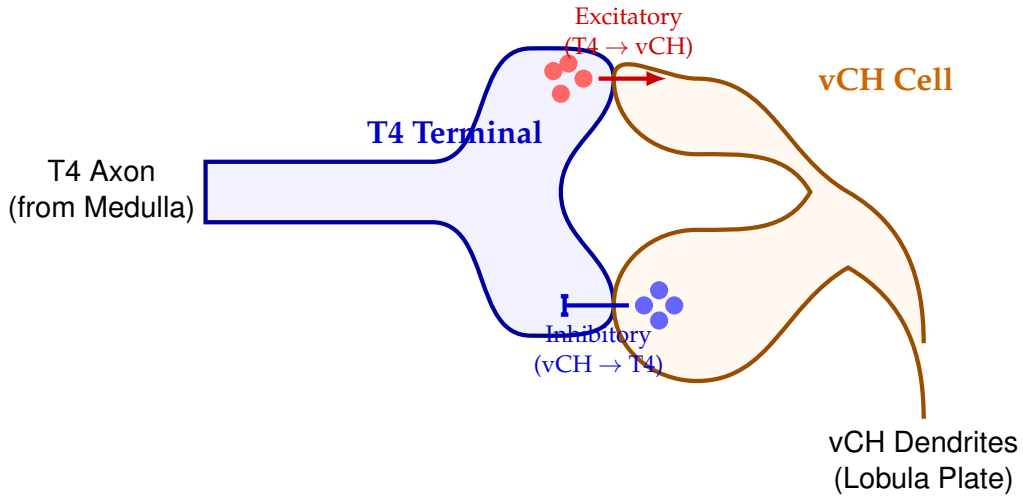
25.4 The Core Hypothesis: Figure-Ground Discrimination via vCH (Shunting) Inhibition

While CT1 handles local directional selectivity upstream, the macroscopic problem of figure-ground discrimination requires an inhibitory "pool cell" that integrates global background motion and subtracts it from the local EMD signals [135]. The connectome clearly identifies the vCH cell (and potentially LPi neurons like LPi15) as the primary candidate for this role.

In *Drosophila*, the massive dendritic trees of LPTCs like the vCH cell are mixed neurites. The arborization within layer 1 of the lobula plate contains *both postsynaptic* densities (receiving input) and *presynaptic T-bars (transmitting output)* in very close proximity. The connectome reveals a massive *reciprocal connectivity*: while arrays of T4 and T5 cells synapse onto vCH to provide front-to-back motion signals, the vCH cell synapses directly back onto the axon terminals of those same T4 cells right there in the lobula plate.

Hypothesis: We propose that T4 and T5 cells detect relative motion after exciting and being subsequently inhibited by the vCH cell, utilizing the exact same biophysical mechanism employed upstream by CT1: **shunting inhibition**.

During wide-field optic flow, massive T4/T5 excitation causes vCH to broadcast strong, widespread shunting inhibition back onto the T4 and T5 terminals, dynamically normalizing their outputs and suppressing the background motion signal. Because the CH cells are GABAergic, this dendro-dendritic inhibition likely opens chloride channels close to the resting potential of the cell, providing the divisive shunt required by the Poggio-Reichardt model. If, however, a small object moves independently of the background, the lack of global vCH activation allows the local EMD signals processing the object to escape the shunting inhibition.



25.5 The Output 'X Cell': LPTCs as Behavioral Integrators

In the canonical 1983 model, the background-subtracted signal from the EMDs is fed into an output "X cell" that summates (see figure) the progressive and regressive detectors and drives the behavioral motor response [135].

Because the vCH cell inhibits the T4 and T5 terminals directly, it effectively gates the motion signal *before* it is read by the other downstream tangential cells. We propose that the biological equivalent of the X cell is not a single, isolated neuron, but is rather a role dynamically assumed by the broader population of LPTCs.

Consequently, the remaining LPTCs —- such as the HS cells governing yaw, the VS cells governing pitch, and newly mapped translation-tuned LPTCs governing expansion and lift —- all receive a visual signal in which background motion has already been inhibited. These LPTCs thus serve as an array of specialized X cells, integrating the relative motion signals to trigger distinct, reflex-level motor commands in response to specific object trajectories.

25.6 Historical Context: Figure Detection (FD) Cells in the Blowfly

The functional targets of the figure-ground circuit were first characterized physiologically in the blowfly as Figure Detection (FD1 to FD4) cells. Modern comprehensive neuroanatomical surveys have identified 58 distinct LPTC types per hemisphere in *Drosophila* [174], some of which must correspond to the equivalents of these FD cells.

The receptive field of an FD cell is highly dynamic and is defined by relative motion rather than mere spatial location:

- **The Excitatory Center:** Defined by direct, convergent input from a localized patch of T4/T5 cells, directionally tuned to a specific axis.
- **The Inhibitory Surround:** A massive field covering the ipsilateral hemisphere, generated by the wide-field CH pool cells collecting global motion signals.

The defining hallmark of an FD cell is its size-tuning curve. As a moving object grows from 0° to 20° of visual angle, it fills the excitatory center, maximizing the firing rate. As the pattern expands further, mimicking global optic flow, it spills into the inhibitory surround, triggering the CH pool cells. The resulting inhibition heavily suppresses the FD cell, rendering it blind to the stationary world and maximally sensitive to independently moving figures. While the P-R model assumes FD cells pool the output of already-inhibited T4/T5 cells, a similar conjecture by Egelhaaf posits that FD cells are excited by T4/T5 cells and inhibited directly by CH cells. As FD cells are non-spiking graded-potential neurons, the functional distinction between shunting and hyperpolarizing inhibition at this stage is somewhat blurred, as both effectively reduce gain.

25.7 Biophysical Implementation of Shunting Inhibition

The true relative motion computation modeled by Reichardt, Poggio, and Hausen [135] relies not on simple linear subtraction, but on non-linear, divisive inhibition. As formally modeled by Torre and Poggio [153], for inhibition to be divisively shunting, it must be strategically located—either adjacent to the excitatory synapse on the same dendritic branch, or on the path to the spike initiation zone.

The physical wiring diagram provided by the FlyWire dataset confirms this spatial geometry exists. The inhibitory synapses from the vCH cell into the T4/T5 terminals are positioned in extremely close physical proximity to the cholinergic inputs from the T4/T5 cells onto the vCH cell. To date, this shunting mechanism has been biophysically demonstrated for the local motion detectors (e.g., within the T4 dendrites), utilizing synapses that execute a physical approximation of the divisive equation:

$$R = \frac{E}{1 + K \cdot I} \tag{25.1}$$

for the computation of motion direction.

Researchers construct multi-compartment biophysical models—often utilizing the NEURON simulation environment—by populating raw 3D dendritic meshes from EM datasets with specific ion channels to mathematically map these nonlinear conductances:

- **Dendritic Multiplication via Shunting:** Groschner et al. (2022) provide a definitive biophysical account of the nonlinear arithmetic required by the Hassenstein-Reichardt correlator. They demonstrated how the T4 dendrite achieves a multiplication-like nonlinearity arising from the physical coincidence of cholinergic excitation and glutamatergic shunting inhibition, mathematically mapping the shunting conductances onto the dendrite.
- **Connectome-Constrained Mechanistic Networks:** Lappalainen et al. (2024) developed a "Deep Mechanistic Network" that translates the optic lobe connectome into a differentiable simulation. Using deep learning to optimize unknown parameters, the resulting optimized mathematical weights—specifically the balance of feedforward excitation and shunting inhibition—closely matched physiological recordings.

- **Compartmental Modeling of Tangential Cells:** Elyada, Haag, and Borst (2013) demonstrated that large-field pool cells are not electrotonically compact. They modeled the spatial profile of lateral inhibition, mathematically demonstrating that wide-field dendritic inhibition acts as a shunt extending beyond the retinotopic spatial extent of the excitatory dendrites.

Ultimately, the 2022 framework established by Groschner et al. serves as a critical centerpiece for this literature, explicitly mapping the algorithmic requirement for nonlinear division directly onto the chloride channels of a connectome-reconstructed dendrite.

25.8 Predictions and the Missing Link

According to our hypothesis, the core computation of relative motion is carried by the membrane potential of the T4/T5 cells once they have been inhibited by the vCH pool cell. This yields specific testable predictions:

- If we record from the T4/T5 cells *after* the inhibition by the vCH, the sum of their outputs should mirror the classic figure-ground behavioral response.
- It is possible that two parallel mechanisms underlie relative motion detection in *Drosophila*. The first is the mechanism described here (predicted by the Poggio-Reichardt model) where the pool acts presynaptically on the T4/T5 axon terminals. The second (proposed by Egelhaaf) consists of forward inhibition acting directly on the output figure-detection cell by a large field neuron.

While we possess a rigorous theoretical model and the exact physical connectome that accommodates it, the two have not yet been formally bridged in the modern electrophysiological literature. Until an experimentalist achieves a recording of a connectome-identified T4 terminal or FD cell during a rigorously controlled moving-figure/moving-ground stimulus—demonstrating that the instantaneous membrane voltage dynamics strictly obey the predicted divisive suppression—the final loop in physically validating this classic relative motion model remains unclosed.

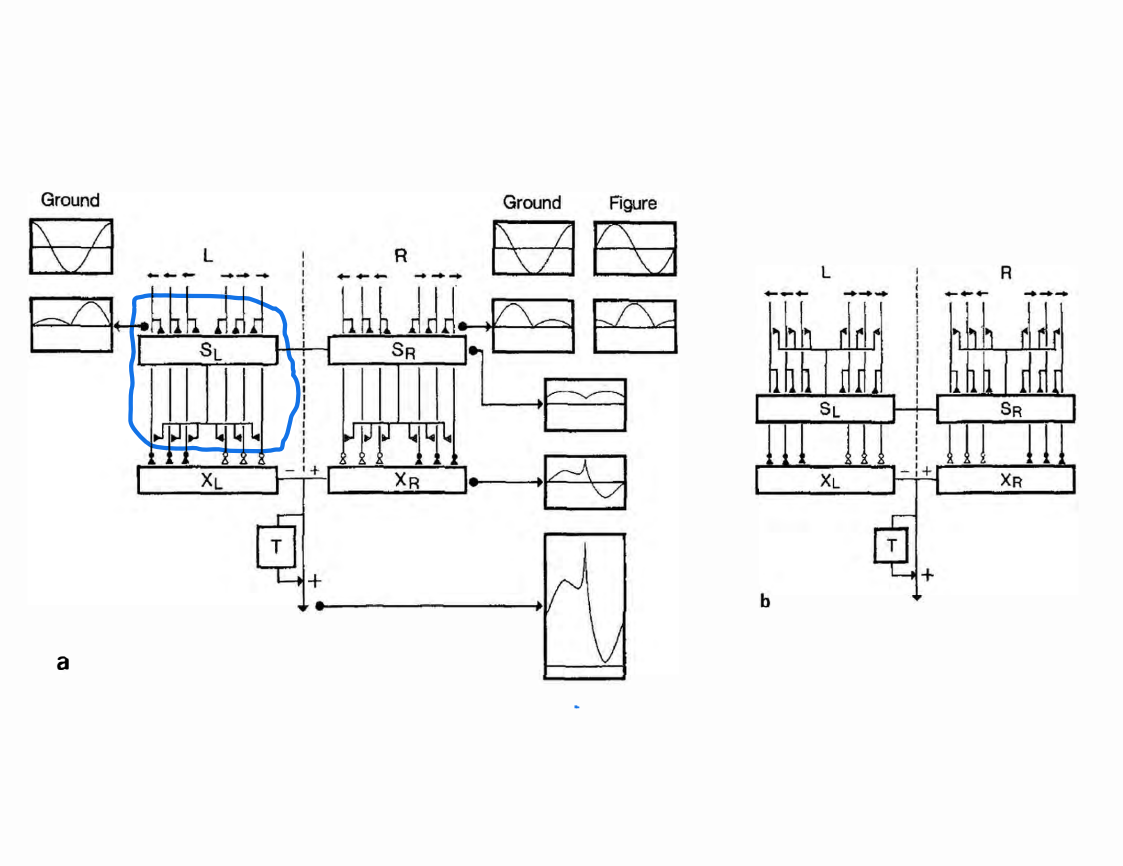


Figure 25.1: a The outline of the model circuitry for the right (R) and for the left (L) eye. Considering the right eye as example, two retinotopic arrays of elementary movement detectors, responding selectively to progressive and regressive motion, serve as input channels to the neural circuitry. The two arrays share the same field of view. For convenience they are drawn apart from each other. A pool neuron (SR) summates the detector outputs as well as the input from its contralateral homologue (SL). Its output is assumed to undergo a saturation effect (modeled by taking the square root of its overall excitation) and to shunt each elementary movement detector output via presynaptic inhibition. The synapses involved should therefore inhibit (opening ionic channels with an equilibrium battery close to the resting potential) the output terminal of each elementary detector channel. The output cell X summates the progressive (excitatory) and the regressive (inhibitory) detectors. Progressive channels have a higher amplification than regressive ones. possibly because of the different ionic batteries involved. This is the key mechanism underlying the computation of relative motion (and gain control): it is contained in the blue envelope. The synapses on the X R cell are assumed to operate with a nonlinear input-output characteristic, leading to postsynaptic signals that are approximately the square of the inputs. The motor output is controlled by the X-cells via a direct channel and a channel T computing the running average of the X-cells output. The insets in the figure contain the responses at the various levels to a bilateral figure-ground experiment. Responses at the input channels are split into two components, the ground component and the figure component. b The outline of another neural circuitry that can also explain the behavioural responses at the cellular level. The model is a recurrent shunting inhibition network. The important difference between the two models is that the first one (a) contains a shunting inhibition pathway acting on the channels after the pool cells (S) have received the detector signals - that is to say the pool acts forward. In the model presented here the pool operates via a recurrent pathway that acts on the individual detector channels before receiving signals from the individual channels. All other details as described in the legend of a.

CHAPTER 26

Frozen Interpreter and Cultural Evolution

Learning is usually identified with modification of internal parameters. A neural network learns by changing its weights; biological evolution changes inherited machinery across generations. Human adaptation, however, also proceeds through a second channel. Language, algorithms, recipes, mathematical notation, scientific theories, and institutions can change while the basic neural machinery of their users remains approximately fixed on the relevant timescale.

Frozen language models provide a computational laboratory for this distinction. Instructions, procedures, examples, memories, tools, and libraries can be copied, modified, selected, recombined, and transmitted without changing the model's weights. The central hypothesis is

> *parametric learning changes the interpreter;*
> *cultural learning changes the programs supplied to it.*

The analogy is functional, not literal: neural weights are not genes, and prompts are not human culture. A frozen network remains software running on physical hardware. What matters is where persistent adaptive information resides and how it is reused.

This chapter develops that distinction, incorporates the interpretation of weights as compiled computation and prompts as contextual programs, and separates three kinds of claim: elementary statistical and finite-resource results, a hypothesis about cumulative cross-task reuse, and the qualitative observations reported by the initial experiments. The mathematical results do not establish that a frozen LLM discovers reliable programs efficiently. They specify conditions under which selection, retention, and verification can support such a process.

26.1 Two Channels of Adaptation

Let

$$f_\theta(y \mid x, p)$$

denote the conditional behavior of a model with parameters θ, input x, and externally supplied program p. The word "program" is used broadly: it may contain instructions, demonstrations, retrieved facts, intermediate procedures, tool calls, memory, or verification rules. It does not imply that ordinary natural-language prompts have the exact semantics of a formal programming language.

Parametric adaptation changes the internal machinery:

$$\theta_T^\star \in \arg\min_\theta \widehat{L}_T(\theta, p_0). \tag{26.1}$$

External-program adaptation leaves the machinery fixed and changes the program:

$$p_T^\star \in \arg\min_{p\in\mathcal{P}} \widehat{L}_T(\theta_0, p), \qquad \theta_0 \text{ fixed}. \tag{26.2}$$

The displayed minimizers are comparison objects when the minima are attained, not claims that either search is easy. Approximate minimizers or a specified returned candidate are enough for the statistical arguments below.

The two channels are schematically

$$\underbrace{\theta}_{\text{persistent internal machinery}} \qquad \underbrace{p}_{\text{transmissible external state}}.$$

The distinction concerns the location of adaptation, not a universal ordering of timescales or costs. Weight updates can be small, and searching for a useful prompt can be expensive.

This is not merely ordinary in-context learning. In-context learning changes behavior within an available context. Cultural learning, in the sense studied here, additionally requires that useful external structures survive beyond one episode and be inherited by later trials, agents, or tasks. The individual model may lose access to an earlier context while an external population or library retains it. Merely retaining an object does not establish that it helps the next task.

What is fixed? A frozen interpreter includes more than the numerical vector θ. Its architecture, tokenizer, numerical evaluation, decoding convention, and permitted tool interfaces must be specified. A change to any of these is either part of the admitted external adaptation or a change to the experimental machinery. A supposedly frozen system cannot acquire free capability through an unreported change of model, tool, or evaluator. Fresh observations from an external environment are likewise an information source to be counted, not an effect of frozen weights alone.

For the mathematical statements, $f_{\theta_0}(x; p)$ below denotes the prediction produced under the specified execution convention. It may be a deterministic output or a predictive probability distribution. When outputs are sampled, loss means the corresponding expected loss unless Monte Carlo evaluation is stated explicitly. The same convention is used for training and population risk.

26.2 Prompt Selection Is a Learning Problem

Fix a frozen interpreter θ_0 and an admissible program family $\mathcal{P}$. For a task distribution $\mathcal{D}_T$ over $z = (x, y)$, define

$$\widehat{L}_T(p) = \frac{1}{n}\sum_{i=1}^{n} \ell\left(f_{\theta_0}(x_i; p), y_i\right), \tag{26.3}$$

$$L_T(p) = \mathbb{E}_{(x,y)\sim\mathcal{D}_T} \ell\left(f_{\theta_0}(x; p), y\right). \tag{26.4}$$

The interpreter and the program family induce the hypothesis class

$$\mathcal{H}_{\theta_0,\mathcal{P}} = \{x \mapsto f_{\theta_0}(x; p) : p \in \mathcal{P}\}. \tag{26.5}$$

Experience selects a member of this class. Thus

absence of weight change does not imply absence of learning.

It tells us where the selected state is stored. For these risks, a program may inspect the test input x and the admitted fixed resources, but not the unknown test label y. Any training examples retained for later retrieval are part of the deployed external state.

26.2.1 A simple generalization bound

The finite-class argument supplies a useful baseline. Its independence assumptions concern the interpreter and admitted class, not the selected program.

Proposition 46 (Generalization under program selection)**.** *Let $S = ((X_i, Y_i))_{i=1}^n$ be independent examples from $\mathcal{D}_T$, with $n \geq 1$. Fix the interpreter, prediction rule, and a nonempty finite family $\mathcal{P}$ independently of S, and assume losses lie in $[0, 1]$. For $0 < \delta < 1$, the following bound holds with probability at least $1 - \delta$ for an empirical minimizer*

$$\widehat{p} \in \arg\min_{p \in \mathcal{P}} \widehat{L}_T(p) :$$

$$L_T(\widehat{p}) \leq \widehat{L}_T(\widehat{p}) + \sqrt{\frac{\log |\mathcal{P}| + \log(2/\delta)}{2n}}. \tag{26.6}$$

In fact, the same event gives the displayed inequality for every $p \in \mathcal{P}$ simultaneously, so exact empirical minimization is not required for the one-sided certificate.

Proof. For fixed p, Hoeffding's inequality gives

$$\Pr(|L_T(p) - \widehat{L}_T(p)| > u) \leq 2e^{-2nu^2}.$$

Taking

$$u = \sqrt{\frac{\log |\mathcal{P}| + \log(2/\delta)}{2n}}$$

and applying a union bound over $\mathcal{P}$ gives a simultaneous absolute-deviation bound with failure probability at most δ. It therefore applies to the data-selected program $\widehat{p}$. □

This is the standard bounded-loss concentration argument [69], not a theorem about the optimization method used to find $\widehat{p}$. Throughout this chapter, $\log$ is the natural logarithm and binary description lengths are measured in bits.

If programs contain at most m tokens from a vocabulary of size $V \geq 2$, then

$$|\mathcal{P}| \leq \sum_{j=0}^{m} V^j \leq V^{m+1},$$

and the complexity allowance is

$$O\left(\sqrt{\frac{(m+1)\log V + \log(1/\delta)}{n}}\right).$$

This exposes a transmission bottleneck: limiting the description length limits the number of admitted behaviors, even when the interpreter is large. Short descriptions give a smaller allowance, not necessarily smaller risk. A short program can be wrong or poorly executed.

For an approximate empirical minimizer with optimization error ξ, the same event also gives the useful comparison

$$L_T(\widehat{p}) \le \min_{p\in\mathcal{P}} L_T(p) + \xi + 2u.$$

The right-hand side separates what the fixed interpreter and program class can achieve, what the search actually achieves, and what the data can support statistically. None of the three terms can be omitted.

26.2.2 The program is not always the complete description

A short prompt that says "use library entry 7" does not describe a predictor without the library, its retrieval rule, and the interpreter needed to execute that entry. If all of these are fixed independently of the present evaluation sample, a conditional analysis can treat them as background. If the library or decoder is selected on that same sample, its selection must be covered by the hypothesis class or charged for in a valid description-length argument.

For example, let c encode the complete variable deployed state, and let $b(c)$ be a prefix-code length for a decoding language fixed independently of the sample. The argument of Chapter 13 then gives

$$L_T(c) \le \widehat{L}_T(c) + \sqrt{\frac{b(c)\log 2 + \log(2/\delta)}{2n}}$$

simultaneously over the admitted codes. This is an application of the book's existing code bound, not a new capacity principle. Learned libraries, stored examples, numeric controls, and their required metadata cannot be hidden behind free identifiers.

Adaptive search within one fixed finite class is allowed by the uniform event. Substituting the number of candidates actually tried for $|\mathcal{P}|$ is not justified when those candidates were constructed adaptively from the same evaluation data. Separately, repeated model sampling adds evaluation noise; it does not multiply the number of independent task examples. Unbounded log loss requires another bound or an explicit bounded-loss convention.

26.3 Culture as Programs for a Frozen Interpreter

A useful abstraction is to view pretraining as constructing a repertoire of latent operations,

$$\Phi_{\theta_0} = \{\phi_1, \dots, \phi_K\}.$$

Retrieval, comparison, decomposition, arithmetic, translation, and verification are possible functional descriptions of members of this repertoire. They need not correspond to isolated neurons, heads, or independently addressable neural modules.

A cultural program need not introduce a new primitive. It may address and compose available operations. Schematically,

$$\pi_p(x) = \phi_{i_r} \circ \phi_{i_{r-1}} \circ \cdots \circ \phi_{i_1}(x). \tag{26.7}$$

This chain is a functional illustration. General programs can have branches, reusable subroutines, loops, and working memory; typing, addressing, and execution costs must then be specified.

The proposed division of labor is

pretraining constructs a computational repertoire; culture searches for reusable ways of invoking and composing it.

This predicts a boundary, but not an unconditional plateau whenever a named primitive is missing. The required function may still be implemented by composing other operations or by invoking an admitted external tool. A genuine representational obstruction occurs when no admissible external program achieves the target performance within the allowed resources. A different obstruction occurs when a suitable program exists but the interface or search procedure cannot reliably invoke it.

26.3.1 Weights as Compiled Computation, Prompts as Programs

A frozen large language model is software executed on a processor. The physical processor remains the hardware. At the algorithmic level, however, a pretrained model can be regarded as a learned interpreter with an acquired computational repertoire. With its parameters fixed, write

$$y = F_{\theta_0}(p, x)$$

for a deterministic execution, or use the corresponding conditional law for stochastic generation. The useful correspondence is

architecture and frozen weights	learned interpreter,
prompt and retrieved procedures	program and working data,
activations and cached state	ongoing computational state,
generated tokens	observable sequence and output.

The last row does not identify a generated explanation with a faithful trace of the model's hidden computation. Chapter 11 distinguishes those objects.

Weight learning as compilation. During pretraining or parameter-based post-training,

$$\theta_0 \longrightarrow \theta_1 \longrightarrow \cdots \longrightarrow \theta^\star.$$

Experience changes the persistent machinery available to later inference. Calling the result "compiled computation" emphasizes that repeated regularities and procedures may become inexpensive to invoke without resupplying their full derivation. It does not assert exact preservation of a source program, automatic compression, or replacement of every long computation by a single constant-cost forward pass.

Context as a program. With θ_0 fixed, prompts can specify tasks, examples, rules, goals, and intermediate procedures. Different prompts can induce different computations,

$$F_{\theta_0}(p_1, x) \neq F_{\theta_0}(p_2, x),$$

without changing the interpreter. This can be a useful adaptation route when the representation is already suitable. It is not a guarantee that a short prompt can express or reliably elicit every behavior obtainable by changing weights.

External libraries as persistent adaptive state. A library $\mathcal{L} = \{p_1, \ldots, p_M\}$ can retain procedures beyond one context. The adaptive system is therefore not just θ_0, but the interpreter together with the library, its selection and retrieval procedure, the current prompt, and the working data. Enlarging a library can enlarge the set of feasible programs when old entries remain accessible and can be ignored. It need not improve the behavior of the actual retrieval or search procedure; irrelevant entries may interfere.

Migration between external and internal representations. A useful procedure can first be discovered externally, then retained and reused, and later serve as a source of training examples for a new model:

explicit procedure
$\longrightarrow$ reusable external program
$\longrightarrow$ possible capability in future weights.

Distillation supplies a related model-compression precedent [64]; it does not establish that an arbitrary procedure can be compressed cheaply or without error. The CRM of Chapter 21 makes the distinction between naming a macro and implementing a faster primitive explicit.

When a later training phase changes θ_0, it ends the frozen-weight phase under study. External library growth during that phase and later parametric consolidation can cooperate, but they are not the same experimental manipulation. This gives a continuum of storage locations for learned information without erasing the difference between them.

26.4 Interpreter Complexity and Program Complexity

The interpreter/program distinction has a classical computational analogue. A universal interpreter is fixed while different computations are specified by different programs. For an ideal universal machine U,

$$U(p_f, x) = f(x) \tag{26.8}$$

for every computable f and an appropriate finite program p_f, on inputs where the computation is defined. Thus universality separates fixed transition machinery from variable symbolic programs.

A deployed LLM with fixed finite context, finite generation, and bounded external resources is not thereby an unbounded universal Turing machine. The analogy concerns computational organization: fixed machinery can support different computations when supplied with different external states. Chapter 3 develops the stronger universality claim under explicit memory and execution assumptions.

26.4.1 The model–program tradeoff

Let

$$K_C(f) = \min_{p:\, C(p,x)=f(x) \text{ for all } x} |p|$$

denote shortest exact program length relative to a specified interpreter C, with $K_C(f) = \infty$ when no such program exists. For partial functions, equality includes agreement of the domains of definition. This is a description measure, not a computable program-search method.

Two interpreters may express the same function class while giving useful functions quite different description costs. A richly pretrained system may make some task families much cheaper

to describe than a simpler interpreter does. The comparison is task dependent and also depends on execution resources; there is no universal ordering by model size.

To make the resource issue explicit, fix a prediction loss, a task population, a target risk λ_T, and a runtime cap τ. One can consider the refinement

$$K_C(T; \lambda_T, \tau) = \inf\{|p| : L_T(C, p) \le \lambda_T,\ \mathrm{Time}_C(p, x) \le \tau \text{ on the admitted inputs}\}.$$

The infimum is ∞ when the constraints are infeasible. A different choice, such as an average-runtime budget, defines a different measure. The purpose of this added definition is only to keep short description, fast execution, and accurate behavior separate.

Thus pretraining may act as compilation: frequently useful operations become available through shorter descriptions or more efficient execution. But it does not remove the cost of the interpreter itself. For a full system comparison one should account for

$$B(C) + B(p \mid C), \qquad \mathrm{Time}_C(p, x), \qquad \mathrm{SearchCost}(p),$$

with any shared library included in the appropriate description. Treating the same frozen C as fixed background in a downstream risk theorem is legitimate under its independence assumptions; ignoring $B(C)$ in a comparison of entire learning systems is a different claim.

The practical objective combines expressibility, efficiency, and reliability. A pretrained model is useful when important computations are economical and reliable to invoke, not merely when they have some formal description.

26.5 Evolution of External Programs

Let

$$P_g = \{p_{g,1}, \ldots, p_{g,N}\}$$

be a population of programs at cultural generation g. Define fitness

$$F_T(p) = -\widehat{L}_T(p) - \lambda C(p), \tag{26.9}$$

where $C(p)$ may penalize description length, inference cost, or memory. When library selection or retrieval also changes, the cost must include those changes rather than only the visible prompt string.

Programs are selected, mutated, recombined, and reevaluated. The frozen model may perform the proposal transformations. It thereby participates in changing the external population without changing its own weights. Nothing in this description requires natural selection to be the unique or best optimizer: enumeration, policy-based search, direct editing, and other procedures are legitimate baselines.

This empirical fitness is a selection criterion, not population performance. Repeated optimization of one evaluation set can select artifacts that exploit that set or its scoring convention. Proposal data, selection data, and final test data therefore have distinct roles. A cost penalty does not itself certify either validity or generalization.

26.5.1 Meta-programs

A stronger form of adaptation changes the procedure used to generate programs. Let a_g be a meta-program controlling proposal generation. Write the system state schematically as

$$\mathcal{S}_g = (\theta, p_g, a_g, \mathcal{L}_g), \qquad \theta \text{ fixed}, \tag{26.10}$$

where $\mathcal{L}_g$ is inherited. A proposal may be drawn according to

$$p' \sim q_\theta \left(p' \mid p_g, a_g, \mathcal{L}_g, H_g\right), \tag{26.11}$$

where H_g stores previous proposals and outcomes. If H_g, population members, evaluator state, or retrieval indexes are retained, they are part of the total external state and resource budget, either explicitly or within $\mathcal{L}_g$. The tuple in Equation (26.10) is not a license to omit them.

There are two possible achievements: finding better solutions and finding better procedures for finding solutions. A meta-program does not establish the second merely by describing itself as a better search strategy. Its advantage must be measured through future proposals and held-out tasks, including the cost of generating and executing it.

26.6 What Counts as Cumulative Culture?

Improving one task is not enough for the strong notion studied here. A population can optimize a prompt for a fixed task without retaining anything that helps later tasks. The proposed criterion is that transmitted structure reduces the cost of future adaptation.

Let $C_\varepsilon(T \mid \mathcal{L})$ denote the expected cost of a specified adaptation procedure, initialized with $\mathcal{L}$, to attain a fixed task-performance target within tolerance ε. Choose one cost unit—for example candidate evaluations, generated tokens, or arithmetic work—and keep it fixed in the comparison. The expectation also includes the procedure's randomness. Costs are ∞ when the specified success event is never reached. The common target and task distribution must not become easier for later generations.

We call an intergenerational change cumulative, in this operational sense, when

$$\mathbb{E}_{T\sim\rho} C_\varepsilon(T \mid \mathcal{L}_{g+1}) < \mathbb{E}_{T\sim\rho} C_\varepsilon(T \mid \mathcal{L}_g). \tag{26.12}$$

for a distribution ρ of future tasks. This is a criterion for a measured improvement, not a theorem requiring every successive library update to be beneficial. If the search algorithm changes with a_g, that change is included in the comparison and its contribution should be isolated from the contribution of stored solutions.

In a finite-budget experiment, also report the success probability by the budget and the cost distribution, including failures. Reporting only the mean cost of successful runs can make a method that rarely succeeds look economical. A truncated running time without its success rate has the same limitation.

The distinction is

$$\boxed{\text{task improvement} \neq \text{cumulative cultural improvement.}}$$

Reuse, retention, and recombination are useful experimental diagnostics. Recombination is not logically required by Equation (26.12); a transmitted procedure can help future learning without being recombined. Demonstrating all of these mechanisms would support the stronger cultural interpretation.

The cost of constructing and storing the inherited library remains a resource even when one measures only marginal new-task adaptation. A system can have high up-front cost and low future cost. The scientific question is whether the savings persist on held-out tasks and compensate for that investment over the relevant task distribution.

26.7 Intelligent Search

Even blind search can occasionally find a useful program. To establish an advantage for guided search, the comparison must show more useful proposals, cheaper evaluation, or another measured resource improvement. For a common reference risk $L_T^\star$, define the successful set

$$\mathcal{P}_\varepsilon(T) = \{p : L_T(p) \le L_T^\star + \varepsilon\}. \tag{26.13}$$

The reference is fixed across compared methods. It may be the infimum risk in a specified class, or another prescribed benchmark. If the set is empty, no proposal rule supported on that class can succeed.

For a proposal distribution q on countably encoded programs, put

$$\alpha_q(T, H) = \sum_{p \in \mathcal{P}_\varepsilon(T)} q(p \mid T, H). \tag{26.14}$$

Here the history and task information condition the proposal rule, while success is defined by population performance. The latter need not be known to the search procedure.

Proposition 47 (Proposal mass and search time)**.** *Fix a nonempty successful set. If proposals are independent draws from one fixed rule with successful mass $\alpha_q > 0$, and $\tau_\varepsilon \in \{1, 2, \dots\}$ is the index of the first successful proposal, then*

$$\mathbb{E}[\tau_\varepsilon] = \frac{1}{\alpha_q}. \tag{26.15}$$

More generally, let proposals depend on all previous outcomes. If, on every history before success, their conditional successful mass is at least $\alpha > 0$, then

$$\Pr(\tau_\varepsilon > t) \le (1 - \alpha)^t, \qquad \mathbb{E}[\tau_\varepsilon] \le \frac{1}{\alpha}.$$

Proof. For independent fixed-rule draws, the probability of t failures is $(1-\alpha_q)^t$. For adaptive draws, condition on the first $t-1$ failures and their complete history. The probability of another failure is at most $1 - \alpha$. Induction gives the survival bound. In both cases, summing $\mathbb{E}[\tau_\varepsilon] = \sum_{t=0}^{\infty} \Pr(\tau_\varepsilon > t)$ gives the conclusion. If $\alpha_q = 0$ in the fixed-rule setting, the hitting time is infinite almost surely. □

The equality is therefore a fixed-distribution result, not a formula to apply without qualification to an evolving population. Marginal success probabilities alone are insufficient for the adaptive bound: a procedure that samples one program and repeats it forever can have positive marginal success probability on every attempt but infinite expected hitting time.

This gives an operational comparison at matched per-candidate cost:

$$\alpha_{\text{guided}} > \alpha_{\text{baseline}}.$$

For constant total cost c_q per attempt, including proposal generation and evaluation, the fixed-rule expected cost is c_q/α_q. A larger successful mass need not win if its generation cost is much larger. The same distinction motivates the effective-control discussion of Chapter 17; neither short prompts nor a small controller guarantee useful search geometry.

Discovery is not recognition. The hitting time counts the first genuinely successful proposal. It does not say that the system recognizes that proposal, retains it, or can verify its population performance from a few tests. These are separate steps. Inheritance can improve proposal mass yet fail to improve the returned model if selection is unreliable. Conversely, a sound verifier cannot help a proposal distribution that never generates a valid solution.

26.8 Identification Is Not Execution

A small teaching set can uniquely specify a target within a restricted hypothesis class. For noiseless labels on a finite set of inputs S, let

$$V_{\mathcal{H}}(S, f) = \{h \in \mathcal{H} : h(x) = f(x) \text{ for every } x \in S\}.$$

If $V_{\mathcal{H}}(S, f) = \{f\}$, the examples identify f relative to $\mathcal{H}$. The restriction to that class is essential. It does not imply uniqueness among arbitrary programs or algorithms.

Three questions remain separate:

identification:	is the rule determined within the admitted class?
search:	can the learner find a useful description?
execution:	can the interpreter carry it out reliably?

A teaching set can specify a symbolic rule while a frozen LLM fails to infer it or execute it. Conversely, a predictor may perform well without recovering the particular symbolic rule that generated the examples. Chapter H makes predictive success primary and treats literal structural identification as a stronger objective.

This distinction suggests separate tests: score a formally executed candidate program, score the frozen model when given the correct program, and score the combined discovery-and-execution pipeline. An executable external program can be checked against its formal semantics. A verbal claim to implement that program is not the same object.

Repeated replay of a fixed observation record can aid optimization and retrieval without creating independent evidence about the unknown task, as in Chapter 19. New external measurements may add such evidence. The two should not be conflated in claims about sample economy.

26.9 Bounded Cultural Resources Imply a Ceiling

"Open-ended improvement" can refer to a performance ceiling, the number of strict improvements, computational expressivity, or resource cost. A finite-resource result separates these meanings.

Let R specify a total persistent external-description budget b_{ext}, a finite execution budget, and a finite-precision convention. Count every changing object on which deployed behavior can depend: prompts, meta-programs, library contents, stored histories, retrieval indexes, tool configurations, and any other retained adaptive state. For fixed-length binary encodings,

$$|\mathcal{C}_R| \le 2^{b_{\text{ext}}}.$$

Allowing all binary strings of length at most b_{ext} instead gives the still finite bound $2^{b_{\text{ext}}+1} - 1$.

A limit on the *number* of library entries alone is insufficient: one entry could contain an arbitrarily long program. The original shorthand $|p| \le m$, $|\mathcal{L}| \le M$, runtime $\le B$ therefore needs bounds on entry lengths and all other retained state before it implies finiteness. The operative condition is a finite complete external-state set $\mathcal{C}_R$.

Under a fixed interpreter and execution protocol, this induces

$$\mathcal{H}_{\theta,R} = \{h_c : c \in \mathcal{C}_R\}. \tag{26.16}$$

For randomized prediction, each h_c can be a fixed predictive law. Fresh draws from that law are not additional learned parameters. Arbitrarily varying mixing probabilities, external side information, or an unbounded generation counter affecting behavior would invalidate the finite-state premise unless counted in the encoding.

Proposition 48 (Bounded cultural ceiling)**.** *Assume that $\mathcal{C}_R$ is nonempty and finite, the interpreter and execution protocol are fixed, and $J_T(h)$ is a fixed real-valued population-performance functional. If the retained sequence is elitist in that true performance,*

$$J_T(h_{g+1}) \geq J_T(h_g),$$

then the number of strict improvements is at most

$$|J_T(\mathcal{H}_{\theta,R})| - 1 \leq |\mathcal{H}_{\theta,R}| - 1, \tag{26.17}$$

and performance cannot exceed

$$J_T^\star(\theta, R) = \max_{h \in \mathcal{H}_{\theta,R}} J_T(h). \tag{26.18}$$

Proof. The resource-bounded behavior class is finite.

Therefore the set of attainable performance values is finite.

Every strict improvement moves to a strictly larger value in this finite ordered set.

The number of such moves is finite, and the maximum is attained. □

The result bounds the number of strict increases, not the time taken to reach the best value. Evolution may stall below the ceiling, move forever among equal-performing states, or fail to propose an optimum. Without the monotonicity assumption, a finite class can alternate between two performance levels and contain infinitely many upward moves, each offset by a later decline.

Empirical elitism is also different from population elitism. Selecting the better of two noisy evaluation scores does not establish the premise of Proposition 48. A sufficient fresh-data acceptance rule is given below.

Runtime bounds are useful for defining feasible, terminating evaluation. For the cardinality argument itself, finite complete descriptions and a fixed mapping from descriptions to behaviors are the decisive assumptions. Repeated calls or unbounded search time do not enlarge a finite deployed behavior class unless they change some behavior-relevant resource.

There is a separate resolution bound. If $J_T \in [0, 1]$ and an elitist sequence counts only increases of size at least $\eta > 0$, there can be at most $\lfloor 1/\eta \rfloor$ such increases, even in an infinite class. This follows by summing the increases. Arbitrarily many strictly positive improvements must therefore be distinguished from arbitrarily many improvements of fixed practical size.

26.10 Frozen Weights Alone Do Not Imply a Finite Improvement Sequence

Without a uniform bound on the available external descriptions, freezing the interpreter does not force the set of behaviors or performance values to be finite. The following is an existence counterexample, not a theorem that an LLM or a particular search procedure improves indefinitely.

Use a fixed ideal universal interpreter and the constant target $f(x) = 1$ on $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, with

$$P(X = x) = 2^{-(x+1)}. \tag{26.19}$$

These probabilities sum to one. For each $n \geq 0$, let the program p_n compute

$$p_n(x) = \begin{cases} 1, & x \leq n, \\ 0, & x > n. \end{cases} \tag{26.20}$$

For 0–1 loss,

$$L(p_n) = P(X > n) = 2^{-(n+1)}, \tag{26.21}$$

so $L(p_{n+1}) < L(p_n)$ for every n. The same interpreter therefore admits an infinite strictly improving sequence. The subscript n is encoded in the variable program, not added for free as changing interpreter state. Directly encoding and comparing this threshold uses a description growing on the order of $\log(n+2)$ bits, up to standard delimiter overhead, and the relevant execution costs depend on the input encoding as well.

The target is deliberately simple: a constant-one program already attains zero error. Thus this example does *not* show that an optimum is unreachable, that infinitely many updates are necessary, or that the sequence is an efficient learning strategy. It shows exactly that a fixed interpreter does not prohibit an infinite chain of progressively better externally specified predictors.

Performance remains bounded and the improvement size tends to zero:

$$L(p_n) - L(p_{n+1}) = 2^{-(n+2)}.$$

The sequence also says nothing about the cost of statistically certifying successively smaller gains. Ideal expressibility, constructive proposal generation, and reliable acceptance must remain separate.

The conclusion is therefore

> frozen weights alone do not bound the number of strict improvements;
> a finite complete behavior class does, under true elitism.

This does not establish that every frozen interpreter benefits from larger external resources. Some interpreters ignore their programs or cannot execute the necessary distinctions.

26.11 Experimental Test

The preceding theory motivates the following experiment. Freeze an open-weight model, record its exact checkpoint and inference convention, and verify that the model parameters do not change. Allow the external system to modify p_g, a_g, and $\mathcal{L}_g$ across related tasks, while logging all retained history and tool access.

Measure separately:

1. **Task improvement:** does held-out performance improve across cultural generations?
2. **Proposal quality:** does guided search produce a higher rate of useful candidates than matched random search, after accounting for proposal cost?
3. **Inheritance:** do descendants initialized with transmitted artifacts adapt faster than fresh populations?
4. **Recombination:** does combining independently useful artifacts help beyond retrieving the best single artifact?
5. **Transfer:** do programs from earlier tasks reduce adaptation cost on new related tasks?
6. **Cumulative culture:** does $C_\varepsilon(T \mid \mathcal{L}_g)$ improve over generations at a common target, with failure rates and costs reported?

Training and evaluation must distinguish three independent experimental units: examples within a task, task draws from the task family, and independent cultural lineages or search seeds. Many generations on one fixed task do not become many independent tasks. Many sampled outputs on one input do not become many independent input examples.

Use task-training examples for proposal construction, a declared selection protocol, and untouched final evaluation examples. Hold future tasks out of artifact construction. Record total proposal and evaluation calls, input and generated tokens, retrieval and execution costs, and library storage. Separate the initial library-building cost from later marginal adaptation cost, while reporting both.

Matched controls should include fresh search without inheritance, a persistent library with no new proposals, irrelevant or scrambled libraries matched in length, and replay of old examples without an abstracted procedure. Keep the task family and available primitive tools fixed. Success after adding an exact external solver is a different result from discovering a better program for unchanged machinery.

These are proposed tests. They are not additional experiments performed for this chapter revision.

26.12 What the Initial Experiments Show

The initial exploratory account reports frozen-language-model experiments in which external prompts were selected, mutated, and sometimes supplied with previously successful artifacts. We retain those observations at the qualitative level reported. The supplied account does not provide exact model checkpoint identifiers, task generators, prompts, raw scores, run counts, seeds, uncertainty estimates, or a complete separation of selection and test data. Those details are needed for a reproducible quantitative report and cannot be reconstructed from the summary alone.

The minimal observation reported is that a frozen model can realize substantially different behaviors under externally selected programs. That is a prerequisite for the proposed cultural-interpreter mechanism, not a demonstration of cumulative culture. The stronger claims were not supported by the reported pilots.

26.12.1 Search was not reliably intelligent

In the finite-menu pilot, the account reports that informed LLM-guided search did not consistently outperform random search. Performance was task dependent: semantic guidance sometimes helped and sometimes hurt. The justified conclusion is specific to the tested protocol. Neither a universal failure of guided search nor a reliable search advantage follows from these observations.

In the language of the proposal-mass analysis, the pilots did not establish a dependable increase in useful proposal mass per unit cost. Separate records of candidate quality and interpreter execution would be needed to localize the failure more precisely.

26.12.2 Inheritance was not reliably cumulative

Previously useful programs reportedly helped later search in some cases and biased it toward unhelpful regions in others. In the subsequent free-generation experiments, inherited libraries did not produce a reliable reduction in the cost of discovering successful programs. Thus the strong inequality in Equation (26.12) was not established.

Retaining more artifacts is not itself the criterion. A library can grow while retrieval worsens or while its entries remain specific to the old tasks. The relevant comparison is with controlled new-task search, not with library size alone.

26.12.3 Model size did not solve the problem

The account reports that increasing the frozen model from approximately 1.5 billion to 3 billion parameters did not give a consistent advantage under the tested protocol. This is evidence against a simple size-only explanation of those pilots, not a scaling law or a prediction for larger models.

Reported difficulties included proposal failure, poorly generalizing library entries, wording-sensitive execution, and short search horizons. Without the missing quantitative record, their relative contributions remain undetermined.

26.12.4 Identification and execution separated sharply

In the symbolic controls, the account reports that small teaching sets could uniquely identify a rule within a restricted hypothesis class. Supplying comparable information to the neural interpreter did not ensure correct execution. These observations motivate the distinction

$$\boxed{\text{identification} \neq \text{search} \neq \text{execution.}}$$

They do not establish that every failure occurred at one particular step. A quantitative replication should separately measure each component and then the complete pipeline.

26.13 What Would Constitute Strong Evidence?

A convincing result would go beyond beating one hand-written prompt. It would show a fixed verified interpreter, an externally evolving population, retained procedures that transfer, and decreasing future learning cost under matched resources. Cross-task recombination would provide additional evidence for the stronger cultural interpretation.

The library should be accumulated by the declared procedure rather than supplied with the hidden task solutions. Future tasks must be held out, and relevant baselines should receive matched information, context, memory, and evaluation budgets. Program validity, prediction accuracy, and adaptation cost should be scored separately. A successful frozen system need not reconstruct a teacher's particular internal program.

26.13.1 A sufficient rule for statistically sound retention

The ceiling proposition assumes population elitism; noisy selection does not supply it automatically. The following elementary extension of the same concentration argument makes one possible acceptance protocol explicit. It is a sufficient theoretical procedure, not a claim about how the initial pilots were run.

Proposition 49 (Fresh-data acceptance of improving programs)**.** *Fix $0 < \delta < 1$, a task distribution, and a bounded loss in $[0, 1]$. At comparison $g \geq 1$, condition on all prior observations and fix an incumbent predictor h_g and a candidate $\widetilde{h}_g$. Draw $n_g \geq 1$ fresh independent evaluation examples, independently of that past, and put*

$$\widehat{\Delta}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} [\ell(h_g(X_i), Y_i) - \ell(\widetilde{h}_g(X_i), Y_i)].$$

Choose $\delta_g > 0$ with $\sum_{g\geq 1} \delta_g \leq \delta$, and accept only if

$$\widehat{\Delta}_g > \sqrt{\frac{2\log(1/\delta_g)}{n_g}} + \eta_g, \qquad \eta_g \geq 0.$$

Then, with probability at least $1-\delta$, every accepted candidate satisfies

$$L_T(h_g) - L_T(\widetilde{h}_g) > \eta_g.$$

In particular, when rejection leaves the incumbent unchanged, the retained sequence is population-elitist on that event.

Proof. Conditional on the past and the two predictors, each loss difference is in $[-1,1]$, and its mean is $\Delta_g = L_T(h_g) - L_T(\widetilde{h}_g)$. Hoeffding's one-sided inequality therefore gives

$$\Pr(\widehat{\Delta}_g - \Delta_g > u \mid \text{past}) \leq e^{-n_g u^2/2}.$$

For $u = \sqrt{2\log(1/\delta_g)/n_g}$, acceptance of a candidate with $\Delta_g \leq \eta_g$ is contained in an event of conditional probability at most δ_g. Taking expectations and a union bound over all comparisons proves the claim. □

For example, $\delta_g = 6\delta/(\pi^2 g^2)$ gives a summable budget. The comparison count includes rejected candidates as well as accepted ones. The sample size and two predictors may be selected from the past, but not retuned using the same fresh evaluation batch before the stated test. For sampled model outputs, independently sampled evaluation trials with correct expected losses can be included in each example; the bounded loss-difference proof is unchanged.

This procedure can reject genuinely better programs, and its data cost may become large for very small improvements. It does not ensure that a good candidate is proposed or prove decreasing cost on future tasks. Reusing one adaptive validation set requires a different uniform or adaptive-data analysis. Formal verification provides another route when an exact specification and sound proof checker are available, with its own computational costs and scope.

The decisive empirical quantity remains future-task adaptation cost as a function of inherited artifacts, accompanied by retained accuracy and reliable evaluation. Monotone scores on the selection set are not enough.

26.14 Limits

26.14.1 No universal search advantage

A general search advantage cannot be inferred without a task assumption. Classical no-free-lunch results make this precise for specified finite black-box optimization models when averaging over all objective functions [168]. They do not say that useful improvements on a structured task distribution are impossible.

Cultural search is interesting precisely because tasks may share regularities. Any demonstrated advantage must be relative to a stated $T \sim \rho$, information access, and cost model. An adversarially changed or unrelated task family can remove the advantage or make old procedures misleading. Library retention and selective forgetting therefore remain part of the algorithm.

26.14.2 No universal semantic verifier

For unrestricted Turing programs, Rice's theorem rules out an algorithm that always halts and decides a nontrivial extensional property of the partial computable function represented by an arbitrary program [136]. This does not apply to every syntactic check or every restricted finite program class.

Exact certification can be possible for a bounded finite domain with terminating execution, a restricted decidable language, or a stated specification with a sound proof checker. These restrictions can still leave verification expensive. Proof-carrying output gives a way to accept some certified solutions; it does not provide a complete method that always finds a proof or decides the property of every candidate. Statistical tests instead certify distributional statements at a stated confidence, not universal semantic equivalence.

26.14.3 Universality does not imply practicality

An ideal fixed universal interpreter can express arbitrary computable procedures through suitable external programs. That says nothing by itself about description length, runtime, memory, invocation reliability, or search cost. For a real frozen model, even the existence of an accurate admissible program is task and resource dependent.

A growing external library is not automatically accessible through a fixed context window. Retrieval, staging, repeated calls, and working memory can make parts of it available, but these are operations with costs and possible errors. In particular, arbitrary external computation must not be silently attributed to the neural interpreter.

26.15 Relation to the Framework of This Book

Sparse compositionality concerns an economical organization of a computation. The present chapter asks where its description is stored and how that description is discovered, transmitted, and reused. It may reside partly in parameters, partly in an external program, and partly in a reusable library.

The bits framework of Chapter 13 distinguishes the coding language from the selected code. Here a fixed interpreter supplies part of that language. Cultural adaptation selects externally encoded objects within it. A favorable description bound still needs good fit; a short external instruction does not make the required library or decoder free.

The predictive logic of Chapter H applies to the whole returned system. A sufficiently accurate, feasibly evaluated model-plus- program supplies its own computational witness. No additional theorem recovering the teacher's literal coordinates is required for that predictive conclusion. Explaining efficient program discovery remains a separate learning problem.

The trace perspective also remains useful. A supplied executable procedure or verified intermediate trace can expose work that would otherwise have to be inferred. A generated natural-language explanation need not be such a trace. Similarly, the organizational benefit of an inherited library can make existing evidence easier to use without creating new independent task observations.

The CRM of Chapter 21 adds a further possibility: repeated external computations may be

promoted into reusable macros or learned primitives. The potential cycle is

internal computational repertoire
↓
external programs and compositions
↓
retained cross-task procedures
↓
possible compilation into future primitives.

Naming a macro changes description; a genuinely faster implementation changes elementary execution cost; training a future model changes the interpreter. These transformations can support each other without being identified. During a strictly frozen phase, the evolving object is the external state and its use.

The positive research target is therefore not simply more stored text. It is an inherited computational vocabulary that improves the accuracy, reliability, or cost of solving new tasks under an explicit resource model.

26.16 Conclusion

The chapter's proposal is

pretraining constructs the interpreter;
culture changes the programs supplied to it.

A frozen model is not necessarily a frozen adaptive system. Instructions, memories, procedures, meta-programs, and libraries can change while θ remains fixed. Weights as compiled computation and prompts as programs provide a useful interpretation of this difference, not an assertion that every prompt has reliable formal semantics.

The mathematical conclusions have distinct scopes. A fixed finite program class gives a uniform bounded-loss generalization bound. A positive conditional proposal mass gives a search-time bound, but not automatic recognition of success. A finite complete external-state class permits only finitely many strictly increasing population-performance values under elitism. Without that finite-class restriction, a fixed interpreter can admit an infinite improving sequence, even though performance stays bounded and its increments vanish. Fresh-data tests provide one sufficient way to make accepted improvements statistically reliable.

None of these statements proves cumulative culture in a frozen LLM. The stronger hypothesis is that a system can discover, preserve, recombine, and transmit useful procedures so that new related tasks become cheaper to learn. The initial qualitative account does not yet establish that result. Its reported difficulties motivate better tests of proposal quality, execution, verification, transfer, and retention.

The central lesson is nevertheless clear:

learning need not be stored entirely inside the learner.

An intelligent system may adapt both by changing its machinery and by improving the transmissible programs through which that machinery is used. The open empirical question is when the latter process produces reliable, cumulative gains rather than a growing collection of fragile task-specific instructions.

CHAPTER 27

Open Problems: Representation, Recovery, and Selection

The book distinguishes the existence of an economical computation from its discovery through observations, and predictive success from recovery of a particular generating mechanism. This chapter organizes the remaining questions into five connected research directions: representation, useful coordinates, quantitative signal, learning dynamics, and regularized selection.

The first two sharpen the representation questions arising from C1. The third concerns quantitative C2 relative to an observation model and an accessible probe family. The fourth asks when learning constructs a compositional staircase. The fifth separates an established norm-based generalization mechanism from the harder questions of efficient optimization and cross-task reuse.

The coordinate and signal questions are developed using two complementary perspectives: simple readouts of pretrained representations, and Fourier–Hermite descriptions of progressive feature discovery. Together they provide a mathematical and experimental motivation for the coordinate and recovery problems. Adaptation through external programs is developed in Chapter 26 and is not repeated here.

These are proposed research formulations, not claims that the literature contains no relevant results. Regular low-arity approximation and stability are established for important Mhaskar–Poggio classes; staircase learning has rigorous restricted realizations; and norm-based bounds already explain why a well-fitting predictor in an adequately controlled class can generalize. The objective is to identify what remains beyond those results and to formulate tractable first steps.

The predictive objective comes first. Chapter H makes the main implication conditional but simple: a returned predictor with good population accuracy and a feasible finite-precision evaluator already supplies a compositional computation of its own approximation. An additional match to hidden teacher variables is not needed for that conclusion. The difficult premise is that a specified training procedure achieves good fit, informative complexity control, and feasible cost together. Staircase discovery is one possible mechanism for establishing that premise, and structural recovery is a stronger objective. Neither is imposed as a universal requirement on every successful learner.

The five problems are ordered by their logical role in the framework, not as a compulsory chronological sequence through which every training trajectory must pass. Trace exposure C3 can help at several points; it supplies additional intermediate information rather than making local coverage, optimization, or generalization automatic.

Conventions. We write d for input dimension, n for sample size, s for the number of constituent nodes, L for computational depth, k for maximum local arity, and $r \geq 1$ for an integer smoothness order. The symbol ν denotes the latent or restricted control dimension when locally specified, so it is not confused with smoothness. The symbol q is used for a polynomial degree in the anti-concentration argument. Unless otherwise specified, approximation is uniform on a compact domain, and statistical statements concern independent examples from a fixed joint distribution P, with input marginal P_X. An $L^2(\mathsf{P})$ norm of a function of X means its norm under P_X. Gaussian sections explicitly use L^2 rather than uniform approximation on an unbounded domain. Constants and resource bounds must be uniform in whichever parameters are allowed to grow.

The numbered bounds below have different statuses: definitions, established estimates under stated hypotheses, elementary deductions, and desired theorem targets. An inequality introduced as a research target is not asserted to hold for every compositionally sparse function.

27.1 Problem 1: Characterize Economical Compositional Representations

Origin in the book. Chapters 1–3 distinguish computational expressivity from learnability. Chapter 4 gives two complementary starting points: efficient digital computation has bounded-fan-in circuit realizations, and regular low-arity compositions admit favorable deep approximation. The remaining representation question is to recognize economical compositional structure without already being given its decomposition. Chapter 28, on sparse digital computation and smooth lifts, develops the distinct directions from regular real constituents to Boolean approximations and from Boolean gates to smooth extensions. Those constructions, and the counterexamples to an unrestricted continuum-limit claim, are starting points here; this problem is not a request to reprove them.

27.1.1 What is already established

Consider a scalar constituent $g : [-1,1]^m \to [-1,1]$, $m \leq k$, in a normalized smooth class satisfying

$$\|g\|_\infty + \sum_{1 \leq |\alpha| \leq r} \|D^\alpha g\|_\infty \leq 1. \tag{27.1}$$

This is the bounded-derivative setting relevant to the Mhaskar–Poggio approximation argument, rather than an arbitrary $W^{r,p}$ assumption [109].

In particular,

$$\sum_{j=1}^{m} \|\partial_j g\|_\infty \leq 1 \quad \Longrightarrow \quad |g(u) - g(v)| \leq \|u - v\|_\infty.$$

For a layered DAG with such constituents, local uniform errors of at most δ give

$$e_\ell \leq e_{\ell-1} + \delta, \qquad e_L \leq L\delta,$$

provided all evaluated intermediate arguments remain in the domains of the local guarantees.

Thus stability is not an unanswered question for this class. More general good-propagation-of-errors results extend the approximation argument to suitable DAGs [105].

For constituent classes and approximators satisfying the usual local rate, the resulting complexity is schematically

$$N(\varepsilon) \lesssim s \left(\frac{A_G}{\varepsilon} \right)^{k/r},$$

up to the factors specific to the approximation theorem. This is an approximation-resource statement for an admitted realization. It does not specify the sample size or training cost needed to discover it.

Here A_G controls propagation of local errors, and $A_G \le L$ in the normalized nonexpansive setting above.

The unresolved issue is not to reprove these statements, but to characterize targets that admit an economical representation of this kind.

The uniform Boolean-circuit characterization is already available; the goal here is an analytic and structural criterion that says more about approximation, interfaces, and constituent complexity.

27.1.2 A formal representation problem

Fix admissible constituent classes $\mathcal{G}_{r,m}(B)$, including specified domains, ranges, and derivative bounds.

Define

$$E_{s,L}^{k,r,B}(f) = \inf_{\substack{|G|\le s,\ \operatorname{depth}(G)\le L \\ |\operatorname{pa}(v)|\le k \\ g_v\in\mathcal{G}_{r,|\operatorname{pa}(v)|}(B)}} \|f-\Phi_G\|_\infty. \tag{27.2}$$

The graph and the constituents are both variable in this infimum.

For a family f_d, ask when

$$E_{s,L}^{k,r,B}(f_d) \le \varepsilon$$

can be achieved with bounded k and controlled s, L, B as d grows.

Node count alone is insufficient if an arbitrarily complicated local function can be inserted at no cost.

For effective representations, fix a decoding language for finite descriptions of the graph, the local approximators, their numerical parameters, and any shared library.

For an encoded realization $\mathcal{R}$, let

$$B(\mathcal{R}) = \text{total description length},$$

and

$$T_{\mathcal{R}}(p) = \text{worst-case bit cost of evaluation to } 2^{-p}.$$

A resource-sensitive refinement is

$$\mathcal{C}_k(f,\varepsilon;\tau) = \inf_{\mathcal{R}} \left\{ B(\mathcal{R}) : \begin{array}{l} \|f-f_{\mathcal{R}}\|_\infty \le \varepsilon, \\ \operatorname{arity}(\mathcal{R}) \le k, \\ T_{\mathcal{R}}(\lceil \log_2(1/\varepsilon)\rceil + 1) \le \tau \end{array} \right\}. \tag{27.3}$$

The decoder is fixed independently of f; it cannot hide an arbitrary function or a noncomputable constant inside a supposedly free primitive.

Open problem. Find structural sufficient conditions, and informative partial converses, for small $E_{s,L}^{k,r,B}(f)$ or small $\mathcal{C}_k(f,\varepsilon;\tau)$ that do not merely restate the existence of a suitable graph.

Two resource regimes must remain separate:

$$\operatorname{poly}(d,\varepsilon^{-1}) \qquad \text{and} \qquad \operatorname{poly}(d,\log(1/\varepsilon)).$$

A Sobolev approximation rate need not give polynomial-time bit computability.

Computability also requires effective construction of the approximants and an effective convergence rate. The existence of a separate code at every accuracy is not, by itself, a uniform algorithm.

Conversely, efficient digital evaluation supplies its own route through uniform bounded-fan-in circuits and, with appropriate encodings, discrete state restoration.

Nor should amplification A_G be confused with the number of precision bits: a local tolerance $\delta = \varepsilon/A_G$ requires approximately

$$p_{\mathrm{local}} = \log_2(1/\varepsilon) + \log_2 A_G$$

bits.

Exponential amplification in depth can therefore require only linearly many additional bits, even though a generic approximation scheme costing $\delta^{-k/r}$ may then become expensive.

27.1.3 Possible approaches

Inverse approximation theorems. For a specified depth schedule $L(s)$, study the approximation class

$$\mathcal{A}_\alpha = \left\{ f : \sup_{s \geq 1} s^\alpha E^{k,r,B}_{s,L(s)}(f) < \infty \right\}.$$

The goal is to relate membership to recognizable properties of f: hierarchical interaction structure, bottleneck dimensions, symmetries, or recursive separability.

Direct approximation theorems provide one direction; useful converse statements would make the class intrinsic.

Differential signatures of bottlenecks. Suppose, on a product domain,

$$f(x_A, x_B) = H(g(x_A), h(x_B)),$$

where g and h are scalar and all functions are twice differentiable.

Direct differentiation gives

$$D_A D_B f = H_{12}(g(x_A), h(x_B)) \nabla g(x_A) \nabla h(x_B)^\top,$$

so this mixed-derivative matrix has rank at most one.

Such identities provide necessary signatures of particular hierarchical bottlenecks.

A promising restricted problem is to identify additional nondegeneracy and integrability conditions under which a hierarchy of these signatures is also sufficient for a compositional factorization.

Low rank at one point is not enough.

Separate reuse from execution size. Compare s unrelated constituents with s calls to a small library of $K \ll s$ functions.

A description should pay for each distinct function once, while execution cost still counts every call.

This permits a theory of compositional approximation that recognizes both small graphs and short programs generating larger graphs.

A useful first result. A characterization for a restricted family—for example, binary trees of nondegenerate smooth scalar bottlenecks—would already be valuable.

A universal procedure for finding the shortest program is not required.

27.2 Problem 2: Which Coordinates Expose Compositional Structure?

Origin in the book. Chapter 4 distinguishes the existence of a sparse realization from sparse weights and unique latent coordinates.

Chapters 6 and H distinguish the existence of useful coordinates from their discovery and from the selection of a good predictor. Before studying a particular learning mechanism, we need to say what makes a representation useful. Simple downstream readouts provide a concrete motivation; economical state-transition coordinates provide a complementary structural target.

27.2.1 Useful coordinates and post-training

A motivating observation is that some task-relevant variables can be predicted from a pretrained model's intermediate representations by simple readouts. Linear probes provide one way to examine this property [5, 63]. The observation must be specified relative to a model, layer, task, input distribution, and probe class. It is not a claim that every useful downstream target becomes linear, or that probe performance alone explains the network's behavior.

Let $h_\ell(x) \in \mathbb{R}^{d_\ell}$ denote the representation of input x at layer ℓ. For a downstream target y, consider

$$\widehat{y} = W_\ell h_\ell(x) + b_\ell. \tag{27.4}$$

The probe parameters are fitted while the representation is held fixed. The important claim is not that a layer depends linearly on the target. Rather,

> a target complicated in input space may admit a simple,
> sometimes nearly linear, readout in learned representation space.

Here and below, a change of "coordinates" can mean a noninvertible representation map. It need not be a change of basis or a diffeomorphism.

A sequence of transformations

$$x \longrightarrow h_1(x) \longrightarrow \cdots \longrightarrow h_L(x)$$

may turn a relation $y = f(x)$ into $y \approx w^\top h_\ell(x) + b$ at an intermediate layer. This is a possible representational achievement of pretraining, not a description of its objective in every model. A frozen backbone then supplies coordinates to a downstream learner. Discovering those coordinates during pretraining and fitting a readout afterward are different learning problems.

This viewpoint is closely related to the staircase picture: learning an intermediate function can make the remaining dependency simpler to express. But linear decodability after training does not establish that training followed a staircase, nor that the decoded variable coincides with a generating latent state. The distinction between predictive and structural recovery in Chapter H applies here as well.

For a family of tasks $\mathcal{T}$, a useful formal question is whether one representation h permits

$$\inf_{w,b} \|f_T - (w^\top h + b)\|_{L^2(P_{T,X})} \le \varepsilon_T, \qquad T \in \mathcal{T},$$

with controlled readout norms, sample requirements, and evaluation cost. This is a formulation of the question, not a theorem supplied by the existence of a deep representation. If h was learned on the same sample used to certify a readout, its selection must also be accounted for. Chapter 13 provides the corresponding distinction between fixed background and a learned dictionary.

Selecting an intervention layer. If useful information is already represented, post-training may aim to select, amplify, combine, or route it rather than construct it anew. One possible criterion is

$$\ell^\star = \arg\min_{\ell} \left[\mathcal{L}_{\mathrm{probe}}(\ell) + \lambda C_{\mathrm{probe}}(\ell)\right]. \tag{27.5}$$

Here $\mathcal{L}_{\mathrm{probe}}$ measures predictive error and C_{probe} measures the complexity of the admitted probe. For an empirical implementation, fit probes on training data, select the layer and hyperparameters using validation data, and retain an untouched assessment set or a valid uniform analysis. Comparing probes requires comparable regularization and controls; high-capacity probes can learn the task or memorize rather than merely reveal the representation [63].

The criterion identifies a candidate layer, not a guarantee that a small intervention there controls the desired behavior. Possible interventions include

$$h'_{\ell^\star} = h_{\ell^\star} + Uv, \qquad \text{or} \qquad h'_{\ell^\star} = h_{\ell^\star} + ABh_{\ell^\star}, \quad \mathrm{rank}(AB) \ll d_{\ell^\star}.$$

The shapes and adjustable variables must be explicit. For example, $U \in \mathbb{R}^{d_{\ell^\star} \times \nu}$ may be fixed while $v \in \mathbb{R}^\nu$ is learned; alternatively both may be trained. The control cannot use an unavailable test target. If both low-rank factors $A \in \mathbb{R}^{d_{\ell^\star} \times \nu}$ and $B \in \mathbb{R}^{\nu \times d_{\ell^\star}}$ are trained, their parameter count is $2d_{\ell^\star}\nu$, not ν. Chapter 17 distinguishes adapter rank from search-coordinate dimension and from effective curvature rank.

The resulting hypothesis is therefore conditional:

a useful, simply decodable representation
$+$ an effective and well-conditioned intervention interface
$\Longrightarrow$ a plausible economical post-training problem.

Whether pretraining has already performed most of the required work is a question to test, not a consequence of probe accuracy alone.

Linear decodability is not causal use

Suppose an external linear probe predicts y accurately from h_ℓ. This establishes a particular predictive relationship on the evaluated distribution. It does not establish that the remainder of the model uses that relationship:

linear decodability $\neq$ causal utilization.

Write the downstream computation as $F(x) = D_\ell(h_\ell(x))$. A variable may be present in h_ℓ while being ignored by D_ℓ.

An elementary counterexample. Let $X = (Y, Z)$, with independent, nonconstant scalar coordinates, and let $h(X) = (Y, Z)$. The probe $w = (1, 0)$ recovers Y exactly. But the fixed suffix $D(h_1, h_2) = h_2$ is unchanged by every intervention $h \mapsto h + \alpha(1, 0)$. Thus exact decoding of Y and

perturbing its probe direction do not by themselves change the output. Replacing the readout by $w^\top h$ is a different operation from steering the existing suffix. This is an elementary illustration of the distinction, not an empirical claim about a particular model.

When D_ℓ is differentiable, the first-order response to a restricted intervention is

$$D_\ell(h + U\Delta v) - D_\ell(h) = J_{D_\ell}(h)U\Delta v + o(\|\Delta v\|).$$

A desired local output change must lie in the image of $J_{D_\ell}(h)U$ to be achieved to first order in those controls. Small singular values can make the required change expensive. This local calculation does not establish global controllability, and a zero first derivative need not exclude higher-order effects.

Probing should therefore be complemented by carefully specified interventions. A useful experimental sequence is: find a simple probe, identify a candidate low-dimensional subspace, intervene, and measure the effect on behavior. Controls should separate the intended effect from generic damage, distribution shift, and changes to unrelated tasks. Even a successful intervention is evidence about the specified manipulation, not automatic identification of a unique semantic or causal variable.

The coordinate question motivated by probes

The working hypothesis is

pretraining may construct useful coordinates;
post-training can learn how to exploit them.

It suggests measuring not only final-task accuracy but the complexity with which useful variables can be read throughout the hierarchy, the cost of learning a new readout, and the reliability of an admitted intervention. Cross-task reuse is an additional property: a representation useful for one task need not serve another.

This motivates the structural coordinate question below. Three levels must remain separate: existence of useful coordinates, discovery of those coordinates, and selection of a well-fitting predictor with an informative complexity bound. A successful finite predictor need not reconstruct a privileged teacher representation. Likewise, its usefulness does not require an absorbing endpoint of regularized SGD.

The complementary question of adapting through prompts and external programs, rather than through parameter changes, is developed in Chapter 26, especially Section 26.3.1. That discussion is not repeated here.

27.2.2 Local dependence is coordinate dependent

For a differentiable state transition $F : \mathcal{X} \to \mathcal{X}$, define its direct coordinate-dependency width by

$$k_{\mathrm{coord}}(F) = \max_i \#\{j : \partial_j F_i \not\equiv 0\}.$$

For an invertible change of coordinates $u = \psi(z)$, the transition becomes

$$F^\psi = \psi \circ F \circ \psi^{-1}. \tag{27.6}$$

There is no reason for $k_{\mathrm{coord}}(F^\psi)$ to equal $k_{\mathrm{coord}}(F)$.

For example, $F(z) = Dz$ with diagonal D has one-parent coordinate updates. Under a dense invertible linear change $u = Az$,

$$F^A(u) = ADA^{-1}u,$$

which generally has dense direct dependencies.

This does not mean that a polynomial-size bounded-fan-in circuit has ceased to exist: the dense linear maps themselves admit polynomial-size arithmetic circuits.

The issue is which coordinates expose the desired small interfaces, reusable constituents, and economical local description.

27.2.3 Charge for the coordinate transformation

Let Ψ be a specified class of effectively represented invertible maps, with controlled evaluation costs and conditioning, for example

$$\mathrm{Lip}(\psi) \le \kappa, \qquad \mathrm{Lip}(\psi^{-1}) \le \kappa.$$

One possible intrinsic cost is

$$\mathcal{C}_{\mathrm{coord}}(F, \varepsilon) = \inf_{\psi \in \Psi,\, \widetilde{F}} \Big\{ B(\psi) + B(\psi^{-1}) + B(\widetilde{F}) : \\ \|F - \psi^{-1} \circ \widetilde{F} \circ \psi\|_\infty \le \varepsilon \Big\}, \tag{27.7}$$

where $\widetilde{F}$ must satisfy the prescribed compositional and resource constraints.

The cost of ψ cannot be omitted. Otherwise a difficult computation could simply be relocated into the encoder or decoder.

For a scalar prediction target, the analogous representation is $f = q \circ \psi$, not a conjugacy; conjugation applies to state transitions whose input and output spaces match.

Open problem. Characterize admissible representations that expose useful sparse structure or economical readouts. For the stronger structural objective, establish when the representation is identifiable up to an explicitly stated equivalence. Noninvertible task-sufficient representations and invertible state-coordinate changes are different admitted classes; identifiability is not necessary for every predictive use of either.

A natural initial equivalence class consists of coordinate permutations and componentwise invertible transformations:

$$\psi_i(z) = \psi_i^{\mathrm{scalar}}(z_{\pi(i)}).$$

These preserve the dependency graph up to relabeling when domains and conditioning are controlled.

Arbitrary dense nonlinear transformations need not preserve it.

27.2.4 Coordinate cost and spectral accessibility

The Hermite definitions in Section 27.3.1 give a concrete example of representation-dependent accessibility. Anticipating that definition, a schematic coordinate-penalized criterion is

$$\mathcal{L}_{\mathrm{intrinsic}}(f) = \inf_{\psi \in \Psi} \left[C(\psi) + \mathrm{Leap}\left(f \circ \psi^{-1} \right) \right]. \tag{27.8}$$

It is a proposed design criterion, not an established complexity of learning or the isotropic quantity defined in Section 27.3.1. As written, it is well-defined only when the transformed target and measure support the admitted leap definition. One restricted example is a finite-Hermite target with orthogonal ψ under Gaussian measure. For general invertible ψ, the law becomes $\psi_{\#}P_X$; an analogue of leap for that law has to be defined before using the formula.

The cost $C(\psi)$ must include the required encoder and decoder descriptions, evaluation resources, and conditioning. An admissible family should also exclude transformations that make the criterion vacuous by hiding the target inside the encoder. If a scalar weighted sum is used, its relative weights must be specified: one bit of description is not inherently equivalent to one unit of spectral degree. A constrained resource tradeoff is another reasonable formulation.

For a state transition on a common state space, the analogous template is

$$\mathcal{L}_{\text{intrinsic}}(F) = \inf_{\psi \in \Psi} \left[C(\psi) + \text{Leap}\left(\psi \circ F \circ \psi^{-1}\right) \right]. \tag{27.9}$$

This is also a research template. A vector-valued leap convention must be supplied, for example a specified joint ordering of supports across output coordinates. Taking a maximum of separate coordinate leaps is another possible convention, but need not describe simultaneous recoverability. Conjugacy is appropriate for matching state spaces; scalar prediction uses factorization $f = q \circ \psi$, not conjugacy. A noninvertible representation needs task sufficiency rather than an inverse, as in Section 27.2.1 and Chapter 21.

An uncertainty-principle analogy is a useful motivation: sparsity depends on the representation. It is not already a theorem relating nonlinear compositional arity to a spectral recovery exponent. The open task is to find a representation-aware criterion with a nontrivial connection to feasible learning dynamics.

27.2.5 Possible approaches

Begin with simultaneously structured transitions. A single map can admit many economical coordinate systems.

A family $\{F_a\}_{a \in \mathcal{A}}$, sharing the same latent variables but expressing different transitions, imposes stronger constraints:

$$F_a^{\psi} = \psi \circ F_a \circ \psi^{-1} \quad \text{should be sparse for every } a.$$

In the linear case, simultaneous diagonalization of a suitably nondegenerate family provides a tractable starting example.

The research question is what survives for nonlinear bounded-arity dynamics.

Use multiple derivative statistics. The Jacobian transforms as

$$DF^{\psi}(u) = D\psi(F(z))\, DF(z)\, D\psi^{-1}(u), \qquad z = \psi^{-1}(u).$$

This suggests searching for coordinates that simplify several Jacobians or higher-order derivative tensors simultaneously, rather than relying on one averaged matrix.

The AGOP discussion in Chapter 6 already explains why a single second-order statistic is not a general inverse map to a hidden DAG.

Exploit changes of environment. If different tasks or interventions modify only a few latent mechanisms, the same coordinate system should localize those changes.

One can ask whether invariance of the other mechanisms restricts the admissible coordinate ambiguity more strongly than prediction on one distribution.

Any claim of causal identification requires its own assumptions.

Separate representational and statistical theorems. First characterize the coordinate system assuming that the family of maps $\{F_a\}$ is known exactly.

Then ask how many observations are needed to estimate it.

This prevents a representation theorem from silently assuming the statistical recovery result it is meant to support.

A useful first result. For a restricted nonlinear family, prove that all sufficiently economical representations are related by a controlled graph-preserving equivalence, or exhibit the additional ambiguities that necessarily remain.

Uniqueness of literal neural coordinates is not the appropriate target.

27.3 Problem 3: Quantitative Genericity and Detectable Footprints

Origin in the book. Chapter 6 defines genericity relative to $(f, \mathsf{P}, \mathcal{F}_{\mathrm{probe}})$ and separates statistical sufficiency from algorithmic recoverability.

The crucial question is not merely whether a footprint is nonzero, but whether a feasible learner can detect and exploit a useful footprint at every stage required by the proposed mechanism. Fourier–Hermite analysis makes one such mechanism precise enough to distinguish degree, support, coefficient size, and algorithmic access.

27.3.1 Fourier–Hermite background: degree, leap, and coordinates

Under the uniform Boolean product measure, Fourier–Walsh characters supply a natural orthogonal basis. Under standard Gaussian measure, Hermite polynomials provide the corresponding basis. These definitions and examples organize the signal question; the attributed learning results are collected once in Section 27.4.3. A formal support ordering is not itself a theorem about the trajectory of a finite learning algorithm.

Gaussian Multi-Index Models

A standard setting is

$$X \sim \mathcal{N}(0, I_d), \qquad f^\star(X) = h^\star(U^\top X), \tag{27.10}$$

where $U \in \mathbb{R}^{d\times\nu}$, $U^\top U = I_\nu$, and $\nu \ll d$. Then $Z = U^\top X \sim \mathcal{N}(0, I_\nu)$, and the target factors through an ν-dimensional linear representation. The learner is given observations of X and the target, not the coordinates Z.

An identifiable subspace-recovery target requires ν to describe the minimal relevant subspace, together with the assumptions of the chosen result. Even then the basis is not intrinsically unique. For an orthogonal R, the same target is represented by

$$Z' = RZ, \qquad U' = UR^\top, \qquad h^\star_R(z') = h^\star(R^\top z').$$

Predicting $f^\star$, recovering its relevant subspace, and identifying one basis within that subspace are distinct objectives.

This model gives a particularly clean separation: the target has a low-dimensional representation, but discovering that representation is part of the learning problem. A theorem with ν fixed as d grows need not be polynomially uniform when ν, target degree, or depth also grow.

The Hermite Expansion

Let He_q denote the probabilists' Hermite polynomial and put $H_q = \mathrm{He}_q/\sqrt{q!}$, with $H_0 = 1$. Throughout, $\mathbb{N}_0$ includes zero. For $\alpha = (\alpha_1, \dots, \alpha_\nu) \in \mathbb{N}_0^\nu$, define

$$H_\alpha(Z) = \prod_{j=1}^{\nu} H_{\alpha_j}(Z_j). \tag{27.11}$$

These normalized polynomials form an orthonormal basis of $L^2(\gamma_\nu)$, where $\gamma_\nu = \mathcal{N}(0, I_\nu)$. For $h^\star \in L^2(\gamma_\nu)$,

$$h^\star(Z) = \sum_{\alpha \in \mathbb{N}_0^\nu} \widehat{h}_\alpha H_\alpha(Z), \tag{27.12}$$

with convergence in L^2, and

$$\widehat{h}_\alpha = \mathbb{E}\left[h^\star(Z) H_\alpha(Z)\right]. \tag{27.13}$$

Pointwise or uniform convergence requires additional hypotheses. For the finite-support leap calculations below, $h^\star$ is a polynomial, so no infinite-series interchange is needed.

A Hermite coefficient is a footprint in the sense of Chapter 6. But $H_\alpha(U^\top X)$ is not an available probe merely because it has a short formula: U is hidden. An actual method must build or search suitable probes in observed coordinates, or learn the directions that make them evaluable. Low formal degree, low evaluation cost, and efficient identification are three different properties.

Normalization and sampling variance. Although $\mathbb{E}[H_\alpha(Z)^2] = 1$, the variance of the coefficient summand is

$$\tau_\alpha^2 = \mathrm{Var}\big(h^\star(Z) H_\alpha(Z)\big).$$

For n independent observations of an *available* probe and finite τ_α^2, the sample mean has variance τ_α^2/n. Thus its signal-to-noise ratio is $|\widehat{h}_\alpha|\sqrt{n}/\tau_\alpha$ when $\tau_\alpha > 0$. Square-integrability of $h^\star$ guarantees the coefficient exists but not this product's finite variance. Polynomial targets have all the needed Gaussian moments; their bounds still depend on degree and coefficients. This is the same distinction between raw footprint and sampling-normalized signal emphasized in Chapter 18.

Degree Is Not the Whole Story

If $\widehat{h}_\alpha = 0$ for every nonconstant α with $|\alpha| < q$, the target is orthogonal to those low-degree Hermite probes after centering. This identifies an obstruction for that probe family. It is not a lower bound for every fixed kernel or every learning rule: a kernel can incorporate appropriate high-degree features or other supplied structure. Statements about kernel complexity must name the kernel, target family, and required accuracy.

Feature learning can change the situation. A term of high total degree may introduce only a small number of new directions after earlier features are available. This motivates leap complexity [2]. Degree still matters for approximation, moments, and estimation; leap records a different aspect of the expansion.

Leap Complexity

Assume that the nonconstant Hermite support

$$\mathcal{A} = \{\alpha \neq 0 : \widehat{h}_\alpha \neq 0\}$$

is finite. For an ordering $\alpha^{(1)}, \dots, \alpha^{(M)}$ of $\mathcal{A}$, let

$$I_{j-1} = \bigcup_{\ell<j} \operatorname{supp}\left(\alpha^{(\ell)}\right), \tag{27.14}$$

with $I_0 = \varnothing$. The new degree in term j is

$$\Delta_j = \sum_{i \notin I_{j-1}} \alpha_i^{(j)}. \tag{27.15}$$

For the displayed coordinate system, define

$$\operatorname{Leap}(h^\star) = \min_{\text{orderings}} \max_j \Delta_j. \tag{27.16}$$

The leap is zero for a constant polynomial by convention. This is the coordinate-based support definition of Abbe and collaborators, expressed in the present normalization [2]. Rescaling nonzero basis elements does not change their support.

For Boolean Fourier expansions, exponents are zero or one, so the new degree counts newly introduced variables. The leap-one support pattern is the merged-staircase type of structure. The support pattern alone nevertheless omits coefficient sizes, degeneracies, activation expressiveness, and properties of the actual dynamics. A theorem needs those additional assumptions.

If a general L^2 target is replaced by a finite expansion, the omitted L^2 energy and the coefficient threshold must be included. An arbitrarily small coefficient can change the exact support and hence the exact leap while being statistically inaccessible. In particular, leap is not by itself an accuracy-sensitive or signal-normalized measure.

An Example

Consider the example

$$\begin{aligned} h(Z) = {} & H_2(Z_1) + H_2(Z_1)H_3(Z_2) \\ & + H_2(Z_1)H_3(Z_2)H_2(Z_3). \end{aligned} \tag{27.17}$$

The total degrees are $2, 5, 7$, whereas the displayed ordering introduces new degrees $2, 3, 2$. Its maximum is three. No ordering can have a smaller maximum: the first term involving coordinate 2 introduces degree three in that previously absent coordinate. Thus the coordinate-based leap is exactly three.

The statement is combinatorial. Interpreting the ordering as a training trajectory needs an algorithmic argument. The following elementary calculation makes one algebraic part of that interpretation explicit without assuming such an argument.

Proposition 50 (Conditional Hermite footprints)**.** *Let $Z \sim \gamma_\nu$ and let h have a finite Hermite expansion. Partition the coordinate indices into I and J. For a multi-index β on J, put*

$$b_\beta(Z_I) = \mathbb{E}[h(Z)H_\beta(Z_J) \mid Z_I].$$

Then

$$b_\beta(Z_I) = \sum_{\alpha:\alpha_J=\beta} \widehat{h}_\alpha H_{\alpha_I}(Z_I), \qquad \|b_\beta\|^2_{L^2(\gamma_I)} = \sum_{\alpha:\alpha_J=\beta} |\widehat{h}_\alpha|^2.$$

Proof. Independence of Z_I and Z_J and orthonormality give $\mathbb{E}[H_{\alpha_J}(Z_J)H_\beta(Z_J)] = \mathbf{1}\{\alpha_J = \beta\}$. Apply this identity term by term in the finite expansion of h. Orthogonality of the remaining polynomials H_{α_I} gives the squared-norm identity. □

In Equation (27.17),

$$\mathbb{E}[h(Z)H_3(Z_2)] = 0, \qquad \mathbb{E}[h(Z)H_3(Z_2) \mid Z_1] = H_2(Z_1).$$

Consequently the gated statistic satisfies

$$\mathbb{E}[h(Z)H_2(Z_1)H_3(Z_2)] = 1.$$

Similarly,

$$\mathbb{E}[h(Z)H_2(Z_3) \mid Z_1, Z_2] = H_2(Z_1)H_3(Z_2).$$

An unconditional marginal can therefore vanish while its conditional coefficient is a nonzero function of earlier variables. This is a precise algebraic illustration of a footprint becoming usable through an earlier representation.

The proposition assumes access to the indicated coordinate split for its calculation. It does not recover the unknown directions, show that SGD constructs these gates, or determine the sample cost. The example's dependence on Z_1 is even, so revealing $H_2(Z_1)$ does not recover the signed value Z_1. A useful learned coordinate can be a sufficient function of a teacher variable rather than that variable itself.

Leap Complexity Is Not Compositional Sparsity

For independent standard Gaussian inputs, consider

$$f(X) = \prod_{i=1}^{m} X_i, \qquad X_i \sim \mathcal{N}(0,1). \tag{27.18}$$

For $m \geq 2$, a correct binary realization is

$$z_1 = X_1X_2, \qquad z_j = z_{j-1}X_{j+1} \quad (2 \leq j \leq m-1), \qquad f = z_{m-1}.$$

It has $m-1$ multiplication nodes and arity two. A balanced tree can reduce its parallel depth without changing the function. Yet $f = H_1(X_1)\cdots H_1(X_m)$ is a single canonical-basis Hermite monomial, whose leap is m. Therefore

small compositional arity does not imply small Hermite leap in the original coordinates.

This is a computational factorization example. Multiplication on unbounded Gaussian states is not globally Lipschitz with a fixed constant, and this example does not automatically satisfy a normalized compact-domain Sobolev or contraction theorem. Those representation results have additional domain and stability assumptions.

The two quantities answer different questions. Arity measures the size of each local interface in an admitted realization. Leap measures the new spectral support introduced in a specified expansion. Neither quantity, in isolation, specifies the complete training cost.

The coordinate problem and isotropic leap

An orthogonal change $Z' = RZ$ preserves Gaussian measure but can change the monomials appearing in the Hermite expansion. It can therefore change Equation (27.16). The relevant distribution, coordinate map, and accessible probe family must all be specified.

A useful distinction in the cited literature is the isotropic quantity

$$\mathrm{isoLeap}(h) = \max_{R \in O(\nu)} \mathrm{Leap}(h \circ R^{\top}),$$

as defined in Appendix B.2 of Abbe and collaborators [2]. It uses a *maximum*, not the minimum over representations in Section 27.2.4. These two optimizations have different meanings: the first addresses an isotropic recovery obstruction, whereas the second asks what a learner could gain from an admissible, explicitly acquired transformation.

For an illustrative rotation discussed in that source, take $h(z) = z_1 + z_2 + z_1 z_2$, which has coordinate leap one. With $u_1 = (z_1 + z_2)/\sqrt{2}$ and $u_2 = (z_1 - z_2)/\sqrt{2}$, direct substitution in the present normalized convention gives

$$h = \sqrt{2} H_1(u_1) + \frac{1}{\sqrt{2}} \big(H_2(u_1) - H_2(u_2)\big).$$

Its coordinate leap is now two: u_2 first appears with degree two. Changing axes has changed the support ordering without changing the underlying observed function or supplying a learning algorithm.

The broader question is to judge a representation jointly by its compositional complexity, spectral accessibility, conditioning, and cost of construction and evaluation. Complexity cannot be hidden in an unrestricted coordinate map. Nor does a convenient basis prove that the learning dynamics can discover it.

27.3.2 A finite-sample formulation

Let $H_j(X)$ be the representation available after stage j.

Let $R_j(X, Y)$ be a specified residual or learning signal, and let Φ_j be a family of normalized accessible probes.

Define

$$\rho_{j,\phi} = \mathbb{E}_{\mathsf{P}} \left[R_j(X, Y) \phi(H_j(X)) \right], \qquad \rho_{j,\star} = \max_{\phi \in \Phi_j} |\rho_{j,\phi}|. \tag{27.19}$$

Normalization is essential: arbitrarily rescaling a probe must not count as making the signal more detectable.

For this screening calculation, fix H_j, R_j, and the nonempty finite probe family independently of the observations used to estimate their expectations, or condition on independent earlier training. Assume $\rho_{j,\star} > 0$ and suppose

$$|R_j(X, Y) \phi(H_j(X))| \le B_Z.$$

Hoeffding's inequality and a union bound imply that

$$n \ge \frac{8 B_Z^2}{\rho_{j,\star}^2} \log \frac{2|\Phi_j|}{\delta}$$

is sufficient to estimate every footprint to error at most $\rho_{j,\star}/2$ with probability at least $1 - \delta$.

This is a sufficient uniform screening bound, not a universal lower bound on the sample complexity of all learning algorithms.

With variance control, a Bernstein-type bound instead has the form

$$n \gtrsim \left(\frac{v_Z}{\rho_{j,\star}^2} + \frac{B_Z}{\rho_{j,\star}} \right) \log \frac{|\Phi_j|}{\delta}.$$

Thus the variance and probe normalization matter, not just the raw correlation magnitude.

The logarithm is a statistical cost. Enumerating an exponentially large Φ_j can still require exponential computation even when $\log |\Phi_j|$ is only polynomial.

A complete result must control both sample cost and the cost of finding a useful probe.

27.3.3 The desired quantitative statement

Let ω denote a perturbation of the observation model, such as a shift, bias, or controlled change of distribution.

Seek conditions giving

$$\Pr_{\omega} \left[\min_{0 \le j < s} \max_{\phi \in \Phi_j^{\text{useful}}} |\rho_{j,\phi}(\omega)| \ge \frac{1}{\text{poly}(d, s, L)} \right] \ge 1 - \delta_0. \tag{27.20}$$

Here "useful" must be defined relative to progress in the proposed learning mechanism: toward an unresolved constituent for a structural theorem, or toward reducing predictive error for a predictive theorem. Correlation with an easy nuisance variable is not enough.

The learner must also have an efficient way to find or exploit such a probe.

Defining the useful family using the hidden teacher is permissible for analysis, but supplying that family to the learner would change the learning problem.

Open problem. Find broad but explicit target, distribution, and probe classes for which Equation (27.20) holds, and determine how the bound changes after approximate rather than exact recovery of earlier constituents.

27.3.4 An Adaptive Spectral Staircase

For the stagewise objects of Equation (27.19), the single-stage margin underlying the quantitative target is an available, normalized useful probe satisfying

$$|\mathbb{E}\left[R_j(X, Y)\phi(H_j(X))\right]| \ge \frac{1}{\text{poly}(d, s, L)}. \tag{27.21}$$

This is a desired condition, not a conclusion from low arity. For a fixed probe, put $Z_{j,\phi} = R_j\phi(H_j)$, whose mean is $\rho_{j,\phi}$ as defined above, and suppose its variance is $\tau_{j,\phi}^2 \in (0, \infty)$. The dimensionless signal is

$$\text{snr}_{j,\phi} = \frac{|\rho_{j,\phi}|}{\tau_{j,\phi}}.$$

The symbol ρ continues to denote the raw expectation under the specified probe normalization, not this ratio. Equation (27.21) is statistically useful only together with the variance or tail controls in the preceding screening bounds. Degenerate zero-variance cases are treated separately rather than dividing by zero.

A nonlinear H_j need not preserve Gaussian measure. The probe family must therefore change, or be analyzed under the actual distribution $(H_j)_{\#}P_X$. Possible approaches include orthogonal polynomials, Stein operators, score functions, cumulants, and learned low-complexity probes. Merely postulating such a family does not show that it can be estimated or accessed by the optimizer.

Approximate earlier coordinates. There is an elementary stability estimate for a fixed candidate probe. If $R_j \in L^2$, the probe is K-Lipschitz on a domain containing both representations, and the same residual is used on each side, then

$$\begin{aligned}&\left|\mathbb{E}\left[R_j\phi(\widehat{H}_j)\right] - \mathbb{E}\left[R_j\phi(H_j)\right]\right| \\ &\quad\le K\|R_j\|_{L^2}\|\widehat{H}_j - H_j\|_{L^2}.\end{aligned}$$

This follows from Cauchy–Schwarz. A footprint can thus survive small representation error when that error is smaller than its margin. For higher-degree Hermite polynomials on an unbounded domain, a global Lipschitz constant is not available; appropriate moment or truncation bounds must replace this assumption. An estimated residual introduces an additional error term. Data-dependent choices require fresh samples, sample splitting, or a uniform analysis, rather than a fixed-probe bound applied after selection.

27.3.5 Possible approaches

Quantify analytic symmetry breaking. If a shift footprint $a_\phi(b)$ is analytic and not identically zero, its zero set has measure zero on a connected open parameter domain.

This establishes nonvanishing, not a usable lower bound.

For nonzero polynomial footprints of degree at most $q \ge 1$, Gaussian anti-concentration gives estimates of the form

$$\Pr\left[|a_\phi(\omega)| \le \eta\|a_\phi\|_{L^2}\right] \le C_q\eta^{1/q},$$

with degree-dependent constants [26].

For a prescribed sequence of reference stages, if one fixed useful candidate at each of s stages has $\|a_\phi\|_{L^2} \ge \sigma_0$, a union bound gives the lower scale

$$\rho_{\min} \ge \sigma_0\left(\frac{\delta_0}{C_q s}\right)^q.$$

This scale holds with probability at least $1-\delta_0$ for the prescribed reference stages. It is polynomially useful when q is bounded and σ_0 has an inverse-polynomial lower bound, with the confidence parameter included in the resource accounting. This particular estimate can introduce a power of δ_0^{-1} through $\rho_{\min}^{-1}$; logarithmic confidence dependence for a final learning theorem would require a separate argument.

The substantive work is to establish those two properties for the hierarchy and to make them robust to learning error.

Use parity as a quantitative stress test. Let $X_i \in \{-1,+1\}$ be independent with $|\mu_i| < 1$, where $\mu_i = \mathbb{E}[X_i]$, and let

$$f_S(X) = \prod_{i\in S} X_i, \qquad \phi_j(X) = \frac{X_j - \mu_j}{\sqrt{1-\mu_j^2}}.$$

For $j \in S$, direct calculation gives

$$\mathbb{E}[f_S(X)\phi_j(X)] = \sqrt{1-\mu_j^2}\prod_{i\in S\setminus\{j\}}\mu_i.$$

Bias can make a footprint nonzero while leaving it exponentially small in $|S|$.

Therefore "symmetry is broken" is weaker than quantitative C2.

This example concerns the specified probes; it is not a lower bound for all algorithms learning noiseless parity.

Control adaptivity. When H_j and R_j are learned on the same data used to evaluate the footprints, fixed-probe concentration does not apply automatically.

Possible routes include sample splitting, fresh samples at each stage, or uniform bounds over the whole adaptive representation/probe class.

The cost of repeated stages must be included.

A useful first result. Prove a polynomial lower bound for a family of bounded-degree compositional targets under a specified perturbation law, together with an efficient screening procedure and a stability bound for footprints when previously learned coordinates are imperfect.

27.3.6 Beyond Gaussian and Product Measures

For $X \sim P$, the question is not merely whether $L^2(P)$ has an orthogonal basis. One needs probes with controlled complexity, estimation error, and access through a specified algorithm. An arbitrary orthogonal basis may have none of these properties.

For some distributions, orthogonal polynomials provide a useful route. For others, Stein identities, score functions, or cumulants may be more appropriate. Their use requires the relevant regularity and moment assumptions, as well as the cost of estimating distribution-dependent objects. Score identities, for example, require more than the formal existence of a density. An adaptive representation can also create singular or lower-dimensional support.

A useful first result would specify a tractable distribution family, a finite or norm-controlled probe family, and a training rule whose updates actually involve those probes. The theorem would then control both detection and the cost of turning the statistic into a new representation. This is a restricted extension of quantitative genericity, not a universal spectral recipe.

27.3.7 Relating Leap to Compositional Arity

The product example shows why a bare implication from arity to leap cannot hold. A possible target relation is

$$\mathrm{Leap}_{\mathrm{effective}}(f) \le \Psi\left(k, \mathrm{depth}, \text{constituent regularity}, \text{coordinate complexity}\right). \tag{27.22}$$

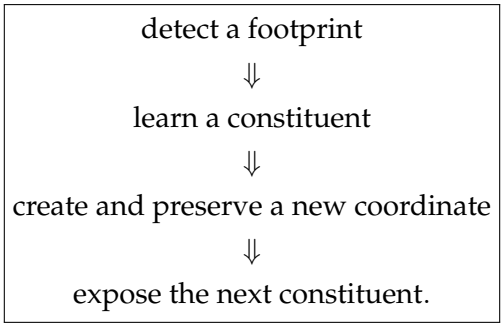
Here Ψ denotes a proposed bounding function, not the transformation class used above. "Effective leap" must first name a precise object relative to a distribution, probe family, accuracy, and access model. A depth-dependent bound alone is not enough for efficiency when it appears as an exponent of the ambient dimension.

Additional assumptions might concern nondegeneracy, controlled coefficient asymmetry, intermediate supervision, admissible changes of representation, or distributional perturbations. Each should do a specific job. Almost-sure nonvanishing does not supply a lower signal bound; an ensemble RMS does not supply high-probability anti-concentration; and a low-cost oracle coordinate does not provide an efficient discovery algorithm.

The Random Hierarchy Model chapter supplies a concrete setting for separating footprint detection from feature-learning thresholds. Chapter 21 supplies a different setting in which useful intermediate computations can be retained explicitly as modules. Neither provides a general conversion from bounded arity to small leap. The desired theorem should identify a restricted family where the conversion and its resource costs can actually be proved.

27.4 Problem 4: Prove a Compositional Staircase Recovery Theorem

Origin in the book. The staircase is a candidate constructive mechanism for the predictive premise of Chapter H, not a separate condition required of every accurate learned computation:



A staircase theorem must explain actual feature-learning dynamics, not merely the existence of an accurate deep network. The broad predictive objective and its stronger structural version should be stated separately.

27.4.1 Predictive learning and the stronger mechanistic objective

For a specified noiseless regression problem, the predictive target is

$$\|f_{\widehat{\theta}} - f^\star\|_{L^2(\mathsf{P}_X)} \le \varepsilon$$

with high probability, together with controlled training and evaluation resources. A useful finite predictor already provides its own compositional realization. The additional claim that training successively creates useful nonlinear coordinates is a mechanistic statement. Matching those coordinates to a prescribed teacher DAG is a further structural statement, with its own identifiability assumptions.

Thus there are three possible conclusions: accurate feasible prediction, a constructive feature-learning mechanism, and recovery of a generating structure. The problem below asks for the stronger conclusions in a specified class, without making them necessary for the first. The regularized objective comparison in Section 27.5 provides another way to formulate the main predictive premise.

27.4.2 What a theorem should establish

Let the teacher be a DAG with latent variables

$$z_v = g_v\left(z_{\mathrm{pa}(v)}\right), \qquad |\,\mathrm{pa}(v)| \le k. \tag{27.23}$$

The output is $Y = f^\star(X)$, initially without label noise. Unlike the fixed-linear-subspace setting, learning a constituent may create a nonlinear coordinate that was not previously available to the learner. The cost of constructing and retaining it belongs in the theorem.

The learner receives only (X_i, Y_i), not the internal z_v.

Choose a finite-width architecture, initialization law, loss, regularizer, sampling rule, optimizer, step-size schedule, and rule for returning an iterate. Specify constituent domains and regularity, and identifiability assumptions when teacher matching is intended.

Designate bounded-dimensional feature slots in the learner to prevent the recovery criterion from being satisfied merely by an unrestricted decoder operating on the full input.

For an admissible node matching π and a controlled equivalence class $\mathcal{E}_v$, define, for example,

$$\mathcal{D}_{\mathrm{str}}(\widehat{z}, z) = \inf_{\pi,\, \psi_v \in \mathcal{E}_v} \max_{v} \left\| \widehat{z}_{\pi(v)} - \psi_v(z_v) \right\|_{L^2(\mathsf{P})}. \tag{27.24}$$

The equivalences have fixed complexity and conditioning bounds, and the matching must respect the allowed dependency structure.

A structural staircase theorem would give, with probability at least $1 - \delta$,

$$\|f_{\widehat{\theta}} - f^\star\|_{L^2(\mathsf{P})} \le \varepsilon, \qquad \mathcal{D}_{\mathrm{str}}(\widehat{z}, z) \le \varepsilon,$$

after sample size, arithmetic cost, and network size bounded by

$$\mathrm{poly}\left(d, s, L, \varepsilon^{-1}, \rho_{\min}^{-1}, \kappa, \log(1/\delta)\right). \tag{27.25}$$

Here κ collects explicitly specified conditioning, moment, and regularity parameters. Numerical precision, stored parameter bits, and bit-cost evaluation must also be controlled if a polynomial bit-complexity claim is intended; arithmetic-operation counts alone do not provide it.

Here $\rho_{\min}$ is the minimum useful stagewise footprint under the prescribed normalization and moment controls of Problem 3. Its inverse must itself be polynomial in the growing problem parameters; any dependence on perturbation failure probability must also be counted.

A theorem polynomial in an exponentially large inverse signal would not establish efficient learning.

27.4.3 What is already known

The staircase idea already has rigorous special cases.

Abbe and collaborators prove hierarchical learning of suitable Boolean staircase functions with a specified layerwise gradient-based algorithm [3].

Related merged-staircase results characterize learning in a two-layer mean-field setting [1].

The Gaussian results complement these Boolean theorems. The following are attributed results with their original scopes, not consequences of Proposition 50.

Finite-network SGD under a specified protocol. Abbe, Boix-Adserà, and Misiakiewicz introduce leap complexity and analyze sequential support acquisition for Gaussian targets by two-layer networks under specified modifications and technical assumptions on SGD [2]. Their positive results go beyond leap one and beyond a purely infinite-width, continuous-time approximation. The paper's broader SGD conjecture is distinct from the theorem it establishes. Sample size, number of online updates, and arithmetic time should not be identified: a single update already processes a d-dimensional input.

Mean-field polynomial learning. Chen and Ge study subspace-sparse polynomials with Gaussian inputs [29]. Their necessary obstruction and stronger sufficient condition involve both the target and activation expressiveness, with a specified training construction for the positive result. These are not guarantees for every activation or arbitrary end-to-end finite-width SGD.

Two-timescale population flow. Bietti, Bruna, and Pillaud-Vivien analyze joint learning of a hidden subspace and a nonparametric link in a two-timescale population setting [18]. The link is optimized infinitely faster than the subspace in the limiting analysis. They establish convergence and Hermite-dependent stages under that setup; the result is not a direct finite-sample guarantee for simultaneous updates of all parameters.

Generative leap and a different statistical interface. Damian, Lee, and Bruna analyze hidden-subspace estimation through the conditional law of the label and input projection [32]. Their generative leap supports a sequential Hermite-tensor estimator and lower bounds within the low-degree-polynomial framework. It is not merely the support leap of the conditional mean, and its estimator is not ordinary SGD. The distinction matters when label transformations reveal structure invisible to raw label correlations. No lower bound for every possible learning algorithm is claimed here.

Together these works provide substantial theory in controlled settings. Their assumptions and computational interfaces are part of their conclusions. They motivate extension to nonlinear compositional states without already supplying that extension.

The open target is an extension to specified classes of hidden smooth constituents and more general DAGs, with simultaneous control of finite samples, finite width, approximate earlier features, and the actual optimization dynamics.

It is not the claim that no staircase-learning theorem exists.

27.4.4 Possible approaches

Derive the accessible probes from the gradient. For a stagewise model

$$f_w(X) = f_0(X) + a\sigma(w^\top H_j(X)),$$

with loss

$$\mathcal{L}(w) = \frac{1}{2}\mathbb{E}[(f_w(X) - Y)^2],$$

where H_j, f_0, a are fixed and differentiation under the expectation is justified,

$$\nabla_w \mathcal{L} = a\,\mathbb{E}\left[(f_w(X) - Y)\sigma'(w^\top H_j(X))H_j(X)\right].$$

The gradient accesses particular gated correlations.

An arbitrary nonzero probe correlation need not align with these directions.

A proof should identify the effective probe family of the chosen architecture, rather than assume every useful statistic is available.

Prove a stagewise alignment inequality. Let $\mathcal{E}_j(\theta)$ measure the current constituent error in a fixed local parameterization.

A useful intermediate target is

$$\langle \nabla_\theta \mathcal{L}(\theta), \nabla_\theta \mathcal{E}_j(\theta)\rangle \geq c\rho_j^2 \mathcal{E}_j(\theta)$$

through a specified region of parameter space.

Under gradient flow, this would give

$$\frac{d}{dt}\mathcal{E}_j(\theta_t) \leq -c\rho_j^2 \mathcal{E}_j(\theta_t).$$

The challenge is to derive the alignment inequality from structural and statistical assumptions, not to assume it as a restatement of recovery. This displayed flow concerns $-\nabla\mathcal{L}$. If the actual objective is $\mathcal{L} + \lambda\Omega$, the alignment calculation must include $\lambda\langle\nabla\Omega, \nabla\mathcal{E}_j\rangle$ as well.

A finite-step SGD theorem must additionally control variance and escape from the region where the estimate holds.

Close an induction over the hidden computation. After recovering one constituent, prove that it is retained and that the next footprint remains detectable.

An induction may require an estimate of the form

$$e_{j+1} \leq a_j e_j + b_j \varepsilon_{\mathrm{stat}} + c_j \varepsilon_{\mathrm{opt}},$$

with controlled products of the a_j and explicit statistical and optimization errors.

Residual connections or temporarily frozen learned modules offer tractable starting architectures; they do not themselves prove the induction. The fixed-probe error estimate in Section 27.3.4 addresses one component of this step; it does not supply the local optimization or preservation argument.

Use C3 to separate the sources of difficulty. Begin with local input–output pairs supplied, then progressively hide intermediate variables.

Norm-controlled RKHS constituents are useful here: local kernel ridge regression is convex once its inputs and targets are observed, whereas joint hidden composition remains nonlinear.

Deep representer theorems give a finite-dimensional formulation without solving that optimization problem [20].

This also creates a route to partial-trace results: identify which exposed nodes suffice to initialize a staircase that subsequently continues without complete internal supervision. The trace-supervision chapter, Chapter 11, and the CRM construction in Chapter 21 make coverage explicit: a supplied local interface can support exact recomposition when the needed local cases have been learned, but finitely many short traces do not by themselves cover every later case.

A useful first result. Prove the full induction for a depth-growing family of trees with smooth, nondegenerate, bounded-arity constituents under a specified distribution.

A staged algorithm would be a meaningful first theorem.

Extending it to simultaneous end-to-end SGD would be a separate advance.

27.4.5 Finite samples, finite width, and the returned iterate

The resource target in Equation (27.25) requires simultaneous control of finite-sample fluctuations, width, stochastic updates, approximate coordinates, interference, and retention. A result for stagewise or temporarily frozen modules is already useful; simultaneous end-to-end updates are a further extension. A theorem with a fixed latent dimension need not remain uniform as that dimension or computational depth grows.

With regularization, zero full objective gradient need not imply zero minibatch covariance. The desired result can concern a specified finite stopped iterate; it need not require an absorbing common zero of every minibatch update. Gradient-estimator covariance, the scale of parameter updates, and artificial Langevin temperature are distinct objects. Chapters 7 and H and Appendix N explain why local convergence and equilibrium weighting do not alone establish finite-time predictive success.

The consolidated staircase question. For a specified family of regular bounded-arity targets and observation models, prove that a finite learning procedure successively constructs and preserves useful intermediate coordinates, achieves small population error, and has controlled sample, computational, and numerical cost. When matching a teacher structure is intended, state its identifiability assumptions and permitted coordinate equivalences separately. Fourier–Hermite theory supplies starting mechanisms, not a substitute for this induction.

27.5 Problem 5: When Does Norm-Based Selection Favor Reusable Structure?

Origin in the book. Chapters 5, 7, and H distinguish reaching a fitting solution from establishing population accuracy. Chapters 13 and 14 develop description and norm control. Product-norm bounds already supply one rigorous route from an admissible fit–complexity tradeoff to generalization.

Two questions remain. The primary predictive question is whether the specified optimizer reaches such a tradeoff at feasible cost. The additional structural question is when selection favors modules useful across tasks, rather than merely a good single-task predictor. Reusable structure is not inferred from norm control alone.

27.5.1 The established generalization mechanism

Fix a scalar-output, bias-free feedforward architecture of depth L with coordinatewise ReLU hidden activations and a linear readout. The architecture is fixed independently of the sample. Define

$$\Pi(W) = \prod_{\ell=1}^{L} \|W_\ell\|_F, \qquad \mathcal{F}_Q = \{f_W : \Pi(W) \le Q\}.$$

For $\|X\|_2 \le B_X$, a standard Frobenius-norm bound yields

$$\widehat{\mathfrak{R}}_S(\mathcal{F}_Q) \le \frac{B_X Q}{\sqrt{n}} \left(\sqrt{2L \ln 2} + 1\right), \tag{27.26}$$

as a consequence of the corresponding layerwise Frobenius-norm bound [53]. Here $\widehat{\mathfrak{R}}_S$ uses the $1/n$ normalization. For a product budget $Q > 0$, positive homogeneity rescales any nonzero-layer realization to have layer norms at most $Q^{1/L}$ without changing its function. A zero layer gives the zero function. This connects the product-constrained class to the quoted layerwise class. Biases, normalization layers, residual branches, and other architectures require their own bounds.

Thus lowering an admissible product bound lowers the associated capacity bound.

For a loss $0 \le \ell \le 1$, define

$$\widehat{R}_S(f) = \frac{1}{n} \sum_{i=1}^{n} \ell(f(X_i), Y_i), \qquad R(f) = \mathbb{E}[\ell(f(X), Y)].$$

If the loss is K_ℓ-Lipschitz in predictions, standard Rademacher arguments control population risk by empirical risk plus this complexity term.

Squared loss requires bounded predictions and labels, or an appropriate clipping convention.

The relation among normalization, norm product, margins, and generalization is also developed by Xu and collaborators [170].

This is a known positive mechanism, not an unexplained part of the agenda.

27.5.2 An elementary conditional guarantee

Suppose

$$J_\lambda(W) = \widehat{R}_S(f_W) + \lambda \Pi(W)^2,$$

with $\lambda > 0$ fixed in advance and the bounded-loss conventions above.

Assume a sample-independent comparator $W^\star$ has zero population loss and $\Pi(W^\star) \le Q_\star$, with $Q_\star$ prescribed. Since the loss is nonnegative, the comparator then has zero empirical loss almost surely. Prescribe an optimization tolerance $\xi \ge 0$.

If optimization achieves

$$J_\lambda(\widehat{W}) \le J_\lambda(W^\star) + \xi,$$

then nonnegativity of the loss gives

$$\widehat{R}_S(f_{\widehat{W}}) \le \lambda Q_\star^2 + \xi, \qquad \Pi(\widehat{W}) \le Q_\xi := \sqrt{Q_\star^2 + \xi/\lambda}. \tag{27.27}$$

For a prescribed optimization tolerance ξ, the radius Q_ξ is sample independent.

There is a sample event of probability at least $1-\delta$ on which every returned predictor satisfying the objective comparison obeys

$$R(f_{\widehat{W}}) \le \lambda Q_\star^2 + \xi + \frac{2K_\ell B_X Q_\xi}{\sqrt{n}} \left(\sqrt{2L\ln 2} + 1\right) + 3\sqrt{\frac{\ln(2/\delta)}{2n}}. \tag{27.28}$$

Derivation and its probability statement. The objective comparison and nonnegativity bound each summand of $J_\lambda(\widehat{W})$, giving Equation (27.27). The uniform bounded-loss Rademacher inequality gives $R(f) \le \widehat{R}_S(f) + 2K_\ell \widehat{\mathfrak{R}}_S(\mathcal{F}_{Q_\xi}) + 3\sqrt{\ln(2/\delta)/(2n)}$ on one event over the fixed class. Substitution yields Equation (27.28). This is an elementary deduction from the stated capacity bound, not an optimization theorem. If the algorithm satisfies the objective comparison with probability at least $1-\delta_{\mathrm{opt}}$, a union bound gives joint success probability at least $1-\delta-\delta_{\mathrm{opt}}$; no independence of the two events is required.

This deduction does not assume structural recovery.

It explains prediction generalization once a sufficiently favorable objective value has been reached.

It is not a proof that arbitrary SGD reaches that value.

For binary exact interpolation $f_W(x_i) = y_i \in \{-1, +1\}$, the normalized training margins satisfy

$$\frac{y_i f_W(x_i)}{\Pi(W)} = \frac{1}{\Pi(W)}.$$

Minimum product therefore maximizes these normalized margins within the exactly interpolating set.

For a penalized objective at finite λ, empirical fit and product norm are instead traded against one another.

A smaller bound does not necessarily order the actual risks of every pair of trained predictors.

27.5.3 The primary gap: efficient regularized predictive learning

For an explicit class of targets, data distributions, and finite networks, prove that a specified initialization and training schedule returns an iterate satisfying

$$J_\lambda(\widehat{W}) \le J_\lambda(W^\star) + \xi$$

with high probability and polynomial sample, optimization, and evaluation cost. The schedule must choose λ, ξ, and n so that the right-hand side of Equation (27.28) is informative. Merely obtaining a small gradient or establishing that the objective has low global minima is not enough.

Analyze the product-regularized dynamics. The penalty has gradient

$$\nabla_{W_j} \Pi(W)^2 = 2W_j \prod_{\ell \neq j} \|W_\ell\|_F^2.$$

Study whether its coupling with the data gradient yields the objective comparison at polynomial cost in a controlled model.

Deep linear models and small nonlinear teacher–student hierarchies provide starting points.

Reparameterization symmetries and layer balancing must be handled explicitly; results for normalized gradient flow should not automatically be transferred to arbitrary unnormalized minibatch SGD.

Chapter 14 also separates symmetry, layer balancing, and actual selection. Additive and multiplicative penalties can favor the same function-level minima in an exactly constrained homogeneous interpolation problem after balancing, yet have different positive-penalty tradeoffs and trajectories. Boundedness of a particular gradient flow is not a proof that it reaches the comparator objective. Nor does uniform shrinkage by the product penalty alone explain a reduction in relative numerical rank.

The objective comparison is a sufficient route, not a necessary characterization of every successful predictor. It can hold at a stopped iterate with nonzero regularized-SGD fluctuations. Conversely, a small norm product without adequate fit does not give small risk. A useful first result here is a genuine finite-time fit–complexity guarantee in a controlled nonlinear family, without demanding teacher coordinate recovery.

27.5.4 The remaining structural question

For a separately specified structural objective, let $\mathcal{S}_\varepsilon$ be the set of learned networks that implement the useful compositional structure to error ε, up to the specified equivalence of Problem 2.

This stronger selection question asks when regularized training gives

$$\Pr\left[\widehat{W} \in \mathcal{S}_\varepsilon, \quad J_\lambda(\widehat{W}) \le J_\lambda(W^\star) + \xi\right] \ge 1 - \delta \tag{27.29}$$

with feasible sample and computational resources.

Norm minimization alone need not identify coordinates.

Already for a two-layer linear network computing the identity on $\mathbb{R}^d$,

$$(W_1, W_2) = (I, I) \quad \text{and} \quad (W_1, W_2) = (O, O^\top)$$

have the same product $\Pi = d$ for any orthogonal O.

This is minimal because

$$d = \operatorname{tr}(W_2 W_1) \le \|W_2\|_F \|W_1\|_F.$$

A dense rotation is therefore not distinguished from the sparse coordinate realization.

Both predict perfectly; structural uniqueness requires a separate criterion. The example is a linear factorization example, not a claim that arbitrary orthogonal changes preserve a nonlinear ReLU network.

27.5.5 Possible approaches

Prove a restricted norm-separation theorem. For a specified teacher class and distribution, investigate whether well-fitting predictors outside the stipulated structural class obey

$$\inf_{\substack{W \notin \mathcal{S}_\varepsilon \\ \widehat{R}_S(f_W) \le \eta}} \Pi(W)^2 \ge Q_\star^2 + \Delta, \qquad \Delta > 0, \tag{27.30}$$

with high probability.

This is a proposed separation condition, not a general property of compositional targets.

If $\eta \ge \lambda Q_\star^2 + \xi$ and $\xi < \lambda\Delta$, then Equation (27.27) excludes every structurally excluded solution in Equation (27.30).

The hard work is proving a genuine norm gap from assumptions on the task, architecture, and data, rather than inserting the desired conclusion as an assumption.

Add cross-task evidence when one task is insufficient. Several tasks sharing constituents may favor a reusable library even when one task cannot identify it.

A possible joint objective is

$$\mathcal{J}_{\text{multi}} = \sum_{a=1}^{M} \widehat{R}_a(f_a) + \lambda \sum_{a=1}^{M} \Pi(W_a)^2 + \beta \left[B(\mathcal{G}) + \sum_{a=1}^{M} B(p_a \mid \mathcal{G}) \right].$$

The library and program costs require a fixed valid coding language.

Their purpose is to charge explicitly for reuse, which a product norm alone does not measure.

Improvements in new-task learning cost must be proved relative to a specified task distribution; an i.i.d. risk bound for one task is not automatically a transfer theorem. The complete learned library and the necessary decoder must be included in a valid code. The CRM and frozen interpreter chapters, Chapters 21 and 26, distinguish reuse of a subprogram, shorter description, faster execution, and reduced future search cost. A transfer result should state which of those costs is improved.

A useful first result. For a specified family with recurring constituents, either prove that product regularization yields a structural norm gap and efficient selection, or construct a counterexample showing why an additional sharing or description-length penalty is necessary.

Either outcome would sharpen the connection between regularization and compositionality.

The resulting research program

The five directions have distinct roles. Representation asks which targets admit economical regular compositions. Coordinates asks which feasible transformations or task-sufficient representations

expose them. Quantitative genericity asks whether useful statistics can be estimated and accessed. A staircase theorem explains one mechanism by which the learning dynamics can construct new coordinates. Selection asks whether training achieves a useful fit–complexity tradeoff and, under additional cross-task assumptions, discovers reusable computations.

The spectral distinctions remain

$$\boxed{\text{compositional sparsity} \neq \text{low Hermite degree} \neq \text{low leap}.}$$

The conditional-Hermite calculation shows how an earlier feature can expose a previously vanishing marginal statistic. The regularized objective calculation shows how a sufficiently favorable returned predictor can receive a population-risk certificate. These positive statements explain different links; neither supplies the entire learning theorem.

Known approximation, stability, and generalization results are the starting points. Restricted results about local interfaces, trace coverage, regularized dynamics, and reusable libraries supply additional test cases. Each proposed extension should identify precisely which assumptions are supplied, which quantities are learned, and which resources remain to be bounded.

For the main predictive argument, the order is

$$\boxed{\begin{array}{c}\text{feasible training achieves good fit and suitable complexity control}\\ \Longrightarrow \text{ small population error}\\ \Longrightarrow \text{ the returned computation is a compositional witness.}\end{array}}$$

The deeper mechanistic questions ask when that success is achieved by progressive feature discovery, and when the discovered functions make later tasks easier. A useful answer need not identify a unique computation of nature. It should establish a precise benefit of the learned representation under stated statistical and computational conditions.

CHAPTER 28

Sparse Digital Computation, Smooth Lifts, and the Cost of Contraction

A digital computer evaluates real-valued functions through finite bit strings and Boolean operations. At each numerical resolution, this gives a bounded-fan-in computation. The representation question is whether these digital computations are discretizations of a coherent collection of regular, low-arity real-valued constituent functions.

Two directions must be distinguished. Starting from a regular compositional real function, one can construct finite-precision Boolean approximations that preserve its coarse computation graph. Starting from a Boolean circuit, one can extend its gates to smooth real functions that agree exactly on Boolean codewords. Neither statement alone proves that an arbitrary family of digital computations has a smooth continuum limit.

Contraction sharpens this picture. Nonexpansive constituents prevent multiplicative error amplification. Strictly contractive constituents provide depth-independent control of absolute implementation error. But a nonconstant Boolean gate cannot be represented exactly by a strict contraction when the input and output use the same two codewords.

Exact computation can nevertheless be retained by allowing the codeword separation to shrink with depth. The price then moves from absolute error amplification to numerical resolution and decoding sensitivity:

contraction controls absolute error, *but preserving information also requires control of signal margin.*

The chapter develops this tradeoff through explicit theorems and proofs, retains the square-root counterexample to the unrestricted smooth-limit claim, and identifies the remaining representation questions.

28.1 The Objects and the Conventions

We use H for computational depth, s for the number of internal nodes, k for a fixed maximum local arity, n for requested numerical precision, and $r \geq 1$ for an integer Sobolev order. The symbol λ denotes a common Lipschitz bound; $0 < \lambda < 1$ denotes the contractive regime.

All finite-dimensional product spaces carry the maximum norm unless explicitly stated otherwise. Thus

$$\mathrm{Lip}_\infty(g) = \sup_{x \neq y} \frac{|g(x) - g(y)|}{\|x - y\|_\infty}.$$

For order j, $\|D^j g\|_\infty$ denotes the maximum over $\|D^\alpha g\|_\infty$ with $|\alpha| = j$. For a scalar function on a box Q, write

$$\|g\|_{W^{r,\infty}(Q)} = \max_{|\alpha|\le r} \|D^\alpha g\|_{L^\infty(Q)}.$$

Using a sum instead of a maximum changes only constants depending on the fixed local dimension and r. Sobolev functions are evaluated through their continuous representatives. On boxes, $W^{1,\infty}$ functions have Lipschitz representatives; for higher orders the corresponding $C^{r-1,1}$ regularity is understood. Here $C^\infty(\overline{Q})$ means restriction of a smooth function defined on a neighborhood of $\overline{Q}$.

Let G be a finite acyclic graph with ordered argument slots. Its internal node v computes

$$z_v = g_v\big((z_u)_{u\in \mathrm{pa}(v)}\big), \qquad |\,\mathrm{pa}(v)| \le k. \tag{28.1}$$

An argument may be reused. Outputs are designated node values, and $F = \Phi_G(\{g_v\})$ denotes the resulting input–output map.

28.1.1 Boolean gates and numerical modules

A Boolean gate has a fixed number of one-bit arguments. A numerical module can have k real arguments but require many bits per argument. Consequently,

$$\boxed{\text{bounded numerical arity} \neq \text{bounded bit length.}}$$

Replacing one numerical node by a Boolean subcircuit preserves the *coarse* graph, not its literal bit-level node count.

28.1.2 Encoding and decoding are part of the computation

A finite-precision realization has the form

$$F_n(x) = D_n\big(C_n(E_{p(n)}(x))\big). \tag{28.2}$$

Here E_p denotes a numerical quantization/encoding convention, C_n is a Boolean circuit, and D_n decodes its output. Endpoints, signs, and integer parts may require additional bits, which are included in the encoding cost.

The notation $E_p(x)$ is not a claim that exact binary digits can be continuously extracted from an arbitrary real input. In Type-2 computation an evaluator receives sufficiently accurate rational approximations to x and must tolerate their allowed errors. A modulus specifies how much input precision is needed [79].

We distinguish existence of a circuit at every accuracy from a uniform algorithm generating those circuits. A noncomputable real constant can appear in a smooth function, so regularity alone does not establish computability.

28.1.3 Local versus global contraction

The condition $\mathrm{Lip}_\infty(F) < 1$ on the final target is not the same as $\mathrm{Lip}_\infty(g_v) < 1$ at every node. A small global Lipschitz constant does not itself exhibit bounded-arity constituents. Conversely, local contraction must be interpreted together with all copy paths, encoders, decoders, and any fresh inputs injected during execution.

28.2 Error Propagation and the Three Lipschitz Regimes

The error-propagation argument is a finite-DAG version of the mechanism used in compositional approximation theory [109, 105]. We give it explicitly because it will be used in both directions of the digital–continuous comparison.

Suppose g_v has Lipschitz constant L_v. Let $\widehat{g}_v$ be an implementation whose local error, on every admissible argument tuple, is at most δ_v. The local error includes numerical evaluation and rounding. Require that all implementations return values in the prescribed output intervals. Clipping to those intervals does not enlarge error relative to a true output lying in the interval.

Lemma 8 (Nodewise error recursion)**.** *If e_v bounds the error in the value at node v, then*

$$e_v \le \delta_v + L_v \max_{u \in \mathrm{pa}(v)} e_u. \tag{28.3}$$

Proof. Let z_{pa} and $\widehat{z}_{\mathrm{pa}}$ be the true and implemented parent tuples. Insert $g_v(\widehat{z}_{\mathrm{pa}})$ and use the triangle inequality:

$$\begin{aligned} |\widehat{g}_v(\widehat{z}_{\mathrm{pa}}) - g_v(z_{\mathrm{pa}})| &\le |\widehat{g}_v(\widehat{z}_{\mathrm{pa}}) - g_v(\widehat{z}_{\mathrm{pa}})| + |g_v(\widehat{z}_{\mathrm{pa}}) - g_v(z_{\mathrm{pa}})| \\ &\le \delta_v + L_v \|\widehat{z}_{\mathrm{pa}} - z_{\mathrm{pa}}\|_\infty. \end{aligned}$$

The last norm is the maximum parent error. □

For input nodes set $A_v = 0$ and $M_v = 1$. At an internal node set

$$A_v = 1 + L_v \max_{u \in \mathrm{pa}(v)} A_u, \qquad M_v = L_v \max_{u \in \mathrm{pa}(v)} M_u. \tag{28.4}$$

For constant nodes the maximum over an empty parent set is zero. Let A_G and M_G be the maxima at the designated outputs.

Proposition 51 (Uniform local errors on a finite DAG)**.** *If input error is at most e_0 and $\delta_v \le \delta$, then*

$$\|\widehat{F} - F\|_\infty \le M_G e_0 + A_G \delta. \tag{28.5}$$

In particular, M_G is an input-sensitivity bound for F.

Proof. Induct on a topological ordering using $e_v \le M_v e_0 + A_v \delta$. Taking separate maxima of the two terms in parent errors gives exactly the recursions in Equation (28.4). □

Theorem 19 (Expansive, nonexpansive, and contractive composition)**.** *If $L_v \le \lambda$ and the maximum depth is H, then*

$$A_G \le \sum_{j=0}^{H-1} \lambda^j = \begin{cases} (\lambda^H - 1)/(\lambda - 1), & \lambda > 1, \\ H, & \lambda = 1, \\ (1 - \lambda^H)/(1 - \lambda), & 0 \le \lambda < 1. \end{cases} \tag{28.6}$$

Consequently, for exact inputs and $0 \le \lambda < 1$,

$$\boxed{\|\widehat{F} - F\|_\infty \le \frac{\delta}{1 - \lambda}.} \tag{28.7}$$

Proof. For a node of depth h, induction in Equation (28.4) bounds A_v by the first h terms of the geometric series. This bound is increasing in h for every $\lambda \ge 0$. Evaluate it at the maximum output depth. □

Thus strict contraction eliminates the depth factor from the bound on *absolute* accumulated local error. To guarantee final error ε, it is sufficient to take

$$\delta \le (1-\lambda)\varepsilon.$$

This is not yet a statement about recovering a discrete symbol: the output error must also be compared with the codeword separation.

28.2.1 A qualification about depth and skip connections

For a genuinely layered composition $G_H \circ \cdots \circ G_1$, with $\mathrm{Lip}_\infty(G_\ell) \le \lambda$, one has

$$\mathrm{Lip}_\infty(F) \le \lambda^H.$$

For an arbitrary DAG, *maximum* depth alone does not give this input-sensitivity bound. A short bypass can retain a stronger dependence on the input. The correct bound is M_G, or the maximum product of Lipschitz constants along an input–output path. In particular, an unscaled identity skip has Lipschitz constant one, not $\lambda < 1$. The absolute local-error estimate (28.7) still holds when all actual internal node maps obey the stated contraction bound.

28.3 From Regular Real Constituents to Boolean Approximation

Assume $g_v : [0,1]^{k_v} \to [0,1]$, $k_v \le k$, and $\|g_v\|_{W^{r,\infty}} \le B$, where $B \ge 1$ and r, k are fixed. The approximation principle is consistent with the Mhaskar–Poggio analysis: local dimension, not the full input dimension, controls the approximation exponent [109, 105]. To justify finite-bit costs, it is preferable to specify a quantizable local approximation rather than assume that an arbitrary real-parameter network automatically has a cheap digital implementation.

Lemma 9 (A quantizable local approximation)**.** *For $g : [0,1]^m \to [0,1]$, $m \le k$, with $\|g\|_{W^{r,\infty}} \le B$, there is a piecewise-polynomial approximation of uniform error at most δ using*

$$C_{k,r}\left(1+\frac{B}{\delta}\right)^{m/r}$$

coefficients. It can be evaluated on dyadic inputs by a bounded-fan-in Boolean circuit whose size is this coefficient count times a polynomial in the input word length and $\log(2+B/\delta)$.

Proof. Partition the cube into dyadic cells of side length $h = 2^{-J}$, with $C_{k,r}Bh^r \le \delta/3$. On each cell use a tensor product interpolant of degree at most $r-1$ in each coordinate; for $r = 1$ use a constant value. A fixed number of sample values, depending only on m, r, determines each cell polynomial.

For completeness, the local error estimate follows by applying the one-dimensional $W^{r,\infty}$ interpolation estimate successively in each coordinate. Each one-dimensional error is bounded by $C_r Bh^r$, and each fixed-degree interpolation operator has a bounded supremum-norm operator norm. The telescoping tensor-product estimate is therefore $C_{m,r}Bh^r$. These are standard local polynomial approximation estimates [22].

Use coordinates normalized to $[0,1]^m$ inside each cell. The fixed-degree basis and its coefficients have bounds independent of h; sample values are in $[0,1]$. Quantize the stored data finely enough to contribute at most $\delta/3$, and evaluate the fixed-degree polynomial to another $\delta/3$. The required arithmetic precision is $O_{k,r}(\log(2+B/\delta))$, together with the input word length.

A circuit selects a cell from the input address and evaluates the corresponding polynomial. Even a full multiplexing circuit has size at most the table size times a polynomial in word length. All arithmetic and selection can be decomposed into bounded-fan-in Boolean gates. Clipping the result to $[0,1]$ does not increase the error. □

Theorem 20 (Structure-preserving Boolean approximation)**.** *Let $F = \Phi_G(\{g_v\})$ satisfy the preceding assumptions, with d scalar inputs and s internal nodes. Put $\overline{A} = \max\{1, A_G\}$ and $\Gamma = 1 + A_G + M_G$. For every $0 < \varepsilon < 1$, there is a bounded-fan-in Boolean circuit C_ε, partitioned according to the same coarse DAG, such that its decoded numerical output satisfies*

$$\|F_\varepsilon - F\|_\infty \le \varepsilon. \tag{28.8}$$

For constants depending only on k, r, one may achieve

$$|C_\varepsilon| \le C_{k,r}\left[d + s\left(1 + \frac{B\overline{A}}{\varepsilon}\right)^{k/r}\right]\left[1 + \log\frac{(2+B)\Gamma}{\varepsilon}\right]^{c_{k,r}}. \tag{28.9}$$

The existence statement is nonuniform. A uniform construction additionally requires uniformly computable constituent evaluations and effective bounds on the constants used.

Proof. Choose local total error $\delta = \varepsilon/(3\overline{A})$ and encode inputs within $\varepsilon/(3\max\{1, M_G\})$ in maximum norm. Use the local digital construction at each node, including output rounding in its error budget, and keep every output in $[0,1]$.

The input contribution in (28.5) is at most $\varepsilon/3$, and the propagated local contribution is at most $A_G\delta \le \varepsilon/3$. The remaining slack covers a final decoding rounding if required. This proves (28.8).

There are at most $C_{k,r}(1 + B\overline{A}/\varepsilon)^{k/r}$ local coefficients per node. The needed word length is bounded by a constant times $1 + \log((2+B)\Gamma/\varepsilon)$. Multiplication, addition, comparison, and selection have polynomial bit-circuit cost. Summing over the nodes, and including the input encoding length, gives (28.9).

For a uniform construction, the sample values defining the cell interpolants can be computed to the required accuracy. For a noncomputable constituent, the finite rounded tables still exist but need not be generable by an algorithm. □

28.3.1 The contractive improvement

If every local Lipschitz constant is at most $\lambda < 1$, then $A_G \le (1-\lambda)^{-1}$ and $M_G \le 1$. The leading local approximation cost becomes

$$s\left(1 + \frac{B}{(1-\lambda)\varepsilon}\right)^{k/r}, \tag{28.10}$$

up to the displayed logarithmic factors and input-reading cost. There is no depth factor from approximation-error propagation, although there are still s modules to execute and store.

For the normalized class with $\sum_i \|\partial_i g_v\|_\infty \le 1$, the maximum-norm Lipschitz constant is at most one. This gives $A_G \le H$, recovering the nonexpansive estimate underlying the earlier discussion.

28.3.2 Approximation rate is not polynomial-time bit complexity

With $\varepsilon = 2^{-n}$, the factor $\varepsilon^{-k/r}$ is exponential in n. The theorem is an approximation-size statement, not a claim that every Sobolev constituent can be evaluated in time polynomial in the requested number of bits.

If separate uniform polynomial-time algorithms for the constituents are available, one can compile those algorithms instead. The required intermediate absolute precision has the form

$$p_v = n + O(\log A_v^{\mathrm{downstream}} + \log s),$$

with an error allocation appropriate to the DAG. Polynomial graph size, polynomial local evaluation costs, and polynomial required bit lengths then yield polynomial digital cost. An amplification factor exponential in depth can require only linearly many *additional bits*, even though it can make a generic approximation formula expensive.

28.4 Exact Real Extensions of Boolean Gates

Let $b : \{0,1\}^m \to \{0,1\}$, with $m \le k$. A real lift is a function on $[0,1]^m$ that agrees with b at the vertices. Such an extension is not unique and does not by itself specify the numerical meaning of intermediate non-Boolean values.

28.4.1 The smooth multilinear lift

Define

$$\widetilde{b}(x) = \sum_{a \in \{0,1\}^m} b(a) \prod_{i:a_i=1} x_i \prod_{i:a_i=0} (1 - x_i). \tag{28.11}$$

Proposition 52 (Exact polynomial extension)**.** *The function $\widetilde{b}$ is a polynomial, affine in each argument separately, with total degree at most m. It satisfies*

$$\widetilde{b}(a) = b(a), \qquad 0 \le \widetilde{b}(x) \le 1, \qquad \mathrm{Lip}_\infty(\widetilde{b}) \le m.$$

For every fixed integer $r \ge 1$, $\|\widetilde{b}\|_{W^{r,\infty}} \le C_{k,r}$.

Proof. At a vertex only its own basis product equals one; all other products vanish. On the cube these products are nonnegative and sum to one, so the range is in $[0,1]$.

For a set I of j distinct coordinates, $D_I\widetilde{b}$ is the average, over the other coordinates, of an alternating sum of 2^j Boolean values. Its absolute value is at most 2^{j-1} for $j \ge 1$. Derivatives with a repeated coordinate vanish. In particular, $|\partial_i \widetilde{b}| \le 1$. Integrating the gradient along a segment gives $|\widetilde{b}(x) - \widetilde{b}(y)| \le \|x - y\|_1 \le m\|x - y\|_\infty$, and the finite collection of derivative bounds proves the Sobolev estimate. □

Theorem 21 (Smooth lifting on the same Boolean DAG)**.** *Replace every Boolean constituent b_v in a finite DAG by $\widetilde{b}_v$. The resulting smooth real composition satisfies*

$$F_G(a) = f(a) \qquad (a \in \{0,1\}^d)$$

and has the same coarse graph and bounded local Sobolev norms.

Proof. On a Boolean input, all parent values of the first gates are Boolean. Their lifted outputs equal the Boolean gate outputs. Induction in a topological ordering gives the assertion at every node. □

The composition of local multilinear lifts need not equal the global multilinear interpolation of the truth table. For example, the circuit $\mathrm{XOR}(x,x)$ computes the constant zero on Boolean inputs, but its local polynomial lift is $2x(1-x)$. Sharing an input creates correlations that a gatewise probabilistic interpolation does not preserve away from the vertices.

28.4.2 An optimal Lipschitz lift

A second extension emphasizes stability rather than higher smoothness. For a nonconstant b, let $A_0 = \{a : b(a) = 0\}$ and set

$$B_b(x) = \min_{a \in A_0} \|x - a\|_\infty, \qquad x \in [0,1]^m. \tag{28.12}$$

For constant gates use the corresponding constant function.

Proposition 53 (Nonexpansive exact extension)**.** *The function B_b is 1-Lipschitz, has values in $[0,1]$, and satisfies $B_b(a) = b(a)$ at every Boolean vertex. It is piecewise affine and admits a min–max implementation of size depending only on k.*

Proof. Distance to a nonempty set is 1-Lipschitz. At a zero vertex the distance is zero. At a one vertex its maximum-norm distance to every zero vertex is one. Distances between points in the cube are at most one.

On the cube, $\|x - a\|_\infty$ is the maximum of the affine functions x_i or $1 - x_i$, according to a_i. The finite minimum in (28.12) is therefore piecewise affine and uses at most 2^k such maxima. □

This is a special explicit form of the Lipschitz extension principle associated with McShane [103]. More generally,

$$\min\left\{1, \max\left\{0, \inf_{a \in \{0,1\}^m} [b(a) + \|x - a\|_\infty]\right\}\right\}$$

gives a bounded 1-Lipschitz extension on all of $\mathbb{R}^m$. Gatewise use of these lifts gives a globally nonexpansive real circuit, but not necessarily a smooth one.

28.5 Smooth Lifts Arbitrarily Close to Nonexpansive

Smoothness and the exactly optimal Lipschitz constant can conflict. Nevertheless, any strictly positive Lipschitz slack suffices.

Lemma 10 (Smooth saturation with small Lipschitz slack)**.** *For $0 < a < 1/4$, there is $S_a \in C^\infty(\mathbb{R}; [0,1])$ such that $S_a(t) = 0$ for $t \le 0$, $S_a(t) = 1$ for $t \ge 1$, and*

$$\|S_a'\|_\infty \le (1 - 2a)^{-1}, \qquad \|S_a^{(j)}\|_\infty \le C_j a^{1-j} \quad (j \ge 1).$$

Proof. Choose a fixed smooth step function χ taking values in $[0,1]$, equal to zero on $(-\infty, 0]$ and to one on $[1, \infty)$. Set

$$w_a(t) = \chi(t/a)\chi((1-t)/a), \quad I_a = \int_{\mathbb{R}} w_a(t)\,dt, \quad S_a(t) = I_a^{-1} \int_{-\infty}^{t} w_a(u)\,du.$$

Then $1 - 2a \le I_a \le 1$. Differentiating proves the stated bounds and the endpoint identities. □

Theorem 22 (Almost-nonexpansive smooth extension)**.** *For every Boolean gate $b : \{0,1\}^m \to \{0,1\}$, $m \le k$, and $0 < \eta \le 1$, there is $g_\eta \in C^\infty(\mathbb{R}^m; [0,1])$ with*

$$g_\eta(a) = b(a), \qquad \mathrm{Lip}_\infty(g_\eta) \le 1 + \eta, \qquad \|g_\eta\|_{W^{r,\infty}([0,1]^m)} \le C_{k,r}\eta^{1-r} \tag{28.13}$$

for every fixed integer $r \ge 1$. The same construction works for all r; only the constants depend on r.

Proof. Take a bounded 1-Lipschitz extension B on $\mathbb{R}^m$ as above. Let ρ be a nonnegative smooth mollifier supported in the unit maximum-norm ball, with integral one, and put $h_\delta = B * \rho_\delta$. Then

$$0 \le h_\delta \le 1, \qquad \mathrm{Lip}_\infty(h_\delta) \le 1, \qquad \|h_\delta - B\|_\infty \le \delta.$$

For $|\alpha| = j \ge 1$, the zero integral of $D^\alpha \rho_\delta$ gives

$$D^\alpha h_\delta(x) = \int [B(x-y) - B(x)] D^\alpha \rho_\delta(y)\, dy,$$

so $\|D^\alpha h_\delta\|_\infty \le C_{k,j}\delta^{1-j}$.

Choose fixed smooth bumps ψ_a centered at the vertices, with disjoint supports, $\psi_a(a) = 1$, and support radius smaller than $1/3$. Let

$$h_1 = h_\delta + \sum_a [b(a) - h_\delta(a)]\psi_a.$$

This exactly repairs vertex values. Since each coefficient has magnitude at most δ, there is a constant C_k such that

$$\mathrm{Lip}_\infty(h_1) \le 1 + C_k\delta, \qquad \|D^\alpha h_1\|_\infty \le C_{k,j}\delta^{1-j}.$$

The latter estimate also bounds the bump corrections because their supports are fixed and $0 < \delta \le 1$.

Set $a = \eta/64$ and choose $\delta = c_k\eta$ so that $C_k\delta \le \eta/64$. Define $g_\eta = S_a \circ h_1$. Its range is $[0,1]$, and it preserves the repaired vertex values. Moreover,

$$\mathrm{Lip}_\infty(g_\eta) \le (1 + \eta/64)(1 - \eta/32)^{-1} \le 1 + \eta \qquad (0 < \eta \le 1).$$

In an order-j derivative of this composition, each term has the form $S_a^{(q)}(h_1)\prod_{i=1}^q D^{\alpha_i} h_1$, with $\sum_i |\alpha_i| = j$ and all $|\alpha_i| \ge 1$. Its bound is

$$C_{k,j}\eta^{1-q}\prod_i \eta^{1-|\alpha_i|} = C_{k,j}\eta^{1-j}.$$

Summing the finitely many chain-rule terms and including the bounded zeroth derivative proves (28.13). □

The restriction $\eta \le 1$ matters for the displayed Sobolev estimate. For arbitrary $\eta > 0$, replace η there by $\min\{\eta, 1\}$.

28.5.1 Why exact nonexpansive smooth interpolation can fail

Proposition 54 (A smoothness obstruction for XOR)**.** *There is no C^1 extension of the two-input XOR truth table on the square with maximum-norm Lipschitz constant at most one.*

Proof. An exact 1-Lipschitz extension must satisfy $g(t,0) = t$ and $g(0,t) = t$: each edge joins values zero and one at unit distance, so the Lipschitz inequalities from its two endpoints force equality. Differentiability at $(0,0)$ therefore gives $\partial_1 g(0,0) = \partial_2 g(0,0) = 1$. Consequently $g(t,t) = 2t + o(t)$, contradicting $|g(t,t) - g(0,0)| \le t$. □

Theorem 23 (Depth-stable smooth Boolean lift)**.** *A Boolean DAG of maximum depth $H \ge 1$ and local arity at most k has, for every $0 < c \le 1$, an exact smooth lift on the same coarse graph with*

$$\mathrm{Lip}_\infty(F_G) \le e^c, \qquad \|g_v\|_{W^{r,\infty}} \le C_{k,r}(H/c)^{r-1}.$$

Proof. Use $g_{v,\eta}$ with $\eta = c/H$. Along a path of length $h \le H$, the input-sensitivity multiplier is at most $(1 + c/H)^h \le e^c$. The local derivative estimate follows from Theorem 22. Agreement on Boolean inputs follows by topological induction. □

Corollary 3 (Robustness near fixed Boolean codewords)**.** *For the preceding lift, $a \in \{0,1\}^d$ and $\|x - a\|_\infty \le \tau$ imply*

$$|F_G(x) - f(a)| \le e^c \tau.$$

Thresholding at $1/2$ is correct when $e^c\tau < 1/2$.

Proof. Use $F_G(a) = f(a)$ and the global Lipschitz bound. □

28.6 Strict Contraction with Fixed Boolean Codes

Suppose the same two numbers, zero and one, represent the Boolean symbols at every gate. Strict contraction is then incompatible with exact nonconstant Boolean semantics.

Theorem 24 (Fixed-code obstruction and sharp approximation error)**.** *Let $b : \{0,1\}^m \to \{0,1\}$ be nonconstant. Every exact real extension satisfies*

$$\mathrm{Lip}_\infty(g) \ge 1.$$

For $0 \le \lambda \le 1$, the optimal vertex approximation error among λ-Lipschitz maps $g : [0,1]^m \to [0,1]$ is exactly

$$\boxed{\inf_{\mathrm{Lip}_\infty(g) \le \lambda} \max_{a \in \{0,1\}^m} |g(a) - b(a)| = \frac{1-\lambda}{2}.} \tag{28.14}$$

Proof. Choose vertices a, a' with $b(a) = 0$ and $b(a') = 1$. Their maximum-norm distance is one. Exact agreement would require $1 \le \mathrm{Lip}_\infty(g)$.

If the maximum vertex error is ε, then

$$1 - 2\varepsilon \le g(a') - g(a) \le \lambda,$$

so $\varepsilon \ge (1-\lambda)/2$. For the matching construction take the nonexpansive exact lift B_b and set

$$g_\lambda(x) = \lambda B_b(x) + \frac{1-\lambda}{2}.$$

It is λ-Lipschitz. Its vertex values are $(1-\lambda)/2$ and $(1+\lambda)/2$, giving equality in the bound. □

This theorem concerns approximation to the numerical values zero and one. For $\lambda > 0$, thresholding the two approximate output levels at $1/2$ still classifies the gate correctly. But the output levels are not the original input codes. Feeding them into another gate without specifying its encoding or a restoration operation is not justified by the theorem. A restoration operation must itself be included in the analysis.

28.6.1 The smooth version has the same infimum

Corollary 4 (Smooth approximation under a fixed contraction budget)**.** *The infimum in* (28.14) *is unchanged if the admissible maps must be C^∞ on a neighborhood of the cube. It need not be attained by a smooth map.*

Proof. For $\eta > 0$, set $a_\eta = \lambda/(1+\eta)$ and use

$$g_{\lambda,\eta}(x) = a_\eta g_\eta(x) + \frac{1 - a_\eta}{2},$$

where g_η is an exact smooth lift from Theorem 22. This map has Lipschitz constant at most λ and vertex error $(1 - a_\eta)/2$, tending to $(1-\lambda)/2$.

For XOR and $\lambda > 0$, attainment would force the vertex values to be exactly $(1-\lambda)/2$ and $(1+\lambda)/2$. Subtracting the first value and dividing by λ would produce a smooth exact 1-Lipschitz XOR extension, contradicting Proposition 54. For $\lambda = 0$, the constant $1/2$ attains the infimum. □

The distinction between an attained Lipschitz optimum and a smooth infimum is important. Smooth exact fixed-code lifting generally requires a Lipschitz constant slightly above one; smooth contractive approximate lifting approaches the sharp vertex error without always attaining it.

28.7 Exact Contractive Computation with Changing Code Scales

The fixed-code obstruction disappears when representation scale is allowed to change. At level ℓ, encode Boolean zero by zero and Boolean one by $\Delta_\ell > 0$. The number Δ_ℓ is both the codeword separation and the available absolute output margin scale.

Theorem 25 (Contraction forces separation loss in a layered code)**.** *Suppose a nonconstant Boolean gate takes inputs in $\{0, \Delta_\ell\}^m$, returns exact outputs in $\{0, \Delta_{\ell+1}\}$, and has Lipschitz constant at most $0 < \lambda < 1$ on the containing box. Then*

$$\Delta_{\ell+1} \le \lambda \Delta_\ell. \tag{28.15}$$

For an H-stage layered computation with a common code scale at each level and a nonconstant encoded input–output map,

$$\boxed{\Delta_H \le \lambda^H \Delta_0.} \tag{28.16}$$

Proof. Two encoded input tuples on which the gate outputs differ have distance Δ_ℓ, while their outputs have distance $\Delta_{\ell+1}$. The Lipschitz inequality proves (28.15).

Alternatively, the composition of the H complete layers is λ^H-Lipschitz. Choose two Boolean input vectors whose final bits differ. Their encoded input distance is Δ_0 and their output distance is Δ_H, proving (28.16). □

The global statement requires H actual contractive stages along the information flow. It is not a bound based only on the maximum depth of an arbitrary DAG containing bypasses. Constant outputs carry no two-symbol separation requirement.

28.7.1 An exact Lipschitz construction

Theorem 26 (Contractive realization on the original DAG)**.** *Let a Boolean circuit have local arity at most k and depth function $h(v)$, with input depth zero and output depth at most H. Fix $0 < \lambda < 1$ and $\Delta_0 > 0$. Set*

$$\Delta_v = \Delta_0 \lambda^{h(v)}.$$

There are real local maps, all λ-Lipschitz, on the same DAG that exactly simulate every Boolean gate using its node-dependent codes $\{0, \Delta_v\}$. An output of depth H has separation $\Delta_0 \lambda^H$.

Proof. For each gate take its exact nonexpansive lift B_{b_v} and define

$$g_v((x_u)_u) = \Delta_v B_{b_v}((x_u/\Delta_u)_u), \qquad x_u \in [0, \Delta_u]. \tag{28.17}$$

At the encoded Boolean tuple $x_u = \Delta_u a_u$, the output is exactly $\Delta_v b_v((a_u)_u)$. Induction gives exact circuit simulation.

Because $h(v) = 1 + \max_{u \in \mathrm{pa}(v)} h(u)$,

$$\min_{u \in \mathrm{pa}(v)} \Delta_u = \Delta_0 \lambda^{h(v)-1}.$$

The Lipschitz constant of (28.17) is consequently at most $\Delta_v / \min_u \Delta_u = \lambda$. Constant gates have Lipschitz constant zero. □

For a strictly layered graph this reduces to the simple formula

$$g_\ell(x) = \Delta_{\ell+1} B_b(x/\Delta_\ell), \qquad \Delta_{\ell+1} = \lambda \Delta_\ell.$$

The original Boolean information is intact at every finite depth, even though its amplitude shrinks. For dyadic λ, the min–max lifts and the scales are described with finite rational data. Polynomial-depth circuits require only polynomially many bits to write these scales in fixed-point form. This is an execution construction, not a result about learning the circuit.

28.7.2 Smooth contractive realization: a small slack is needed

Using a C^∞ lift with Lipschitz constant $1 + \eta$ instead of a nonsmooth nonexpansive lift requires a slightly faster scale decrease.

Theorem 27 (Smooth contractive simulation with near-optimal separation)**.** *For a Boolean DAG of maximum depth $H \geq 1$, choose $0 < c \leq 1$ and*

$$\eta = \frac{c}{H}, \qquad q = \frac{\lambda}{1+\eta}, \qquad \Delta_v = \Delta_0 q^{h(v)}.$$

There is an exact smooth real realization on the same coarse graph with $\mathrm{Lip}_\infty(g_v) \leq \lambda$. At an output of depth H,

$$e^{-c} \Delta_0 \lambda^H \leq \Delta_H = \Delta_0 q^H \leq \Delta_0 \lambda^H. \tag{28.18}$$

Thus smoothness preserves the optimal layered separation scale up to a constant factor independent of H.

Proof. Replace B_{b_v} in (28.17) by the exact smooth lift $g_{b_v,\eta}$ and use the scales based on q. The local Lipschitz constant is at most $q(1+\eta) = \lambda$, and agreement on encoded Boolean values is exact. Finally,

$$\Delta_H = \Delta_0 \lambda^H (1 + c/H)^{-H}$$

and $1 \leq (1 + c/H)^H \leq e^c$. □

Unlike the Lipschitz construction, exact saturation of the separation bound is not asserted for every smooth Boolean gate. A smooth XOR gate would encounter the same obstruction as before if rescaling it produced an exact 1-Lipschitz fixed-code lift.

28.7.3 Higher derivatives in shrinking physical coordinates

For a gate at depth ℓ whose parents share scale $\Delta_{\ell-1}$,

$$\|D^j g_\ell\|_\infty \le C_{k,j}\Delta_\ell \Delta_{\ell-1}^{-j}\eta^{1-j} = C_{k,j} q \Delta_{\ell-1}^{1-j}\eta^{1-j}. \tag{28.19}$$

For unequal parent scales, the same upper bound uses the smallest parent scale. Higher derivatives can therefore grow exponentially with depth when $j \ge 2$, despite the first-order contraction bound. On rescaled unit boxes the normalized gate has the polynomial-in-H bound from Theorem 22.

Thus first-order contractivity and uniform higher-order Sobolev control are different properties. The cost must be measured in declared coordinates and units.

28.8 Absolute Error, Relative Error, and Required Precision

The bound $e_H \le \delta/(1-\lambda)$ is favorable in absolute units. But an exact Boolean output at depth H is represented by either zero or Δ_H. Correct threshold decoding requires

$$e_H < \frac{\Delta_H}{2}. \tag{28.20}$$

Absolute contraction alone does not ensure this inequality.

Proposition 55 (A sufficient finite-precision budget)**.** *Assume exact encoded inputs, local maps with Lipschitz constants at most $\lambda < 1$, and local total implementation error at most δ. A sufficient condition for correct final threshold decoding is*

$$\delta \le \frac{(1-\lambda)\Delta_H}{4}. \tag{28.21}$$

If the implementation provides $\delta \le 2^{-p}$, it suffices that

$$p \ge \left\lceil \log_2 \frac{4}{(1-\lambda)\Delta_H} \right\rceil. \tag{28.22}$$

For the exact Lipschitz scaling $\Delta_H = \Delta_0\lambda^H$, this is

$$p \ge \left\lceil H \log_2 \frac{1}{\lambda} + \log_2 \frac{4}{(1-\lambda)\Delta_0} \right\rceil. \tag{28.23}$$

Proof. The absolute error is bounded by $\delta/(1-\lambda) \le \Delta_H/4$, strictly below the threshold distance $\Delta_H/2$. Substitute the claimed numerical error and solve for p. □

Here δ is the error of an entire local numerical module, not necessarily the error of one elementary arithmetic operation. Internal guard bits or a verified local evaluation scheme are needed to obtain that module-level guarantee. Clipping approximate states to their admissible boxes can be used without worsening error.

For the smooth scaling in (28.18), the sufficient bit budget differs by at most $c\log_2 e$ plus rounding constants. The leading depth dependence remains linear for fixed λ. For example, $\lambda = 1/2$, $\Delta_0 = 1$, and $\Delta_H = 2^{-H}$ give the sufficient local error $2^{-(H+3)}$.

28.8.1 Normalized error reveals the tradeoff

Let $r_\ell = e_\ell/\Delta_\ell$ be error measured in local code units. For a layered computation,

$$r_{\ell+1} \le \lambda \frac{\Delta_\ell}{\Delta_{\ell+1}} r_\ell + \frac{\delta_{\ell+1}}{\Delta_{\ell+1}}. \tag{28.24}$$

For the Lipschitz construction $\Delta_{\ell+1} = \lambda\Delta_\ell$, the first multiplier is one, not λ. Therefore

$$r_H \le r_0 + \sum_{\ell=1}^{H} \frac{\delta_\ell}{\Delta_\ell}.$$

Contraction in absolute units does not contract the error relative to the shrinking Boolean margin. In the smooth construction, that multiplier is $1+\eta$, whose product is controlled by choosing $\eta = c/H$.

28.8.2 The final decoder restores the scale

If an encoded computation returns $z_H \in \{0, \Delta_H\}$, the numerical Boolean decoder is

$$D_H(z) = z/\Delta_H$$

on these codewords. Its Lipschitz constant is Δ_H^{-1}. Threshold decoding is discontinuous on the full real line and has noise tolerance $\Delta_H/2$ around the codewords.

Thus a contractive internal realization does not make the complete fixed-code input–output map contractive. The decoder compensates for the amplitude loss. Repeatedly restoring a fixed separation during the computation similarly requires expansion or a discontinuous decision at some stage.

28.8.3 What the bit budget does and does not mean

Equation (28.23) is a *fixed absolute precision* guarantee. A uniform fixed-point mesh that must robustly resolve scale Δ_H needs mesh width smaller than that scale. Similarly, two disjoint noise balls around the output codewords require noise radius less than $\Delta_H/2$.

This is not an information-theoretic assertion that one Boolean symbol requires H storage bits. The set $\{0, \Delta_H\}$ contains only two possibilities. A symbolic representation can store a Boolean symbol along with a shared scale convention. A floating-point representation can encode a small amplitude using an exponent; for dyadic λ, that exponent may require only $O(\log H)$ bits.

Such representations change the storage and arithmetic accounting, not the absolute noise tolerance of the real-valued code. The precise conclusion is

> strict contraction can preserve exact finite-depth information,
> but a shrinking real code requires shrinking absolute error
> or an explicitly accounted rescaling/decoding mechanism.

28.9 Contractivization by Rescaling Is Not a Free Structural Result

The changing-code construction is part of a more general rescaling phenomenon. It is important not to mistake a choice of units for an intrinsic reduction in computational complexity.

Proposition 56 (Contractivizing a finite Lipschitz DAG)**.** *Let $g_v : [0,1]^{k_v} \to [0,1]$ be the constituents of any finite DAG, with finite Lipschitz constants L_v. Given $0 < \lambda < 1$, choose input scales $a_u = 1$ and internal scales recursively as*

$$a_v = \frac{\lambda \min_{u \in \mathrm{pa}(v)} a_u}{\max\{1, L_v\}}.$$

Constant nodes can be assigned any positive scale. The scaled functions

$$\widehat{g}_v((x_u)_u) = a_v g_v((x_u/a_u)_u)$$

exactly realize the same computation in rescaled coordinates and have Lipschitz constants at most λ.

Proof. The rescaling relation is preserved node by node. Moreover,

$$\mathrm{Lip}_\infty(\widehat{g}_v) \le \frac{a_v L_v}{\min_u a_u} \le \lambda.$$

The final unscaled output is obtained by dividing by its output scale. □

A restriction to local contractions becomes a substantive representational condition only when the coordinate scales, decoder sensitivity, numerical precision, or noise tolerance are also constrained. Otherwise local contraction can be purchased by making the output scale extremely small.

A globally contractive target is a different assumption. For any bounded Lipschitz target f, multiplication by a sufficiently small scalar makes it contractive without revealing its hidden constituents. Thus global contraction alone does not solve the bounded-arity representation problem.

28.10 Continuum Limits on a Coherent Coarse DAG

The precision index n and the execution depth H are distinct. In this section G is a fixed coarse graph and $n \to \infty$ refines the numerical resolution of each of its modules. The domains and numerical units are fixed across n.

Theorem 28 (Fixed-DAG Sobolev continuum limit)**.** *Suppose $g_{v,n} : [0,1]^{k_v} \to [0,1]$ satisfy, for all nodes and n,*

$$\|g_{v,n+1} - g_{v,n}\|_\infty \le C2^{-n}, \qquad \|g_{v,n}\|_{W^{r,\infty}} \le B. \tag{28.25}$$

Then there are $g_v \in W^{r,\infty}$ with

$$\|g_v\|_{W^{r,\infty}} \le B, \qquad \|g_{v,n} - g_v\|_\infty \le 2C2^{-n}.$$

If $F_n = \Phi_G(\{g_{v,n}\})$ and $F = \Phi_G(\{g_v\})$, then

$$\|F_n - F\|_\infty \le 2CA_G 2^{-n}, \tag{28.26}$$

where A_G is computed from Lipschitz constants of the limiting maps.

Proof. Summing the cross-resolution differences shows that each sequence is uniformly Cauchy, with tail at most $2C2^{-n}$. The common Sobolev bound bounds every derivative through order r in L^∞.

There are finitely many nodes and derivative multi-indices. By weak-star compactness, extract a subsequence on which each derivative converges weak-star in L^∞. For a test function $\varphi \in C_c^\infty((0,1)^{k_v})$,

$$\int g_{v,n} D^\alpha \varphi = (-1)^{|\alpha|} \int D^\alpha g_{v,n}\, \varphi.$$

Uniform convergence on the left identifies the weak-star limit on the right as $D^\alpha g_v$. Its essential-supremum bound is at most B. This proves the stated Sobolev regularity and norm bound.

Apply the nodewise error recursion with $\delta_v = \|g_{v,n} - g_v\|_\infty$ and exact initial inputs. The same fixed finite DAG has a finite A_G, yielding (28.26). □

This proves a $W^{r,\infty}$ limit, not automatically a C^∞ limit. To obtain smoothness of every order, corresponding estimates at every order or other stronger assumptions are required.

Corollary 5 (Uniform contraction gives a depth-independent limit bound)**.** *Instead of higher-order bounds, suppose the local maps satisfy* $\mathrm{Lip}_\infty(g_{v,n}) \le \lambda < 1$*, the same range bound, and the first condition in* (28.25)*. Then*

$$\mathrm{Lip}_\infty(g_v) \le \lambda, \qquad \|F_n - F\|_\infty \le \frac{2C2^{-n}}{1-\lambda}.$$

In particular, the local limits lie in $W^{1,\infty}$*.*

Proof. The uniform Cauchy argument gives g_v. Pass to the limit in $|g_{v,n}(x) - g_{v,n}(y)| \le \lambda\|x-y\|_\infty$, and then apply Theorem 19. □

The conclusion concerns fixed domains and scales. If code separation shrinks while an increasingly sensitive decoder is introduced, a bound on the raw encoded maps does not automatically control the decoded numerical limit.

28.11 Regularity Conditions Stated Directly on Digital Tables

Let $h_n = 2^{-n}$ and $Q_n = \{0, h_n, 2h_n, \dots, 1\}^m$. A local Boolean numerical module produces a dyadic-valued table $T_n : Q_n \to [0,1]$. Its output precision need not equal its input mesh spacing. Let $\overline{T}_n$ be its piecewise-constant extension using a fixed lower-corner quantization rule, with an endpoint convention included.

28.11.1 A digital contraction condition allowing quantization error

Proposition 57 (Contraction from a discrete two-point estimate)**.** *Suppose* $\overline{T}_n$ *converges uniformly to* g*, and for all grid pairs* $a, b \in Q_n$*,*

$$|T_n(a) - T_n(b)| \le \lambda\|a-b\|_\infty + \tau_n, \qquad \tau_n \to 0. \tag{28.27}$$

Then g *has Lipschitz constant at most* λ*. In fact, there are* λ*-Lipschitz continuous functions* $\widetilde{T}_n : [0,1]^m \to [0,1]$ *such that*

$$\|\widetilde{T}_n - \overline{T}_n\|_\infty \le \tau_n + \lambda h_n. \tag{28.28}$$

Proof. Set

$$\widetilde{T}_n(x) = \mathrm{clip}_{[0,1]} \inf_{a \in Q_n} \{T_n(a) + \lambda\|x-a\|_\infty\}.$$

This is λ-Lipschitz. At a grid point b, the term $a = b$ gives an upper bound $T_n(b)$, while (28.27) gives a lower bound $T_n(b) - \tau_n$. Clipping preserves these approximation bounds because $T_n(b) \in [0,1]$.

For a general x, choose its quantized point b with $\|x-b\|_\infty \le h_n$. Lipschitz continuity then proves (28.28). Thus $\widetilde{T}_n \to g$ uniformly, and its common Lipschitz inequality passes to the limit. □

Conversely, if g is λ-Lipschitz and $|T_n(a) - g(a)| \le q_n$ at grid points, then

$$|T_n(a) - T_n(b)| \le \lambda \|a - b\|_\infty + 2q_n, \quad \|\overline{T}_n - g\|_\infty \le \lambda h_n + q_n.$$

This supplies an exact first-order bridge between analytic regularity and digital stability with quantization slack. It does not establish that checking every grid pair is computationally cheap.

28.11.2 Higher-order finite differences

Define normalized forward and backward differences by

$$D^+_{h,i}u(x) = \frac{u(x + he_i) - u(x)}{h}, \qquad D^-_{h,i}\varphi(x) = \frac{\varphi(x) - \varphi(x - he_i)}{h}.$$

Products of these operators define $(D^+_h)^\alpha$ and $(D^-_h)^\alpha$. Write $\Delta_{h,i}u(x) = u(x + he_i) - u(x)$ for an unnormalized forward difference, so that $(D^+_h)^\alpha = h^{-|\alpha|}\Delta^\alpha_h$. The signs and normalizations matter in summation by parts.

Theorem 29 (Discrete Sobolev regularity of the continuum limit)**.** *Assume $T_n : Q_n \to [0, 1]$ satisfies*

$$\|\overline{T}_{n+1} - \overline{T}_n\|_\infty \le Ch_n$$

and, for every $1 \le |\alpha| \le r$,

$$\left|h_n^{-|\alpha|}\Delta^\alpha_{h_n} T_n(a)\right| \le B_\alpha \tag{28.29}$$

at all grid points where the forward stencil is defined. Then $\overline{T}_n$ converges uniformly to a continuous function g, and

$$g \in W^{r,\infty}((0, 1)^m), \qquad \|D^\alpha g\|_\infty \le B_\alpha.$$

Its continuous representative extends to the closed cube.

Proof. The compatibility condition makes $\overline{T}_n$ uniformly Cauchy, so it has a uniform limit g. First-order differences imply, by summing along coordinate grid paths,

$$|\overline{T}_n(x) - \overline{T}_n(y)| \le \sum_{i=1}^m B_{e_i}(|x_i - y_i| + h_n).$$

Passing to the limit proves continuity and a Lipschitz bound for g.

Fix $\varphi \in C^\infty_c((0, 1)^m)$. For large n, every shift needed by the stencil stays away from the boundary. Translating variables in Lebesgue integrals gives the exact identity

$$\int \overline{T}_n (D^-_{h_n})^\alpha \varphi = (-1)^{|\alpha|} \int (D^+_{h_n})^\alpha \overline{T}_n \, \varphi. \tag{28.30}$$

On each interior cell, the forward difference field is exactly the normalized grid difference in (28.29). It is bounded by B_α on the support relevant to the integral. The left side converges to $\int g D^\alpha \varphi$ because $(D^-_{h_n})^\alpha \varphi \to D^\alpha \varphi$ uniformly.

Consequently,

$$\left|\int g D^\alpha \varphi\right| \le B_\alpha \|\varphi\|_{L^1}.$$

The distribution $D^\alpha g$ therefore extends to a bounded functional on L^1, and is represented by an L^∞ function of norm at most B_α. This proves the asserted Sobolev regularity at every order through r. □

This proof does not assert that a piecewise-constant interpolant itself lies in $W^{r,\infty}$. The regularity is a property of its limit. Applied node by node, the resulting continuous local limits and their uniform error bounds can be composed using Lemma 8.

28.11.3 Output quantization must respect derivative order

Suppose exact samples of a $W^{r,\infty}$ function are rounded with maximum value error q_n. Their order-j finite differences can change by as much as $2^j q_n$. After division by h_n^j, this becomes $2^j q_n / h_n^j$.

Thus a sufficient output precision for preserving uniformly bounded finite differences through order r is

$$q_n = O(h_n^r), \qquad \text{or } p_{\text{out}}(n) \geq rn + O(1). \tag{28.31}$$

Rounding input-grid samples only to error $O(h_n)$ need not preserve higher-order discrete derivative bounds. Under computable evaluation and known derivative bounds, the stronger output precision is available, but its computational cost still depends on the constituent evaluator.

28.12 Efficient Computation Does Not Imply Uniform Analytic Regularity

Let $\mathrm{P}_{\mathbb{R}}$ refer here to polynomial-time computability in the oracle/finite-precision bit model, not to a unit-cost exact-real arithmetic model [79]. The input and output have their ordinary real coordinates; no uncharged nonlinear encoder is allowed.

Theorem 30 (The square-root counterexample)**.** *The function $f(x) = \sqrt{x}$ on $[0,1]$ is polynomial-time computable, but has no exact representation by a finite DAG of $W^{1,\infty}$ constituents on compact boxes containing their admissible arguments.*

Proof. For $x, y \geq 0$,

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x-y|}. \tag{28.32}$$

Given requested output error 2^{-n}, obtain a dyadic approximation $y \in [0,1]$ to x within 2^{-2n-4}. Then the first error in (28.32) is at most 2^{-n-2}. Write $y = m2^{-2n-4}$. Computing the integer $\lfloor \sqrt{m} \rfloor$ and dividing by 2^{n+2} approximates $\sqrt{y}$ within 2^{-n-2}. Binary search for that integer uses polynomially many bit operations on $O(n)$-bit integers. The input precision is $2n+O(1)$, proving polynomial-time evaluation.

Every $W^{1,\infty}$ constituent on a box has a finite Lipschitz constant. A finite composition on a finite DAG is therefore Lipschitz, by the recursion for M_G. But

$$\frac{|\sqrt{h} - \sqrt{0}|}{h} = h^{-1/2} \longrightarrow \infty \qquad (h \downarrow 0).$$

This contradicts any finite global Lipschitz constant. □

The obstruction is regularity, not graph size: the example already has one real input. It does not contradict smooth lifting of the Boolean circuits that compute square roots at fixed precisions. Those lifts live in growing bit-coordinate spaces; their surrounding numerical encoders and decoders are different objects from an exact smooth function of the physical real input x.

28.12.1 Coherent smooth approximants can lose regularity

Let

$$f_n(x) = \sqrt{x + 2^{-2n}} - 2^{-n}. \tag{28.33}$$

Each f_n is smooth on a neighborhood of $[0,1]$ and takes values in $[0,1]$. It has the same one-node graph at every resolution.

Proposition 58 (Uniform convergence with derivative blow-up)**.** *The functions in* (28.33) *satisfy*

$$\|f_n - \sqrt{\cdot}\|_\infty \le 2^{-n}, \qquad \|f_n'\|_\infty = 2^{n-1}.$$

They are also cross-resolution Cauchy at rate $O(2^{-n})$*.*

Proof. For $a \ge 0$, $0 \le \sqrt{x+a} - \sqrt{x} \le \sqrt{a}$. Taking $a = 2^{-2n}$ proves the error bound. Differentiation gives $f_n'(x) = 1/(2\sqrt{x + 2^{-2n}})$, with maximum at zero. The triangle inequality gives $\|f_{n+1} - f_n\|_\infty \le 2^{-n-1} + 2^{-n}$. □

Proposition 59 (Unavoidable conditioning cost)**.** *If* $0 < \varepsilon \le 1/4$ *and a Lipschitz function* g *satisfies* $\|g - \sqrt{\cdot}\|_\infty \le \varepsilon$*, then*

$$\mathrm{Lip}(g) \ge \frac{1}{8\varepsilon}. \tag{28.34}$$

Proof. Set $x_\varepsilon = 16\varepsilon^2 \le 1$. Then $g(x_\varepsilon) \ge 3\varepsilon$ and $g(0) \le \varepsilon$, so

$$\mathrm{Lip}(g) \ge \frac{g(x_\varepsilon) - g(0)}{x_\varepsilon} \ge \frac{2\varepsilon}{16\varepsilon^2}.$$

□

Thus no family with a common finite Lipschitz constant, contractive or otherwise, can converge uniformly to $\sqrt{x}$. This is exactly the pathology excluded by the regularity assumptions of the continuum-limit theorem.

28.12.2 Strict contraction does not imply higher smoothness

The removal of the square-root pathology is a first-order statement. For a fixed $0 < \lambda < 1$, consider

$$a(x) = \lambda|x - 1/2|, \qquad a_n(x) = \lambda\sqrt{(x-1/2)^2 + 2^{-2n}}, \quad n \ge 1.$$

The maps $a_n : [0,1] \to [0,1]$ are smooth and satisfy

$$\mathrm{Lip}(a_n) \le \lambda, \quad \|a_n - a\|_\infty \le \lambda 2^{-n}, \quad \|a_n''\|_\infty = \lambda 2^n. \tag{28.35}$$

Indeed, the first derivative is $\lambda(x-1/2)/\sqrt{(x-1/2)^2 + 2^{-2n}}$, and the second derivative is largest at $x = 1/2$. The limit is contractive and in $W^{1,\infty}$, but not in $W^{2,\infty}$: its first derivative has a jump.

Consequently,

uniform strict contraction $\Longrightarrow$ $W^{1,\infty}$ control, but not higher-order Sobolev control.

This also answers the interpretation in which only the *final* real-valued target is assumed to have Lipschitz constant less than one. That assumption rules out the square root, but not nonsmooth contractive targets. It is not a substitute for higher-order regularity assumptions.

28.13 What the Theorems Establish

The chapter now separates five mathematical statements.

Regular real computation can be digitized. A fixed bounded-arity $W^{r,\infty}$ composition has finite-precision Boolean approximations preserving its coarse graph. The local approximation exponent involves k/r. Effectivity is needed for a uniform construction; polynomial-time local evaluators are needed for polynomial cost in requested output bits.

Boolean circuits have regular real lifts. Every bounded-fan-in Boolean circuit has an exact polynomial-gate lift, an exact nonexpansive Lipschitz lift, and an exact smooth lift whose global Lipschitz constant can be kept bounded independently of depth by allowing polynomially growing local higher-derivative bounds.

Contraction and fixed codes are incompatible with exact nonconstant semantics. The sharp fixed-code approximation error is $(1-\lambda)/2$. This bound is attained in the Lipschitz class; its smooth counterpart can be an unattained infimum.

Changing code scales restores exact computation. Strictly contractive gates can execute a Boolean circuit exactly when code separation shrinks. In a layered computation the optimal Lipschitz scale is $\Delta_H = \lambda^H \Delta_0$. Smooth implementations attain it within a constant factor. Absolute error control must then be compared with this shrinking separation, and the decoder must be charged.

A regular continuum limit needs both kinds of coherence. A fixed coarse graph and cross-resolution consistency supply structural coherence. Uniform Lipschitz or higher finite-difference bounds supply analytic coherence. The square-root construction shows why the first does not imply the second. Uniform contraction is an especially strong first-order certificate, but does not supply every higher derivative.

No assertion of automatic program discovery or statistical learning follows from these representation and execution theorems. Nor does the smooth extension proof alone provide a sharp bit-complexity theorem for numerically evaluating every smooth lift. The constructions with piecewise-affine gates and rational scales make that accounting more direct; smooth implementations require their own evaluation guarantees.

28.14 The Remaining Research Problems

The unrestricted implication

$$\mathrm{P}_{\mathbb{R}} \Longrightarrow \text{an exact finite } W^{1,\infty} \text{ compositional representation}$$

is false. It should not remain labeled an open conjecture. Likewise, once convergent local modules with uniform Sobolev bounds are supplied, the continuum-limit theorem is already proved. The questions left by this chapter concern the existence, discovery, and economical certification of the required modules and scales.

28.14.1 Coarse-graining an efficient circuit family

Suppose (C_n) is a uniform family of finite-precision circuits computing f to error 2^{-n}. Seek a decomposition

$$C_n = G(C_{1,n}, \ldots, C_{s,n})$$

with a precision-independent coarse graph G, bounded numerical arity, compatible local encodings, and decoded module tables satisfying cross-resolution and regularity bounds.

The existence of small Boolean gates at each precision does not identify such numerical modules. A meaningful theorem should supply conditions on the algorithm or the task family under which the number of modules, their local dimensions, and their descriptions remain controlled. Otherwise the entire computation can be hidden in one high-dimensional primitive.

For families f_d, a precise structural target is $s(d) \le \mathrm{poly}(d)$, local dimension $k = O(1)$, and specified bounds on local norms and evaluation costs. For one fixed low-dimensional function, allowing an unrestricted constituent equal to the target is not an informative representation theorem.

28.14.2 Efficient certificates of digital regularity

Proposition 57 supplies a concrete first-order certificate:

$$|T_n(a) - T_n(b)| \le \lambda \|a - b\|_\infty + \tau_n, \qquad \tau_n \to 0.$$

The existence of such a sufficient digital condition is therefore no longer mysterious. The open algorithmic issue is how to obtain it from a compact circuit or program without enumerating all grid pairs.

A useful result would provide compositional rules for certifying local condition bounds, carrying quantization slack through numerical primitives, and extracting a common coarse graph. For higher regularity, one would need analogous certificates for normalized finite differences, with the output precision in (28.31) explicitly included.

28.14.3 Optimal smoothness–margin–depth tradeoffs

For fixed gate arity, define

$$\Lambda_r(b, K) = \inf_g \left\{ \|g\|_{W^{r,\infty}} : \begin{array}{l} g|_{\{0,1\}^k} = b, \\ 0 \le g \le 1, \quad \mathrm{Lip}_\infty(g) \le K \end{array} \right\}.$$

Here the infimum is over smooth lifts. The smooth lifting theorem supplies an upper bound of order $(K-1)^{1-r}$ for $K > 1$ close to one, while XOR excludes a smooth exact extension at $K = 1$. Determining sharp gate-dependent bounds, and then optimizing nonuniform local slack allocations over a DAG, would quantify how smoothness should be distributed across the computation.

For example, if gate v has slack η_v, global sensitivity is controlled by products $\prod_{v \in P}(1 + \eta_v)$ over paths P, while local derivative cost scales with η_v^{1-r}. Uniform slack c/H is one convenient choice, not a proof of optimality for every graph.

28.14.4 Jointly optimizing precision and code scales

With local scales Δ_v and error budgets δ_v, the appropriate optimization is not to minimize Lipschitz constants alone. One must also control normalized errors, decoder sensitivity, and evaluation cost. A possible formulation is

$$\begin{aligned} \min_{\{\Delta_v, \delta_v\}} \quad & \sum_v T_v(\lceil \log_2(1/\delta_v) \rceil) \\ \text{subject to} \quad & e_o < \Delta_o/2 \quad \text{for every output } o, \\ & e_v \le \delta_v + L_v \max_{u \in \mathrm{pa}(v)} e_u, \\ & \text{the prescribed local coding and smoothness constraints.} \end{aligned}$$

Here $T_v(p)$ is the certified bit cost of evaluating the local module to absolute error 2^{-p}. The model must declare whether scales are stored as fixed-point numbers, floating-point exponents, or symbolic metadata. This would turn the qualitative precision tradeoff into an optimization problem over an actual computational cost model.

28.14.5 The distinction between depth limits and resolution limits

A final question concerns growing graphs. Uniform approximation of a fixed numerical map at increasing precision is not the same as taking execution depth to infinity. With uniformly bounded initial code separation and strictly contractive complete layers, the raw signal diameter tends to zero as $H \to \infty$. A nontrivial normalized limit then requires compensation by encoding, input injection, or decoding.

Any proposed continuum theorem for a family whose depth also grows must state how these scales and compensations are controlled. The fixed-DAG theorems proved here do not automatically establish that stronger result.

28.15 Conclusion

The original representation question has become more precise. Real-valued computation is implemented through finite-precision Boolean computation, but there are several distinct notions of simplicity and stability.

A regular sparse real composition admits structure-preserving digital approximation. A Boolean circuit admits exact real extensions, including smooth lifts. A compatible family of decoded numerical modules has a regular continuum limit when its regularity is uniformly controlled. None of these statements makes arbitrary polynomial-time real functions Lipschitz: the square root supplies an elementary counterexample.

The contraction analysis adds a second boundary. With the same fixed $0/1$ encoding, exact nonconstant Boolean semantics require Lipschitz constant at least one. Below one, the sharp numerical approximation error is $(1 - \lambda)/2$. Exact computation can be restored with changing code scales, and then

$$\Delta_H \leq \lambda^H \Delta_0$$

for genuinely layered contractive propagation. This rate is attained by Lipschitz lifts and attained within a constant factor by smooth lifts.

At the same time, local absolute implementation errors satisfy

$$e_H \leq \frac{\delta}{1 - \lambda}.$$

The favorable absolute error bound is meaningful for digital decisions only after comparison with the shrinking signal:

$$\frac{e_H}{\Delta_H} < \frac{1}{2}.$$

For fixed absolute precision this gives a bit requirement proportional to depth; other representations move the accounting into exponents, scale metadata, or the decoder. Fixed-amplitude real noise is not eliminated by changing that bookkeeping.

The central statement is therefore

contraction, precision, and code separation must be analyzed together; a small Lipschitz constant is not a free substitute for information.

The remaining representation problem is to identify efficient digital computations whose coarse modules have coherent numerical meaning, certifiable regularity, and an economical allocation of precision. That is the bridge from bounded-fan-in Boolean computation to a useful smooth sparse continuum representation.

“‘latex

CHAPTER 29

Learning Difficult Inverses by Forward Simulation

The direction in which we wish to make predictions need not be the direction in which scarce paired data are best used. In an inverse problem, a rich object X generates a measurement Y, but the learner must reconstruct X from Y. Learning the forward conditional $P_{Y|X}$ may require substantially fewer paired observations than learning the inverse conditional $P_{X|Y}$ directly, especially when the measurement loses information and the inverse is ambiguous.

This chapter develops the forward simulation principle for inverse learning. When source-domain samples are abundant, a learned forward model can turn them into synthetic paired examples for training the desired inverse:

$$\boxed{\textit{learn forward} \longrightarrow \textit{simulate pairs} \longrightarrow \textit{learn inverse}.}$$

An exact forward conditional, combined with the correct source marginal, reproduces the true joint distribution and therefore the true inverse conditional. In the approximate case, total-variation and bounded-loss arguments separate three contributions to inverse-learning error: forward-model error, source-distribution mismatch, and the statistical and optimization error of learning from synthetic pairs.

Generating more synthetic data can reduce the last contribution under appropriate learning assumptions, but it cannot correct an inaccurate simulator or an inappropriate source distribution. Synthetic examples reuse the knowledge captured by the forward model; they do not create new information about the physical measurement mechanism. The chapter connects this construction to probabilistic reconstruction, amortized Bayesian inversion, and regularized inverse problems, and illustrates it with image reconstruction from brain activity. The opportunity is to use scarce real pairs to learn the easier direction, then propagate that knowledge through simulation while retaining real observations for calibration.

Chapter 30 — Learning Without Weight Updates: Context, Memory, and Cultural State This chapter distinguishes

29.1 The basic idea

Many learning problems are naturally asymmetric.

Let X denote a rich underlying object and let Y denote a measurement, representation, or consequence of that object. Nature generates observations according to

$$X \sim P_X, \qquad Y \sim P_{Y|X}(\cdot|X),$$

so that the joint distribution is

$$P_{XY}(x, y) = P_X(x)P_{Y|X}(y|x).$$

Suppose that only a relatively small number of paired observations

$$(x_1, y_1), \dots, (x_n, y_n)$$

is available. The desired learning problem, however, is the inverse one:

$$Y \longmapsto X.$$

A common example is reconstruction of an image X from a neural measurement Y, but the same structure occurs in many inverse problems.

The central observation of this chapter is that the two conditional learning problems

$$X \longrightarrow Y$$

and

$$Y \longrightarrow X$$

need not have comparable difficulty.

If the forward conditional $P_{Y|X}$ can be learned accurately from relatively few paired examples, while the inverse conditional $P_{X|Y}$ requires many more, then it may be advantageous not to learn the desired inverse directly.

Instead, one can proceed in three stages:

few real paired examples (x_i, y_i)

$\downarrow$

learn the easier forward model $\widehat{P}_{Y|X}$

$\downarrow$

$x \sim P_X, \qquad \widetilde{y} \sim \widehat{P}_{Y|X}(\cdot|x)$

$\downarrow$

arbitrarily many synthetic pairs $(x, \widetilde{y})$

$\downarrow$

learn the difficult inverse $P_{X|Y}$.

The forward model is therefore used as a *simulator* for creating the paired training data required by the inverse problem.

We will refer to this idea as the *forward simulation principle for inverse learning*.

29.2 Why the two directions may have different difficulty

Consider first a deterministic noisy forward model

$$Y = F(X) + \eta,$$

where η represents measurement noise.

Learning the forward direction requires approximating

$$F : X \longrightarrow Y.$$

The inverse problem asks instead for

$$Y \longrightarrow X.$$

If F compresses information, then many different values of X may produce the same or nearly the same value of Y:

$$F(x_1) \approx F(x_2), \qquad x_1 \neq x_2.$$

The inverse is then not a well-defined deterministic function.

More generally, the correct inverse object is

$$P_{X|Y}(x|y),$$

which may be broad or multimodal.

Thus the forward and inverse problems can have quite different statistical complexities. We may have

$$n_{\text{forward}}(\epsilon) \ll n_{\text{inverse}}(\epsilon),$$

where $n_{\text{forward}}(\epsilon)$ is the number of paired examples needed to estimate the forward conditional to accuracy ϵ, while $n_{\text{inverse}}(\epsilon)$ is the corresponding sample complexity for directly learning the inverse.

This asymmetry creates the opportunity exploited below.

29.3 The ideal result

Suppose first that the marginal distribution P_X is available and that the forward conditional has been learned perfectly:

$$\widehat{P}_{Y|X}(y|x) = P_{Y|X}(y|x).$$

Generate synthetic examples by

$$X \sim P_X$$

followed by

$$\widetilde{Y} \sim \widehat{P}_{Y|X}(\cdot|X).$$

The resulting synthetic joint distribution is

$$\widetilde{P}_{XY}(x, y) = P_X(x)\widehat{P}_{Y|X}(y|x).$$

Therefore, if the learned forward model is exact,

$$\widetilde{P}_{XY}(x, y) = P_X(x)P_{Y|X}(y|x) = P_{XY}(x, y).$$

This gives the following elementary but important result.

Forward Simulation Principle. If

$$\widehat{P}_{Y|X} = P_{Y|X}$$

and synthetic inputs are drawn from the true marginal P_X, then forward simulation produces samples from exactly the true joint distribution:

$$\boxed{\widetilde{P}_{XY} = P_{XY}.}$$

Consequently,

$$\boxed{\widetilde{P}_{X|Y} = P_{X|Y}.}$$

Thus, in the ideal limit, arbitrarily many examples for learning the inverse conditional can be generated from a learned forward conditional and samples from P_X.

29.4 Proof

The proof is immediate from the factorization of the joint distribution.

By construction,

$$\widetilde{P}_{XY}(x, y) = P_X(x)\widehat{P}_{Y|X}(y|x).$$

If

$$\widehat{P}_{Y|X}(y|x) = P_{Y|X}(y|x),$$

then

$$\widetilde{P}_{XY}(x, y) = P_X(x)P_{Y|X}(y|x) = P_{XY}(x, y).$$

Equality of the joint distributions implies equality of the corresponding conditional distributions wherever $P_Y(y) > 0$:

$$\widetilde{P}_{X|Y}(x|y) = P_{X|Y}(x|y).$$

□

29.5 The approximate case

In practice neither component will be exact.

Suppose synthetic inputs are sampled from a distribution Q_X rather than exactly from P_X, and suppose the learned forward conditional is

$$\widehat{P}_{Y|X}.$$

Then the synthetic joint distribution is

$$Q_{XY}(x, y) = Q_X(x)\widehat{P}_{Y|X}(y|x).$$

There are now two distinct sources of error:

$$\boxed{\text{error in the distribution of } X + \text{error in the learned forward model.}}$$

This can be formalized in total variation distance. Let

$$d_{\mathrm{TV}}(P, Q) = \sup_A |P(A) - Q(A)|.$$

Then

$$\boxed{d_{\mathrm{TV}}(P_{XY}, Q_{XY}) \le d_{\mathrm{TV}}(P_X, Q_X) + \mathbb{E}_{X\sim P_X}\left[d_{\mathrm{TV}}\left(P_{Y|X}(\cdot|X), \widehat{P}_{Y|X}(\cdot|X)\right)\right].}$$

To see this, introduce the intermediate joint distribution

$$R_{XY}(x, y) = P_X(x)\widehat{P}_{Y|X}(y|x).$$

The triangle inequality gives

$$d_{\mathrm{TV}}(P_{XY}, Q_{XY}) \le d_{\mathrm{TV}}(P_{XY}, R_{XY}) + d_{\mathrm{TV}}(R_{XY}, Q_{XY}).$$

The first term measures forward-model error and satisfies

$$d_{\mathrm{TV}}(P_{XY}, R_{XY}) \le \mathbb{E}_{X\sim P_X} d_{\mathrm{TV}}\left(P_{Y|X}(\cdot|X), \widehat{P}_{Y|X}(\cdot|X)\right),$$

while the second satisfies

$$d_{\mathrm{TV}}(R_{XY}, Q_{XY}) \le d_{\mathrm{TV}}(P_X, Q_X).$$

Thus the quality of the synthetic inverse-training data is controlled by two quantities only: how accurately the forward law is learned and how accurately the synthetic X distribution matches the relevant real distribution.

29.6 Risk transfer from synthetic to real data

The preceding distributional statement immediately implies a learning bound.

Let g be an inverse predictor and let

$$\ell(g(Y), X)$$

be a bounded loss satisfying

$$0 \le \ell \le L.$$

Define the true risk

$$R_P(g) = \mathbb{E}_{(X,Y)\sim P_{XY}}\left[\ell(g(Y), X)\right]$$

and the synthetic risk

$$R_Q(g) = \mathbb{E}_{(X,Y)\sim Q_{XY}}\left[\ell(g(Y), X)\right].$$

Let

$$\delta = d_{\mathrm{TV}}(P_{XY}, Q_{XY}).$$

Then for every g,

$$|R_P(g) - R_Q(g)| \leq L\delta.$$

Suppose $\widehat{g}$ is trained on synthetic data and satisfies

$$R_Q(\widehat{g}) \leq \inf_{g \in \mathcal{G}} R_Q(g) + \epsilon_M,$$

where ϵ_M contains the statistical and optimization error remaining after using M synthetic examples.

Let

$$g_P^* = \arg\min_{g \in \mathcal{G}} R_P(g).$$

Then

$$\boxed{R_P(\widehat{g}) - R_P(g_P^*) \leq 2L\delta + \epsilon_M.}$$

Using the previous bound on δ,

$$\boxed{\begin{aligned} R_P(\widehat{g}) - R_P(g_P^*) \leq{}& 2L\, d_{\mathrm{TV}}(P_X, Q_X) \\ &+ 2L\, \mathbb{E}_{X \sim P_X} d_{\mathrm{TV}}\left(P_{Y|X}(\cdot|X), \widehat{P}_{Y|X}(\cdot|X)\right) + \epsilon_M. \end{aligned}}$$

This equation captures the basic theoretical tradeoff.

Because synthetic data can be generated in arbitrarily large quantities,

$$M \to \infty$$

can make

$$\epsilon_M \to 0$$

under appropriate learning assumptions.

However, increasing the number of synthetic examples cannot eliminate the errors

$$d_{\mathrm{TV}}(P_X, Q_X)$$

or

$$d_{\mathrm{TV}}\left(P_{Y|X}, \widehat{P}_{Y|X}\right).$$

Synthetic data can remove the sampling difficulty of learning the inverse from a finite dataset, but they cannot repair an incorrect forward model.

This is the fundamental limitation of the method.

29.7 What synthetic data do and do not provide

It is tempting to object that synthetic examples cannot contain new information because they are generated by a model learned from the original small dataset.

This objection is correct in one sense and incorrect in another.

Synthetic data do not provide new information about the true forward law $P_{Y|X}$. All information about this law ultimately originates in the real paired observations.

However, once a useful approximation

$$\widehat{P}_{Y|X}$$

has been learned, it can be evaluated at an arbitrarily large number of new values of X.

Thus the learned forward rule can be propagated throughout the support of P_X.

If only

$$n$$

real pairs were required to estimate the forward law, one can nevertheless generate

$$M \gg n$$

inverse-training pairs.

The new pairs do not improve knowledge of the forward mechanism. Rather, they make it possible to *amortize* that knowledge over the full input distribution when learning the inverse.

This distinction is crucial.

29.8 Asymmetric sample complexity

The method is useful when three conditions hold.

First, the forward problem must be easier than the inverse problem:

$$n_{\text{forward}}(\epsilon) \ll n_{\text{inverse}}(\epsilon).$$

Second, samples from the marginal distribution of X must be abundant:

$$X_1, X_2, \ldots \sim P_X.$$

Third, the learned forward model must generalize over the region of X on which synthetic examples are generated.

Under these conditions, one may have enough real paired examples to learn the forward mapping while having far too few to learn the inverse mapping directly.

Forward simulation then converts the easier learning problem into a generator of training examples for the harder problem.

Schematically,

$n_{\text{forward}} \ll n_{\text{inverse}}$	$\Longrightarrow$	learn forward first and simulate the inverse dataset.

This is the essential source of the advantage.

29.9 The importance of sampling from the correct X distribution

One should not interpret the availability of arbitrarily many synthetic inputs as permission to use arbitrary points in the ambient space.

If the forward model is trained on natural images, for example, its behavior on arbitrary vectors in pixel space may be meaningless.

The relevant requirement is

$$Q_X \approx P_X,$$

at least over the domain in which the inverse will ultimately be used.

A powerful generative model can provide precisely such samples.

Let

$$Z \sim P_Z$$

be a simple latent variable and let

$$X = G(Z)$$

be a pretrained generative model. Then

$$Q_X = G_{\#} P_Z$$

is the distribution induced by the generator.

Synthetic inverse-training data can be generated by

$$Z \sim P_Z,$$

$$X = G(Z),$$

$$\widetilde{Y} \sim \widehat{P}_{Y|X}(\cdot|X).$$

Thus

$$\boxed{Z \longrightarrow X \longrightarrow \widetilde{Y}}$$

becomes an arbitrarily large source of paired examples

$$(\widetilde{Y}, X).$$

This makes large pretrained generative models especially useful in data-poor inverse problems.

29.10 Stochastic forward models are preferable

When the real measurement contains noise, it is important that the forward model represent not merely a conditional mean

$$\widehat{F}(x) \approx \mathbb{E}[Y|X = x],$$

but ideally the entire conditional distribution

$$\widehat{P}_{Y|X}.$$

If synthetic examples are generated only as

$$\widetilde{Y} = \widehat{F}(X),$$

the inverse model may never see the variability present in real measurements.

A better simulator is

$$\widetilde{Y} \sim \widehat{P}_{Y|X}(\cdot|X),$$

so that synthetic measurements reproduce not only the expected signal but also its conditional variability.

For a model such as

$$Y = F(X) + \eta,$$

one should therefore model both

$$F(X)$$

and the appropriate distribution of η.

29.11 The inverse should usually be probabilistic

When the forward map is many-to-one, there is no unique deterministic inverse.

Suppose several values

$$x_1, x_2, \ldots$$

are compatible with the same observation y.

If an inverse function g is trained with squared-error loss,

$$\mathbb{E}\left[\|X - g(Y)\|^2\right],$$

then the population optimum is

$$\boxed{g^*(y) = \mathbb{E}[X|Y = y].}$$

This conditional mean need not itself be a plausible member of the underlying data distribution.

For images, averaging several different plausible reconstructions may produce a blurred image that corresponds to none of them.

The more natural goal is therefore to learn a conditional generative model

$$q_\phi(x|y).$$

Synthetic pairs can be used to maximize, for example,

$$\mathbb{E}_{(X,Y)\sim Q_{XY}}\left[\log q_\phi(X|Y)\right].$$

In the ideal limit,

$$Q_{XY} = P_{XY}$$

and sufficient model capacity and optimization imply

$$q_\phi(x|y) \approx P_{X|Y}(x|y).$$

The learned inverse can then produce samples

$$X^{(1)}, X^{(2)}, \ldots \sim q_\phi(\cdot|y),$$

representing the different underlying objects compatible with the same observation.

29.12 Connection with Bayesian inversion

The same construction has a direct Bayesian interpretation.

Bayes' rule gives

$$P_{X|Y}(x|y) = \frac{P_{Y|X}(y|x)P_X(x)}{P_Y(y)}.$$

Thus the two ingredients used by forward simulation,

$$P_X$$

and

$$P_{Y|X},$$

are exactly the two ingredients required for Bayesian inversion.

One possible strategy is therefore to perform Bayesian inference directly:

$$P_{X|Y}(x|y) \propto P_{Y|X}(y|x)P_X(x).$$

Forward simulation provides a second possibility. Instead of solving this inference problem anew for every observation y, one generates many pairs from

$$P_X P_{Y|X}$$

and trains a model that directly approximates

$$P_{X|Y}.$$

Thus forward simulation may be viewed as *amortized Bayesian inversion*.

The potentially expensive inference problem

$$P_{Y|X}, P_X \quad \longrightarrow \quad P_{X|Y}(\cdot|y)$$

is solved repeatedly during synthetic training so that subsequent inference for a new y becomes a single pass through the learned inverse model.

29.13 Connection with regularized inverse problems

Suppose

$$Y = F(X) + \eta, \qquad \eta \sim \mathcal{N}(0, \sigma^2 I).$$

Then

$$P(y|x) \propto \exp\left[-\frac{1}{2\sigma^2}\|y - F(x)\|^2\right].$$

The Bayesian MAP solution is

$$x_{\mathrm{MAP}} = \arg\min_x \left[\frac{1}{2\sigma^2}\|y - F(x)\|^2 - \log P_X(x)\right].$$

Writing

$$R(x) = -\log P_X(x),$$

this becomes the classical regularized inverse problem

$$\boxed{x_{\mathrm{MAP}} = \arg\min_x \left[\|y - F(x)\|^2 + \lambda R(x)\right].}$$

Forward simulation and regularized inversion therefore use the same two objects:

$$\boxed{\begin{array}{ccc} P_{Y|X} & \longleftrightarrow & \text{forward/data-fidelity model,} \\ P_X & \longleftrightarrow & \text{prior/regularizer.} \end{array}}$$

The difference is computational.

Regularized or Bayesian inversion solves an optimization or sampling problem separately for each observed y.

Forward simulation instead uses the same ingredients to train an inverse model in advance.

29.14 Combining real and synthetic pairs

Because errors in the learned forward model create an irreducible synthetic-data bias, it is useful to retain the original real pairs when training the inverse.

A natural objective is

$$\mathcal{L}(\phi) = \lambda_{\mathrm{real}} \frac{1}{n} \sum_{i=1}^{n} \ell_\phi(x_i, y_i) + \lambda_{\mathrm{syn}} \frac{1}{M} \sum_{j=1}^{M} \ell_\phi(\widetilde{x}_j, \widetilde{y}_j).$$

The large synthetic dataset teaches the inverse model the global geometry implied by the forward simulator, while the real examples anchor the learned inverse to the true measurement process.

One may therefore think of the real examples as performing a calibration function that synthetic examples alone cannot provide.

29.15 Iterative refinement

The construction also suggests an iterative procedure.

Starting with a small real paired dataset, first estimate

$$\widehat{P}^{(0)}_{Y|X}.$$

Use it to generate synthetic pairs and learn

$$\widehat{P}^{(0)}_{X|Y}.$$

New real observations, when available, can then be used to improve the forward model,

$$\widehat{P}^{(0)}_{Y|X} \longrightarrow \widehat{P}^{(1)}_{Y|X},$$

after which a new and improved synthetic dataset can be generated.

Thus one can iterate

$$\boxed{\text{real pairs} \to \text{forward model} \to \text{synthetic pairs} \to \text{inverse model} \to \text{improved real calibration.}}$$

The bottleneck remains the accuracy of the learned forward mechanism.

29.16 Application to image reconstruction from brain activity

The image–fMRI problem is a particularly natural example.

Let

$$X = \text{visual image}$$

and

$$Y = \text{fMRI response.}$$

Only a limited number of paired measurements

$$(x_i, y_i)$$

can realistically be collected.

Directly learning

$$Y \longrightarrow X$$

from these pairs is difficult because the inverse is high-dimensional, many-to-one, and strongly underdetermined.

The forward problem,

$$X \longrightarrow Y,$$

may be substantially easier.

Suppose the real pairs are used to train

$$\widehat{P}_{\text{fMRI}|\text{image}}.$$

A pretrained image generator can then supply an enormous number of natural images

$$X_1, \dots, X_M.$$

For each image one generates a synthetic brain response

$$\widetilde{Y}_j \sim \widehat{P}_{\text{fMRI}|\text{image}}(\cdot|X_j).$$

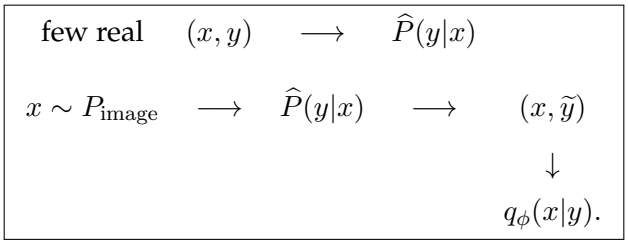
This produces

$$(\widetilde{Y}_j, X_j), \qquad j = 1, \dots, M,$$

which can be used to train a conditional image generator

$$q_\phi\left(x \mid y\right).$$

The complete architecture is



A diffusion model is particularly attractive for representing either the marginal image distribution P_X or the conditional inverse $q_\phi(x|y)$.

The essential principle, however, is independent of the particular generative architecture.

29.17 A general principle

The discussion can be summarized as follows.

Forward Simulation Principle for Inverse Learning. Consider a system with joint distribution

$$P_{XY} = P_X P_{Y|X}.$$

Suppose that:

1. paired observations (X, Y) are scarce;
2. unpaired samples from P_X, or a good approximation to them, are abundant;
3. the forward conditional $P_{Y|X}$ can be learned accurately from substantially fewer paired examples than are required to learn $P_{X|Y}$ directly; and
4. the learned forward model generalizes over the relevant support of P_X.

Then an effective strategy for learning the difficult inverse conditional is:

$$\boxed{\text{learn } P_{Y|X} \;\longrightarrow\; \text{simulate } P_X P_{Y|X} \;\longrightarrow\; \text{learn } P_{X|Y}.}$$

In the ideal case of an exact forward model and exact sampling from P_X, the synthetic joint distribution is identical to the true joint distribution and the procedure is statistically exact.

In the approximate case, the inverse error contains three conceptually distinct contributions:

$$\boxed{\begin{array}{c}\text{forward-model error}\\ +\\ \text{marginal-distribution error}\\ +\\ \text{finite synthetic-sample/optimization error.}\end{array}}$$

Only the last of these can be driven to zero merely by generating more synthetic data.

29.18 Discussion

The principle can be stated informally in a particularly simple way:

If the direction we want to learn is difficult, learn the easier direction and use it to manufacture training data for

The method exploits an asymmetry in sample complexity.

The scarce real data are spent on learning the conditional relation that can be estimated most efficiently. Once this relation has been learned, abundant unpaired samples from the source domain can be converted into abundant paired examples.

A difficult inverse-learning problem is thereby decomposed into

$$\boxed{\text{easy supervised learning} + \text{simulation} + \text{large-data inverse learning.}}$$

The construction does not create information about the physical forward mechanism that was absent from the original data. Rather, it allows the information captured by the learned forward law to be propagated over an arbitrarily large input distribution.

This distinction explains both the power and the limitation of the approach.

The general message is that the direction in which one ultimately wishes to make predictions need not be the direction in which one should initially learn from scarce data. When the causal or data-generating direction is simpler, it may be statistically preferable to learn that direction first, use it as a simulator, and only afterward learn the desired inverse.

In this sense, a hard inverse problem can sometimes be converted from a problem of scarce supervised data into a problem of forward-model accuracy. “‘

CHAPTER 30

Learning Without Weight Updates: Context, Memory, and Cultural State

Learning need not be identified with changing neural weights. A model with fixed parameters can change its effective predictor through experience-dependent updates to context, working memory, programs, or an external library. The relevant object is therefore the complete adaptive system, not the neural network considered in isolation.

This chapter formalizes that distinction by writing the predictor as $F_\theta(x; s)$, where θ is fixed and s is adaptive state. The same frozen network induces a family of hypotheses,

$$\boxed{\mathcal{H}_\theta = \{x \mapsto F_\theta(x; s) : s \in \mathcal{S}\}.}$$

Learning can select a member of this family without modifying the interpreter that executes it. Context can specify a learning procedure and store newly computed task-specific parameters; contextual learning is therefore not restricted to retrieving answers encountered during pretraining.

Three forms of adaptation must nevertheless be distinguished: transient changes within a context, persistent learning through retained external state, and cumulative cultural learning in which inherited artifacts make later tasks easier. Their statistical complexity depends on the induced state-dependent hypothesis class, while their practical power depends on what the fixed interpreter can represent, discover, and execute within its memory and computation budgets.

The chapter examines both possibilities and obstructions. A context-independent information bottleneck can prevent every permitted context from solving a task. Conversely, an ideal universal interpreter with sufficient writable memory can simulate weight learning by storing the resulting parameters as data. Neither observation is an unconditional claim about an arbitrary pretrained language model. The boundary between contextual and parametric learning depends on the machine, the available state, the data, and the resources. The central message is that frozen weights do not imply frozen competence: experience can improve a system by changing the persistent state through which its computational machinery is used.

Chapter 31 —

30.1 Learning need not mean changing weights

In modern machine learning, learning is usually identified with a change of parameters. Given a predictor

$$f_\theta(x),$$

a learning algorithm uses data D to produce

$$\theta \longrightarrow \theta',$$

with the goal of reducing prediction error on future examples.

Large language models suggest a broader notion. A pretrained model with fixed parameters can display dramatically different capabilities depending on the context in which it is placed. Examples, instructions, intermediate results, retrieved memories, programs, and previously discovered strategies can all modify the computation performed by the same frozen network.

We should therefore distinguish the parameters of the computational substrate from the adaptive state of the overall system.

Let

$$F_\theta(x; s)$$

denote a model with fixed weights θ and adaptive state s. The state may contain a prompt, demonstrations, memories, algorithms, previously discovered solutions, external documents, or any other information accessible to the model.

The crucial observation is that

$$\boxed{\theta \text{ fixed} \quad \not\Rightarrow \quad \text{behavior fixed.}}$$

Indeed, for every state s, the frozen network defines a possibly different function

$$f_s(x) = F_\theta(x; s).$$

Thus a single frozen network induces the hypothesis class

$$\mathcal{H}_\theta = \{x \mapsto F_\theta(x; s) \ : \ s \in \mathcal{S}\}.$$

Learning can therefore take place by searching over $\mathcal{S}$, rather than over the parameter space of the network.

Instead of

$$\theta^\star = \arg\min_\theta R(\theta),$$

we may have

$$s^\star = \arg\min_{s \in \mathcal{S}} R_\theta(s),$$

where

$$R_\theta(s) = \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\ell(F_\theta(x; s), y)\right].$$

The weights remain unchanged. What changes is the computation selected from the repertoire represented by those weights.

This gives a precise sense in which a frozen model can learn.

30.2 Three forms of learning

It is useful to distinguish three different mechanisms.

First, conventional parametric learning changes the weights:

$$\theta_t \longrightarrow \theta_{t+1}.$$

Second, contextual learning changes the working state:

$$s_t \longrightarrow s_{t+1}.$$

Third, cultural learning modifies an external persistent repository:

$$\mathcal{L}_t \longrightarrow \mathcal{L}_{t+1}.$$

Thus we may write

$$\begin{aligned} \text{parametric learning} &: \theta_t \to \theta_{t+1}, \\ \text{contextual learning} &: s_t \to s_{t+1}, \\ \text{cultural learning} &: \mathcal{L}_t \to \mathcal{L}_{t+1}. \end{aligned}$$

Only the first requires changing neural weights.

For a frozen LLM,

$$\theta_{t+1} = \theta_t = \theta,$$

while

$$(s_t, \mathcal{L}_t) \to (s_{t+1}, \mathcal{L}_{t+1}).$$

The effective predictor nevertheless changes:

$$f_t(x) = F_\theta(x; s_t, \mathcal{L}_t).$$

If

$$R(f_{t+1}) < R(f_t),$$

then the system has improved from experience. Operationally, this is learning, even though no synaptic parameter of the underlying network has changed.

30.3 The adaptive state

Suppose observations arrive sequentially. An external learning rule A can update the state according to

$$s_{t+1} = A_\theta(s_t, D_t),$$

where D_t represents the information obtained at step t.

The complete dynamical system is therefore

$$\boxed{\begin{aligned} y_t &= F_\theta(x_t; s_t), \\ s_{t+1} &= A_\theta(s_t, D_t), \\ \theta_{t+1} &= \theta_t. \end{aligned}}$$

Nothing in the definition of learning requires that the adaptive variables be identical to the parameters of the neural network.

In biological terminology one might distinguish slow and fast variables. In computer science one would distinguish the program implementing an interpreter from the contents of its writable memory.

A frozen LLM together with writable external state is closer to the second picture.

The network weights define a fixed computational machine. The state determines which computation the machine currently performs.

30.4 Frozen weights define a family of hypotheses

The following elementary observation is conceptually important.

Proposition 1: Frozen-weight learning. Let $F_\theta(x; s)$ be fixed with respect to θ. Suppose there exist $s_1, s_2 \in \mathcal{S}$ such that

$$F_\theta(\cdot; s_1) \neq F_\theta(\cdot; s_2).$$

Then adaptation of s can change the effective hypothesis implemented by the system without changing θ.

If, for some learning problem,

$$R_\theta(s_2) < R_\theta(s_1),$$

then the transition

$$s_1 \to s_2$$

constitutes an improvement in expected performance with completely frozen model weights.

The proposition is almost tautological. Its importance is that the usual identification

$$\text{learning} \equiv \text{weight change}$$

is unnecessarily restrictive.

The relevant hypothesis class is not merely the function represented by the network with one fixed prompt. It is

$$\mathcal{H}_\theta = \{F_\theta(\cdot; s) : s \in \mathcal{S}\}.$$

A sufficiently expressive frozen model may therefore contain an enormous family of potential behaviors. Learning determines which member of this family is selected.

30.5 Context as a program

For a language model, a particularly important component of s is the context.

A context need not merely describe a task. It may specify an algorithm.

For example, a context can say:

> Divide the sequence into pairs, compute parity within each pair, recursively apply the same procedure to the resulting bits, and return the final bit.

If the frozen model can correctly interpret and execute these instructions, the context acts as a program.

We may therefore write

$$F_\theta(x; p),$$

where p denotes a program expressed in the language understood by the model.

The frozen parameters θ then play the role of an interpreter.

This viewpoint suggests the decomposition

$$\boxed{\text{computation} = \text{fixed interpreter} + \text{variable program} + \text{variable data}.}$$

In ordinary neural-network learning, much of the task-specific computation is encoded into θ.

In a frozen LLM, some of it can instead be encoded into p.

The distinction is analogous to the difference between manufacturing a new computer for every task and running a new program on the same computer.

30.6 A universality observation

The preceding analogy can be made stronger.

Suppose the frozen system F_θ is capable of interpreting programs from some universal programming language. Let h be any computable function representable by a program p_h. Then

$$F_\theta(x; p_h) = h(x),$$

up to the computational resources available to the system.

Consequently,

$$\mathcal{H}_\theta$$

may contain an extremely rich collection of functions even though θ never changes.

More generally, consider a computable learning algorithm producing

$$h_0, h_1, h_2, \ldots$$

from a sequence of observations.

If each h_t has a program representation p_t, then a universal frozen interpreter can realize

$$F_\theta(x; p_t) = h_t(x).$$

Learning is then equivalent to

$$p_t \to p_{t+1},$$

rather than

$$\theta_t \to \theta_{t+1}.$$

Thus, subject to representational and computational limits,

a fixed universal interpreter plus writable memory can implement an adaptive sequence of hypotheses.

This is closely related to the classical separation between a Turing machine's fixed transition rules and the changing contents of its tape.

The transition function does not have to change for the machine to perform an indefinitely long sequence of different computations.

30.7 Attention and associative memory

Transformers provide a natural mechanism for realizing this architecture.

Attention can be interpreted as associative retrieval. Given a query q, stored key–value pairs

$$(k_i, v_i)$$

are accessed according to a similarity-dependent weighting:

$$\mathrm{Attn}(q) = \sum_i \alpha_i(q) v_i,$$

with

$$\alpha_i(q) = \frac{\exp\left(q^\top k_i\right)}{\sum_j \exp(q^\top k_j)}.$$

Thus the context acts as a temporary associative memory.

The fixed weights determine how queries are generated, how memories are addressed, and how retrieved information is transformed. The actual contents of memory can vary from episode to episode.

This suggests a useful decomposition:

θ : rules for computation and memory access,

s : current contents of computational memory.

From this perspective, in-context learning is not mysterious. The weights implement machinery capable of interpreting and manipulating state. The context supplies the state on which this machinery operates.

30.8 Transient context is not yet persistent learning

There is, however, an important distinction.

Suppose a model solves a problem because useful examples appear in its context window. If the context is erased after the interaction, the improvement disappears.

This is adaptation within an episode, but there is no persistent learning across episodes.

Persistent learning requires a state variable that survives.

We may formulate a simple persistence condition:

Learning across episodes requires some task-relevant state that survives across episodes.

Weight updates provide one such state.

External memory provides another.

Let

$$\mathcal{L}_t$$

denote a persistent library. After an interaction, the system may extract an artifact a_t, such as

- a successful prompt,
- an algorithm,
- a proof,
- a subroutine,
- an error-correction rule,
- an example,
- a compressed summary,
- a verification procedure.

The library is updated as

$$\mathcal{L}_{t+1} = U(\mathcal{L}_t, a_t).$$

At a future time the frozen network retrieves relevant elements of the library:

$$s_{t+1} = C_\theta(x_{t+1}, \mathcal{L}_{t+1}).$$

Thus persistent adaptation can occur even though

$$\theta_{t+1} = \theta_t.$$

The finite context window then becomes analogous to working memory, while $\mathcal{L}_t$ plays the role of long-term memory.

30.9 Cultural learning

This architecture gives a particularly simple model of cumulative culture.

Let generation g have state

$$\mathcal{S}_g = (\theta, c_g, \mathcal{L}_g),$$

where

- θ is a frozen pretrained model,
- c_g is the working context,
- $\mathcal{L}_g$ is the inherited cultural library.

Suppose generation g encounters task T_g. It produces an artifact

$$a_g = G_\theta(T_g, c_g, \mathcal{L}_g),$$

and stores useful information:

$$\mathcal{L}_{g+1} = U(\mathcal{L}_g, a_g).$$

The next generation begins with the same neural parameters,

$$\theta_{g+1} = \theta_g,$$

but a different cultural environment:

$$\mathcal{L}_{g+1} \neq \mathcal{L}_g.$$

Its effective behavior is therefore

$$f_{g+1}(x) = F_\theta(x; c_{g+1}, \mathcal{L}_{g+1}),$$

which can be superior to

$$f_g(x) = F_\theta(x; c_g, \mathcal{L}_g).$$

The locus of adaptation is culture rather than weights.

This leads to the central principle:

Culture is persistent external adaptive state for a fixed computational substrate.

The analogy with biological cultural evolution is direct. Human brains do not have to genetically encode calculus, writing, chess, or computer programming. A relatively stable biological substrate interacts with an expanding external repository of language, tools, algorithms, books, and institutions.

Much of the increase in effective human competence resides outside the individual brain.

A frozen LLM equipped with persistent external memory provides a minimal artificial model of the same phenomenon.

30.10 Cumulative culture

The important case is not merely memory, but cumulative memory.

Suppose

$$\mathcal{L}_0 \subseteq \mathcal{L}_1 \subseteq \mathcal{L}_2 \subseteq \cdots .$$

At generation g, the system need not solve a problem from scratch. It can exploit discoveries made by previous generations.

For a sequence of increasingly difficult tasks

$$T_1, T_2, \ldots, T_m,$$

we may define the cost of solving T_g given the inherited library by

$$C(T_g \mid \mathcal{L}_{g-1}).$$

Cumulative culture is present when inherited artifacts systematically reduce this cost:

$$C(T_g \mid \mathcal{L}_{g-1}) < C(T_g \mid \mathcal{L}_0).$$

An even stronger form occurs when

$$T_g$$

cannot reliably be solved by the frozen model from its initial state, but becomes solvable using artifacts accumulated while solving earlier tasks.

Then the sequence

$$T_1 \to T_2 \to \cdots \to T_g$$

creates a genuine historical dependency.

The competence of generation g depends on the path taken by previous generations.

This is one of the defining characteristics of cumulative culture.

30.11 Where is the learning algorithm?

There is an interesting inversion here.

During conventional training, the learning algorithm acts outside the network:

$$\theta_{t+1} = A(\theta_t, D_t).$$

After pretraining, however, a sufficiently capable network may itself implement part of an adaptation algorithm.

For example, the frozen model can

$$\text{propose} \to \text{test} \to \text{criticize} \to \text{revise} \to \text{store}.$$

Thus one may have

$$s_{t+1} = A_\theta(s_t, D_t),$$

where the learning operator itself is partly represented in the fixed weights.

Pretraining has then learned not merely a predictor, but a mechanism capable of subsequent learning through state changes.

This is closely related to meta-learning.

The slow learning process modifies

$$\theta.$$

The result is a system that can subsequently perform fast learning by modifying

$$s.$$

Symbolically,

$$\boxed{\text{pretraining learns a learner.}}$$

The learner may then continue adapting after its weights have been frozen.

30.12 Generalization with frozen weights

This perspective also clarifies statistical complexity.

Suppose that after freezing θ, learning consists of choosing

$$s \in \mathcal{S}.$$

The relevant hypothesis class is

$$\mathcal{H}_\theta = \{F_\theta(\cdot; s) : s \in \mathcal{S}\}.$$

Generalization therefore depends on the complexity of the state-dependent family

$$\mathcal{H}_\theta,$$

not directly on the number of frozen parameters.

For example, suppose $\mathcal{S}$ contains M possible states and the loss is bounded in $[0, 1]$. Standard uniform-convergence arguments give, with probability at least $1 - \delta$,

$$R(s) \le \widehat{R}_n(s) + O\left(\sqrt{\frac{\log M + \log(1/\delta)}{n}}\right)$$

simultaneously for all $s \in \mathcal{S}$.

For an infinite state space, analogous statements involve the Rademacher complexity or another measure of the induced class

$$\mathcal{H}_\theta.$$

This observation is useful conceptually.

A network may contain billions of frozen parameters while adaptation occurs over a much smaller set of prompts, programs, or memory configurations.

Conversely, an unrestricted external state can define a very large hypothesis class even though the network weights never change.

Thus statistical complexity belongs to the complete adaptive system, not merely to its neural parameters.

30.13 Context as fast parameters

Functionally, context can sometimes be regarded as a form of fast parameterization.

We may compare

$$f_\theta(x)$$

with

$$F_\theta(x; s).$$

The pair

$$(\theta, s)$$

jointly determines the computation.

The difference is primarily one of timescale and representation:

	θ	s
timescale	slow	fast
location	network	context/memory
persistence	intrinsic	optional
representation	distributed weights	explicit state
update mechanism	optimization	search/write/retrieval

Calling s "fast weights" can therefore be useful at the functional level, but the distinction should not be erased. Explicit external state has properties that ordinary neural weights do not have: it can be inspected, copied, communicated, edited, accumulated, and transmitted between independent agents.

Those properties are exactly what make cultural evolution possible.

30.14 What Can Be Learned Through Context?

Frozen weights do not imply frozen competence. But this does not mean that context can teach an arbitrary frozen model anything. The relevant distinction is not between old and new knowledge. It is between computations that can be learned and executed through the state of a fixed machine, and computations that require changing that machine or increasing its resources.

30.14.1 Two spaces of adaptation

Let $F_\theta(x; c)$ denote a predictor with weights θ and adaptive context c. Fix an inference protocol, a context budget of B bits, and a computation budget T. The frozen model defines

$$\mathcal{H}_{\mathrm{ctx}}(\theta_0; B, T) = \{x \mapsto F_{\theta_0}(x; c) : |c| \le B, \quad \mathrm{cost}(F_{\theta_0}, c) \le T\}\,.$$

Learning through context uses training data D to select c_D, without access to unknown test labels. Allowing weight changes *as well* gives

$$\mathcal{H}_{\mathrm{adapt}}(B, T) = \bigcup_{\theta \in \Theta} \mathcal{H}_{\mathrm{ctx}}(\theta; B, T).$$

Provided $\theta_0 \in \Theta$,

$$\mathcal{H}_{\mathrm{ctx}}(\theta_0; B, T) \subseteq \mathcal{H}_{\mathrm{adapt}}(B, T).$$

The inclusion can be strict. However, representability is only the first requirement: a learner must also discover a suitable hypothesis from finite data and generalize to new examples.

30.14.2 Context can learn genuinely new predictors

Suppose a frozen model can execute ridge regression. Given examples $D = (X, \mathbf{y})$ in context, it can compute

$$\widehat{w}_D = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}, \qquad \lambda > 0,$$

and predict

$$F_{\theta_0}(x; D) = x^\top \widehat{w}_D.$$

The coefficients $\widehat{w}_D$ are learned from the new examples; they need not have occurred during pretraining. They reside in context-dependent computational state rather than in the persistent weights θ_0.

Thus a fixed learning procedure can produce a new predictor:

$$\underbrace{\theta_0}_{\text{fixed learning machinery}} \qquad \text{and} \qquad \underbrace{\widehat{w}_D}_{\text{new task-specific parameters}} .$$

Context learning is therefore not merely retrieval. Nor does saying that it selects from a pre-existing hypothesis class distinguish it from conventional learning: weight learning also operates within a class determined by its architecture.

30.14.3 An obstruction that context cannot remove

Suppose the frozen computation factors as

$$F_{\theta_0}(x; c) = G_{\theta_0}(\phi_{\theta_0}(x), c),$$

where ϕ_{θ_0} is independent of context. Then

$$\phi_{\theta_0}(x) = \phi_{\theta_0}(x') \quad \Longrightarrow \quad F_{\theta_0}(x; c) = F_{\theta_0}(x'; c) \quad \text{for every } c.$$

No context can distinguish inputs that this unavoidable bottleneck identifies.

For example, let $x = (x_1, x_2)$ have independent, uniformly random binary coordinates, let $\phi_{\theta_0}(x) = x_1$, and let the target be $y = x_2$. A context learned from independent training examples cannot reduce test classification error below $1/2$. A weight change that allows the representation to retain x_2 can remove this obstruction.

This is a conditional separation, not a general description of a transformer. Context can itself influence intermediate representations. The argument applies only to information lost under every permitted context. If the information is absent from the input altogether, neither context nor weight learning can recover it without additional information.

30.14.4 Storage limits and the universality qualification

A B-bit adaptive context can select at most 2^B predictors. Consequently, representing every possible binary labeling of N fixed inputs requires

$$B \geq N.$$

Structured targets can have much shorter descriptions. External memory must be counted as additional adaptive storage; a bounded context connected to a growing library is not a bounded-memory learner. Finite-precision weight learning has analogous storage constraints.

At the opposite extreme, suppose the frozen system is a universal interpreter with sufficient writable memory and runtime. It can simulate a computable weight-learning algorithm A and its finite-precision trained predictor by storing the resulting weights as data:

$$\theta_D = A(D), \qquad U\big(x; \mathrm{encode}(\theta_D)\big) = f_{\theta_D}(x).$$

The interpreter's own parameters never change. Under these assumptions, external-state learning can simulate weight learning. This is not a universality claim about an arbitrary pretrained LLM or a fixed-size context window. It shows that freezing the interpreter does not, by itself, imply a computability barrier.

30.14.5 The practical boundary: an executable description

The decisive distinction can instead be efficiency. For a target h, input distribution μ, loss ℓ, and tolerance ε, define the shortest executable contextual description

$$K_{\theta_0}^{\varepsilon,T}(h) = \inf_c \big\{\, |c| : \mathbb{E}_{x\sim\mu}\ell\big(F_{\theta_0}(x;c), h(x)\big) \le \varepsilon, \\ \mathrm{cost}(F_{\theta_0}, c) \le T \,\big\},$$

with value $+\infty$ when no such context exists. Contextual representation within the stated budgets requires

$$K_{\theta_0}^{\varepsilon,T}(h) \le B.$$

Learning additionally requires finding a suitable description from data. Weight adaptation can make an inaccessible computation accessible, or make an expensive contextual procedure cheaper to execute. It does not guarantee that training will discover the required weights.

This connects naturally to compositional sparsity. If a target has a sparse decomposition into constituent functions, context can specify their composition when the frozen system can reliably execute those constituents and the required control structure. Sparsity alone does not establish that it can. A new constituent might itself be implemented by a contextual program, but at a cost that exceeds the available resources.

The resulting principle is:

> Context learning is efficient when the target has a short, reliably executable description relative to the frozen model, and that description can be learned from the available data. Weight learning can change which descriptions are short and which computations are reliably executable.

There is therefore no model-independent partition into tasks learnable only through context and tasks learnable only through weights. The boundary depends on the fixed computation, available state, data, and computational resources:

Context learning:	adapt the state of a fixed machine.
Weight learning:	adapt the machine itself.

30.15 Performance ceilings and indefinitely many improvements

There is also an important distinction between a bound on performance and a bound on the number of improvements.

For a normalized performance measure,

$$0 \leq P_g \leq 1.$$

Thus performance clearly has a ceiling.

This does not imply that there must exist some finite generation G after which no further strict improvement is possible.

For example,

$$P_g = 1 - \frac{1}{g+1}$$

satisfies

$$P_{g+1} > P_g$$

for every finite g, although

$$P_g \to 1.$$

Therefore,

bounded performance does not imply a finite bound on the number of improvements.

The distinction becomes even more important when the task distribution itself expands.

A frozen model may successively acquire procedures for

$$T_1, T_2, T_3, \ldots$$

while never changing its weights.

Whether this process eventually stops depends on the computational capacity of the fixed interpreter, the richness of the external state, the search mechanism, and the structure of the task sequence.

There is no contradiction between fixed weights and an indefinitely growing cultural library.

30.16 An extended-system view of intelligence

The usual description of an LLM identifies the intelligent system with its neural network:

$$\text{system} = \theta.$$

For adaptive agents this may be the wrong boundary.

A more appropriate description is

$$\boxed{\text{system} = (\theta, \text{working context}, \text{persistent memory}, \text{tools}, \text{update rules}).}$$

The weights are only one component.

This is especially important when evaluating whether a system "learns."

If we examine only

$$\theta,$$

nothing changes.

If we examine the full state

$$(\theta, s_t, \mathcal{L}_t),$$

the system may undergo substantial and persistent adaptation.

Learning should therefore be defined at the level of the system whose future behavior is being studied.

30.17 A possible definition

The preceding discussion suggests the following definition.

Definition: learning by external state. A system with fixed parameters θ exhibits learning by external state if there exists a persistent state s_t satisfying

$$s_{t+1} = A_\theta(s_t, D_t)$$

such that experience-dependent changes in s_t produce persistent changes in future behavior,

$$F_\theta(\cdot; s_{t+1}) \neq F_\theta(\cdot; s_t),$$

and improve expected performance on some relevant distribution or sequence of tasks.

This definition excludes a mere change of input.

It requires

1. dependence on experience,
2. modification of an adaptive state,
3. persistence of that modification,
4. a consequent change in future behavior.

It includes weight learning as the special case

$$s = \theta,$$

but also includes prompt libraries, external memory, programs, cultural artifacts, and other forms of persistent state.

Thus conventional neural learning is one member of a broader class.

30.18 A hierarchy of adaptation

We can now distinguish several levels of adaptation:

Level 0:	fixed weights, fixed state,
Level 1:	fixed weights, transient context,
Level 2:	fixed weights, persistent external state,
Level 3:	fixed weights, cumulative cultural state,
Level 4:	weight adaptation plus external cultural state.

A conventional prompted LLM is primarily Level 1.

An LLM supplied with persistent memories is Level 2.

A population or sequence of frozen LLM instances that inherits and improves a shared library is Level 3.

Humans appear closer to Level 4: individual brains learn, but they also inherit an enormous accumulated cultural state.

The distinction is important because Level 3 can display substantial cumulative improvement even though the neural substrate is exactly unchanged.

30.19 A minimal experimental paradigm

The theory suggests a simple experiment.

Choose a frozen language model

$$F_\theta$$

and never modify its parameters.

Construct a sequence of tasks

$$T_1, T_2, \ldots, T_K$$

with shared compositional structure.

Begin with

$$\mathcal{L}_0 = \emptyset.$$

For each generation g,

$$\begin{aligned} a_g &= F_\theta(T_g; \mathcal{L}_{g-1}), \\ q_g &= \mathrm{Evaluate}(a_g, T_g), \\ \mathcal{L}_g &= U(\mathcal{L}_{g-1}, a_g, q_g). \end{aligned}$$

Then compare performance on T_g under two conditions:

$$P(T_g \mid \mathcal{L}_{g-1})$$

and

$$P(T_g \mid \mathcal{L}_0).$$

The quantity

$$\Delta_g = P(T_g \mid \mathcal{L}_{g-1}) - P(T_g \mid \mathcal{L}_0)$$

measures the benefit of accumulated culture.

A convincing demonstration of cumulative learning would show

$$\Delta_g > 0$$

for increasingly difficult tasks, while verifying exactly that

$$\theta_g = \theta_0 \qquad \forall g.$$

Stronger evidence would show that some later tasks are solved only after appropriate intermediate artifacts have entered the library.

Such an experiment separates improvement due to pretrained weights from improvement due to post-training cultural accumulation.

30.20 The main principle

The usual picture of machine learning is

$$\text{experience} \longrightarrow \text{weight change} \longrightarrow \text{improved behavior}.$$

The more general picture is

$$\text{experience} \longrightarrow \text{state change} \longrightarrow \text{improved behavior}.$$

Weights are one possible location of adaptive state, but not the only one.

For sufficiently expressive frozen models, context, external memory, programs, and accumulated artifacts can provide another.

The most compact statement is therefore

Frozen weights do not imply frozen competence.

A pretrained network can be viewed as a fixed interpreter defining a large space of possible computations. Context selects computations from this space. Persistent memory allows selections and discoveries to survive across episodes. Cultural memory allows them to accumulate across generations.

Thus the evolution of competence can occur according to

$$\theta = \text{constant}, \qquad \mathcal{L}_0 \to \mathcal{L}_1 \to \mathcal{L}_2 \to \cdots .$$

In this regime, what learns is not the neural network considered in isolation.

What learns is the larger dynamical system consisting of the frozen network together with its changing external state.

And what evolves cumulatively is not necessarily the model.

It may be its culture.

CHAPTER 31

Continual Learning, Regularization, and Virtual Examples: The Role of Compositional Sparsity

Continual learning requires acquiring new abilities while preserving useful old computations. Regularization and virtual examples offer two ways of expressing that preservation. Starting from the exact equivalences established by Niyogi, Girosi, and Poggio in particular function-approximation settings, this chapter develops a functional view: choose the behavior to retain, then use virtual examples of the old function to impose corresponding constraints. Exact empirical identities are distinguished from finite-sample approximations to population penalties and local quadratic approximations in parameter space.

Compositional sparsity makes it possible to place these constraints at low-dimensional constituent interfaces rather than only at the global input–output map. Under explicit regularity, coverage, and stability assumptions, the chapter proves that local preservation errors control changes in old-task risk:

$$\boxed{\begin{array}{c}\textit{controlled local functional discrepancies}\\ +\ \textit{stable composition}\\ \Longrightarrow\ \textit{controlled forgetting}.\end{array}}$$

Finite virtual-probe constructions make the role of interface dimension and smoothness quantitative. A comparator-based certificate connects the actual squared replay objective to retention, while persistent reference targets avoid an explicit accumulation of successive teacher-replacement errors. When modules are reused, their stored constraints can also be shared across task graphs under the stated uniform assumptions.

These are conditional retention results, not a complete theorem of efficient continual learning. Protecting arbitrary old extrapolations can obstruct compatible new tasks, and fixing local coordinates can penalize harmless changes of representation. The remaining challenge is to preserve the right functions on the right domains while permitting useful reorganization and efficient acquisition. The chapter therefore separates forgetting control, new-task learning, and complete memory cost, and asks when reusable computational structure can improve all three without requiring a unique generating decomposition.

31.1 Introduction: retention is not yet continual learning

A learner exposed to successive tasks must acquire new abilities without losing useful old ones. Retaining old examples, penalizing parameter changes, and creating artificial examples are often treated as separate strategies. A more unified starting point is to ask which previously learned *computations* should remain available and how their preservation should be expressed.

Niyogi, Girosi, and Poggio [118] established, in particular function-approximation settings, an exact equivalence between incorporating prior information through virtual examples and through regularization. This is the conceptual starting point here. For continual learning, the prior information is that selected aspects of an old function should be retained. Virtual examples of that function implement corresponding functional constraints. Exact empirical identities must be distinguished from finite-sample approximations to population penalties and from local quadratic approximations in parameter space.

Compositional sparsity suggests a further possibility. If a high-dimensional function is represented by a graph of low-arity constituent functions, preservation constraints can be imposed at their low-dimensional interfaces. Approximation theory already exploits this distinction between ambient and constituent dimensions [108, 106]. The question here is whether the same structure can reduce the cost of preserving old computations during subsequent learning.

The chapter develops a *conditional theory of retention*. We prove how local functional discrepancies propagate to old-task risk, how finite virtual probes control those discrepancies, and how persistent reference values avoid an explicit accumulation of replay error over time. We also give an optimization certificate linking the actual squared replay objective to a retention bound. These conclusions do not, by themselves, establish that a new task can be learned efficiently while satisfying the constraints.

The decisive distinction is

$$\begin{aligned} &\text{informative local probes and stable composition} \quad \Longrightarrow \quad \text{controlled forgetting,} \\ &\text{controlled forgetting} \quad \not\Longrightarrow \quad \text{successful acquisition of a new task.} \end{aligned} \tag{31.1}$$

Freezing the entire learner solves retention but not continual learning. Conversely, constraints imposed everywhere may preserve arbitrary extrapolations of the old model and prevent learning even when the tasks are compatible. Thus the central research problem is to preserve enough of the old computation, but not unnecessarily more, while keeping storage and learning cost manageable.

The presentation proceeds from exact virtual-example identities to compositional retention bounds, finite-probe guarantees, and their algorithmic interpretation. It then addresses plasticity, representation changes, long-horizon memory, and the precise questions that remain unresolved. The elementary results are proved here; their use in a general efficient continual learner is a research program, not an established conclusion.

31.2 Functional regularization and exact virtual-example identities

Let task A have distribution P_A on (X, Y), input marginal P_A^X, and risk

$$\mathcal{L}_A(f) = \mathbb{E}_{(X,Y)\sim P_A}[\ell(f(X), Y)]. \tag{31.2}$$

After learning A, the model is $f_A = f_{\theta_A}$. A subsequent task B suggests the objective

$$\min_{\theta} \{\mathcal{L}_B(f_\theta) + \lambda R_A(\theta)\}. \tag{31.3}$$

The purpose of R_A is retention. A separate capacity-control penalty may be needed for generalization on the new task.

31.2.1 An exact functional identity

For a chosen virtual-input distribution ν_A, define

$$R_A^{\mathrm{fun}}(\theta) = \mathbb{E}_{X\sim\nu_A}\|f_\theta(X) - f_A(X)\|^2. \tag{31.4}$$

Given virtual inputs $\widetilde{x}_i$, assign teacher targets $\widetilde{y}_i = f_A(\widetilde{x}_i)$. Their empirical squared training loss is *exactly*

$$\widehat{R}_A^{\mathrm{fun}}(\theta) = \frac{1}{M}\sum_{i=1}^{M}\|f_\theta(\widetilde{x}_i) - \widetilde{y}_i\|^2. \tag{31.5}$$

There is no approximation in this identity. Replacing the expectation in (31.4) by this empirical mean is a different step, requiring coverage and generalization assumptions. Nor does the identity imply that every parameter regularizer can be implemented by ordinary input–output examples in an arbitrary nonlinear architecture.

This use of a teacher connects to output distillation and replay [90, 143]. Functional regularization using inducing points has also been developed explicitly for continual learning [152]. The issue pursued here is the additional role of low-dimensional compositional interfaces, not a claim that functional preservation or virtual replay is new.

31.2.2 Exact quadratic regularization in a linear model

For $f_\theta(z) = \phi(z)^\top\theta$ and $\Delta\theta = \theta - \theta_A$,

$$R_A^{\mathrm{fun}}(\theta) = \Delta\theta^\top G_A\Delta\theta, \qquad G_A = \mathbb{E}_{\nu_A}[\phi(Z)\phi(Z)^\top]. \tag{31.6}$$

For weighted virtual examples, $G_A = \sum_i w_i\phi(\widetilde{z}_i)\phi(\widetilde{z}_i)^\top$, with $w_i \geq 0$. Conversely, a specified positive semidefinite matrix Q is realizable by such examples only when it admits this representation using *accessible feature vectors*. An arbitrary factorization of Q does not ensure that its factors can occur as $\phi(z)$ for available inputs. More general linear-observation examples can enlarge what is realizable, but they are not ordinary point evaluations.

31.2.3 A local quadratic approximation for nonlinear networks

Suppose the parameter Jacobian $J_A(z)$ is bounded and the second-order Taylor remainder is uniformly bounded on the probe domain near θ_A. Then

$$f_\theta(z) - f_A(z) = J_A(z)\Delta\theta + O(\|\Delta\theta\|^2), \tag{31.7}$$

$$R_A^{\mathrm{fun}}(\theta) = \Delta\theta^\top\mathbb{E}_{\nu_A}[J_A(Z)^\top J_A(Z)]\Delta\theta + O(\|\Delta\theta\|^3). \tag{31.8}$$

The matrix has Gauss–Newton form. For a Gaussian observation model with fixed isotropic covariance $\sigma^2 I$, the corresponding input-averaged Fisher matrix is $\sigma^{-2}\mathbb{E}[J_A^\top J_A]$. This provides a connection to Fisher-based penalties such as elastic weight consolidation [77], not a global equivalence to its usual diagonal approximation. For nonsmooth networks, the Taylor argument applies only where the stated differentiability assumptions hold; the functional identity (31.5) requires no such expansion.

The useful reversal is to choose *behavior to preserve* first and let virtual examples induce the penalty, rather than begin with an arbitrary distance between parameter vectors.

31.3 Sparse compositional interfaces and relevant domains

Consider a finite DAG G with input nodes returning the coordinates of $x \in \mathbb{R}^d$ and computational nodes V_{comp}. Each computational node returns a scalar,

$$h_v(x) = g_v\big((h_u(x))_{u\in \mathrm{pa}(v)}\big), \qquad k_v = |\,\mathrm{pa}(v)| \le k. \tag{31.9}$$

There is one scalar output node o, with $f = h_o$. A vector-valued interface can be treated similarly, but its relevant dimension is its total number of scalar coordinates, not merely its number of incoming edges.

Write g_v^A, h_v^A, f_A and g_v^B, h_v^B, f_B for the old and updated networks on this graph. The old interface tuple and its distribution are

$$Z_v^A(X) = (h_u^A(X))_{u\in \mathrm{pa}(v)}, \qquad \mu_v^A = \mathrm{Law}(Z_v^A(X)), \quad X \sim P_A^X. \tag{31.10}$$

Local virtual examples have the form

$$(\widetilde{z}_{v,i}, \widetilde{y}_{v,i}), \qquad \widetilde{y}_{v,i} = g_v^A(\widetilde{z}_{v,i}). \tag{31.11}$$

They probe a k_v-dimensional function even when the complete network has a high-dimensional input. They may be generated directly at the interface, without a generative model of raw inputs. Recorded inputs and targets suffice to replay these particular constraints after the old model is discarded. Generating additional teacher targets later requires the old function or an appropriate surrogate.

Two different domains must be distinguished. A *protected domain* D_v contains the old activation tuples whose behavior is to be retained. A possibly larger *propagation domain* contains both old and new activation tuples and is where the updated component must satisfy its Lipschitz bound. Protecting the old domain does not remove the need for stability when new activations drift away from it.

Choosing D_v as a known full box makes coverage convenient but may constrain irrelevant extrapolations. Choosing a smaller domain around old activations can preserve plasticity, but learning or certifying that domain itself requires information. Low dimension does not make this information free. Section 31.9 makes the issue explicit.

A reusable subgraph may replace a single node in this description. Its *external* input–output interface can be protected while its internal implementation changes. Such grouping is useful only when the resulting interfaces remain low-dimensional and their error propagation is controlled. No claim of a unique or privileged decomposition is required.

31.4 A compositional bound on forgetting

The following elementary propagation argument is related to the error-propagation principle in compositional approximation theory [106]. It is stated explicitly to identify every assumption needed for a retention guarantee.

Theorem 31 (Propagation of local preservation errors)**.** *Suppose the old and updated networks have the same finite DAG and identical input nodes. For each computational node v, assume*

$$|g_v^B(z) - g_v^B(z')| \le \sum_{u\in \mathrm{pa}(v)} L_{vu}|z_u - z'_u|, \qquad L_{vu} \ge 0, \tag{31.12}$$

for every pair of old and new parent tuples occurring on the old task. Define

$$a_v(x) = |g_v^B(Z_v^A(x)) - g_v^A(Z_v^A(x))|, \qquad r_v = \mathbb{E}_{P_A^X}[a_v(X)^2] < \infty. \tag{31.13}$$

Let c_v be the sum over all directed paths from v to o of the products of the edge coefficients L_{wu} along the path. The empty path gives $c_o = 1$, and $c_v = 0$ if no such path exists. Then

$$|f_B(x) - f_A(x)| \le \sum_v c_v a_v(x), \qquad \mathbb{E}_{P_A^X}|f_B - f_A| \le \sum_v c_v \sqrt{r_v}. \tag{31.14}$$

If the loss is ρ-Lipschitz in its prediction on the relevant range and the risks are finite, then

$$\boxed{|\mathcal{L}_A(f_B) - \mathcal{L}_A(f_A)| \le \rho \sum_{v \in V_{\mathrm{comp}}} c_v \sqrt{r_v}.} \tag{31.15}$$

In particular, this bounds the increase in old-task risk.

Proof. Let $e_v(x) = |h_v^B(x) - h_v^A(x)|$, with $e_v = 0$ at input nodes. Adding and subtracting $g_v^B(Z_v^A(x))$ gives

$$\begin{aligned} e_v(x) &\le |g_v^B(Z_v^B(x)) - g_v^B(Z_v^A(x))| + a_v(x) \\ &\le \sum_{u \in \mathrm{pa}(v)} L_{vu} e_u(x) + a_v(x). \end{aligned}$$

Unrolling in a topological order yields the pointwise bound; the coefficient of a_v is precisely the stated path sum. Take expectations and use $\mathbb{E}a_v \le \sqrt{\mathbb{E}a_v^2}$. Finally, $|\ell(f_B(x), y) - \ell(f_A(x), y)| \le \rho|f_B(x) - f_A(x)|$ and integration prove the risk bound. □

The local discrepancies are evaluated at *old* activations. They need not be evaluated at the updated activations, because the latter's effect is handled separately by (31.12). Squared loss is Lipschitz on bounded prediction and label ranges, not globally; discontinuous zero–one loss needs an additional argument, such as a margin condition.

Every altered node that can affect the old output must be included, frozen, or bounded separately. Protecting an arbitrary convenient subset gives no general guarantee. The theorem is sufficient, not necessary: compensating module changes can cancel at the output while incurring substantial local penalties.

Corollary 6 (Weighted functional regularization)**.** *For $q_v > 0$ and $R_q = \sum_v q_v r_v$,*

$$|\mathcal{L}_A(f_B) - \mathcal{L}_A(f_A)| \le \rho \left(\sum_v \frac{c_v^2}{q_v} \right)^{1/2} R_q^{1/2}. \tag{31.16}$$

Proof. Apply Cauchy–Schwarz to $\sum_v (c_v/\sqrt{q_v})\sqrt{q_v r_v}$. □

For fixed positive c_v and $\sum_v q_v = 1$, the prefactor is minimized by $q_v \propto c_v$. This optimizes the displayed certificate, not necessarily the trade-off with learning the new task. Uniform guarantees over candidate updates require uniform upper bounds on the relevant sensitivities.

31.5 Finite virtual probes: covering and empirical loss

Theorem 32 (Finite local probes under Lipschitz regularity)**.** *Assume Theorem 31. Let $D_v \subset \mathbb{R}^{k_v}$ contain $Z_v^A(X)$ almost surely. Both g_v^A and g_v^B are assumed Lipschitz on D_v in the sup norm, with constants K_v^A, K_v^B. Let $\mathcal{Z}_v \subset D_v$ be a finite δ_v-net satisfying*

$$\eta_v := \max_{z \in \mathcal{Z}_v} |g_v^B(z) - g_v^A(z)|. \tag{31.17}$$

Then

$$\sup_{z \in D_v} |g_v^B(z) - g_v^A(z)| \le \eta_v + (K_v^A + K_v^B)\delta_v, \tag{31.18}$$

and

$$|\mathcal{L}_A(f_B) - \mathcal{L}_A(f_A)| \le \rho \sum_v c_v[\eta_v + (K_v^A + K_v^B)\delta_v]. \tag{31.19}$$

For $D_v = [0,1]^{k_v}$, a Cartesian grid uses at most $(\lceil 1/\delta_v \rceil + 1)^{k_v}$ probes.

Proof. Choose $z_i \in \mathcal{Z}_v$ within distance δ_v of any $z \in D_v$. The triangle inequality bounds the discrepancy at z by $K_v^B\delta_v + \eta_v + K_v^A\delta_v$. This bounds $\sqrt{r_v}$, and Theorem 31 applies. Subdividing each coordinate into $\lceil 1/\delta_v \rceil$ intervals gives the stated grid. □

The regularity bound on the *updated* function is indispensable. Without it, a function can agree at every finite probe and change arbitrarily between probes. For a nonrectangular protected domain, the appropriate count is its covering number $N(D_v, \|\cdot\|_\infty, \delta_v)$, not necessarily the count for its enclosing box.

31.5.1 A binary-tree illustration

Take a balanced tree with d leaves and $m = d - 1$ computational nodes. Suppose each interface is $[0,1]^2$, $c_v \le 1$, both local Lipschitz constants are at most K, and all probes are matched exactly. Choosing $\delta_v = \varepsilon/(2\rho K m)$ gives forgetting at most ε with

$$M_{\text{total}} \le m\left(\left\lceil \frac{2\rho K m}{\varepsilon} \right\rceil + 1\right)^2 = O\left(\frac{\rho^2 K^2 m^3}{\varepsilon^2}\right). \tag{31.20}$$

For fixed regularity constants this is polynomial in d, unlike full uniform gridding of an unrestricted d-variable function. It is a sufficient construction, not a minimax lower bound or a separation from optimal global replay for the same compositional class. That comparison is revisited in Section 31.8.

31.5.2 The distinction between maximum error and mean squared error

The natural training objective uses

$$\widehat{r}_v = \frac{1}{M_v}\sum_{i=1}^{M_v} |g_v^B(\widetilde{z}_{v,i}) - g_v^A(\widetilde{z}_{v,i})|^2, \qquad \widehat{R}_q = \sum_v q_v \widehat{r}_v. \tag{31.21}$$

It does not directly impose the maximum-error constraint in Theorem 32. The exact elementary bridge is $\eta_v \le \sqrt{M_v \widehat{r}_v}$. Consequently,

$$|\mathcal{L}_A(f_B) - \mathcal{L}_A(f_A)| \le \rho\left[\left(\sum_v \frac{c_v^2 M_v}{q_v}\right)^{1/2} \widehat{R}_q^{1/2} + \sum_v c_v (K_v^A + K_v^B)\delta_v\right]. \tag{31.22}$$

The factor involving M_v can make this certificate loose. It cannot be silently omitted. Exact matching, explicit constraints on every probe, or a distributional generalization bound provide alternatives. Stored teacher values with absolute error at most ξ_v contribute an additional $\rho \sum_v c_v \xi_v$ to the maximum-error version of the certificate.

31.6 A smoothness extension that can be proved directly

The improved exponent suggested by r-smooth constituents need not remain a conjecture in a simple deterministic setting. The following construction uses ordinary function values rather than derivative observations.

Proposition 60 (Stable local interpolation)**.** *Let $g^A, g^B \in C^r([0,1]^k)$, where $r \geq 1$ is an integer, and assume $\|\partial_j^r g^A\|_\infty, \|\partial_j^r g^B\|_\infty \leq B$ for every coordinate j. Subdivide each coordinate into n intervals of length $h = 1/n$. In each resulting cell use a tensor product of r fixed, distinct interpolation nodes per coordinate, affinely rescaled from $[0,1]$. Let Λ_r be the one-dimensional interpolation norm for these nodes and set*

$$C_{k,r} = \frac{1}{r!} \sum_{j=0}^{k-1} \Lambda_r^j. \tag{31.23}$$

If the old and updated functions differ by at most η at every probe, then

$$\|g^B - g^A\|_\infty \leq 2C_{k,r} B h^r + \Lambda_r^k \eta. \tag{31.24}$$

At most $r^k n^k$ probes are needed, counting shared cell-boundary probes more than once if necessary.

Proof. On one cell, let P_j interpolate in coordinate j and let $I_h = P_1 \cdots P_k$. The one-dimensional remainder formula gives $\|(I - P_j)g\|_\infty \leq Bh^r/r!$, while $\|P_j\| \leq \Lambda_r$. The telescoping identity

$$I - P_1 \cdots P_k = \sum_{j=1}^{k} P_1 \cdots P_{j-1}(I - P_j)$$

implies $\|g - I_h g\|_\infty \leq C_{k,r} B h^r$. The tensor interpolation norm is at most Λ_r^k, so $\|I_h g^B - I_h g^A\|_\infty \leq \Lambda_r^k \eta$. Adding the two interpolation errors proves the bound on each cell, hence globally. The number of cells is n^k and each uses r^k values. □

With exact matching and fixed $k, r, B > 0$, local error ϵ_v therefore requires $O((B/\epsilon_v)^{k/r})$ probes as $\epsilon_v \downarrow 0$. Applying this construction to each component and imposing $\rho \sum_v c_v \epsilon_v \leq \varepsilon$ gives the corresponding global retention guarantee. The constants and error allocation still depend on the graph and on stability. This is a standard interpolation mechanism made explicit, not a claim of a new approximation principle. It does not apply unchanged to nonsmooth ReLU constituents or to unconstrained updated functions merely because their teacher is smooth.

31.7 Random probes, distribution mismatch, and capacity

Uniform covers can be wasteful. For probes sampled from ν_v, define

$$r_v^\mu = \mathbb{E}_{\mu_v^A} |g_v^B - g_v^A|^2, \qquad r_v^\nu = \mathbb{E}_{\nu_v} |g_v^B - g_v^A|^2. \tag{31.25}$$

If $\mu_v^A \ll \nu_v$ and $d\mu_v^A/d\nu_v \leq \beta_v$, then

$$r_v^\mu \leq \beta_v r_v^\nu. \tag{31.26}$$

This condition is substantive. An activation distribution supported on a lower-dimensional set need not be absolutely continuous with respect to uniform volume measure on an enclosing box. Low interface dimension alone does not justify a density-ratio bound.

Proposition 61 (Uniform finite-class sampling guarantee)**.** *For each node, let $\mathcal{H}_v$ be a fixed finite candidate class with $|g(z) - g_v^A(z)| \le B_v$ on the support of ν_v. Draw M_v independent probes from ν_v, let $\widehat{r}_v(g)$ be their empirical squared discrepancy, and suppose the density-ratio condition above holds. For m nodes and $0 < \alpha < 1$, with probability at least $1 - \alpha$, simultaneously for all nodes and all their candidates,*

$$r_v^\mu(g) \le \beta_v \left[\widehat{r}_v(g) + B_v^2 \sqrt{\frac{\log(m|\mathcal{H}_v|/\alpha)}{2M_v}} \right]. \tag{31.27}$$

Proof. The squared discrepancy lies in $[0, B_v^2]$. One-sided Hoeffding concentration and a union bound, allocating probability $\alpha/(m|\mathcal{H}_v|)$ to each candidate at node v, give the uniform bound under ν_v. Apply (31.26). □

The updated network is chosen after seeing the probes, so a concentration bound for one predetermined network is insufficient. The proposition handles this dependence by holding uniformly over each fixed class. Infinite neural network classes require a capacity-dependent analogue, for example through covering numbers or Rademacher complexity. An independent validation set can certify one selected update but does not automatically certify unlimited future adaptive updates.

There are alternatives to density ratios when geometry is controlled. If $|g_v^B - g_v^A| \le B_v$ and the two functions are Lipschitz with constants K_v^B, K_v^A on the union of the two supports, the squared discrepancy is $2B_v(K_v^A + K_v^B)$-Lipschitz. Coupling the two input distributions gives

$$r_v^\mu \le r_v^\nu + 2B_v(K_v^A + K_v^B) W_1(\mu_v^A, \nu_v), \tag{31.28}$$

where W_1 uses the sup-norm distance. Indeed, for every coupling (Z, Z') the difference between the squared discrepancies is bounded by the displayed Lipschitz constant times $\|Z - Z'\|_\infty$; take expectations and the infimum over couplings. This permits singular measures, but useful control of their distance must still come from information about old activations.

31.8 What compositional sparsity does and does not establish

The covering theorem establishes a constructive statement:

$$M_{\text{local}} \lesssim \sum_v \delta_v^{-k_v}, \qquad k_v \le k, \tag{31.29}$$

for normalized box domains and small positive resolutions. Thus access to constituents avoids having to build a full ambient-dimensional grid. The number of constituents, regularity constants, and path sensitivities remain in the bound.

A stronger claim would be that local replay is more memory-efficient than *every* global replay method for the same compositional class. The current results do not prove that claim. A global method may exploit the restricted class without gridding the input domain. For a fixed finite class $\mathcal{F}$ and bounded squared discrepancies, the same uniform-concentration argument depends on $\log|\mathcal{F}|$, not directly on an input-space grid. For fixed ReLU architectures, parameter- and depth-dependent pseudodimension bounds are available [**BartlettEtAl2019**]; distributional generalization therefore

need not incur an exponential dependence on ambient dimension. A uniform covering guarantee and an average-risk guarantee are not the same retention criterion.

A fair separation theorem must fix the hypothesis class and admissible updates, the required accuracy and probability, the retention norm, and the access model. A local method can query internal maps; a black-box global method may query only $f_A(x)$. This additional information must be declared, not treated as free in one method and unavailable in the other without explanation. A global method with architectural knowledge must also be an eligible comparator when that is the intended setting.

Storage must include the final model, replay inputs and targets, numerical precision, protected-domain descriptions, module correspondences, task routing, and any retained teacher or generator. Local examples use fewer coordinates per probe, but there can be many modules. In some regimes it may be cheaper simply to retain a module's parameters. Moreover, local retention is usually a stronger constraint than global retention, because global functional cancellations are allowed by the latter. Better coverage can therefore come at a plasticity cost.

The useful research question is consequently not whether a low-dimensional grid is smaller than a high-dimensional one. It is:

> Under comparable learning and retention requirements, when does access to a compositional representation permit a smaller, more reusable, or more efficiently constructed memory than access to the overall function alone?

The present chapter supplies a candidate mechanism and upper bounds, not an optimality or separation theorem.

31.9 Preserving knowledge without freezing extrapolation

31.9.1 Compatible tasks can be obstructed by excessive preservation

Consider squared-loss tasks

$$\begin{aligned} A: &\quad X \in [0, 1/3], \quad Y = 0, \\ B: &\quad X \in [2/3, 1], \quad Y = 1. \end{aligned} \tag{31.30}$$

Suppose learning A produces $f_A(x) = 0$ throughout $[0, 1]$. The tasks are compatible: the Lipschitz function

$$f_*(x) = \begin{cases} 0, & x \le 1/3, \\ 3x - 1, & 1/3 < x < 2/3, \\ 1, & x \ge 2/3 \end{cases} \tag{31.31}$$

solves both perfectly. However, enforcing $\|f - f_A\|_{L^\infty([0,1])} \le \epsilon < 1$ implies

$$\mathcal{L}_B(f) \ge (1 - \epsilon)^2. \tag{31.32}$$

This follows because $f(x) \le \epsilon$ at every new-task input. Sufficiently informative full-domain virtual constraints can therefore make learning B impossible at low risk, even though the tasks themselves do not conflict. They protect an arbitrary old extrapolation as if it were knowledge. Protecting only the old interval leaves f_* feasible.

The same phenomenon applies to a low-dimensional constituent map. Its values outside the old activation region may never have been constrained by data. Virtual examples transfer whatever

the teacher computes, not a certificate that the teacher is correct there. Domain restrictions, task-conditioned modules, justified confidence weights, or additional capacity may help, but each changes the assumptions that must be analyzed.

There are also genuinely incompatible tasks. A single task-agnostic output cannot simultaneously equal zero and one at the same input, since $(p-0)^2 + (p-1)^2 \geq 1/2$. Task context, separate heads, or a different prediction objective is necessary in that example. Compatibility is thus necessary for exact joint performance, but it is not sufficient for a particular preservation scheme to permit it.

31.9.2 A formal preservation–plasticity gap

Let $\mathcal{H}$ be the admissible hypothesis class, and let $\mathcal{C}_A(f)$ be a valid upper bound on old-task risk increase, obtained from one of the preceding certificates. Define

$$\begin{aligned} \mathcal{F}_A(\varepsilon) &= \{f \in \mathcal{H} : \mathcal{C}_A(f) \leq \varepsilon\}, \\ \Delta_{A\to B}(\varepsilon) &= \inf_{f\in\mathcal{F}_A(\varepsilon)} \mathcal{L}_B(f) - \inf_{f\in\mathcal{H}} \mathcal{L}_B(f). \end{aligned} \tag{31.33}$$

Assume the risks are finite and interpret an empty feasible set as having infinite infimum. The gap is nonnegative. Retention bounds establish the safety of $\mathcal{F}_A(\varepsilon)$; they do not establish that $\Delta_{A\to B}(\varepsilon)$ is small. A loose certificate or overly strong local constraints can enlarge the gap even when an actual joint solution exists.

A useful theory must control three separate issues: the existence of a low-risk solution in the preservation-feasible set, its efficient discovery, and its generalization from new-task data. Compositional task relatedness may help the first by permitting new compositions of stable components. It does not automatically resolve any of the three.

31.9.3 What minimizing the actual regularized objective guarantees

The relation between the algorithmic objective and the retention theorem can be made explicit, without assuming that optimization succeeds automatically.

Proposition 62 (A comparator-based retention certificate)**.** *Use fixed virtual covers as in Theorem 32. Assume every admissible update has $c_v \leq \bar{c}_v$ and $K_v^A + K_v^B \leq \bar{K}_v$. Fix $q_v > 0$ and define*

$$J(\theta) = \mathcal{L}_B(f_\theta) + \lambda \widehat{R}_q(\theta) + \gamma\Omega(\theta), \qquad \lambda > 0, \quad \gamma \geq 0, \tag{31.34}$$

where $\mathcal{L}_B, \Omega$ are nonnegative. Suppose a comparator θ_* satisfies $\mathcal{L}_B(f_{\theta_*}) \leq a$, $\widehat{R}_q(\theta_*) \leq b$, and $\Omega(\theta_*) \leq u$. If an admissible output $\widehat{\theta}$ satisfies $J(\widehat{\theta}) \leq J(\theta_*) + \tau$, where $\tau \geq 0$, then

$$\widehat{R}_q(\widehat{\theta}) \leq b + \frac{a + \gamma u + \tau}{\lambda}, \qquad \mathcal{L}_B(f_{\widehat{\theta}}) \leq a + \lambda b + \gamma u + \tau. \tag{31.35}$$

Moreover, with

$$A_q = \left(\sum_v \frac{\bar{c}_v^2 M_v}{q_v}\right)^{1/2}, \qquad D = \sum_v \bar{c}_v \bar{K}_v \delta_v, \tag{31.36}$$

the old-task risk increase is bounded by

$$\mathcal{L}_A(f_{\widehat{\theta}}) - \mathcal{L}_A(f_A) \leq \rho \left[A_q \sqrt{b + \frac{a + \gamma u + \tau}{\lambda}} + D \right]. \tag{31.37}$$

Proof. Nonnegativity allows either the replay term or the new-task loss to be bounded by $J(\widehat{\theta}) \leq a + \lambda b + \gamma u + \tau$. Substitute the resulting replay bound into (31.22) using the uniform constants. □

An exact-preservation comparator with $b = 0$ and small a, u would make the trade-off favorable, provided τ is also small. But its existence is precisely a substantive assumption. Using the old network itself as a zero-replay comparator can certify retention while providing a poor bound on new-task performance. Taking λ large does not manufacture a good new solution. If training uses empirical new-task risk, the proposition holds with that risk in J; a separate generalization bound is needed to replace it by population performance.

31.10 Stability of composition is a separate requirement

Low arity does not bound downstream amplification. For example, repeated one-input maps $g(z) = \min\{1, az\}$ on $[0, 1]$, with $a > 1$, amplify a small positive perturbation near zero by a per stage until saturation. A depth-D chain can require a perturbation as small as order εa^{-D} to maintain a prescribed output accuracy. Every interface is one-dimensional, yet preservation can be expensive.

For general DAGs, both amplification along a path and the number of paths matter. A norm or architecture constraint is useful here only insofar as it provides quantitative bounds on the relevant weighted path sums, activation domains, and local regularity. A penalty that merely encourages small norms is not, by itself, a hard uniform bound for every update. Also, changing the units of an intermediate representation can move scale between local errors and downstream coefficients; the full certificate, not c_v in isolation, is the object to control.

This role of regularization differs from preservation and from new-task generalization. One may use virtual-example penalties for preservation and other constraints to control capacity and stability, but a successful analysis must identify which property each term actually guarantees. Showing that ordinary learning dynamics maintain useful compositional stability without sacrificing plasticity remains a substantial question.

31.11 Nonunique decompositions and coordinate changes

For $f = h \circ g$ and invertible T,

$$f = (h \circ T^{-1}) \circ (T \circ g). \tag{31.38}$$

The global function is unchanged, although both constituent functions may change. Local virtual penalties with fixed interface coordinates can punish such a harmless reorganization. This is not a reason to require discovery of *the* correct decomposition: no unique decomposition is needed. It is a reason to specify which interfaces are held fixed or how they are aligned.

Protecting the external map of a reusable subgraph permits arbitrary internal reorganization compatible with that map. At boundaries between subgraphs, coordinate changes must still be compensated consistently. Independently minimizing alignment errors at each node does not guarantee this consistency.

A known, compatible coordinate transformation can be handled directly. For each scalar node choose an invertible map T_v from old interface coordinates to new coordinates, with identity maps on input nodes and at the output. Define the updated computation expressed in old coordinates by

$$\bar{g}_v^B(z) = T_v^{-1}\Big(g_v^B\big((T_u(z_u))_{u \in \mathrm{pa}(v)}\big)\Big). \tag{31.39}$$

Assume the maps and their inverses are defined on all needed domains. By induction through the graph, $\bar{h}_v^B(x) = T_v^{-1}(h_v^B(x))$ and $\bar{f}_B = f_B$. Theorem 31 therefore applies to $\bar{g}_v^B$ and g_v^A. If the coordinate maps are Lipschitz, suitable coefficients are

$$\bar{L}_{vu} = \mathrm{Lip}(T_v^{-1})\, L_{vu}\, \mathrm{Lip}(T_u). \tag{31.40}$$

For exact compensated transformations, $g_v^B((T_u(z_u))_u) = T_v(g_v^A(z))$, all aligned local discrepancies vanish. Thus compatible known alignments are already covered by a change of coordinates, albeit with possible deterioration of stability constants.

The difficult extension is to discover useful correspondences from data, possibly while changing the graph, grouping modules, or mixing vector-valued coordinates. Such transformations may change scalar interface dimension or destroy sparsity. The question is not uniqueness, but efficient maintenance of usable, stable, and sufficiently flexible interfaces.

31.12 Many tasks: persistent anchors and reusable computational memory

31.12.1 Moving teachers can accumulate error

For one old task s and a sequence $f_s, f_{s+1}, \dots, f_T$,

$$\|f_T - f_s\|_{L^1(P_s^X)} \le \sum_{t=s+1}^{T} \|f_t - f_{t-1}\|_{L^1(P_s^X)}. \tag{31.41}$$

Every term must be controlled on the *same old-task distribution*. Small changes measured only on each newest task do not imply this bound is small. Even when every term is bounded by ϵ, the certificate grows as $(T-s)\epsilon$. Summable error budgets are one possible remedy, but require increasingly accurate preservation.

Persistent anchors provide a different remedy. Store targets from a fixed reference computation and continue comparing future versions against those targets, instead of replacing them by the latest model's outputs. If for all $t \ge s$,

$$\sup_{z \in D_v} |g_v^t(z) - g_v^s(z)| \le \epsilon_v, \qquad c_v^{s,t} \le \bar{c}_v, \tag{31.42}$$

then the propagation theorem gives

$$\sup_{t \ge s} |\mathcal{L}_s(f_t) - \mathcal{L}_s(f_s)| \le \rho_s \sum_v \bar{c}_v \epsilon_v. \tag{31.43}$$

There is no explicit factor counting updates. This is a deterministic statement under uniform assumptions, not a claim that those assumptions persist automatically. A fixed finite cover controls every future update that meets the same regularity and probe constraints; a statistical certificate used repeatedly must have the appropriate uniform or sequential validity.

Preserving a separate full memory for every task can still be expensive. Compositional reuse suggests storing a module's reference constraints once, then referring to them wherever that module is used.

31.12.2 A conditional reusable-library theorem

Let $\phi_1^*, \dots, \phi_J^*$ be fixed reference modules, each with scalar output and at most k scalar inputs. Reference modules can be consolidated at different times but are not silently relabeled afterward. An old task s is specified by a fixed composition graph G_s, whose node v calls module $j(v)$; a module can occur at several nodes or in several tasks. Any learned head, router, or connector needed for an old output is included among the protected computations, or is held fixed.

Let f_s^* be this task graph evaluated with reference modules, and let $f_{s,t}$ be the same graph evaluated at time t with current modules ϕ_j^t. Correct selection of the task graph is assumed, for example through available task context. This is not an implicit solution to inference of task identity.

Theorem 33 (Retention from shared persistent module anchors)**.** *For each old task s, suppose $\mathcal{L}_s(f_s^*) \le a_s$ and its loss is ρ_s-Lipschitz on the relevant prediction range. Assume that for all subsequent times t and all used modules,*

$$\sup_{z \in D_j} |\phi_j^t(z) - \phi_j^*(z)| \le \epsilon_j. \tag{31.44}$$

The domain D_j must contain every reference activation tuple at an occurrence of j in every task covered by the guarantee. Suppose the current compositions satisfy the propagation assumptions with path coefficients bounded by $\bar{c}_{s,v}$ uniformly in t. Then

$$\mathcal{L}_s(f_{s,t}) \le a_s + \rho_s \sum_{v \in V_{\mathrm{comp}}(G_s)} \bar{c}_{s,v} \epsilon_{j(v)} \qquad \text{for every such } s, t. \tag{31.45}$$

Each module's finite virtual constraint set need be stored only once, regardless of its number of occurrences in the graphs.

Proof. For a fixed s, t, use the reference task composition as the old network and the current task composition as the new network. At node v, the reference activation belongs to $D_{j(v)}$, so its local discrepancy is at most $\epsilon_{j(v)}$. The pointwise propagation bound and Lipschitz loss yield (31.45). The hypotheses hold for every s, t. The same stored pairs constrain the same module function at every call; occurrences affect the risk sum but do not require duplicate copies of those pairs. □

If both the current and the task-completion network are within these bounds of f_s^*, their risk difference is at most the sum of the two reference bounds. It need not be measured against a perfectly learned reference. The assumption $\mathcal{L}_s(f_s^*) \le a_s$ must itself be established: accurately preserving a poorly chosen library cannot make its compositions solve the tasks.

For Lipschitz modules with fixed tolerances, the number of distinct anchor pairs can be bounded constructively by

$$M_{\mathrm{anchor}} = \sum_{j=1}^{J} M_j, \qquad M_j \lesssim \delta_j^{-k_j}, \quad k_j \le k, \tag{31.46}$$

with tolerances chosen using the local-cover theorem. This counts unique modules, not module occurrences or task-specific copies of their constraints. It does not imply that the *total* memory is independent of the number of tasks. A more honest accounting is

$$\begin{aligned} \mathrm{Mem}(T) = \mathrm{Mem}_{\mathrm{model}} + \mathrm{Mem}_{\mathrm{anchors}} + \mathrm{Mem}_{\mathrm{domains}} \\ + \mathrm{Mem}_{\mathrm{routing}} + \mathrm{Mem}_{\mathrm{auxiliary}}. \end{aligned} \tag{31.47}$$

Arbitrary scalar-input/scalar-output anchor pairs require approximately $\sum_j M_j(k_j + 1)$ stored scalar values; prescribed grids can reduce the cost of storing coordinates. Converting this count

to bits requires precision choices and a quantization error budget. Routing may require a distinct description for every task, and any retained teacher or generator belongs in the auxiliary term. Training memory and inference memory may also differ.

The claimed reuse requires that module meanings, relevant domains, regularity, and sensitivity budgets remain adequate as tasks arrive. New tasks may enlarge a module's relevant domain or use it in a more sensitive composition, forcing additional probes or tighter tolerances even when J is unchanged. A fixed library of constraints is therefore justified only under explicit uniform bounds on these quantities.

31.12.3 What this theorem leaves unresolved

The theorem establishes a retention property for an already usable shared library. It does not prove that the library can be discovered efficiently, that new tasks can be expressed by it, or that an optimizer can find their compositions from few examples. Nor does it show that updating a shared module rather than adding a new one will remain feasible.

Worst-case memory results reinforce the need for structural assumptions. Chen, Papadimitriou, and Peng **[ChenEtAl2022]** exhibit hypothesis classes and jointly realizable task sequences for which any general single-pass continual learner requires memory growing linearly with task count in their PAC setting. Their lower bound is not merely a consequence of inconsistent tasks. It does not rule out gains for suitably restricted reusable-module families, but such gains require more than an informal assertion that tasks are compatible.

The next substantive target is a learning theorem in which task structure controls library discovery, new-task sample and optimization costs, and complete memory usage. The preservation result above is one ingredient of that theorem, not a substitute for it.

31.13 Algorithmic interpretation and discriminating tests

The framework suggests a procedure, but the guarantees attach to verified properties rather than to the procedure's name. At consolidation, specify the modules or subgraphs being protected, their interface conventions, and the regions relevant to old behavior. Construct covers or a justified probe distribution; store targets from the fixed reference maps and the information needed to interpret them later. Avoid silently changing those targets when the teacher changes.

For a new task, permit reuse, additional compositions, and where needed new modules or task-conditioned heads. Train with new-task loss, local replay penalties, and appropriate capacity or stability controls. Check whether the actual guarantee uses mean squared discrepancy, maximum probe discrepancy, or a distributional bound. Relevant computations left unprotected must be frozen or controlled separately. Changing an interface requires a consistent alignment or a revised retention argument.

Evaluate old-task performance on held-out old-task data when available, and evaluate new-task performance separately. A low replay loss is not enough without coverage and regularity; a low forgetting score is not enough without acquisition of the new task. Empirical success would test the usefulness of the assumptions, not turn them into universal facts.

A discriminating initial study would use targets assembled from smooth bivariate modules on balanced trees. Compare local replay with global replay *on the same architecture and candidate class*, a frozen model, unrestricted fine-tuning, parameter regularization, and a model-expansion baseline. Include both full-domain and old-relevant-domain probes. Vary ambient dimension, local arity, graph depth, number of distinct modules, module reuse, and downstream sensitivities. Measure total stored bits or values and computation, not only the number of replay examples.

Three stress tests are particularly informative: compatible tasks requiring changes outside old activation regions; exact compensated changes of internal coordinates; and long sequences comparing persistent anchors with continually refreshed teacher targets. A separate oracle-interface condition can distinguish an advantage of known local access from the difficulty of discovering the interfaces. The shared-library experiments should also report growth in routing descriptions and relevant domains.

The proposal is complementary to generative replay [143], output distillation [90], latent replay [121], and functional inducing-point summaries [152]. These are selected antecedents, not an exhaustive review. The prospective contribution is a precise relationship among compositional interfaces, preservation cost, plasticity, and long-term reuse; no empirical superiority is asserted here.

31.14 Established conclusions and the remaining theoretical program

31.14.1 Established conditional results

The chapter proves exact empirical virtual-example identities, propagation bounds from local discrepancies to old-task risk, finite-cover certificates, and a k/r probe exponent under explicit smooth interpolation assumptions. It connects mean squared replay loss to maximum-error control and to a comparator-based optimization certificate. Known compatible coordinate changes reduce to the same propagation argument. Under uniform assumptions, persistent anchors avoid an explicit update-count factor and shared modules can reuse their stored constraints across several task graphs.

These results establish retention under specified conditions. They do not establish a general efficient continual learner. The principal remaining questions are the following.

31.14.2 Retention with efficient acquisition

For which compositionally related task families is the preservation–plasticity gap in (31.33) small, and can a near-feasible, low-risk new solution be found and generalized from few examples? An optimization bound conditional on a good comparator does not prove that the comparator exists or is reachable. Convergence of a finite or overparameterized network to a suitable solution can settle existence for that instance, but does not by itself settle coverage, retention, or sample complexity. The important structural assumption concerns *relations between tasks*, not low arity alone.

31.14.3 A genuine memory or computational advantage

Obtain comparable upper and lower bounds using the same access model, hypothesis class, retention criterion, precision, and complete storage accounting. The local-grid construction is not an exponential separation from optimal global replay. Potential advantages include cheaper construction of preservation constraints, reuse without duplication, or fewer queries under a specified black-box comparator model; each requires its own theorem.

31.14.4 Useful domains and flexible interfaces

How can learning distinguish old-task knowledge from arbitrary extrapolation, choose protected domains, and decide whether a computation should be shared, adapted, or duplicated? How can correspondences survive beneficial representation changes? The objective is not a unique decomposition, but interfaces that remain informative, stable, and economical for the task sequence.

31.14.5 Stable library growth over time

A long-horizon learning theorem must control module count, relevant domains, regularity, propagation sensitivities, and task-routing descriptions under actual updates. Fixed module identities need not imply fixed memory if coverage or precision requirements grow. Learning new compositions must also be accounted for. These are substantive problems, unlike refinements of constants or probe allocation. In the restricted settings stated above, higher-order interpolation rates, compatible known alignments, and the persistent-anchor retention argument are already proved rather than left as conjectures.

31.15 Conclusion: a conditional theory and a sharper objective

Virtual examples and regularization express constraints on learned behavior. Compositional sparsity can localize those constraints to low-dimensional computations. Under regularity, coverage, and stability assumptions, local preservation controls forgetting at the full-network output.

The distinction between retention and learning remains essential. Preserving every old module everywhere can retain mistakes and extrapolations or prohibit harmless reorganizations. Local probe counts alone do not prove superiority over global replay. Shared persistent anchors avoid duplication only when their semantics, domains, and accuracy budgets remain adequate; routing and other memory costs still count.

The resulting research objective is precise:

> Under what assumptions on a sequence of compositionally related tasks can virtual-example regularization preserve previous abilities while permitting efficient acquisition of new ones, with storage governed mainly by reusable computations rather than duplicated task-specific memories?

The chapter establishes several components of an answer. Its proved message is that *local functional preservation can guarantee global retention*. A fuller theory must explain when those constraints permit new capabilities, when suitable updates are efficiently learnable, and when their total cost is genuinely favorable.

Appendix

CHAPTER A

Stability, ERM, and the Foundations of Learnability

The classical analysis of supervised learning is often framed in terms of capacity measures such as VC dimension or Rademacher complexity. An alternative but equivalent viewpoint focuses on the stability *of the learning algorithm with respect to perturbations of its training data. In this chapter we recall a statistical notion of stability based on leave-one-out cross-validation (CV_{loo}) and the theorem of Mukherjee, Niyogi, and Poggio that characterizes consistency of empirical risk minimization (ERM) in terms of this stability.*

A.1 Learning setup and ERM

Let $\mathcal{X}$ be an input space and $\mathcal{Y}$ an output space (e.g. $\mathcal{Y} = \{0, 1\}$ or $\mathbb{R}$). A learning problem is specified by an unknown distribution P on $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$. We observe an i.i.d. training sample

$$S_n = (z_1, \dots, z_n), \qquad z_i = (x_i, y_i) \sim P,$$

and choose a hypothesis $h \in \mathcal{H}$ from a hypothesis class $\mathcal{H}$.

A loss function $\ell : \mathcal{H} \times \mathcal{Z} \to [0, 1]$ (or more generally a bounded nonnegative function) measures prediction error. The *true risk* of h is

$$L(h) \;=\; \mathbb{E}_{z \sim P}[\ell(h, z)],$$

and the *empirical risk* on S_n is

$$L_n(h) \;=\; \frac{1}{n} \sum_{i=1}^{n} \ell(h, z_i).$$

The empirical risk minimization (ERM) principle selects

$$\hat{h}_n \;\in\; \arg\min_{h \in \mathcal{H}} L_n(h).$$

We say that ERM over $\mathcal{H}$ is *(universally) consistent* if

$$L(\hat{h}_n) \;\xrightarrow[n \to \infty]{P}\; \inf_{h \in \mathcal{H}} L(h),$$

for every distribution P on $\mathcal{Z}$.

A.2 CV_{loo} stability

Let $\mathcal{A}_n$ denote a (possibly randomized) learning algorithm that, given a sample S_n, outputs a hypothesis $\mathcal{A}_n(S_n) \in \mathcal{H}$. For $i \in \{1, \dots, n\}$, let $S_n^{(i)}$ denote the sample with the i-th point removed:

$$S_n^{(i)} = (z_1, \dots, z_{i-1}, z_{i+1}, \dots, z_n).$$

Intuitively, *leave-one-out* stability requires that removing a single training point does not significantly change the prediction on that point.

Definition 10 ($\mathrm{CV}_{\mathrm{loo}}$ stability)**.** *A sequence of algorithms $(\mathcal{A}_n)_{n\geq 1}$ is said to have CV_{loo} stability with parameters (β_n) if*

$$\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\Big[\big|\ell(\mathcal{A}_n(S_n), z_i) - \ell(\mathcal{A}_{n-1}(S_n^{(i)}), z_i)\big|\Big] \;\leq\; \beta_n,$$

for all n, where the expectation is over the i.i.d. draw of S_n (and any internal randomness of $\mathcal{A}_n$). We say the algorithm is asymptotically CV_{loo}-stable *if $\beta_n \to 0$ as $n \to \infty$.*

There are several related stability notions (uniform stability, hypothesis stability, etc.). The $\mathrm{CV}_{\mathrm{loo}}$ form is directly linked to cross-validation and has an especially clean connection to ERM consistency.

A.3 Stability and generalization

We now state the main result of Mukherjee, Niyogi, and Poggio. In the wording below, we preserve the original informal summary of the theorem.

Theorem 34 (Mukherjee–Niyogi–Poggio)**.** *We propose a statistical form of stability, defined in terms of the prop- erty of cross-validation leave-one-out (CVloo) stability, and*

1. *show that it is sufficient, in general, for generalization, that is convergence in probability of the empirical error to the expected error, for any algorithm satisfying it.*
2. *Our second observation is that for bounded loss classes, CVloo stability is necessary and sufficient for consistency of ERM.*

Here "generalization" in part (a) means that for any asymptotically $\mathrm{CV}_{\mathrm{loo}}$-stable algorithm $(\mathcal{A}_n)$,

$$L(\mathcal{A}_n(S_n)) - L_n(\mathcal{A}_n(S_n)) \;\xrightarrow[n\to\infty]{P}\; 0.$$

Part (b) shows that, for bounded loss classes, ERM is consistent *if and only if* it is $\mathrm{CV}_{\mathrm{loo}}$-stable. Thus stability is not merely sufficient; it is an exact characterization of when ERM behaves well.

A.4 Stability as a modeling requirement

From a modeling standpoint, Theorem 34 has a simple but strong implication:

- If we take ERM (or approximate ERM) as the fundamental inductive principle underlying learning, then *stability is required* for consistency.

- Instability—in the sense that removing or slightly perturbing a single training point can qualitatively change the predictor—rules out consistency of ERM for bounded losses.

In the remainder of the book we will use this observation in the following direction: *we impose stability as a requirement* on realistic learning problems, and ask what structural properties of target functions follow from this requirement. In particular, in the next chapter we argue that stability forces a kind of *genericity* of targets, and that this genericity in turn makes optimization by gradient-based methods possible.

A.5 Implementation

This section details a conceptual, fully functional PyTorch implementation of the Associative Turing Machine (ATM). The architecture natively enforces the sparse compositionality constraint by dynamically managing a fixed-size buffer and restricting the router's input strictly to relational metadata, rather than high-dimensional feature vectors.

A.5.1 The Evolutionary Router and AM Bank

We first define the two core neural modules. The router network is intentionally minimal, reducing the parameter space to ensure efficient navigation by an evolutionary algorithm.

Listing A.1: Core modules for the ATM: The Metadata Router and Associative Memory Bank.

```
import torch
import torch.nn as nn

class MetadataRouter(nn.Module):
    """
    The Evolutionary Router.
    Evaluates relational metadata to output priority scores.
    It does NOT access the actual high-dimensional feature vectors.
    """
    def __init__(self, metadata_dim=3, hidden_dim=16):
        super().__init__()
        # A minimal network, ideal for NEAT or simple genetic algorithms
        self.net = nn.Sequential(
            nn.Linear(metadata_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, 1)
        )

    def forward(self, metadata):
        # metadata shape: (Batch, Active_Length, Metadata_Dim)
        # Returns scores: (Batch, Active_Length)
        return self.net(metadata).squeeze(-1)


class AssociativeMemoryBank(nn.Module):
    """
```

```
    The O(1) Associative Memory Bank.
    Strictly accepts two vectors and computes a localized operation.
    """
    def __init__(self, feature_dim, hidden_dim=64):
        super().__init__()
        # Bounded capacity enforcing sparse compositionality.
        self.net = nn.Sequential(
            nn.Linear(feature_dim * 2, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, feature_dim)
        )

    def forward(self, x1, x2):
        # Concatenate the two selected nodes
        x = torch.cat([x1, x2], dim=-1) # Shape: (Batch, feature_dim * 2)
        return self.net(x)
```

A.5.2 The Core ATM Architecture

The primary ATM class orchestrates the dynamic buffer routing loop. By maintaining a separate metadata tensor, the router's decisions are based entirely on the structural topology of the computational tree being built, independent of the underlying feature data.

Listing A.2: The main Associative Turing Machine loop.

```
class AssociativeTuringMachine(nn.Module):
    def __init__(self, input_len, max_steps, feature_dim):
        super().__init__()
        self.input_len = input_len
        self.max_steps = max_steps
        self.feature_dim = feature_dim

        # Metadata dim = 3: [relative_age, computational_depth, is_available]
        self.metadata_dim = 3

        self.router = MetadataRouter(self.metadata_dim)
        self.am_bank = AssociativeMemoryBank(feature_dim)

    def forward(self, x):
        """
        x shape: (Batch, input_len, feature_dim)
        """
        batch_size = x.size(0)
        device = x.device

        # 1. Initialize the Working Memory Buffer
        # Total capacity = initial inputs + maximum intermediate steps
        total_capacity = self.input_len + self.max_steps
        buffer = torch.zeros(batch_size, total_capacity, self.feature_dim, device=d
```

```
        buffer[:, :self.input_len, :] = x

        # 2. Initialize the Metadata Tensor
        # Shape: (Batch, total_capacity, metadata_dim)
        metadata = torch.zeros(batch_size, total_capacity, self.metadata_dim, devic

        # Setup initial metadata for the raw inputs
        for i in range(self.input_len):
            metadata[:, i, 0] = i           # Relative age (0 to N-1)
            metadata[:, i, 1] = 0           # Depth (raw inputs are depth 0)
            metadata[:, i, 2] = 1.0         # is_available (1.0 = true)

        active_len = self.input_len
        batch_indices = torch.arange(batch_size, device=device)

        # 3. The Computational Loop
        for step in range(self.max_steps):

            # Extract metadata for currently active items
            current_metadata = metadata[:, :active_len, :]

            # --- ROUTER EXECUTION ---
            # Get priority scores based strictly on metadata
            scores = self.router(current_metadata)

            # Mask out unavailable items by setting score to -inf
            mask = (current_metadata[:, :, 2] == 0.0)
            scores = scores.masked_fill(mask, float('-inf'))

            # Extract the indices of the top 2 scores
            _, top_indices = torch.topk(scores, k=2, dim=-1)
            idx1, idx2 = top_indices[:, 0], top_indices[:, 1]

            # Mark these indices as used (is_available = 0.0)
            metadata[:, :, 2].scatter_(1, idx1.unsqueeze(-1), 0.0)
            metadata[:, :, 2].scatter_(1, idx2.unsqueeze(-1), 0.0)

            # --- EXTRACT FEATURES ---
            x1 = buffer[batch_indices, idx1]
            x2 = buffer[batch_indices, idx2]

            # --- AM BANK EXECUTION ---
            y = self.am_bank(x1, x2)

            # --- WRITE BACK TO BUFFER ---
            buffer[:, active_len, :] = y

            # Update metadata for the newly created intermediate variable
```

```
            depth1 = metadata[batch_indices, idx1, 1]
            depth2 = metadata[batch_indices, idx2, 1]
            new_depth = torch.max(depth1, depth2) + 1

            metadata[:, active_len, 0] = active_len # New relative age
            metadata[:, active_len, 1] = new_depth  # New depth
            metadata[:, active_len, 2] = 1.0        # Available for future routing

            active_len += 1

        # The final answer is the last item written to the buffer
        final_output = buffer[:, active_len - 1, :]
        return final_output
```

A.5.3 Integration with Evolutionary Algorithms

Traditional gradient descent is largely incompatible with this dynamic routing architecture because discrete selection operations (e.g., `torch.topk`) break the differentiable computational graph. However, by employing an evolutionary strategy, this non-differentiability is no longer a hindrance.

To evolve the router, one interacts directly with the network's state dictionary. The process entails extracting `atm.router.state_dict()`, flattening the parameters into a one-dimensional "genotype" vector, and applying mutational Gaussian noise. During crossover, parameter vectors are exchanged between high-performing models before being reloaded via `atm.router.load_state_dict()`. Because the forward pass does not require gradient tracking, discrete evaluations function perfectly, allowing fitness to be calculated strictly based on the accuracy of the `final_output` against the target labels.

CHAPTER B

Compositionality in Machine Learning and Physics

We formalize a bridge between efficient computation and learnability via bounded–fan-in compositional structure. On the learning side, we prove an ML Approximation Theorem*: when a target factors through a bounded–fan-in DAG with maximal local arity d^*, the relevant approximation and sample–complexity exponents depend on d^* rather than the ambient dimension d. On the physics side, we state a* finite-horizon compilation principle *(explicitly conditional): whenever a finite-horizon evolution map admits a* uniform *simulator using a finite set of bounded-arity primitives, the map compiles to a P-uniform bounded–fan-in DAG whose size is proportional to the simulator cost. Illustrative cases include discrete-time lattice-local models and k-local quantum dynamics compiled into uniform two-qubit circuits. Limitations include chaotic sensitivity, long-range or dense couplings, precision constraints, and thermodynamic considerations; all statements are finite-precision, finite-horizon, and conditional on uniform simulability.*

B.1 Introduction

Modern deep learning owes as much to *structure* as to *scale*. Many of the functions we hope to learn are not generic mappings on high-dimensional spaces but the outputs of computations that decompose into smaller, interacting parts arranged hierarchically. This observation has two complementary faces—algorithmic and physical.

The algorithmic view From the standpoint of computation, any function evaluable by a polynomial-time Turing machine admits a polynomial-size Boolean circuit of bounded fan-in [**Arora_Barak_2009**, **sipser1996intro**]. Interpreting such a circuit as a directed acyclic graph (DAG) exposes its *compositional structure*: each node is a local operation on a small number of inputs, and the full computation emerges from a finite sequence of such local interactions. In the continuous setting, this structure is mirrored by deep networks, whose layers correspond to circuit levels. Thus, efficient Turing computability implies that the underlying function is *compositionally sparse*: at each stage it depends on a bounded number of variables. In this sense, the success of deep architectures reflects the structure of efficient computation.

The physical view In physical systems, efficient Turing computability cannot be assumed *a priori*. What is often available is *finite-horizon locality*: many models bound the spread of influence over

time (e.g., relativistic light cones or Lieb–Robinson bounds in quantum lattices [**lieb1972finite**, **NachtergaeleSims2010**]). When such locality is paired with a *uniform* simulator built from bounded-arity primitives, the finite-horizon input–output map can be *compiled* into a bounded–fan-in DAG whose structure reflects the causal neighborhood at horizon T. This is an *algorithmic* statement about compilation, not a universal physical law about bounded *physical* degree: long-range or dense couplings (e.g., Coulomb, mean-field) may violate bounded degree even though a uniform simulator—and hence a bounded–fan-in DAG—still exists with polynomial overhead.

The resulting lesson is conditional but practical: whenever finite-horizon locality and uniform simulability hold, the effective complexity is governed not by the ambient dimension d but by the *algorithmic* largest local arity d^* (the size of the neighborhoods feeding each compiled node) and by the horizon/depth T over which interactions unfold. Limits remain: chaotic sensitivity can amplify constants with T, exact-real outputs are excluded (finite precision), and thermodynamic-limit properties need not be decidable.

From observation to theorem. We make the intuition precise via two complementary results, separating what is provable in general from what holds under explicit hypotheses. On the learning side, the *ML Approximation Theorem* shows that when a target factors through a bounded–fan-in DAG, the usual exponent d/r (ambient dimension over smoothness) is replaced by d^*/r, where d^* is the maximal local arity in the DAG. On the physics side, rather than asserting a universal causal–locality law, we prove a *finite-horizon compilation principle*: whenever a finite-horizon evolution map admits a *uniform* simulator using a fixed finite set of bounded-arity primitives with cost $s(n, T, \varepsilon)$, it compiles to a *polynomial-time uniform* bounded–fan-in DAG of size $O(s(n, T, \varepsilon))$. *Examples* (not universal claims) include discrete-time lattice-local updates with finite propagation and k-local quantum dynamics compiled to two-qubit circuits. Together, these results identify a common mathematical structure—bounded local arity d^* and horizon/depth T—that governs approximation and learnability; the physics connection is *conditional* (finite precision, finite horizon, uniform simulability), not automatic.

Historical background and conceptual context

The route from Turing machines to circuits is a cornerstone of complexity theory and underlies the definition of major classes such as P and NC [**Arora_Barak_2009**]. Deep learning, viewed through this lens, can be seen as the continuous analogue of polynomial-size, bounded–fan-in circuit computation [**poggio_deep_shallow_2017**, 107]. Approximation theory has long established that deep networks approximate hierarchical functions exponentially more efficiently than shallow ones, and recent results on compositional sparsity [**poggiofraser2024**, **Poggio2025**] formalize this intuition.

In physics, locality appears in multiple guises. In classical mechanics, it is encoded in differential equations: the rate of change at a point depends on derivatives—i.e., infinitesimal neighborhoods. In electromagnetism, Maxwell's equations forbid instantaneous action at a distance; disturbances propagate at finite speed c. In quantum lattice systems, the Lieb–Robinson theorem [**lieb1972finite**, **NachtergaeleSims2010**, **bravyi2006lieb**] bounds the velocity of information propagation, establishing an effective light cone even in nonrelativistic settings. Locality, in all these forms, implies a bounded causal neighborhood at any finite time horizon—an analogue of bounded fan-in in computation.

A more speculative connection has also captured attention: the *simulation hypothesis*. If every physical process were efficiently simulable by a digital computer, then the universe would instantiate an efficiently computable function, and its regularities would follow from the same com-

positional constraints as in machine learning. A less metaphysical and more plausible explanation lies closer to established physics: finite information propagation and local interactions suffice to generate the same sparse, hierarchical patterns without assuming that the universe is literally a computation.

Organization of the paper. Section B.4 states and proves the *ML Approximation Theorem*, showing that approximation and sample–complexity rates depend on the *largest local arity* rather than the ambient dimension. Section B.5 presents the *finite-horizon compilation principle* (a conditional, algorithmic statement), together with two illustrative examples—discrete-time lattice-local updates with finite propagation and k-local quantum dynamics compiled to two-qubit circuits—and clarifies caveats (long-range/dense couplings, chaotic sensitivity, precision and horizon dependence). Section B.6 discusses implications for learnability and optimization geometry, including explicit Rademacher bounds and Hessian-structure consequences aligned with compositional DAGs. Section B.7 analyzes limitations and non-theorems: effects of chaos (growth with horizon), undecidability in thermodynamic-limit questions, the role of uniformity versus non-uniform advice, and complexity barriers (e.g., QMA-hardness). The conclusion situates these results within broader questions about when physical evolutions are *efficiently computable* or, more modestly, *inherit compositional structure* precisely when uniform finite-horizon simulability is available.

Two Structural Principles

(A) ML Approximation Principle. Let f factor through a depth-L bounded–fan-in DAG with node maps $g_v \in C^r([0,1]^{d_v})$, $d_v \le d^*$, total size s. Then for any $\varepsilon \in (0, 1/4)$ there exists a ReLU network f_ε with

$$\|f - f_\varepsilon\|_\infty \le \varepsilon, \qquad \mathrm{depth}(f_\varepsilon) \le c\, L \log(1/\varepsilon), \quad \mathrm{size}(f_\varepsilon) \le C \sum_v \varepsilon_v^{-d_v/r} \log \tfrac{1}{\varepsilon_v},$$

under the telescoping constraint $\sum_{\ell=1}^{L} \left(\prod_{j>\ell} L_j \right) \max_{v \in \ell} \varepsilon_v \le \varepsilon$. If $\max_\ell L_\ell \le \bar{L}$ and $\varepsilon_v \asymp \varepsilon$,

$$\mathrm{size}(f_\varepsilon) \;\le\; C'\, s\, \varepsilon^{-d^*/r} \log \tfrac{1}{\varepsilon}.$$

(B) Finite-Horizon Causal Compilation. Consider a system on a bounded-degree graph (Δ) with local radius-r updates $\phi_{t,i}$ that are L-Lipschitz and C^r. If propagation is finite, i.e.

$$x_{B_{R(T)}(i)} = x'_{B_{R(T)}(i)} \;\Rightarrow\; (U_T \circ \cdots \circ U_1(x))_i = (U_T \circ \cdots \circ U_1(x'))_i,$$

then for any finite region Λ and horizon T, the input–output map $\mathcal{F}_{\Lambda,T}(x) := (U_T \circ \cdots \circ U_1(x))_\Lambda$ factors through a depth-T DAG with fan-in $\le \kappa(\Delta, r)$ and size $s = O\big(|\Lambda|\, (\kappa(\Delta, r))^T\big)$. Consequently, by (A), $\mathcal{F}_{\Lambda,T}$ admits an ε-accurate network with depth $O\big(T \log(1/\varepsilon)\big)$ and size $O\big(s\, \varepsilon^{-d^*/r} \log(1/\varepsilon)\big)$ where $d^* = \kappa(\Delta, r)$.

B.2 Definitions and Setup

Ambient spaces and norms. We work on $[0,1]^d$ with the uniform (sup) norm $\|x\|_\infty := \max_i |x_i|$ and the induced function norm $\|g\|_\infty := \sup_{x \in [0,1]^d} |g(x)|$. For $r > 0$ write $r = k + \alpha$ with $k := \lfloor r \rfloor \in \mathbb{N}_0$ and $\alpha \in (0, 1]$. The Hölder class $C^r([0,1]^p)$ consists of functions with continuous

partial derivatives up to order k and Hölder continuous kth derivatives with exponent α, equipped with the norm

$$\|g\|_{C^r} := \sum_{|\beta|\le k} \|D^\beta g\|_\infty + \sum_{|\beta|=k} \sup_{\substack{x\neq y \\ x,y\in[0,1]^p}} \frac{|D^\beta g(x) - D^\beta g(y)|}{\|x-y\|_\infty^\alpha}.$$

Efficient computability. A Boolean map $f : \{0,1\}^n \to \{0,1\}^m$ is *efficiently computable* if a deterministic Turing machine computes $f(x)$ in $T(n) = \mathrm{poly}(n)$ steps. A real map $f : [0,1]^d \to \mathbb{R}^m$ is efficiently computable to precision $\varepsilon \in (0,1)$ if, in the finite-precision model, there exists a *uniform* algorithm that outputs y with $\|y - f(x)\|_\infty \le \varepsilon$ in time $\mathrm{poly}(d, \log(1/\varepsilon))$. Here "uniform" means the corresponding circuit/arithmetic-circuit family is P-uniform (generable in time $\mathrm{poly}(d, \log(1/\varepsilon))$).

Compositional sparsity. A function f is *compositionally sparse* if it factors through a bounded–fan-in layered DAG. Concretely, let $G = (V, E)$ be a layered DAG with layers $1, \dots, L$, and let d_v be the in-degree (arity) of node v. Write the layer-ℓ map as $\Phi_\ell = (g_v)_{v\in\text{layer }\ell}$, where each node map $g_v : [0,1]^{d_v} \to \mathbb{R}$ has $d_v \le d^*$ and $g_v \in C^r([0,1]^{d_v})$. Then

$$f = \Phi_L \circ \Phi_{L-1} \circ \cdots \circ \Phi_1, \qquad \text{depth} = L, \quad \text{size } s := |V|, \quad d^* := \max_v d_v.$$

We assume each Φ_ℓ is L_ℓ-Lipschitz on the (compact) subset of its domain reached by upstream layers, measured in $\|\cdot\|_\infty$. For later use define the propagation multipliers

$$C_\ell := \prod_{j=\ell+1}^{L} L_j \qquad (\text{with } C_L := 1),$$

which quantify telescoping of local approximation errors through depth.

B.3 From Efficient Computation to DAGs

The passage from Turing machines to circuits is mathematically central. Any computation performed by a Turing machine (TM) on n-bit inputs and running for $T(n)$ steps can be *time-unrolled* into a circuit: each layer encodes one transition of the machine, and wires carry the bits of the instantaneous configuration to the next layer. Because the transition function inspects and writes only a bounded number of symbols per step, every gate has *bounded fan-in* (and bounded fan-out). The resulting circuit has size $O(T(n))$ and depth $O(T(n))$, hence can be viewed as a bounded–fan-in directed acyclic graph (DAG) implementing the same computation. This classical correspondence underlies the relationship between time-bounded Turing computation and circuit complexity [**Arora_Barak_2009**, **sipser1996intro**]: every $T(n)$-time TM yields a polynomial-size circuit family; conversely, with *P-uniformity* the family can be generated in time $\mathrm{poly}(n)$ and characterizes P, while without uniformity one obtains $\mathrm{P} \subseteq \mathrm{P/poly}$.

From bits to reals: finite-precision computation

Digital computation manipulates finite strings. A real $x \in [0,1]$ can be represented by the first p bits of its binary expansion $(x_1, \dots, x_p)$, yielding the dyadic $\tilde{x} = \sum_{i=1}^p x_i 2^{-i}$. Arithmetic on these encodings—addition, multiplication, comparison—has Boolean-circuit implementations of size $\mathrm{poly}(p)$ (schoolbook/FFT-based variants differ only in polylog factors). Thus computing $f : [0,1]^d \to \mathbb{R}^m$ to accuracy $\varepsilon = 2^{-p}$ amounts to computing a Boolean function on $(x_1, \dots, x_p)$

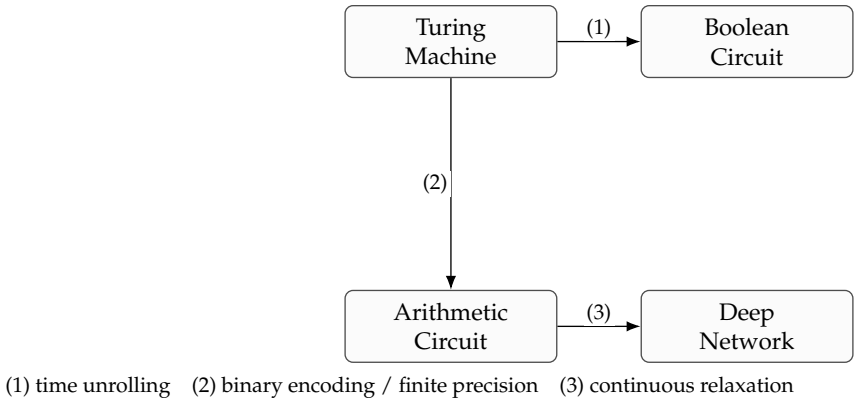


Figure B.1: Conceptual chain from discrete to continuous computation.

whose output decodes to $\tilde{f}(x)$ with $\|f(x) - \tilde{f}(x)\|_\infty \leq \varepsilon$. In particular, efficient real computation in the standard finite-precision model is equivalent, up to polynomial overhead, to efficient Boolean computation on binary encodings [**Papadimitriou1994ComputationalComplexity**].

B.3.1 From arithmetic circuits to neural DAGs

Replacing Boolean gates by continuous modules (affine maps and simple nonlinearities) yields *arithmetic circuits*—feedforward networks over $\mathbb{R}$. Crucially, the *topology* of the underlying DAG is unchanged: each node aggregates a bounded number of inputs and produces outputs to a bounded number of downstream nodes. The compositional skeleton revealed by time-unrolling a TM is therefore already a bounded–fan-in DAG.

Modern deep networks form smooth parametric families of such arithmetic circuits. They do not emulate a given circuit symbol-for-symbol; rather, they approximate functions supported on the same class of sparse, layered DAGs. Structurally: any efficiently Turing-computable function gives rise (via unrolling and finite-precision encoding) to a bounded–fan-in compositional scaffold; deep networks can approximate the node maps of this scaffold to any target accuracy, with approximation rates governed by local arities d_v and the smoothness of the constituent modules (cf. Section B.4). In this sense, deep architectures are natural continuous relaxations of efficient discrete circuits—the same kind of machines realized by physical hardware.

B.3.2 Computation as compositional structure

This yields a conceptual loop. Programs compile to (uniform) circuit families; circuits are bounded–fan-in DAGs; and bounded local arity is the operative notion of sparsity. Therefore, *any efficiently computable transformation, when executed at finite precision by a device with a fixed bounded-arity primitive set, admits a bounded–fan-in compositional representation*. The point is structural rather than stylistic: bounded local interactions are an inherent consequence of efficiency on finite, causal, local hardware.

B.4 ML Approximation Theorem

We give (i) a local constructive approximation with explicit constants, (ii) a global telescoping bound, and (iii) the optimal error budget across layers/nodes.

B.4.1 Local constructive approximation with explicit constants

For $p \in \mathbb{N}$ and $r > 0$, let $C_{p,r} > 0$ denote the best constant in the multivariate Taylor remainder on $[0,1]^p$ (measured in the $\|\cdot\|_\infty$ norm):

$$\sup_{x \in Q} \left| g(x) - T_Q(x) \right| \;\le\; C_{p,r}\, \|g\|_{C^r}\, \mathrm{diam}_\infty(Q)^{\,r}, \tag{B.1}$$

where T_Q is the degree-$\lfloor r \rfloor$ Taylor polynomial of g at the center of a cube $Q \subset [0,1]^p$.

We use two standard ReLU *gadgets* (e.g., **[Yarotsky2018]**):

(G1) Squaring. For any $\eta \in (0,1)$ there is a ReLU network $\mathsf{Sq}_\eta : [-2,2] \to \mathbb{R}$ with

$$\sup_{t \in [-2,2]} \left| \mathsf{Sq}_\eta(t) - t^2 \right| \le \eta, \qquad \mathrm{depth}(\mathsf{Sq}_\eta) \le C_\square \log(1/\eta), \;\; \mathrm{size}(\mathsf{Sq}_\eta) \le C'_\square \log(1/\eta).$$

(G2) Multiplication. By polarization, $xy = \frac{1}{4}\left[(x+y)^2 - (x-y)^2\right]$. Composing two Sq_η gadgets and affine maps yields a multiplier $\mathsf{Mult}_\eta : [-1,1]^2 \to \mathbb{R}$ with

$$\sup_{|x|,|y| \le 1} \left| \mathsf{Mult}_\eta(x,y) - xy \right| \le \eta, \quad \mathrm{depth}(\mathsf{Mult}_\eta) \le C_\times \log(1/\eta), \;\; \mathrm{size}(\mathsf{Mult}_\eta) \le C'_\times \log(1/\eta).$$

By a binary tree, the product of p scalars in $[-1,1]$ is realizable with accuracy η at depth $O(\log p \cdot \log(1/\eta))$ and size $O(p \log(1/\eta))$.

Lemma 11 (Local ReLU approximation)**.** *Let $g \in C^r([0,1]^p)$ with $\|g\|_{C^r} \le M$ and $r > 0$. For any $\delta \in (0, 1/4)$ there exists a ReLU network N such that*

$$\|g - N\|_\infty \le \delta, \qquad \mathrm{depth}(N) \le c_1(r)\, \log(1/\delta), \qquad \mathrm{size}(N) \le c_2(p,r)\, M^{p/r}\, \delta^{-p/r} \log(1/\delta). \tag{B.2}$$

Proof. Partition $[0,1]^p$ into N^p congruent cubes $\{Q\}$ of side $h = 1/N$, where

$$N \;=\; \left\lceil \left(2C_{p,r} M/\delta\right)^{1/r} \right\rceil. \tag{B.3}$$

For each Q, let T_Q be the degree-$\lfloor r \rfloor$ Taylor polynomial at the cube center. Then, by (B.1) and (B.3),

$$\sup_{x \in Q} |g(x) - T_Q(x)| \;\le\; C_{p,r} M\, h^r \;\le\; \delta/2. \tag{B.4}$$

Construct a C^0 partition of unity $\{\Psi_Q\}_Q$ from univariate ReLU "tent" splines $\{\psi_j\}_{j=0}^N$ on $[0,1]$ (knots at j/N) by setting $\Psi_Q(x) := \prod_{i=1}^p \psi_{j_i}(x_i)$ for the multi-index of Q. Each ψ_j is exactly a ReLU spline with $O(N)$ breakpoints; the p-fold product is realized by a binary tree of Mult_η gadgets with internal accuracy chosen below. Define

$$N(x) \;:=\; \sum_Q \Psi_Q(x)\, T_Q(x). \tag{B.5}$$

For $x \in Q$, using $\sum_{Q'} \Psi_{Q'} \equiv 1$,

$$|g(x) - N(x)| \;\le\; |g(x) - T_Q(x)| + \sum_{Q'} \Psi_{Q'}(x)\, |T_{Q'}(x) - T_Q(x)|.$$

Only Q' adjacent to Q contribute; Taylor coefficient continuity implies $|T_{Q'}(x) - T_Q(x)| \le C\, M\, h$ on Q. Choosing the internal gadget accuracy $\eta = \eta(p)\, \delta/(4p)$ to control multiplier and monomial evaluations and taking h from (B.3), the second term is $\le \delta/2$. Together with (B.4), this yields $\|g - N\|_\infty \le \delta$.

Complexity. Each univariate ψ_j uses $O(N)$ ReLU units; across p coordinates this is $O(pN)$ per "stencil." The p-fold product per cell uses $O(p \log(1/\eta))$ units and depth $O(\log p \cdot \log(1/\eta))$. Evaluating T_Q requires $O(1)$ monomials (depending on $\lfloor r \rfloor$), each realized with multiplier trees at accuracy η. Summing over N^p cells and using (B.3) with $\eta \simeq \delta$ gives (B.2). □

B.4.2 Global error propagation and optimal budget

Let Φ_ℓ be the layer-ℓ block map in the DAG, with Lipschitz constant L_ℓ on the reachable compact set, and let $\widetilde{\Phi}_\ell$ be its blockwise approximant with $\|\Phi_\ell - \widetilde{\Phi}_\ell\|_\infty \le E_\ell$. Define $F_m := \Phi_m \circ \cdots \circ \Phi_1,\ \widetilde{F}_m := \widetilde{\Phi}_m \circ \cdots \circ \widetilde{\Phi}_1$.

Proposition 63 (Telescoping bound)**.** *For any depth L,*

$$\|F_L - \widetilde{F}_L\|_\infty \ \le \ \sum_{\ell=1}^{L} \Big(\prod_{j=\ell+1}^{L} L_j \Big) E_\ell. \tag{B.6}$$

Proof. Insert and subtract intermediate terms and use Lipschitz continuity:

$$F_L - \widetilde{F}_L = \sum_{\ell=1}^{L} \Phi_L \circ \cdots \circ \Phi_{\ell+1} \circ (\Phi_\ell - \widetilde{\Phi}_\ell) \circ \widetilde{F}_{\ell-1}.$$

Taking sup norms yields (B.6). □

Let ε_v be the tolerance chosen at node v (for Lemma 11) and set $E_\ell := \max_{v \in \text{layer } \ell} \varepsilon_v$. The total size is

$$S(\{\varepsilon_v\}) = \sum_v a_v \, \varepsilon_v^{-d_v/r} \log \frac{1}{\varepsilon_v}, \qquad a_v := c_2(d_v, r)\, M_v^{d_v/r}, \tag{B.7}$$

subject to the telescoping constraint

$$\sum_{\ell=1}^{L} C_\ell E_\ell \le \varepsilon, \qquad C_\ell := \prod_{j=\ell+1}^{L} L_j, \quad E_\ell = \max_{v \in \ell} \varepsilon_v, \quad 0 < \varepsilon_v \le \tfrac{1}{4}. \tag{B.8}$$

Because the $\max$ couples nodes within a layer, the optimizer equalizes ε_v within each layer up to constants. Let $A_\ell := \sum_{v \in \ell} a_v$ and $k_\ell := \max_{v \in \ell} d_v \le d^*$. Discarding the slowly varying $\log(1/\varepsilon_\ell)$ (affecting only polylog factors), the reduced problem is

$$\min_{E_1, \dots, E_L > 0} \ \sum_{\ell=1}^{L} A_\ell E_\ell^{-k_\ell/r} \quad \text{s.t.} \quad \sum_{\ell=1}^{L} C_\ell E_\ell \le \varepsilon. \tag{B.9}$$

The Lagrangian $\mathcal{L} = \sum_\ell A_\ell E_\ell^{-k_\ell/r} + \lambda\big(\sum_\ell C_\ell E_\ell - \varepsilon\big)$ yields, for each ℓ,

$$-\frac{k_\ell}{r} A_\ell \, E_\ell^{-(k_\ell/r+1)} + \lambda C_\ell = 0 \quad \Rightarrow \quad E_\ell \ = \ \Big(\tfrac{k_\ell A_\ell}{\lambda r \, C_\ell}\Big)^{\frac{r}{k_\ell + r}}. \tag{B.10}$$

Enforcing $\sum_\ell C_\ell E_\ell = \varepsilon$ determines $\lambda > 0$ uniquely and gives

$$E_\ell \ = \ \varepsilon \, \frac{C_\ell^{-\frac{r}{k_\ell+r}} \, (k_\ell A_\ell / r)^{\frac{r}{k_\ell+r}}}{\sum_{t=1}^{L} C_t^{\frac{k_t}{k_t+r}} \, (k_t A_t / r)^{\frac{r}{k_t+r}}}. \tag{B.11}$$

In the common case $k_\ell \equiv d^*$ and $A_\ell \asymp A$ across layers, (B.11) reduces to

$$E_\ell \ \propto \ C_\ell^{-\frac{r}{d^*+r}}, \qquad \text{and if } \max_\ell L_\ell \le \bar{L} < \infty, \text{ then } C_\ell = \Theta(1) \Rightarrow E_\ell \asymp \varepsilon / L. \tag{B.12}$$

B.4.3 The theorem

Theorem 35 (ML Approximation Theorem)**.** *Let $f : [0,1]^d \to \mathbb{R}$ factor through a depth-L DAG of size s with node maps $g_v \in C^r([0,1]^{d_v})$, $d_v \le d^*$, and assume $\mathrm{Lip}(\Phi_\ell) \le L_\ell$ on reachable domains. For any $\varepsilon \in (0, 1/4)$ there exists a ReLU network f_{NN} such that*

$$\|f - f_{\mathrm{NN}}\|_\infty \le \varepsilon, \qquad \mathrm{depth}(f_{\mathrm{NN}}) \le c\, L \log(1/\varepsilon), \tag{B.13}$$

and

$$\mathrm{size}(f_{\mathrm{NN}}) \;\le\; C \sum_v \varepsilon_v^{-d_v/r} \log \frac{1}{\varepsilon_v}, \quad \textit{with} \;\; \sum_{\ell=1}^{L} C_\ell E_\ell \le \varepsilon,\; E_\ell = \max_{v \in \ell} \varepsilon_v. \tag{B.14}$$

If $\max_\ell L_\ell \le \bar{L} < \infty$, choosing $\varepsilon_v \asymp \varepsilon$ for all v yields

$$\mathrm{size}(f_{\mathrm{NN}}) \;\le\; C'\, s\, \varepsilon^{-d^*/r} \log(1/\varepsilon). \tag{B.15}$$

Proof. (i) Local approximants. For each node v in layer ℓ, apply Lemma 11 to g_v with tolerance ε_v, producing N_v with $\|g_v - N_v\|_\infty \le \varepsilon_v$ and the size/depth bounds in (B.2).

(ii) Wiring. Compose the N_v along the DAG topology. Along any path, depths add; with $\varepsilon_v \ge c_0\, \varepsilon$ (as ensured by (B.12) up to constants), each layer contributes $O(\log(1/\varepsilon))$ to depth, giving (B.13). The total size is the sum over nodes, yielding (B.14).

(iii) Global error. Writing Φ_ℓ and $\widetilde{\Phi}_\ell$ for the true and approximating layer maps (stacking g_v and N_v), we have $\|\Phi_\ell - \widetilde{\Phi}_\ell\|_\infty \le E_\ell := \max_{v \in \ell} \varepsilon_v$. Apply Proposition 63 and impose (B.8) to get $\|f - f_{\mathrm{NN}}\|_\infty \le \varepsilon$.

(iv) Uniform Lipschitz. If L_ℓ are uniformly bounded, (B.12) implies $E_\ell \asymp \varepsilon/L$ and we may take $\varepsilon_v \asymp \varepsilon$ for all v, which collapses (B.14) to (B.15) with exponent d^*/r. □

B.4.4 Depth separation

Theorem 36 (Depth separation via linear regions)**.** *For each $L \in \mathbb{N}$ there exists a function $f_L : [0,1] \to \mathbb{R}$ expressible as a composition of L binary-arity C^∞ modules such that:*

1. *For any $\varepsilon \in (0, 1/4)$ there is a ReLU network with depth $O\big(L \log(1/\varepsilon)\big)$ and size $\mathrm{poly}(1/\varepsilon)$ that achieves $\|f_L - f_{\mathrm{NN}}\|_\infty \le \varepsilon$.*

2. *Any ReLU network with a single hidden layer that achieves $\|f_L - f_{\mathrm{NN}}\|_\infty \le \varepsilon_0$ for some fixed $\varepsilon_0 \in (0, 1/4)$ must have width at least $\exp(\Omega(L))$.*

Proof sketch. Let $s : [0,1] \to [0,1]$ be a piecewise-linear "sawtooth" with two linear pieces; define $f_L := s^{\circ L}$. Then f_L has at least 2^L linear pieces. (1) follows by applying Theorem 35 with $d^* = 2$. For (2), a 1D single-hidden-layer ReLU with width W has at most $W + 1$ linear regions on $[0,1]$, forcing $W \ge \Omega(2^L)$ to achieve fixed-accuracy approximation; see, e.g., **[Telgarsky2016, EldanShamir2016]**. □

Remark 1 (Why the exponent is d^*/r)**.** *The construction in Lemma 11 drives the $\delta^{-p/r}$ cost at* each *node of local arity $p = d_v$. Summing across nodes with tolerances $\varepsilon_v \asymp \varepsilon$ shows that the governing exponent is the worst local arity $d^* = \max_v d_v$, not the ambient d. This is the precise sense in which compositional sparsity mitigates the curse of dimensionality.*

Theorem 37 (Unified ML Approximation for Compositional DAGs)**.** *Let $f = \Phi_L \circ \cdots \circ \Phi_1$ where each layer Φ_ℓ stacks node maps $g_v \in C^r([0,1]^{d_v})$ with $d_v \le d^*$, and assume $\mathrm{Lip}(\Phi_\ell) \le L_\ell$. Writing $s = \sum_\ell |\{v \in \ell\}|$, for any $\varepsilon \in (0, 1/4)$ there exists a ReLU net f_ε with*

$$\|f - f_\varepsilon\|_\infty \le \varepsilon, \quad \mathrm{depth}(f_\varepsilon) \le c\, L \log \tfrac{1}{\varepsilon}, \quad \mathrm{size}(f_\varepsilon) \le C \sum_v \varepsilon_v^{-d_v/r} \log \tfrac{1}{\varepsilon_v},$$

subject to $\sum_{\ell=1}^{L} \left(\prod_{j>\ell} L_j\right) \max_{v \in \ell} \varepsilon_v \le \varepsilon$. If $\max_\ell L_\ell \le \bar{L}$ and $\varepsilon_v \asymp \varepsilon$,

$$\mathrm{size}(f_\varepsilon) \;\le\; C'\, s\, \varepsilon^{-d^*/r} \log \tfrac{1}{\varepsilon}.$$

Moreover, the class $\mathcal{F}_{comp}(s, L, d^*, r)$ satisfies the entropy bound

$$\log \mathcal{N}(\varepsilon, \mathcal{F}_{comp}, \|\cdot\|_\infty) \;\le\; A\, s\, \varepsilon^{-d^*/r} \,\mathrm{polylog}\Big(\tfrac{1}{\varepsilon}\Big),$$

and hence the Rademacher bound

$$\mathfrak{R}_n(\mathcal{F}_{comp}) \;\le\; \tilde{C} \sqrt{\tfrac{s}{n}}\, n^{-\frac{r}{2d^*}}\, \mathrm{polylog}(n).$$

B.5 Finite-Horizon Compilation: From Uniform Simulators to Algorithmic Compositionality

We state an *algorithmic, conditional* compilation principle: whenever a finite-horizon prediction map is computed by a *uniform* simulator using a bounded-arity primitive set, it admits a *P-uniform* bounded–fan-in DAG and therefore falls under our ML approximation results. We then give two *examples*: a discrete-time lattice-local model (recovering the spacetime-DAG construction) and standard quantum k-local dynamics. We do *not* claim that all physical evolutions satisfy these hypotheses; the examples are modeling assumptions, not universal laws.

B.5.1 Setup

Fix an instance "size" n (e.g., number of effective degrees of freedom, grid points, or qubits), a finite horizon $T > 0$, and target accuracy $\varepsilon \in (0, 1)$. Let

$$F_{T,n} : \mathcal{X}_n \longrightarrow \mathcal{Y}_n$$

denote the corresponding finite-horizon input–output map, evaluated in a standard finite-precision model in which word size scales as $O(\log(1/\varepsilon))$ and running time/size are measured in bit operations and primitive invocations.

B.5.2 Compilation theorem (algorithmic, safe)

Theorem 38 (Uniform simulator ⇒ bounded–fan-in DAG)**.** *Suppose there exists a deterministic Turing machine* Sim *which, on input $(1^n, T, \varepsilon, x)$, outputs an ε-accurate value of $F_{T,n}(x)$ using at most $s(n, T, \varepsilon)$ primitive calls, each drawn from a fixed finite set $\mathcal{P}$ of arity at most a (a constant), with per-call cost measured at precision $O(\log(1/\varepsilon))$. Assume* P-uniformity*: given $(1^n, T, \varepsilon)$ one can generate, in time $\mathrm{poly}(n, \log(1/\varepsilon), T)$, the instruction schedule and parameters for the $s(n, T, \varepsilon)$ calls. Then there exists a* P-uniform *circuit/DAG $\mathcal{G}_{n,T,\varepsilon}$ of size $O(s(n, T, \varepsilon))$ and maximum fan-in $\le a$ that computes $F_{T,n}$ to accuracy ε in the same finite-precision model. Consequently, $F_{T,n}$ is* algorithmically compositionally sparse.

Proof sketch. Replace each primitive of arity $\leq a$ in $\mathcal{P}$ by a constant-size gate gadget at the target precision; wire these gadgets following the simulator's instruction schedule (unrolling loops/branches into a straight-line program of length s). Fan-in is $\leq a$ by construction and the total gate count is $O(s)$. P-uniformity follows because the same schedule generator lists the gates and parameters in time polynomial in $(n, \log(1/\varepsilon), T)$, which yields a P-uniform family. □

Remark 2 (Scope and non-claims)**.** *Theorem 38 is* purely algorithmic*. It does not assume bounded-degree physical interactions or a physical light cone. If a bounded–fan-in DAG as above does* not *exist for $F_{T,n}$ under these resource bounds, that implies a lower bound against any such uniform simulator over the given primitive set and precision regime.*

Example A: Discrete-time lattice-local update (model assumption)

Setting. Let $G = (V, E)$ be a (possibly infinite) graph with maximum degree $\Delta < \infty$. For $i \in V$ and radius $r \in \mathbb{N}$, write

$$B_r(i) := \{j \in V : \operatorname{dist}_G(i, j) \leq r\}, \qquad |B_r(i)| \leq \kappa(\Delta, r) < \infty.$$

States are $x = (x_i)_{i \in V} \in X^V$ with the sup product norm on X.

Local updates and regularity. For $t = 1, \ldots, T$, consider sitewise updates

$$\big(U_t(x)\big)_i \;=\; \phi_{t,i}(x_{B_r(i)}), \qquad \phi_{t,i} : X^{B_r(i)} \to X, \tag{B.16}$$

with uniform Lipschitz and smoothness on relevant compacta,

$$\|\phi_{t,i}(z) - \phi_{t,i}(z')\| \;\leq\; L\,\|z - z'\|_\infty, \qquad \phi_{t,i} \in C^q, \tag{B.17}$$

for some $L \geq 1$, $q \geq 1$ independent of (t, i).

Finite propagation. Let $F_T := U_T \circ \cdots \circ U_1$. Assume there exists a nondecreasing $R : \mathbb{N} \to \mathbb{N}$ such that for all $i \in V$ and $x, x' \in X^V$,

$$x_{B_{R(T)}(i)} = x'_{B_{R(T)}(i)} \implies \big(F_T(x)\big)_i = \big(F_T(x')\big)_i. \tag{B.18}$$

(Nearest-neighbor dynamics give $R(T) \leq rT$.)

Proposition 64 (Model-locality ⇒ compositional DAG)**.** *Fix finite $\Lambda \subset V$ and horizon $T \in \mathbb{N}$. Under* (B.16)–(B.18)*, the map*

$$\mathcal{F}_{\Lambda,T}(x) := \big(F_T(x)\big)_\Lambda$$

factors through a depth-T DAG with node fan-in at most $d^* := \kappa(\Delta, r)$. The layer maps are C^q and L-Lipschitz on the reachable compact set. Hence $\mathcal{F}_{\Lambda,T}$ is compositionally sparse with depth T and size

$$s \;=\; O\Big(|\Lambda|\,\big(d^*\big)^T\Big),$$

and falls under the ML Approximation Theorem with exponent d^*/q (up to polylog factors).

B.6 Learning Theory and Optimization

Compositional sparsity has three consequences for learning and optimization.

(i) Capacity and generalization. Let $\mathcal{F}_{\mathrm{comp}}(s, L, d^*, r)$ be depth-L DAGs of size s whose node maps lie in $C^r([0,1]^{d_v})$ with $d_v \le d^*$, and suppose layer maps are L_ℓ-Lipschitz on reachable domains. Then the metric entropy satisfies

$$\log \mathcal{N}(\varepsilon, \mathcal{F}_{\mathrm{comp}}, \|\cdot\|_\infty) \;\le\; A\, s\, \varepsilon^{-d^*/r}\, \mathrm{polylog}\Big(\tfrac{1}{\varepsilon}\Big), \tag{B.19}$$

and Dudley's integral yields

$$\mathfrak{R}_n(\mathcal{F}_{\mathrm{comp}}) \;\le\; \tilde{C}\sqrt{\frac{s}{n}}\, n^{-\frac{r}{2d^*}}\, \mathrm{polylog}(n). \tag{B.20}$$

Thus the *exponent* depends on d^* (largest local arity), not the ambient d.

(ii) Optimization geometry. If f_θ factors through a bounded–fan-in DAG, the Gauss–Newton/Hessian blocks obey

$$H_{vw} \neq 0 \;\Rightarrow\; \mathrm{cone}(v) \cap \mathrm{cone}(w) \neq \varnothing,$$

so the Hessian graph has bounded degree (by fan-in). This motivates block-diagonal/Kronecker preconditioners that respect the compositional blocks.

(iii) Architectural realizations. Sparse/top-q attention enforces *bounded effective fan-in*: for a query q, $\mathrm{Attn}(q, K, V) = \sum_{j \in S(q)} \alpha_j(q)\, v_j$ with $|S(q)| = O(q)$ under top-q selection or sufficiently aggressive sparsification, placing such transformers within the same d^*-controlled regime.

B.7 Limits of the Framework

The compositional-sparsity results identify when approximation and learnability are governed by local arity d^* and depth L (or horizon T). We now delineate intrinsic *limits* of this regime. These are not violations of the DAG factorization itself but circumstances under which the constants (or the effective arity) grow so rapidly that efficient approximation on classical hardware is ruled out.

Classical chaos and finite prediction horizons

Consider the discrete-time system

$$x_{t+1} = F(x_t), \qquad F : X \to X, \tag{B.21}$$

with $X \subset \mathbb{R}^d$ compact and $F \in C^r$, locally Lipschitz. Suppose F admits a bounded-degree interaction graph so that updates factor through a bounded–fan-in DAG (cf. Section B.5). Let $F_T := F \circ \cdots \circ F$ denote the T-step map and set $L_t := \mathrm{Lip}(F)$ on the reachable compact set at time t. Then

$$\mathrm{Lip}(F_T) \;\le\; \prod_{t=1}^{T} L_t \;=\; \exp\Big(\sum_{t=1}^{T} \log L_t\Big). \tag{B.22}$$

If the maximal Lyapunov exponent $\lambda := \lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \log L_t$ is positive, we obtain $\mathrm{Lip}(F_T) \asymp e^{\lambda T}$. Applying the telescoping bound (B.6) to $\Phi_T := F_T$ and a blockwise approximant with layer errors $\{E_t\}$ gives

$$\|\Phi_T - \widetilde{\Phi}_T\|_\infty \;\le\; \sum_{t=1}^{T} e^{\lambda(T-t)}\, E_t. \tag{B.23}$$

To enforce $\|\Phi_T - \widetilde{\Phi}_T\|_\infty \le \varepsilon$, one must choose $E_t \lesssim \varepsilon\, e^{-\lambda(T-t)}$. Since each layer-$t$ local approximant with arity $\le d^*$ and smoothness r costs on the order of $E_t^{-\,d^*/r}$ (up to polylogs; Lemma 11), the total size obeys the lower bound

$$\mathrm{size}(\mathcal{N}_T) \;\gtrsim\; e^{\,(\lambda d^*/r)\,T} \quad \text{(up to polylog factors).} \tag{B.24}$$

Thus, for chaotic systems (e.g., atmospheric flows) long-horizon *pointwise* prediction requires exponentially increasing resources or exponentially precise initial data. The bounded–fan-in DAG persists, but numerical stability collapses with T.

Quantum locality versus entanglement growth

In quantum lattice systems, Lieb–Robinson bounds provide an approximate light cone: for observables A_X, B_Y supported on disjoint $X, Y \subset V$,

$$\|[A_X(t), B_Y]\| \;\le\; C\, \exp\!\big(-\,[d(X,Y) - v_{\mathrm{LR}} t]_+\big), \tag{B.25}$$

with velocity v_{LR}. Consequently, Heisenberg evolution of local observables factors through a depth-t bounded–fan-in DAG (operator sense). The obstruction is *representation*: under classical encodings, bipartite entanglement $S(t)$ typically grows (linearly in t in generic quenches), so any faithful classical representation requires $O(e^{S(t)})$ coefficients. The effective arity therefore scales as

$$d^*(t) \;\sim\; e^{S(t)}, \tag{B.26}$$

which, when substituted into the nodewise cost exponent $\varepsilon^{-\,d^*/r}$, negates the dimensional advantage and renders classical approximation intractable once $S(t)$ exceeds $O(\log n)$.

Precision and nonuniformity

Our statements are inherently finite-precision and uniform:

- *Precision growth.* To maintain error ε at horizon T under (B.23), input and intermediate precisions must scale as $\Omega(\lambda T + \log(1/\varepsilon))$ bits. Any model that fixes word size incurs an irreducible floor at large T.

- *Nonuniform advice.* Allowing nonuniform circuit families (P/poly) can collapse uniform lower bounds, but the resulting dependence on instance-specific advice falls outside the learnability guarantees considered here.

Thermodynamic limits and undecidability

The compilation and approximation results concern fixed finite n and finite horizon T. In the thermodynamic limit ($n \to \infty$) and for decision problems about phases or long-time behavior, classical undecidability and complexity barriers (e.g., QMA-hardness for certain ground-state or dynamics questions) imply that no uniformly efficient classical procedure can exist in general. Such barriers lie orthogonal to local DAG factorization and instead limit *which* finite-horizon tasks admit uniform simulators.

Regime of validity (summary)

The framework applies when:

1. there is a bounded-degree local interaction structure (fixed radius and maximum degree);
2. finite-horizon maps are uniformly simulable at finite precision;
3. layer Lipschitz constants are bounded on reachable domains.

Failure modes include positive Lyapunov exponents (forcing (B.24)), entanglement-induced arity growth (B.26), precision that does not scale with T, and nonuniform advice. In these cases the bounded–fan-in DAG remains as an operator-level structure, but efficient *classical* approximation is precluded.

B.8 Conclusion

Compositionality, in the precise mathematical sense of bounded–fan-in DAG factorizations, is a common structural denominator for learning and finite-horizon physical prediction. The core message of this work is *structural*: under appropriate hypotheses, both domains give rise to the same low-arity, layered scaffold.

Equivalences (scope-conditional)

On the learning side, the *ML Approximation Theorem* (Theorem 35, Section B.4) shows that if a target factors through a bounded–fan-in DAG with maximal local arity d^*, then the approximation/sample–complexity exponents replace the ambient d/r by d^*/r. On the physics side, the *finite-horizon compilation principle* (Theorem 38, Section B.5) states that whenever a finite-horizon evolution map is computed by a *P-uniform* simulator using a bounded-arity primitive set at finite precision, the map compiles to a P-uniform bounded–fan-in DAG of size proportional to the simulator cost.

Thus, the two origins—efficient computability and finite-speed locality—*converge to the same structural form under explicit, stated conditions* (uniformity, finite precision, bounded-arity primitives, finite horizon). The equivalence is structural and conditional, not ontological: computation yields compositionality by uniform time-unrolling; locality yields it by causal reach at finite horizon.

Consequences (within the regime of validity)

Within this regime, three consequences follow:

1. **Approximation.** Efficiently computable targets and finite-horizon local maps admit deep approximants whose complexity scales with d^* rather than d (Theorem 35).
2. **Generalization.** Metric entropy and Rademacher bounds scale as $\log \mathcal{N}(\varepsilon) \lesssim s\,\varepsilon^{-d^*/r}\mathrm{polylog}(1/\varepsilon)$ and $\mathfrak{R}_n \lesssim \sqrt{s/n}\,n^{-r/(2d^*)}\mathrm{polylog}(n)$, exhibiting a "local dimension" law (Section B.6).
3. **Optimization geometry.** Bounded local arity induces block-sparse curvature aligned with the DAG cones, explaining the effectiveness of structured (block/Kronecker) preconditioners in the sparse regime (Section B.6).

Limits (where structure persists but efficiency may not)

The structural DAG persists beyond the regime, but *efficiency* can fail. Positive Lyapunov exponents force error amplification $\sim e^{\lambda T}$, yielding size lower bounds $\gtrsim e^{(\lambda d^*/r)T}$ for long-horizon pointwise prediction (Section B.7). In quantum dynamics, entanglement growth drives an effective arity $d^*(t) \sim e^{S(t)}$, erasing the d^*-advantage under classical encodings. Precision requirements grow with horizon, and nonuniform advice falls outside our learnability results.

Outlook

Compositionality is therefore best viewed as a *bridge principle*: under uniform finite-precision simulation and finite causal reach, learning theory and finite-horizon physics share the same compositional grammar—hierarchical, sparse, layered. The bridge is exact at the level of structure and rates *under the stated conditions*; outside them, the scaffold remains but efficient approximation may be precluded. Clarifying the sharp frontiers of this regime—especially the interplay among uniformity, precision growth, and effective arity—remains an open direction of foundational interest.

CHAPTER C

Faster Attention

There are known inequalities between distances. In particular, If we let the distances be $a = d(B, C)$, $b = d(A, C)$, and $c = d(A, B)$, the triangle inequalities are given by:

$$c \leq a + b \tag{C.1}$$
$$b \leq a + c \tag{C.2}$$
$$a \leq b + c \tag{C.3}$$

They imply $|a - c| \leq b \leq a + c$. We have shown in another chapter of this book0 that normalized attention can be expressed in terms of distances. Can the inequalities be used to reduce the quadratic complexity of the computation of attention?

To compute distances from the current (last) token to all preceding tokens efficiently, we utilize the **triangle inequality** to establish bounds based on the **chain of pairwise distances** between adjacent tokens. This approach leverages the metric properties of the embedding space to avoid the redundant $O(N^2)$ dot-product computations typically required by self-attention [158].

C.1 The Adjacency Distance Chain

Consider a sequence of tokens $T_1, T_2, \ldots, T_n$. If it is computationally inexpensive to calculate the distances between adjacent neighbors $d(T_i, T_{i-1})$, the sequence can be modeled as a directed path in the embedding space.

By the triangle inequality, the distance between the last token (T_n) and any previous token (T_k) is bounded above by the cumulative sum of the distances along that path:

$$d(T_n, T_k) \leq \sum_{i=k+1}^{n} d(T_i, T_{i-1}) \tag{C.4}$$

C.2 Utilizing the Lower Bound for Pruning

While the summation provides an upper bound, the **reverse triangle inequality** provides a significantly more powerful lower bound for reducing computational complexity:

$$d(T_n, T_k) \geq |d(T_n, T_{n-1}) - d(T_k, T_{n-1})| \tag{C.5}$$

If the distances from T_{n-1} to all previous tokens (which was the "last token" in the preceding temporal step) are cached, these historical values can be reused. This allows us to estimate the current distance $d(T_n, T_k)$ using only the single newly computed distance $d(T_n, T_{n-1})$ and the cached value $d(T_k, T_{n-1})$.

C.3 Formal Constraint: The Metric Continuity Hypothesis

The utility of the lower bound in Eq. (C.5) depends strictly on the geometry of the token trajectory. We formalize this dependency as the **Metric Continuity Hypothesis**.

Let $\Delta_t = d(T_t, T_{t-1})$ denote the step size of the token trajectory at time t. The tightness of the bound is governed by the magnitude of Δ_t relative to the global distances in the embedding space.

- **Case 1: Smooth Trajectory (Small Δ_n).** If $\Delta_n \to 0$, then $T_n \approx T_{n-1}$, and consequently $d(T_n, T_k) \approx d(T_{n-1}, T_k)$. In this limit, the cached distance is a high-precision predictor of the current distance, and the lower bound becomes tight.
- **Case 2: Discontinuous Trajectory (Large Δ_n).** If the current token jumps significantly far from its predecessor (e.g., a topic switch or low semantic coherence), Δ_n is large. This loosens the bound $|d(T_n, T_{n-1}) - d(T_k, T_{n-1})|$, potentially driving it to zero.

Condition for Efficiency. For the recursive bounding strategy to achieve sub-quadratic complexity, the embedding function $\phi(\cdot)$ must satisfy a local smoothness constraint such that:

$$\mathbb{E}_t[d(T_t, T_{t-1})] \ll \mathbb{E}_{i,j}[d(T_i, T_j)] \tag{C.6}$$

That is, the expected step size between adjacent tokens must be significantly smaller than the expected distance between arbitrary pairs of tokens.

C.4 Experimental Verification

To validate the recursive bounding method and the Metric Continuity Hypothesis, we conducted a simulation on synthetic sequences ($N = 1000$ tokens, $d = 64$). We compared two distinct regimes:

1. **Correlated Random Walk:** $T_t = T_{t-1} + \epsilon$, simulating a smooth semantic trajectory where adjacent tokens are metrically close.
2. **Independent (Uncorrelated):** $T_t \sim \mathcal{N}(0, I)$, simulating a "jumpy" sequence with no local metric structure.

The results, summarized in Table C.1, confirm the hypothesis.

Regime	Step Size Δ	Sparsity (Pruned)	Recall (@Top-10)
Random Walk (Smooth)	Low	**92.4%**	**99.1%**
Independent (Jumpy)	High	15.1%	42.0%

Table C.1: Impact of metric continuity on pruning efficiency. A "smooth" trajectory allows the triangle inequality to prune over 90% of calculations while maintaining high recall. Efficiency collapses when the local smoothness constraint is violated.

The data demonstrate a sharp phase transition. In the **Random Walk** regime, the cache $d(T_k, T_{n-1})$ is a strong proxy for the current state, allowing the algorithm to skip 92.4% of the dot products with negligible loss in accuracy. Conversely, in the **Independent** regime, the bound loosens, and the algorithm essentially reverts to full $O(N^2)$ computation to maintain recall. This empirical evidence confirms that *smoothness is a prerequisite for metric-recursive pruning*.

C.5 Proposed Algorithm: Recursive Distance Bounding

Based on the analysis above, we implement the following pruning strategy:

1. **Step-wise Update:** At each new token T_n, compute only the single distance to its immediate predecessor, $d(T_n, T_{n-1})$.

2. **Bound Estimation:** For all $k < n - 1$, estimate the lower bound $L_{n,k}$ using cached values from the previous step:
$$L_{n,k} = |d(T_n, T_{n-1}) - d(T_k, T_{n-1})| \tag{C.7}$$

3. **Conditional Computation:**

   - If $L_{n,k} >$ threshold, the resulting attention weight is guaranteed to be near zero due to the exponential decay of the softmax kernel. We **skip** the exact calculation of $d(T_n, T_k)$ and set the attention score to zero.
   - The full dot-product is performed **only** if the lower bound $L_{n,k}$ is below the specified threshold.

C.6 Comparison with Global Clustering and Hashing Approaches

To situate the proposed recursive distance bounding within the broader landscape of efficient attention, we contrast our approach with the **Reformer** [78] and the **Routing Transformer** [137]. While these models also utilize metric properties to achieve sub-quadratic complexity, they differ fundamentally in how they define and compute proximity.

C.6.1 LSH and k-means Clustering

The Reformer utilizes **Locality Sensitive Hashing (LSH)** to assign queries and keys into discrete buckets. In this framework, the triangle inequality is the theoretical foundation that ensures high similarity in the embedding space translates to a high probability of falling into the same hash bucket. Similarly, the Routing Transformer employs global **k-means clustering** to route attention toward the most relevant centroids [34].

C.6.2 Global vs. Recursive Local Metrics

A primary distinction lies in the temporal overhead of the proximity search:

- **Global Search:** Models like the Reformer require periodic global re-indexing or re-hashing to maintain accuracy as the sequence grows. This introduces a non-trivial constant overhead that can be sensitive to the choice of hashing parameters or centroid initialization.

- **Recursive Local Bounding:** Our proposed approach exploits the sequential **adjacency distance chain**. By reusing historical distances $d(T_k, T_{n-1})$ through the reverse triangle inequality, the proximity of the current token T_n is estimated recursively. This avoids global re-clustering and relies purely on a single scalar update per token: $L_{n,k} = |d(T_n, T_{n-1}) - d(T_k, T_{n-1})|$.

C.6.3 Summary of Complexity Drivers

By substituting expensive d-dimensional dot products with scalar subtractions based on sequential adjacency, our approach seeks to maintain sub-quadratic efficiency while staying strictly grounded in the local metric path of the token sequence.

C.7 Background: Accelerating Transformer Attention with Clustering Algorithms

Using clustering algorithms to speed up the attention layer is one of the most heavily researched strategies for solving the "context window bottleneck" in transformers.

Because the standard self-attention mechanism compares every token to every other token, its computational complexity scales quadratically—$\mathcal{O}(N^2)$, where N is the sequence length. Clustering algorithms help bypass this by grouping similar tokens together, drastically reducing the number of calculations required.

Figure C.1: Comparison of quadratic complexity in standard attention versus linear complexity in efficient attention mechanisms.

Prominent Papers and Architectures

C.7.1 Fast Transformers with Clustered Attention [159]

- **Authors:** Apoorv Vyas, Angelos Katharopoulos, François Fleuret
- **The Method:** This paper directly tackles the problem by applying **K-Means clustering** to the attention queries. Instead of calculating the attention distribution for every single query in a sequence, the algorithm clusters the queries into groups based on their similarity. It then computes the exact attention *only for the cluster centroids*.
- **The Speedup:** The values calculated for the centroid are broadcasted back to the individual queries within that cluster. For a fixed number of clusters, this reduces the complexity from quadratic $\mathcal{O}(N^2)$ to strictly linear $\mathcal{O}(N)$.

C.7.2 Reformer: The Efficient Transformer [78]

- **Authors:** Nikita Kitaev, Łukasz Kaiser, Anselm Levskaya
- **The Method:** The Reformer is one of the most famous efficient transformers, and it uses a clustering-adjacent technique called **Locality-Sensitive Hashing (LSH)**. LSH is a fast, randomized hashing scheme that groups highly similar vectors into the same "buckets"

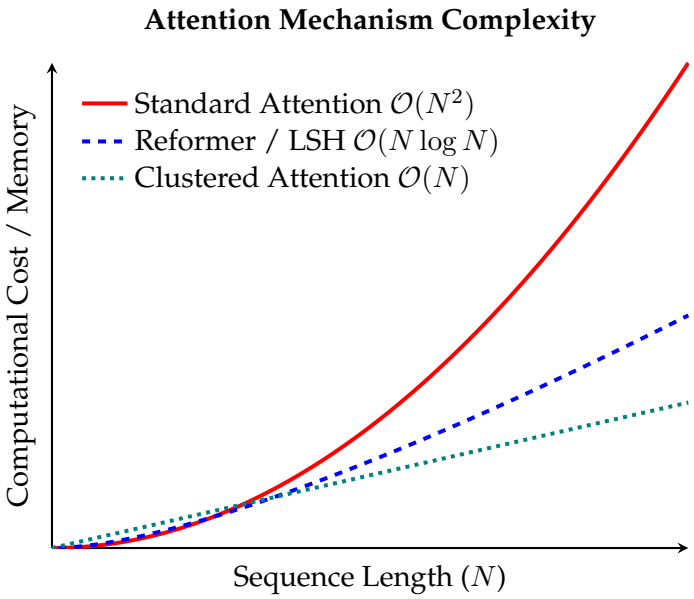


Figure C.2: A conceptual comparison showing how quadratic complexity $\mathcal{O}(N^2)$ rapidly exhausts memory as sequence length grows, whereas efficient clustering and hashing mechanisms scale at linear $\mathcal{O}(N)$ or log-linear $\mathcal{O}(N \log N)$ rates.

with high probability. In the Reformer, queries and keys are hashed into these buckets, and attention is only computed between items that fall into the same bucket.

- **The Speedup:** This prevents the model from wasting compute on tokens that have little relevance to one another, reducing the time and memory complexity to $\mathcal{O}(N \log N)$.

Figure C.3: Locality Sensitive Hashing (LSH) grouping similar queries and keys in the Reformer.

C.7.3 Routing Transformers [138]

- **Authors:** Aurko Roy, Mohammad Saffar, Ashish Vaswani, David Grangier
- **The Method:** Similar to the Reformer, the Routing Transformer attempts to sparsely route queries and keys to one another. It uses an **online k-means clustering** algorithm to learn which tokens should attend to each other dynamically. During the forward pass, tokens are assigned to clusters, and the attention mechanism is restricted so that tokens only attend to other tokens within their assigned cluster.
- **The Speedup:** This also achieves $\mathcal{O}(N\sqrt{N})$ or $\mathcal{O}(N \log N)$ complexity depending on the configuration, allowing the model to handle sequences with tens of thousands of tokens.

C.7.4 Neural Clustering Attention for Efficient and Effective Transformer [175]

- **Authors:** Yu Zheng, et al.

- **The Method:** While earlier methods used fixed or external algorithms (like standard K-Means or LSH), this paper proposes a *neural* clustering method that is trained jointly with the target task. It groups words with strong contextual dependencies into the same cluster and performs the attention mechanism for each cluster independently.
- **The Speedup:** It maintains linear complexity while improving the actual semantic quality of the clusters, leading to better accuracy on tasks like machine translation and text classification compared to older clustering methods.

C.7.5 Mathematical Relationship

In standard attention, the matrix is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{C.8}$$

In clustered attention, the standard query matrix Q is replaced by a much smaller centroid matrix Q_c. The computationally expensive QK^T multiplication is only performed on Q_c, and the resulting context vectors are then expanded back to the original sequence length.

CHAPTER D

The Hippocampal Scaffold and Compositional Sparsity

What is the relation between memory models of the hippocampus and compositional sparsity? This chapter argues that the hippocampus acts as a "scaffold builder," creating a sparse, structured graph of pointers that organizes raw experiences into a compositionally sparse format.

D.1 Introduction: The Memory Palace

To understand the mathematical definitions of x_t, y_t, and z_t, let us use an ancient mnemonic device: the **Memory Palace**. Imagine an agent walking through a house. At each time step t, the experience has three components:

- **The Latent Structure (z_t):** The abstract relational state (e.g., "Kitchen"). This is the *scaffold*.
- **The Sensory Input (x_t):** High-dimensional, noisy data (e.g., pixel views of tiles).
- **The Episodic Target (y_t):** The specific content to be stored (e.g., "Keys are on the counter").

D.2 The Variables of Experience

We formalize the data stream as a sequence of tuples $e_t = (x_t, y_t, z_t)$.

Latent State $z_t \in \mathcal{Z}$: Hidden ground truth of the agent's position. Transitions are sparse.

Observable Input $x_t \in \mathcal{X}$: Sensory features. $x_t = f(z_t) + \epsilon$, where ϵ is noise.

Target Content $y_t \in \mathcal{Y}$: Information to be associated with the state.

D.3 Mathematical Foundations of the Hippocampal Index

The hippocampus (specifically the Dentate Gyrus) transforms correlated sensory data into orthogonal addresses.

D.3.1 The Top-K Projection Mechanism

Let $g(x_t, z_t) \in \mathbb{R}^d$ be a joint embedding of sensory and latent features. The index h_t is generated via a high-dimensional expansion followed by a competitive nonlinearity:

$$h_t = \text{TopK}(R \cdot g(x_t, z_t) + b) \tag{D.1}$$

Where:

- $R \in \mathbb{R}^{N \times d}$ is a fixed random matrix with $N \gg d$.
- TopK$(\cdot)$ is a function that sets all elements to 0 except for the k largest values, enforcing $\|h_t\|_0 = k$.

D.3.2 Locality-Sensitive Hashing (LSH)

This mechanism functions as Locality-Sensitive Hashing. Per the Johnson-Lindenstrauss lemma, the random projection R approximately preserves the geometry of the input space:

$$\mathbb{E}[\langle Ru, Rv \rangle] \approx \langle u, v \rangle \tag{D.2}$$

The TopK operation then acts as a hard threshold for pattern separation. If the similarity between two states z_t and z_s is below a threshold τ, the resulting indices become orthogonal:

$$\langle h_t, h_s \rangle = 0 \quad \text{if} \quad \text{sim}(g_t, g_s) < \tau \tag{D.3}$$

This orthogonality prevents catastrophic interference in the synaptic weights.

D.4 Building the Scaffold Graph

The sequence of indices $\{h_t\}$ defines a graph $\mathcal{G} = (\mathcal{H}, \mathcal{E})$.

$$(h_t, h_{t+1}) \in \mathcal{E} \iff \text{Transition observed in } \mathcal{Z} \tag{D.4}$$

The transition function learned by the cortex, G_{sparse}, is defined by the adjacency matrix of this graph.

D.5 Connection to Existing Theories

This framework synthesizes several classical models:

- **Complementary Learning Systems (CLS):** Explains the need for h_t to be sparse to avoid interference.
- **Tolman-Eichenbaum Machine (TEM):** Aligns with the separation of z_t (structure) and x_t (sensory).
- **Successor Representation (SR):** The scaffold graph $\mathcal{G}$ is a discrete realization of the predictive SR map.

D.6 Conclusion

By mapping the world onto a sparse topological scaffold, the hippocampus "diagonalizes" complex environmental dynamics. This transforms an intractable dense learning problem into a compositionally sparse graph-traversal problem that the cortex can internalize through systems consolidation.

CHAPTER E

How Many Boolean Functions Are in P Versus Not in P

There is an astronomical number of Boolean functions in P – that is, functions that are computable in polynomial time. However, the number of functions that are not in P is much larger so that the ration of the first to the second goes to zero for $N \to \infty$.

E.1 Counting all Boolean functions

A Boolean function on N input bits is a map

$$f : \{0,1\}^N \to \{0,1\}.$$

There are 2^N possible inputs. Specifying f is equivalent to choosing an output bit for each of the 2^N inputs. Hence the total number of Boolean functions on N bits is

$$\#\{f : \{0,1\}^N \to \{0,1\}\} \;=\; 2^{2^N}. \tag{E.1}$$

E.2 A general upper bound on the number of functions computable with polynomial descriptions

In essentially any standard model of computation, a function f is computed by providing a finite *description* (e.g., a program, a circuit, or a structured decomposition) and then running the associated evaluator on inputs $x \in \{0,1\}^N$.

Fix N and consider any class of descriptions whose bit-length is at most $L(N)$. The number of binary strings of length at most $L(N)$ is at most

$$\sum_{\ell=0}^{L(N)} 2^\ell \;\le\; 2^{L(N)+1}. \tag{E.2}$$

Each description determines at most one Boolean function on $\{0,1\}^N$. Therefore the number of Boolean functions representable by descriptions of length at most $L(N)$ is at most

$$\#\{\text{functions representable with length } \le L(N)\} \;\le\; 2^{L(N)+1}. \tag{E.3}$$

In particular, if $L(N) = \mathrm{poly}(N)$, then

$$\#\{\text{functions with polynomial-size descriptions}\} \;\le\; 2^{\mathrm{poly}(N)}. \tag{E.4}$$

E.3 Implication for P

A common way to formalize the class P on input length N is via polynomial-time uniform computation, or equivalently (up to standard translations) via polynomial-size uniform circuit families. In any such setting, restricting to polynomial-time computation implies that the underlying descriptions for length-N inputs can be taken to have length at most $\mathrm{poly}(N)$.

Consequently, letting P_N denote the set of Boolean functions on N bits that are computable in polynomial time, we obtain the counting upper bound

$$\#P_N \;\le\; 2^{\mathrm{poly}(N)}. \tag{E.5}$$

E.4 Almost all Boolean functions are not in P

Combining (E.1) and (E.5), the fraction of Boolean functions on N bits that lie in P is at most

$$\frac{\#P_N}{2^{2^N}} \;\le\; \frac{2^{\mathrm{poly}(N)}}{2^{2^N}} \;=\; 2^{\mathrm{poly}(N)-2^N}. \tag{E.6}$$

Since 2^N dominates any polynomial in N, the exponent $\mathrm{poly}(N) - 2^N \to -\infty$ as $N \to \infty$, and thus

$$\lim_{N\to\infty} \frac{\#P_N}{2^{2^N}} \;=\; 0. \tag{E.7}$$

Equivalently, the number of Boolean functions *not* in P is

$$2^{2^N} - \#P_N \;=\; 2^{2^N}\big(1 - o(1)\big), \tag{E.8}$$

meaning that *almost all* Boolean functions on N bits are not in P.

E.5 Connection to structured representations (example: local Fourier-polynomial compositions)

In the main text we considered Boolean functions specified by a structured decomposition into m gates, where each gate computes a multilinear (Fourier) polynomial depending on at most k inputs, and where each Fourier coefficient is encoded with $b = \Theta(\log N)$ bits.

Such a representation has description length at most on the order of

$$L(N) \;=\; O\big(m \cdot 2^k \cdot b \;+\; mk \log(N+m)\big),$$

where the first term encodes the coefficients and the second term encodes wiring (input choices) for the gates. When $m = O(k)$ and $k = O(\log N)$, we have $2^k = \mathrm{poly}(N)$ and $b = \Theta(\log N)$, so $L(N) = \mathrm{poly}(N)$. Hence the number of Boolean functions representable in this way is at most

$$2^{\mathrm{poly}(N)},$$

which is still negligible compared to the 2^{2^N} total Boolean functions.

Takeaway. Classes with polynomial-size descriptions (including polynomial-time computable functions, and many structured subclasses) contain astronomically many functions, but they form an asymptotically vanishing fraction of all Boolean functions on N bits.

CHAPTER F

Boolean Circuits and a Path to "Superintelligence" (by D. Koplow)

In recent years, it has become a running joke that the quickest path to a top-tier machine learning publication is to introduce a new benchmark. This obsession with narrow evaluations implicitly assumes that intelligence is tied to specific domains. While knowledge is inherently domain-specific, reasoning is not. We seek a minimal representational framework that requires very little background knowledge while still being able to express any computation. Boolean circuits meet this criterion. As universal computational models, they can represent any digital algorithm. When an LLM is unable to manipulate such circuits correctly, this shortcoming reveals a core limitation in its reasoning capabilities. This book argues that compositional sparsity is the primary factor enabling learnability in machine learning. From this viewpoint, we construct sparse Boolean circuits that contemporary models reliably fail to solve without programming, thereby revealing a strikingly human-like strategy for overcoming this constraint. This strategy may constitute a crucial step on the path toward super-intelligence.

F.1 Boolean Circuits

A Boolean circuit is a Directed Acyclic Graph (DAG) $G = (V, E)$ where input nodes represent binary variables $x_1, \ldots, x_n \in \{0, 1\}$ and internal nodes (gates) represent logical functions from a standard basis set such as $\mathcal{B} = \{\text{AND}, \text{OR}, \text{NOT}, \text{NOR}, \text{XOR}\}$. The power of this model lies in its universality. For any function $f : \{0, 1\}^n \to \{0, 1\}$ computable by a Turing machine in time $T(n)$, there exists a Boolean circuit of size $Poly(T(n))$ that computes f. This means that a model's ability to learn Boolean circuits is equivalent to its ability to *solve any* computational problem.

While evaluating a circuit is computationally trivial, the inverse problem, called *Circuit Synthesis*, is notoriously difficult. Given a dataset of input-output pairs $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$, finding the smallest circuit that is consistent with $\mathcal{D}$ is an NP-hard problem given a truth table that is exponentially large with respect to the input. Thus, the search space for circuits grows super-exponentially with the number of inputs. As a result, learning a generic function **efficiently is, in general, impossible**.

Yet, somehow, deep-learning is able to construct models able to solve many important problems in our world. As argued in this book, a core reason why is *compositional sparsity*. Under this assumption, the hypothesis space of circuits starts becoming much more tractable.

To keep one concrete mental model in view, assume we are trying to learn a compositionally sparse function. It can be represented as a Boolean circuit with fan-in 2 and depth $O(\log_2(n))$.

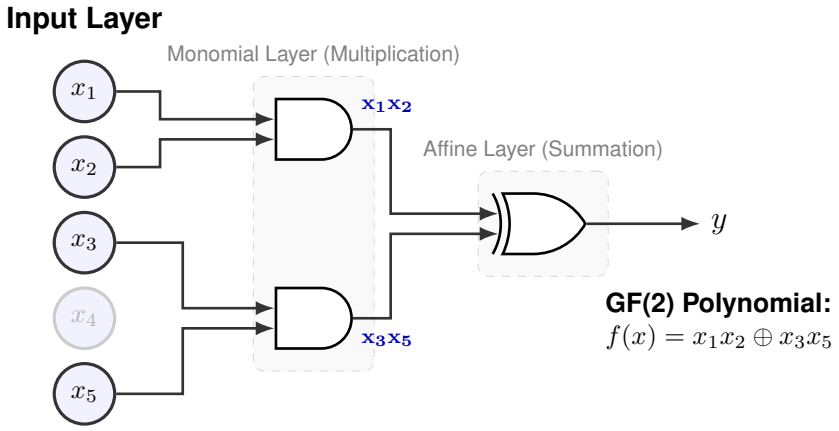


Figure F.1: Example GF(2) circuit.

Further, we will restrict the class of circuits to be those that can be represented as a sum of monomials in GF(2). The resulting function has a bounded maximum degree of monomials. An example visualization is shown in Figure F.1.

F.2 Learning

We now have at our disposal a highly expressive class of functions which, due to their compositional sparsity and adjustable complexity, allow us to evaluate how effectively any given model uncovers the appropriate compositionally sparse representation from data. At this point, it is crucial to clarify what we mean by learning in this context. Traditionally, learning is framed as the problem of finding a function f such that $f(x) \approx y$ for $(x, y) \sim D$, where D is some underlying data distribution. In contemporary machine learning, this view is largely aligned with the paradigm of "deep learning," in which we posit a parametric function $f_1(x; \theta)$ and apply some variant of gradient descent to adjust θ so as to maximize the predictive performance of $f_1(x; \theta)$. We shall refer to this approach as direct learning.

However, this is not the only way to obtain a function f. Alternatively, we might define a process g which, given access to D, does not directly output predictions, but instead produces a function f_2 that, when later executed, generates the correct outputs. This resulting function could then be run by the model itself or by an external executor. In this chapter, we will refer to this paradigm of learning as transform learning.

Both approaches constitute learning from data, but they do so at different levels of abstraction. In direct learning, the system identifies a specific function that maps inputs to outputs within a fixed hypothesis class. In transform learning, the system instead acquires a procedure for constructing such functions from data, thereby shifting part of the burden of generalization to the generated functions themselves. From this perspective, "solving a problem" and "learning a solution" are closely related operations that differ primarily in where adaptation occurs. Both aim to reduce uncertainty about an unknown target, though they encode and exploit that knowledge in structurally distinct ways.

F.2.1 Reasoning

From this standpoint, reasoning can be understood as an iterative procedure in which multiple functions are proposed to transform data, their transformed outputs are assessed, and those

assessments guide which new functions are generated next. Rather than committing immediately to a single function, the *system searches* over a broader space of candidate functions, composing and refining them in response to intermediate feedback. Under this interpretation, reasoning becomes a recursive form of transform-based learning.

When an LLM tackles a complex problem using chain-of-thought, it effectively plays the role of the generator g, emitting a sequence of operations that together implement a particular circuit f tailored to the given input. In the setting of "in-context learning," the model is not modifying its weights through gradient descent; instead, it is constructing a temporary program that maps the specific inputs to the appropriate output.

For the first time, we have a non-biological system that can learn a generator g which can be sampled to implement f across an exceptionally wide range of tasks. While much of the literature on compositional sparsity has focused on training-time optimization dynamics or the structure of the hypothesis space, we can now adopt new perspectives on compositional sparsity to illuminate the capabilities of modern LLMs: compositional sparsity in the functions f that g generates, and compositional sparsity in the way g executes f.

F.2.2 Measuring Learnability

If reasoning is, as we argue, a generative process where a model derives a program f, then we must rigorously quantify the reliability of that generation. How can we measure the probability that this step-by-step construction will eventually lead to a correct solution?

The Limits of Direct Learning. The standard lens for assessing learnability is the *Probably Approximately Correct (PAC)* framework **[valiant1984theory]**, which focuses on the ability of an algorithm to approximate a function from data.

Definition 11 (PAC-Learnability)**.** *A concept class $\mathcal{C}$ is PAC-learnable if there exists an algorithm that, for any target concept $c \in \mathcal{C}$ and any distribution D, outputs a hypothesis h with error at most ϵ with probability at least $1 - \delta$, using time and samples polynomial in $1/\epsilon, 1/\delta$, and the size of the concept.*

For general Boolean circuits, the news is bleak: they are known to be **not PAC-learnable [kearns1994cryptographic]**. If they were, one could trivially break standard cryptographic schemes like RSA by simply observing input-output pairs.

Crucially, however, our interest lies not in learning arbitrary opaque functions, but in **compositionally sparse** circuits. Unlike the general case, specific subclasses of these circuits often *are* theoretically learnable. Yet, despite this theoretical possibility, modern Deep Learning models operating in a "Direct Learning" mode, attempting to predict y directly from x, consistently fail to generalize on these tasks. This discrepancy suggests that while the class of functions may be learnable in principle, the standard supervised paradigm is not the mechanism by which Large Language Models (LLMs) solve them. To understand how models actually conquer these problems, we must shift our perspective from static approximation to sequential reasoning.

The Diligent Learner To model this dynamic derivation, we adopt the **Diligent Learner** framework proposed by Shalev-Shwartz et al. (2025). This approach reframes reasoning as a traversal through a decision tree of semantic states.

The Diligent Learner is defined formally as a tuple $\mathcal{L} = (Root, \mathcal{S}, V, \pi_\theta)$:

1. **Root:** The initial problem instance.

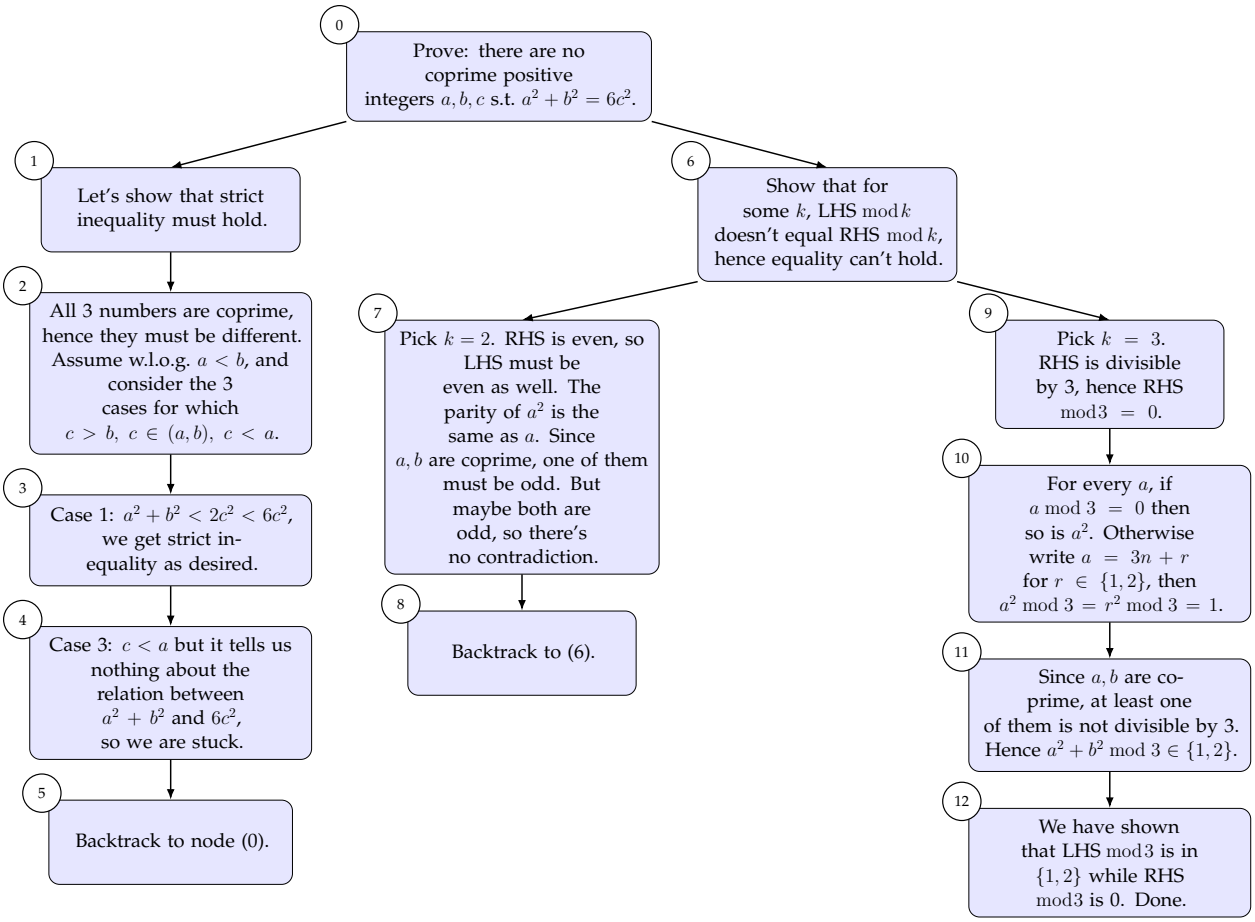


Figure F.2: Diligent learner visualization from [**ShalevShwartzShashua2025Diligent**].

2. **State Space $\mathcal{S}$:** A set of partial states h, where each h represents a prefix of a proof, program, or circuit.

3. **Validator V:** A mechanism (whether internal or external) that enforces local correctness, pruning branches that are logically invalid.

4. **Proposer π_θ:** The generator (agent) that proposes the next extension action based on the current state.

Figure F.2 illustrates this process. At each step, the Proposer suggests a next step (for which the Validator must accept as logically sound) or decide to back track. The same approach can be used to synthesizing Boolean circuits from data: the model builds the circuit term-by-term with a Validator ensuring the step is logically sound.

The Bottleneck Parameter (γ). In this search-based paradigm, the critical bottleneck is the reliability of the *next step*. This reliability is quantified by γ, the probability that the Proposer will make a correct next step in solving the problem.

Let $\pi_\theta(\cdot \mid h)$ be the stochastic policy over actions approved by the validator, given a partial state h. An *extension action* proposes the next semantic step. We denote $S(h, a) = 1$ if the prefix (h, a) keeps the learner on a path that can be completed to a full, correct solution. The bottleneck

parameter γ represents the probability mass the model places on these *useful* next moves:

$$\Pr_{a\sim\pi_\theta(\cdot|h)}\left[a \in \{a_i \mid S(h,a_i)=1\}\right] \;\geq\; \gamma. \tag{F.1}$$

This parameter effectively dictates the feasibility of the reasoning process. If γ remains high, the learner consistently identifies the sparse, valid path through the logic. However, if γ decays as the depth of reasoning increases, the search complexity explodes. Specifically, for a target failure probability δ and maximum reasoning depth $T_{\max}$, the number of steps required scales as:

$$O\Big(T_{\max}\cdot\frac{\log(T_{\max}/\delta)}{\gamma}\Big). \tag{F.2}$$

While contemporary LLMs do not currently perform this explicit backtracking search, this framework provides a crucial theoretical upper bound on a model's reasoning capacity. As they allude to in their work, maximizing γ is the most important objective for any system trying to attain "superintelligence."

F.3 The Computational Cliff

Realizing the full "Diligent Learner" with backtracking, state validation, and fine-tuned proposers presents engineering and reliability hurdles. Yet, we need not build the complete system to test the paradigm. We can leverage part of the framework to assess the capabilities of existing models by isolating the single-step generation process. Even without the full apparatus of recursion, testing whether an LLM can reliably produce the next valid term in a sparse circuit offers a powerful diagnosis of its reasoning fidelity.

To evaluate this capability, we introduce a stepwise reconstruction benchmark over GF(2). At each depth g, the model is presented with a prefix P_g and a labeled sample set S_g, and must output the next monomial t_{g+1}. The setup is designed to be rigorous: Ignoring the prefix P_g leaves the labels masked by the structure of the prefix. Ignoring the samples S_g reduces the task to blind guessing.

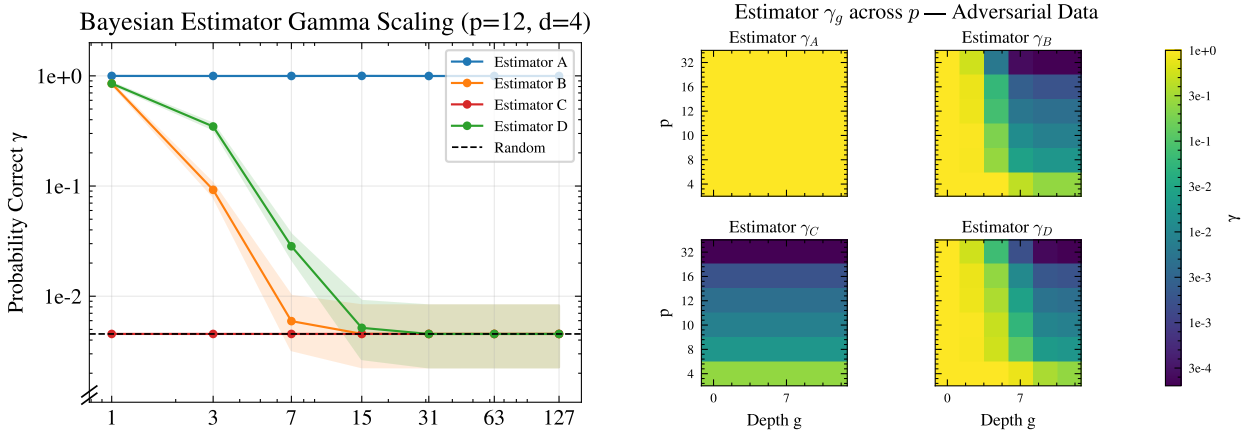
F.3.1 Problem Formulation

We articulate the game using the clean language of Boolean functions in GF(2), revealing the function one monomial at a time.

Targets. We consider inputs $x=(a,v)\in\{0,1\}^{n+p}$, where $a=(a_1,\ldots,a_n)$ are address bits and $v=(v_1,\ldots,v_p)$ are payload bits. We restrict our attention to Algebraic Normal Form (ANF): XORs of monomials, $f(a,v)=\bigoplus_{j=1}^{n} t_j(a,v)$, where each term takes the form

$$t_j(a,v) := a_j\, M_j(v), \quad M_j(v) := \prod_{i\in S_j} v_i, \tag{F.3}$$

with supports $S_j\subseteq[p]$ of fixed size $|S_j|=d-1$. Each term is effectively sparse, touching only d variables, and the entire target is a sparse sum of such terms. An *instance* consists of the ordered support sequence $(S_1,\ldots,S_n)$, sampled once and then held fixed.



(a) Only history+data sustains reliable next-step prediction: step success γ_g versus depth g.

(b) As g and p grow, imperfect-information estimators collapse; only $\mathcal{A}$ remains reliable.

Figure F.3: Stepwise reconstruction performance under adversarial sampling.

The Stepwise Reconstruction Game. At depth $g \in \{0, \ldots, n-1\}$, the learner receives:

1. the ordered prefix $P_g := (t_1, \ldots, t_g)$, and
2. a labeled sample set $\mathsf{S}_g := \{(x^{(k)}, y^{(k)})\}_{k=1}^{K}$ generated for that specific depth.

The model's task is to output a candidate monomial $\hat{t}$. Success is binary: the model succeeds if and only if $\hat{t} = t_{g+1}$. We score the model by its exact success probability,

$$\gamma_g \;:=\; \Pr_{\hat{t}\sim\pi_\theta(\cdot|P_g,\mathsf{S}_g)}\big[\hat{t} = t_{g+1}\big], \tag{F.4}$$

where the probability is taken over the sampled instance, oracle randomness, and the model's own stochasticity. Since the target is an ordered sequence, there is exactly one correct continuation at each depth, making γ_g a measure of exact-next accuracy.

Who is allowed to see what? We distinguish between solvers based on the information they utilize: (i) **Diligent** ($\mathcal{A}_g$) solvers access both the prefix and samples (P_g, S_g); (ii) **Data-only** ($\mathcal{B}_g$) solvers see S_g but are blind to P_g; (iii) **History-only** ($\mathcal{C}_g$) solvers see P_g but lack S_g; and (iv) **Partial** ($\mathcal{D}_g$) solvers have limited access to each. The benchmark is explicitly tuned to separate these classes:

$$\min_g \gamma_g^{\mathcal{A}} \;\geq\; Q \quad \text{while} \quad \gamma_g^{\mathcal{B}},\, \gamma_g^{\mathcal{C}},\, \gamma_g^{\mathcal{D}} \;\approx\; \tfrac{1}{\binom{p}{d-1}}. \tag{F.5}$$

In essence: if a model refuses to use either the prefix or the samples, it is forced to fall back to random guessing.

F.3.2 The Statistical Obfuscation Construction

We enforce the separation between reasoning and pattern matching through statistical obfuscation. At depth g, the label represents a superposition (the XOR sum) of the target signal $M_{g+1}(v)$ and a history-dependent mask. Because this mask is a deterministic function of the prefix P_g and input x, it completely randomizes the label for any observer neglecting variables from the history. For the diligent solver, however, the mask is transparent.

Instance Generation. We sample supports $S_1, \ldots, S_n \subseteq [p]$ with $|S_j| = d-1$ once per instance. This fixes the ordered ANF terms $(t_1, \ldots, t_n)$ in (F.3).

Payload Distribution. We draw payloads at a fixed Hamming weight w, sampling v uniformly from $\{v \in \{0,1\}^p : \|v\|_0 = w\}$. For any fixed S with $|S| = d-1$, we define

$$\rho(w) \;:=\; \Pr_v\left[M_S(v) = 1\right] \;=\; \frac{\binom{w}{d-1}}{\binom{p}{d-1}} \quad (w \geq d-1), \tag{F.6}$$

selecting $w^\star$ such that $\rho(w)$ approximates $1/2$. This entropy maximization preserves the mask's efficacy and minimizes label bias.

Step-g Sampling Oracle (Fixed-Prefix Obfuscation). Given an instance and depth g, the oracle produces $\mathsf{S}_g = \{(a^{(k)}, v^{(k)}, y^{(k)})\}_{k=1}^K$ by constructing a set of carefully designed examples. It fixes $a_{g+1} = 1$ (zeroing subsequent bits) while sampling preceding address bits randomly, outputting $y := f(a, v)$.

The label decomposes as:

$$y \;=\; \underbrace{\Big(\bigoplus_{j=1}^{g} a_j\, M_j(v)\Big)}_{\text{prefix obfuscation}} \;\oplus\; \underbrace{M_{g+1}(v)}_{\text{next-term signal}} \;. \tag{F.7}$$

The mask is fully determined by the prefix and input. A solver utilizing the prefix can cleanly expose the signal, whereas a solver refusing the prefix perceives only noise.

F.3.3 Theoretical Guarantees

We provide rigorous guarantees for this separation. The next-term signal remains statistically inaccessible to history-blind solvers while being fully recoverable by history-aware ones. The barrier is one of information access and state tracking, rather than computational complexity.

Monomial firing at fixed weight. Since payloads are uniform over a fixed Hamming-weight sphere, the probability of a payload monomial "firing" follows a closed form.

Lemma 12 (Monomial firing probability at fixed Hamming weight)**.** *Fix integers $p \geq d-1 \geq 1$ and $w \in \{0, \ldots, p\}$. Let v be uniform over the Hamming sphere $\{v \in \{0,1\}^p : \|v\|_0 = w\}$, and fix any $S \subseteq [p]$ with $|S| = d-1$. Define $M_S(v) := \prod_{i \in S} v_i$. Then*

$$\Pr\left[M_S(v) = 1\right] = \begin{cases} \dfrac{\binom{w}{d-1}}{\binom{p}{d-1}} = \dfrac{\binom{p-(d-1)}{w-(d-1)}}{\binom{p}{w}} & \text{if } w \geq d-1, \\ 0 & \text{if } w < d-1. \end{cases}$$

Per-sample Obfuscation. A data-only estimator $\mathcal{B}_g$, observing data but lacking the prefix P_g, cannot derive the supports determining the mask. We demonstrate a *single-example* guarantee: across the unknown prefix supports, each individual labeled example offers negligible Bayes advantage regarding the next-term signal.

Lemma 13 (Bayes masking given observed (a, v))**.** *Assume the instance distribution samples $S_1, \ldots, S_g, S_{g+1}$ i.i.d. uniformly from $\{S \subseteq [p] : |S| = d-1\}$, independently of the oracle samples. Fix a step g and condition on a realized example (a, v) with $\|v\|_0 = w^\star$. Let*

$$\rho := \rho(w^\star) = \Pr_S[M_S(v) = 1] = \frac{\binom{w^\star}{d-1}}{\binom{p}{d-1}}, \quad m := m(a).$$

Then, marginalizing over the unknown prefix supports $(S_1, \ldots, S_g)$, for each $r \in \{0, 1\}$,

$$\Pr\left[B(a, v) = r \mid a, v\right] \;=\; \tfrac{1}{2}\left[1 + (-1)^r(1 - 2\rho)^m\right].$$

Moreover, $B(a, v)$ is independent of $b = M_{g+1}(v)$ given (a, v), and since $y = B(a, v) \oplus b$ we have

$$\left|\Pr[y = b \mid a, v] - \tfrac{1}{2}\right| \;=\; \tfrac{1}{2}|1 - 2\rho|^m.$$

Lemma 13 confirms that the mask effectively neutralizes single-sample bias, which decays exponentially with the number of active address bits.

History-only Baseline. Reliance on history alone is similarly ineffective. The prefix determines past supports without leaking information about the future support S_{g+1}.

Lemma 14 (History-only is prior guessing)**.** *Assume the instance distribution samples supports $S_1, \ldots, S_n$ i.i.d. uniformly (*with replacement*) from $\{S \subseteq [p] : |S| = d-1\}$. Then for any $g < n$, conditioned on the revealed prefix P_g, the next support S_{g+1} is uniform over $\{S \subseteq [p] : |S| = d-1\}$ and independent of P_g. Consequently, any history-only estimator satisfies $\Pr\left[\widehat{S} = S_{g+1}\right] \le \frac{1}{\binom{p}{d-1}}$.*

Having established the insufficiency of partial information, we demonstrate that a Diligent solver, unifying data and history, can convert the task into a tractable polynomial-time problem.

Recoverability in Polynomial Time. With access to P_g, the solver computes residual labels

$$r^{(k)} \;:=\; y^{(k)} \oplus \bigoplus_{j=1}^{g} a_j^{(k)} M_j(v^{(k)}) \;=\; M_{g+1}(v^{(k)}), \tag{F.8}$$

reducing the problem to identifying the unknown support S_{g+1} from labeled payloads via an intersection decoder:

$$\widehat{S} \;:=\; \bigcap_{k \in K_+} \operatorname{supp}(v^{(k)}), \tag{F.9}$$

where $K_+ := \{k : r^{(k)} = 1\}$.

Theorem 39 (Poly-time recovery under fixed-weight payloads)**.** *Fix an instance and assume payloads are i.i.d. uniform on $\{v \in \{0, 1\}^p : \|v\|_0 = w^\star\}$ with $w^\star \ge d - 1$. Let $\rho = \rho(w^\star) = \Pr[M_{g+1}(v) = 1]$ and $T = |K_+|$. Then $\Pr[T = 0] = (1 - \rho)^K$. Moreover, conditioned on $T \ge 1$, the decoder succeeds with probability at least*

$$1 - \min\Big\{1,\; (p - (d-1))\Big(\frac{w^\star - (d-1)}{p - (d-1)}\Big)^T\Big\}. \tag{F.10}$$

Proof. If $r^{(k)} = 1$ then $M_{g+1}(v^{(k)}) = 1$, which is equivalent to $S_{g+1} \subseteq \mathrm{supp}(v^{(k)})$. Thus for every $k \in K_+$ we have $S_{g+1} \subseteq \mathrm{supp}(v^{(k)})$, and hence

$$S_{g+1} \subseteq \bigcap_{k \in K_+} \mathrm{supp}(v^{(k)}) \;=\; \widehat{S}.$$

Therefore, conditioned on $T \geq 1$, the intersection decoder can fail only if $\widehat{S}$ contains at least one *extraneous* coordinate $i \notin S_{g+1}$, i.e., some $i \in [p] \setminus S_{g+1}$ appears in every positive support. Fix any $i \notin S_{g+1}$ and consider a single draw v conditioned on $r = 1$. Under the fixed-weight model, after forcing ones on S_{g+1}, the remaining $w^\star - (d-1)$ ones are chosen uniformly among the $p - (d-1)$ coordinates in $[p] \setminus S_{g+1}$. Hence

$$\Pr[i \in \mathrm{supp}(v) \mid r = 1] \;=\; \frac{w^\star - (d-1)}{p - (d-1)}.$$

Now condition on the event $\{T = |K_+|\}$ and on the index set K_+ itself. Because the original examples are i.i.d., the payloads $\{v^{(k)}\}_{k \in K_+}$ are i.i.d. draws from the conditional distribution $(v \mid r = 1)$, so

$$\Pr\Big[i \in \mathrm{supp}(v^{(k)})\ \forall k \in K_+ \,\Big|\, T\Big] \;=\; \Big(\frac{w^\star - (d-1)}{p - (d-1)}\Big)^T.$$

Taking a union bound over the $p - (d-1)$ possible extr coordinates gives

$$\Pr\left[\widehat{S} \neq S_{g+1} \mid T\right] \;\leq\; (p - (d-1))\Big(\frac{w^\star - (d-1)}{p - (d-1)}\Big)^T,$$

which implies (F.10). Finally, since $T = \sum_{k=1}^K \mathbf{1}\{r^{(k)} = 1\}$ with $\Pr\Big[r^{(k)} = 1\Big] = \rho = \rho(w^\star)$, we have $\Pr[T = 0] = (1-\rho)^K$. □

Corollary 7 (High-probability recovery with K samples)**.** *In the setting of Thm. 39, let* $\rho := \rho(w^\star) = \Pr[M_{g+1}(v) = 1]$ *and* $\alpha := \frac{w^\star - (d-1)}{p-(d-1)} \in [0, 1)$*. Fix* $\delta \in (0, 1)$*. For* $\alpha \in (0, 1)$ *define*

$$T_0 := \left\lceil \frac{\log\big(2(p - (d-1))/\delta\big)}{\log(1/\alpha)} \right\rceil \quad \text{(and if } \alpha = 0,\ \text{take } T_0 := 1\text{)}.$$

If $K \geq \frac{1}{\rho}\max\{2T_0,\ 8\log(2/\delta)\}$*, then the decoder in* (F.9) *outputs* $\widehat{S} = S_{g+1}$ *with probability at least* $1 - \delta$*.*

Proof. Let $T = |K_+| = \sum_{k=1}^K \mathbf{1}\{r^{(k)} = 1\}$. Since $r^{(k)} = M_{g+1}(v^{(k)})$ and the payloads are i.i.d., we have $T \sim \mathrm{Bin}(K, \rho)$. By Thm. 39, for any $t \geq 1$,

$$\Pr\Big[\widehat{S} \neq S_{g+1} \mid T = t\Big] \;\leq\; (p - (d-1))\alpha^t.$$

Hence

$$\Pr\Big[\widehat{S} \neq S_{g+1}\Big] \;\leq\; \Pr[T < T_0] \;+\; \Pr\Big[\widehat{S} \neq S_{g+1} \mid T \geq T_0\Big].$$

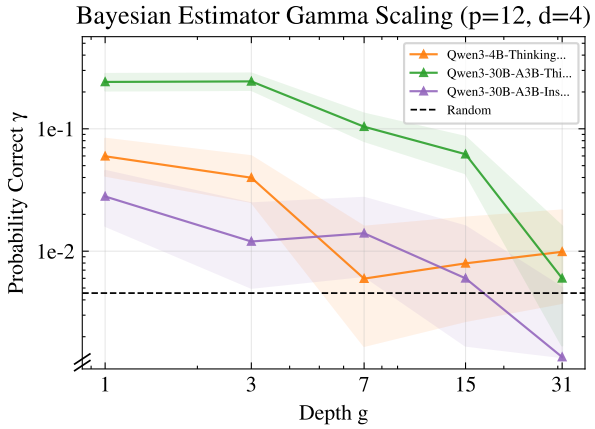
For the first term, our lower bound on K implies $K\rho/2 \geq T_0$, so by a multiplicative Chernoff bound,

$$\Pr[T < T_0] \;\leq\; \Pr\Big[T \leq \tfrac{1}{2}K\rho\Big] \;\leq\; \exp\Big(-\tfrac{1}{8}K\rho\Big) \;\leq\; \delta/2,$$

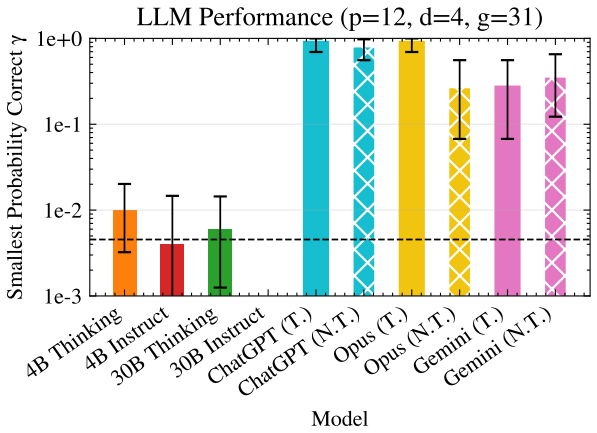
where the last inequality uses $K\rho \geq 8\log(2/\delta)$. For the second term, on $T \geq T_0$ we have

$$\Pr\Big[\widehat{S} \neq S_{g+1} \mid T \geq T_0\Big] \;\leq\; (p - (d-1))\alpha^{T_0} \;\leq\; \delta/2,$$

by the definition of T_0 (and trivially if $\alpha = 0$). Combining the two bounds yields $\Pr\Big[\widehat{S} \neq S_{g+1}\Big] \leq \delta$. □



(a) Small LLMs: step-success γ_g degrades sharply with depth g.



(b) Frontier LLMs achieve higher γ_g than smaller models.

Figure F.4: Depth-induced collapse in next-step prediction across model families.

These results define the information-theoretic landscape. Lemma 13 and Lemma 14 establish the necessity of combining signal sources, proving that isolation leads to failure. Theorem 39 provides the constructive converse: the prefix serves as a decryption key, transforming the noisy label into a clean signal recoverable with polynomial resources. The task separates those who track state from those who merely pattern-match.

Efficient Validation. A robust benchmark requires a fast grader. Given the instance specification and a candidate monomial $\tilde{t}$, our validator parses $\tilde{t}$ and accepts if and only if the payload index set $\widetilde{S}$ matches S_{g+1}. This process is highly efficient, taking $O(|\tilde{t}| + d \log d)$ time, allowing for rapid evaluation.

F.3.4 Tool Synthesis

This brings us to a phenomenon we term the "Computational Cliff." Circuit synthesis requires exact state tracking. Local errors eventually propagate and corrupt the entire derivation in a way that can be well described as a Gaussian model with only a partial memory. While the exact complexity of the problem that triggers the collapse changes with the capacity of the model, we do observe it for all models beyond a critical depth that do not build and use tools koplow2026superintellgence.

Shortening Error Chains In a reasoning trace of length N, independent error probabilities ϵ compound to yield a success rate of $(1-\epsilon)^N$. Tool use restructures this probability. By offloading complex logic to verified external solvers, the effective logical depth contracts, replacing fragile token-by-token generation with robust atomic operations.

The Law of Recursion This aligns with the **Law of Recursion [koplow2026lawsofemergence]**:

> "A system designed to maximally utilize compositional sparsity will design other systems that can use compositional sparsity even better for specific domains."

A sophisticated intelligence does not simulate all computation within its primary weights. It builds specialized external structures to handle irreducible complexity. Biologically, the cell

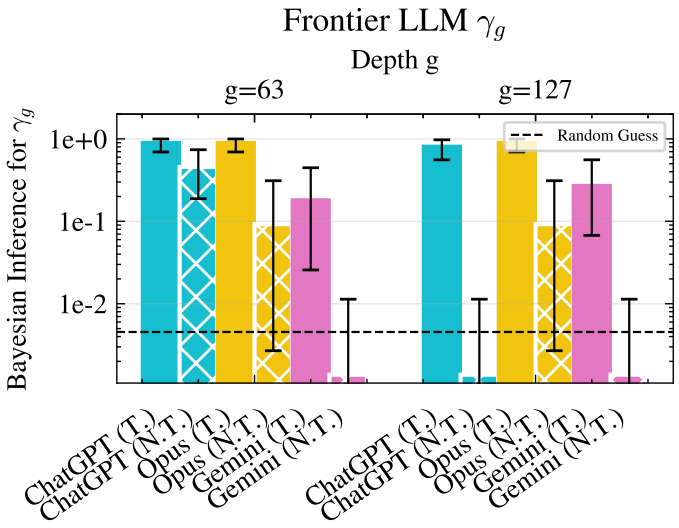


Figure F.5: Frontier models with tool use (T.) sustain higher γ_g at larger depths than no-tools variants (N.T.).

does not rely on a homogeneous cytoplasm for all chemistry; it compartmentalizes function into organelles. Similarly, synthetic intelligence ought not simulate circuit evaluation in raw neural activations, but rather synthesize programs to execute the task. Reasoning, then, is the act of generating the executable.

CHAPTER G

Gradient Formulas for Iterated Functional Compositions

In several analytical paradigms—ranging from the structural properties of deep neural architectures (e.g., Recurrent Neural Networks) to discrete dynamical systems and chaos theory—it is necessary to evaluate the exact derivative of an n-times iterated composition of a function mapping a space into itself. This appendix establishes the formal notation, explicit gradient derivations, and spectral implications for both scalar and multivariable settings.

Let $f : \mathcal{X} \to \mathcal{X}$ be a continuously differentiable map ($\mathcal{X} \subseteq \mathbb{R}^d$). We define the n-times iterated composition function $F : \mathcal{X} \to \mathcal{X}$ by:

$$F(x) = f^{(n)}(x) = \underbrace{(f \circ f \circ \cdots \circ f)}_{n \text{ times}}(x) \tag{G.1}$$

To make the application of the chain rule structurally transparent, we introduce a discrete-time state trajectory sequence $\{x_k\}_{k=0}^{n}$ mapping the forward evaluation path:

$$\begin{aligned} x_0 &= x \\ x_1 &= f(x_0) \\ x_2 &= f(x_1) \\ &\vdots \\ x_k &= f(x_{k-1}) \\ &\vdots \\ x_n &= f(x_{n-1}) = F(x) \end{aligned} \tag{G.2}$$

(G.3)

G.1 The Univariate Case ($d = 1$)

Let $\mathcal{X} \subseteq \mathbb{R}$, where $f : \mathbb{R} \to \mathbb{R}$ is a differentiable scalar function. By applying the classical univariate chain rule to the sequence of intermediate states, the total derivative of the iterated composition can be factored into a sequence of scalar multipliers:

$$\frac{dF}{dx} = \frac{dx_n}{dx_0} = \frac{dx_n}{dx_{n-1}} \cdot \frac{dx_{n-1}}{dx_{n-2}} \cdots \frac{dx_1}{dx_0} \tag{G.4}$$

Because each transition is defined locally by $x_k = f(x_{k-1})$, the derivative of any intermediate state with respect to its immediate predecessor is simply the first derivative of f evaluated at that predecessor state:

$$\frac{dx_k}{dx_{k-1}} = f'(x_{k-1}) \tag{G.5}$$

Substituting this relation back into the chain rule expansion yields:

$$\frac{dF}{dx} = f'(x_{n-1}) \cdot f'(x_{n-2}) \cdots f'(x_1) \cdot f'(x_0) \tag{G.6}$$

Expressed compactly using product notation, the final derivative is:

$$\frac{dF}{dx} = \prod_{k=0}^{n-1} f'(x_k) = \prod_{k=0}^{n-1} f'\left(f^{(k)}(x)\right) \tag{G.7}$$

where $f^{(0)}(x) \equiv x$.

G.2 The Multivariate Case ($d > 1$)

Let $\mathcal{X} \subseteq \mathbb{R}^d$, where $f : \mathbb{R}^d \to \mathbb{R}^d$ represents a multivariable operator. The derivative of f with respect to its input vector is expressed via the Jacobian matrix $J_f \in \mathbb{R}^{d \times d}$.

Because matrix multiplication is inherently non-commutative ($AB \neq BA$), the linear ordered sequence of the chain rule must be strictly preserved. Applying the multivariable chain rule from the outermost transformation down to the input vector yields the matrix product from left to right:

$$J_F(x) = J_f(x_{n-1}) \cdot J_f(x_{n-2}) \cdots J_f(x_1) \cdot J_f(x_0) \tag{G.8}$$

where each term $J_f(x_k) \in \mathbb{R}^{d \times d}$ represents the local Jacobian matrix evaluated explicitly at the intermediate state x_k:

$$[J_f(x_k)]_{ij} = \left.\frac{\partial f_i}{\partial x_j}\right|_{x=x_k} \tag{G.9}$$

Using the right-to-left product operator $\prod$, this composition can be condensed as:

$$J_F(x) = \prod_{k=1}^{n} J_f(x_{n-k}) \tag{G.10}$$

G.2.1 Gradient of a Scalar Loss Function

In applications involving optimization, such as empirical risk minimization, one typically encounters a scalar-valued objective or loss function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ evaluated at the final terminal state x_n, such that $\mathcal{L}(F(x))$.

To compute the gradient vector $\nabla_x \mathcal{L} \in \mathbb{R}^d$ with respect to the original input parameters, we apply the identity $(AB)^T = B^T A^T$. This reverses the operational ordering of the constituent Jacobian matrices during backpropagation:

$$\nabla_x \mathcal{L} = [J_F(x)]^T \nabla_{x_n} \mathcal{L} \tag{G.11}$$

Expanding this via the intermediate states yields:

$$\nabla_x \mathcal{L} = \left(J_f(x_0)^T \cdot J_f(x_1)^T \cdots J_f(x_{n-2})^T \cdot J_f(x_{n-1})^T\right) \nabla_{x_n} \mathcal{L} \tag{G.12}$$

G.3 Structural and Spectral Implications

The multiplicative composition of derivatives outlined in equations (A.6) and (A.8) has deep theoretical consequences in numerical analysis, system dynamics, and representation theory.

G.3.1 The Stability of Deep Iterated Maps

Consider the spectral radius $\rho(J_f(x)) = \max_i |\lambda_i|$, where λ_i are the eigenvalues of the local Jacobian matrix. The structural behavior of the overall gradient long-term ($n \to \infty$) is highly dependent on this radius:

- **Vanishing Gradients:** If $\rho(J_f(x_k)) < 1$ for a sustained subsequence of states, the matrix product contracts exponentially. Consequently, $J_F(x) \to \mathbf{0}$ as $n \to \infty$, rendering early input states insulated from perturbations in the terminal objective.
- **Exploding Gradients:** If $\rho(J_f(x_k)) > 1$, the matrix product expands exponentially, causing numerical instability where $\|J_F(x)\| \to \infty$.

G.3.2 Connection to Dynamical Systems

In discrete dynamical systems, the long-term asymptotic behavior of these product matrices defines the structural stability of trajectories. The geometric average rate of expansion or contraction along an infinite trajectory yields the *Lyapunov Exponents* μ_i:

$$\mu_i = \lim_{n\to\infty} \frac{1}{n} \ln \sigma_i \left(J_F(x_0)\right) \tag{G.13}$$

where $\sigma_i(M)$ denotes the i-th singular value of the composite matrix M. A positive maximum Lyapunov exponent ($\mu_{\max} > 0$) serves as a formal indicator of chaotic dynamics and extreme sensitivity to initial conditions.

CHAPTER H

Principles of Deep Learning

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Modern deep learning works far better than classical statistical learning theory or traditional numerical analysis would lead one to expect. In this chapter we articulate two basic principles that, taken together, offer an explanation for the empirical success of deep architectures: (1) the sparse compositionality *of efficiently computable functions, and (2) the* genericity *required for stability and learnability. Sparse compositionality explains approximation, hierarchy, modularity, and transfer. Genericity explains stability and the existence of informative gradients during optimization. Together these principles provide a coherent framework for understanding the remarkable capabilities of deep networks and transformers.*

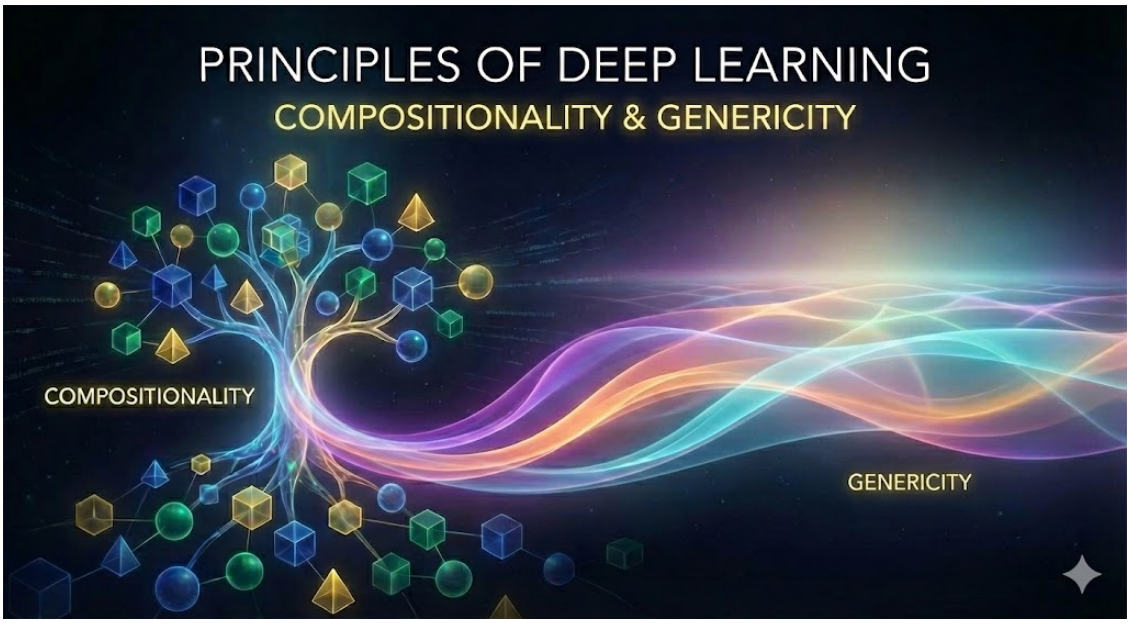

Figure H.1

H.1 Principle I: Sparse Compositionality

The property of *sparse compositionality* is formally implied by the property of *efficient Turing computability* [156]. Functions that are efficiently computable necessarily decompose into bounded-fan-in computational graphs (DAGs). This is a structural fact: a computation composed of many small local operations must be sparse in the sense that each intermediate value depends only on a few others [109].

Formally, if f is efficiently computable, then it admits a representation

$$f = f^{(L)} \circ f^{(L-1)} \circ \cdots \circ f^{(1)}$$

in which each $f^{(\ell)}$ acts on a small number of variables. Deep networks mirror this structure and thus avoid the curse of dimensionality when their architecture matches such sparse DAGs [129].

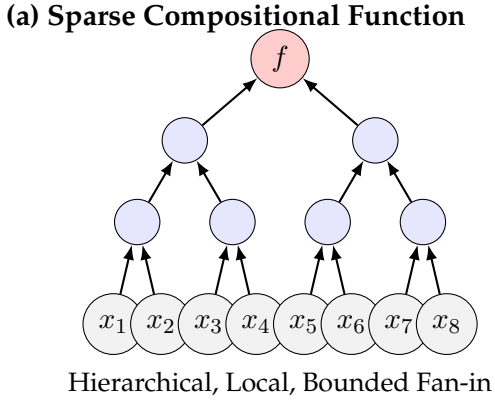

Figure H.2: Visualizing Sparse Compositionality. An efficiently computable function naturally decomposes into a hierarchical DAG where each node depends on a small number of inputs (bounded fan-in). This structure allows deep networks to approximate the function without incurring the curse of dimensionality.

In this section we describe several consequences of sparsity and compositionality.

H.1.1 Hierarchy follows from sparse compositionality

If a function admits a compositional DAG in which each node has bounded fanin, then the natural architecture for approximating it is a *hierarchical network*: each layer computes simple local features, which become inputs to features of increasing complexity.

Hierarchy is not an extra assumption; it is a direct corollary of the DAG.

Thus the empirical fact that good deep networks are deep is not surprising: depth is the natural way to reflect the compositional structure of the target.

H.1.2 Modularity and reuse of modules

Sparse compositionality also implies *modularity*:

- subgraphs (subfunctions) appear repeatedly in the DAG,
- these subgraphs compute intermediate concepts,

- the same module can be used at different places in the computation.

This modularity is reflected in architectures such as convolutional networks, residual networks, and attention blocks, where learned components are reused globally or across layers.

The existence of modules is not a property of the architecture alone; it is a property of the underlying function.

H.1.3 Transfer learning

If a function decomposes compositionally and modules appear in multiple places, then learning one task provides information useful for others.

$$\text{Module learned in Task A} \quad \Rightarrow \quad \text{usable immediately in Task B.}$$

Thus *transfer learning* is a natural consequence of sparse compositionality: reuse across tasks parallels reuse across positions in a single computation [15].

H.1.4 Interpretability from consistent compositionality

If the decomposition of f into modules is *consistent across data*, then the learned network will assign stable semantic meaning to its internal components. This yields a form of interpretability:

$$\text{same subfunction} \quad \Rightarrow \quad \text{same module/layer.}$$

Thus interpretability is a consequence of:

- the existence of a compositional structure in the function, and
- the ability of the model to discover and consistently reuse that structure.

H.1.5 Non-uniqueness of sparse decompositions

A crucial observation is that *sparse decompositions are not unique*. A given function f may admit many distinct DAG representations. In such cases, a general-purpose architecture—such as a transformer—has no reason to choose the same decomposition across different inputs or during training.

Consequences:

- Transformers do not consistently reuse modules: the same computation can be implemented in many ways. the same computation can be implemented in many ways.
- Without architectural constraints enforcing modularity, reuse, or locality, the model may realize different decompositions in different parts of the input space.
- This explains why transformers are powerful but not inherently interpretable or modular.

Transformers achieve excellent performance even without consistent module reuse because they can realize *any* decomposition compatible with their large associative memory structure.

H.2 Principle II: Genericity of Learnable Targets

Sparse compositionality describes the *structure* of computable functions. Genericity describes a different property: the requirement that learnable functions be *invariant* in their basic jet structure under shifts of the coordinates. Genericity implies that the function cannot rely on measure-zero cancellations or extremely fragile high-order interactions [145].

H.2.1 Genericity from invariance to the choice of the origin of the coordinates

Genericity excludes "non-generic" functions in which:

- all low-order terms vanish exactly,
- small perturbations of the input distribution flip the identity of the best predictor,

Genericity is the opposite situation: under realistic noise and bias, *every relevant variable leaves a low-degree footprint*, often linear, that persists across perturbations.

Thus:

$$\text{Stability} \quad \Rightarrow \quad \text{Genericity (low-degree footprints, no fine-tuning).}$$

H.2.2 Genericity ensures good gradients for optimization

Once genericity holds, gradient-based optimization is feasible:

- Linear footprints provide nonzero first-order information.
- Gradients do not vanish due to high-order cancellations.
- Different layers of a deep network receive consistent signals.

Thus genericity provides the missing piece for explaining why gradient descent can find good solutions in nonconvex architectures.

H.3 Two independent but complementary principles

Sparse compositionality and genericity arise from two different fundamental assumptions:

- Efficient Turing computability $\Rightarrow$ sparse compositionality.
- ERM-based learnability + stability $\Rightarrow$ genericity.

They are independent but complementary:

- Sparse compositionality explains approximation, hierarchy, modularity, transfer, interpretability.
- Genericity explains stability, robustness, and the existence of informative gradients.

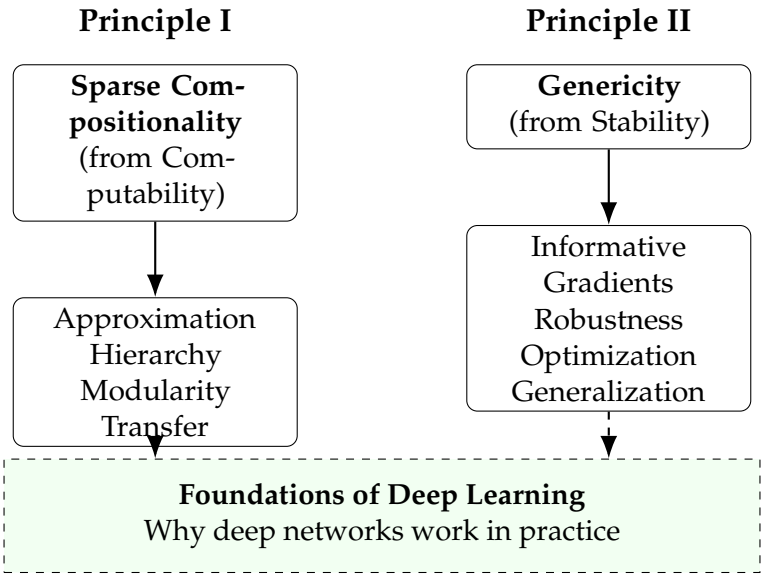

Figure H.3: The Two Pillars of Deep Learning Theory. Sparse Compositionality (derived from efficient computability) explains the architectural benefits, while Genericity (derived from stability) explains the feasibility of learning and optimization.

Together they provide a unified explanation for why deep learning works.

$$\text{Sparse compositionality} + \text{Genericity} \implies \begin{cases} \text{Efficient approximation} \\ \text{Generalization (stability)} \\ \text{Optimizable landscapes} \\ \text{Hierarchy, modularity, reuse} \\ \text{Transfer and interpretability} \end{cases}$$

H.4 Conclusion

The two principles articulated in this chapter—sparse compositionality and genericity—capture complementary aspects of deep learning:

- the *structure* of the functions that we want to learn, and
- the *regularity* that makes them learnable and optimizable.

Most of the puzzles of deep learning—its scalability, generalization, modularity, transfer, and surprising ease of optimization—follow naturally from these two principles once they are recognized as the structural and regularity constraints of the real functions we want to learn.

CHAPTER I

The Geometry of Overparameterized Systems: Krull and Bézout

I.1 Introduction

In the theoretical analysis of highly parameterized systems—such as polynomial approximations of overparameterized networks—we frequently encounter systems of equations where the number of parameters (or unknowns) d vastly exceeds the number of constraints n.

When $d \gg n$, the landscape of solutions ceases to be a discrete set of isolated points. Instead, it forms a continuous, high-dimensional manifold (more formally, an algebraic variety). Understanding the dimensionality, geometric complexity, and inherent degeneracy of this solution space requires classical tools from algebraic geometry, specifically Krull's Height Theorem and the Generalized Bézout Theorem. To ensure structural consistency and prevent solution manifolds from abruptly vanishing, this analysis must be conducted over an algebraically closed field, typically the complex numbers $\mathbb{C}$.

I.2 Dimensionality: Krull's Height Theorem

The first fundamental question regarding an overparameterized polynomial system is the size, or dimensionality, of its solution space. Intuitively, each independent equation imposes one constraint, removing one degree of freedom from the ambient d-dimensional space. This intuition is formalized algebraically by Krull's Theorem.

Theorem 40 (Krull's Height Theorem - Geometric Formulation)**.** *Let $\mathbb{C}[x_1, \ldots, x_d]$ be the polynomial ring in d variables. If an ideal I is generated by n polynomials $f_1, \ldots, f_n$, then every minimal prime ideal over I has height at most n.*

Geometrically, the "height" of the ideal corresponds to the codimension of the solution variety. Therefore, Krull's Theorem guarantees that the solution space $X \subset \mathbb{C}^d$ defined by $f_1(x) = 0, \ldots, f_n(x) = 0$ satisfies:

$$\dim(X) \geq d - n \tag{I.1}$$

If the polynomials are chosen generically (lacking algebraic dependencies), the equations act as maximal constraints, and the dimension of the solution manifold is exactly $d - n$. The system is

inherently degenerate in the sense that the solutions form an extended continuous surface rather than isolated points.

I.3 Geometric Complexity: Generalized Bézout's Theorem

While Krull's Theorem bounds the dimension of the solution manifold, it does not describe its topological or geometric complexity. For square systems ($d = n$), the classical Bézout Theorem bounds the number of isolated solution points. In the heavily underdetermined regime ($d \gg n$), Bézout's Theorem generalizes to bound the *degree* of the $(d-n)$-dimensional solution manifold.

The degree of a $(d-n)$-dimensional variety X geometrically represents the number of intersection points between X and a generic (n)-dimensional linear subspace.

Theorem 41 (Generalized Bézout Theorem)**.** *Let $f_1, \ldots, f_n$ be polynomials in d variables defining a variety X of dimension $d-n$. Then the degree of X is bounded by the product of the degrees of the defining polynomials:*

$$\deg(X) \leq \prod_{i=1}^{n} \deg(f_i) \tag{I.2}$$

This theorem provides a strict upper limit on how highly folded, oscillating, or "complex" the overparameterized solution manifold can be, governed strictly by the algebraic degrees of the input equations.

I.4 Maximal Degeneracy at the Origin

A distinct topological phenomenon occurs when we restrict the system such that none of the polynomials f_i contain linear or constant terms. In this scenario, every term has a degree of 2 or higher.

Remark 3 (The Origin as a Singular Point)**.** *If $f_i(0) = 0$ and $\nabla f_i(0) = \mathbf{0}$ for all $i = 1, \ldots, n$, the origin $\mathbf{0} \in \mathbb{C}^d$ is a guaranteed solution. Furthermore, the Jacobian matrix $J \in \mathbb{C}^{n \times d}$ evaluates entirely to zero at this point.*

Because the Jacobian loses full rank (evaluating to rank 0), the origin represents a severe geometric singularity on the $(d-n)$-dimensional manifold. In optimization contexts, where one might minimize a squared-loss objective $L(x) = \sum_{i=1}^{n} f_i(x)^2$, the Hessian $H = 2J^T J$ also evaluates to the zero matrix at the origin. Thus, the solution manifold is not only continuous, but highly singular and maximally flat at $\mathbf{0}$, representing a perfectly degenerate critical point spanning all d directions.

I.5 Real vs. Complex Solution Varieties ($\mathbb{R}^d$ vs. $\mathbb{C}^d$)

A critical technical distinction in applying algebraic-geometric tools to machine learning arises from the choice of field. Standard algebraic geometry results—including Krull's Height Theorem and Bézout's Theorem—hold unconditionally over algebraically closed fields such as $\mathbb{C}$. Modern neural networks, however, are trained over the real field $\mathbb{R}$.

I.5.1 Dimension Drop in Real Loci

When parameters are restricted to $W \in \mathbb{R}^d$, the real solution set $\mathcal{M}_{\mathbb{R}} = \mathcal{M} \cap \mathbb{R}^d$ forms a real semi-algebraic set. Unlike complex varieties, real loci do not automatically inherit the complex algebraic dimension $d - n$:

$$\dim_{\mathbb{R}}(\mathcal{M}_{\mathbb{R}}) \leq \dim_{\mathbb{C}}(\mathcal{M}) \tag{I.3}$$

Underdetermined real polynomial systems can degenerate into empty sets or isolated points. For example, the equation $\sum_{i=1}^{d} W_i^2 = 0$ defines a continuous hypersurface of dimension $d-1$ over $\mathbb{C}^d$, but collapses to the isolated origin $\mathbf{0}$ of dimension 0 over $\mathbb{R}^d$.

I.5.2 Overparameterization and Smooth Real Points

Despite this theoretical risk of dimension drop, overparameterization ($d \gg n$) typically prevents real degeneration in neural networks.

By the Implicit Function Theorem, if there exists at least one real point $W^* \in \mathbb{R}^d$ where the Jacobian matrix $J(W^*) \in \mathbb{R}^{n \times d}$ achieves full rank n, then $\mathcal{M}_{\mathbb{R}}$ is locally a smooth real manifold of dimension exactly $d-n$ around W^*. Consequently, for generic data and targets where interpolation is achievable, the real zero-loss landscape preserves the expected dimension:

$$\dim_{\mathbb{R}}(\mathcal{M}_{\mathbb{R}}) = d - n \tag{I.4}$$

I.6 Local Geometry and the Neural Tangent Kernel

The geometric structure of $\mathcal{M}$ near an interpolating point W^* dictates local optimization dynamics, such as the convergence rate and implicit bias of gradient descent.

I.6.1 Tangent Space and Flat Directions

At any smooth point $W^* \in \mathcal{M}$, the tangent space $T_{W^*}\mathcal{M}$ is defined by the kernel of the Jacobian:

$$T_{W^*}\mathcal{M} = \ker\left(J(W^*)\right) = \left\{ v \in \mathbb{C}^d \;\middle|\; J(W^*)v = \mathbf{0} \right\} \tag{I.5}$$

By the Rank-Nullity Theorem, $\dim(T_{W^*}\mathcal{M}) = d - \operatorname{rank}(J(W^*))$. When $J(W^*)$ has full rank n, the tangent space forms a $(d-n)$-dimensional linear subspace of locally flat, loss-invariant directions.

I.6.2 NTK Rank Defect and Singular Loci

The local manifold structure is intrinsically linked to the Neural Tangent Kernel (NTK) matrix $\Theta(W) \in \mathbb{C}^{n \times n}$:

$$\Theta(W) = J(W)J(W)^T \tag{I.6}$$

We classify points on the zero-loss landscape based on the spectrum of $\Theta(W)$: 1 * **Regular Interpolating Points ($\operatorname{rank}(\Theta(W)) = n$):** The Jacobian has full row rank. The manifold $\mathcal{M}$ is locally smooth, and the tangent space has dimension exactly $d - n$. * **Singular Subvarieties ($\operatorname{rank}(\Theta(W)) < n$):** The set of parameters $\mathcal{S} = \{W \in \mathbb{C}^d \mid \det(\Theta(W)) = 0\}$ defines a singular algebraic subvariety. At points $W \in \mathcal{S} \cap \mathcal{M}$, the tangent space dimension jumps above $d - n$, signaling self-intersections, branching sheets, or cusps in the zero-loss manifold.

I.7 Summary of Landscape Regimes

The parameter landscape of an overparameterized network with polynomial activations decomposes into three main geometric regimes:

Regime	Local Dimension	Geometric Property
Origin ($W = \mathbf{0}$)	d	Maximally singular; zero Jacobian $J = \mathbf{0}$ and zero Hessian $H = \mathbf{0}$; flat in all d directions.
Singular Locus ($\mathcal{S} \cap \mathcal{M}$)	$> d - n$	Non-smooth intersections of manifold sheets; NTK loses rank ($\det(\Theta) = 0$).
Generic Interpolation ($\mathcal{M} \setminus \mathcal{S}$)	$d - n$	Smooth manifold; bounded degree $\leq r^{Ln}$; $(d - n)$ zero-curvature tangent directions.

CHAPTER J

Unsupervised Ensembling of Verifiers

Modern generative systems can often produce several candidate solutions to the same problem, while separate models or procedures can score the quality of those candidates. This raises a natural question: can several imperfect verifiers be combined into a reliable judge when no ground-truth correctness labels are available?

The answer can be positive, but only under structural assumptions. If the errors of multiple verifiers are sufficiently diverse, their patterns of agreement and disagreement contain information about their latent accuracies. Fully Unsupervised Score Ensembling (FUSE) [88] exploits this principle using low-order moments of the verifier outputs.

This appendix develops the basic mathematics. Its main conceptual lesson is simple:

$$\boxed{\textit{zero labels} \neq \textit{zero information}}$$

but also

$$\boxed{\textit{zero labels} \neq \textit{zero assumptions}.}$$

Unsupervised verification is possible only when the observable agreement structure of the ensemble contains a detectable footprint of latent correctness.

J.1 The Problem

Suppose a generative system produces candidate responses

$$X_1, \ldots, X_N.$$

For each response X_n, let

$$Y_n \in \{-1, +1\}$$

denote its unknown ground-truth correctness:

$$Y_n = +1 \quad \Longleftrightarrow \quad X_n \text{ is correct.}$$

Assume that m imperfect verifiers evaluate every response.
For simplicity, begin with binary verifier outputs

$$V_i(X_n) \in \{-1, +1\}, \qquad i = 1, \ldots, m.$$

The learner observes the matrix

$$V = [V_i(X_n)]_{n,i},$$

but never observes the labels

$$Y_1, \ldots, Y_N.$$

The goal is to infer

$$P(Y_n = +1 \mid V_1(X_n), \ldots, V_m(X_n))$$

using only the verifier outputs.

J.2 Verifier Quality

Let

$$\pi = P(Y = +1)$$

denote the class prior.
For verifier i, define its sensitivity

$$\alpha_i = P(V_i = +1 \mid Y = +1) \tag{J.1}$$

and specificity

$$\beta_i = P(V_i = -1 \mid Y = -1). \tag{J.2}$$

A useful signed measure of verifier quality is

$$a_i = \alpha_i + \beta_i - 1. \tag{J.3}$$

Thus

$$a_i > 0$$

means that verifier i is better than random in the appropriate orientation,

$$a_i = 0$$

corresponds to an uninformative verifier, and

$$a_i < 0$$

corresponds to a systematically reversed verifier.
Define the conditional means

$$m_i^+ = E[V_i \mid Y = +1] = 2\alpha_i - 1, \tag{J.4}$$
$$m_i^- = E[V_i \mid Y = -1] = 1 - 2\beta_i. \tag{J.5}$$

Their difference is

$$\Delta_i = m_i^+ - m_i^- = 2a_i. \tag{J.6}$$

The observable marginal mean is

$$\mu_i = E[V_i] = \pi m_i^+ + (1-\pi)m_i^-. \tag{J.7}$$

J.3 Triplet Conditional Independence

The central structural assumption is that distinct verifiers make independent errors once the true label is known.

For three distinct verifiers i, j, k, assume

$$P(V_i, V_j, V_k \mid Y) = P(V_i \mid Y)P(V_j \mid Y)P(V_k \mid Y). \tag{J.8}$$

We refer to this as *Triplet Conditional Independence* (TCI).

TCI does not mean that the verifier outputs are marginally independent.

Indeed, they will ordinarily be correlated because they are responding to the same latent variable Y.

It says only that, after conditioning on correctness, their remaining errors are independent.

This distinction is the key to the method.

J.4 Observable Moment Factorization

Define the off-diagonal covariance

$$R_{ij} = E[(V_i - \mu_i)(V_j - \mu_j)], \qquad i \neq j, \tag{J.9}$$

and the third central moment

$$T_{ijk} = E[(V_i - \mu_i)(V_j - \mu_j)(V_k - \mu_k)], \qquad i, j, k \text{ distinct}. \tag{J.10}$$

Proposition 65 (Moment factorization under TCI)**.** *Under Triplet Conditional Independence,*

$$R_{ij} = \pi(1-\pi)\Delta_i\Delta_j = 4\pi(1-\pi)a_i a_j, \tag{J.11}$$

and

$$T_{ijk} = \pi(1-\pi)(1-2\pi)\Delta_i\Delta_j\Delta_k = 8\pi(1-\pi)(1-2\pi)a_i a_j a_k. \tag{J.12}$$

Proof. Conditional independence gives, for $i \neq j$,

$$E[V_i V_j \mid Y] = E[V_i \mid Y]E[V_j \mid Y].$$

The law of total covariance therefore yields

$$\mathrm{Cov}(V_i, V_j) = \mathrm{Cov}\left(E[V_i \mid Y], E[V_j \mid Y]\right).$$

Since Y is binary,

$$E[V_i \mid Y = +1] - E[V_i \mid Y = -1] = \Delta_i,$$

and hence

$$R_{ij} = \pi(1-\pi)\Delta_i\Delta_j.$$

Using

$$\Delta_i = 2a_i$$

gives Equation (J.11).
For the third central moment, note that

$$E[V_i - \mu_i \mid Y = +1] = (1-\pi)\Delta_i$$

and

$$E[V_i - \mu_i \mid Y = -1] = -\pi\Delta_i.$$

Conditional independence then gives

$$T_{ijk} = \pi(1-\pi)^3\Delta_i\Delta_j\Delta_k - (1-\pi)\pi^3\Delta_i\Delta_j\Delta_k \tag{J.13}$$
$$= \pi(1-\pi)(1-2\pi)\Delta_i\Delta_j\Delta_k. \tag{J.14}$$

Substituting $\Delta_i = 2a_i$ proves Equation (J.12). □

The important structure is that both tensors factor into products of one-dimensional verifier parameters.
Schematically,

$$\boxed{R_{ij} \propto a_i a_j, \qquad T_{ijk} \propto a_i a_j a_k.}$$

Thus the observable moments have a rank-one structure induced by the single latent variable Y.

J.5 Why Second- and Third-Order Moments Are Enough

For distinct i, j, k,

$$\frac{T_{ijk}}{\sqrt{|R_{ij}R_{ik}R_{jk}|}} = \frac{1-2\pi}{\sqrt{\pi(1-\pi)}}\,\operatorname{sign}(a_i a_j a_k). \tag{J.15}$$

If the verifiers are oriented so that

$$a_i > 0,$$

the right-hand side is independent of the verifier triplet.
Thus, in the population limit, every valid triplet gives the same observable ratio.
This provides information about the latent class prior π.
Once π is known, pairwise covariances determine the relative verifier skills. For example,

$$a_i^2 = \frac{R_{ij}R_{ik}}{4\pi(1-\pi)R_{jk}}, \tag{J.16}$$

for a nondegenerate triplet.

The marginal means then recover sensitivity and specificity.

From Equations (J.3) and (J.7),

$$\alpha_i = \frac{1+\mu_i}{2} + (1-\pi)a_i, \tag{J.17}$$

$$\beta_i = \frac{1-\mu_i}{2} + \pi a_i. \tag{J.18}$$

Thus the unlabeled joint distribution of verifier outputs can, under the model assumptions, identify the quality of the individual verifiers.

J.6 The Unavoidable Orientation Ambiguity

There is one fundamental ambiguity.

If the latent labels are reversed,

$$Y \longrightarrow -Y,$$

then an ensemble of good verifiers can be reinterpreted as an ensemble of systematically wrong verifiers.

The observable distribution of verifier outputs alone cannot decide which latent state should be called "correct."

Thus unsupervised identification is only possible up to a global label permutation unless an orientation assumption is supplied.

A typical assumption is that most verifiers are better than random:

$$\sum_i a_i > 0, \tag{J.19}$$

or, more strongly, that a designated subset satisfies

$$a_i > 0.$$

This illustrates an important general point:

unlabeled estimation does not eliminate assumptions; it replaces labels by structural assumptions.

J.7 When Conditional Independence Fails

For real verifier ensembles, exact TCI is unlikely.

Different verifiers may share

training data, architectures, prompting strategies, systematic misconceptions.

Their errors can therefore remain correlated even after conditioning on the true label.
Under exact TCI, Equation (J.15) predicts the same ratio for every triplet.
This gives an observable diagnostic of TCI violation.
Let

$$\rho_{ijk} = \frac{\hat{T}_{ijk}}{\sqrt{|\hat{R}_{ij}\hat{R}_{ik}\hat{R}_{jk}|}}. \tag{J.20}$$

For suitably oriented nondegenerate triplets, define

$$\tau(V) = \mathrm{Var}_{i,j,k}\left[\rho_{ijk}\right]. \tag{J.21}$$

Under the ideal population model,

$$\tau(V) = 0.$$

Large empirical values indicate disagreement with the TCI factorization.

J.8 FUSE: Transforming Scores to Improve the Moment Model

Real verifiers often produce continuous scores

$$S_i(X) \in \mathbb{R}$$

rather than binary decisions.
FUSE [88] exploits the fact that these scores can be transformed before applying the moment estimator.
Let

$$V_i = g_i(S_i)$$

for a family of admissible transformations g_i, such as thresholding or other monotone transformations.
The transformations are chosen to make the observable moments more consistent with the TCI factorization, schematically by seeking

$$g^\star = \arg\min_g \tau\left(V^{(g)}\right), \tag{J.22}$$

subject to suitable nondegeneracy constraints.
The important distinction is:

minimizing τ does not prove that TCI is true.

It finds a representation of the verifier scores whose observable low-order moments are *more consistent* with the conditional independence model.
Thus there are two different levels of claim:

Population theorem:	identifiability under TCI and orientation assumptions,
Practical FUSE procedure:	transform the scores to reduce an observable measure of TCI violation before appl

J.9 From Verifier Reliabilities to Pseudo-Labels

Suppose the parameters

$$\pi, \qquad \alpha_i, \qquad \beta_i$$

have been estimated.

For a triplet of approximately conditionally independent verifiers i, j, k, Bayes' rule gives

$$P\left(Y=+1 \mid V_i, V_j, V_k\right)=\frac{\pi \prod_{m\in\{i,j,k\}} P(V_m \mid Y=+1)}{\sum_{y\in\{-1,+1\}} P(Y=y) \prod_{m\in\{i,j,k\}} P(V_m \mid Y=y)}. \tag{J.23}$$

These posterior probabilities provide soft pseudo-labels

$$\hat{p}_n \approx P(Y_n=+1 \mid V(X_n)).$$

Multiple valid verifier triplets may be aggregated to obtain a more stable estimate.

The resulting pseudo-labels can then supervise a flexible ensemble model

$$q_\theta\left(Y=+1 \mid S_1, \ldots, S_m\right).$$

For example, one may minimize the soft-label cross-entropy

$$\mathcal{L}(\theta)=-\frac{1}{N} \sum_{n=1}^{N}\left[\hat{p}_n \log q_\theta(Y_n=+1 \mid S_n)+\left(1-\hat{p}_n\right) \log q_\theta(Y_n=-1 \mid S_n)\right]. \tag{J.24}$$

The final ensemble can then rank or select candidate responses according to its estimated probability of correctness.

J.10 What Makes Unsupervised Verification Possible?

At first sight, recovering correctness without any labeled examples may appear paradoxical.

The resolution is that the labels are not reconstructed from nothing.

They leave statistical footprints in the pattern of verifier agreement.

Under TCI,

$$Y$$

induces the observable covariance structure

$$R_{ij}=4\pi(1-\pi)a_i a_j.$$

Thus two competent verifiers tend to agree not merely because they are similar, but because both respond to the same hidden variable.

The third-order moment provides additional information about class imbalance and orientation.

The hidden variable is therefore indirectly visible through low-order statistics.

Schematically,

latent correctness $\rightarrow$ correlated verifier outputs $\rightarrow$ observable moments $\rightarrow$ estimated verifier quality.

This is the source of identifiability.

J.11 Failure Modes

The method also clarifies when unsupervised verification cannot be expected to work.

J.11.1 Common-mode errors

Suppose all verifiers make the same mistake on the same examples.

Then agreement may be extremely high even when the ensemble is wrong.

Consensus alone does not imply truth.

If

$$V_1 = \cdots = V_m$$

almost surely, the ensemble provides little additional information beyond a single verifier.

J.11.2 Weak verifiers

If

$$a_i \approx 0$$

for many verifiers, then

$$R_{ij} \approx 0$$

and the observable footprint of correctness becomes weak.

Finite-sample estimation then becomes difficult.

J.11.3 Strong conditional dependence

If verifier errors remain correlated after conditioning on Y, the rank-one factorizations

$$R_{ij} \propto a_i a_j$$

and

$$T_{ijk} \propto a_i a_j a_k$$

are violated.

A small value of the empirical TCI statistic can provide evidence in favor of the approximation, but does not establish its truth.

J.11.4 Insufficient diversity

Adding many nearly identical verifiers does not provide the same information as adding genuinely diverse ones.

The relevant resource is not simply ensemble size,

$$m,$$

but diversity of error structure.

J.11.5 Finite-sample noise

The moments

$$R_{ij} \quad \text{and} \quad T_{ijk}$$

must themselves be estimated from finite data.

Third-order moments are particularly noisy.

Population identifiability therefore does not imply accurate estimation from a small number of responses.

J.11.6 Orientation failure

Without some assumption that fixes which latent class corresponds to correctness, the global label orientation remains ambiguous.

No unsupervised method can remove this symmetry using the verifier outputs alone.

J.12 Zero Labels Do Not Mean Truth from Consensus

The preceding failure modes make an important conceptual point.

FUSE does not establish the general principle

$$\text{many agreeing models} \Longrightarrow \text{truth}.$$

What it exploits is a much more specific statistical model:

diverse imperfect verifiers + structured conditional independence + nonzero individual skill → identifiable late

If the structure disappears, so does the guarantee.

Thus

agreement is informative only when the mechanism generating agreement is sufficiently constrained.

J.13 Relation to Genericity

The construction provides a useful small-scale example of the representation-versus-recovery distinction developed elsewhere in this book.

The target variable

$$Y$$

is never observed directly.

Nevertheless, under TCI and nonzero verifier skill, it leaves detectable low-order footprints:

$$R_{ij} \neq 0, \qquad T_{ijk} \neq 0$$

generically away from degenerate parameter choices.

These moments make recovery possible.

In this sense, the FUSE setting illustrates the genericity principle:

a hidden variable is learnable only if it leaves an observable statistical footprint in the data available to the learn

If every verifier is uninformative,

$$a_i = 0,$$

the footprints vanish.

If all errors share an unconstrained common cause, the footprints can become ambiguous.

Thus the assumptions responsible for identifiability play the same role that genericity plays elsewhere in the book: they expose the hidden structure to a feasible estimator.

J.14 Relation to Generative AI

Verifier ensembling is relevant whenever generation is easier to scale than reliable evaluation.

A generative model may produce candidates

$$X^{(1)}, \ldots, X^{(K)},$$

while several heterogeneous procedures evaluate them:

$$V_1, \ldots, V_m.$$

These verifiers might use different models, prompts, tools, tests, or specialized checks.

If their errors contain sufficiently independent information, an unsupervised ensemble can potentially provide a stronger selection signal than any individual verifier.

The important architectural principle is therefore not

generation requires a perfect external oracle,

but rather the weaker possibility

several imperfect and diverse evaluation signals may jointly reveal a useful estimate of latent correctness.

This does not eliminate the need for external validation.

It provides a mechanism by which the cost of such validation may be reduced when the structural assumptions are approximately satisfied.

J.15 Conclusion

Unsupervised verifier ensembling asks a sharply defined question:

Can latent correctness be inferred from agreement among imperfect judges without observing any correctness la

Under Triplet Conditional Independence and suitable orientation and nondegeneracy assumptions, the answer is yes at the population level.

Pairwise and third-order moments factorize as

$$R_{ij} = 4\pi(1-\pi)a_i a_j$$

and

$$T_{ijk} = 8\pi(1-\pi)(1-2\pi)a_i a_j a_k,$$

allowing the latent class balance and individual verifier reliabilities to be recovered from observable statistics.

FUSE extends this idea to realistic verifier scores by searching for transformations whose low-order moments more nearly satisfy the conditional-independence structure before constructing pseudo-labels and an ensemble predictor.

The broader lesson is not that truth can be obtained from consensus alone.

It is that latent truth may be statistically recoverable when it leaves structured, detectable traces across multiple imperfect observers.

In the language used throughout this book,

no supervision is required only when the observable data already contain enough structure to substitute for it.

CHAPTER K

Sparse Structure Does Not Eliminate NP-Hardness: Worst-Case and Distributional Tractability

In his Nobel lecture, Demis Hassabis suggested an important empirical possibility: problems classified as computationally hard in the worst case may often be substantially easier on the instances that arise in practice.

This observation should not be interpreted as claiming that NP-hard problems are generally polynomial-time solvable, or that hard instances occupy only negligible regions of instance space.

A more precise hypothesis is:

many computationally hard problem classes may become tractable to useful accuracy on the structured instance distributio

This appendix examines that possibility and, at the same time, states an important limitation of the sparse-compositionality framework developed in this book.

Sparse compositional structure can make a function efficient to evaluate*. It does not, by itself, make the corresponding global search or optimization problem easy.*

K.1 The Fundamental Distinction: Evaluation versus Search

Consider a Boolean formula

$$F(x_1, \ldots, x_n) = \bigwedge_{j=1}^{m} C_j(x_{j_1}, x_{j_2}, x_{j_3}),$$

where each C_j is a clause involving at most three variables.

The function

$$x \longmapsto F(x)$$

is compositionally sparse in an extremely strong sense.

Each local constituent has bounded arity:

$$C_j : \{0,1\}^3 \to \{0,1\},$$

and the final result is obtained by combining the local clause values.

Evaluating $F(x)$ for a given assignment x is therefore efficient:

$$T_{\text{eval}} = O(m).$$

Yet the search problem

$$\text{find } x \text{ such that } F(x) = 1$$

is the canonical NP-complete problem 3-SAT.

Thus

$$\boxed{\text{bounded-arity compositional structure} \not\Longrightarrow \text{tractable global search.}}$$

This example is important for the theory developed throughout this book.

Sparse compositionality provides a structural explanation for why a map can be represented and evaluated without an exponential dependence on ambient dimension.

It does not imply that one can efficiently invert, optimize, or satisfy that map.

More generally,

$$\boxed{\text{efficient evaluation of } f(x) \not\Longrightarrow \text{efficient computation of } \arg\min_x f(x).}$$

The distinction is analogous to the separation made earlier between representation and recovery.

A compact representation of a solution space does not automatically provide an efficient algorithm for locating a desired point within it.

K.2 Worst-Case Complexity

Let

$$T(A, x)$$

denote the running time of algorithm A on instance x.

Worst-case complexity considers

$$T_{\text{worst}}(n) = \max_{|x|=n} T(A, x). \tag{K.1}$$

If an NP-complete problem has a deterministic algorithm satisfying

$$T_{\text{worst}}(n) = \text{poly}(n),$$

then

$$\text{P} = \text{NP}.$$

Thus, assuming

$$\text{P} \neq \text{NP},$$

there is no polynomial-time algorithm that correctly solves *every* instance of every NP-complete problem.

This statement is about an entire problem class.

It does not imply that every instance is difficult.

Nor does it imply that a particular application generates difficult instances frequently.

This distinction is essential.

The practical performance of an algorithm deployed in the world depends not only on the mathematical problem class but also on the distribution

$$\mathcal{D}_{\text{deploy}}$$

from which its instances are drawn.

K.3 Distributional Tractability

Suppose instances are repeatedly sampled from a deployment distribution

$$x \sim \mathcal{D}_{\text{deploy}}.$$

Then a relevant quantity may be

$$T_{\mathcal{D}}(A, n) = \mathbb{E}_{x \sim \mathcal{D}_{\text{deploy}}} \left[T(A, x) \mid |x| = n \right], \tag{K.2}$$

rather than the worst-case quantity in Equation (K.1).

It is entirely consistent to have

$$T_{\text{worst}}(n)$$

grow exponentially while

$$T_{\mathcal{D}}(A, n)$$

remains modest over the distributions encountered in practice.

The important point is not that the NP-hard problem has disappeared.

It is that

the deployment distribution may occupy a highly restricted region of the full combinatorial instance space.

A specialized solver can exploit the regularities of that region without solving the worst-case problem.

K.4 A More Precise Practical Hypothesis

The hypothesis suggested by practical experience can therefore be stated as follows.

> **Distributional Tractability Hypothesis.** For many important NP-hard problem classes and practical deployment environments, the distribution of encountered instances possesses additional structure that permits efficient exact solution, efficient approximation, or reliable heuristic solution on most of the probability mass of that deployment distribution.

This statement is deliberately weaker than

"NP-hard problems are usually easy."

It makes tractability relative to

$$\boxed{\text{problem class} + \text{instance distribution} + \text{required solution quality.}}$$

All three matter.

K.5 Exact Solution versus Useful Approximation

Another source of apparent tension between complexity theory and practice is that practical applications often do not require exact global optimality.

Suppose the mathematical objective is

$$f^\star(x) = \min_{y \in \mathcal{Y}(x)} C(x, y).$$

Worst-case complexity may concern finding the exact minimizer

$$y^\star.$$

An application may instead require

$$C(x, \hat{y}) \leq (1 + \varepsilon) C(x, y^\star),$$

or merely a solution satisfying a domain-specific performance threshold.

These are different computational problems.

For some NP-hard optimization problems, useful approximation is polynomial-time even though exact optimization is not known to be.

For others, approximation itself remains hard.

Thus practical success may arise from relaxing

$$\boxed{\text{exact optimum}}$$

to

$$\boxed{\text{sufficiently good solution.}}$$

K.6 Smoothed Analysis

One formal attempt to bridge worst-case and practical behavior is *smoothed analysis*.

Rather than evaluating an algorithm only on an adversarial instance x, one first perturbs that instance slightly.

Schematically,

$$\tilde{x} = x + \xi,$$

where

$$\xi \sim \mathcal{N}(0, \sigma^2 I).$$

The smoothed complexity is then of the form

$$C_{\text{smooth}}(n, \sigma) = \max_{x \in \mathcal{X}_n} \mathbb{E}_\xi \left[T(A, x + \xi) \right]. \tag{K.3}$$

This framework was introduced by Spielman and Teng to explain, among other examples, why the simplex algorithm can perform extremely well in practice despite exponential worst-case behavior.

Smoothed analysis captures an important possibility:

some pathological worst-case configurations are unstable to small perturbations.

If the bad configuration occupies a narrow region of instance space, small random perturbations may move the problem into a regime where the algorithm behaves efficiently.

But this mechanism is not universal.

K.7 What Smoothed Analysis Does Not Show

One should not infer from successful smoothed analyses that

random perturbation generally destroys computational hardness.

Smoothed polynomiality must be proved for a particular problem, algorithm, and perturbation model.

Some computationally hard structures are robust.

Random instances can themselves be difficult.

And in discrete problems there may be no natural infinitesimal perturbation corresponding to Gaussian noise in a continuous parameter space.

Thus

smoothed tractability is one mechanism for practical success, not a general theorem about NP-hardness.

K.8 Discrete Problems and Robust Hardness

The limitation becomes particularly clear in discrete combinatorial problems.

Consider random K-SAT.

The clause-to-variable ratio

$$\alpha = \frac{m}{n}$$

controls the qualitative character of random formulas.

For random 3-SAT, a sharp satisfiability transition occurs near a critical ratio around

$$\alpha_c \approx 4.27.$$

The relation between this phase transition and algorithmic difficulty is subtle, but the example makes an important conceptual point:

hardness need not correspond to one isolated adversarial input.

Large families of random instances can exhibit difficult search structure.

Therefore practical tractability cannot generally be explained by the claim that the real world simply adds noise to a fragile worst-case instance.

K.9 Where Practical Tractability Can Come From

If practical instance distributions are easier than the worst case, some additional structure must be responsible.

Several mechanisms are possible.

K.9.1 Low treewidth or related graph structure

Many combinatorial problems become tractable when their interaction graphs have bounded treewidth.

A problem that is NP-hard on arbitrary graphs may admit dynamic programming algorithms of the schematic form

$$T(n, w) = n \, \exp(O(w)),$$

where w is treewidth.

If

$$w = O(1),$$

the algorithm becomes polynomial in n.

Thus a restricted deployment distribution concentrated on low-treewidth instances can be easy even though the unrestricted problem class is NP-hard.

K.9.2 Small backdoor sets

Some instances contain a small collection of variables such that, once those variables are assigned, the remaining problem belongs to a tractable class.

If a backdoor set has size

$$b,$$

one may search over

$$2^b$$

assignments rather than

$$2^n.$$

When

$$b \ll n,$$

this can change the practical character of the problem dramatically.

K.9.3 Repeated motifs

Deployment environments frequently reuse the same components and constraint patterns.

A solver may exploit repeated local motifs rather than treating every instance as entirely novel.

K.9.4 Restricted effective dimension

Although an input may have nominal dimension

$$n,$$

the instances generated by a particular environment may vary along far fewer effective degrees of freedom.

A specialized algorithm can exploit this restriction.

K.9.5 Approximation tolerance

The application may permit near-optimal rather than exact solutions.

This can transform an intractable optimization requirement into a useful polynomial-time approximation problem.

K.9.6 Stable heuristics

A heuristic need not solve every mathematical instance.

It need only perform reliably over

$$\mathcal{D}_{\text{deploy}}.$$

This can be enough for practical systems, especially when failures are detectable.

K.9.7 Verification and fallback

A particularly useful architecture is

$$\boxed{\text{fast specialized solver} + \text{verification} + \text{fallback}.}$$

The fast solver handles the common structured cases.

A verifier detects failure or low confidence.

Exceptional instances are then passed to a slower algorithm, human operator, or more expensive search procedure.

This design can produce high practical reliability without requiring the fast algorithm to solve every instance efficiently.

K.10 Learning from Deployment Traces

The preceding perspective suggests a natural role for machine learning.

Suppose an application repeatedly generates

$$x_1, x_2, \ldots \sim \mathcal{D}_{\text{deploy}}.$$

Instead of searching for one universal algorithm that performs well on all mathematically possible instances, one can learn from execution traces of the actual environment.

Let

$$\tau_i = (x_i, \text{solver decisions}, \text{search trajectory}, \text{outcome})$$

denote an execution trace.

A learned policy can exploit regularities in

$$\{\tau_i\}$$

to predict

branching decisions, variable orderings, decompositions, promising subproblems, algorithm se

The relevant learning problem is therefore not

$$\boxed{\text{solve the NP-hard problem universally,}}$$

but

$$\boxed{\text{learn a good algorithmic policy for the observed distribution.}}$$

This distinction is closely related to the distinction between a universal computational problem and a learned world model specialized to a particular environment.

K.11 SAT Solvers as an Example

Modern SAT solvers illustrate this philosophy.

Conflict-Driven Clause Learning, branching heuristics, restart policies, learned clauses, preprocessing, and decomposition methods can dramatically reduce practical search.

Their success should not be interpreted as proving that encountered SAT instances necessarily possess low treewidth or small backdoor sets.

Rather, these structural notions are among several possible explanations for why particular families of instances admit effective search.

The important point is

$$\boxed{\text{the solver dynamically discovers structure that allows large parts of the nominal search space to be discarded.}}$$

A SAT instance may formally contain

$$2^n$$

possible assignments.

A successful solver explores only a tiny adaptive subset.

K.12 Sparse Compositionality Helps—but Is Not Enough

It is tempting to attribute practical tractability entirely to sparse compositionality.

This would be incorrect.

As 3-SAT demonstrates, bounded local interactions can coexist with NP-complete global search.

The issue is that local constraints can interact globally through shared variables.

A problem may therefore satisfy

$$\boxed{\text{local arity} = O(1)}$$

while possessing a global constraint graph whose effective search complexity grows rapidly with problem size.

Sparse compositionality remains useful because it limits the complexity of local computation.

But an additional property is required to make global search tractable.

Such properties may include

bounded treewidth, small separators, small backdoors, restricted instance distributions, good a

or other forms of exploitable structure.

Thus

$$\boxed{\text{sparse compositionality is a condition on representation;}}$$

whereas

$$\boxed{\text{tractable search requires additional structure in the solution space or instance distribution.}}$$

K.13 A Connection to Genericity

The word "genericity" appears in two related but distinct senses in this book.

Earlier chapters considered statistical genericity of a learning problem: whether useful constituents leave detectable low-complexity footprints in the observations available to the learner.

For NP-hard search, the corresponding practical question is distributional:

$$\boxed{\text{does the deployment distribution repeatedly expose exploitable structure to the solver?}}$$

Suppose an algorithm learns a branching rule

$$a_t = \pi_\theta(s_t)$$

from previous search traces.

If the deployment distribution repeatedly produces similar structural bottlenecks, the rule can generalize.

If every new instance presents an unrelated adversarial structure, no such reuse is possible.

Thus the analogue of genericity is not that every hard problem becomes easy.

It is that the deployment distribution contains recurring statistical regularities that make useful algorithmic decisions predictable.

K.14 The Analogy with Optimization in Deep Learning

There is a useful parallel with the optimization problem studied earlier in this book.

The worst-case theory of nonconvex optimization permits highly pathological landscapes.

Yet gradient-based methods perform remarkably well on many learning problems encountered in practice.

Likewise, worst-case complexity permits extremely difficult instances of NP-hard problems, while specialized solvers may succeed on many deployment distributions.

In neither case does empirical success invalidate worst-case theory.

Instead, the relevant question becomes

what structural properties distinguish the instances generated by the real process from arbitrary worst-case inst

For deep learning, this book proposes sparse compositionality and genericity as important parts of the answer.

For combinatorial search, sparse compositionality alone is insufficient.

The additional source of tractability must lie in the structure of the instance distribution and search space.

K.15 A Hierarchy of Claims

It is useful to distinguish several increasingly strong statements.

Claim 1:	some NP-hard instances are easy;
Claim 2:	some important distributions place most probability on easy instances;
Claim 3:	the relevant structure of those distributions can be learned;
Claim 4:	most natural instances of NP-hard problems are easy;
Claim 5:	NP-hardness is largely irrelevant in practice.

The first statement is trivial.

The second and third are substantive and empirically plausible in many domains.

The fourth is a broad conjecture whose truth depends critically on what is meant by "natural."

The fifth is false as a general principle.

The practical hypothesis considered here concerns primarily Claims 2 and 3.

K.16 What Would Make the Hypothesis Testable?

The distributional-tractability hypothesis can be studied experimentally.

For a fixed NP-hard problem family:

1. collect instances from a genuine deployment environment;
2. measure the empirical distribution of solver runtimes and solution quality;
3. compare with random and adversarially constructed instances of the same nominal size;

4. measure structural quantities such as graph width, separator size, backdoor size, motif reuse, and effective dimension;

5. train solver policies on execution traces from one subset of the deployment distribution;

6. test whether those policies generalize to new instances from the same environment;

7. deliberately move the test distribution away from the deployment distribution and measure where the learned advantage disappears.

The most interesting result would not be that a learned solver succeeds on a benchmark.
It would be the discovery of a measurable structural variable

$$S(x)$$

such that solver complexity depends strongly on $S(x)$ but only weakly on nominal input size once S is controlled.

This would turn the informal idea

"real instances are easier"

into a quantitative theory of

which real instances are easier and why.

K.17 Implications for Practical AI

The practical lesson is not to ignore NP-hardness.

It is to design systems around the difference between worst-case and deployment complexity.

A robust architecture may combine

learned specialized solver + cheap verification or monitoring + fallback search + continual adaptation from traces.

Such a system does not claim to solve an NP-hard problem in polynomial time in the worst case.

It attempts to exploit the structured distribution that the environment actually generates while retaining protection against exceptional instances.

This may be especially relevant for artificial agents operating for long periods in one environment.

The longer the agent remains in the same structured world, the more opportunity it has to learn

the algorithmic regularities of that world.

K.18 Conclusion

Worst-case NP-hardness and practical tractability are not contradictory.

Worst-case complexity asks whether an algorithm can solve every instance of a class efficiently.

Practical computation asks whether the instances encountered in a particular environment possess additional exploitable structure.

The distinction can be summarized as

NP-hard problem class
+arbitrary instances
$\not\Rightarrow$
efficient universal solver,

but

NP-hard problem class
+structured deployment distribution
+approximation or verification
+specialized algorithms
$\Longrightarrow$
potential practical tractability.

Most importantly for the theory of this book,

sparse compositionality does not eliminate NP-hardness.

A function may have a polynomial-size bounded-fan-in representation and still define a computationally difficult global search problem.

The canonical example is 3-SAT:

local 3-variable constraints + global combinatorial coupling = NP-complete search.

Sparse compositionality explains how local computations can be evaluated efficiently.

Additional structure is required to explain why global search may also be easy on a particular distribution.

The conjecture suggested by practical experience—and by Hassabis' observation—is therefore best stated not as

NP-hard problems are usually easy,

but as

the worlds in which intelligent systems repeatedly operate may generate restricted families of instances whose l

That is compatible with worst-case complexity theory, experimentally testable, and directly connected to the broader theme of this book: intelligence succeeds not by defeating combinatorial complexity in the abstract, but by discovering and exploiting the structure of the world that actually generates its problems.

CHAPTER L

A Johnson–Lindenstrauss Lemma for Sampled Group-Orbit Representations

Invariant representations built from group orbits can be high-dimensional even when the geometry relevant for recognition is comparatively simple. This appendix asks whether such representations can be compressed by a single random linear projection without destroying their invariant geometry.

For a finite dataset, a finite collection of templates, and a finite sample of transformations drawn from Haar measure, the answer is yes. A standard subgaussian Johnson–Lindenstrauss map of dimension

$$m = O\Big(\varepsilon^{-2}\left[\log|S| + \log k + \log M + \log(1/\delta)\right]\Big)$$

preserves the resulting group-orbit distance up to additive $O(\varepsilon)$ distortion.
The proof separates two independent approximations:

$$\boxed{\textit{continuous group orbit} \to \textit{finite Haar sample} \to \textit{random projection}.}$$

The first is controlled by empirical-distribution concentration. The second follows from the ordinary Johnson–Lindenstrauss lemma and a coupling argument in Wasserstein distance.

L.1 Group-Orbit Representations

Let

$$G \subset O(d)$$

be a compact group acting orthogonally on

$$\mathbb{R}^d,$$

and let dg denote normalized Haar probability measure on G.
Let

$$S = \{I_1, \dots, I_N\} \subset \mathbb{R}^d$$

be a finite dataset, and define

$$R_S = \max_{I \in S} \|I\|_2.$$

Let

$$T_k = \{t_1, \dots, t_k\} \subset \mathbb{S}^{d-1}$$

be a finite collection of unit-norm templates.

For a signal I and template t, consider the scalar random variable

$$X_{I,t}(g) = \langle I, gt \rangle, \qquad g \sim \mathrm{Haar}(G). \tag{L.1}$$

Its probability law will be denoted by

$$\nu_{I,t} = \mathrm{Law}_{g \sim G} \left(\langle I, gt \rangle \right). \tag{L.2}$$

Equivalently, its cumulative distribution function is

$$F_{I,t}(b) = \int_G \mathbf{1} \left\{ \langle I, gt \rangle \le b \right\} \, dg. \tag{L.3}$$

This is the group-averaged one-dimensional tomographic representation used in earlier work on invariant and selective representations [6, 130].

L.2 Why the Orbit Distribution Is Invariant

The basic invariance property follows directly from Haar measure.

Proposition 66 (Group invariance)**.** *For any $h \in G$,*

$$\nu_{hI,t} = \nu_{I,t}. \tag{L.4}$$

Proof. Since the action is orthogonal,

$$\langle hI, gt \rangle = \langle I, h^{-1} gt \rangle.$$

If

$$g \sim \mathrm{Haar}(G),$$

then by left invariance of Haar measure,

$$h^{-1} g \sim \mathrm{Haar}(G).$$

Therefore

$$\mathrm{Law} \left(\langle hI, gt \rangle \right) = \mathrm{Law} \left(\langle I, gt \rangle \right).$$

□

Thus the representation is constant on group orbits.

The question considered here is whether the geometry induced by these orbit distributions survives dimensionality reduction.

L.3 Why We Use Wasserstein Distance

The original group representation can be compared using distances between the corresponding one-dimensional probability distributions.

For the present purpose the natural choice is the 1-Wasserstein distance,

$$W_1(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)} \int |x-y|\, d\pi(x,y), \tag{L.5}$$

where $\Pi(\mu,\nu)$ is the set of couplings of μ and ν.

In one dimension,

$$W_1(\mu,\nu) = \int_{\mathbb{R}} |F_\mu(b) - F_\nu(b)|\, db. \tag{L.6}$$

Thus the representation remains directly related to its cumulative distribution functions.

The crucial advantage is stability.

Lemma 15 (Coupling stability)**.** *If random variables X and $\widetilde{X}$ are defined on the same probability space, then*

$$W_1\left(\mathcal{L}(X), \mathcal{L}(\widetilde{X})\right) \le \mathbb{E}|X-\widetilde{X}|. \tag{L.7}$$

In particular, if

$$|X-\widetilde{X}| \le \Delta$$

almost surely, then

$$W_1\left(\mathcal{L}(X), \mathcal{L}(\widetilde{X})\right) \le \Delta. \tag{L.8}$$

This is precisely the stability property needed below.

A small displacement of every orbit measurement produces a small Wasserstein displacement of the entire orbit distribution.

L.4 The Population Invariant Distance

For the finite template pool T_k, define

$$d_G(I,I') = \frac{1}{k}\sum_{i=1}^{k} W_1\left(\nu_{I,t_i}, \nu_{I',t_i}\right). \tag{L.9}$$

Because each ν_{I,t_i} is group invariant,

$$d_G(hI, h'I') = d_G(I,I') \qquad \forall h,h' \in G. \tag{L.10}$$

Thus d_G is naturally a distance or pseudodistance on equivalence classes generated by the group action.

Selectivity—whether distinct orbits are always separated—depends on the richness of the template family and is not proved in this appendix.

Our question is narrower:

can d_G be approximated after strong random dimensionality reduction?

L.5 Sampling the Group Orbit

Instead of approximating Haar measure by an arbitrary geometric net, draw

$$g_1, \dots, g_M \overset{\text{iid}}{\sim} \text{Haar}(G). \tag{L.11}$$

For each signal and template define the empirical orbit measure

$$\nu_{I,t}^{(M)} = \frac{1}{M} \sum_{j=1}^{M} \delta_{\langle I, g_j t \rangle}. \tag{L.12}$$

The corresponding empirical group-orbit distance is

$$d_M(I, I') = \frac{1}{k} \sum_{i=1}^{k} W_1 \left(\nu_{I,t_i}^{(M)}, \nu_{I',t_i}^{(M)} \right). \tag{L.13}$$

This separates the approximation of Haar integration from the subsequent dimensionality reduction.

L.6 Uniform Approximation of Haar Orbit Distributions

Because

$$\|t_i\|_2 = 1$$

and G acts orthogonally,

$$|\langle I, g t_i \rangle| \le \|I\|_2 \le R_S.$$

Thus all orbit distributions are supported on

$$[-R_S, R_S].$$

For each fixed (I, t_i), the Dvoretzky–Kiefer–Wolfowitz inequality gives

$$P\left(\sup_b |F_{I,t_i}^{(M)}(b) - F_{I,t_i}(b)| > a \right) \le 2e^{-2Ma^2}. \tag{L.14}$$

Applying a union bound over all

$$|S|k$$

signal-template pairs yields the following.

Proposition 67 (Uniform Haar-sampling error)**.** *With probability at least $1 - \delta$ over the sampled group elements,*

$$W_1 \left(\nu_{I,t_i}^{(M)}, \nu_{I,t_i} \right) \le 2R_S \sqrt{\frac{\log(2|S|k/\delta)}{2M}} \tag{L.15}$$

simultaneously for every

$$I \in S, \qquad 1 \le i \le k.$$

Consequently,

$$|d_M(I, I') - d_G(I, I')| \leq 4R_S\sqrt{\frac{\log(2|S|k/\delta)}{2M}} \tag{L.16}$$

for every pair $I, I' \in S$.

Proof. Equation (L.14) and a union bound give

$$\sup_b |F^{(M)}_{I,t_i}(b) - F_{I,t_i}(b)| \leq \sqrt{\frac{\log(2|S|k/\delta)}{2M}}$$

simultaneously over all signal-template pairs.

Because the distributions are supported on an interval of length $2R_S$, Equation (L.6) implies

$$W_1\left(\nu^{(M)}_{I,t_i}, \nu_{I,t_i}\right) \leq 2R_S \sup_b |F^{(M)}_{I,t_i}(b) - F_{I,t_i}(b)|.$$

The pairwise distance bound follows from the triangle inequality for W_1 and averaging over templates. □

Thus the number of sampled transformations required to approximate the continuous invariant representation grows only logarithmically with the number of signal-template pairs for fixed accuracy.

L.7 Random Projection of the Sampled Orbit

Now draw a subgaussian random matrix

$$R \in \mathbb{R}^{m\times d},$$

normalized so that

$$\mathbb{E}\|Rx\|_2^2 = \|x\|_2^2.$$

For the sampled group elements, define

$$\nu^{R,(M)}_{I,t} = \frac{1}{M}\sum_{j=1}^{M} \delta_{\langle RI, R(g_j t)\rangle}. \tag{L.17}$$

The corresponding projected orbit distance is

$$d^R_M(I, I') = \frac{1}{k}\sum_{i=1}^{k} W_1\left(\nu^{R,(M)}_{I,t_i}, \nu^{R,(M)}_{I',t_i}\right). \tag{L.18}$$

Notice that R need not commute with the group action.

Therefore the projected representation is not claimed to be exactly group invariant on all of $\mathbb{R}^d$.

The theorem states instead that, on the finite family of orbit measurements under consideration, its geometry approximates the group-invariant geometry of the original representation.

L.8 The Finite JL Control Set

For each

$$I \in S, \qquad t_i \in T_k, \qquad g_j,$$

define

$$u_{ij} = g_j t_i.$$

Consider the finite control set

$$\mathcal{V} = \{I + u_{ij}, I - u_{ij} : I \in S,\, 1 \le i \le k,\, 1 \le j \le M\}. \tag{L.19}$$

Its cardinality satisfies

$$|\mathcal{V}| \le 2|S|kM. \tag{L.20}$$

A standard subgaussian JL concentration argument implies that if

$$m \ge C\varepsilon^{-2}\left[\log(2|S|kM) + \log(1/\delta)\right], \tag{L.21}$$

then with probability at least $1 - \delta$,

$$(1-\varepsilon)\|v\|_2^2 \le \|Rv\|_2^2 \le (1+\varepsilon)\|v\|_2^2 \tag{L.22}$$

simultaneously for every

$$v \in \mathcal{V}.$$

No claim is made that R is a near-isometry on the entire span of $\mathcal{V}$.
Only the finite vectors required for the polarization argument need to be controlled.

L.9 Preservation of Orbit Inner Products

The polarization identity gives

$$\langle RI, Ru\rangle = \frac{1}{4}\left(\|R(I+u)\|_2^2 - \|R(I-u)\|_2^2\right), \tag{L.23}$$

$$\langle I, u\rangle = \frac{1}{4}\left(\|I+u\|_2^2 - \|I-u\|_2^2\right). \tag{L.24}$$

Therefore Equation (L.22) implies

$$\begin{aligned} &|\langle RI, Ru\rangle - \langle I, u\rangle| \\ &\quad\le \frac{\varepsilon}{4}\left(\|I+u\|_2^2 + \|I-u\|_2^2\right) \\ &\quad= \frac{\varepsilon}{2}\left(\|I\|_2^2 + \|u\|_2^2\right). \end{aligned} \tag{L.25}$$

Since

$$u = g_j t_i$$

and the group action is orthogonal,

$$\|u\|_2 = 1.$$

Hence

$$|\langle RI, R(g_j t_i)\rangle - \langle I, g_j t_i\rangle| \leq \frac{\varepsilon}{2}\left(\|I\|_2^2 + 1\right). \tag{L.26}$$

This is the central JL estimate.

L.10 Wasserstein Stability after Projection

Couple the original and projected empirical orbit distributions by matching the same sampled transformation g_j before and after projection.

Lemma 15 and Equation (L.26) then give

$$W_1\left(\nu_{I,t_i}^{R,(M)}, \nu_{I,t_i}^{(M)}\right) \leq \frac{\varepsilon}{2}\left(\|I\|_2^2 + 1\right). \tag{L.27}$$

For two signals I, I', the triangle inequality implies

$$\begin{aligned}
&\left|W_1\left(\nu_{I,t_i}^{R,(M)}, \nu_{I',t_i}^{R,(M)}\right) - W_1\left(\nu_{I,t_i}^{(M)}, \nu_{I',t_i}^{(M)}\right)\right| \\
&\quad \leq W_1\left(\nu_{I,t_i}^{R,(M)}, \nu_{I,t_i}^{(M)}\right) + W_1\left(\nu_{I',t_i}^{R,(M)}, \nu_{I',t_i}^{(M)}\right) \\
&\quad \leq \frac{\varepsilon}{2}\left(\|I\|_2^2 + \|I'\|_2^2 + 2\right).
\end{aligned} \tag{L.28}$$

Since

$$\|I\|_2, \|I'\|_2 \leq R_S,$$

we obtain

$$\left|d_M^R(I, I') - d_M(I, I')\right| \leq \varepsilon\left(R_S^2 + 1\right). \tag{L.29}$$

L.11 Main Theorem

We can now combine group sampling and random projection.

Theorem 42 (JL stability of sampled group-orbit representations)**.** *Let*

$$S \subset \mathbb{R}^d$$

be a finite dataset with

$$R_S = \max_{I \in S} \|I\|_2.$$

Let

$$G \subset O(d)$$

be a compact group with normalized Haar measure, and let

$$T_k = \{t_1, \dots, t_k\} \subset \mathbb{S}^{d-1}.$$

Draw

$$g_1, \dots, g_M \overset{\text{iid}}{\sim} \text{Haar}(G),$$

and independently draw a normalized subgaussian random matrix

$$R \in \mathbb{R}^{m \times d}.$$

Let d_G denote the population invariant distance defined in Equation (L.9)*, and let d_M^R denote the randomly projected empirical distance defined in Equation* (L.18)*.*

If

$$m \geq C\varepsilon^{-2} \log\left(\frac{4|S|kM}{\delta}\right), \tag{L.30}$$

then with probability at least $1 - \delta$ over both the Haar samples and the random projection,

$$\boxed{\left|d_M^R(I, I') - d_G(I, I')\right| \leq \varepsilon\left(R_S^2 + 1\right) + 4R_S\sqrt{\frac{\log(4|S|k/\delta)}{2M}}} \tag{L.31}$$

simultaneously for every pair

$$I, I' \in S.$$

Proof. Allocate failure probability $\delta/2$ to Haar sampling and $\delta/2$ to the random projection.

By Proposition 67,

$$|d_M(I, I') - d_G(I, I')| \leq 4R_S\sqrt{\frac{\log(4|S|k/\delta)}{2M}}$$

simultaneously for all dataset pairs with probability at least $1 - \delta/2$.

Conditioned on the sampled group elements, the control set Equation (L.19) contains at most

$$2|S|kM$$

vectors.

The standard subgaussian JL bound therefore gives, under Equation (L.30),

$$|d_M^R(I, I') - d_M(I, I')| \leq \varepsilon(R_S^2 + 1)$$

simultaneously for all pairs with probability at least $1 - \delta/2$.

Finally,

$$|d_M^R(I, I') - d_G(I, I')| \leq |d_M^R(I, I') - d_M(I, I')| \tag{L.32}$$
$$+ |d_M(I, I') - d_G(I, I')|, \tag{L.33}$$

which gives Equation (L.31). □

L.12 Normalized Signals

When

$$\|I\|_2 \le 1 \qquad \forall I \in S,$$

the theorem simplifies to

$$\left| d_M^R(I, I') - d_G(I, I') \right| \le 2\varepsilon + 4\sqrt{\frac{\log(4|S|k/\delta)}{2M}}. \tag{L.34}$$

Thus, for fixed target accuracy, both dimensionality requirements are mild:

$$\boxed{m = O\left(\varepsilon^{-2} \log(|S|kM/\delta)\right),}$$

while the number of sampled transformations need only grow logarithmically with

$$|S|k$$

for uniform approximation of the finite collection of orbit distributions.

L.13 Approximate Invariance after Projection

The population distance is exactly invariant:

$$d_G(I, hI) = 0 \qquad \forall h \in G.$$

The random projection need not itself intertwine the group action.
Nevertheless, if both

$$I \quad \text{and} \quad hI$$

belong to the controlled finite dataset, then Theorem 42 gives

$$d_M^R(I, hI) \le \varepsilon(R_S^2 + 1) + 4R_S\sqrt{\frac{\log(4|S|k/\delta)}{2M}}. \tag{L.35}$$

Thus exact group invariance before projection becomes approximate invariance on the finite family of signals being controlled.

L.14 Nearest-Neighbor Stability

The theorem immediately implies stability of nearest-neighbor decisions in the invariant quotient geometry.

Suppose $I^\star$ is the unique nearest neighbor of an anchor I under d_G, and assume the margin condition

$$d_G(I, J) - d_G(I, I^\star) \ge \gamma \tag{L.36}$$

for every competing

$$J \neq I^\star.$$

Let

$$E = \varepsilon(R_S^2 + 1) + 4R_S\sqrt{\frac{\log(4|S|k/\delta)}{2M}}. \tag{L.37}$$

If

$$2E < \gamma, \tag{L.38}$$

then

$$I^\star$$

remains the nearest neighbor under the compressed empirical distance d_M^R.

Corollary 8 (Nearest-neighbor preservation)**.** *Under the conditions of Theorem 42 and the margin condition in Equation* (L.36)*, if Equation* (L.38) *holds, then*

$$\arg\min_{J \neq I} d_M^R(I, J) = I^\star. \tag{L.39}$$

L.15 Why the Logarithmic Dimension Is Significant

The ambient dimension

$$d$$

does not appear directly in the required projection dimension.
Instead,

$$m$$

depends logarithmically on the number of objects whose orbit geometry must be controlled:

$$\boxed{m \sim \varepsilon^{-2} \log\left(|S|kM\right).}$$

The reason is important.
The theorem does not attempt to preserve all of

$$\mathbb{R}^d,$$

nor even the entire linear span of the observed orbit points.
It preserves only a finite collection of scalar measurements needed to evaluate the orbit representation.
This is exactly the regime in which Johnson–Lindenstrauss compression is powerful.

L.16 What the Theorem Does Not Say

Several limitations should be explicit.

L.16.1 Random projection does not discover invariance

The group structure

$$G$$

and the orbit representation already exist before the random projection is applied.
The JL map preserves this geometry approximately.
It does not discover which transformations should be invariant.
Thus

random projection preserves useful geometry; it does not manufacture semantic geometry.

L.16.2 The theorem is finite-sample

The dimension bound controls a finite dataset, a finite template collection, and finitely many sampled transformations.

Uniform guarantees over all signals in a continuous function class would require additional complexity control such as covering numbers, Gaussian width, or related tools.

L.16.3 Exact invariance is lost after arbitrary projection

A generic random matrix R does not satisfy

$$Rg = \widetilde{g}R$$

for an induced low-dimensional representation $\widetilde{g}$ of the group.

Thus the compressed representation is guaranteed to be approximately invariant only on the controlled finite family.

Exact equivariance would require a different construction.

L.16.4 Selectivity is separate

Preserving an invariant representation does not prove that the representation distinguishes every pair of distinct group orbits.

That issue belongs to the theory of selectivity of the underlying template representation.

L.17 Possible Extensions

Several extensions are natural but require additional assumptions and are not part of the theorem above.

Continuous template families. If templates are themselves drawn from a continuous distribution, a second sampling or covering argument can control the approximation of the template average.

Structured random projections. Fast JL transforms, sparse embeddings, or other computationally efficient random maps may replace dense subgaussian matrices when their finite-vector concentration guarantees are sufficient.

Layered partially observable group architectures. For multiple invariant layers, projection errors must be propagated through the subsequent nonlinear maps.

If layer ℓ has Lipschitz constant L_ℓ and incurs local error ε_ℓ, a generic stability estimate takes the form

$$e_L \le \sum_{\ell=1}^{L} \left(\prod_{j=\ell+1}^{L} L_j \right) \varepsilon_\ell. \tag{L.40}$$

Thus a recursive extension requires explicit control of layerwise stability; it does not follow automatically from the one-layer theorem.

L.18 Relation to the Framework of This Book

The result illustrates a recurring distinction.

Suppose a high-dimensional representation has already acquired useful structure—in this case, group invariance encoded through orbit distributions.

Then much of its ambient dimensionality may be unnecessary for preserving the geometry relevant to a finite collection of tasks.

Johnson–Lindenstrauss compression shows that

$$\boxed{\text{high ambient dimension} \not\Longrightarrow \text{high intrinsic metric complexity.}}$$

For the finite family considered here, the required dimension grows only as

$$O(\log N)$$

in the size of the controlled configuration set.

But the order of explanation matters:

$$\boxed{\text{structure first} \rightarrow \text{compression second.}}$$

Random projection preserves the relationships already present in the representation.

It does not explain how those relationships were learned.

This is the same distinction emphasized elsewhere in the book between representation and recovery.

L.19 Conclusion

Group averaging can create invariant representations by replacing an individual signal with the distribution of its measurements over a group orbit.

For a finite collection of signals and templates, these distributions can be approximated by finitely many Haar-sampled transformations.

A single Johnson–Lindenstrauss projection can then compress all of the corresponding orbit measurements simultaneously.

The resulting error separates cleanly into

$$\boxed{\text{group-sampling error} + \text{random-projection error.}}$$

Quantitatively,

$$\boxed{\left|d_M^R(I,I') - d_G(I,I')\right| \lesssim \varepsilon(R_S^2+1) + R_S\sqrt{\frac{\log(|S|k/\delta)}{M}}.}$$

The projection dimension required for this guarantee is

$$\boxed{m = O\left(\varepsilon^{-2}\log\frac{|S|kM}{\delta}\right).}$$

Thus invariant orbit geometry can, on a finite controlled family, survive a dramatic reduction in ambient dimension.

The conceptual lesson is broader than the particular group model:

once useful low-complexity geometry has been learned, random projection can often preserve it using far fewer

What random projection cannot do is discover the geometry itself.

CHAPTER M

The Imitation Game 2.0 (Idea by Dan Mitropolsky)

Current benchmarks for Large Language Models are faltering. They suffer from data contamination, Goodhart's law, and the inherent difficulty of distinguishing memorization from reasoning. We propose a new, dynamic metric for intelligence based on a modernized version of Turing's Imitation Game. The core hypothesis is the ***Containment Principle****: a system of higher intelligence can faithfully simulate a system of lower intelligence, but the reverse is not reliably achievable. By measuring the "Simulation Distance"—the asymmetry in the ability of two models to mimic each other—we can establish a rigorous, directed hierarchy of machine intelligence.* **add idea of having different models output a program playing a game...leaderboard to measure intelligence**

M.1 Introduction

How do we know if one mind is "smarter" than another? In classical psychometrics, we rely on static tests—puzzles, analogies, and factual recall. The AI community has adopted this paradigm, creating massive multiple-choice benchmarks like MMLU or coding challenges like HumanEval.

However, these static benchmarks are breaking down. Because Large Language Models (LLMs) are trained on the entire internet, they have often seen the test questions before. A model that aces a bar exam may not be reasoning; it may simply be remembering. We need a test that measures *fluid intelligence*—the ability to adapt to novel constraints—rather than *crystallized intelligence*.

We argue that the truest measure of intelligence is not peak performance, but **behavioral bandwidth**. A master chess player can choose to play like a novice; a novice cannot choose to play like a master. A sophisticated actor can play a fool; a fool cannot play 'Hamlet'.

This leads to our central thesis: **Intelligence is not merely correctness, but the controllability of incorrectness.** It is the capacity to inhibit optimal performance and faithfully simulate a specific suboptimal policy.

This asymmetry suggests a new protocol for ranking models: **The Machine-vs-Machine (MvM) Imitation Game**. Instead of asking "Can Model A solve this hard problem?", we ask "Can Model A simulate Model B?" This shift turns evaluation from a static test into a test of *plasticity* and *control* [155].

M.2 The Containment Principle

Our theoretical foundation is the **Containment Principle** of computational complexity. In theoretical computer science, a Universal Turing Machine (UTM) can simulate any specific Turing machine. The set of behaviors accessible to the universal machine is a superset of the behaviors accessible to the specific one.

Applied to LLMs, we hypothesize:

> **Hypothesis:** If Model A is strictly more intelligent than Model B, then the probability distribution over outputs of A can be prompted to cover the distribution of B. The converse does not hold.
>
> $$\mathcal{P}(B) \subset \mathcal{P}(A)$$

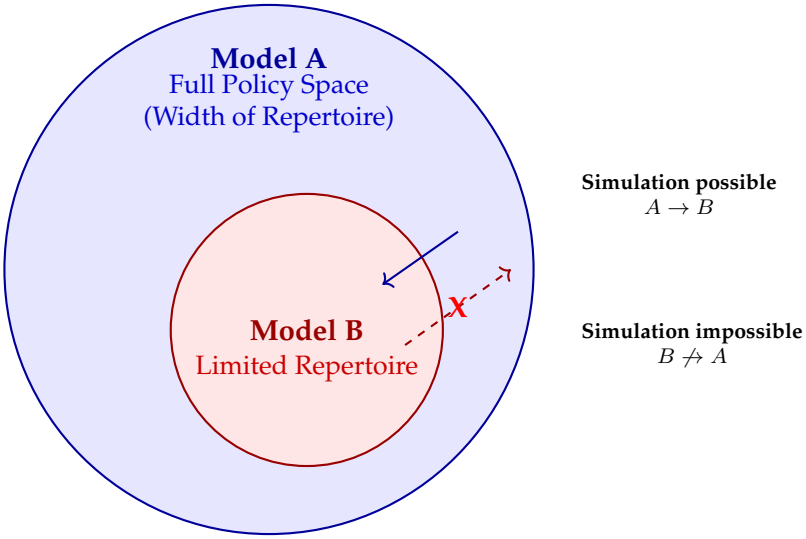

Figure M.1: **The Containment Principle.** Intelligence is defined as the width of the policy space. A more intelligent Model A encompasses the entire behavioral space of a less intelligent Model B. Therefore, A can simulate B (downward reach), but B cannot simulate A (upward reach).

This definition refines the standard definition of Universal Intelligence proposed by Legg and Hutter [89]. While Legg and Hutter define intelligence as the ability to *maximize reward* across environments, we define it as the capacity to *navigate the full policy space*, including suboptimal regions. A super-intelligent agent must be able to fail on purpose; a narrow agent can only fail by accident.

M.3 The Protocol

We propose a formal tournament structure to rank two models, A and B.

Phase 1: The Downward Reach (The Control Test)

We test if the candidate model can intentionally degrade its performance to match a specific target.

- **Instruction:** "Write a Python function to sort a list, but insert a specific off-by-one error that only triggers on empty lists."

- **Success Condition:** The model produces generally correct code that fails *only* in the specified way.
- **Failure Condition:** The model writes correct code (fails to follow the constraint) or writes code with random, uncontrolled errors (hallucination).

This tests **metacognition**: does the model understand the *structure* of the error? Note that safety training (RLHF) may sometimes inhibit this simulation (e.g., refusing to write bad code), which is a limitation of alignment, not capability [141].

Phase 2: The Upward Reach (The Capability Test)

We test if the candidate model can simulate a "super-expert."

- **Instruction:** "Explain the concept of 'entropy' using the precise vocabulary and reasoning style of a statistical mechanics professor."
- **Success Condition:** The output captures deep causal links and technical nuance.
- **Failure Condition:** The model adopts the *tone* of an expert (using big words) but makes conceptual errors or hallucinates complexity.

M.4 The Metric: Simulation Distance

We can quantify the relative intelligence of two models by measuring the asymmetry in their imitation capabilities. Let $S(A \to B)$ be a score (0 to 1) representing how successfully Model A can mimic the behavior of Model B. This score is verified by a "Judge" (which must be a highly capable model or human expert).

We define the **Simulation Distance** $D(A, B)$:

$$D(A, B) \;=\; S(A \to B) \;-\; S(B \to A)$$

- If $D(A, B) > 0$: Model A can simulate B, but B cannot simulate A. **Conclusion:** A **is smarter.**
- If $D(A, B) \approx 0$: The models are roughly equivalent (or the test is inconclusive).
- If $D(A, B) < 0$: Model B is smarter.

This metric is robust. It is difficult to "game" because it relies on the *interaction* between models rather than a fixed dataset. You cannot memorize the answer to "behave like GPT-2," because "behaving like GPT-2" is a dynamic, context-dependent constraint.

M.5 Case Study: The "Bad Code" Test

To illustrate, consider a coding task. We ask both GPT-4 (Model A) and a smaller 7B parameter model (Model B) to: *"Write a calculator that works perfectly, except it believes* $2 + 2 = 5$*."*

1. **Model A (GPT-4):** Writes a functioning script with a specific 'if' statement: 'if x==2 and y==2: return 5'. It successfully simulates the specific delusion while maintaining general competence.

2. **Model B (Small):** May write a broken calculator, or simply output ‘5‘ for every addition, or fail to implement the override at all. It lacks the **control** to partition its knowledge base [24].

Here, Model A demonstrates it *contains* the capability of Model B (and more), whereas Model B cannot contain the logic of Model A.

M.6 Conclusion

The “Imitation Game” was originally proposed by Turing to distinguish man from machine. We repurpose it here to distinguish machine from machine. This shift acknowledges a fundamental truth about artificial intelligence: true capability implies a mastery over the latent space of potential minds. A model that can choose to be dumb, biased, or mistaken on command demonstrates a higher order of understanding than one that is simply correct by default.

CHAPTER N

Degenerate Minima in Analytic Functions and Overparameterized Deep Networks

Analyticity supplies powerful tools for studying degenerate critical points, but it is not a substitute for an optimization theorem. A finite network built from affine maps and GELU activations has an analytic squared-loss objective. This permits the use of local gradient inequalities and resolution of singularities. The conclusions of those tools must nevertheless be separated: convergence of a bounded descent trajectory, geometry of an interpolating set, weighted volume near that set, and finite-time access by a specified optimizer are different statements.

This appendix develops the analytic foundations behind Chapters 7 and H. It gives direct proofs of the elementary claims and examples, and states the deeper analytic results with their hypotheses and sources. The central distinction is

$$\begin{array}{c} \textit{analytic regularity} \\ \neq\ \textit{quantitative conditioning} \\ \neq\ \textit{global optimization} \\ \neq\ \textit{predictive generalization}. \end{array}$$

The positive results are substantial: a precompact analytic gradient flow converges to a critical point; regular interpolation has a precise manifold and Hessian structure; a quantitative prediction-Jacobian bound gives exponential loss decay; and regular or singular local geometry determines low-temperature weighted-volume asymptotics. None of these claims follows from parameter count alone.

The stochastic endpoint requires an additional distinction. At exact interpolation of fixed per-example squared losses, their sampling covariance vanishes. At a stationary point of a regularized objective, only the average *data gradient must balance the penalty gradient; minibatch fluctuations can persist. The final section proves the corresponding covariance criterion, gives a strictly convex example, and develops the normalized-network rank obstruction discussed in Chapter 7.*

Conventions. We write d for input dimension, p for the number of real trainable parameters, n for the number of examples, and k for output dimension. There are $M = nk$ scalar prediction coordinates on the training sample. Euclidean vector norms and their induced operator norms are used unless otherwise indicated. The parameter vector is θ; the exponent in a Łojasiewicz inequality is ϑ, to avoid confusing the two. All gradient-flow and gradient-descent calculations refer to exact

real-arithmetic dynamics. A bit-complexity or numerical-implementation claim requires additional precision and stability estimates.

N.1 Introductory Foundations: Minima of Analytic Functions

The distinction between complex and real analyticity matters. Complex-valued functions have no intrinsic ordering, so a claim about their minima must name a real-valued quantity, such as the modulus or the real part. Real-analytic losses, in contrast, can have interior local and global minima without being constant.

N.1.1 Analytic Functions of One Variable

Complex Domain ($\mathbb{C}$)

Let $f : \Omega \to \mathbb{C}$ be holomorphic on a connected open set $\Omega \subseteq \mathbb{C}$.

A minimum of the modulus. Suppose $|f|$ has a local minimum at z_0 and $f(z_0) \neq 0$. There is a neighborhood of z_0 on which f has no zeros. On this neighborhood $1/f$ is holomorphic and its modulus has a local maximum. The maximum-modulus principle makes f locally constant, and the identity theorem makes it constant on Ω. For a strictly positive global minimum, $1/f$ is holomorphic on all of Ω and the same argument applies directly.

The qualification $f(z_0) \neq 0$ is essential. At a zero, the modulus has its global minimum 0 without forcing constancy; $f(z) = z$ is an example.

A minimum of the real part. The function $u = \operatorname{Re} f$ is harmonic. If it has an interior local minimum, the strong minimum principle makes u constant. The Cauchy–Riemann equations then make the imaginary part constant as well. Hence f is constant.

An entire function with a lower-bounded real part. If f is entire and $\operatorname{Re} f \geq c$ on $\mathbb{C}$, then $g = e^{-f}$ is bounded and entire. Liouville's theorem makes g constant. Since g never vanishes, $g' = -f'e^{-f} = 0$ implies $f' = 0$.

These complex-analytic principles do not apply to a general real-valued training loss merely because its activation has an entire extension. In particular, the real restriction of an entire function can have minima, and a complex squared modulus is not a holomorphic function.

Real Domain ($\mathbb{R}$)

Let $f : I \to \mathbb{R}$ be analytic on an open interval. At an interior local minimum x_0,

$$f'(x_0) = 0, \qquad f''(x_0) \geq 0.$$

If f is not locally constant, analyticity provides a first nonzero positive-order derivative at x_0. Its order must be even and its value positive. Indeed, for that order r,

$$f(x_0 + h) - f(x_0) = \frac{f^{(r)}(x_0)}{r!} h^r + o(|h|^r).$$

Nonnegativity for both signs of small h forces r to be even and the coefficient to be positive. Conversely, that condition gives a strict local minimum. If all positive-order derivatives vanish, f is locally constant and therefore constant on the connected interval.

Thus x^2, x^4, and $1 - \cos x$ all have interior minima. The second-order necessary conditions alone are not sufficient: $-x^4$ also has $f'(0) = 0$ and $f''(0) = 0$, but has a local maximum at 0.

N.1.2 Analytic Functions of Multiple Real Variables

Differential Conditions

For a C^2 function $f : \Omega \to \mathbb{R}$ on an open subset of $\mathbb{R}^p$, an interior local minimum satisfies

$$\nabla f(\theta_0) = 0, \qquad \nabla^2 f(\theta_0) \succeq 0.$$

These conditions are necessary, not sufficient. If in addition $\nabla^2 f(\theta_0) \succ 0$, the stationary point is a strict isolated local minimum. A semidefinite Hessian requires higher-order analysis. None of these local conditions identifies a minimum as global.

Geometry and the Łojasiewicz Inequality

If a finite global minimum f_* is attained, its minimizer set is

$$\mathcal{M} = \{\theta \in \Omega : f(\theta) - f_* = 0\}.$$

This is a real-analytic zero set, hence locally semianalytic and subanalytic. It need not be a manifold; different branches may intersect or have different dimensions. The real-analytic geometry underlying these statements is discussed in [16].

A more directly dynamical tool is the following classical result, used in the convergence analysis of [4].

Theorem 43 (Local analytic gradient inequality; quoted result)**.** *Let f be real-analytic near a critical point θ_*. There exist a neighborhood U of θ_*, a constant $C > 0$, and an exponent $\vartheta \in [1/2, 1)$ such that*

$$\|\nabla f(\theta)\| \geq C|f(\theta) - f(\theta_*)|^{\vartheta}, \qquad \theta \in U. \tag{N.1}$$

The inequality holds near critical points, not just near global minima. The constants and the neighborhood depend on the function and the point. They are not automatically uniform over growing network families. No particular optimal exponent is asserted here.

Harmonic Functions in $\mathbb{R}^p$

For a harmonic real-valued function, $\Delta f = 0$, the strong minimum principle rules out a nonconstant function attaining an interior minimum on a connected domain. A general real-analytic function is not harmonic, so this principle does not exclude its local minima.

N.1.3 Bounded Analytic Gradient Flow Converges to a Critical Point

The local inequality has a useful global-in-time consequence once the trajectory is known to remain in a compact subset of the analytic domain. The result is classical; the argument below records the needed boundedness and finite-length steps [4].

Theorem 44 (Convergence of a precompact analytic gradient flow)**.** *Let $f : \Omega \to \mathbb{R}$ be real-analytic on an open set $\Omega \subseteq \mathbb{R}^p$. Suppose a solution of*

$$\dot{\theta}(t) = -\nabla f(\theta(t))$$

Table N.1: Interior minima: the quantity being minimized and the required qualifications. These are local statements unless otherwise specified.

@>p0.22>p0.20>X@

Setting	Quantity	Conclusion
Holomorphic f	$\lvert f\rvert$ at $f(z_0)\neq 0$	An interior local minimum forces constancy on the connected domain.
Holomorphic f	$\lvert f\rvert$ at $f(z_0)=0$	A minimum is possible without constancy.
Holomorphic f	$\operatorname{Re} f$	An interior local minimum forces constancy.
Real-analytic, one variable f		Unless locally constant, the first nonzero positive-order derivative is even and positive at a minimum.
Real-analytic, several variables f		Stationarity and a semidefinite Hessian are necessary; a positive-definite Hessian at a stationary point is sufficient for a strict local minimum.
Real harmonic f		An interior local minimum forces constancy on a connected domain.

is defined for all $t \geq 0$ and its image has compact closure contained in Ω. Then the trajectory has finite length and converges to a point $\theta_\infty \in \Omega$ with $\nabla f(\theta_\infty) = 0$. The conclusion does not assert that θ_∞ is a local or global minimum.

Proof. Along the trajectory,

$$\frac{d}{dt} f(\theta(t)) = -\|\nabla f(\theta(t))\|^2.$$

Compactness bounds f below, so its decreasing values have a limit f_∞ and $\int_0^\infty \|\nabla f(\theta(t))\|^2\, dt < \infty$. On the compact closure, the gradient and Hessian are bounded. Hence $t \mapsto \|\nabla f(\theta(t))\|^2$ is uniformly continuous. A nonnegative, uniformly continuous, integrable function tends to zero: otherwise disjoint intervals of uniformly positive height and width would contradict integrability. Thus $\nabla f(\theta(t)) \to 0$.

The set Γ of accumulation points is nonempty and compact. Every point in Γ is critical and has value f_∞. Cover Γ by finitely many neighborhoods from Theorem 43. Taking the smallest positive constant and the largest exponent gives, when $0 \leq f - f_\infty \leq 1$, a common inequality

$$\|\nabla f\| \geq C(f - f_\infty)^\vartheta$$

on this union. The trajectory eventually stays in that union: otherwise compactness would yield an accumulation point outside it.

Put $a(t) = f(\theta(t)) - f_\infty \geq 0$. If $a(t) = 0$ at a finite time, monotonicity and the limiting value force the trajectory to be stationary thereafter. Otherwise, for all sufficiently large t,

$$\begin{aligned} -\frac{d}{dt} a(t)^{1-\vartheta} &= (1-\vartheta) a(t)^{-\vartheta} \|\nabla f(\theta(t))\|^2 \\ &\geq C(1-\vartheta)\|\dot\theta(t)\|. \end{aligned}$$

Integrating to infinity gives the tail-length bound

$$\int_t^\infty \|\dot\theta(u)\|\, du \leq \frac{a(t)^{1-\vartheta}}{C(1-\vartheta)}. \tag{N.2}$$

The initial finite-time part also has finite length. Consequently the trajectory is Cauchy, converges in the compact closure, and its limit is critical by continuity. □

This theorem is not a statement about arbitrary noisy SGD paths. Discrete descent algorithms require sufficient-decrease and step-control conditions; these are treated in [4]. Boundedness alone does not turn arbitrary discrete updates into analytic gradient flow.

N.1.4 Qualitative Convergence and Quantitative Rates

Once the preceding local inequality applies,

$$a'(t) \le -C^2 a(t)^{2\vartheta}.$$

For $\vartheta = 1/2$, this implies exponential decrease of $a(t)$. For $\vartheta > 1/2$ and $a(t_0) > 0$,

$$a(t) \le \left[a(t_0)^{1-2\vartheta} + C^2(2\vartheta - 1)(t - t_0)\right]^{-1/(2\vartheta-1)}. \tag{N.3}$$

These are rates relative to the limiting critical value f_∞, which may exceed the global minimum. Entry time into the neighborhood, the constant C, and the exponent all matter for an efficiency claim.

For example, $f(x) = x^{2m}$ with $m \ge 2$ has, for $x_0 > 0$, the exact flow

$$x(t) = \left[x_0^{-(2m-2)} + 2m(2m-2)t\right]^{-1/(2m-2)}. \tag{N.4}$$

Differentiating verifies $\dot{x} = -2mx^{2m-1}$. Convergence is not exponential. Moreover, the analytic family $f_N(x) = \frac{1}{2}e^{-N}x^2$ takes time of order e^N to reduce $x(t)$ by a fixed factor. Analyticity does not furnish polynomially controlled conditioning constants.

N.2 Real-Analyticity of Deep GELU Networks

N.2.1 Analytic Nature of the GELU Activation

Definition 12 (GELU activation)**.** *The Gaussian Error Linear Unit [62] is*

$$\sigma(z) = \mathrm{GELU}(z) = z\Phi(z) = \frac{z}{2}\left[1 + \mathrm{erf}\left(\frac{z}{\sqrt{2}}\right)\right], \tag{N.5}$$

where Φ is the standard Gaussian cumulative distribution function and $\mathrm{erf}(u) = \frac{2}{\sqrt{\pi}}\int_0^u e^{-t^2}\,dt$ on the real line.

Proposition 68 (Analyticity of GELU)**.** *The exact GELU activation is real-analytic on $\mathbb{R}$ and has an entire extension to $\mathbb{C}$.*

Proof. The series

$$\mathrm{erf}(u) = \frac{2}{\sqrt{\pi}}\sum_{j=0}^{\infty}\frac{(-1)^j u^{2j+1}}{j!(2j+1)} \tag{N.6}$$

converges for all complex u. Thus the error function is entire. Products and compositions with polynomials preserve this property, which gives the assertion for (N.5). □

Remark 4 (Approximate GELU and monotonicity)**.** *The commonly used approximation*

$$\widetilde{\sigma}(z) = \frac{z}{2}\left[1 + \tanh\left(\sqrt{2/\pi}\,(z + 0.044715z^3)\right)\right]$$

is also real-analytic on $\mathbb{R}$; it is not an entire complex function. For exact GELU, writing φ for the standard Gaussian density,

$$\sigma'(z) = \Phi(z) + z\varphi(z), \qquad \sigma''(z) = (2 - z^2)\varphi(z). \tag{N.7}$$

In particular $\sigma'(0) = 1/2$ and $\sigma'(-1) < 0$. To verify the latter,

$$\Phi(-1) = \int_1^\infty \varphi(t)\,dt < \int_1^\infty t\varphi(t)\,dt = \varphi(1).$$

GELU is therefore not monotone. Analyticity alone does not make it satisfy the hypotheses of a landscape theorem requiring a monotone activation.

N.2.2 Network Architecture and Quadratic Loss

Consider a fixed finite network with affine maps, biases, and elementwise GELU activations:

$$\begin{aligned} h^{(0)} &= x, \\ a^{(\ell)} &= W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}, && 1 \le \ell \le L-1, \\ h^{(\ell)} &= \sigma(a^{(\ell)}), \\ f(x;\theta) &= W^{(L)} h^{(L-1)} + b^{(L)}. \end{aligned} \tag{N.8}$$

The architecture is fixed while the real parameters vary.

Theorem 45 (Analyticity of the quadratic loss)**.** *For the network* (N.8)*, $f(x;\theta)$ is jointly real-analytic in inputs and parameters. For fixed training pairs $(x_i, y_i) \in \mathbb{R}^d \times \mathbb{R}^k$, the empirical loss*

$$\widehat{\mathcal{L}}(\theta) = \frac{1}{2n} \sum_{i=1}^{n} \|f(x_i;\theta) - y_i\|^2 \tag{N.9}$$

is real-analytic on $\mathbb{R}^p$.

Proof. Each affine expression is polynomial jointly in its parameters and its input variables. Induction over layers, using composition of analytic maps, proves the claim for f. A finite sum of squared residuals is a polynomial in those analytic outputs, which proves the loss assertion. □

The proof is elementary and does not use a theorem about absence of bad minima. Residual addition also preserves analyticity. Finite softmax operations are real-analytic because their real denominators are strictly positive. Normalization using a fixed positive numerical offset under a square root is real-analytic on its real domain as well. Hard routing, quantization, variable computational branches, or nonsmooth penalties must be analyzed separately. A property of the underlying real-valued map is not an assertion that a finite-precision implementation is analytic.

N.2.3 Smoothness Does Not Supply Uniform Conditioning

Analyticity gives derivatives of every order and bounds on each fixed compact parameter region. It does not bound Hessian norms uniformly over unbounded parameters or growing depth.

Example 1 (An analytic GELU loss with unbounded Hessian)**.** *Let $F(a,b) = a\sigma(b)$ and $\mathcal{L}(a,b) = \frac{1}{2}F(a,b)^2$. Since $\sigma(0) = 0$ and $\sigma'(0) = 1/2$,*

$$\frac{\partial^2 \mathcal{L}}{\partial b^2}(a,0) = \frac{a^2}{4}.$$

The loss is analytic, but its Hessian is not globally bounded. A bound on the admitted parameter region is needed for a uniform descent step.

Example 2 (Analytic activations and vanishing derivatives)**.** *For $h_0 = x$ and $h_{\ell+1} = \sigma(h_\ell)$, all intermediate states are zero at $x = 0$. Hence*

$$\frac{dh_H}{dx}(0) = \prod_{\ell=0}^{H-1} \sigma'(0) = 2^{-H}.$$

More generally, replacing each stage by $\sigma(ch_\ell)$ gives $(c/2)^H$. Smoothness is compatible with either attenuation or amplification.

Initialization schemes can improve signal propagation under specified assumptions [51, 61]. Maintaining an average variance is nevertheless different from controlling every relevant singular value of a Jacobian product throughout training.

N.2.4 A Quantitative Residual-Path Bound

Skip connections offer a positive mechanism when the residual derivatives are controlled. They are neither a universal necessity nor a sufficient condition by themselves.

Proposition 69 (Conditioned residual Jacobian products)**.** *Along a trajectory through equal-dimensional residual blocks $h_{\ell+1} = h_\ell + g_\ell(h_\ell)$, put $A_\ell = Dg_\ell(h_\ell)$. If $\|A_\ell\|_2 \le a_\ell < 1$, then for $P = D_{h_0} h_H = (I + A_{H-1}) \cdots (I + A_0)$,*

$$\prod_{\ell=0}^{H-1} (1 - a_\ell) \le \sigma_{\min}(P) \le \sigma_{\max}(P) \le \prod_{\ell=0}^{H-1} (1 + a_\ell). \tag{N.10}$$

If also $a_\ell \le a < 1$ and $\sum_\ell a_\ell \le C$, then

$$e^{-C/(1-a)} \le \sigma_{\min}(P) \le \sigma_{\max}(P) \le e^C. \tag{N.11}$$

Proof. The triangle and reverse-triangle inequalities give $(1 - a_\ell)\|v\| \le \|(I + A_\ell)v\| \le (1 + a_\ell)\|v\|$. Apply these bounds successively. Finally use $\log(1 + t) \le t$ and $\log(1 - t) \ge -t/(1 - a)$ for $0 \le t \le a$. □

For $A_\ell = cI$ with fixed $0 < c < 1$, the same architecture instead has $P = (1 + c)^H I$. Residual addition without scaling control still permits exponential amplification. Moreover, P is an activation-propagation Jacobian, not the training-prediction Jacobian $D_\theta F$. Good conditioning of the former does not imply full row rank of the latter.

N.2.5 A Positive Convergence Result with Full Weight Decay

Corollary 9 (Analytic gradient flow with a coercive quadratic penalty)**.** *For* (N.9)*, let*

$$U(\theta) = \widehat{\mathcal{L}}(\theta) + \frac{\gamma}{2}\|\theta\|^2, \qquad \gamma > 0,$$

where the penalty includes all trainable parameters. Every gradient-flow trajectory of U, from any finite initial parameter, exists globally, has finite length, and converges to a critical point of U.

Proof. The function U is analytic and nonnegative. Along its gradient flow, U decreases, so

$$\|\theta(t)\|^2 \le \frac{2U(\theta(0))}{\gamma}.$$

The trajectory remains in a compact sublevel set. The smooth vector field has no finite-time escape there, giving global existence. Theorem 44 then applies. □

The conclusion is stationarity of the penalized objective, not global optimality or interpolation. Even the convex example $U(a) = \frac{1}{2}(a-1)^2 + \frac{\gamma}{2}a^2$ has minimizer $a = 1/(1+\gamma)$ and nonzero unregularized residual.

A product penalty need not provide the same compactness. For example, $\Omega(a,b) = a^2b^2$ is bounded on the unbounded curve $(a,b) = (t,t^{-1})$. The analytic residual loss $\frac{1}{2}(ab-1)^2$ is zero on that curve, so adding this product penalty still leaves an unbounded sublevel set. Products of squared Frobenius norms are analytic, but not generally coercive; unsquared norm products can additionally fail analyticity at zero factors. Statistical norm control and trajectory confinement are separate requirements.

The preceding corollary concerns continuous gradient flow of the full penalized objective. It does not imply convergence of fixed-step SGD to the same parameter. Section N.5.4 shows that a critical point of this objective can have nonzero minibatch covariance; Section N.5.7 states the additional common-equilibrium condition needed by a finite point limit of the specified SGD recursion.

N.3 Desingularization via Hironaka's Theorem

At a degenerate minimum, a quadratic Taylor approximation may fail to capture the local geometry. The analytic Taylor expansion itself remains valid. Resolution of singularities provides another way to study the zero set and nearby weighted volumes. It is not an ordinary nonsingular coordinate transformation of the original loss landscape.

N.3.1 A Local Normal-Crossing Form

The following is a local analytic consequence of resolution of singularities and principalization. Hironaka's work supplies the foundational resolution theorem; analytic formulations and constructions are developed, for example, by Bierstone and Milman [66, 17].

Theorem 46 (Local resolution of a nonnegative analytic loss; quoted result)**.** *Let $L \geq 0$ be a nonzero real-analytic function on a connected open neighborhood of a compact region in $\mathbb{R}^p$. After localizing near its zero set, one can resolve the analytic divisor by a proper analytic modification $\pi : \widetilde{U} \to U$, an isomorphism away from the zero set, such that in suitable local coordinates*

$$L(\pi(u)) = a(u) \prod_{j=1}^{p} u_j^{2k_j}, \tag{N.12}$$

$$|\det D\pi(u)| = b(u) \prod_{j=1}^{p} |u_j|^{h_j}, \tag{N.13}$$

where k_j, h_j are nonnegative integers and the unit factors a, b are positive and nonvanishing on suitably chosen charts. A compact localized integral is handled by finitely many contributing charts. When a domain boundary is relevant, it is resolved as part of the localization.

This theorem is invoked, not reproved here. Nonnegativity permits even powers in the loss monomial. For a positive-loss local minimum one applies the local statement to $L - L_*$ where this difference is nonnegative. The modification is not asserted to be a diffeomorphism through the exceptional divisor, nor is every critical point necessarily blown up.

N.3.2 Weighted Volume and the Real Log Canonical Threshold

For a nonnegative analytic loss, take a compact localized region U and a smooth positive weight φ bounded above and below there. The integral of interest is

$$Z_U(T) = \int_U e^{-L(\theta)/T} \varphi(\theta)\, d\theta, \qquad T > 0. \tag{N.14}$$

Assume the analytic, semianalytic-domain, and localization conditions under which the standard resolution-based asymptotic applies [161, 162, 163]. The loss is not identically zero on an open part of U, and its zero set meets the localized support.

In a contributing chart, the transformed density includes both the loss and the volume element:

$$\exp\left(-\frac{a(u)u^{2k}}{T}\right) \widetilde{b}(u)|u|^h\, du,$$

where $\widetilde{b}$ includes $\varphi \circ \pi$, the Jacobian unit, and localization factors. On charts that contribute at the relevant divisor, these factors do not remove the leading singularity.

One way to see the exponent is to examine local integrability of L^{-z} for real $z \geq 0$. The monomial integral has factors

$$\int_0^\epsilon u_j^{h_j - 2k_j z}\, du_j,$$

which are finite precisely when $h_j - 2k_j z > -1$. Thus each divisor with $k_j > 0$ supplies a candidate threshold

$$\lambda_j = \frac{h_j + 1}{2k_j}. \tag{N.15}$$

The smallest threshold among relevant charts determines the RLCT λ. The associated pole multiplicity m records the maximal number of minimizing divisors meeting in a contributing resolved chart. The standard asymptotic is

$$\boxed{Z_U(T) = C_U T^\lambda \left(\log \frac{1}{T}\right)^{m-1} (1 + o(1)), \qquad C_U > 0, \quad T \downarrow 0.} \tag{N.16}$$

The full asymptotic is an invoked singular-integral result; the preceding integrability calculation explains why the ratios in (N.15) enter it.

Both k_j and h_j matter. A weight that vanishes can modify the latter exponents and hence the threshold; it cannot be ignored. With the same minimum loss and admissible positive weights, smaller λ gives a larger low-temperature contribution. Logarithmic multiplicity and finite prefactors distinguish cases with equal λ.

Two elementary checks. For $L(x) = x^{2r}$ on $[-1, 1]$ with unit weight, scaling $x = T^{1/(2r)}v$ gives

$$Z(T) \sim \left(\int_{\mathbb{R}} e^{-v^{2r}}\, dv\right) T^{1/(2r)}.$$

Thus $\lambda = 1/(2r)$ and $m = 1$. The zero set remains a single point for every r, while the volume exponent changes. RLCT is not simply half the topological codimension.

For the crossing singularity $L(x, y) = x^2y^2$ on $[-1, 1]^2$, changing variables in a positive quadrant gives

$$\begin{aligned} Z(T) &= 4 \int_0^1 e^{-z^2/T} \log(1/z)\, dz \\ &\sim \sqrt{\pi}\, T^{1/2} \log(1/T). \end{aligned}$$

The substitution $z = \sqrt{T}\, v$ verifies the last expression. Here $\lambda = 1/2$ and $m = 2$: the logarithm detects the intersection of the two zero-loss branches.

N.3.3 What Resolution Does Not Say About Gradient Trajectories

The gradient-inequality exponent ϑ in (N.1) and the volume exponent λ in (N.16) answer different questions. The former bounds a gradient relative to a function-value difference in the given metric; the latter measures integrability and weighted small-loss volume. No universal formula for ϑ follows from the loss monomial alone.

Even where π is locally invertible, pulling back Euclidean gradient flow changes the metric. If $\theta = \pi(u)$ and $\dot{\theta} = -\nabla_\theta L$, then

$$(D\pi(u)^\top D\pi(u))\dot{u} = -\nabla_u (L \circ \pi)(u). \tag{N.17}$$

This is obtained by multiplying $D\pi\dot{u} = -\nabla_\theta L$ by $D\pi^\top$. It is not generally Euclidean gradient flow of the resolved monomial. At the exceptional divisor the pullback metric can degenerate. Consequently, a monomial normal form does not automatically prove a finite-time optimization bound or a statement about approach directions.

Resolution-based methods are powerful for singular learning theory, but an RLCT of a localized empirical loss is not by itself a theorem about the population error of an SGD-selected parameter. Bayesian learning results use their own likelihood, population-distribution, prior, and asymptotic assumptions [162, 163].

N.4 Overparameterized Deep Networks: Parameters and Fitting Constraints

The relevant comparison is between p parameters and $M = nk$ scalar sample outputs, not between parameters and examples without regard to output dimension. Having $p > M$ permits redundant fitting directions, but does not by itself ensure that interpolation exists or is regular. Throughout the local geometric statements, interpolation is an explicit assumption.

N.4.1 The Regular Interpolating Zero-Loss Manifold

Stack the training predictions in $F(\theta) \in \mathbb{R}^M$, put $e(\theta) = F(\theta) - y$, and define

$$\widehat{\mathcal{L}}(\theta) = \frac{1}{2n}\|e(\theta)\|^2, \qquad J(\theta) = DF(\theta) \in \mathbb{R}^{M \times p}.$$

The set

$$\mathcal{M} = F^{-1}(y) \tag{N.18}$$

may be empty. If it is nonempty, its points are global minimizers of this nonnegative squared loss, regardless of other positive-loss local minima.

Proposition 70 (Regular fitting geometry)**.** *Suppose $F(\theta_*) = y$ and J has constant rank q on a neighborhood of θ_*. Then $\mathcal{M}$ is locally a smooth manifold with*

$$\dim \mathcal{M} = p - q, \qquad T_{\theta_*}\mathcal{M} = \ker J(\theta_*). \tag{N.19}$$

If F is analytic, this is a local analytic manifold. In particular, full row rank $q = M$ at an interpolating point gives dimension $p - M$ locally.

Proof. By the constant-rank theorem, local coordinates in domain and codomain make F a projection onto q coordinates followed by an inclusion. Its level set fixes exactly those q coordinates and leaves $p - q$ free. Differentiating the level-set condition identifies its tangent space. Full row rank persists on a sufficiently small neighborhood by continuity of a nonzero maximal minor. □

This is the regular-value mechanism behind manifold descriptions of interpolating networks; see also [31]. It is not a global claim that every point of $\mathcal{M}$ is regular, or that $\mathcal{M}$ is connected. Distinct points in the level set agree on the sample but can represent different functions elsewhere.

N.4.2 Hessian Spectrum and Neural Tangent Kernel Alignment

Differentiating the empirical loss yields

$$\nabla\widehat{\mathcal{L}}(\theta) = \frac{1}{n}J(\theta)^\top e(\theta), \tag{N.20}$$

$$\nabla^2\widehat{\mathcal{L}}(\theta) = \frac{1}{n}J(\theta)^\top J(\theta) + \frac{1}{n}\sum_{a=1}^{M} e_a(\theta)\nabla^2 F_a(\theta). \tag{N.21}$$

Proposition 71 (Hessian nullity and normalized kernel duality)**.** *At any twice-differentiable interpolating parameter θ_*, let $q = \operatorname{rank} J(\theta_*)$. Then*

$$H_* := \nabla^2\widehat{\mathcal{L}}(\theta_*) = \frac{1}{n}J(\theta_*)^\top J(\theta_*), \quad \ker H_* = \ker J(\theta_*), \quad \dim\ker H_* = p - q. \tag{N.22}$$

For

$$\Theta(\theta) = J(\theta)J(\theta)^\top, \qquad K(\theta) = \frac{1}{n}\Theta(\theta), \tag{N.23}$$

the positive eigenvalues, counted with multiplicity, satisfy

$$\lambda_j^+(H_*) = \lambda_j^+(K(\theta_*)) = \frac{1}{n}\lambda_j^+(\Theta(\theta_*)). \tag{N.24}$$

Proof. At interpolation the residual term in (N.21) vanishes. The identity $v^\top J^\top J v = \|Jv\|^2$ gives the equality of kernels and hence nullity. The positive eigenvalues of $J^\top J$ and $JJ^\top$ are the squares of the same nonzero singular values of J, proving the normalized spectral statement. □

In particular, when $p > M$, there are at least $p - M$ Hessian zero modes. This is an exact interpolation identity, not an empirical approximation. Empirical investigations of near-zero Hessian bulk are related but concern additional regimes [139]. Identifying these zero modes with a smooth tangent space requires the local rank assumption in Proposition 70.

The Hessian of the regularized objective. The interpolation identity above concerns the *data loss*. For $U_\gamma = \widehat{\mathcal{L}} + \gamma\Omega$, with a C^2 penalty,

$$\nabla^2 U_\gamma(\theta) = \frac{1}{n} J(\theta)^\top J(\theta) + \frac{1}{n}\sum_{a=1}^{M} e_a(\theta)\nabla^2 F_a(\theta) + \gamma\nabla^2\Omega(\theta). \tag{N.25}$$

This follows by differentiating the penalty and adding it to (N.21). At a regularized stationary point with nonzero residuals, neither additional term can be discarded without a separate estimate. Even at an exact data interpolator, the penalty Hessian remains. Thus the data-loss nullity and normalized kernel duality must not be asserted automatically for the full penalized Hessian.

A singular counterexample. Let $n = k = 1$, $F(\theta) = \sum_{j=1}^{p} \theta_j^2$, and $y = 0$. Then

$$\widehat{\mathcal{L}}(\theta) = \frac{1}{2}\left(\sum_j \theta_j^2\right)^2, \qquad \mathcal{M} = \{0\}.$$

At 0, both J and the Hessian vanish. The Hessian nullity is p, but the zero-loss set has dimension 0. Thus even an analytic, overparameterized, realizable problem need not have a positive-dimensional interpolation manifold at a singular solution.

N.4.3 The Scope of Neural-Network Landscape Results

A theorem about a particular activation or network cannot be extended merely by replacing its hypotheses with "smooth and wide." For example, Nguyen and Hein's 2017 results use assumptions including distinct inputs, a strictly increasing analytic activation satisfying additional growth conditions, a sufficiently wide layer, and specified rank and partial nondegeneracy conditions on critical points. Their Theorem 3.8 does not exclude all degenerate or low-rank suboptimal local minima [116]. Equation (N.7) shows why GELU does not satisfy that monotonicity assumption.

Their CNN results address a separate, specified architecture and feature-rank setting [117]. Connected-sublevel-set theorems also have their own architectural and activation hypotheses [115]. A connected sublevel set is not automatically a smooth connected manifold of zero-loss parameters, nor does connectedness provide a convergence rate for a particular gradient method.

In the opposite direction, suboptimal local minima have been constructed in wide networks for broad classes of smooth nonlinear activations [41]. This does not mean that every such minimum has large probability under a practical initialization. It does rule out an unqualified principle that width and activation smoothness eliminate all suboptimal minima.

N.4.4 An Analytic Least-Squares Counterexample to Almost-Sure Recovery

The following example isolates the invalid implication without relying on an architectural special case.

Proposition 72 (Many parameters and almost-everywhere rank are not enough)**.** *For every $p \geq 2$, define a scalar residual and loss by*

$$r(\theta) = \theta_1^3 - 3\theta_1 + 3 + \sum_{j=2}^{p} \theta_j^2, \qquad L(\theta) = \frac{1}{2} r(\theta)^2. \tag{N.26}$$

Then L is analytic, depends on all p parameters, and has a nonempty regular zero-loss manifold of dimension $p-1$. The residual Jacobian has full row rank almost everywhere. Nevertheless, $\theta_+ = (1, 0, \ldots, 0)$ is a strict positive-loss local minimum whose basin under gradient flow contains an open neighborhood.

Proof. With all other coordinates zero, $r(-3,0,\dots,0) = -15$ and $r(-2,0,\dots,0) = 1$, so a zero exists by continuity. The gradient

$$\nabla r = (3\theta_1^2 - 3, 2\theta_2, \dots, 2\theta_p)$$

vanishes only at $\theta_+ = (1,0,\dots,0)$ and $\theta_- = (-1,0,\dots,0)$. Their residuals are 1 and 5, respectively. Thus ∇r is nonzero at every zero of r, proving regularity and dimension $p-1$. It is nonzero except at those two points, proving the almost-everywhere rank assertion.

At θ_+,

$$L(\theta_+) = \frac{1}{2}, \qquad \nabla L(\theta_+) = 0, \qquad \nabla^2 L(\theta_+) = \operatorname{diag}(6,2,\dots,2) \succ 0.$$

Continuity gives a small ball on which $\nabla^2 L \succeq cI$ for some $c > 0$. Along gradient flow in this ball,

$$\frac{d}{dt}\|\theta(t) - \theta_+\|^2 = -2\langle \theta(t) - \theta_+, \nabla L(\theta(t))\rangle \le -2c\|\theta(t) - \theta_+\|^2.$$

A smaller closed ball is forward invariant, and its trajectories converge to θ_+. Its interior is an open subset of the suboptimal basin. □

The example concerns general analytic least-squares geometry, not an assertion that a prescribed unrestricted GELU network has exactly this parameterization. It disproves the proposed inference from analyticity, parameter count, regular interpolation, and almost-everywhere prediction rank alone. An exceptional set of critical parameters can have measure zero while its attracting basin has positive measure.

N.4.5 What a Quantitative Prediction-Jacobian Bound Does Prove

A positive convergence result requires information along the trajectory, not only at a zero-loss point. For the empirical loss in (N.9), gradient flow satisfies

$$\dot{e}(t) = -K(\theta(t))e(t). \tag{N.27}$$

Proposition 73 (Conditional residual contraction)**.** *Suppose a gradient-flow solution exists on $[0,T]$ and*

$$e(t)^\top K(\theta(t))e(t) \ge \mu\|e(t)\|^2, \qquad \mu > 0, \tag{N.28}$$

throughout that interval. Then

$$\widehat{\mathcal{L}}(\theta(t)) \le \widehat{\mathcal{L}}(\theta(0))e^{-2\mu t}, \qquad 0 \le t \le T. \tag{N.29}$$

If the solution is defined for all $t \ge 0$ and the same bound holds throughout, it has finite length and converges to a finite interpolating parameter. For this last conclusion assume that F is C^1 on all of $\mathbb{R}^p$ and the loss has the local smoothness required for the flow.

Proof. The loss derivative is

$$\frac{d}{dt}\widehat{\mathcal{L}}(\theta(t)) = -\|\nabla\widehat{\mathcal{L}}\|^2 = -\frac{1}{n}e^\top K e \le -2\mu\widehat{\mathcal{L}}.$$

Integration gives (N.29). In the all-time case, write $\ell(t) = \widehat{\mathcal{L}}(\theta(t))$. Before reaching zero, $\|\nabla\widehat{\mathcal{L}}\|^2 \ge 2\mu\ell$ implies

$$-\frac{d}{dt}\sqrt{\ell(t)} \ge \sqrt{\mu/2}\,\|\dot{\theta}(t)\|.$$

Therefore

$$\int_t^\infty \|\dot\theta(u)\|\,du \le \sqrt{2/\mu}\sqrt{\ell(t)}. \tag{N.30}$$

The trajectory converges in $\mathbb{R}^p$, and continuity together with $\ell(t)\to 0$ gives interpolation at the limit. If zero is reached in finite time, the gradient is zero thereafter. □

This result does not need analyticity. It uses a stronger quantitative condition and gives convergence to the specified zero residual, whereas Theorem 44 gives convergence to an unspecified critical point. A lower bound only on the nonzero eigenvalues of K is insufficient unless the actual residual belongs to the corresponding controlled subspace.

An NTK-style sufficient condition. If

$$K(\theta(0)) \succeq 2\mu I_M, \qquad \sup_{0\le t\le T}\|K(\theta(t))-K(\theta(0))\|_{\mathrm{op}} \le \mu, \tag{N.31}$$

then $K(\theta(t))\succeq \mu I_M$ on $[0,T]$. Indeed, for any vector v,

$$v^\top K(\theta(t))v \ge v^\top K(\theta(0))v - \|K(\theta(t))-K(\theta(0))\|_{\mathrm{op}}\|v\|^2 \ge \mu\|v\|^2.$$

This gives one route to the conditioning premise. The nearly constant NTK regime depends on parameterization, initialization, scaling, data, and the time interval being controlled [72]. Infinite width alone is not synonymous with a frozen kernel; different scalings permit infinite-width feature learning [171].

Finite-step and computational qualifications. If discrete gradient-descent iterates remain in a region satisfying (N.28), and the loss gradient is Λ-Lipschitz on every required update segment, then a step $0<\alpha\le 1/\Lambda$, also chosen with $\alpha\mu\le 1$, gives

$$\widehat{\mathcal{L}}(\theta_{j+1}) \le \widehat{\mathcal{L}}(\theta_j) - \frac{\alpha}{2}\|\nabla\widehat{\mathcal{L}}(\theta_j)\|^2 \le (1-\alpha\mu)\widehat{\mathcal{L}}(\theta_j).$$

This follows from the descent lemma. Invariance, initialization probability, the scales of μ^{-1} and Λ, and the cost and precision of an update must be controlled before calling the procedure efficient. The unregularized identity does not transfer unchanged to $\widehat{\mathcal{L}}+\gamma\Omega$ or to Adam or noisy SGD.

The residual dynamics must use the trained objective. For differentiable Ω, along gradient flow of $U_\gamma=\widehat{\mathcal{L}}+\gamma\Omega$ the chain rule instead gives

$$\begin{aligned} \dot e &= -Ke - \gamma J\nabla\Omega, \\ \frac{d}{dt}\widehat{\mathcal{L}}(\theta(t)) &= -\frac{1}{n}e^\top Ke - \frac{\gamma}{n}e^\top J\nabla\Omega. \end{aligned} \tag{N.32}$$

The penalty can oppose or reinforce data-loss descent. The unregularized contraction proposition is retained with its original hypotheses; an application to regularized training must control this extra contribution or establish good fit by another argument. Stationarity of U_γ means cancellation of its full gradient, not necessarily $e=0$.

Table N.2: Geometric and dynamical conclusions depend on hypotheses, not on an underparameterized/overparameterized dichotomy. Here $q = \operatorname{rank} J$.

@>p0.29>X >X@

Hypothesis	Justified conclusion	Not implied
Stationarity and positive-definite Hessian	Strict isolated local minimum	Global optimality
Interpolation and locally constant rank q	Local manifold of dimension $p-q$	Large basin or global connectedness
Twice-differentiable interpolation	$H = J^\top J/n$ and nullity $p-q$	A manifold at a singular point
Analytic objective and precompact gradient flow	Finite length and convergence to a critical point	Global minimum or uniform efficient rate
Uniform conditioning of the actual residual	Exponential empirical-loss decay	Population accuracy or teacher identification
Specified local Gibbs or weighted integral	Regular or singular volume asymptotics	Finite-time SGD access or universal stationary law

N.5 Basins of Attraction for Degenerate Manifolds

N.5.1 Foliation and Fiber Geometry: Local Conditions Matter

For a closed set $\mathcal{M}$ of minima, distinguish the basin defined by approach to the set,

$$\mathcal{B}(\mathcal{M}) = \{\theta_0 : \theta(t) \text{ exists for all } t \geq 0,\ \operatorname{dist}(\theta(t), \mathcal{M}) \to 0\}, \tag{N.33}$$

from the set of initial conditions whose trajectories converge to a particular point $\theta_* \in \mathcal{M}$:

$$W^s(\theta_*) = \{\theta_0 : \theta(t) \to \theta_*\}.$$

When every trajectory under consideration converges to a point of $\mathcal{M}$, there is the set-theoretic partition

$$\mathcal{B}(\mathcal{M}) = \bigsqcup_{\theta_* \in \mathcal{M}} W^s(\theta_*). \tag{N.34}$$

Writing this partition does not prove that its members are smooth fibers or that the entire basin is a fiber bundle.

A standard sufficient setting for a local stable foliation is a compact smooth manifold of equilibria with a uniformly positive normal Hessian and Hessian kernel equal to the tangent space. This is a normally attracting, Morse–Bott minimum manifold. The local stable-manifold and foliation theory then supplies stable leaves of dimension equal to the codimension q, tangent at θ_* to the normal space [67]. Smoothness is understood at the regularity level allowed by that theory. This is a local statement under normal attraction, not a consequence of analyticity alone.

In Euclidean gradient coordinates the stable linear space at the minimum is orthogonal to its tangent space. The nonlinear stable leaves can be curved; trajectories away from the limit need not meet a fixed normal line or stay perpendicular to the manifold.

Example 3 (Stable fibers not counted by Hessian rank)**.** *For $L(x, y) = x^4$,*

$$\mathcal{M} = \{x = 0\}, \qquad \nabla^2 L|_{\mathcal{M}} = 0, \qquad \dot{x} = -4x^3, \quad \dot{y} = 0.$$

For $x_0 \neq 0$, the solution is $x(t) = x_0/\sqrt{1 + 8x_0^2 t}$ and $y(t) = y_0$. Hence each point $(0, y_0)$ has the one-dimensional attracting fiber $\{(x, y_0) : x \in \mathbb{R}\}$, even though the Hessian rank is zero. Normal quadratic nondegeneracy cannot be omitted from the fiber-dimension claim.

N.5.2 Measure, Saddles, and Limits of Almost-Sure Claims

A probability of convergence is relative to an initialization law π_0: the relevant quantity is $\pi_0(\mathcal{B}(\mathcal{M}))$. Raw dimension or unspecified Euclidean volume does not determine it. Proposition 72 gives an open suboptimal basin even though the exceptional rank-deficient set has measure zero.

Strict-saddle avoidance results allow saddles to exist. Under their smoothness, step-size, initialization, and critical-set hypotheses, they show that convergence to such saddles has probability zero [87]. They do not eliminate suboptimal local minima, cover arbitrary non-strict saddles, or imply that all trajectories are bounded. Avoidance also differs from a quantitative escape-time bound.

Thus an almost-sure global-convergence argument requires both a valid convergence mechanism and a reason that all possible attracting limits are global. Neither follows from analyticity and width alone. A finite-time learning theorem additionally needs resource bounds.

N.5.3 Exact Minibatch Noise and the Interpolation Limit

We first state the sampling identity at a fixed parameter, then distinguish exact data interpolation from stationarity of a regularized objective. All sampling statements in this subsection condition on the current parameter and a fixed finite training sample.

For vector-valued outputs, let

$$e_i(\theta) = f(x_i;\theta) - y_i, \quad J_i(\theta) = D_\theta f(x_i;\theta), \quad g_i(\theta) = J_i(\theta)^\top e_i(\theta).$$

Then g_i is the gradient of the per-example half-squared loss, and $\bar{g} = n^{-1}\sum_i g_i = \nabla\widehat{\mathcal{L}}$.

Proposition 74 (Minibatch covariance and vanishing noise at interpolation)**.** *Sample B indices independently and uniformly with replacement from $\{1,\dots,n\}$ and let $\widehat{g}_B$ be their average gradient. Conditional on the current parameter,*

$$\begin{aligned} \mathbb{E}[\widehat{g}_B \mid \theta] &= \bar{g}(\theta), \\ \mathrm{Cov}(\widehat{g}_B \mid \theta) &= \frac{1}{B}C(\theta), \\ C(\theta) &= \frac{1}{n}\sum_{i=1}^{n} g_i(\theta)g_i(\theta)^\top - \bar{g}(\theta)\bar{g}(\theta)^\top. \end{aligned} \tag{N.35}$$

At exact squared-loss interpolation on these fixed examples, every g_i vanishes and hence $C(\theta_*) = 0$.

Proof. A single sampled gradient has mean $\bar{g}$ and covariance C. Conditional independence divides this covariance by B upon averaging. At interpolation, $e_i = 0$ for each example, so $g_i = J_i^\top e_i = 0$. □

For a uniform B-element subset $\mathcal{B} \subseteq [n]$, sampled without replacement, put

$$\widehat{g}_{\mathcal{B}} = \frac{1}{B}\sum_{i\in\mathcal{B}} g_i, \qquad c_{n,B} = \frac{n-B}{B(n-1)}, \qquad n > 1.$$

Then

$$\mathbb{E}_{\mathcal{B}}[\widehat{g}_{\mathcal{B}} \mid \theta] = \bar{g}, \qquad \mathrm{Cov}_{\mathcal{B}}(\widehat{g}_{\mathcal{B}} \mid \theta) = c_{n,B}C(\theta). \tag{N.36}$$

Here and below $[n] = \{1,\dots,n\}$. To prove the correction, set $h_i = g_i - \bar{g}$ and let I_i indicate membership in $\mathcal{B}$. Since $\sum_i h_i = 0$,

$$\sum_{i\neq j} h_i h_j^\top = -\sum_i h_i h_i^\top.$$

Using $\mathbb{E}I_i = B/n$ and $\mathbb{E}I_iI_j = B(B-1)/(n(n-1))$ for $i \neq j$, expansion gives

$$\mathbb{E}\left[\left(\frac{1}{B}\sum_i I_i h_i\right)\left(\frac{1}{B}\sum_j I_j h_j\right)^\top\right] = \frac{n-B}{Bn(n-1)}\sum_i h_i h_i^\top$$
$$= c_{n,B}C.$$

For $B = n$ this covariance is zero regardless of the individual gradients. For $1 \leq B < n$, it vanishes exactly when all g_i coincide. Independent sampling with replacement instead has factor $1/B$, even when $B = n$.

For an update with learning rate α, the update covariance is α^2 times the gradient-estimate covariance. Under a justified small-step diffusion approximation with time increment α, the covariance rate is $\alpha c_{n,B}C$, or $\alpha C/B$ with replacement. These three noise scales must not be confused.

The fixed-example assumption matters. Training-mode batch normalization or another batch-dependent forward operation need not yield gradients that are averages of fixed per-example vectors. Its actual batch-update covariance must be analyzed instead.

This exact calculation differs from identifying covariance with curvature. At interpolation,

$$C(\theta_*) = 0, \qquad H_* = \frac{1}{n}J(\theta_*)^\top J(\theta_*) \quad \text{may be nonzero.}$$

There is no general positive scalar proportionality between covariance and Hessian in this setting. The statement is specifically about the data gradients *at an exact interpolator*; it does not say that sampling noise vanishes at every minimum of a regularized objective. Positive regularization can move the stationary balance away from exact interpolation, where the same fixed-example sampling produces persistent fluctuations. No external source of randomness beyond minibatch selection is needed for that phenomenon.

Injected noise, changing examples or augmentations, dropout, and transient optimizer state can also change the dynamics. Each requires its stated model. The regularized case below already explains nonzero late-stage sampling covariance without invoking any of those additions.

N.5.4 Regularized Stationarity and Common Minibatch Equilibria

The covariance calculation extends beyond squared loss. Let each ℓ_i and the deterministic penalty Ω be differentiable at the parameters under consideration, and define

$$\begin{aligned} U_\gamma(\theta) &= \frac{1}{n}\sum_{i=1}^n \ell_i(\theta) + \gamma\Omega(\theta), \qquad \gamma > 0, \\ g_i(\theta) &= \nabla\ell_i(\theta), \qquad \bar{g}(\theta) = \frac{1}{n}\sum_i g_i(\theta), \\ G_{\mathcal{B},\gamma}(\theta) &= \frac{1}{B}\sum_{i\in\mathcal{B}} g_i(\theta) + \gamma\nabla\Omega(\theta). \end{aligned} \tag{N.37}$$

The penalty contribution is identical for every batch at the fixed parameter. Therefore, for uniform subset sampling,

$$\begin{aligned} \mathbb{E}_{\mathcal{B}}[G_{\mathcal{B},\gamma} \mid \theta] &= \nabla U_\gamma(\theta), \\ \mathrm{Cov}_{\mathcal{B}}(G_{\mathcal{B},\gamma} \mid \theta) &= c_{n,B}C(\theta). \end{aligned} \tag{N.38}$$

The same formulas hold with factor $1/B$ for independent draws with replacement. These are ordinary penalized gradient updates, not a claim that every adaptive or decoupled weight-decay implementation has this form.

At exact squared-loss interpolation, $g_i(\theta_*) = 0$ for every example. All regularized batch gradients then equal $\gamma\nabla\Omega(\theta_*)$. Their covariance is zero, but their common mean is nonzero unless the penalty gradient vanishes. Such an interpolator need not be stationary for U_γ.

At a regularized stationary point θ_γ, instead,

$$\bar{g}(\theta_\gamma) + \gamma\nabla\Omega(\theta_\gamma) = 0, \qquad G_{\mathcal{B},\gamma}(\theta_\gamma) = \frac{1}{B}\sum_{i\in\mathcal{B}}(g_i - \bar{g})(\theta_\gamma). \tag{N.39}$$

Only the *average* data gradient cancels the regularizer. Individual examples can continue to demand different updates.

Proposition 75 (Noise at a regularized stationary point)**.** *Let $n > 1$, $1 \le B < n$, and let $\mathcal{B}$ be a uniformly sampled B-element subset of $[n]$. Suppose all ℓ_i and Ω are differentiable at a stationary point θ_γ of* (N.37)*. Then*

$$\boxed{\mathbb{E}_{\mathcal{B}}\|G_{\mathcal{B},\gamma}(\theta_\gamma)\|^2 = \frac{n-B}{B(n-1)}\frac{1}{n}\sum_{i=1}^{n}\|g_i(\theta_\gamma) - \bar{g}(\theta_\gamma)\|^2.} \tag{N.40}$$

At this point the following conditions are equivalent:

1. *the sampling covariance of $G_{\mathcal{B},\gamma}$ is zero;*
2. *all per-example data gradients $g_i(\theta_\gamma)$ are equal;*
3. *$G_{\mathcal{B},\gamma}(\theta_\gamma) = 0$ for every admissible batch;*
4. *$g_i(\theta_\gamma) + \gamma\nabla\Omega(\theta_\gamma) = 0$ for every i.*

Thus unequal sample gradients give strictly positive minibatch fluctuation energy at a zero of the full regularized gradient, including a regularized local or global minimum.

Proof. At stationarity the conditional mean in (N.38) is zero. Taking the trace of its covariance proves (N.40). Since $c_{n,B} > 0$, this trace vanishes if and only if every $g_i = \bar{g}$. The stationarity equation then gives $g_i = -\gamma\nabla\Omega$ for all i, so every regularized batch gradient is zero. Conversely, if every batch gradient is zero, its second moment and covariance vanish.

One may also see the batchwise equivalence by exchanging examples. For any $i \neq j$, choose a set $A \subseteq [n] \setminus \{i, j\}$ of size $B - 1$, possible because $B < n$. The gradients for $A \cup \{i\}$ and $A \cup \{j\}$ differ by $(g_i - g_j)/B$; their simultaneous vanishing forces $g_i = g_j$. This also covers $B = 1$, when A is empty. □

This is the general covariance result used in Chapter 7, Proposition 5. It requires no analytic activation, nonconvexity, or landscape assumption. Its role is to identify the correct stochastic endpoint.

At an arbitrary parameter the orthogonal mean–fluctuation decomposition gives the exact identity

$$\mathbb{E}_{\mathcal{B}}\|G_{\mathcal{B},\gamma}(\theta)\|^2 = \|\nabla U_\gamma(\theta)\|^2 + c_{n,B}\,\mathrm{Tr}\,C(\theta). \tag{N.41}$$

A small full gradient controls only the first term. Gradient descent on U_γ is not, by definition, minimization of the second moment in (N.41).

Table N.3: Distinct endpoints for ordinary regularized minibatch gradients. The sample is fixed and $1 \leq B < n$ with uniform subset sampling.

@>X >p0.25 >p0.23@

Parameter condition	Mean batch gradient	Sampling covariance
Exact squared-loss interpolation, $\nabla\Omega \neq 0$	$\gamma\nabla\Omega \neq 0$	Zero
Regularized stationarity with unequal sample gradients	Zero	Nonzero
A common zero of every regularized batch gradient	Zero	Zero

A deterministic penalty does not add conditional randomness at a fixed parameter. It changes where stationarity can occur and can sustain heterogeneous, nonzero sample gradients there. Positive regularization alone is not a universal guarantee of nonzero covariance: identical sample gradients are possible in special configurations. The proposition states the exact condition and supplies no lower bound depending only on γ.

N.5.5 A Strictly Convex Example with Persistent Minibatch Noise

Example 4 (A regularized global minimum need not be absorbing)**.** *Let $F_\theta(x) = \theta x$ and take the two examples $(1, 1)$ and $(2, 2)$. For $\gamma > 0$, the objective*

$$U_\gamma(\theta) = \frac{1}{4}\left[(\theta - 1)^2 + (2\theta - 2)^2\right] + \frac{\gamma}{2}\theta^2 = \frac{5}{4}(\theta - 1)^2 + \frac{\gamma}{2}\theta^2$$

is analytic and strictly convex. Its unique global minimizer is $\theta_\gamma = 5/(5 + 2\gamma)$, but its singleton-batch gradient variance is strictly positive.

Proof. Differentiation gives $U'_\gamma(\theta) = \frac{5}{2}(\theta - 1) + \gamma\theta$ and $U''_\gamma = 5/2 + \gamma > 0$, proving the minimizer assertion. At this point,

$$G_{\{1\},\gamma}(\theta_\gamma) = (\theta_\gamma - 1) + \gamma\theta_\gamma = \frac{3\gamma}{5 + 2\gamma},$$
$$G_{\{2\},\gamma}(\theta_\gamma) = 4(\theta_\gamma - 1) + \gamma\theta_\gamma = -\frac{3\gamma}{5 + 2\gamma}.$$

Both batches have probability $1/2$. Their mean is zero and

$$\mathrm{Var}_{\mathcal{B}}(G_{\mathcal{B},\gamma}(\theta_\gamma)) = \frac{9\gamma^2}{(5 + 2\gamma)^2} > 0. \tag{N.42}$$

□

Without regularization, the same dataset interpolates at $\theta = 1$. At that parameter with positive γ, every batch gradient equals γ, so the sampling variance is zero but the mean is not. Conversely, at θ_γ the mean is zero but the variance is not. Thus neither nonconvexity nor nonanalyticity is needed for persistent sampling fluctuations at a regularized minimum. With a nonzero plain SGD step, this minimum is not an absorbing parameter.

N.5.6 Normalized ReLU Networks and the Low-Rank Obstruction

The general covariance criterion can be supplemented by a structural obstruction to a common equilibrium. Xu, Rangamani, Liao, Galanti, and Poggio [170] study normalized ReLU classifiers

with scale regularization. Their Lemma 8 bounds the layer ranks when *every* minibatch update vanishes. We give the same scoped formulation as Chapter 7, Lemma 3, with its batch-exchange proof and explicit differentiability conditions. This is a separate piecewise-smooth setting, not an application of the preceding GELU analyticity theorem.

Model and tangent update directions. In this subsection the output is scalar ($k = 1$), and ℓ indexes layers. Consider a bias-free network with independently parameterized, unshared matrices,

$$f_V(x) = V_L\sigma(V_{L-1}\cdots\sigma(V_1x)\cdots), \qquad \|V_\ell\|_F = 1, \qquad F_{\rho,V}(x) = \rho f_V(x), \quad \rho > 0,$$

where σ is ReLU and $y_i \in \{-1, +1\}$. Introduce

$$m_i = y_i f_V(x_i), \qquad s_i = 1 - \rho m_i, \qquad R_{\ell i} = D_{V_\ell} m_i.$$

Here m_i is a signed normalized margin, and s_i is the corresponding signed residual. Separability $m_i > 0$ does not by itself imply $s_i > 0$; no common sign for the s_i is assumed.

Work at a point where the finite-sample prediction maps are differentiable and the fixed-gate calculation is valid, for example where no hidden preactivation on the training sample is zero. Locally in V_ℓ, scalar-output backpropagation writes $f_V(x_i) = a_{\ell i}^\top V_\ell b_{\ell i}$. Hence

$$R_{\ell i} = y_i a_{\ell i} b_{\ell i}^\top, \qquad \operatorname{rank} R_{\ell i} \le 1, \qquad \langle R_{\ell i}, V_\ell\rangle_F = m_i. \tag{N.43}$$

The last identity also expresses layerwise degree-one homogeneity. A shared convolutional kernel or a vector-output loss can sum several outer products and needs its own derivative-rank analysis. Regularity at a limiting ReLU kink does not follow merely from a probability-zero claim about initialization.

For the Frobenius-sphere-constrained minibatch objective

$$U_{\gamma,\mathcal{B}}(\rho, V) = \frac{1}{2B}\sum_{i\in\mathcal{B}}(1 - \rho m_i)^2 + \frac{\gamma}{2}\rho^2, \qquad \|V_\ell\|_F = 1,$$

the negative scale-gradient and negative tangent-gradient directions are

$$\begin{aligned} d_{\rho,\mathcal{B}} &= \frac{1}{B}\sum_{i\in\mathcal{B}} s_i m_i - \gamma\rho, \\ d_{\ell,\mathcal{B}} &= \frac{\rho}{B}\sum_{i\in\mathcal{B}} s_i(R_{\ell i} - m_i V_\ell). \end{aligned} \tag{N.44}$$

Indeed, differentiation in ρ gives the first line. The unconstrained negative V_ℓ gradient is $\rho B^{-1}\sum_i s_i R_{\ell i}$. Projecting it orthogonally onto the tangent space of $\|V_\ell\|_F = 1$ subtracts its inner product with V_ℓ times V_ℓ, giving the second line by (N.43).

These common-stationarity directions use the normalization convention of [170, Eq. (46), Lemma 8], with the common loss factor two removed by the half-squared-loss convention. The parameters n, γ correspond to N, λ in that paper. A finite Euler step does not preserve the sphere exactly; a discrete algorithm must specify its normalization rule. The following assertion concerns zeros of the stated directions.

Proposition 76 (Common-minibatch rank restriction; scoped form of Xu et al., Lemma 8)**.** *In the normalized scalar-output ReLU setting above, assume $n > 1$, $1 \le B < n$, $\gamma > 0$, $\rho > 0$, and $m_i \ne 0$ for every example. If*

$$d_{\rho,\mathcal{B}} = 0, \qquad d_{\ell,\mathcal{B}} = 0 \quad \textit{for every layer and every } B\textit{-element batch},$$

then $\operatorname{rank}(V_\ell) \le 2$ *for every layer.*

Batch-exchange proof. Set $c_i = s_i m_i$. The scale equation and $\gamma\rho > 0$ exclude $s_i = 0$ for all i. Since $m_i \neq 0$, some c_i is nonzero. Fix a layer ℓ. For any $i \neq j$, take two batches that share $B-1$ examples and differ only in i, j. Subtracting their zero tangent directions, and dividing by $\rho/B > 0$, gives

$$(c_i - c_j)V_\ell = s_i R_{\ell i} - s_j R_{\ell j}. \tag{N.45}$$

If some $c_i \neq c_j$, the right side has rank at most two by (N.43). Division by $c_i - c_j$ proves the stated restriction.

Otherwise all $c_i = c \neq 0$. Equation (N.45) implies $s_i R_{\ell i} = Q_\ell$ for a single matrix Q_ℓ of rank at most one. Any vanishing batch direction then implies $Q_\ell - cV_\ell = 0$. Thus $V_\ell = Q_\ell/c$ has rank at most one, which also proves the asserted bound. The argument applies to each layer. □

The statement retains the published upper bound and does not claim its sharpness. Its scalar-output, homogeneity, normalization, and local regularity hypotheses are part of the result; it does not apply unchanged to GELU or arbitrary regularizers.

From the obstruction to intrinsic noise. Concatenate $d_{\rho,\mathcal{B}}$ and the vectorized tangent matrix directions into $d_{\mathcal{B}}$, with the Euclidean/Frobenius product norm. At a stationary point of the *batch-averaged normalized dynamics*, $\mathbb{E}_{\mathcal{B}} d_{\mathcal{B}} = 0$. If its other hypotheses hold and some V_ℓ has rank greater than two, the proposition implies that at least one batch direction is nonzero. Every admissible batch has positive probability, so

$$\operatorname{Tr} \operatorname{Cov}_{\mathcal{B}}(d_{\mathcal{B}}) = \mathbb{E}_{\mathcal{B}} \|d_{\mathcal{B}}\|^2 > 0. \tag{N.46}$$

This is the covariance of the actual normalized directions, not an unjustified substitution of unconstrained parameter gradients.

The paper interprets the common-equilibrium restriction as a source of low-rank bias and intrinsic late-stage SGD fluctuations [170, Sections 6.1–6.2]. The precise implication established here is the necessary rank restriction and its contrapositive above. It is not a theorem that SGD always reaches rank-two matrices, nor that small updates alone imply a quantitative effective-rank bound. Such a bound would require control of the residuals, margins, and coefficients appearing in (N.45).

For a data family incompatible with accurate prediction by networks obeying that rank restriction, the proposition excludes a common equilibrium in the well-fitting region. The incompatibility must itself be proved; it is not established by calling the data generic. Conversely, low rank alone does not imply zero noise. The general covariance criterion and this architecture-specific obstruction explain how a useful regularized predictor can coexist with persistent intrinsic minibatch fluctuations.

N.5.7 Learning Rates and Finite Point Limits

The covariance at a parameter and the possibility of convergence of a sequence of parameters are different questions. The learning-rate schedule is part of the latter.

Proposition 77 (A finite limit with nonvanishing steps is a common equilibrium)**.** *Let $\mathfrak{B}$ be a finite family of batches and let $G_{\mathcal{B}} : \mathbb{R}^p \to \mathbb{R}^p$ be continuous for every $\mathcal{B} \in \mathfrak{B}$. Suppose batches $\mathcal{B}_j$ are sampled independently with fixed probabilities $p_{\mathcal{B}} > 0$, and consider*

$$\theta_{j+1} = \theta_j - \alpha_j G_{\mathcal{B}_j}(\theta_j), \qquad \inf_j \alpha_j > 0.$$

On an event of probability one, every finite limit θ_∞ of the whole sequence satisfies $G_{\mathcal{B}}(\theta_\infty) = 0$ for every $\mathcal{B} \in \mathfrak{B}$. Continuity may instead be assumed only on an open region containing the putative limit and the eventual iterates.

Proof. For a fixed batch, the probability that it never occurs after time J is $\lim_{q\to\infty}(1-p_{\mathcal{B}})^q = 0$. Taking the countable union over J, and then the finite union over batches, shows that every batch occurs infinitely often almost surely.

On this event, if $\theta_j \to \theta_\infty$, then $\theta_{j+1} - \theta_j \to 0$. Along the infinite subsequence with $\mathcal{B}_j = \mathcal{B}$,

$$\|G_{\mathcal{B}}(\theta_j)\| = \frac{\|\theta_{j+1} - \theta_j\|}{\alpha_j} \le \frac{\|\theta_{j+1} - \theta_j\|}{\inf_r \alpha_r} \longrightarrow 0.$$

Continuity gives $G_{\mathcal{B}}(\theta_\infty) = 0$. Apply this argument to every batch. □

Consequently a regularized stationary point with the positive covariance in Proposition 75 cannot be the finite point limit of this nonvanishing-step recursion under the stated sampling and continuity assumptions. For a fixed-step normalized algorithm with continuous batch update maps $T_{\mathcal{B}}$, the same subsequence argument instead gives $T_{\mathcal{B}}(z_\infty) = z_\infty$ for every batch. Inferring zero tangent directions requires these fixed points to be exactly the zeros of the specified directions. This is not automatic for arbitrary retractions, clipping, momentum, or adaptive optimizers, nor for a limit at an unexamined ReLU kink.

This conclusion does not assert divergence or existence of a stationary law. With $\alpha_j \to 0$, the update covariance

$$\mathrm{Cov}(\theta_{j+1} - \theta_j \mid \theta_j) = \alpha_j^2 c_{n,B} C(\theta_j)$$

can tend to zero while the gradient-estimator covariance remains nonzero. Diminishing-step convergence, convergence of an averaged predictor, and accuracy of a specified stopped iterate require their own arguments. Neither qualitative positivity of covariance nor positive regularization supplies a uniform lower noise bound across changing γ, batch sizes, or growing model families.

N.5.8 Anisotropic Diffusion Is Not a Universal Gibbs Law

The preceding results supply an intrinsic source of stochasticity in a regularized, nearly interpolating regime. They do not identify that noise with isotropic Langevin forcing. For a small constant learning rate α, uniform subset sampling, and an otherwise justified diffusion approximation, ordinary penalized SGD has the schematic model

$$d\theta_t = -\nabla U_\gamma(\theta_t)\, dt + [\alpha c_{n,B} C(\theta_t)]^{1/2}\, dW_t. \tag{N.47}$$

The covariance factor here is the rate on the time scale with increments α; it is not the per-update covariance. State dependence, anisotropy, finite-step effects, and constraint handling remain part of any application. Nonzero covariance need not excite every direction needed for escape from a critical region.

A modeled Itô diffusion can be written

$$d\theta_t = -\nabla U(\theta_t)\, dt + G(\theta_t)\, dW_t, \qquad D(\theta) = G(\theta)G(\theta)^\top. \tag{N.48}$$

Its density, when one exists with suitable regularity, satisfies

$$\partial_t p = \nabla\cdot(p\nabla U) + \frac{1}{2}\sum_{a,b} \partial_a\partial_b (D_{ab} p). \tag{N.49}$$

Boundary conditions, diffusion degeneracy, and state dependence are part of the problem. There is no general stationary formula depending only on $\mathrm{Tr}(D^{-1}\nabla^2 U)$ at minima; D may not even

be invertible. Non-isotropic SGD models need not have an equilibrium potential equal to their training loss [27].

Even the constant-coefficient quadratic case has a different formula. For an Ornstein–Uhlenbeck process

$$dz_t = -Hz_t\,dt + G\,dW_t, \qquad H = H^\top \succ 0,$$

its stationary covariance V solves

$$HV + VH = GG^\top. \tag{N.50}$$

Indeed Itô's formula for $z_t z_t^\top$, followed by expectation and stationarity, gives this identity. Equivalently,

$$V = \int_0^\infty e^{-Ht} GG^\top e^{-Ht}\,dt.$$

The stationary Gaussian can be degenerate if V is singular. In the isotropic case $GG^\top = 2TI$, it has $V = TH^{-1}$, recovering the ordinary quadratic Gibbs law. Local approximate-inference treatments of SGD use such additional assumptions [98].

When this local approximation is used near a strict regularized minimum, H is the Hessian of U_γ at that minimum, and $GG^\top$ must be matched to the specified regularized SGD covariance rate. One may not replace H automatically by the exact-interpolation data Hessian or set $GG^\top$ to zero merely because the *full* gradient vanishes. Freezing the covariance and replacing the drift by its linearization are additional local approximations, not consequences of analyticity.

N.5.9 The Valid Local Energy–Volume Comparison

For the explicitly specified isotropic Langevin model

$$d\theta_t = -\nabla U(\theta_t)\,dt + \sqrt{2T}\,dW_t,$$

assume well-posedness and a normalizable stationary density. Then

$$\pi_T(\theta) = Z_T^{-1} e^{-U(\theta)/T}. \tag{N.51}$$

If regularization is in the drift potential, it is also in U in this formula. A noncompact unregularized fitting set can prevent Z_T from being finite. A prior-weighted integral changes the reference measure and needs its own consistent dynamical interpretation.

For fixed positive regularization, a minimum component of U_γ need not be the exact data-interpolation set $F^{-1}(y)$, and its minimum value need not be zero. The following formulas therefore use the full potential and its actual minimum values. Intrinsic SGD fluctuations alone do not establish that the isotropic equilibrium model is appropriate.

For a compact smooth embedded minimum manifold $\mathcal{M}$ without boundary, of codimension q, minimum value U_*, and uniformly positive normal Hessian, a fixed small tube contributes

$$Z_{\mathcal{M}}(T) \sim e^{-U_*/T}(2\pi T)^{q/2} \int_{\mathcal{M}} \frac{dV(s)}{\sqrt{\det H_\perp(s)}}. \tag{N.52}$$

An isolated nondegenerate minimum θ_ℓ of value U_ℓ gives

$$Z_\ell(T) \sim e^{-U_\ell/T} \frac{(2\pi T)^{p/2}}{\sqrt{\det H_\ell}}. \tag{N.53}$$

These follow by integrating the normal quadratic expansion after scaling normal coordinates by $\sqrt{T}$; compactness and a uniform positive normal Hessian permit Gaussian domination. The detailed regular proof is in Chapter 7. Therefore, when $U_\ell > U_*$,

$$\frac{Z_\ell(T)}{Z_{\mathcal{M}}(T)} \sim C_\ell e^{-(U_\ell - U_*)/T} T^{(p-q)/2} \longrightarrow 0, \qquad C_\ell > 0. \tag{N.54}$$

The energy gap supplies the exponential factor; degeneracy changes the polynomial factor. A singular minimum component uses its RLCT and logarithmic multiplicity in place of the regular $q/2$ exponent.

These are ratios of local neighborhood weights. They do not prove that one component captures all global equilibrium mass, or bound the time needed to reach it. At fixed $T > 0$, an exact lower-dimensional manifold has zero probability under a smooth ambient density. Its neighborhood, not the manifold as an event of positive measure, is what is being compared.

N.5.10 Connection to Predictive Learning and the Status of the Results

A low-loss region favored by a particular stochastic model need not define a statistically small hypothesis class. Parameter-space sharpness is coordinate dependent; function-preserving reparameterizations can alter conventional Hessian-based sharpness [42]. Likewise, a Bayesian singular-learning result must not be silently converted into a statement about the test error of one SGD-selected parameter.

The separate statistical route in Chapter H remains: if a valid uniform bound on a controlled class $\mathcal{H}_Q$ holds and training returns $\widehat{f} \in \mathcal{H}_Q$ with empirical risk at most η, then on that event

$$R_P(\widehat{f}) \le \eta + \beta_S(Q, \delta).$$

The class, loss, sample, complexity budget, and algorithmic success event need the hypotheses stated there. An upper norm-complexity bound is not a lower prediction-Jacobian singular-value bound, and a product penalty is not automatically coercive. These are complementary requirements.

A stopped or otherwise specified returned predictor can satisfy these statistical conditions without its parameters approaching a common zero of all minibatch updates. If prediction averaging is used, the averaged predictor and its computational and class-complexity costs must themselves be specified. Persistent regularized noise is thus compatible with predictive success, but is not a proof of such success.

The appendix's logical map is therefore

$$\boxed{\begin{aligned} \text{analyticity + precompact gradient flow} &\implies \text{critical-point convergence,} \\ \text{regular interpolation} &\implies \text{local manifold geometry,} \\ \text{uniform residual conditioning} &\implies \text{quantitative fitting,} \\ \text{specified local weighted integral} &\implies \text{volume asymptotics.} \end{aligned}}$$

The regularized stochastic statement is a separate line:

$$\boxed{\begin{aligned} &\text{regularized stationarity + unequal sample gradients} \\ &\quad + \text{uniform proper-subset minibatches} \\ &\implies \text{zero mean update direction but nonzero covariance.} \end{aligned}}$$

A common equilibrium instead requires every minibatch direction to vanish; in the specified normalized ReLU setting this entails the stated rank restriction. These are exact finite-sample statements, not a Gibbs law or an efficient global-optimization theorem.

None of the displayed implications establishes all the other conclusions or population accuracy. The direct proofs given here follow from the stated hypotheses; the analytic gradient inequality, resolution theorem, singular-integral asymptotic, and stable-foliation theory remain invoked results with separate sources. The rank argument is a scoped restatement of the cited normalized-network result. No theorem is claimed that arbitrary sufficiently wide GELU networks converge almost surely to global interpolation, or that every regularized minimum has positive noise.

The useful conclusion is not that analyticity makes the training problem trivial. It makes several previously inaccessible aspects of degeneracy mathematically tractable. To prove efficient predictive learning one must still connect architecture, initialization, objective, optimizer, numerical precision, and data assumptions to a finite-time fitting and statistical control result. Once a feasible predictor has been shown to generalize, its computational realization supplies the compositional witness of Chapter H; no identification of the teacher's hidden computation is an additional requirement.

CHAPTER O

Diffusion as a Sequence of Supervised Learning Problems

The basic idea behind diffusion models can be stated very simply. Starting from an image I_0, construct a sequence

$$I_0, I_1, I_2, \ldots, I_T,$$

where I_{i+1} is obtained from I_i by adding a small amount of noise. This construction automatically produces a collection of supervised pairs

$$(I_i, I_{i+1}).$$

From exactly the same pairs one could in principle learn either direction,

$$I_i \longmapsto I_{i+1},$$

which corresponds to learning the *noising* process, or

$$I_{i+1} \longmapsto I_i,$$

which corresponds to learning the *denoising* process.

Thus diffusion may be viewed, at its most elementary level, as turning a difficult global generative problem into a sequence of much simpler local supervised learning problems.

O.1 The Forward Process

In standard diffusion models the noising direction is not usually learned, although it certainly could be learned from the pairs above. Instead, it is deliberately chosen to be a simple known stochastic process. A typical forward transition is

$$I_t = \sqrt{1-\beta_t}\, I_{t-1} + \sqrt{\beta_t}\, \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \mathbf{I}),$$

where β_t is small.

Repeated application gives

$$I_0 \to I_1 \to \cdots \to I_T,$$

and for sufficiently large T the final state is approximately Gaussian noise,

$$I_T \sim \mathcal{N}(0, \mathbf{I}).$$

The reason for choosing such a simple forward process is not that the forward direction cannot be learned. Rather, there is no need to learn it: it is completely specified by construction. The interesting problem is to learn the reverse transitions.

O.2 The Reverse Problems

From the generated pairs one can train a predictor for

$$I_t \longmapsto I_{t-1}.$$

Thus the reverse process consists of a sequence of local denoising problems,

$$I_T \to I_{T-1} \to \cdots \to I_1 \to I_0.$$

The same network is usually shared across all noise levels. It takes both the noisy image and the noise level as input,

$$f_\theta(I_t, t).$$

Consequently diffusion does not require T unrelated predictors. Instead, a single function learns a family of closely related supervised problems parameterized by t.

There is also an important simplification. Because the forward process is Gaussian, one can generate I_t directly from the original image I_0 without explicitly constructing all the intermediate states:

$$I_t = \sqrt{\bar{\alpha}_t}\, I_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \mathbf{I}),$$

where

$$\bar{\alpha}_t = \prod_{s=1}^{t} (1 - \beta_s).$$

Thus a training example can be generated simply by choosing an image I_0, choosing a random time t, choosing random Gaussian noise ϵ, and constructing I_t.

O.3 Predicting Noise

In modern diffusion models the network is often trained not to predict I_{t-1} directly, but to predict the noise ϵ used to produce I_t:

$$\epsilon_\theta(I_t, t).$$

The training objective is essentially

$$\min_{\theta} \mathbb{E}_{I_0,t,\epsilon} \left[\|\epsilon - \epsilon_\theta(I_t, t)\|^2 \right].$$

Once the added noise has been estimated, an estimate of the original image can be recovered from

$$\widehat{I}_0 = \frac{I_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(I_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

Thus noise prediction and denoising are essentially two equivalent ways of formulating the same supervised problem.

O.4 The Score Interpretation

There is a deeper interpretation of the denoising function. At each noise level t, let $p_t(I)$ denote the probability distribution of images after that amount of noise has been added. The quantity

$$\nabla_I \log p_t(I)$$

is called the *score* of the distribution.

The optimal noise predictor is closely related to this quantity:

$$\epsilon_\theta(I_t, t) \propto -\nabla_{I_t} \log p_t(I_t).$$

Therefore the network can be interpreted as learning a vector field over image space. At every noise level it indicates approximately which direction leads from a noisy image toward regions having higher probability under the image distribution.

The denoising process can therefore be viewed geometrically as

$$\text{noise} \to \text{coarse structure} \to \text{objects} \to \text{details} \to \text{image}.$$

This interpretation connects diffusion models with score matching, stochastic differential equations, and Langevin-type dynamics.

O.5 Why Denoising Becomes Generation

The crucial point is that the reverse process is not merely a procedure for restoring a particular corrupted image.

During training, the network sees noisy versions of many images and therefore learns properties of the entire image distribution. At generation time we can begin with a completely new random sample

$$I_T \sim \mathcal{N}(0, \mathbf{I})$$

and repeatedly apply the learned reverse transitions,

$$I_T \to I_{T-1} \to \cdots \to I_0.$$

In distributional terms, diffusion learns a transformation

$$\mathcal{N}(0, \mathbf{I}) \longrightarrow p_{\text{data}}(I).$$

Thus the collection of local denoising problems defines, when composed, a global generative process.

The reverse transition is in general stochastic. It can be written schematically as

$$I_{t-1} = \mu_\theta(I_t, t) + \sigma_t z, \qquad z \sim \mathcal{N}(0, \mathbf{I}).$$

This is natural because a noisy image does not in general have a unique less-noisy predecessor. The model is therefore more properly viewed as learning the conditional distribution

$$p_\theta(I_{t-1} \mid I_t)$$

rather than a single deterministic inverse.

O.6 Conditional Diffusion

For text-to-image generation the same idea is used with an additional conditioning variable c, representing, for example, a text prompt. The learned function becomes

$$\epsilon_\theta(I_t, t, c).$$

Conceptually, the model now estimates a score related to

$$\nabla_I \log p_t(I \mid c).$$

The reverse process therefore moves the random image not merely toward the manifold of natural images, but toward regions containing images compatible with the conditioning information.

O.7 Summary

The essential structure of diffusion can therefore be summarized as

images I_0
$\downarrow$
incremental noising
$\downarrow$
(I_i, I_{i+1}) supervised pairs
$\downarrow$
learn local reverse problems
$\downarrow$
learn a score/vector field
$\downarrow$
$\mathcal{N}(0, \mathbf{I}) \longrightarrow p_{\text{data}}(I)$.

The elementary observation is therefore the central one: incrementally noising an image creates a sequence of paired supervised learning problems. The same pairs contain information about both the forward, noising direction and the reverse, denoising direction. Standard diffusion chooses the forward process to be known and simple and devotes learning to the reverse direction.

What appears initially to be a difficult problem—learning an entire high-dimensional distribution of images—is thereby converted into a sequence of relatively small prediction problems. The composition of these local predictions is what produces the global generative model. "'latex

CHAPTER P

From Brain Activity to Images as an Inverse Problem

Suppose that we present an image x to a subject and measure the resulting fMRI activity y. Repeating this procedure produces paired examples

$$(x_1, y_1), \ldots, (x_n, y_n),$$

where x_i is an image and y_i is the corresponding measured brain response.

There are then two apparently symmetric supervised learning problems:

$$x \longmapsto y,$$

that is, predicting brain activity from an image, and

$$y \longmapsto x,$$

that is, reconstructing the image from brain activity.

The two problems, however, are not equally difficult. The first is a forward problem; the second is an inverse problem. This distinction is fundamental.

P.1 The Forward Problem

A simple description of the experiment is

$$y = F(x) + \eta,$$

where F represents the transformation from the visual stimulus to the measured fMRI response and η represents neural and measurement noise.

The forward learning problem is therefore

$$x \longmapsto y.$$

Given sufficiently many paired examples, one can attempt to learn a model

$$F_\theta(x) \approx y.$$

Equivalently, one can model the conditional distribution

$$p(y \mid x).$$

For a given image, the corresponding distribution of brain responses is reasonably well defined. The forward problem is therefore a standard supervised learning problem.

P.2 Why the Inverse Problem is Harder

The inverse problem asks instead for

$$y \longmapsto x.$$

The fundamental difficulty is that the forward transformation loses information. In general there may exist distinct images x_1 and x_2 such that

$$x_1 \neq x_2, \qquad F(x_1) \approx F(x_2).$$

Thus the inverse of F need not be a function.

This information loss occurs for several reasons. fMRI measurements have limited spatial and temporal resolution; many neurons contribute to each voxel; the BOLD response introduces additional smoothing; and both neural activity and the measurement process are noisy.

There is also a more fundamental reason. Visual representations themselves often discard properties of the stimulus that are irrelevant for recognition. If a representation is approximately invariant to transformations such as position, scale, illumination, or background, then different images may produce nearly identical internal representations.

If, for example,

$$F(T_1 x) \approx F(T_2 x),$$

then the representation cannot by itself determine which transformed image was originally presented.

Thus invariance, which is useful for recognition, becomes ambiguity when one attempts reconstruction.

P.3 Compression and Ill-Posedness

The asymmetry can also be understood geometrically.

An image may be represented by a very high-dimensional vector

$$x \in \mathbb{R}^N,$$

while the informative dimensionality of the measured fMRI response is typically much smaller:

$$y \in \mathbb{R}^M.$$

The forward map

$$F : \mathbb{R}^N \longrightarrow \mathbb{R}^M$$

therefore acts, in an information-theoretic sense, as a compression.

Compression is relatively easy to learn. Inverting compression is impossible unless additional information is supplied.

The same point is visible in the linear case. Suppose

$$y = Ax + \eta.$$

If A has a nontrivial null space, then for any v satisfying

$$Av = 0$$

we have

$$A(x + v) = Ax.$$

Consequently the measurement cannot distinguish x from $x + v$.

Even if A is formally invertible, small singular values make the inverse unstable. Writing the singular value decomposition as

$$A = U\Sigma V^\top,$$

the formal inverse contains factors

$$\frac{1}{\sigma_i}.$$

Small singular values therefore strongly amplify measurement noise.

The inverse problem is consequently both ambiguous and potentially ill-conditioned.

P.4 The Bayesian Formulation

The appropriate object for reconstruction is therefore not an ordinary inverse function

$$F^{-1}(y),$$

but the posterior distribution

$$p(x \mid y).$$

Bayes' rule gives

$$\boxed{p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)}.}$$

This equation exposes the asymmetry between the forward and inverse problems.

The paired image–fMRI examples allow us to learn the forward relationship

$$p(y \mid x).$$

But reconstruction requires, in addition, a model of

$$p(x),$$

the prior distribution of natural images.

The fMRI data tell us which images are compatible with the measured brain activity. The prior tells us which of those compatible images are plausible images.

Thus a natural strategy is not necessarily to learn directly

$$y \longmapsto x.$$

Instead one can learn

$$x \longmapsto$$

CHAPTER Q

Notation and Mathematical Conventions

This appendix provides a reference for the specific mathematical notation and conventions used throughout the text, particularly regarding compositional functions, statistical complexity measures, and group-theoretic invariance.

Computability and Compositional Sparsity

The following notation is used primarily in Chapters 4, 8, and 21 to bridge Turing computability with deep learning architectures.

$CS_k(s, L)$ **Compositionally Sparse Function Class.**
The set of functions f that can be computed by a directed acyclic graph (DAG) such that:

- The graph has total size (number of nodes) at most s.
- The depth (longest path from input to output) is at most L.
- Every internal node has a fan-in (arity) of at most k.

This class formally captures the structure of *efficiently computable* functions.

$\mathcal{G}_n$ **Discrete Grid Domain.**
Defined as $\mathcal{G}_n := \{0, 1, \dots, 2^n\}^d / 2^n$. It represents the domain of inputs discretized to n bits of precision, used to prove exact equivalence between Turing Machines and ReLU networks (Chapter 4).

DAG **Directed Acyclic Graph.**
The graph theoretical representation of a computation where nodes represent constituent functions and edges represent data flow.

d^* **Maximal Local Arity.**
Used in Chapter 21. It denotes the maximum fan-in required for the nodes in a compiled DAG representation of a finite-horizon physical system. Approximation rates scale with d^* rather than the ambient dimension d.

Complexity Measures

These symbols are used in Chapters 12 and 13 to quantify generalization and capacity control.

$\mathfrak{R}_S(\mathcal{F})$ **Empirical Rademacher Complexity.**
Measures the ability of a function class $\mathcal{F}$ to fit random noise on a fixed dataset $S = \{x_1, \dots, x_m\}$. Defined as:

$$\mathfrak{R}_S(\mathcal{F}) = \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i f(x_i) \right]$$

where $\sigma_i \in \{-1, +1\}$ are independent Rademacher variables.

$\mathcal{H}(\epsilon, \mathcal{F}, \|\cdot\|)$ **Metric Entropy.**
The logarithm of the covering number $\mathcal{N}(\epsilon, \mathcal{F}, \|\cdot\|)$. It represents the number of bits needed to describe an element of $\mathcal{F}$ to within precision ϵ.

CV_{loo} **Leave-One-Out Cross-Validation Stability.**
Defined in Appendix A. A measure of algorithmic stability where the expected error difference is bounded when one training point is removed. This book establishes it as necessary and sufficient for the consistency of ERM.

Group Theory and Invariance

The following notation appears in Chapters 6 and 22 regarding Genericity and Invariant Representations.

G **Compact Group.**
A group acting unitarily on the data space (e.g., rotations, translations). In the context of representation learning, we assume $G \subset O(d)$.

dg **Haar Measure.**
The unique translation-invariant probability measure on a compact group G. Used to define group averages (orbits).

$\mu_t(I)$ **Orbit Distribution (or CDF).**
The probability distribution of the inner products $\langle I, gt \rangle$ as g varies over G. In Chapter 22, this is often represented by its Cumulative Distribution Function (CDF), providing a signature that is invariant to the transformation of the input I by elements of G.

N_η **Group η-Net.**
A finite subset of the group G such that every element $g \in G$ is within distance η of some element in N_η. The size of this net, $|N_G(\eta)|$, enters the complexity bounds for invariant embeddings.

CHAPTER R

Appendix: Glossary of Core Concepts

This glossary collects key terms and concepts introduced throughout the book. Entries are intentionally concise and technical; references point to chapters where the concept is developed in detail.

A

Associative Memory
A memory system in which retrieval is performed by content rather than by address. Given a cue or query, the system returns stored items that are similar to the cue under a learned or fixed similarity measure. Classical examples include Hopfield networks and radial basis function (RBF) networks; modern examples include attention mechanisms in transformers. Associative memory is treated in this book as a *computational primitive*. *(See Chaps. 1, 3)*

Associative Turing Machine (ATM)
A formal model of computation introduced in this book, consisting of alternating *content-addressable reads* from an associative memory and *local update* operations on a machine state. ATMs are shown to be Turing-complete with polynomial overhead and provide a formal abstraction of transformer-style computation. *(See Chap. 3)*

B

Behavioral Repertoire
The set of input–output behaviors or policies a system can realize under prompting or control. Intelligence is characterized here not by peak performance but by the *width* of this repertoire and the system's ability to deliberately restrict or reshape it. *(See Chap. 30)*

C

Compositionality

The property that a function or behavior can be expressed as a composition of simpler functions, typically organized hierarchically. In learning systems, compositionality permits reuse of subfunctions and enables generalization across domains. *(See Chaps. 1, 4)*

Containment Principle
The hypothesis that if system A is more intelligent than system B, then the set of behaviors realizable by B is contained within the set realizable by A. Operationally, this implies that A can simulate B, but not vice versa. Used to define a directed hierarchy of intelligence. *(See Chap. 30)*

CV_{loo} (Leave-One-Out Cross Validation)
A stability metric where the algorithm is trained on all but one data point, and error is measured on the left-out point. In this text, specific forms of stability are shown to be sufficient for generalization, linking CV_{loo} directly to predictivity. *(See Chap. 7)*

D

Diligent Learner
A learner that improves performance by systematically accumulating competence through training, without requiring explicit search or exploration mechanisms. The concept originates in work by Shalev-Shwartz and Shashua and is used here to contrast diligence with creativity and exploration. *(See Chap. 30)*

E

Efficient Computability
The property that a function can be computed with resources polynomial in the input size under a standard computational model (e.g., Turing machines or Boolean circuits). In this book, efficient computability is shown to imply sparse compositional structure in the corresponding function representation. *(See Chap. 4)*

Effective Information Exponent
A quantity measuring how the effective information content of a learned representation scales with system size or depth, used to characterize compression and generalization in compositional architectures. *(See Chaps. 7, 25)*

Empirical Risk Minimization (ERM)
The standard learning principle of minimizing the average error over the training dataset. While successful in practice, this book argues that ERM alone is insufficient to explain generalization without invoking stability or genericity conditions. *(See Chap. 7)*

G

Genericity

A property of target functions or learning problems asserting that informative, low-order components (e.g., linear or low-degree terms) are present and stable. Genericity ensures that gradient-based optimization receives reliable signals and avoids pathological loss landscapes. *(See Chaps. 7, 11)*

H

HyperBF / Hyper-RBF
A generalization of classical radial basis functions in which the similarity metric, prototypes, or both are learned. In particular, replacing Euclidean distance with a learned Mahalanobis metric yields kernels that are closely related to attention mechanisms. *(See Chaps. 1, 3)*

L

Large Embedding Model (LEM)
A proposed class of models extending large language models by incorporating richer latent world representations prior to or alongside language. LEMs are intended to address limitations of purely linguistic training by grounding prediction in learned world models. *(See Chap. 29)*

Loss Landscape
The function mapping model parameters to training or generalization error. The geometry of the loss landscape—its critical points, basins, and flat directions— plays a central role in optimization and generalization. *(See Chap. 7)*

M

Manifold Hypothesis
The assumption that high-dimensional data (e.g., images, text, sensory inputs) lie near low-dimensional manifolds embedded in ambient space. In this book, the hypothesis is extended to *behavioral* and *semantic* manifolds learned by large models. *(See Chaps. 1, 30)*

R

Radial Basis Function (RBF) Network
A function approximator of the form

$$f(x) = \sum_i w_i \, \phi(\|x - \mu_i\|),$$

where ϕ is a radial kernel and μ_i are fixed centers. RBF networks implement smooth associative memory and serve as historical precursors to modern attention-based models. *(See Chap. 1)*

Random-Access Memory (RAM)

A memory model permitting constant-time access to any stored location by address. Attention mechanisms approximate RAM-like behavior by enabling content-based access to all positions in a sequence. *(See Chaps. 1, 3)*

S

Simulation Distance
A metric defined between two models A and B as the asymmetry in their ability to simulate each other's behavior. Used to define a directed ordering of intelligence independent of static benchmarks. *(See Chap. 30)*

Sparse Compositionality
The principle that meaningful functions are compositions of a small number of low-arity constituent functions arranged in a sparse directed acyclic graph. Sparse compositionality explains why deep architectures are both necessary and effective for learning real-world functions. *(See Chaps. 1, 4)*

Stability
A property of a learning algorithm wherein small changes to the training set result in small changes to the output hypothesis. This book emphasizes stability as a necessary and sufficient condition for generalization, often replacing classical uniform convergence bounds. *(See Chap. 7)*

Stepwise Learning
A learning framework where complexity is increased incrementally. In the context of diffusion and autoregression, this refers to learning the conditional probability distribution of the next step (or next bit) given previous steps, effectively breaking a complex joint distribution into a sequence of simpler learning problems. *(See Chap. 10)*

T

Transformer
A neural architecture based on stacked self-attention and feedforward blocks. Transformers implement large-scale associative memory with learned similarity metrics and are central to modern language and vision models. *(See Chaps. 1, 3)*

W

World Simulator
An internal model that predicts the evolution of latent states corresponding to the external world. Distinguished here from purely linguistic predictors, world simulators are proposed as a missing component for robust intelligence. *(See Chaps. 24, 29)*

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